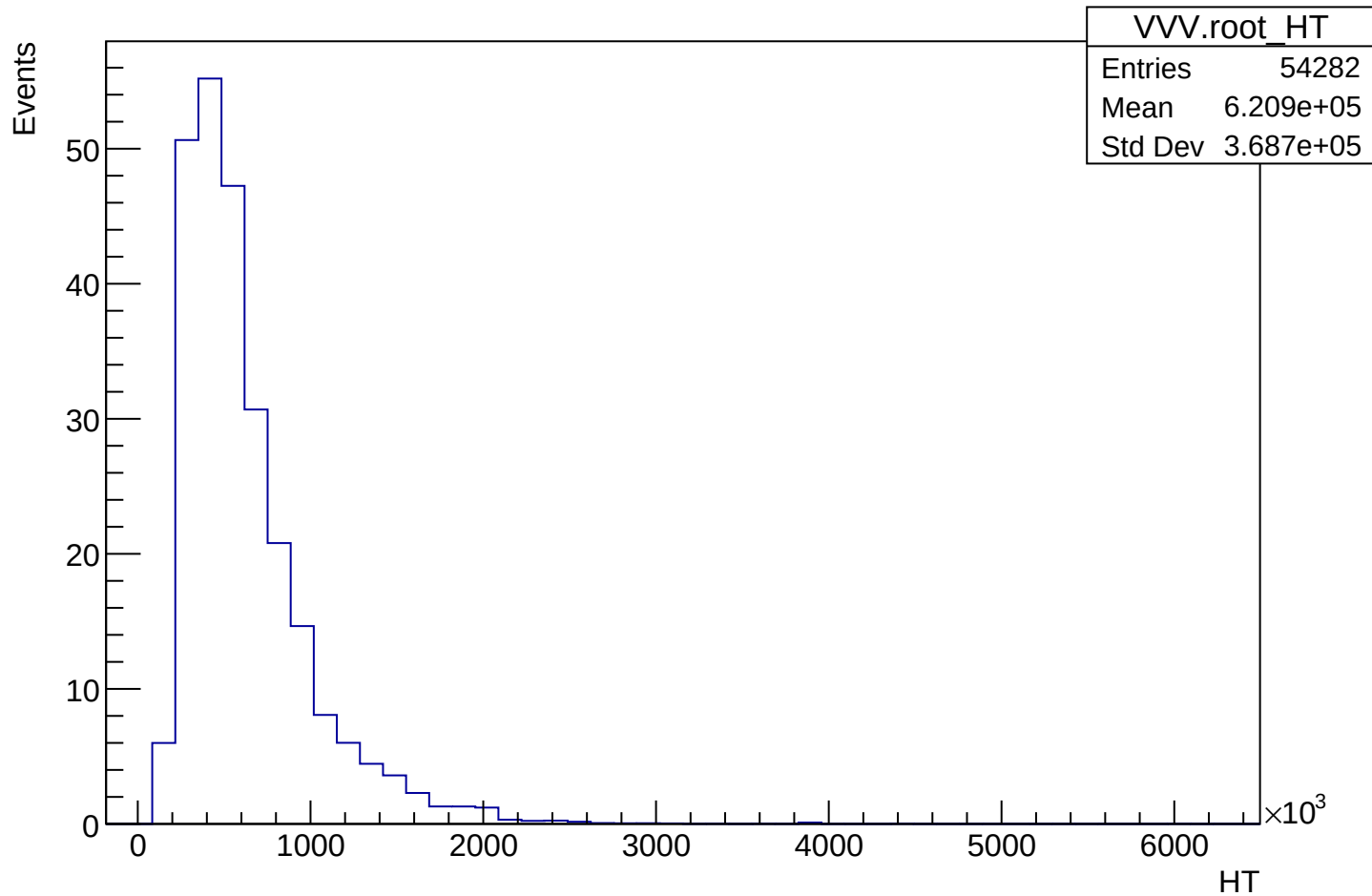
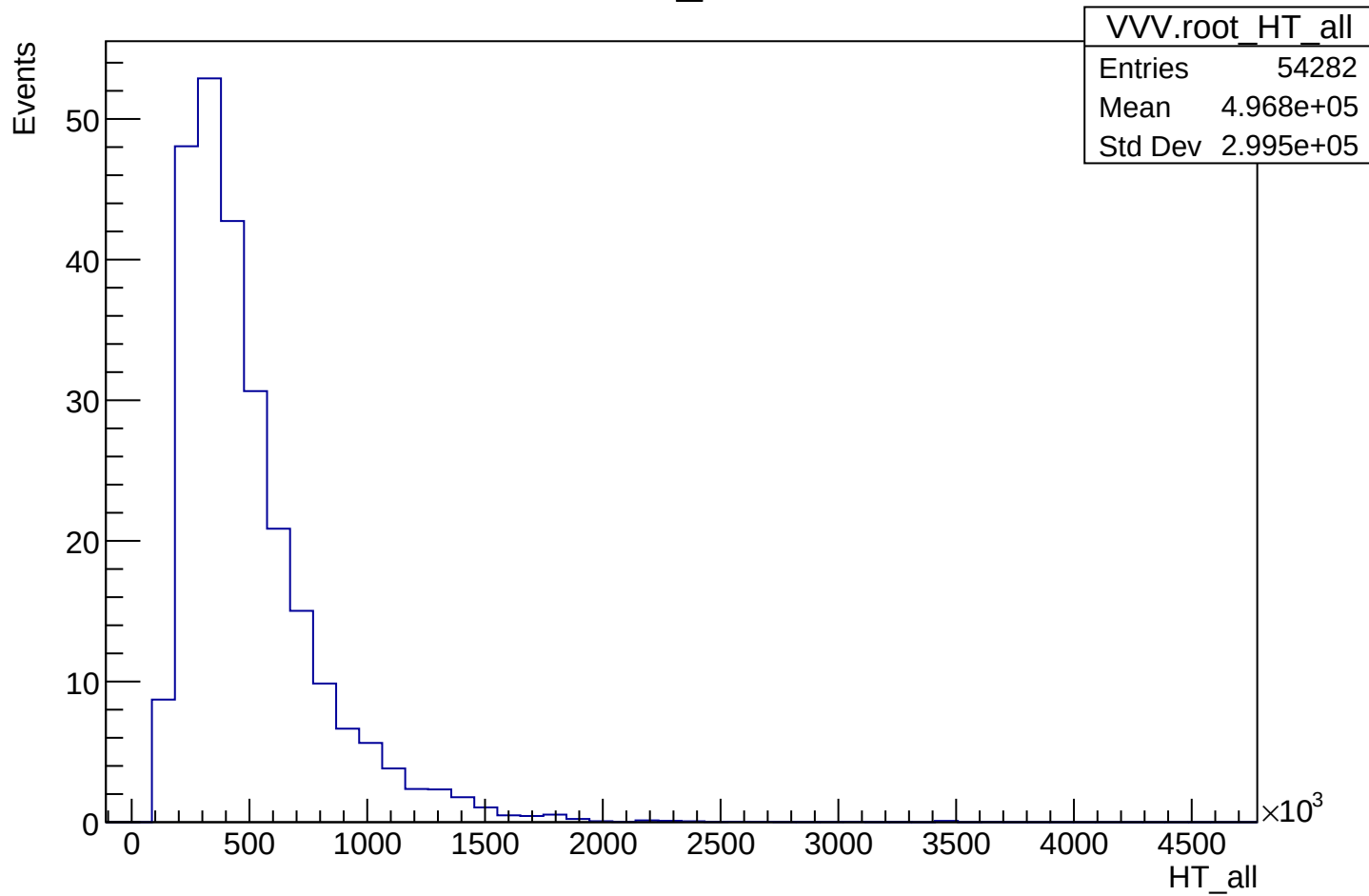


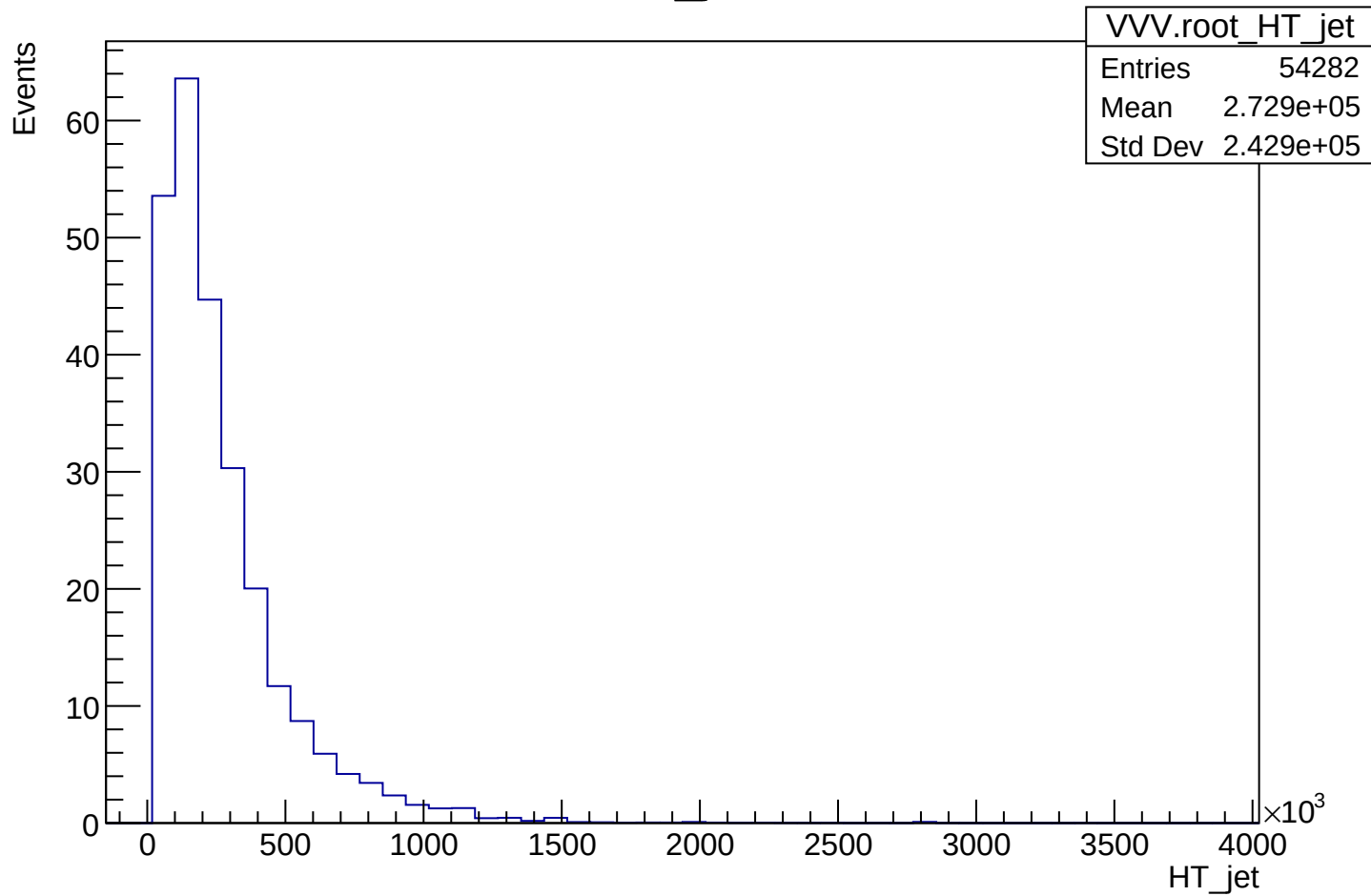
HT



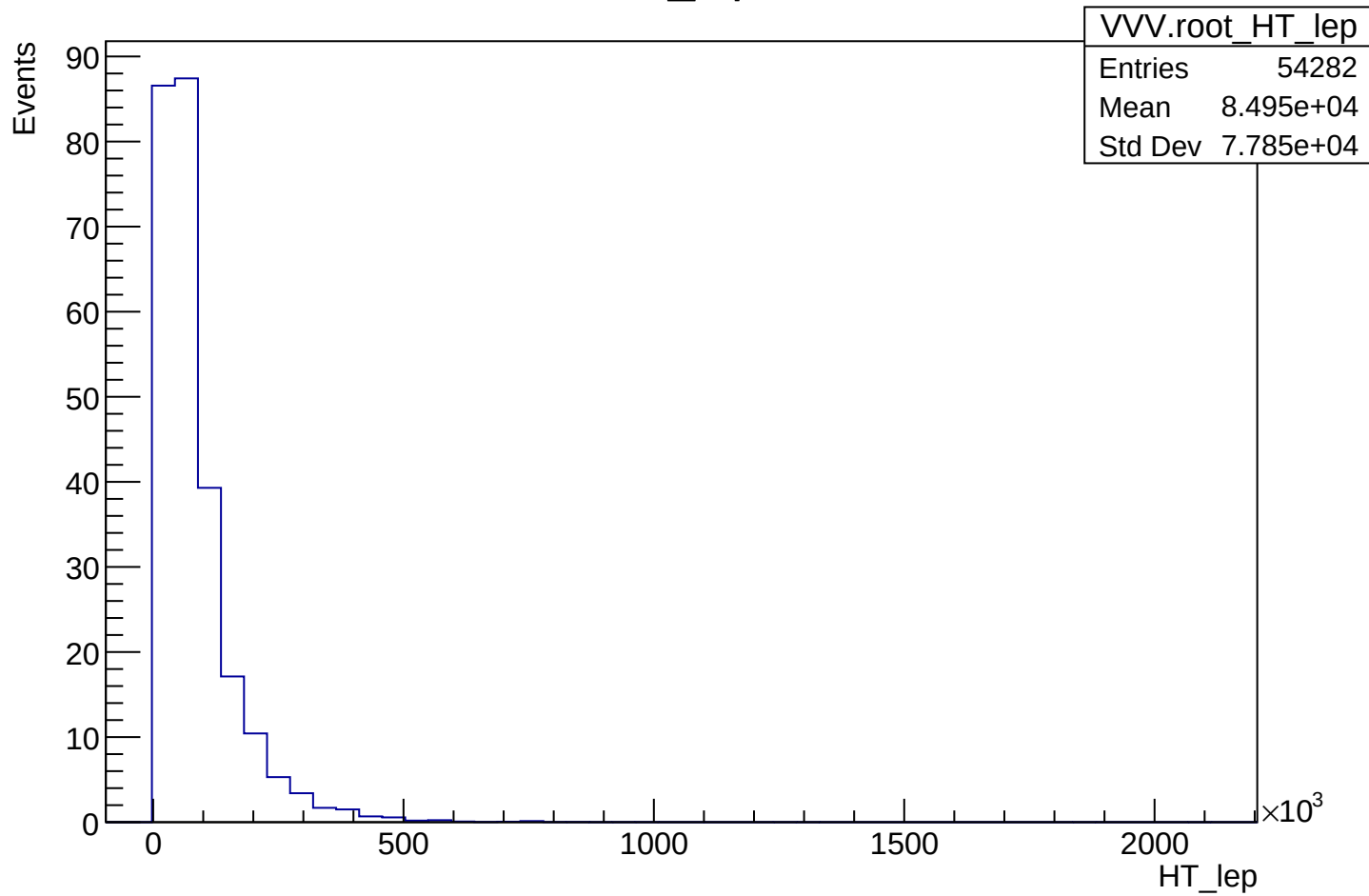
HT_all



HT_jet



HT_lep



HT_lepMET

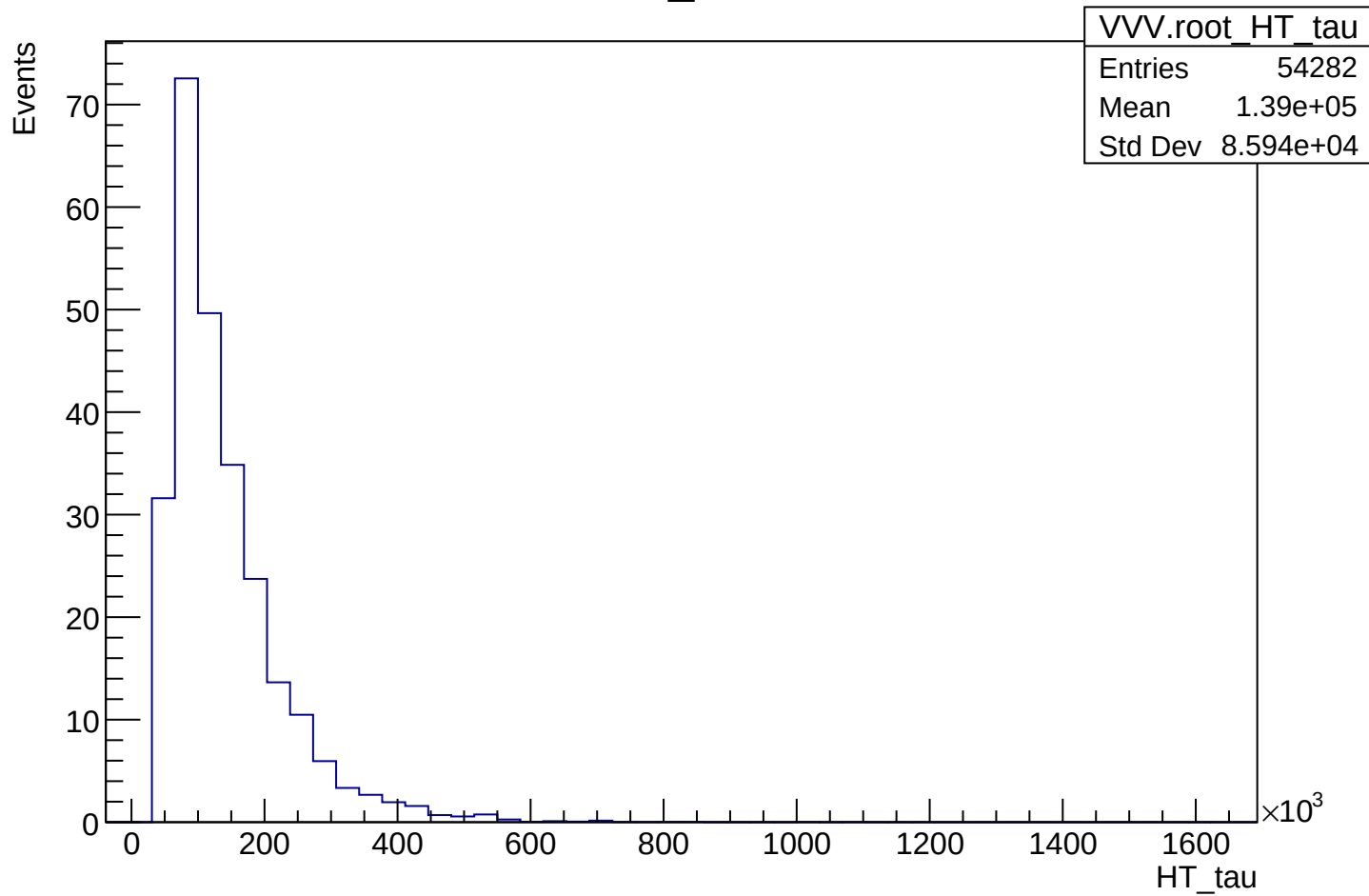
Events

100
80
60
40
20
0

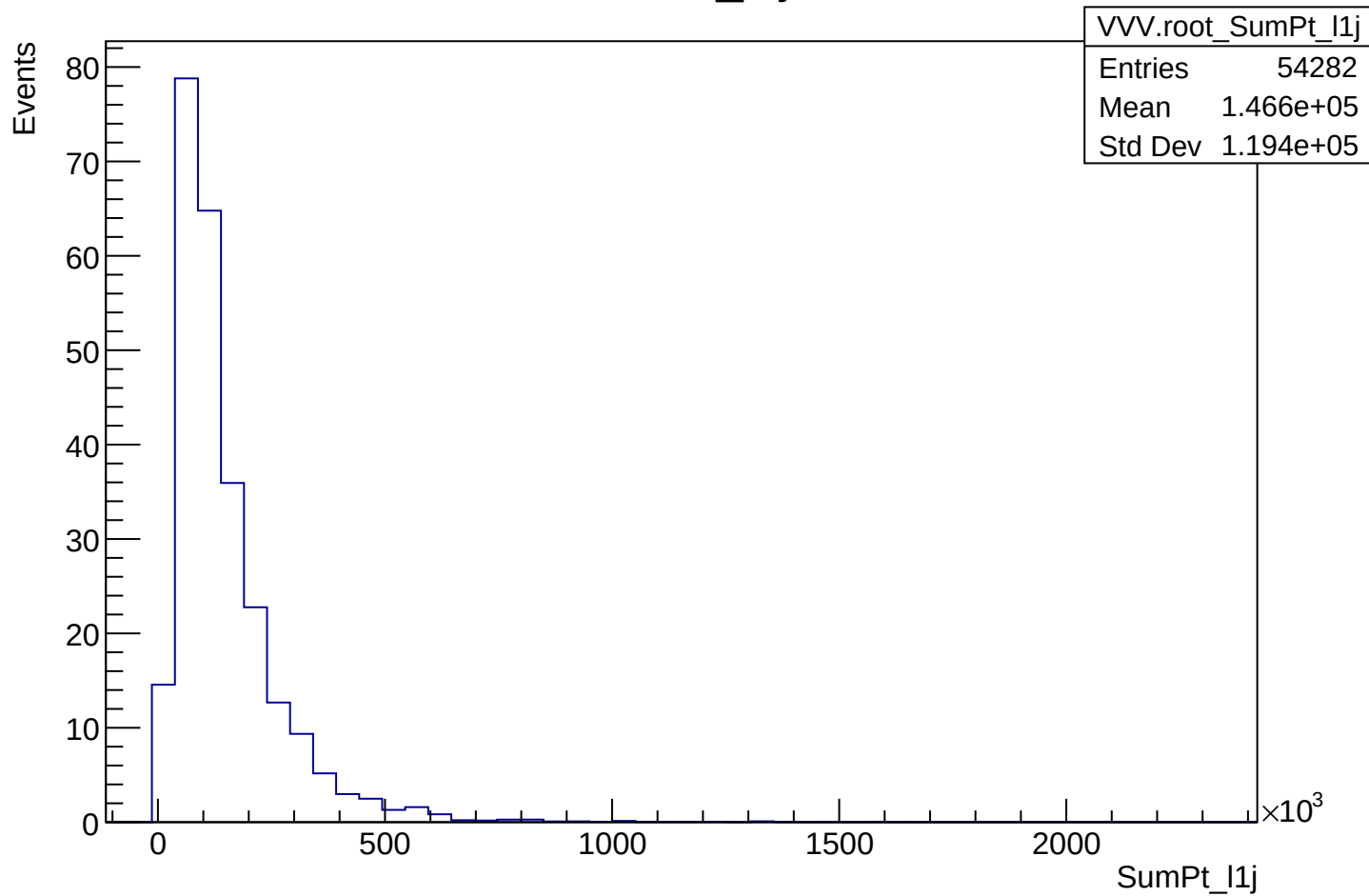
0 500 1000 1500 2000 2500 3000 3500 4000 $\times 10^3$
HT_lepMET

VVV.root_HT_lepMET	
Entries	54282
Mean	2.09e+05
Std Dev	1.417e+05

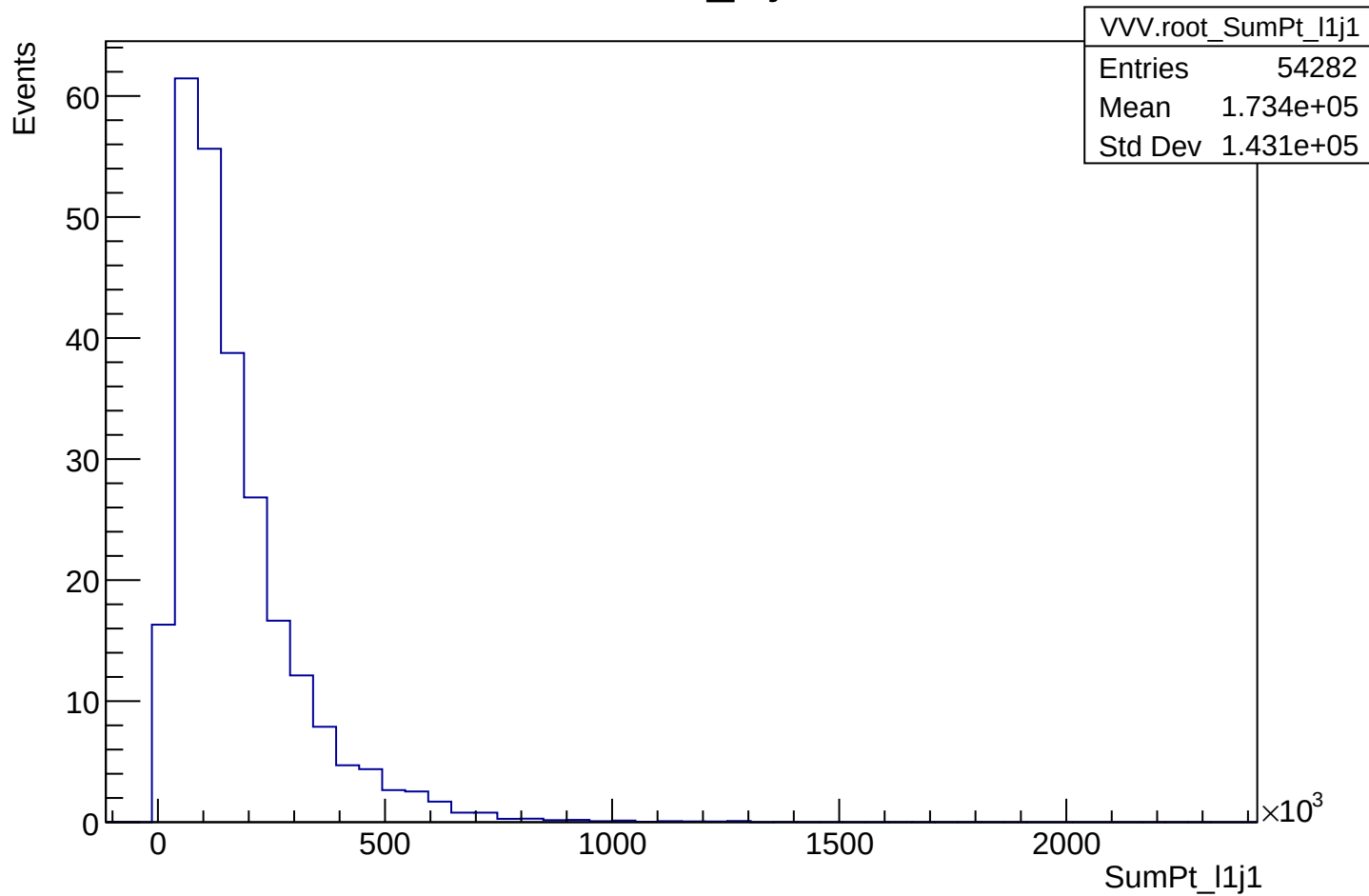
HT_tau



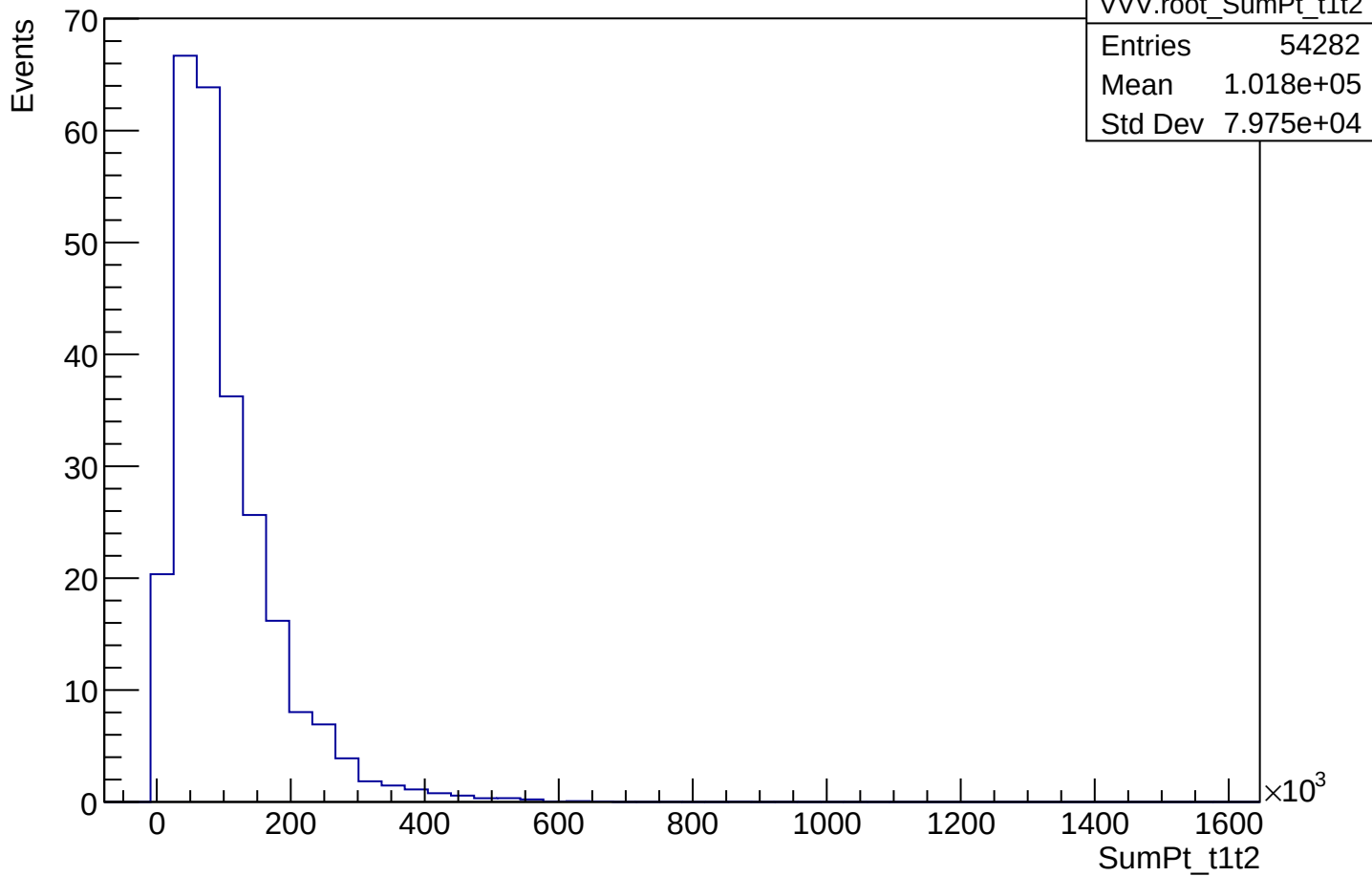
SumPt_l1j



SumPt_l1j1

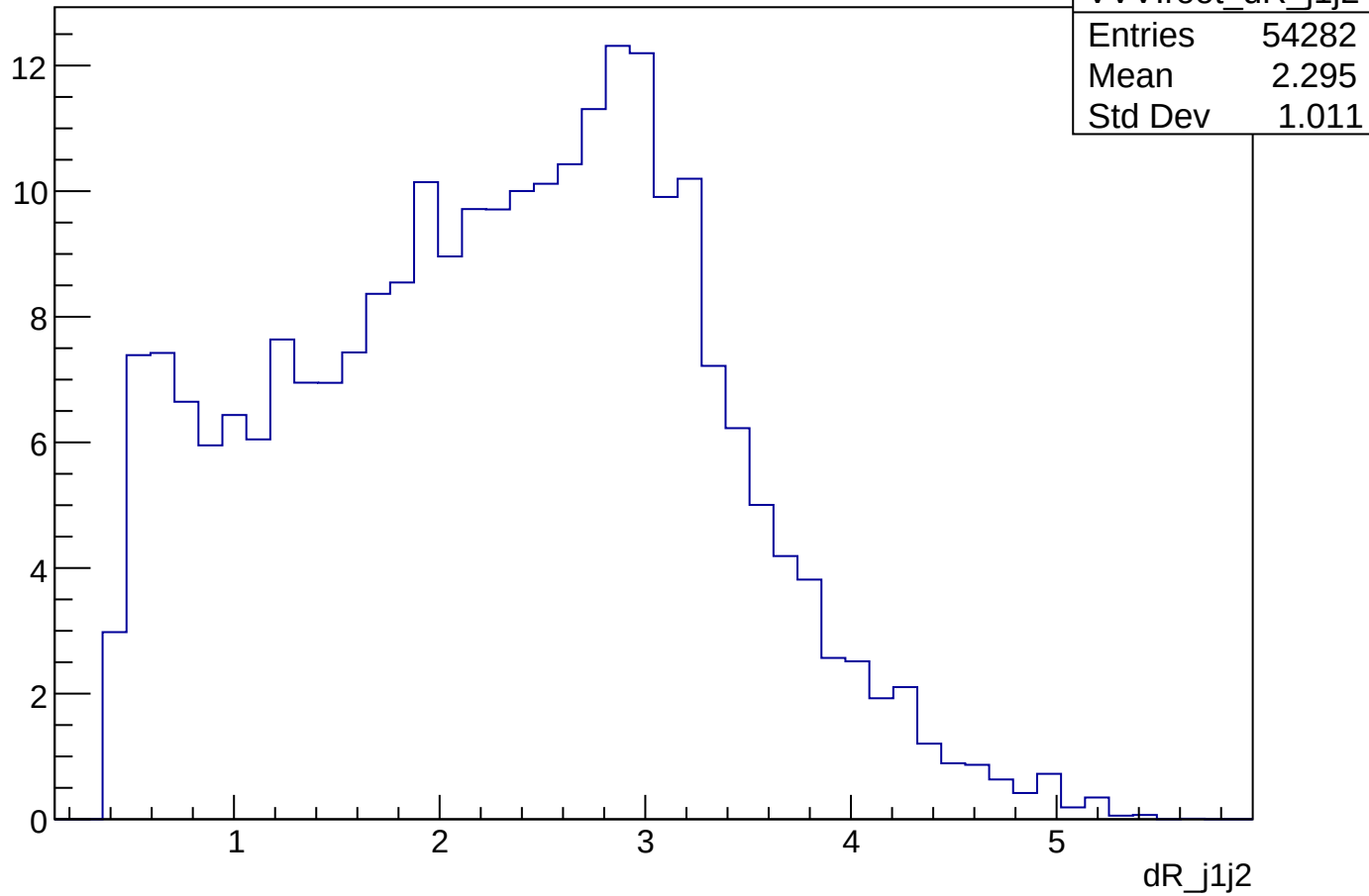


SumPt_t1t2



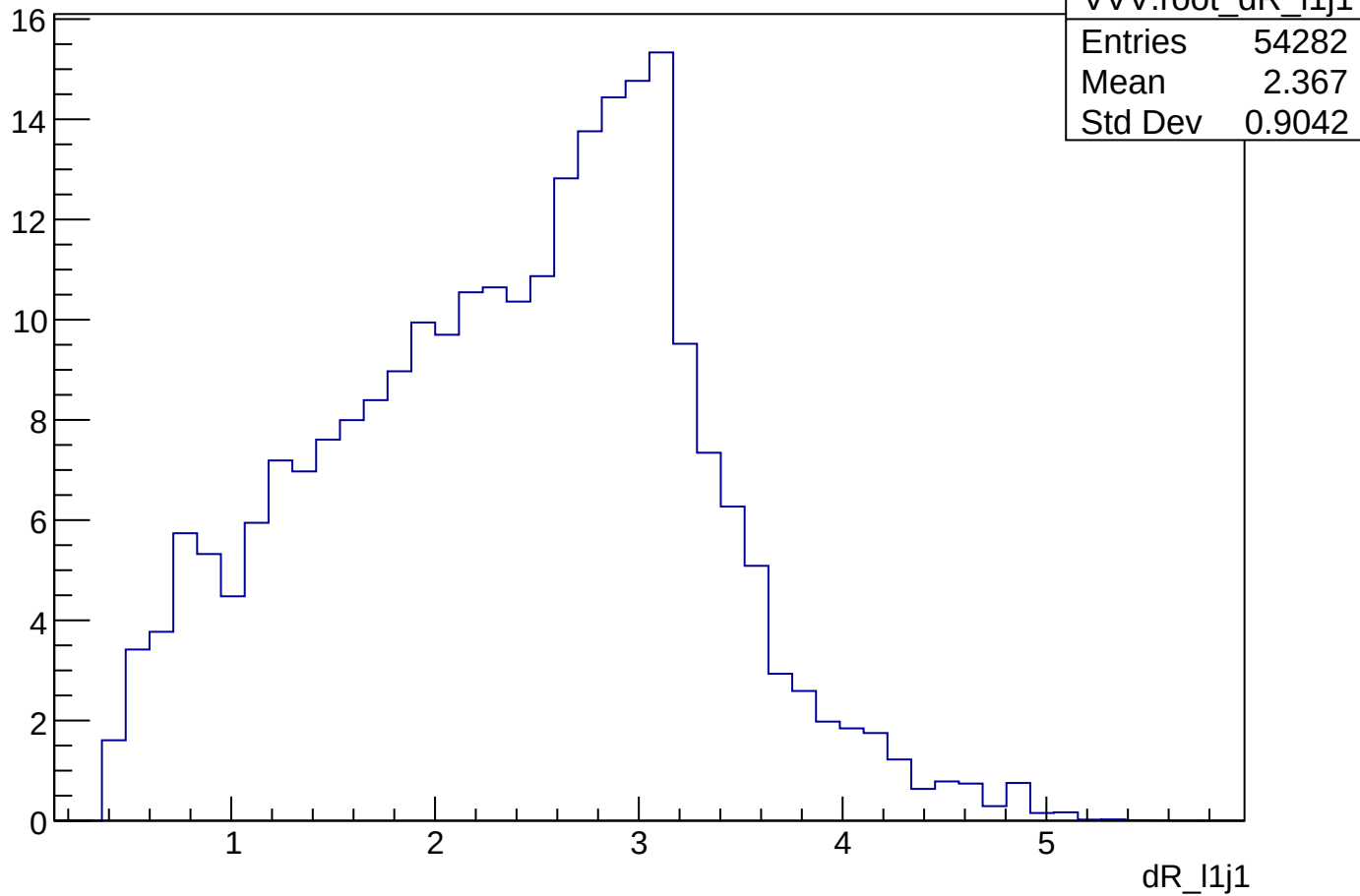
dR_j1j2

Events



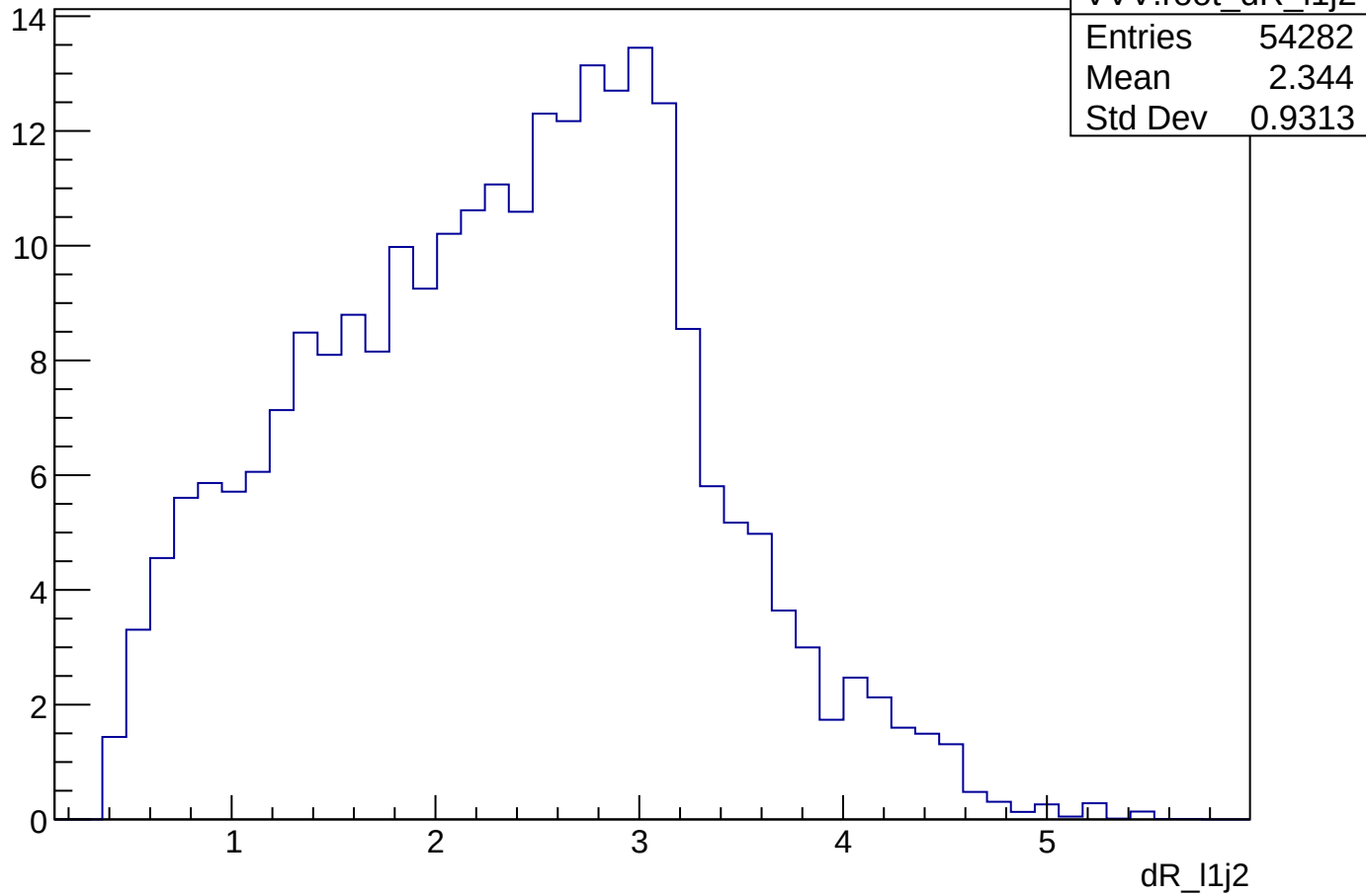
dR_l1j1

Events



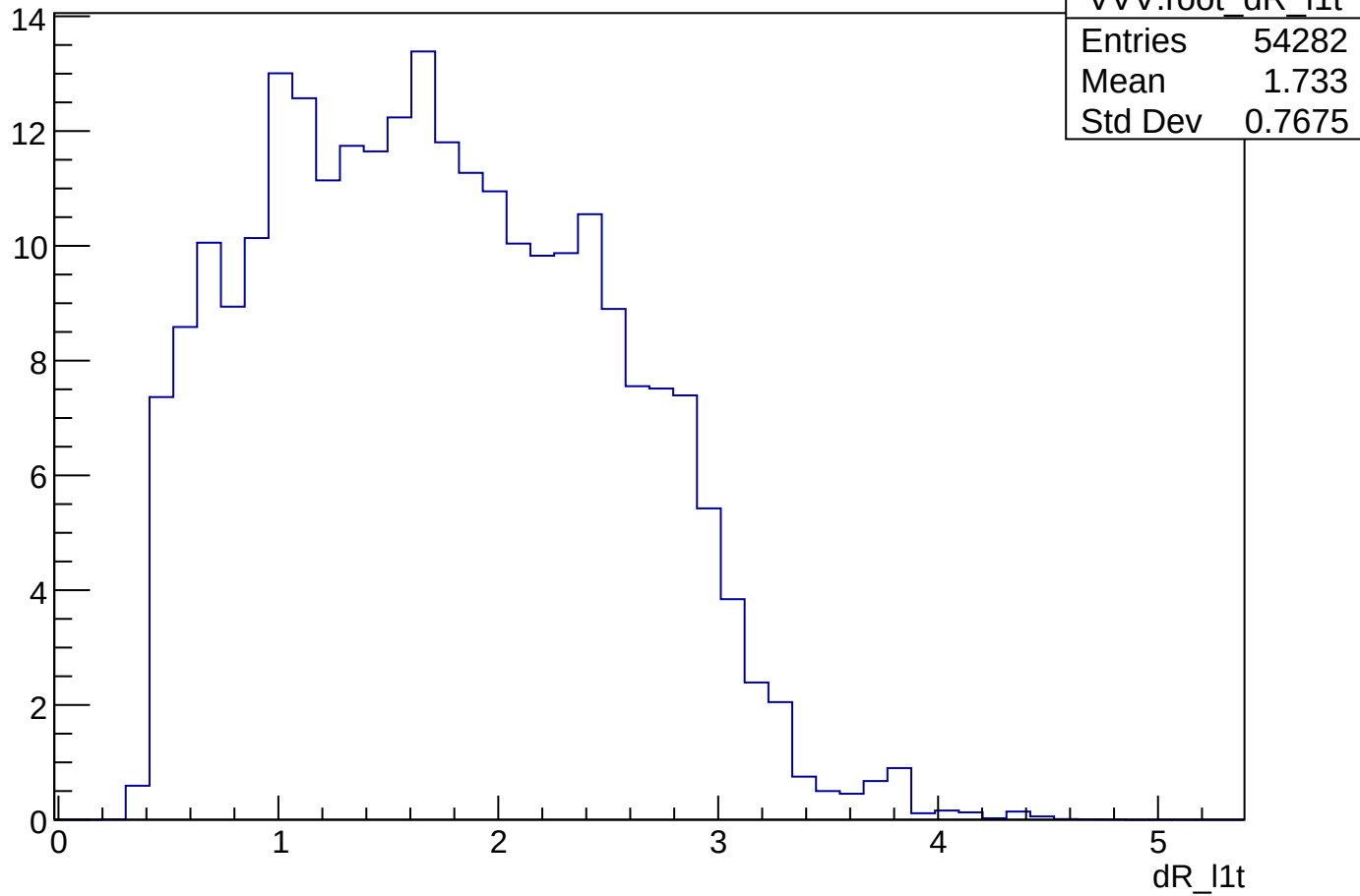
dR_l1j2

Events



dR_l1t

Events



dR_l1t1

Events

16
14
12
10
8
6
4
2
0

0

1

2

3

4

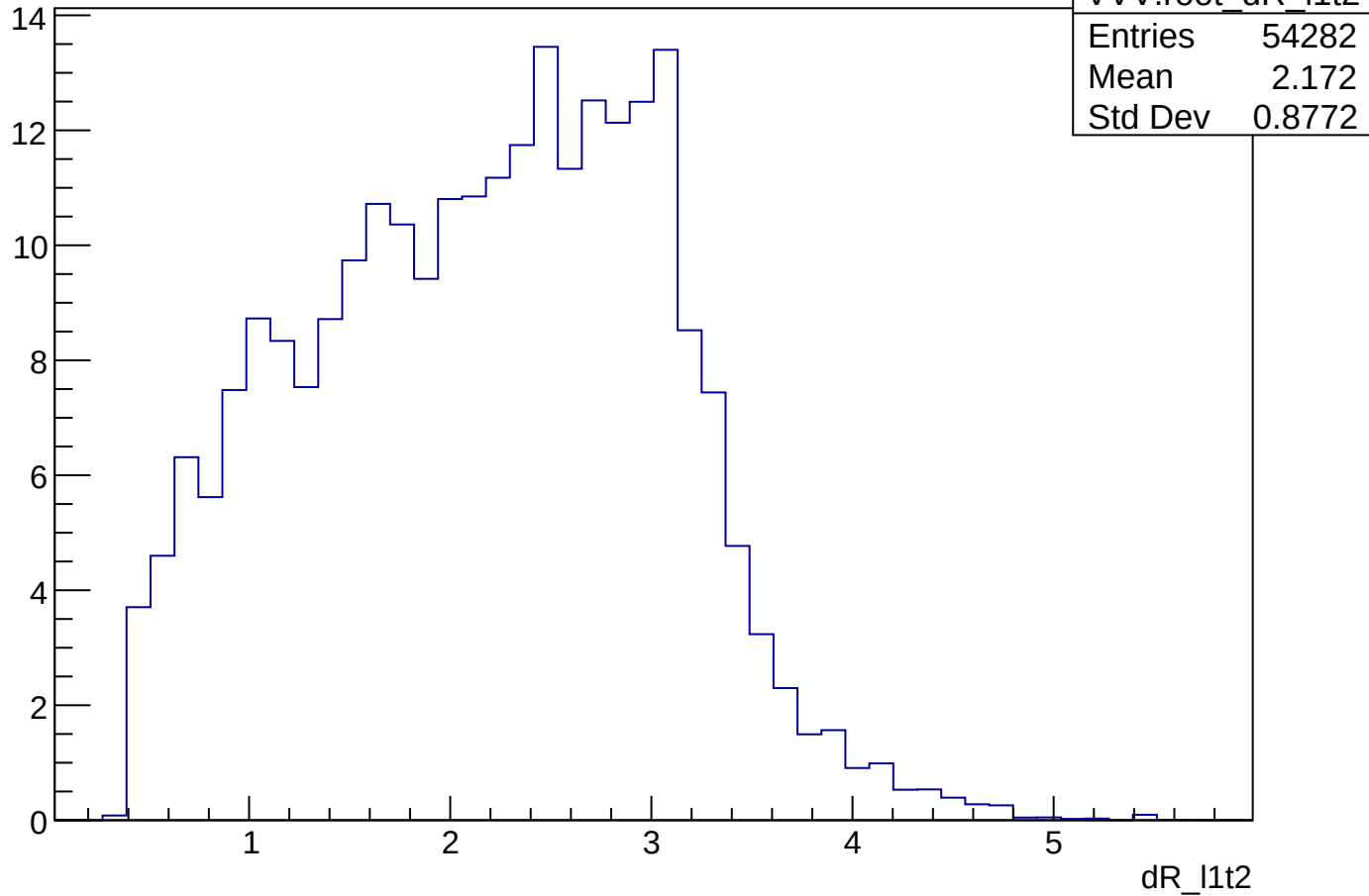
5

dR_l1t1

VVV.root_dR_l1t1	
Entries	54282
Mean	2.238
Std Dev	0.8806

dR_l1t2

Events



dR_t1j1

Events

22
20
18
16
14
12
10
8
6
4
2
0

0

1

2

3

4

5

6

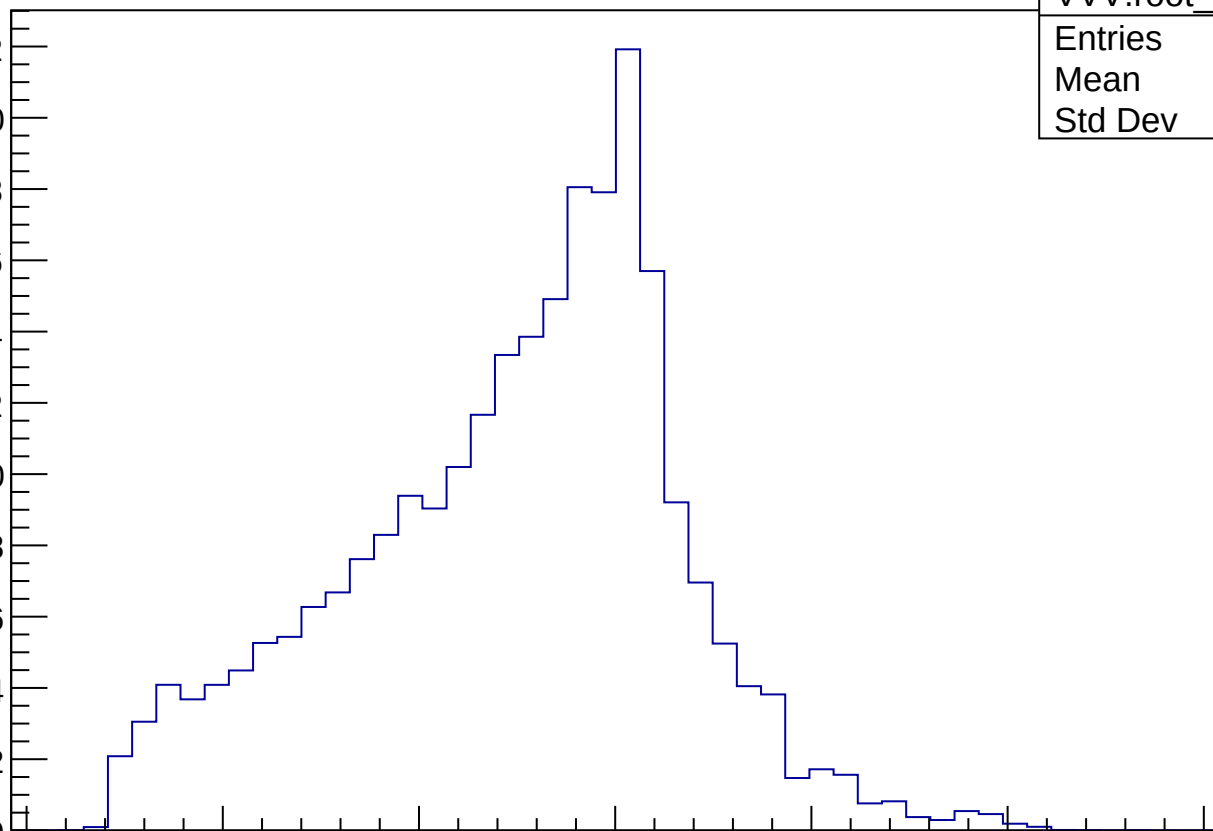
dR_t1j1

VVV.root_dR_t1j1

Entries 54282

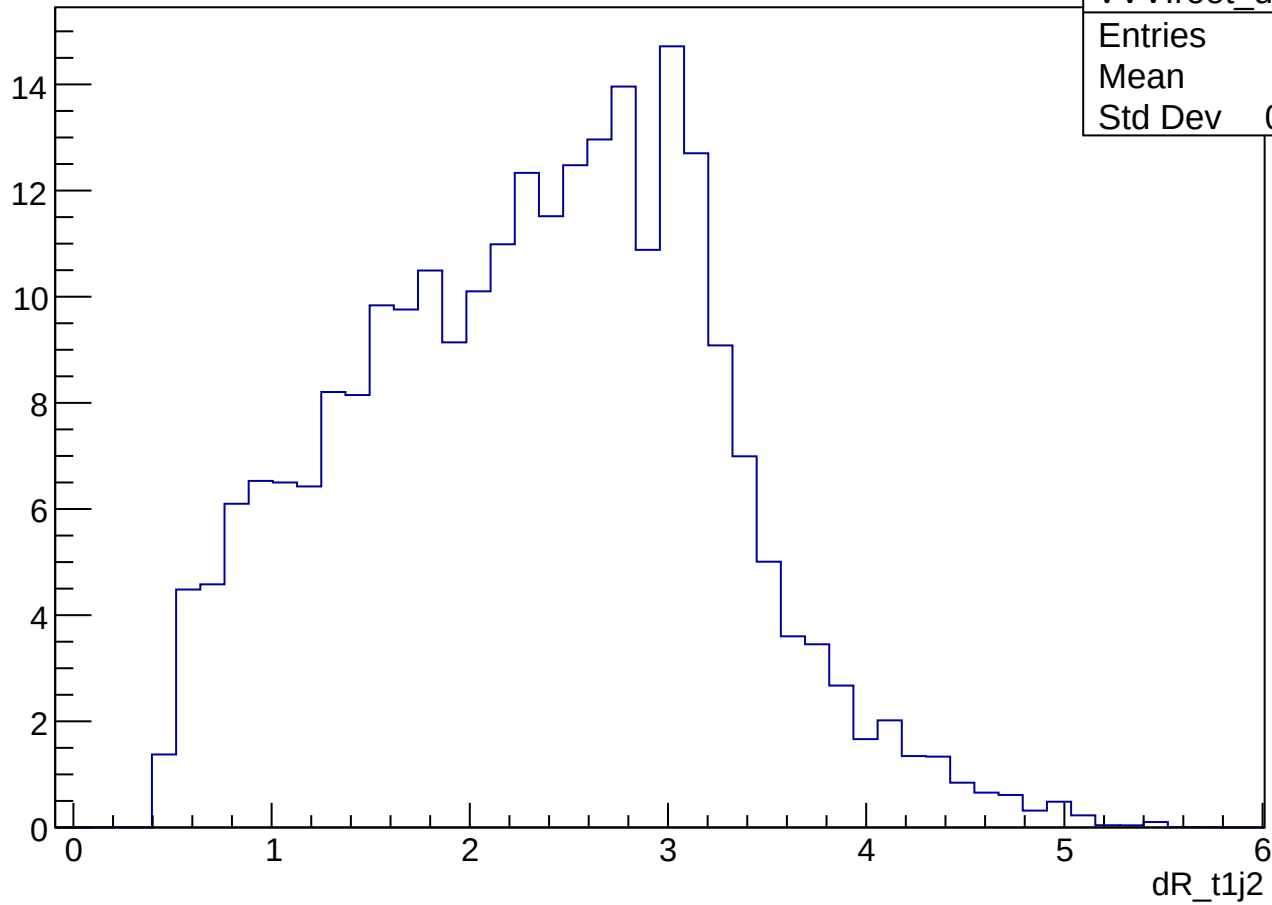
Mean 2.488

Std Dev 0.8456



dR_t1j2

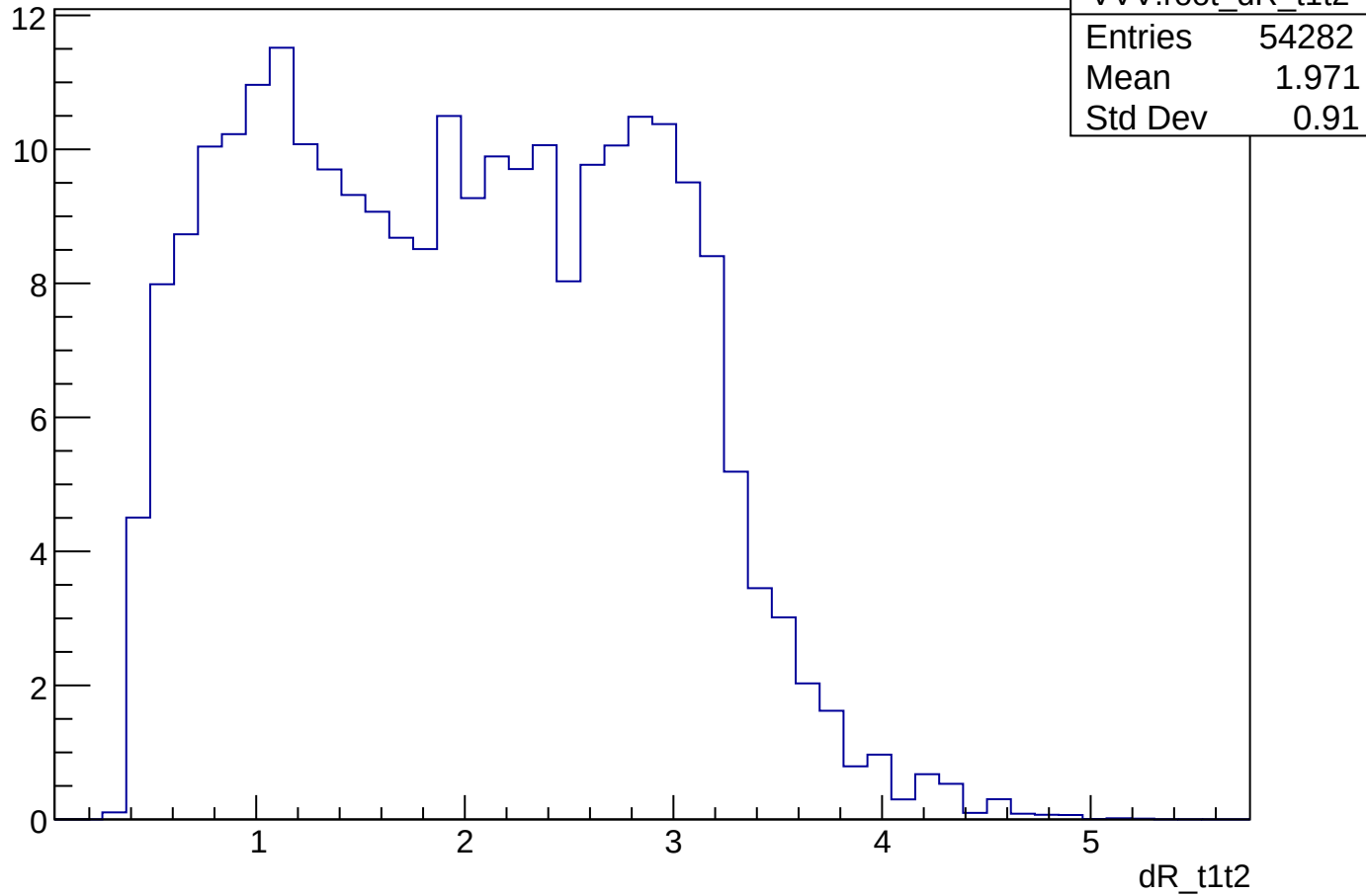
Events



VVV.root_dR_t1j2	
Entries	54282
Mean	2.33
Std Dev	0.9155

dR_t1t2

Events



dR_t1t2_l1

Events

25
20
15
10
5
0

0

2

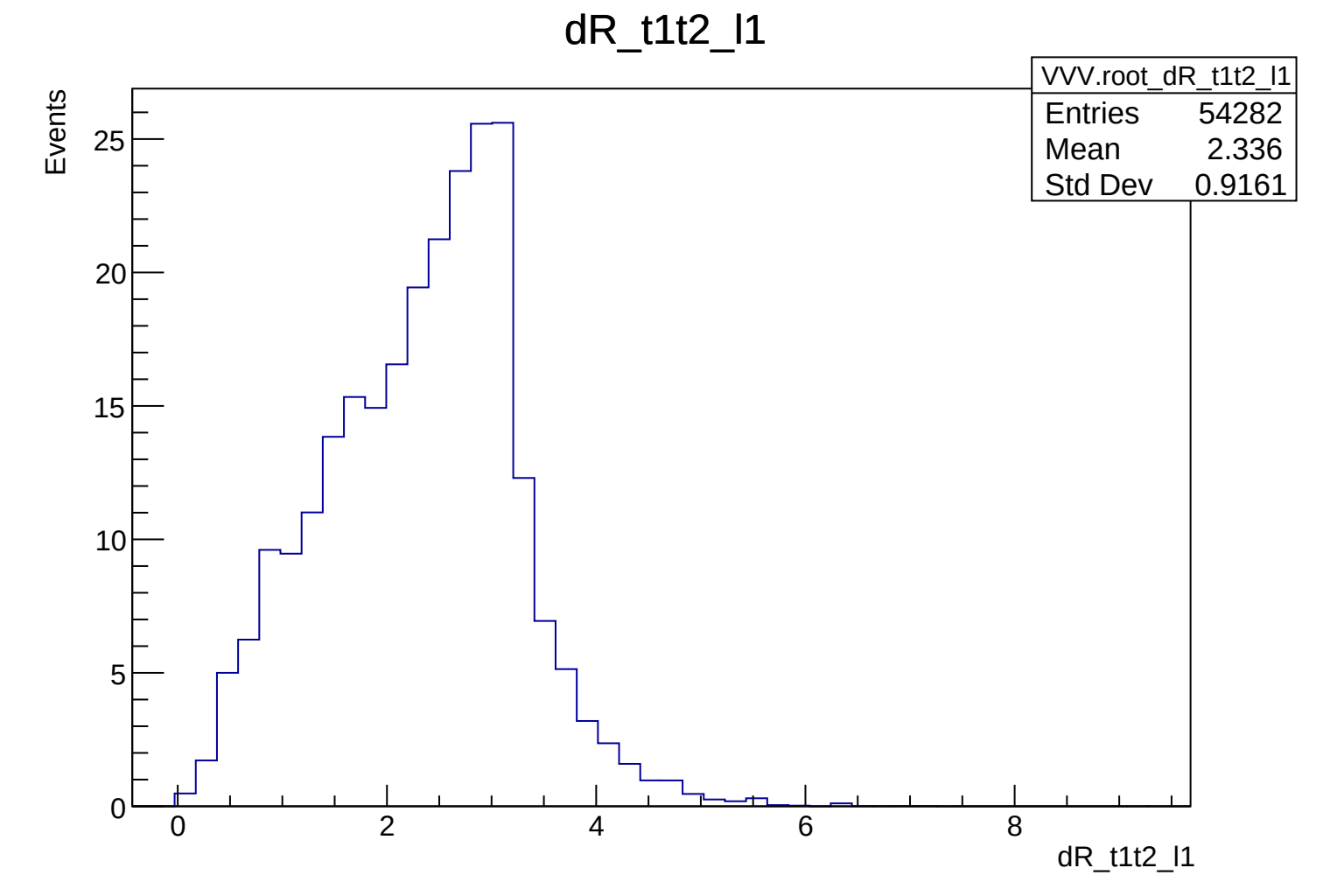
4

6

8

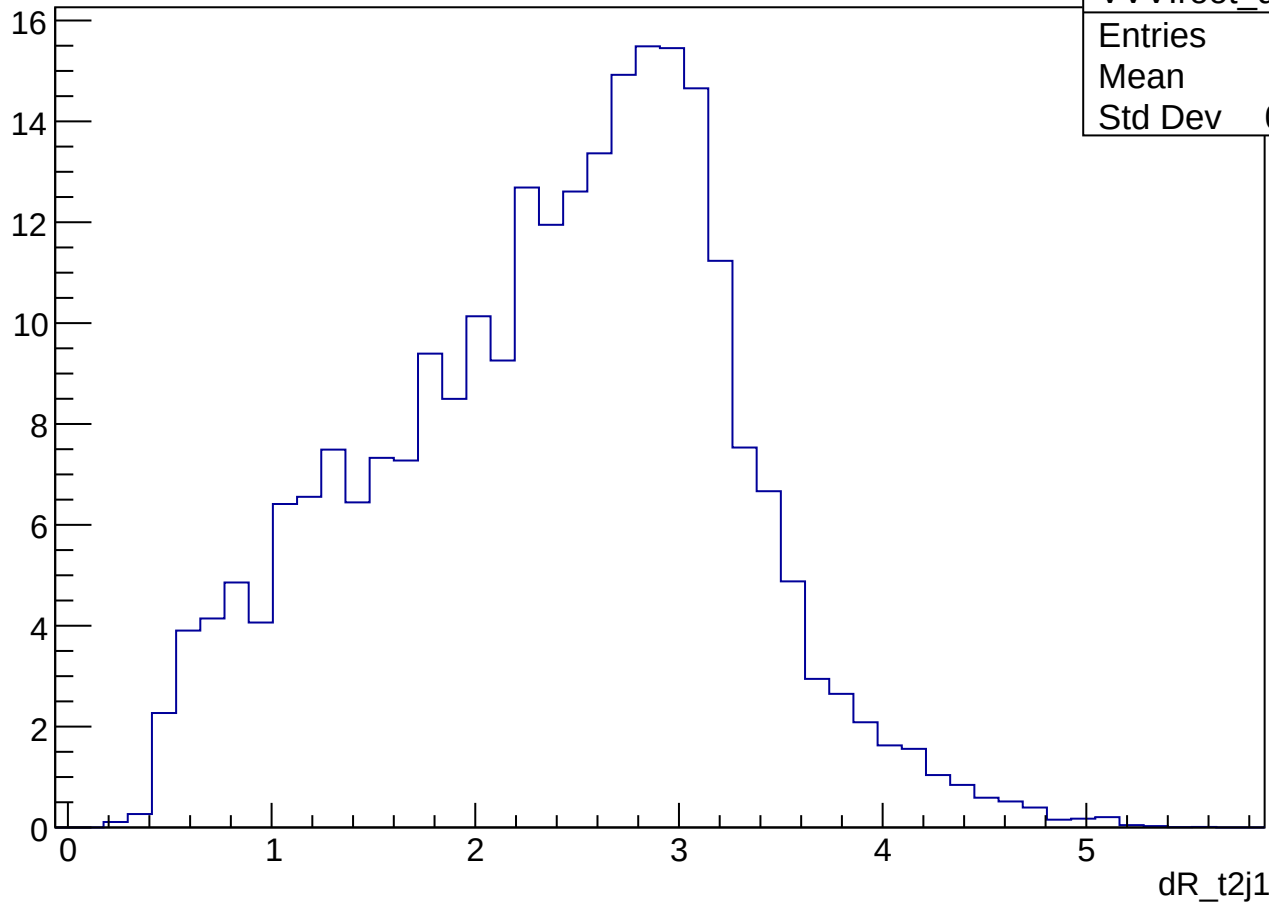
dR_t1t2_l1

VVV.root_dR_t1t2_l1	
Entries	54282
Mean	2.336
Std Dev	0.9161



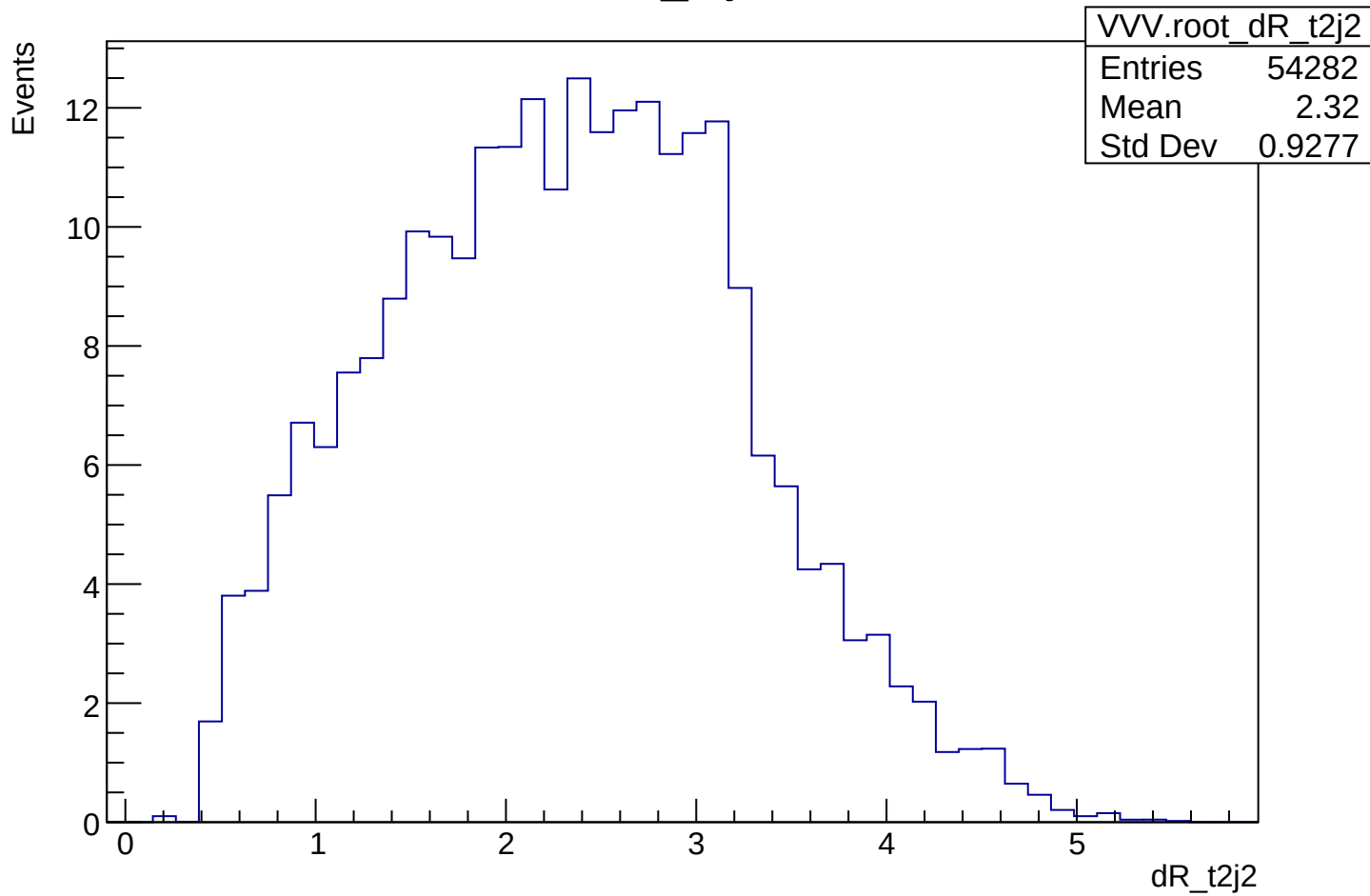
dR_t2j1

Events



VVV.root_dR_t2j1	
Entries	54282
Mean	2.375
Std Dev	0.8777

dR_t2j2



fjvt_effSF

Events

200
180
160
140
120
100
80
60
40
20
0

0.6

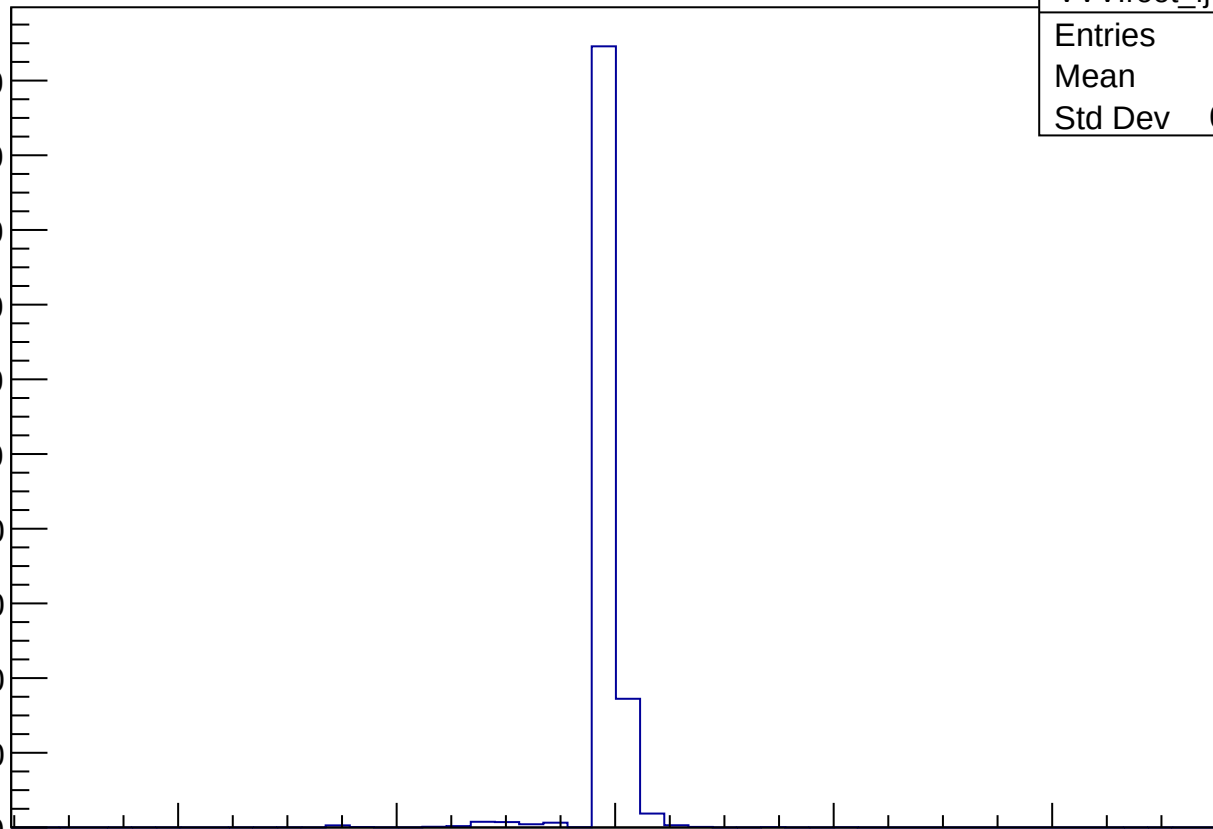
0.8

1.2

1.4

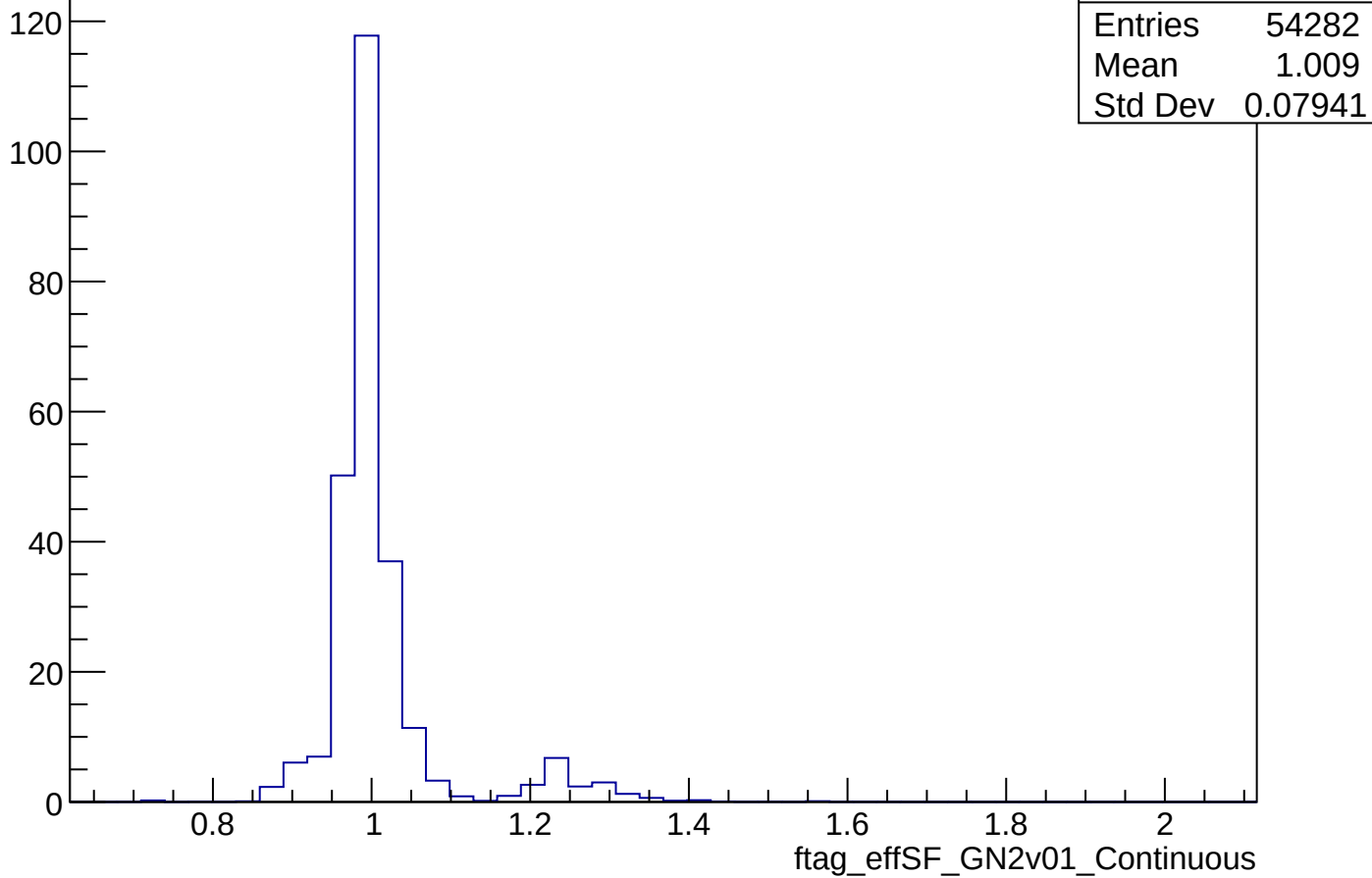
fjvt_effSF

VVV.root_fjvt_effSF	
Entries	54282
Mean	0.999
Std Dev	0.02254

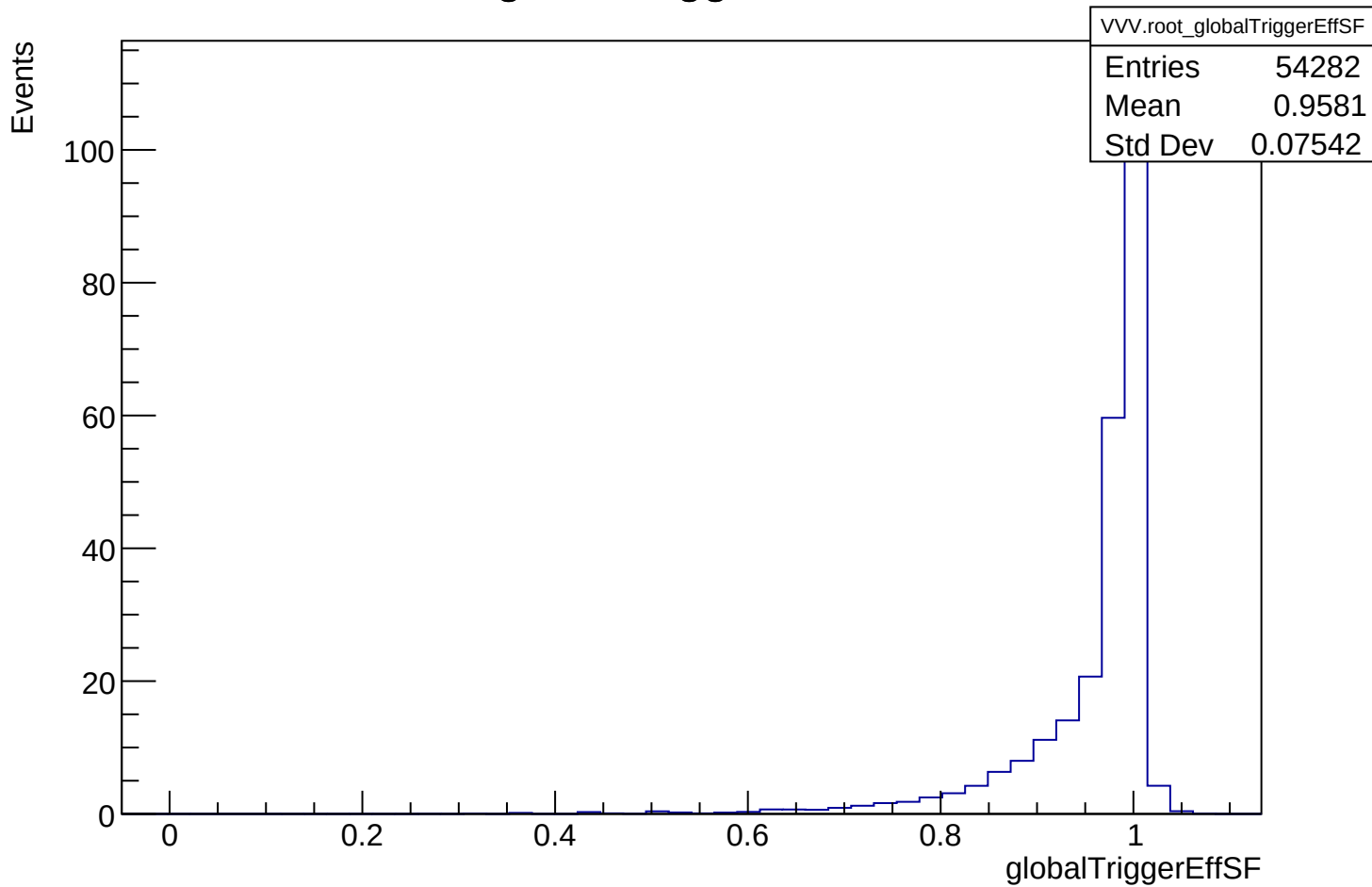


ftag_effSF_GN2v01_Continuous

Events



globalTriggerEffSF



hhml_subChannelFlavor

Events

120

100

80

60

40

20

0

1

1.2

1.4

1.6

1.8

2

hhml_subChannelFlavor

VVV.root_hhml_subChannelFlavor	
Entries	54282
Mean	1.48
Std Dev	0.4996

isDiTauFakes

Events

250
200
150
100
50
0

0

0.2

0.4

0.6

0.8

1

isDiTauFakes

VVV.root_isDiTauFakes	
Entries	54282
Mean	0.002269
Std Dev	0.04758

isSingleFakeTau

Events

240
220
200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

isSingleFakeTau

VVV.root_isSingleFakeTau	
Entries	54282
Mean	0.06715
Std Dev	0.2503

isTrueTauEvents

Events

240
220
200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

isTrueTauEvents

VVV.root_isTrueTauEvents	
Entries	54282
Mean	0.9306
Std Dev	0.2542

j1_e

Events

60
50
40
30
20
10
0

0

500

1000

1500

2000

2500

3000

3500

×10³

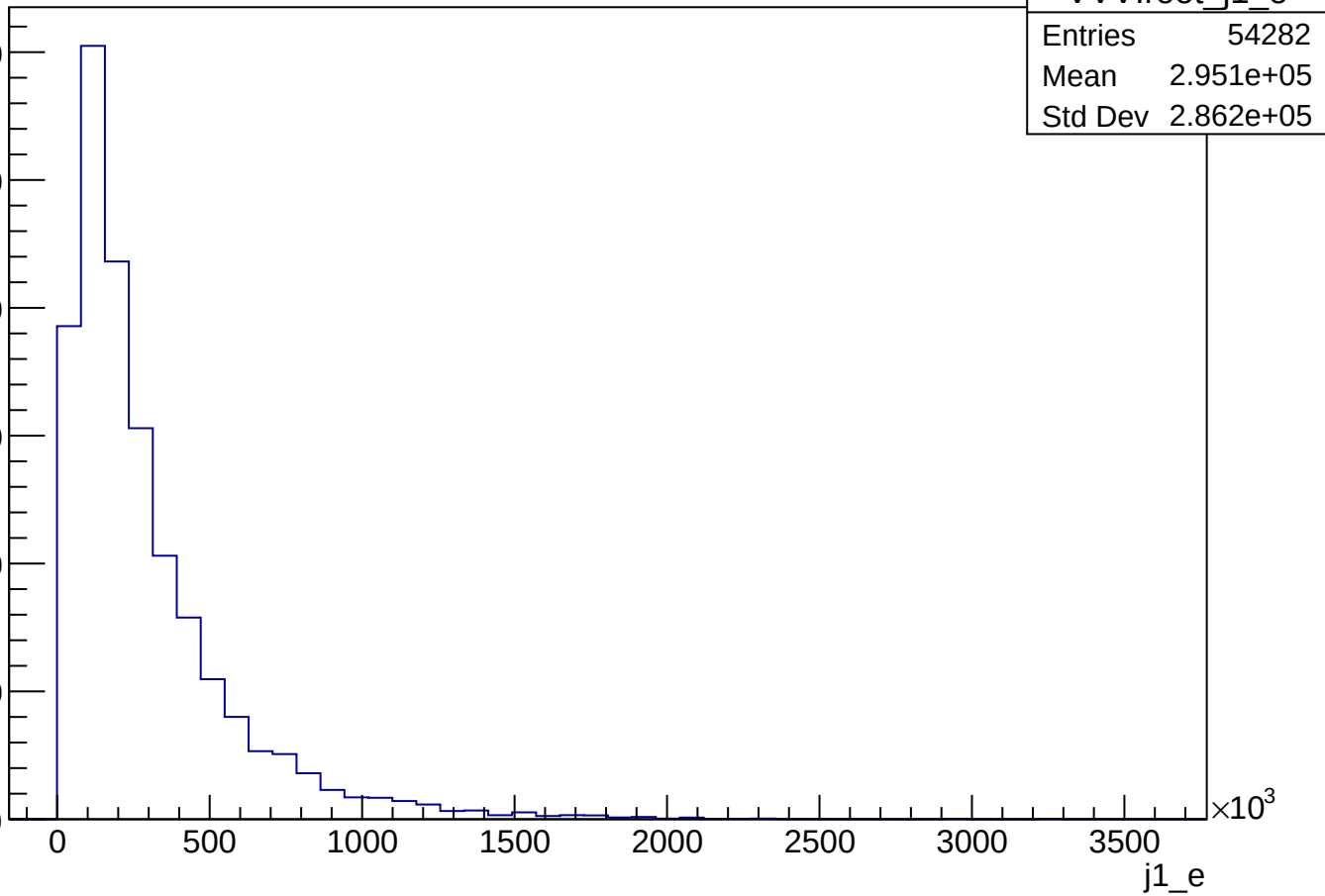
j1_e

VVV.root_j1_e

Entries 54282

Mean 2.951e+05

Std Dev 2.862e+05



j1_eta

Events

9
8
7
6
5
4
3
2
1
0

-2

-1

0

1

2

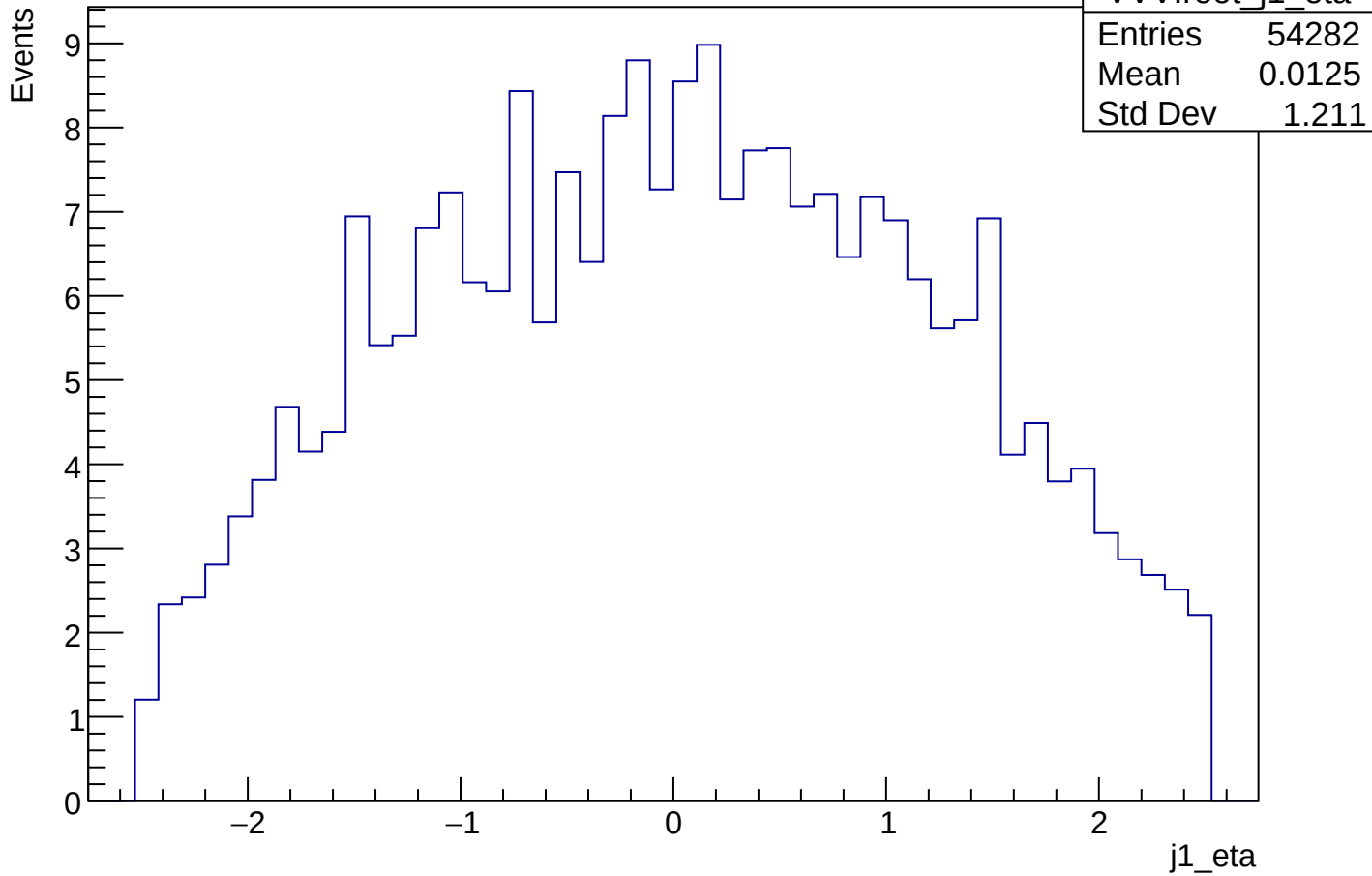
j1_eta

VVV.root_j1_eta

Entries 54282

Mean 0.0125

Std Dev 1.211



j1_phi

Events

7
6
5
4
3
2
1
0

-3

-2

-1

0

1

2

3

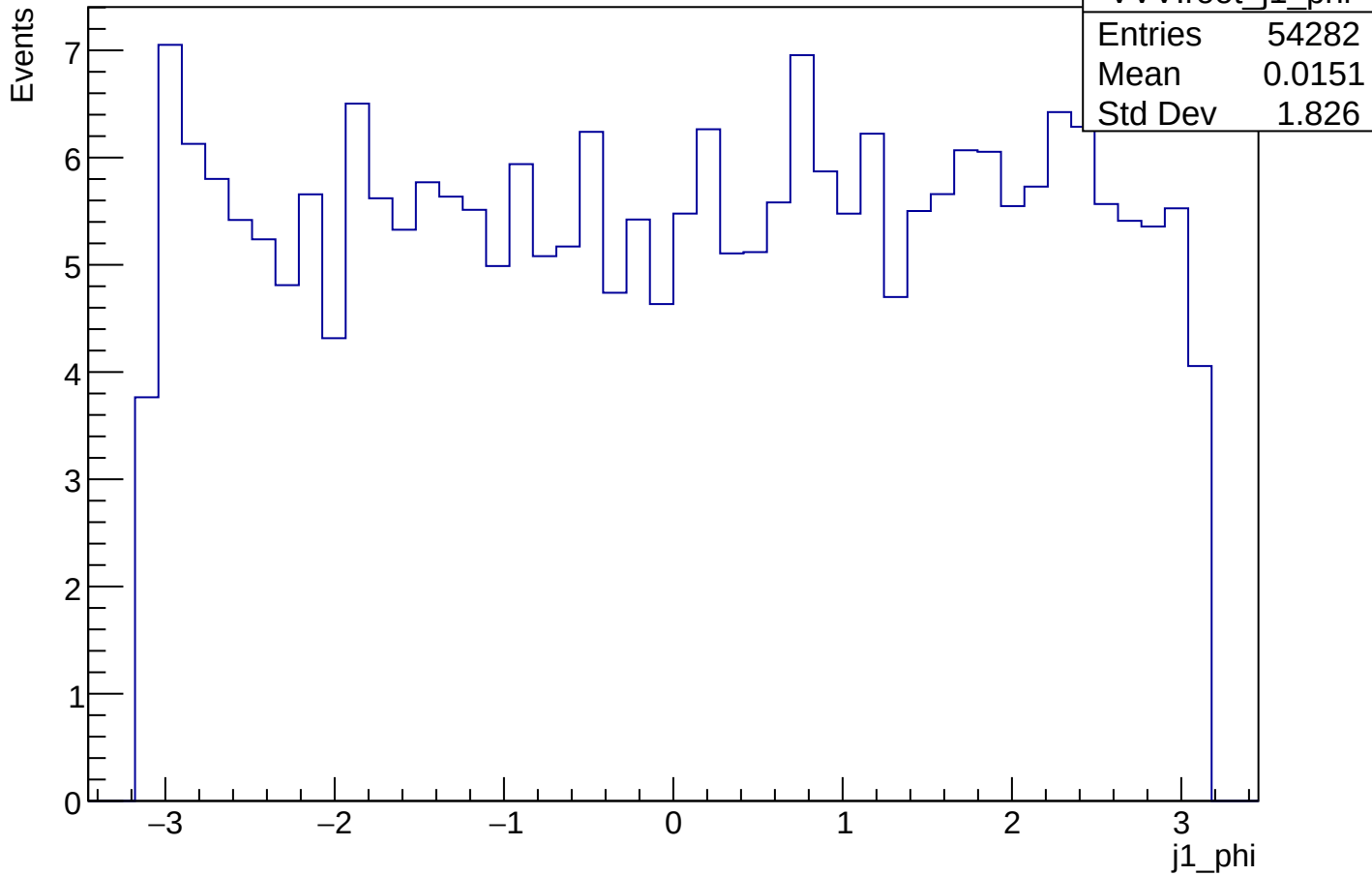
j1_phi

VVV.root_j1_phi

Entries 54282

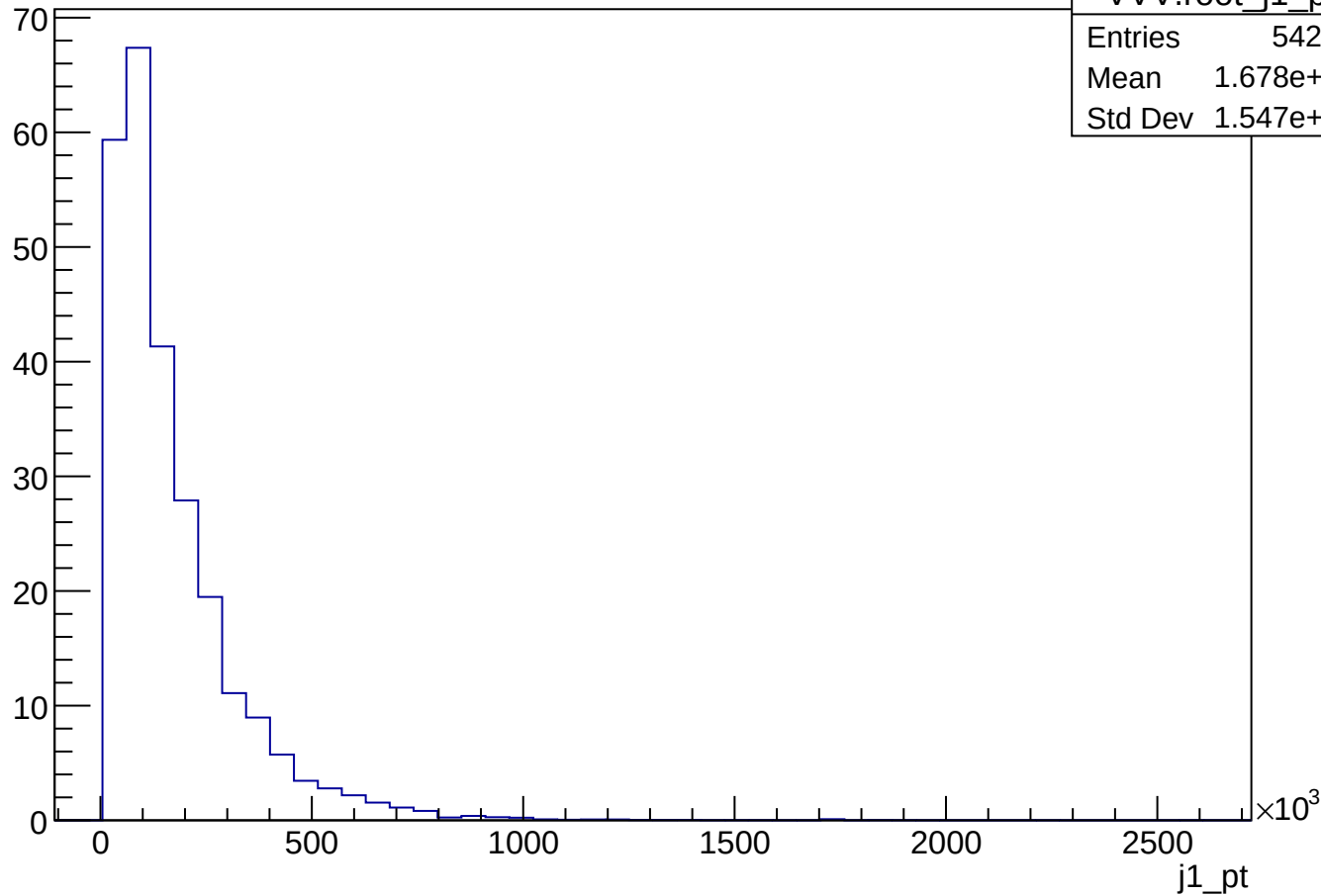
Mean 0.0151

Std Dev 1.826



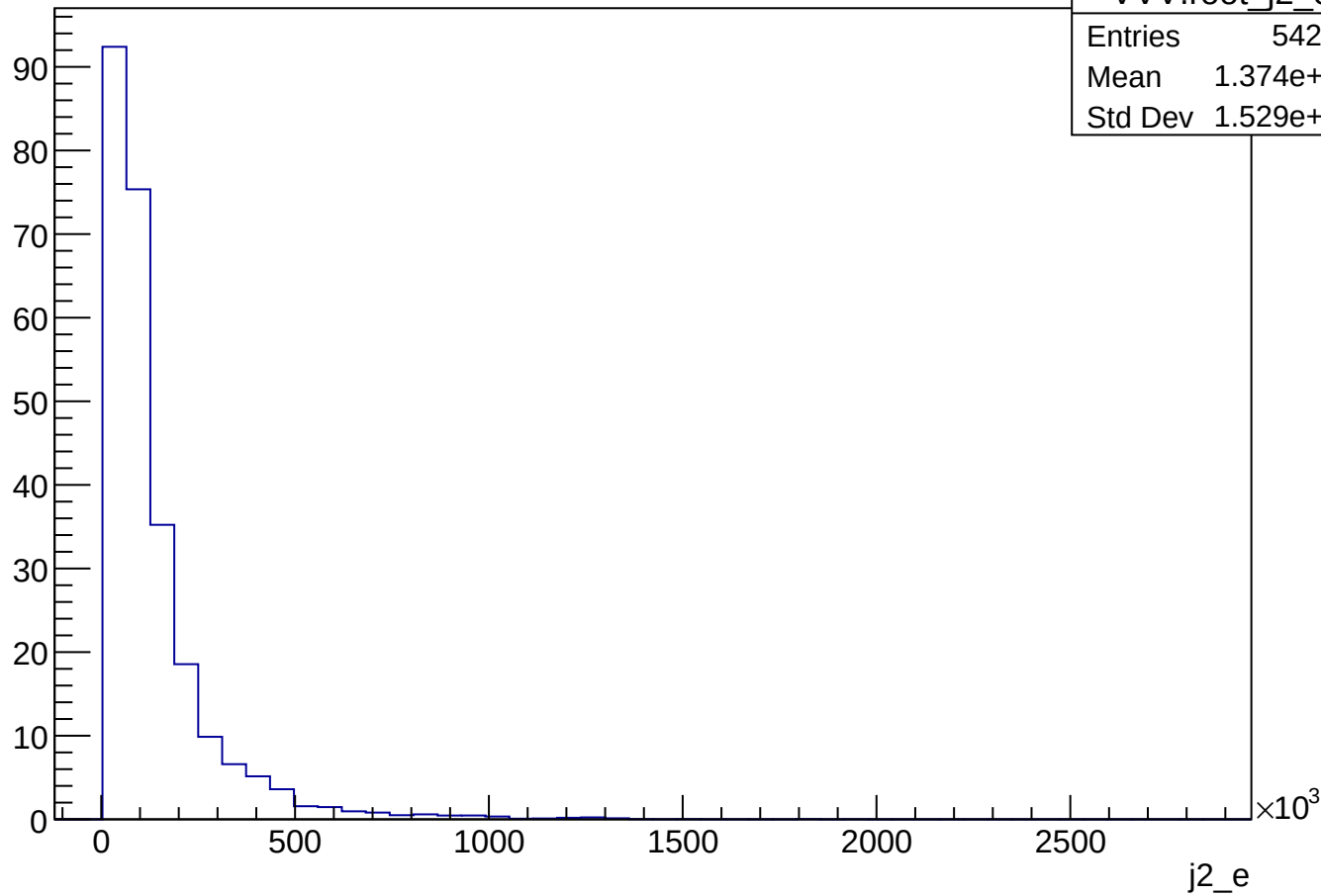
j1_pt

Events



j2_e

Events



j2_eta

Events

VVV.root_j2_eta

Entries 54282

Mean -0.02031

Std Dev 1.301

8
7
6
5
4
3
2
1
0

-2

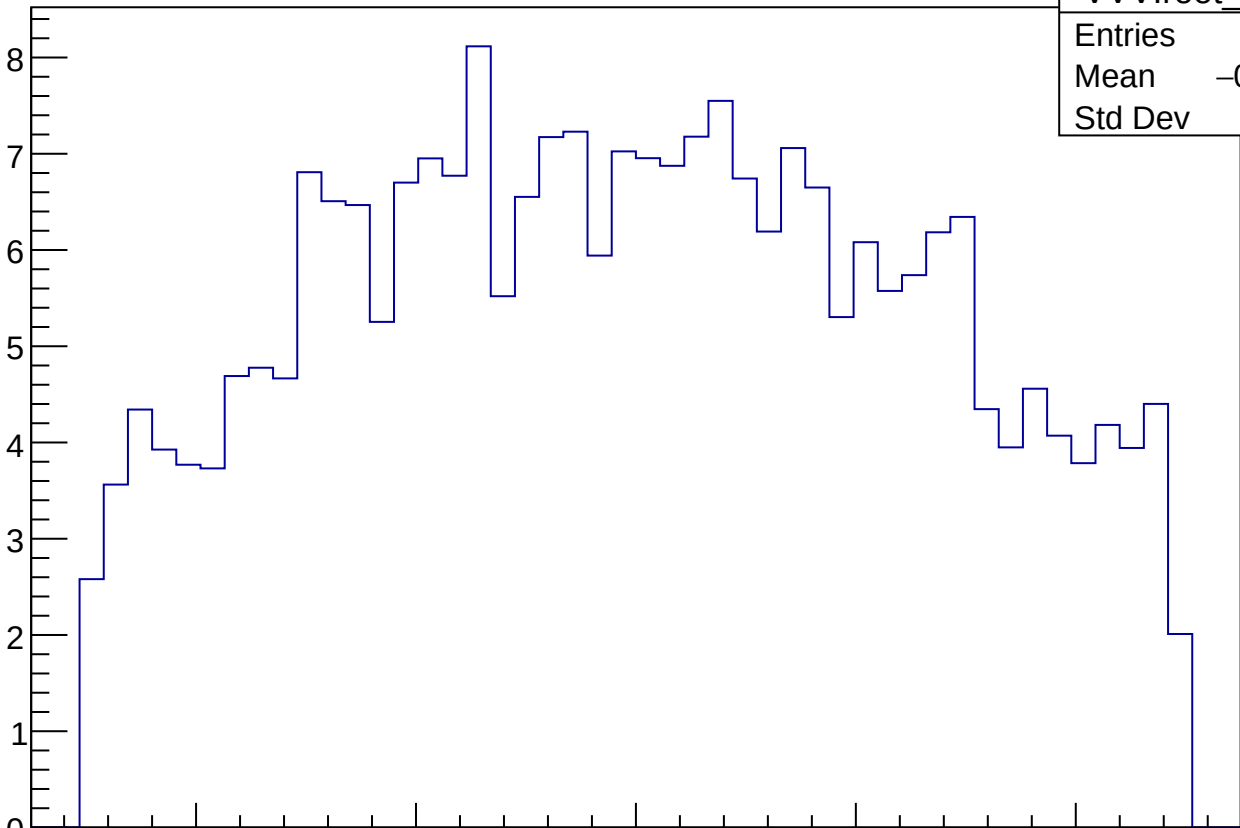
-1

0

1

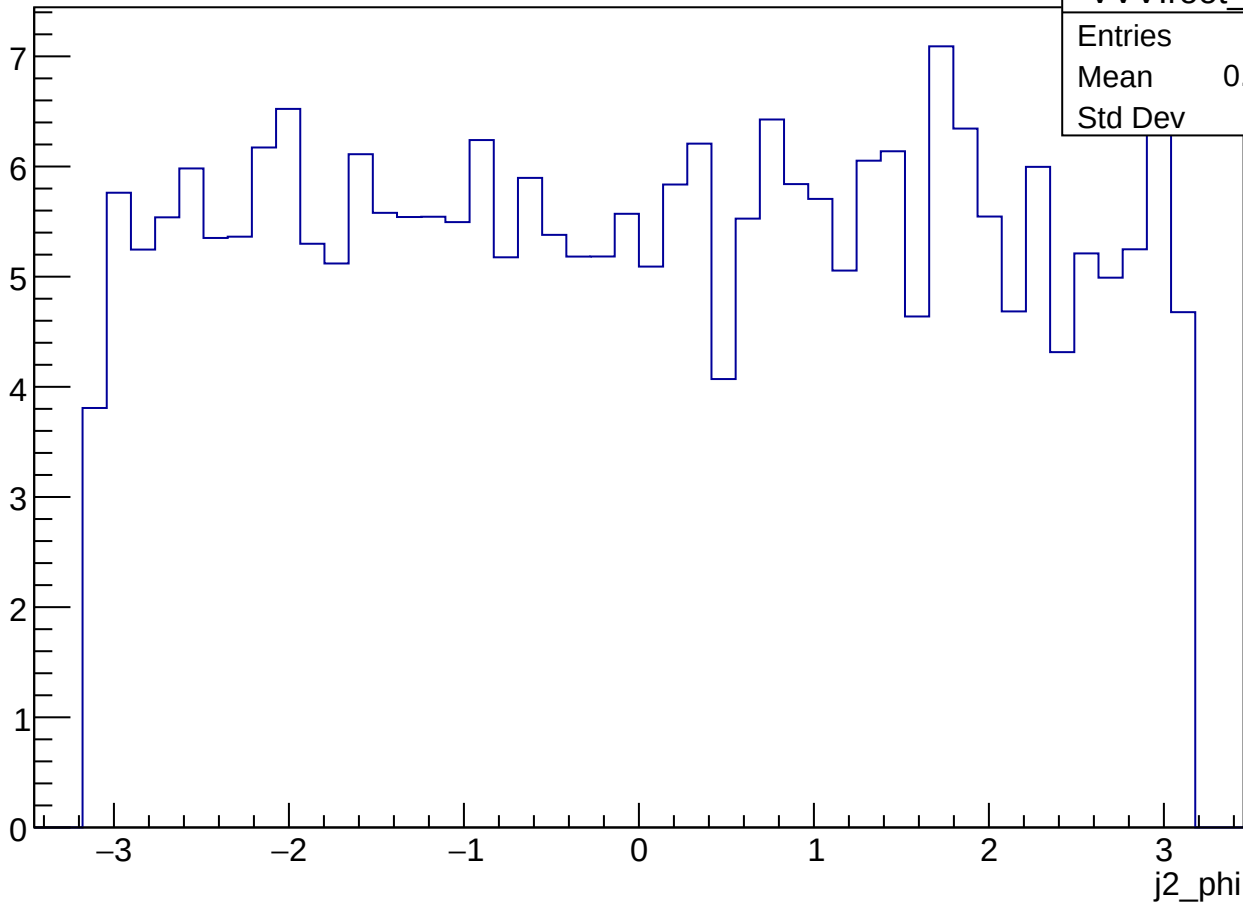
2

j2_eta



j2_phi

Events



VVV.root_j2_phi

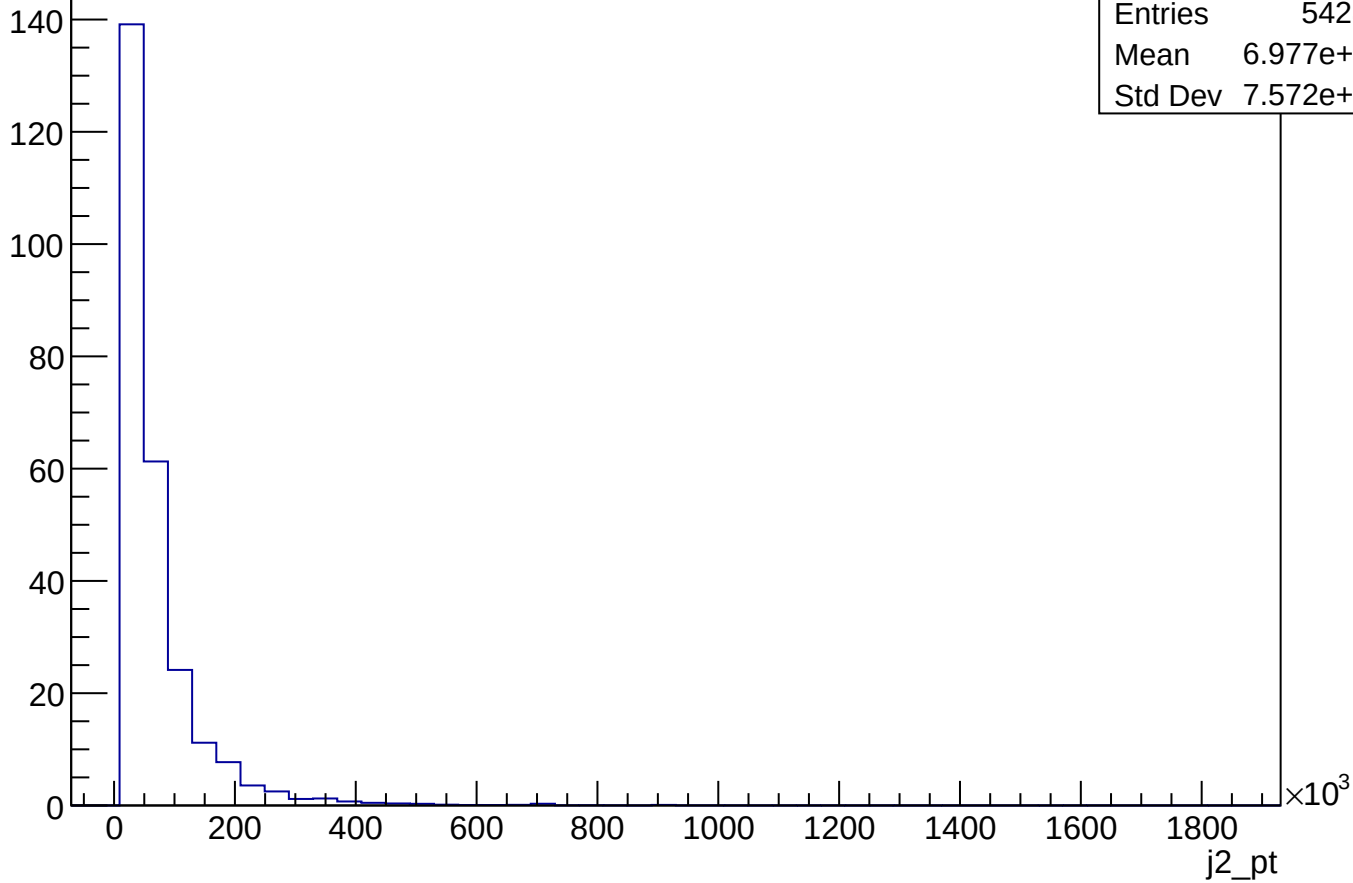
Entries 54282

Mean 0.003372

Std Dev 1.818

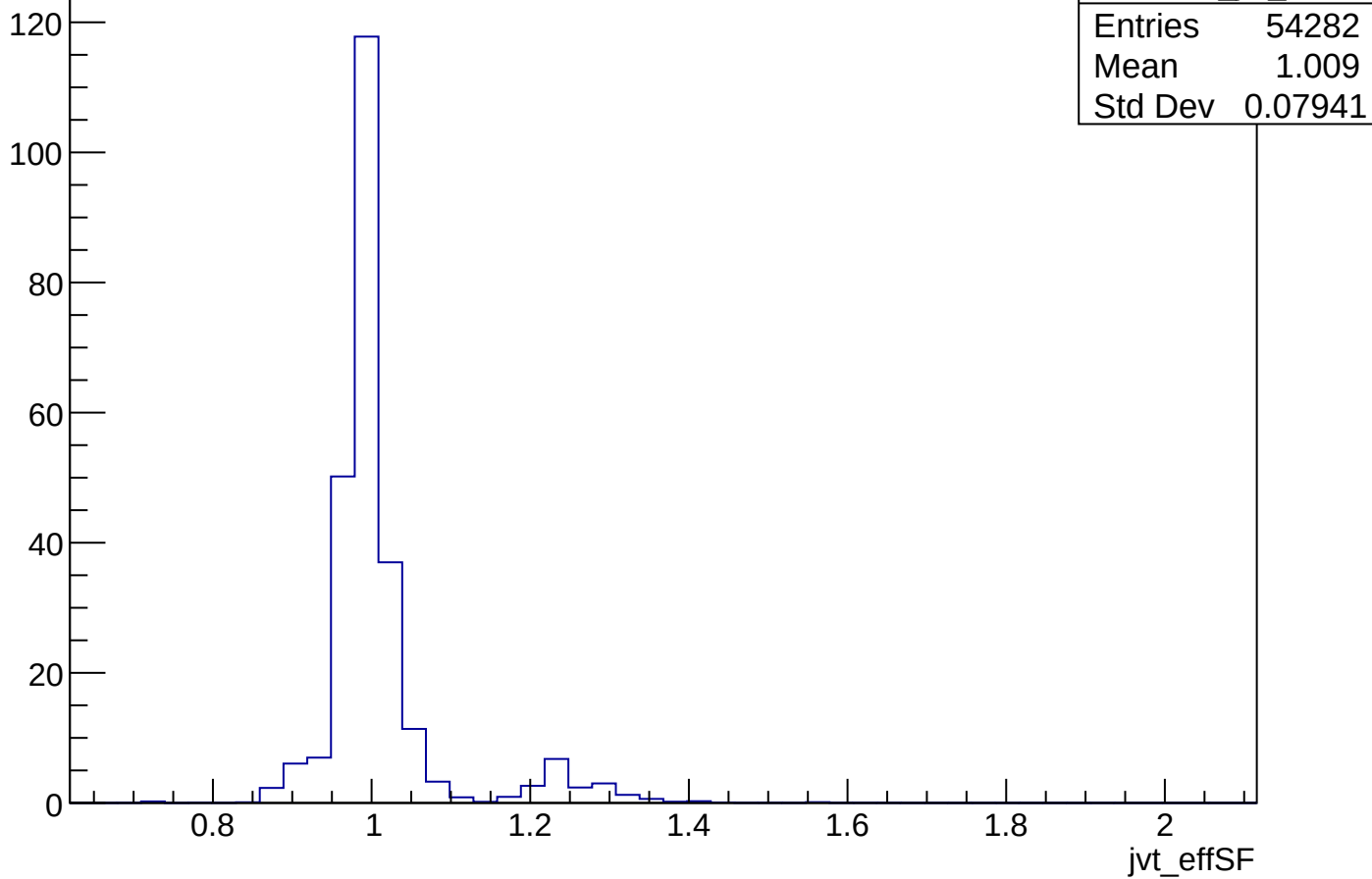
j2_pt

Events



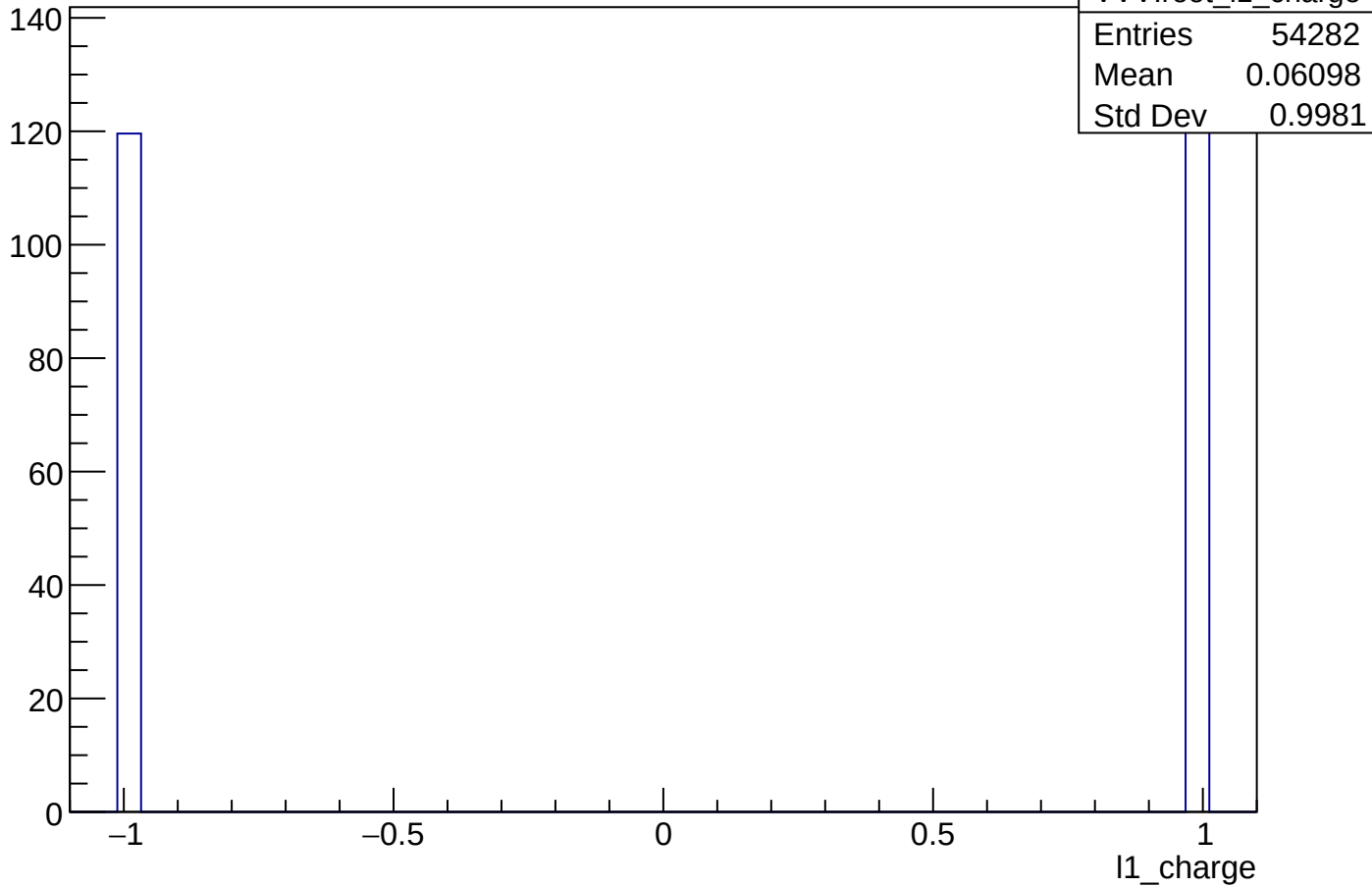
jvt_effSF

Events



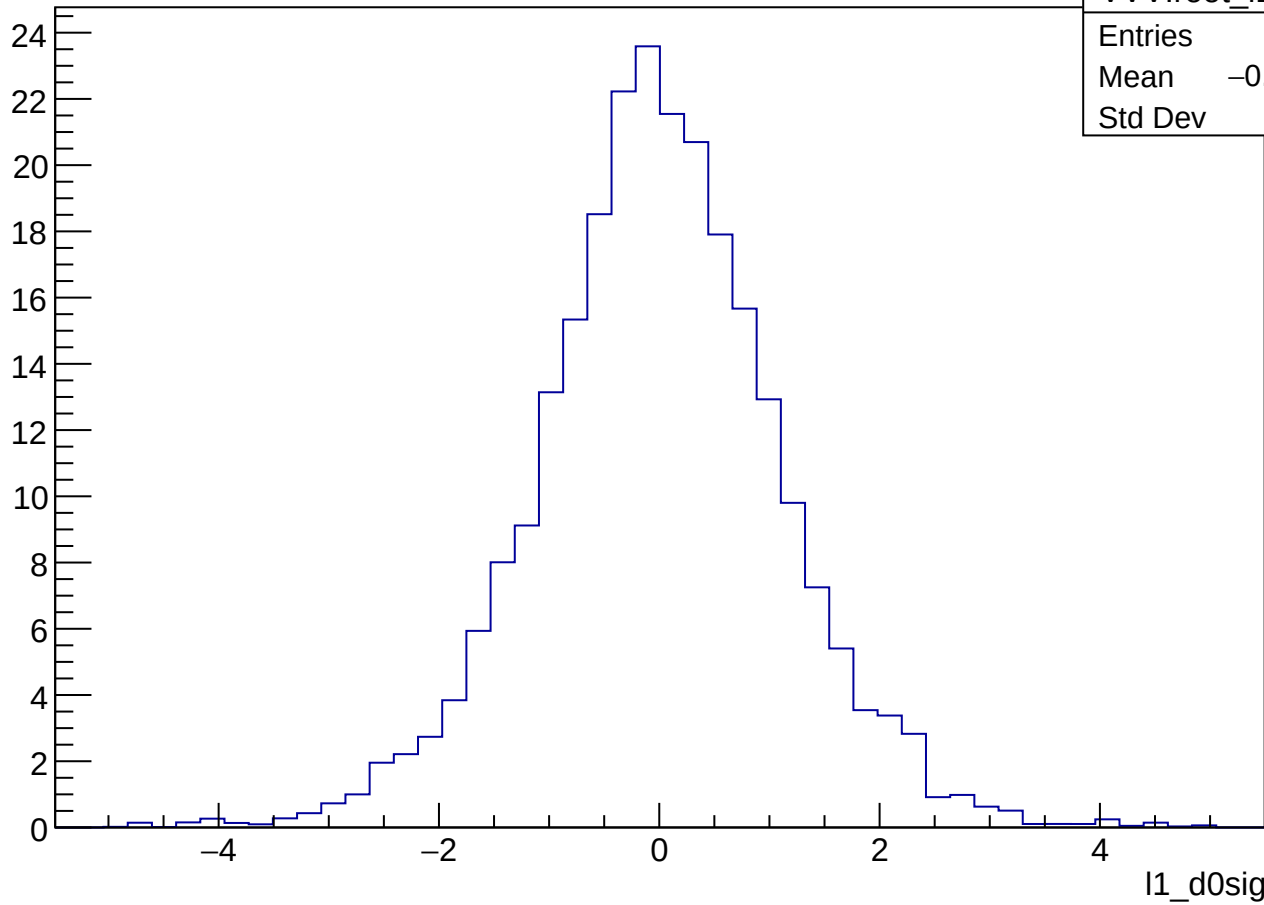
l1_charge

Events



l1_d0sig

Events



VVV.root_l1_d0sig

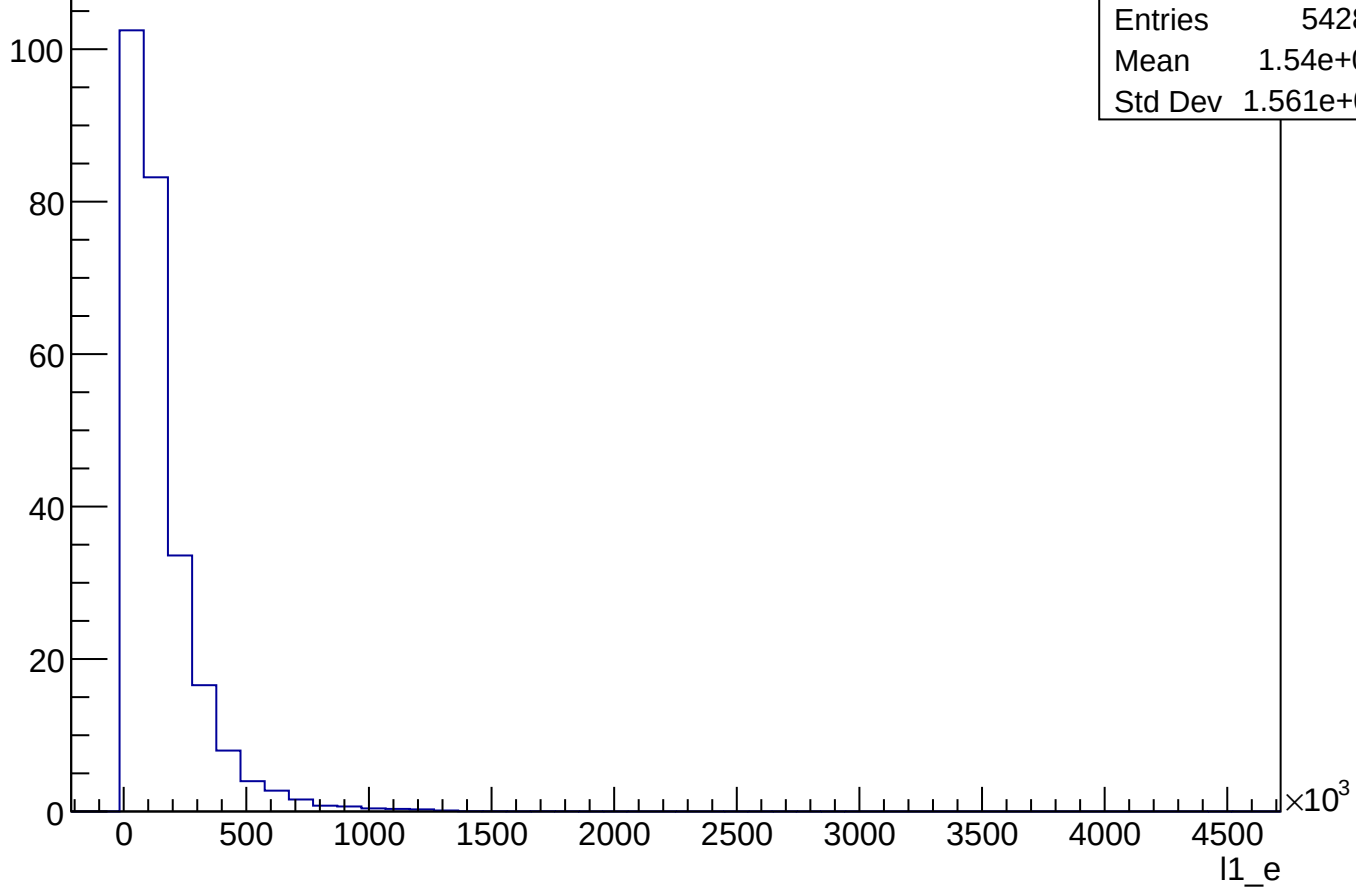
Entries 54282

Mean -0.007527

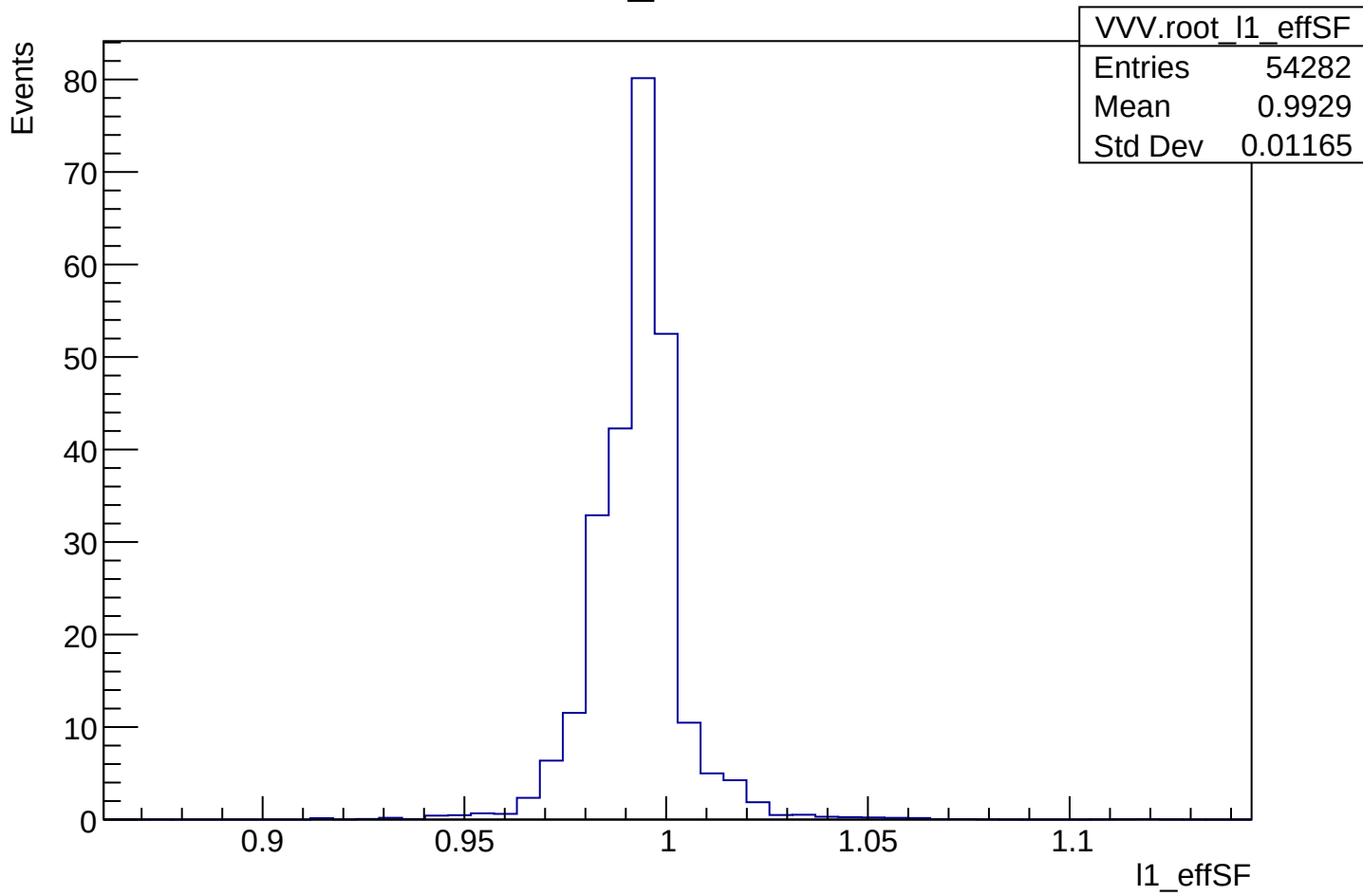
Std Dev 1.11

l1_e

Events

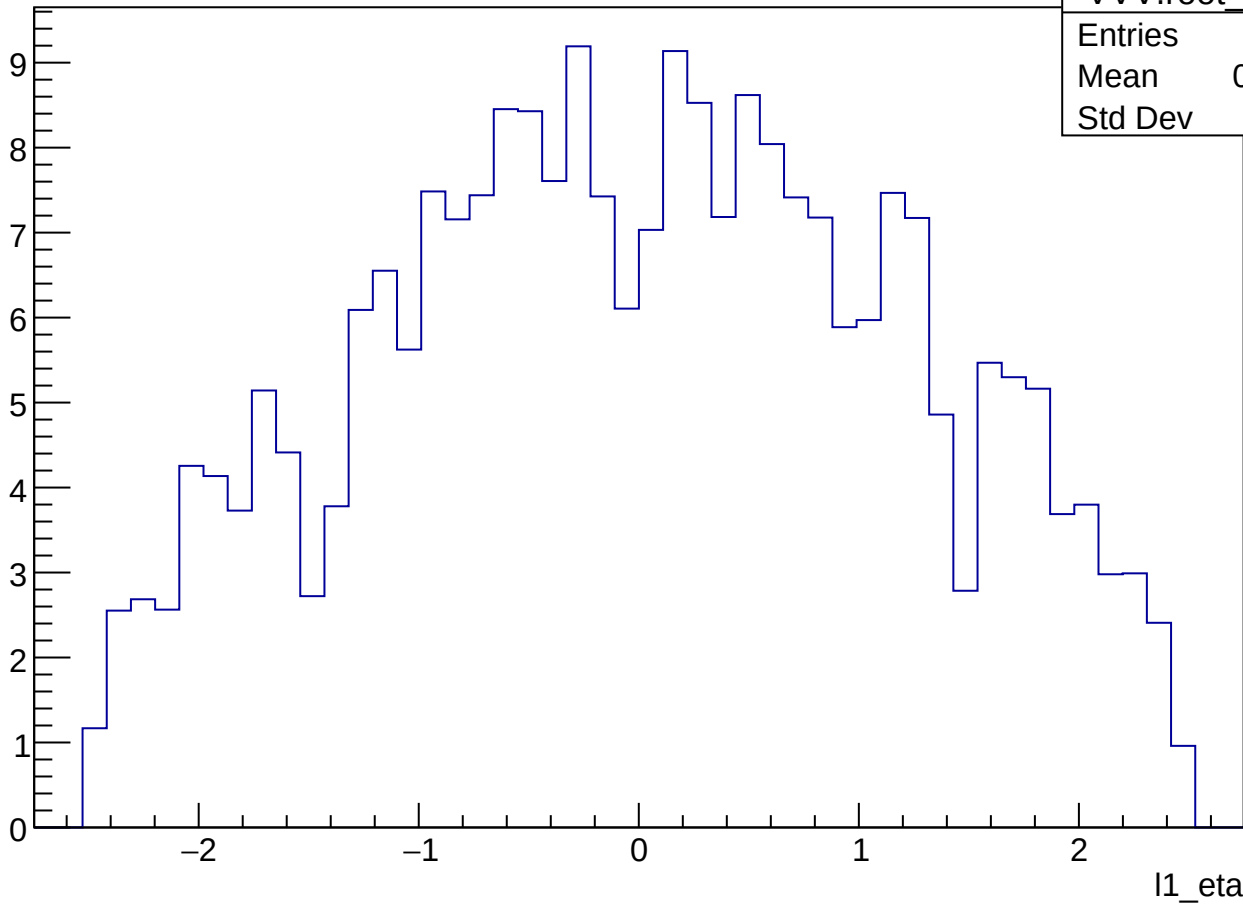


l1_effSF



l1_eta

Events



VVV.root	l1_eta
Entries	54282
Mean	0.02408
Std Dev	1.198

l1_pdg

Events

70
60
50
40
30
20
10
0

-10

-5

0

5

10

l1_pdg

VVV.root_l1_pdg	
Entries	54282
Mean	-0.7401
Std Dev	11.98

70

60

50

40

30

20

10

0

l1_phi

Events

7
6
5
4
3
2
1
0

-3

-2

-1

0

1

2

3

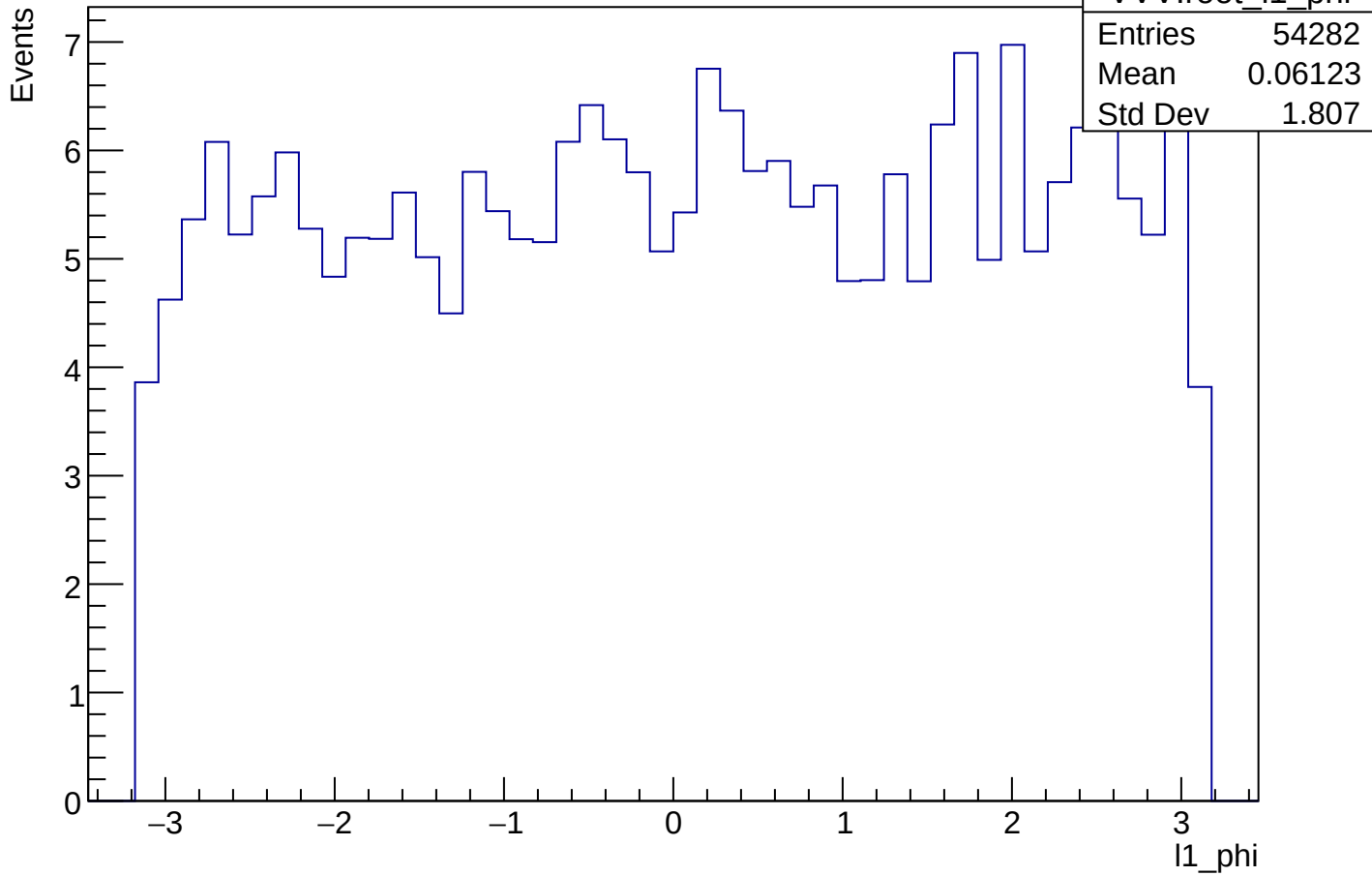
l1_phi

VVV.root_l1_phi

Entries 54282

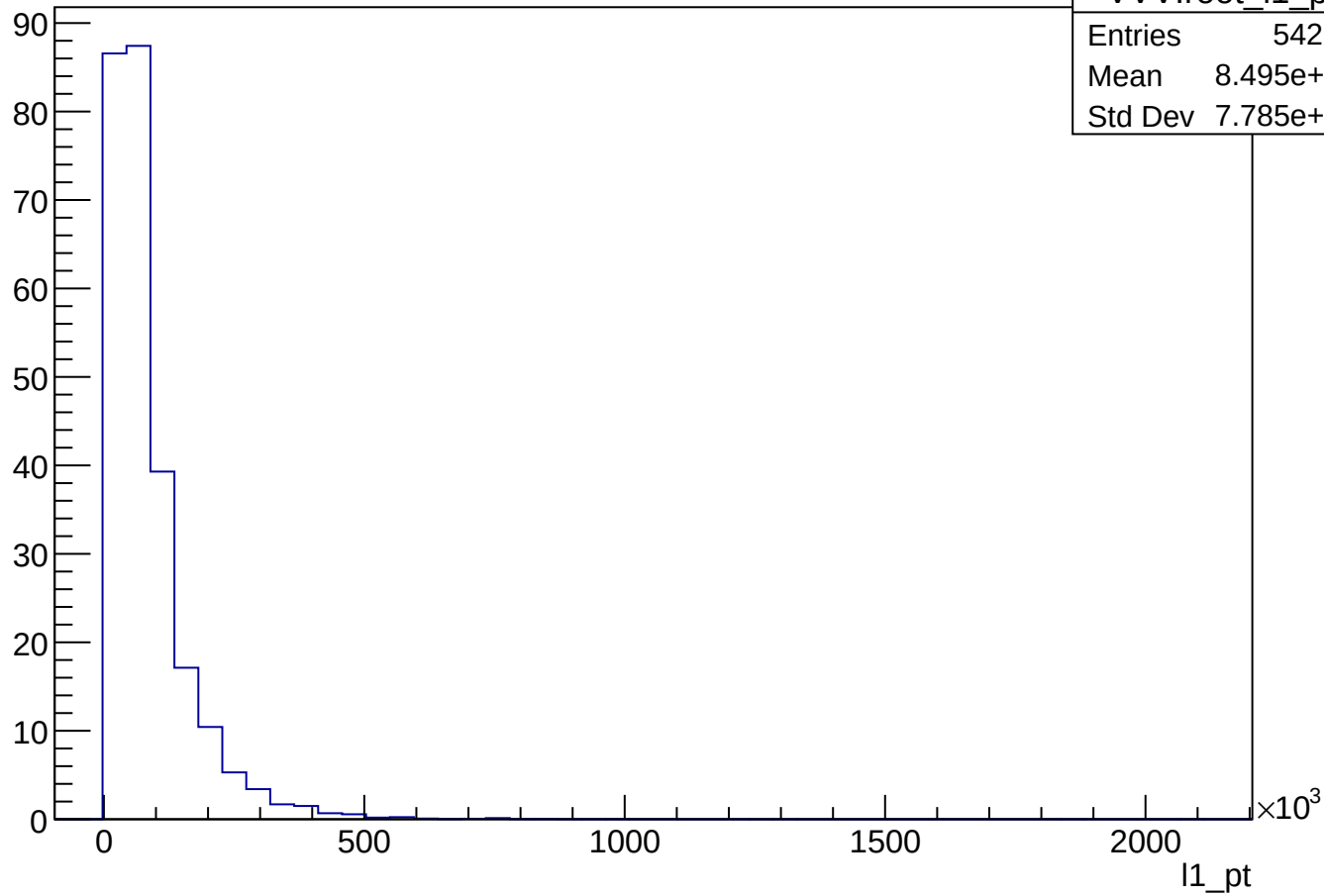
Mean 0.06123

Std Dev 1.807

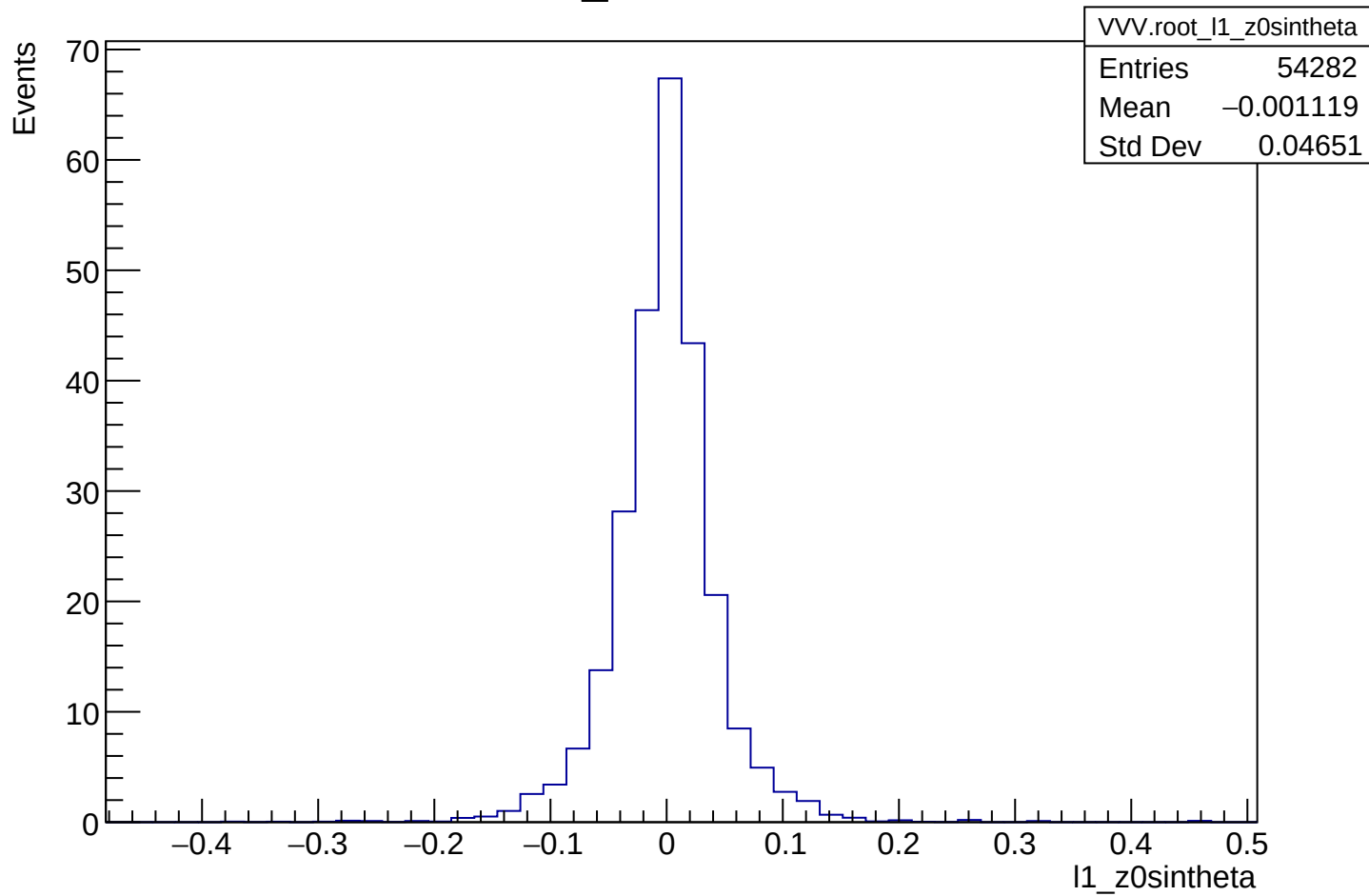


l1_pt

Events

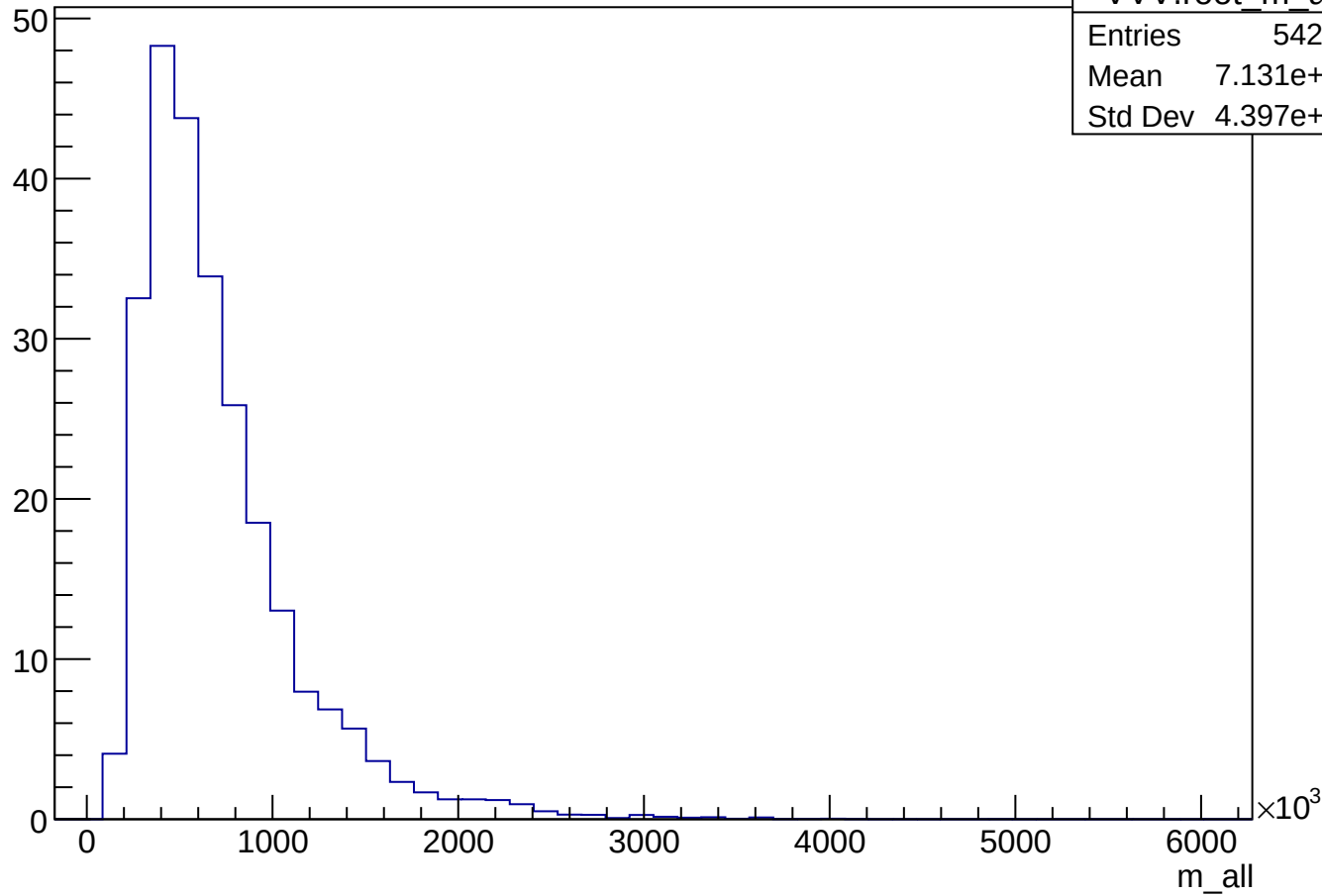


l1_z0sintheta



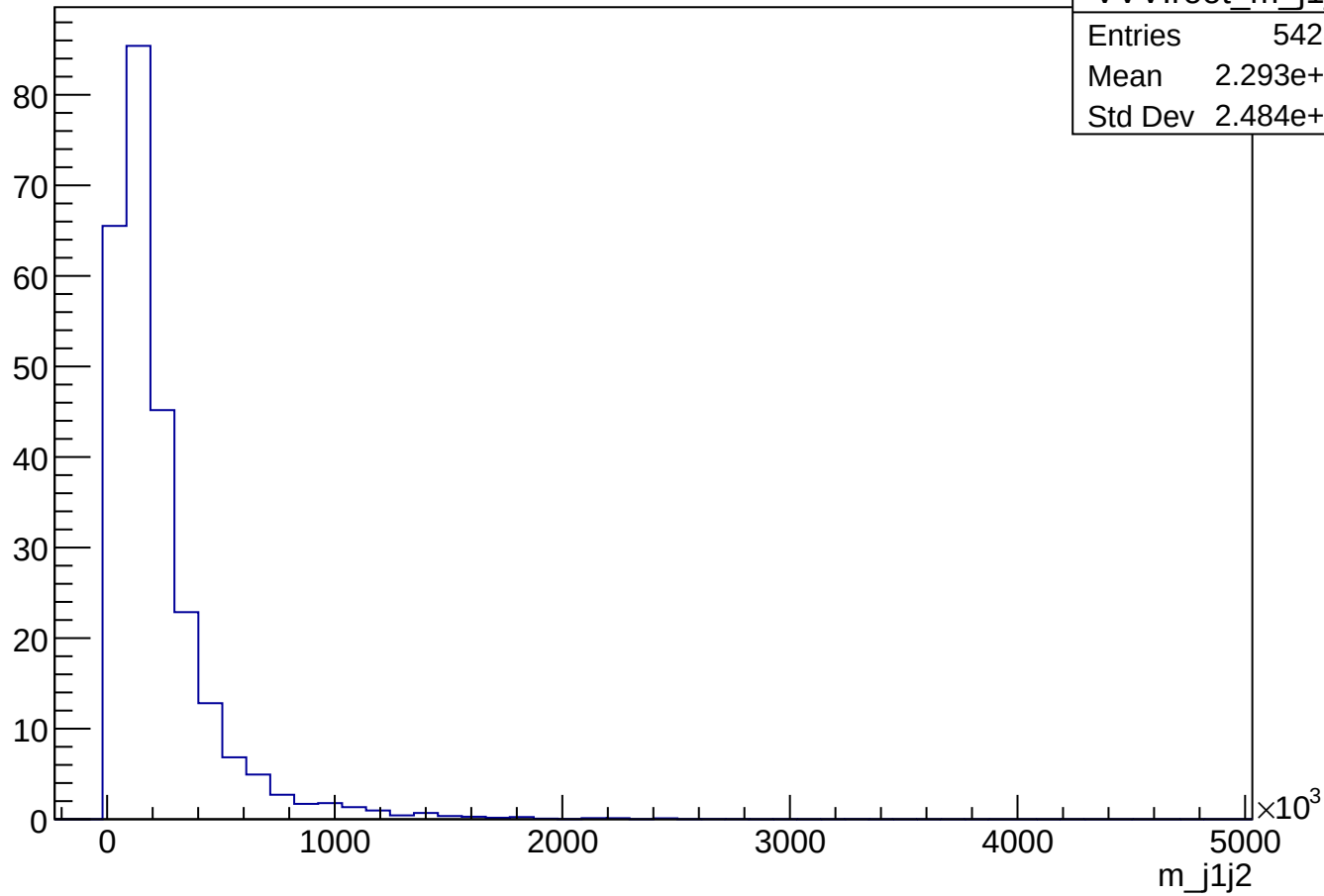
m_all

Events



m_j1j2

Events



m_l1j

Events

100

80

60

40

20

0

0

500

1000

1500

2000

2500

3000

$\times 10^3$

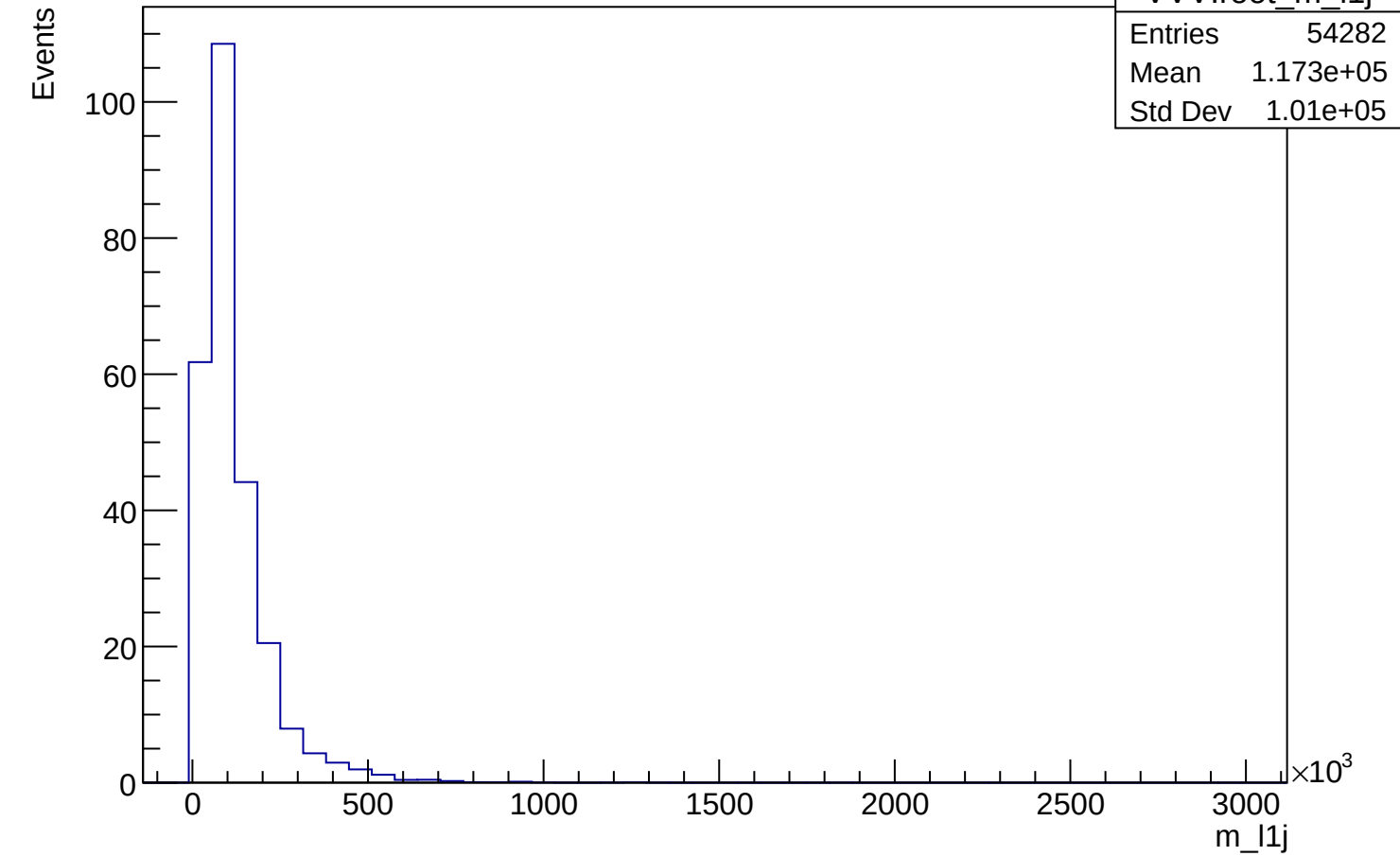
m_l1j

VVV.root_m_l1j

Entries 54282

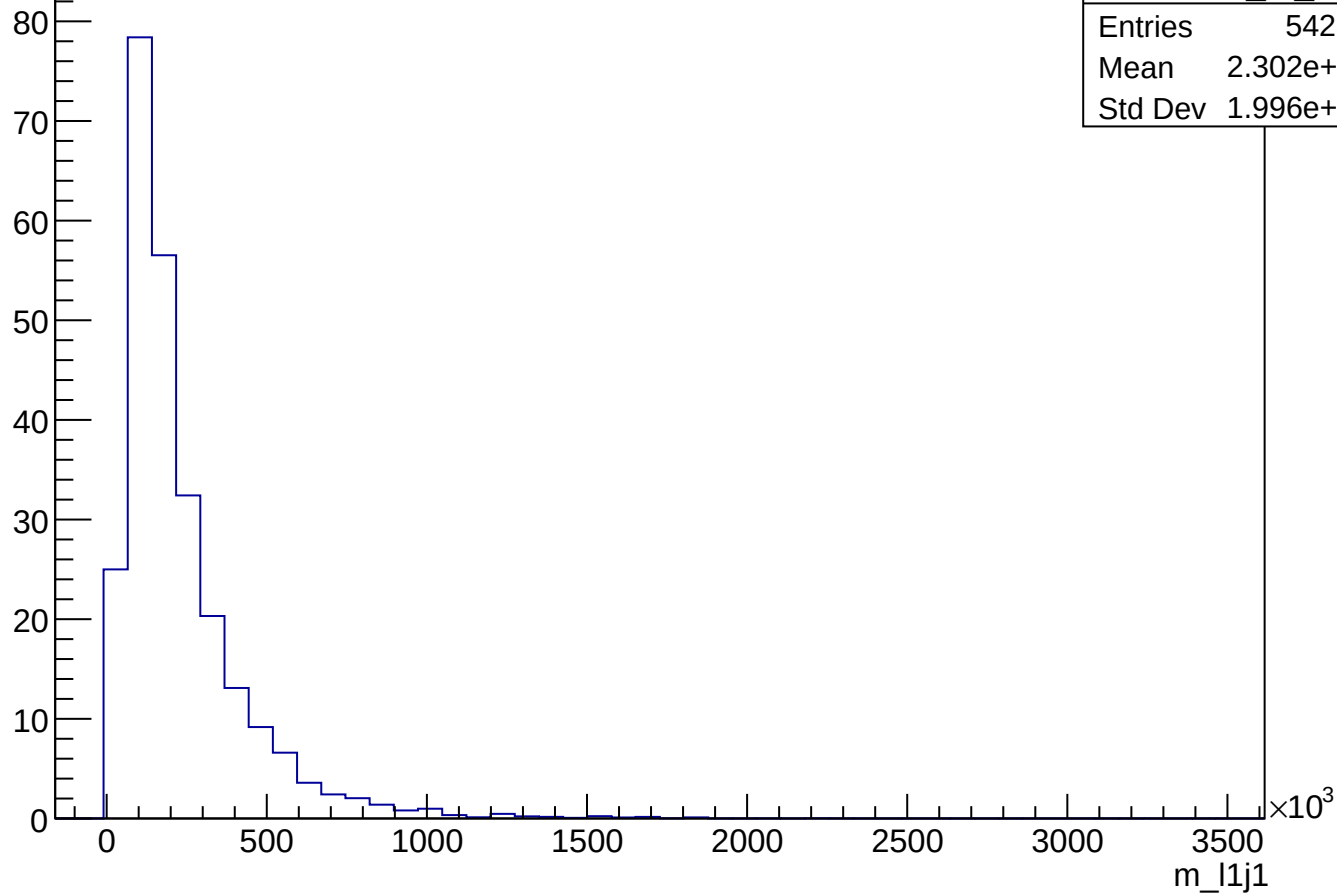
Mean 1.173e+05

Std Dev 1.01e+05



m_l1j1

Events



m_l1j2

Events

100

80

60

40

20

0

0

500

1000

1500

2000

2500

3000

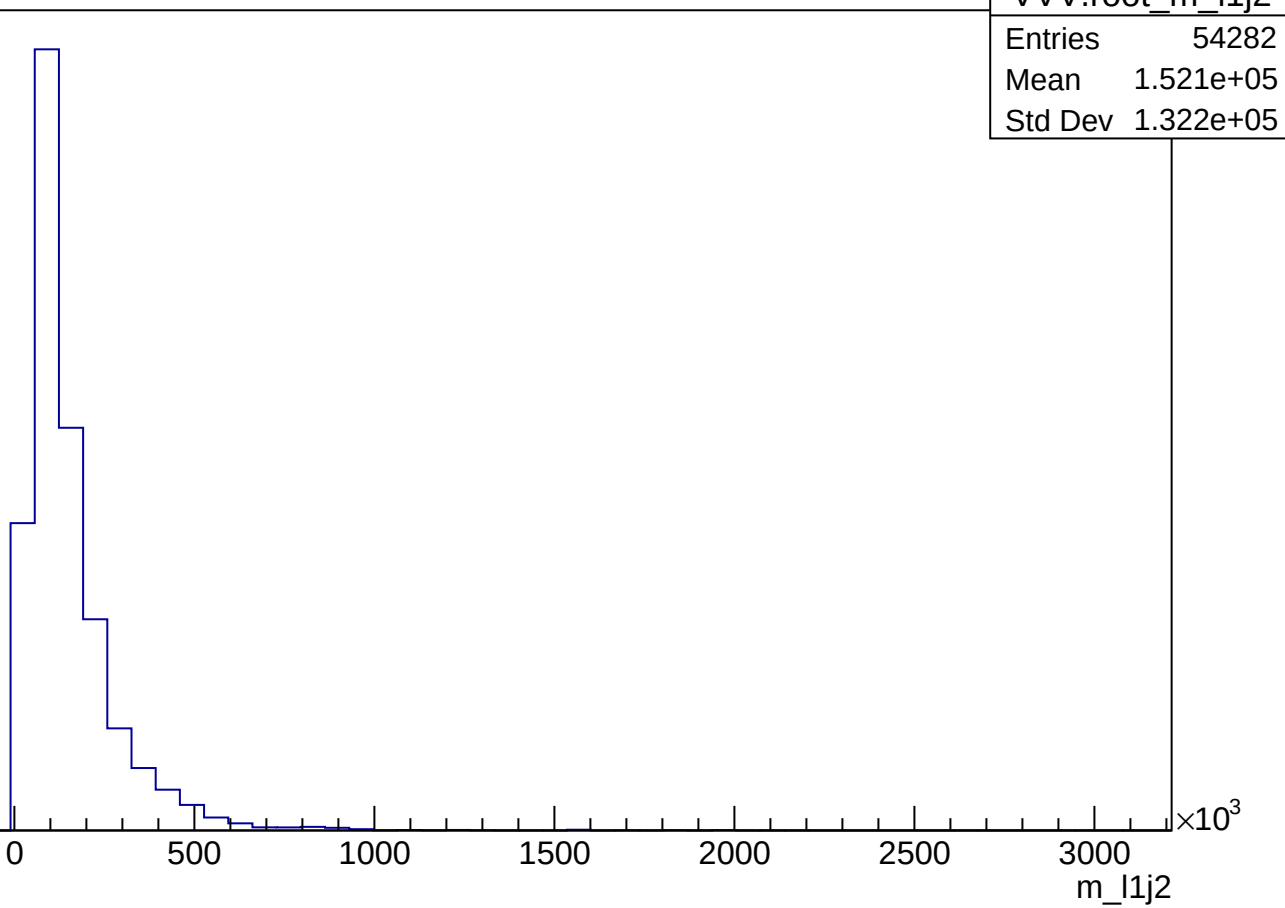
$\times 10^3$
m_l1j2

VVV.root_m_l1j2

Entries 54282

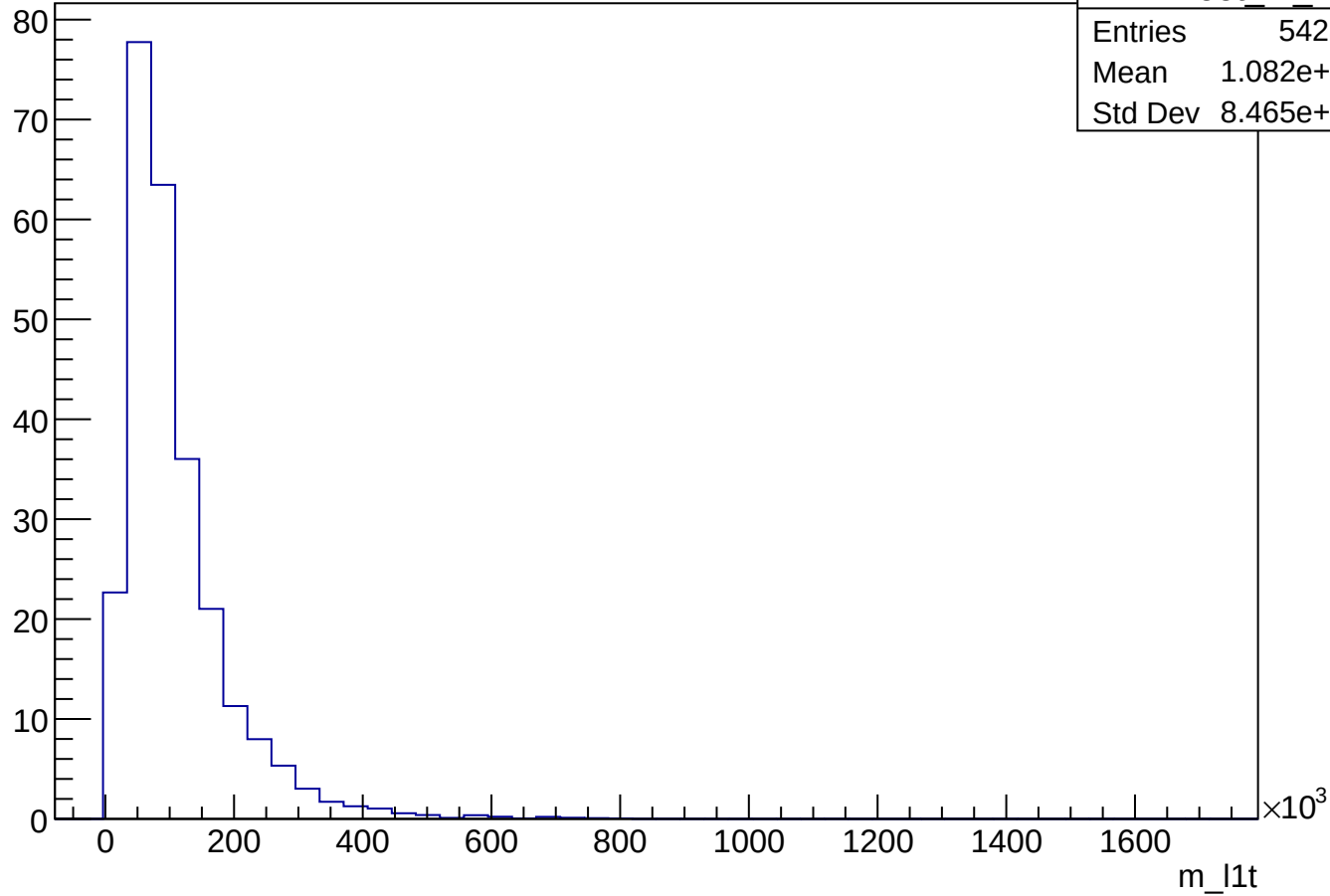
Mean 1.521e+05

Std Dev 1.322e+05



m_l1t

Events



m_l1t1

Events

100

80

60

40

20

0

0

500

1000

1500

2000

2500

3000

3500

4000

m_l1t1

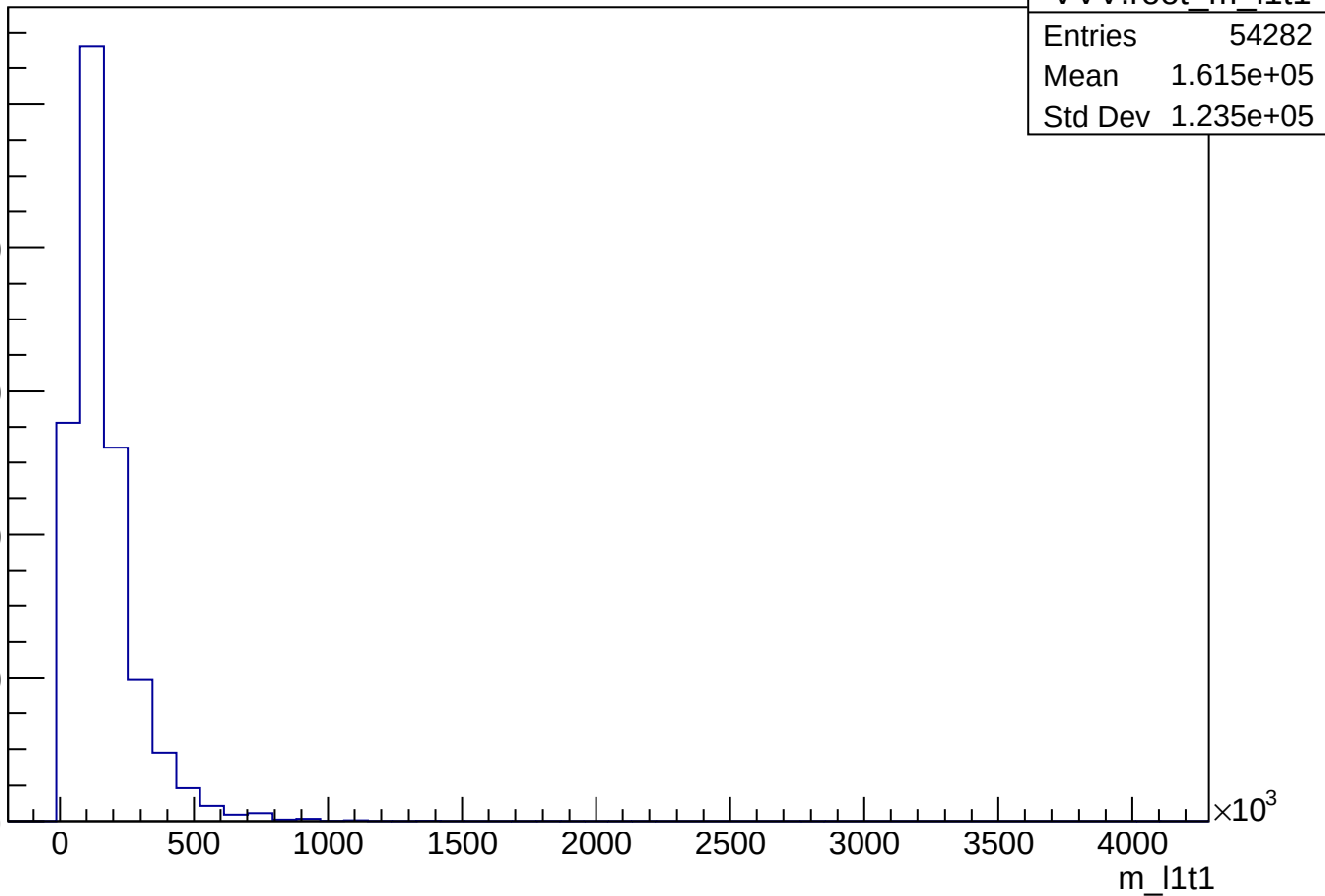
$\times 10^3$

VVV.root m_l1t1

Entries 54282

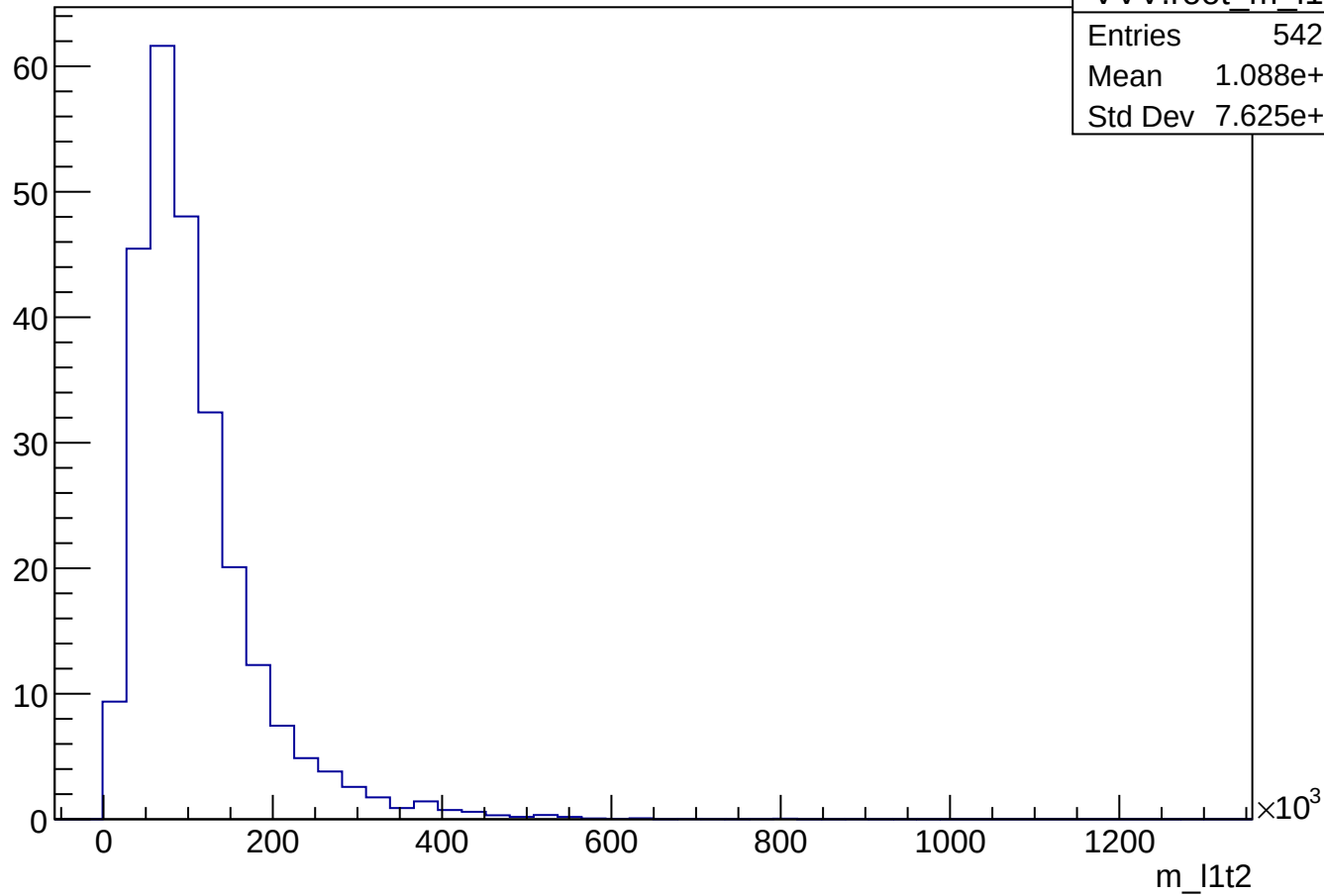
Mean 1.615e+05

Std Dev 1.235e+05



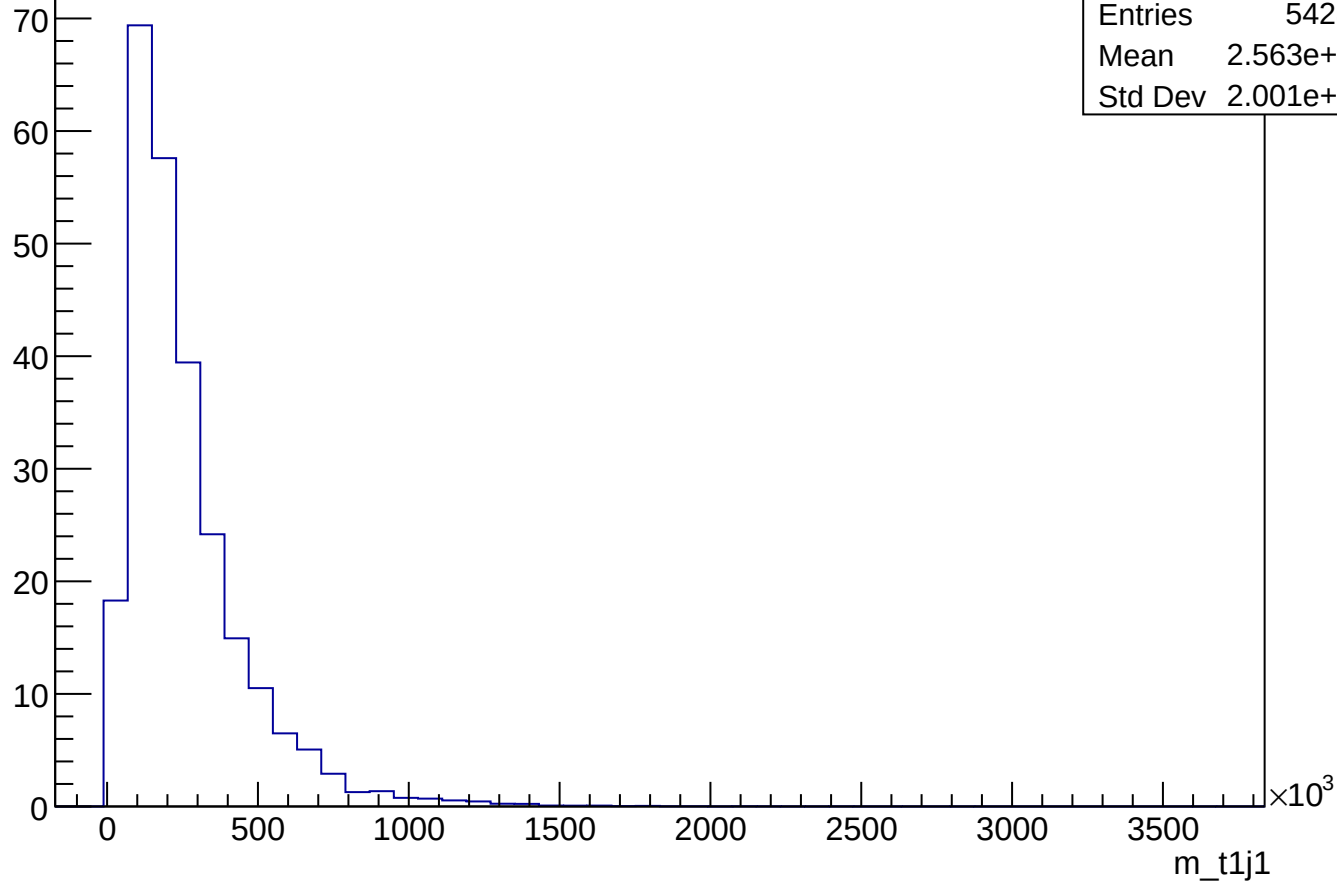
m_l1t2

Events



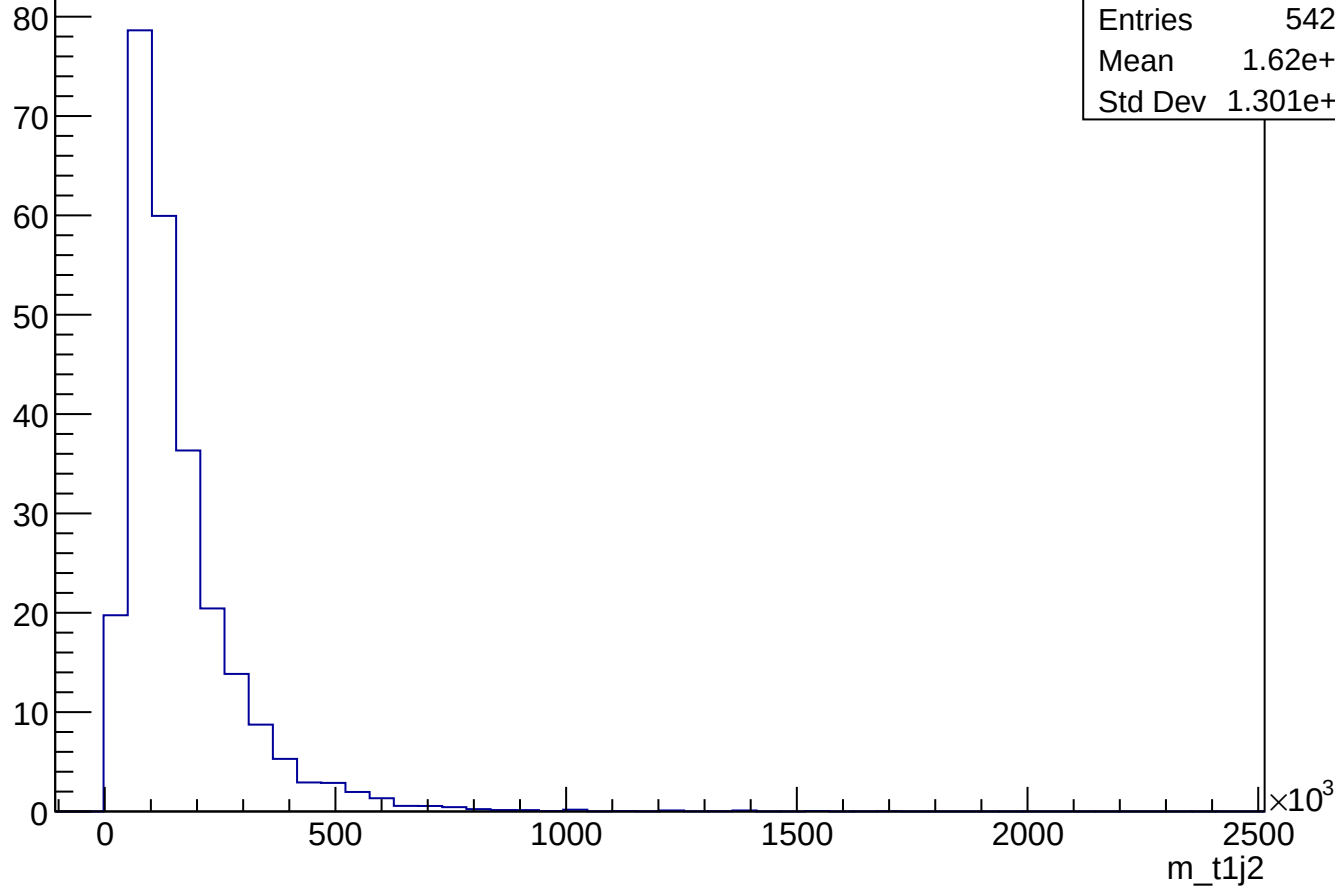
m_t1j1

Events



m_t1j2

Events



m_t1t2

Events

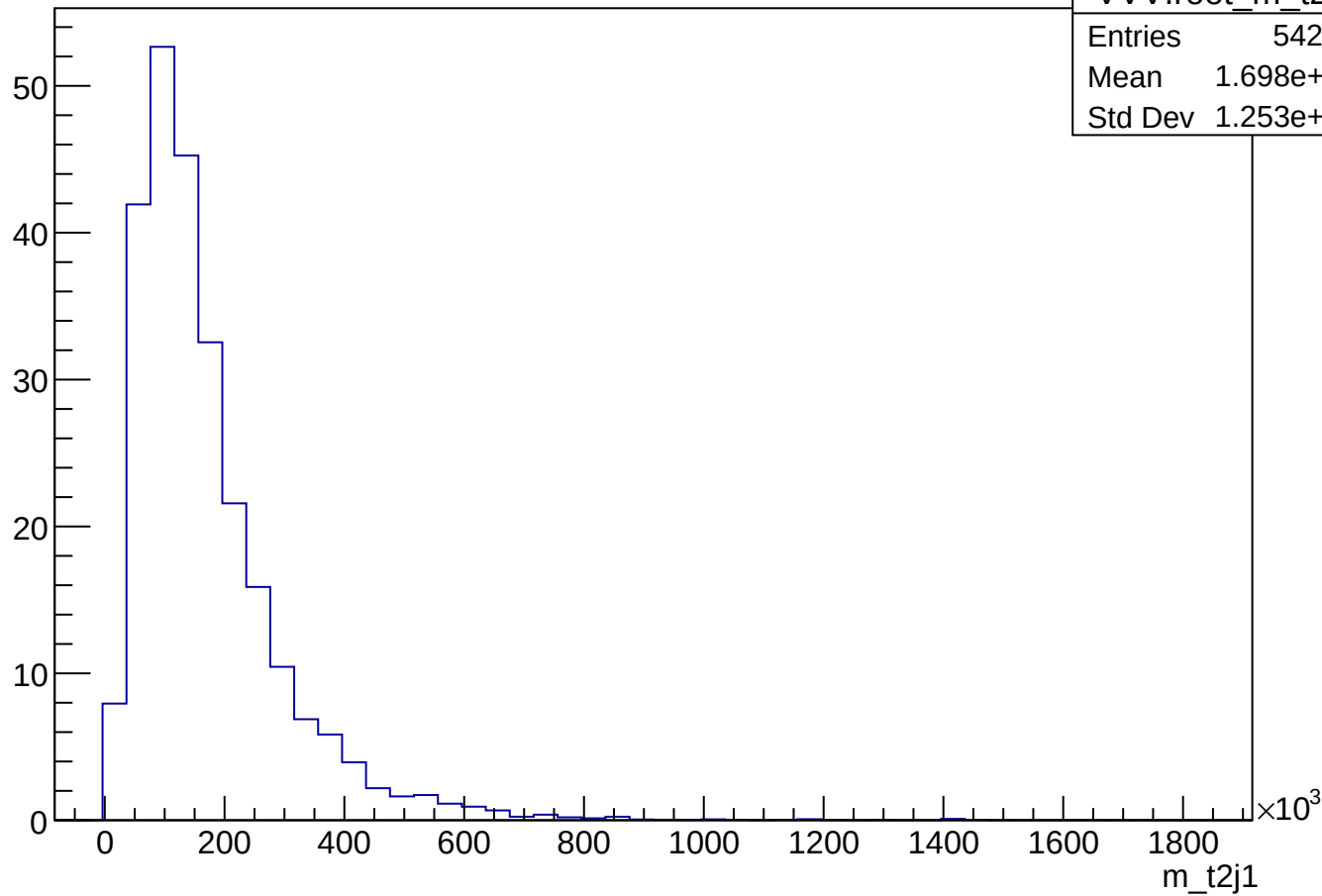
100
80
60
40
20
0

0 200 400 600 800 1000 1200 1400 1600 $\times 10^3$
m_t1t2

VVV.root m_t1t2	
Entries	54282
Mean	1.07e+05
Std Dev	8.096e+04

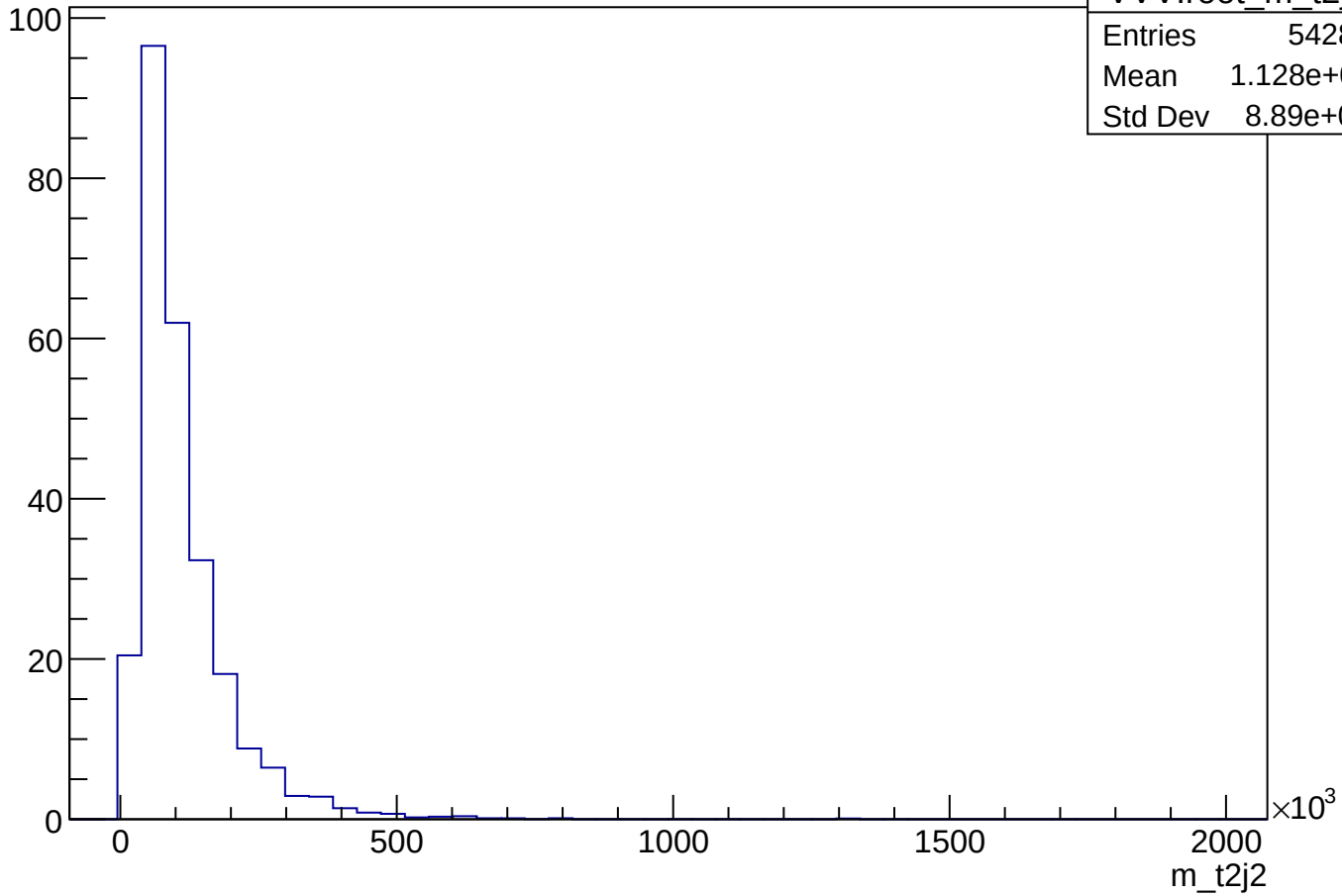
m_t2j1

Events



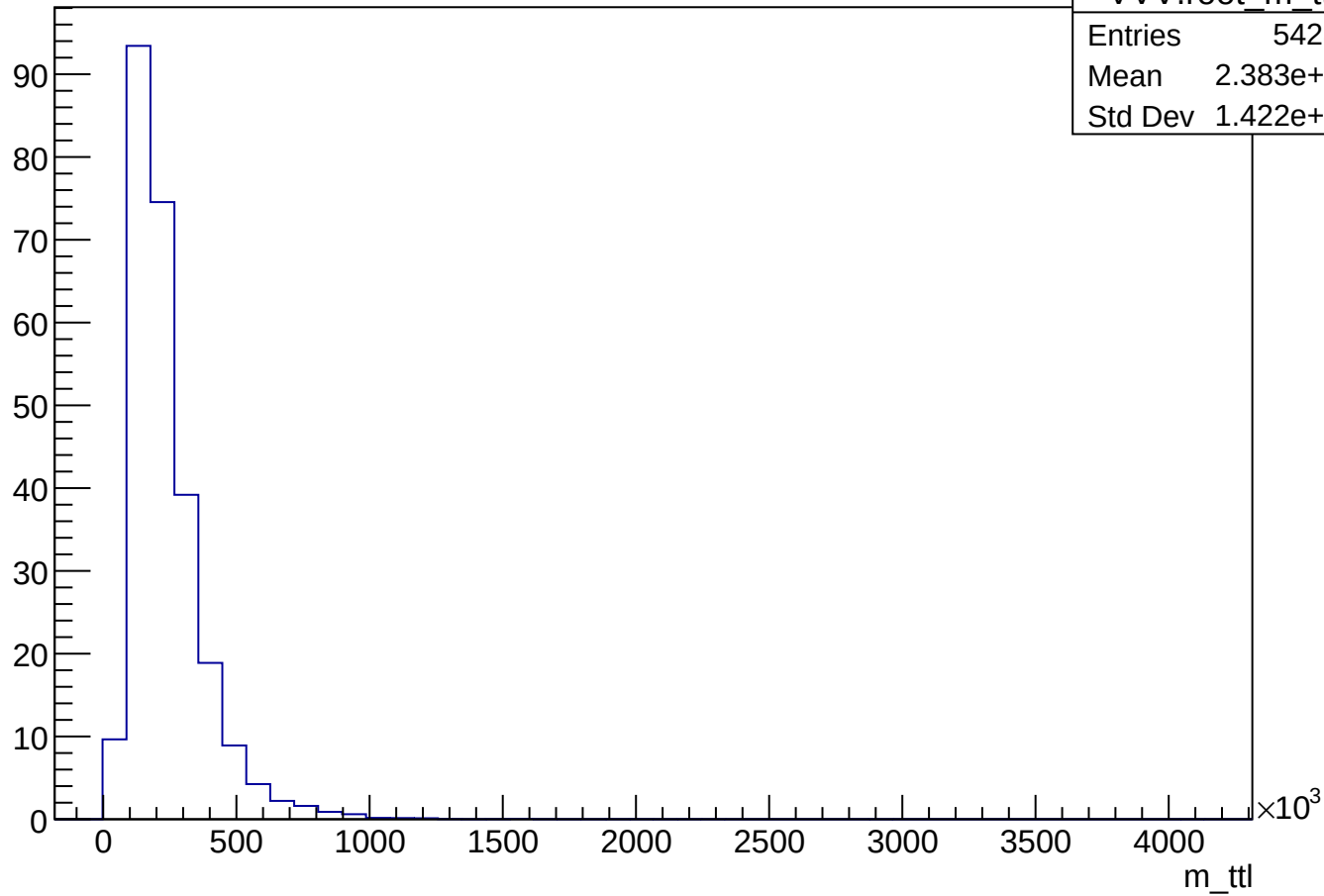
m_t2j2

Events

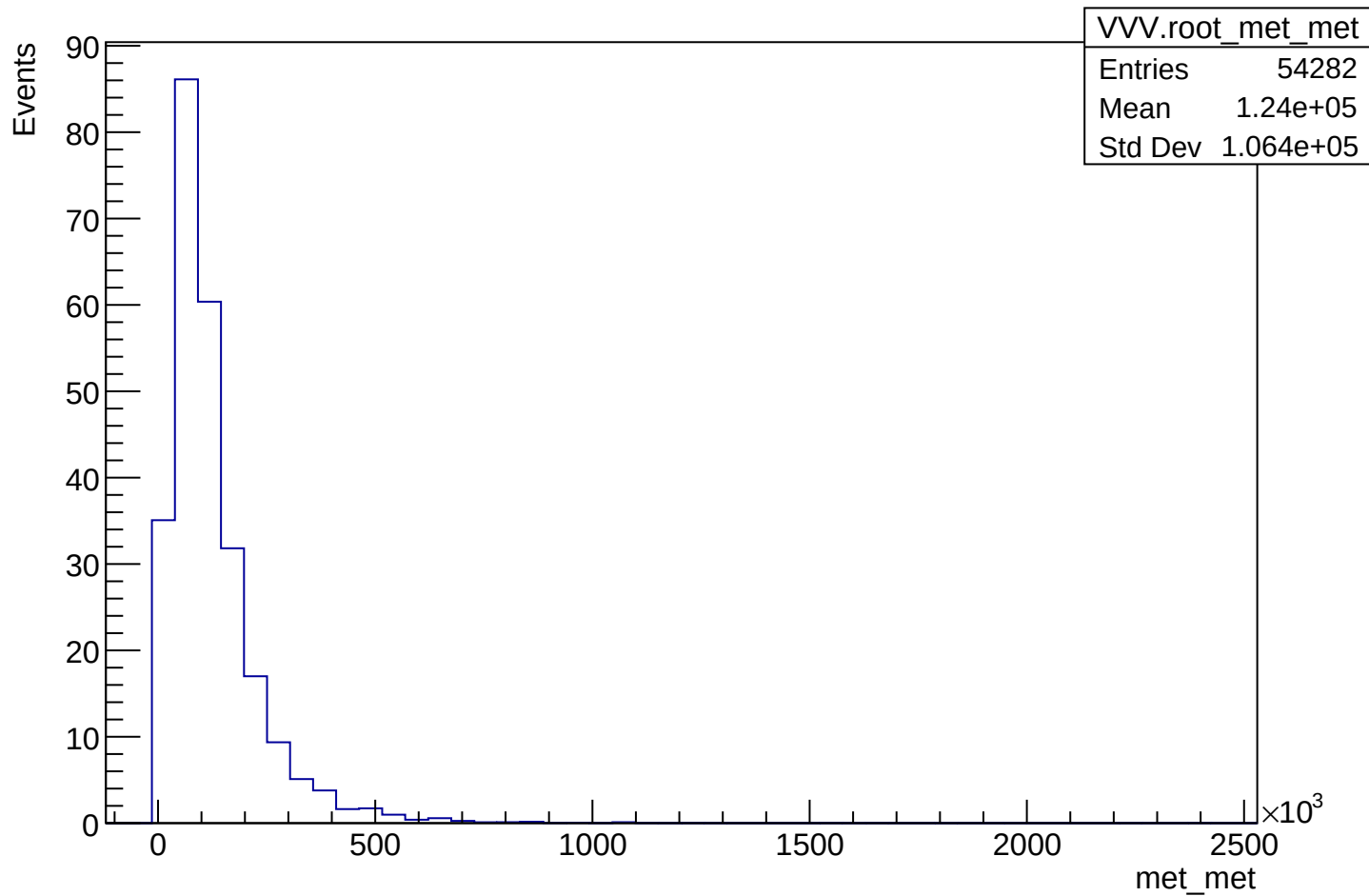


m_ttl

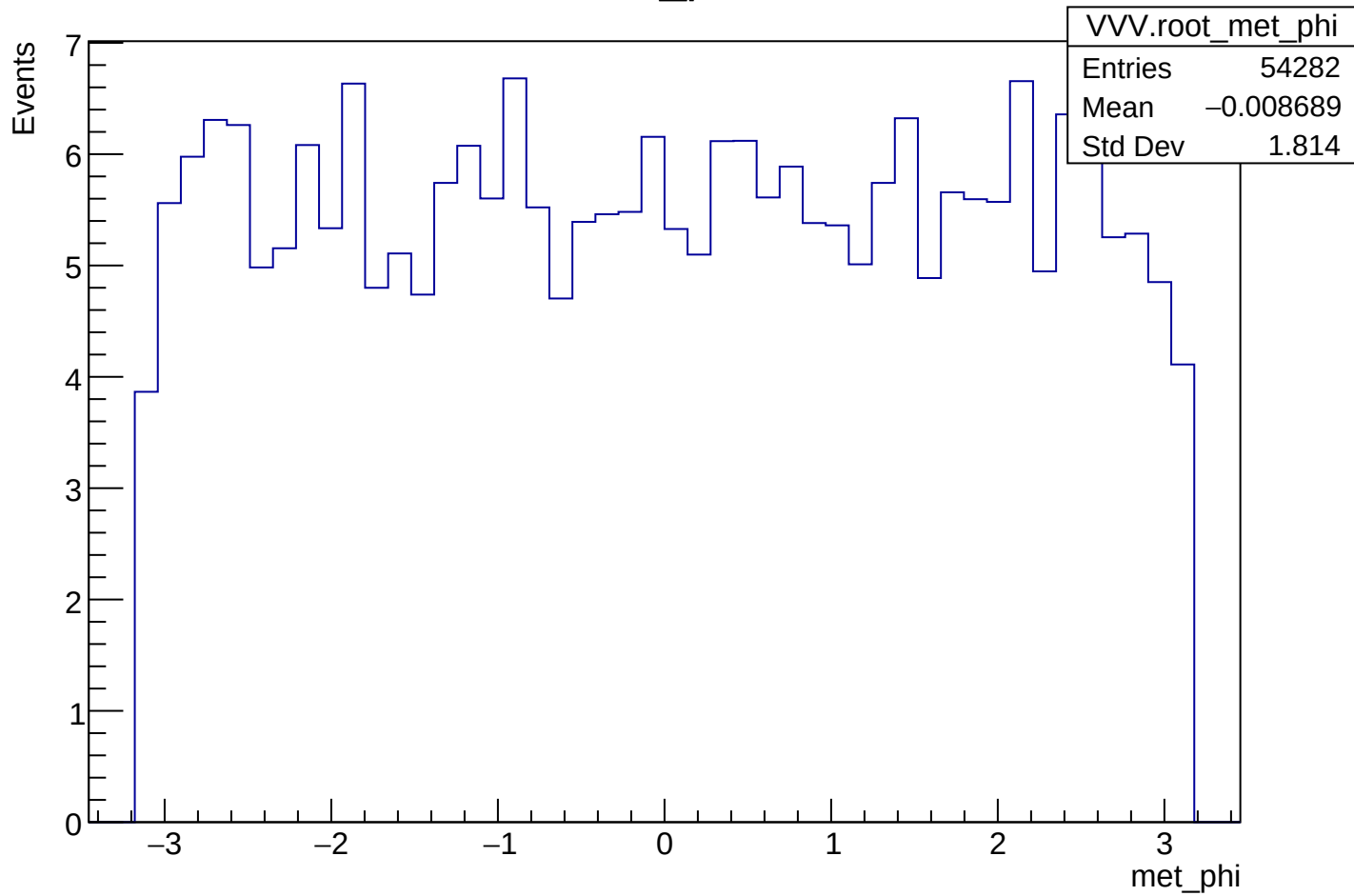
Events



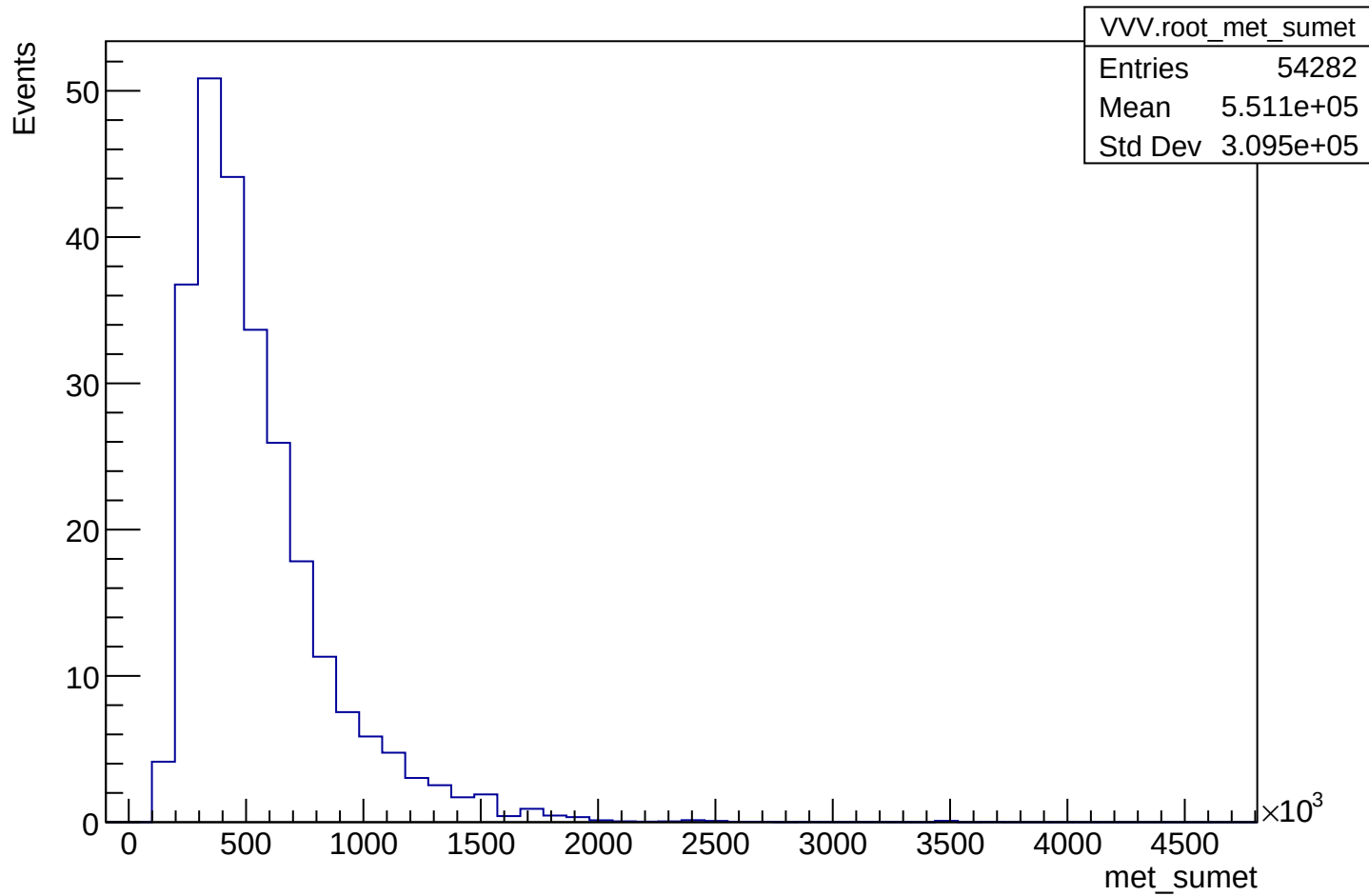
met_met



met_phi



met_sumet



n_jet

Events

120

100

80

60

40

20

0

2

3

4

5

6

7

8

9

10

11

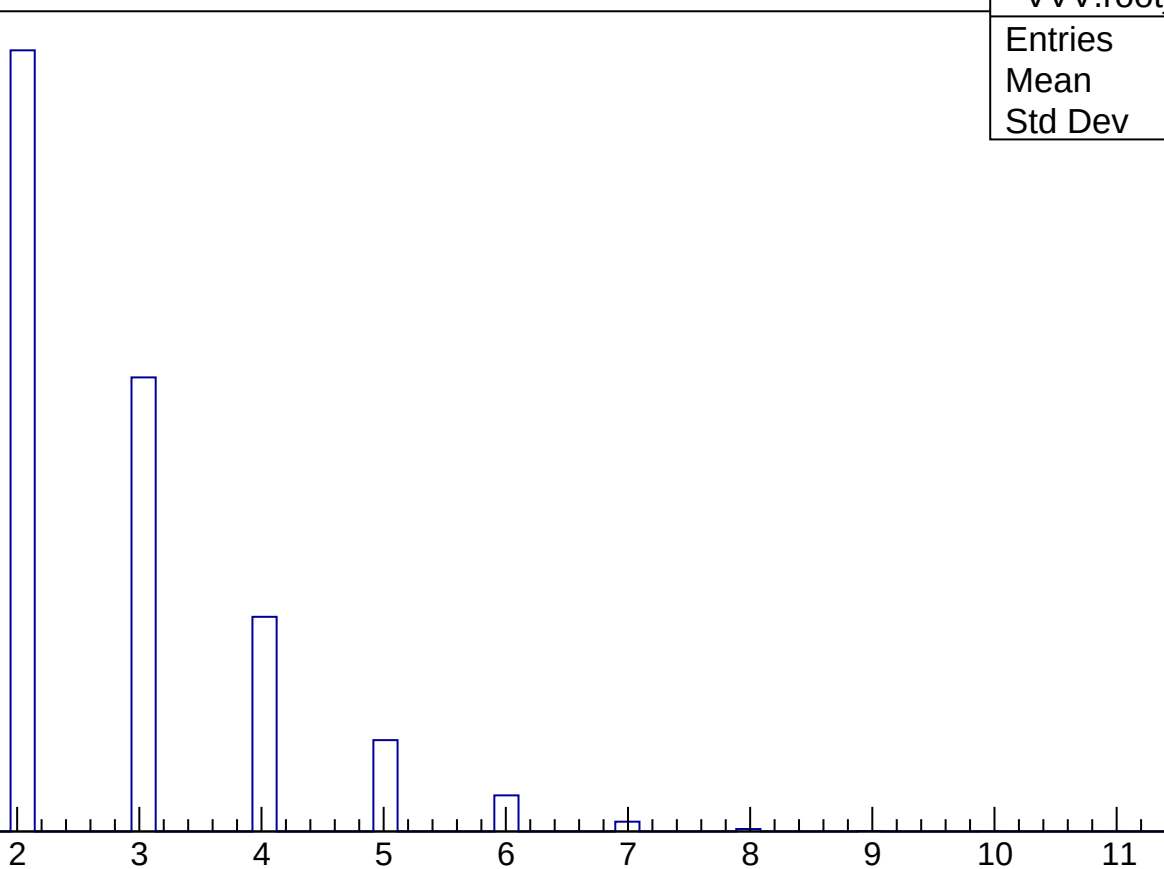
n_jet

VVV.root_n_jet

Entries 54282

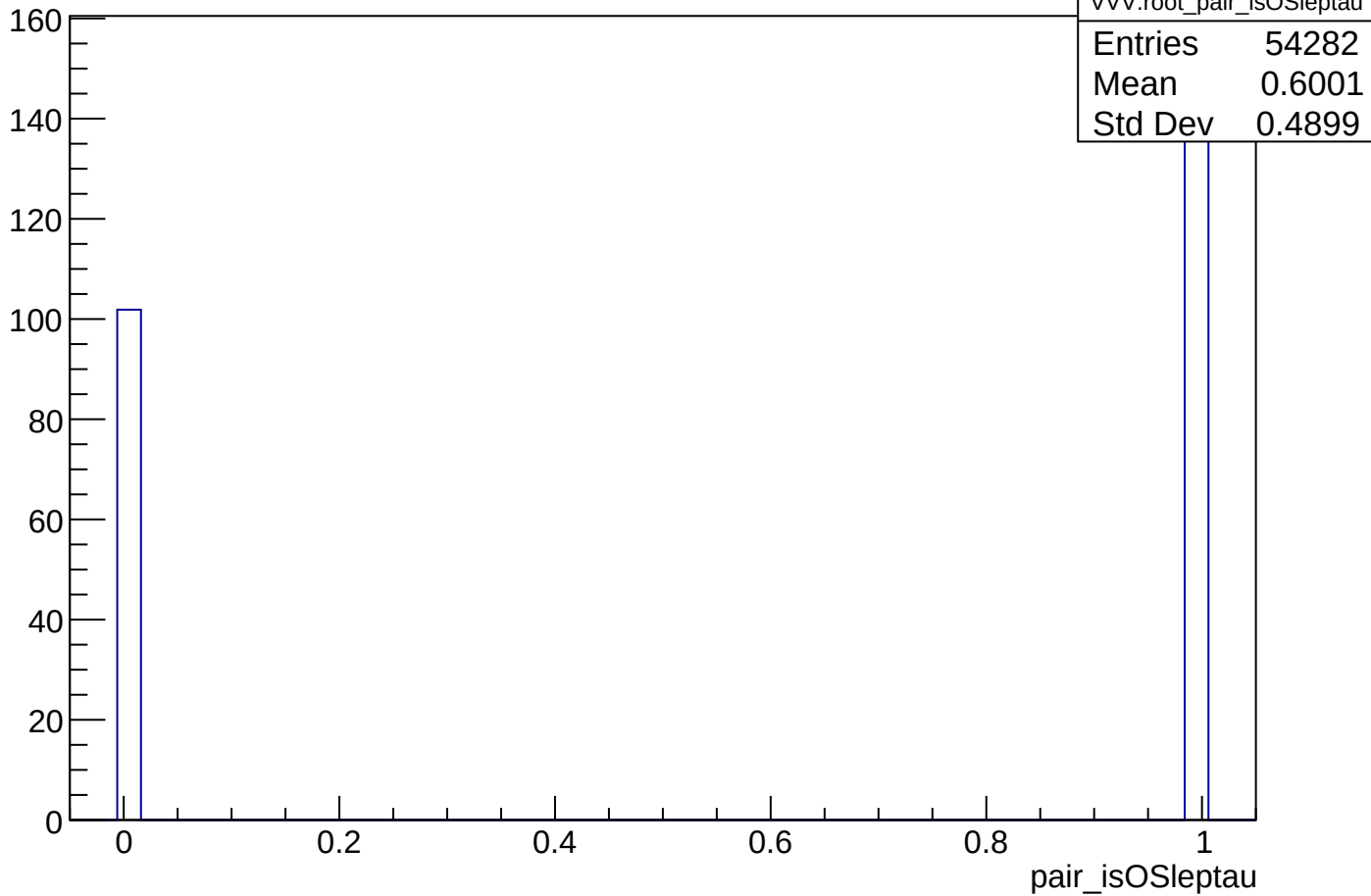
Mean 2.861

Std Dev 1.091



pair_isOSleptau

Events



pair_isOStaus

Events

180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

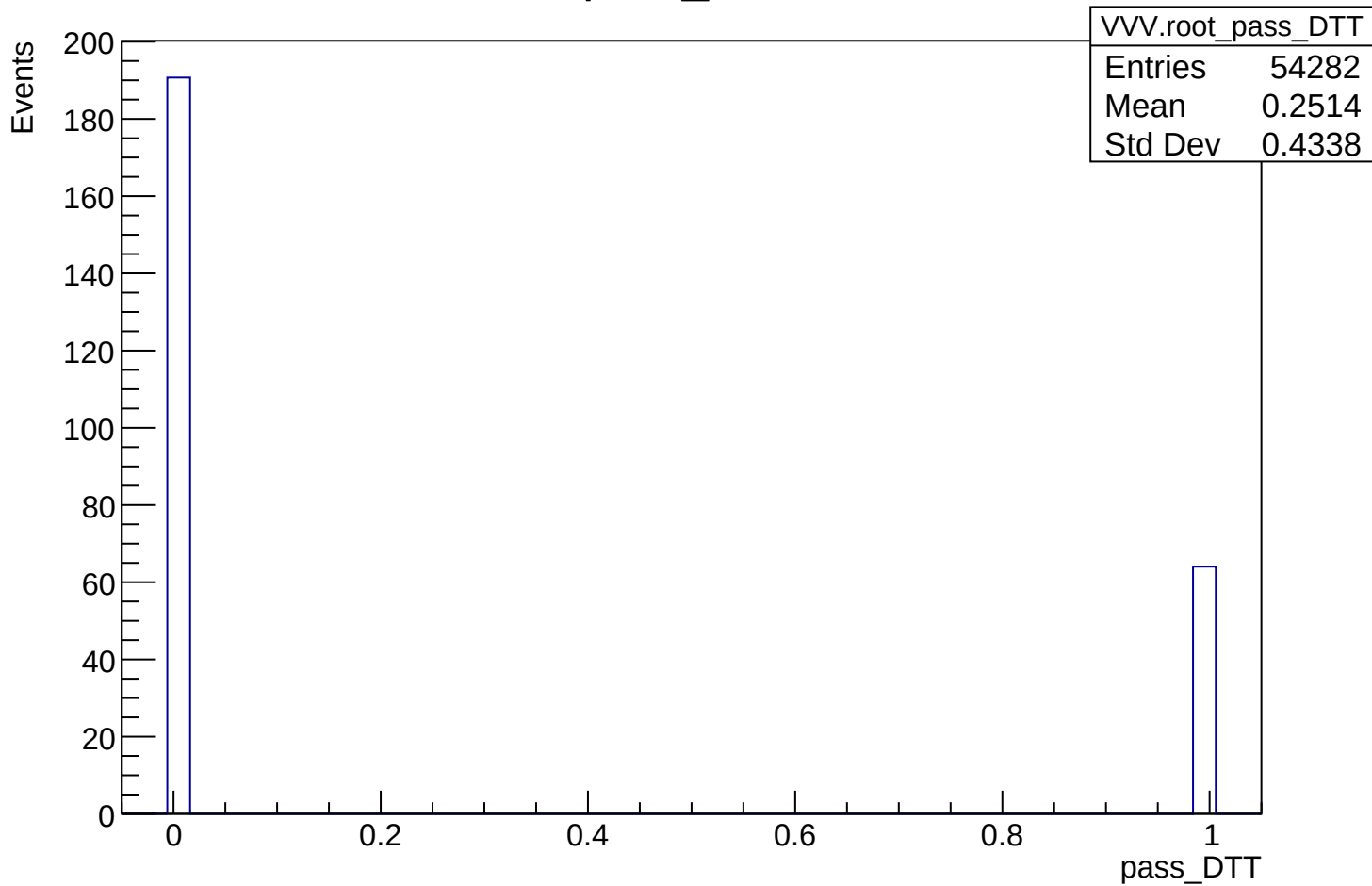
0.8

1

pair_isOStaus

VVV.root_pair_isOStaus	
Entries	54282
Mean	0.706
Std Dev	0.4556

pass_DTT



pass_LTT

Events

220
200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

pass_LTT

VVV.root_pass_LTT	
Entries	54282
Mean	0.8443
Std Dev	0.3626

pass_SLT

Events

200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

pass_SLT

VVV.root_pass_SLT	
Entries	54282
Mean	0.8014
Std Dev	0.399

pass_STT

Events

200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

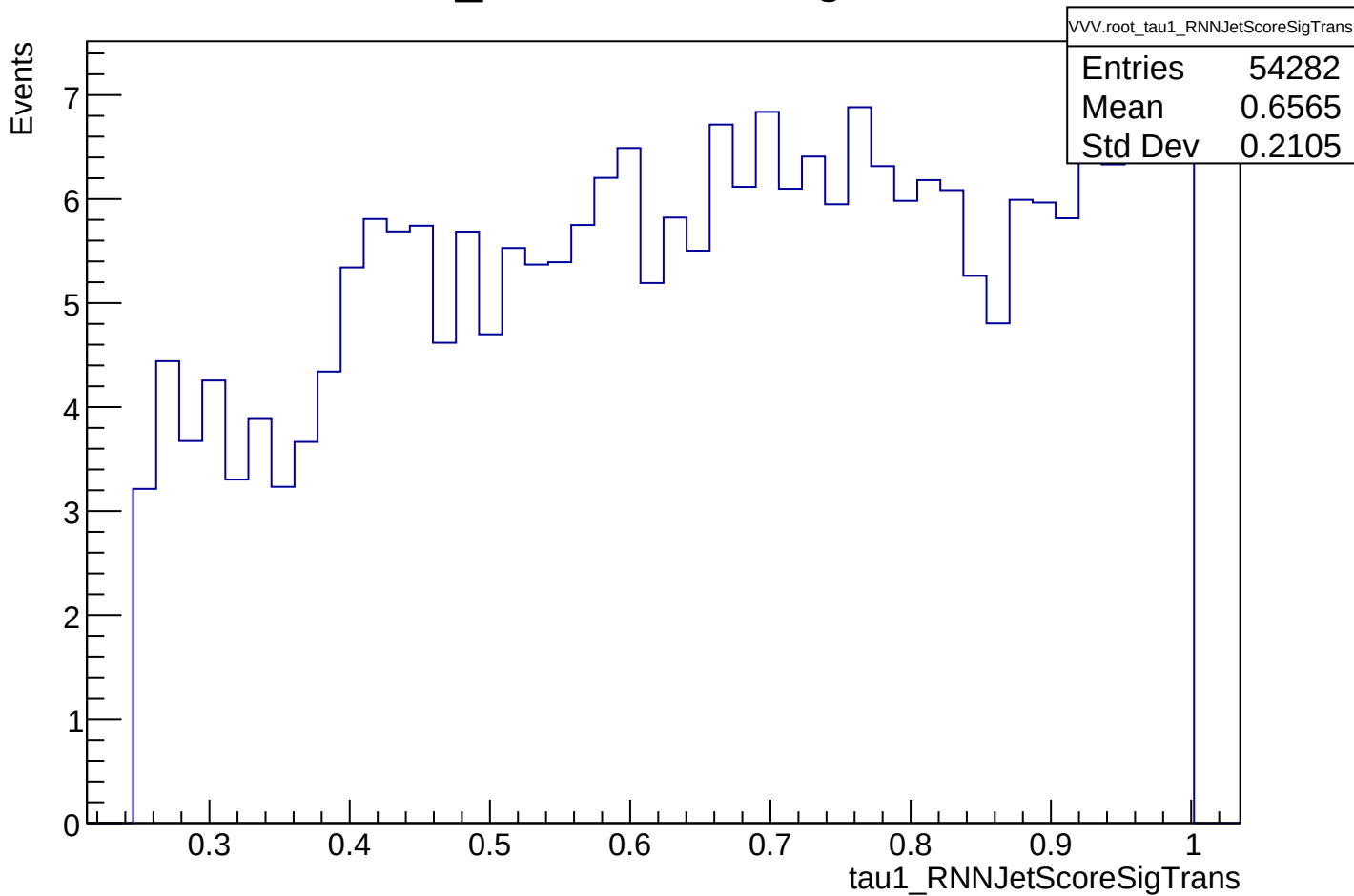
1

pass_STT

VVV.root_pass_STT	
Entries	54282
Mean	0.1953
Std Dev	0.3964

50

tau1_RNNJetScoreSigTrans



tau1_charge

Events

120

100

80

60

40

20

0

-1

-0.5

0

0.5

1

tau1_charge

VVV.root_tau1_charge

Entries 54282

Mean -0.03279

Std Dev 0.9995

120

100

80

60

40

20

0

tau1_effSF

Events

120
100
80
60
40
20
0

0.8

0.9

1.0

1.1

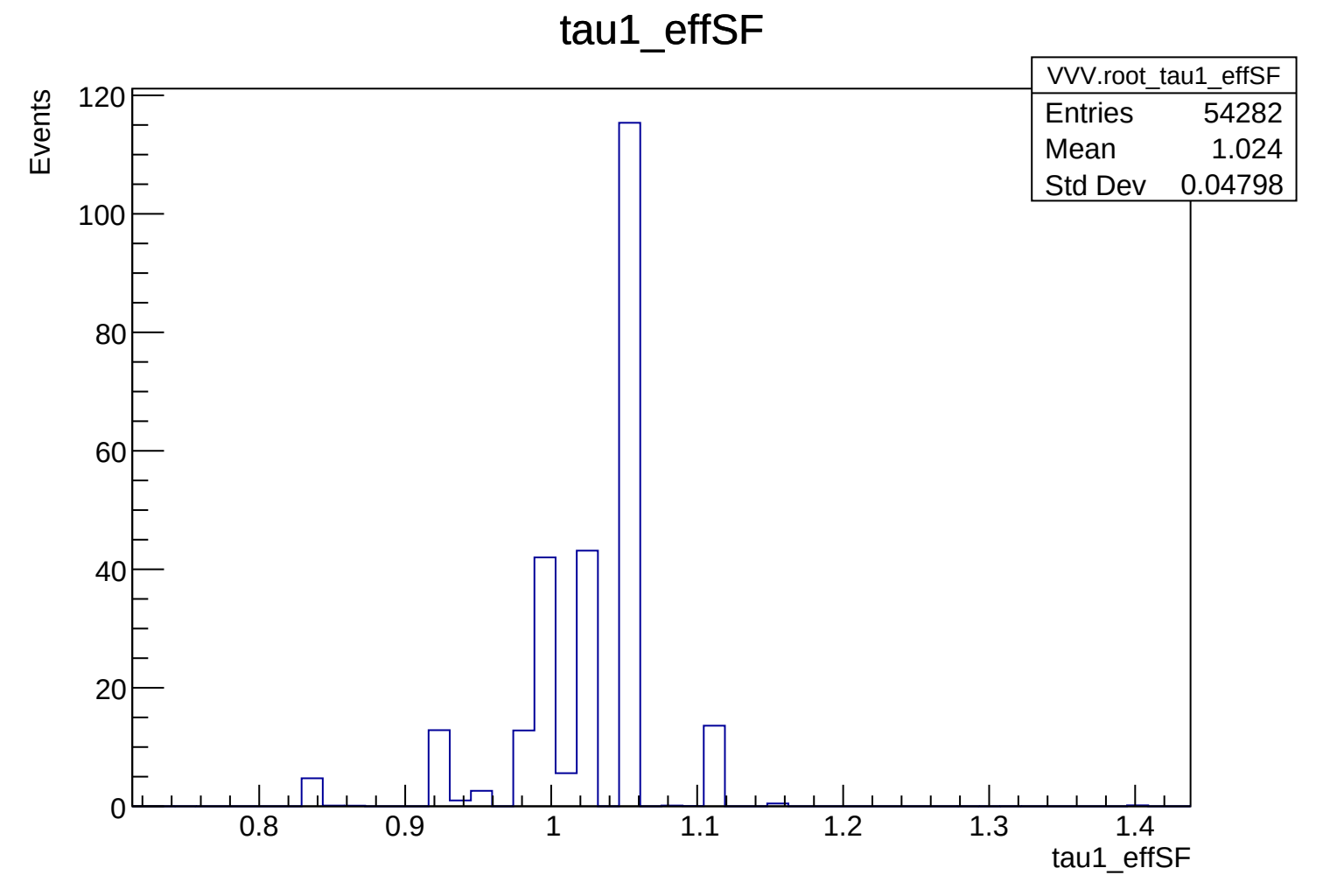
1.2

1.3

1.4

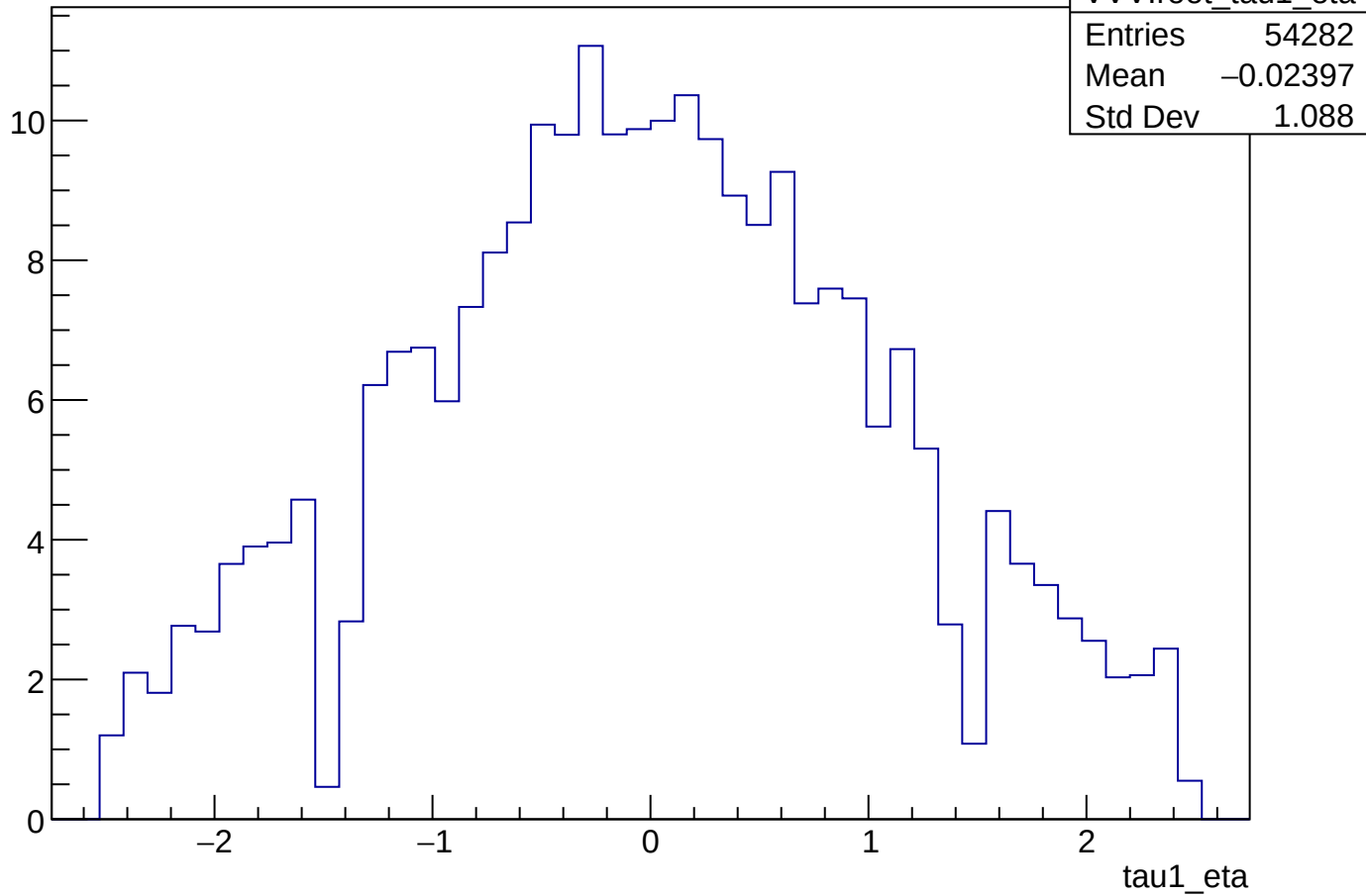
tau1_effSF

VVV.root_tau1_effSF	
Entries	54282
Mean	1.024
Std Dev	0.04798



tau1_eta

Events



tau1_nprong

Events

200
180
160
140
120
100
80
60
40
20
0

1

1.5

2

2.5

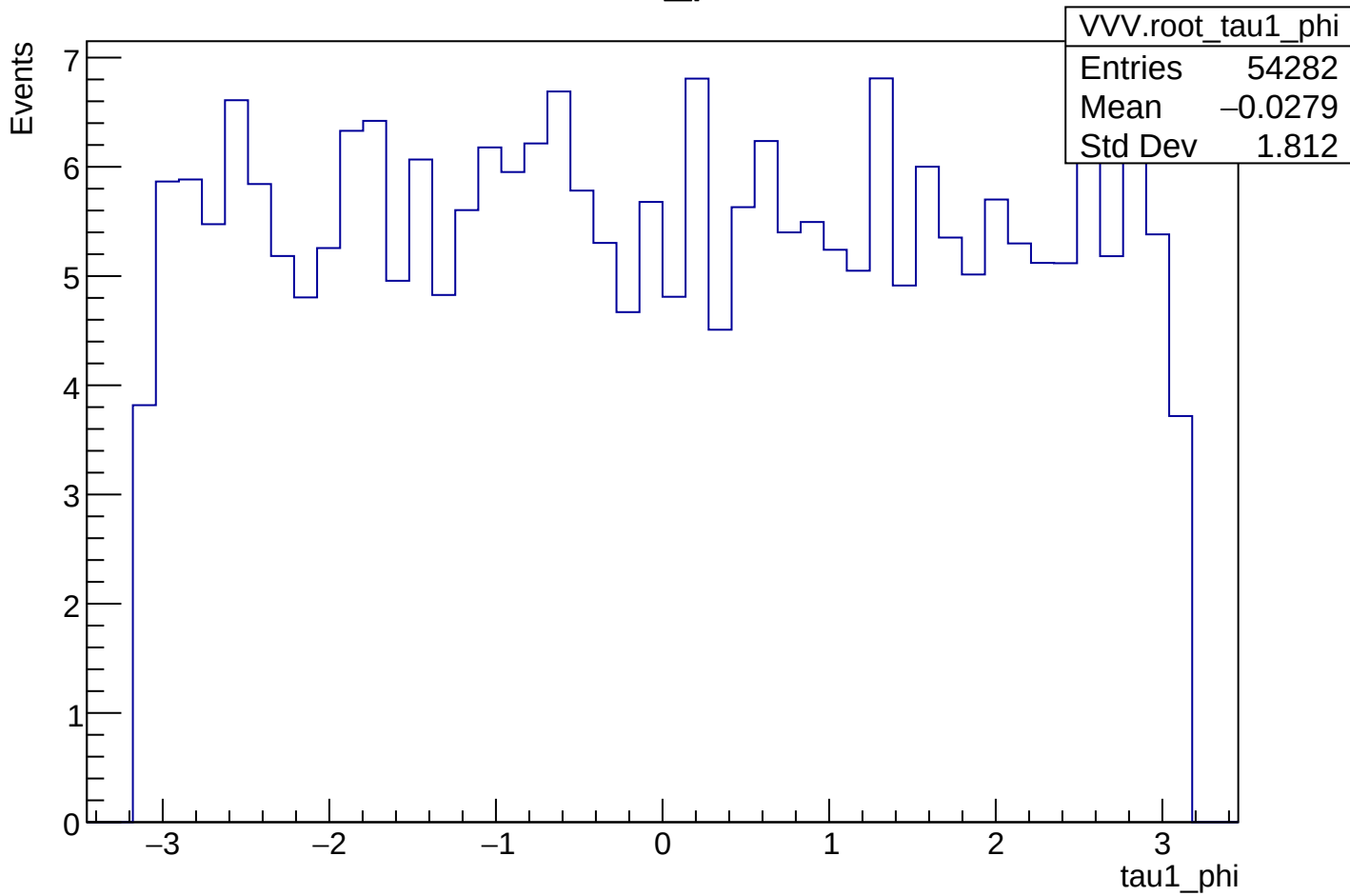
3

tau1_nprong

VVV.root_tau1_nprong	
Entries	54282
Mean	1.44
Std Dev	0.8287

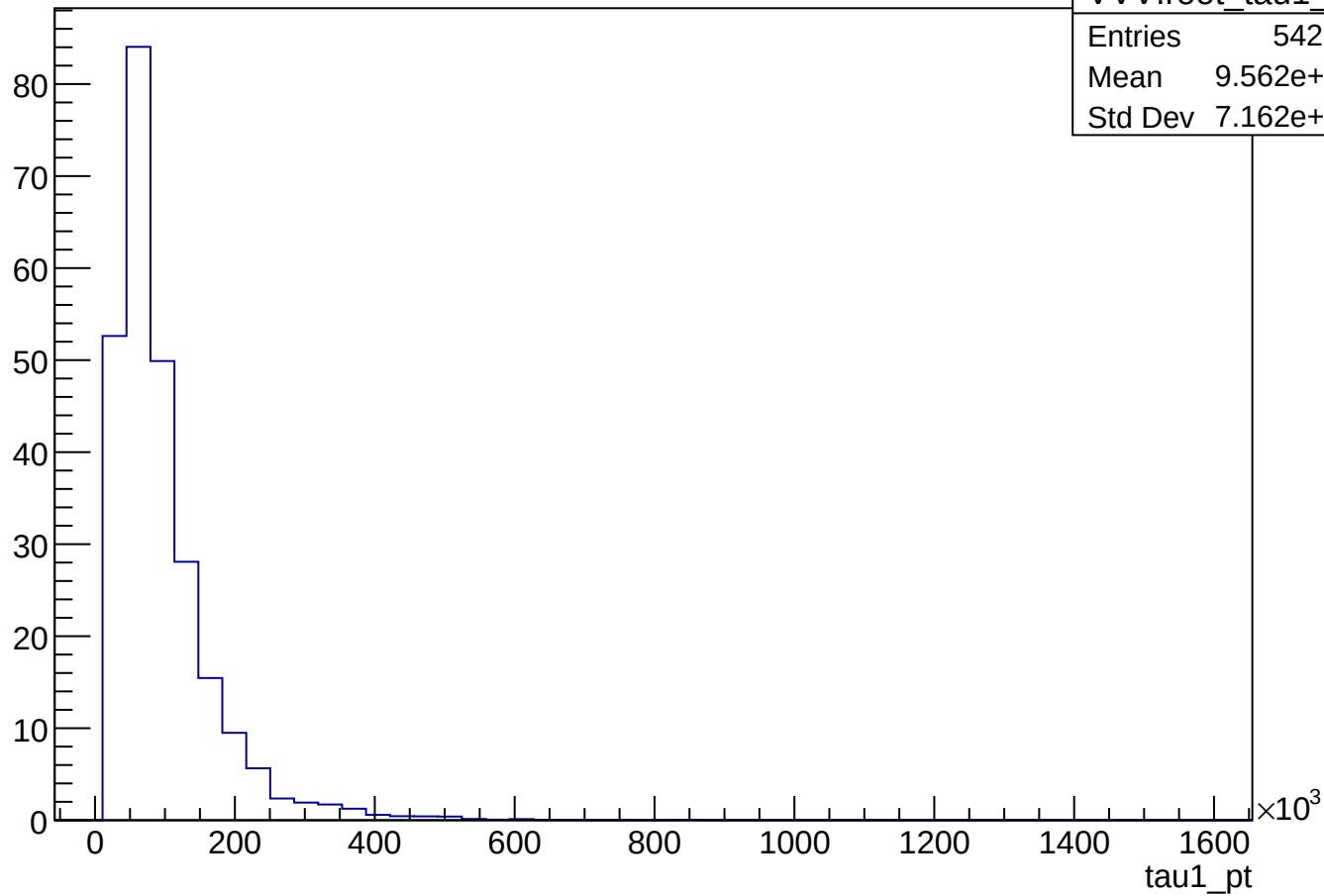
56

tau1_phi



tau1_pt

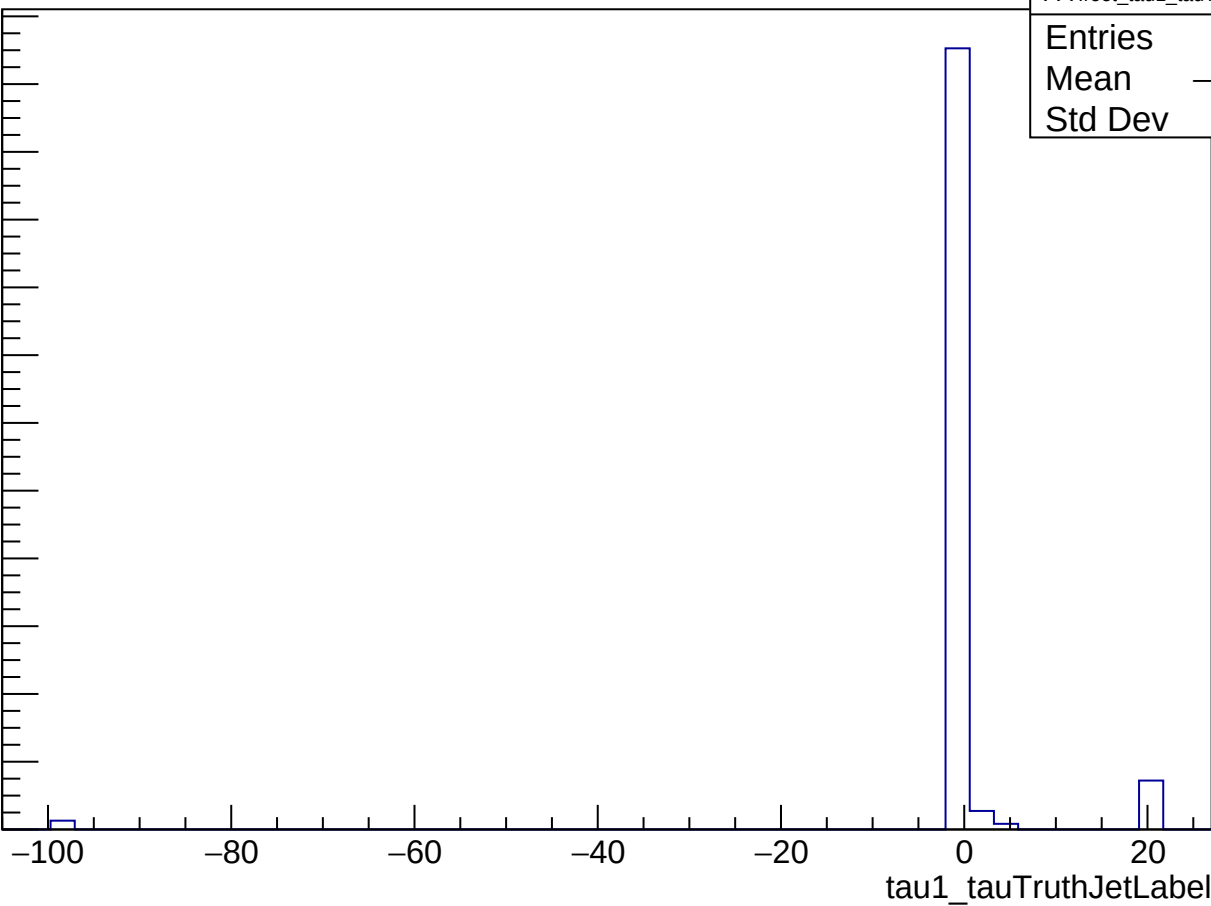
Events



tau1_tauTruthJetLabel

Events

240
220
200
180
160
140
120
100
80
60
40
20
0



VVV.root_tau1_tauTruthJetLabel	
Entries	54282
Mean	-0.6586
Std Dev	11.21

tau1_truthType

Events

250
200
150
100
50
0

0

1

2

3

4

5

tau1_truthType

VVV.root_tau1_truthType	
Entries	54282
Mean	1.062
Std Dev	0.4557

tau1_truth_IsHadronicTau

Events

250
200
150
100
50
0

0

0.2

0.4

0.6

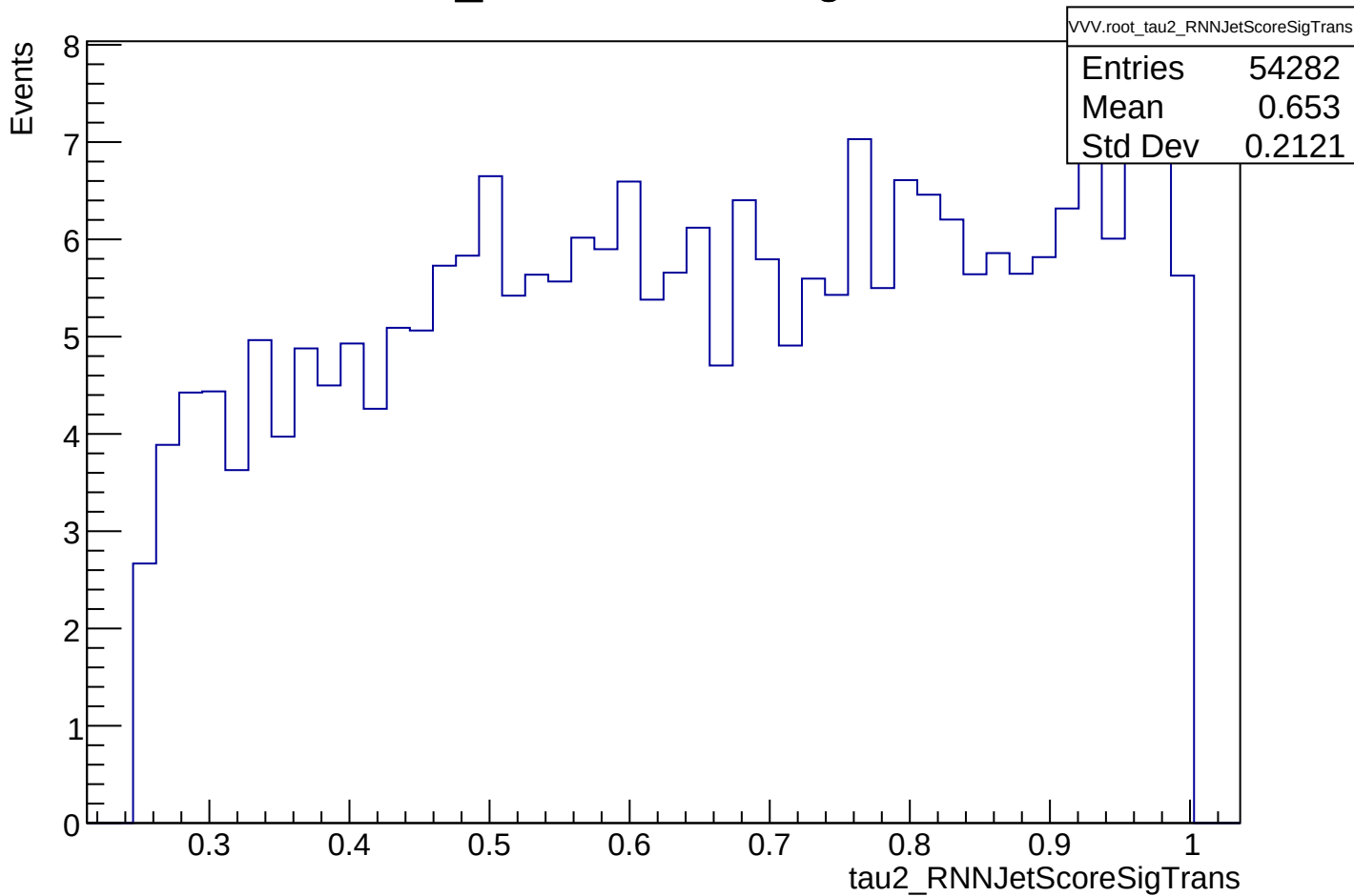
0.8

1

tau1_truth_IsHadronicTau

VVV.root_tau1_truth_IsHadronicTau	
Entries	54282
Mean	0.9779
Std Dev	0.1471

tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

Events

200
180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

tau2_baseline_RNNTight

VVV.root_tau2_baseline_RNNTight	
Entries	54282
Mean	0.8055
Std Dev	0.3958

tau2_charge

Events

120

100

80

60

40

20

0

-1

-0.5

0

0.5

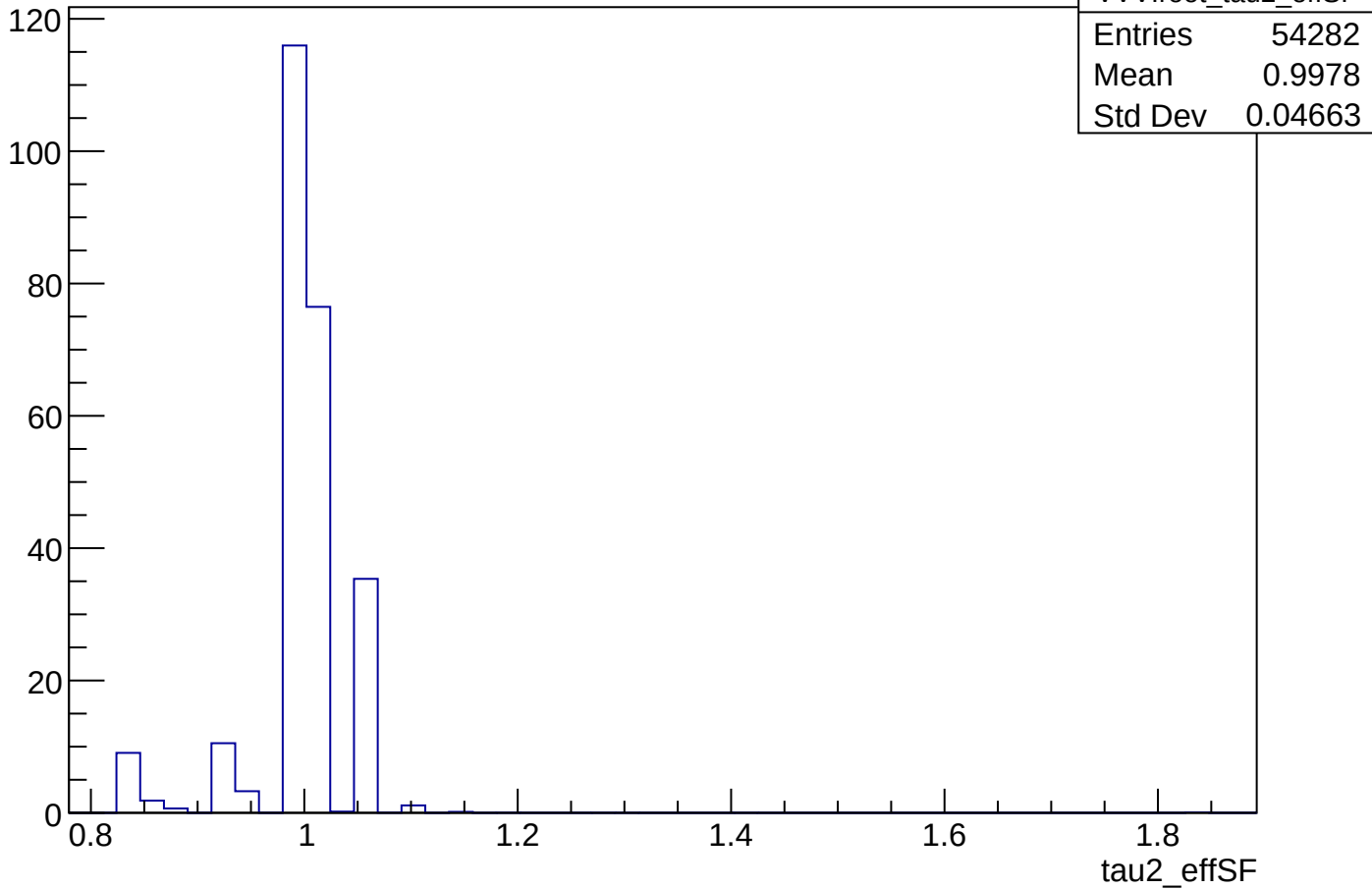
1

tau2_charge

VVV.root_tau2_charge	
Entries	54282
Mean	0.02666
Std Dev	0.9996

tau2_effSF

Events



tau2_eta

Events

10
8
6
4
2
0

-2

-1

0

1

2

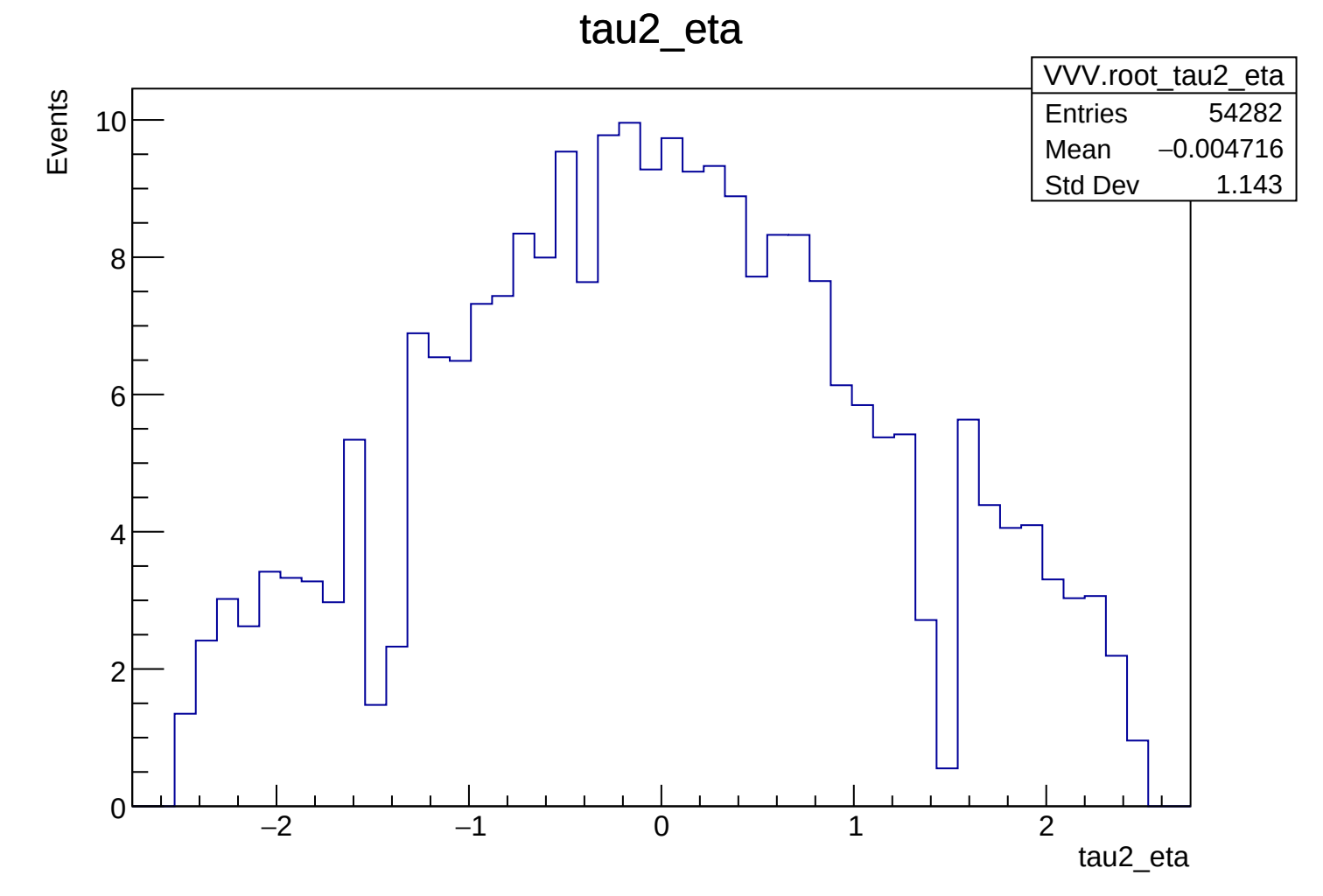
tau2_eta

VVV.root_tau2_eta

Entries 54282

Mean -0.004716

Std Dev 1.143



tau2_nprong

Events

220
200
180
160
140
120
100
80
60
40
20
0

1

1.5

2

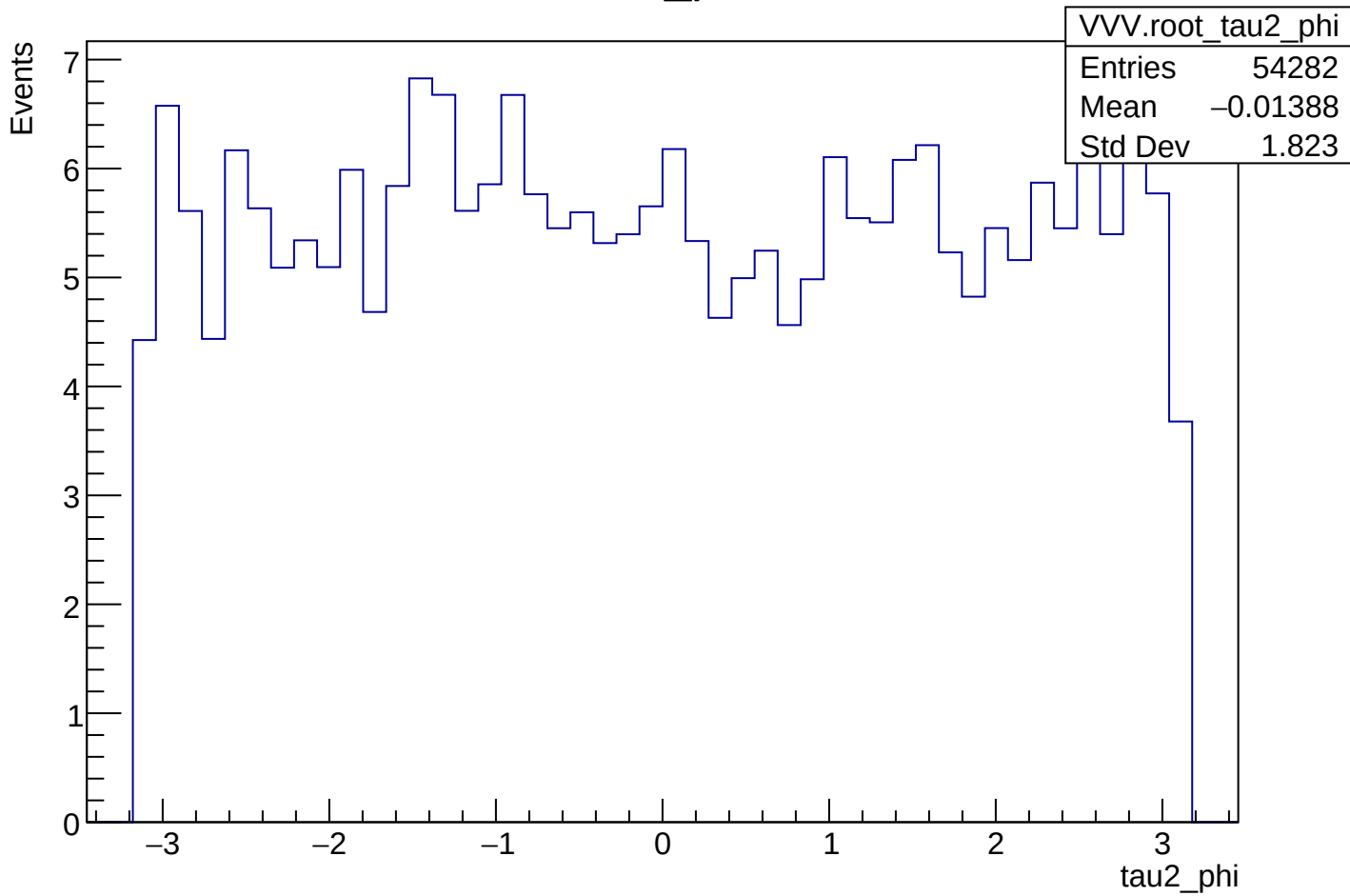
2.5

3

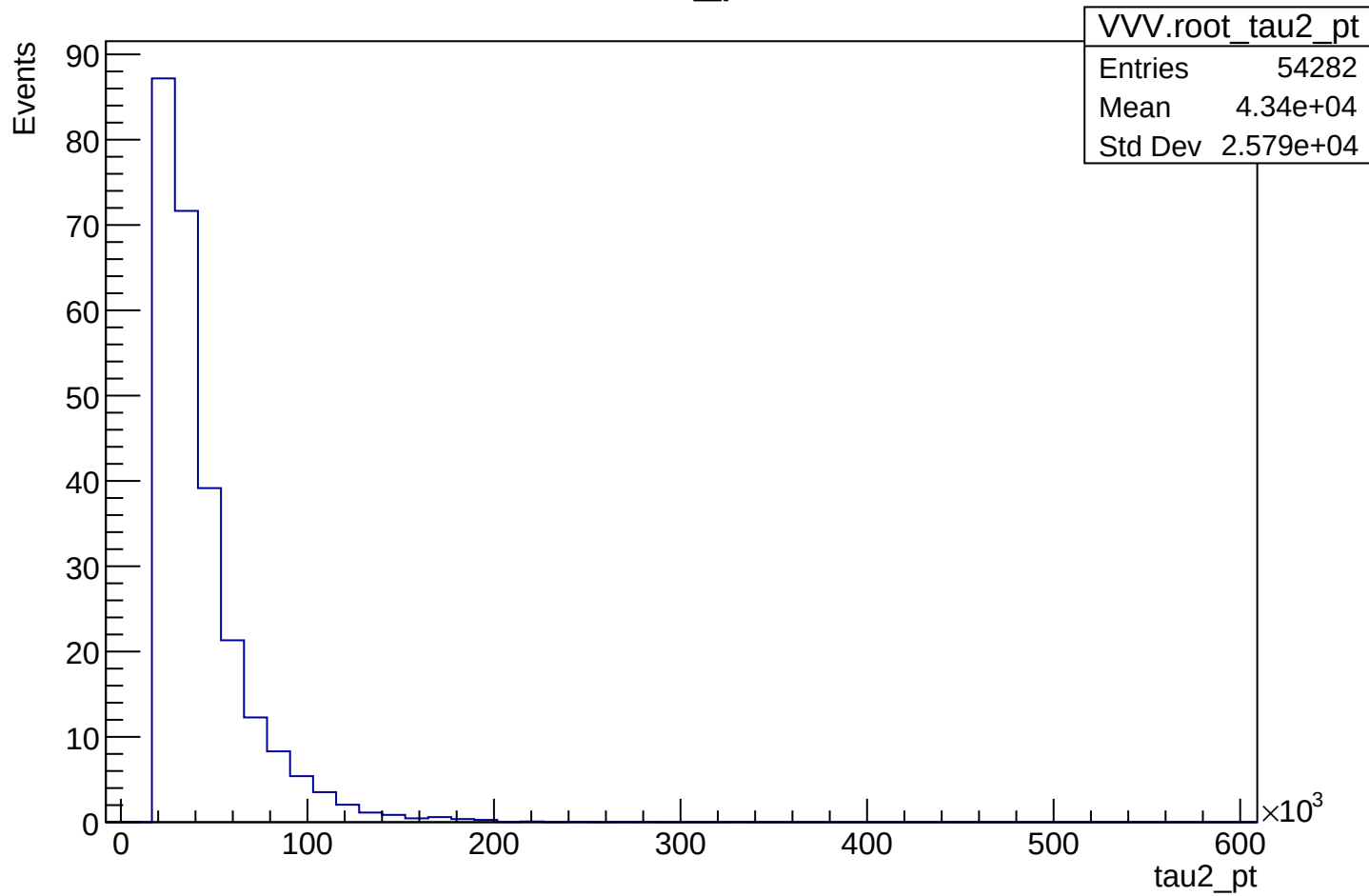
tau2_nprong

VVV.root_tau2_nprong	
Entries	54282
Mean	1.353
Std Dev	0.7628

tau2_phi



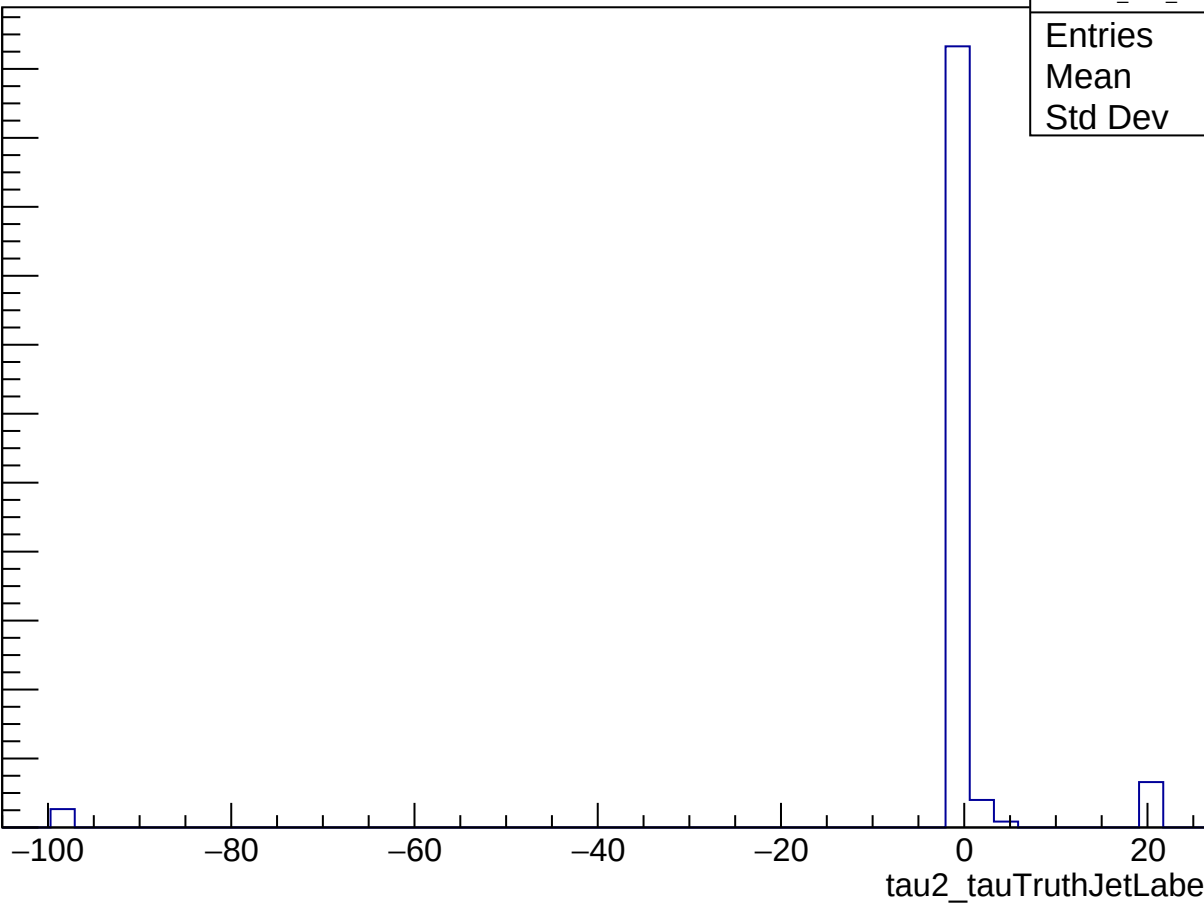
tau2_pt



tau2_tauTruthJetLabel

Events

220
200
180
160
140
120
100
80
60
40
20
0



VVV.root_tau2_tauTruthJetLabel	
Entries	54282
Mean	-1.792
Std Dev	15.02

tau2_truthType

Events

250
200
150
100
50
0

0

1

2

3

4

5

tau2_truthType

VVV.root_tau2_truthType	
Entries	54282
Mean	1.128
Std Dev	0.6942

tau2_truth_IsHadronicTau

Events

250
200
150
100
50
0

0

0.2

0.4

0.6

0.8

1

tau2_truth_IsHadronicTau

VVV.root_tau2_truth_IsHadronicTau	
Entries	54282
Mean	0.9504
Std Dev	0.217

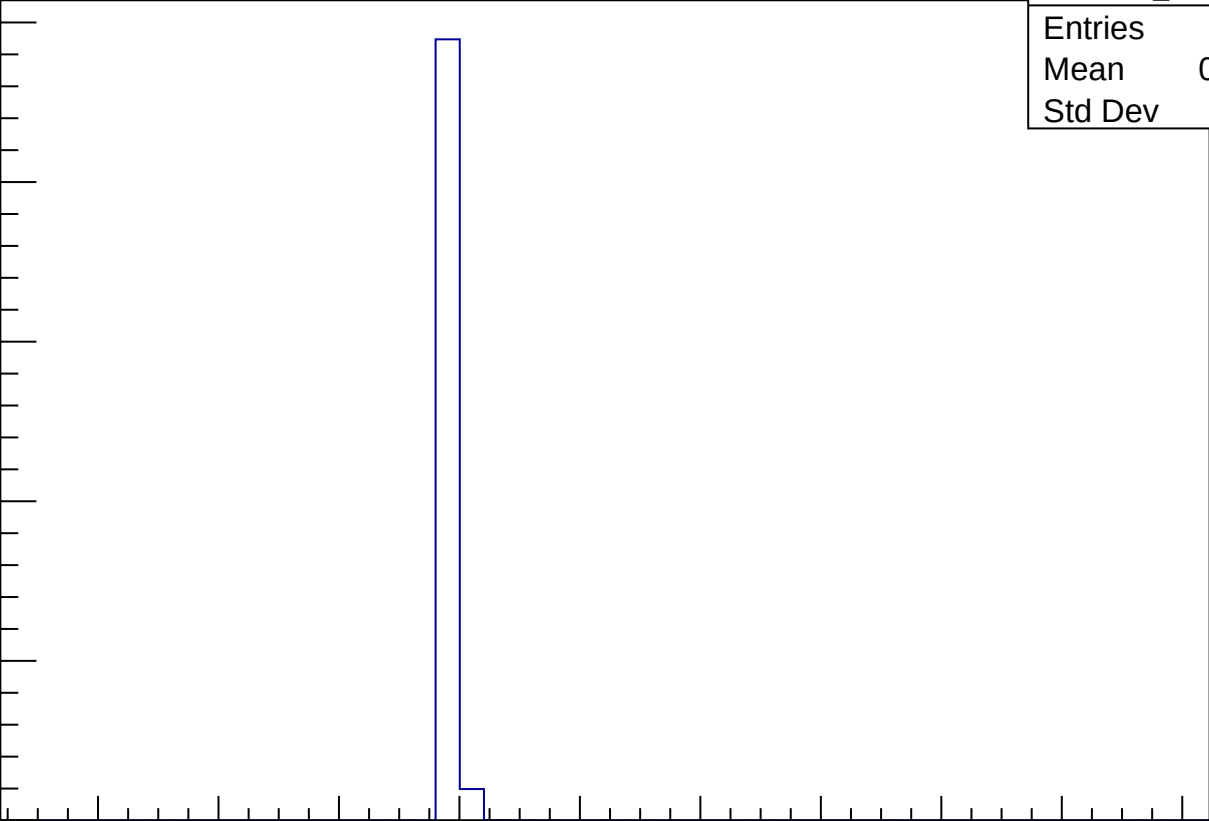
weight_final

Events

250
200
150
100
50
0

-60 -40 -20 0 20 40 60 80 100 120
weight_final

VVV.root_weight_final	
Entries	54282
Mean	0.03698
Std Dev	0.298



weights

Events

250
200
150
100
50
0

-60

-40

-20

0

20

40

60

80

100

120

weights

VVV.root_weights

Entries 54282

Mean 0.03698

Std Dev 0.298



weights_detector

Events

120
100
80
60
40
20
0

0

0.5

1.0

1.5

2.0

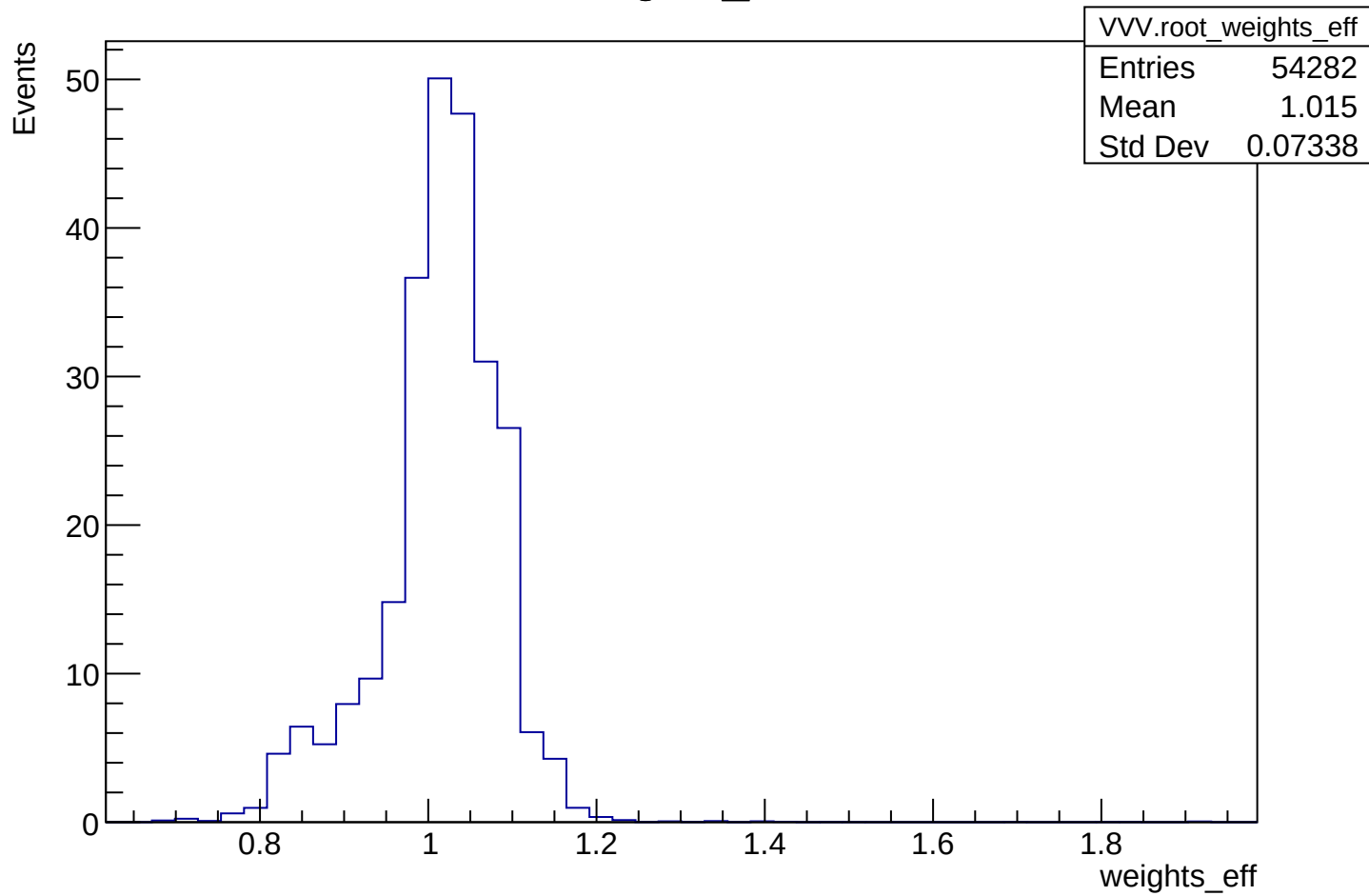
2.5

3.0

weights_detector

VVV.root_weights_detector	
Entries	54282
Mean	0.9983
Std Dev	0.1965

weights_eff



weights_mc

Events

250
200
150
100
50
0

-40

-20

0

20

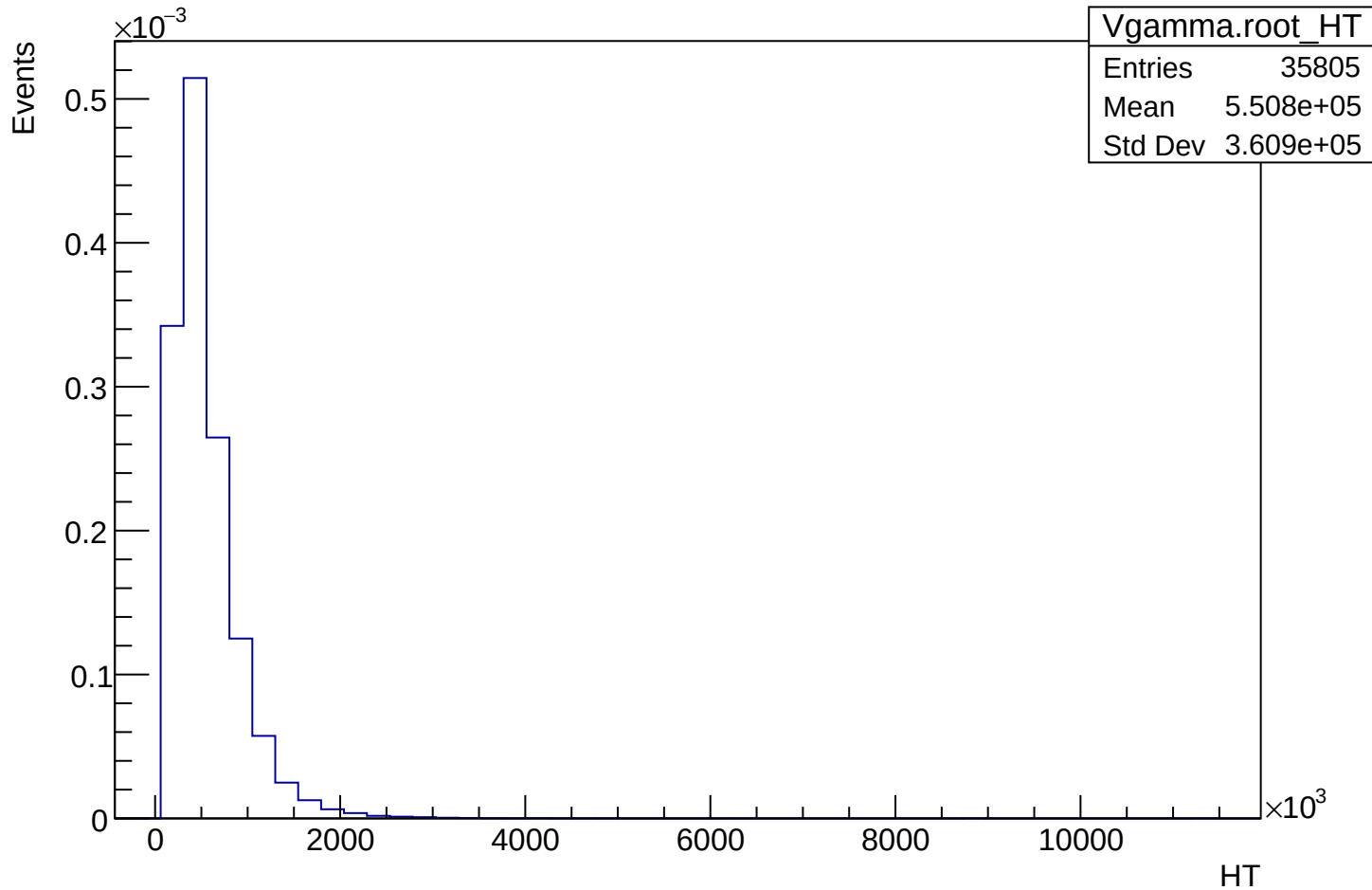
40

60

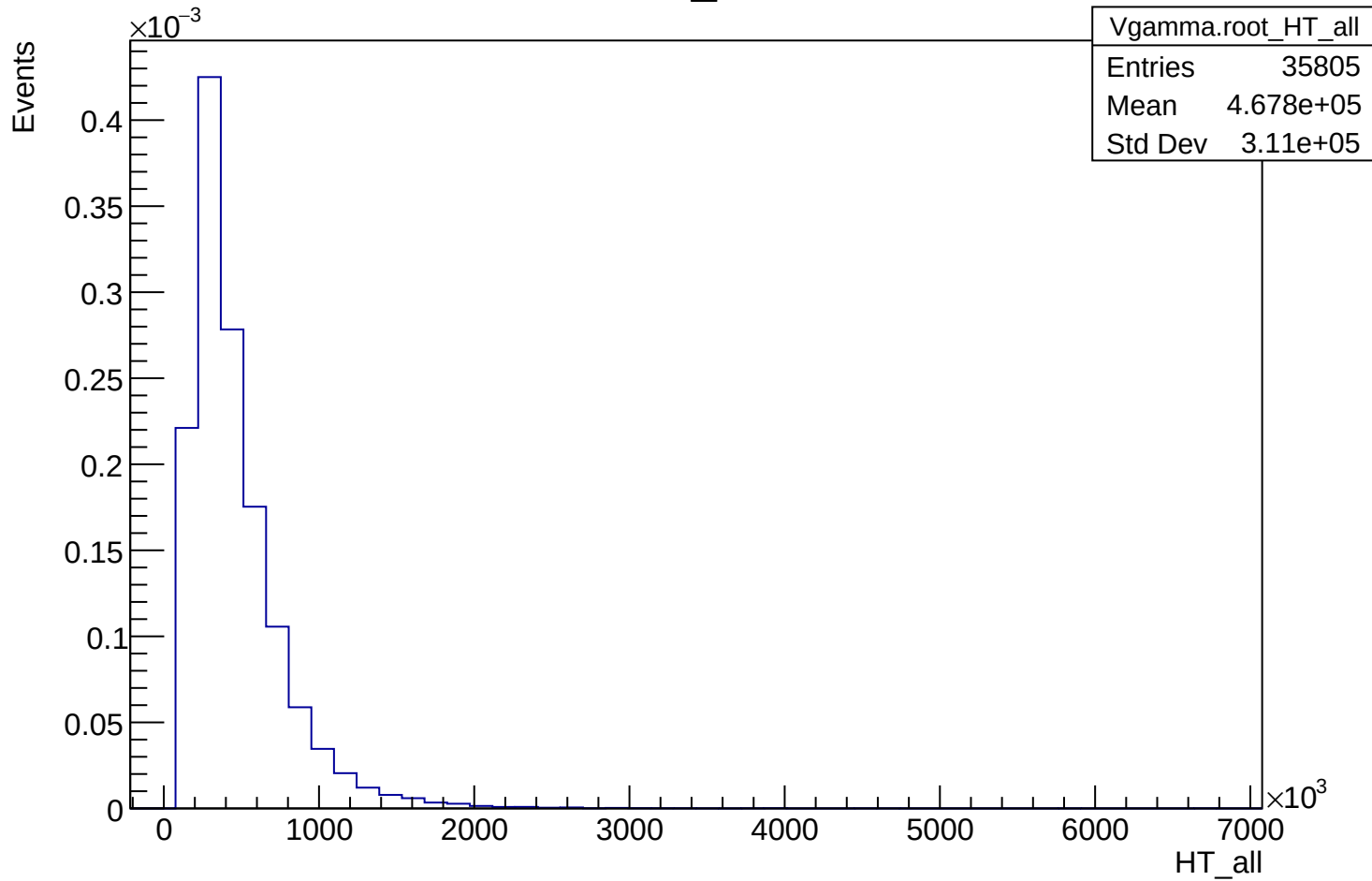
weights_mc

VVV.root_weights_mc	
Entries	54282
Mean	0.03841
Std Dev	0.2705

HT

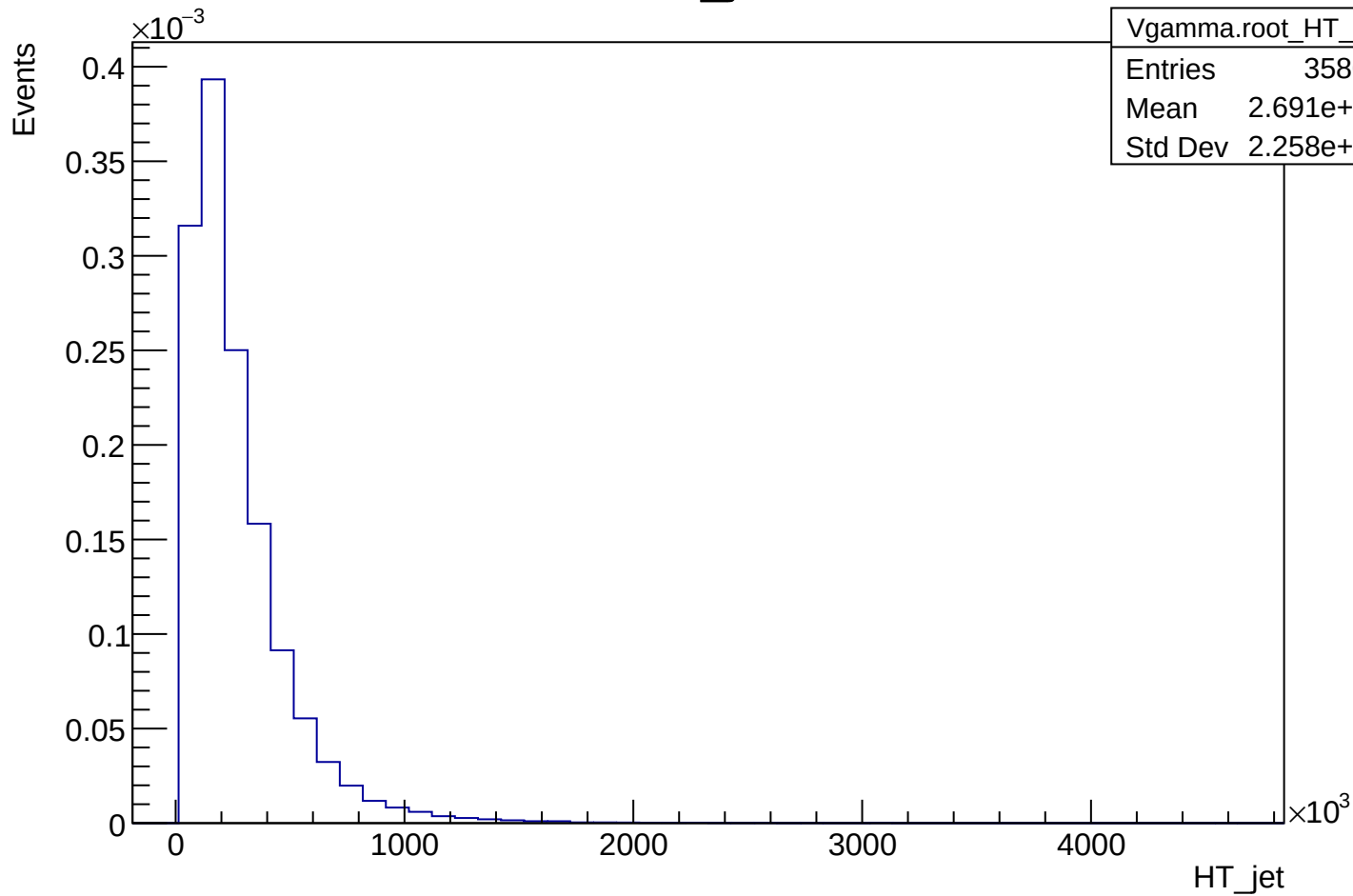


HT_all



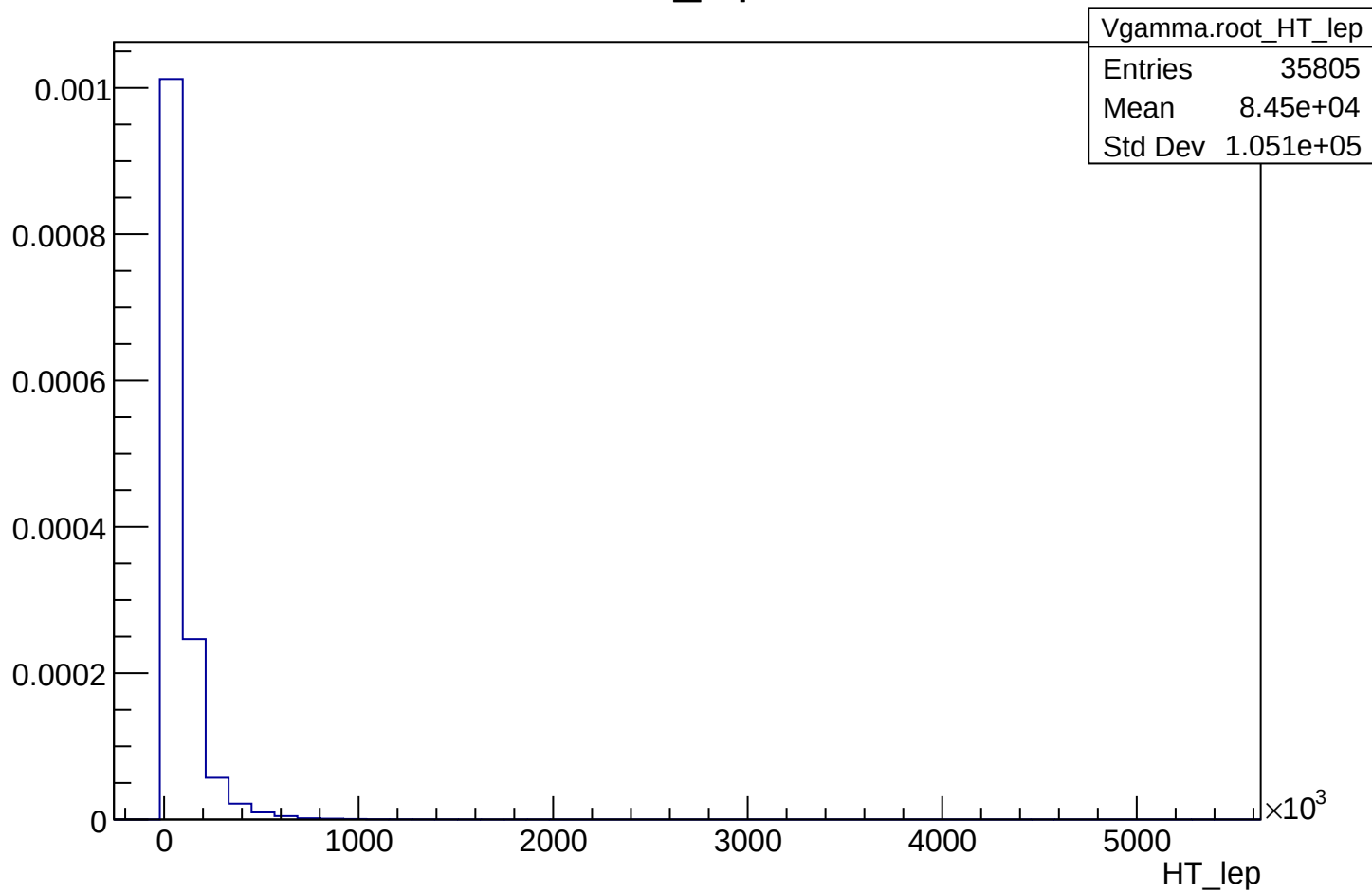
HT_jet

Vgamma.root_HT_jet	
Entries	35805
Mean	2.691e+05
Std Dev	2.258e+05

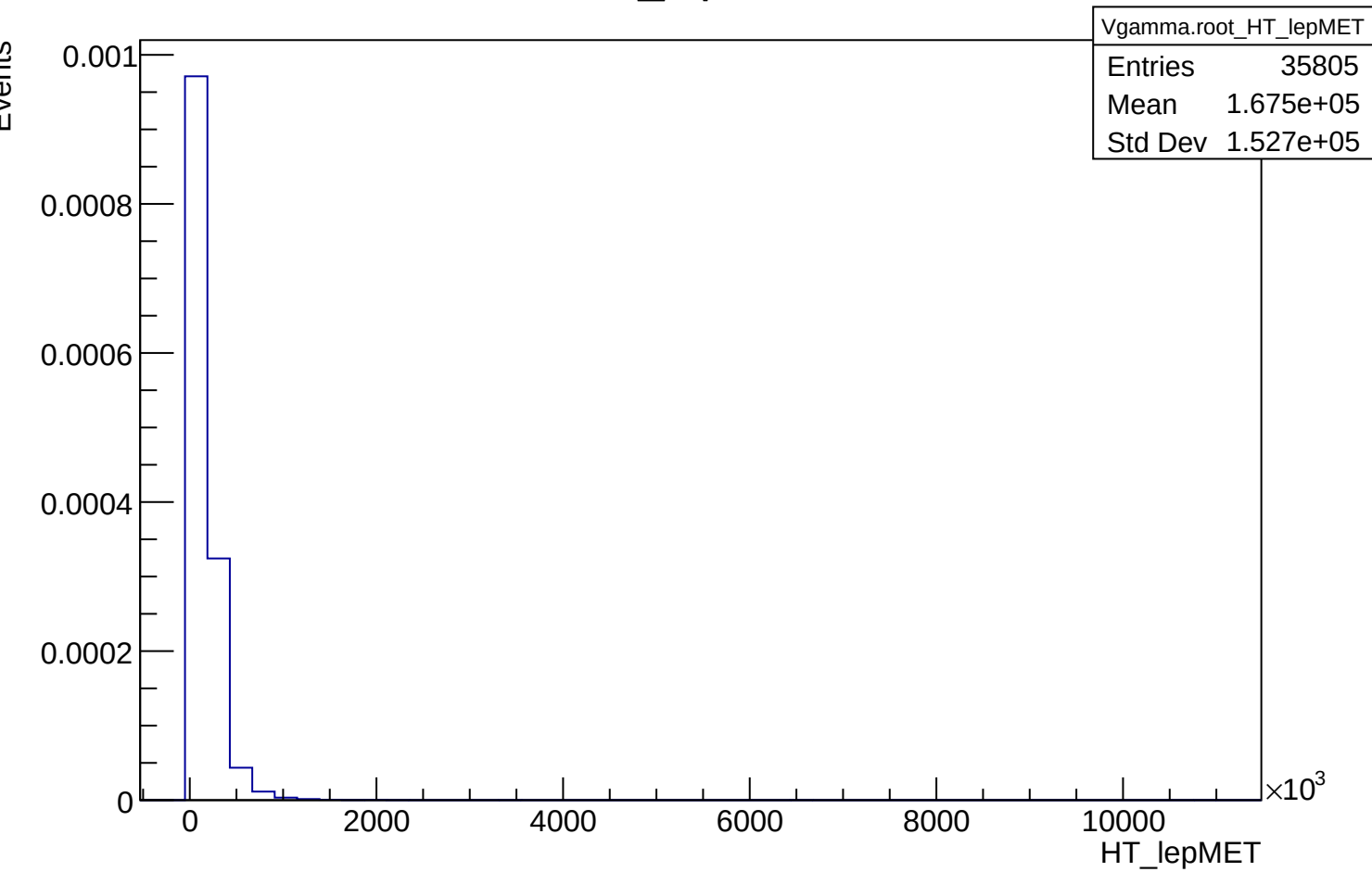


HT_lep

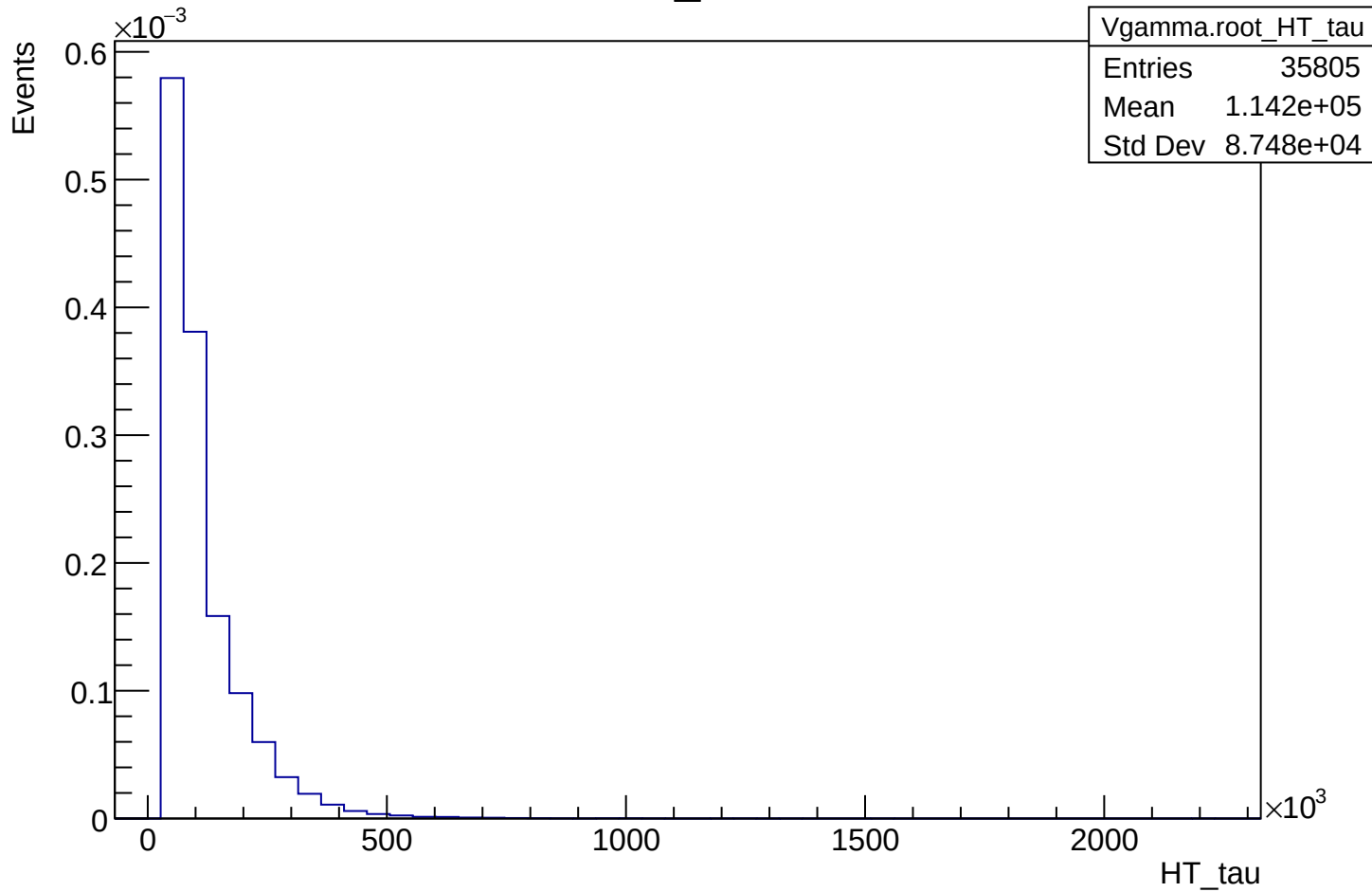
Events



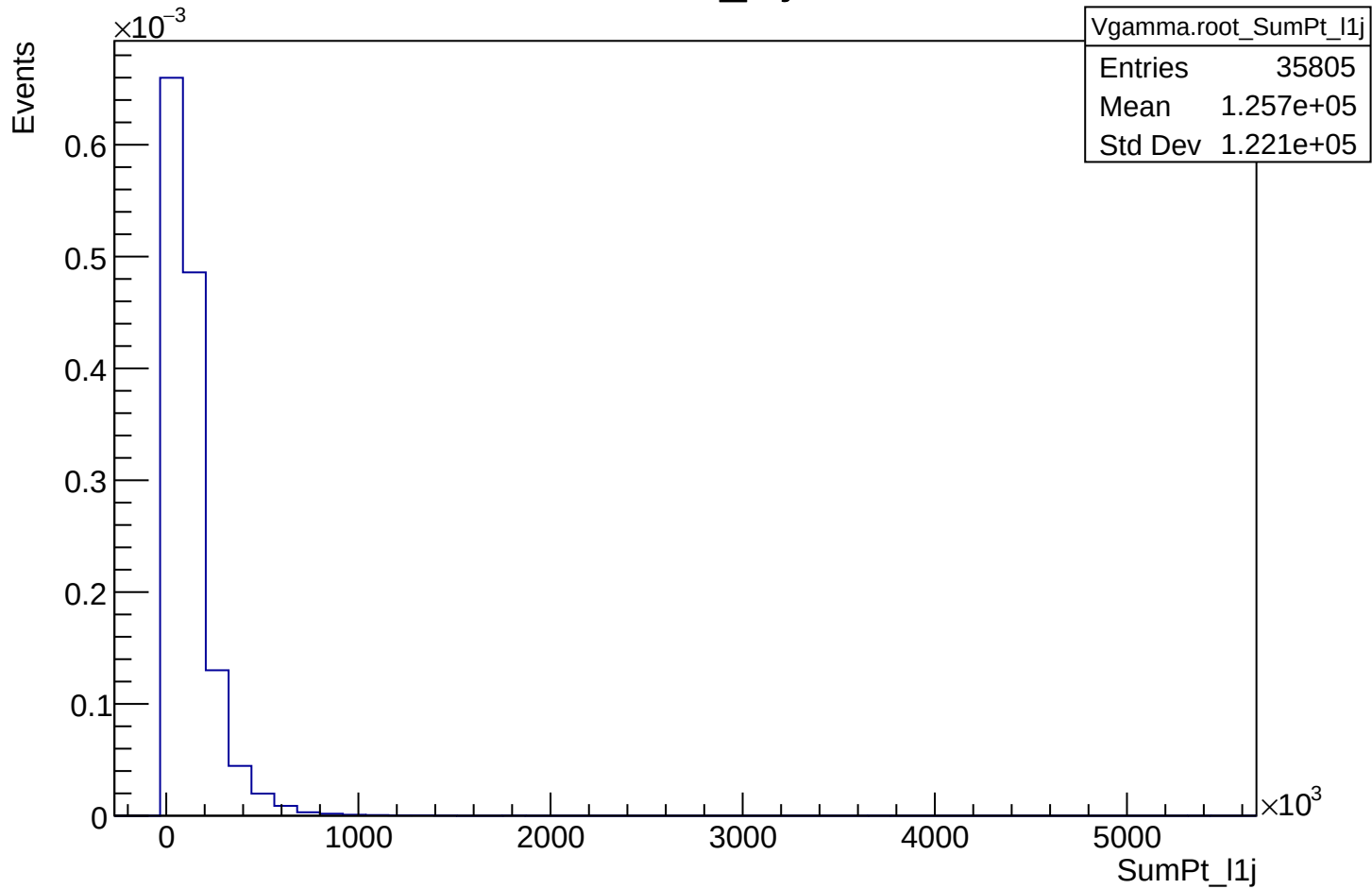
HT_lepMET



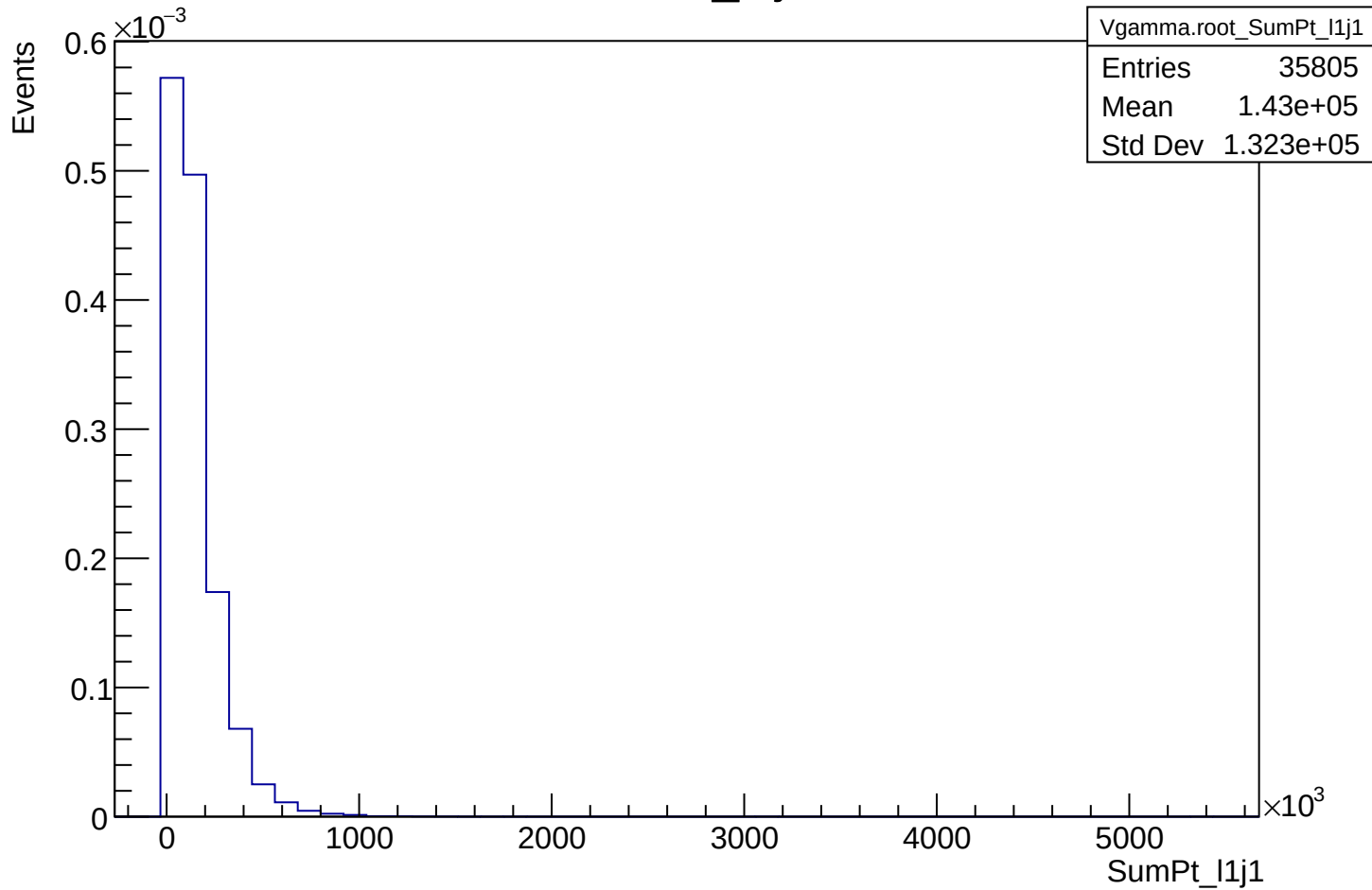
HT_tau



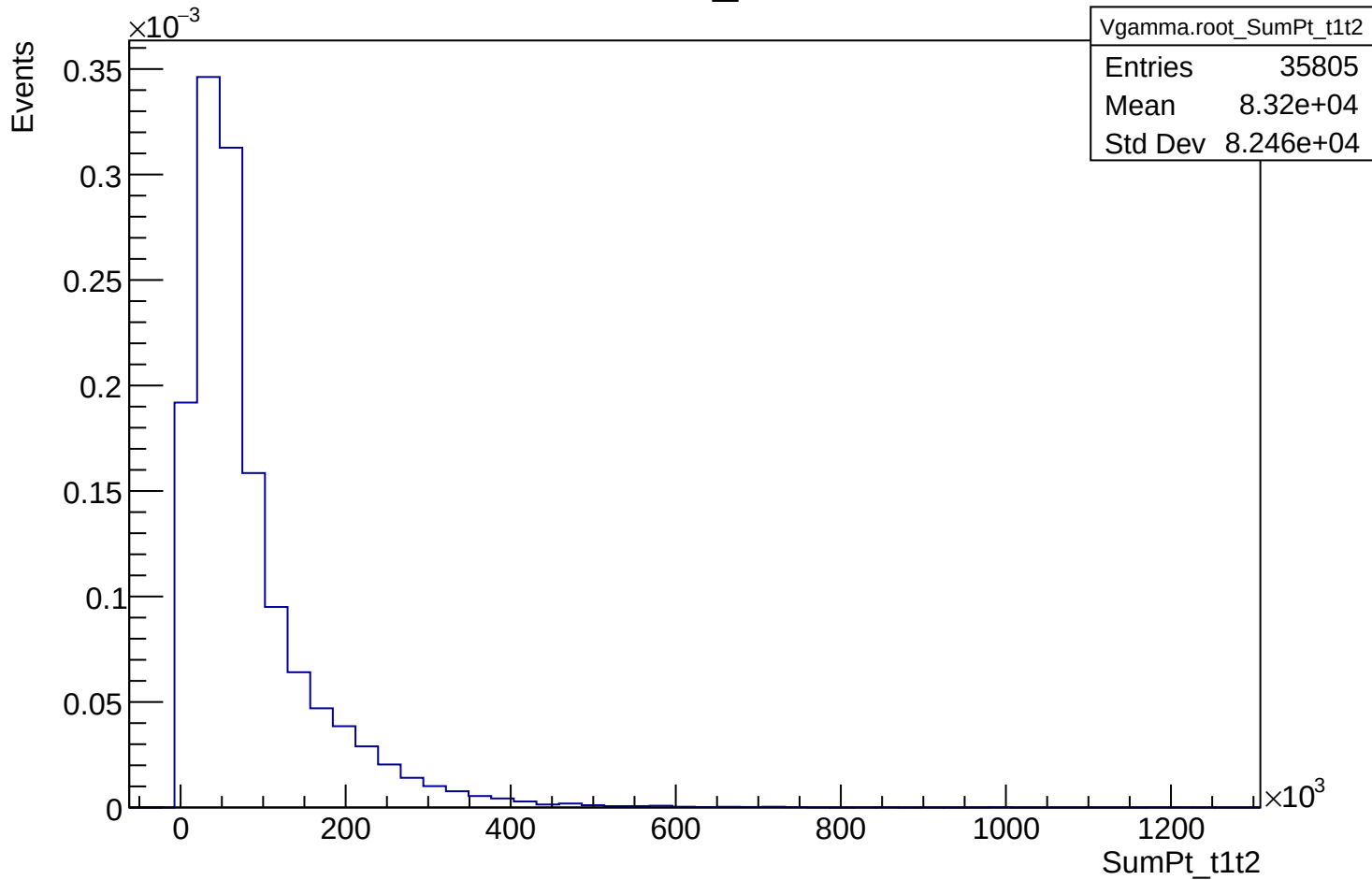
SumPt_l1j



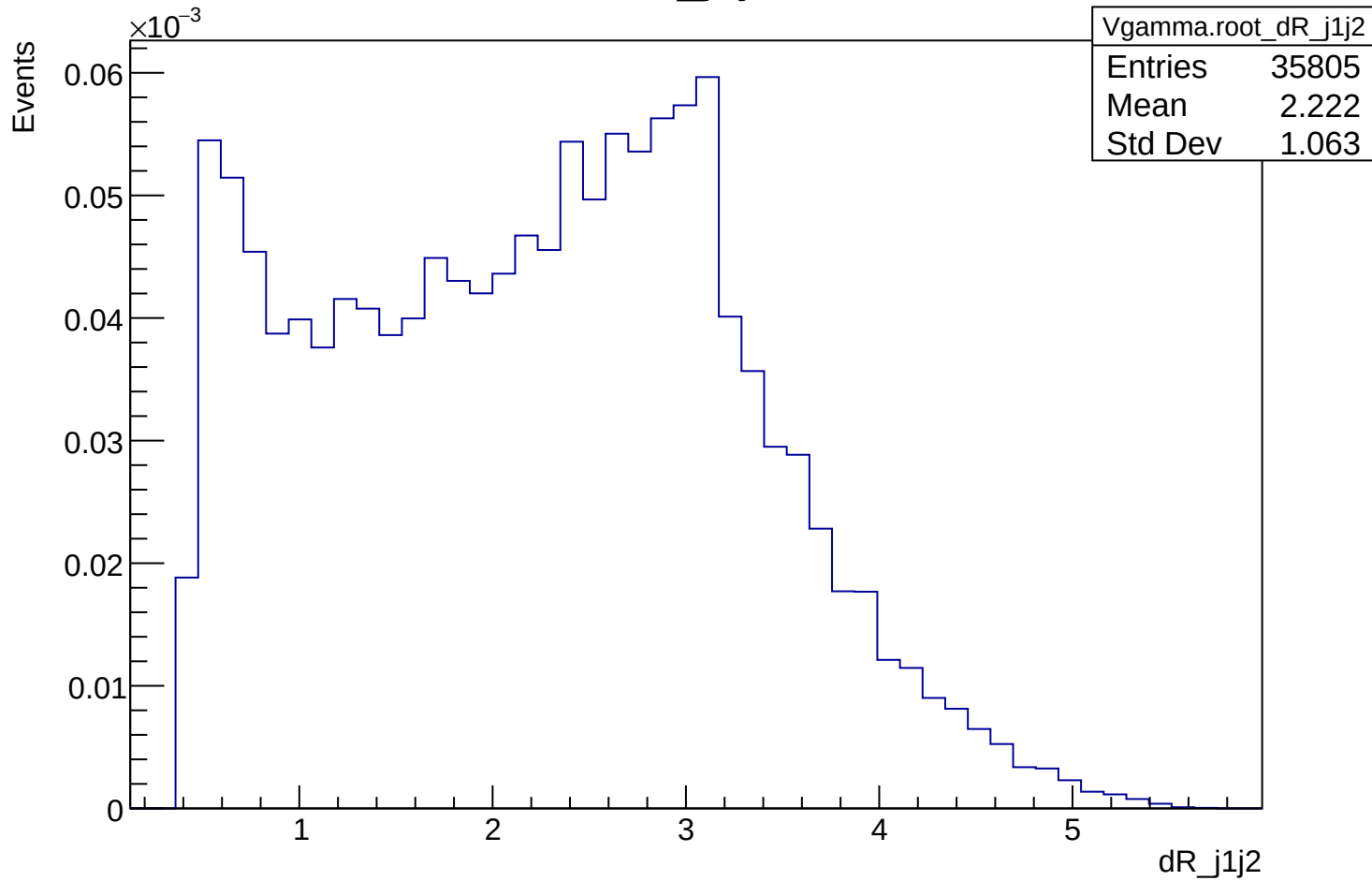
SumPt_l1j1



SumPt_t1t2

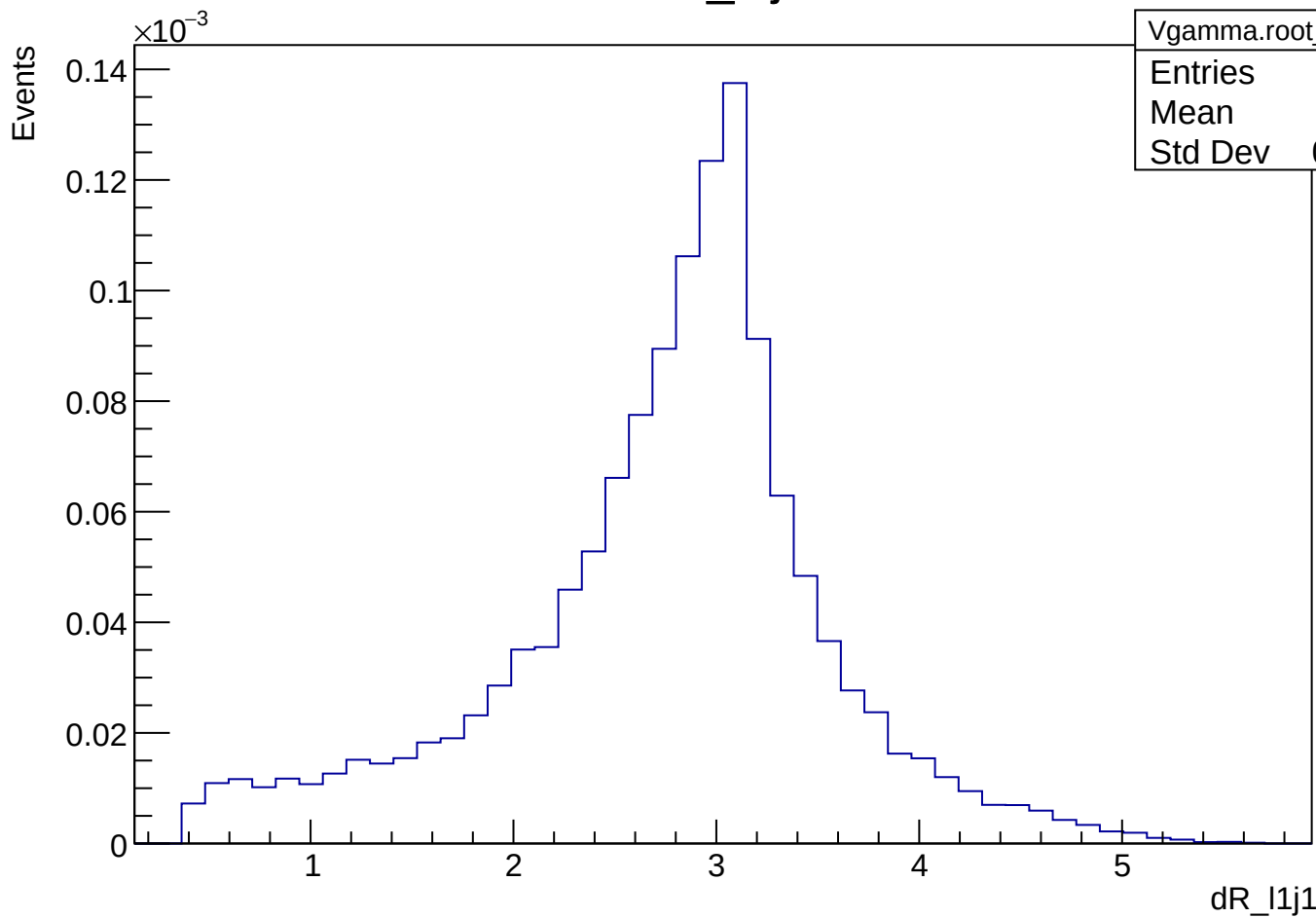


dR_j1j2



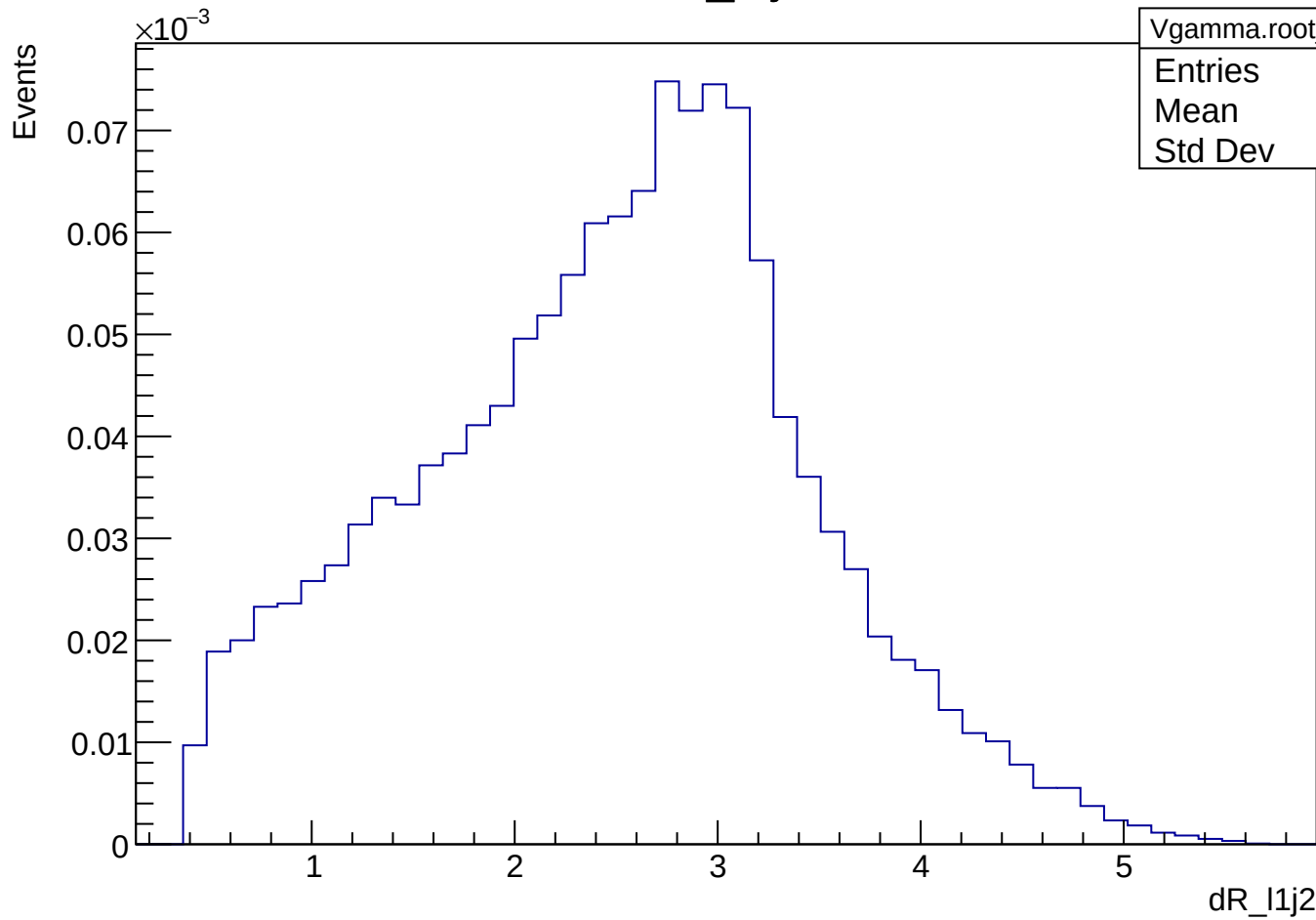
dR_l1j1

Vgamma.root_dR_l1j1	
Entries	35805
Mean	2.753
Std Dev	0.8109

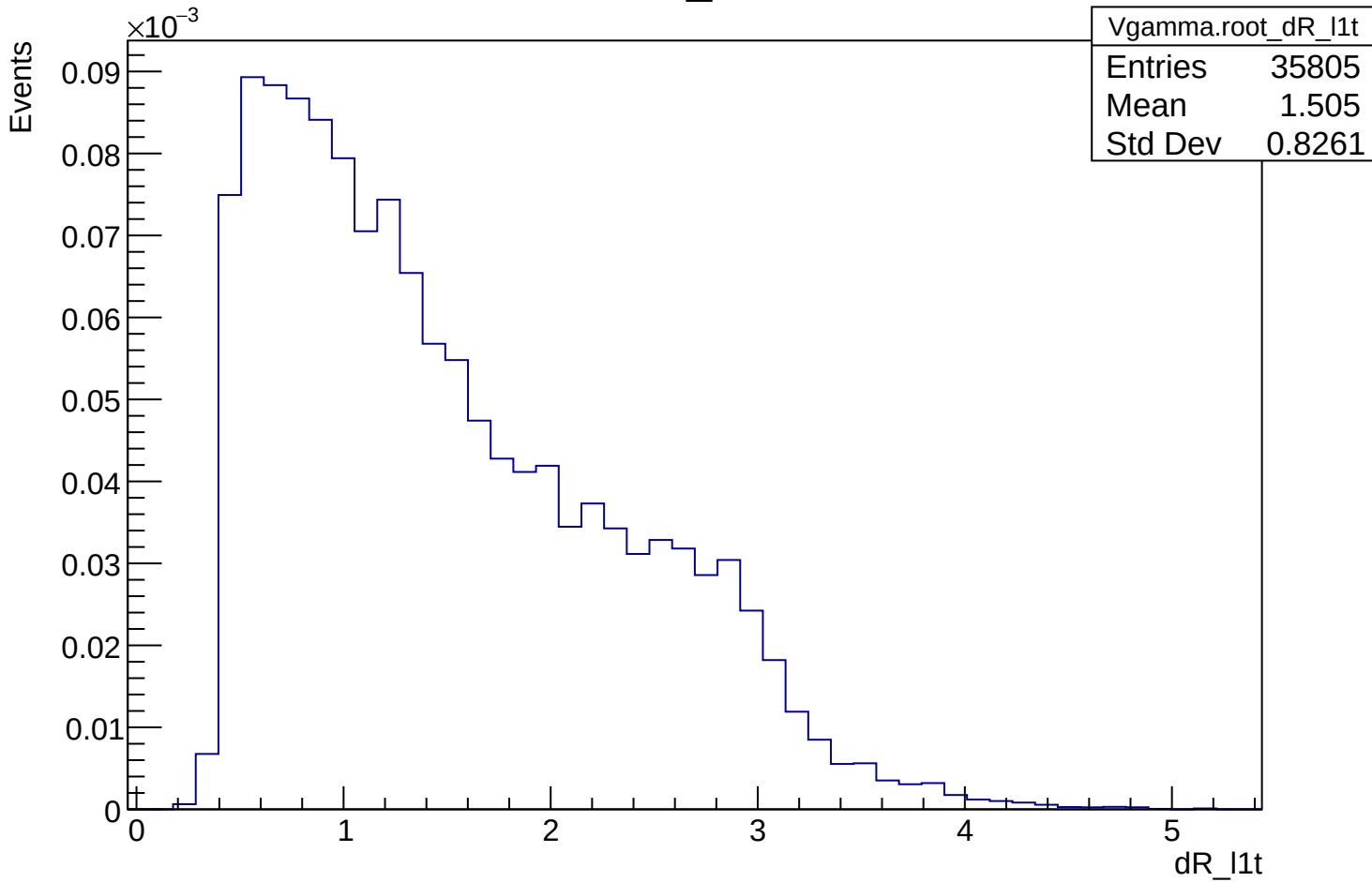


dR_l1j2

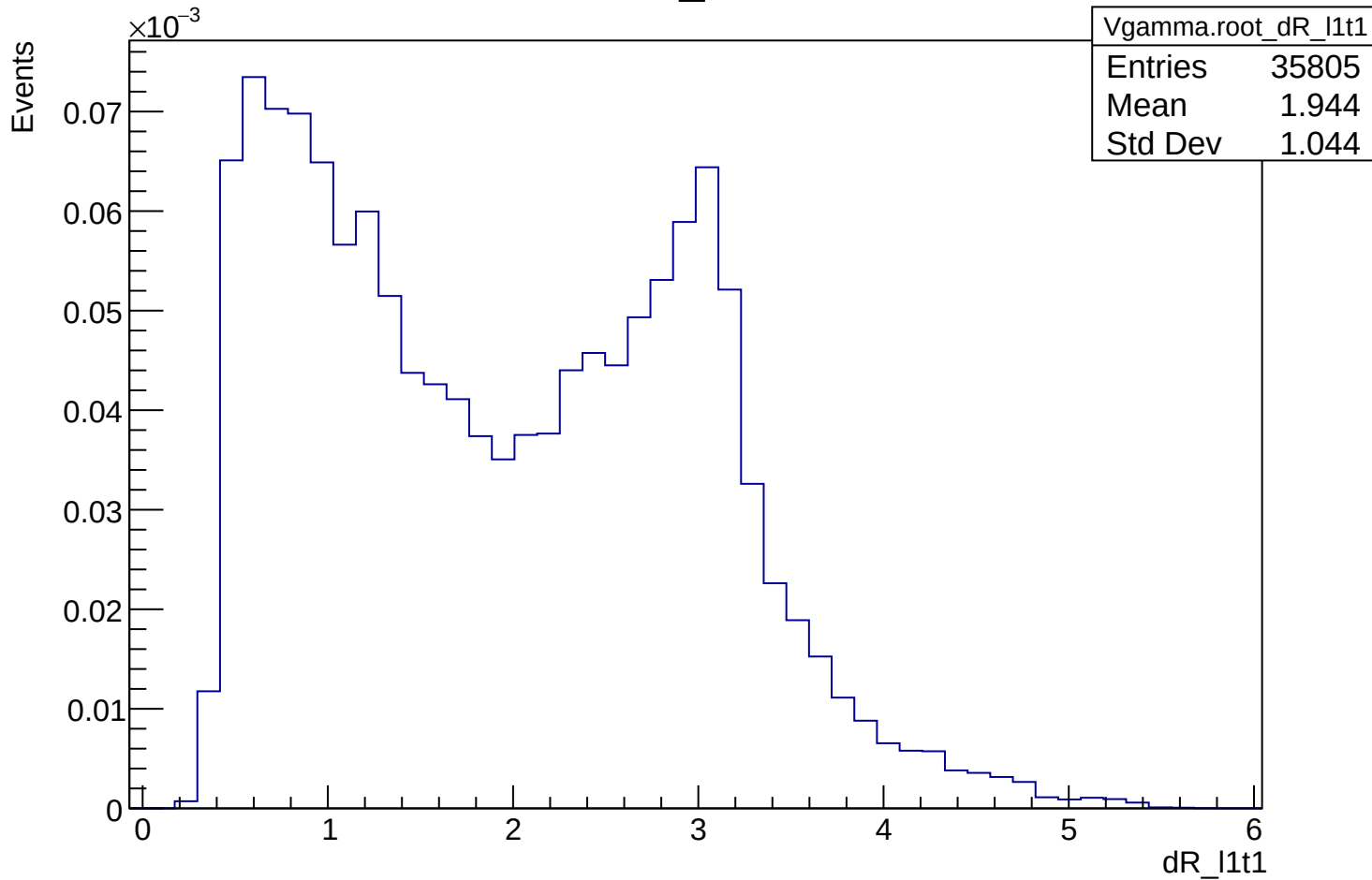
Vgamma.root_dR_l1j2	
Entries	35805
Mean	2.485
Std Dev	0.9596



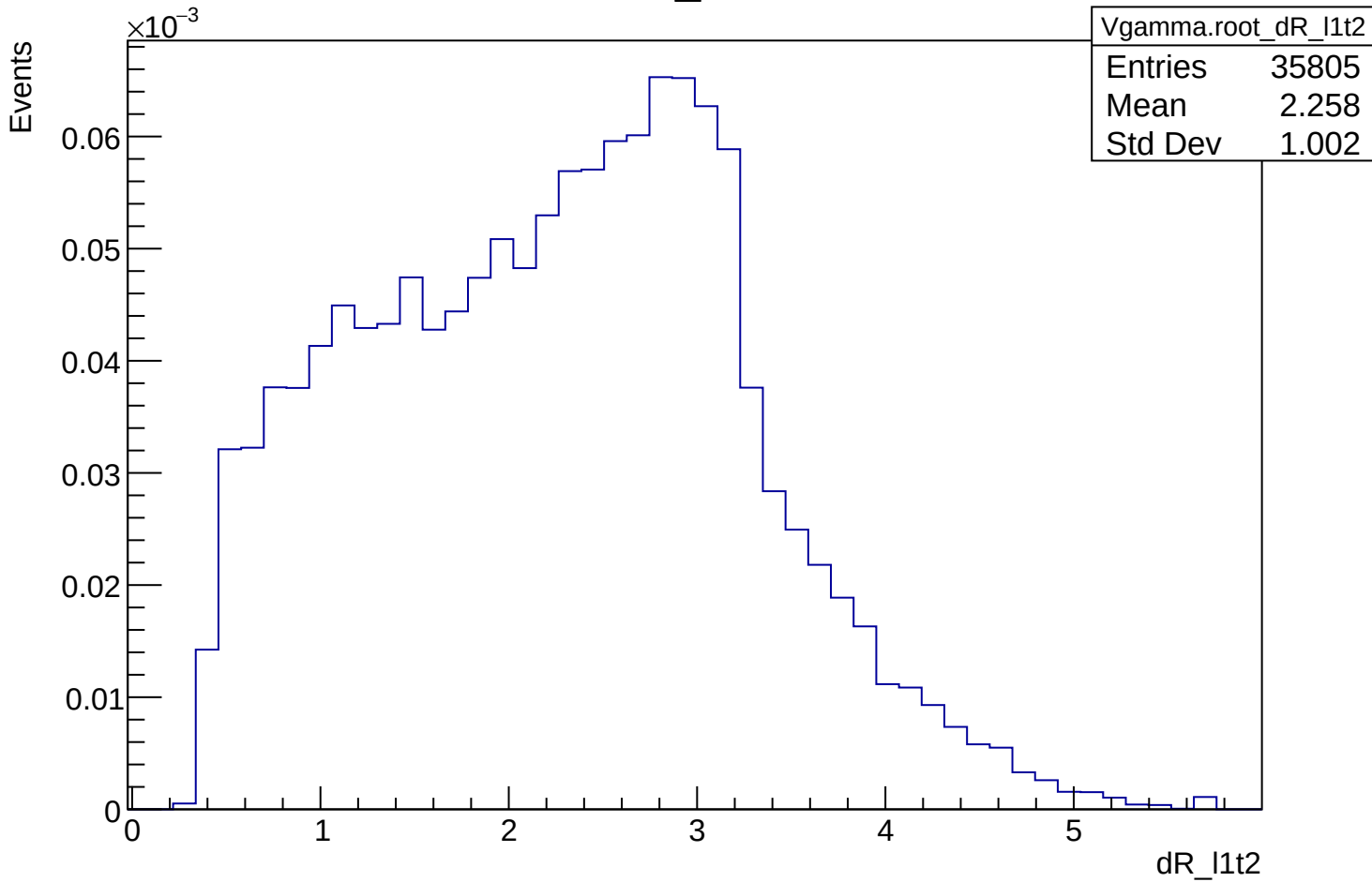
dR_l1t



dR_l1t1

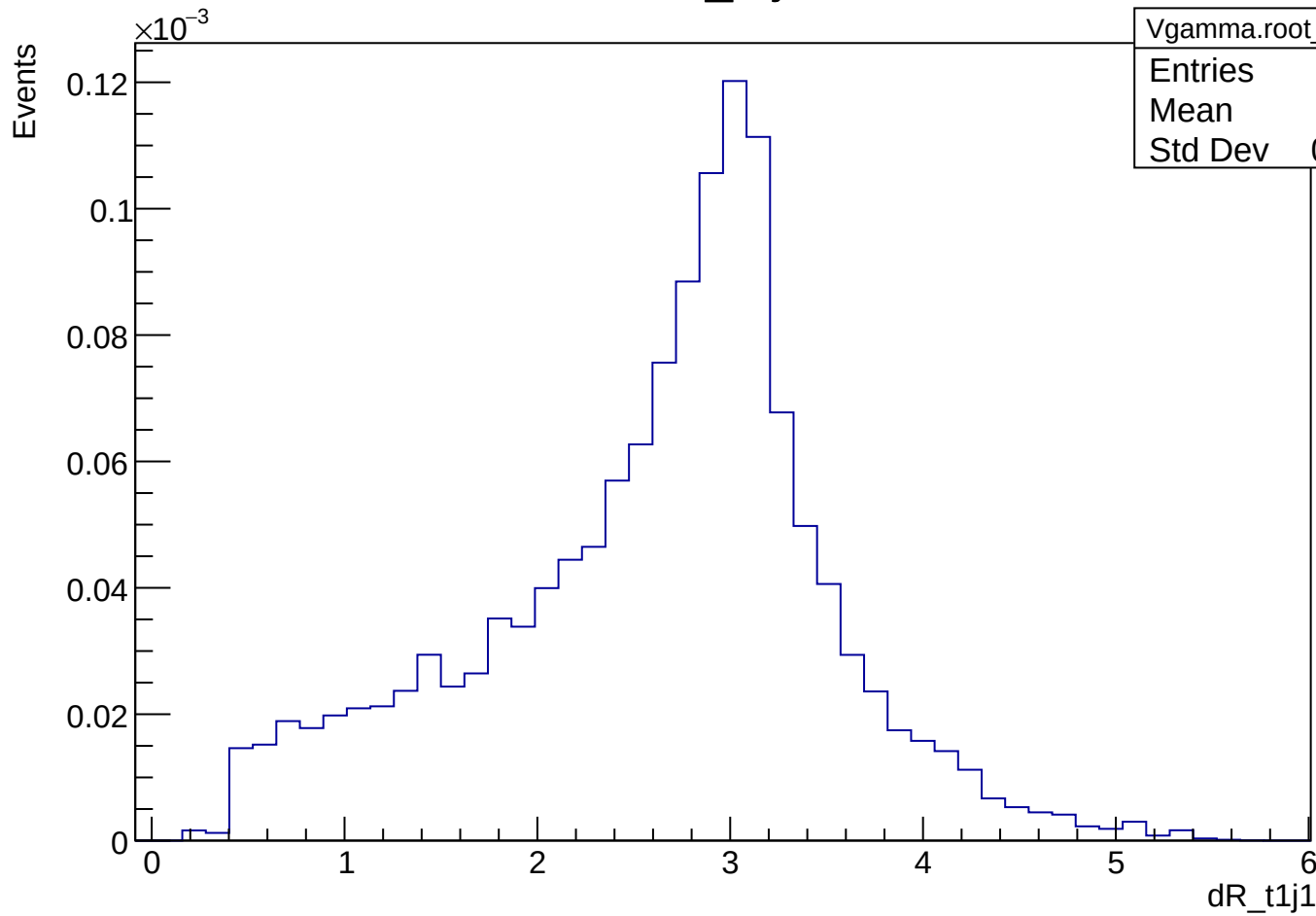


dR_l1t2

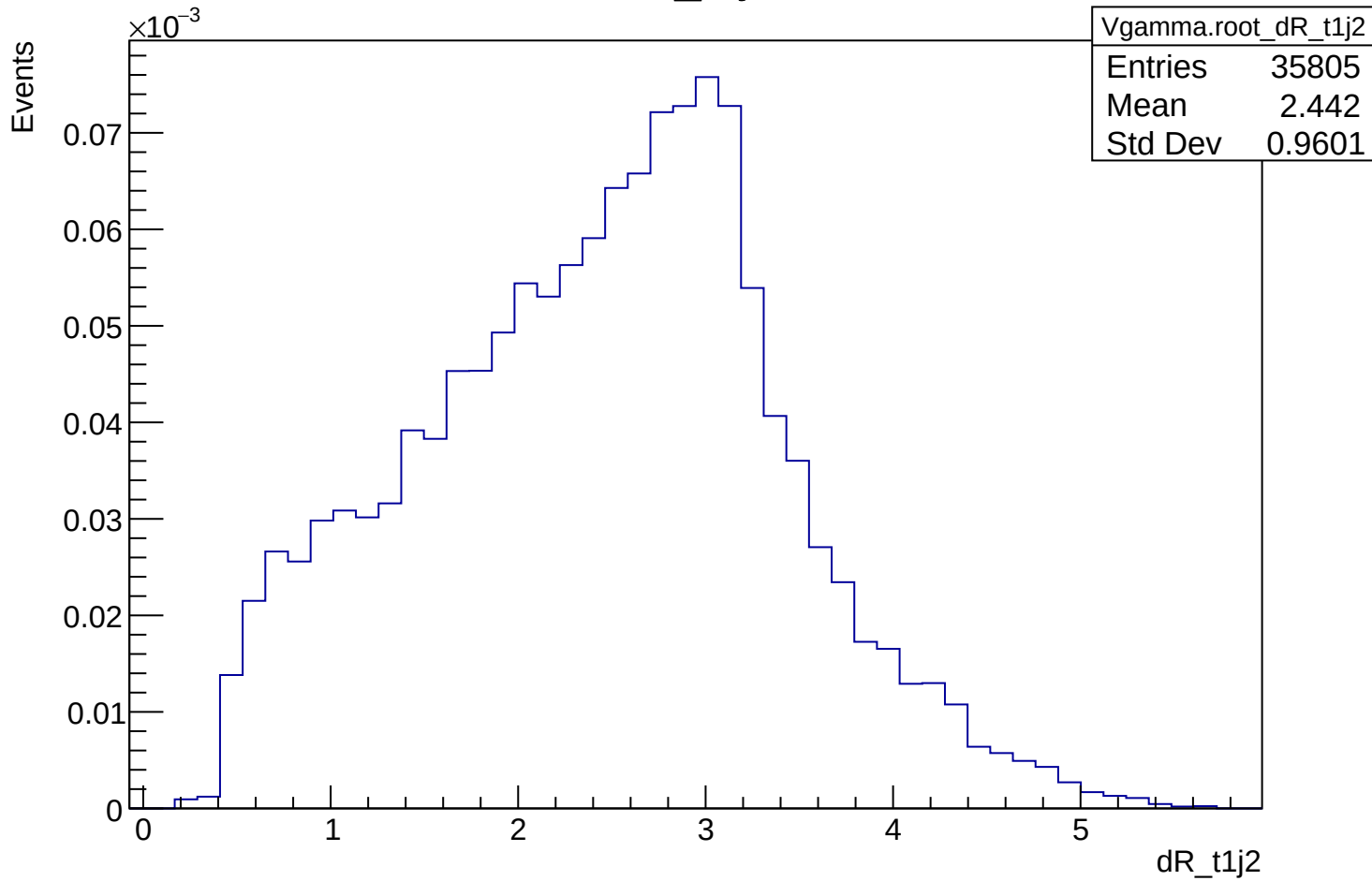


dR_t1j1

Vgamma.root_dR_t1j1	
Entries	35805
Mean	2.62
Std Dev	0.8956

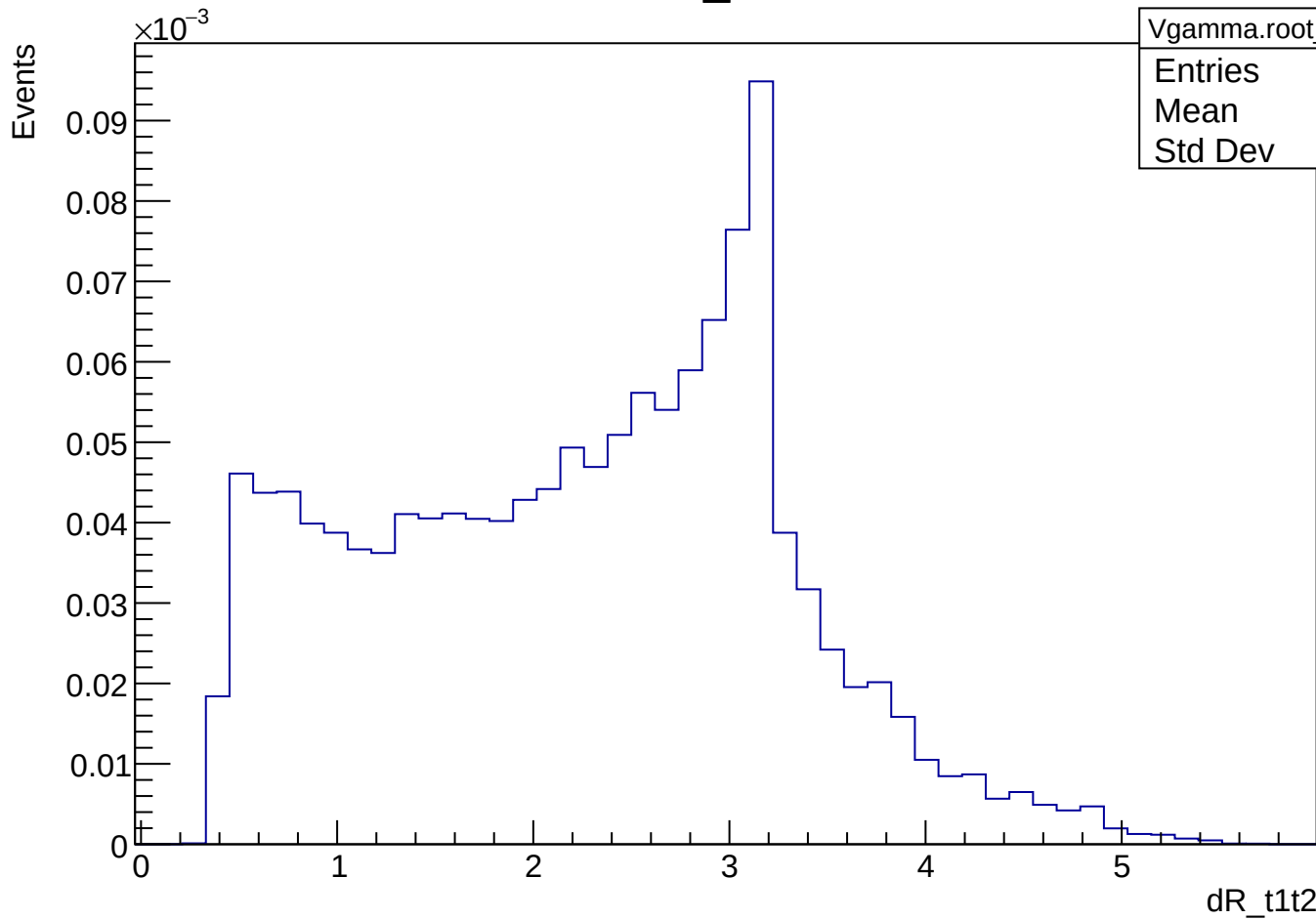


dR_t1j2

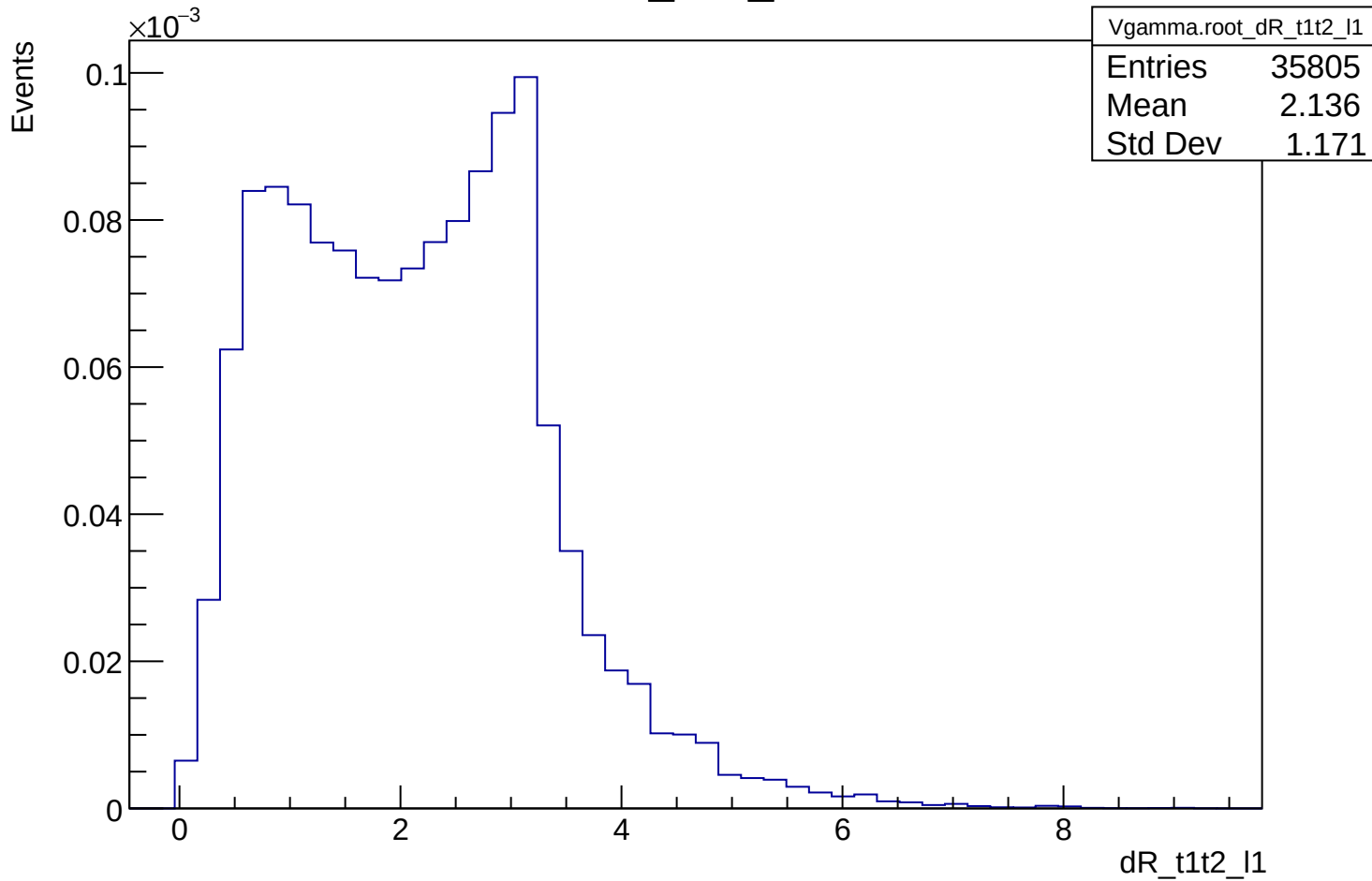


dR_t1t2

Vgamma.root_dR_t1t2	
Entries	35805
Mean	2.255
Std Dev	1.038

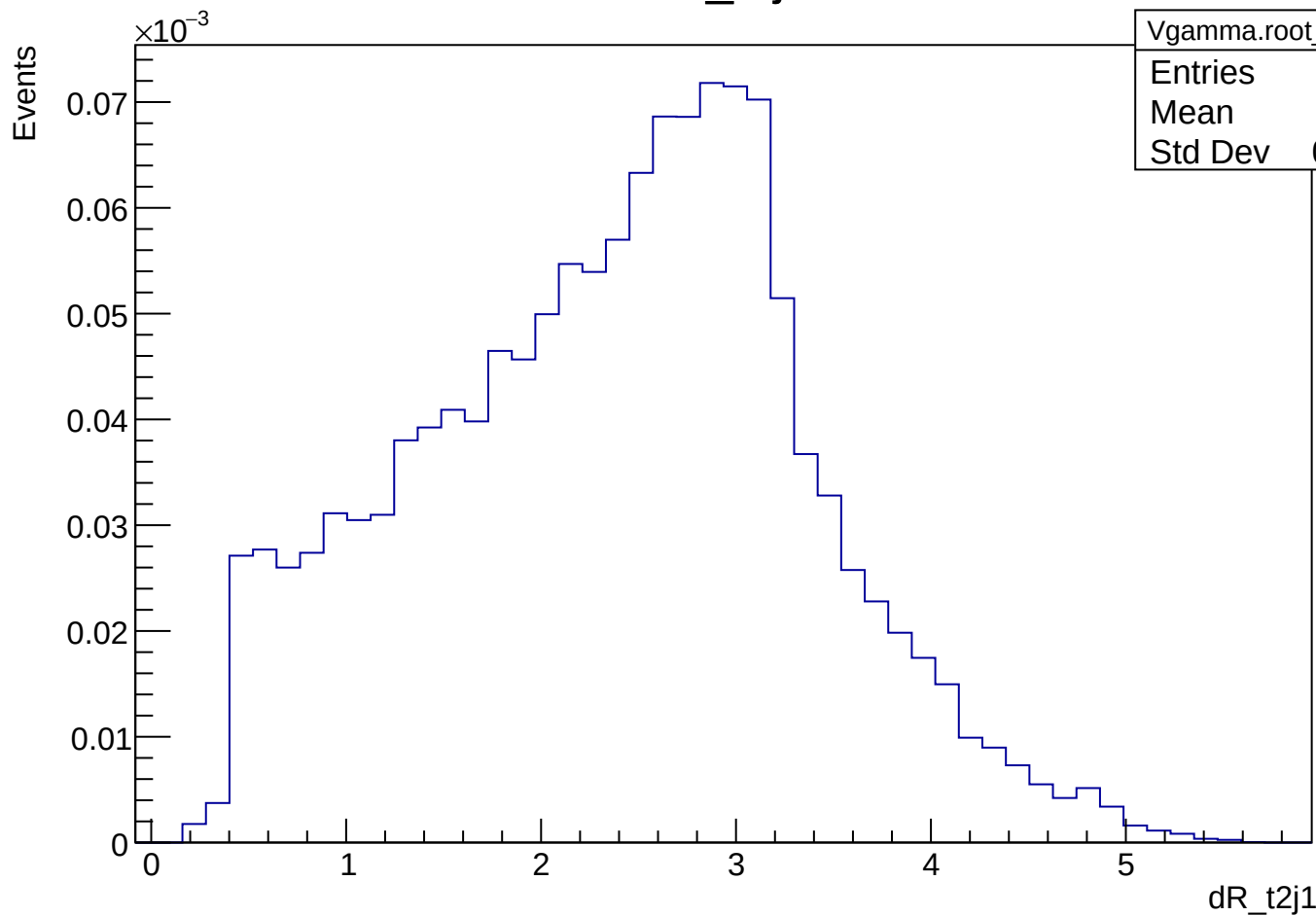


dR_t1t2_l1



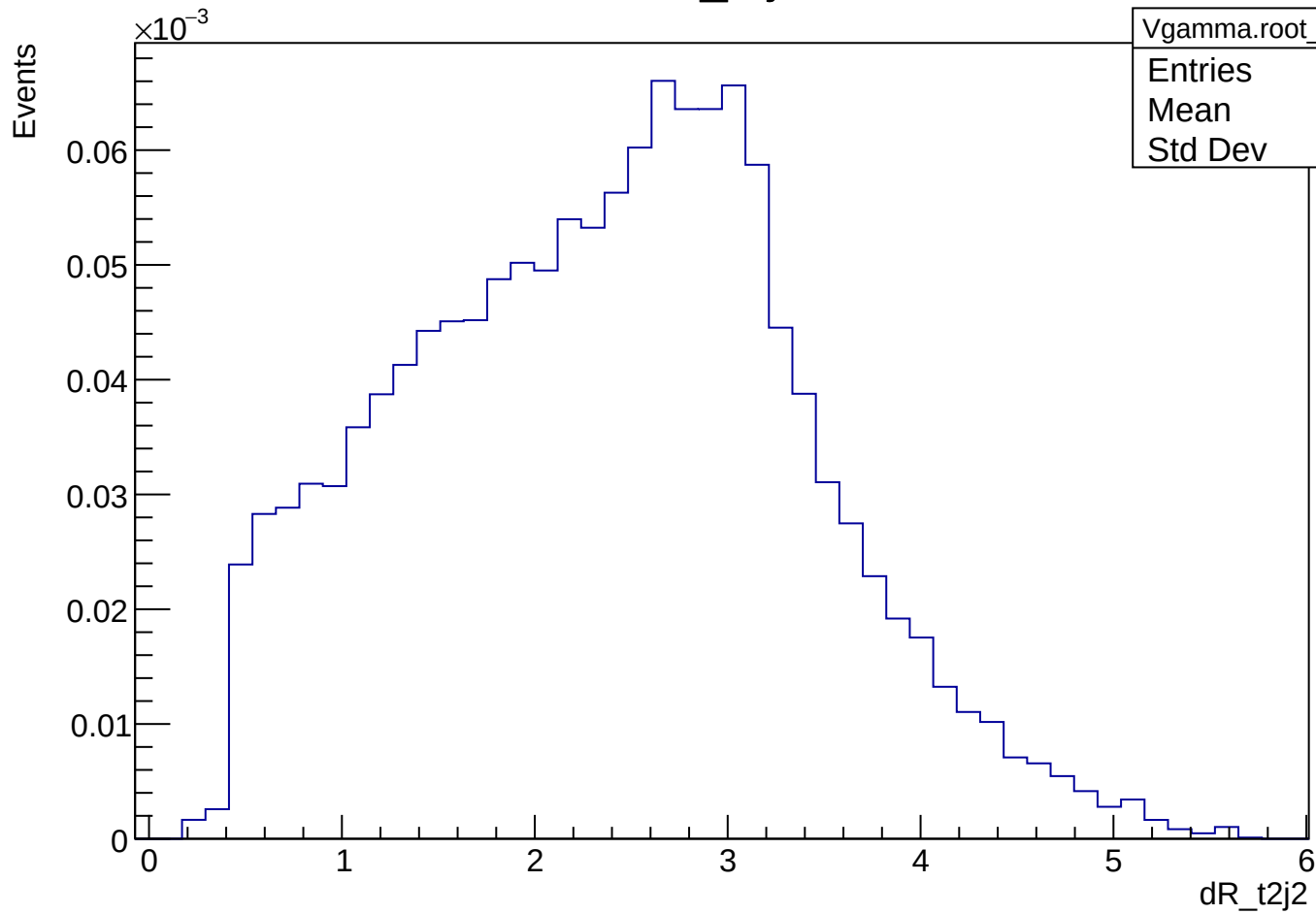
dR_t2j1

Vgamma.root_dR_t2j1	
Entries	35805
Mean	2.383
Std Dev	0.9896

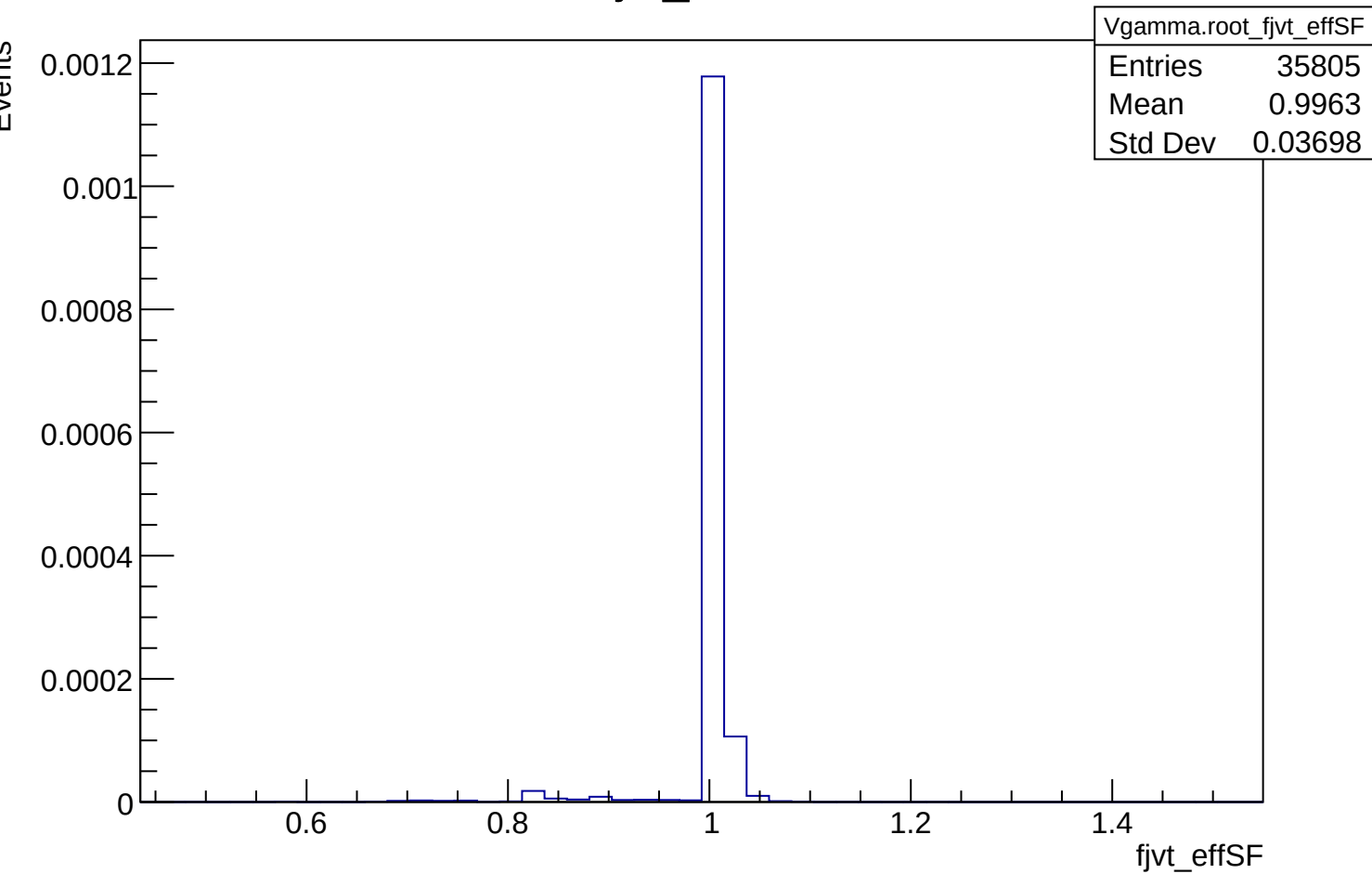


dR_t2j2

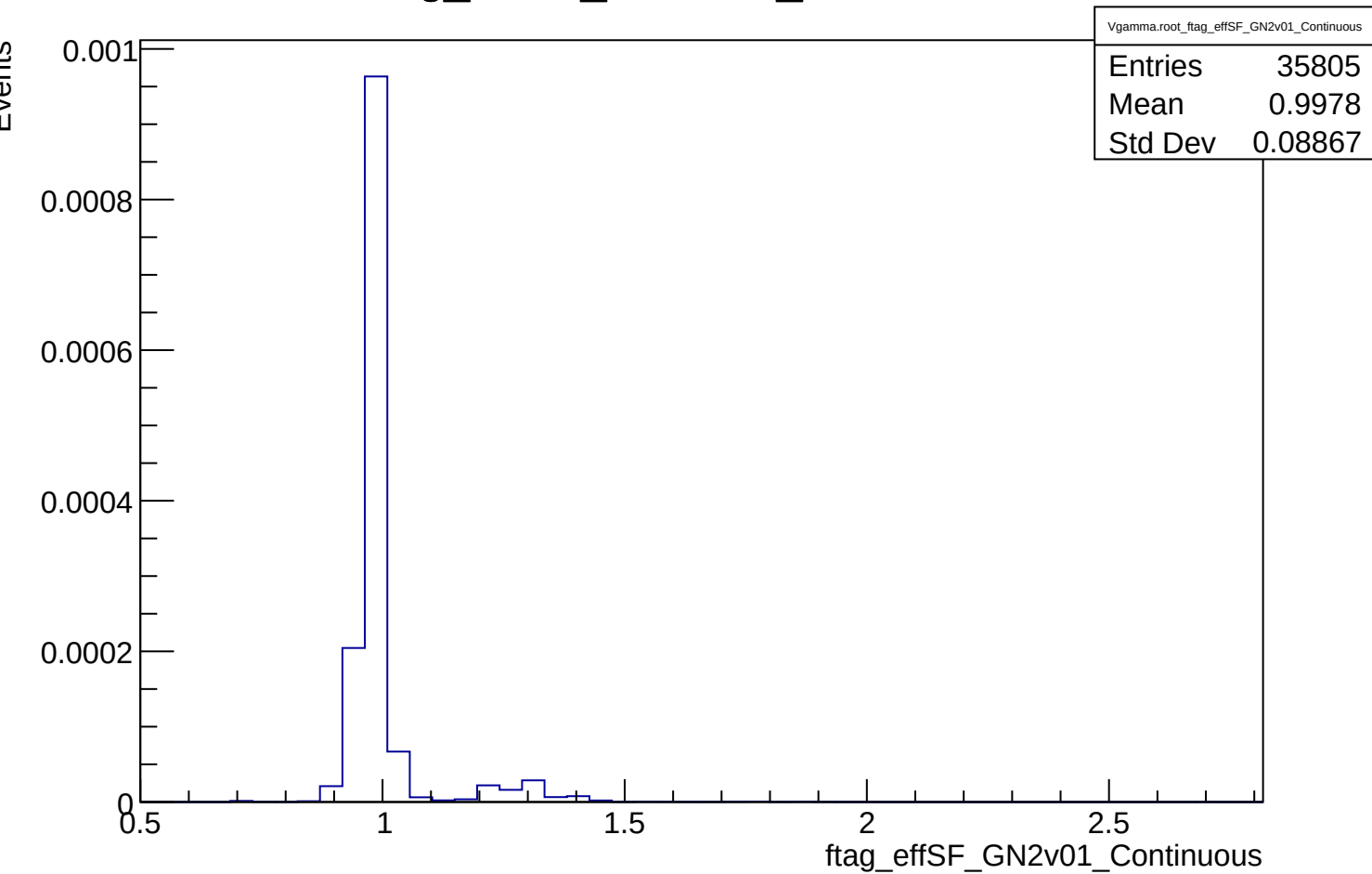
Vgamma.root_dR_t2j2	
Entries	35805
Mean	2.375
Std Dev	1.017



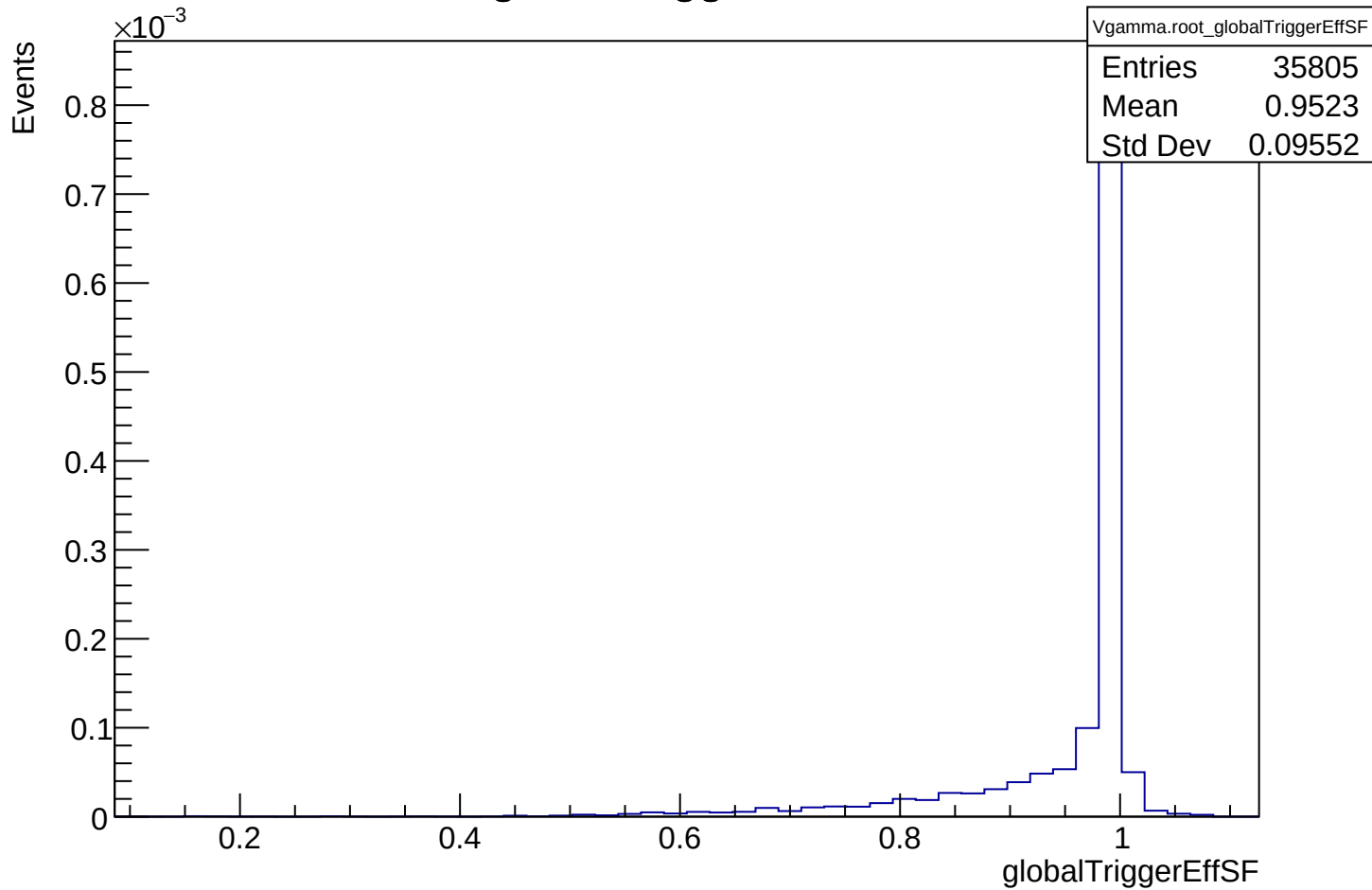
fjvt_effSF



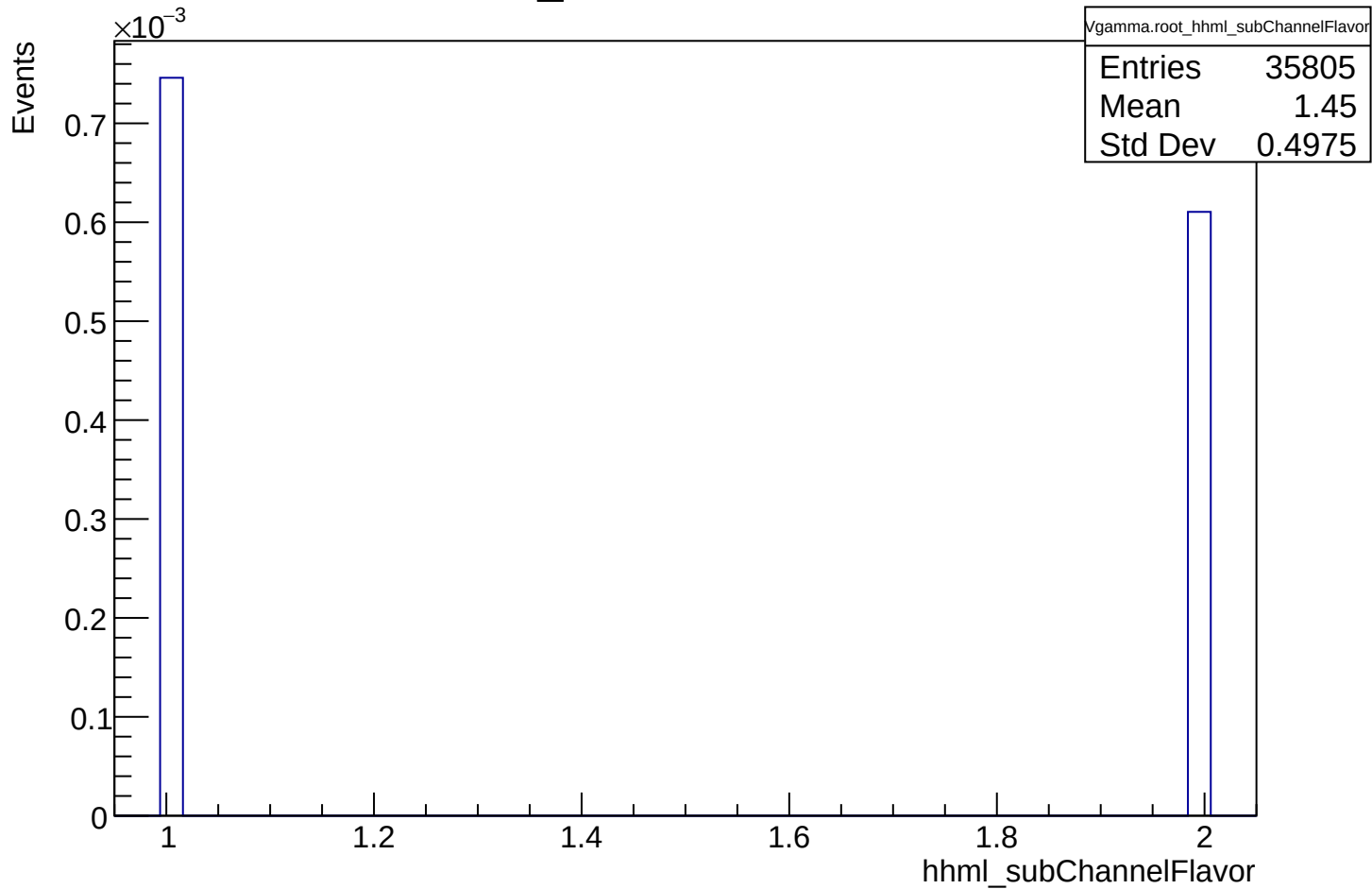
ftag_effSF_GN2v01_Continuous



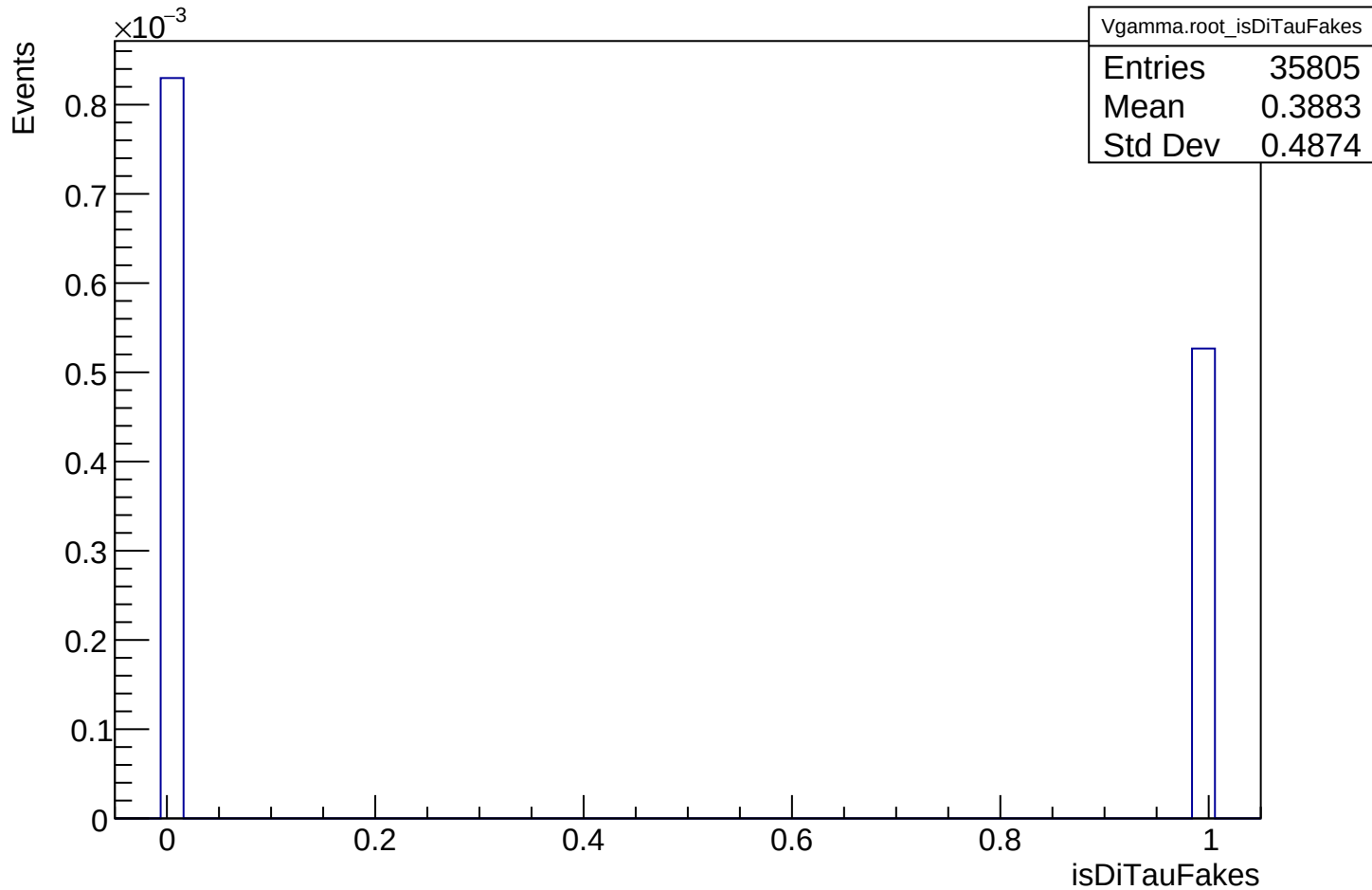
globalTriggerEffSF



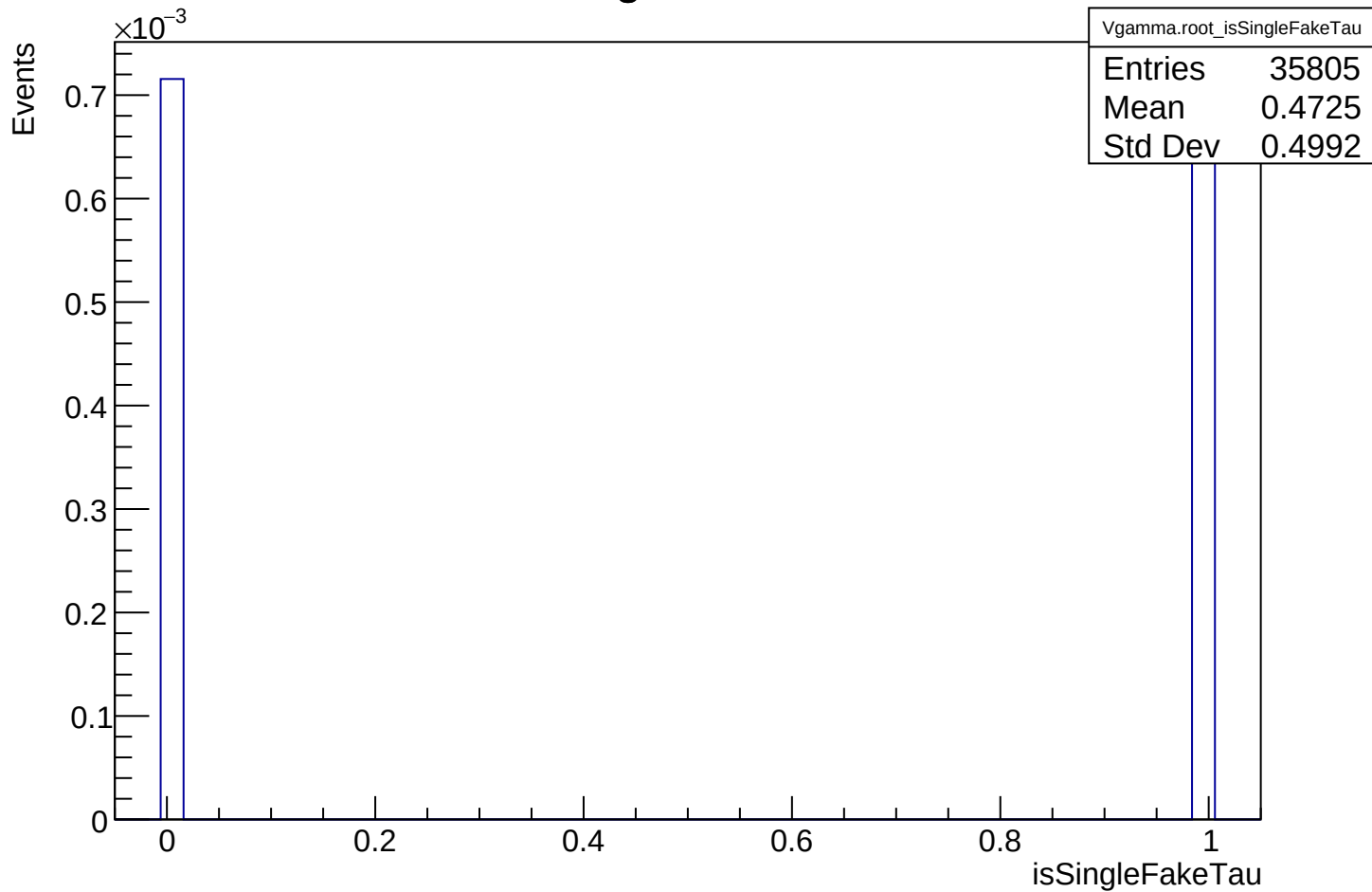
hhml_subChannelFlavor



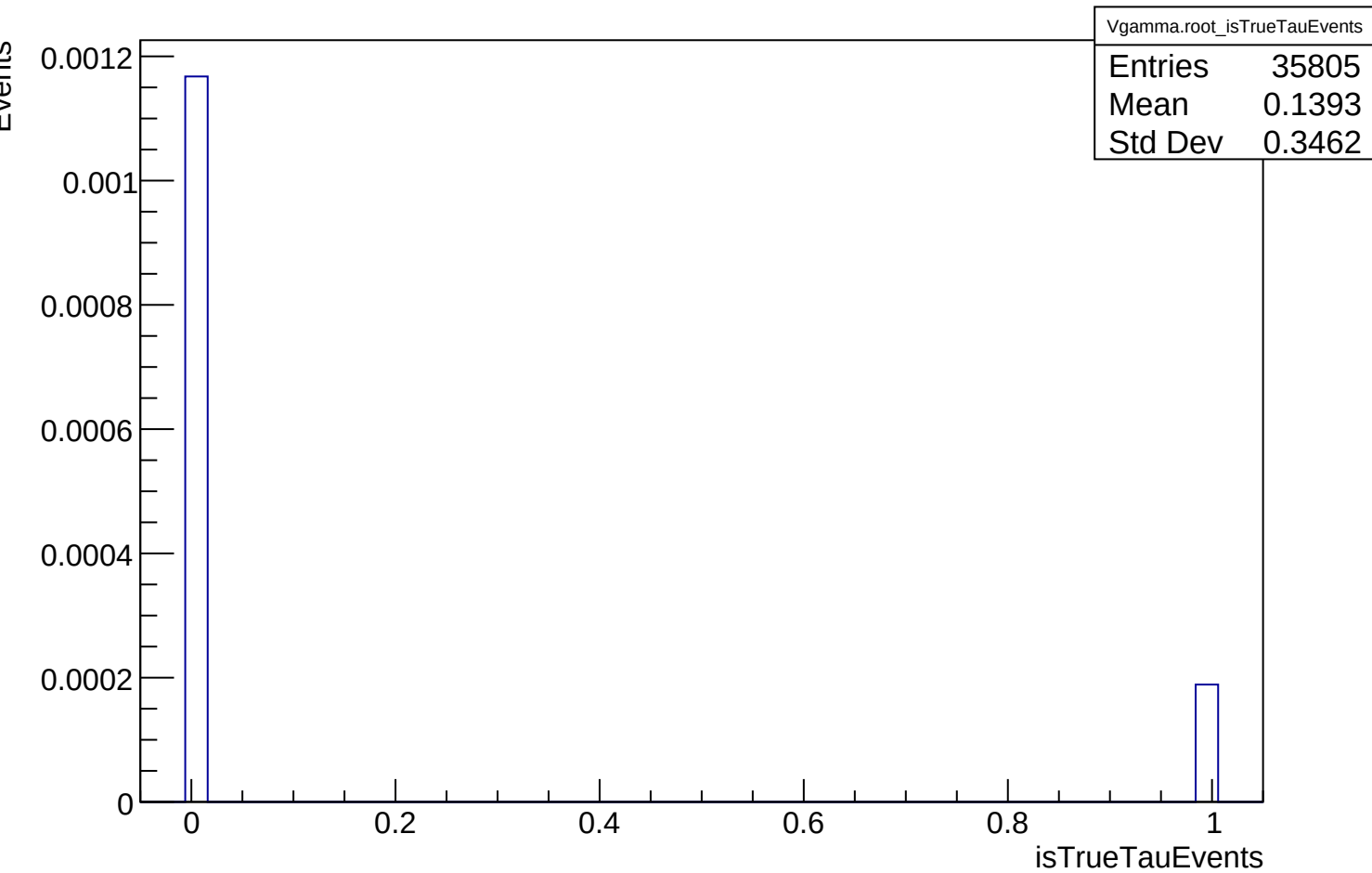
isDiTauFakes



isSingleFakeTau

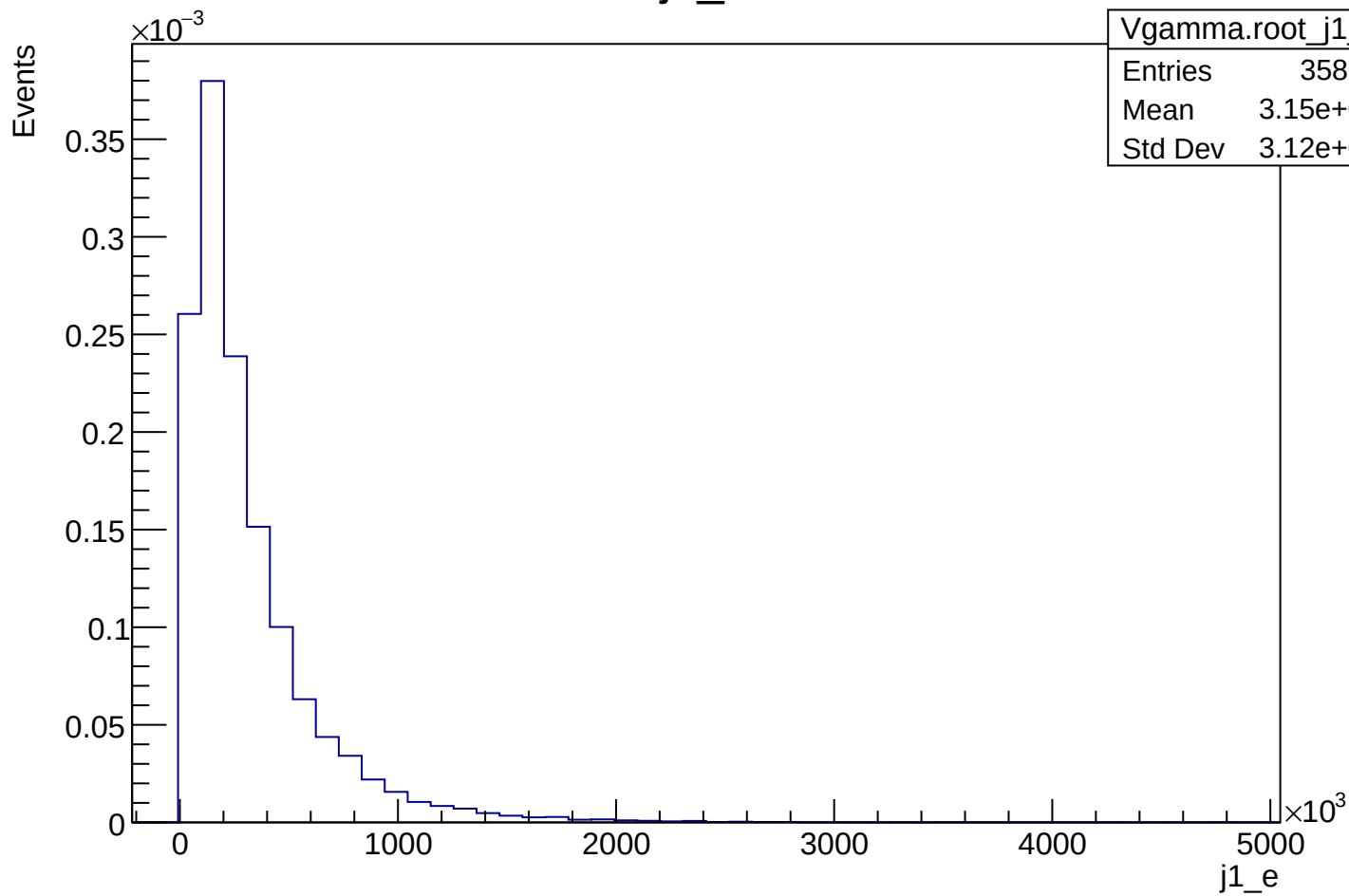


isTrueTauEvents

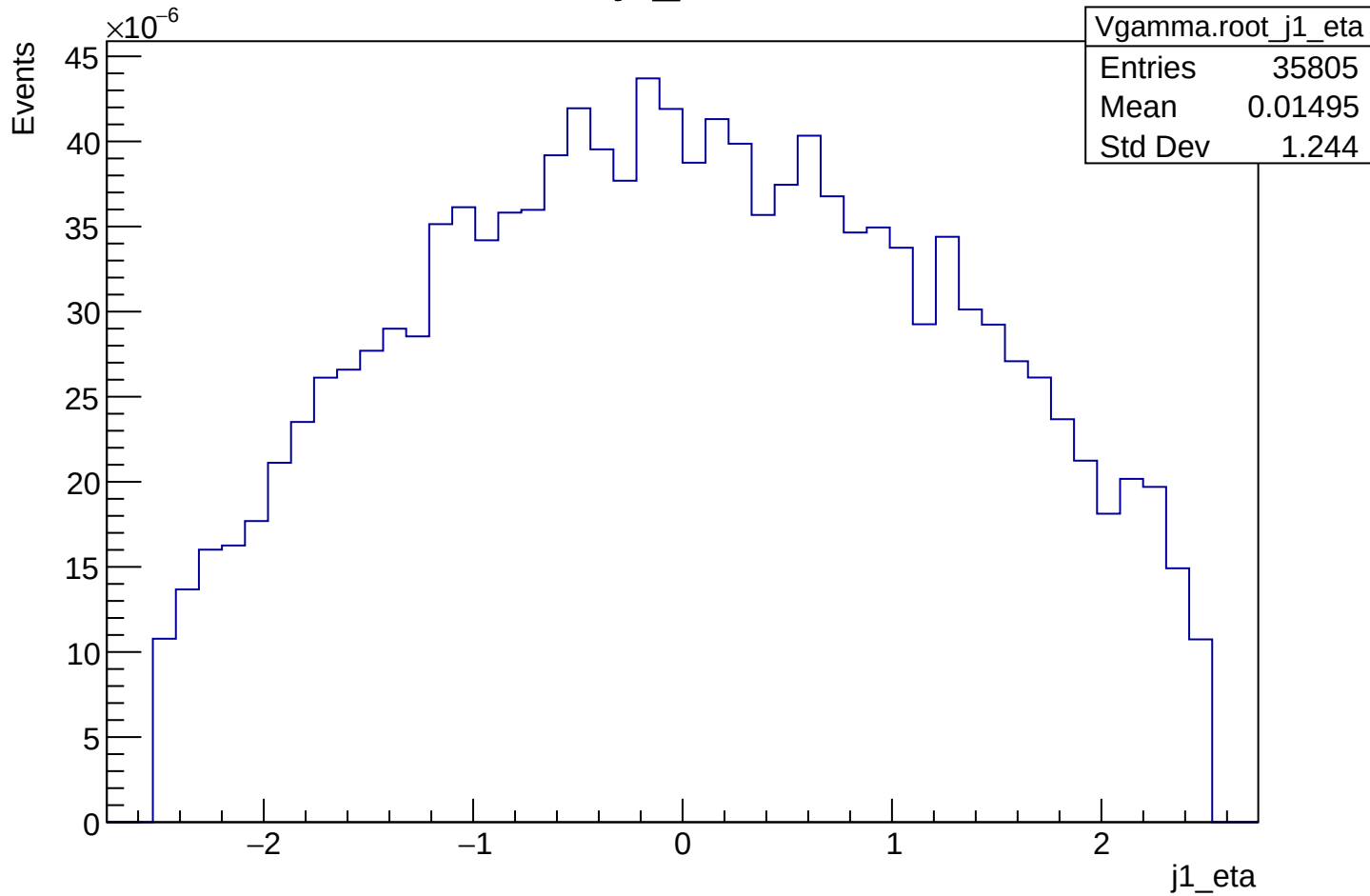


j1_e

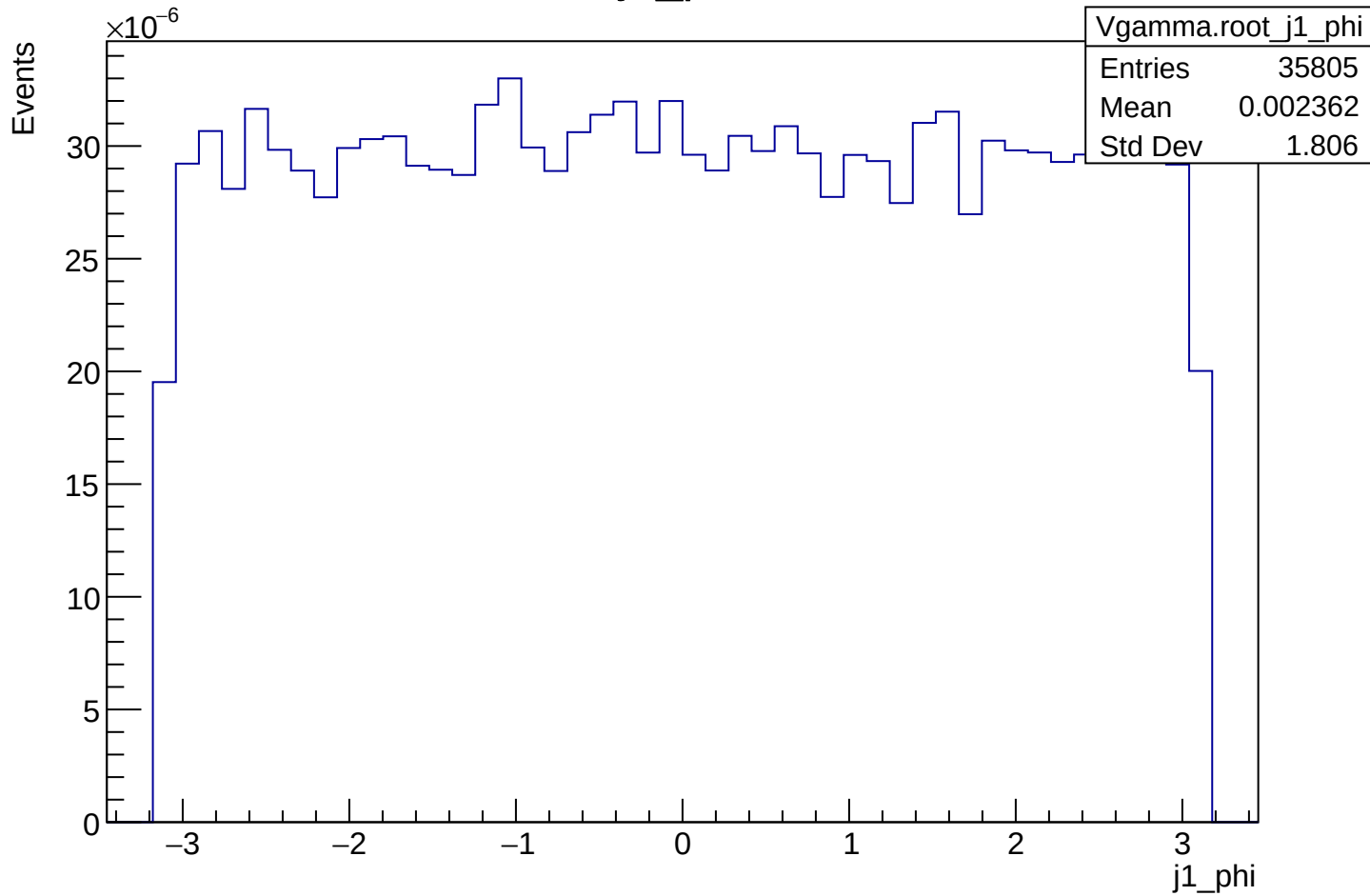
Vgamma.root_j1_e	
Entries	35805
Mean	3.15e+05
Std Dev	3.12e+05



j1_eta

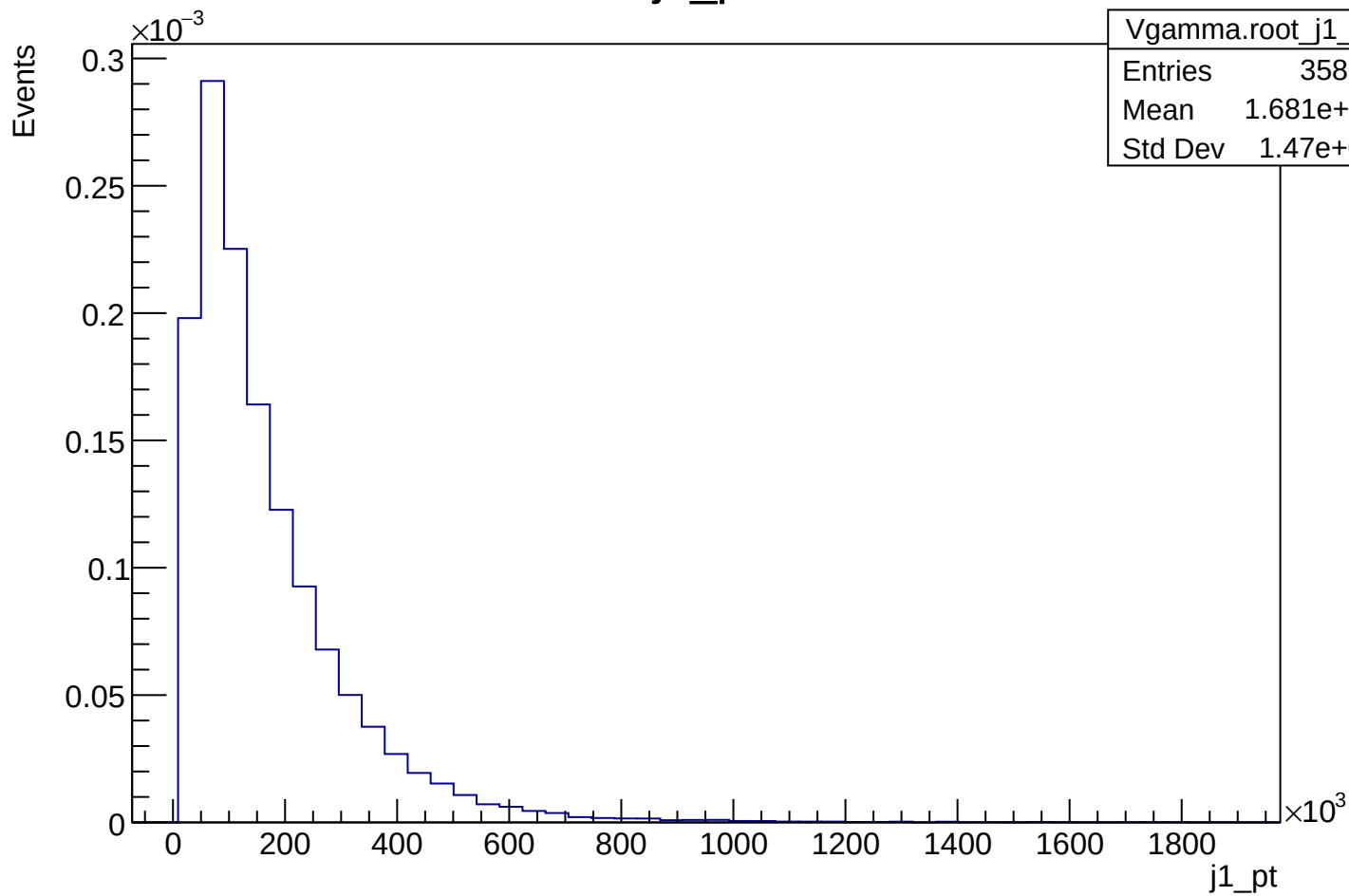


j1_phi

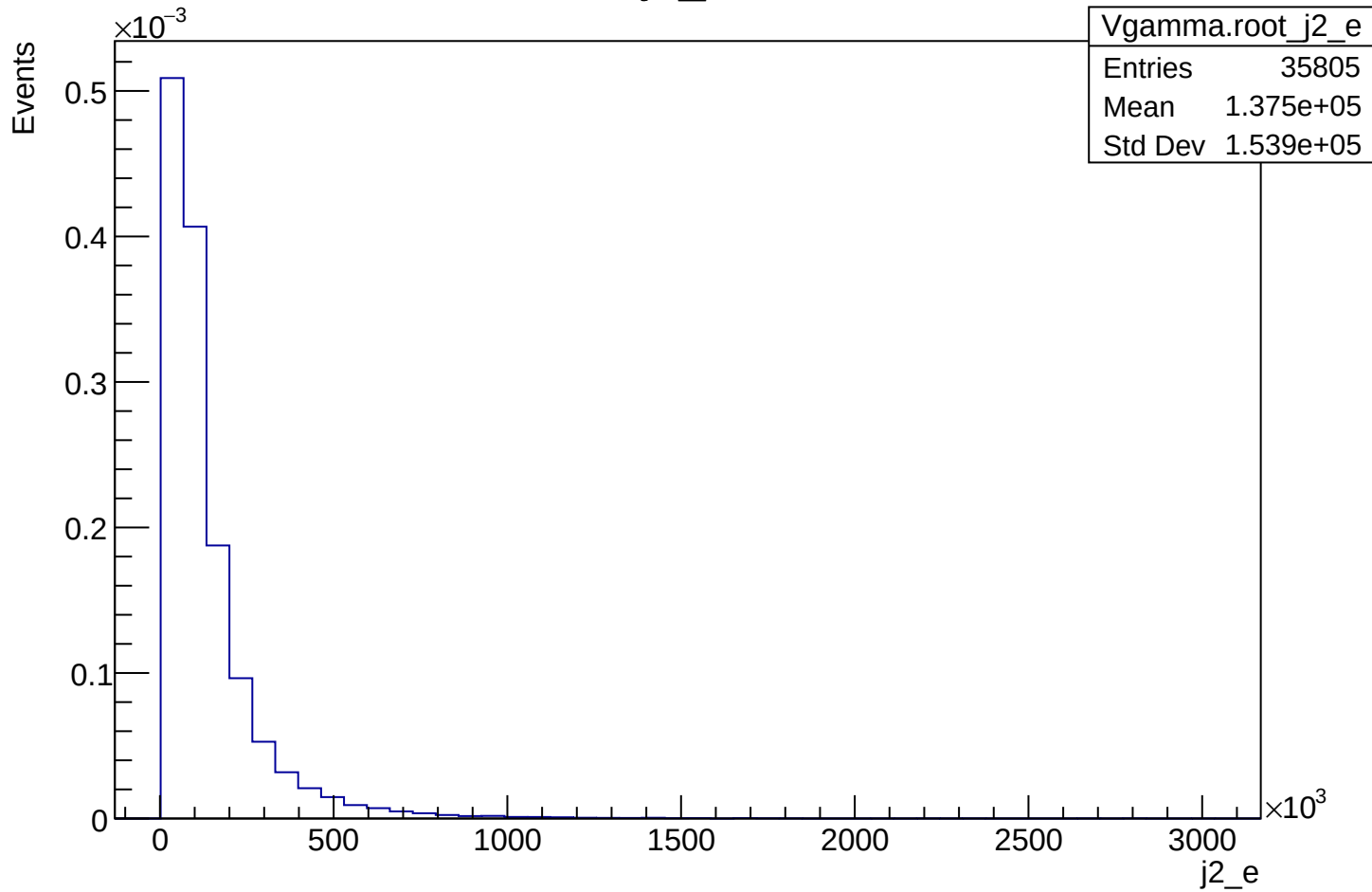


j1_pt

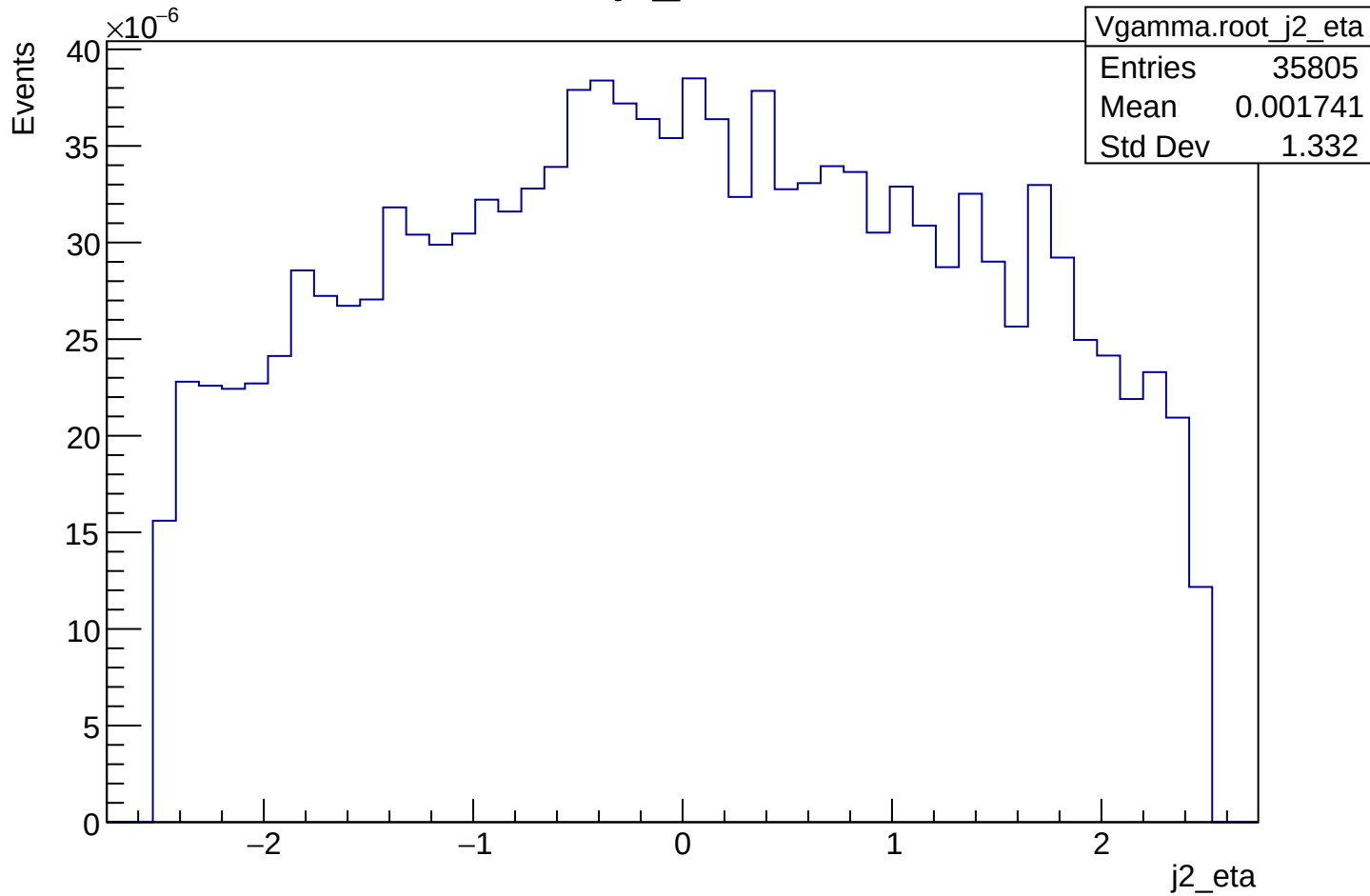
Vgamma.root_j1_pt	
Entries	35805
Mean	1.681e+05
Std Dev	1.47e+05



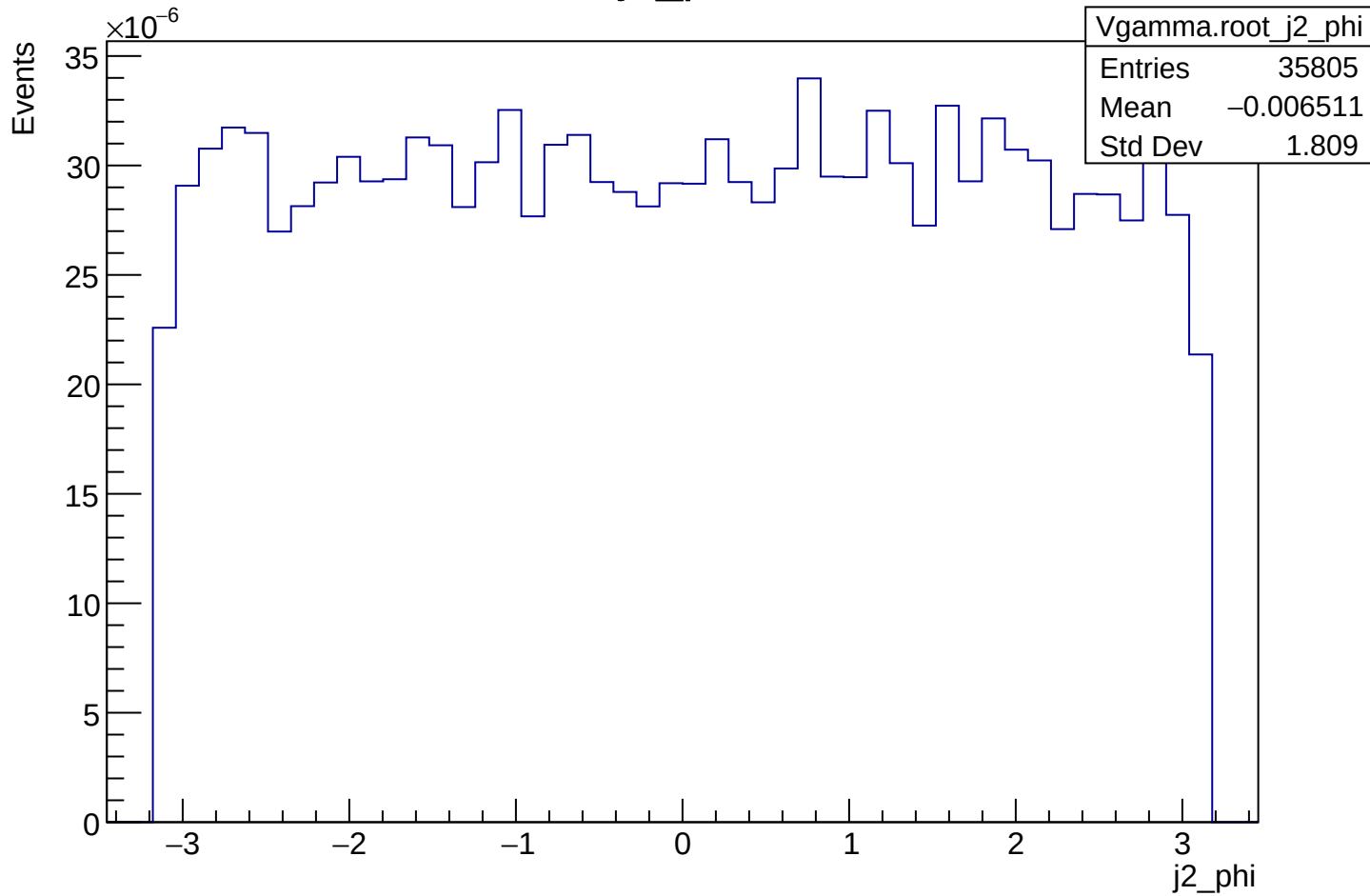
j2_e



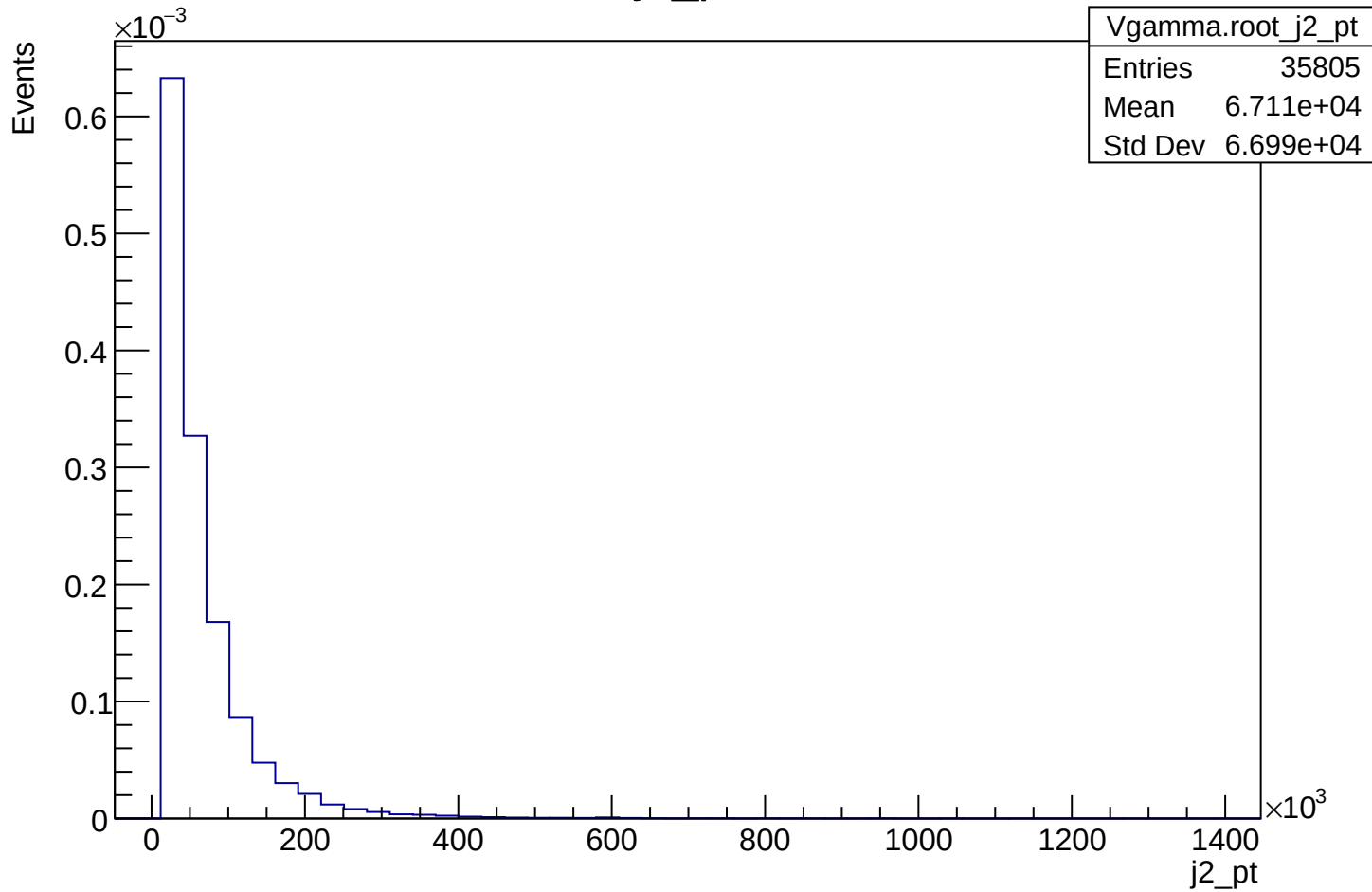
j2_eta



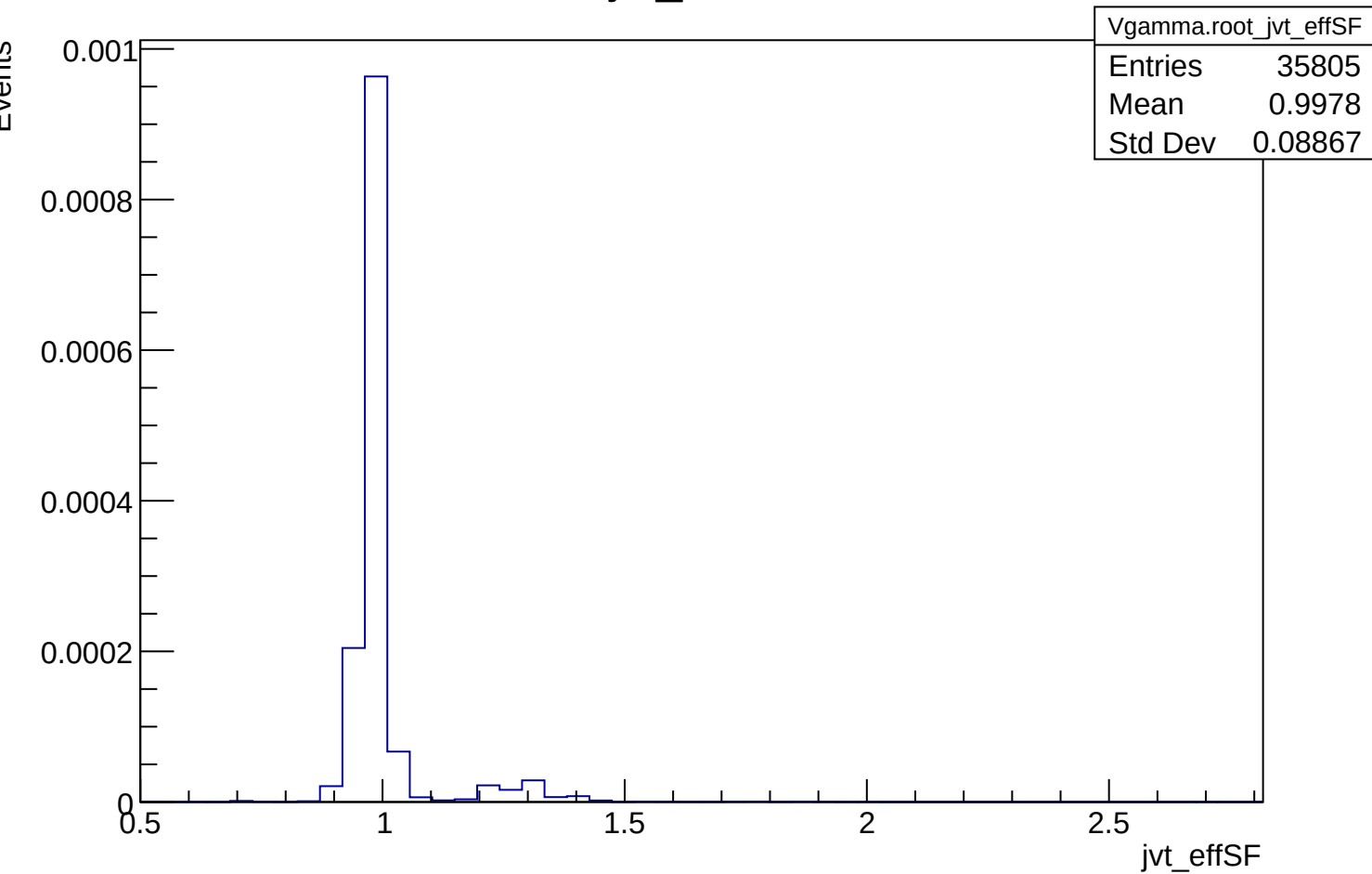
j2_phi



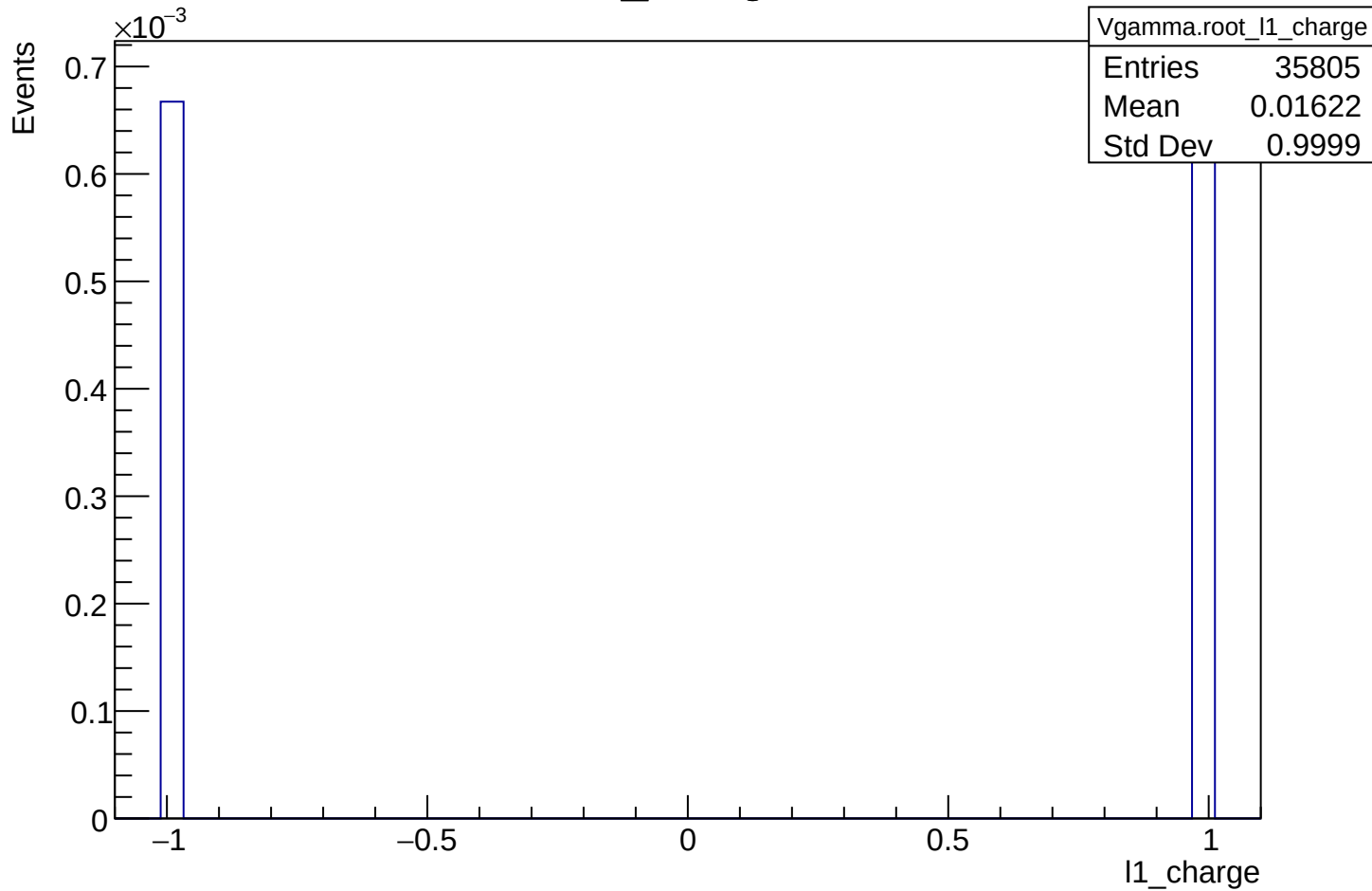
j2_pt



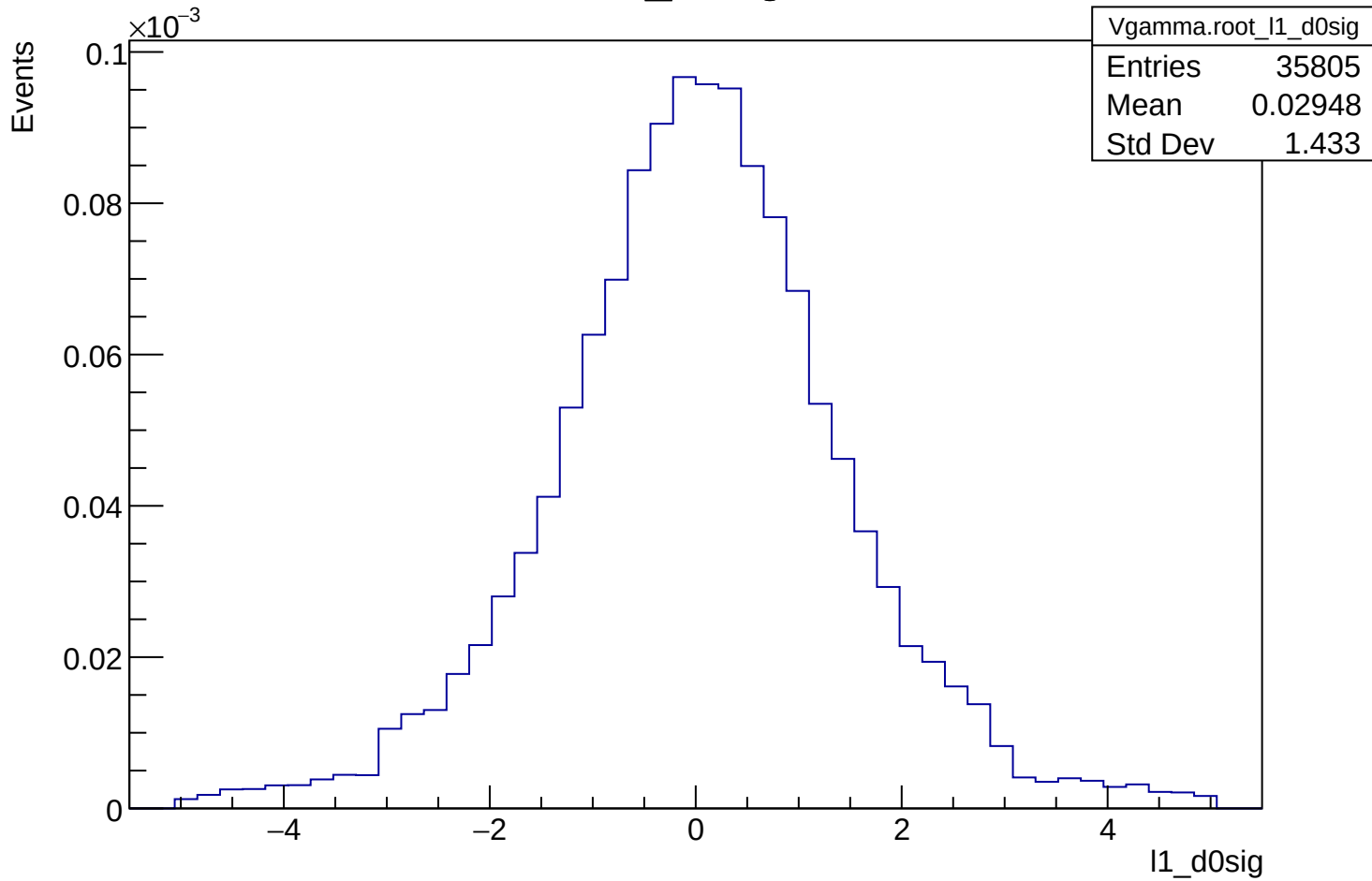
jvt_effSF



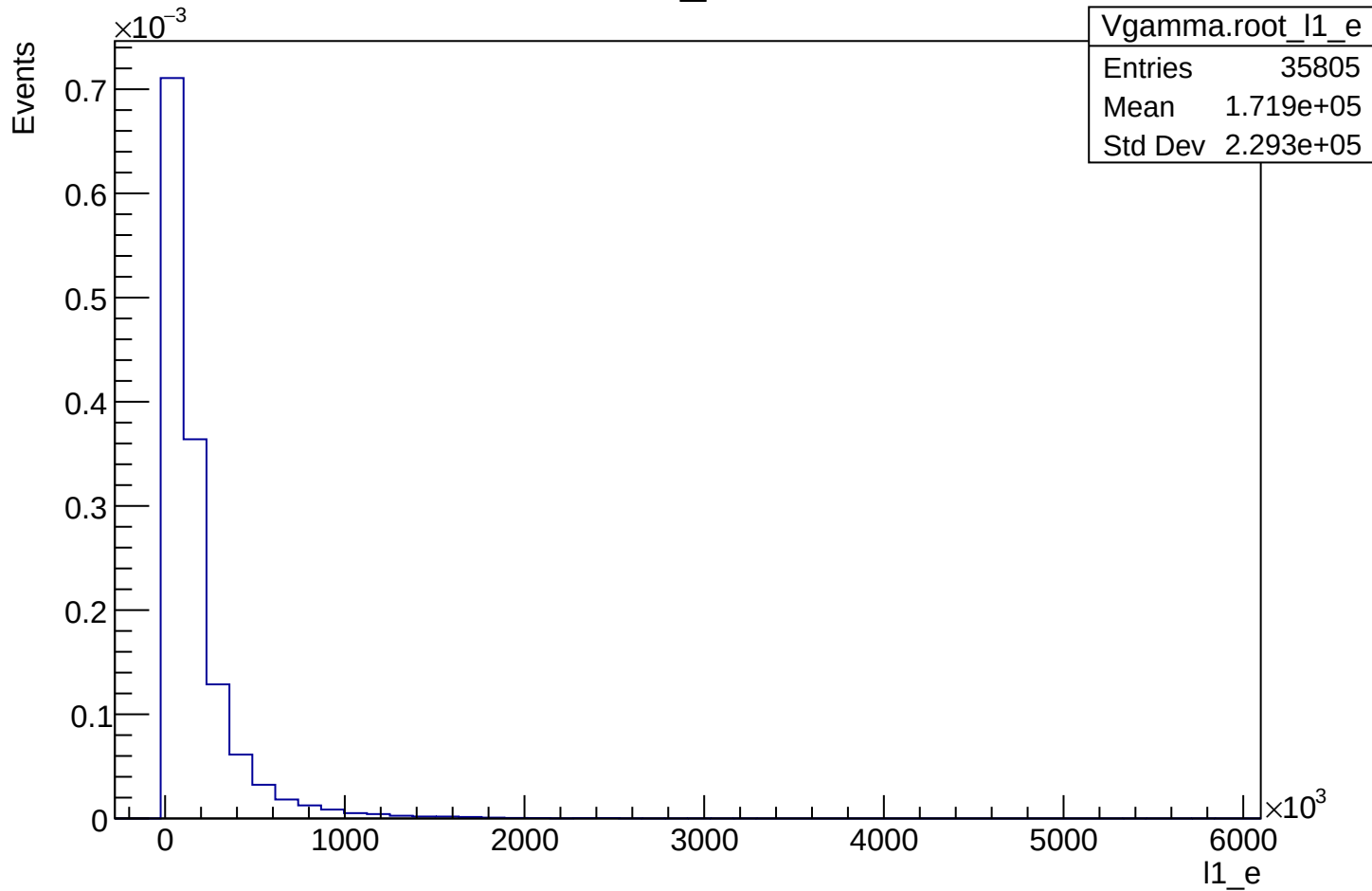
l1_charge



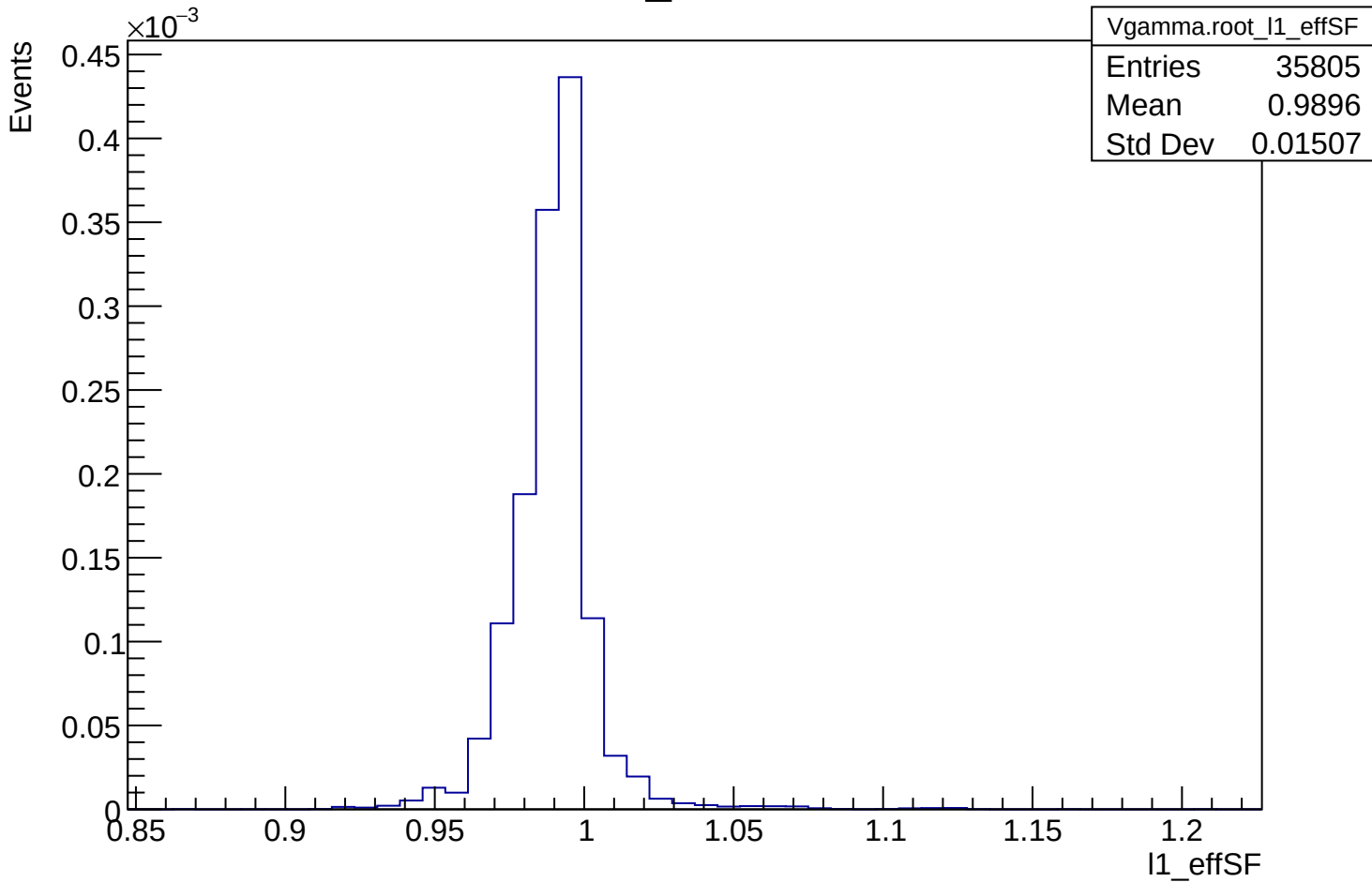
l1_d0sig



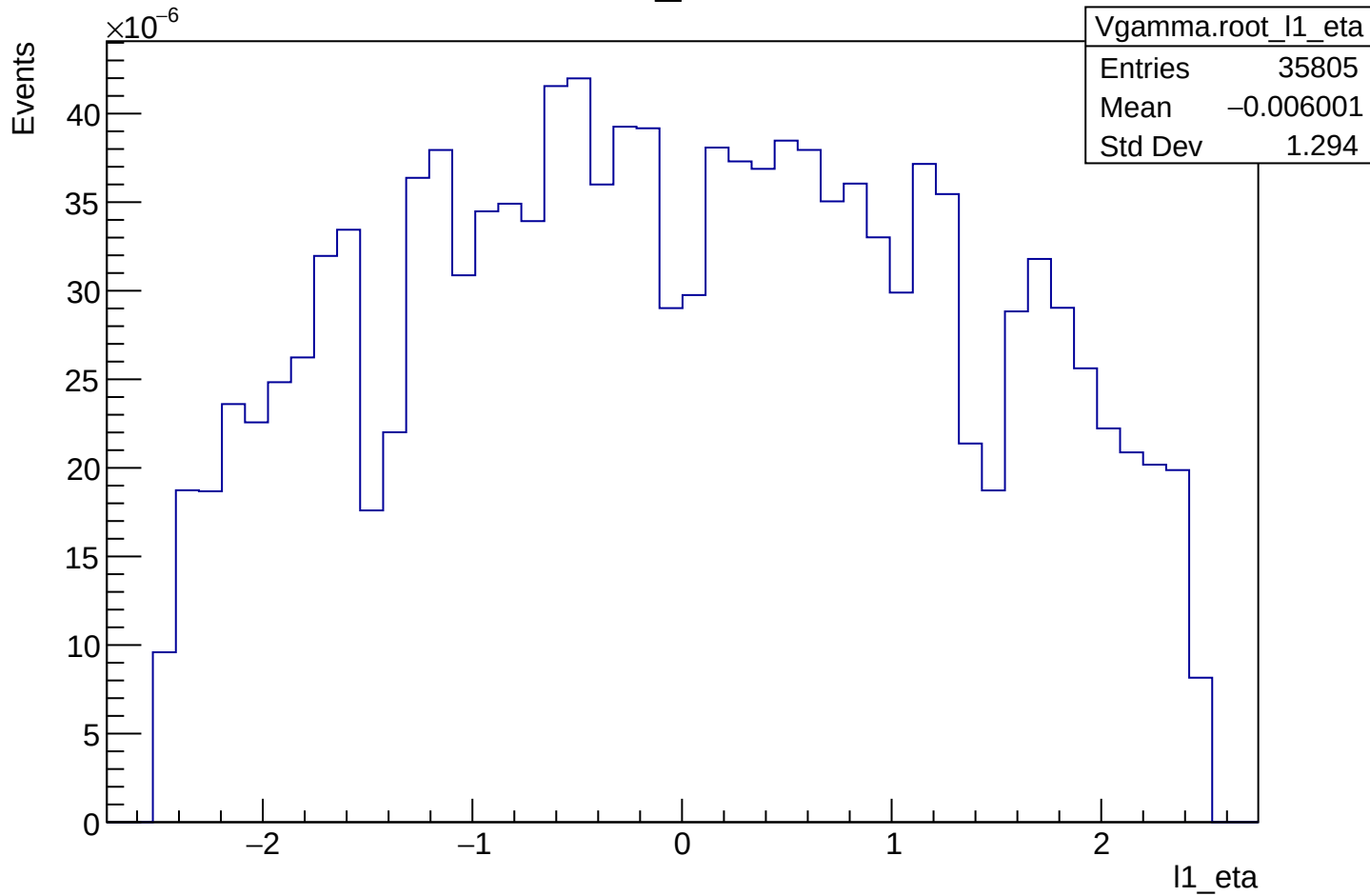
l1_e



l1_effSF

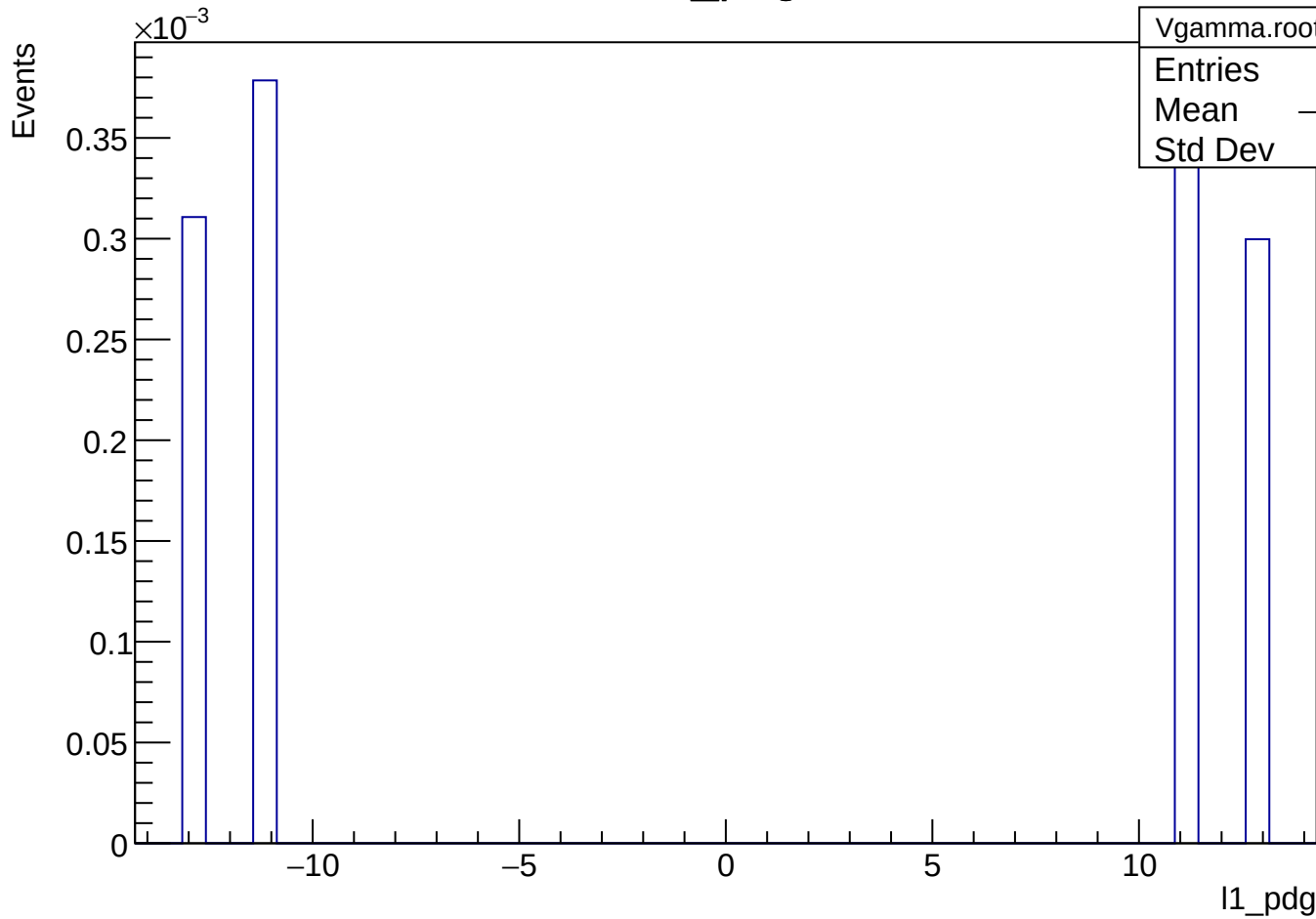


$l1_eta$

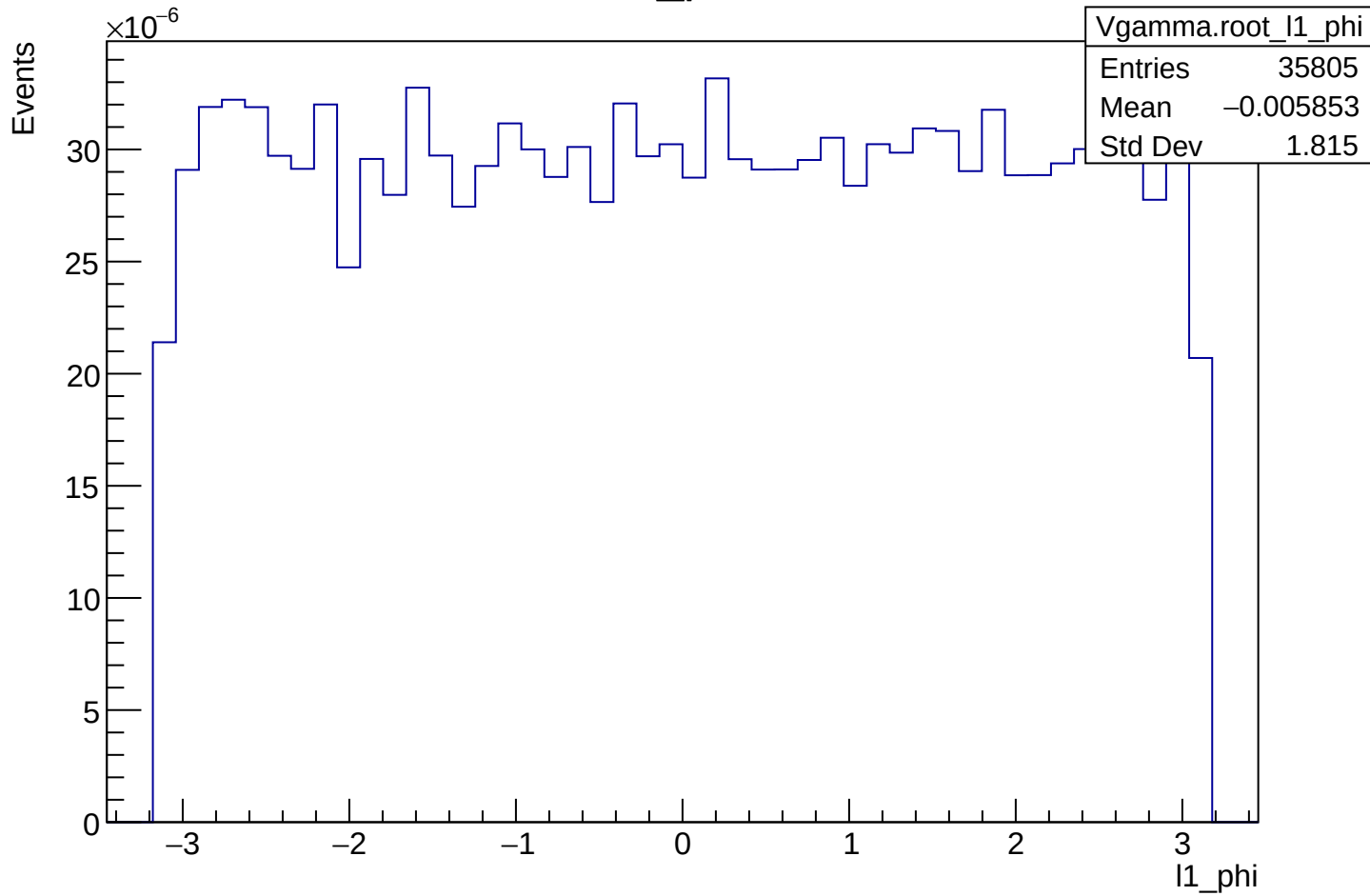


l1_pdg

Vgamma.root_l1_pdg	
Entries	35805
Mean	-0.1947
Std Dev	11.94

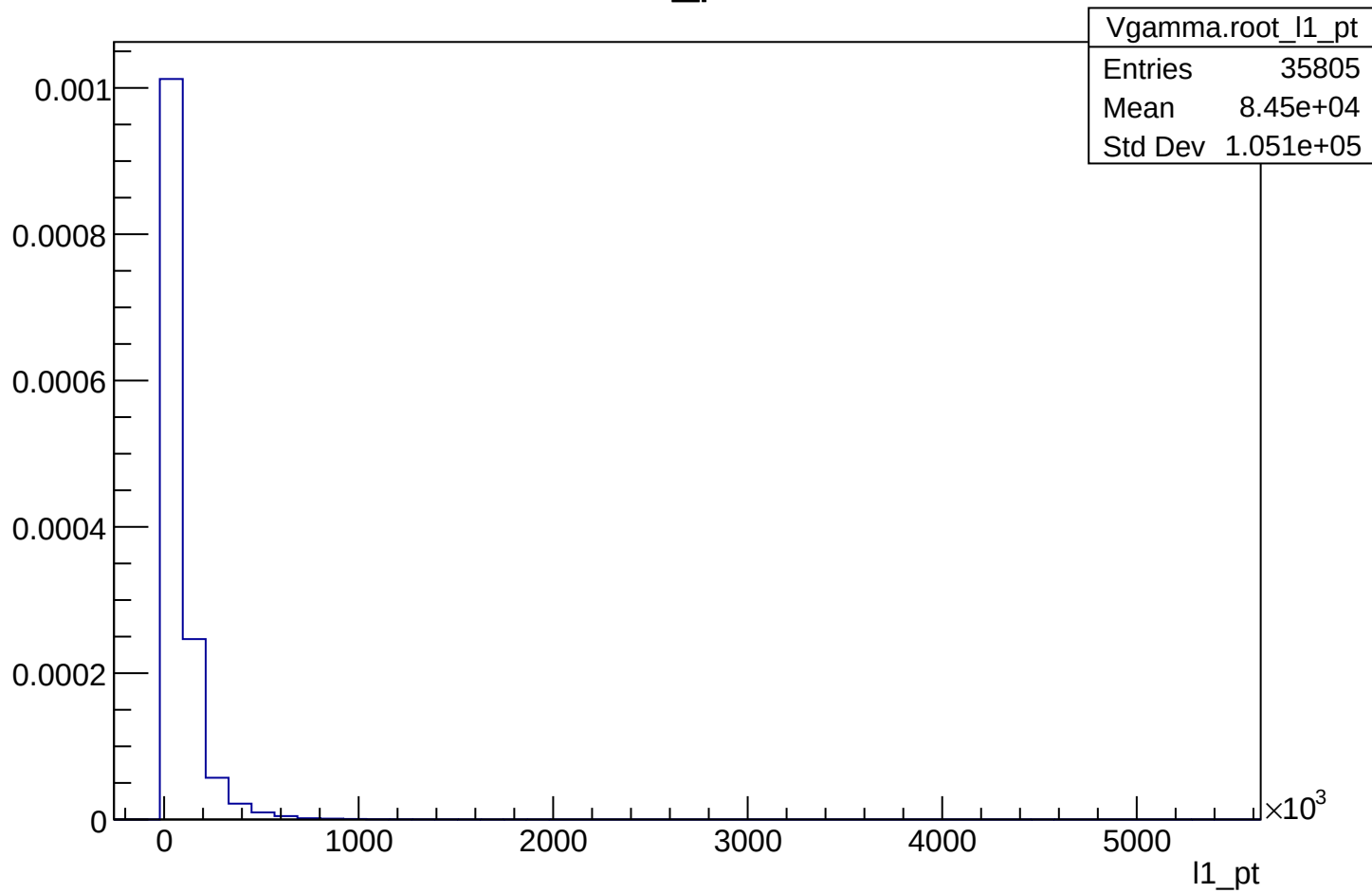


l1_phi

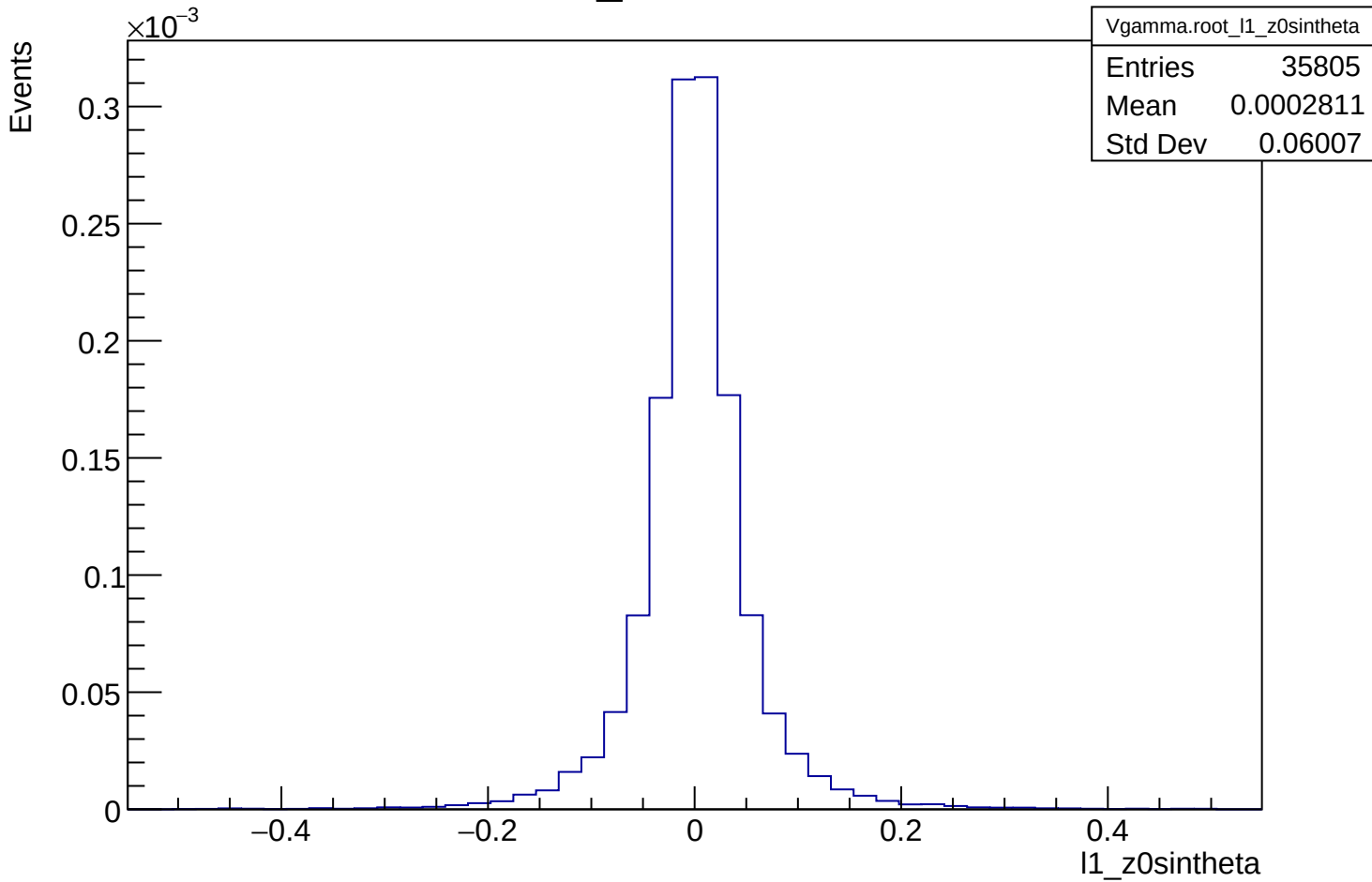


l1_pt

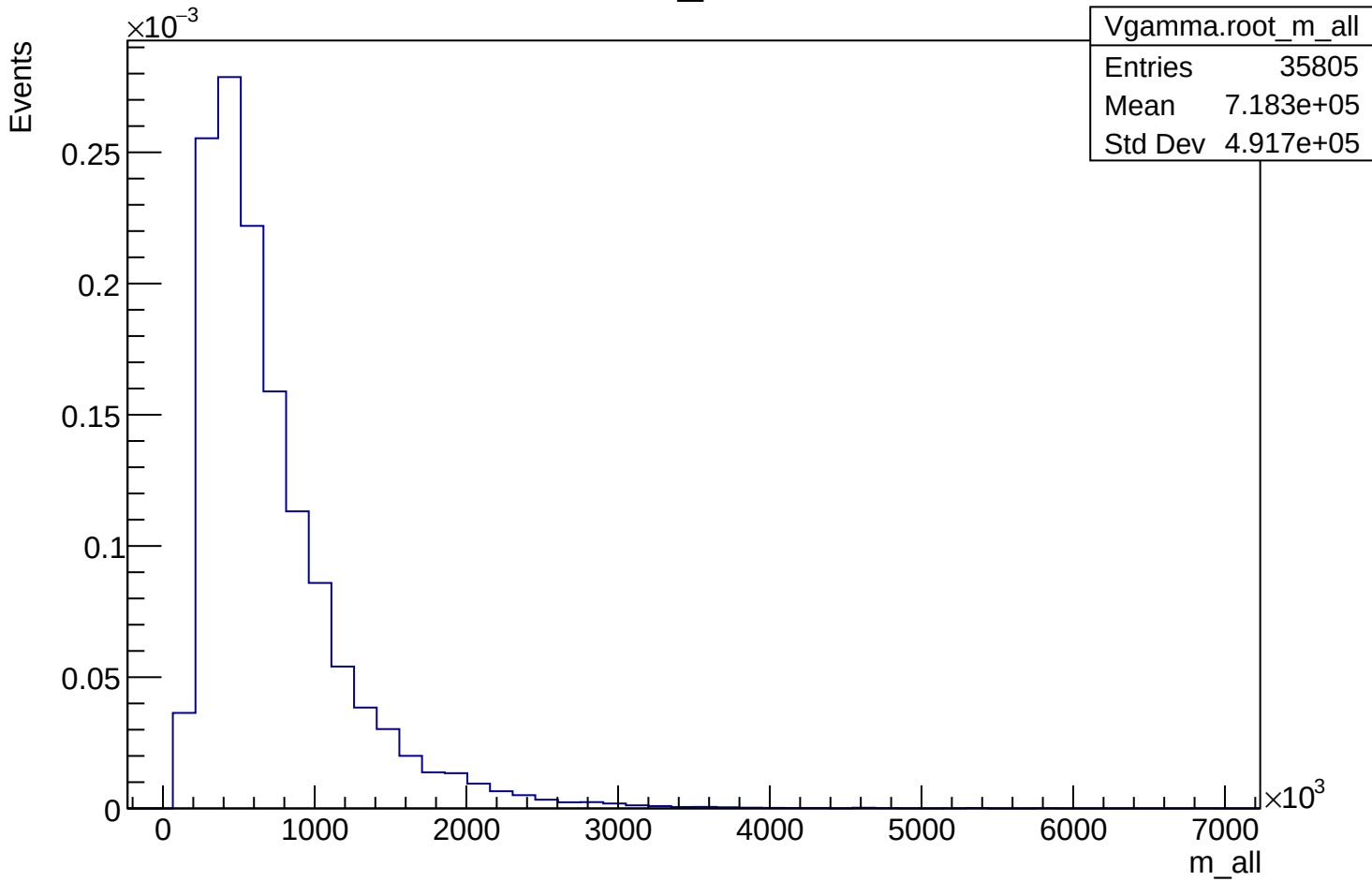
Events



l1_z0sintheta

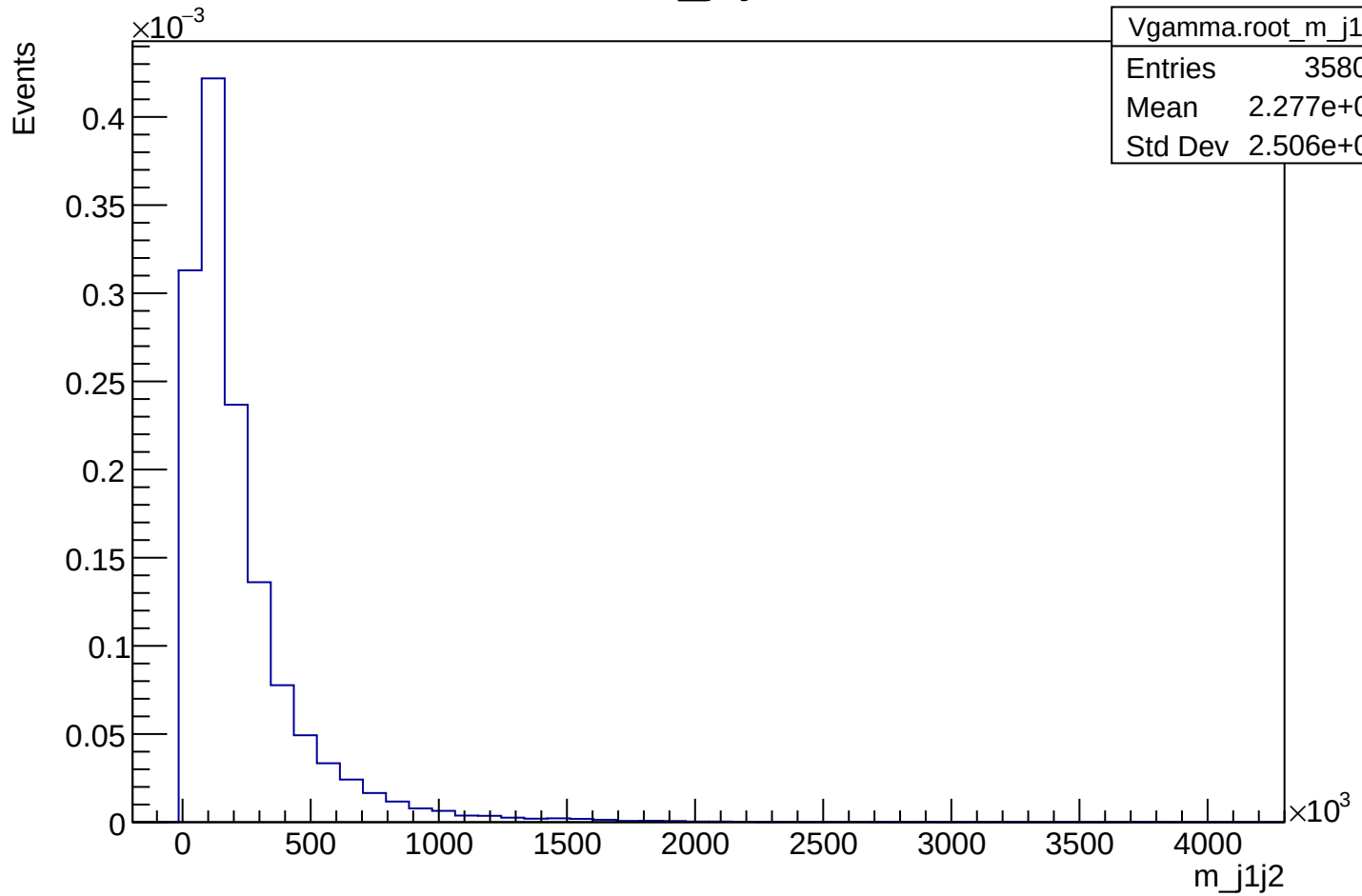


m_all

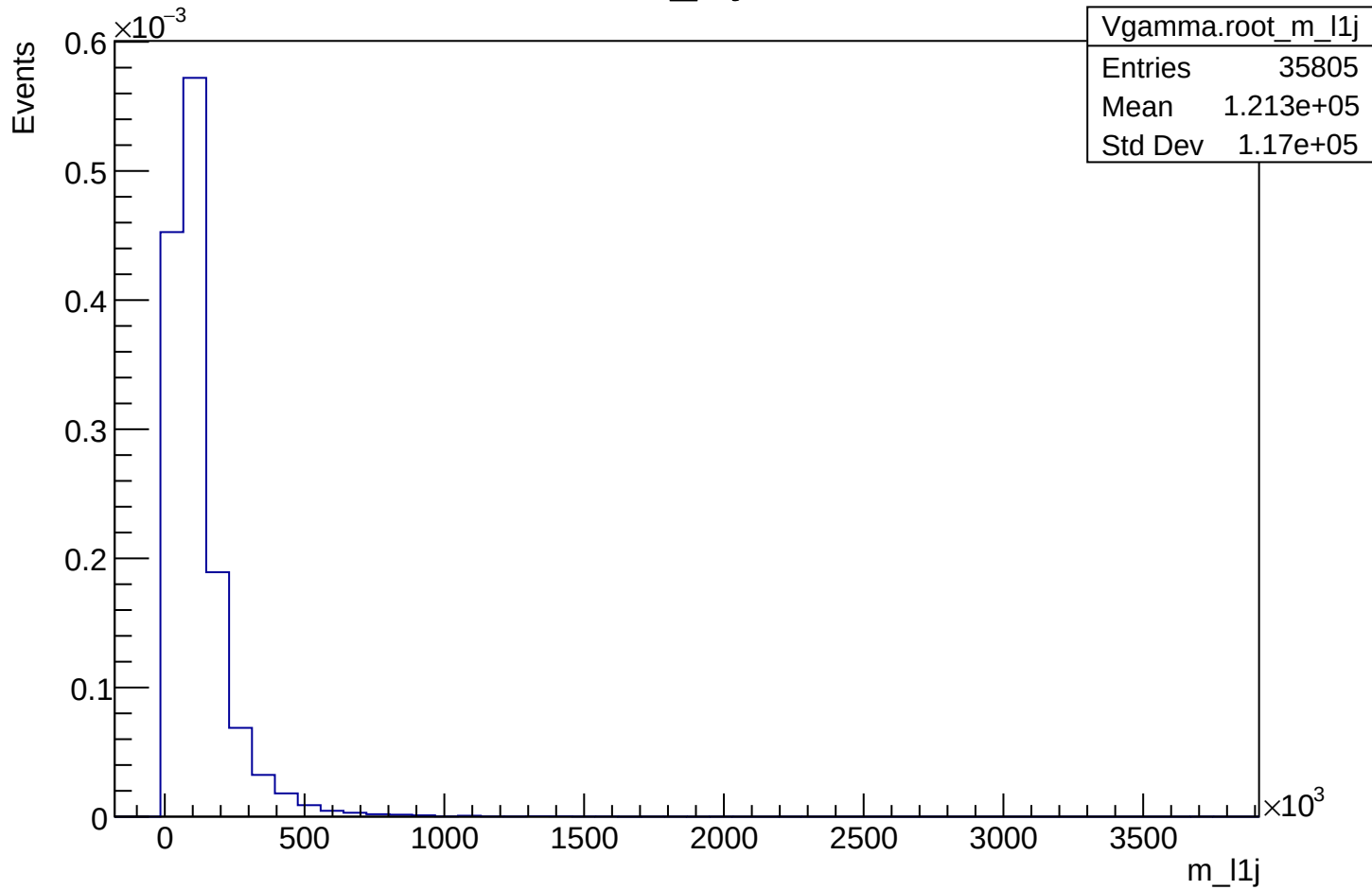


m_j1j2

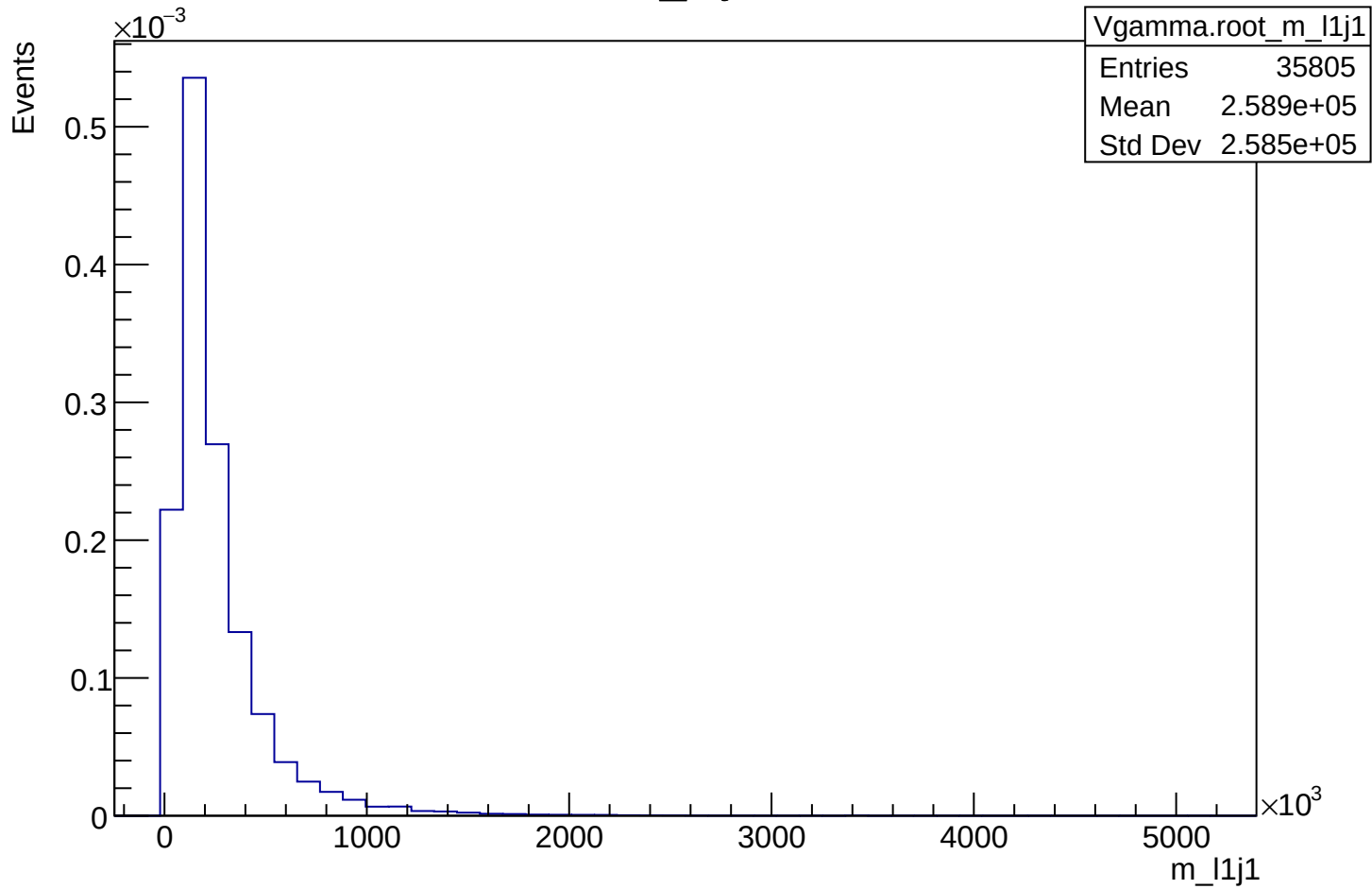
Vgamma.root_m_j1j2	
Entries	35805
Mean	2.277e+05
Std Dev	2.506e+05



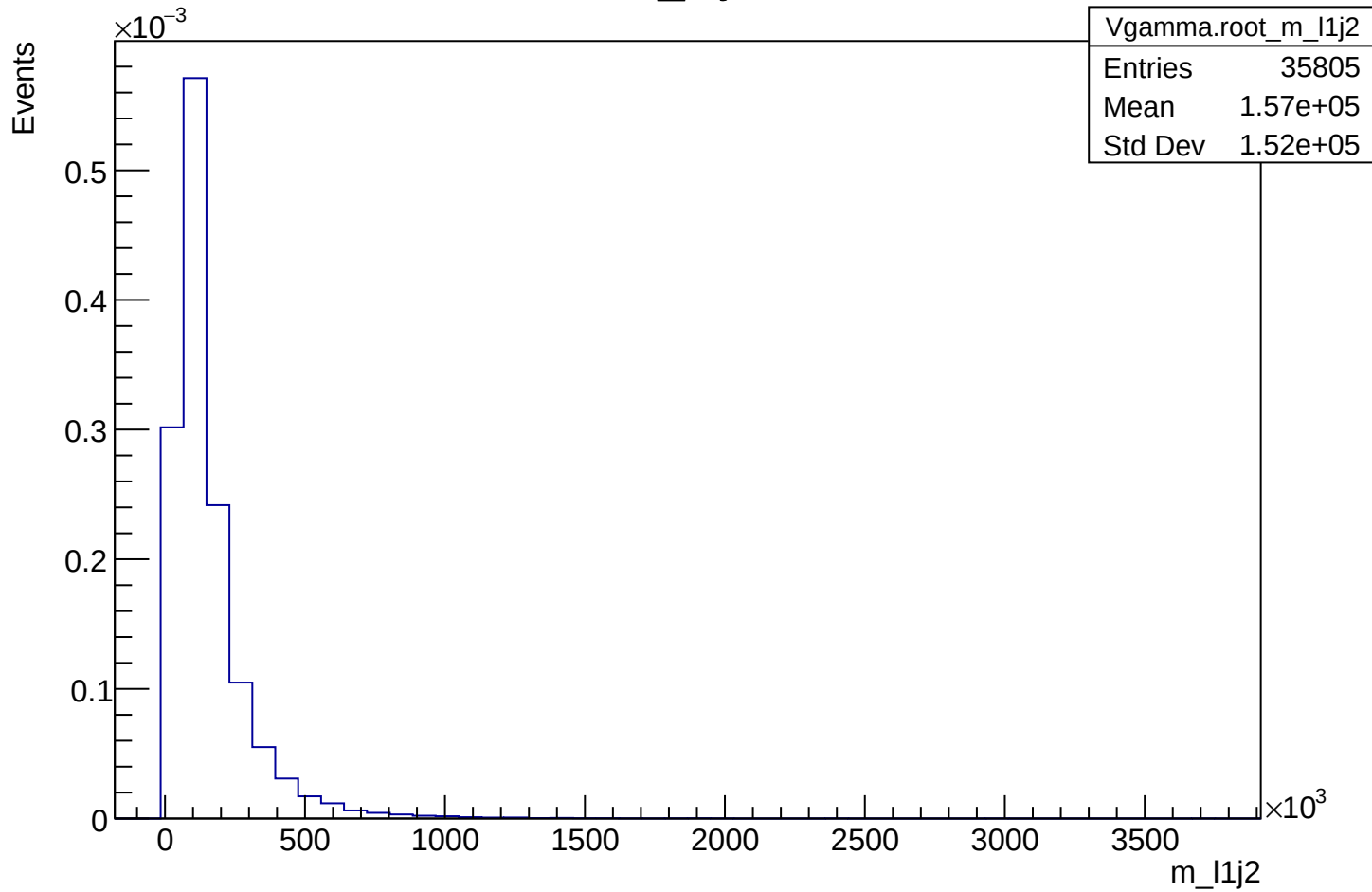
m_l1j



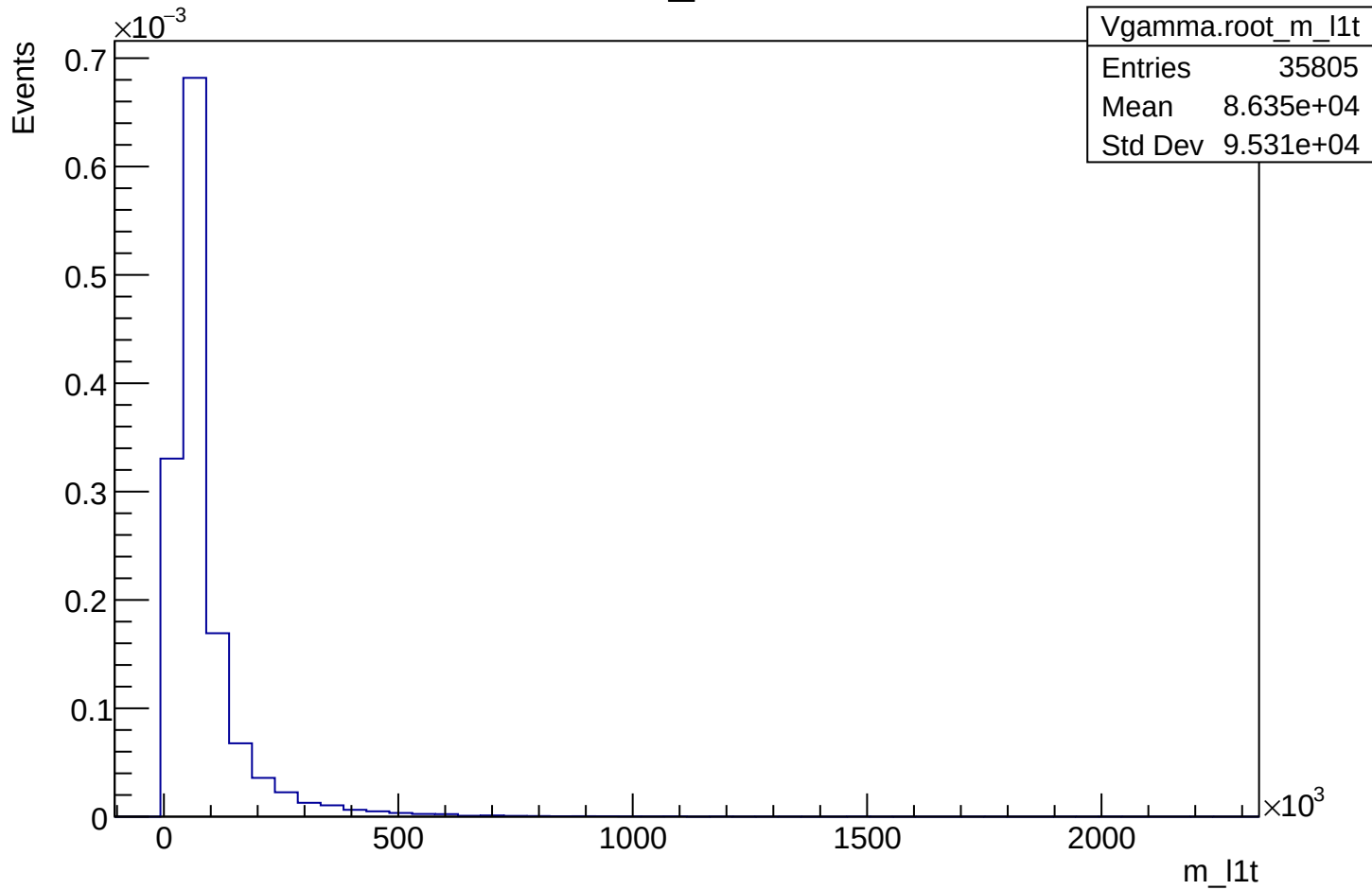
m_l1j1



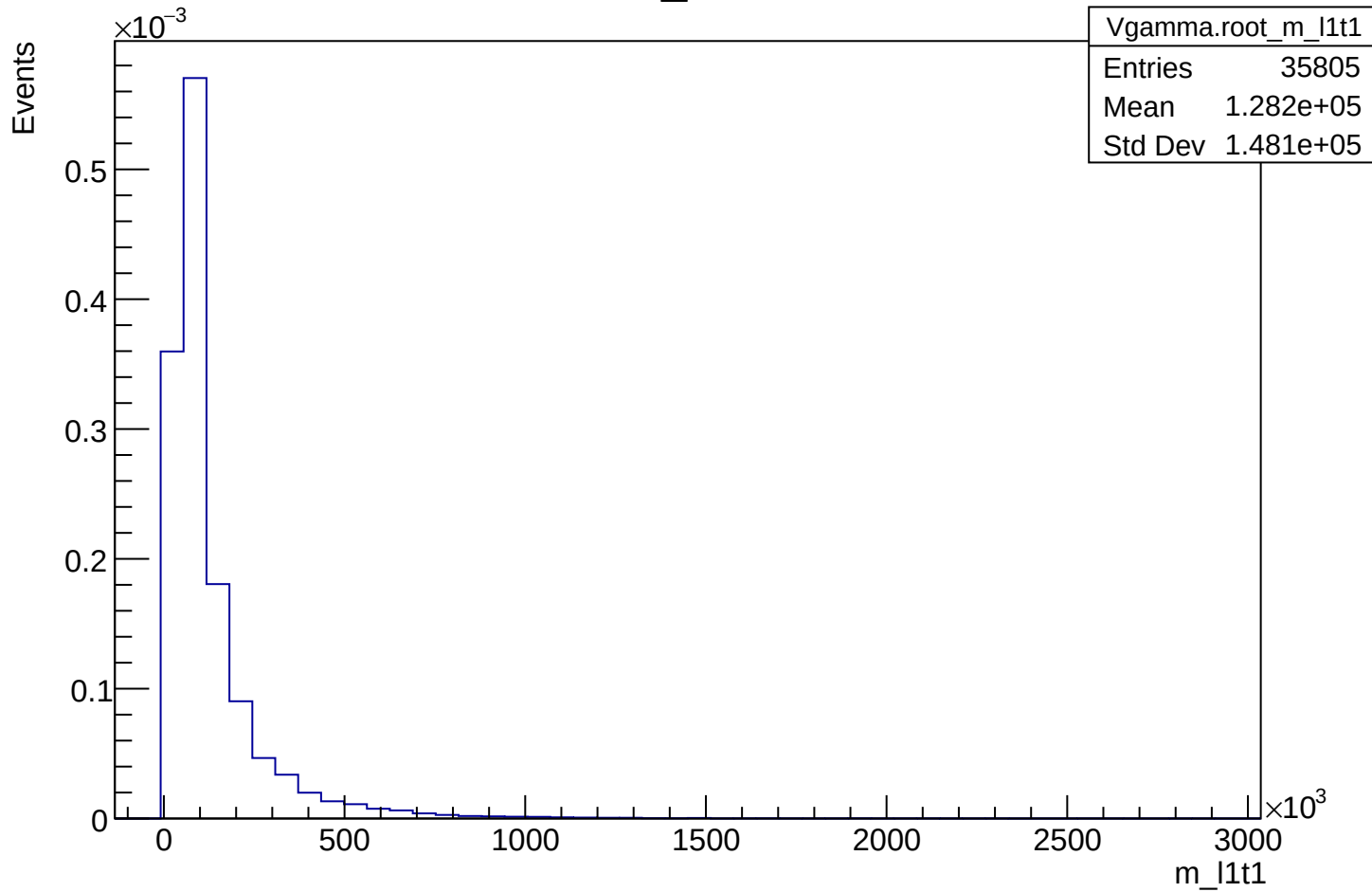
m_l1j2



m_l1t

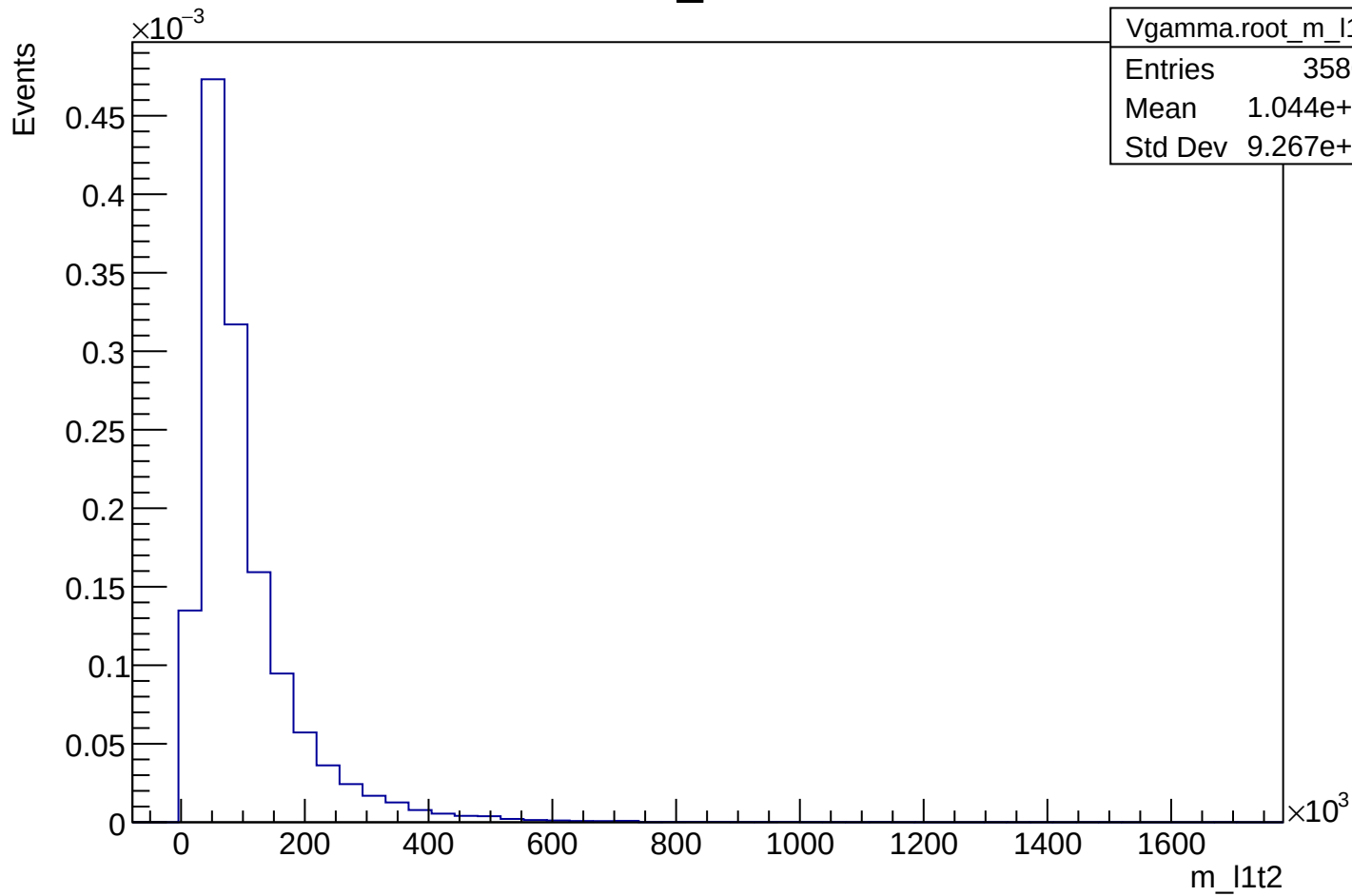


m_l1t1



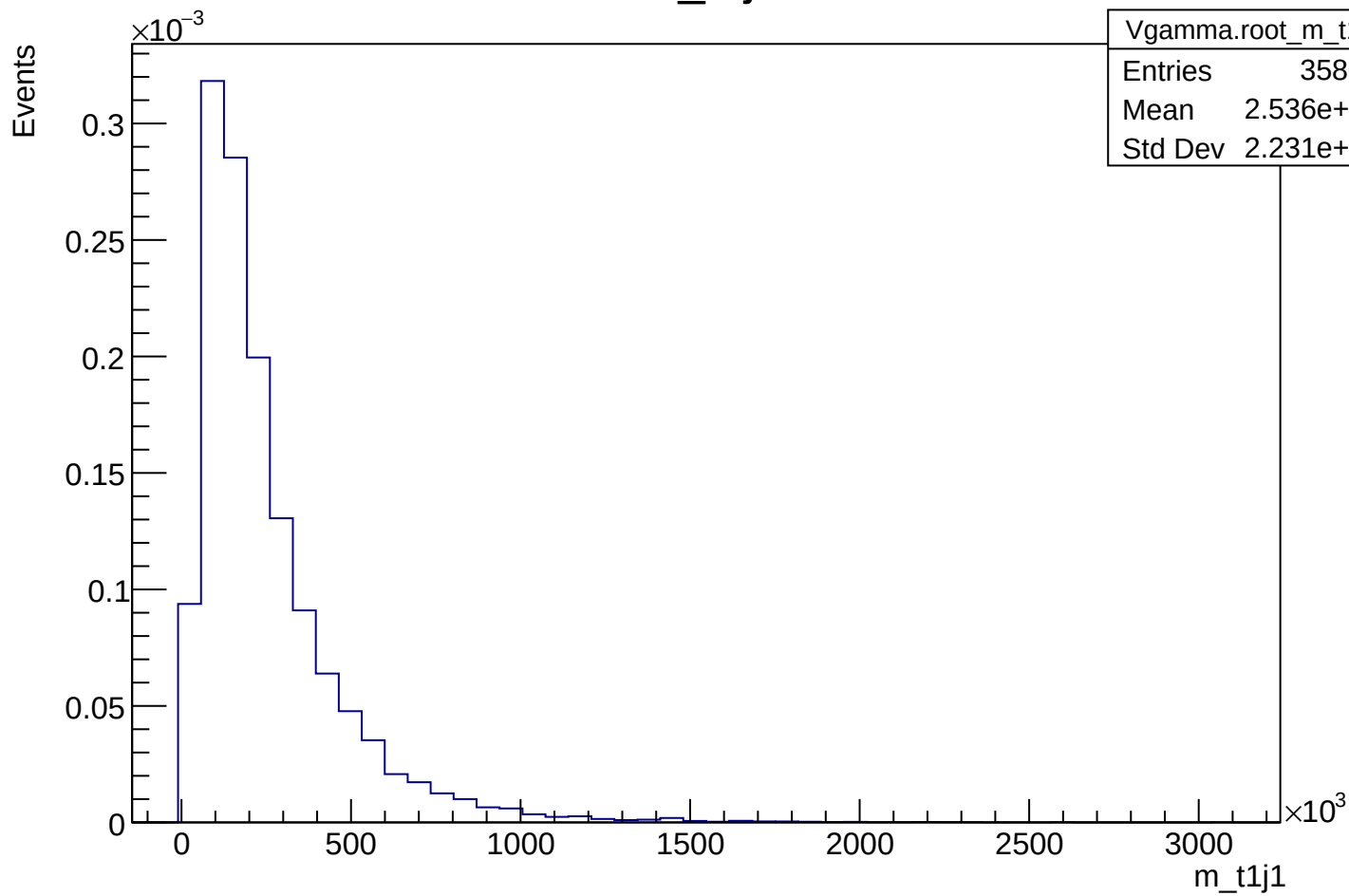
m_l1t2

Vgamma.root_m_l1t2	
Entries	35805
Mean	1.044e+05
Std Dev	9.267e+04

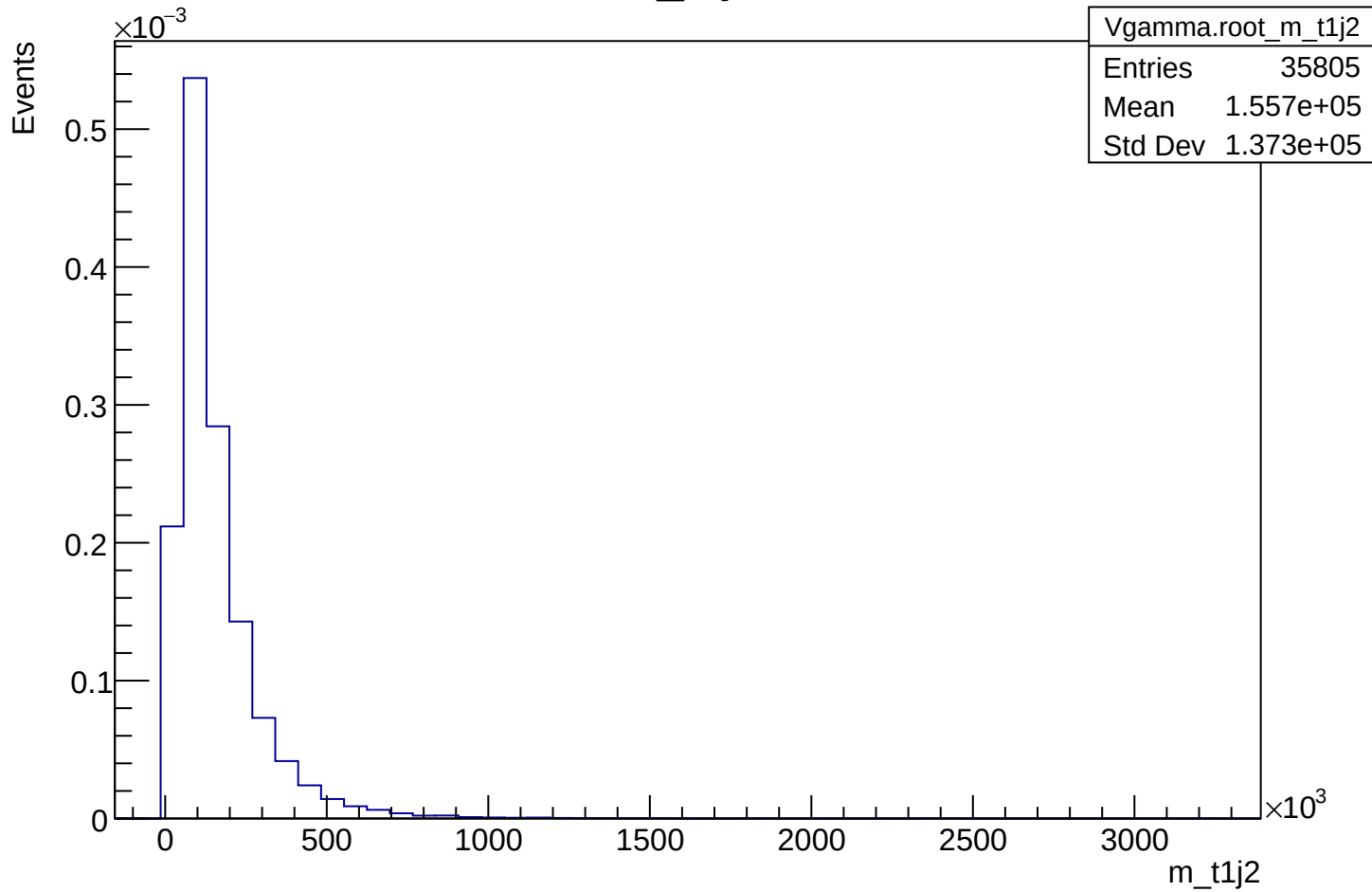


m_t1j1

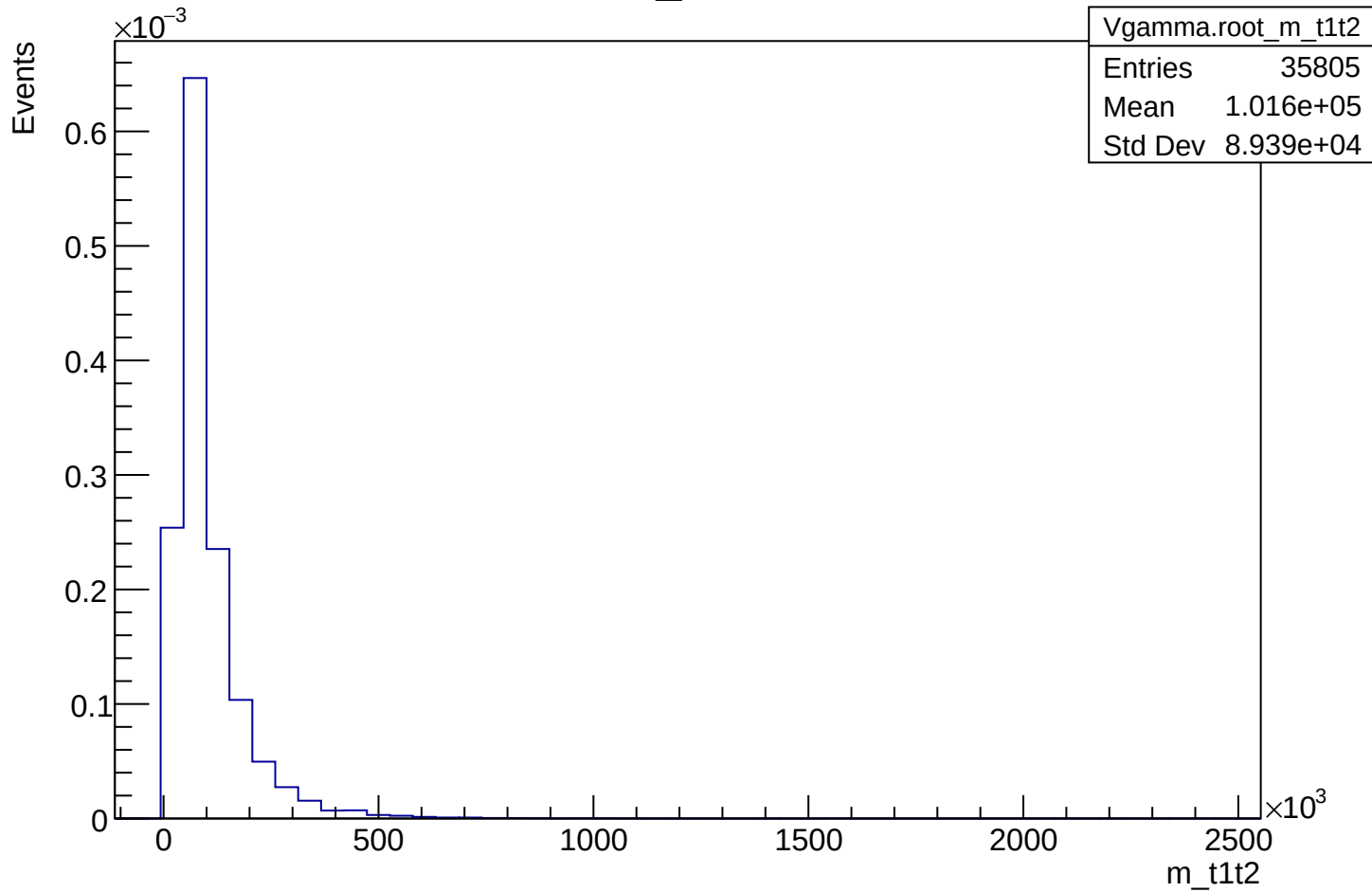
Vgamma.root_m_t1j1	
Entries	35805
Mean	2.536e+05
Std Dev	2.231e+05



m_t1j2

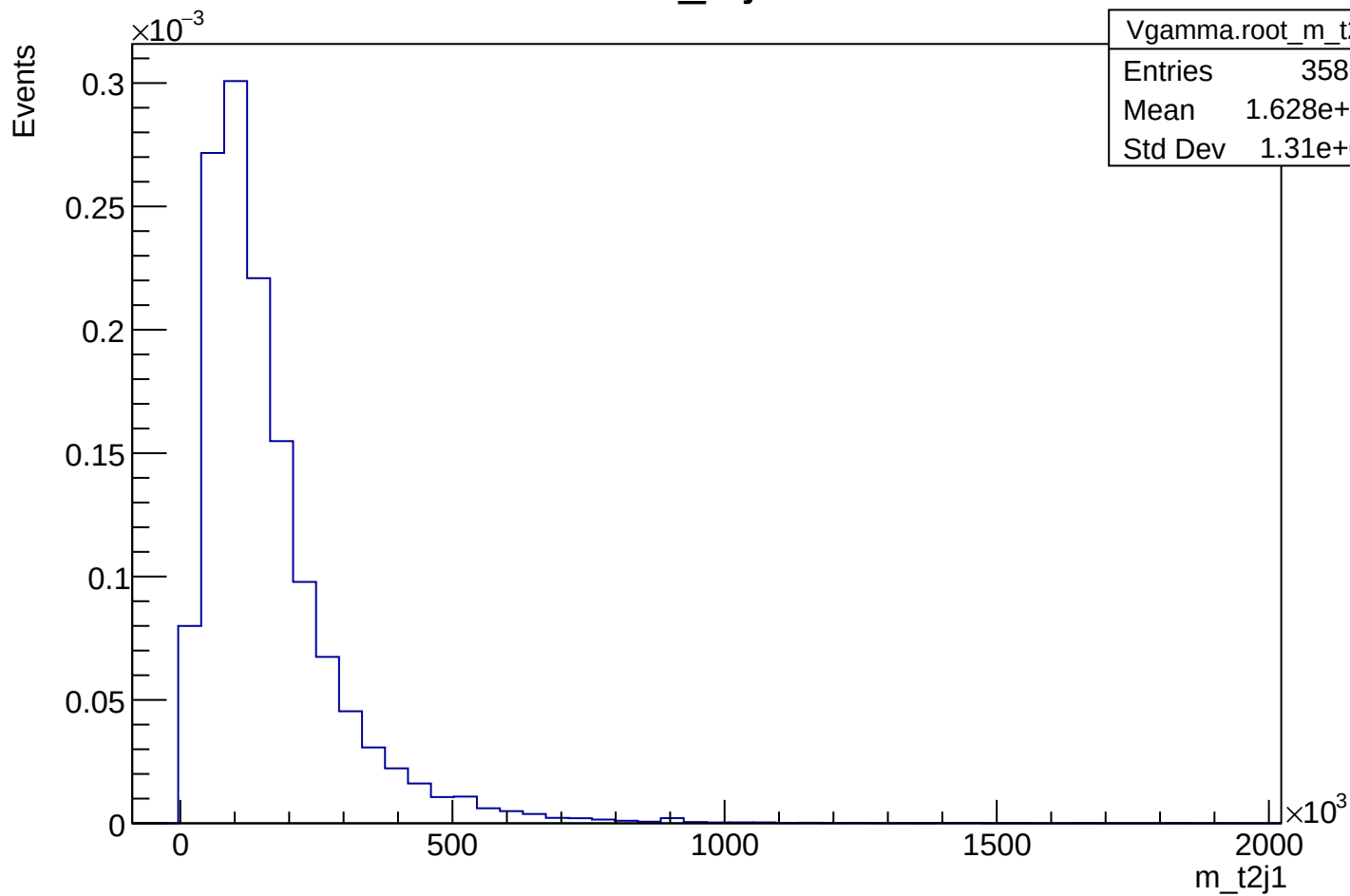


m_t1t2



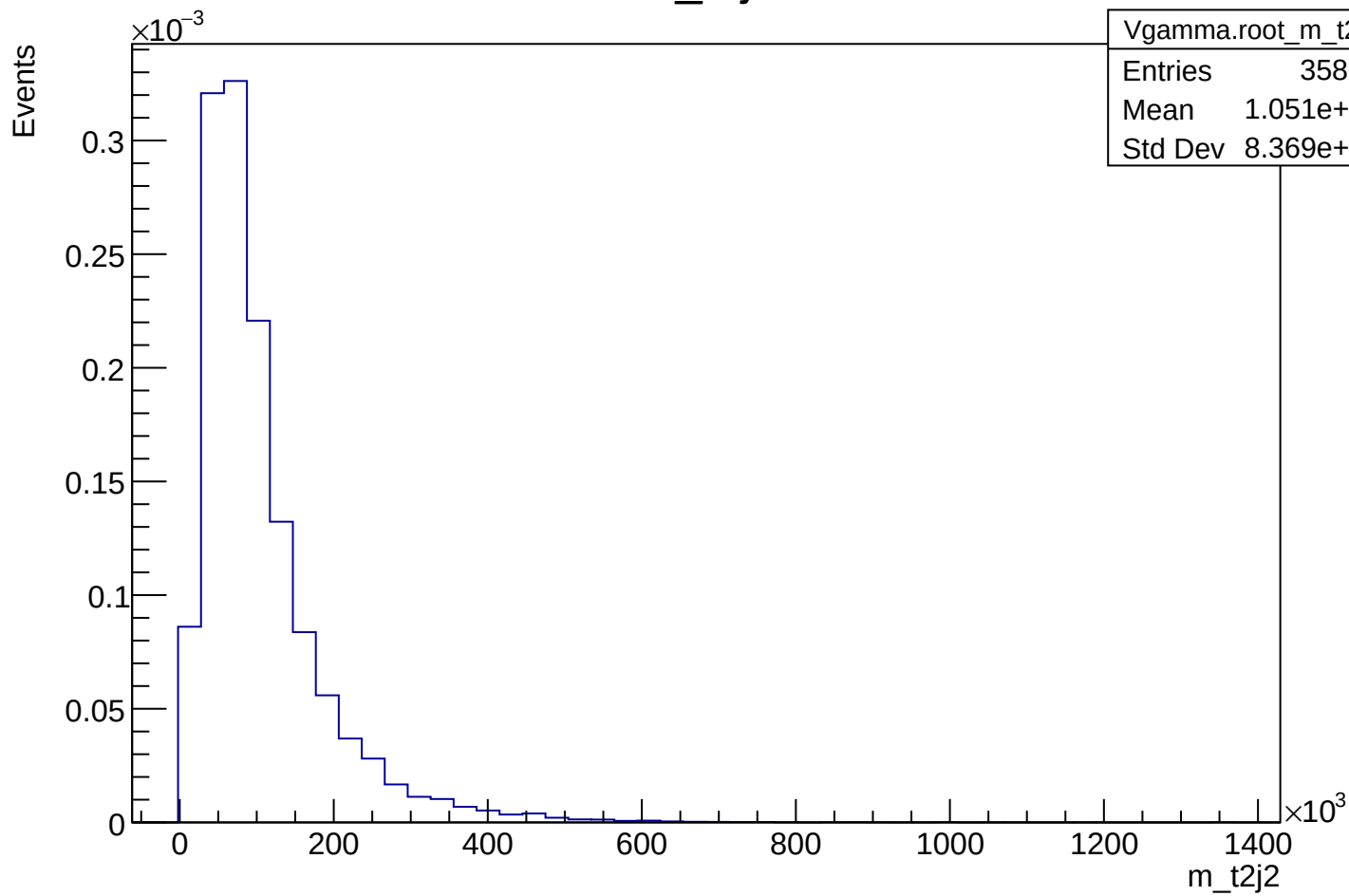
m_t2j1

Vgamma.root_m_t2j1	
Entries	35805
Mean	1.628e+05
Std Dev	1.31e+05

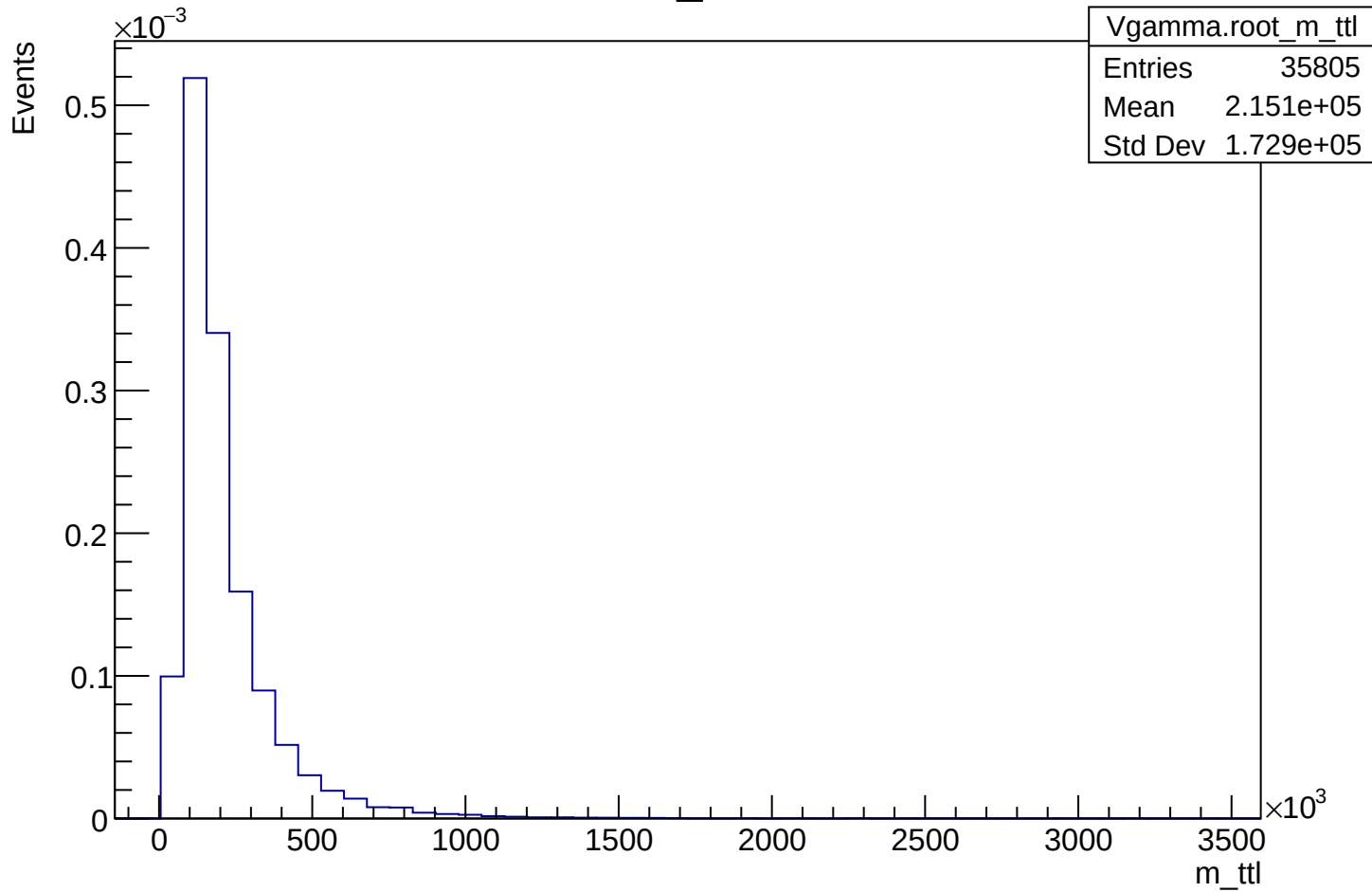


m_t2j2

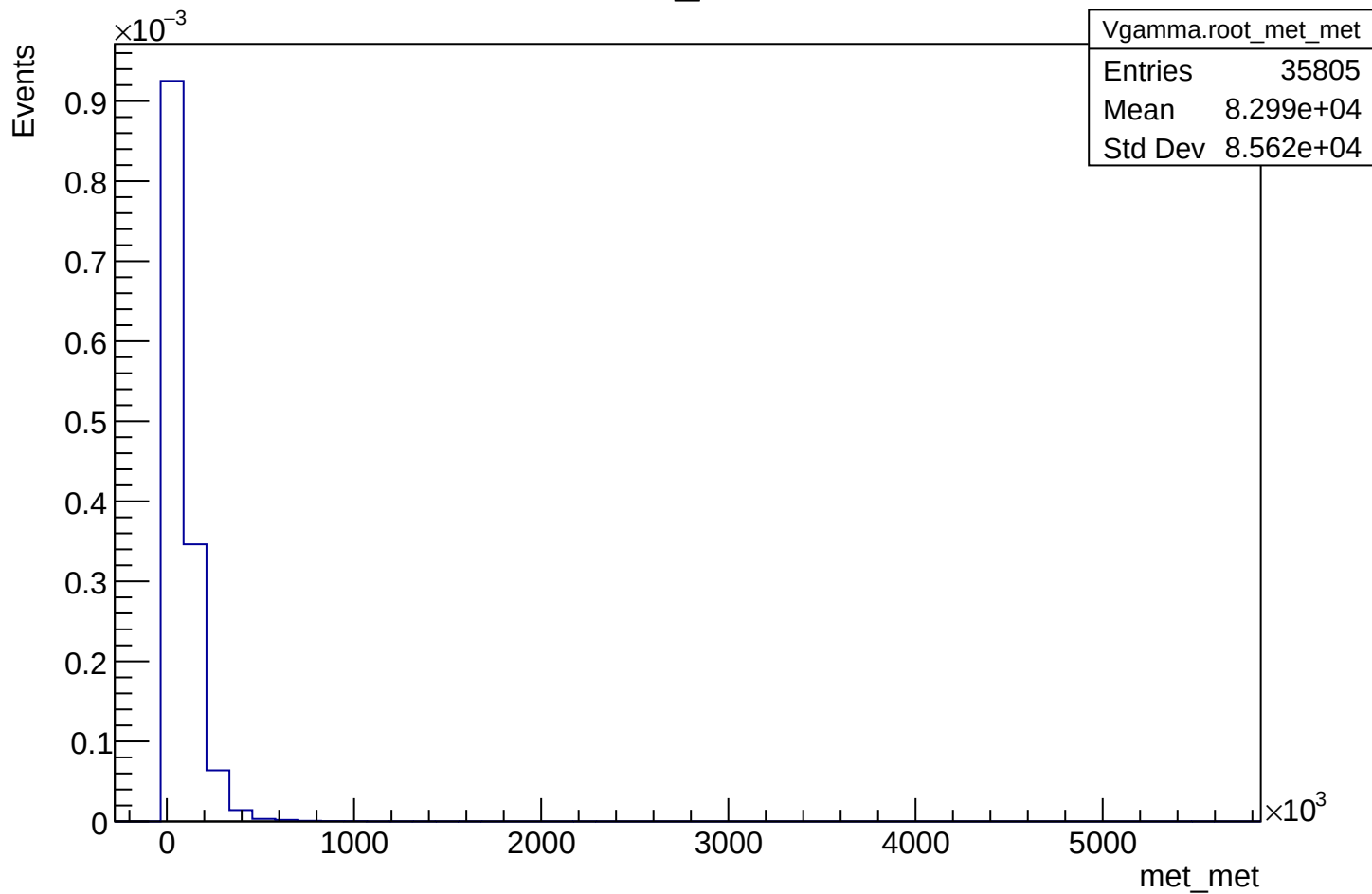
Vgamma.root_m_t2j2	
Entries	35805
Mean	1.051e+05
Std Dev	8.369e+04



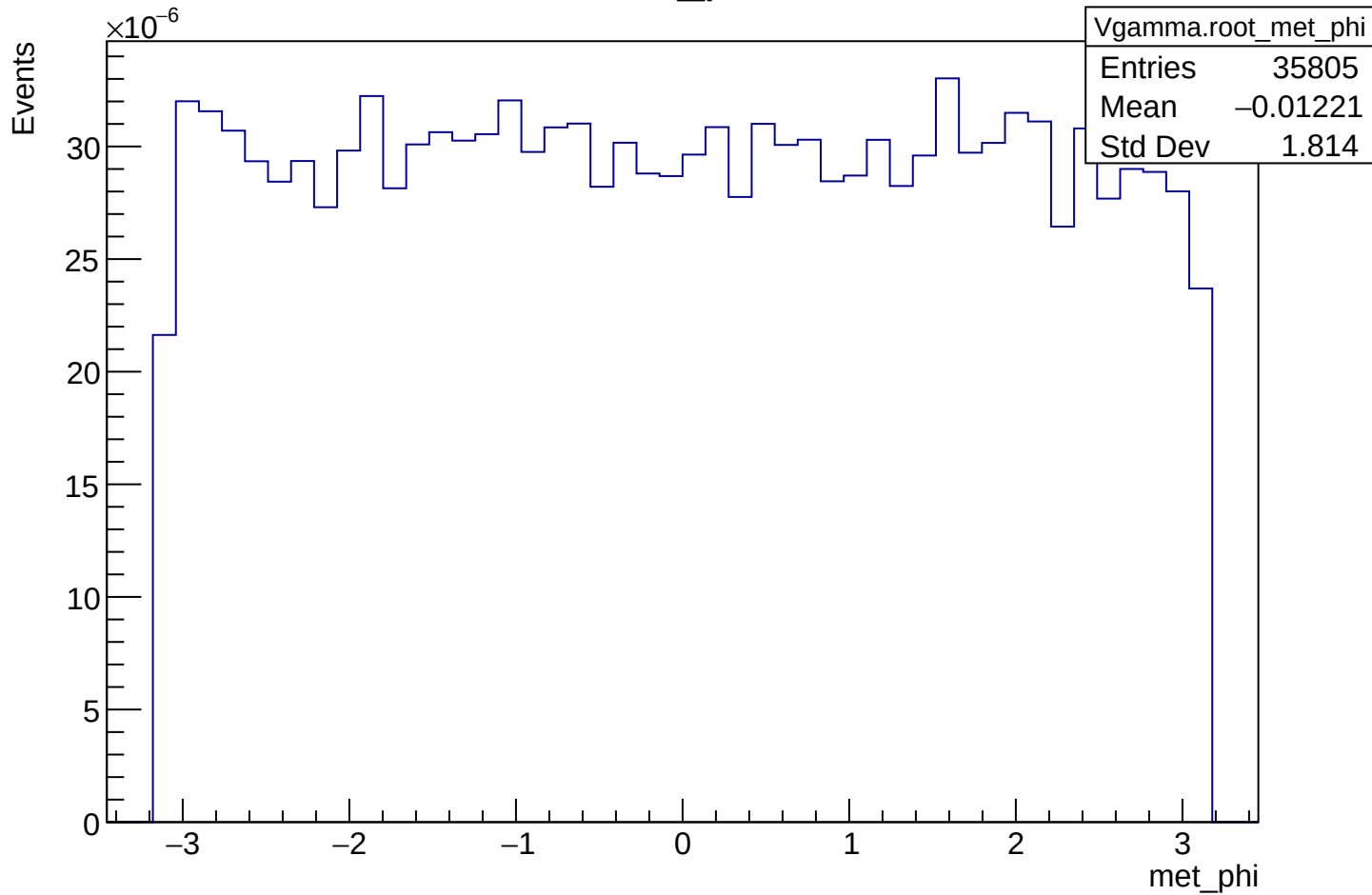
m_ttl



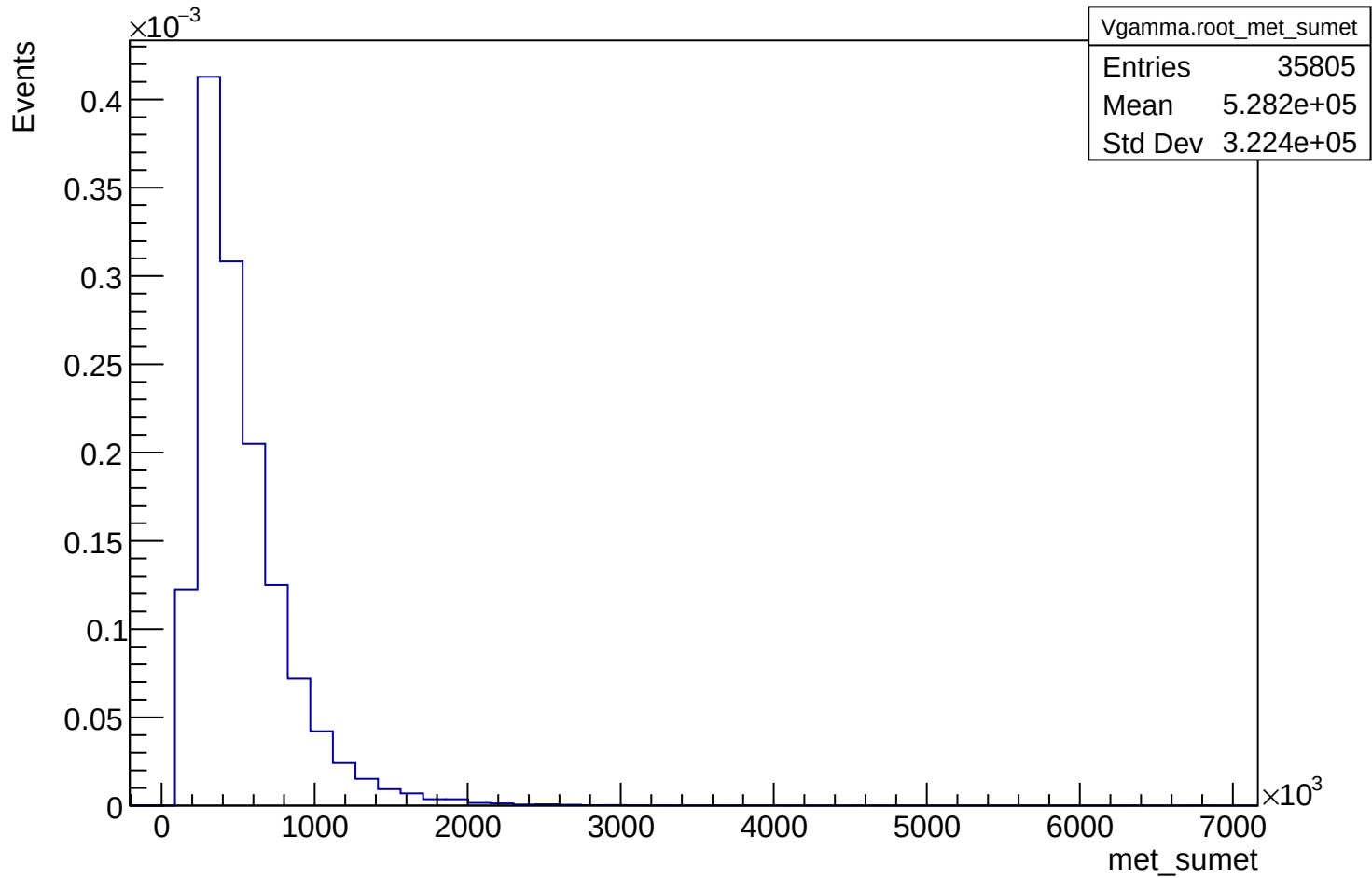
met_met



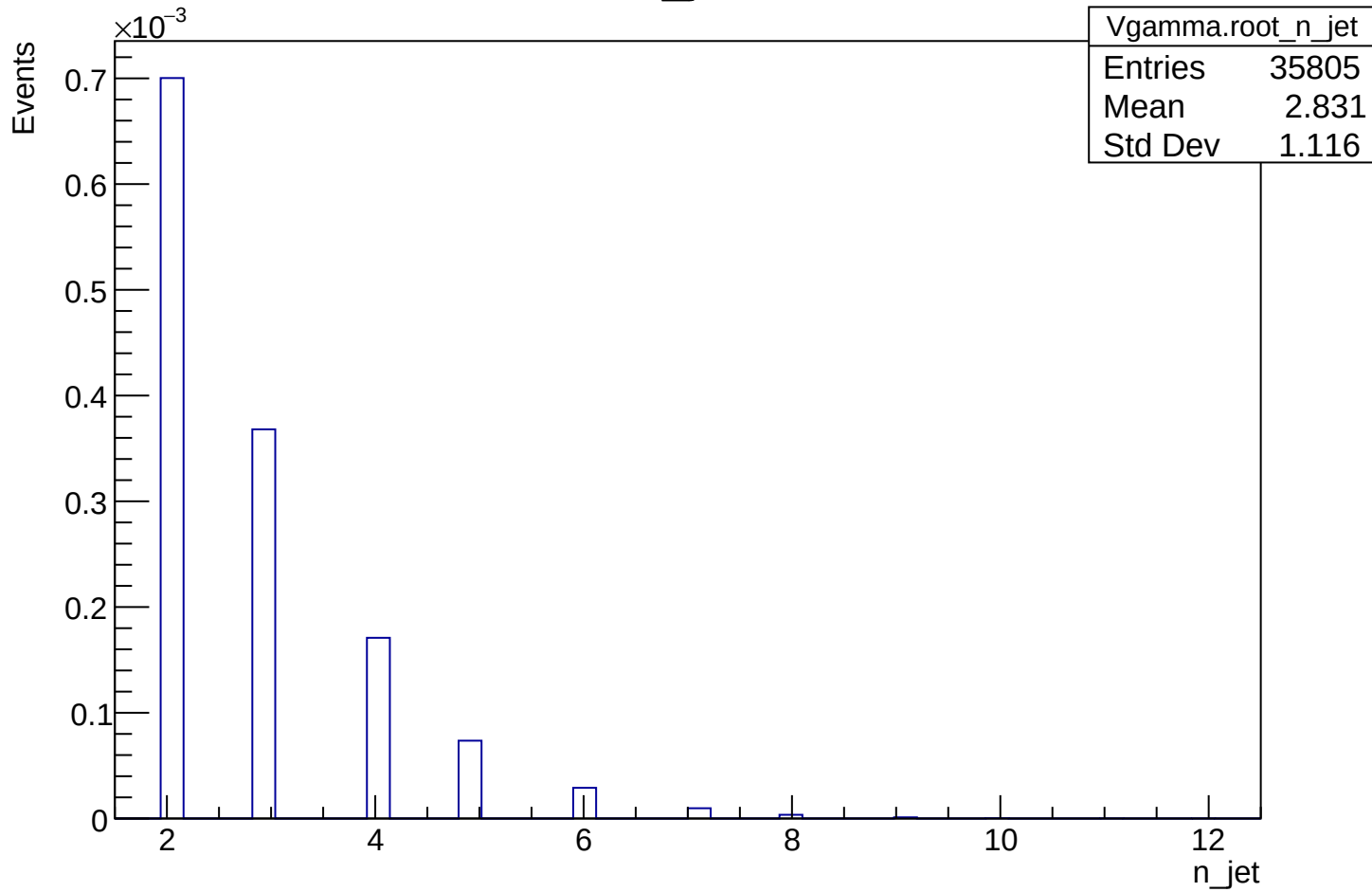
met_phi



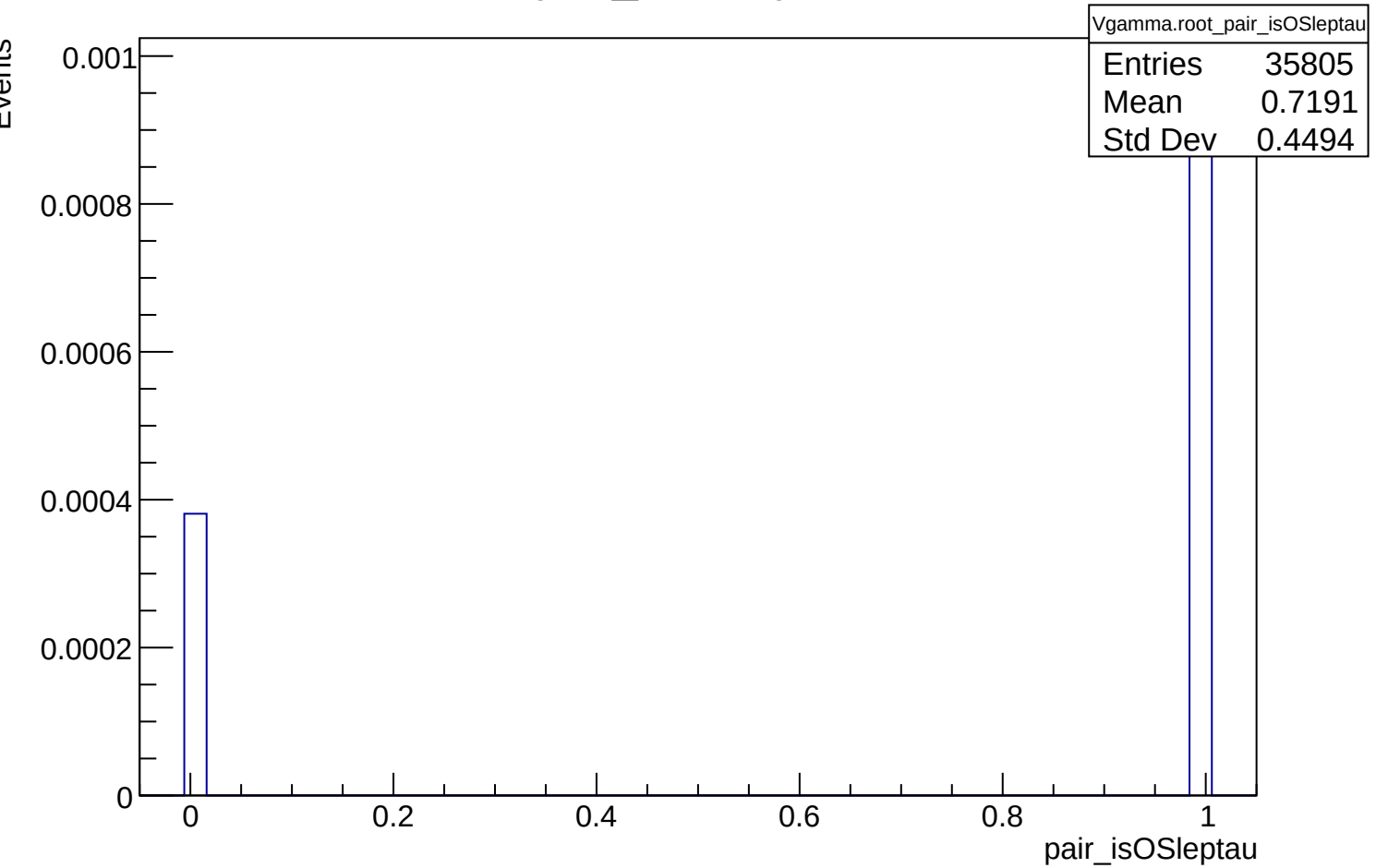
met_sumet



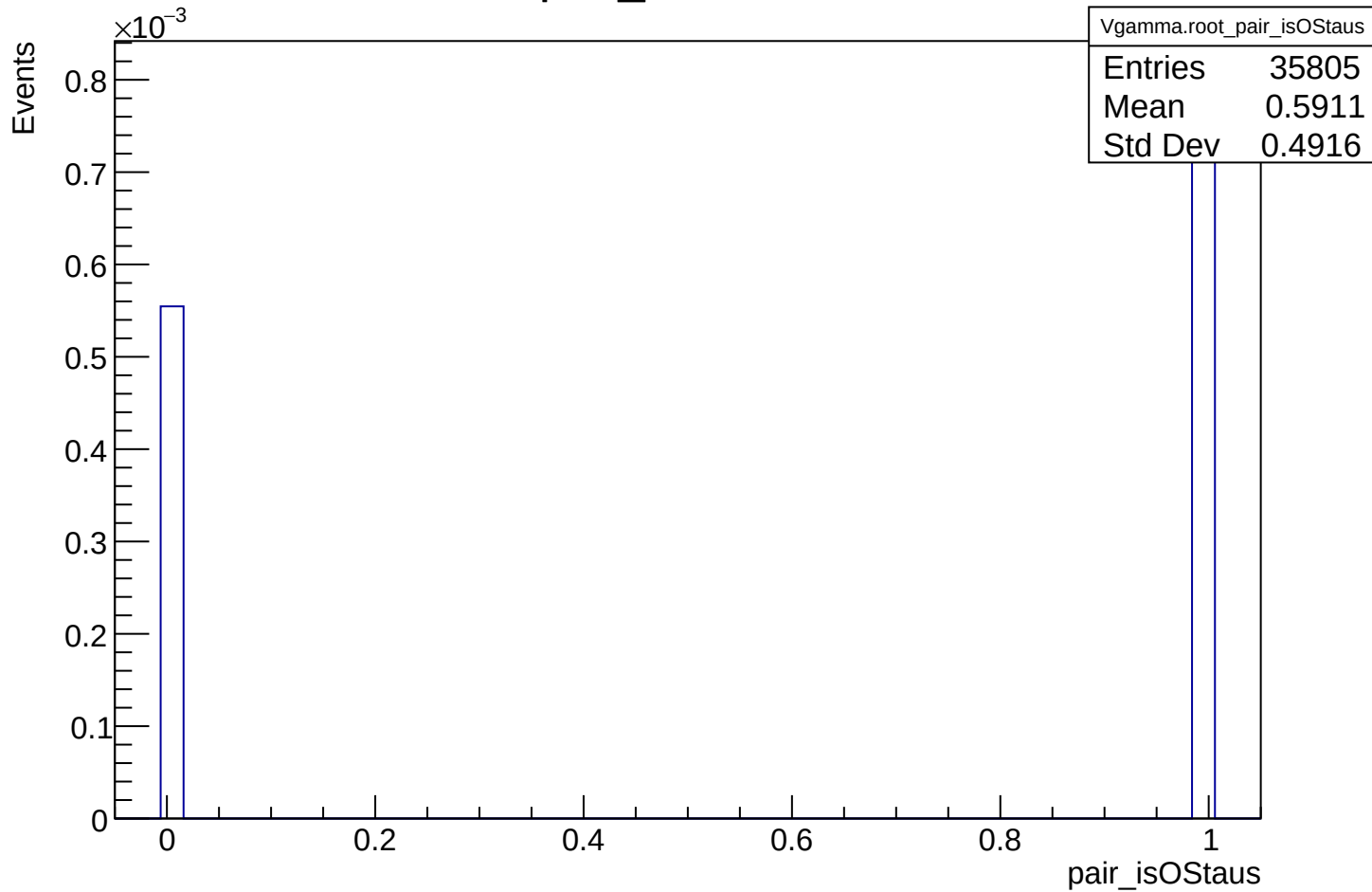
n_jet



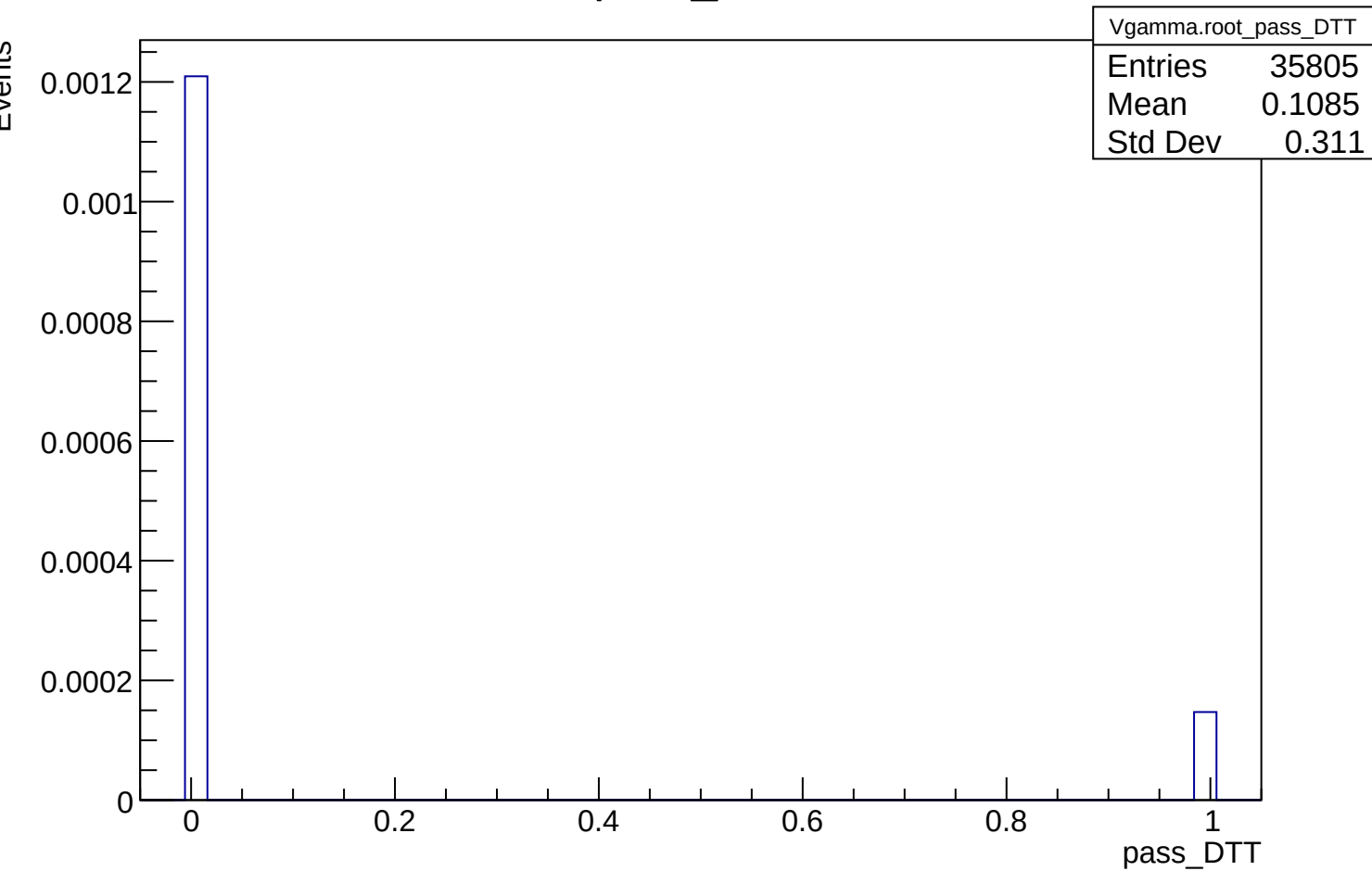
pair_isOSleptau



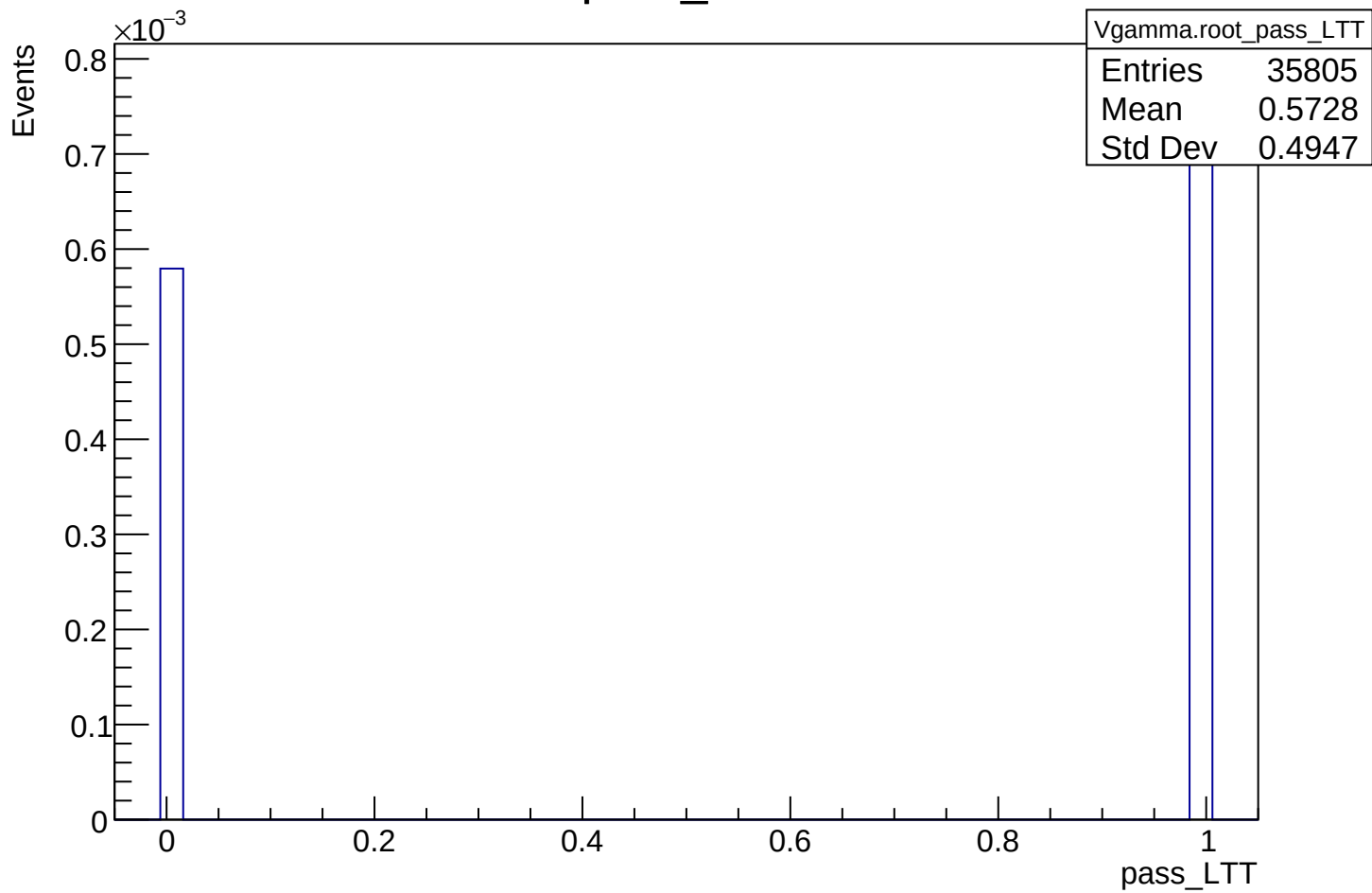
pair_isOStaus



pass_DTT



pass_LTT



pass_SLT

Events

0.001

0.0008

0.0006

0.0004

0.0002

0

0

0.2

0.4

0.6

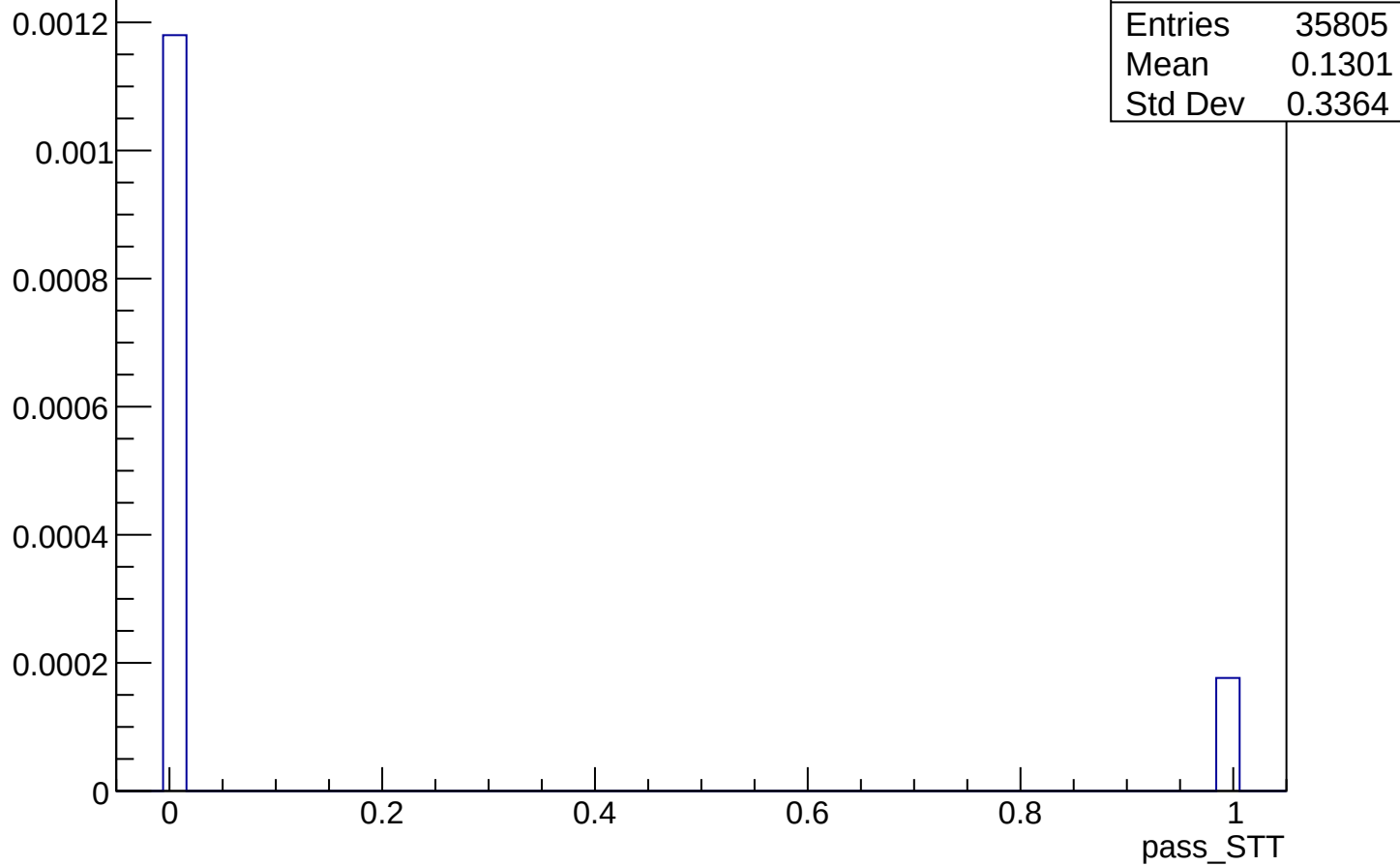
0.8

pass_SLT

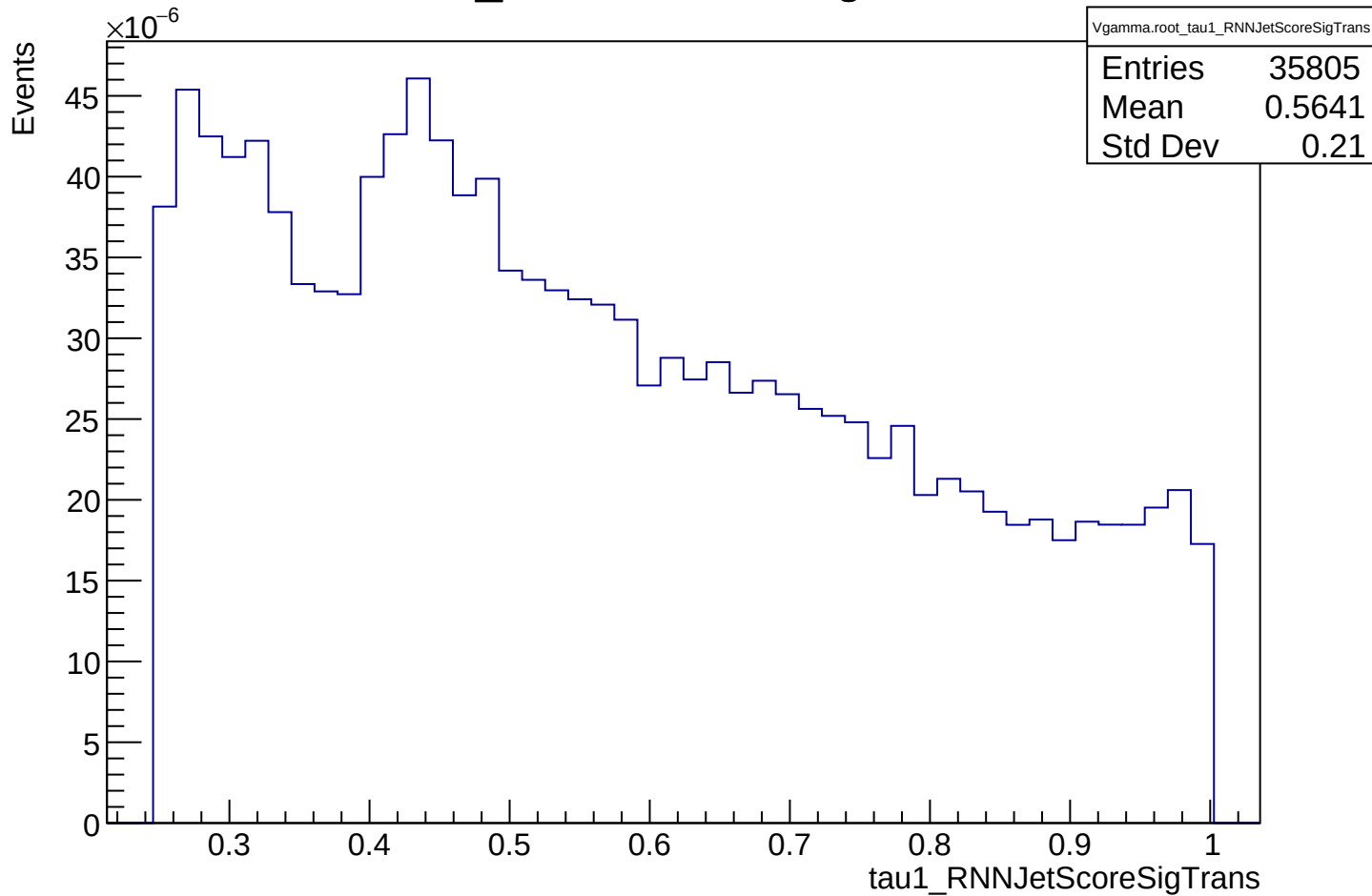
Vgamma.root_pass_SLT	
Entries	35805
Mean	0.7595
Std Dev	0.4274

pass_STT

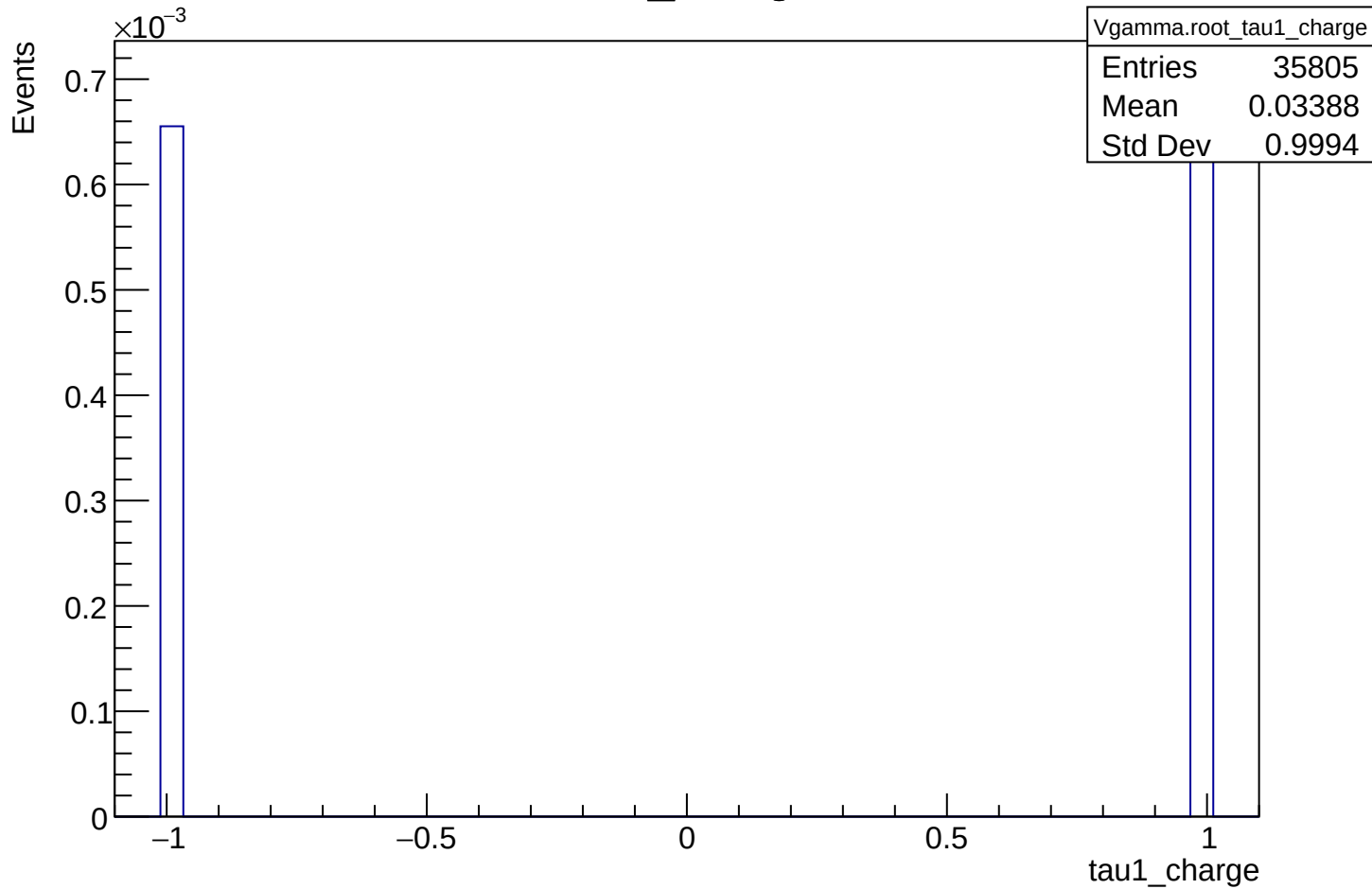
Events



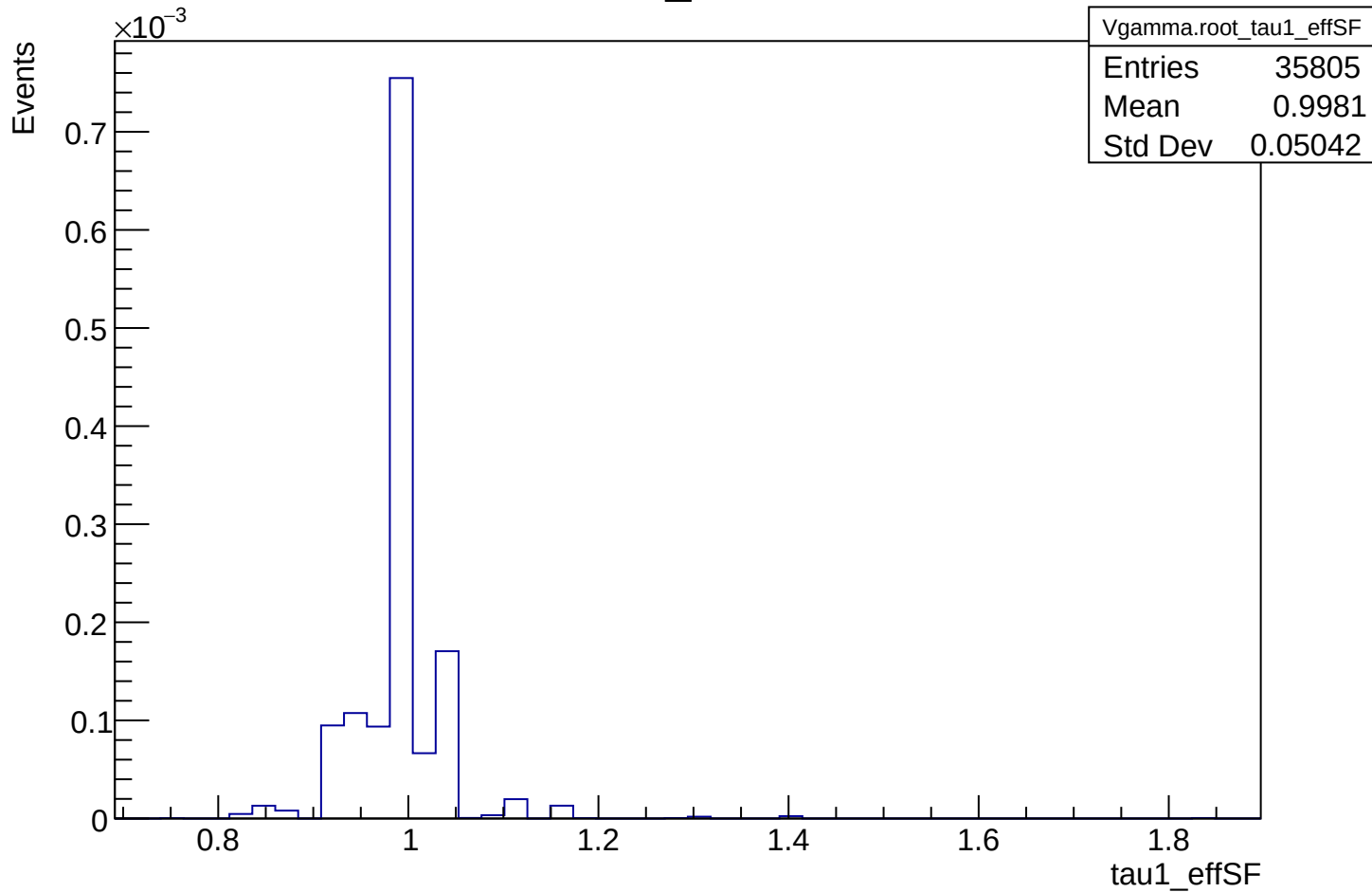
tau1_RNNJetScoreSigTrans



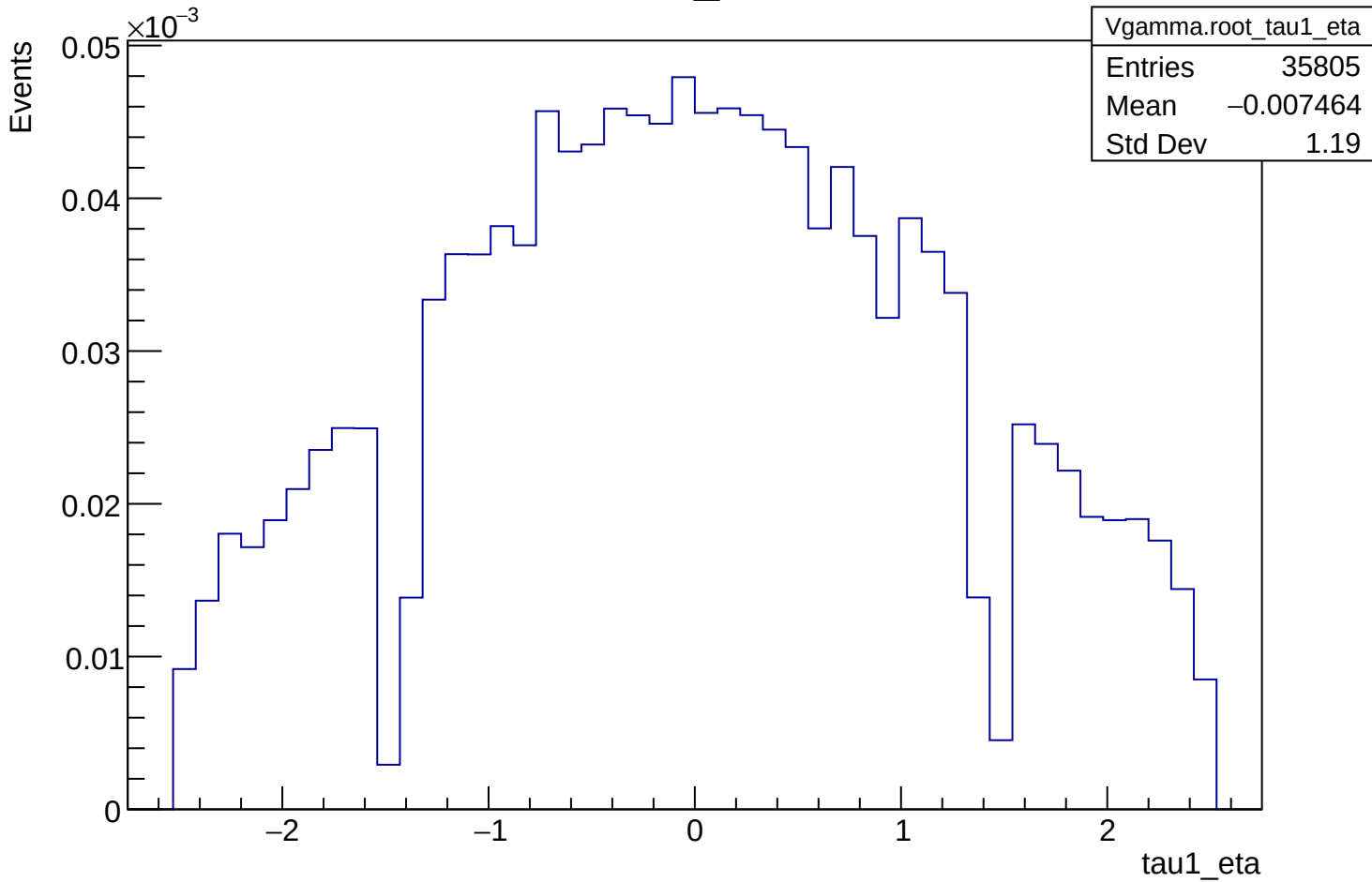
tau1_charge



tau1_effSF

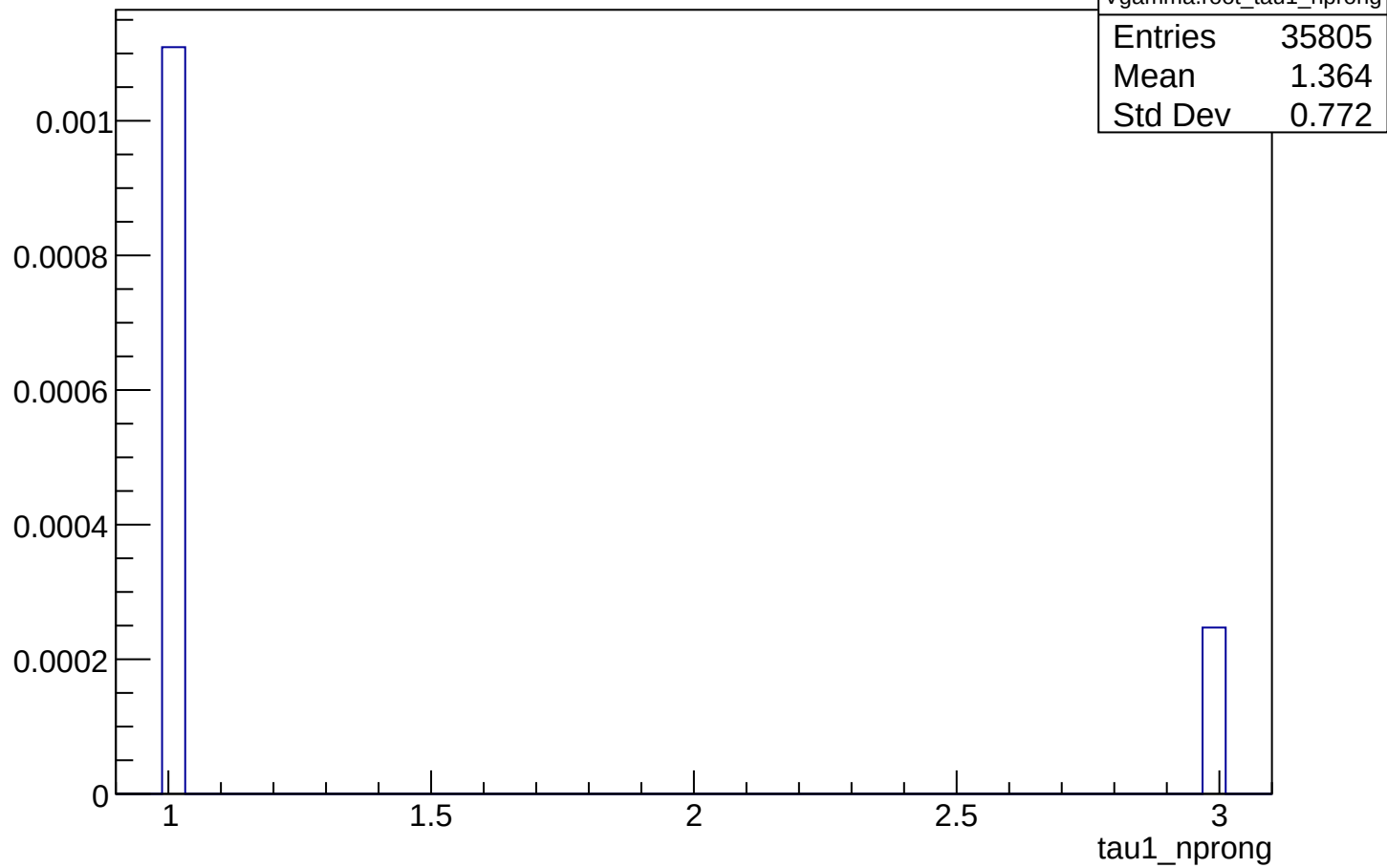


tau1_eta

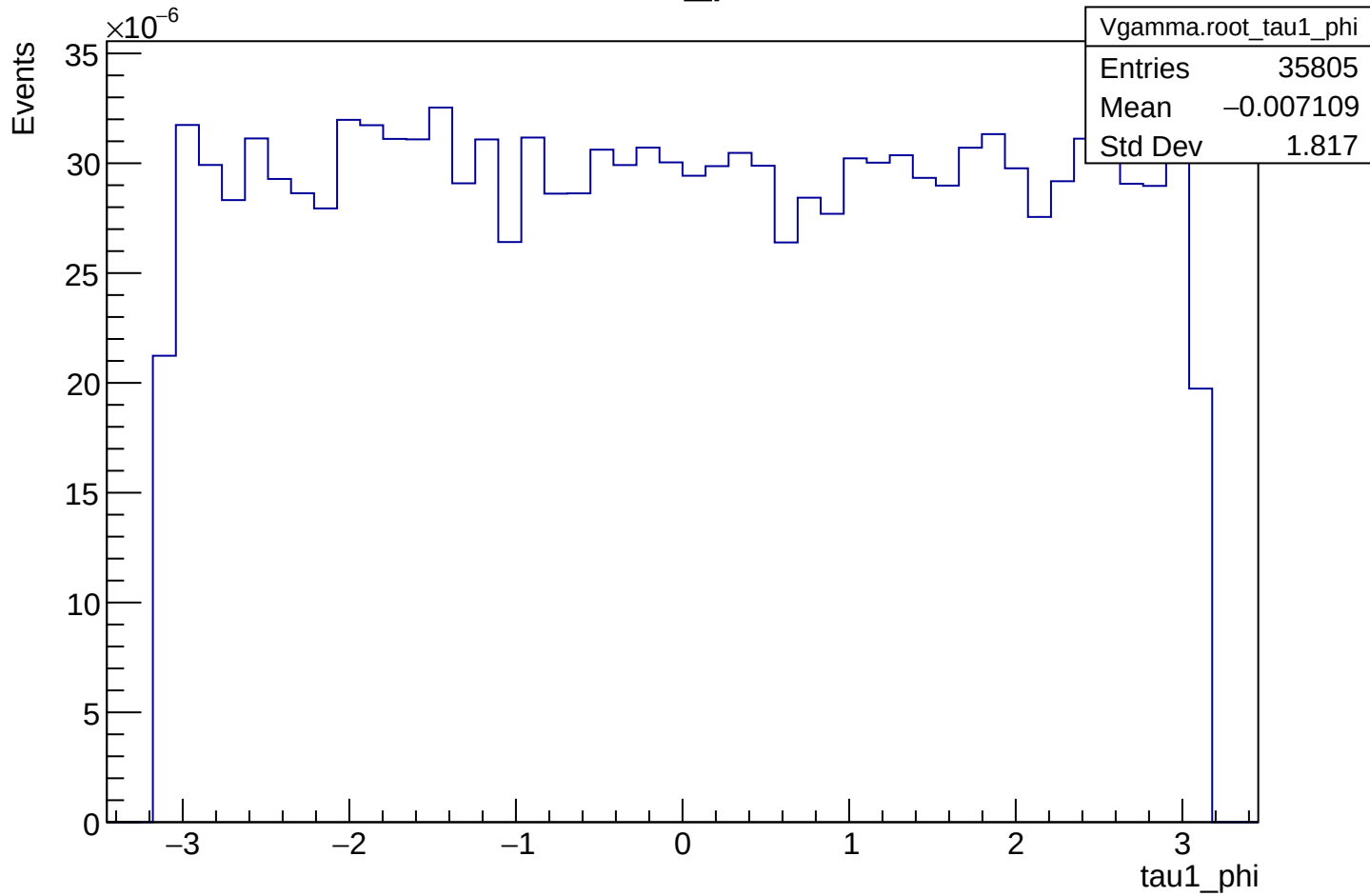


tau1_nprong

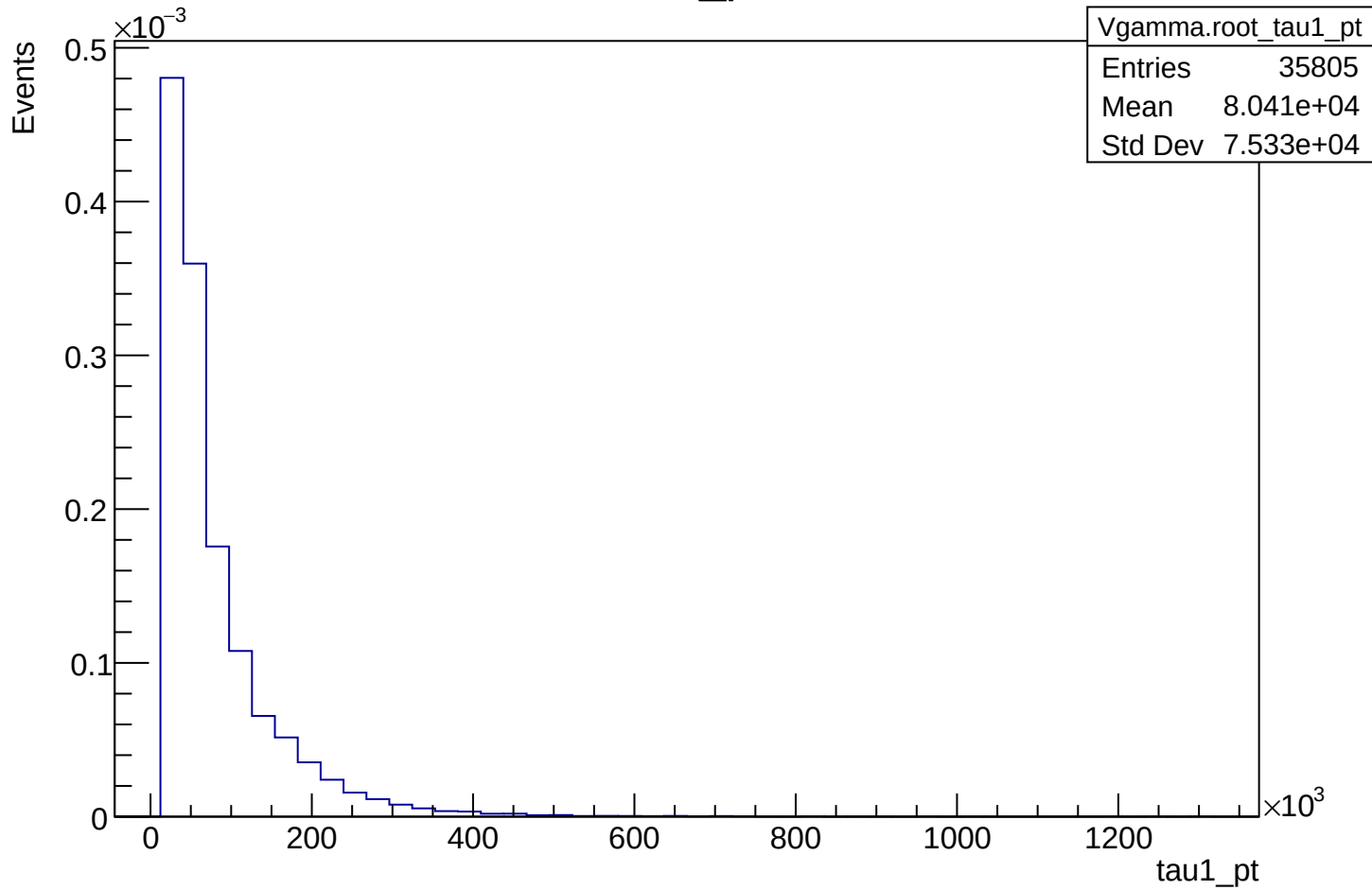
Events



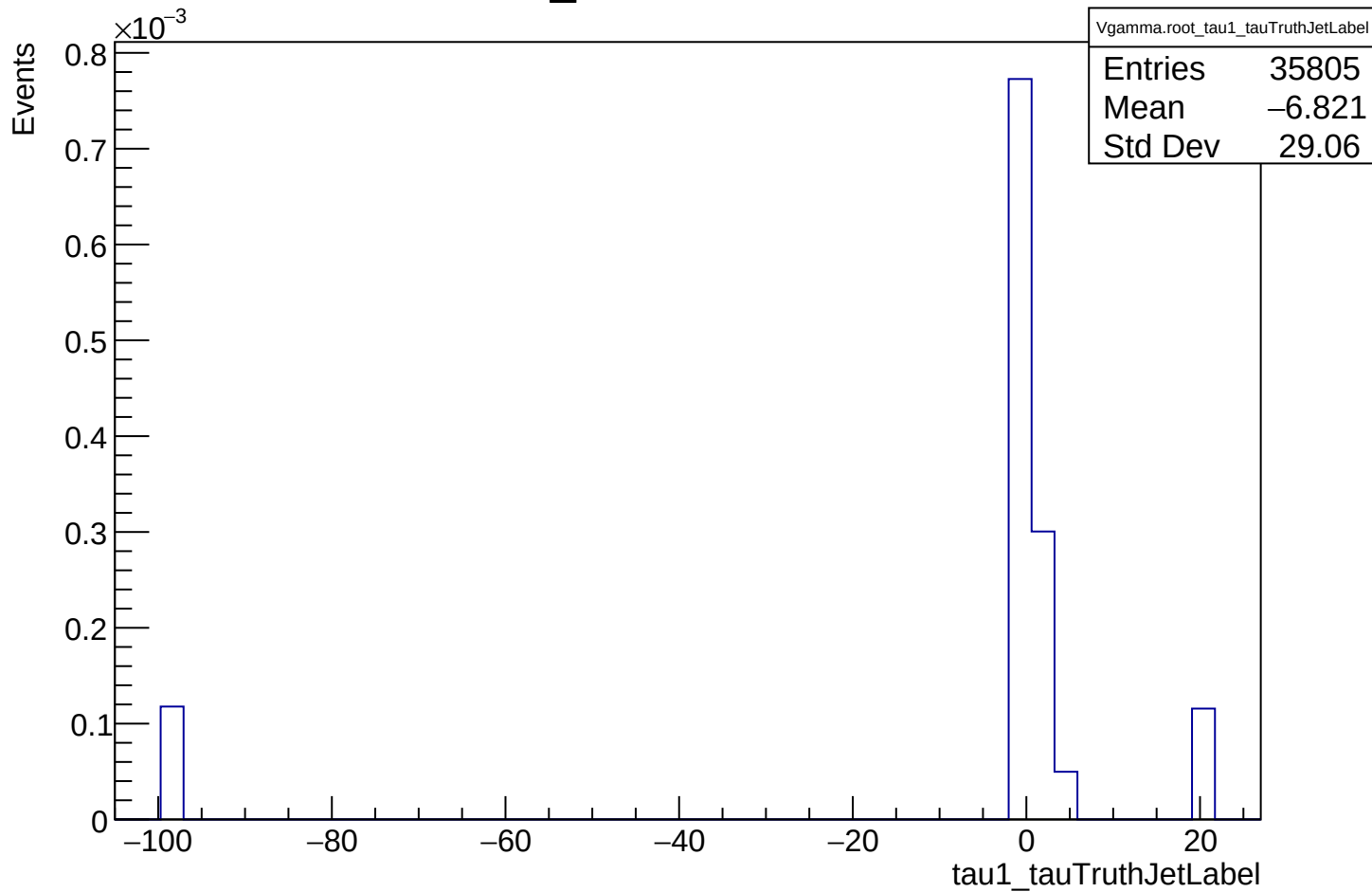
tau1_phi



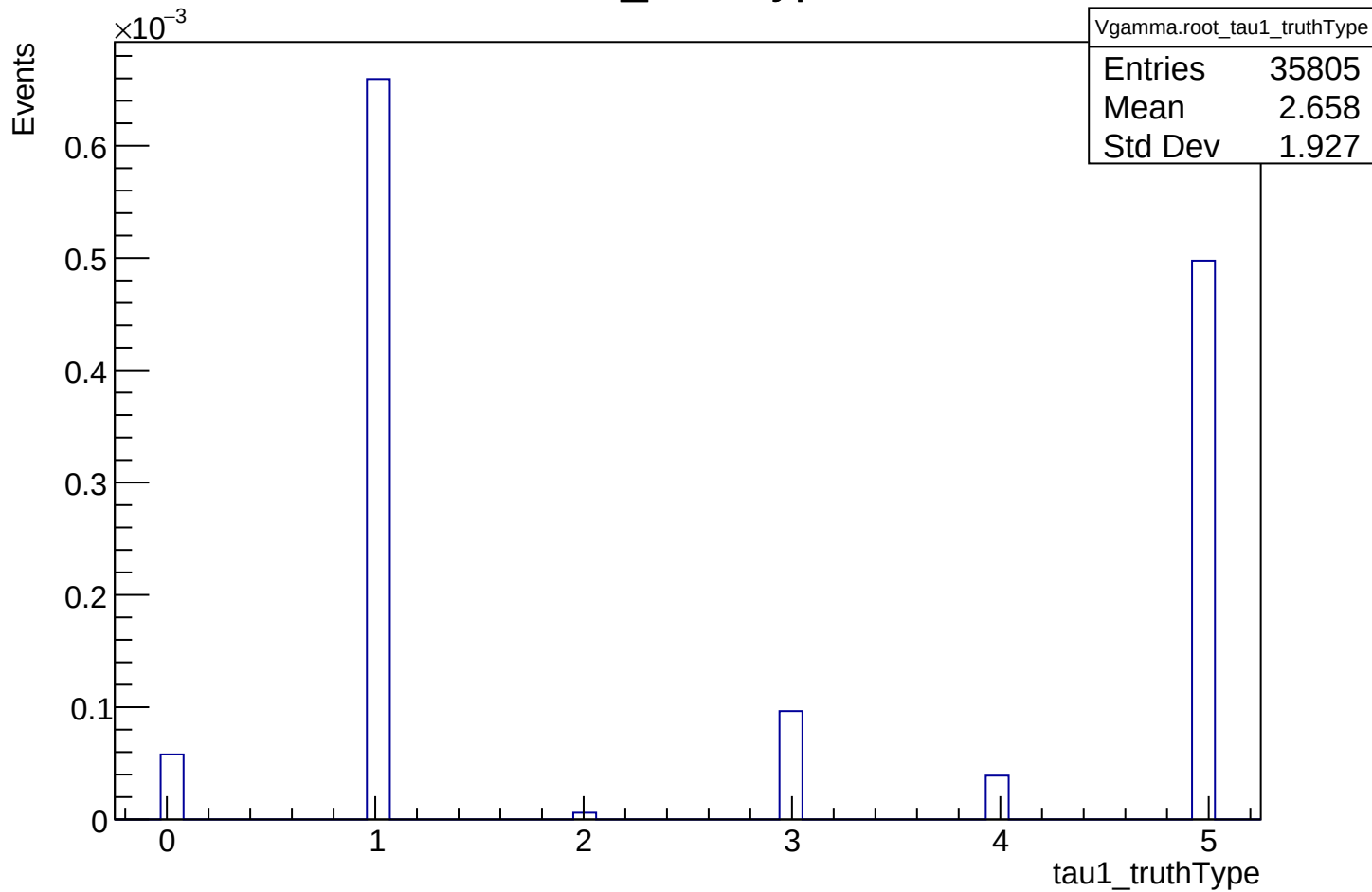
tau1_pt



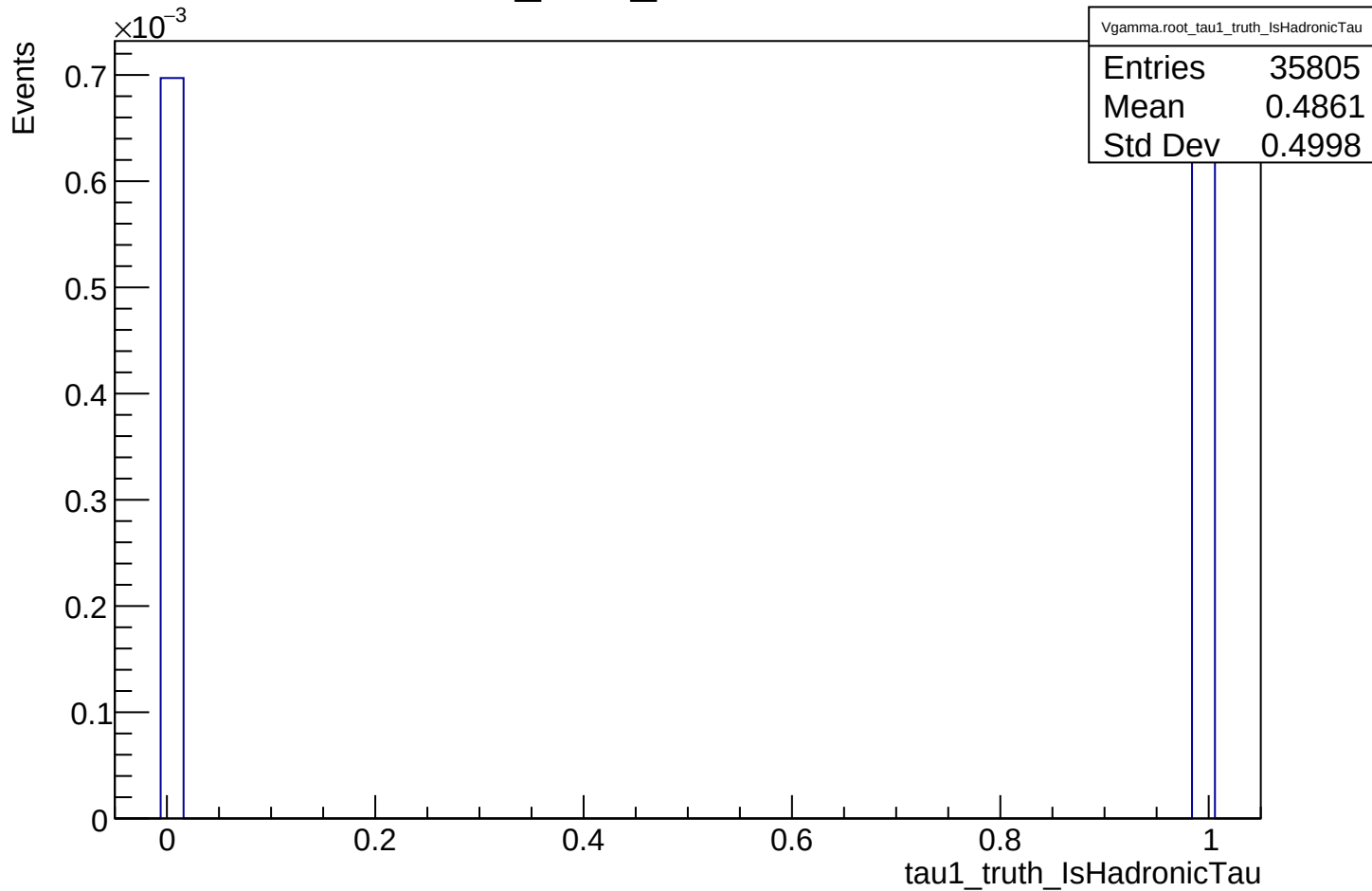
tau1_tauTruthJetLabel



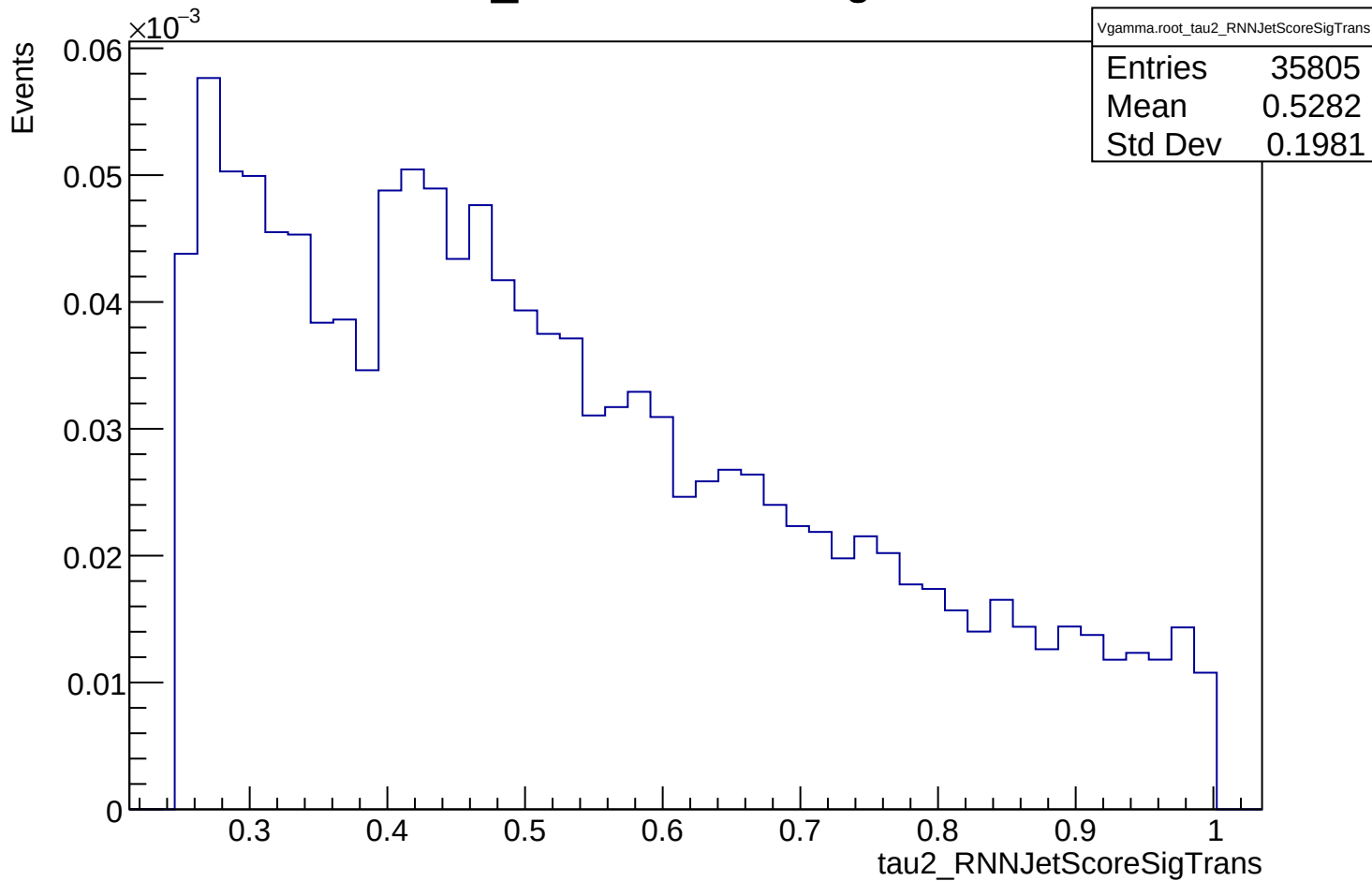
tau1_truthType



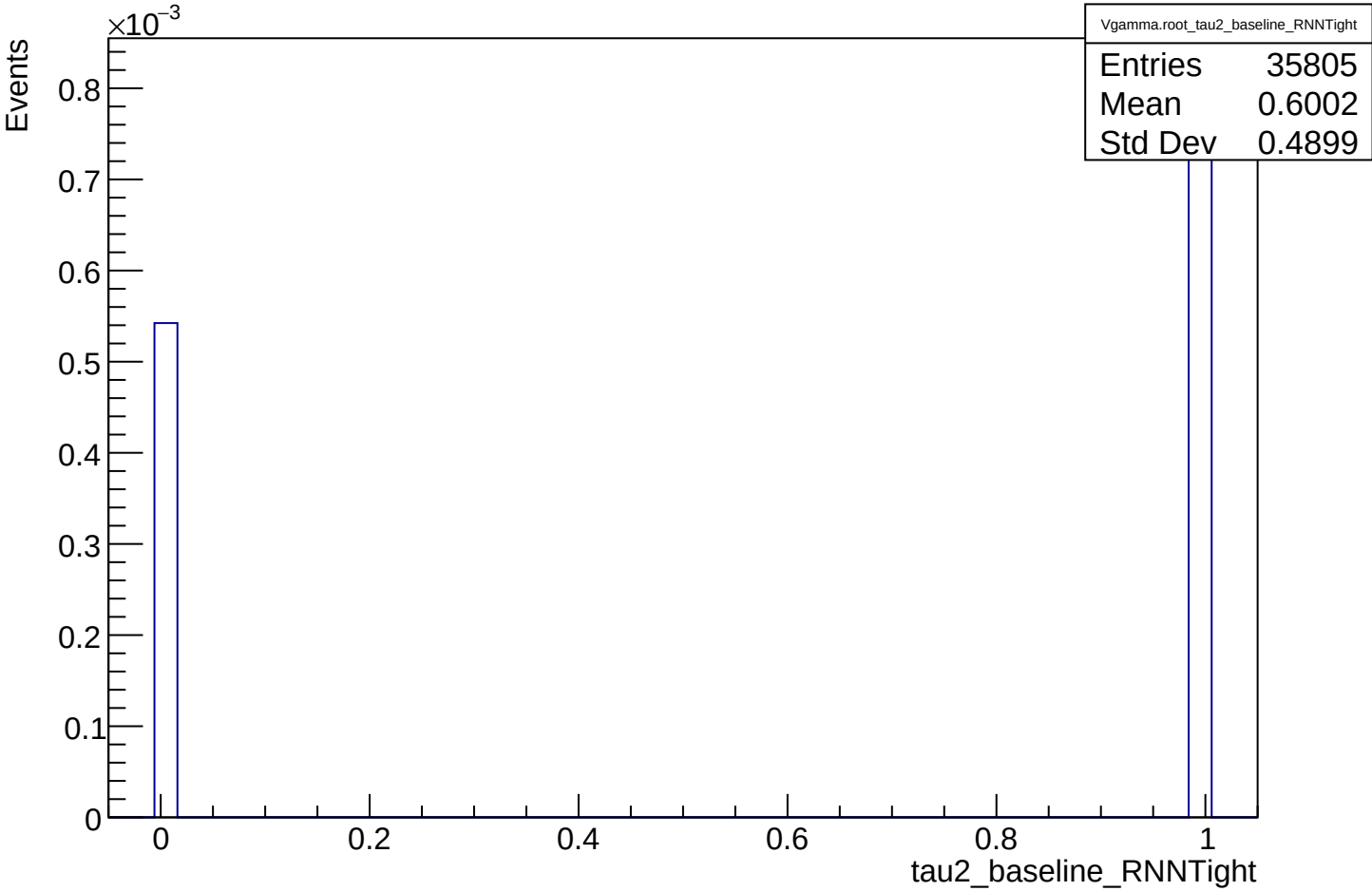
tau1_truth_IsHadronicTau



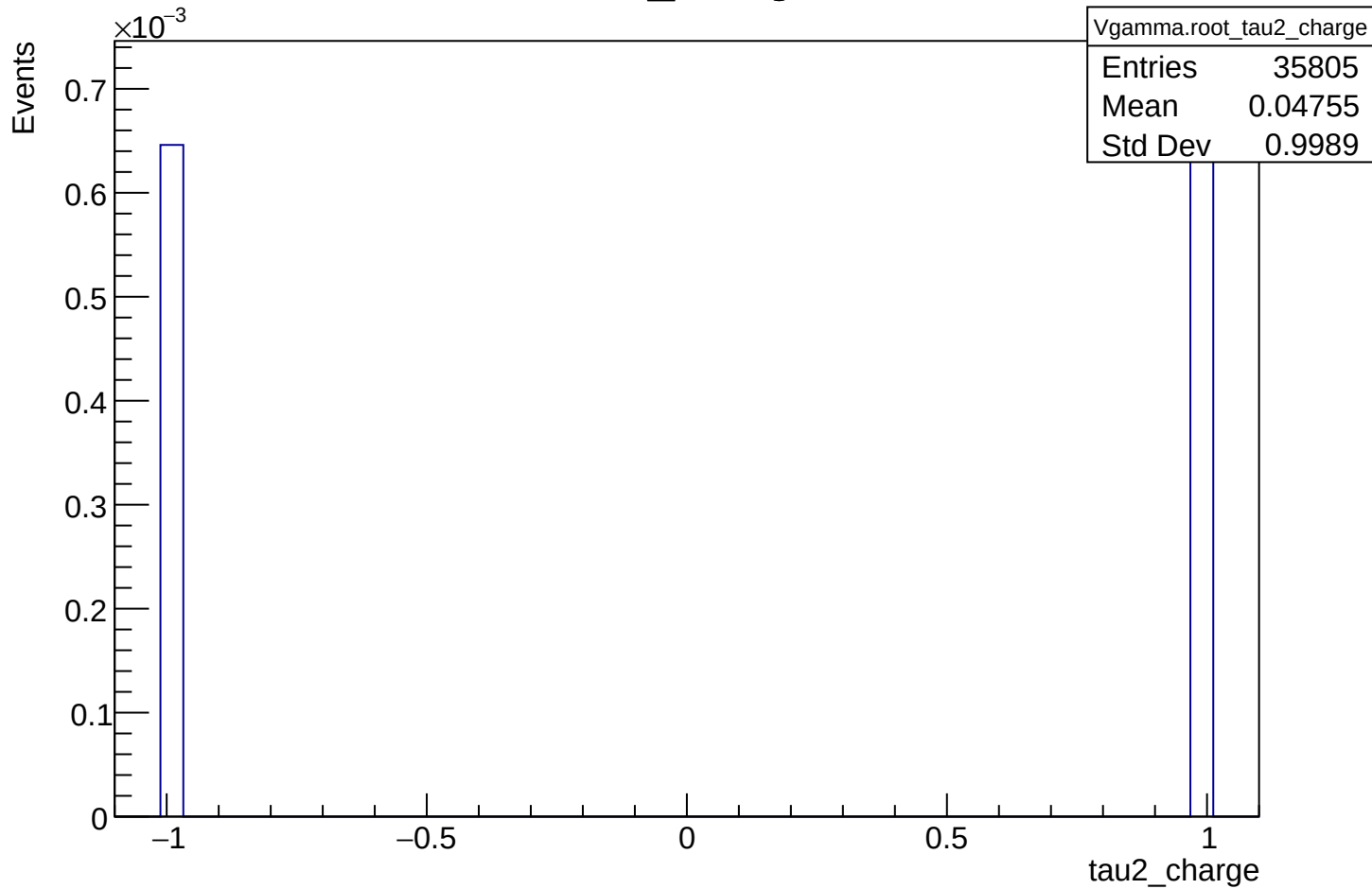
tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

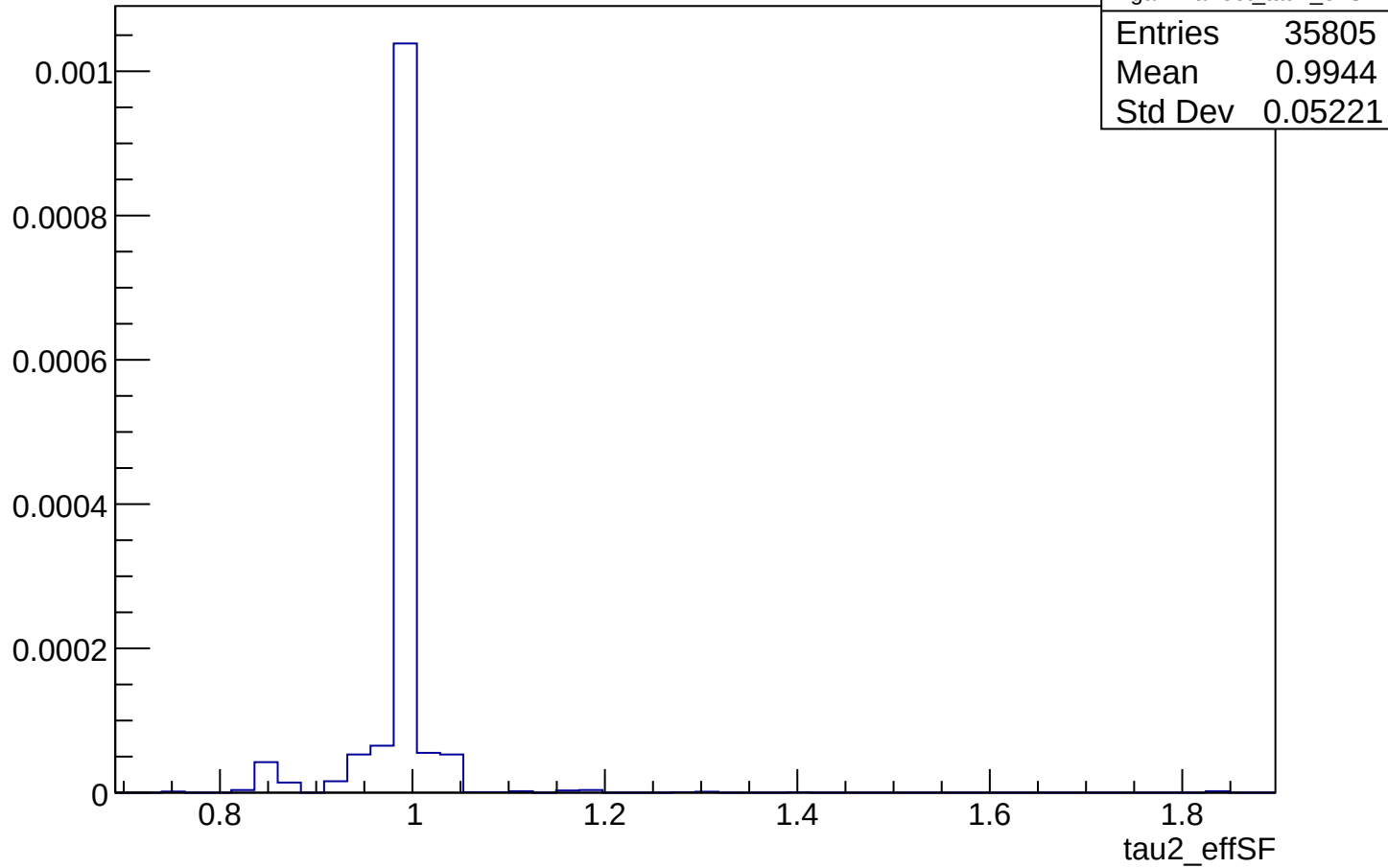


tau2_charge

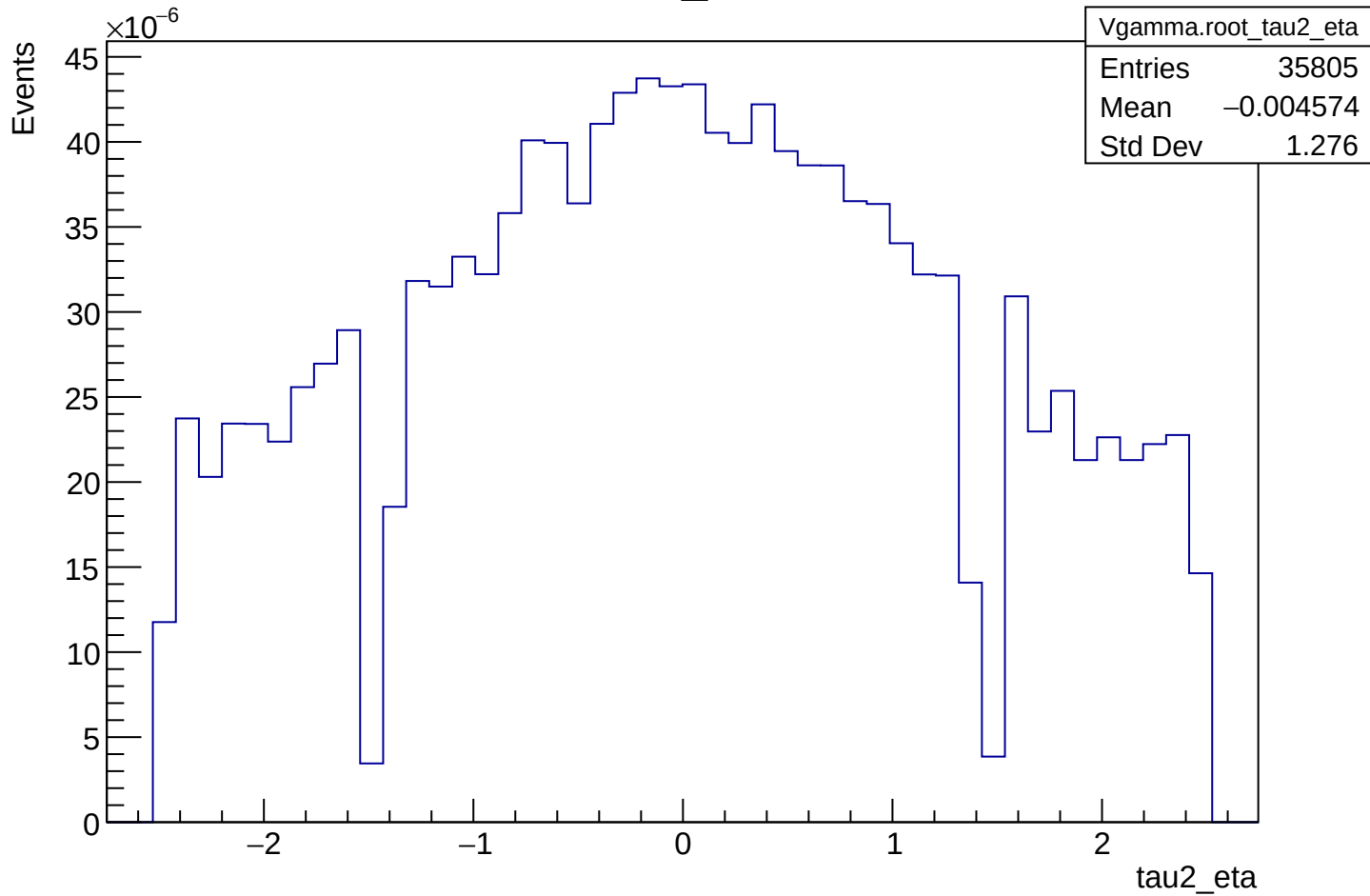


tau2_effSF

Events

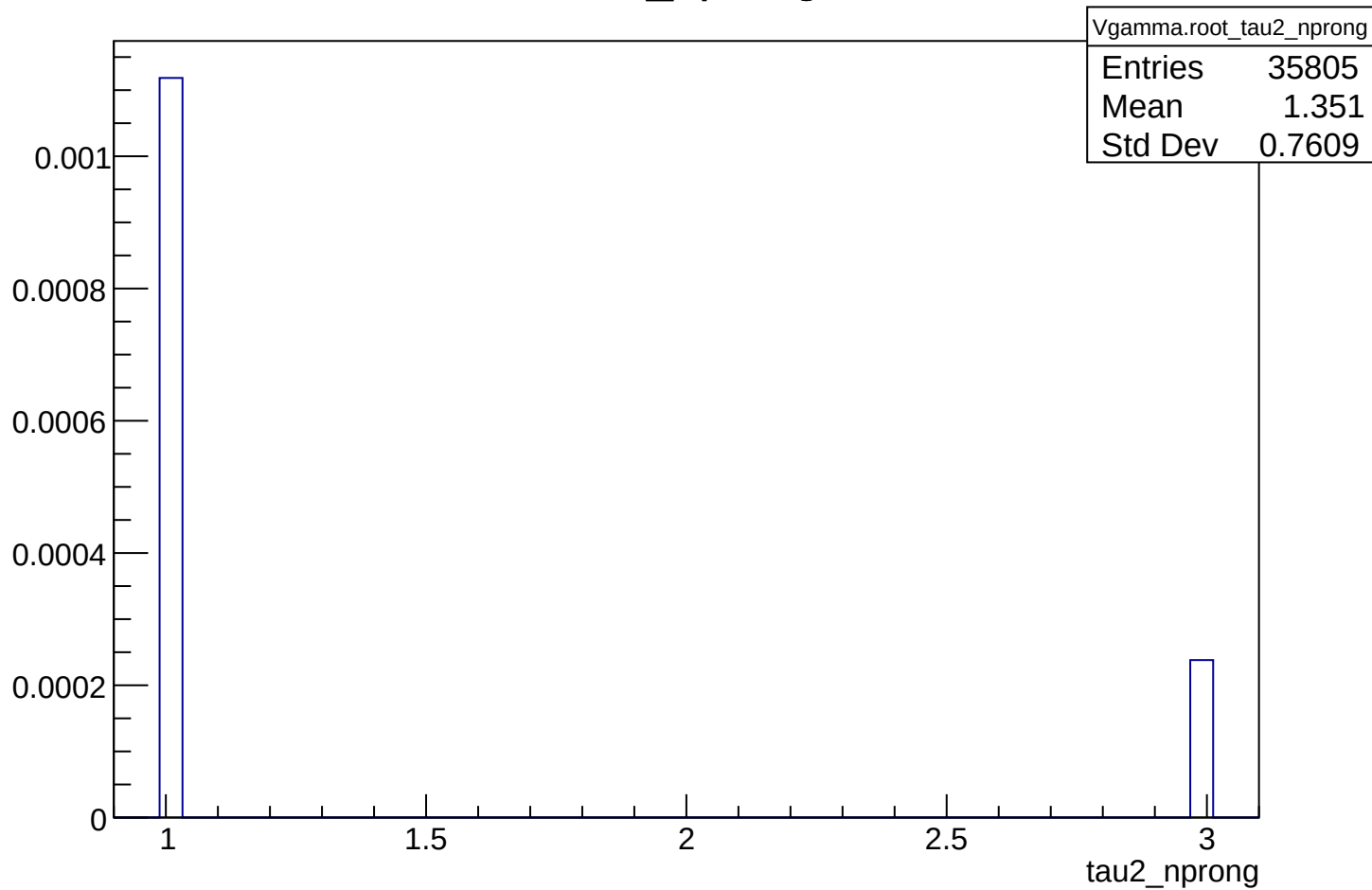


tau2_eta

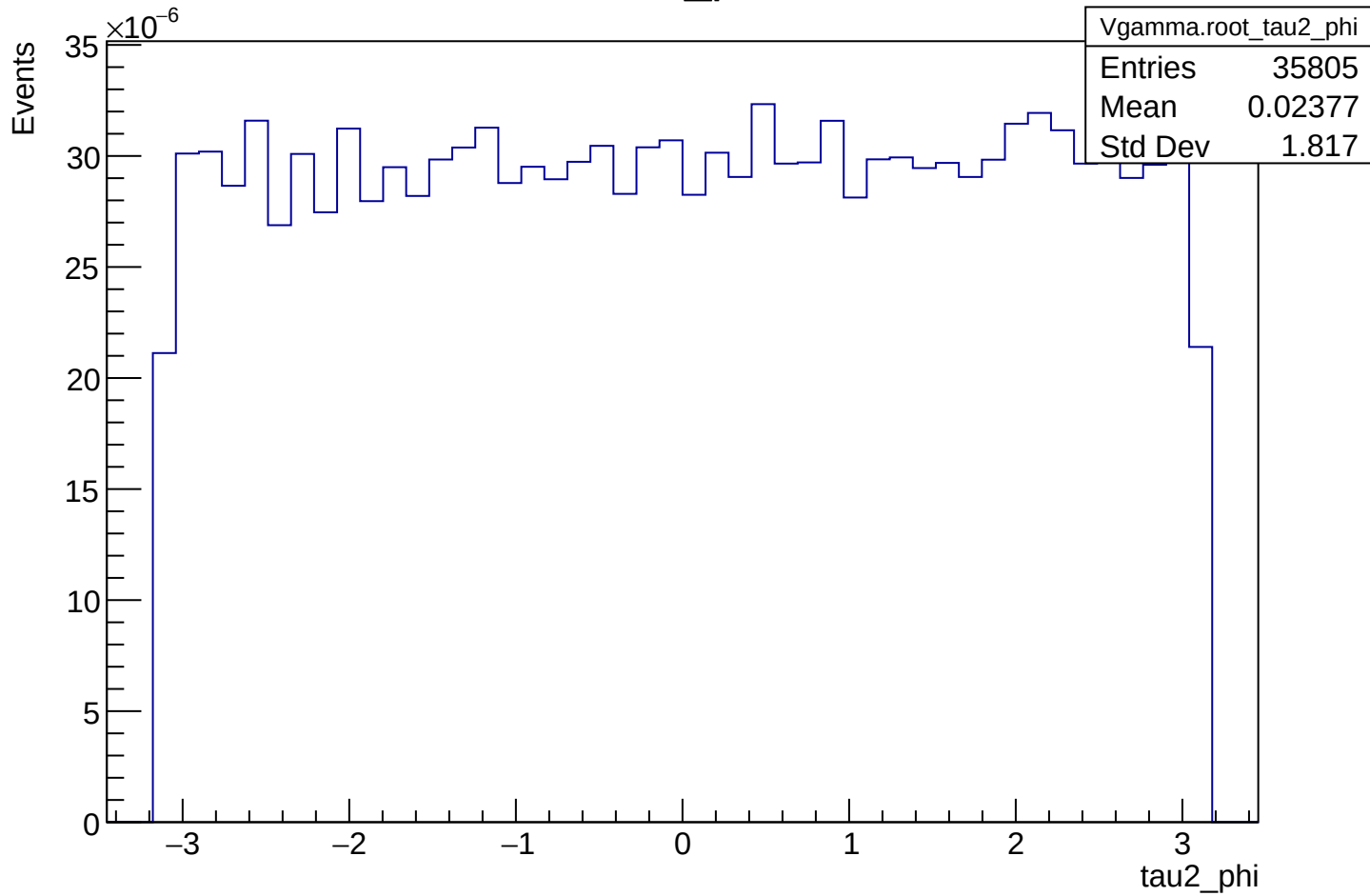


tau2_nprong

Events

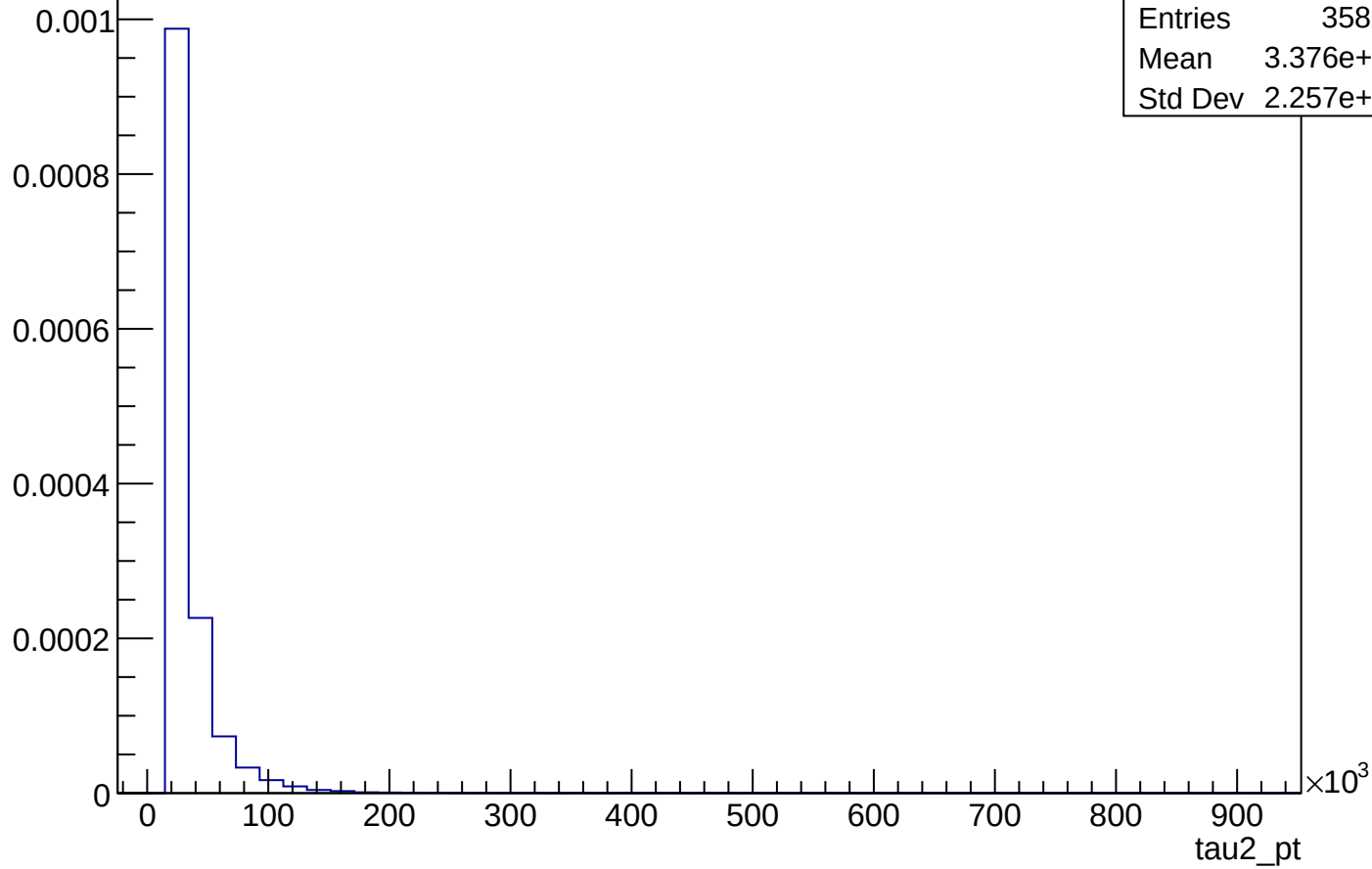


tau2_phi

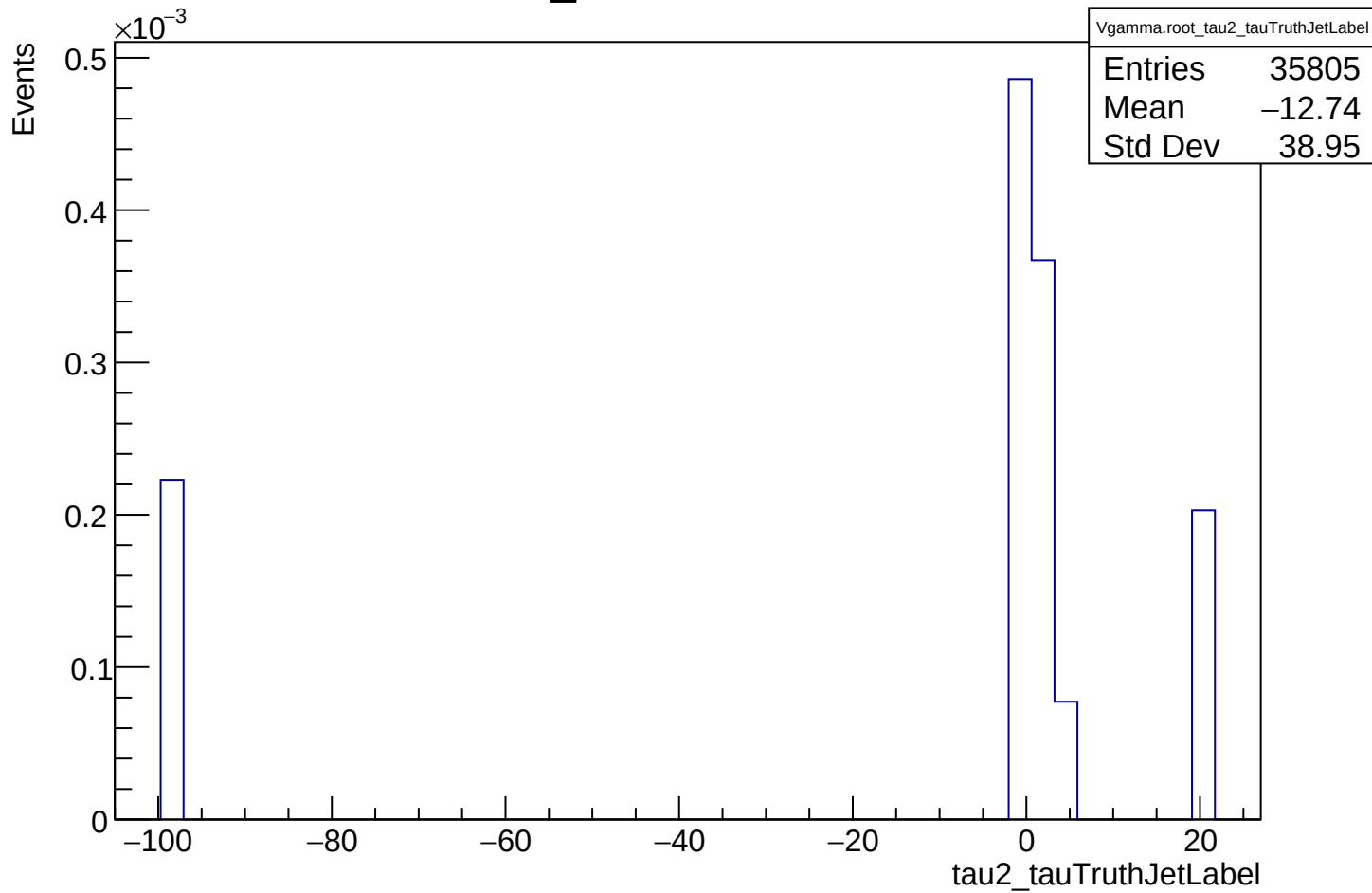


tau2_pt

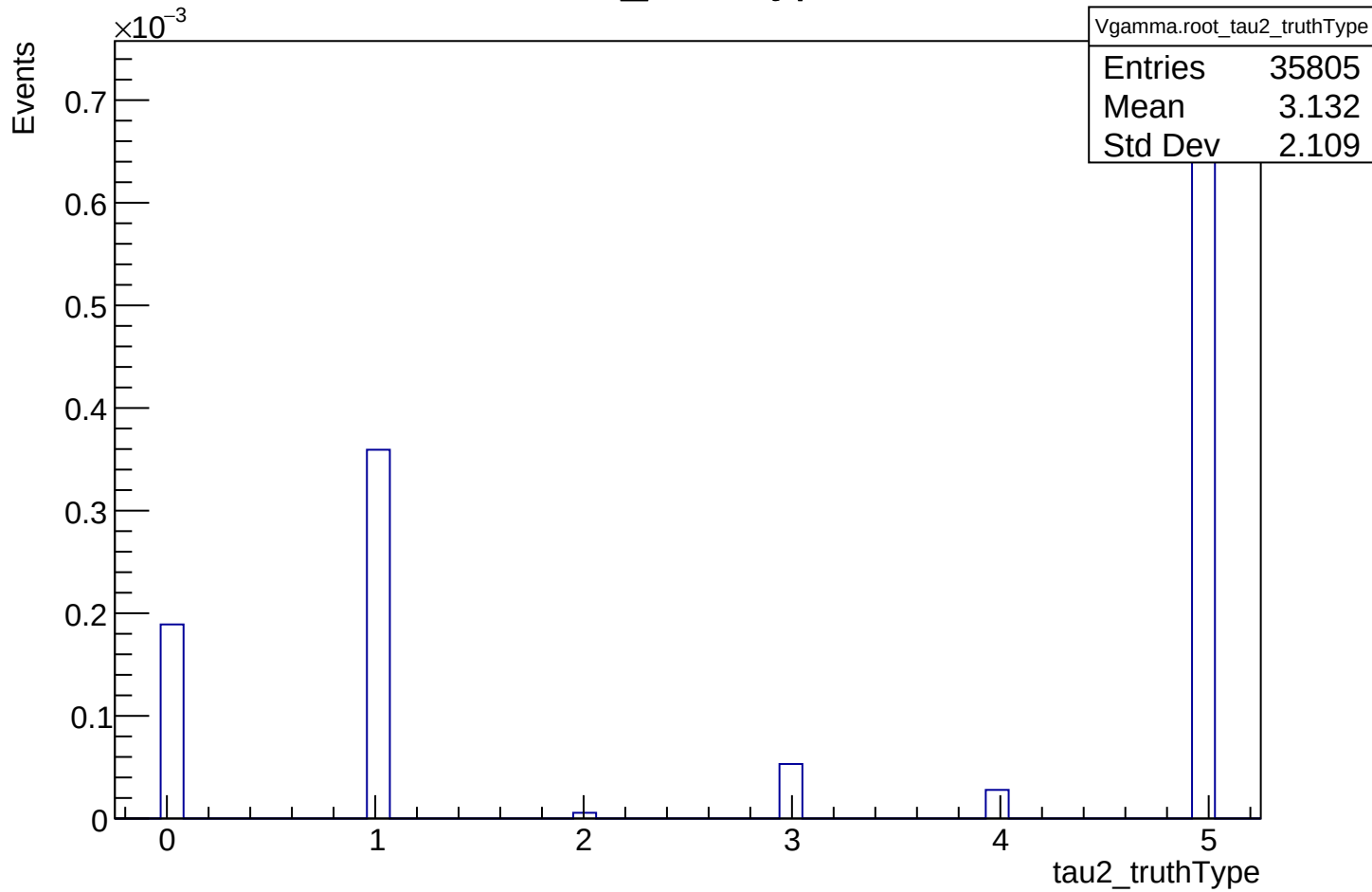
Events



tau2_tauTruthJetLabel



tau2_truthType



tau2_truth_IsHadronicTau

Events

0.001
0.0008
0.0006
0.0004
0.0002
0

0

0.2

0.4

0.6

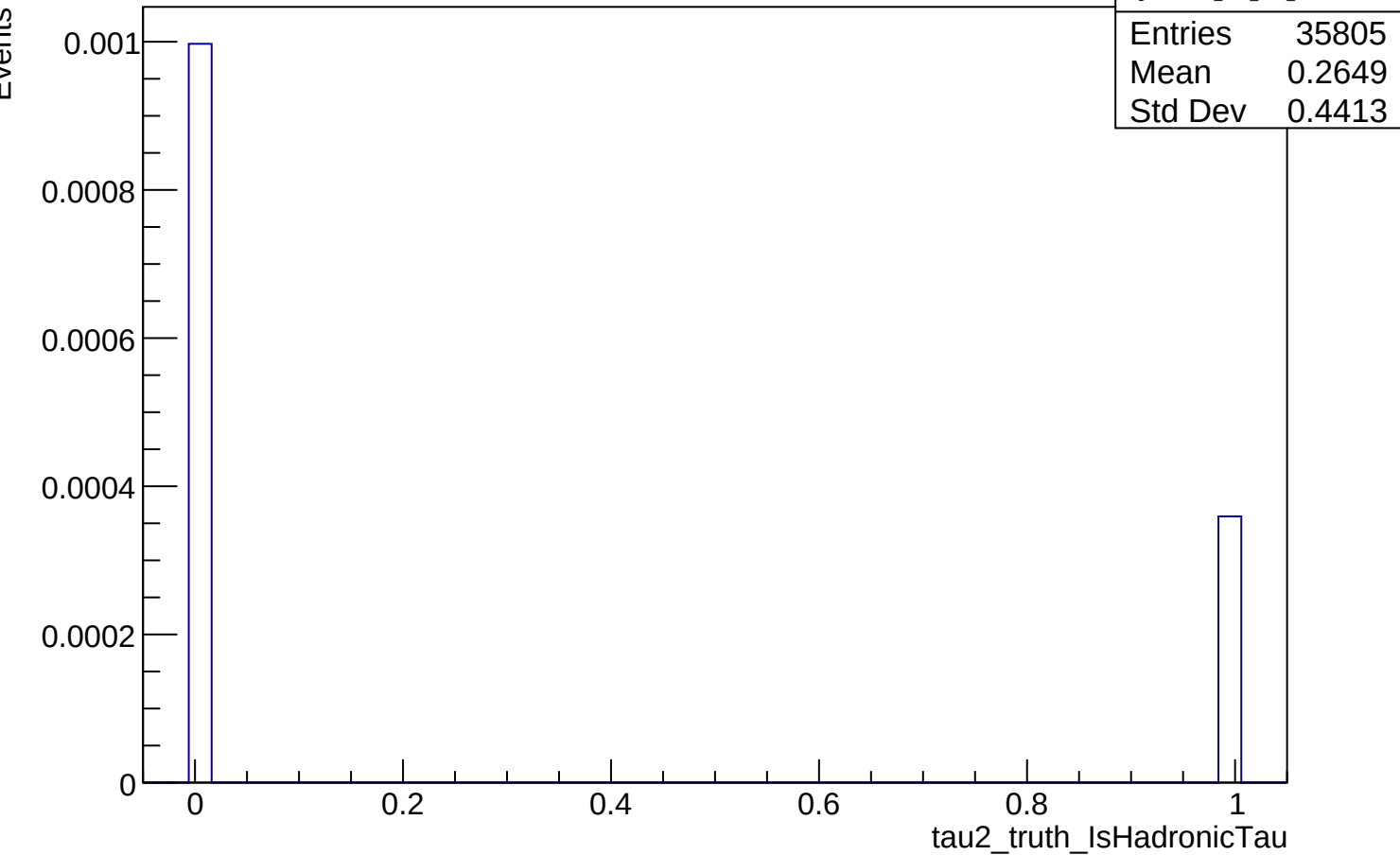
0.8

1

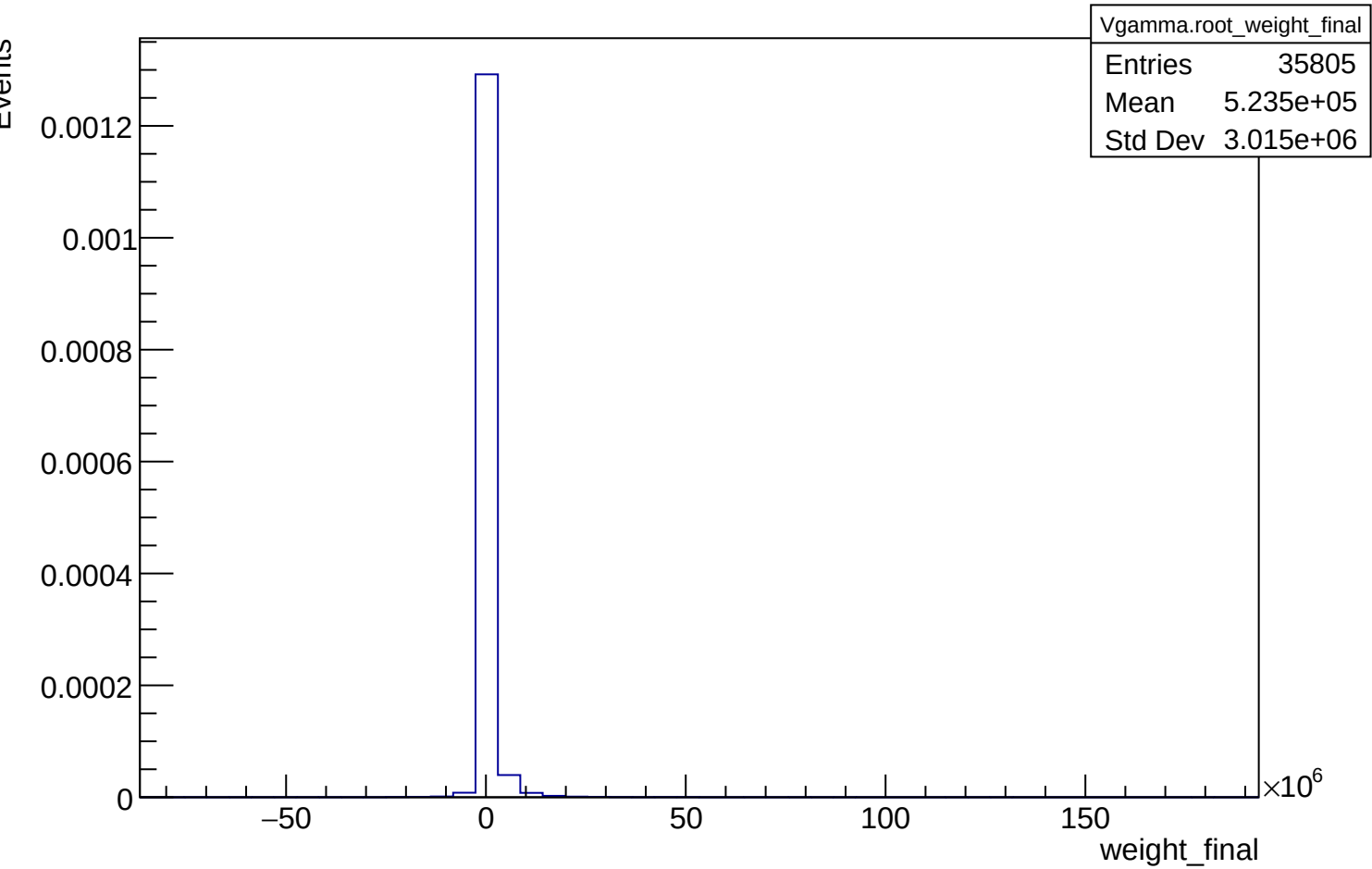
tau2_truth_IsHadronicTau

Vgamma.root_tau2_truth_IsHadronicTau

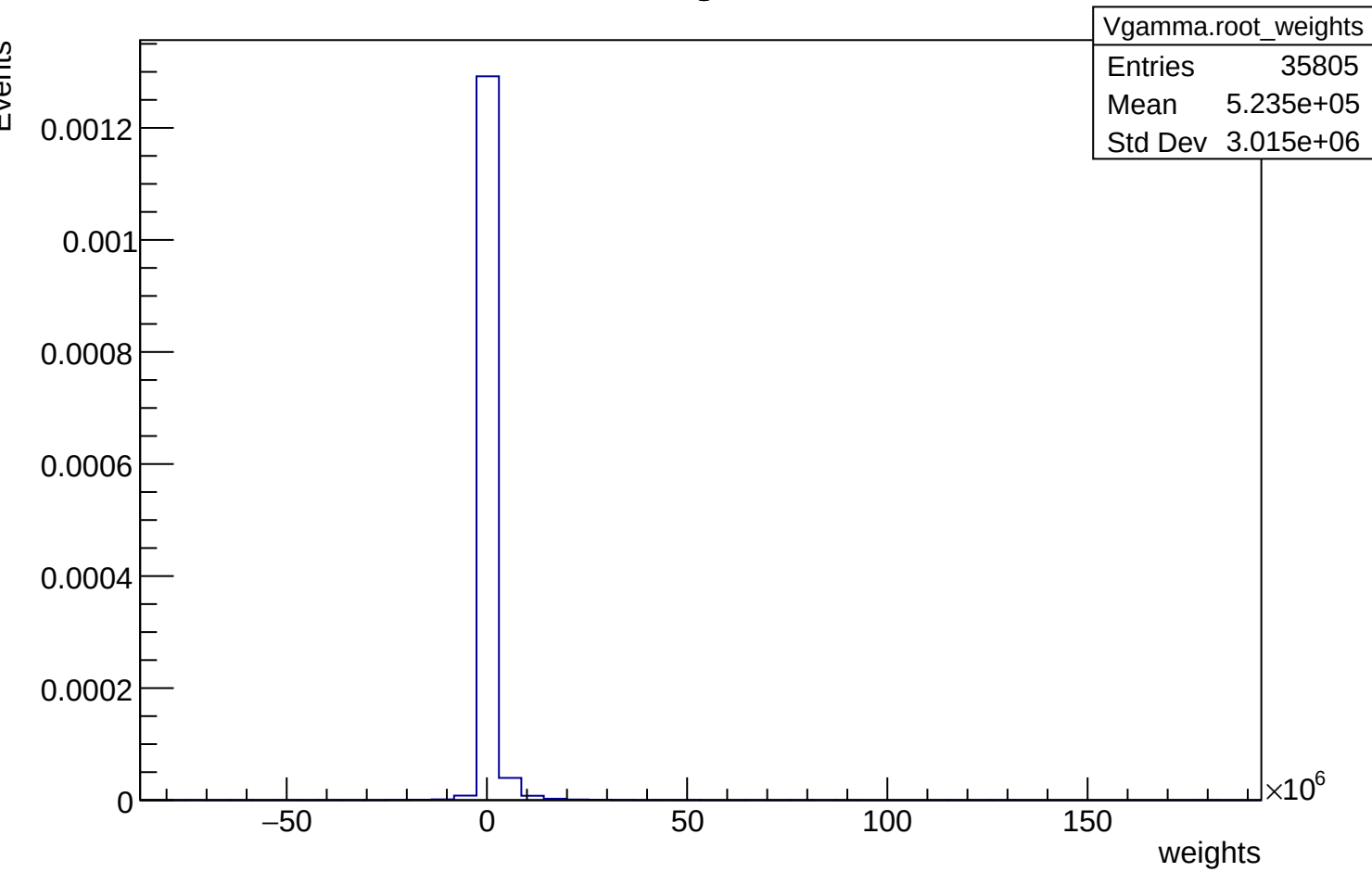
Entries	35805
Mean	0.2649
Std Dev	0.4413



weight_final

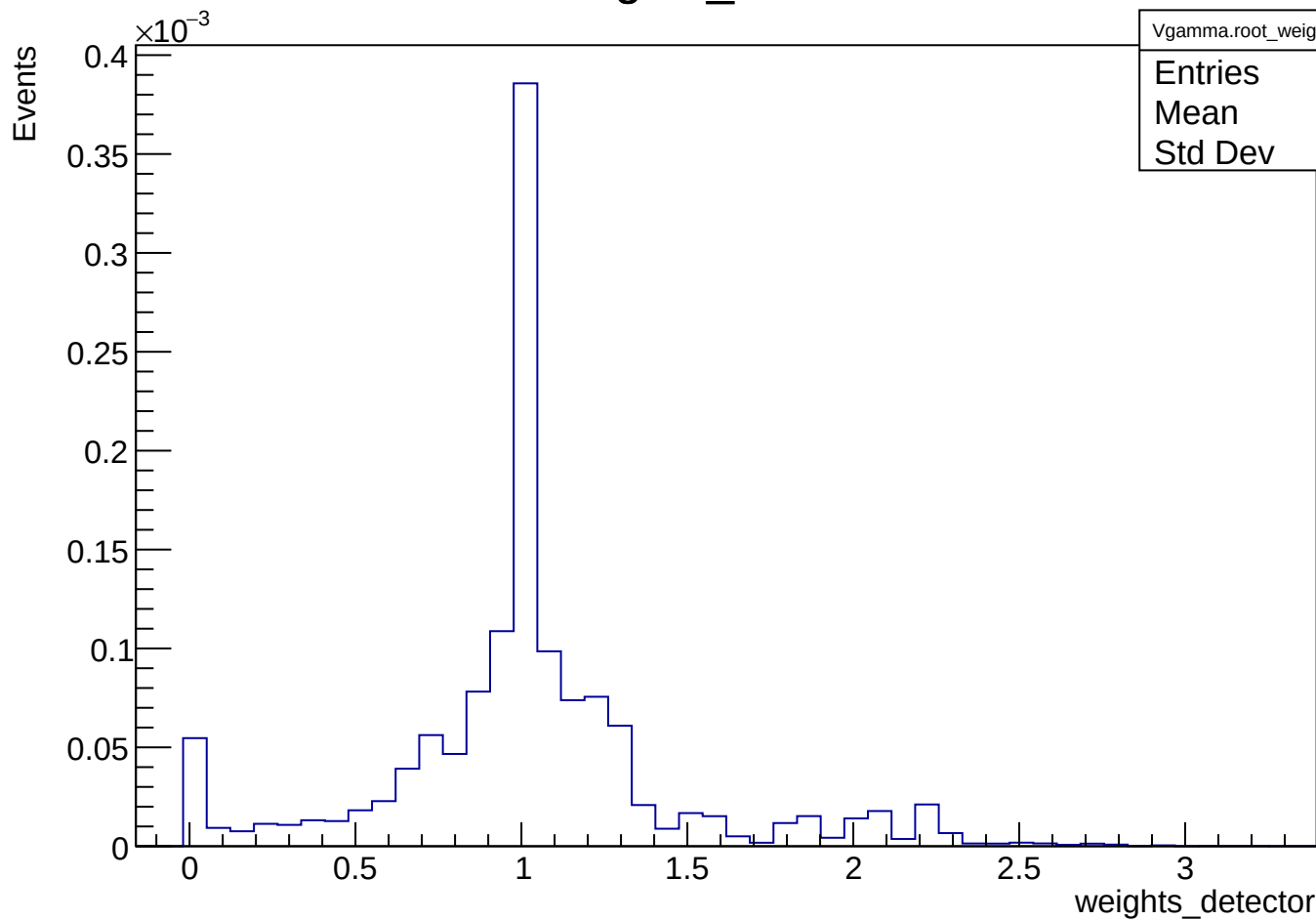


weights

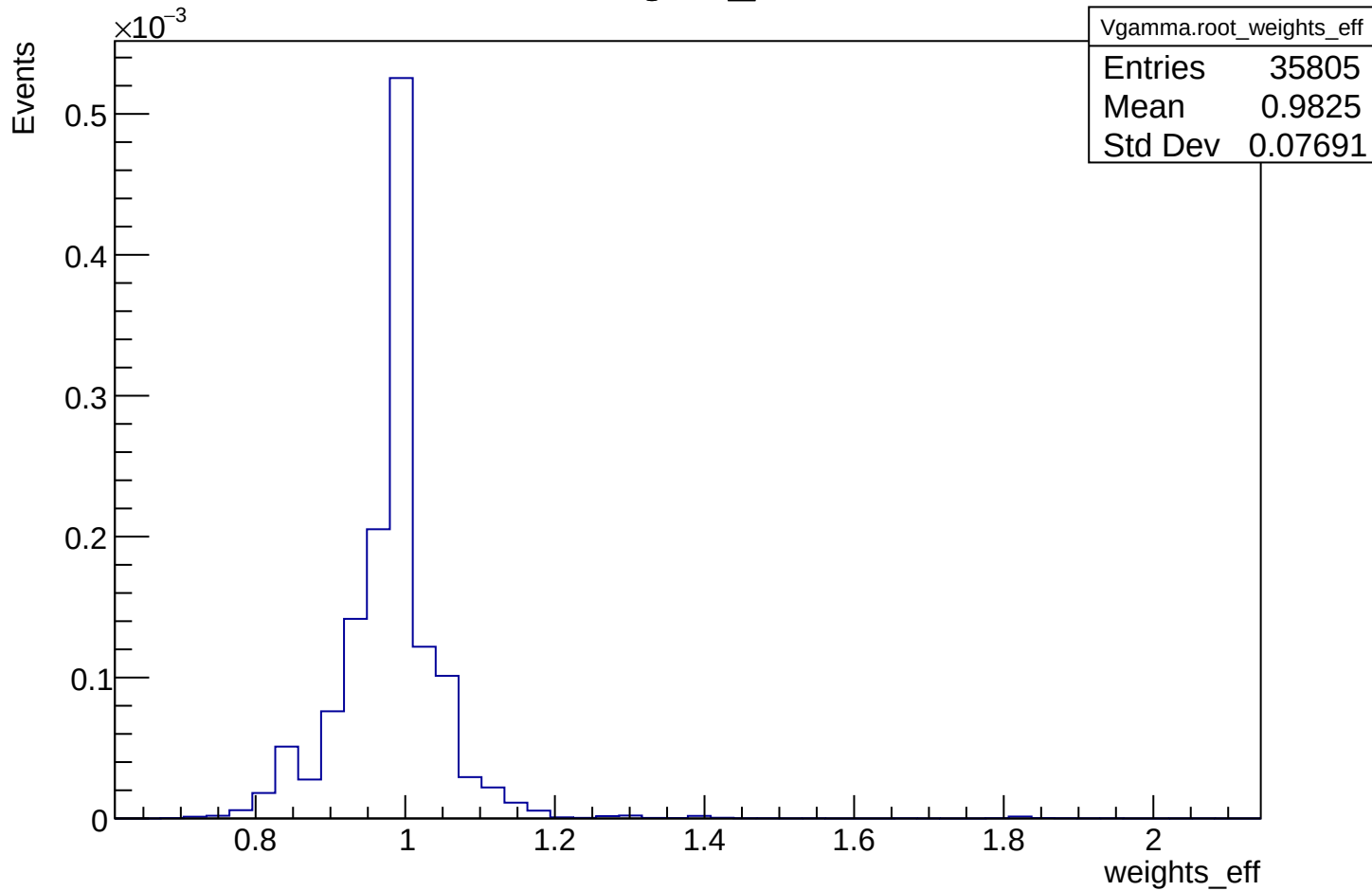


weights_detector

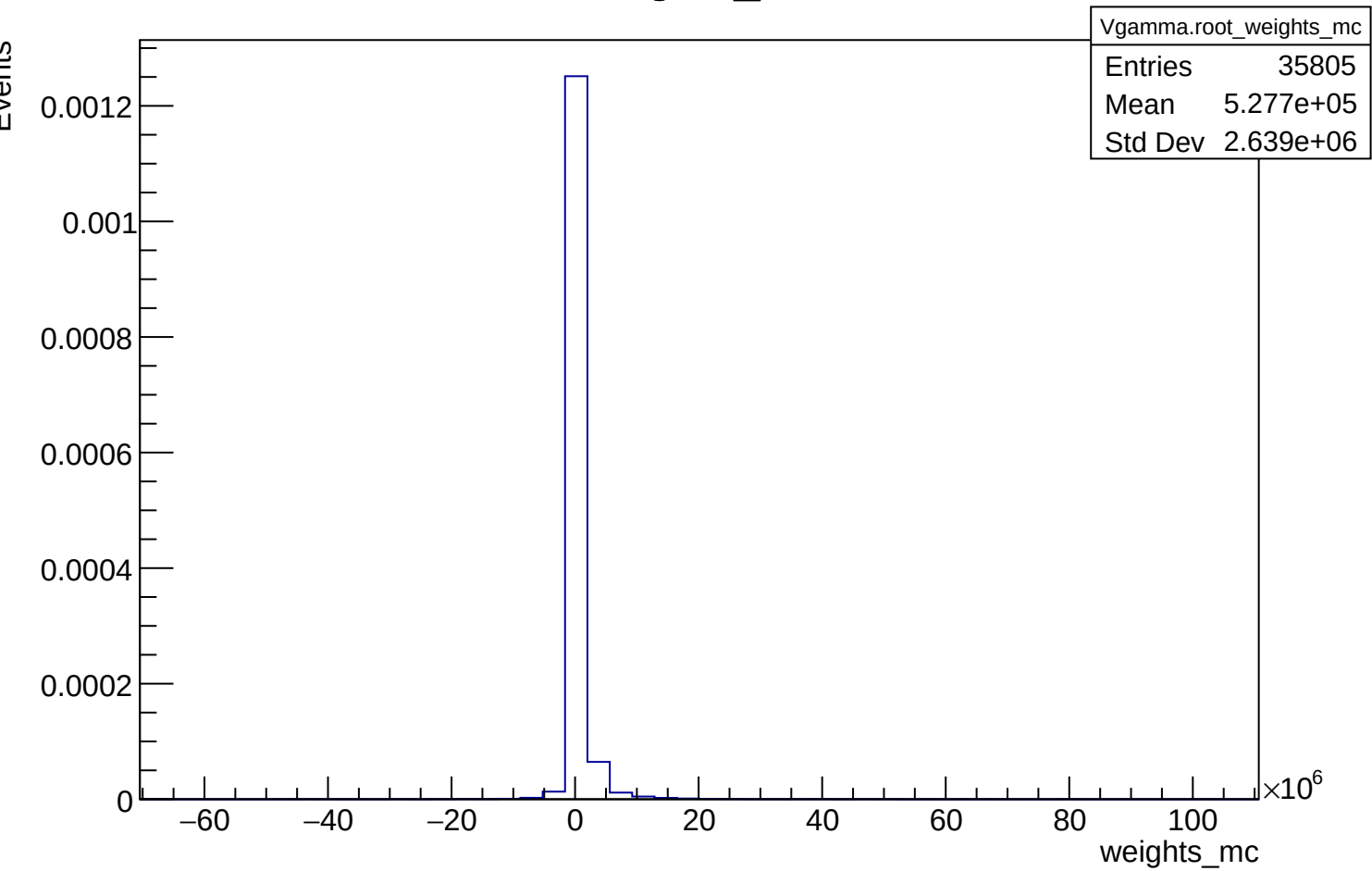
Vgamma.root_weights_detector	
Entries	35805
Mean	1.026
Std Dev	0.4429



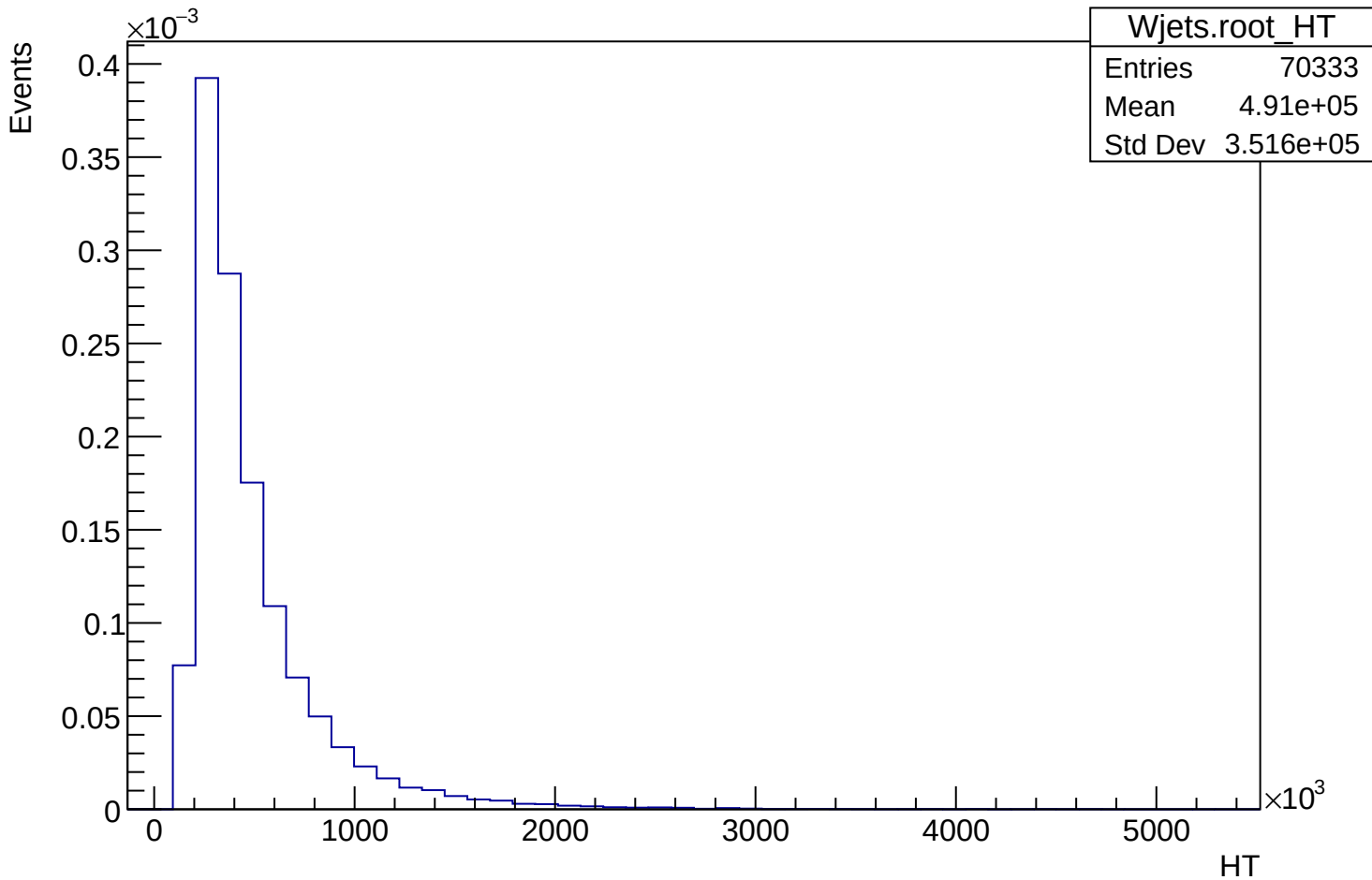
weights_eff



weights_mc

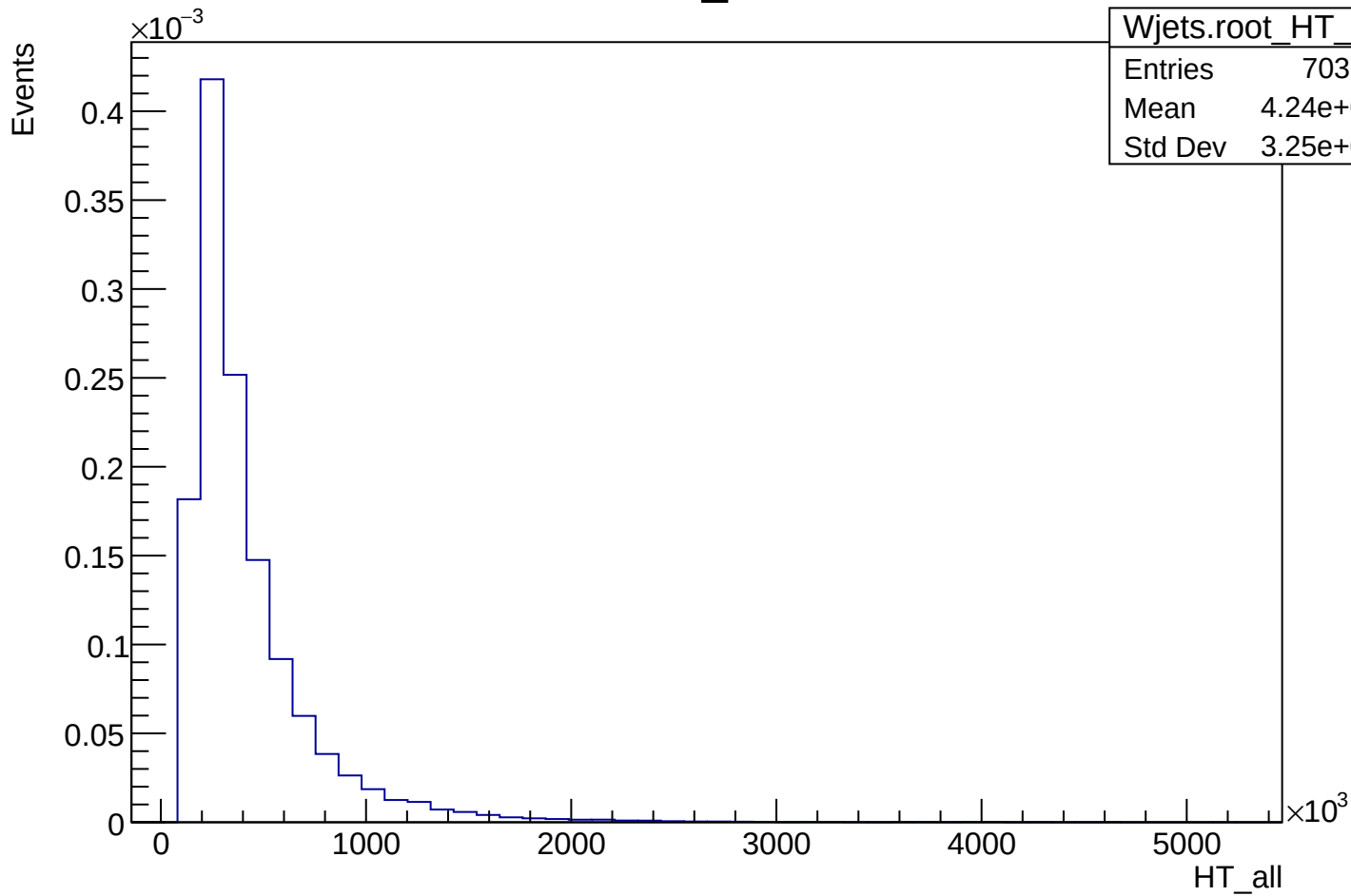


HT



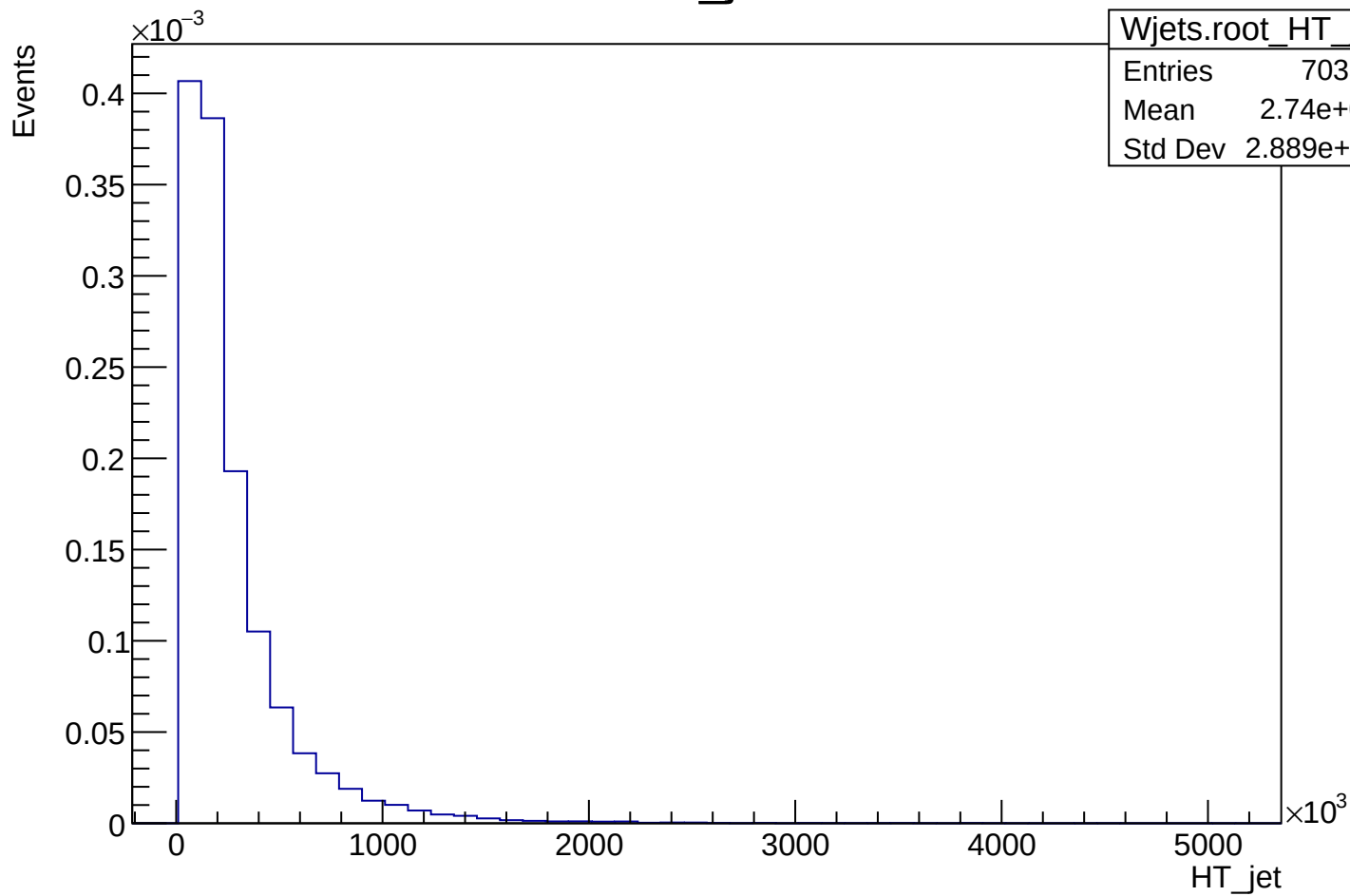
HT_all

Wjets.root_HT_all	
Entries	70333
Mean	4.24e+05
Std Dev	3.25e+05

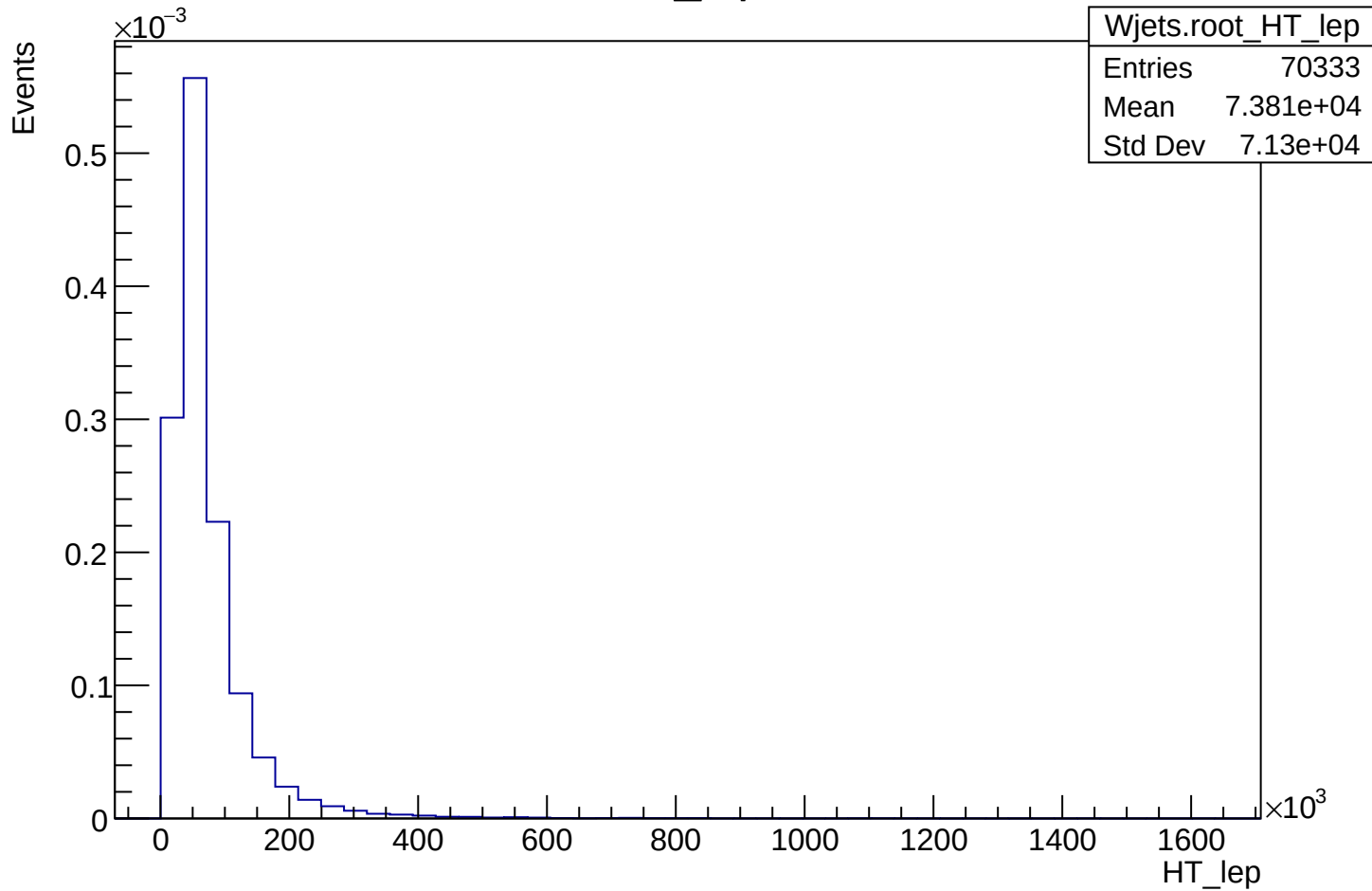


HT_jet

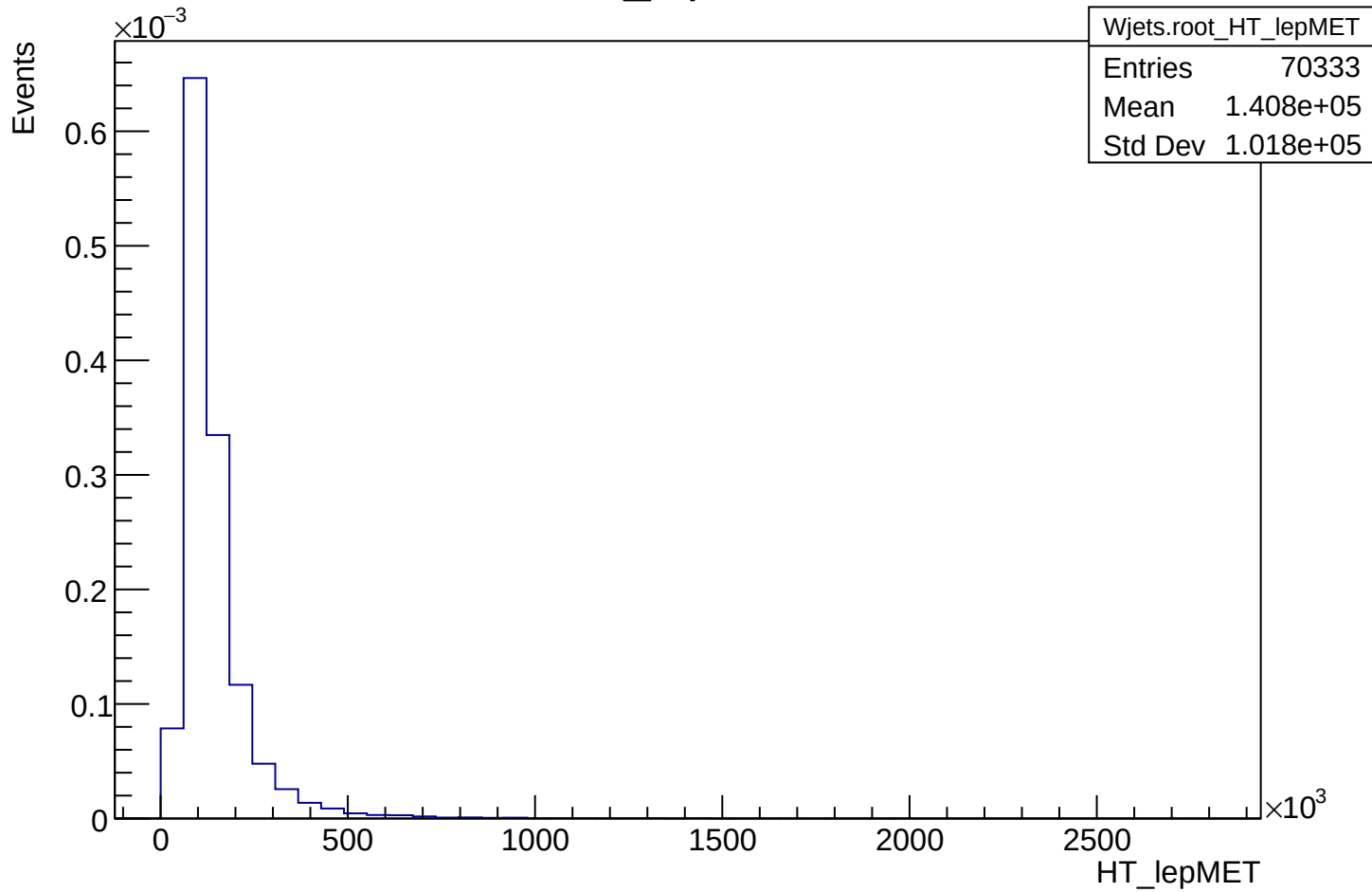
Wjets.root_HT_jet	
Entries	70333
Mean	2.74e+05
Std Dev	2.889e+05



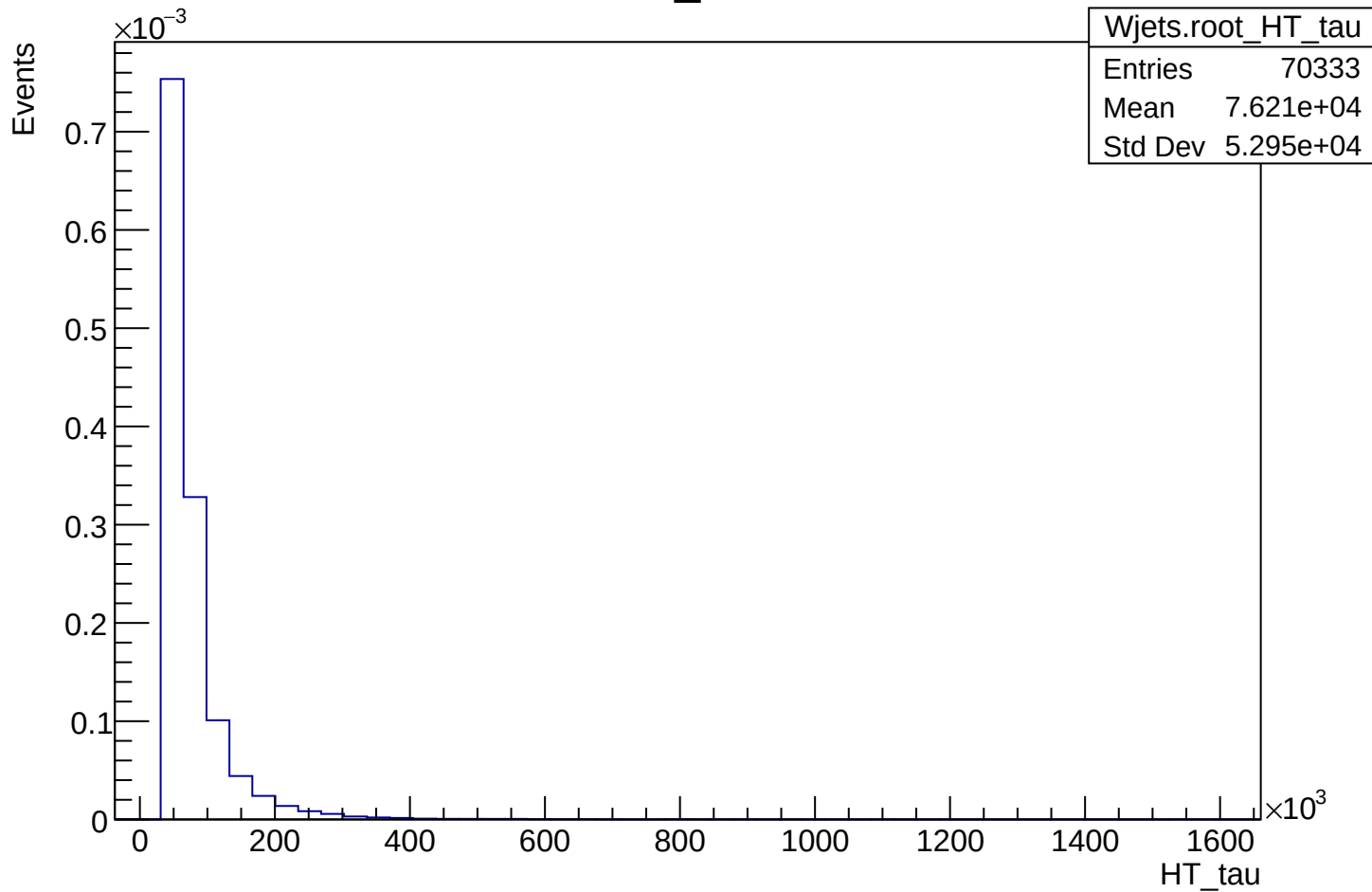
HT_lep



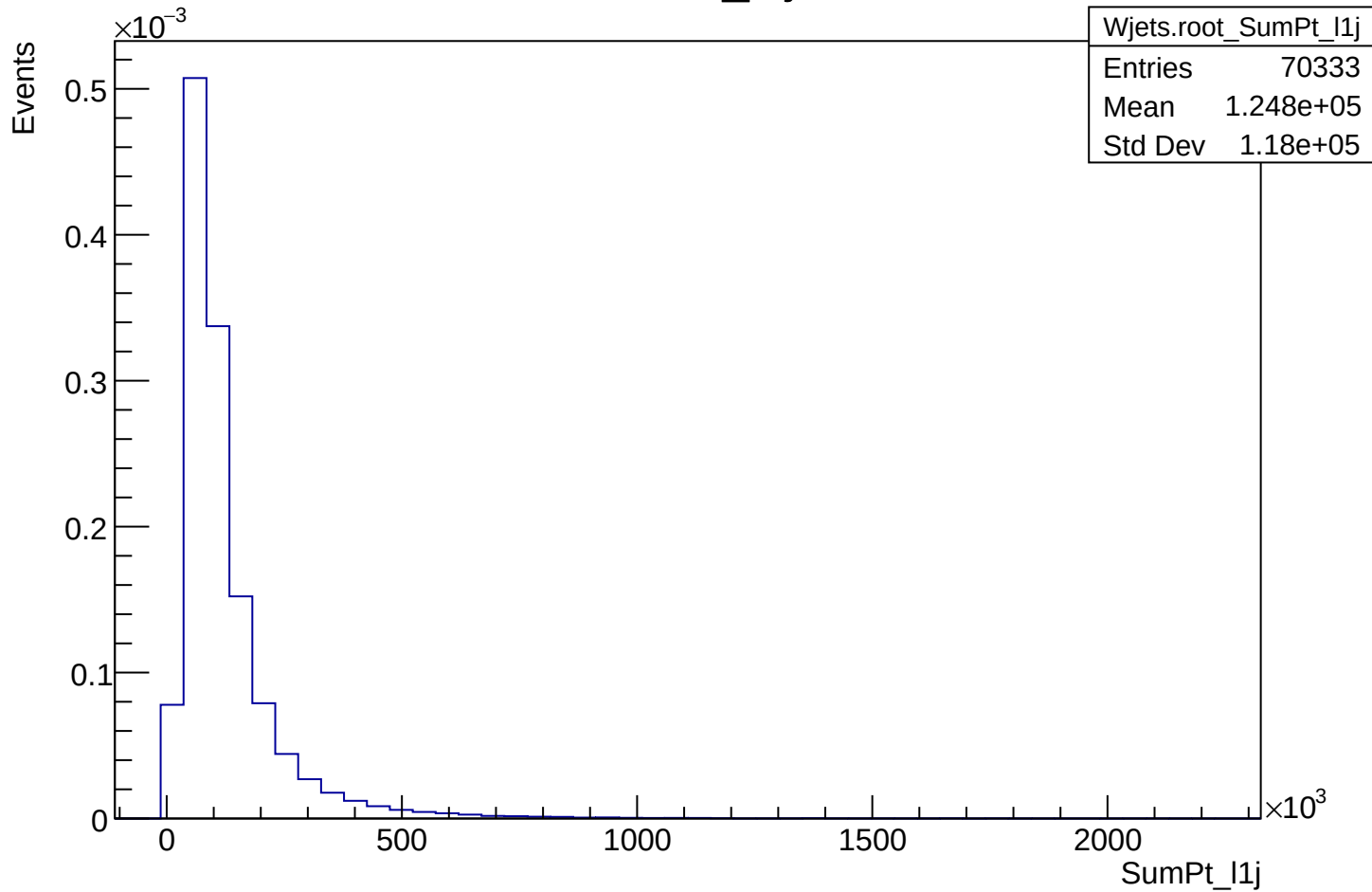
HT_lepMET



HT_tau

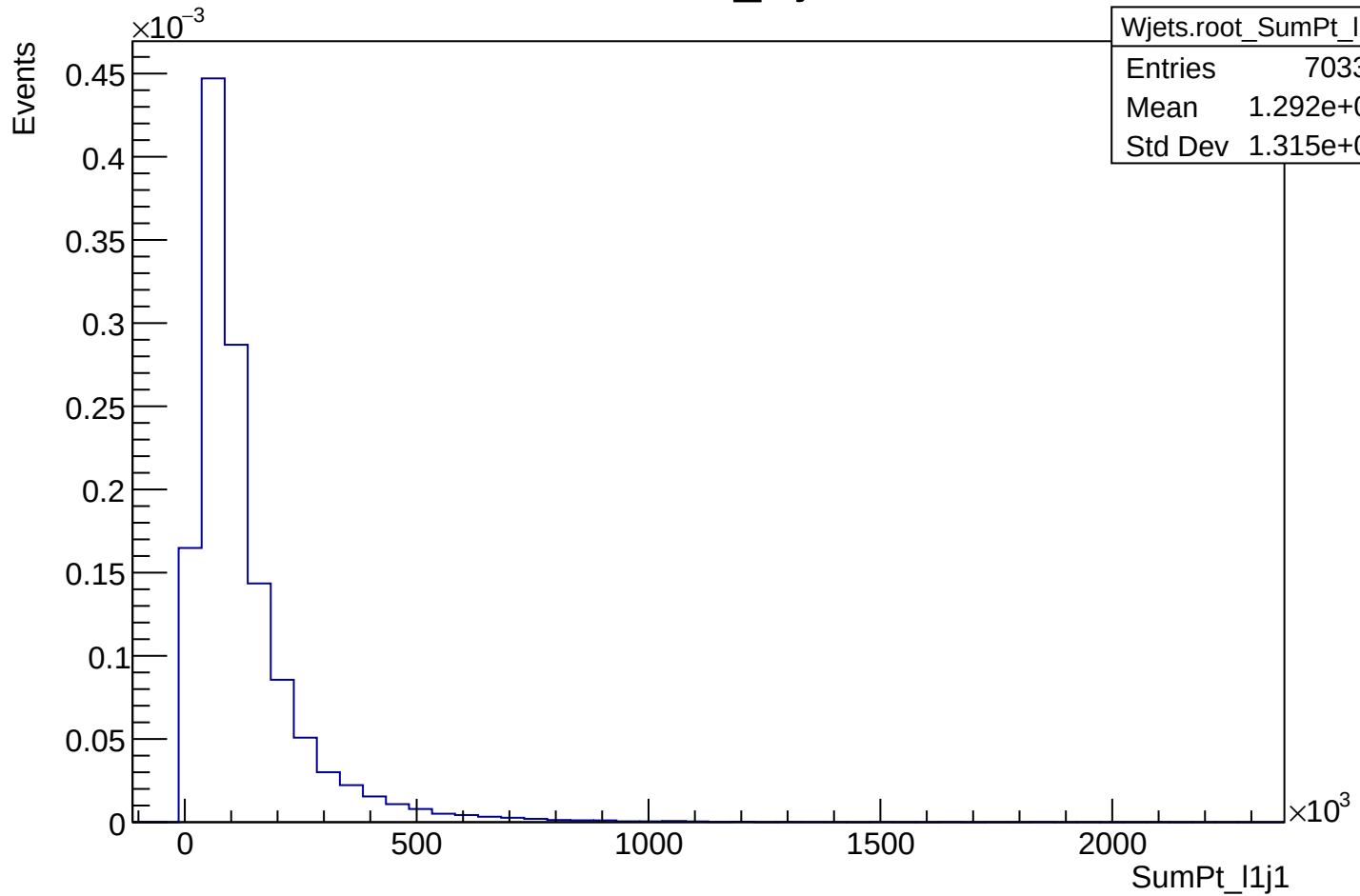


SumPt_l1j

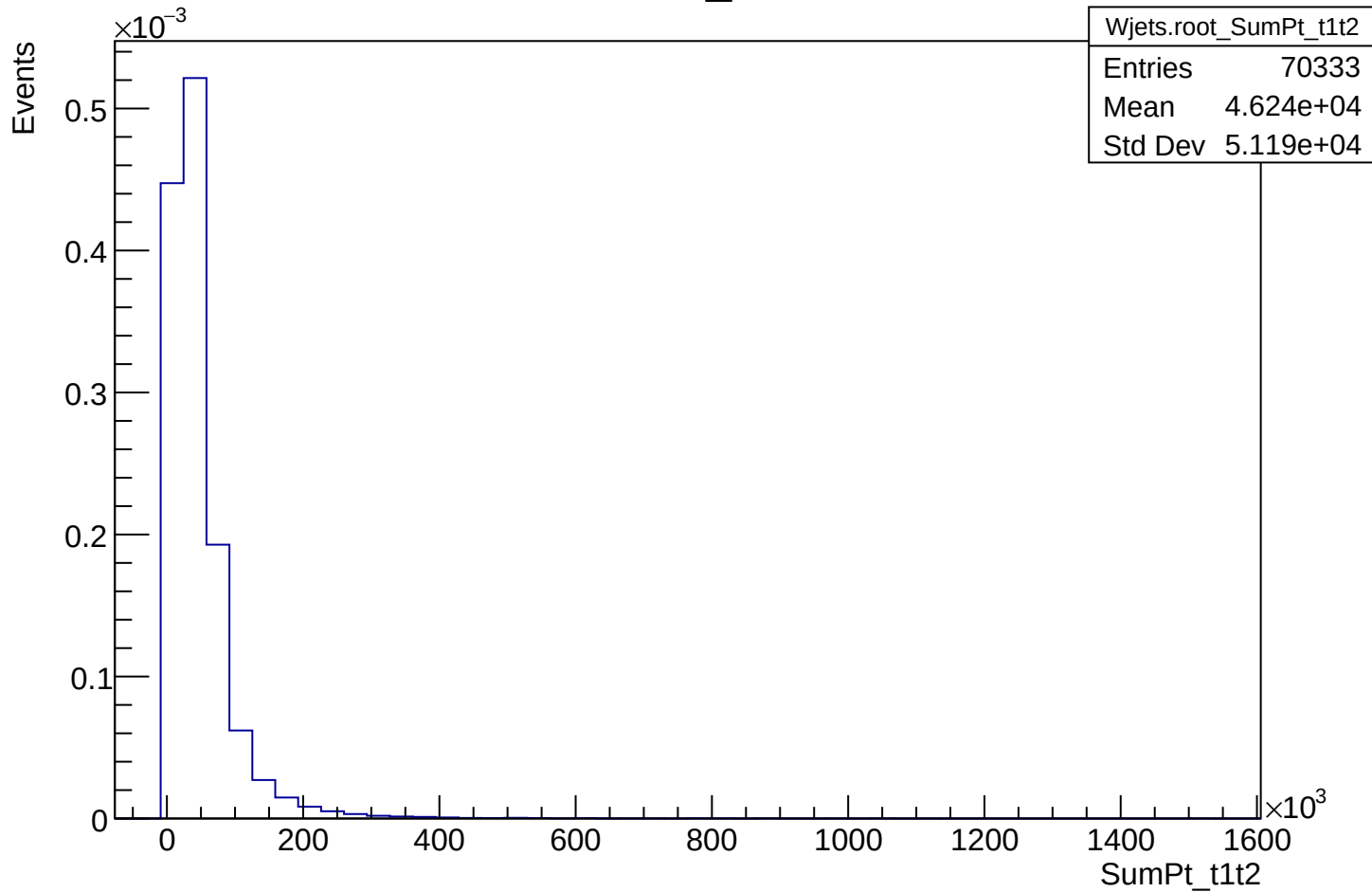


SumPt_l1j1

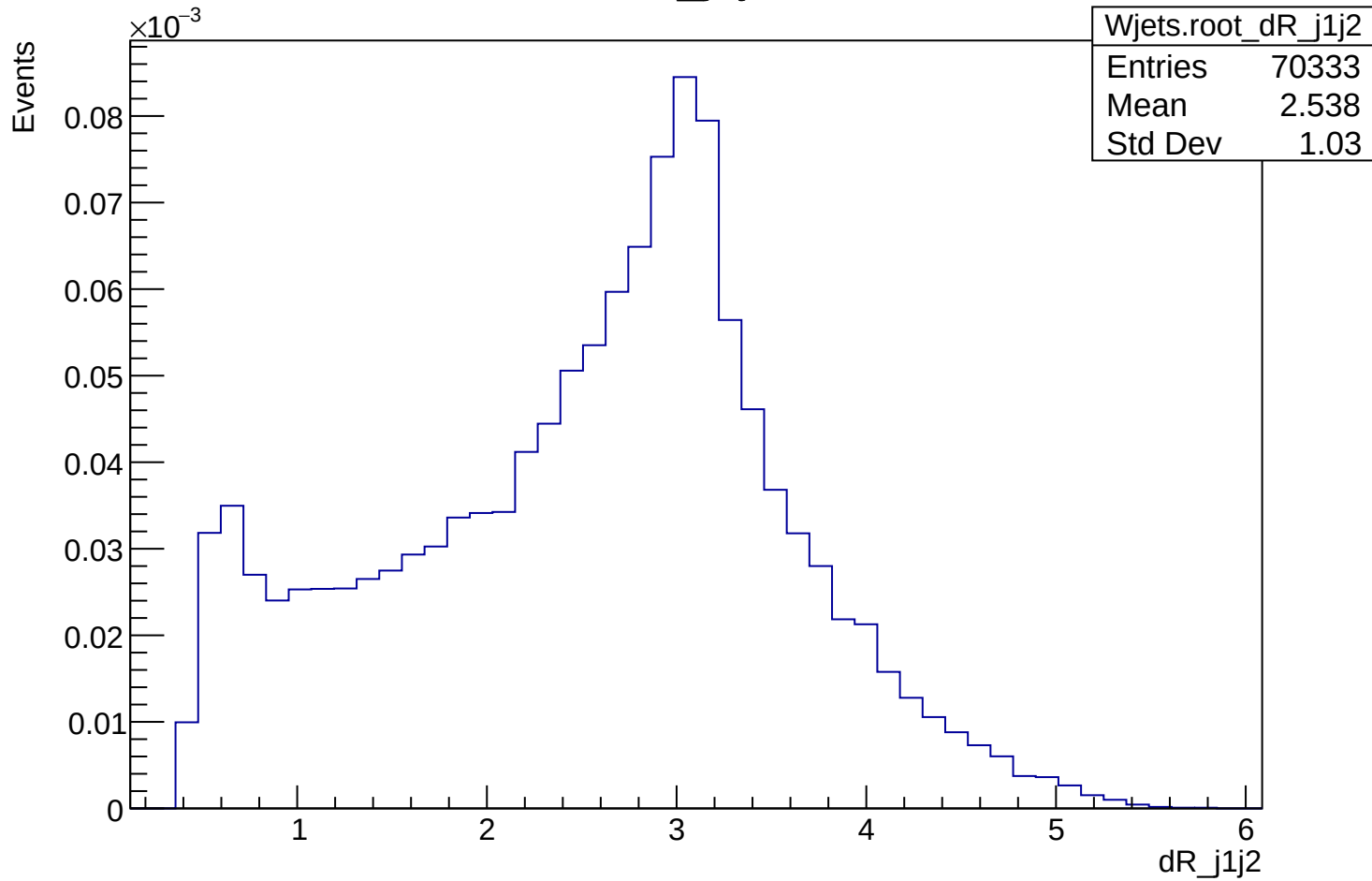
Wjets.root_SumPt_l1j1	
Entries	70333
Mean	1.292e+05
Std Dev	1.315e+05



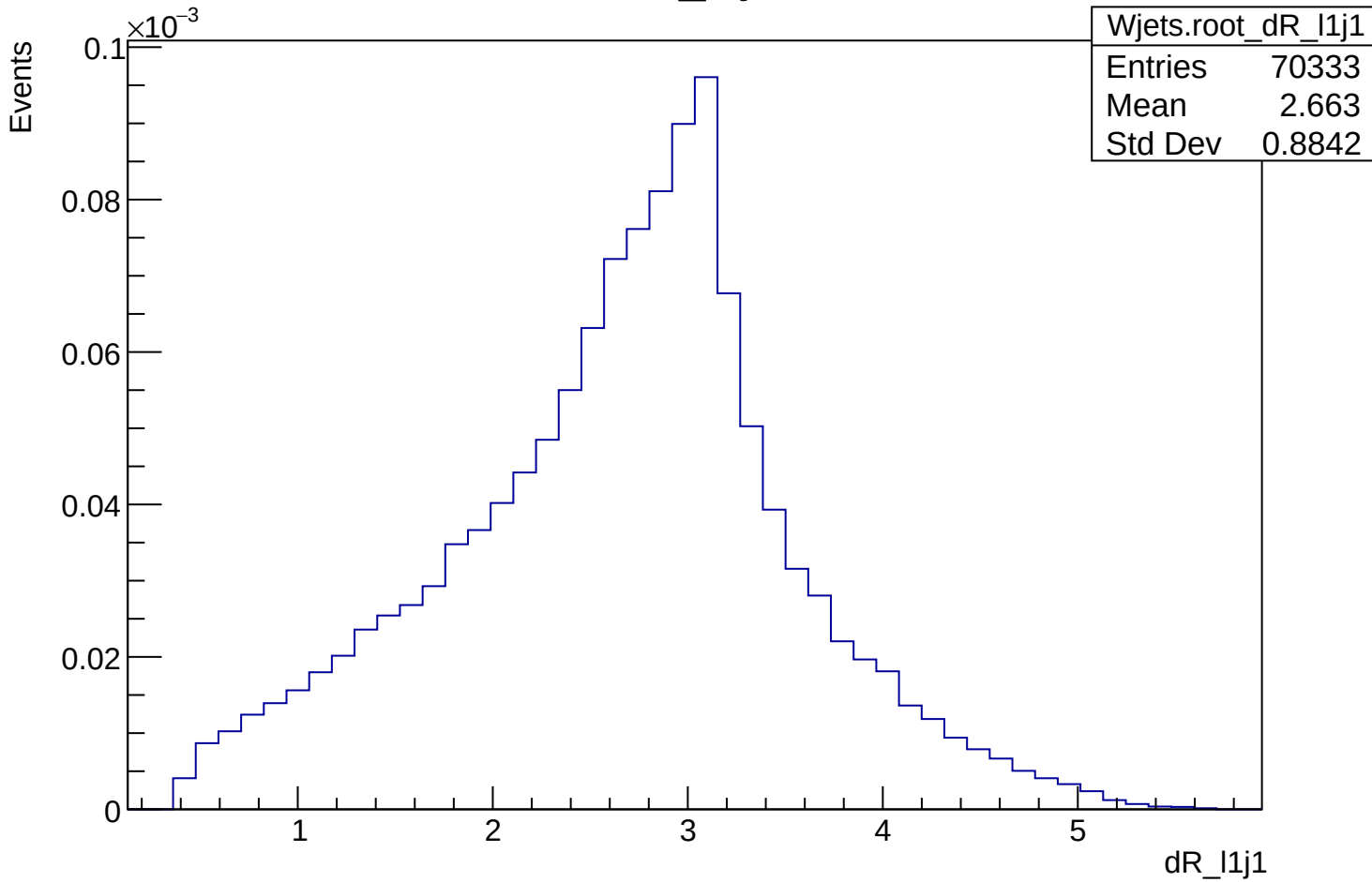
SumPt_t1t2



dR_j1j2

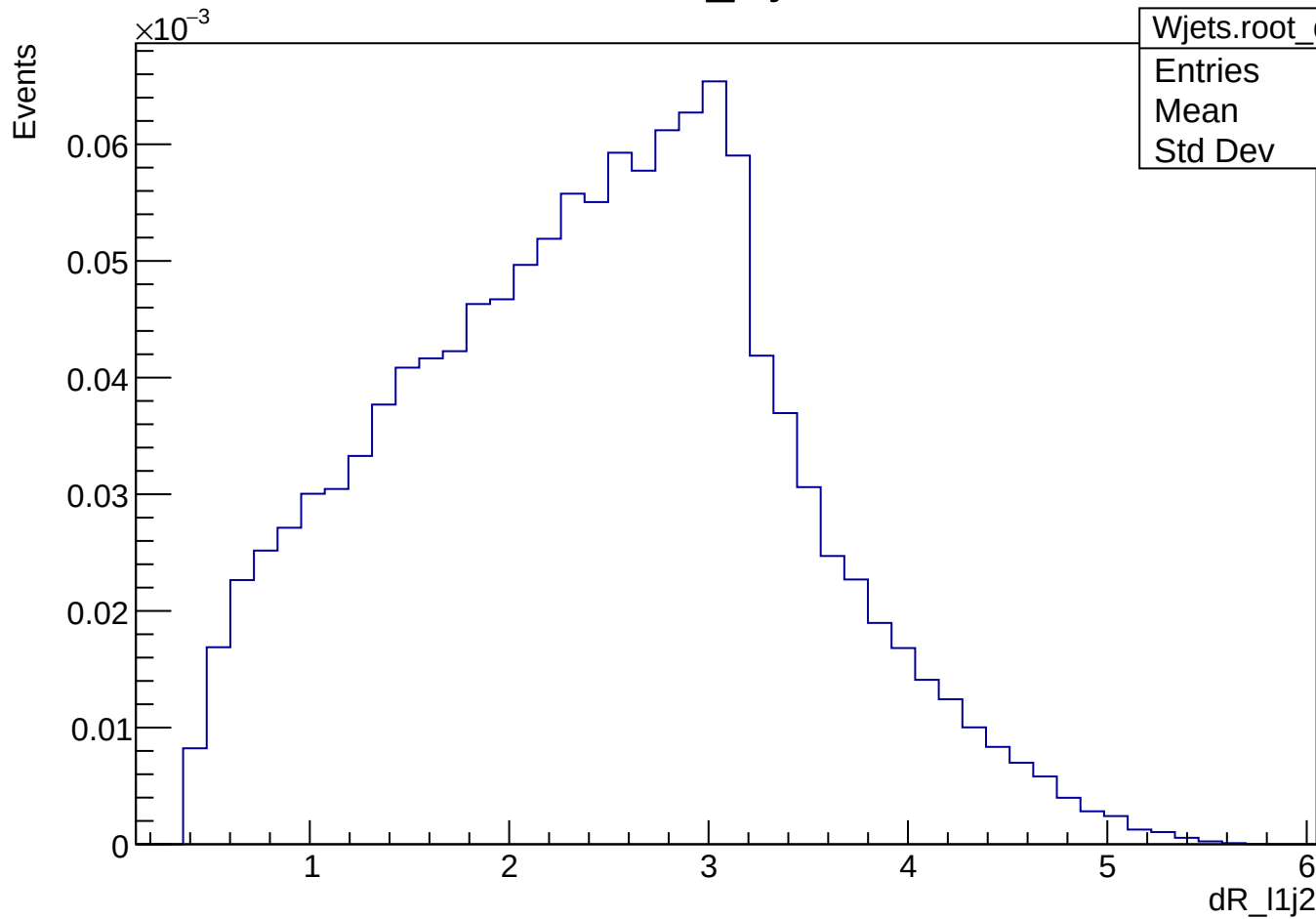


dR_l1j1



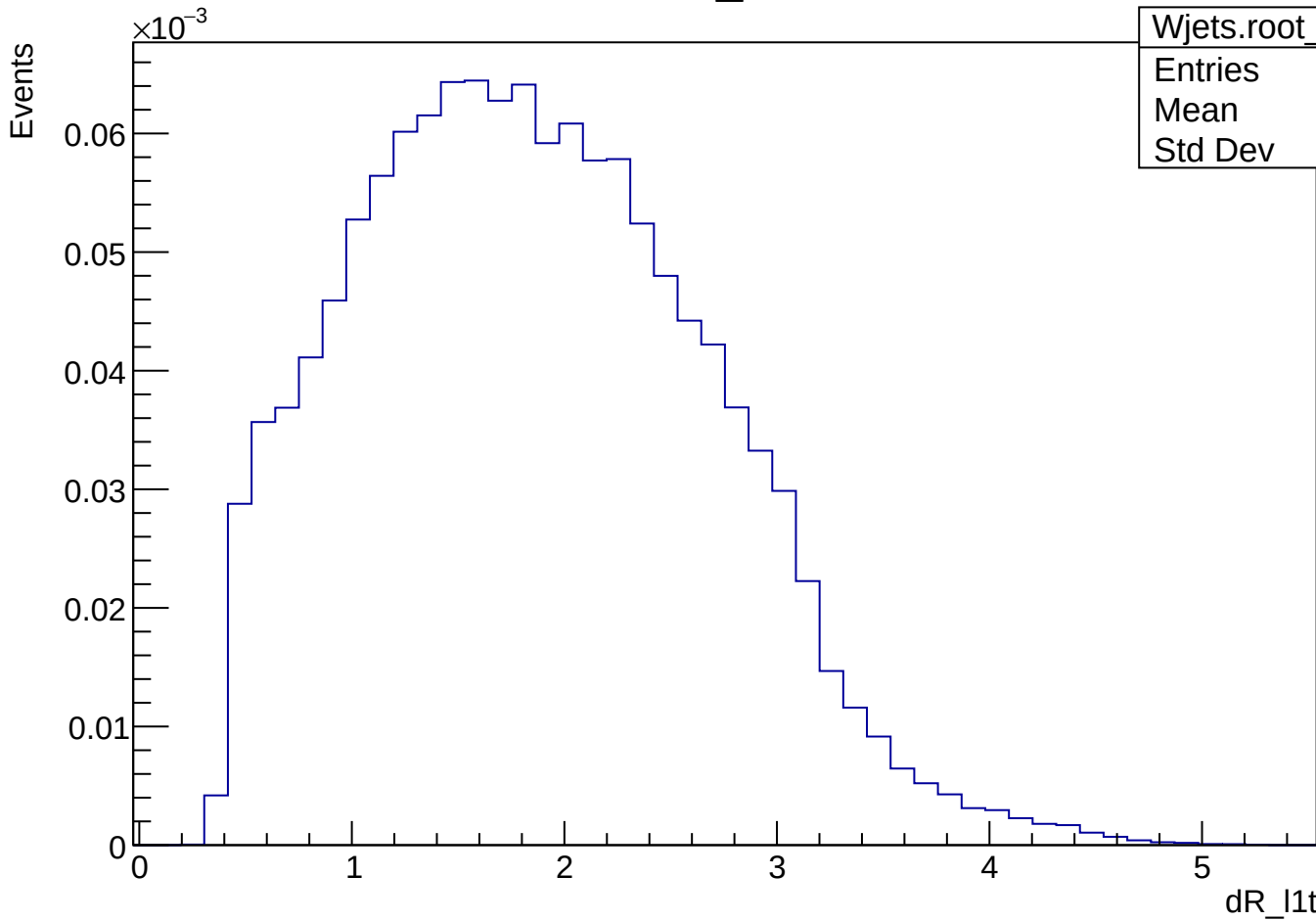
dR_l1j2

Wjets.root_dR_l1j2	
Entries	70333
Mean	2.425
Std Dev	0.9806

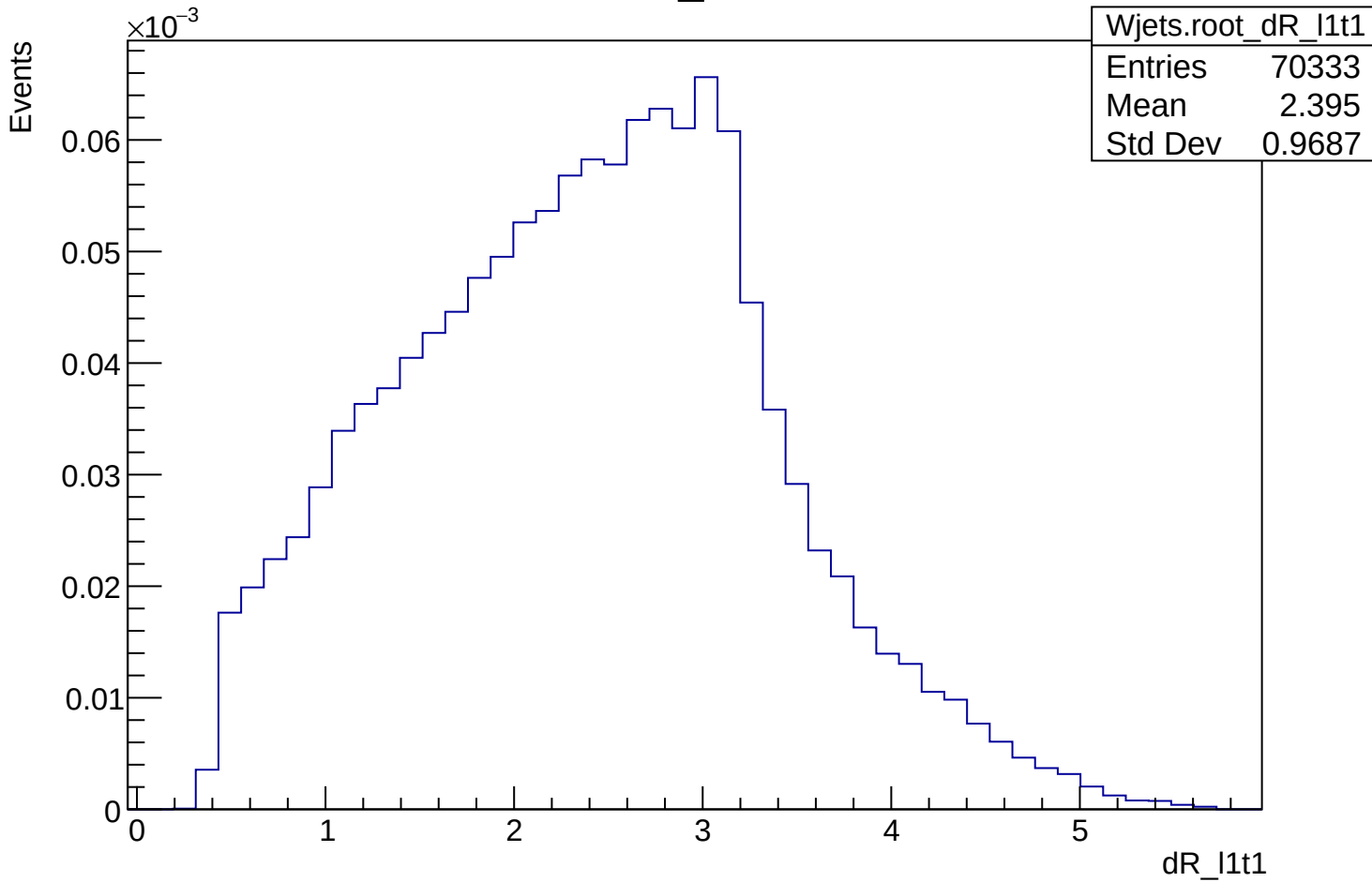


dR_l1t

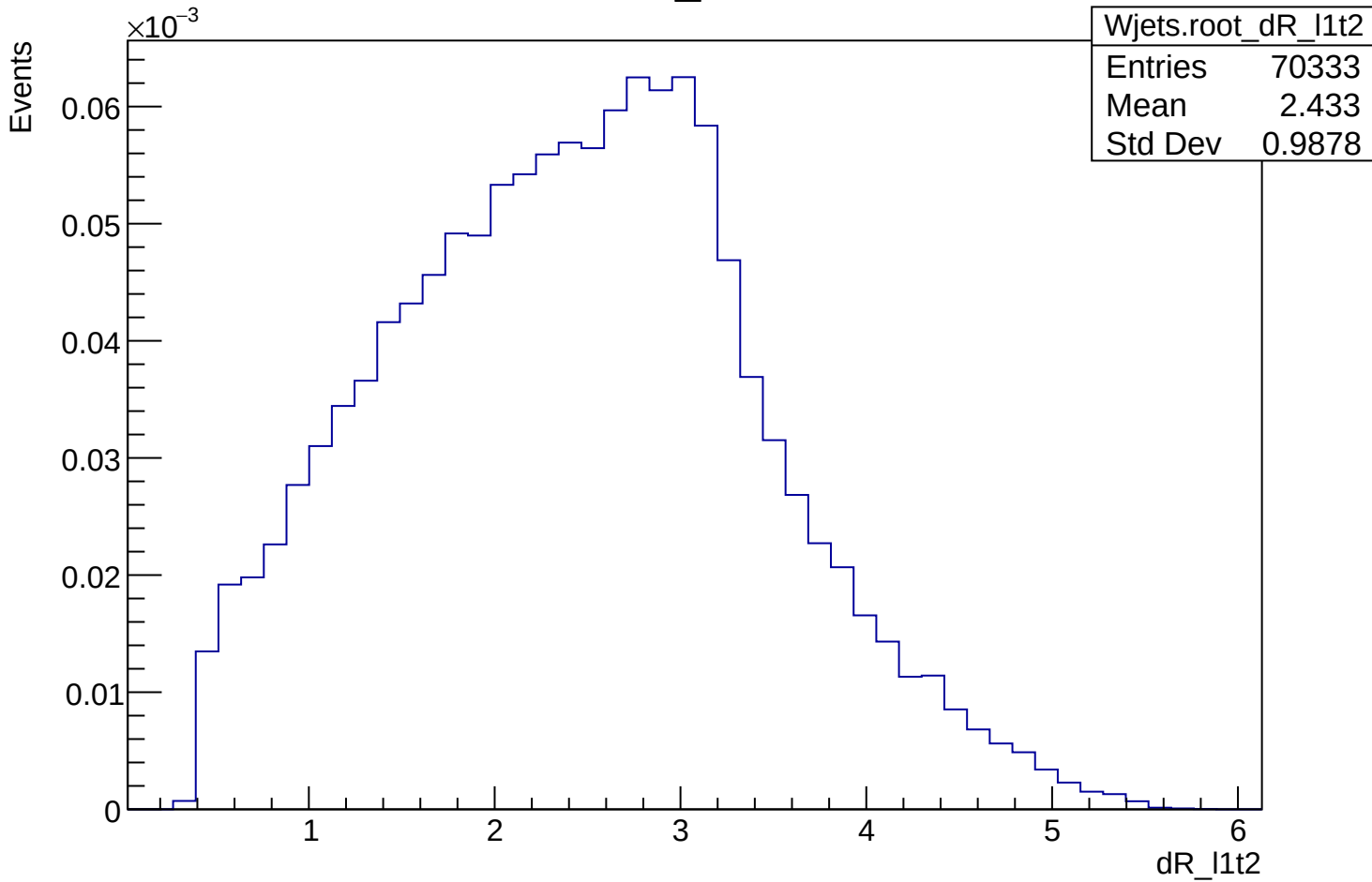
Wjets.root_dR_l1t	
Entries	70333
Mean	1.859
Std Dev	0.8075



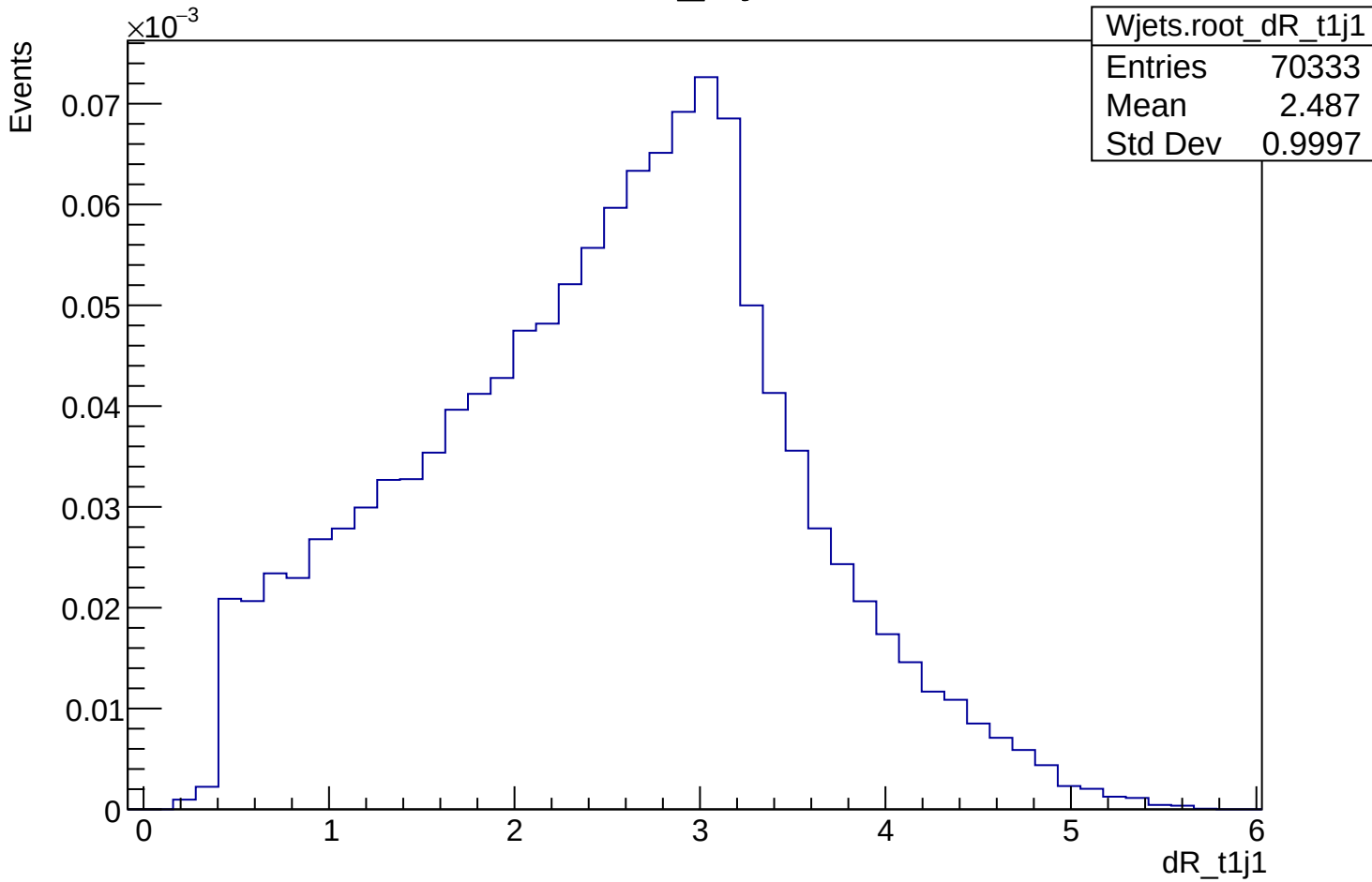
dR_l1t1



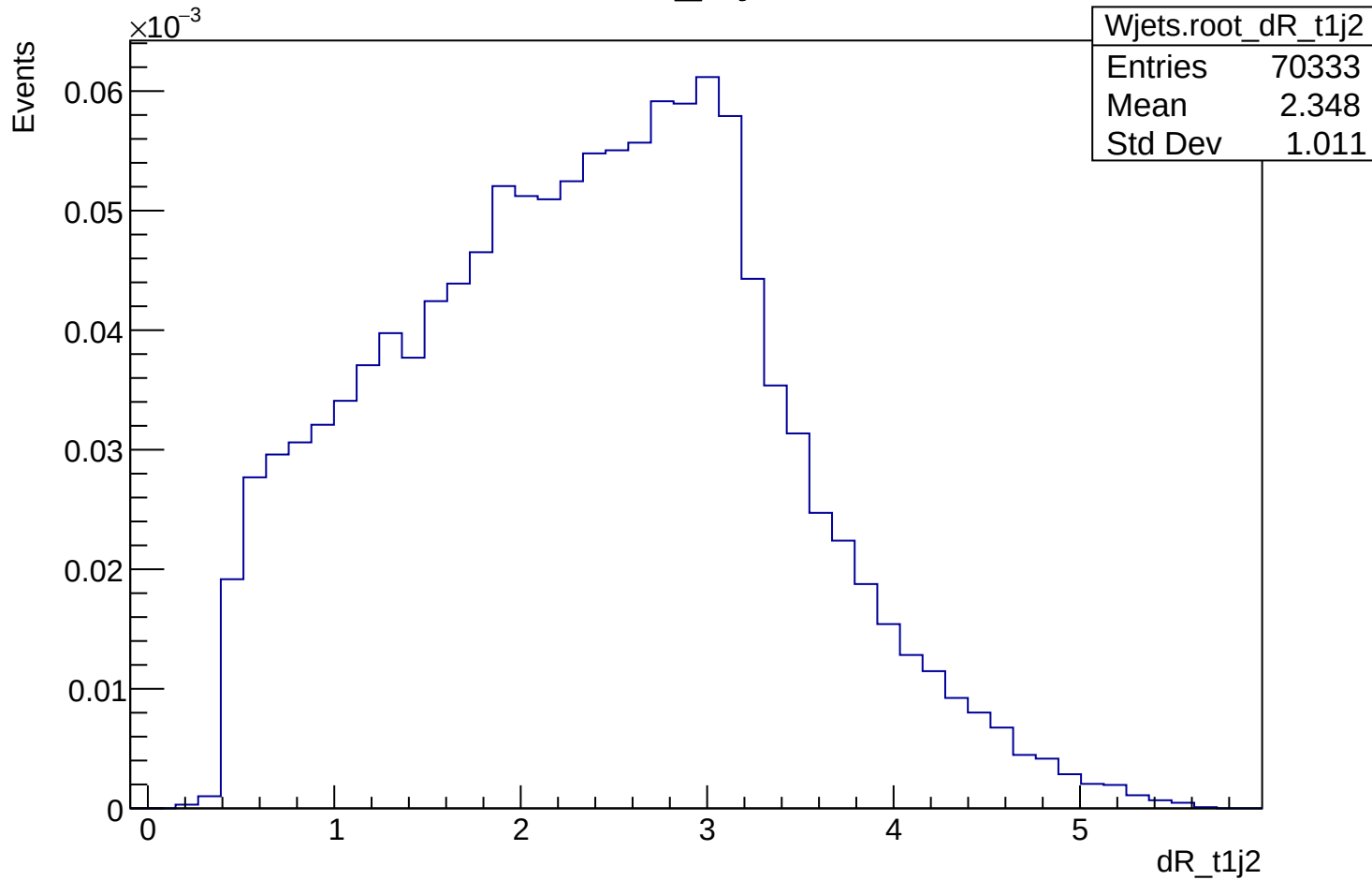
dR_l1t2



dR_t1j1

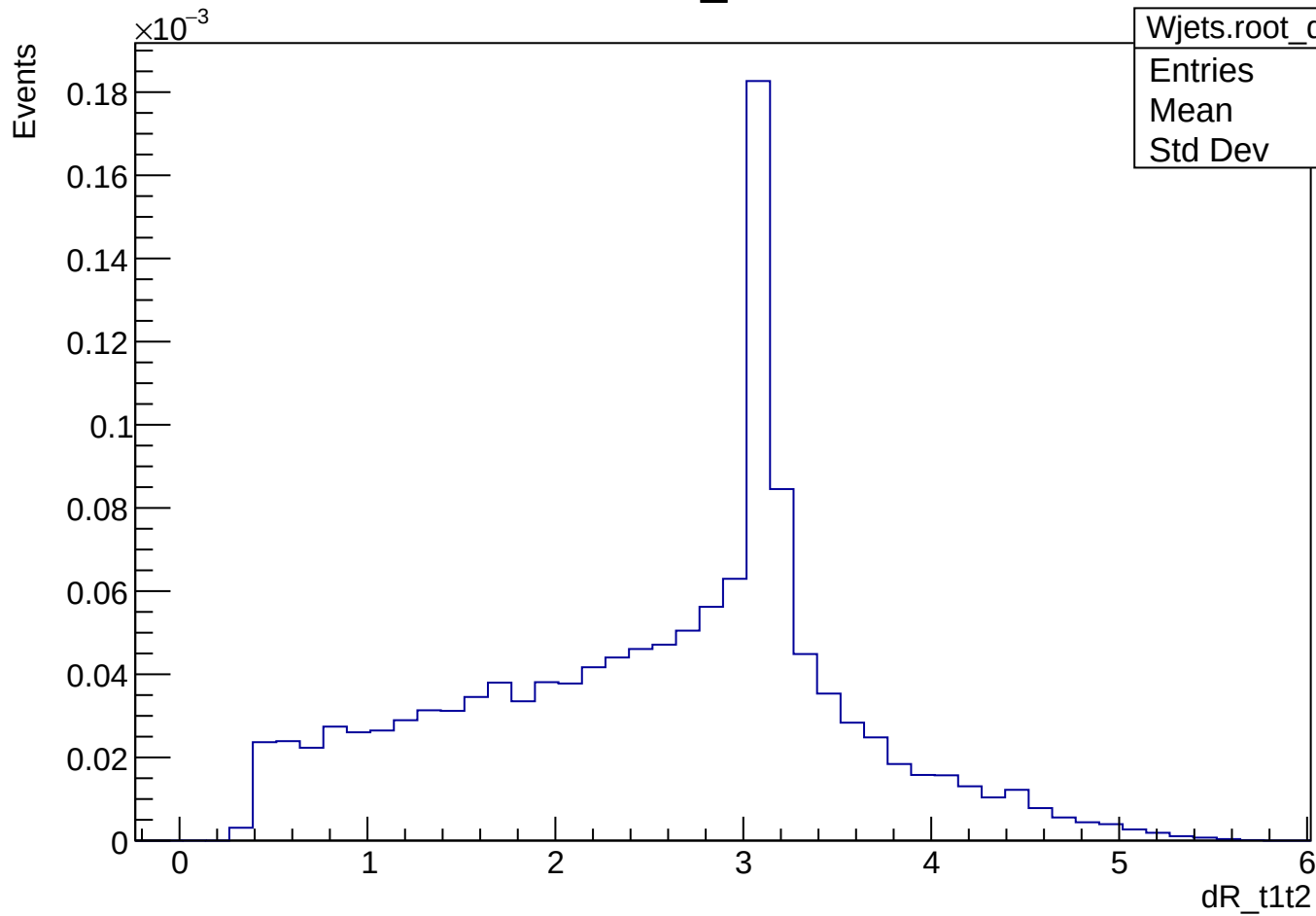


dR_t1j2

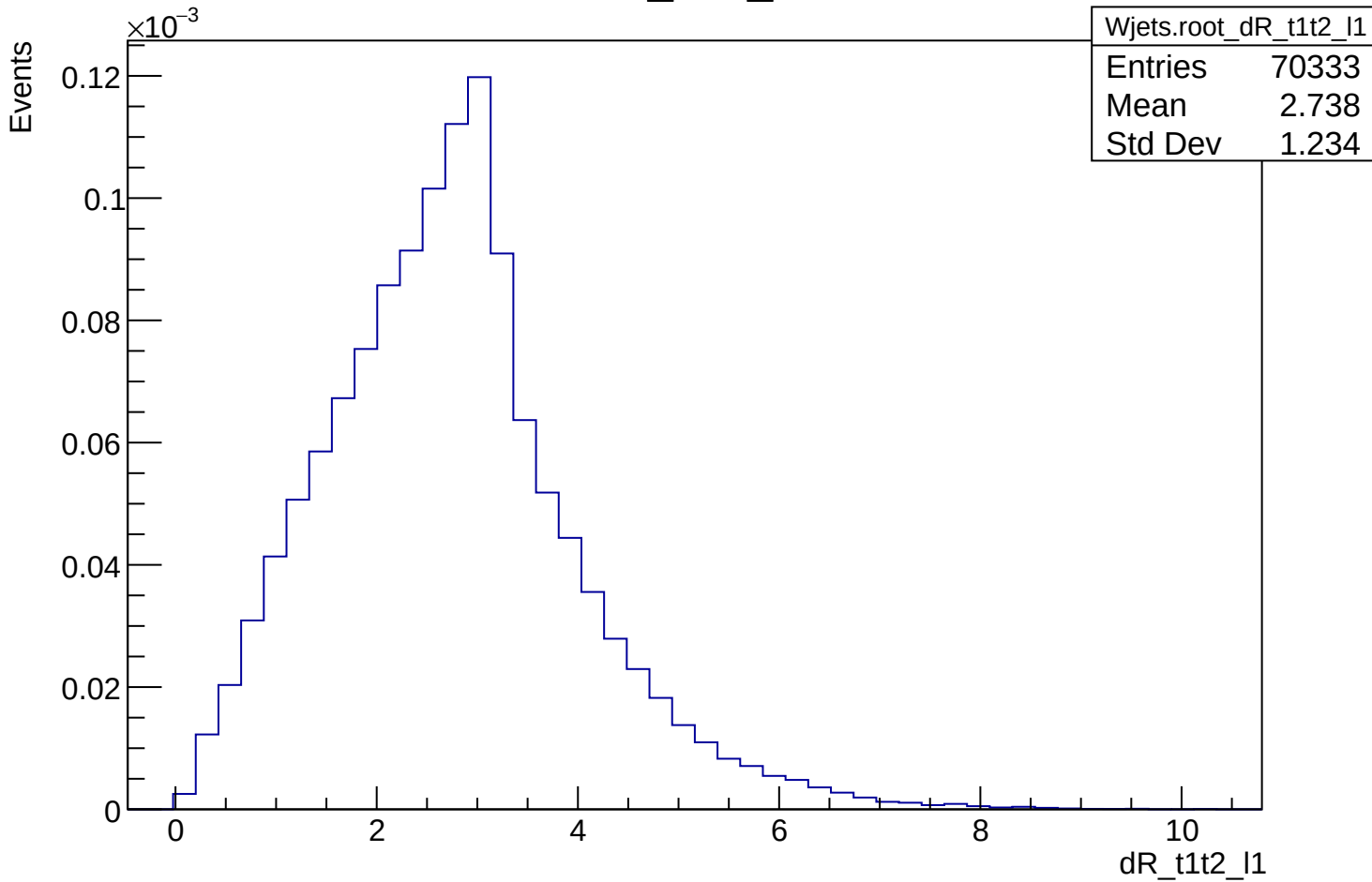


dR_t1t2

Wjets.root_dR_t1t2	
Entries	70333
Mean	2.54
Std Dev	1.015

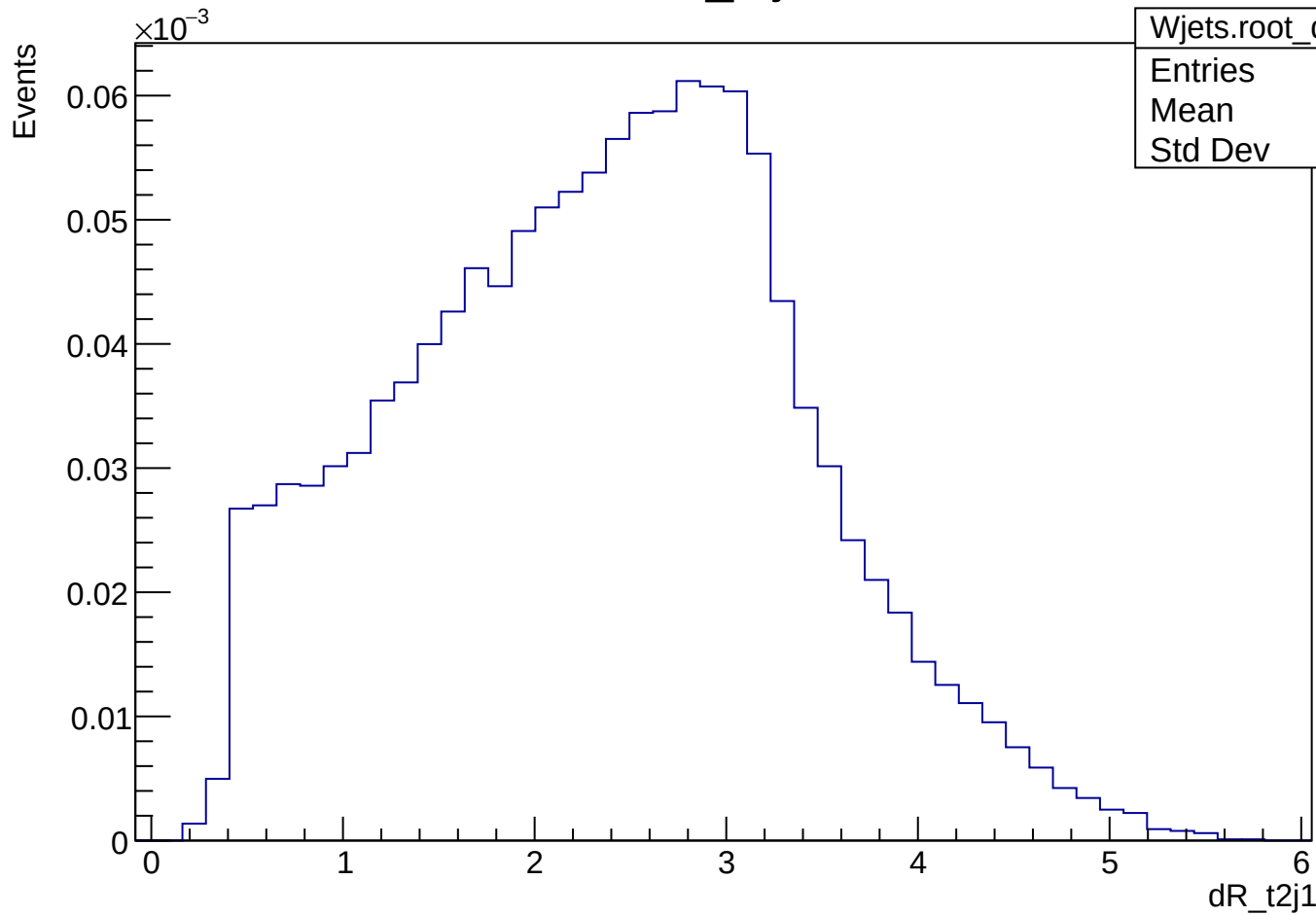


dR_t1t2_l1

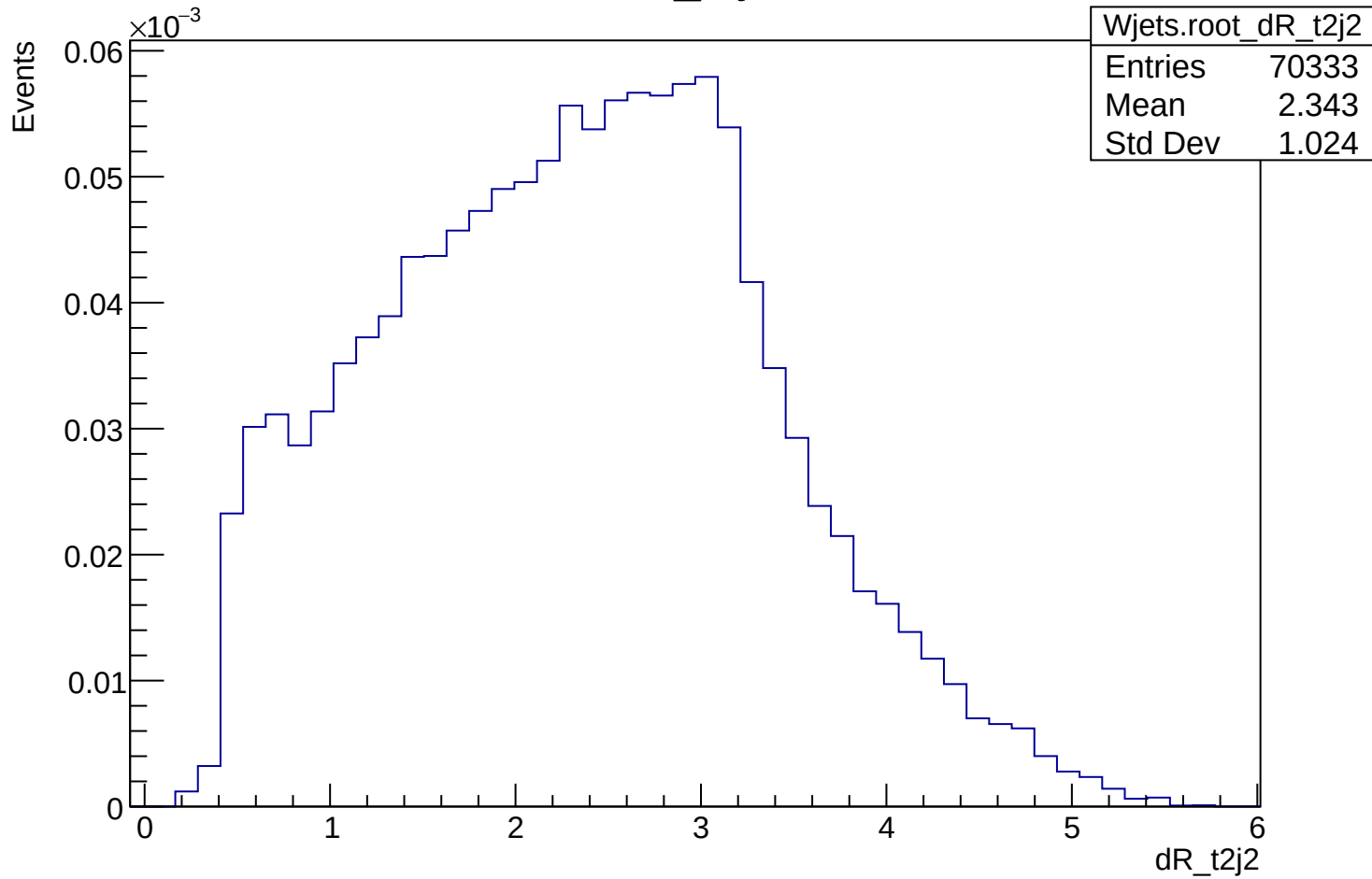


dR_t2j1

Wjets.root_dR_t2j1	
Entries	70333
Mean	2.364
Std Dev	1.017

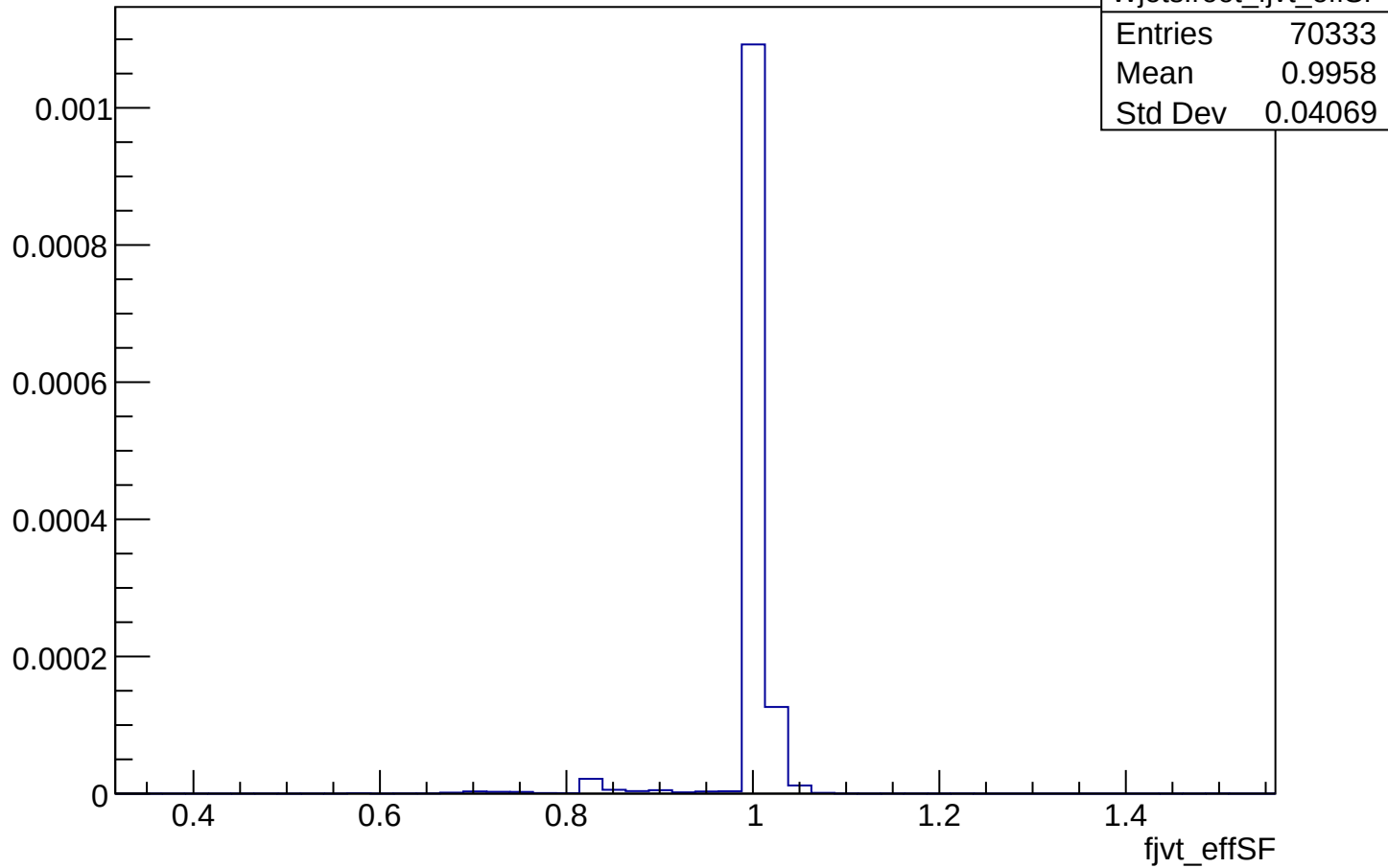


dR_t2j2

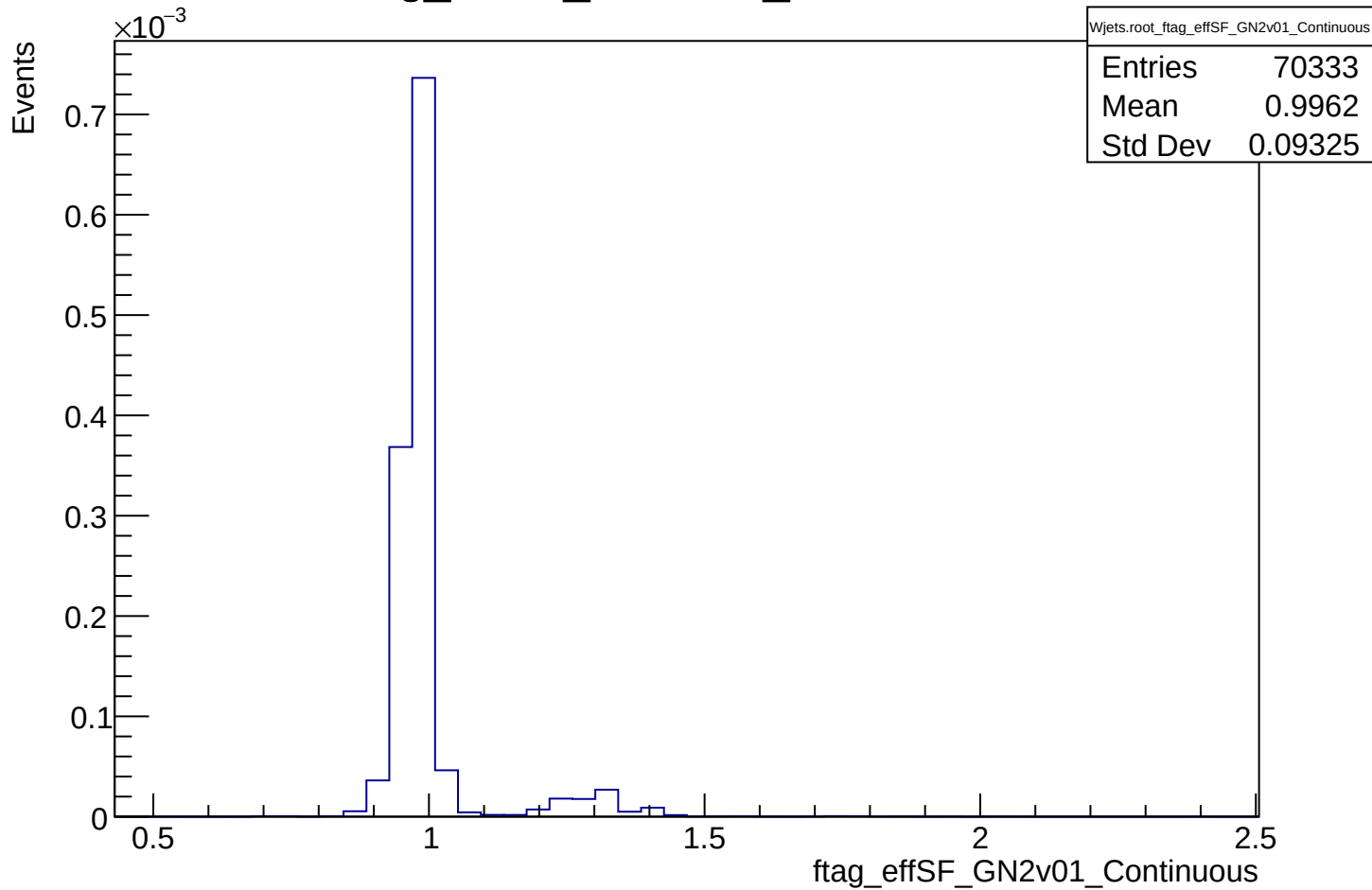


fjvt_effSF

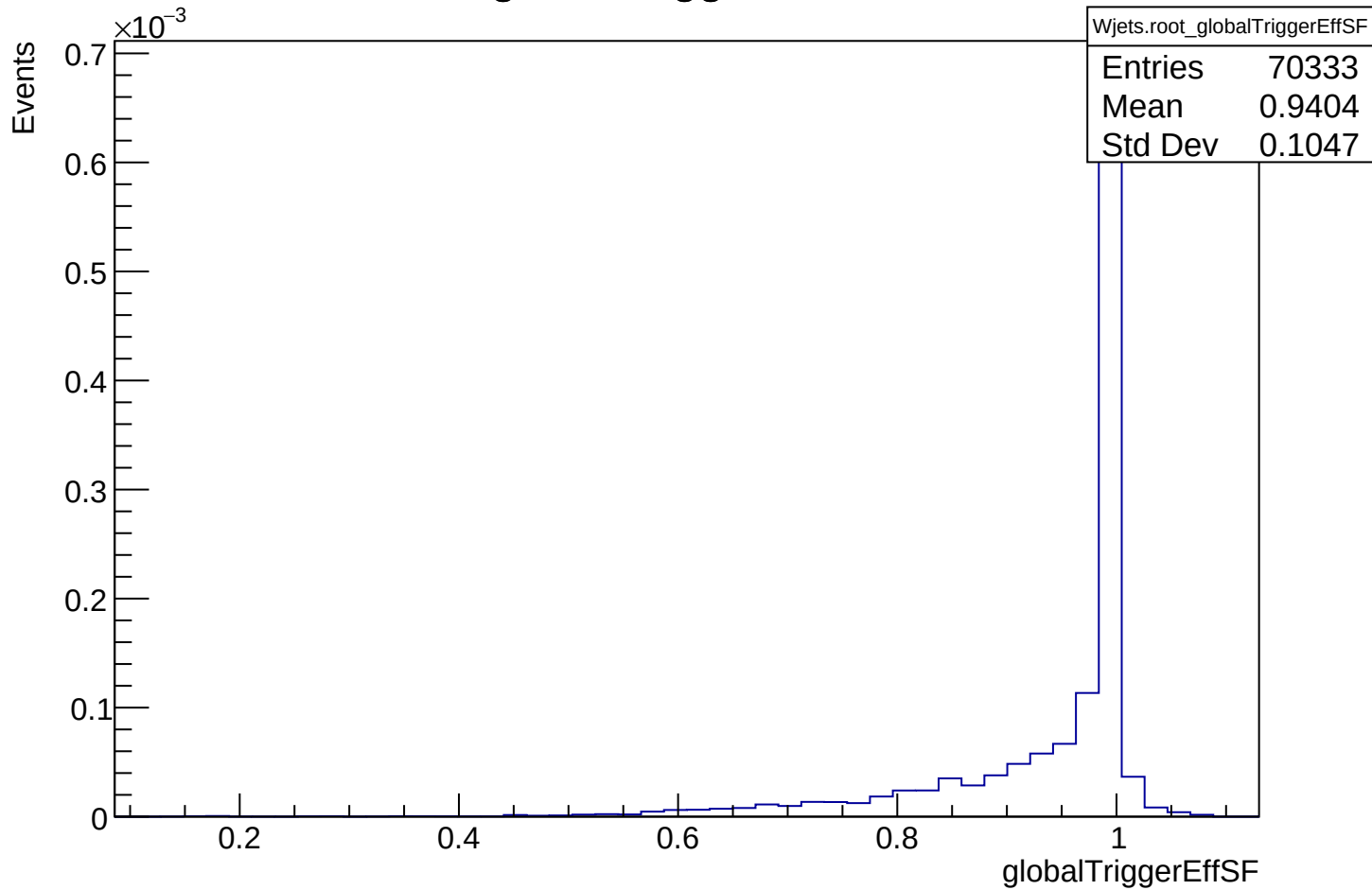
Events



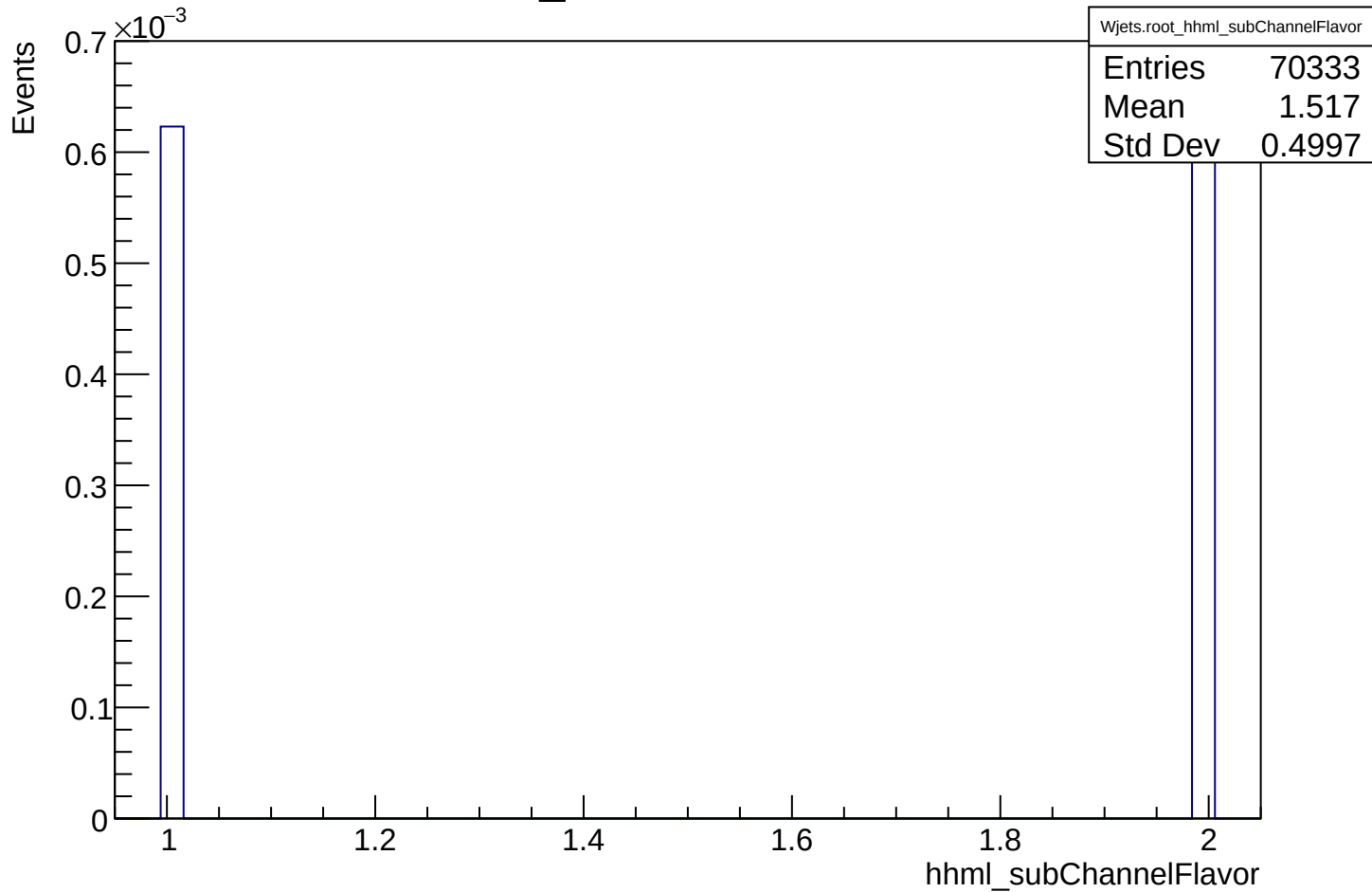
ftag_effSF_GN2v01_Continuous



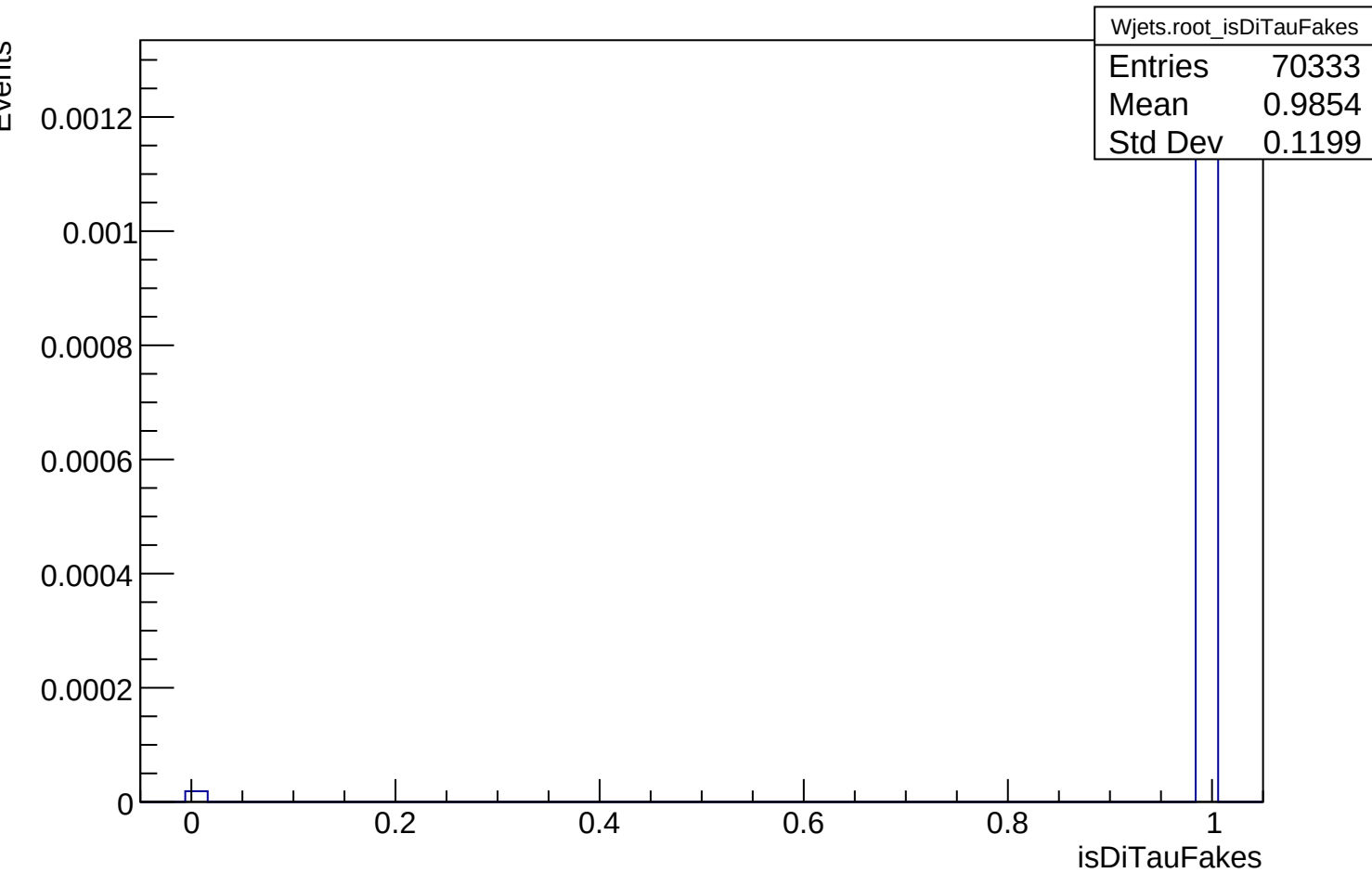
globalTriggerEffSF



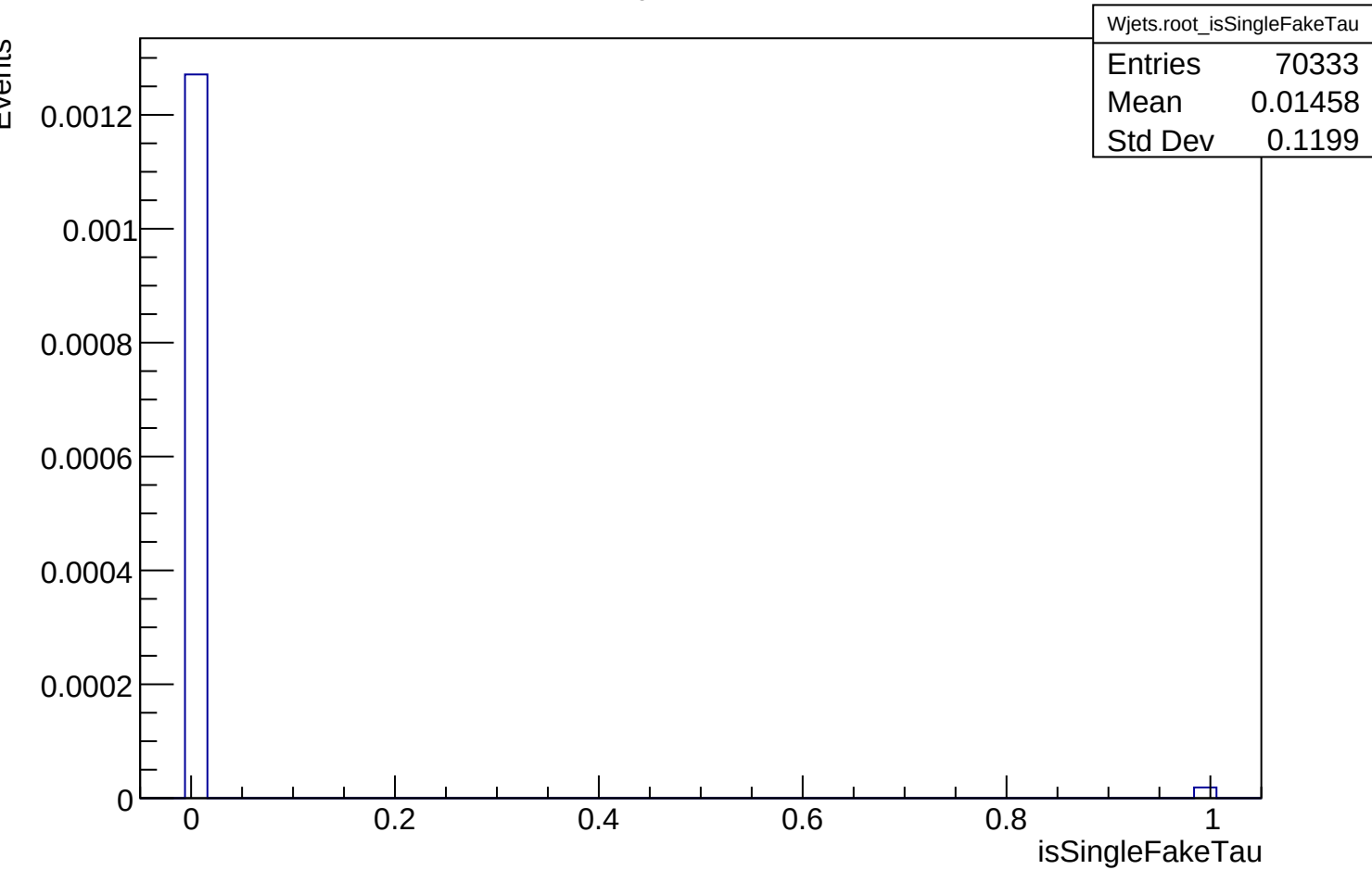
hhml_subChannelFlavor



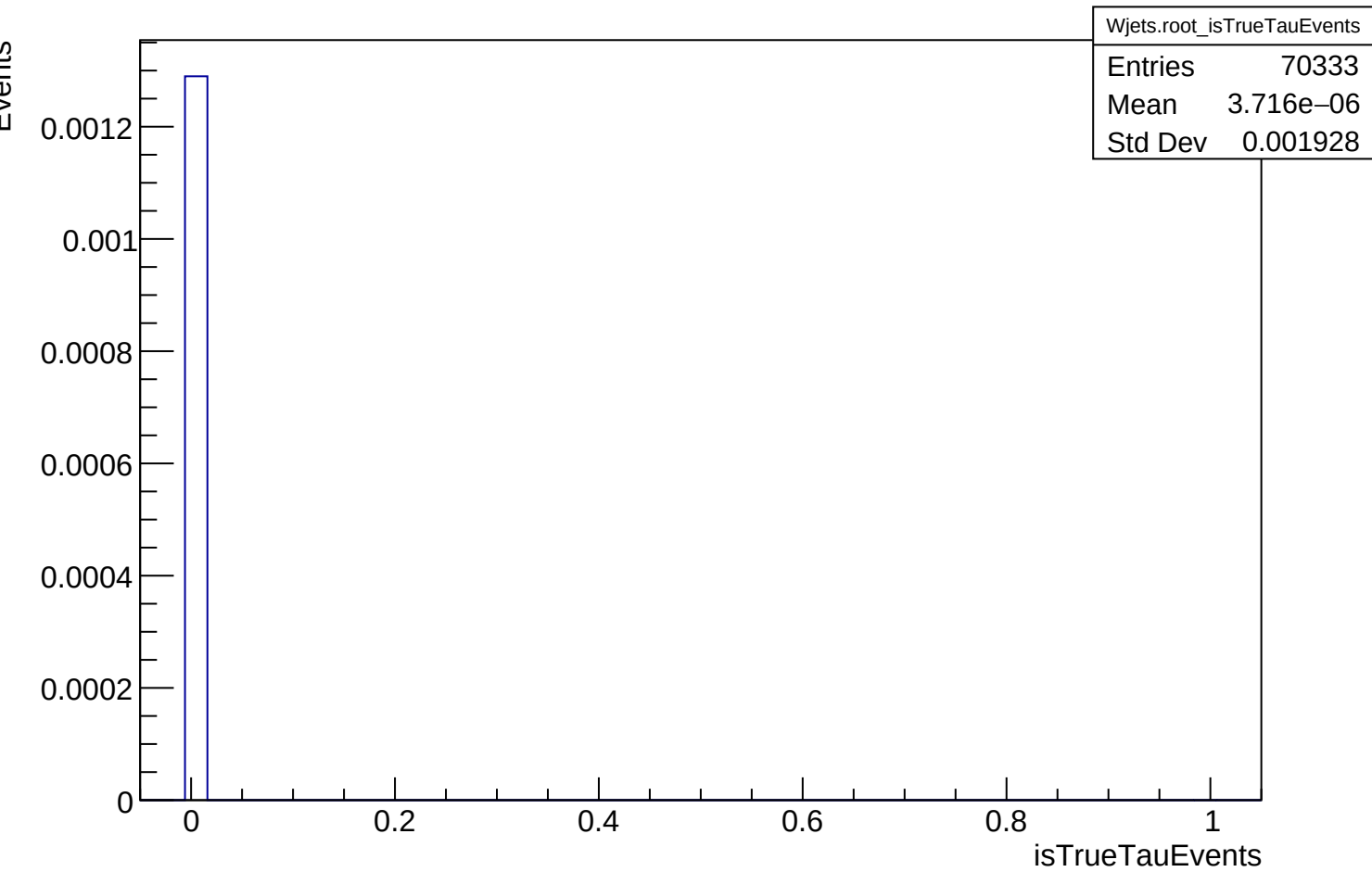
isDiTauFakes



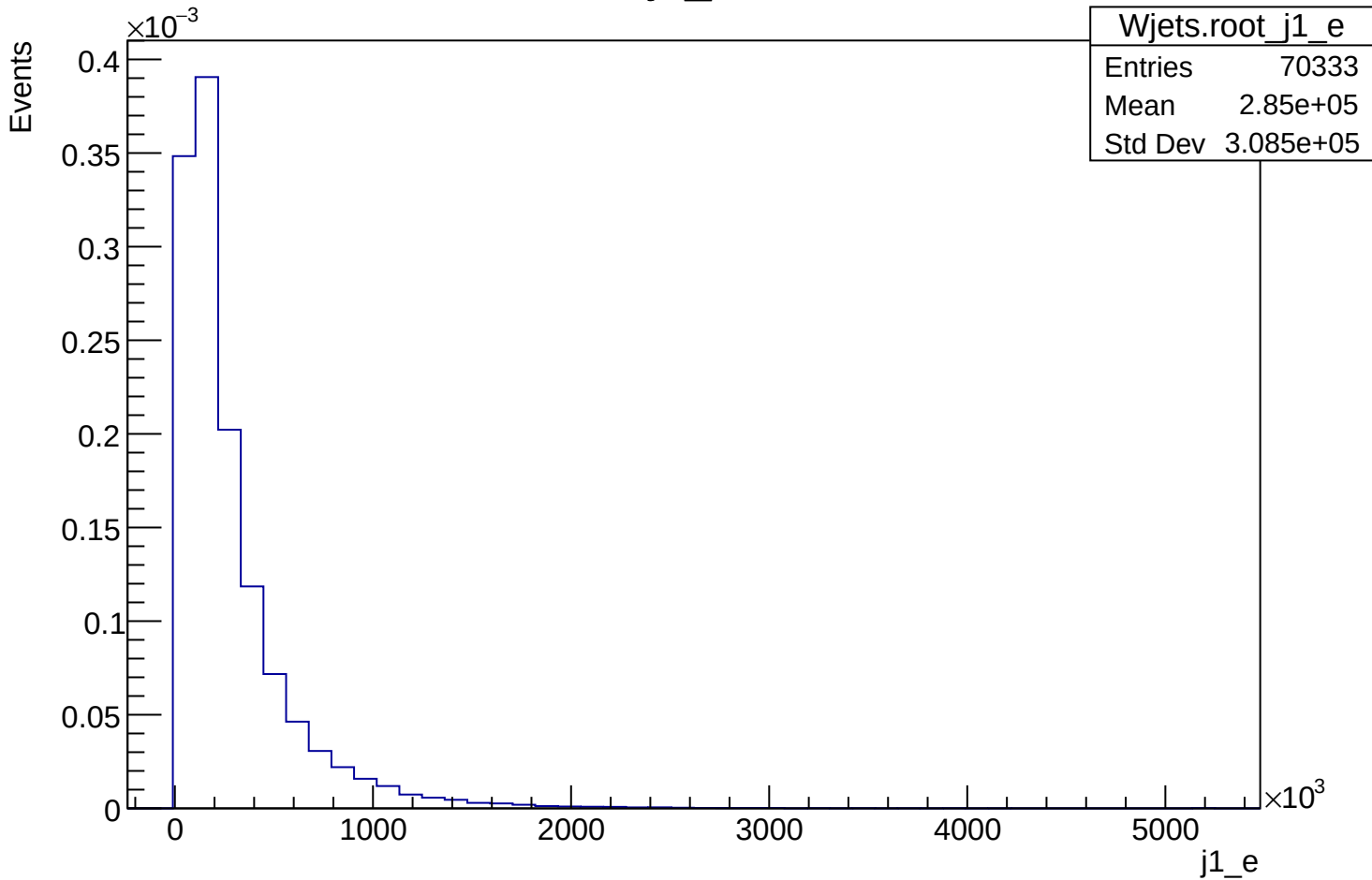
isSingleFakeTau



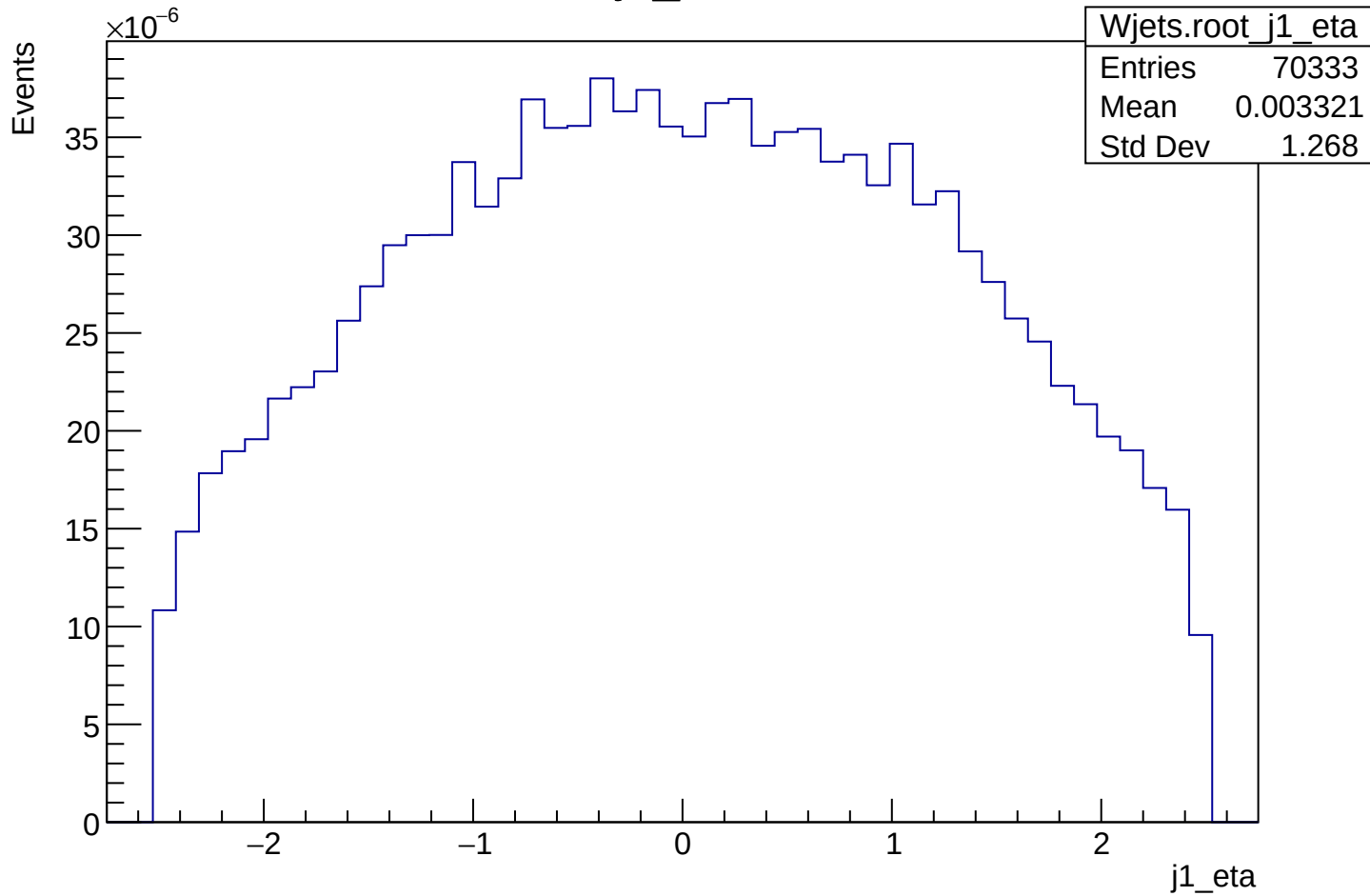
isTrueTauEvents



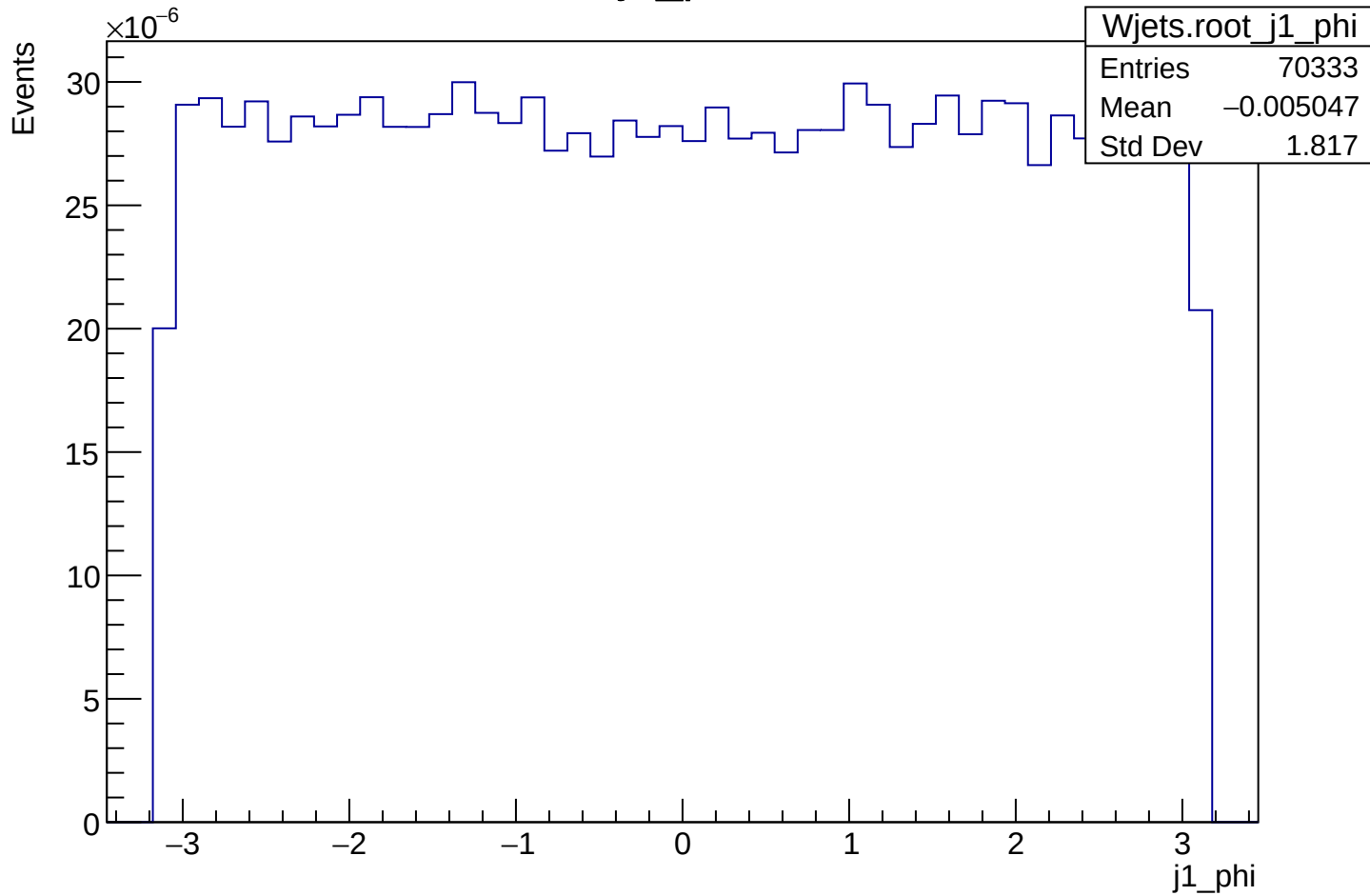
j1_e



j1_eta



j1_phi



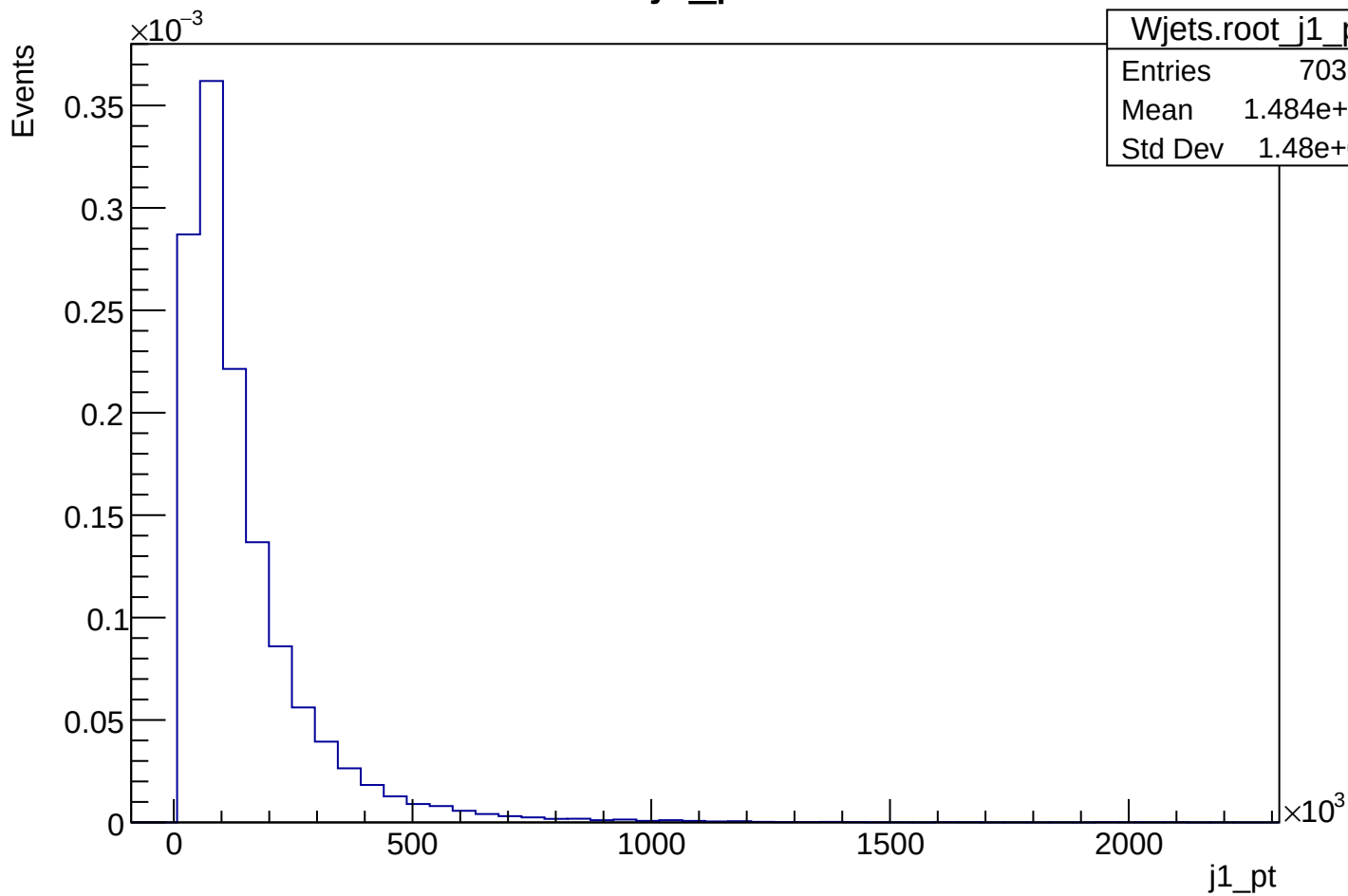
j1_pt

Wjets.root_j1_pt

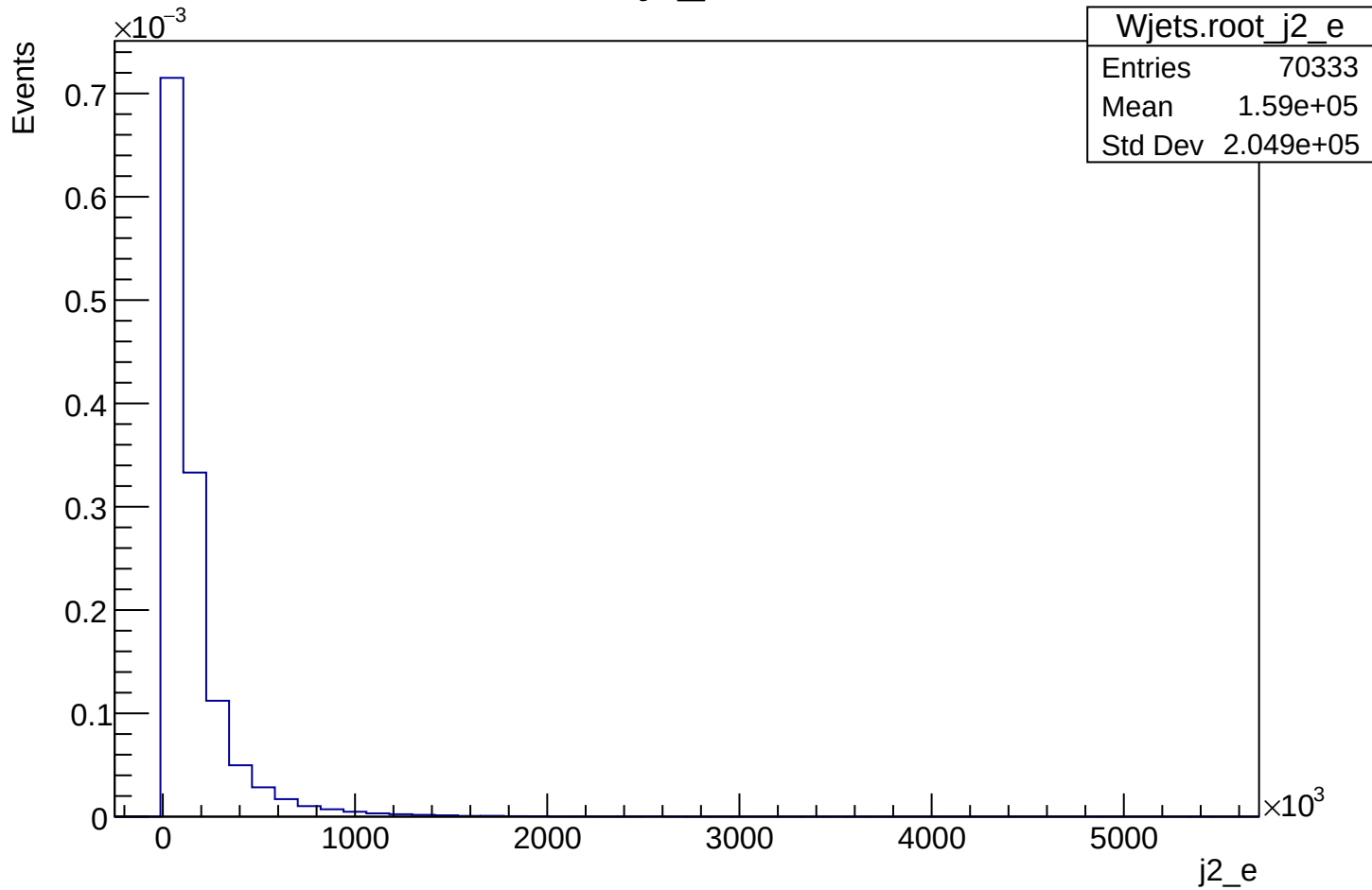
Entries 70333

Mean 1.484e+05

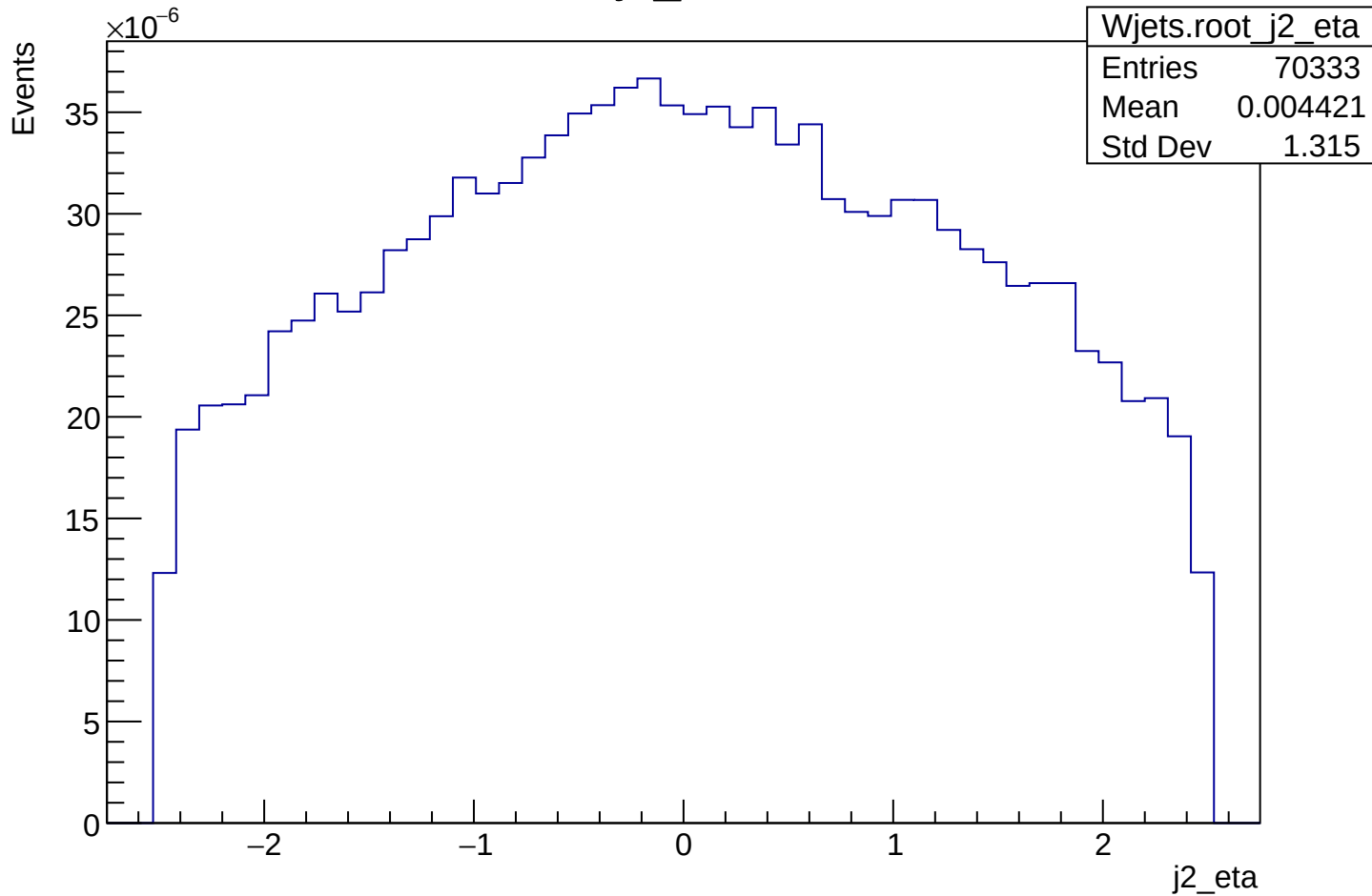
Std Dev 1.48e+05



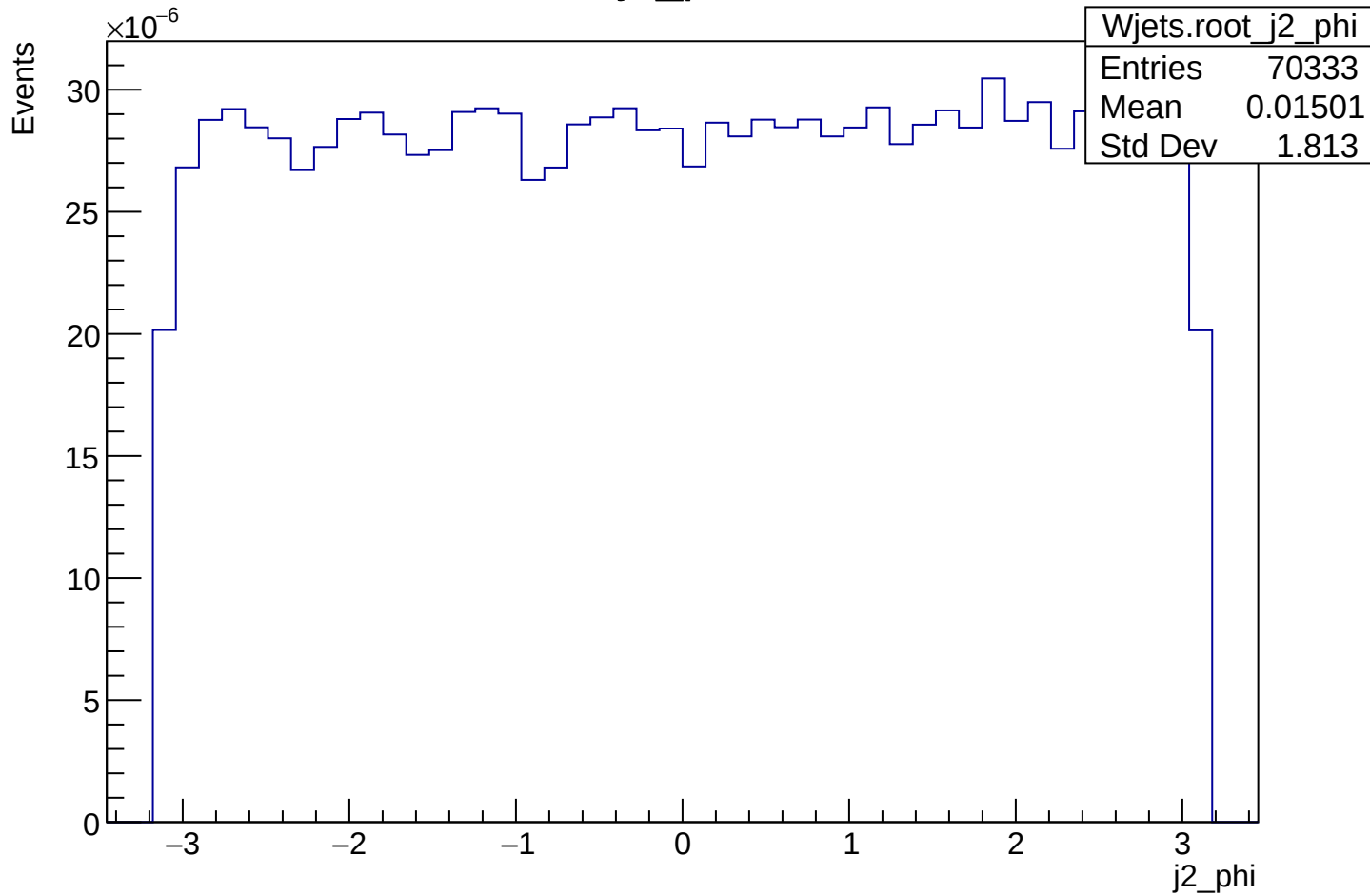
j2_e



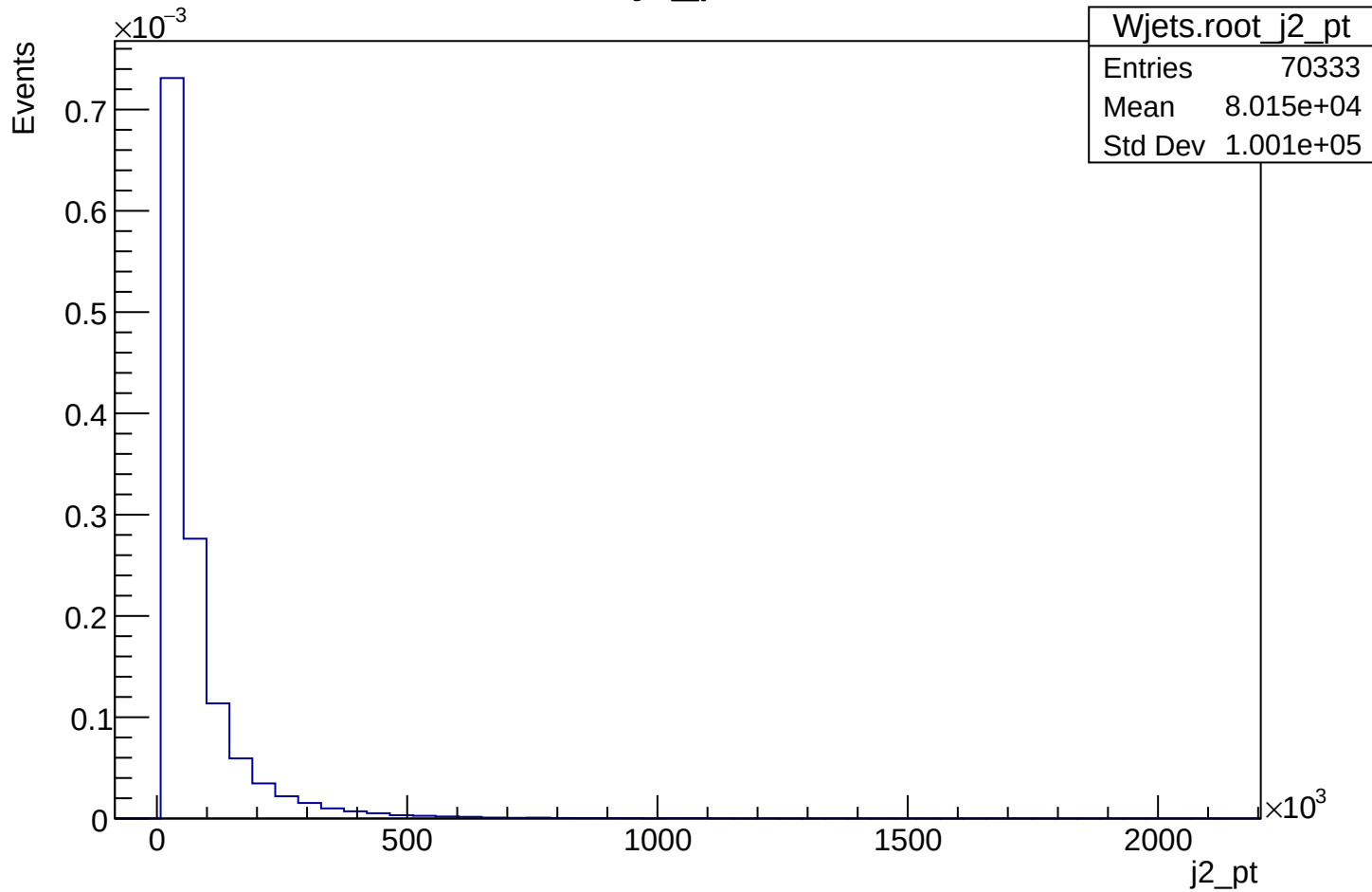
j2_eta



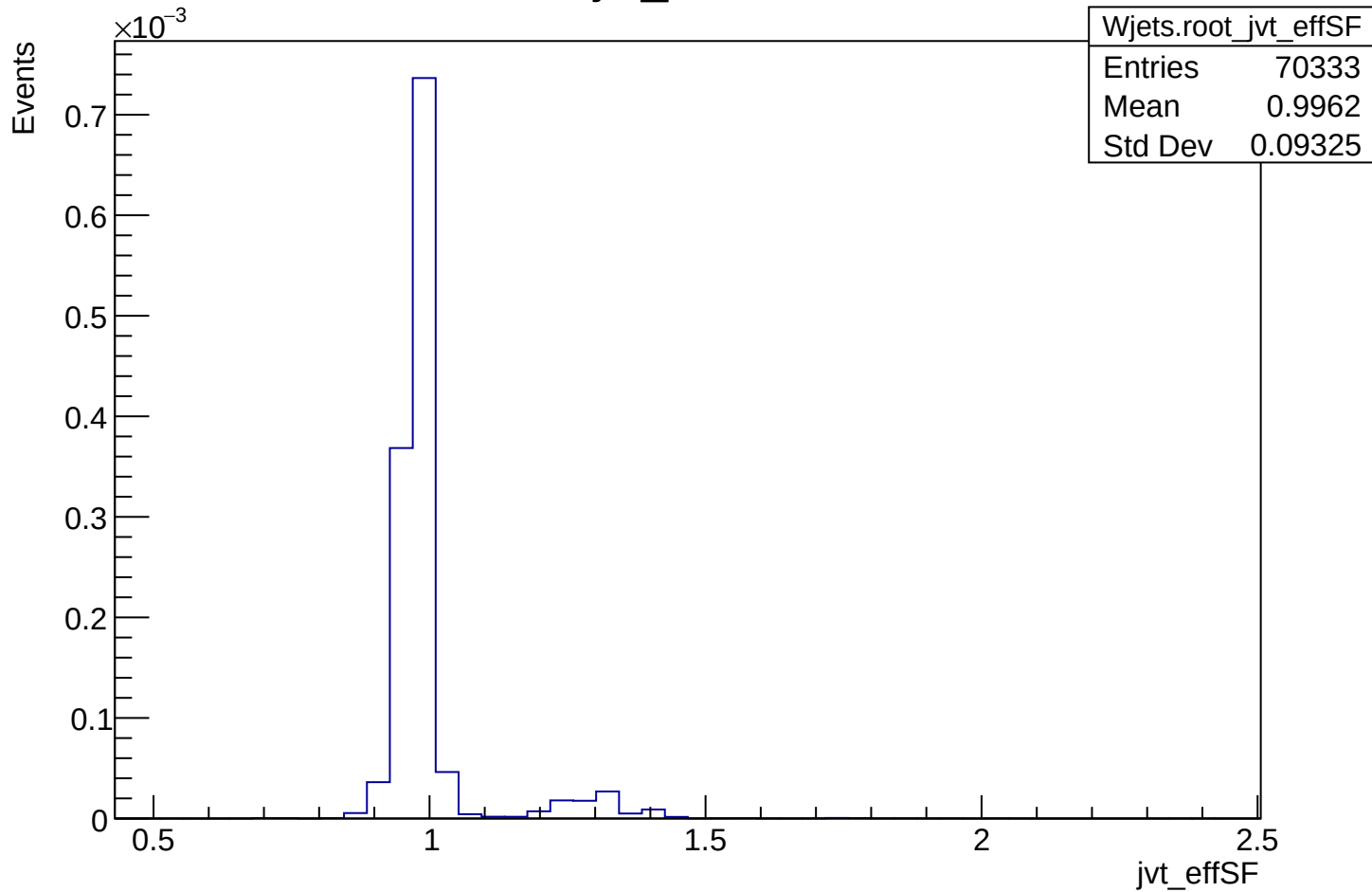
j2_phi



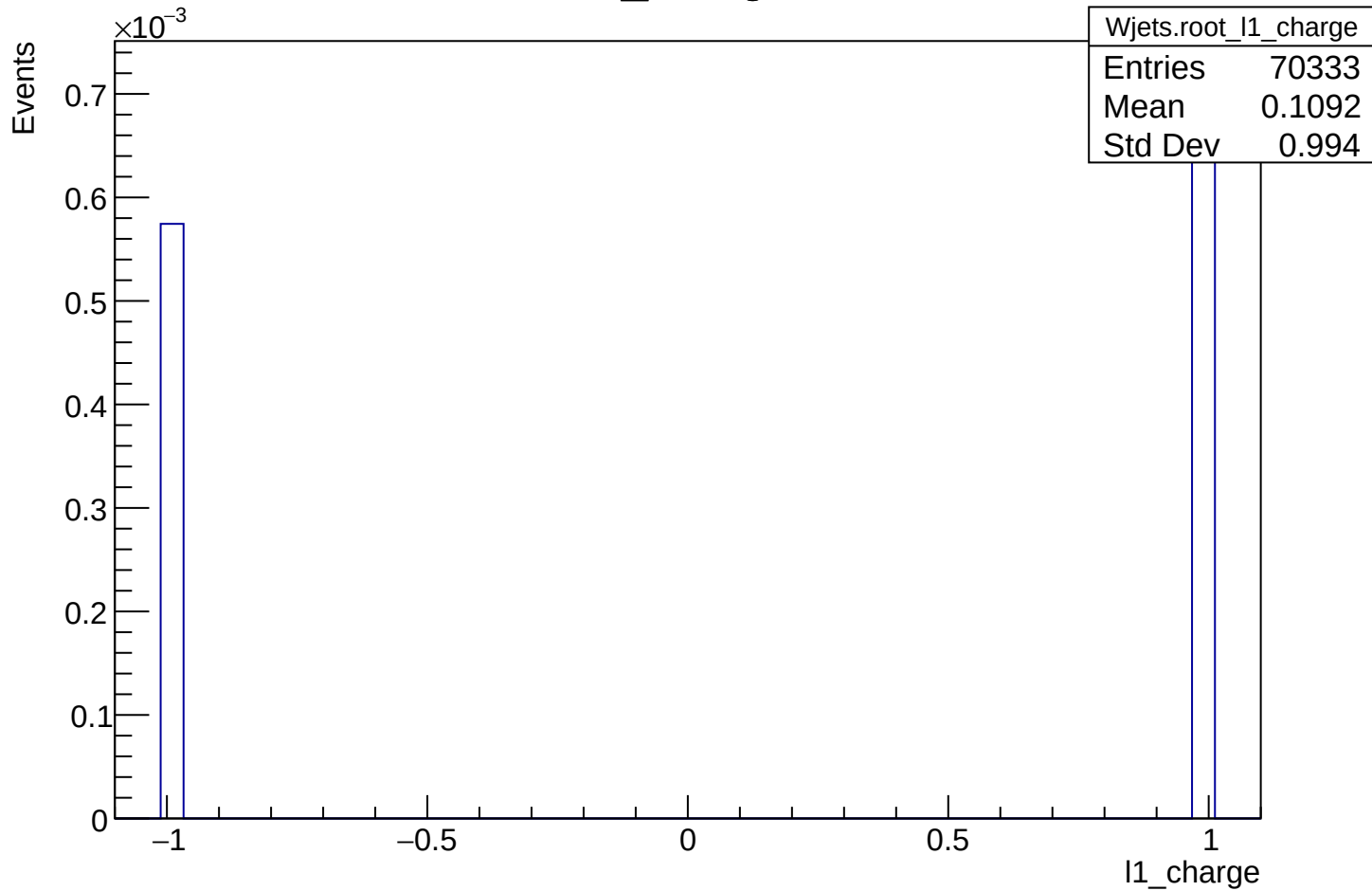
j2_pt



jvt_effSF

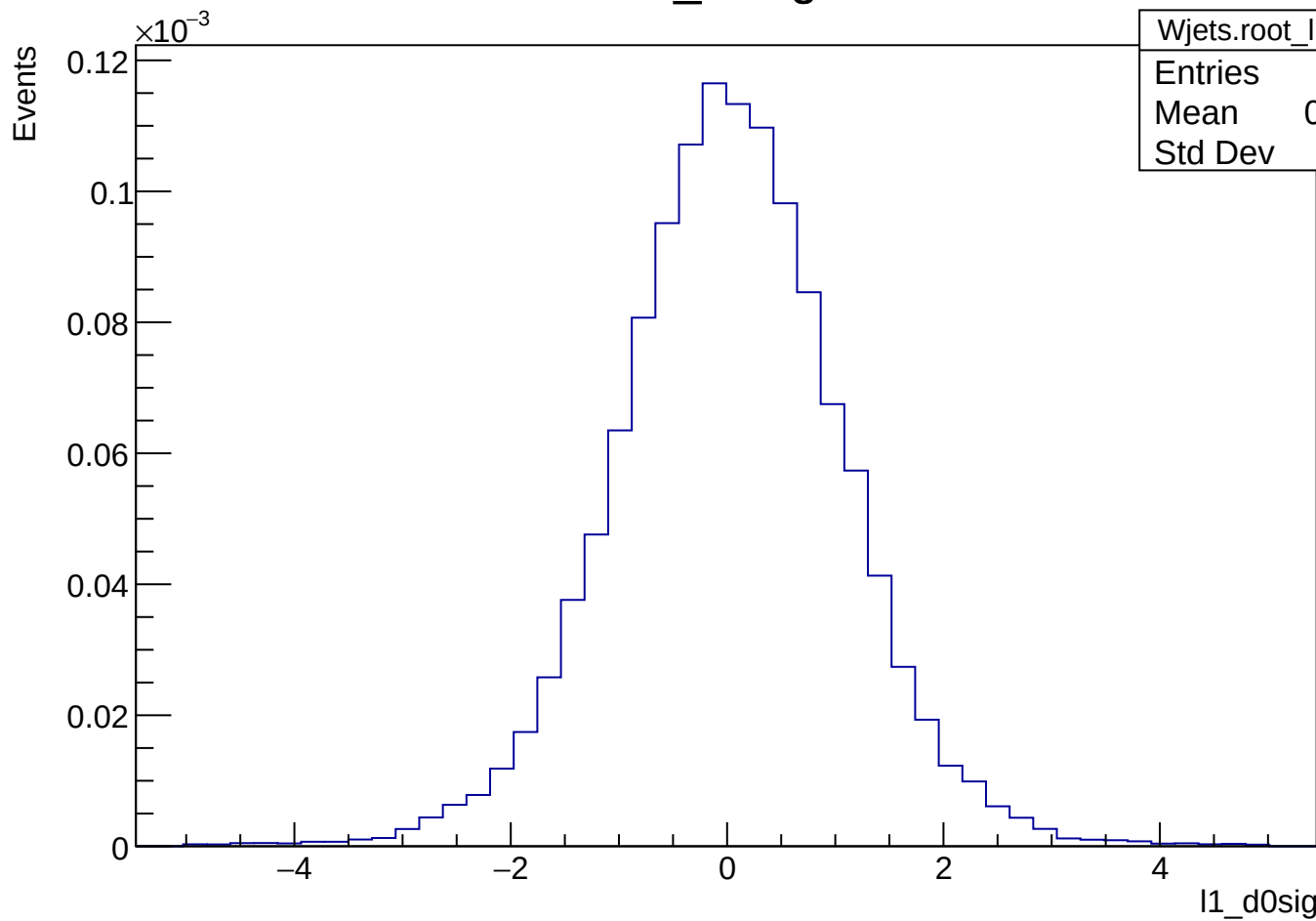


l1_charge

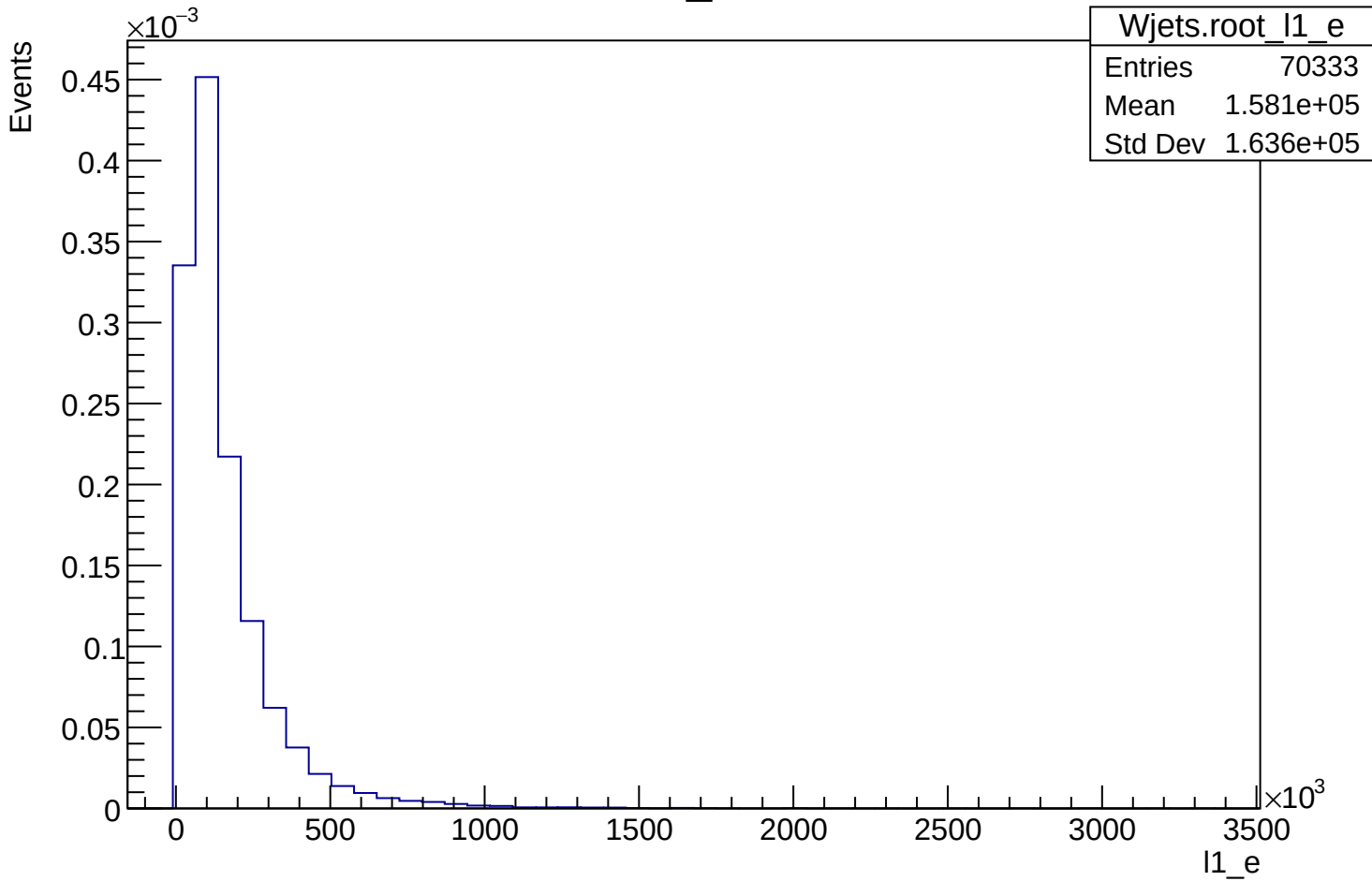


l1_d0sig

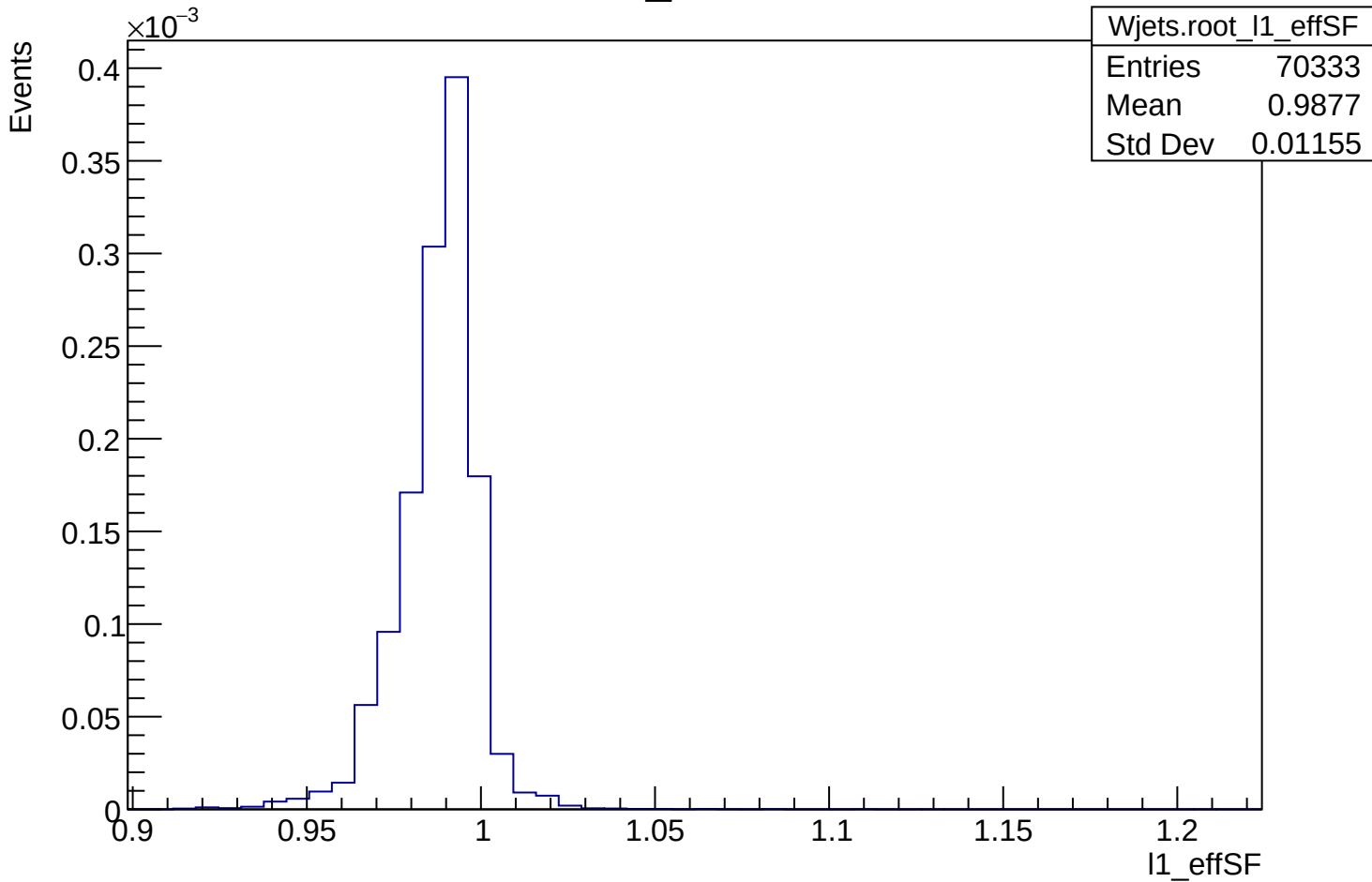
Wjets.root_l1_d0sig	
Entries	70333
Mean	0.02071
Std Dev	1.054



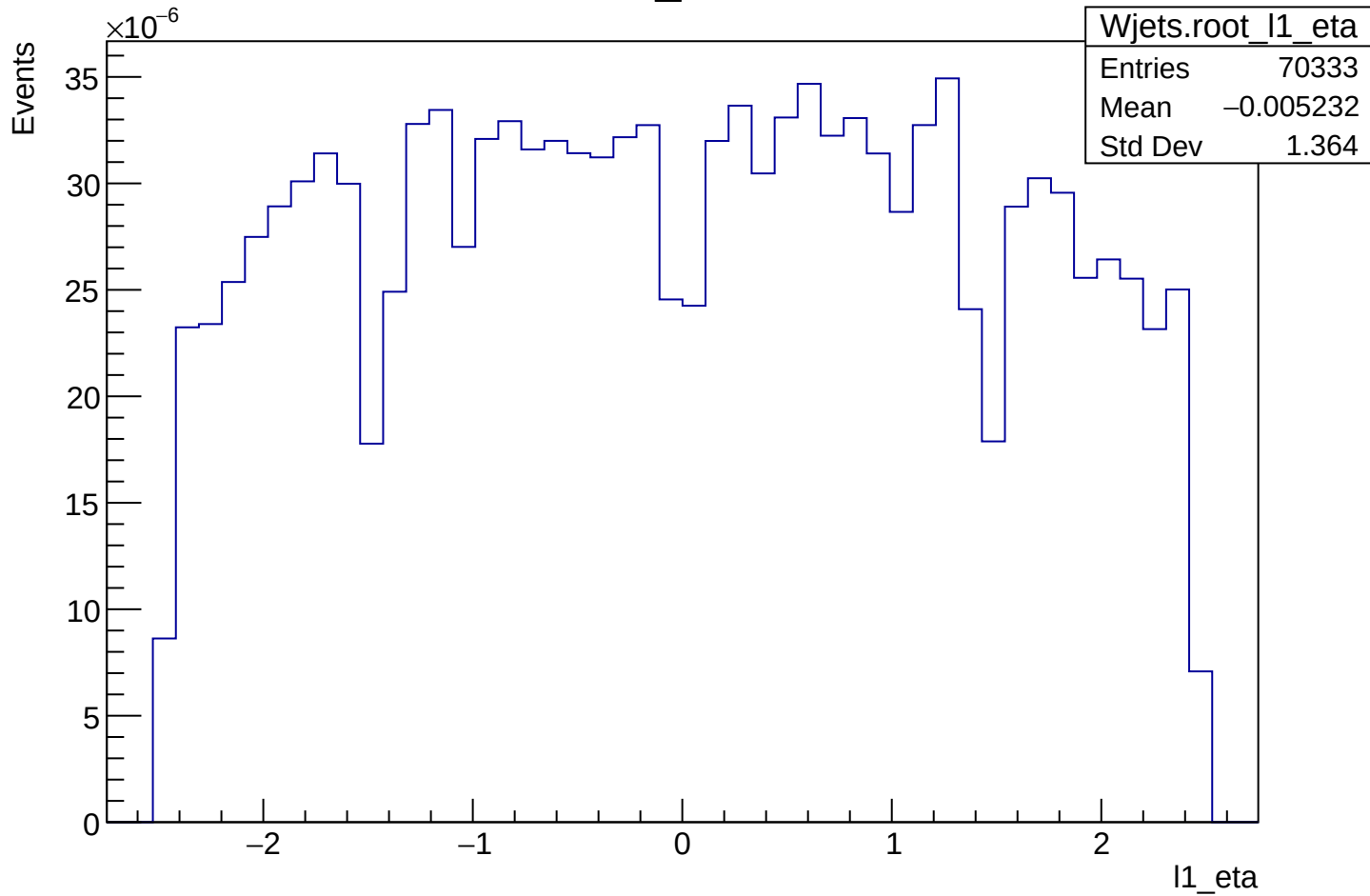
l1_e



l1_effSF

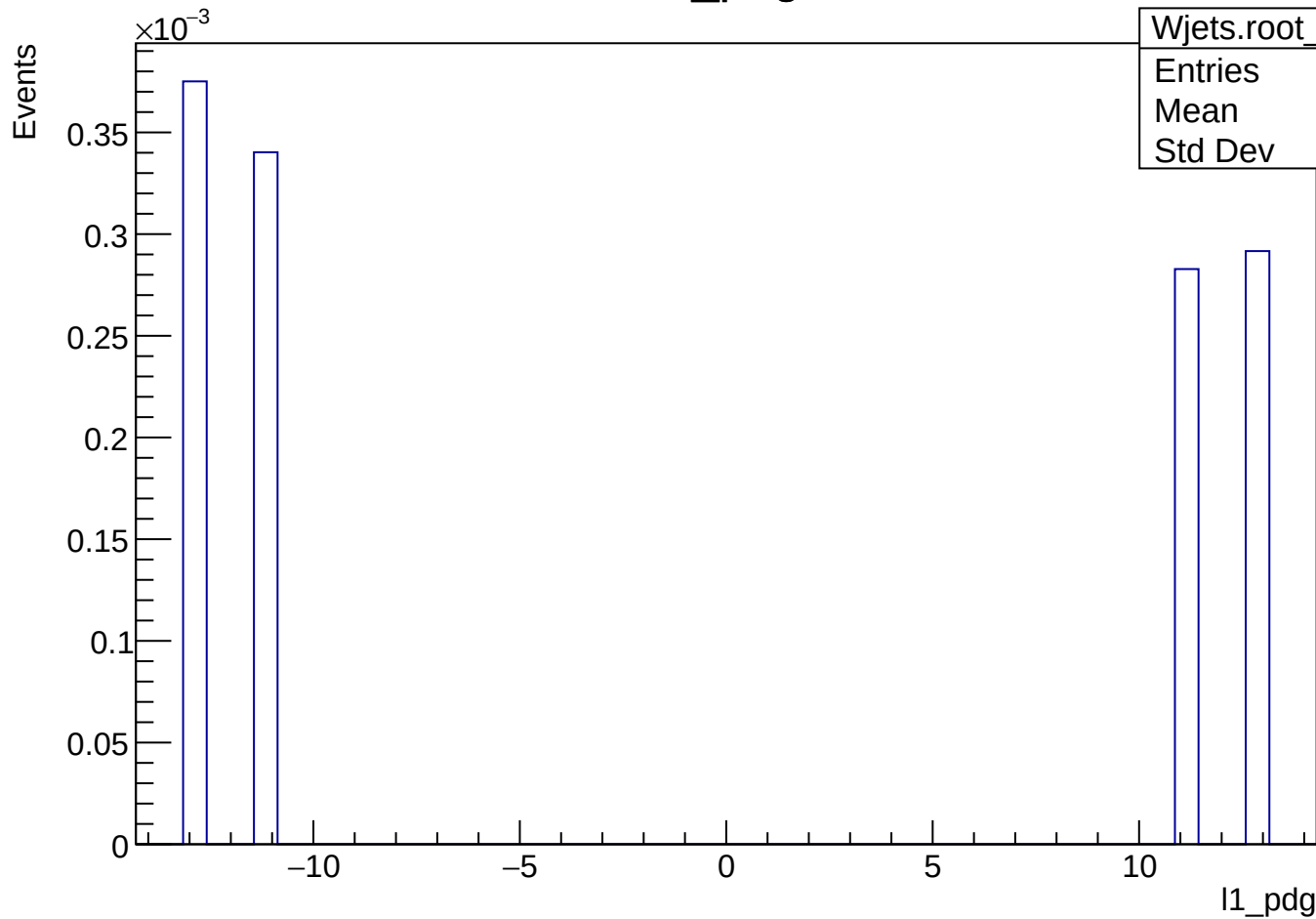


l1_eta

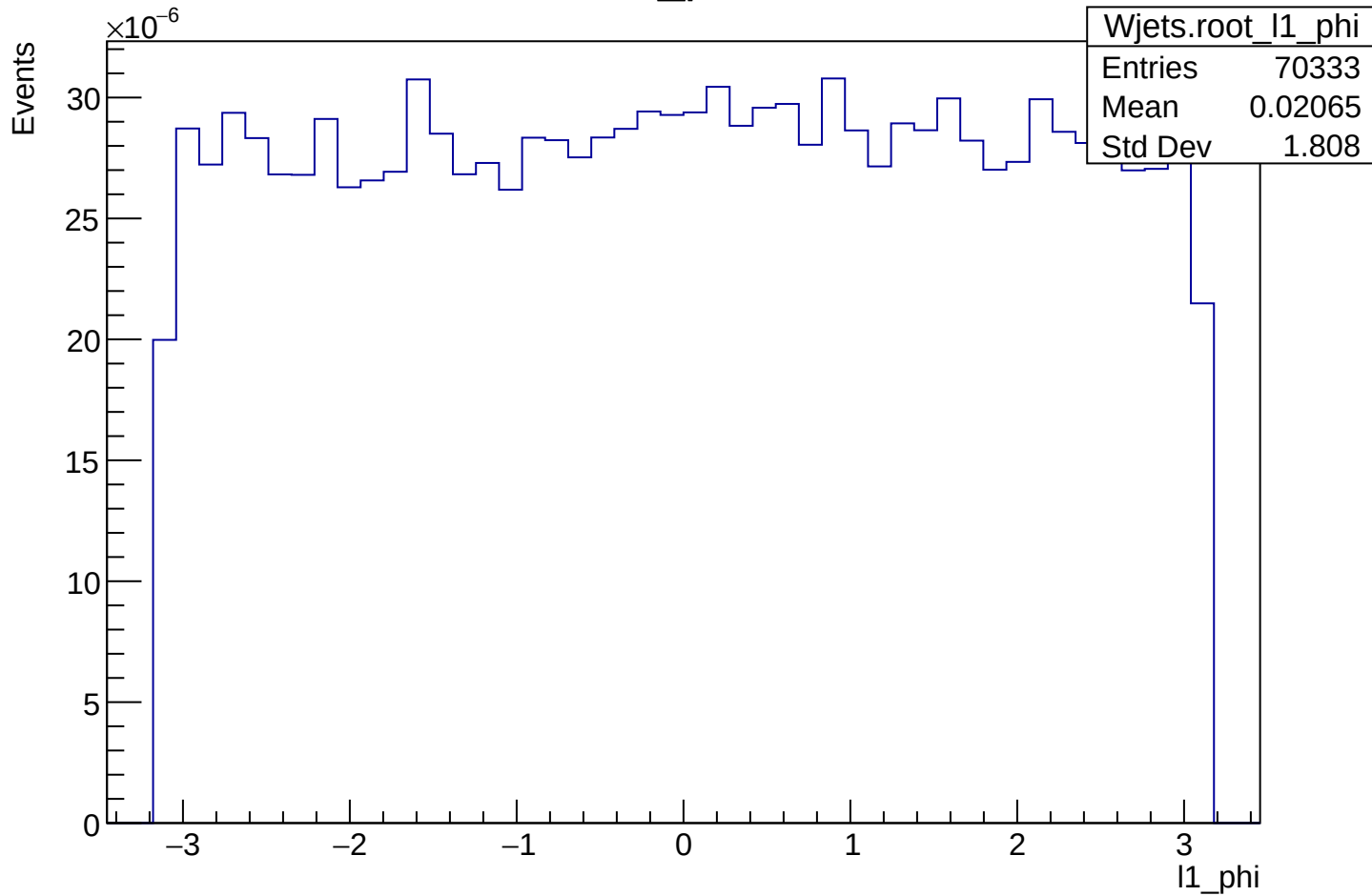


l1_pdg

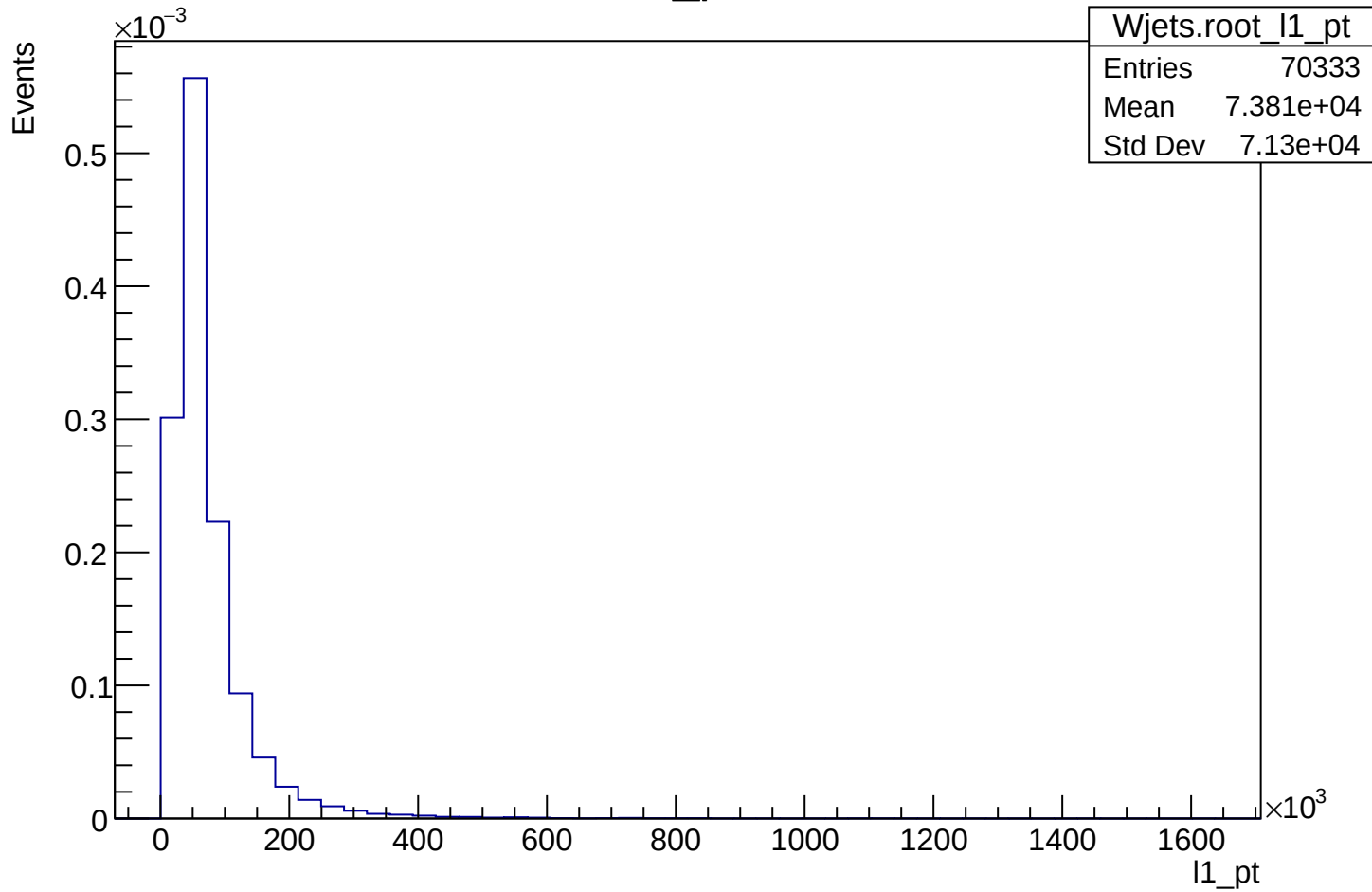
Wjets.root_l1_pdg	
Entries	70333
Mean	-1.331
Std Dev	12



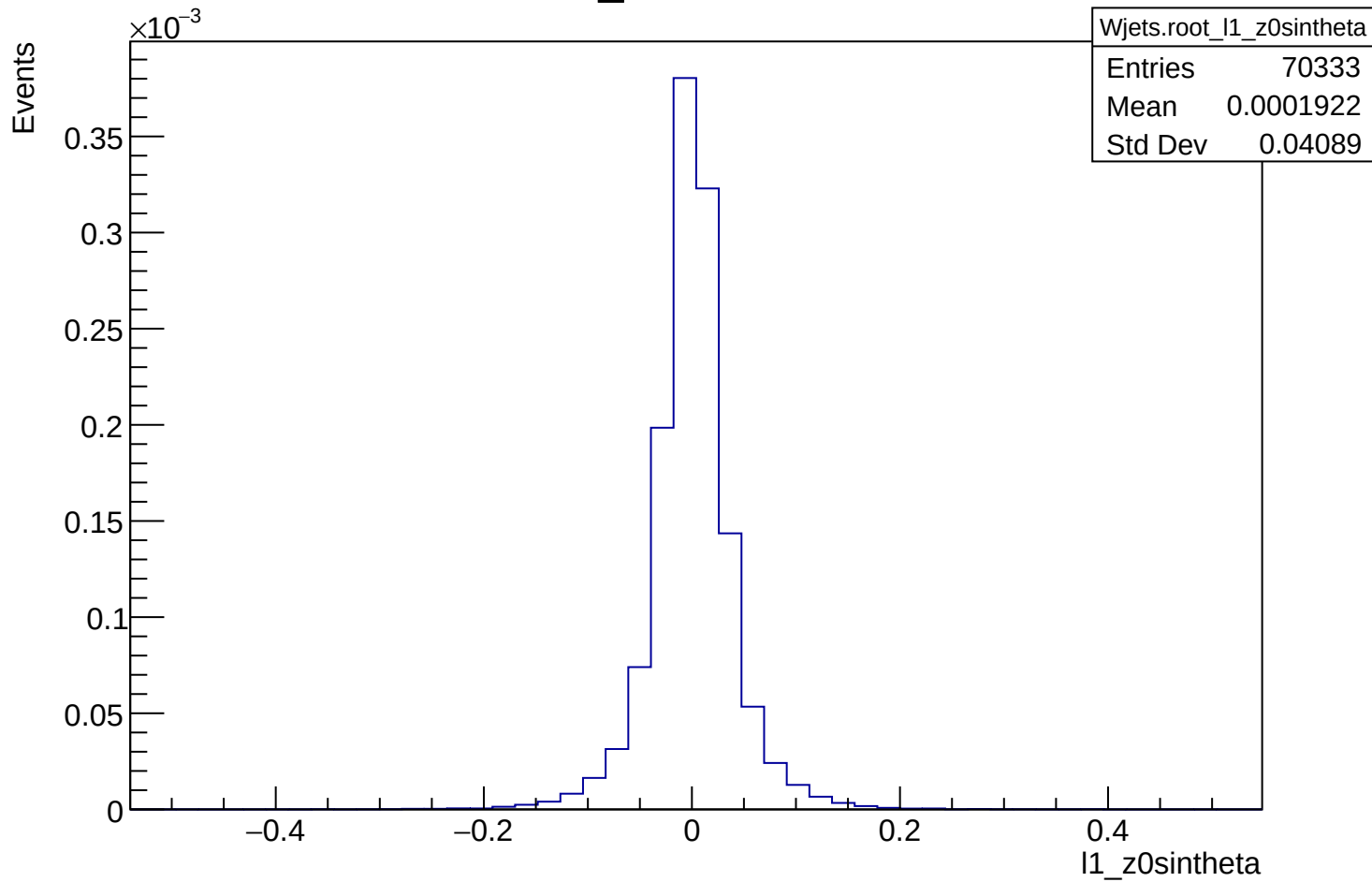
l1_phi



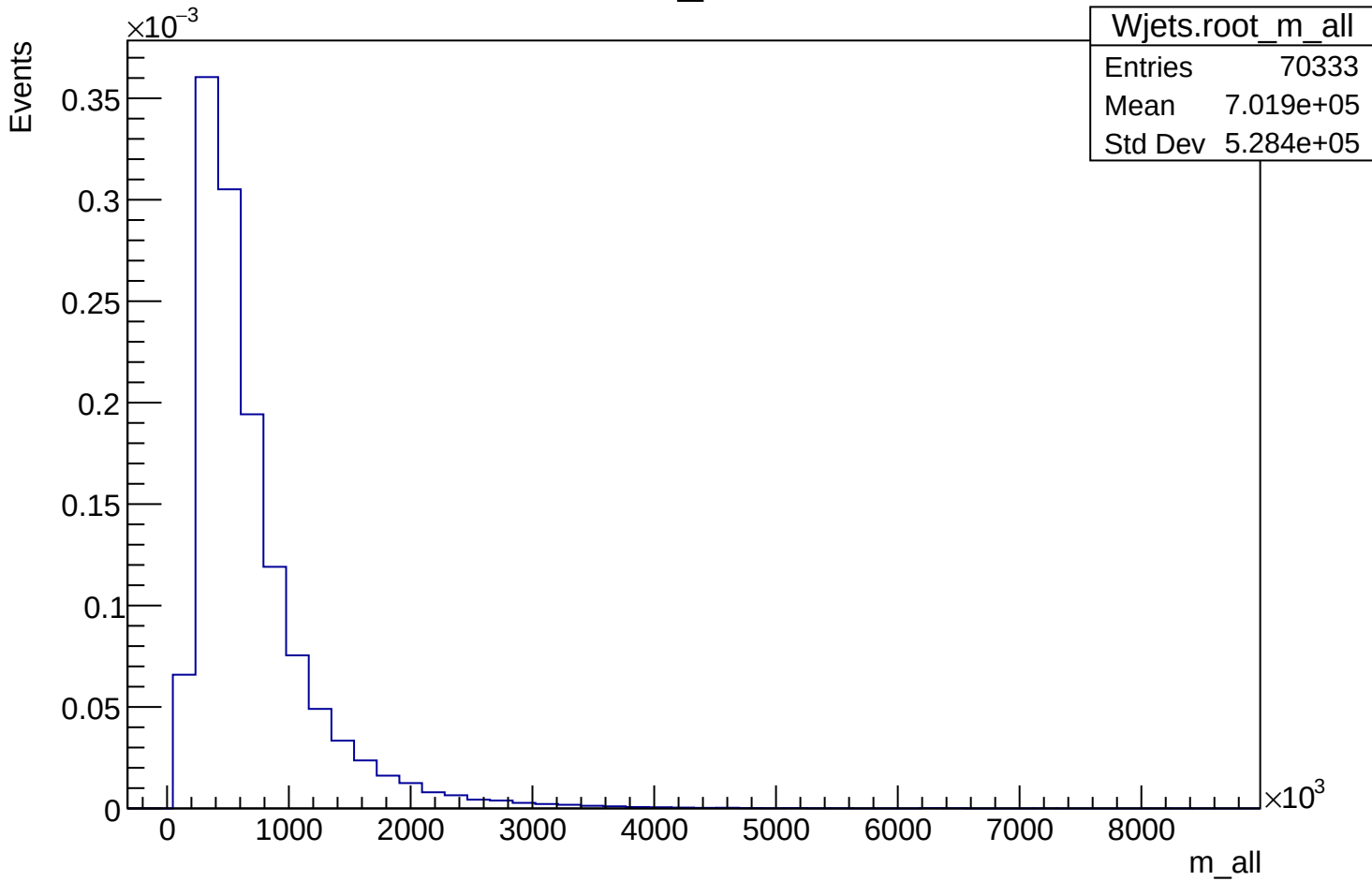
l1_pt



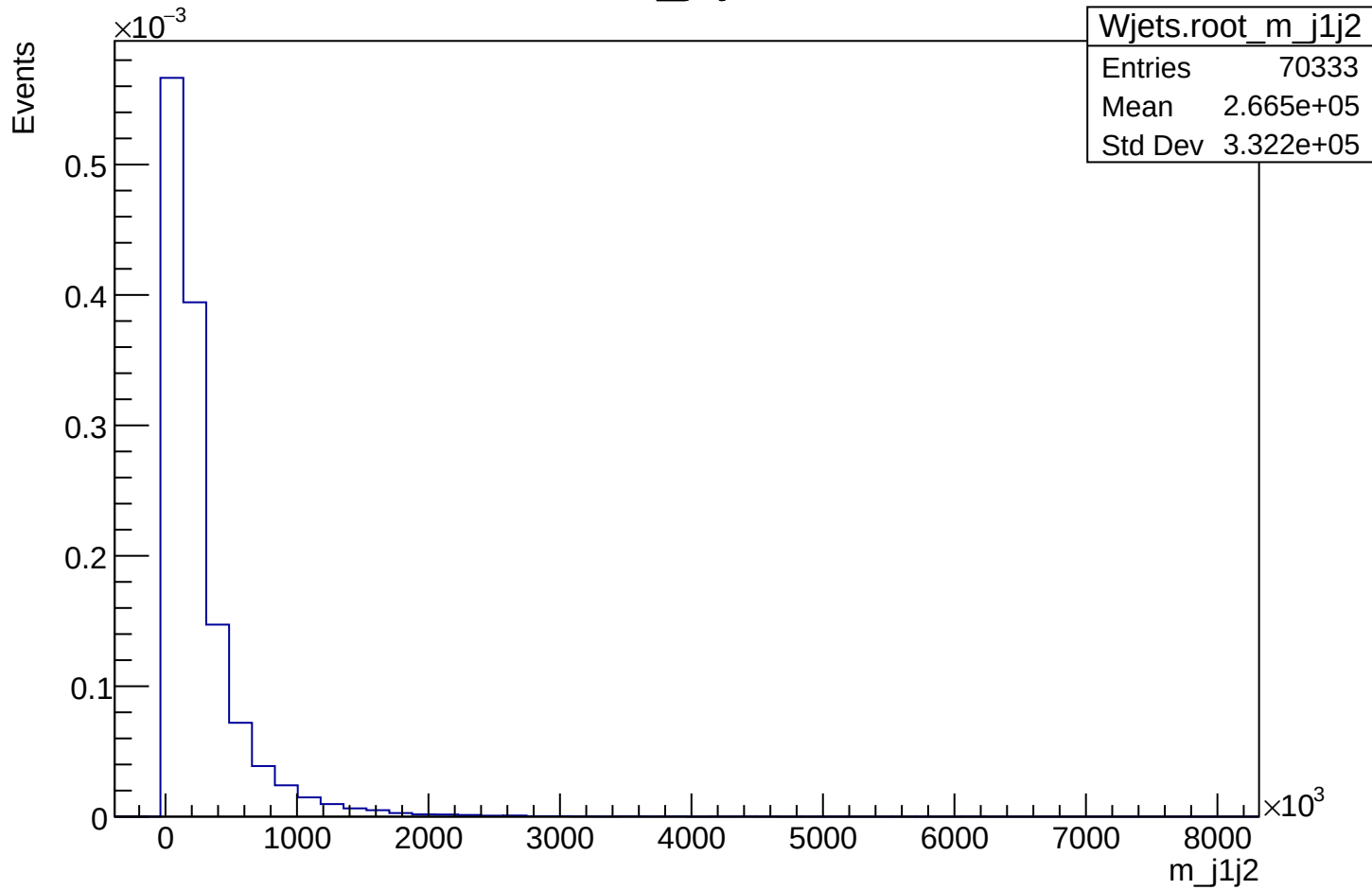
l1_z0sintheta



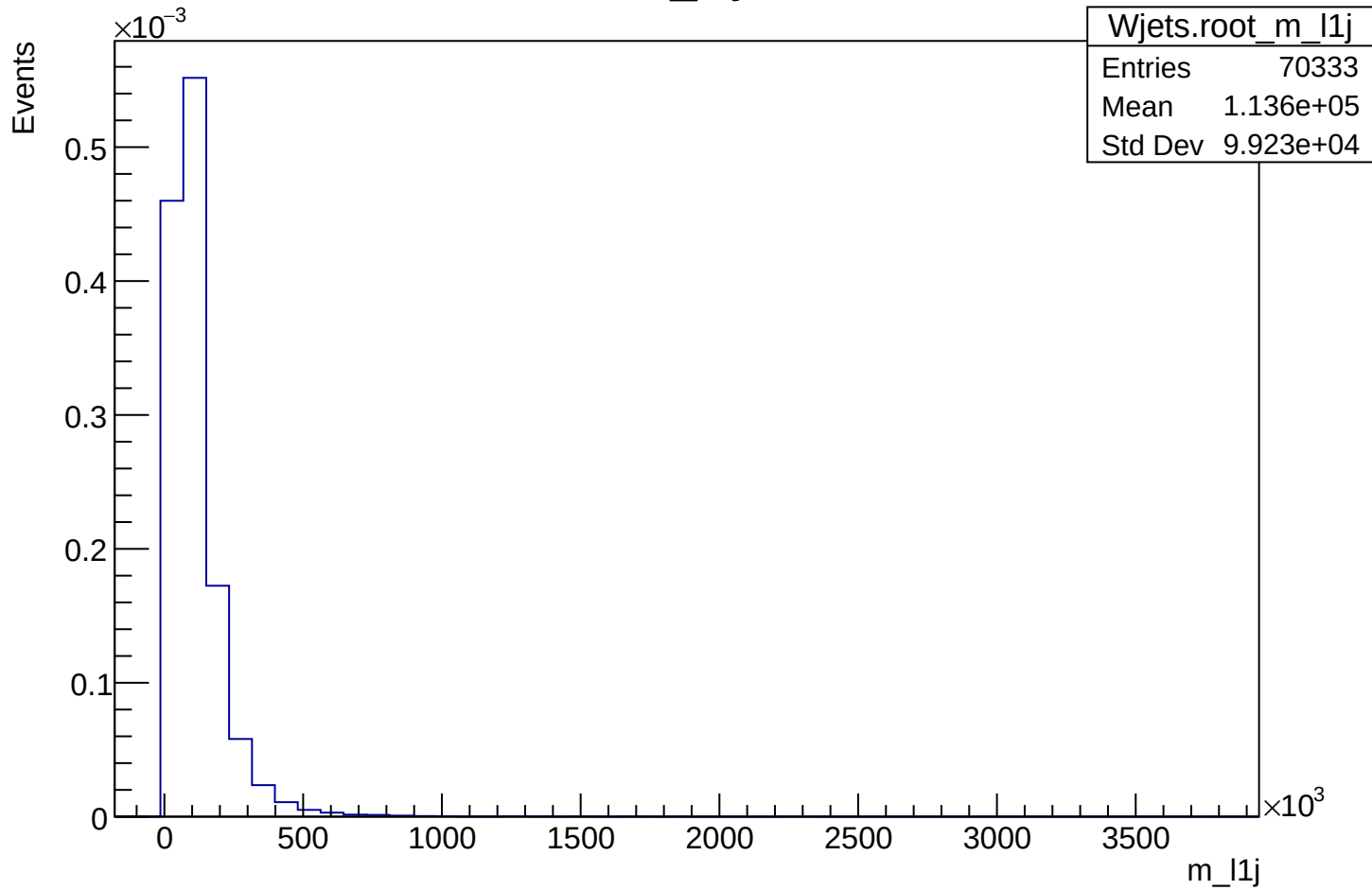
m_all



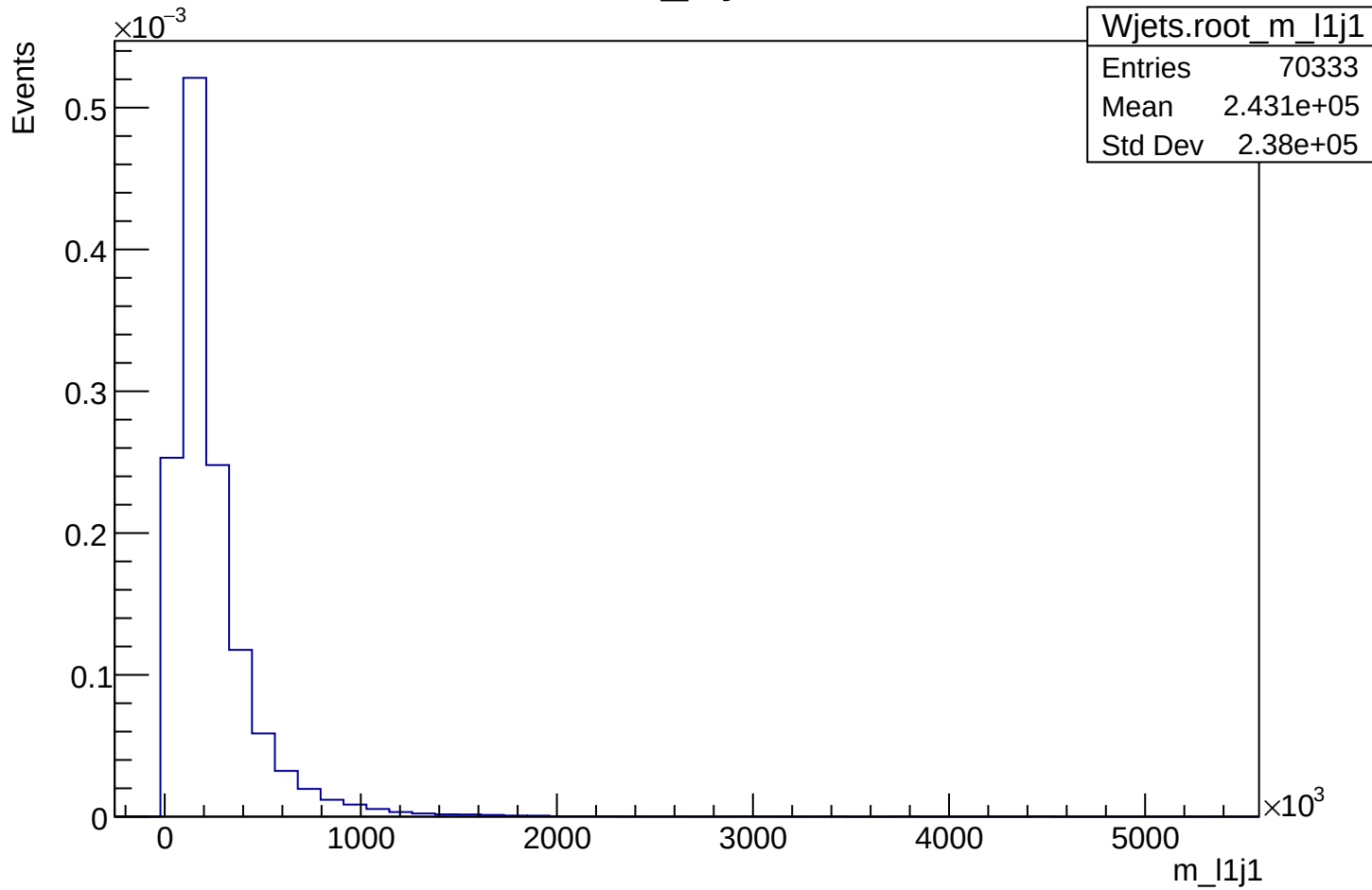
m_j1j2



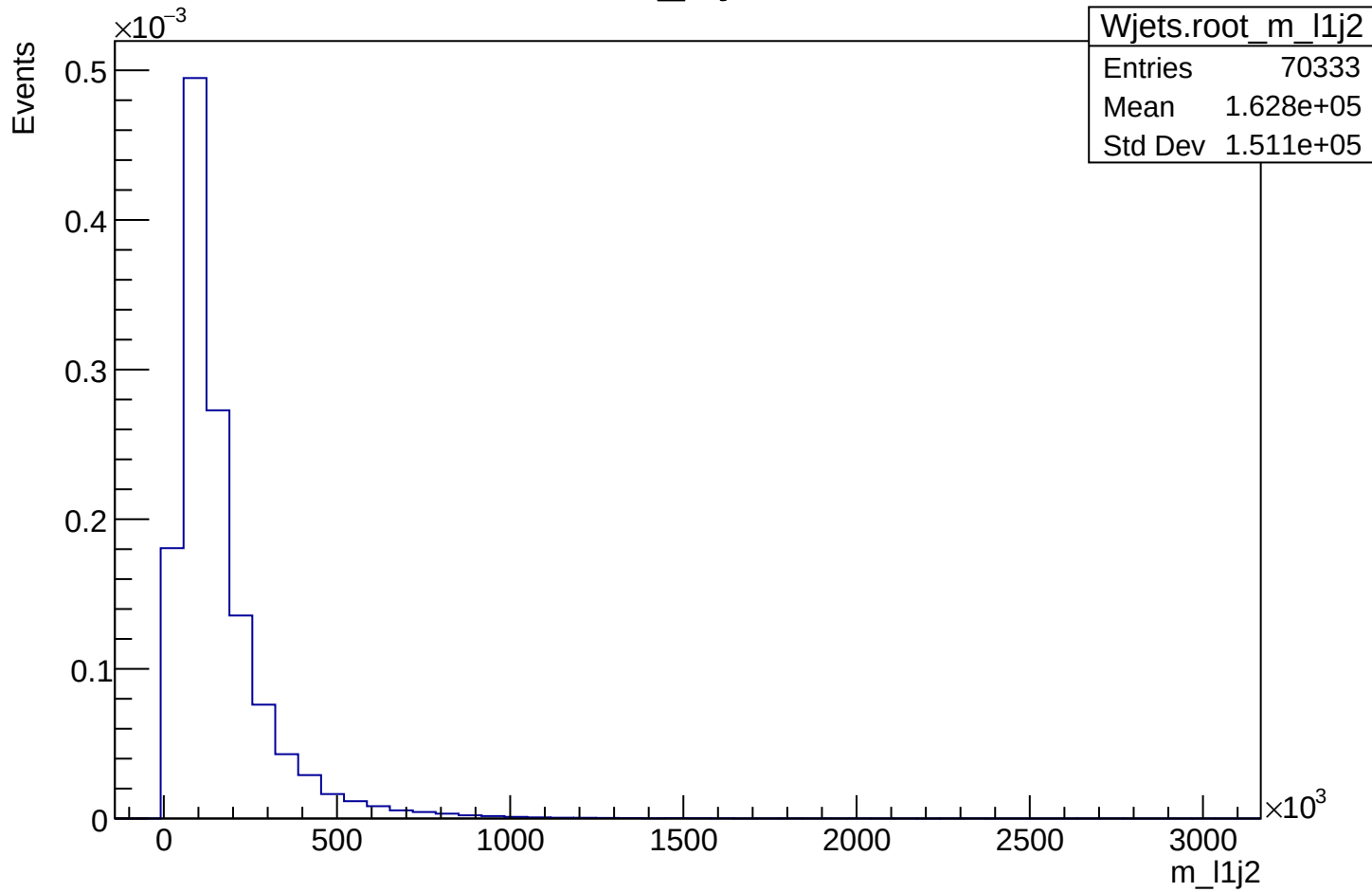
m_l1j



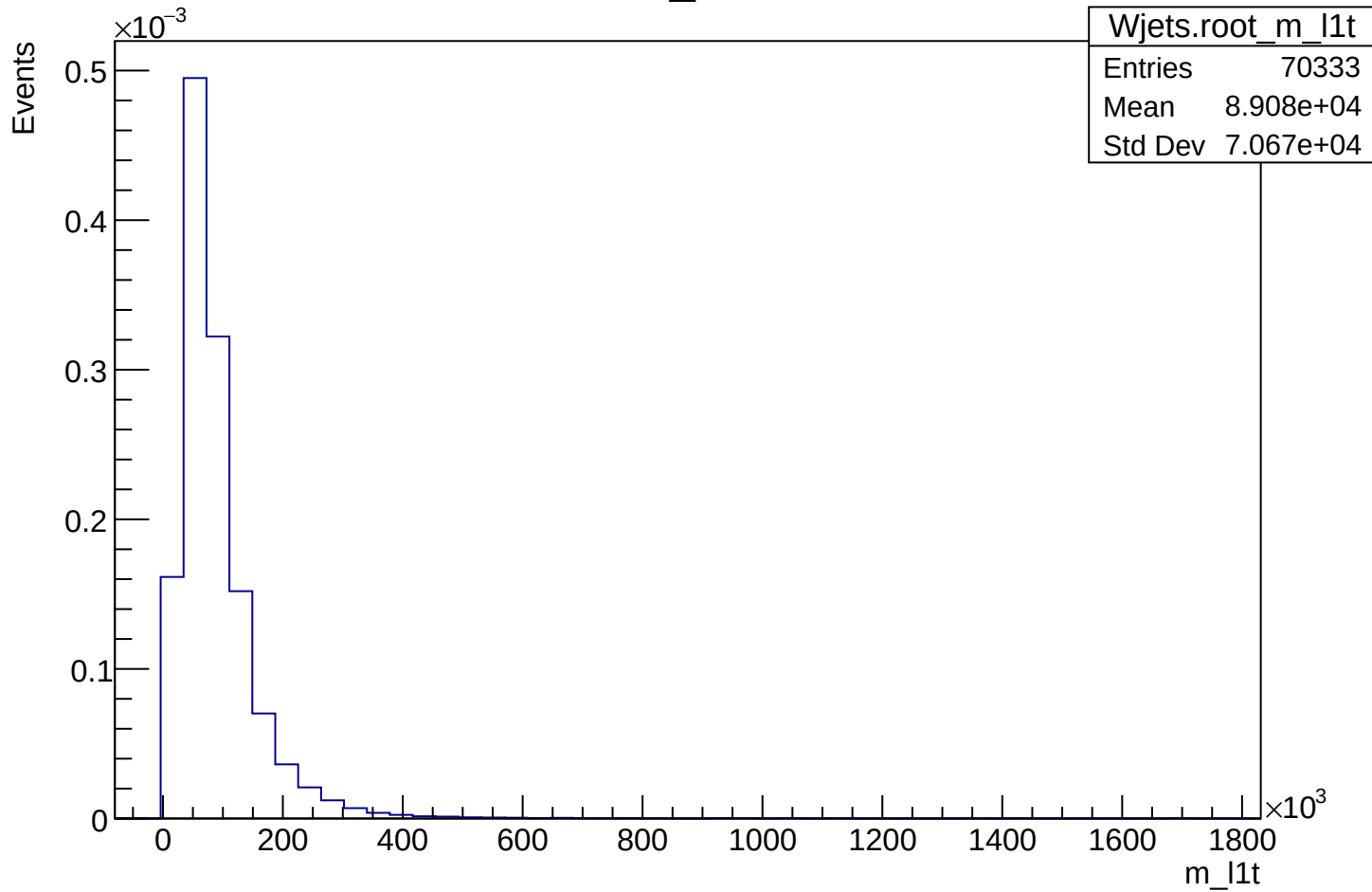
m_l1j1



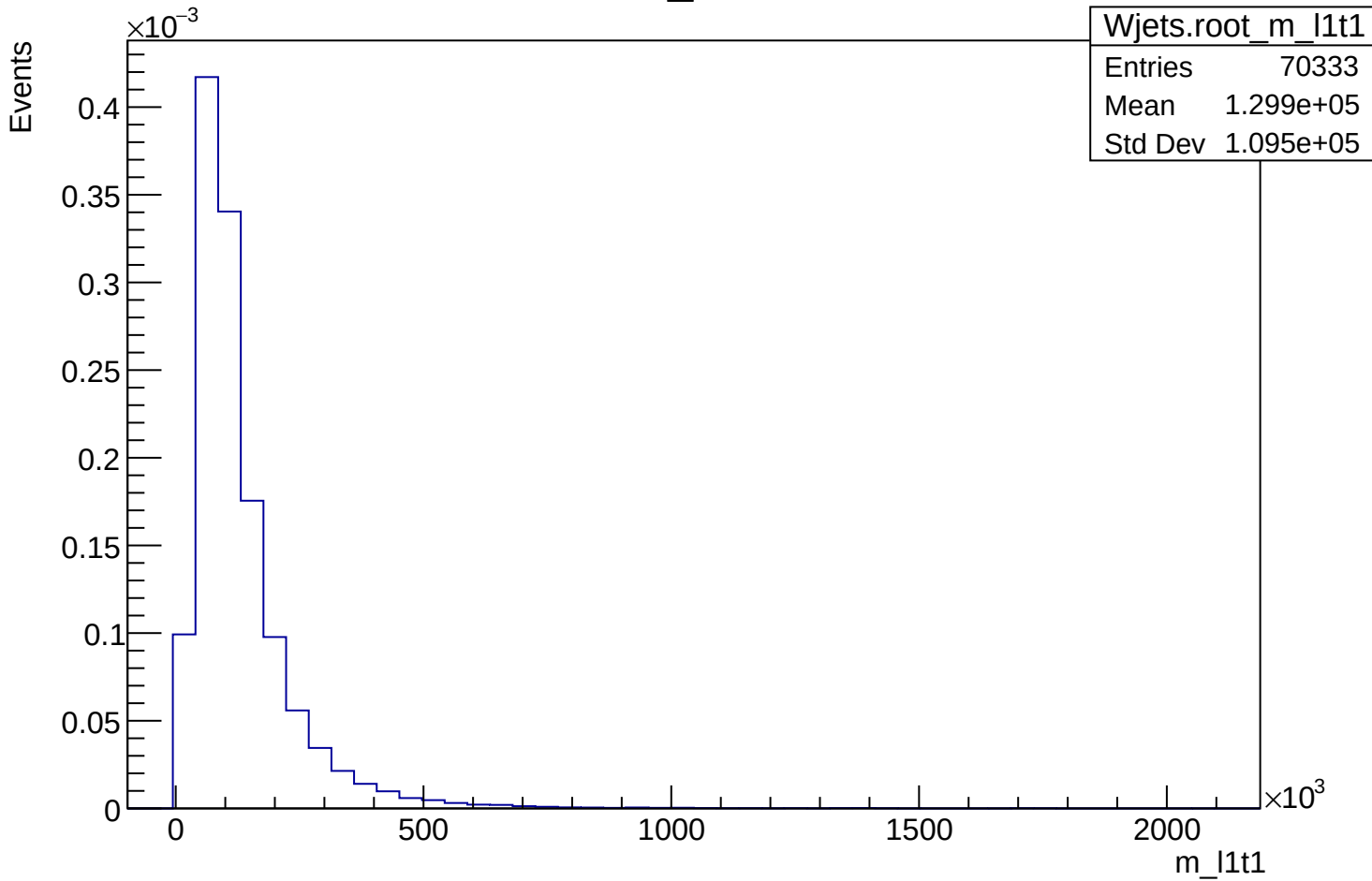
m_l1j2



m_l1t

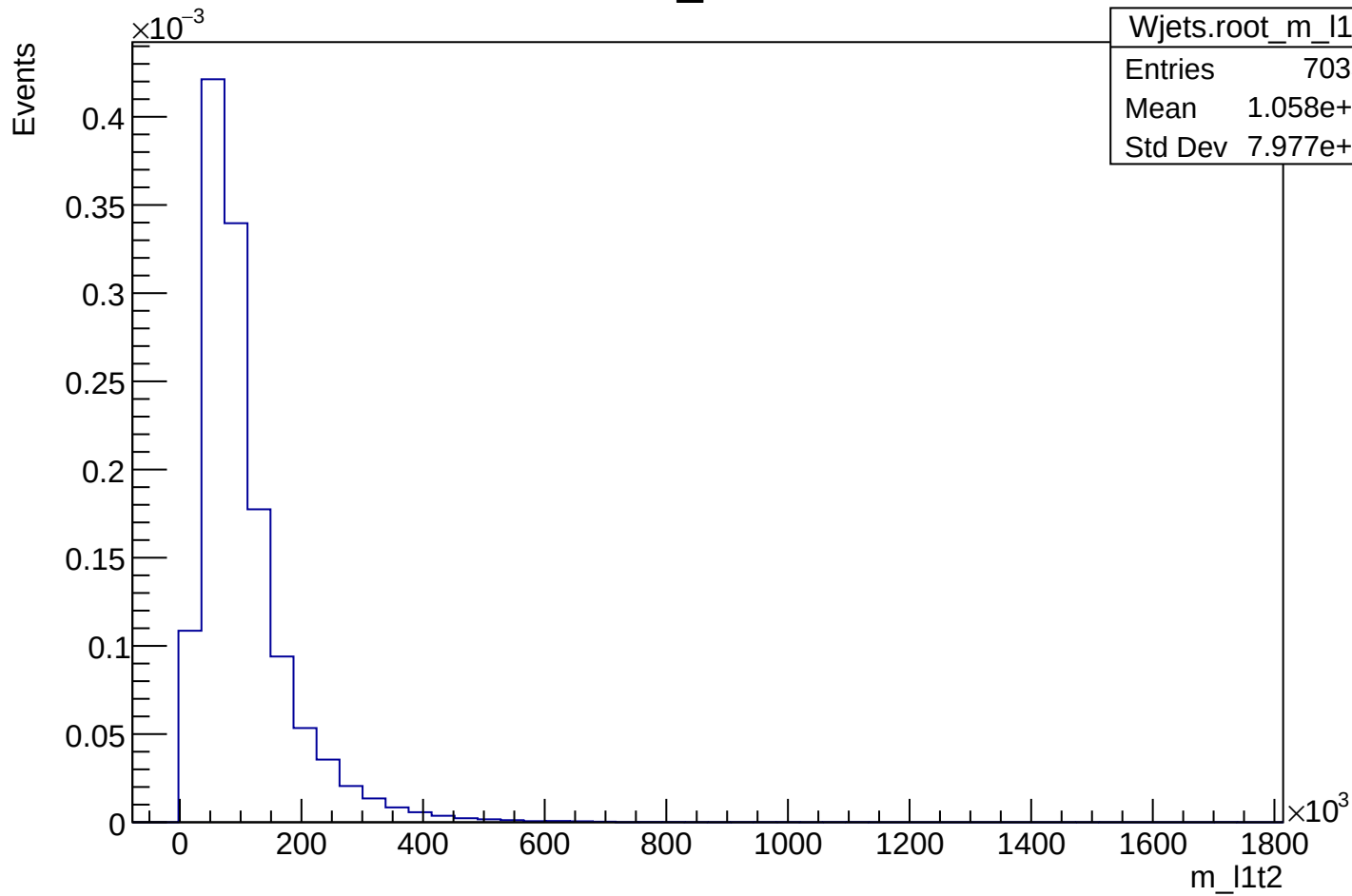


m_l1t1

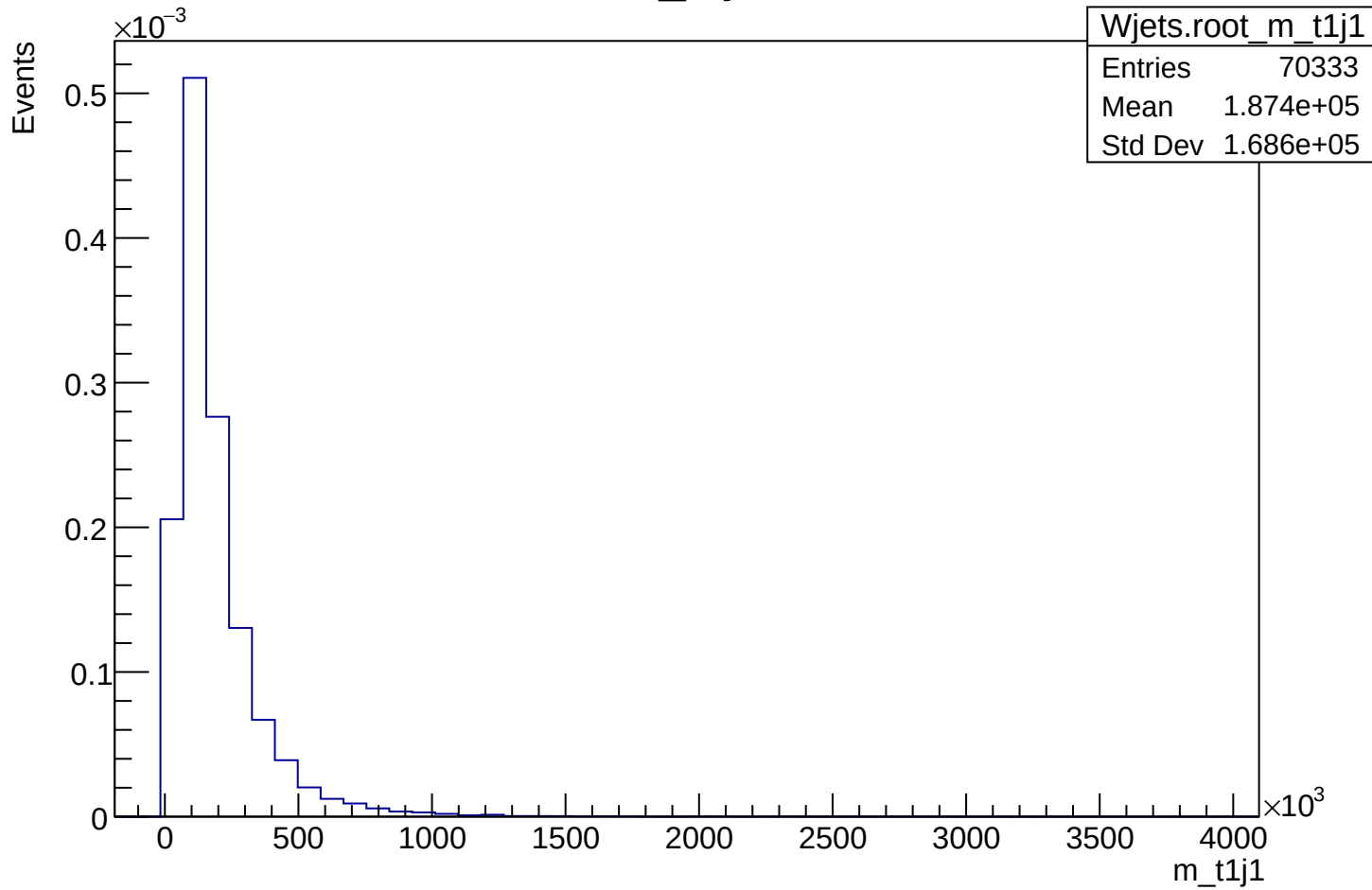


m_l1t2

Wjets.root_m_l1t2	
Entries	70333
Mean	1.058e+05
Std Dev	7.977e+04

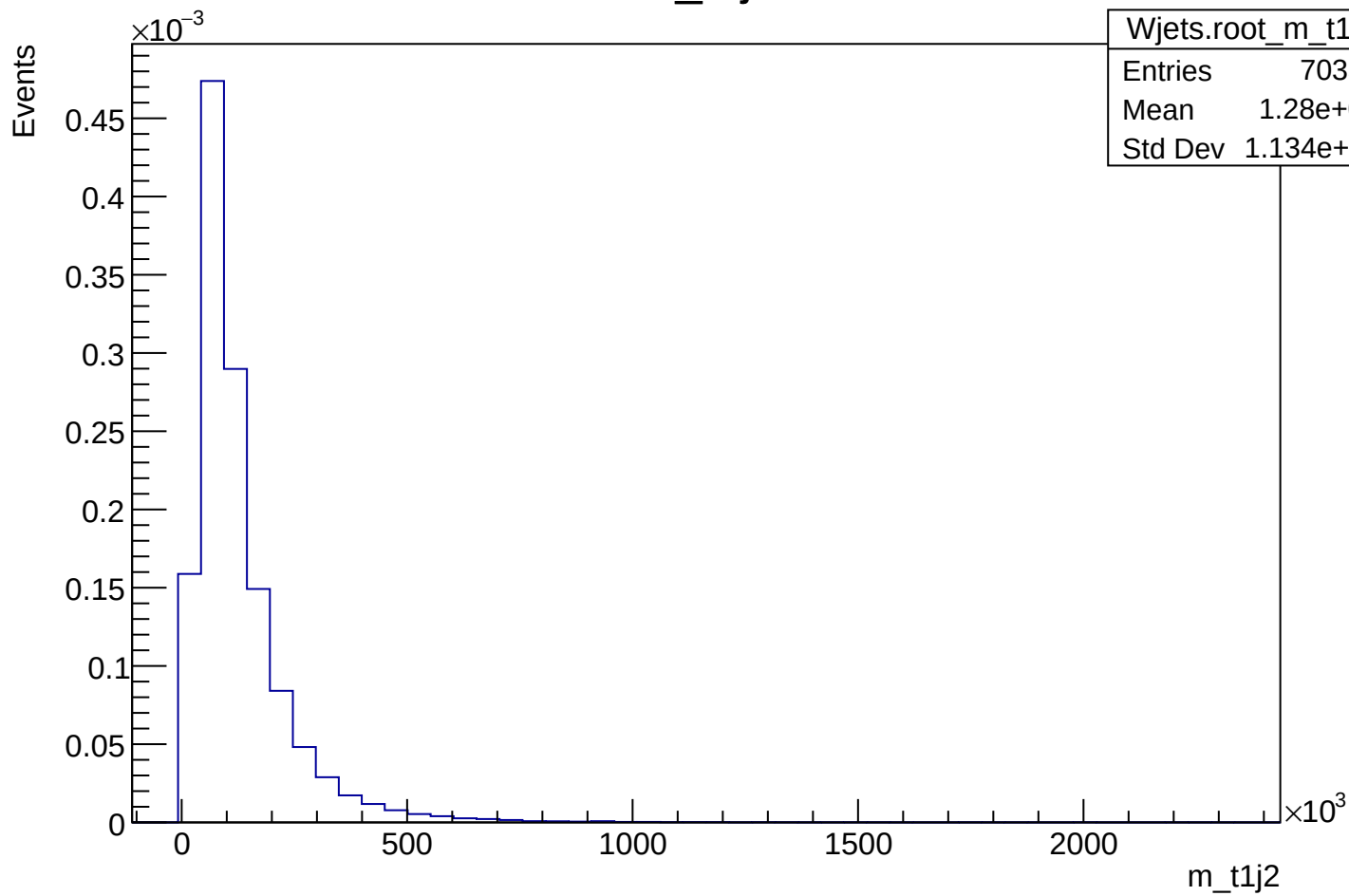


m_t1j1

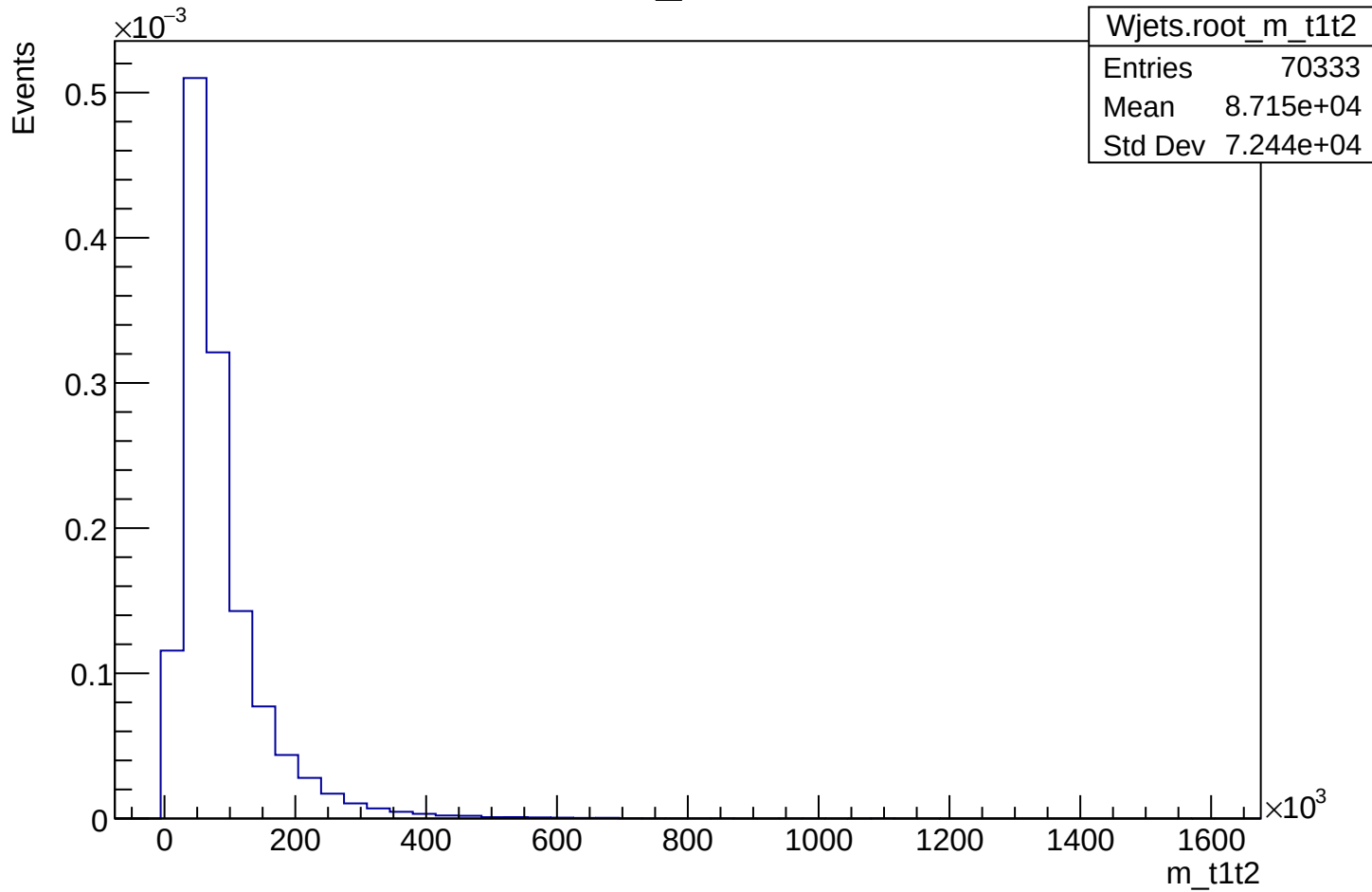


m_t1j2

Wjets.root_m_t1j2	
Entries	70333
Mean	1.28e+05
Std Dev	1.134e+05



m_t1t2



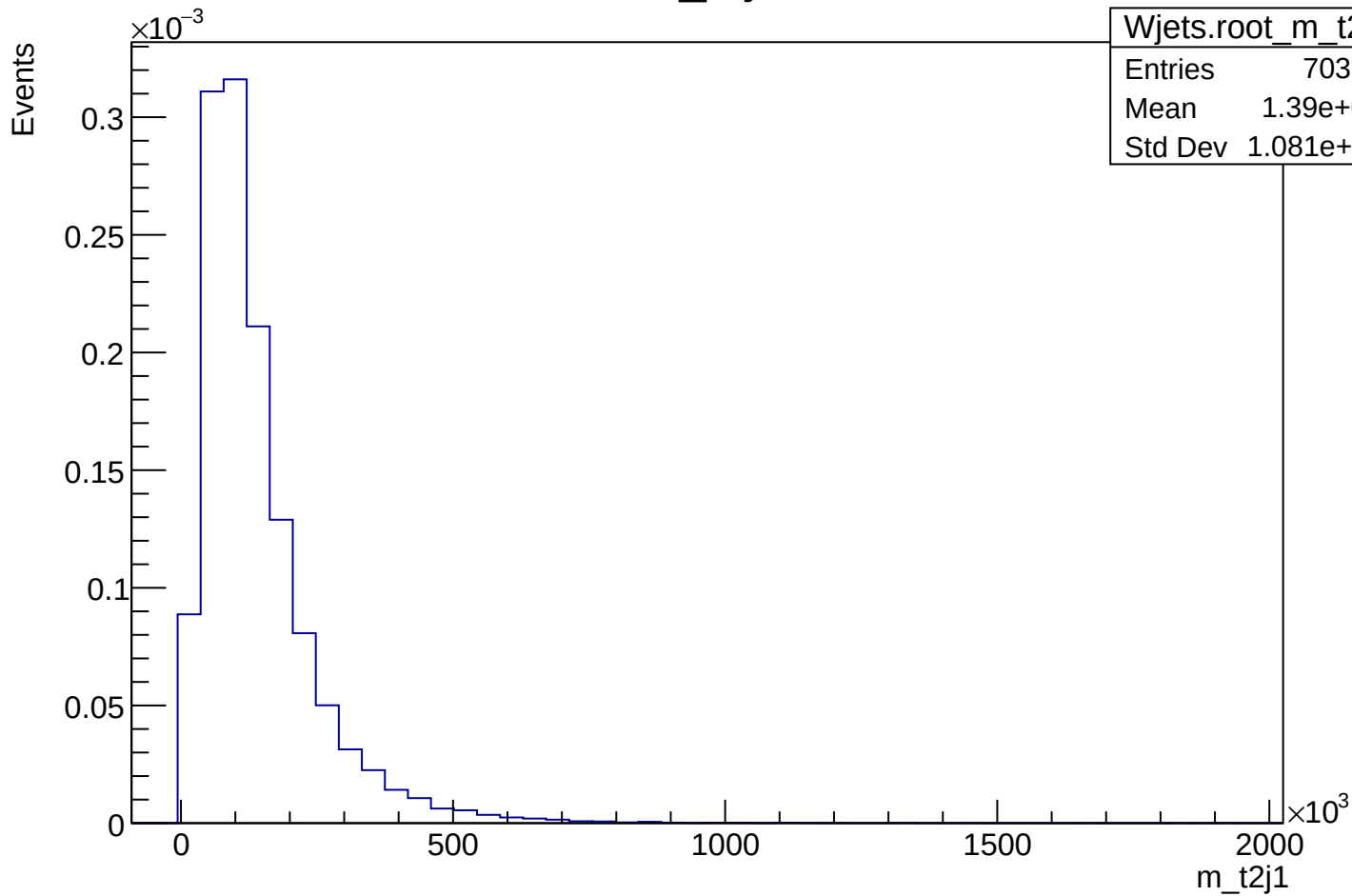
m_t2j1

Wjets.root_m_t2j1

Entries 70333

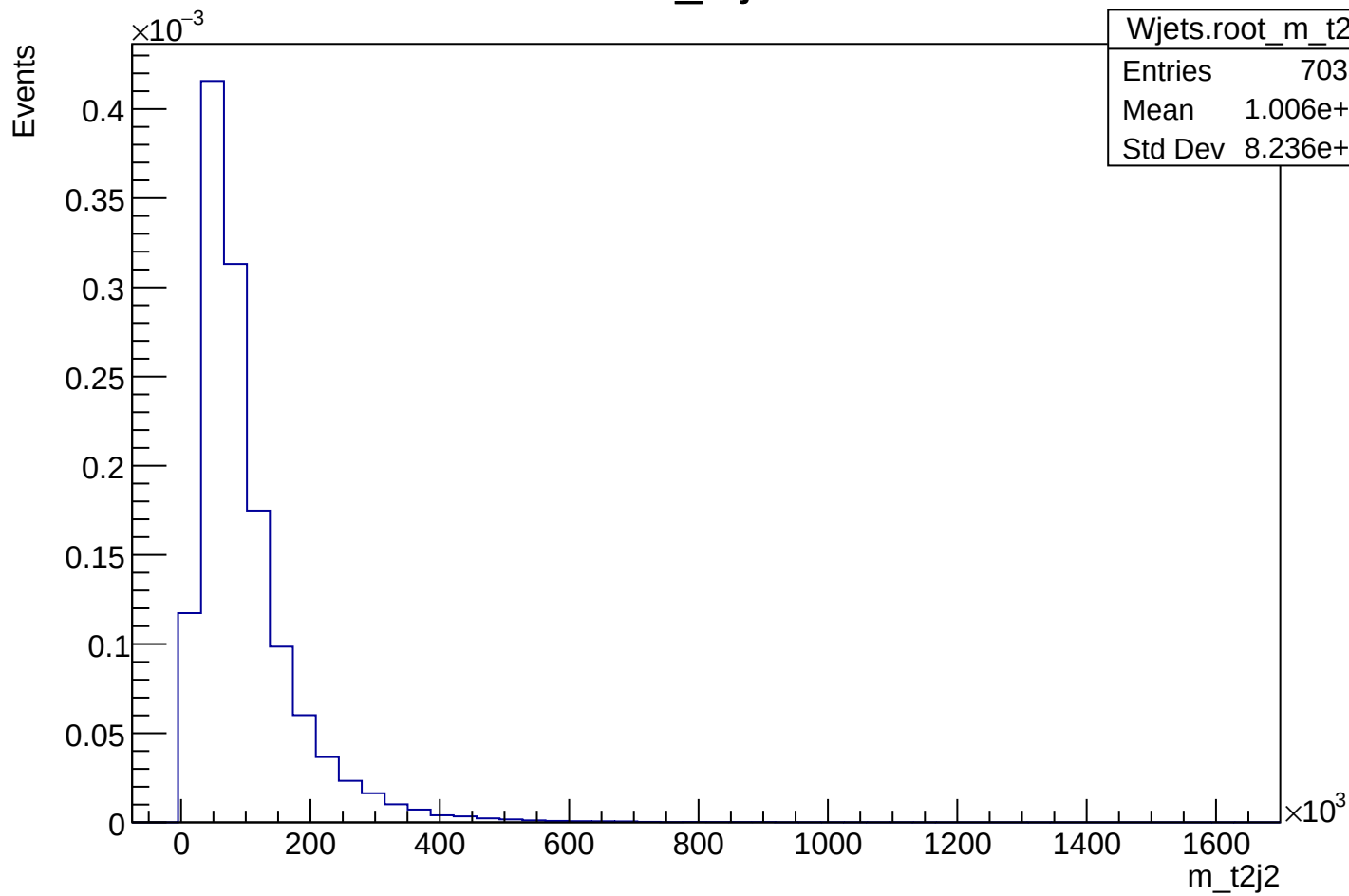
Mean 1.39e+05

Std Dev 1.081e+05



m_t2j2

Wjets.root_m_t2j2	
Entries	70333
Mean	1.006e+05
Std Dev	8.236e+04



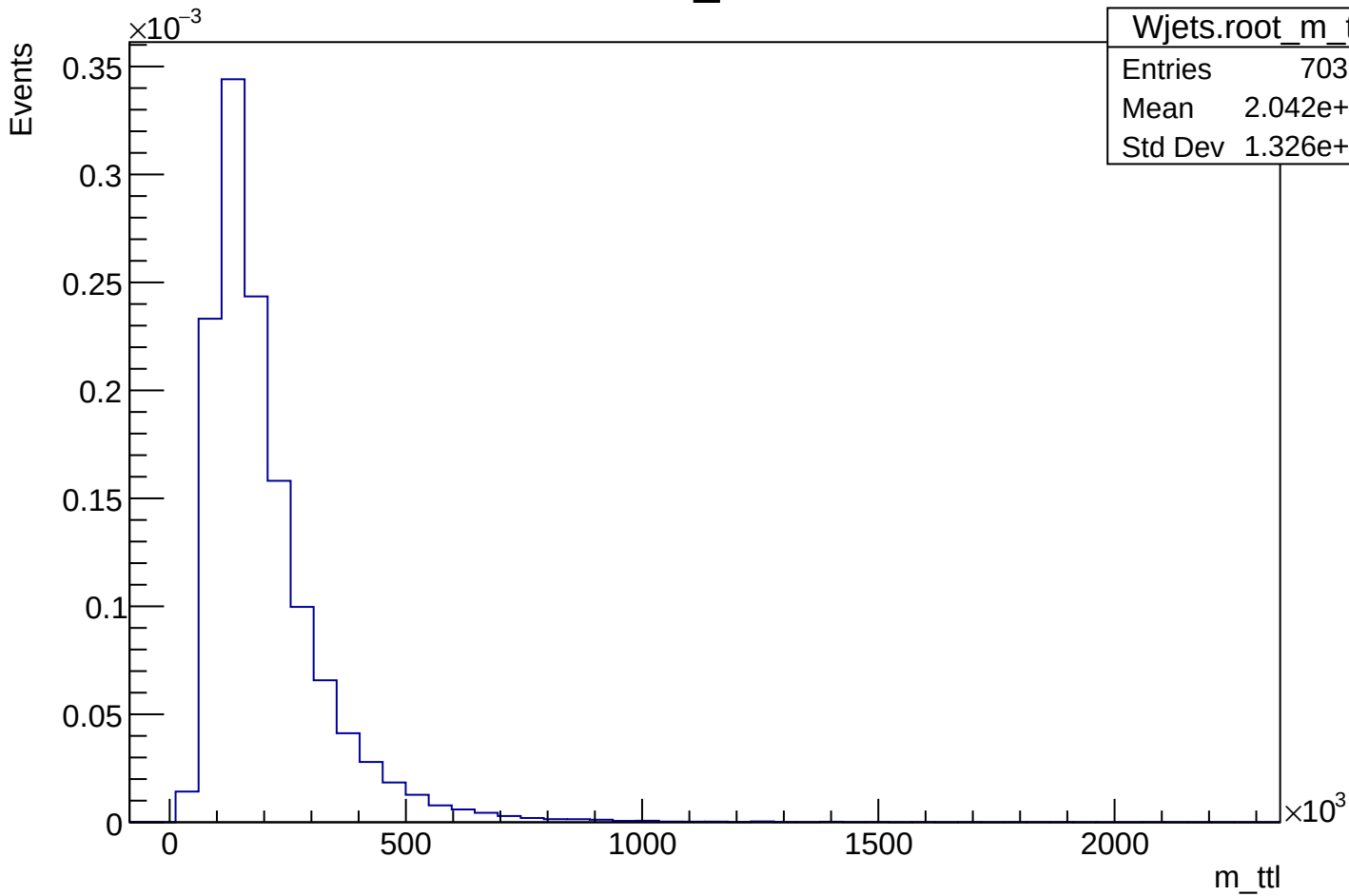
m_ttl

Wjets.root_m_ttl

Entries 70333

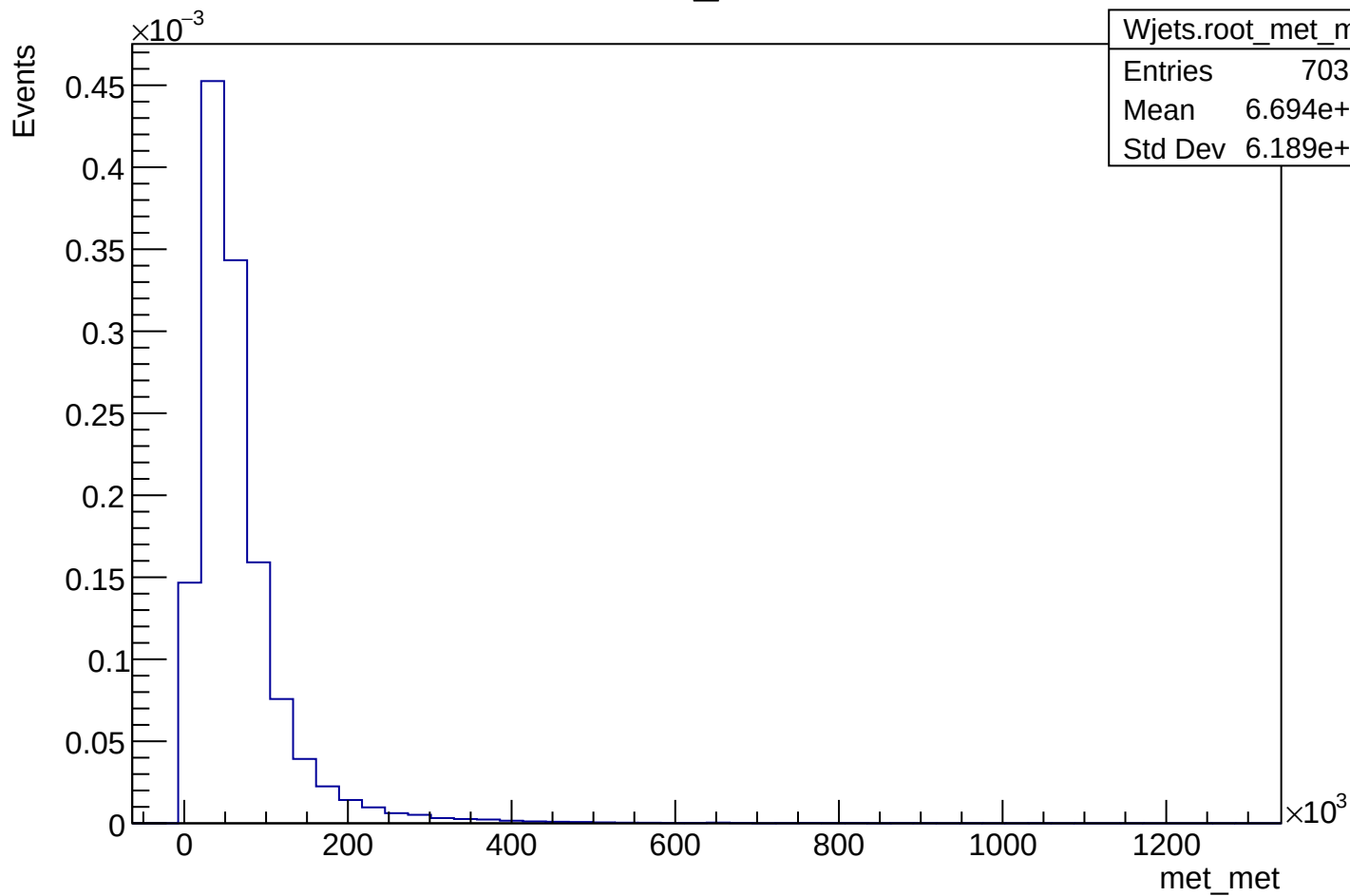
Mean 2.042e+05

Std Dev 1.326e+05

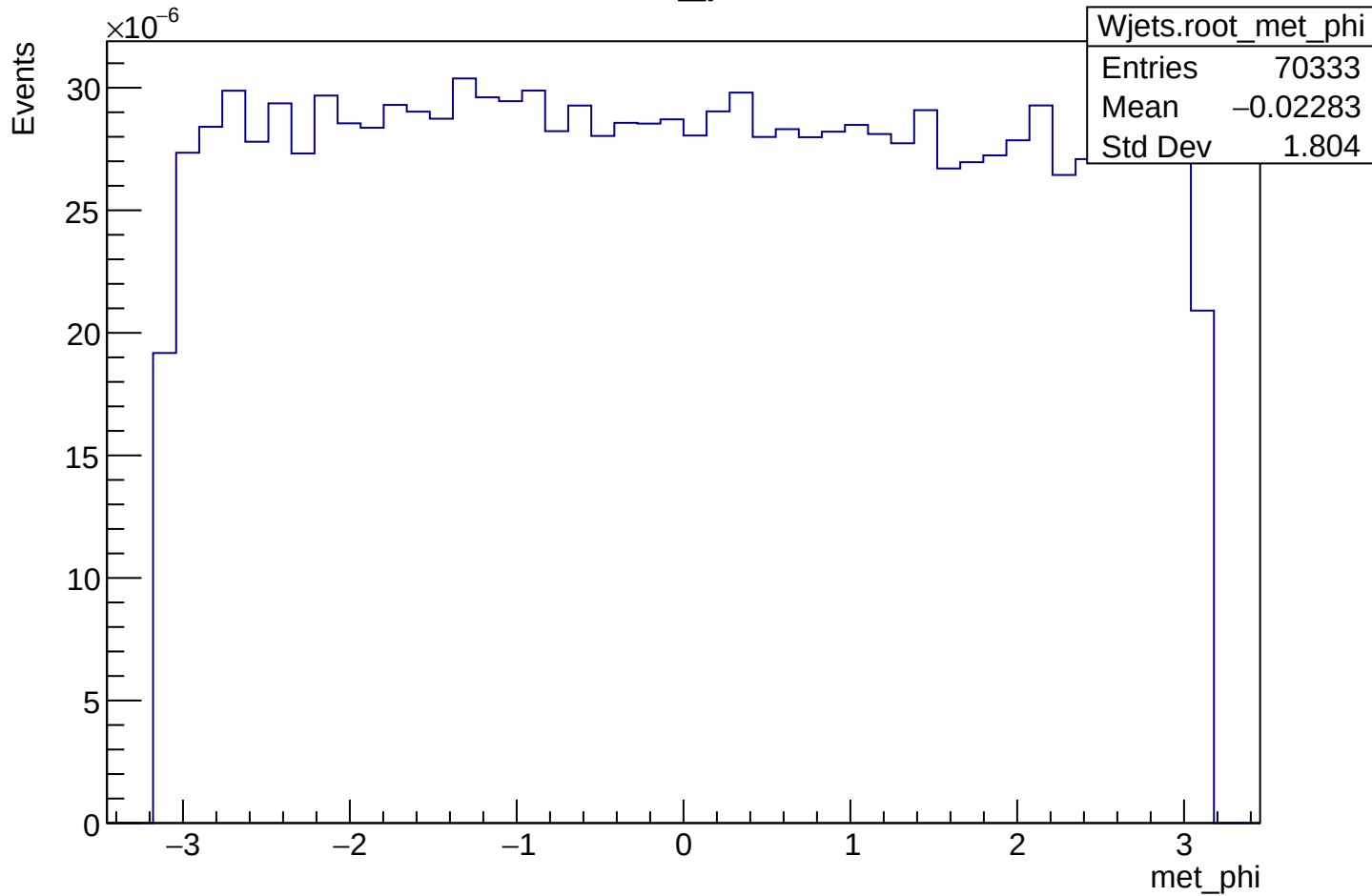


met_met

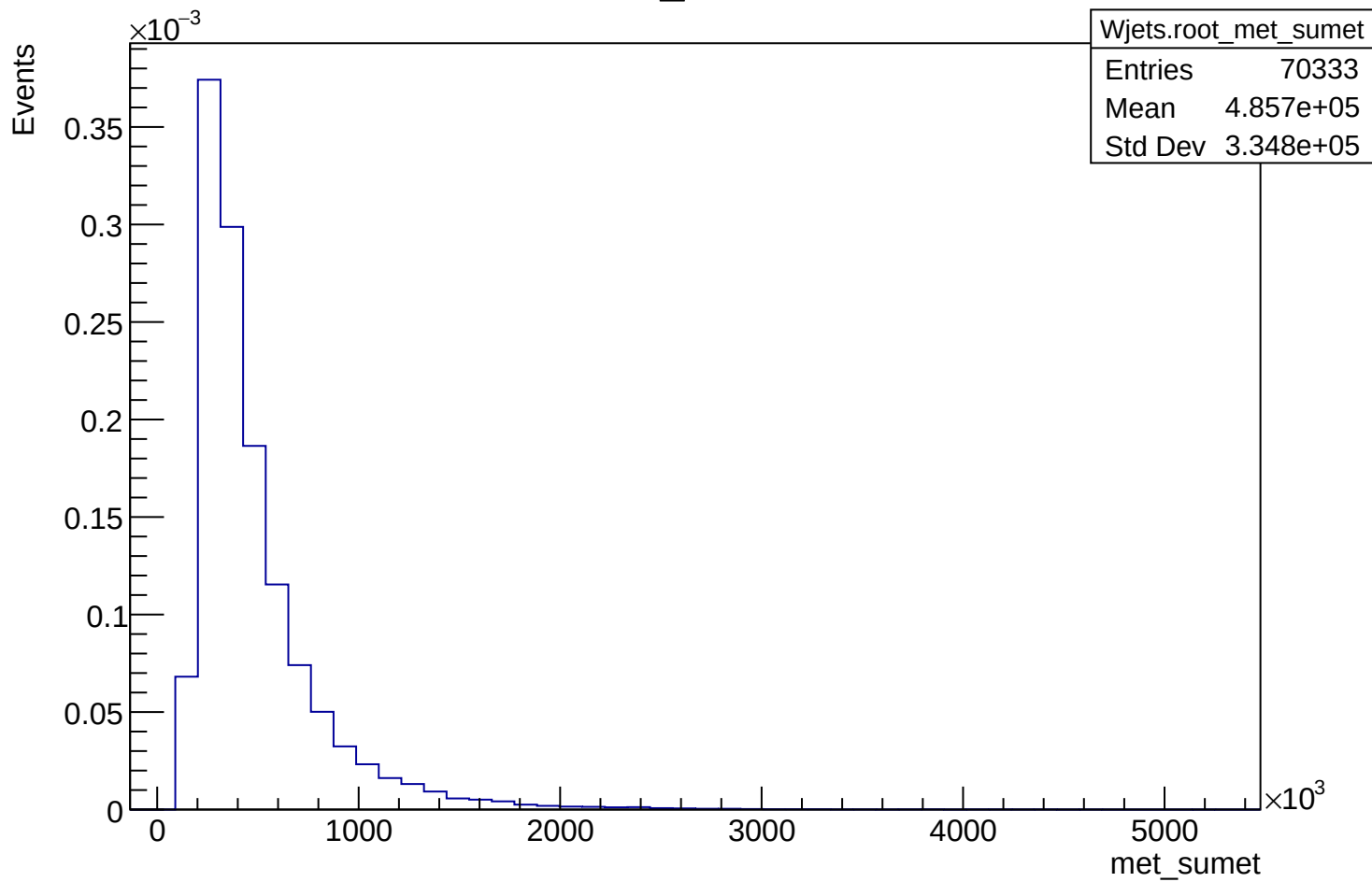
Wjets.root_met_met	
Entries	70333
Mean	6.694e+04
Std Dev	6.189e+04



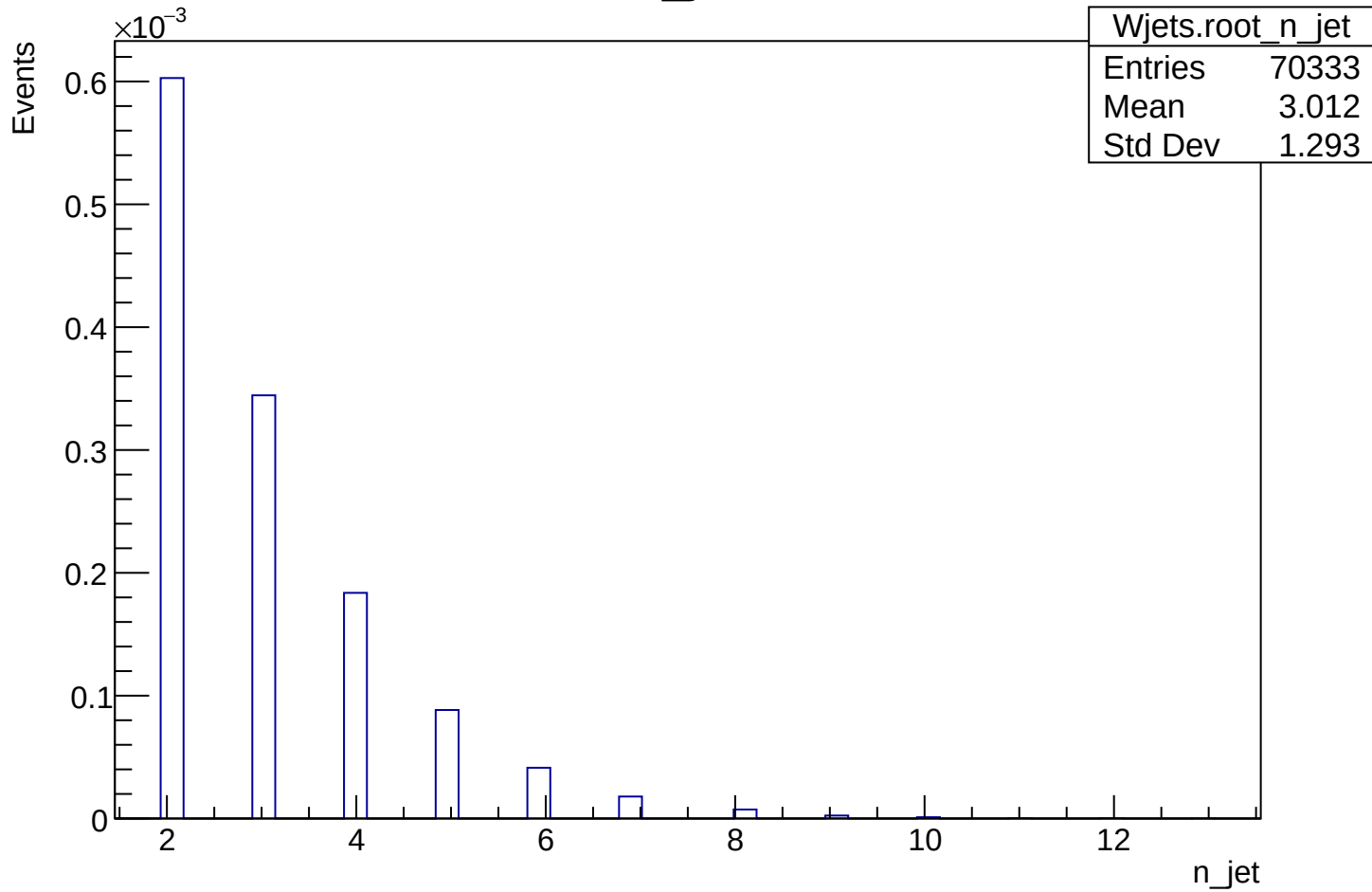
met_phi



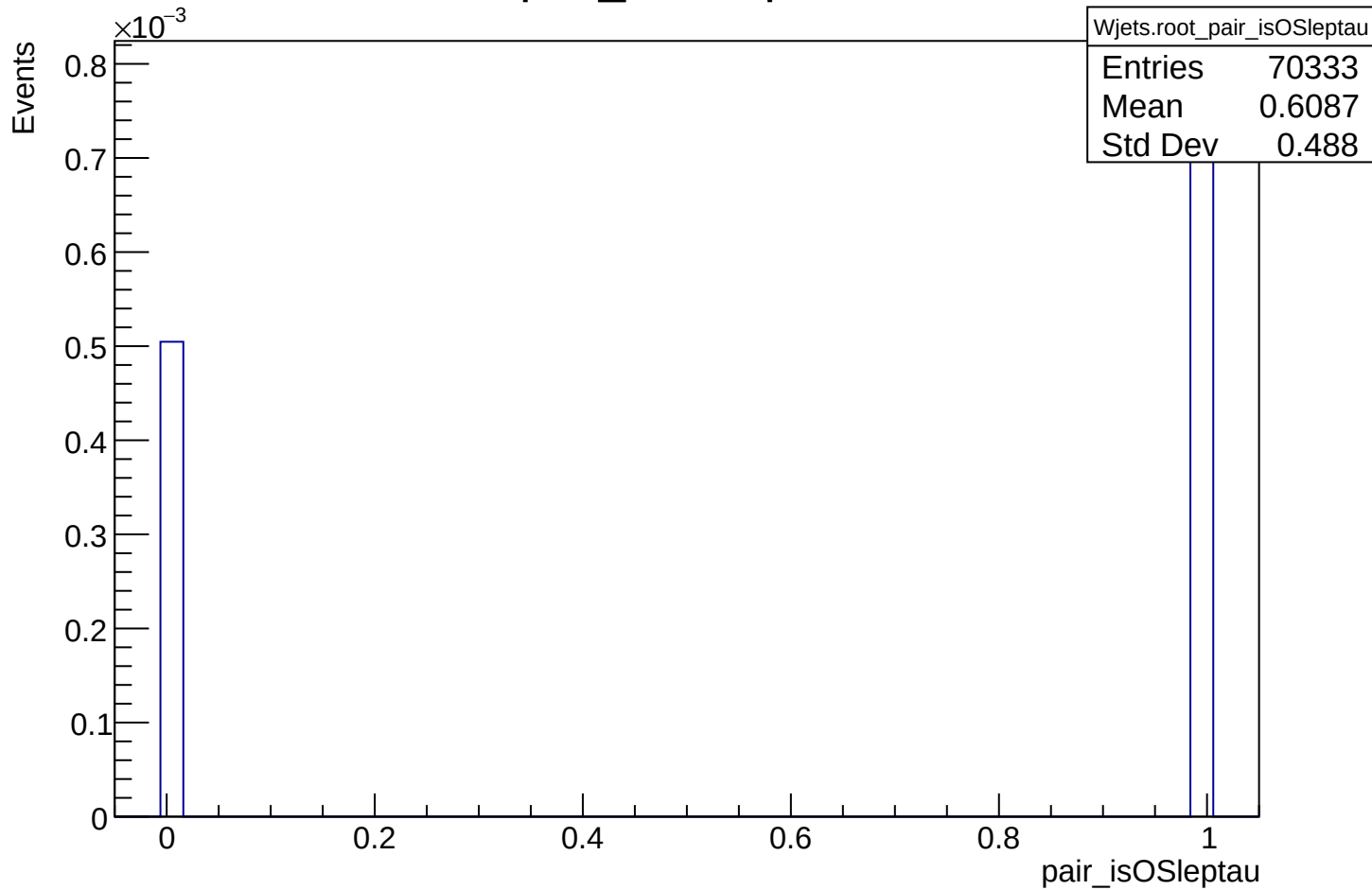
met_sumet



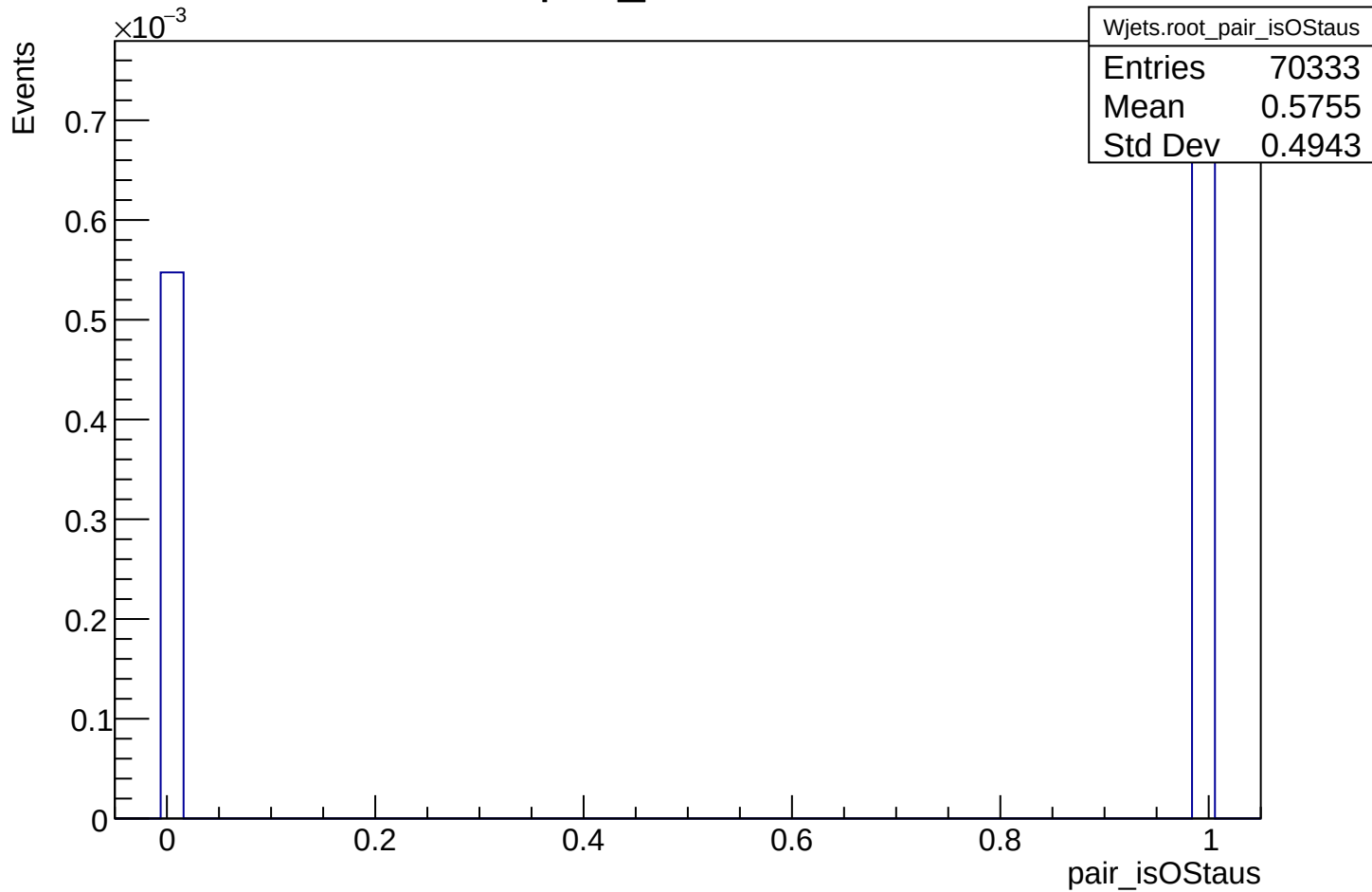
n_jet



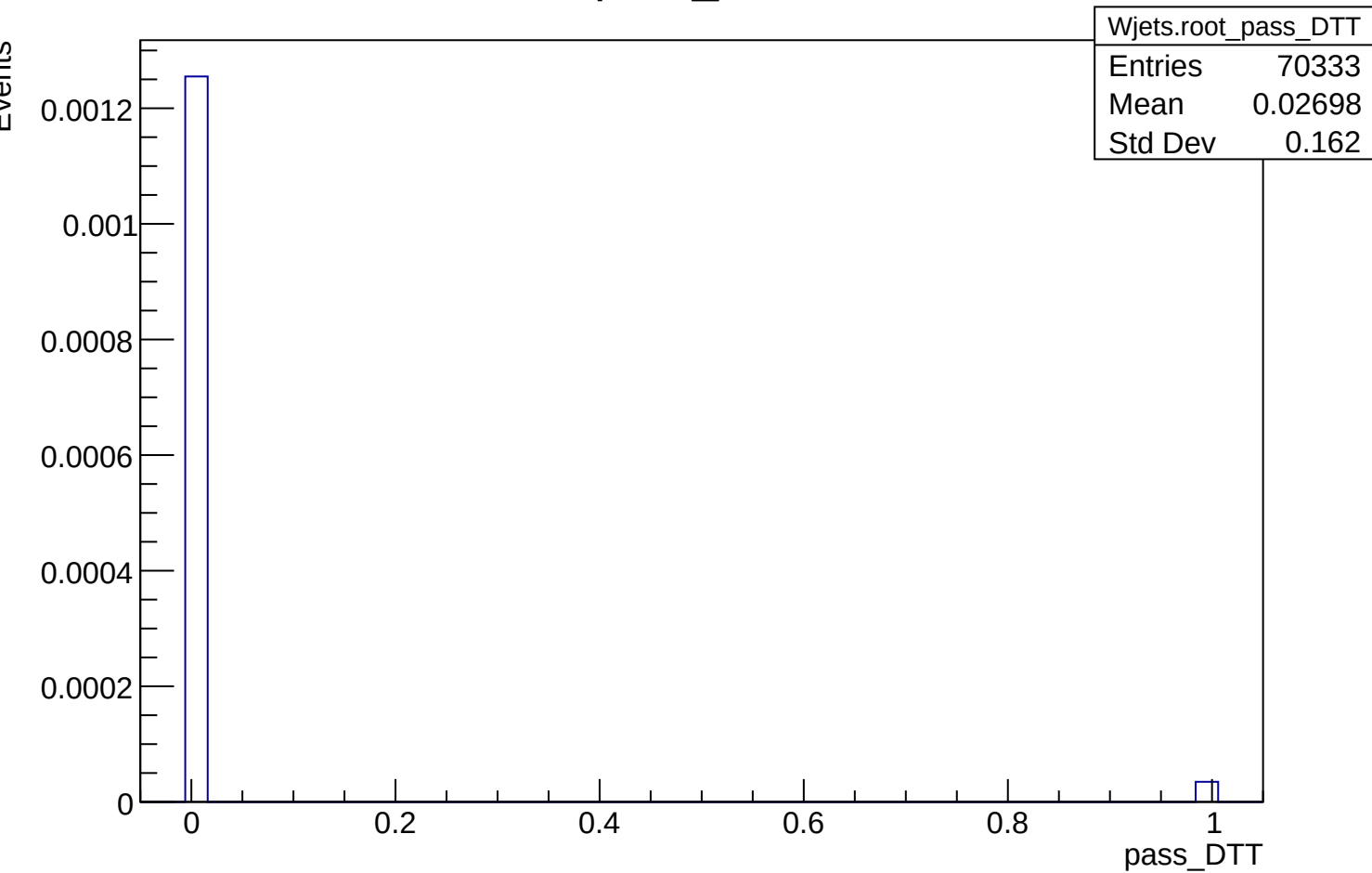
pair_isOSleptau



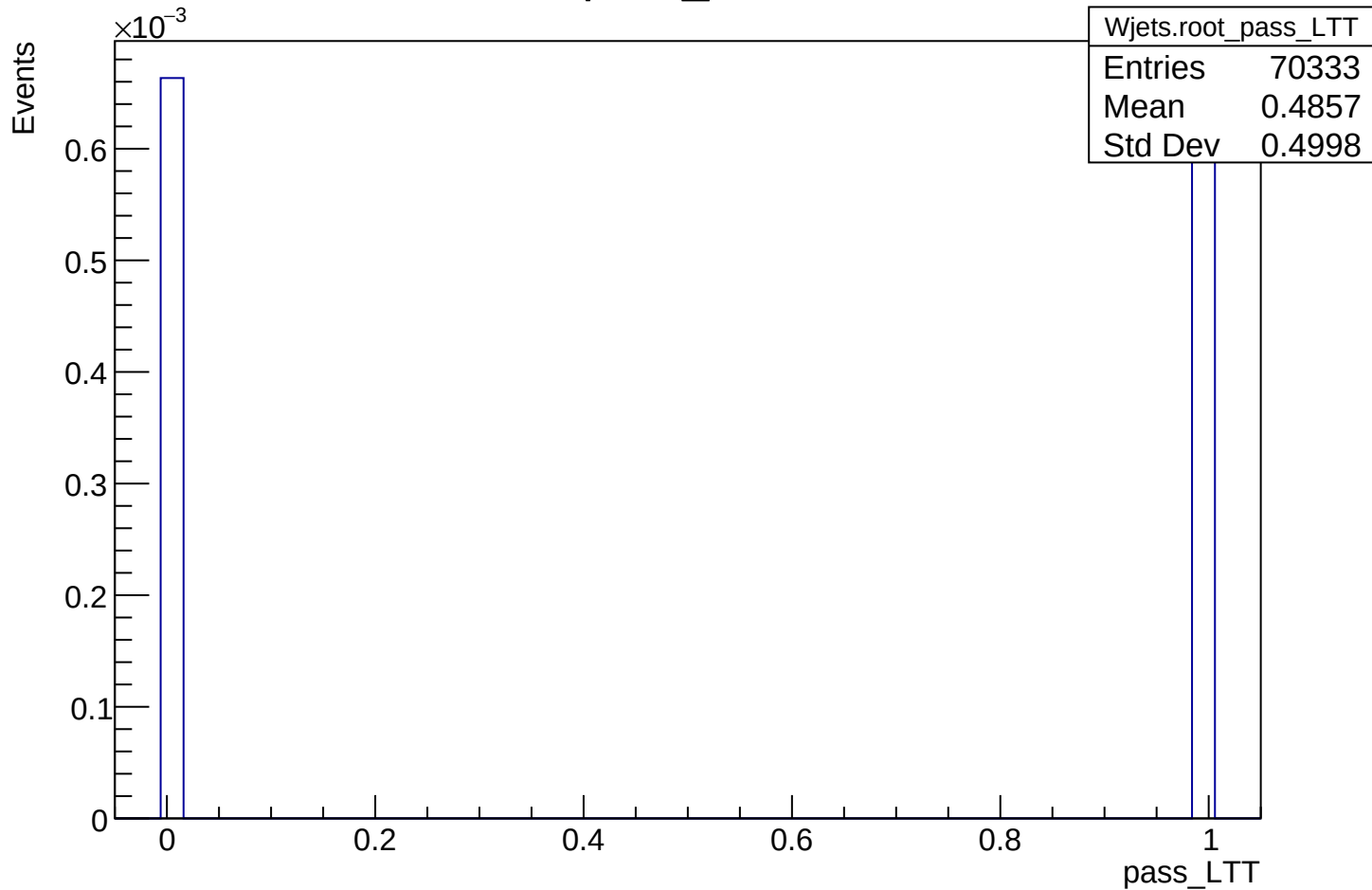
pair_isOStaus



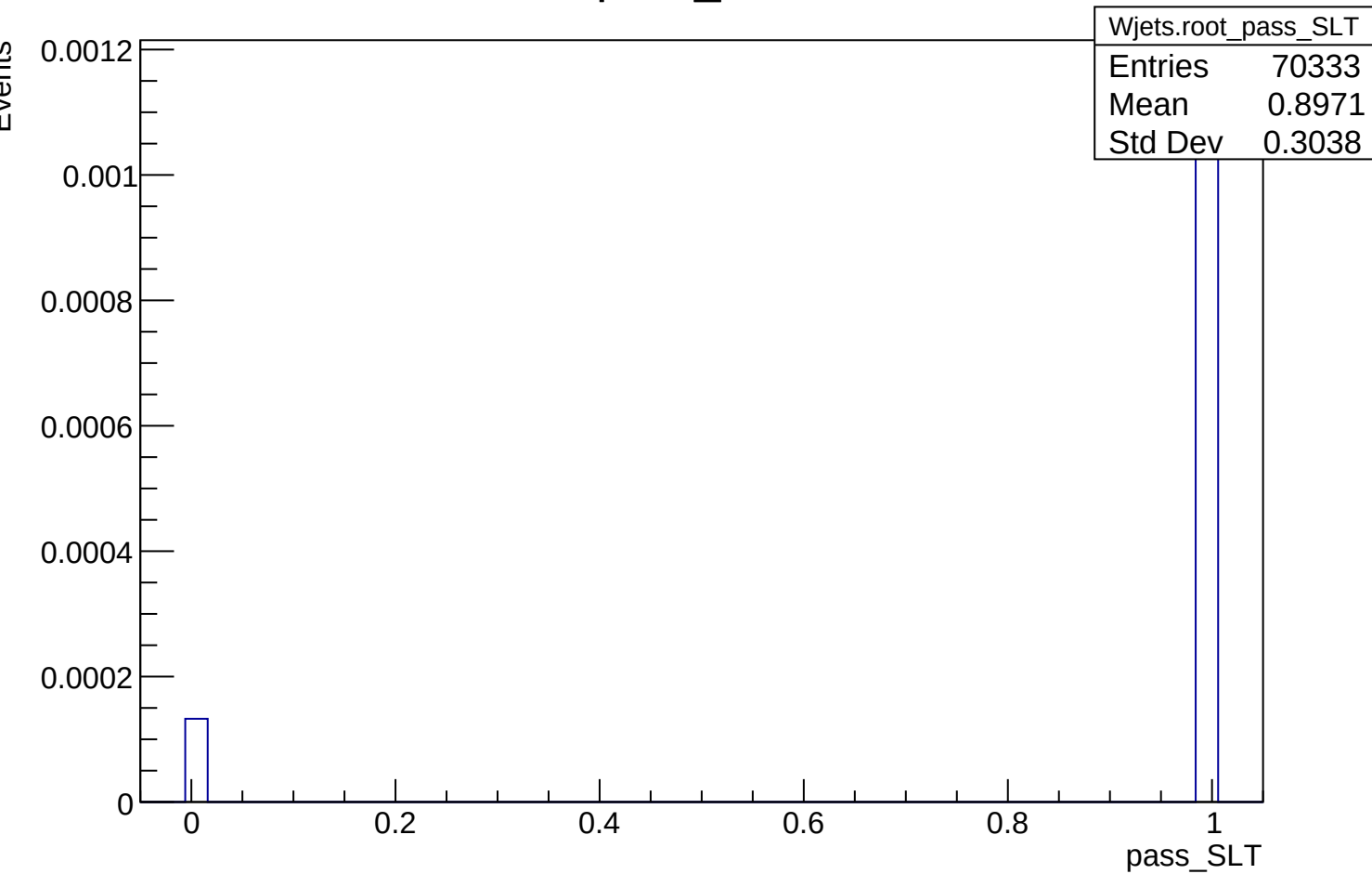
pass_DTT



pass_LTT



pass_SLT



pass_STT

Events

0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0

0

0.2

0.4

0.6

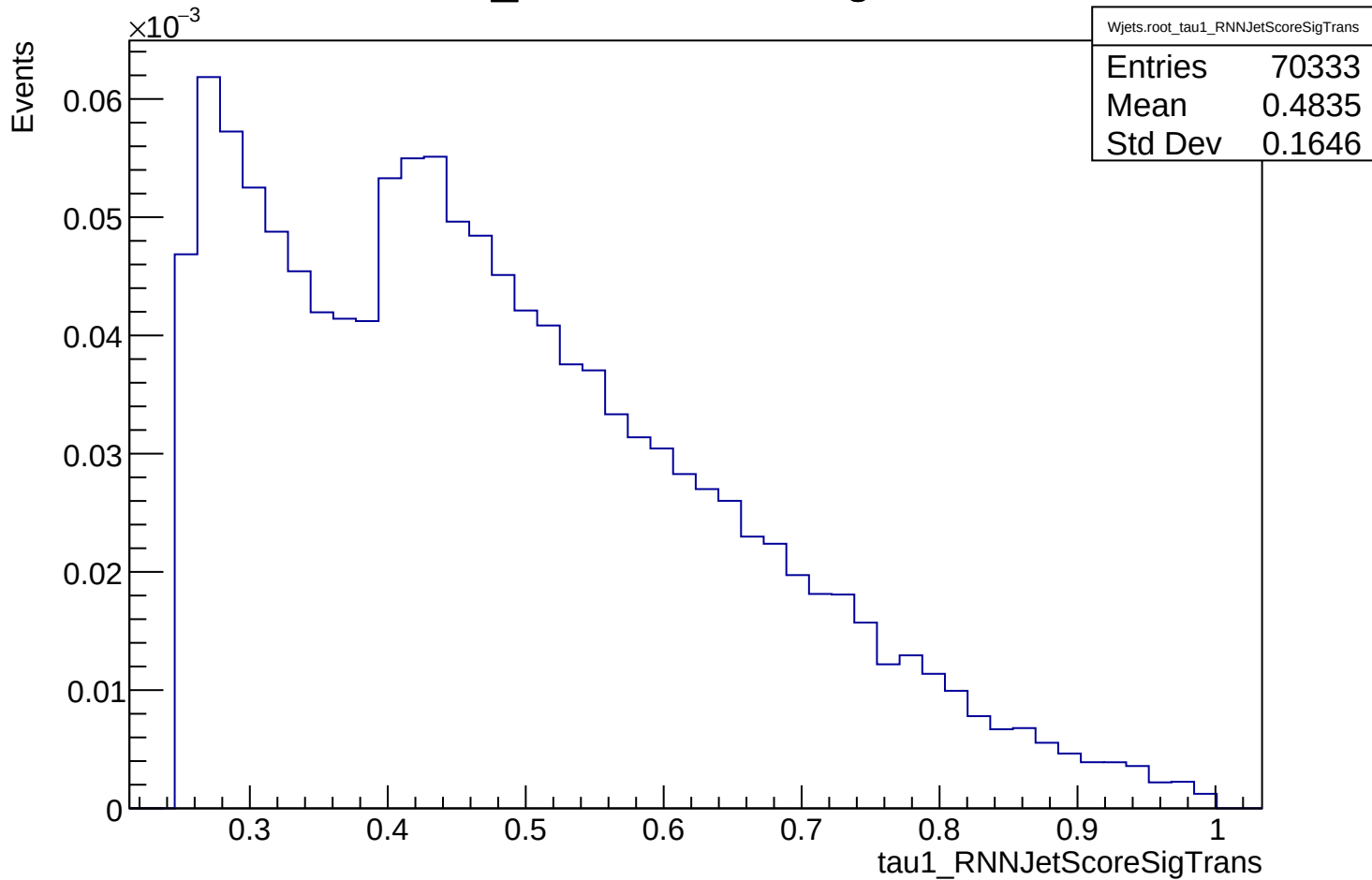
0.8

1

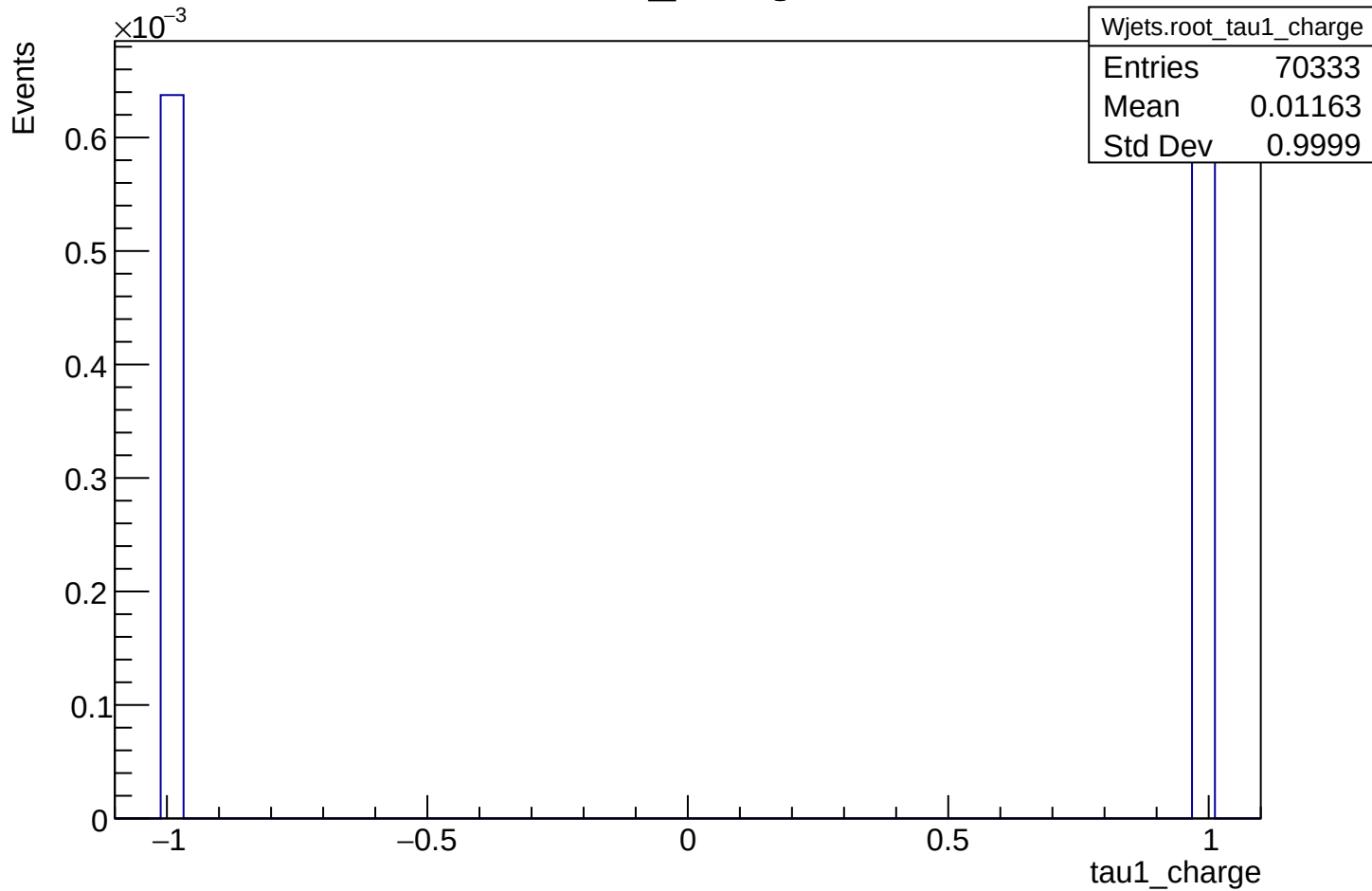
Wjets.root_pass_STT	
Entries	70333
Mean	0.03296
Std Dev	0.1785

pass_STT

tau1_RNNJetScoreSigTrans

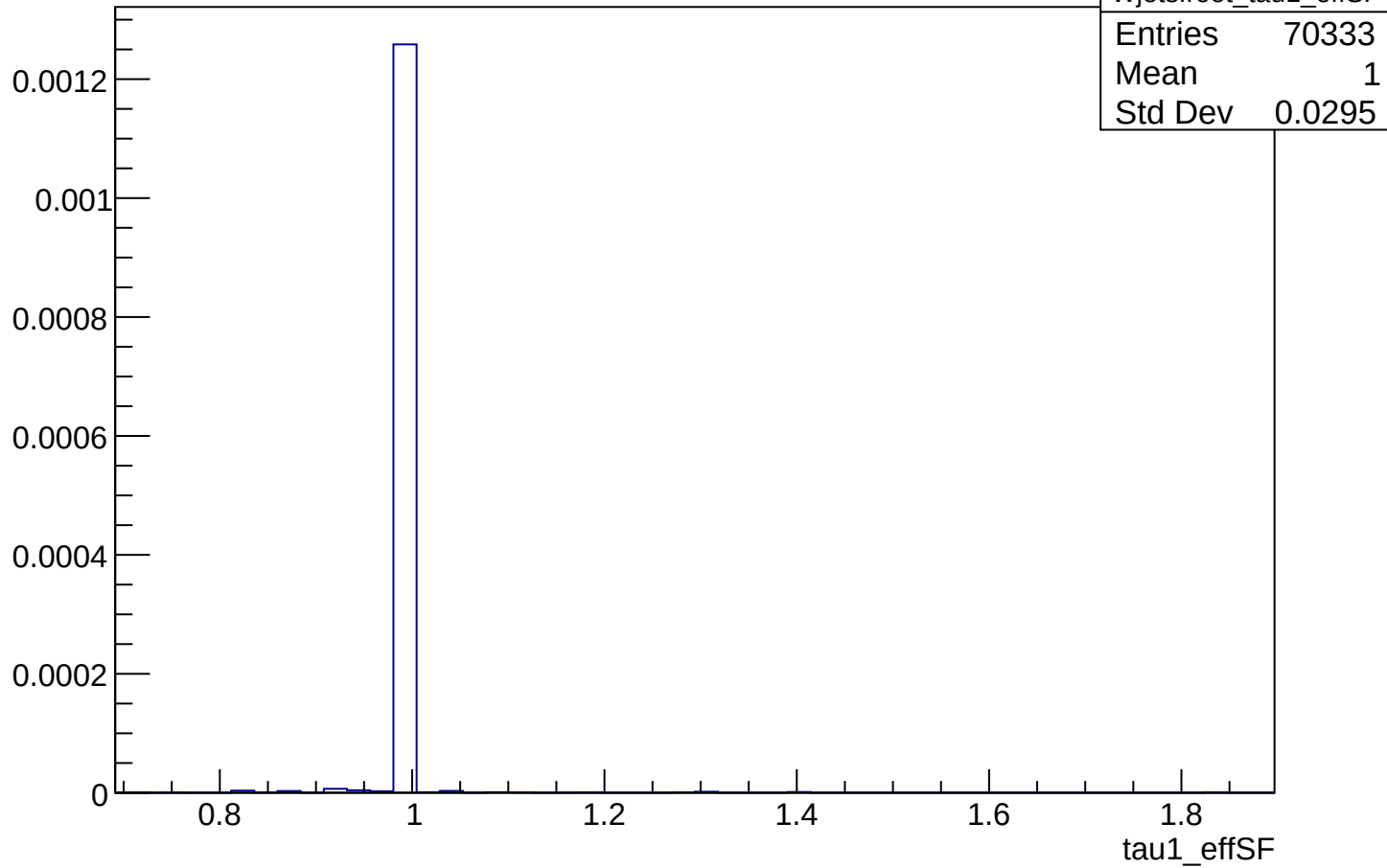


tau1_charge

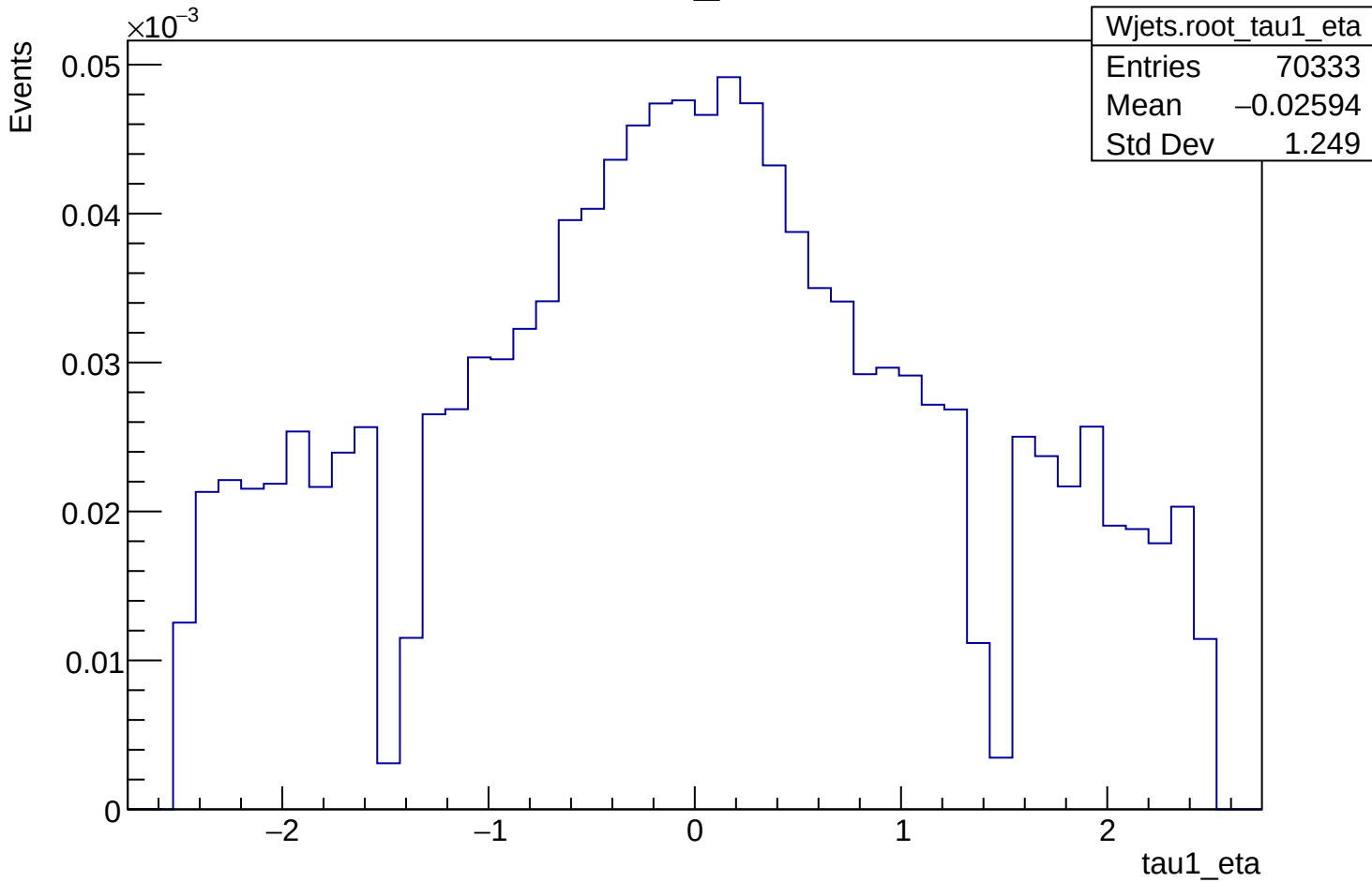


tau1_effSF

Events

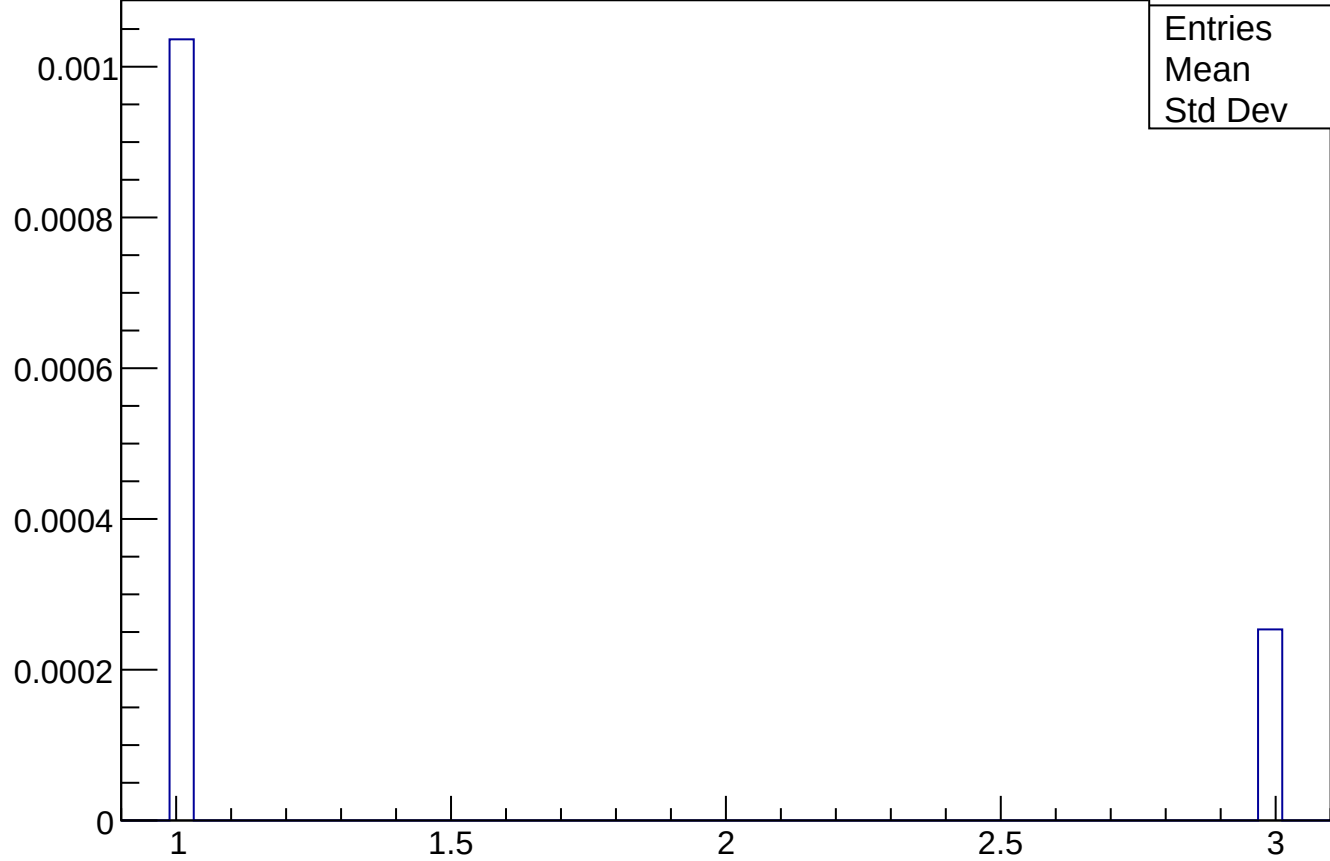


tau1_eta



tau1_nprong

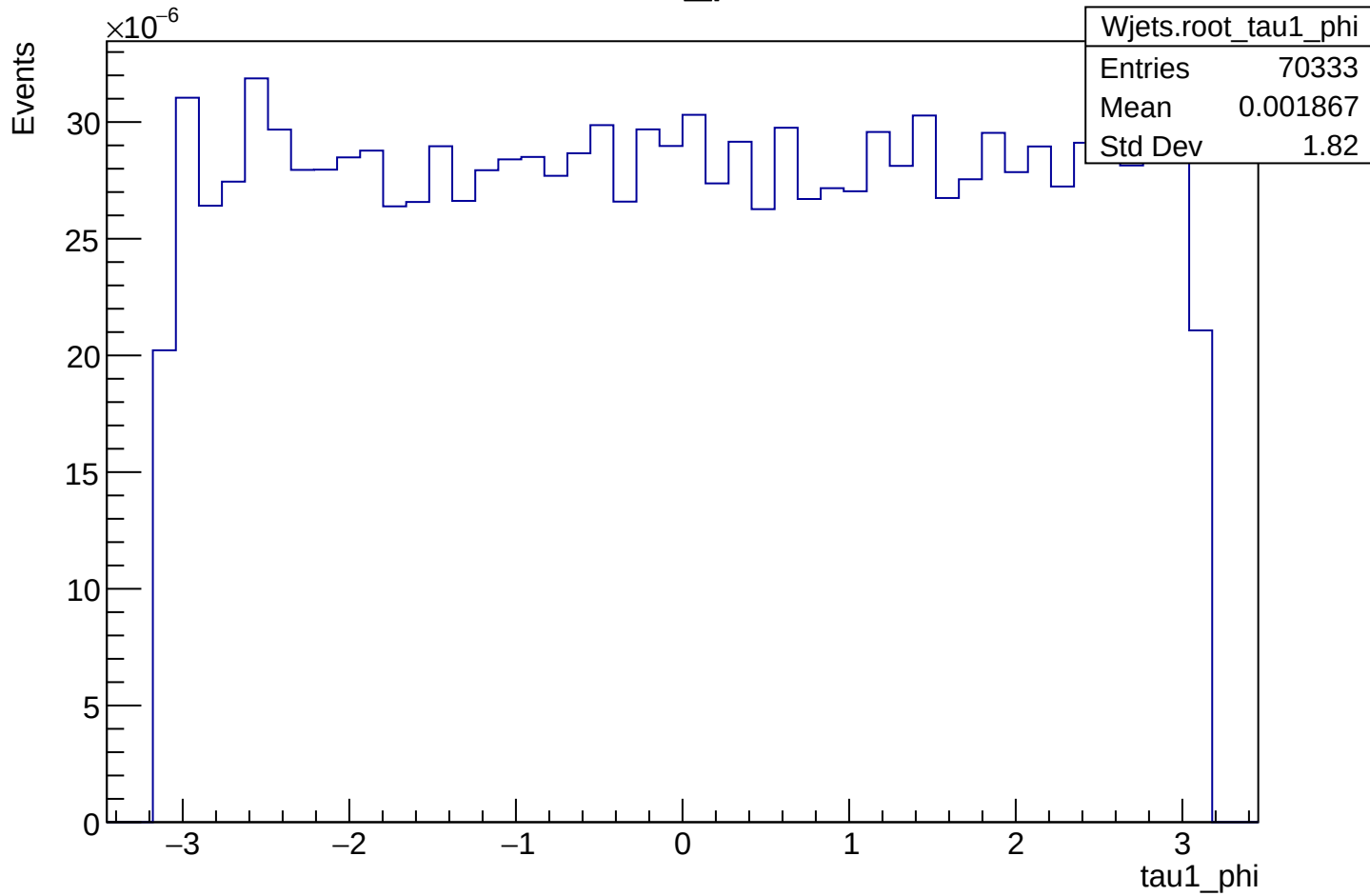
Events



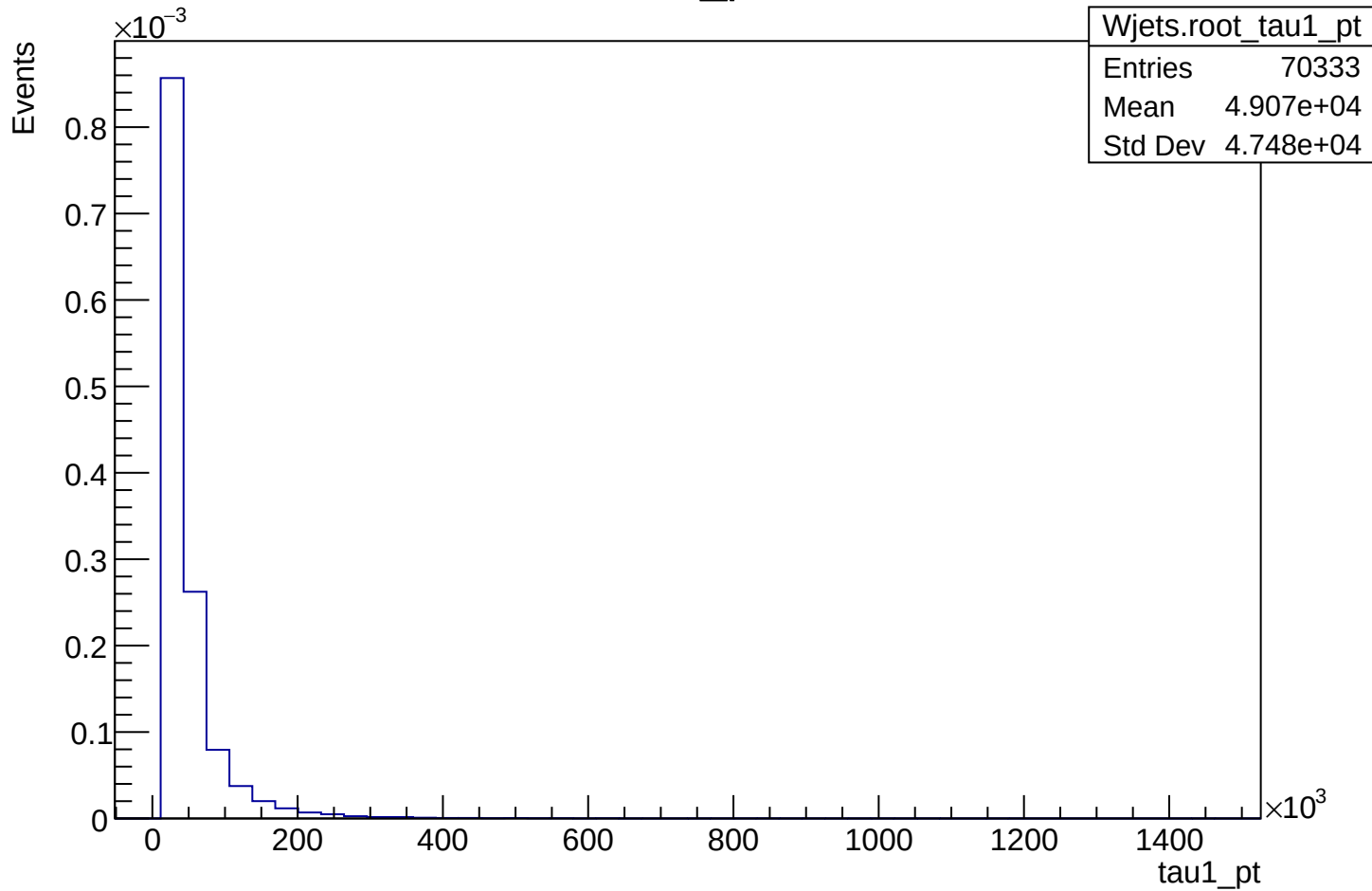
Wjets.root_tau1_nprong	
Entries	70333
Mean	1.393
Std Dev	0.7947

tau1_nprong

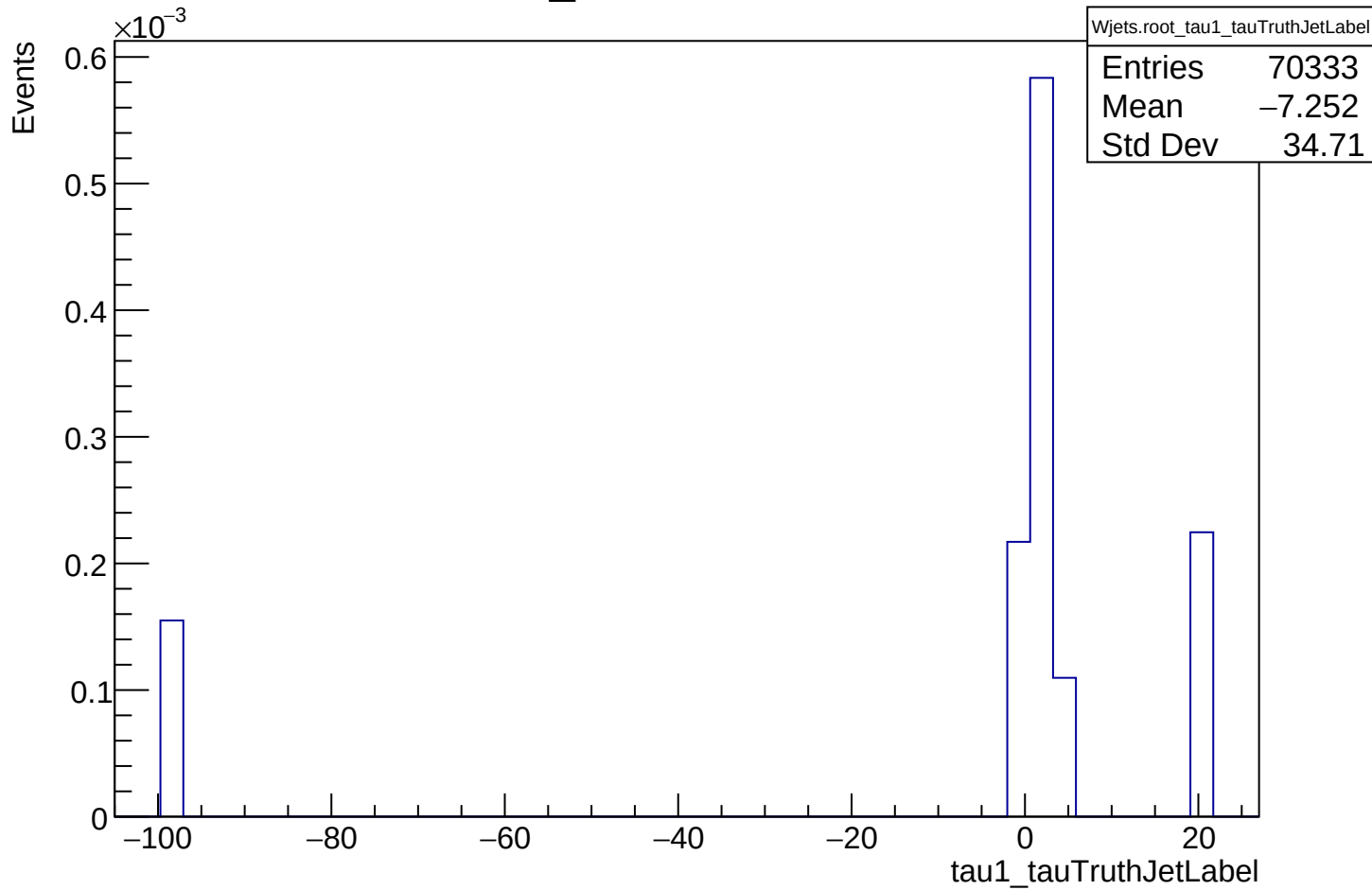
tau1_phi



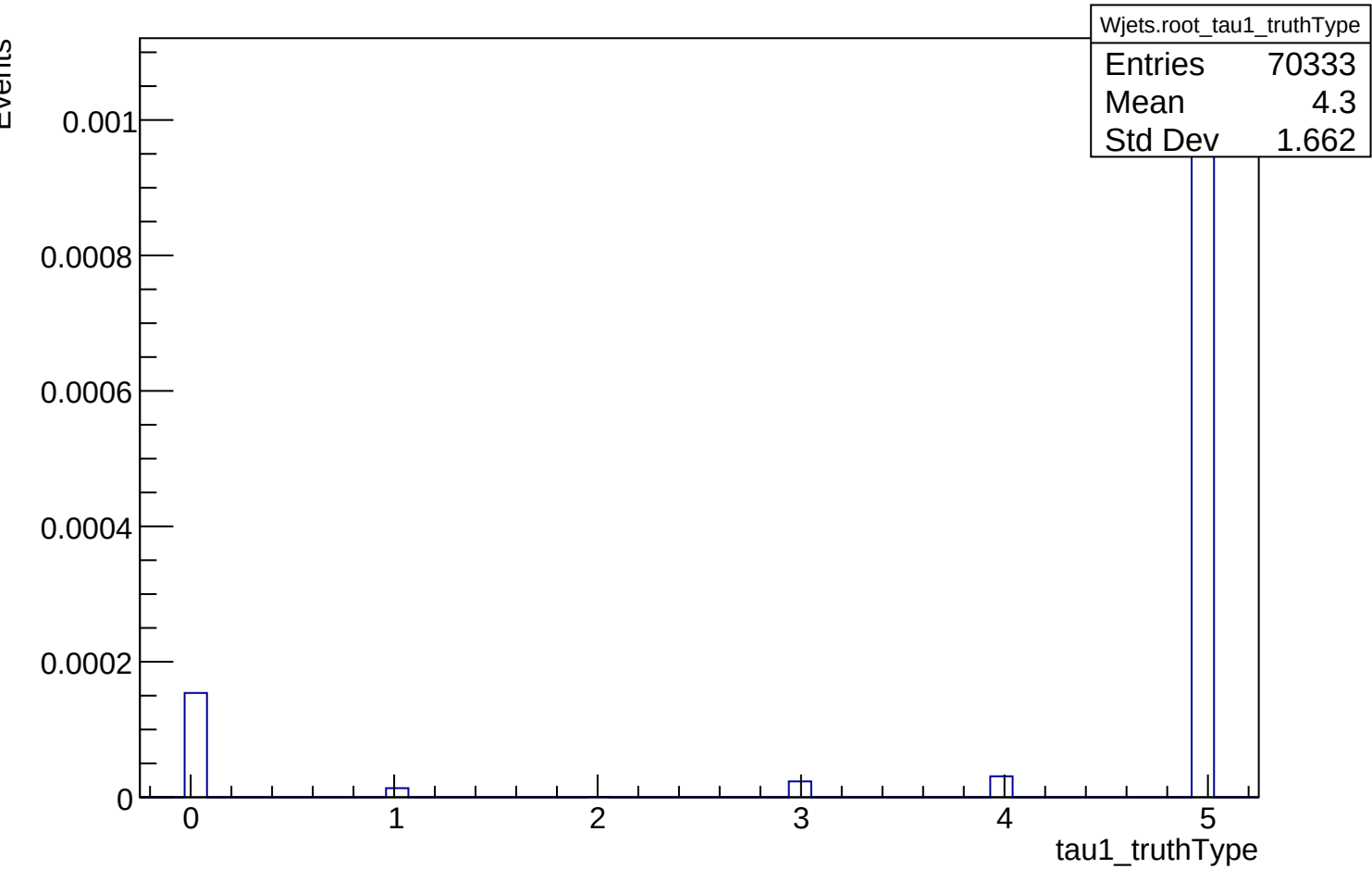
tau1_pt



tau1_tauTruthJetLabel

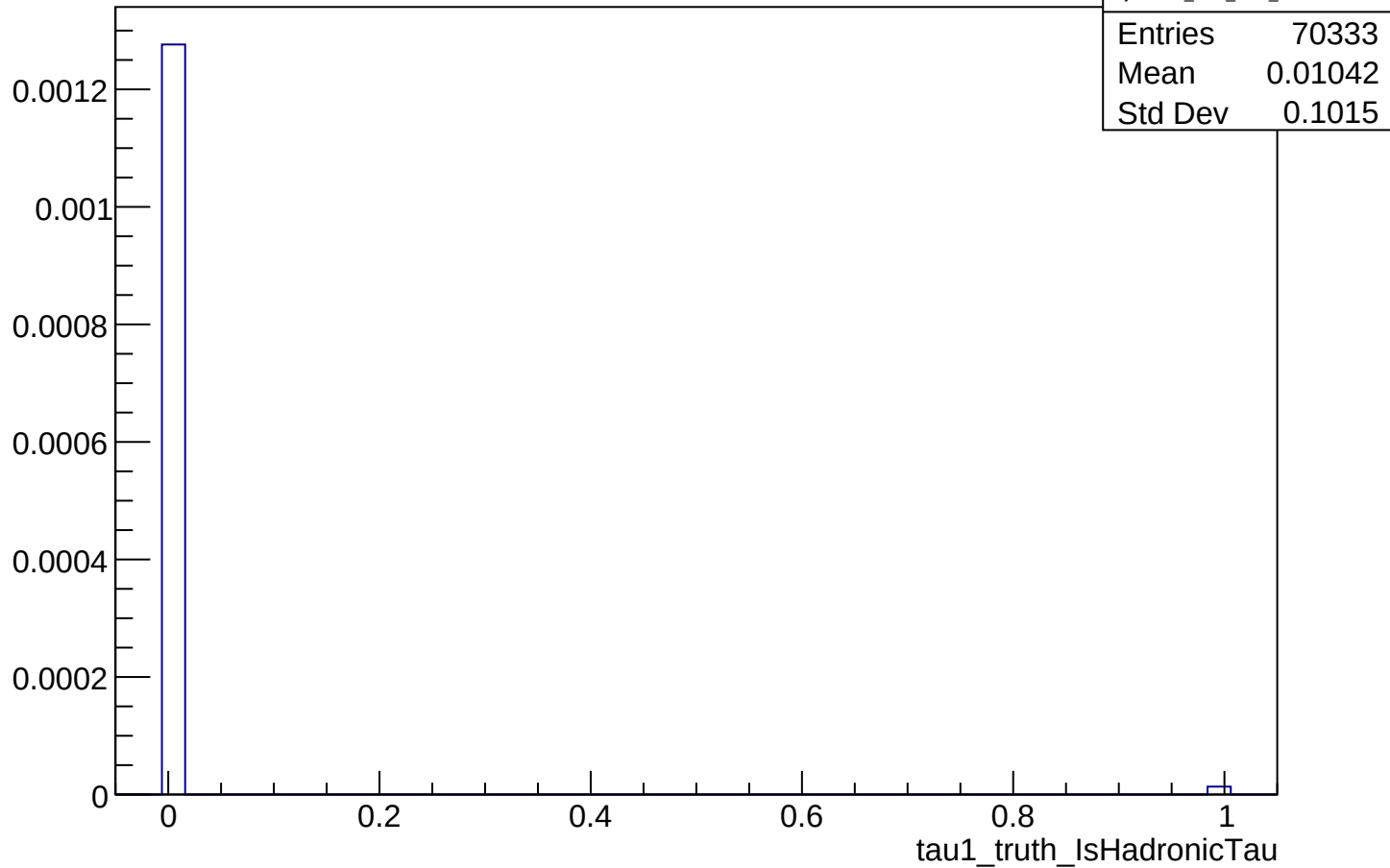


tau1_truthType

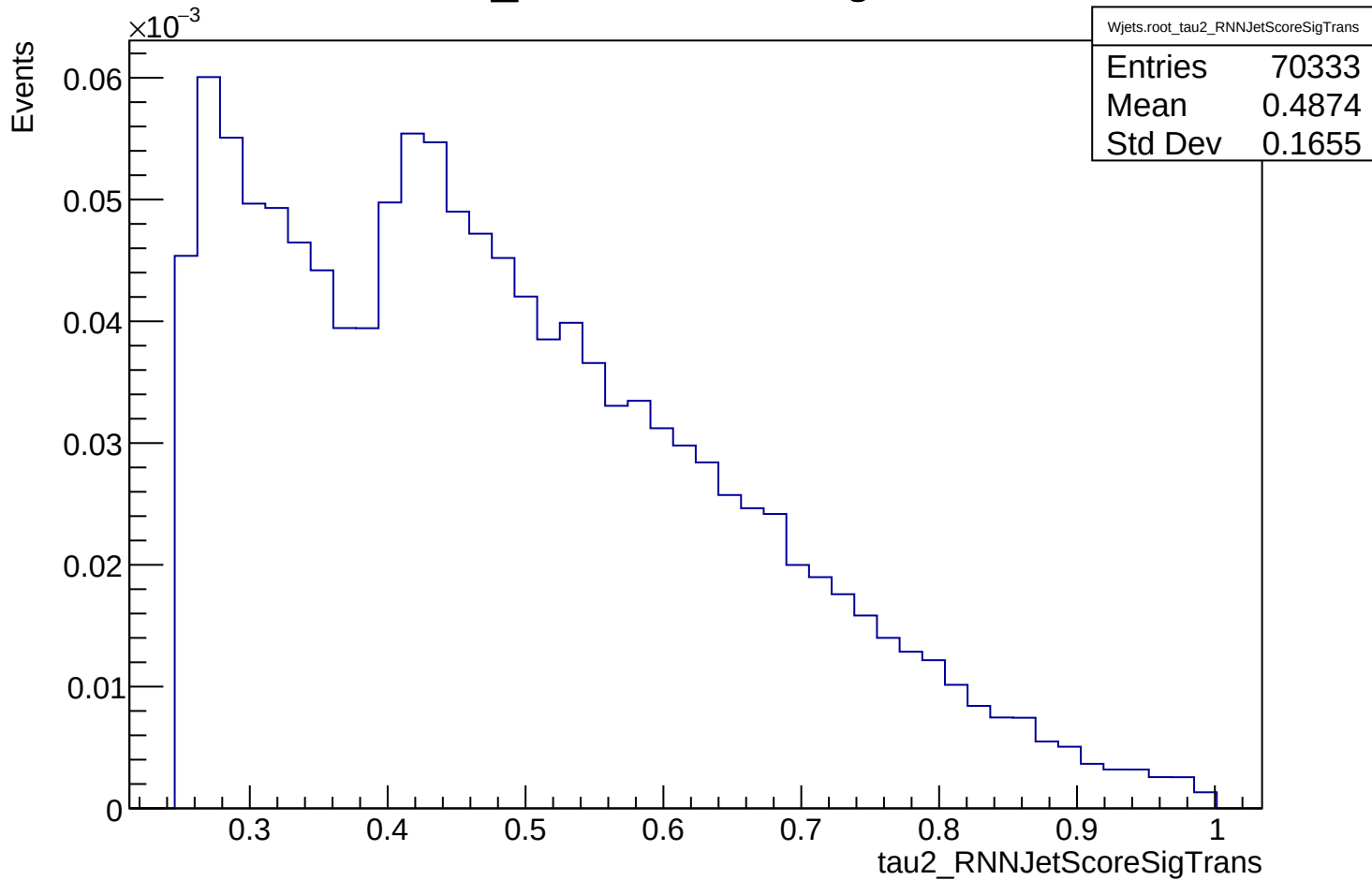


tau1_truth_IsHadronicTau

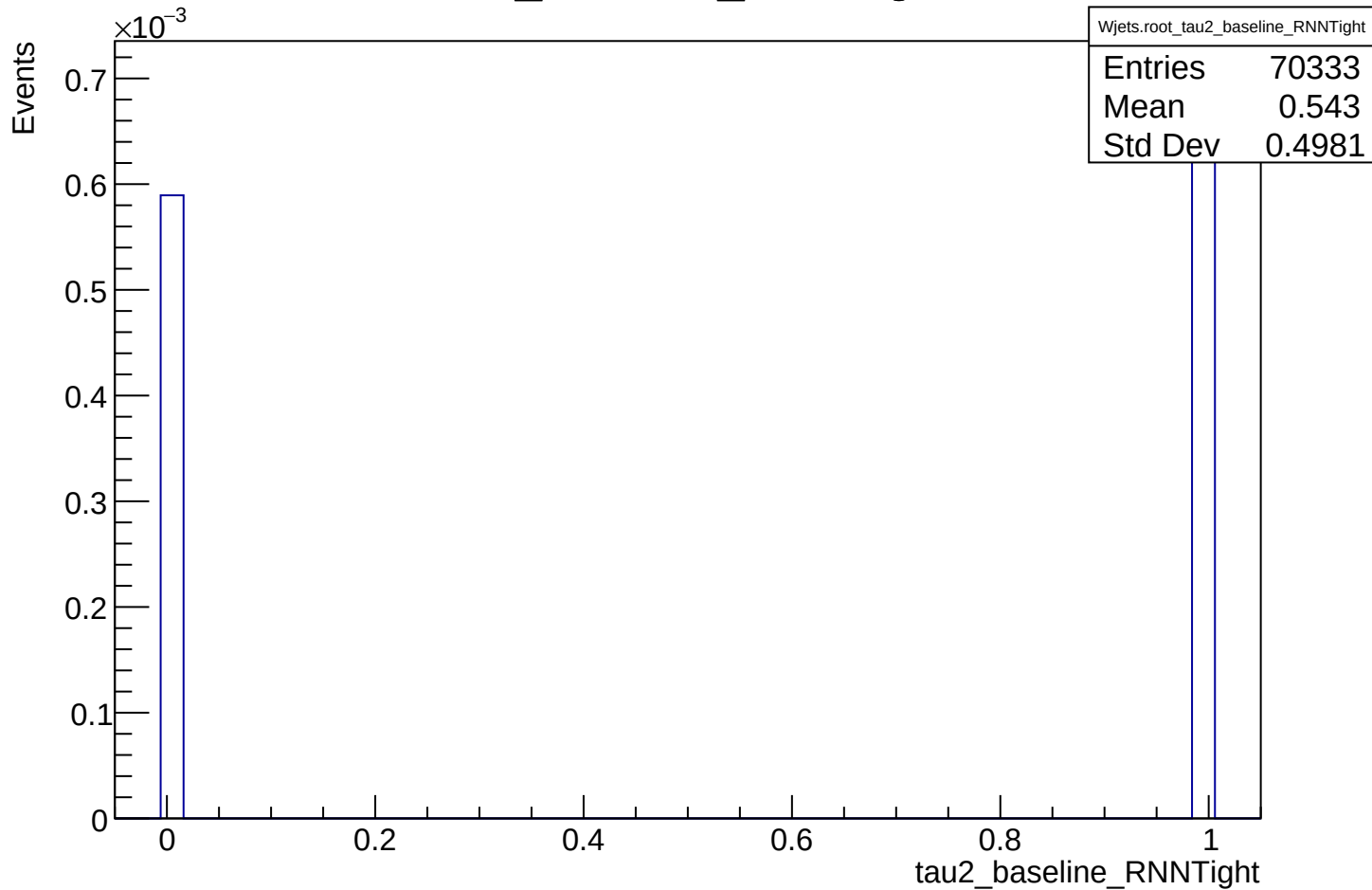
Events



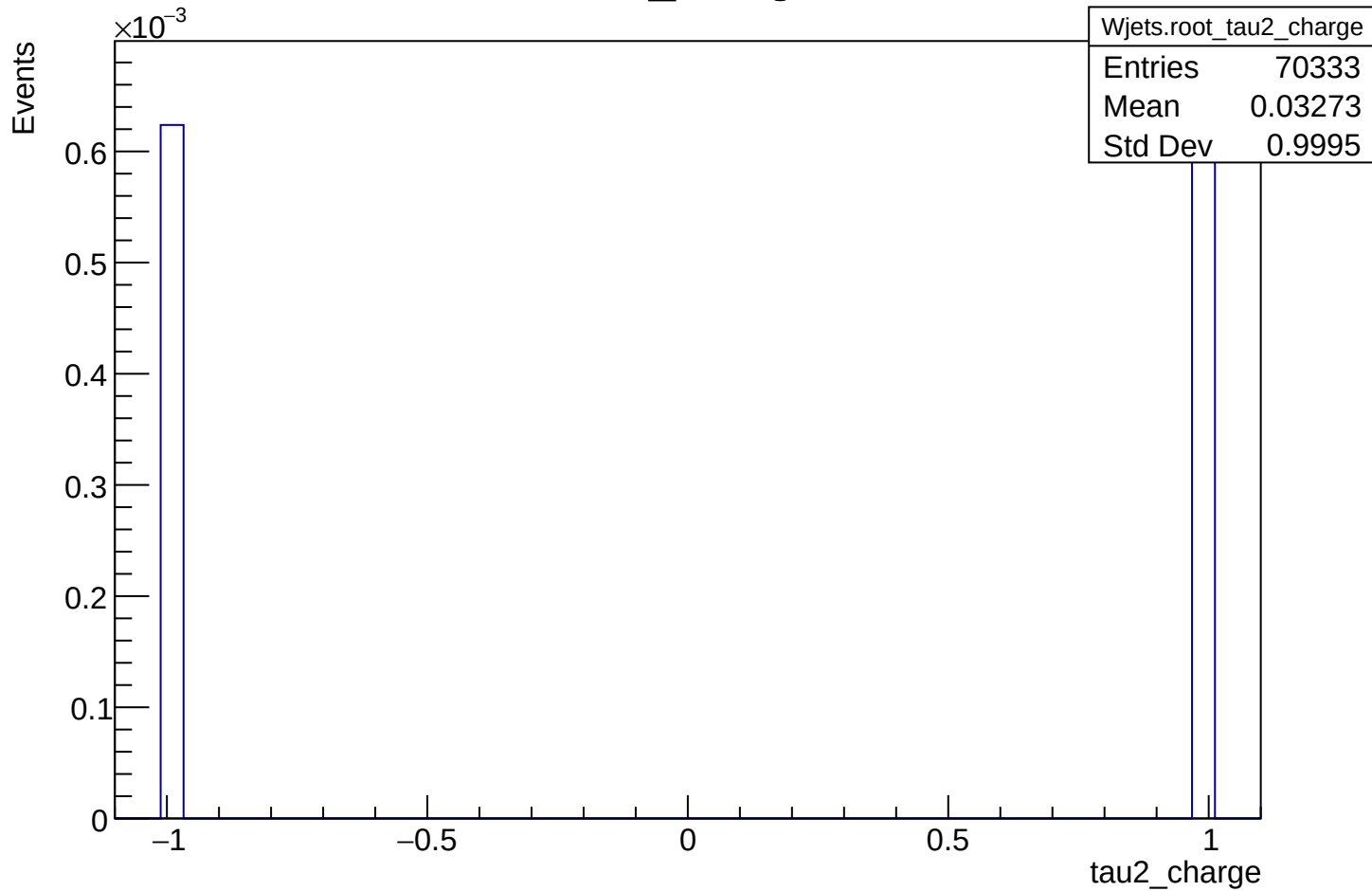
tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

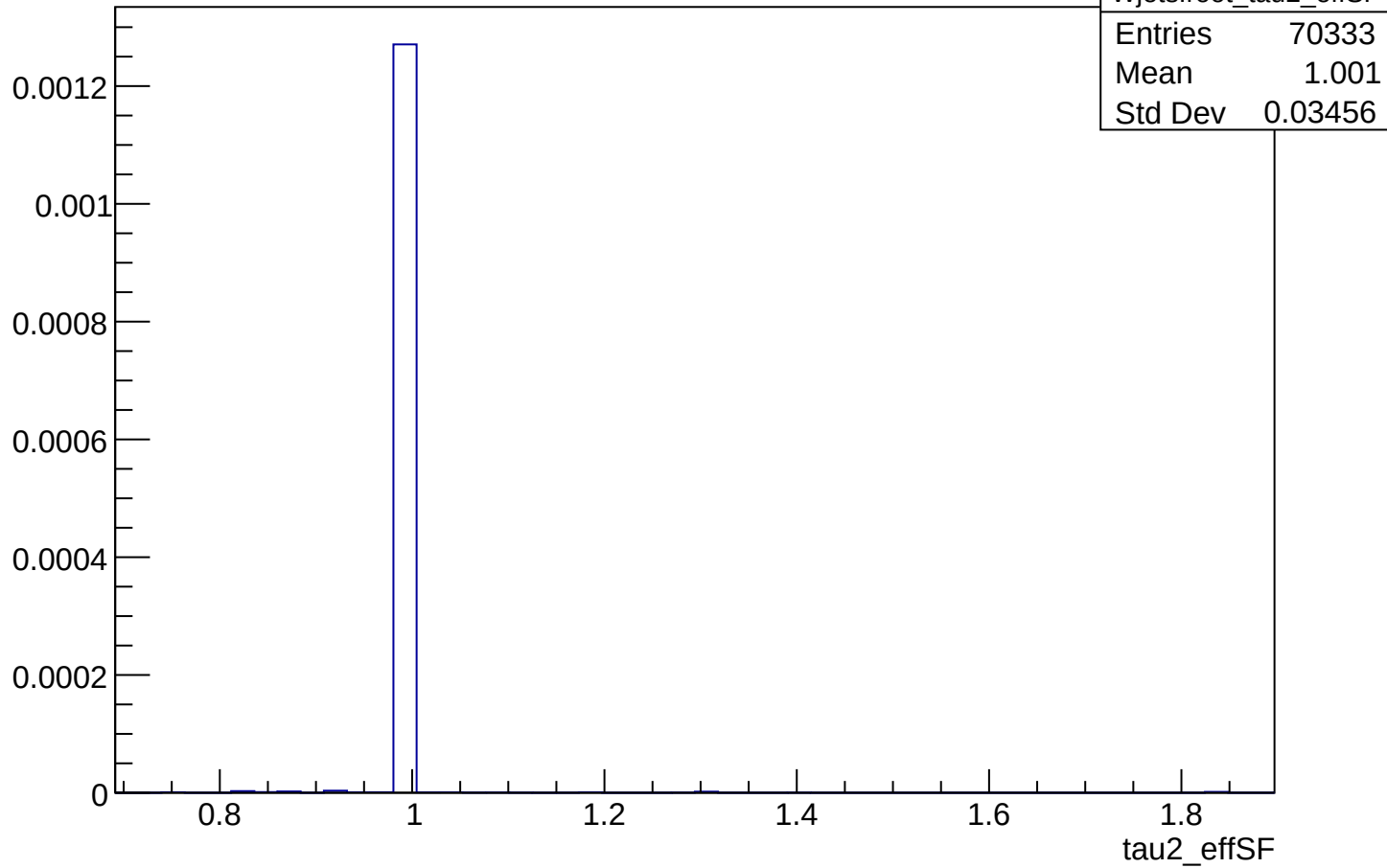


tau2_charge

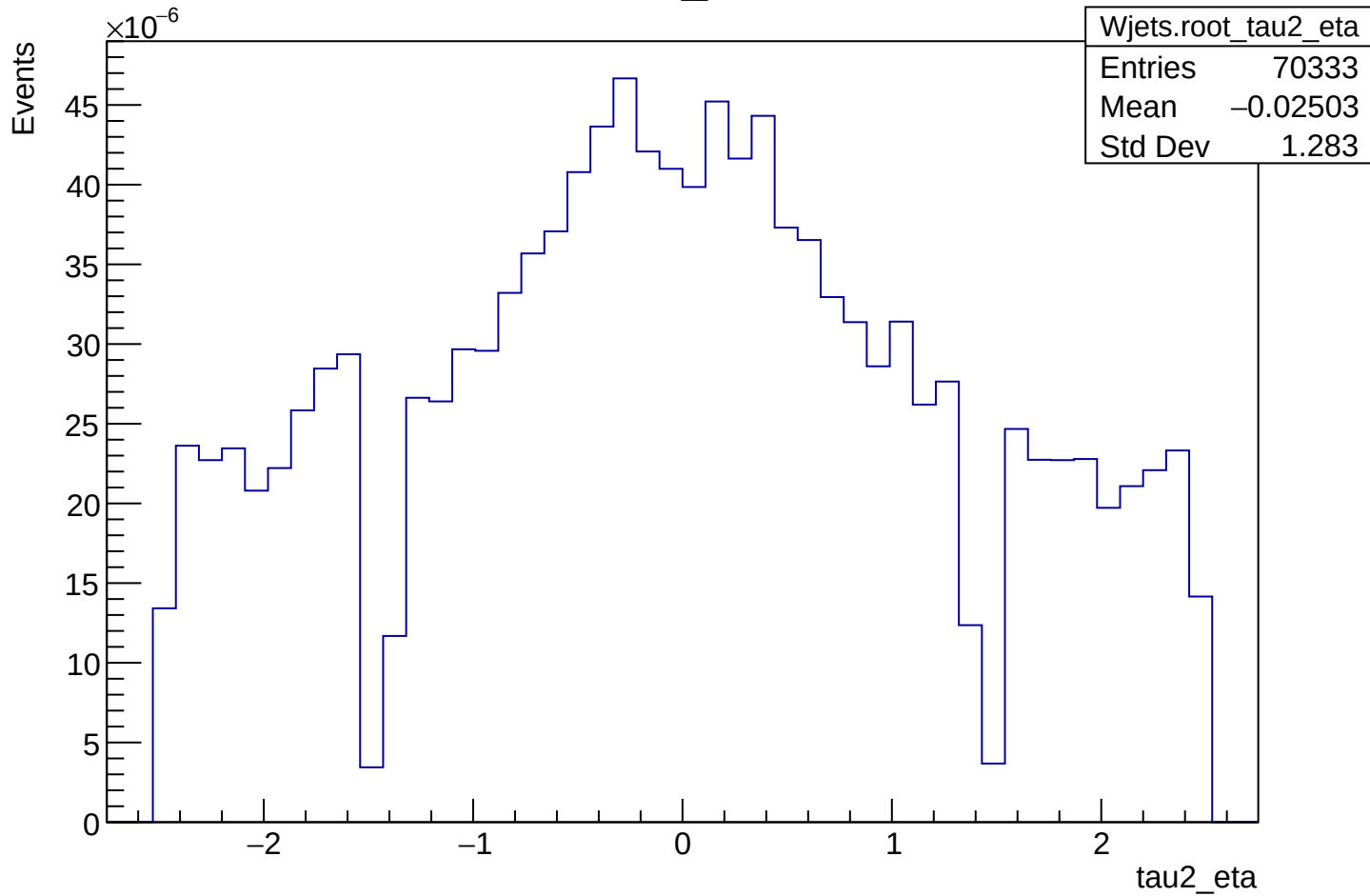


tau2_effSF

Events

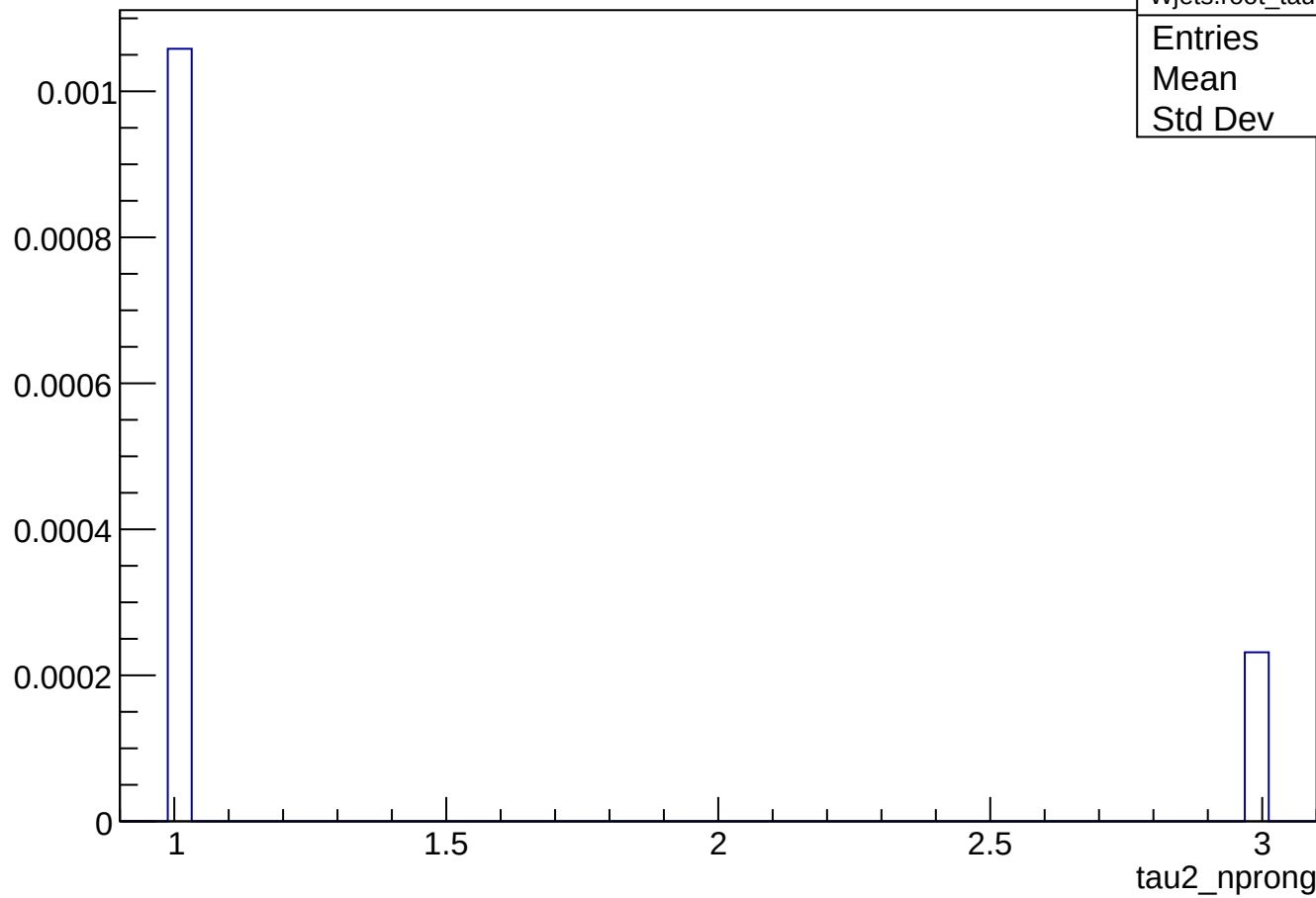


tau2_eta



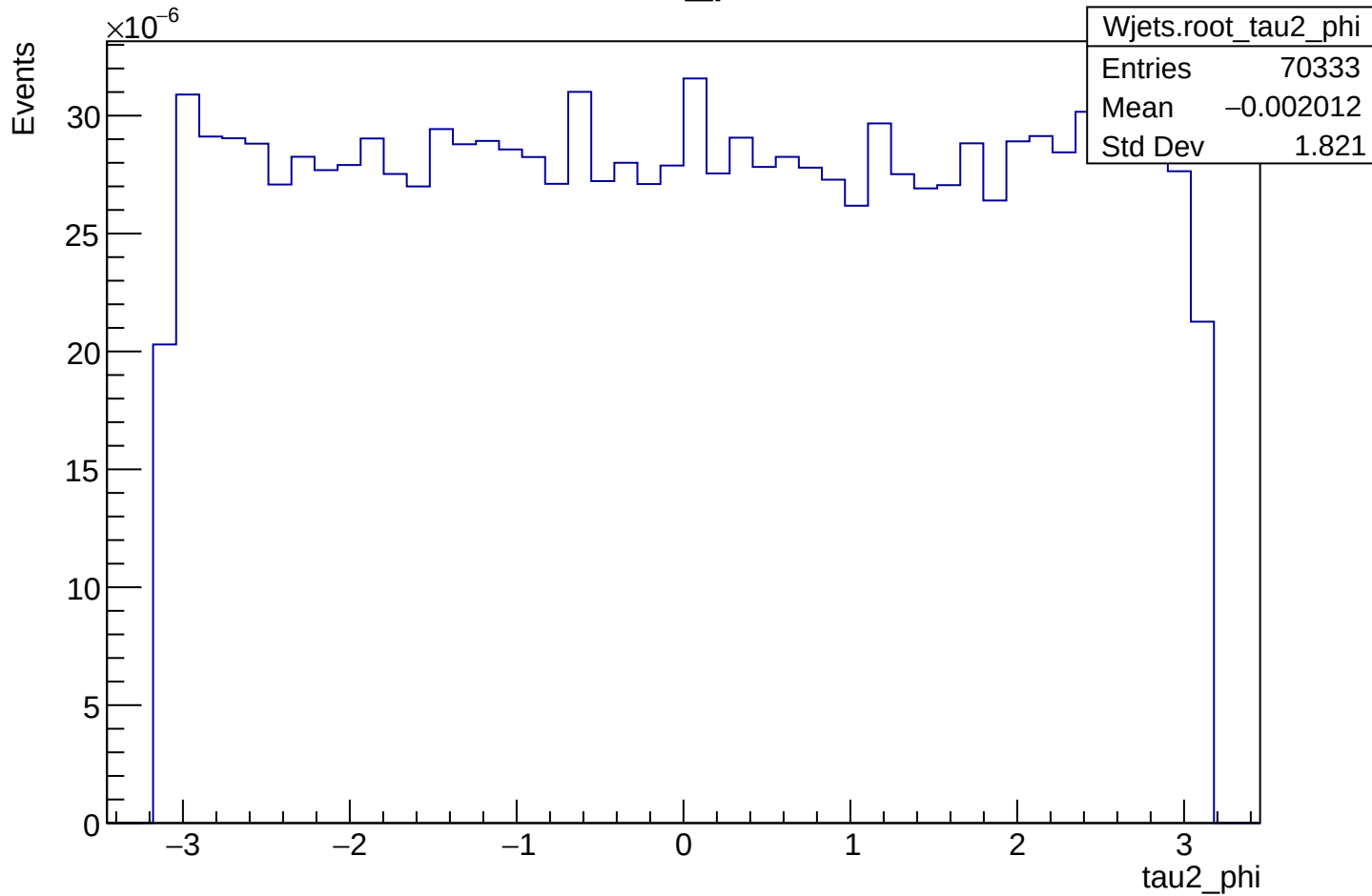
tau2_nprong

Events



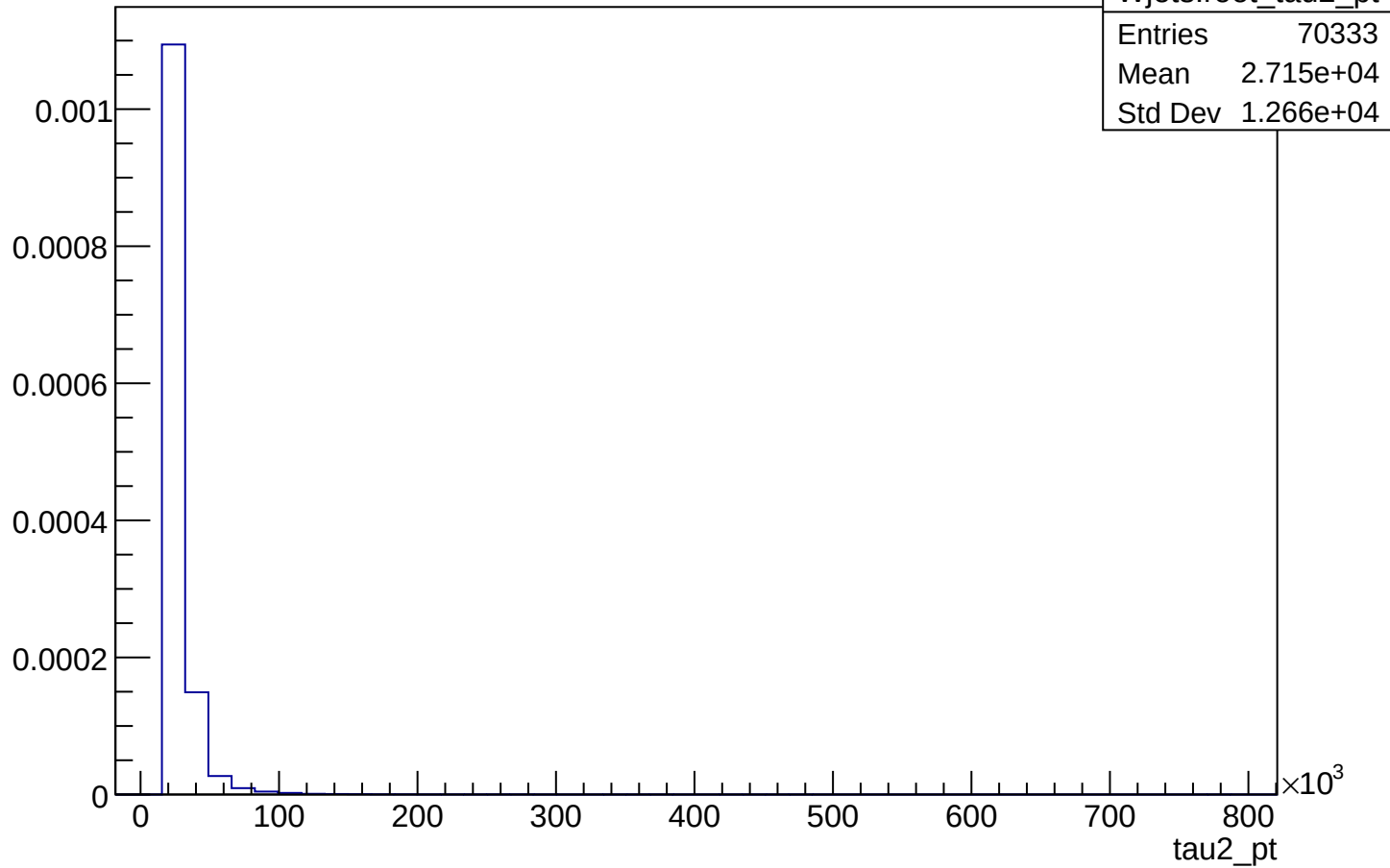
Wjets.root_tau2_nprong	
Entries	70333
Mean	1.359
Std Dev	0.7676

tau2_phi

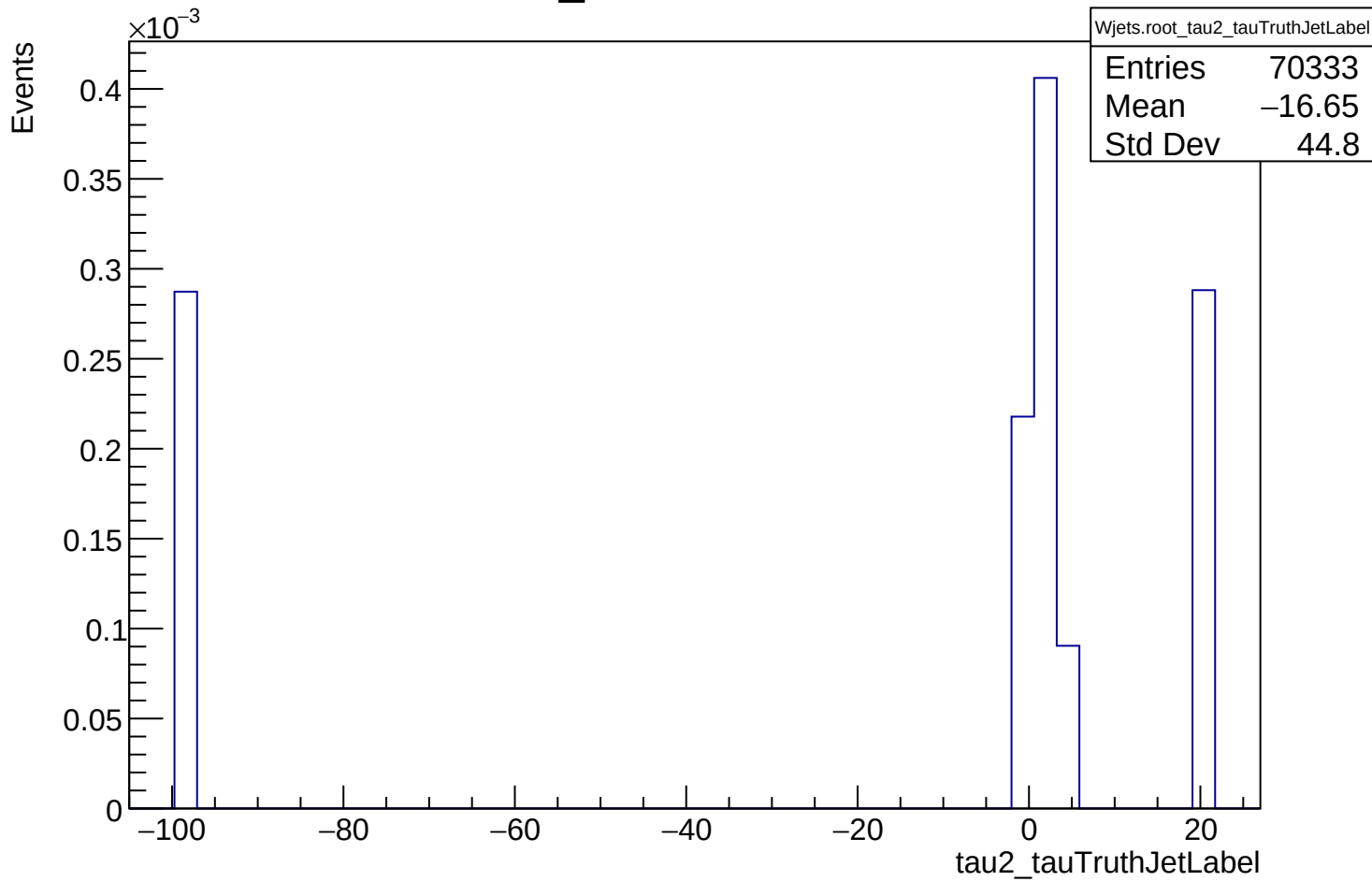


tau2_pt

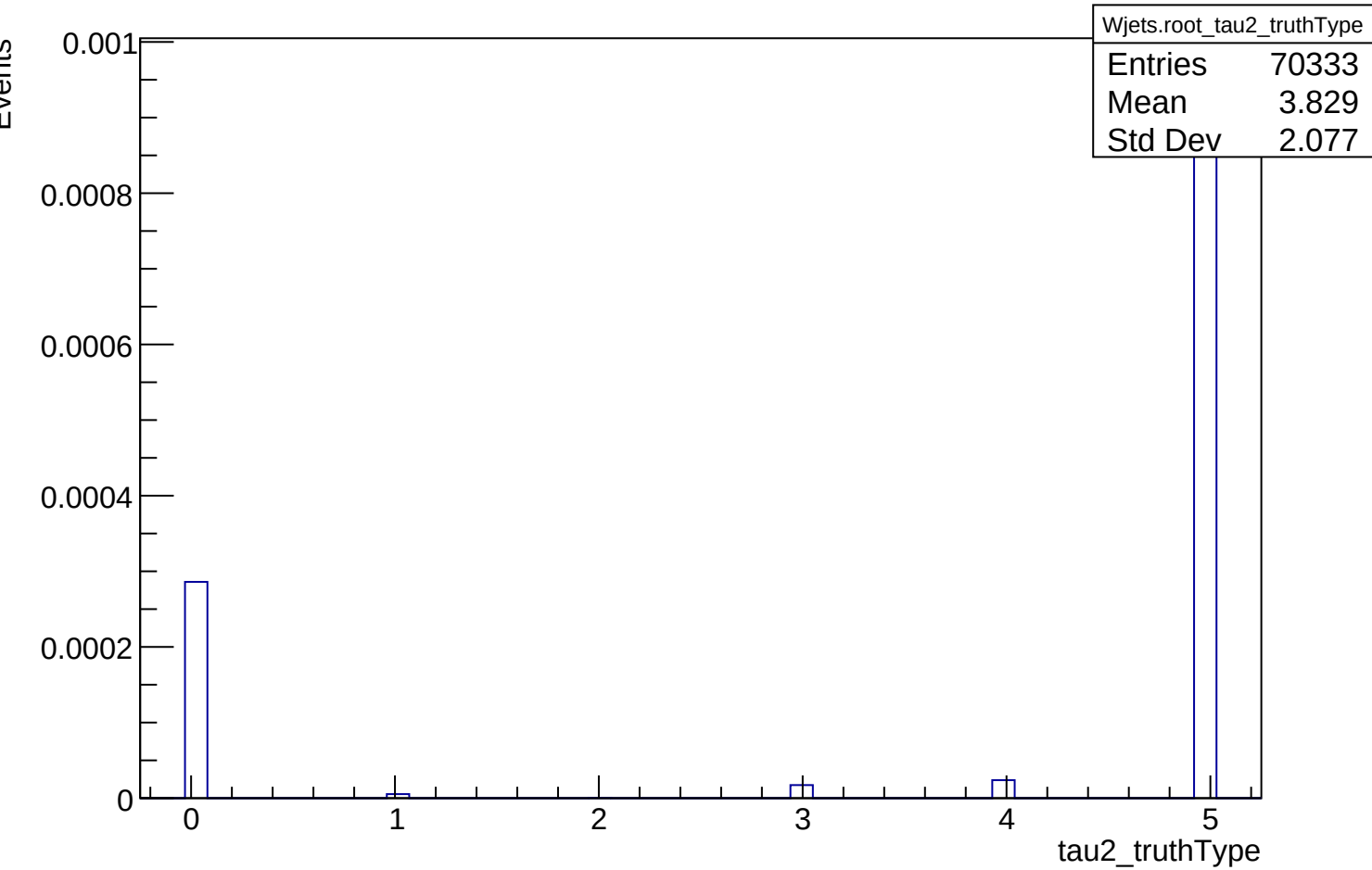
Events



tau2_tauTruthJetLabel

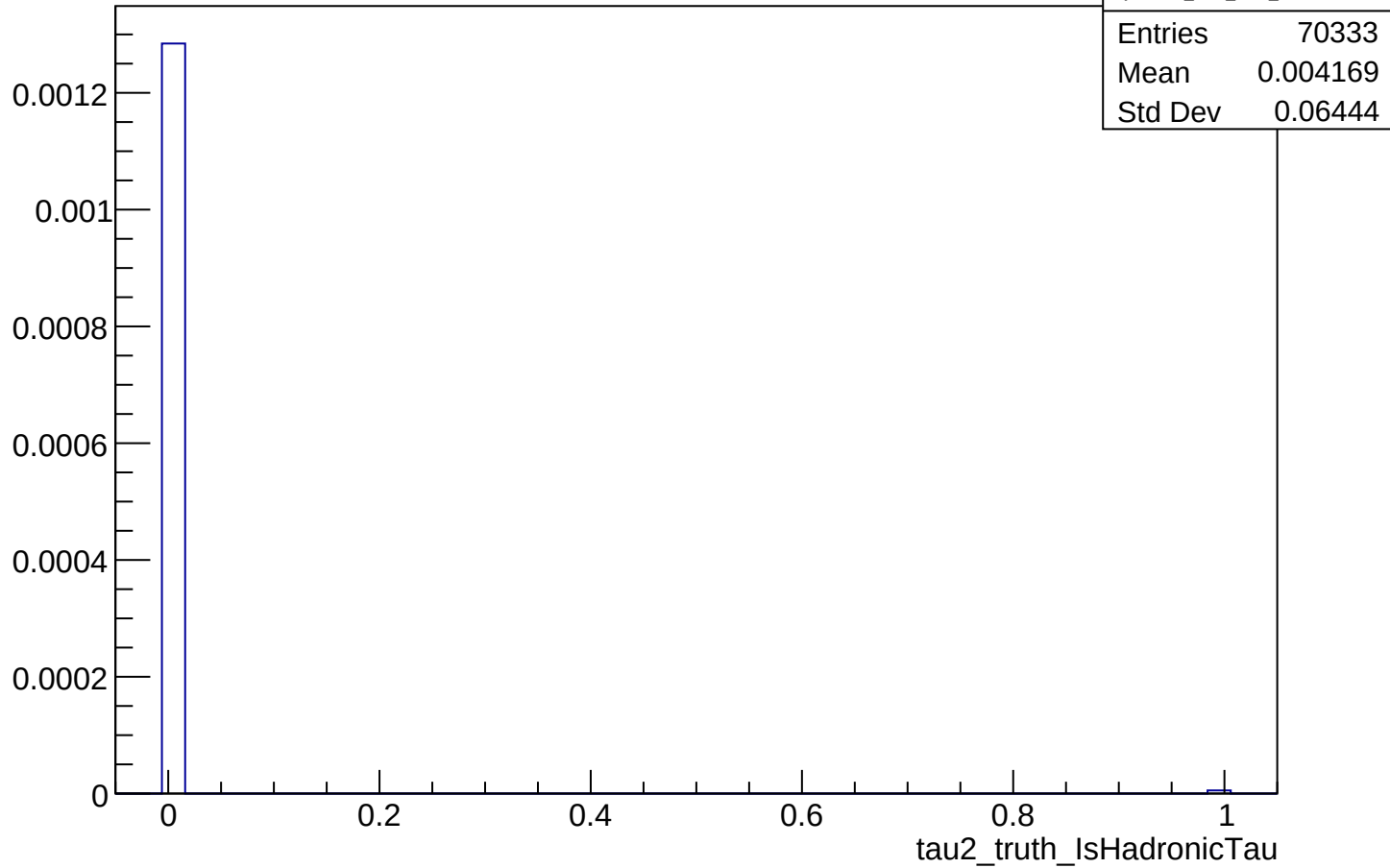


tau2_truthType

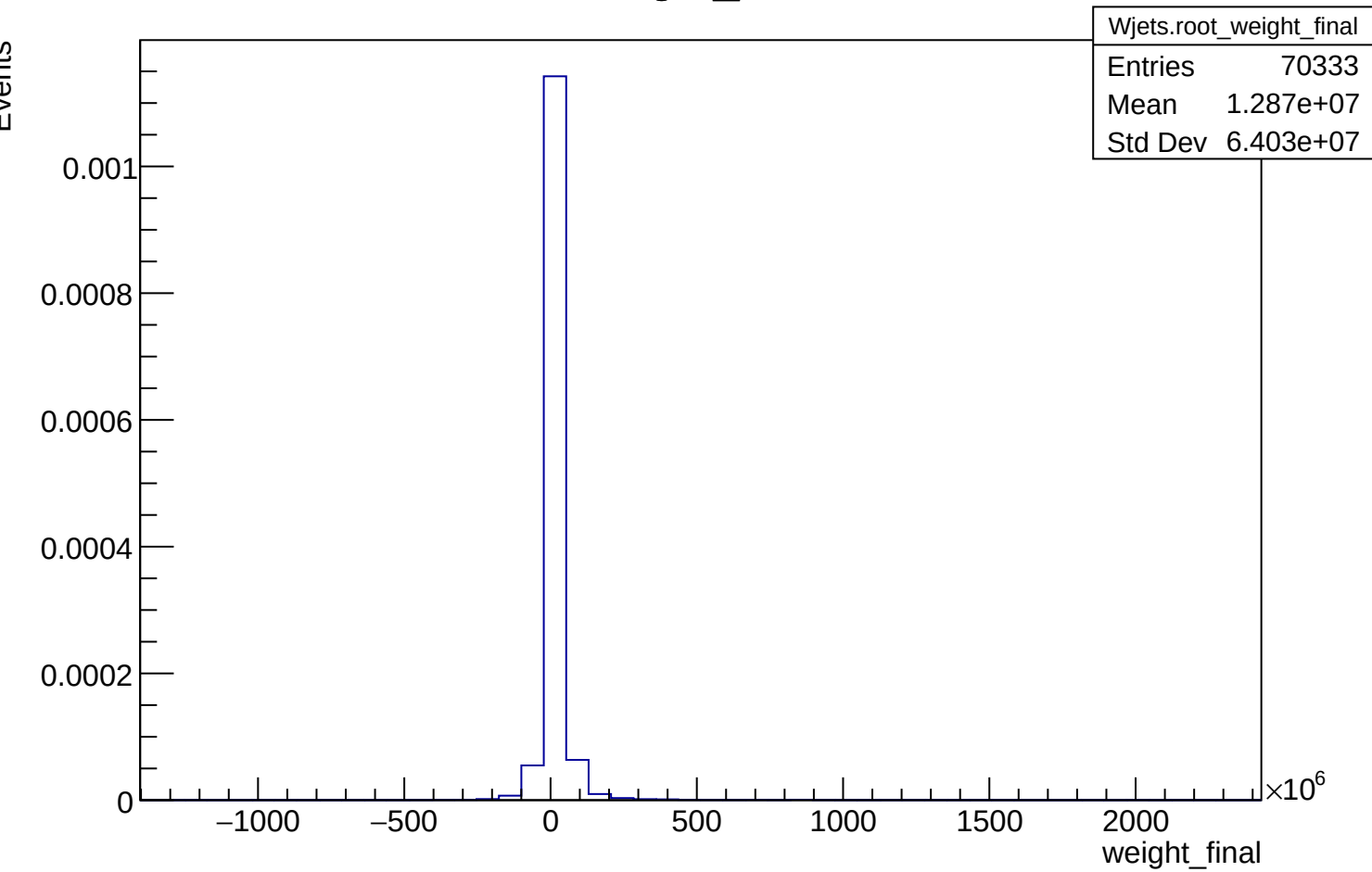


tau2_truth_IsHadronicTau

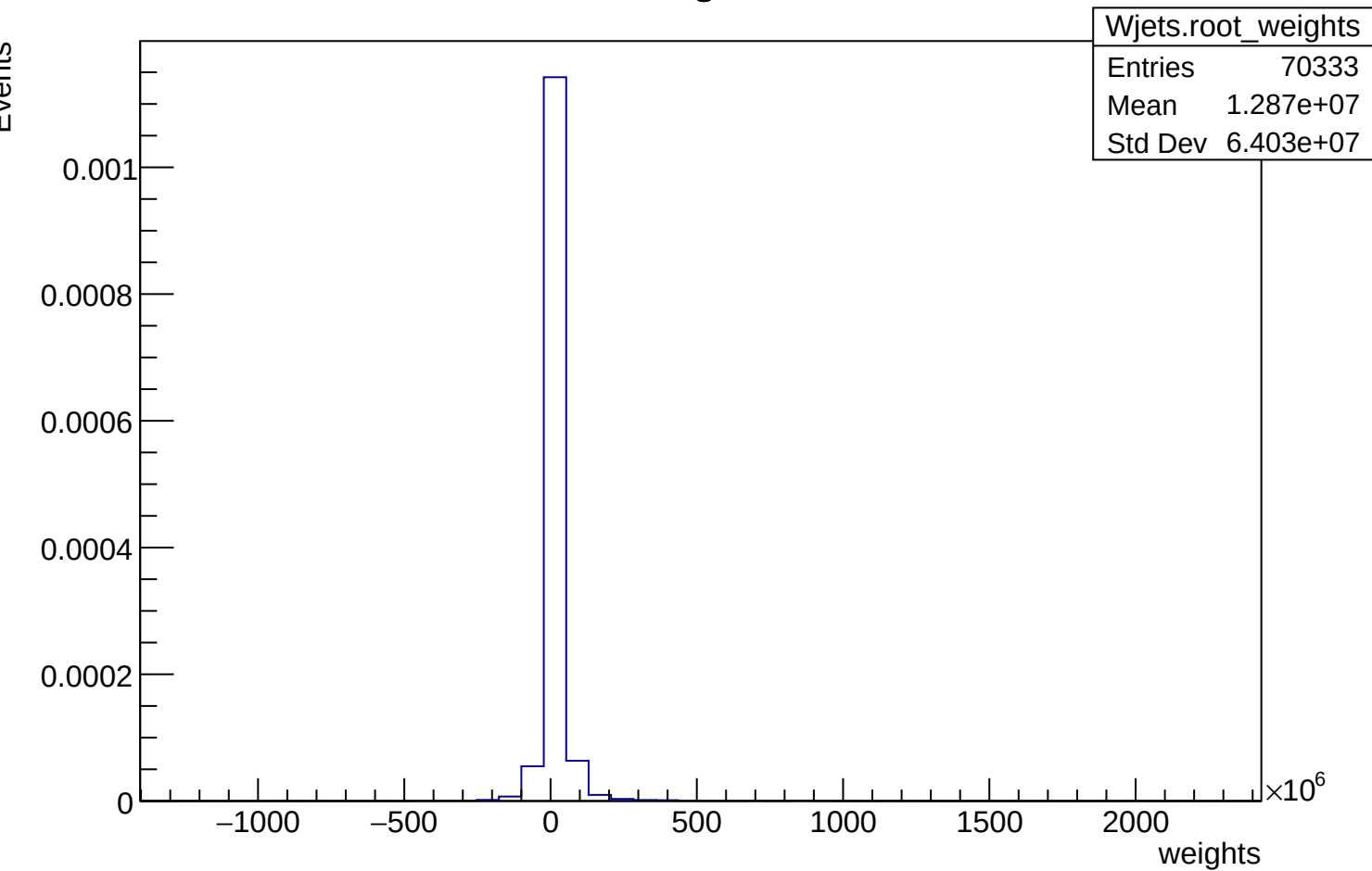
Events



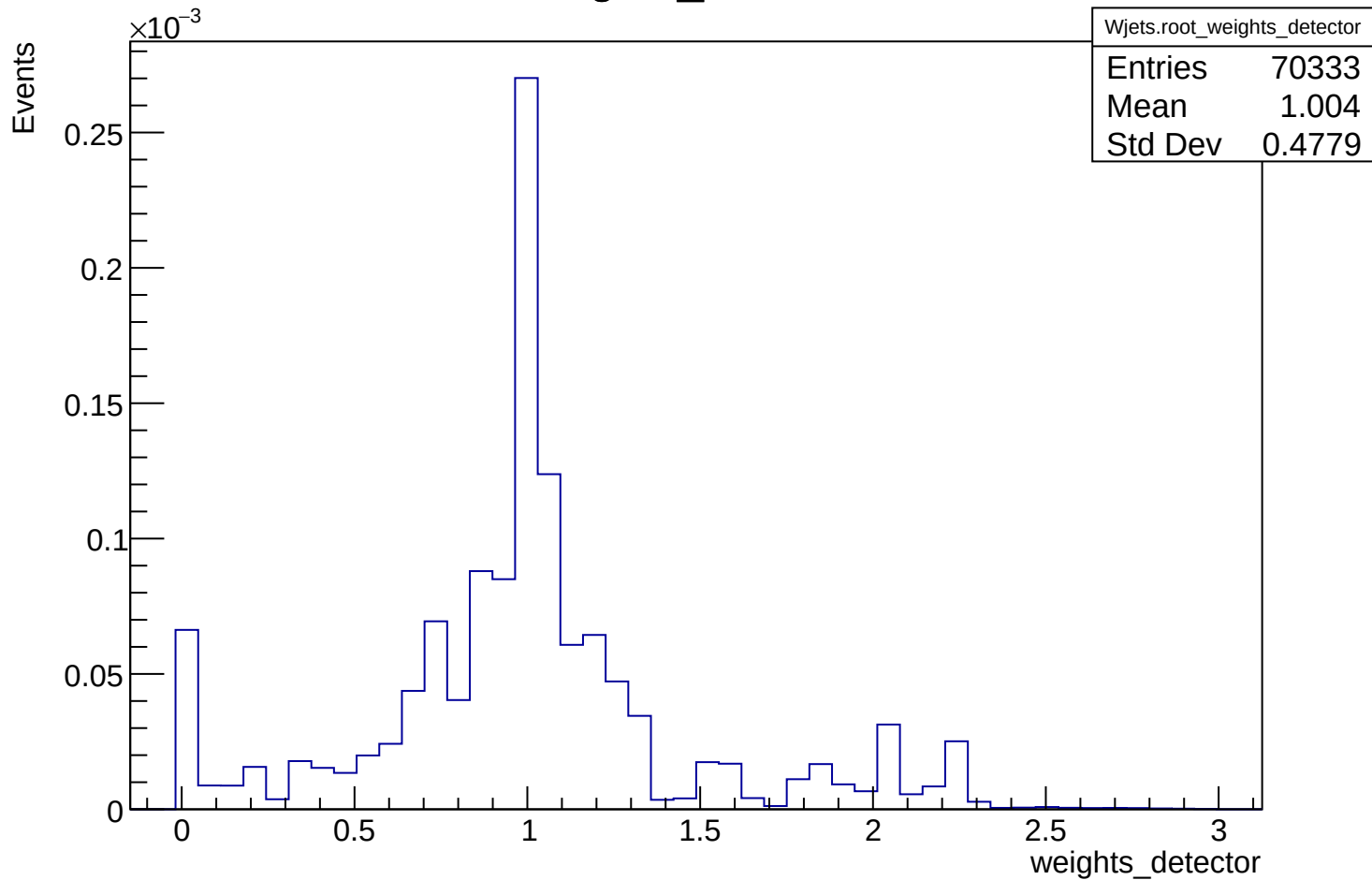
weight_final



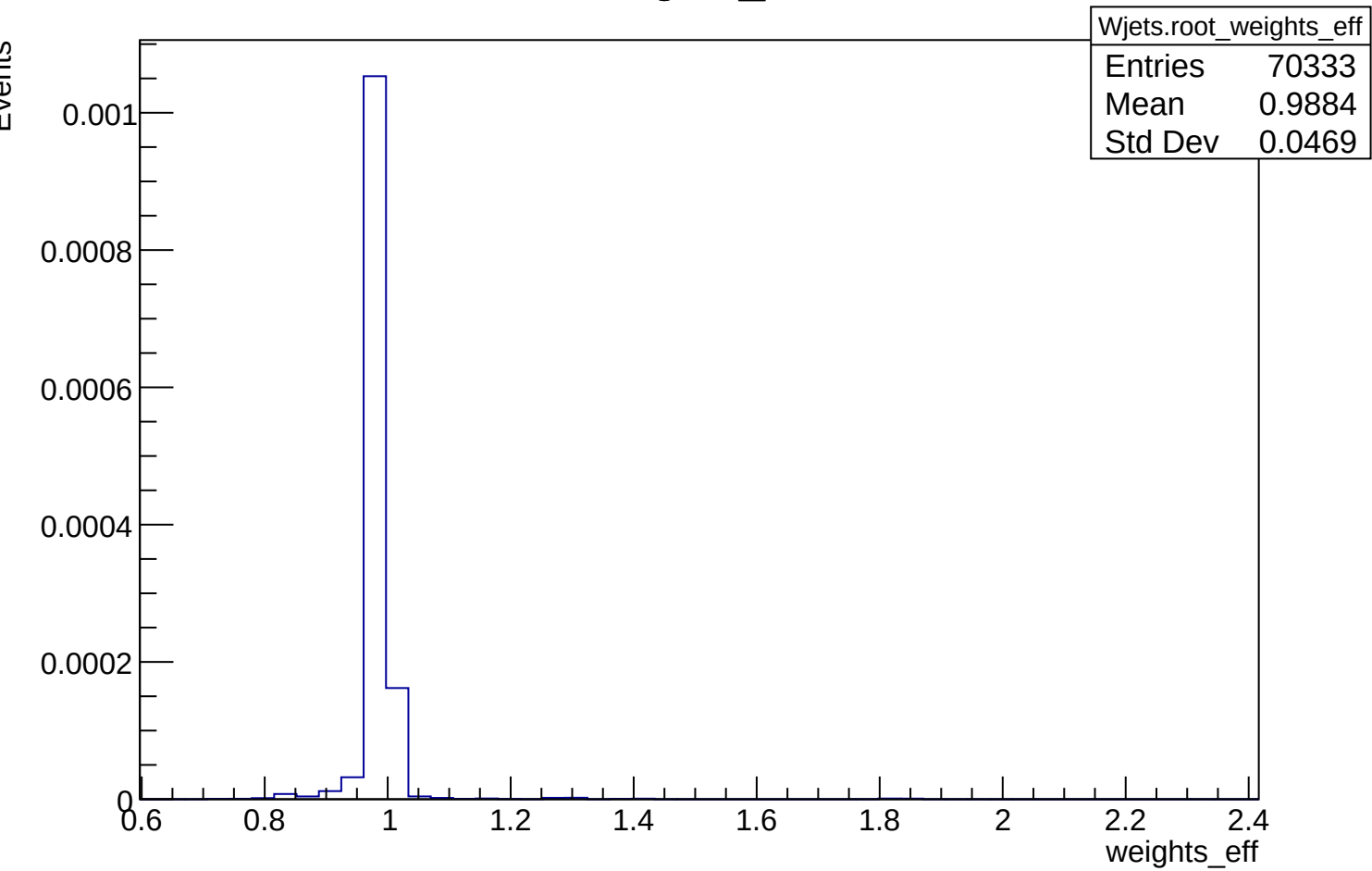
weights



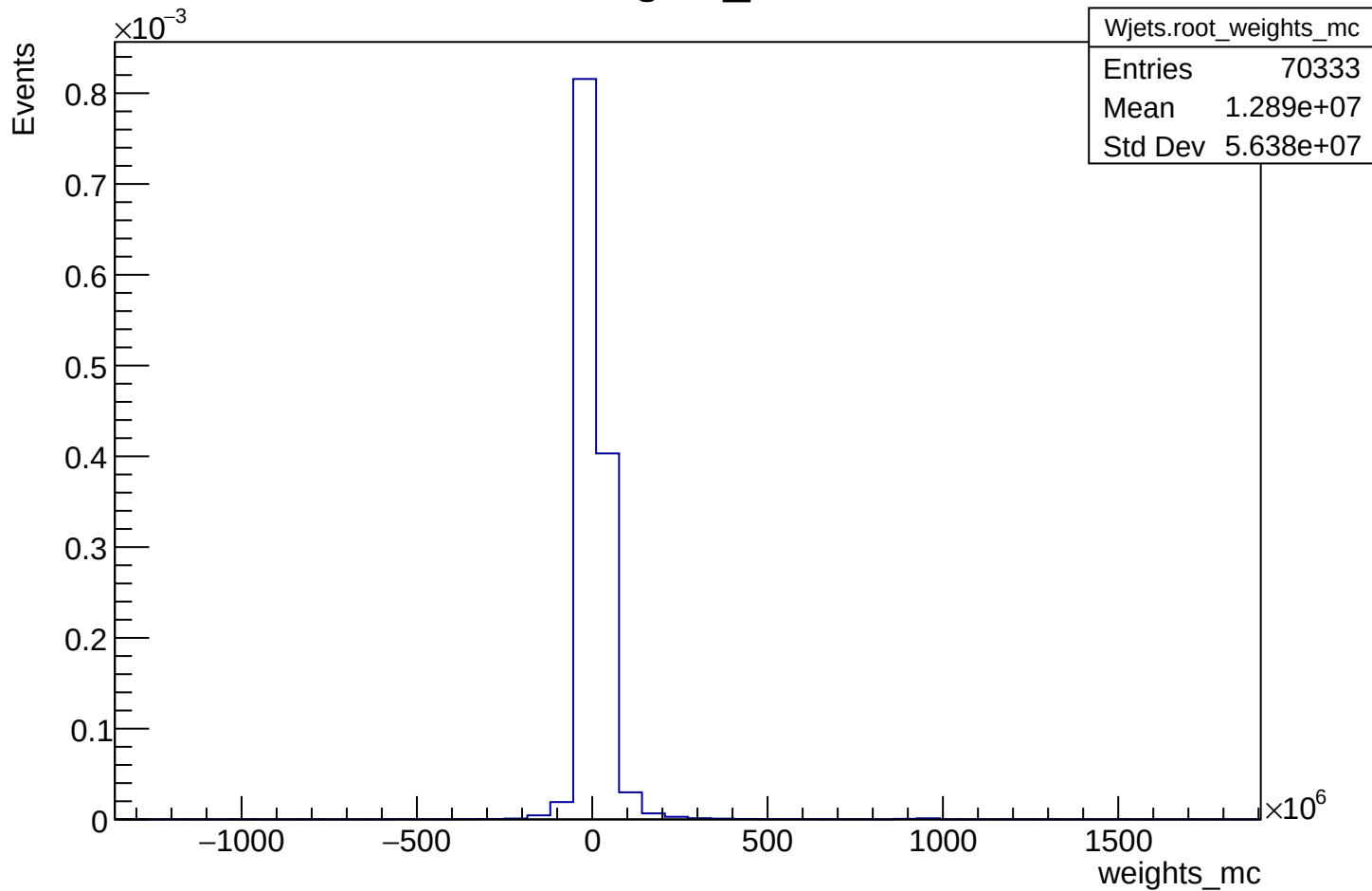
weights_detector



weights_eff

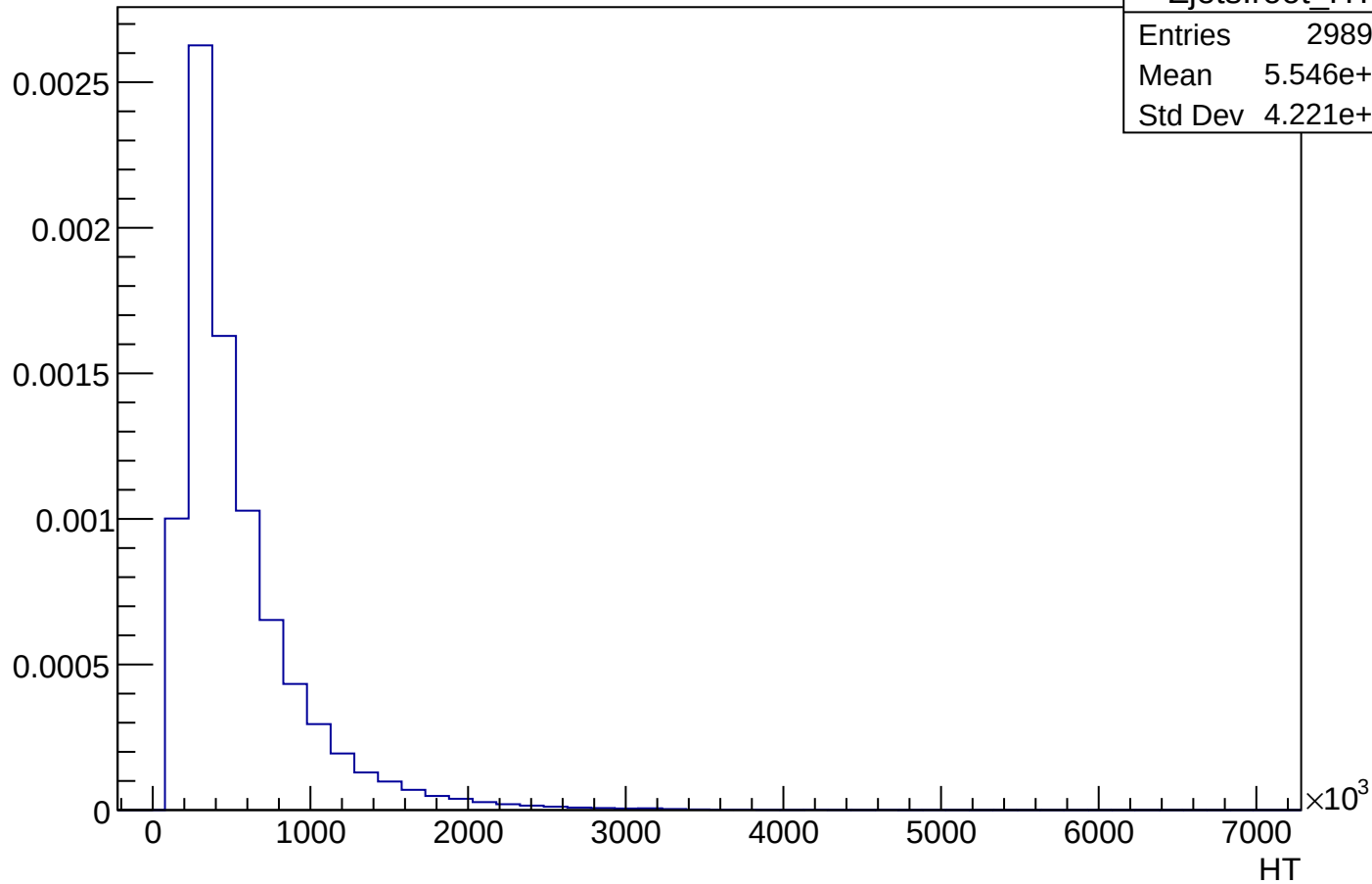


weights_mc

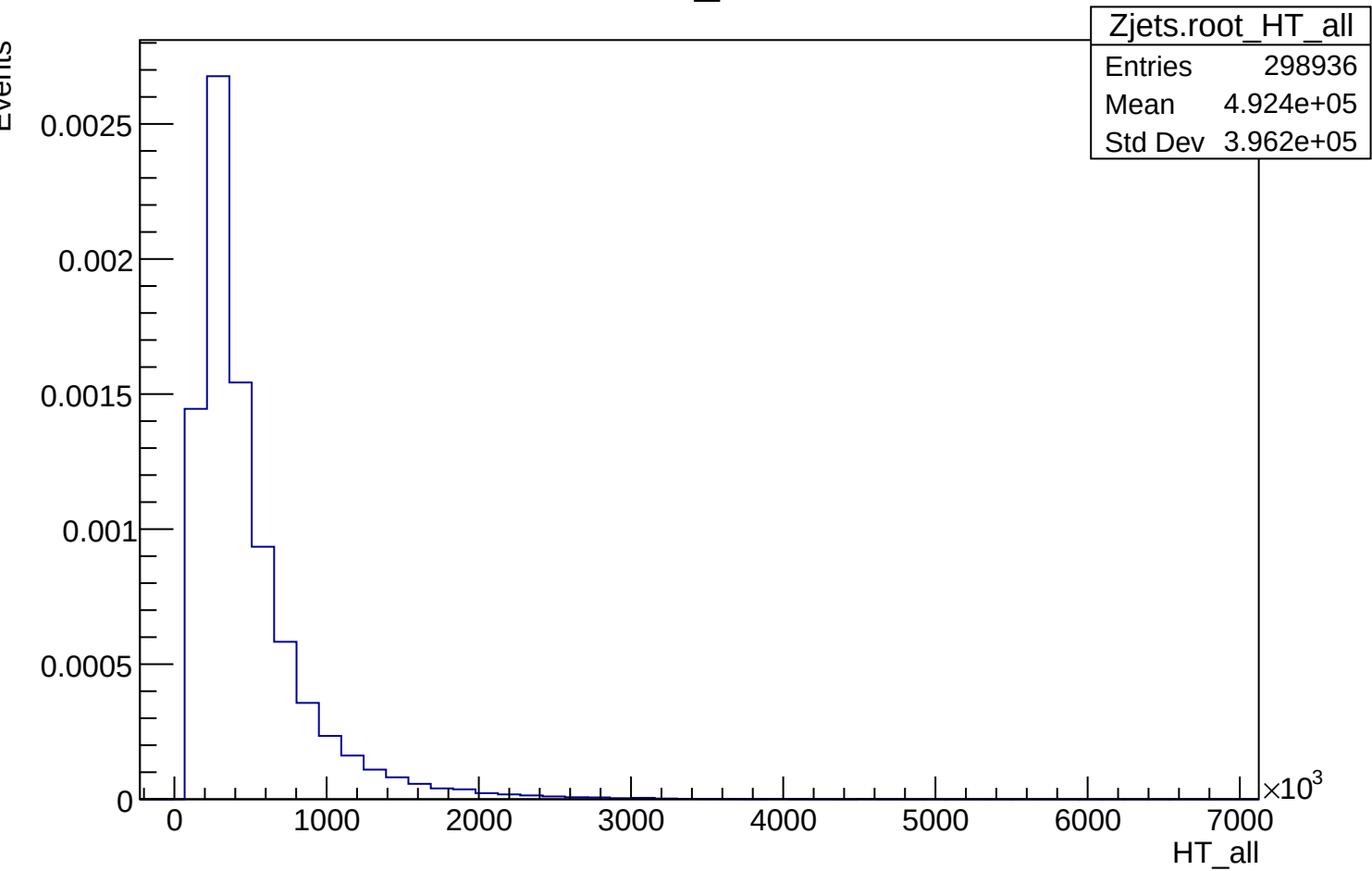


HT

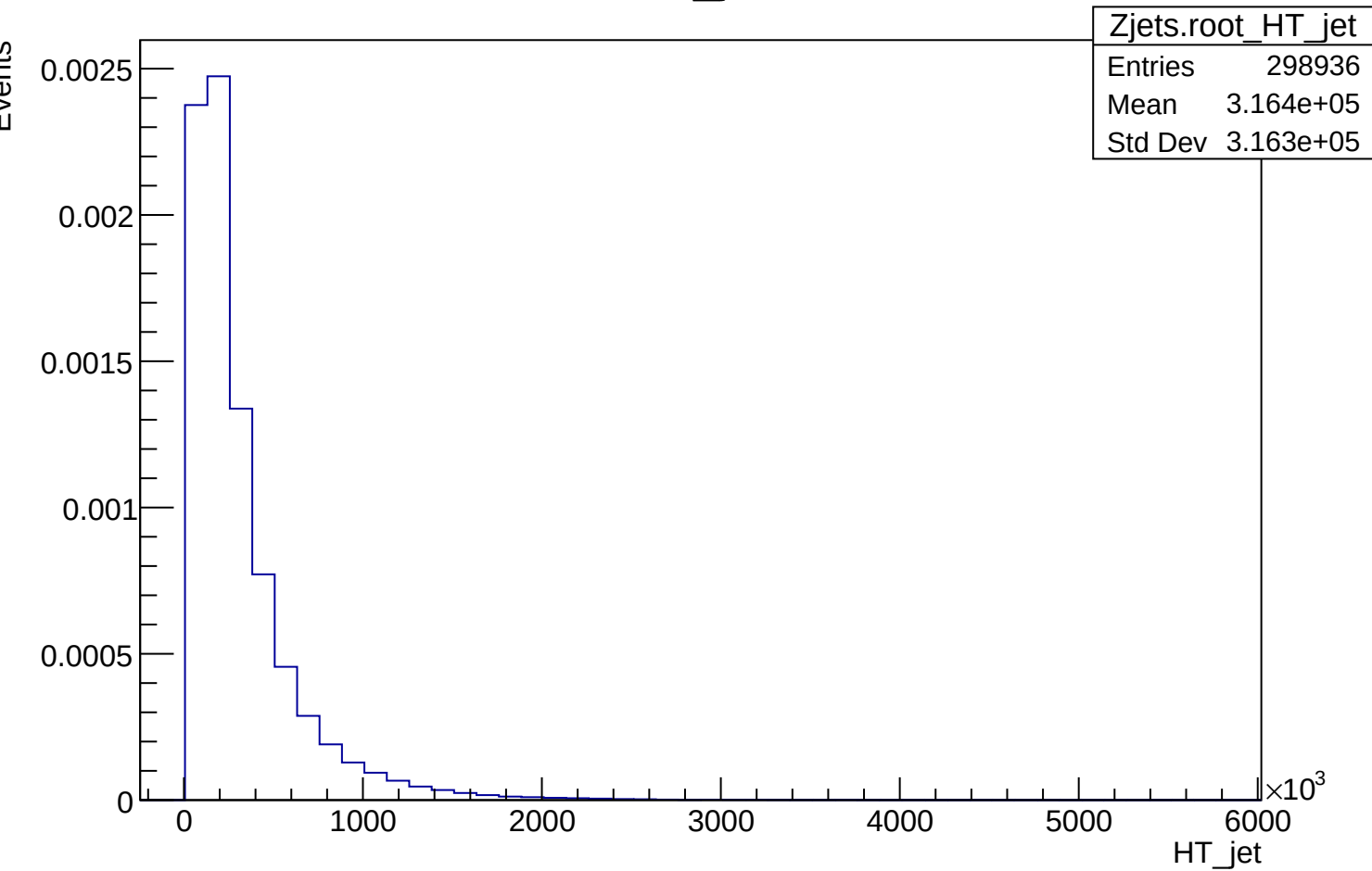
Events



HT_all



HT_jet



HT_lep

Events

0.005

0.004

0.003

0.002

0.001

0

0

500

1000

1500

2000

2500

3000

HT_lep

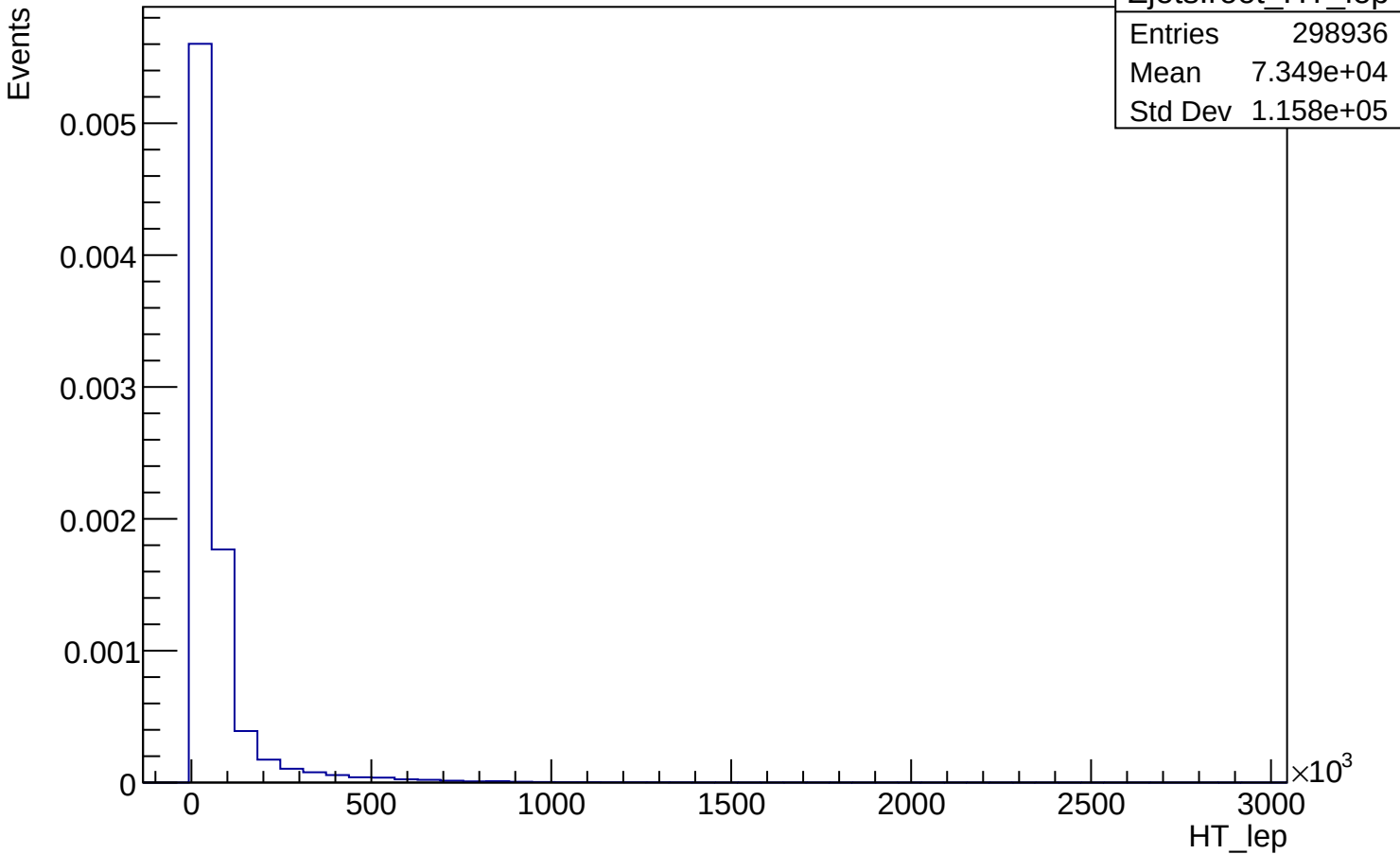
$\times 10^3$

Zjets.root_HT_lep

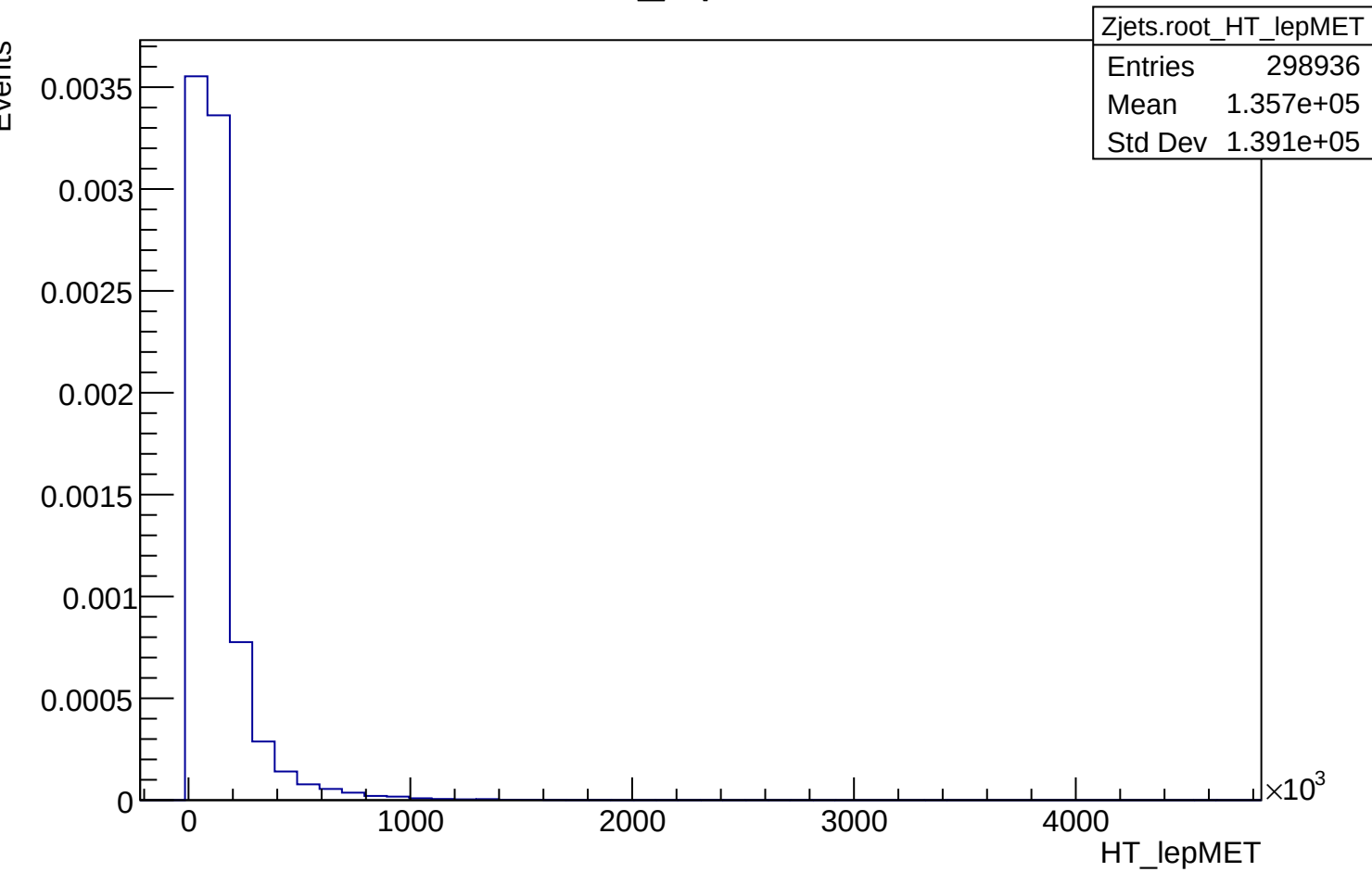
Entries 298936

Mean 7.349e+04

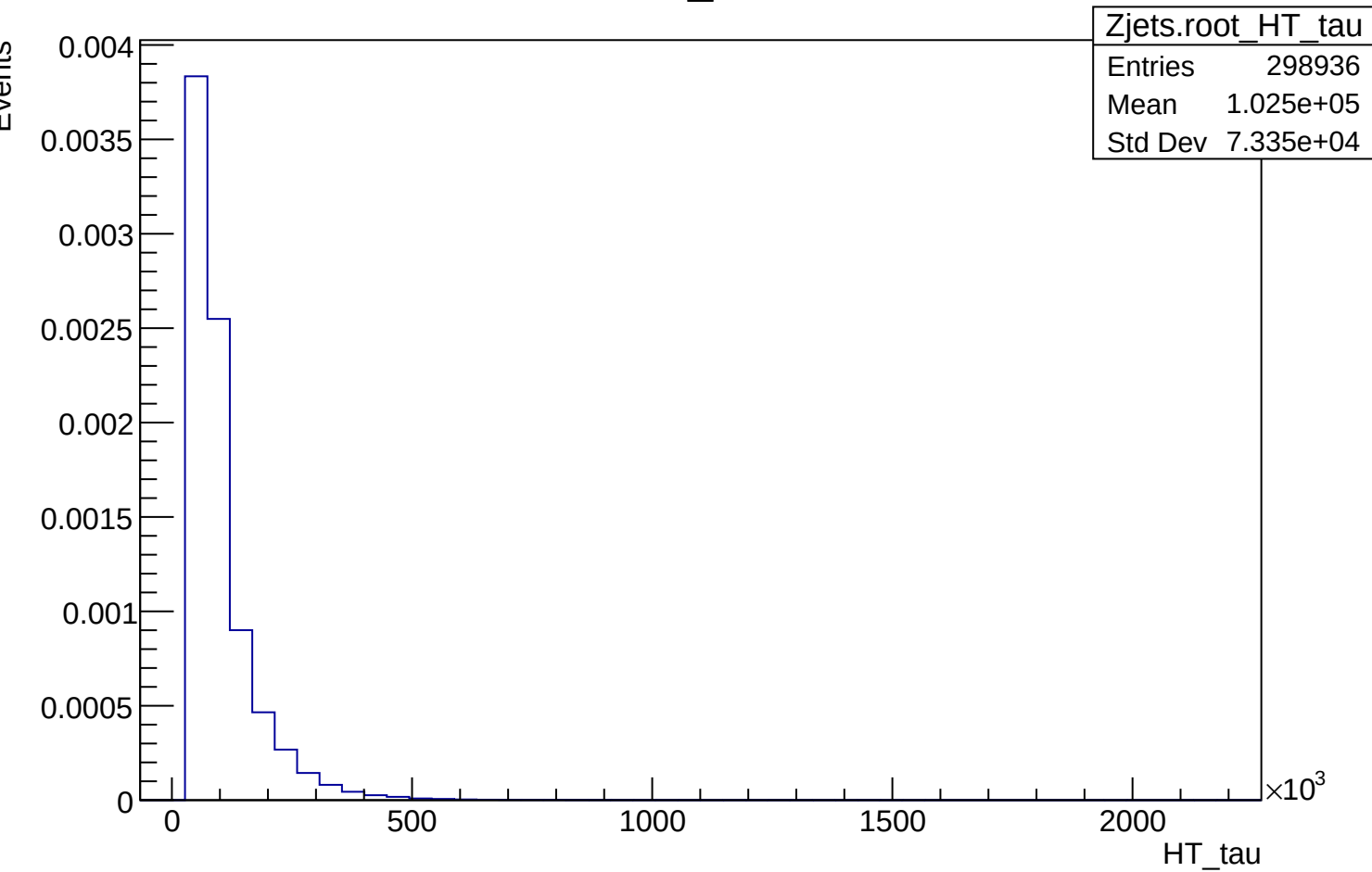
Std Dev 1.158e+05



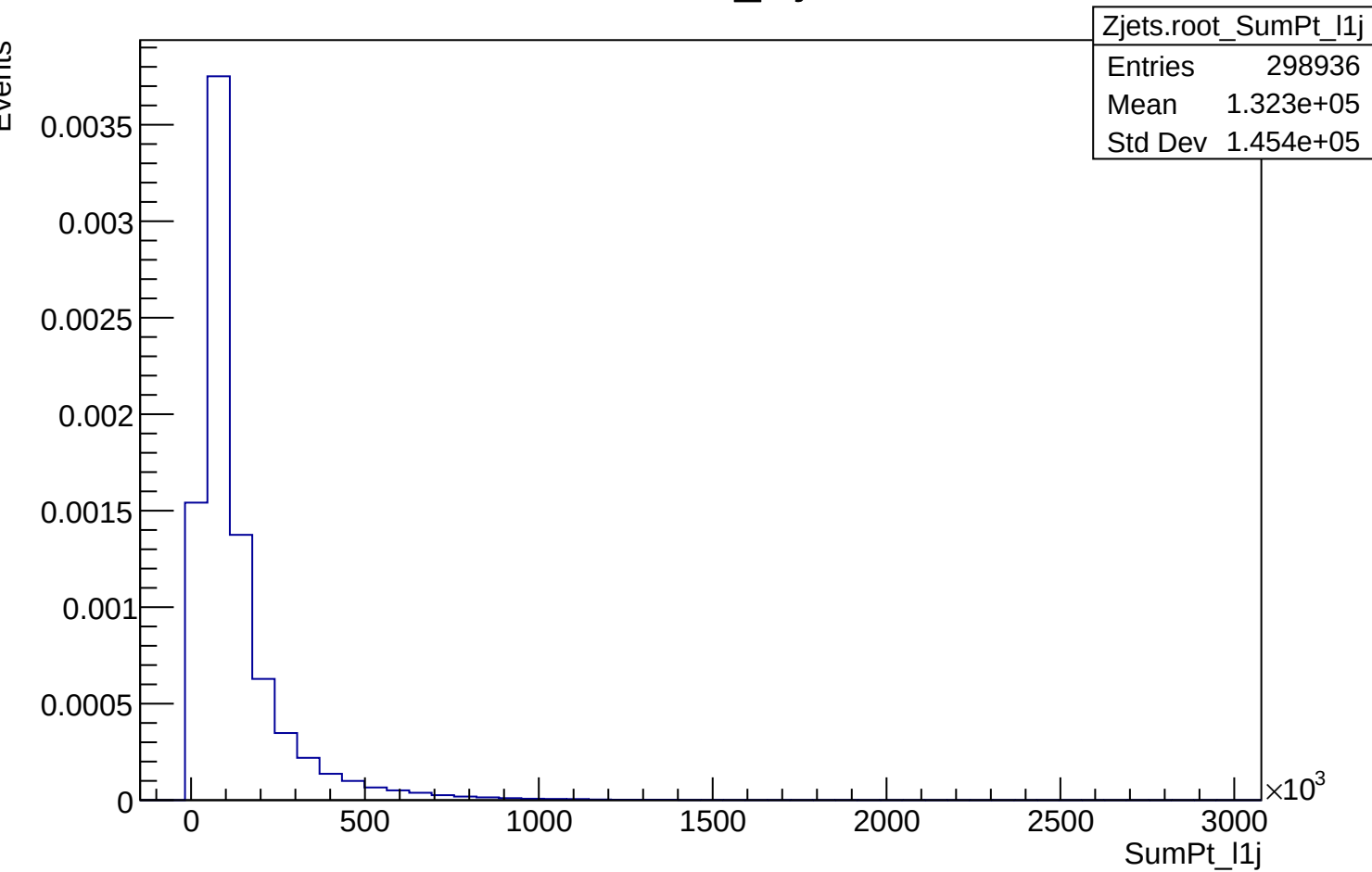
HT_lepMET



HT_tau

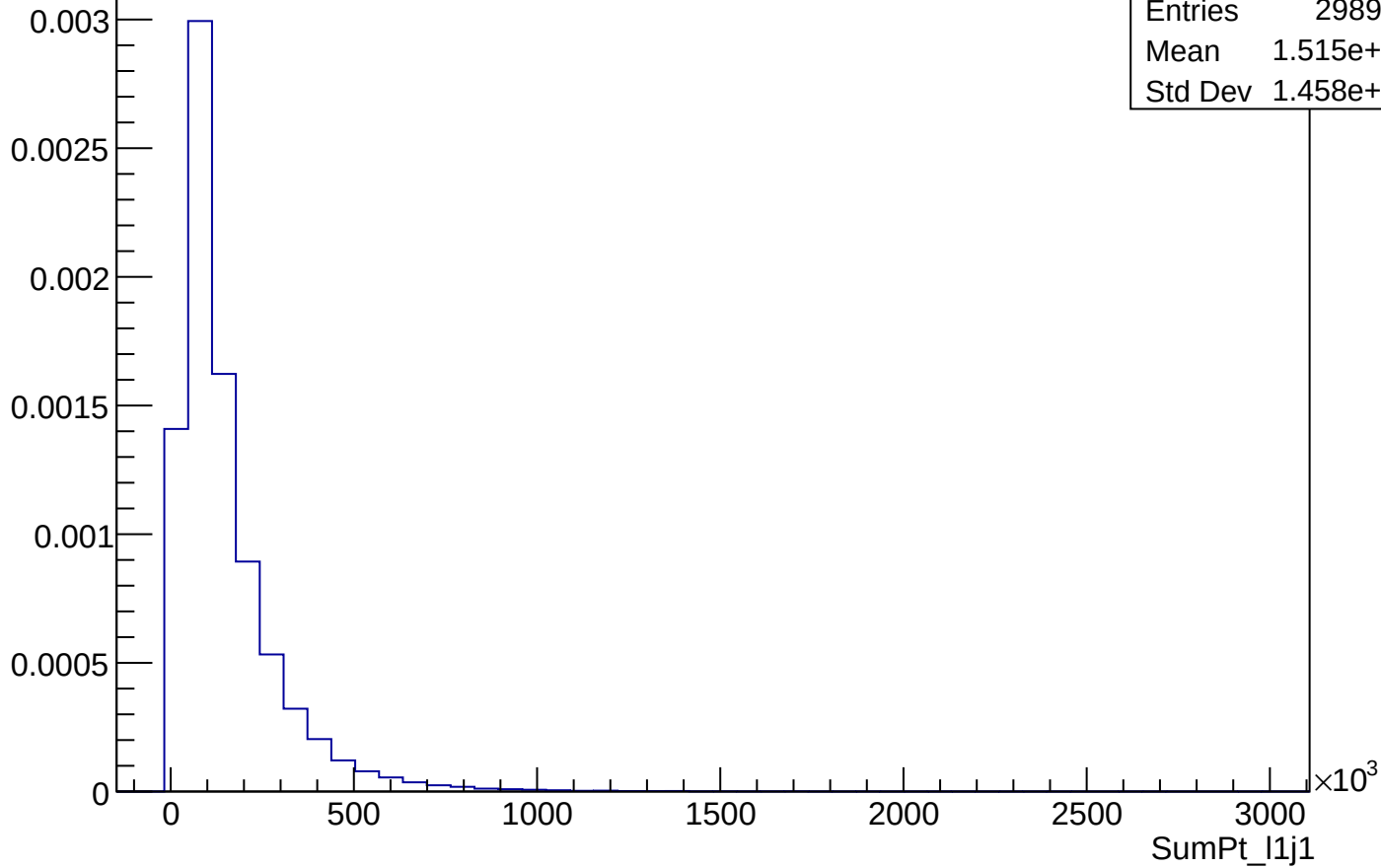


SumPt_l1j

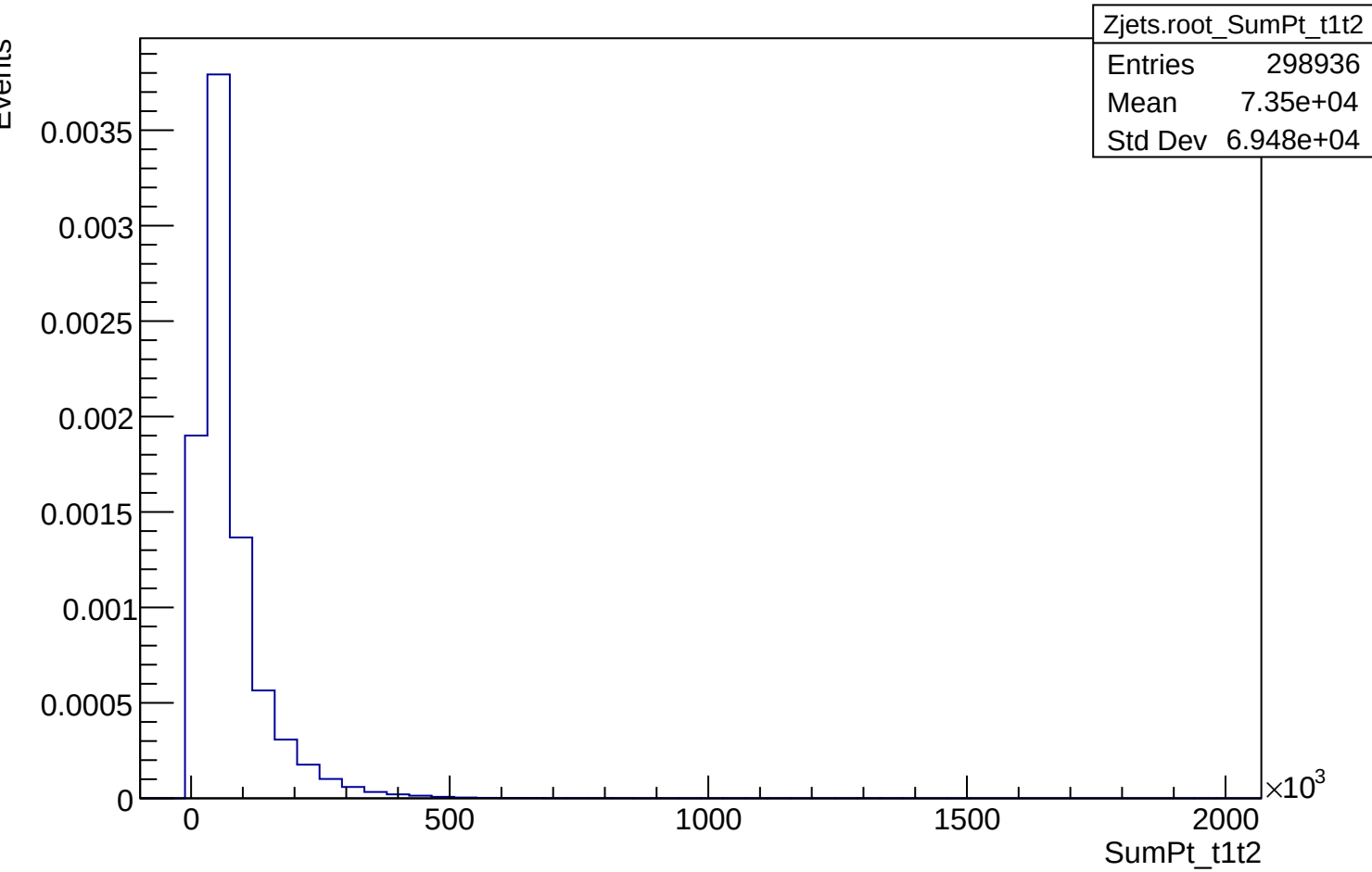


SumPt_l1j1

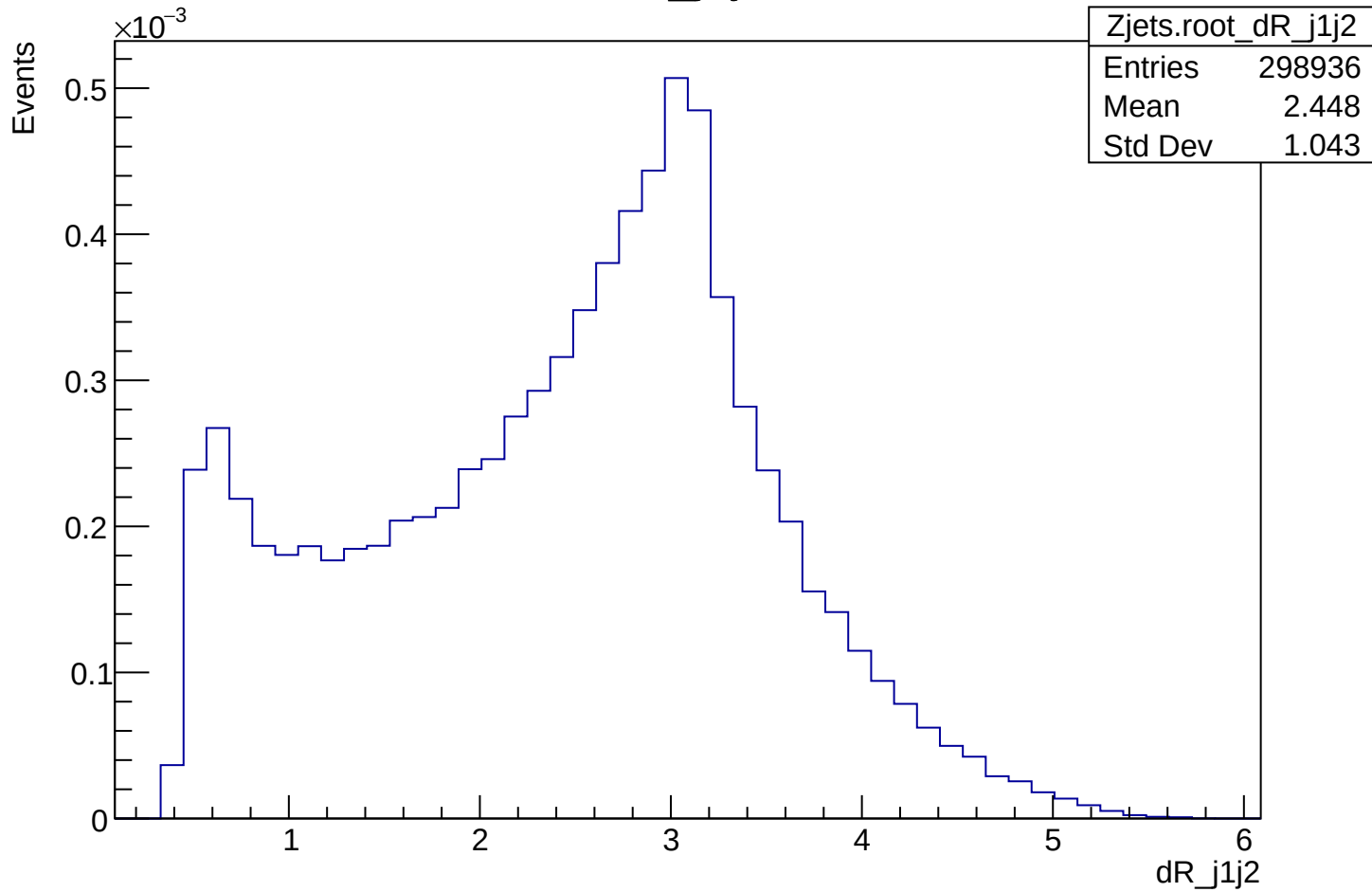
Events



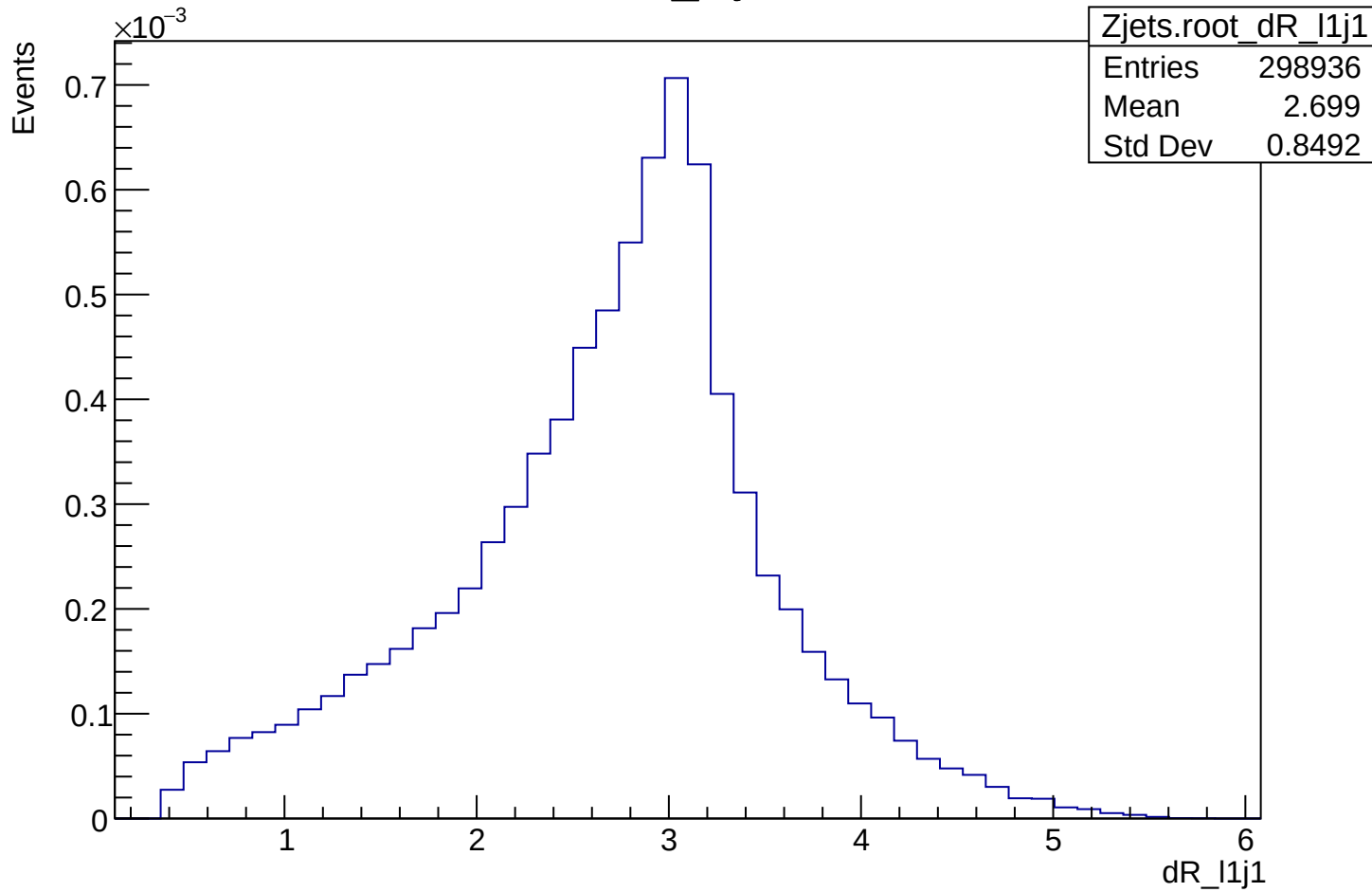
SumPt_t1t2



dR_j1j2

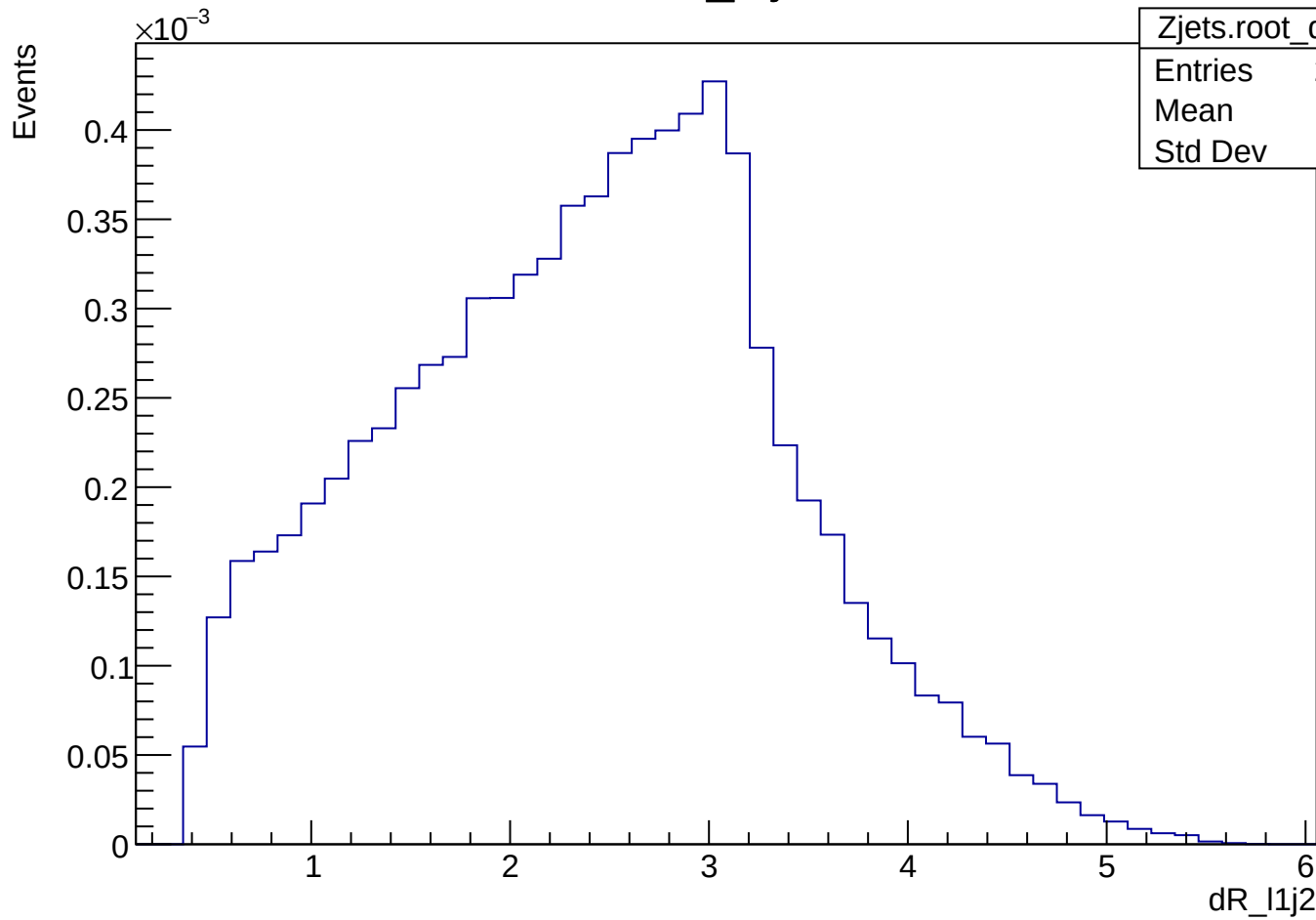


dR_l1j1

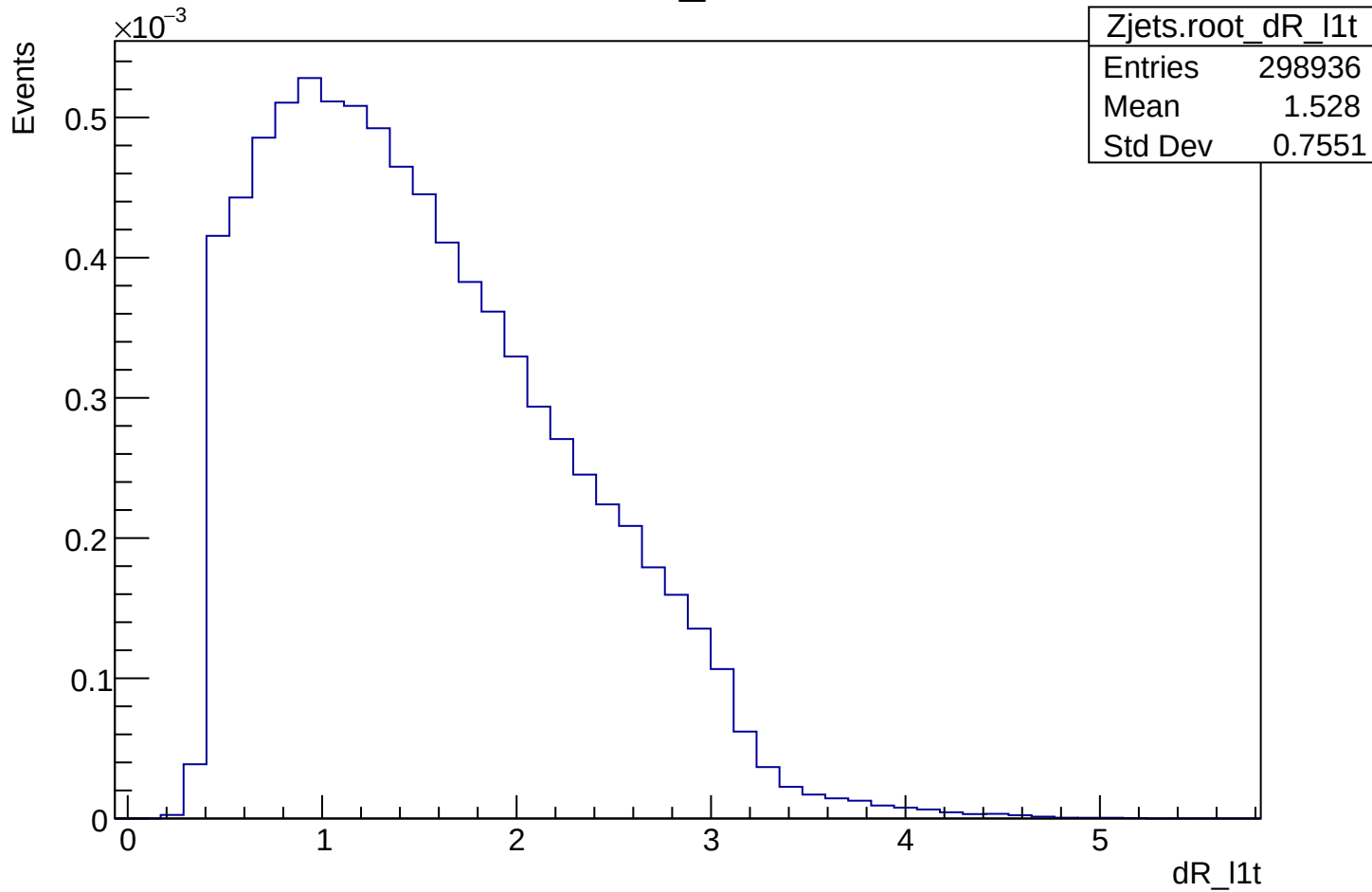


dR_l1j2

Zjets.root_dR_l1j2	
Entries	298936
Mean	2.406
Std Dev	0.9793

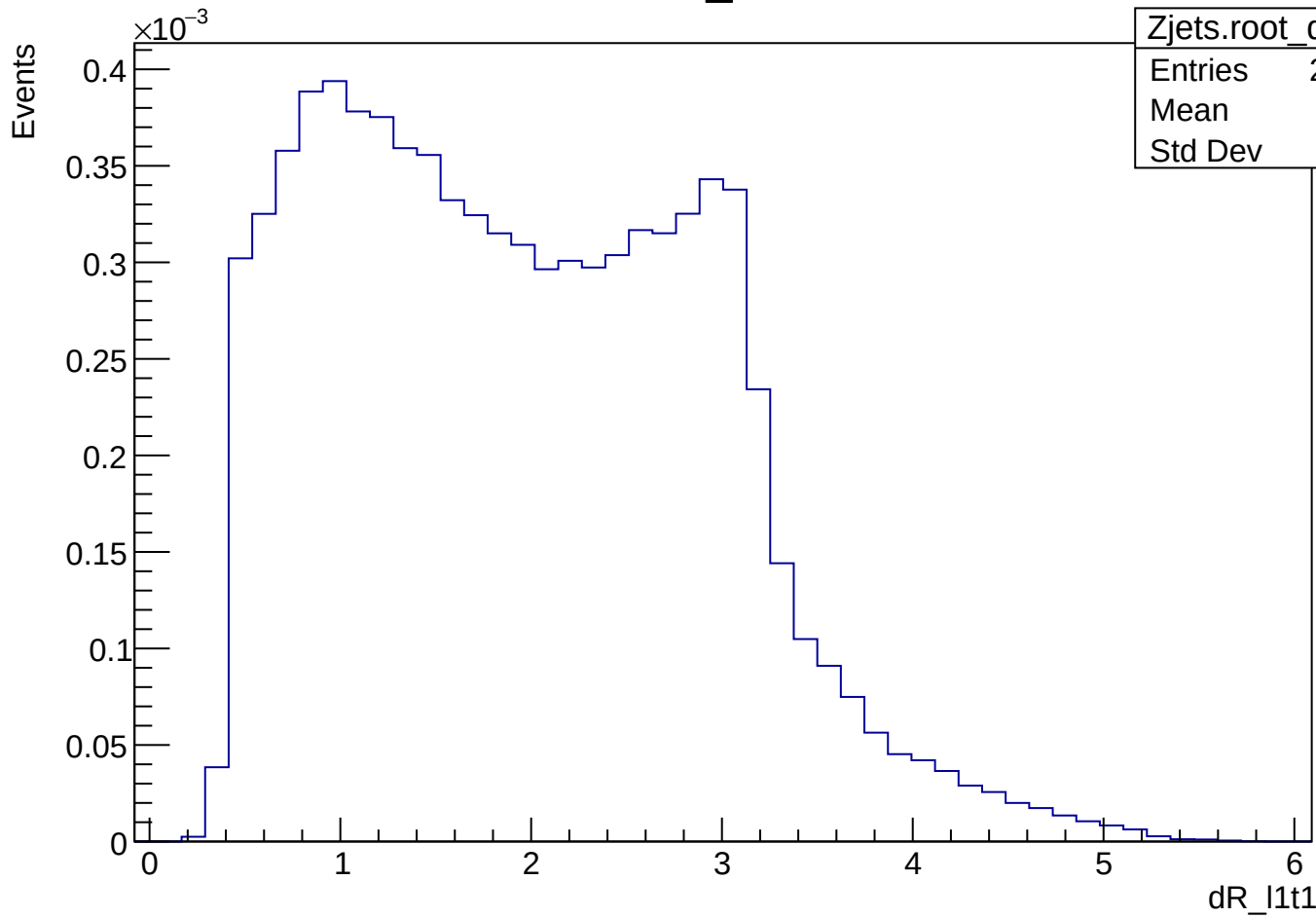


dR_l1t

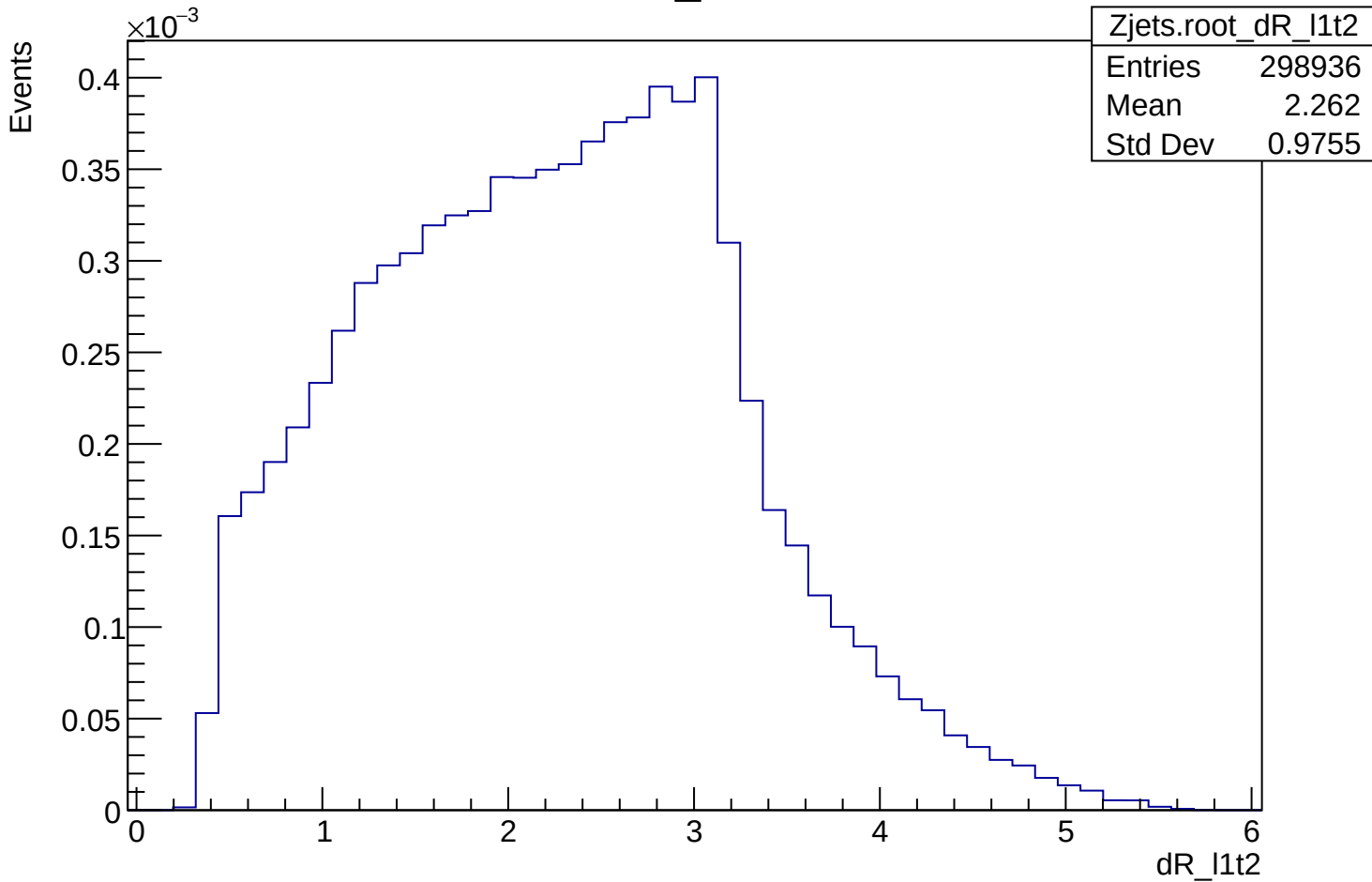


dR_l1t1

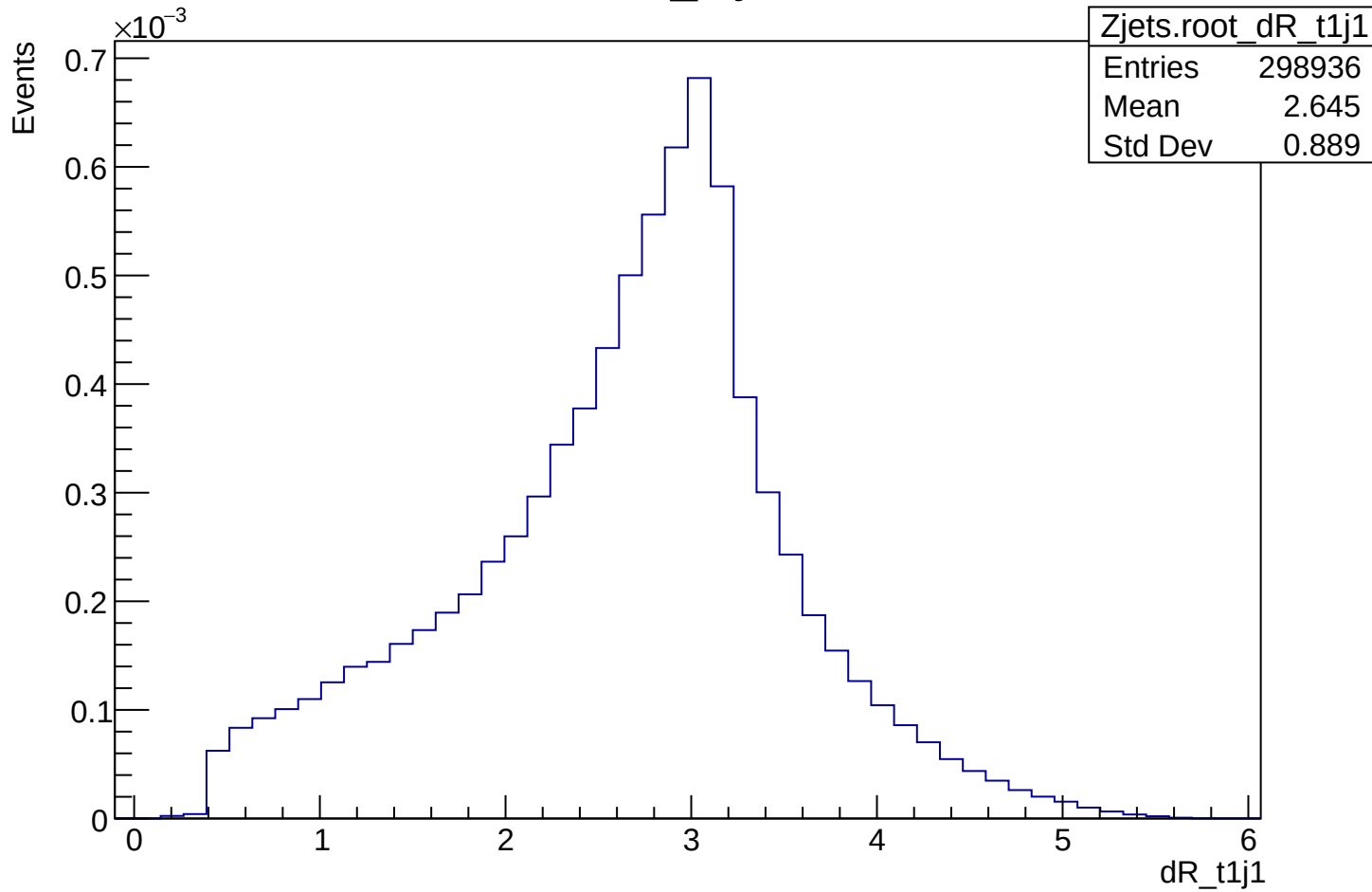
Zjets.root_dR_l1t1	
Entries	298936
Mean	1.955
Std Dev	0.9788



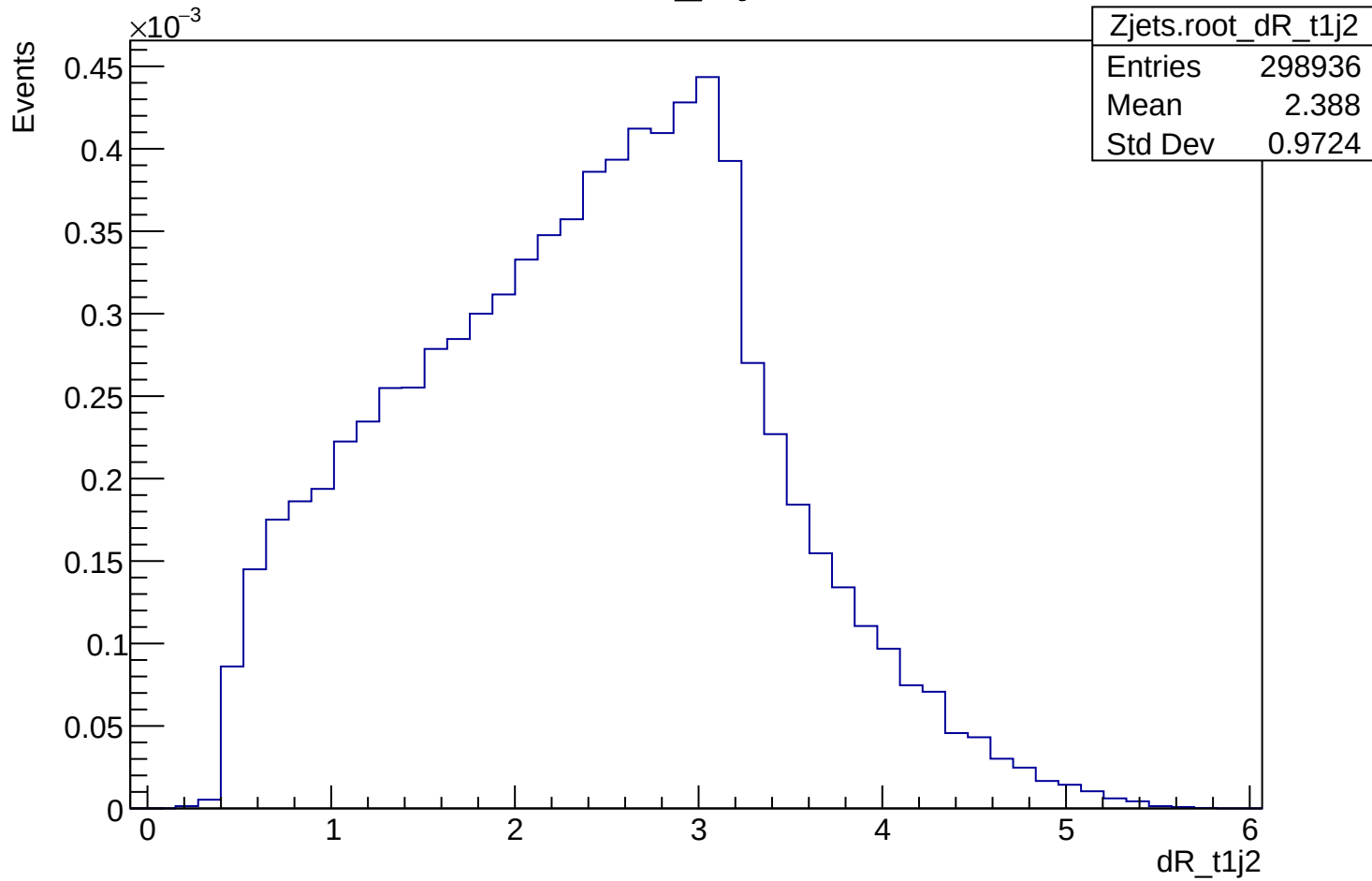
dR_l1t2



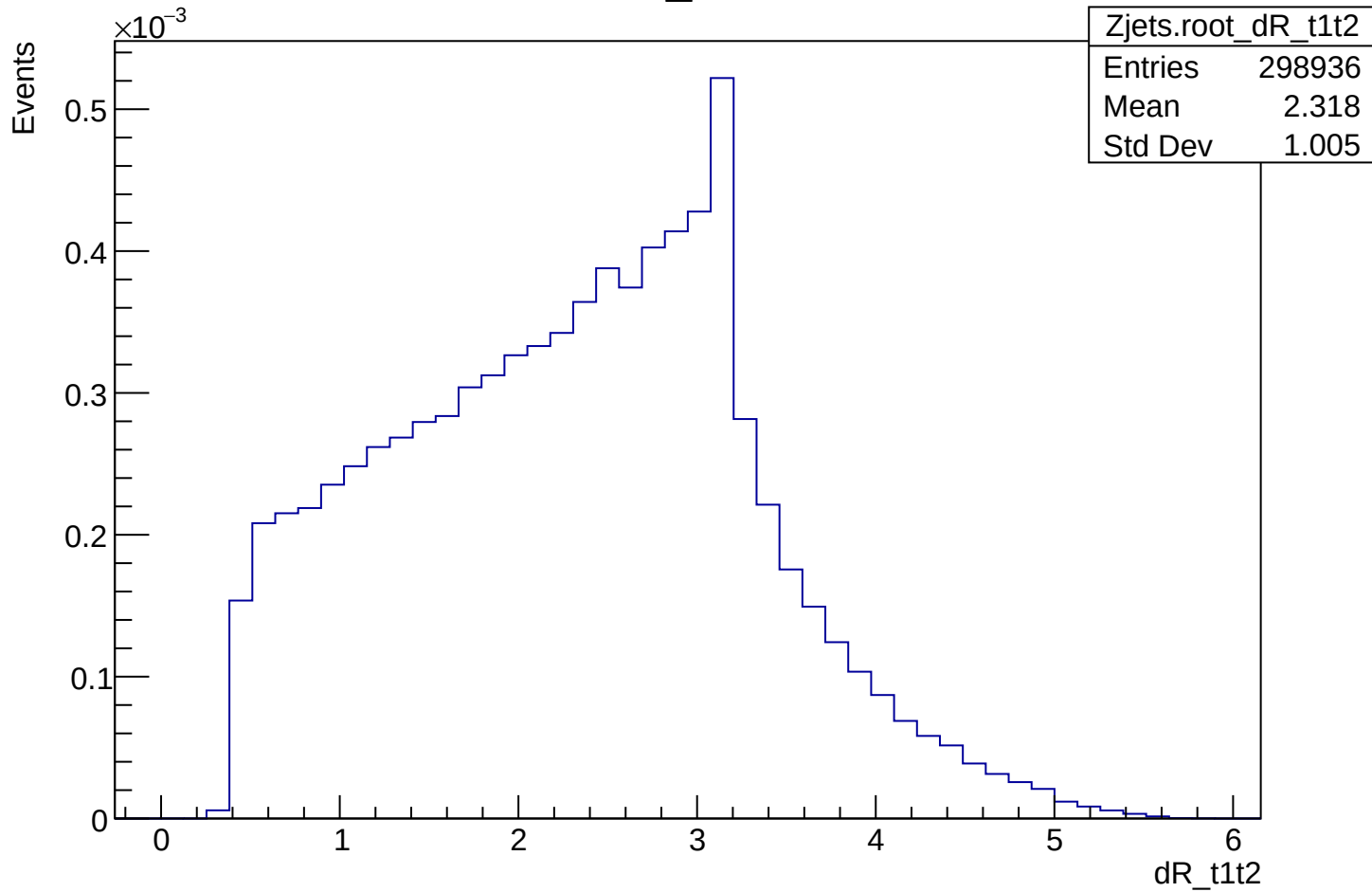
dR_t1j1



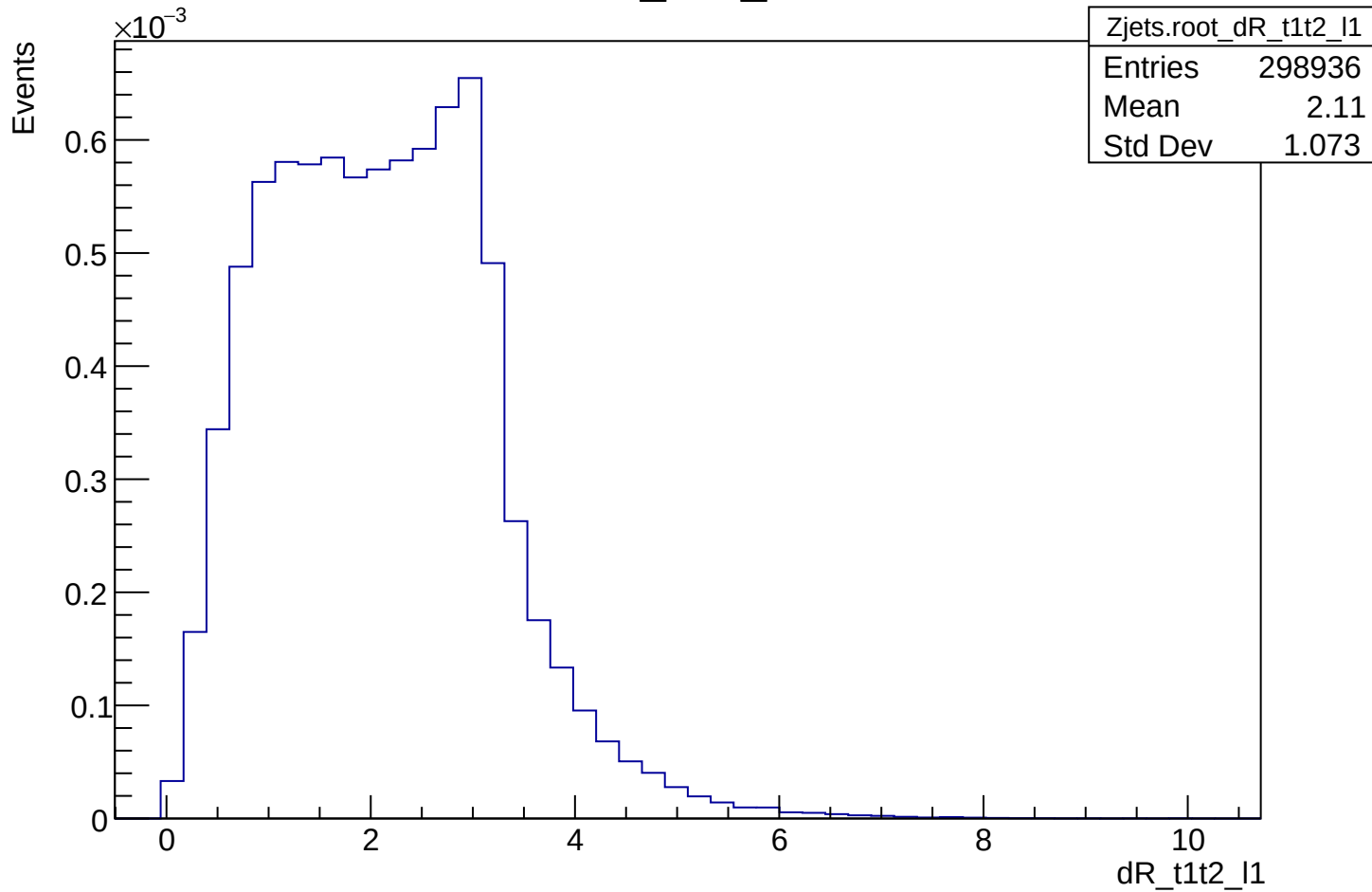
dR_t1j2



dR_t1t2

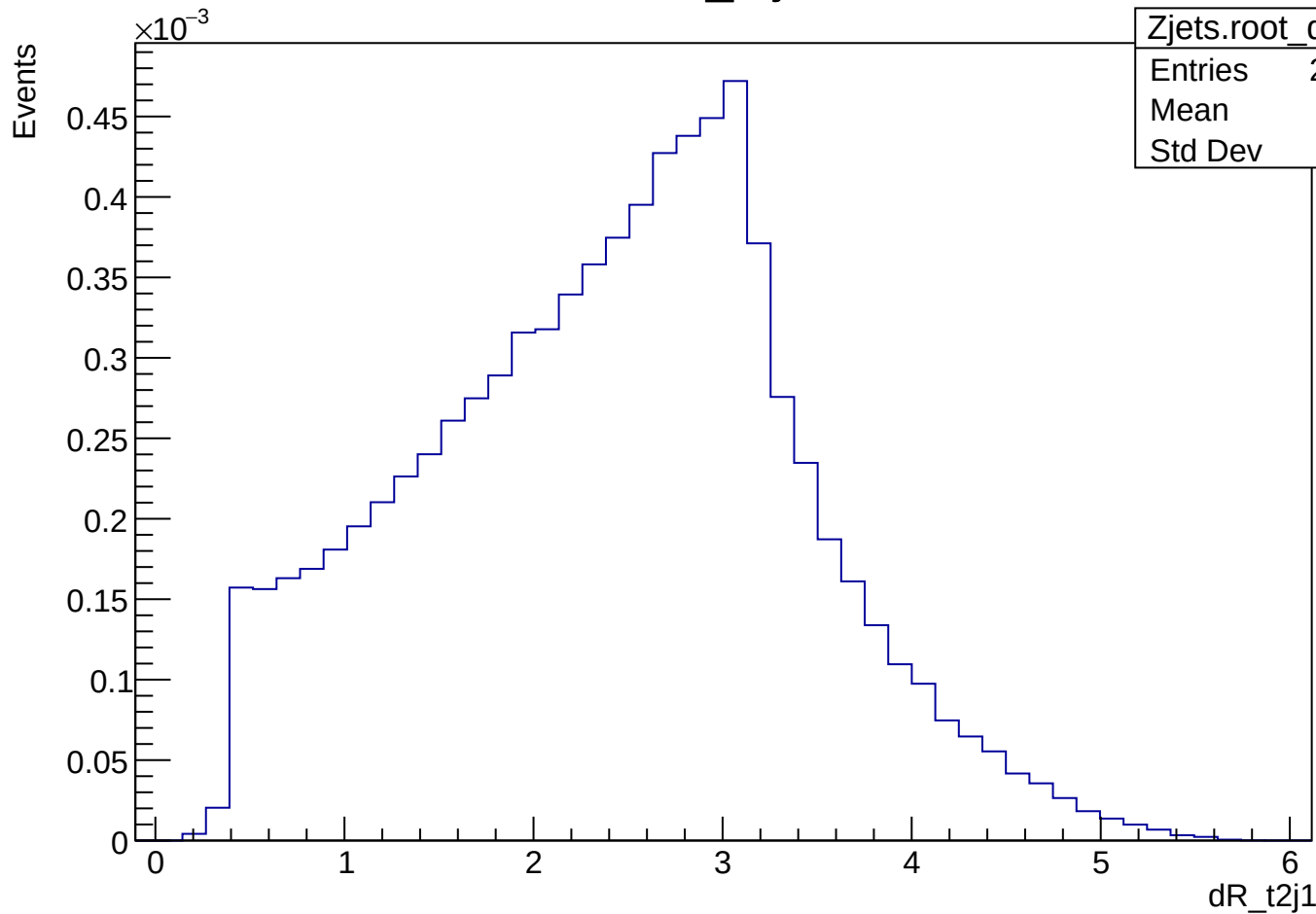


dR_t1t2_l1



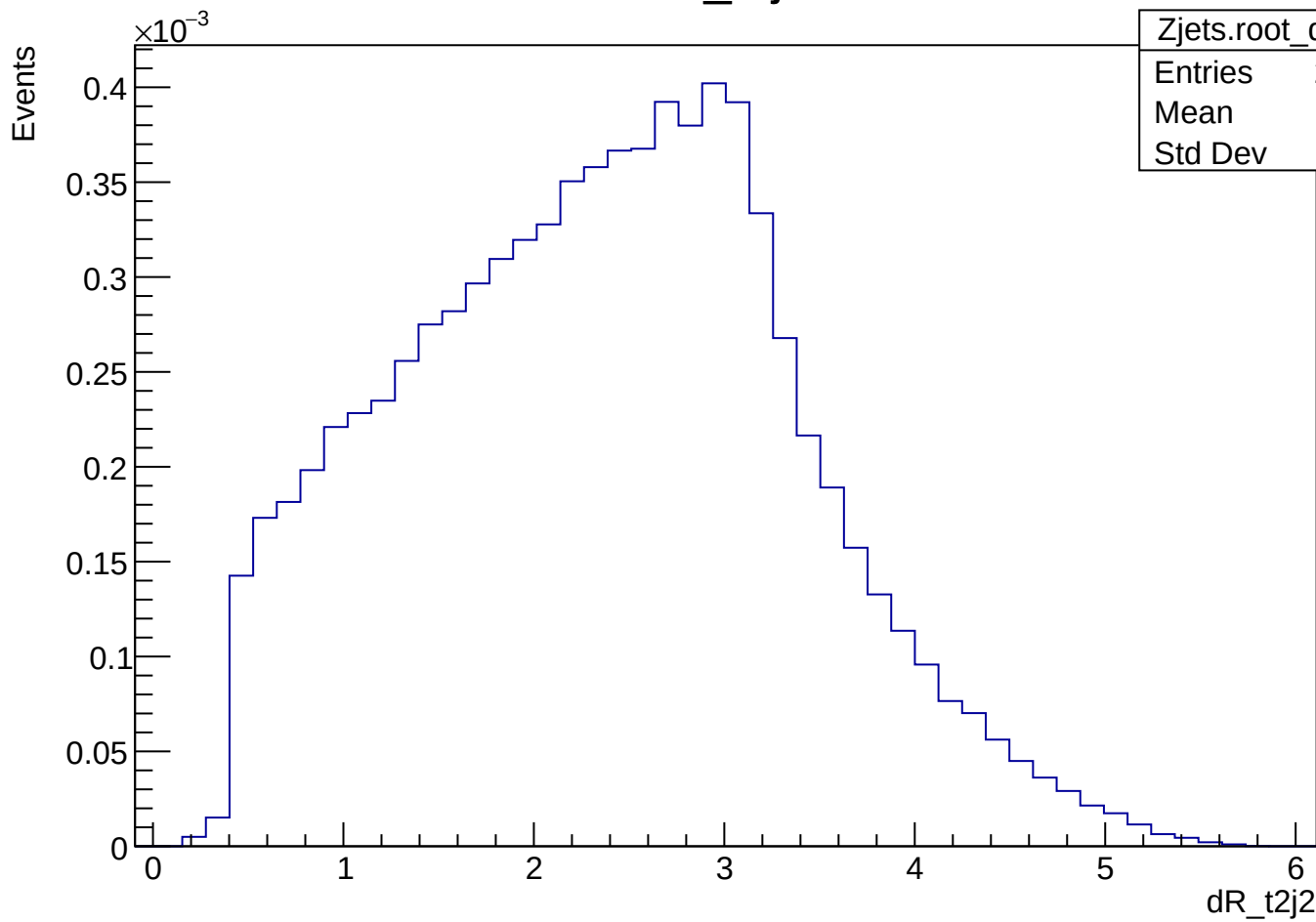
dR_t2j1

Zjets.root_dR_t2j1	
Entries	298936
Mean	2.411
Std Dev	0.9948



dR_t2j2

Zjets.root_dR_t2j2	
Entries	298936
Mean	2.363
Std Dev	1.015



fjvt_effSF

Events

0.007

0.006

0.005

0.004

0.003

0.002

0.001

0

0.4

0.6

0.8

1.0

1.2

1.4

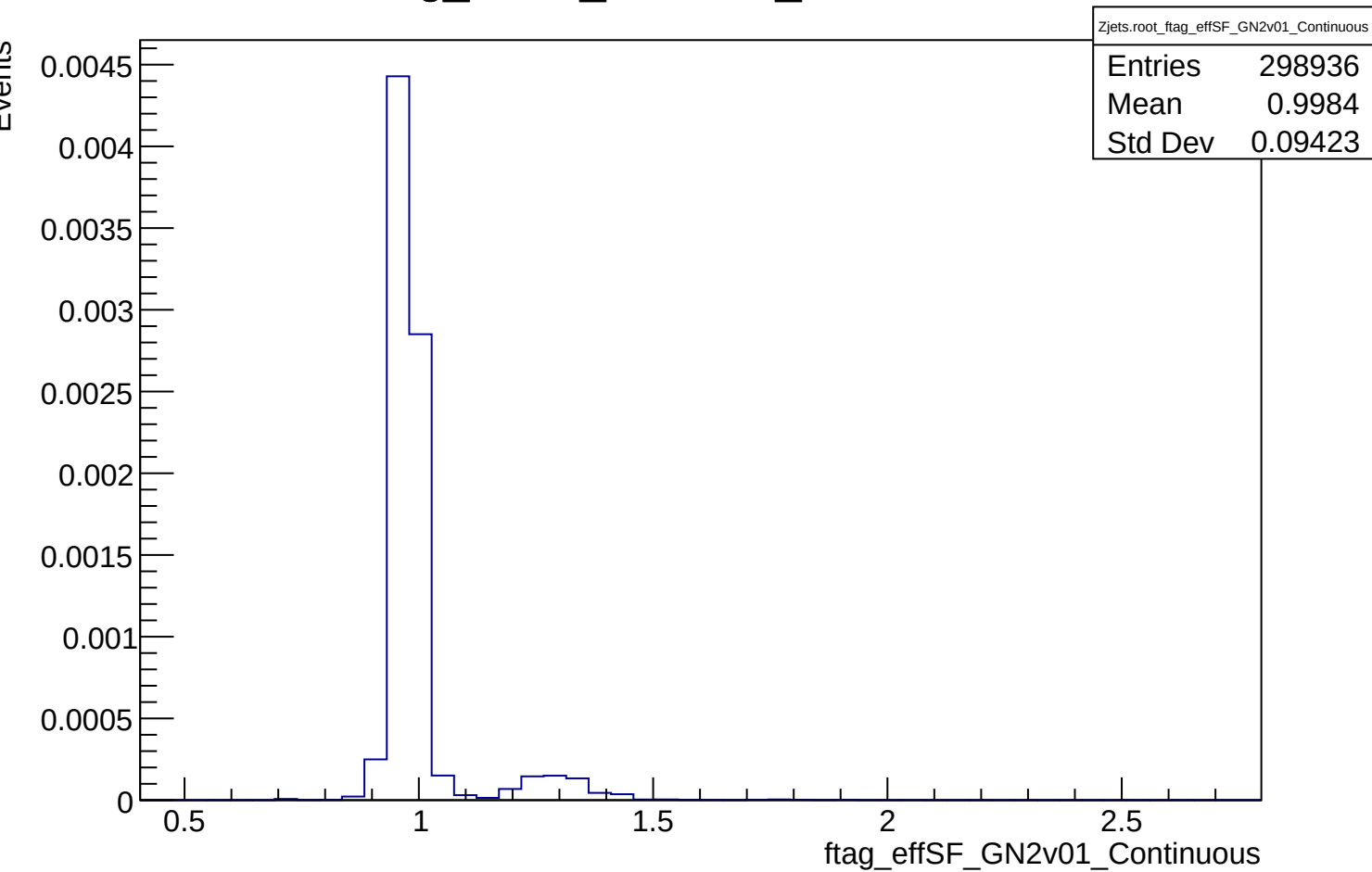
1.6

1.8

fjvt_effSF

Zjets.root_fjvt_effSF	
Entries	298936
Mean	0.997
Std Dev	0.03585

ftag_effSF_GN2v01_Continuous



globalTriggerEffSF

Events

0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

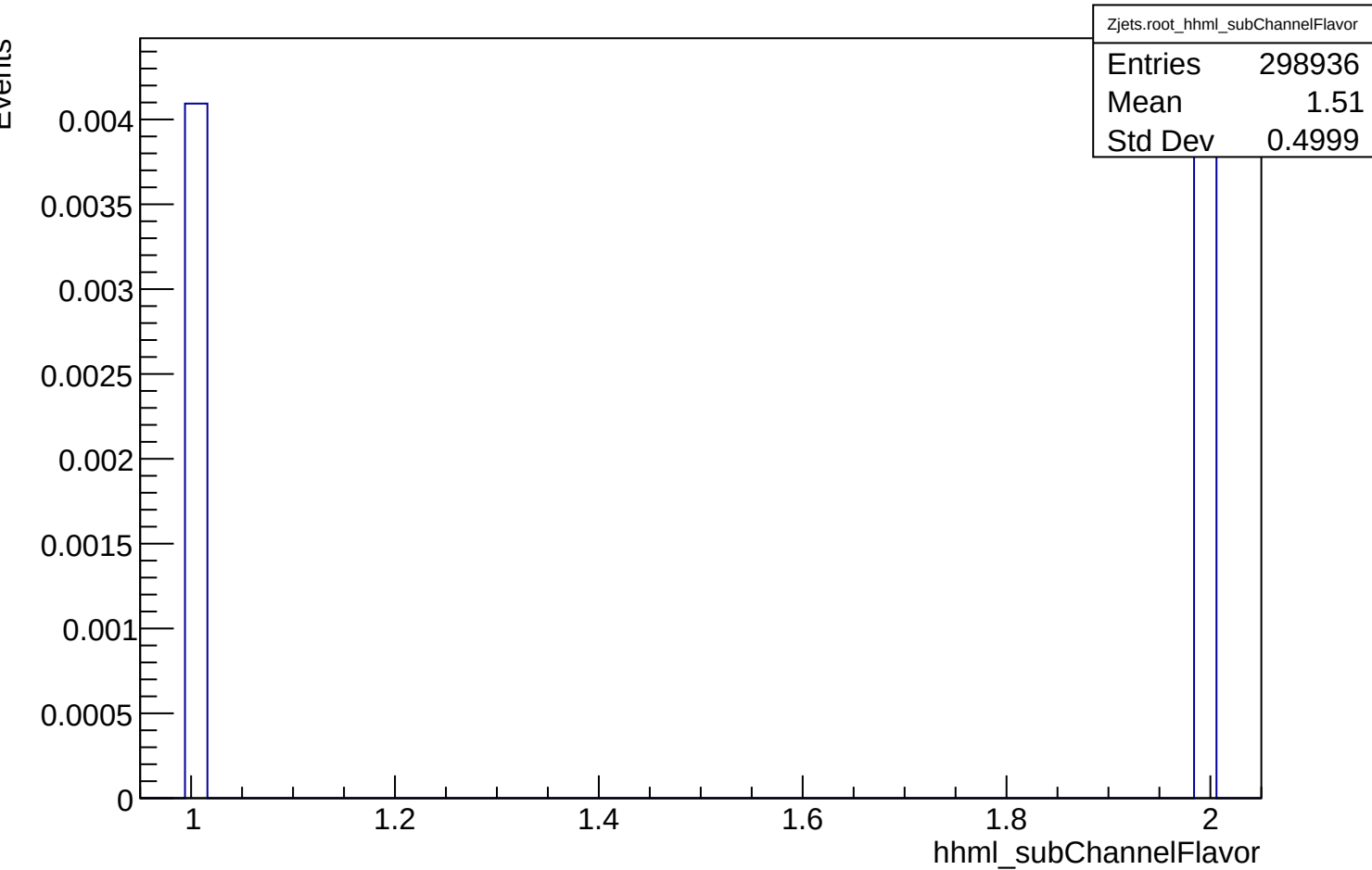
0.6

0.8

globalTriggerEffSF

Zjets.root_globalTriggerEffSF	
Entries	298936
Mean	0.9516
Std Dev	0.09718

hhml_subChannelFlavor



isDiTauFakes

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

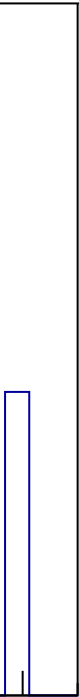
0.6

0.8

1

isDiTauFakes

Zjets.root_isDiTauFakes	
Entries	298936
Mean	0.2799
Std Dev	0.4489



isSingleFakeTau

Events

0.005

0.004

0.003

0.002

0.001

0

0

0.2

0.4

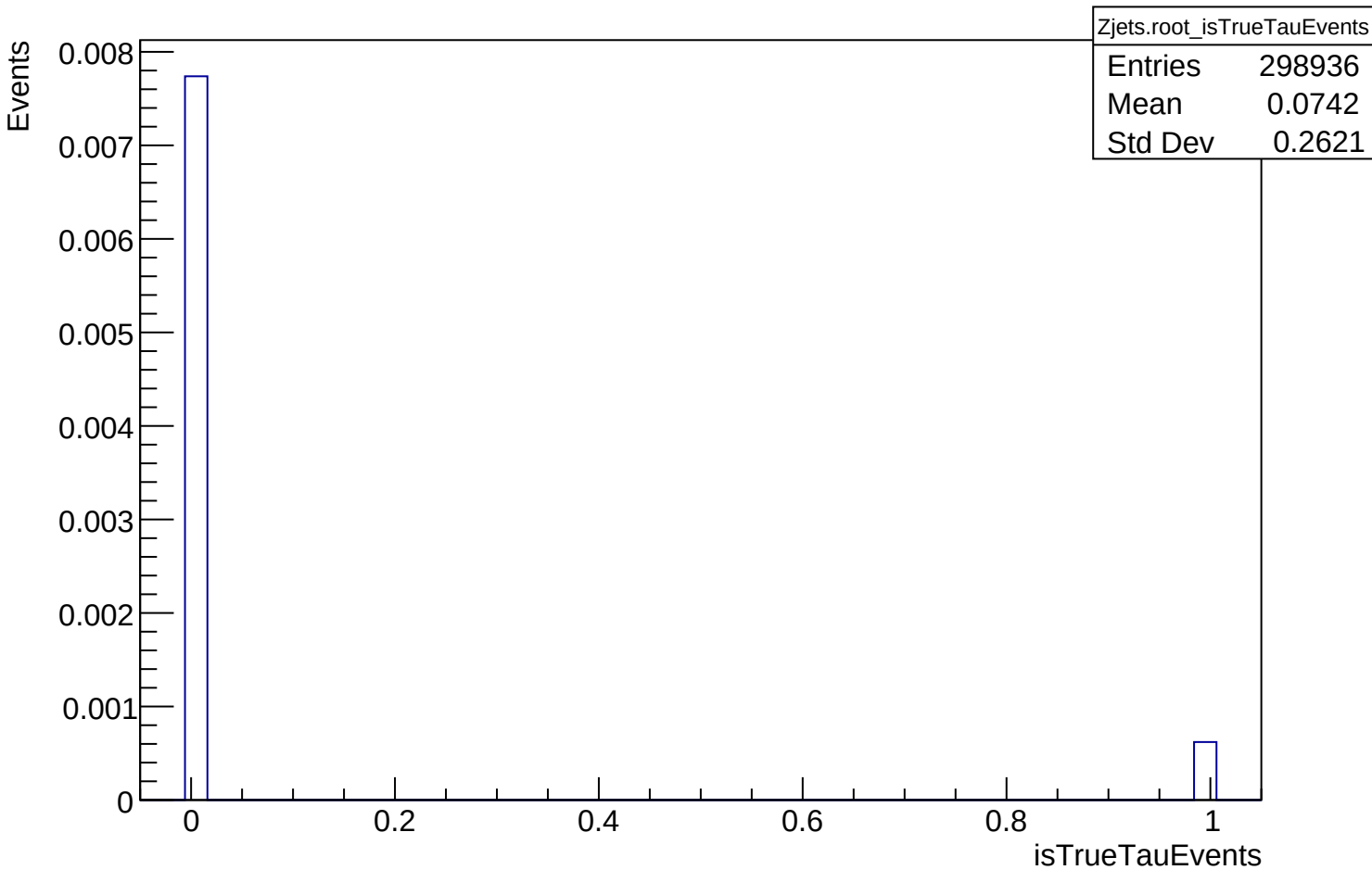
0.6

0.8

isSingleFakeTau

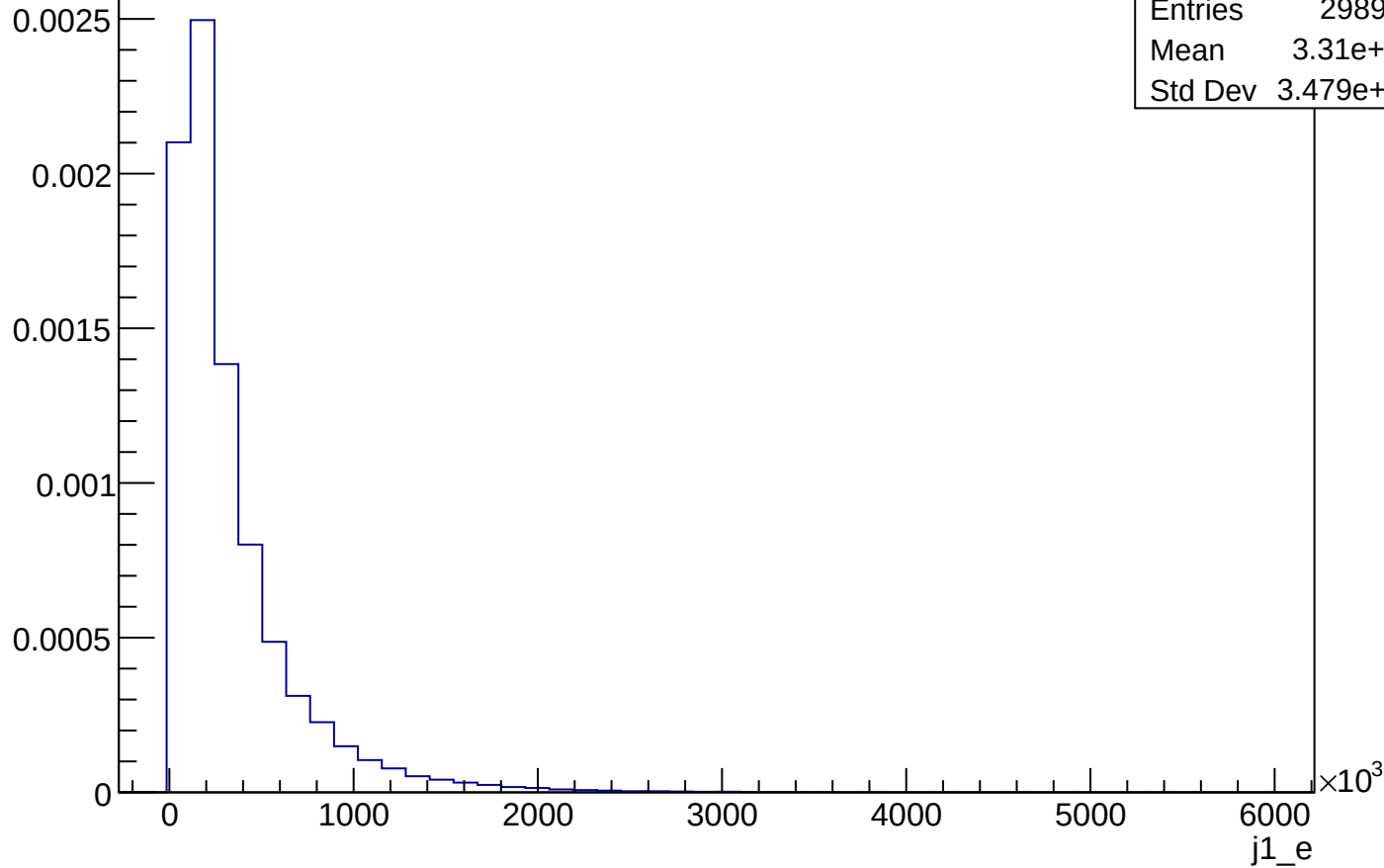
Zjets.root_isSingleFakeTau	
Entries	298936
Mean	0.6459
Std Dev	0.4782

isTrueTauEvents



j1_e

Events



j1_eta

Zjets.root_j1_eta

Entries 298936

Mean 0.003279

Std Dev 1.232

Events

$\times 10^{-3}$

0.25

0.2

0.15

0.1

0.05

0

-2

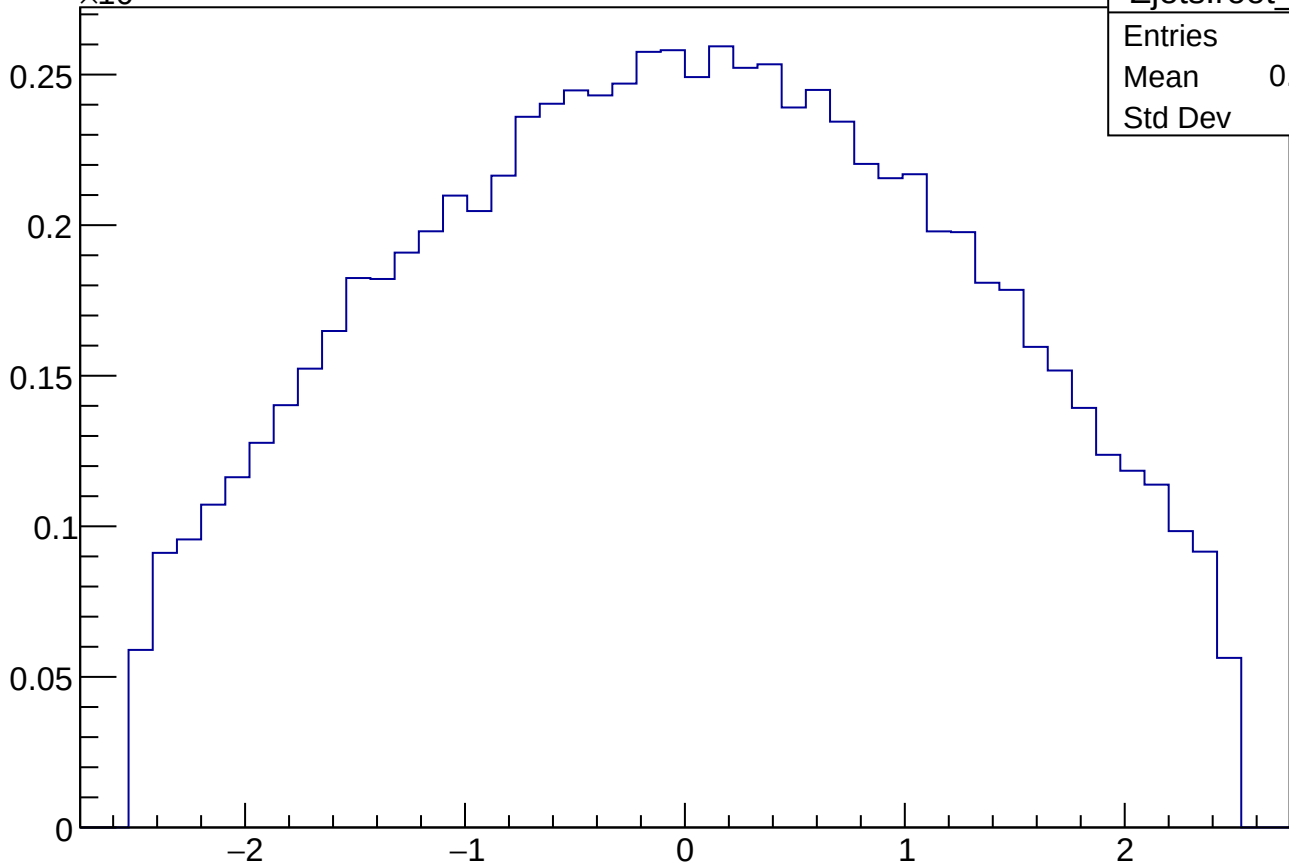
-1

0

1

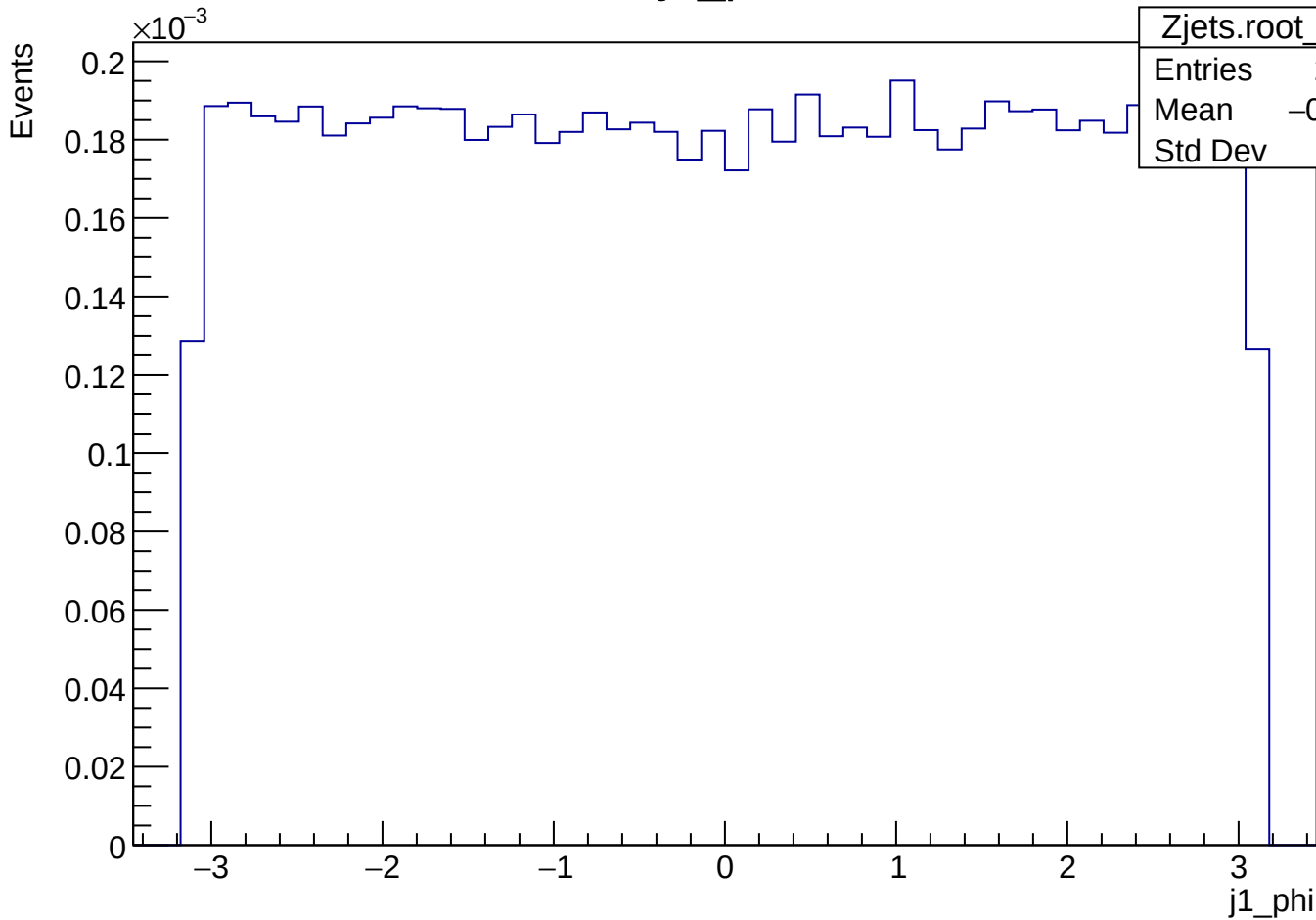
2

j1_eta



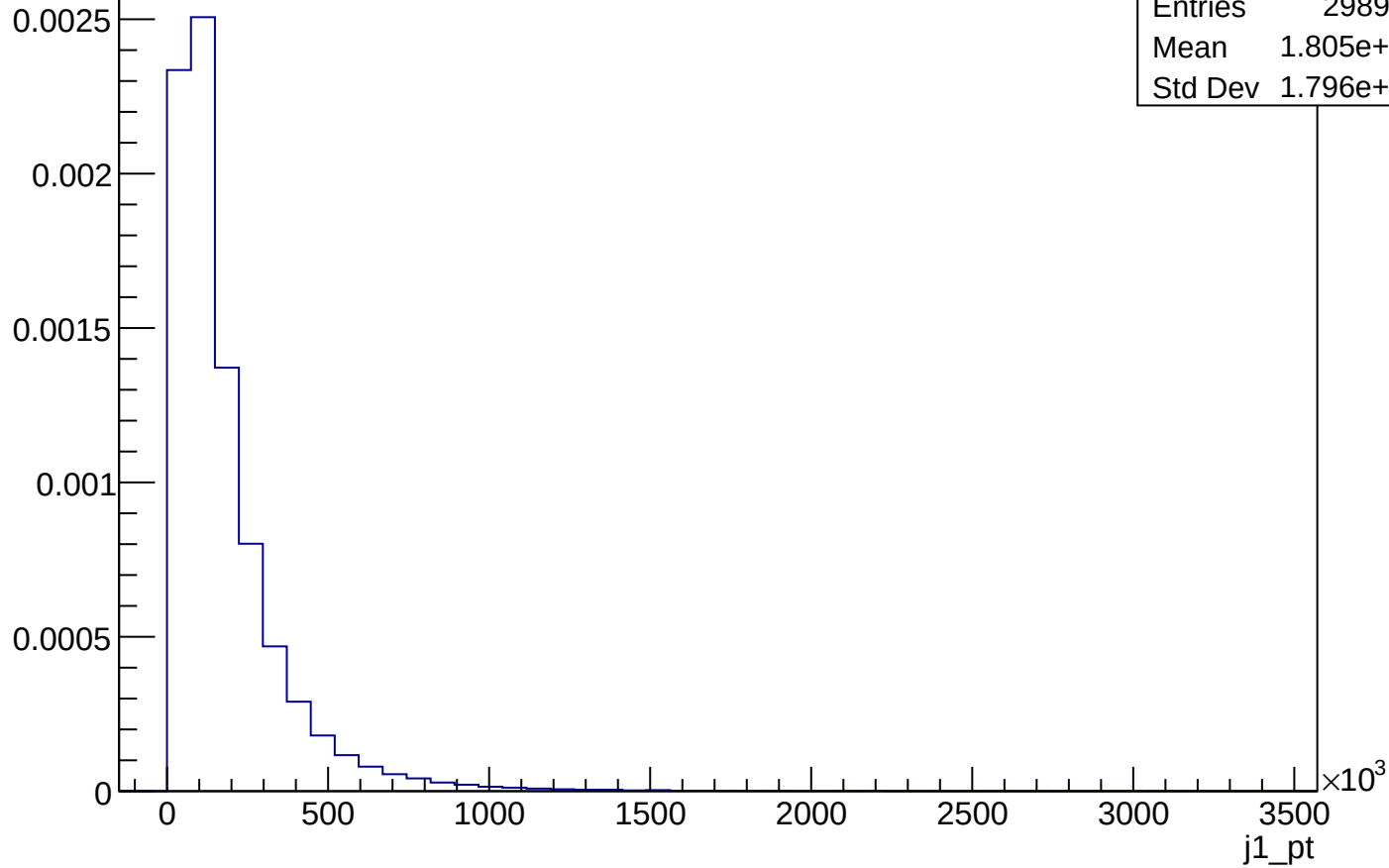
j1_phi

Zjets.root_j1_phi	
Entries	298936
Mean	-0.00574
Std Dev	1.815



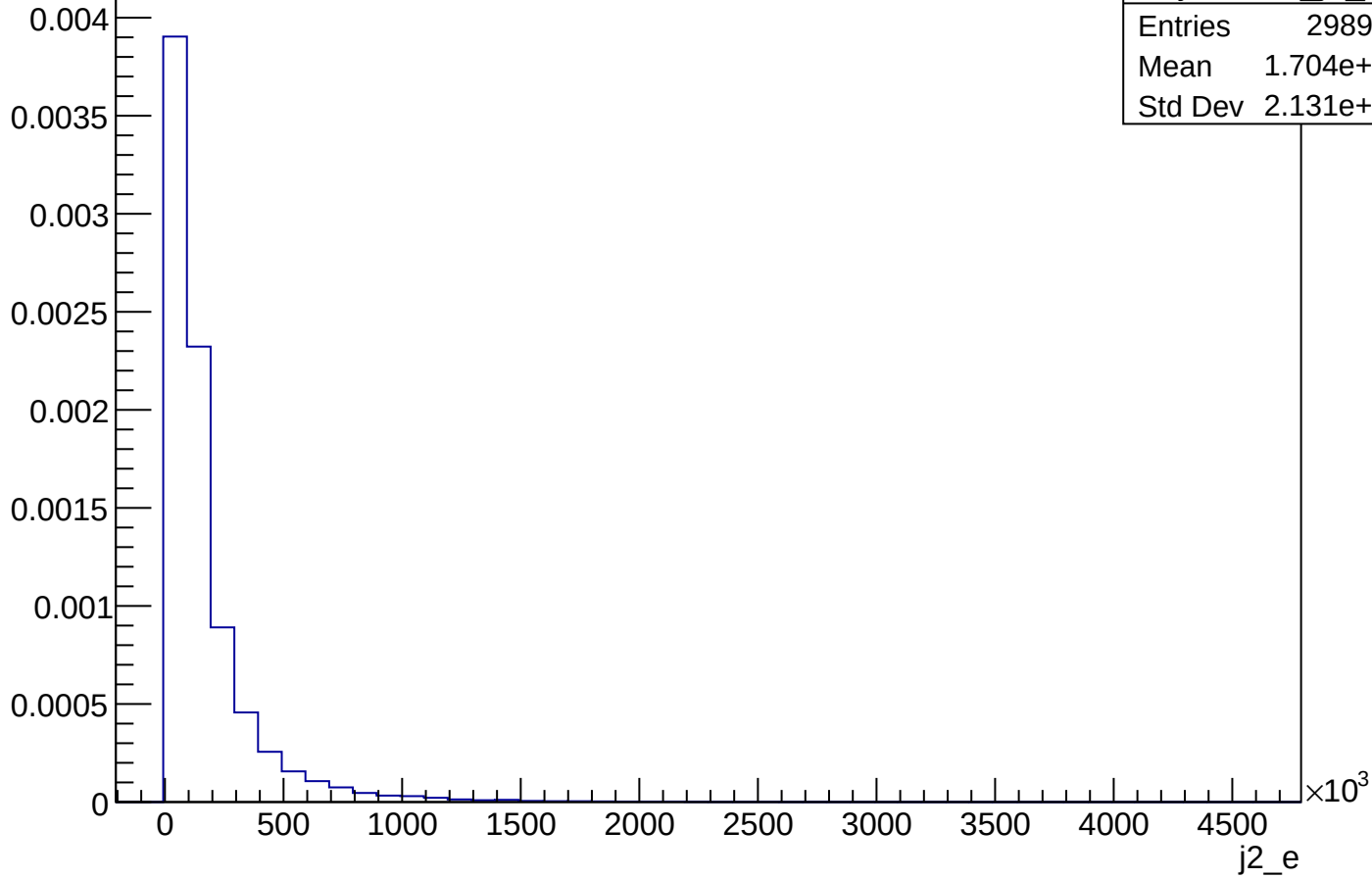
j1_pt

Events

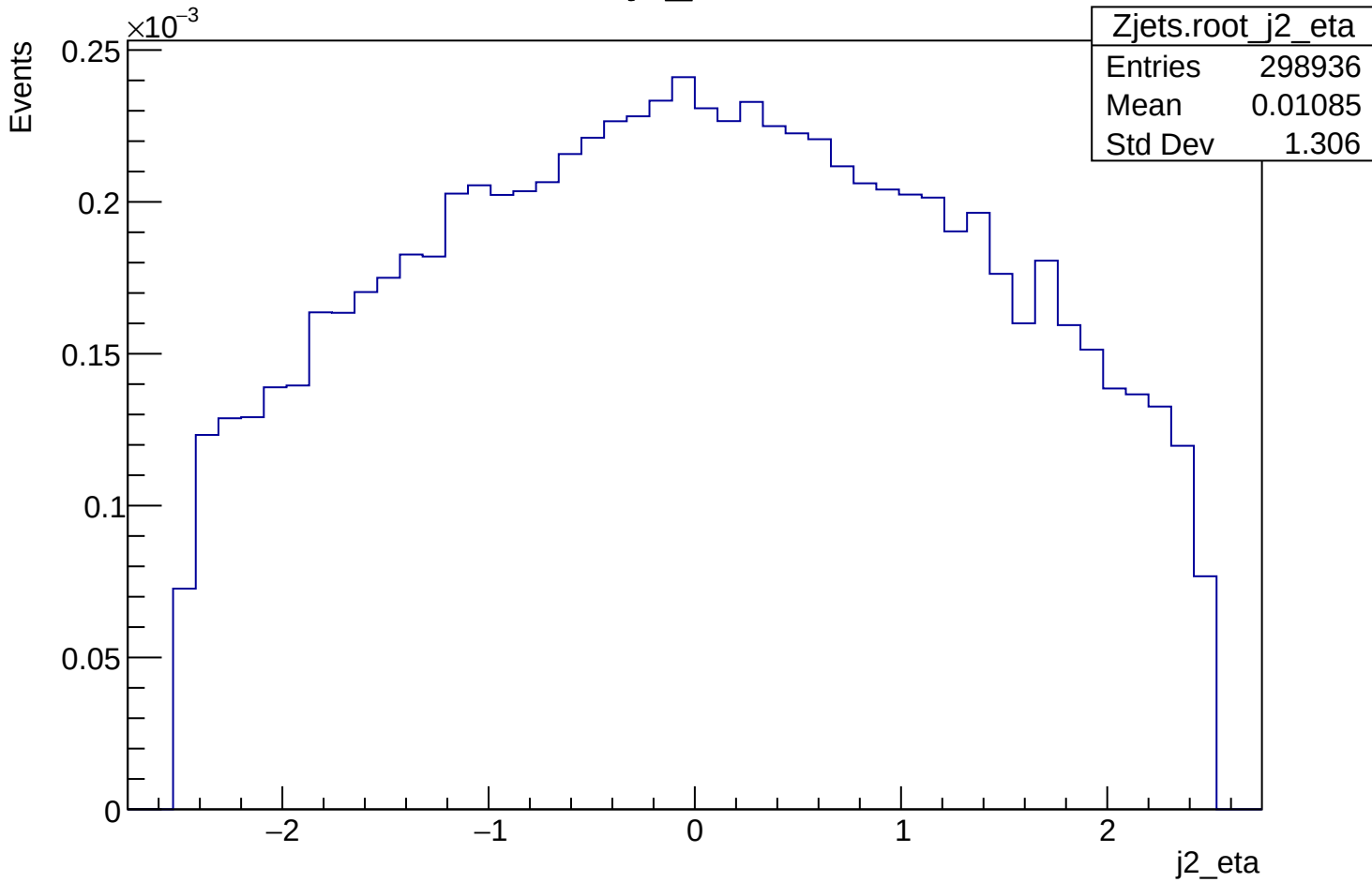


j2_e

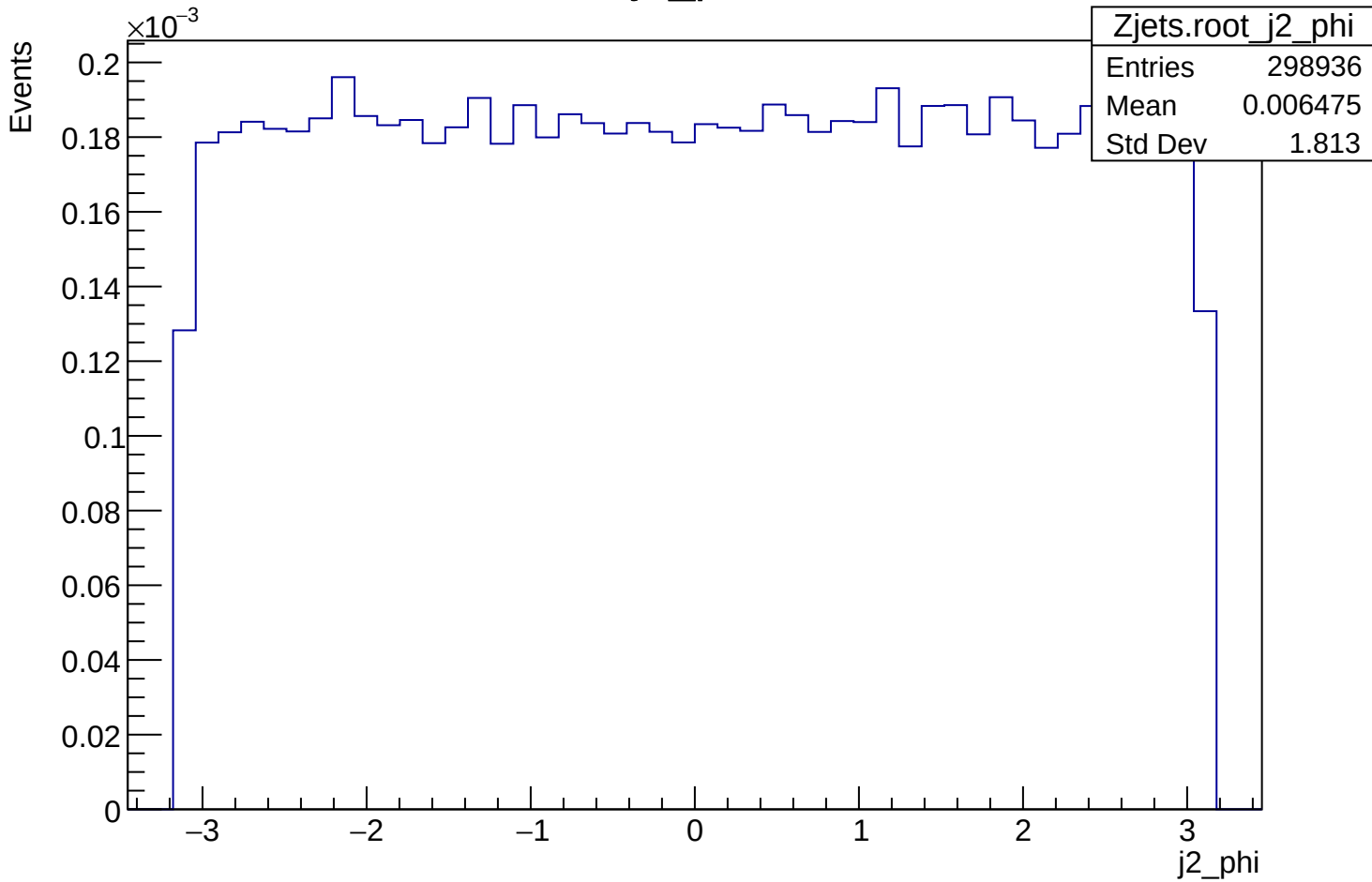
Events



j2_eta



j2_phi



j2_pt

Events

0.005
0.004
0.003
0.002
0.001
0

0

500

1000

1500

2000

2500

j2_pt

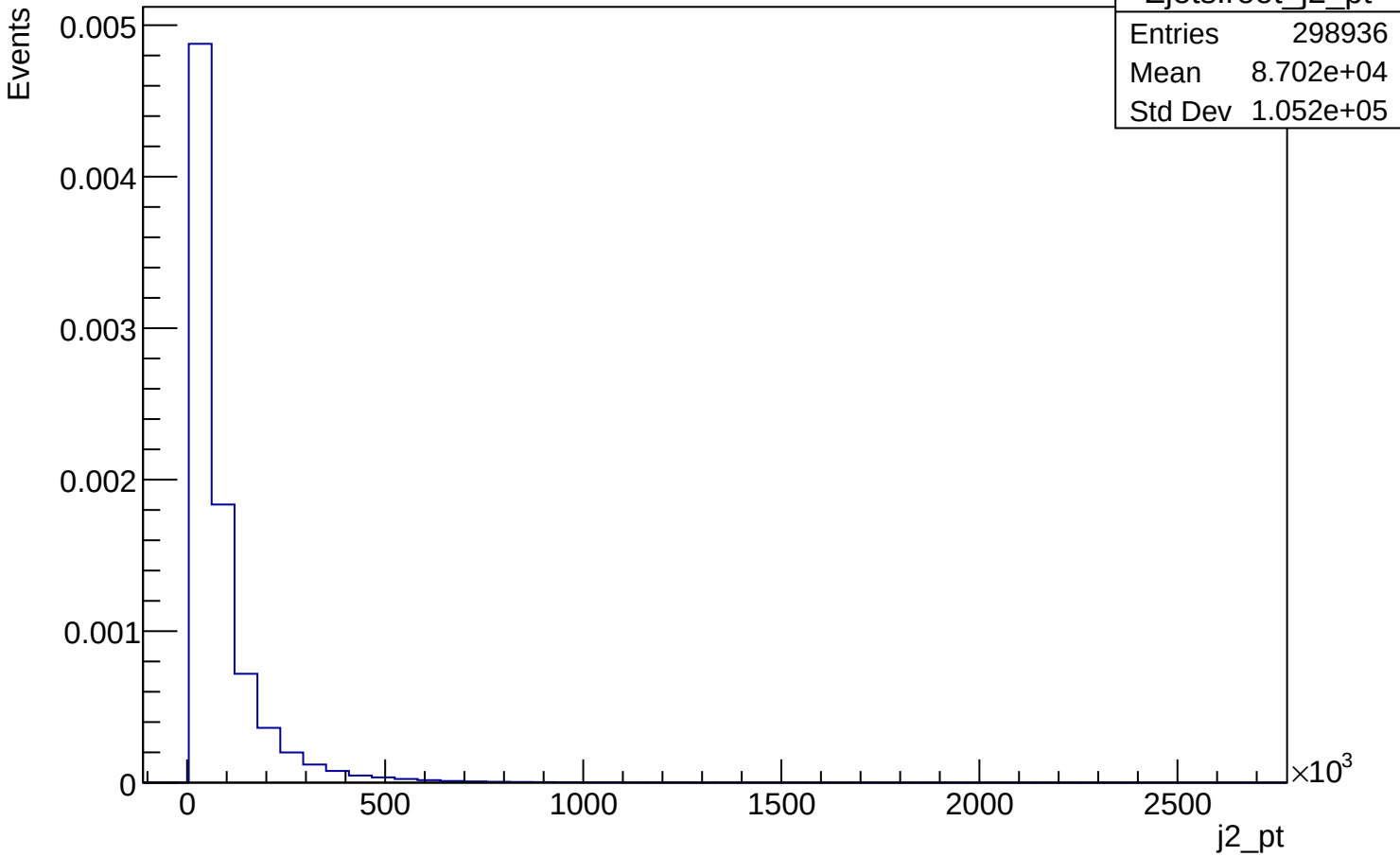
$\times 10^3$

Zjets.root_j2_pt

Entries 298936

Mean 8.702e+04

Std Dev 1.052e+05



jvt_effSF

Events

0.0045
0.004
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0

0.5

1

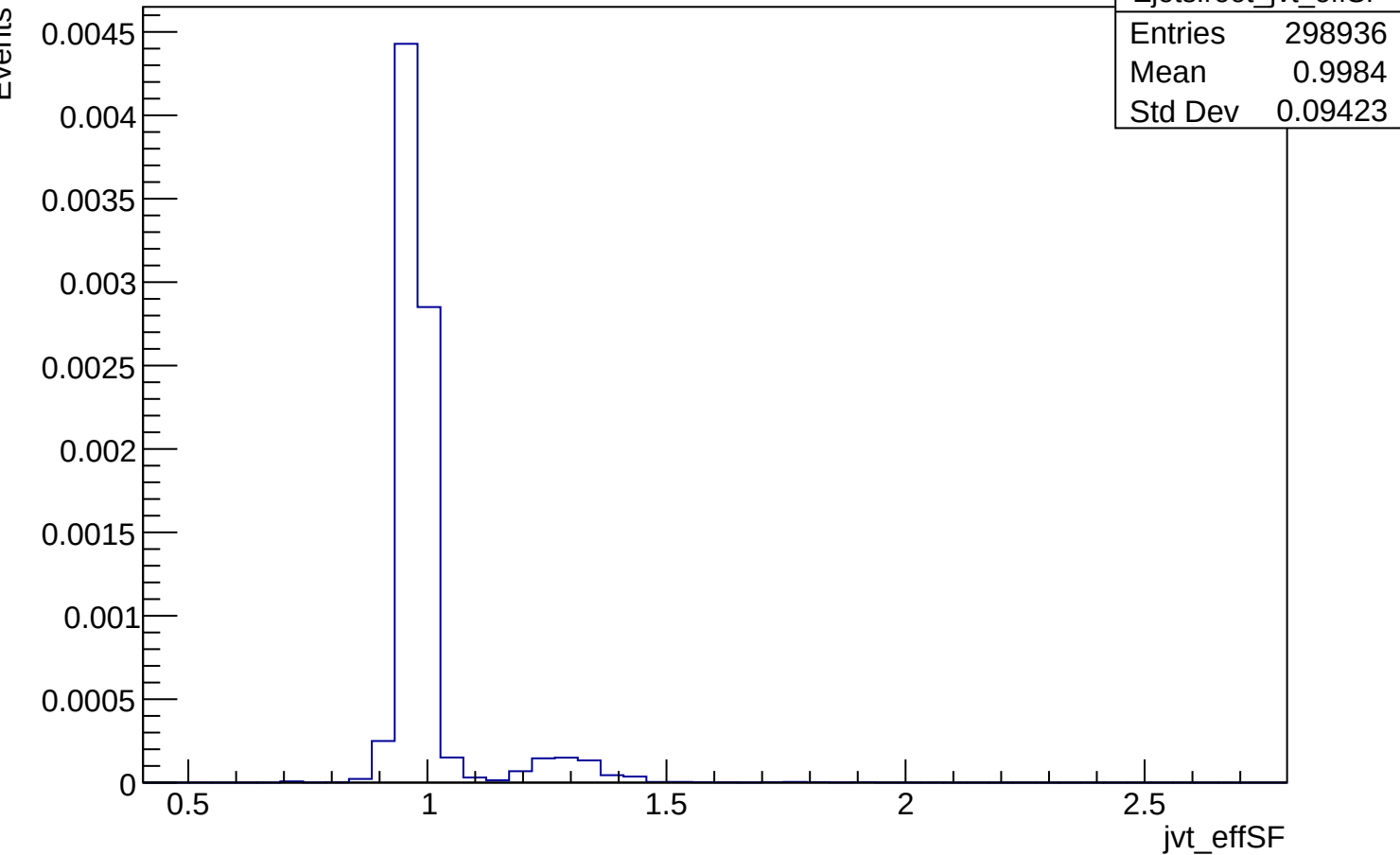
1.5

2

2.5

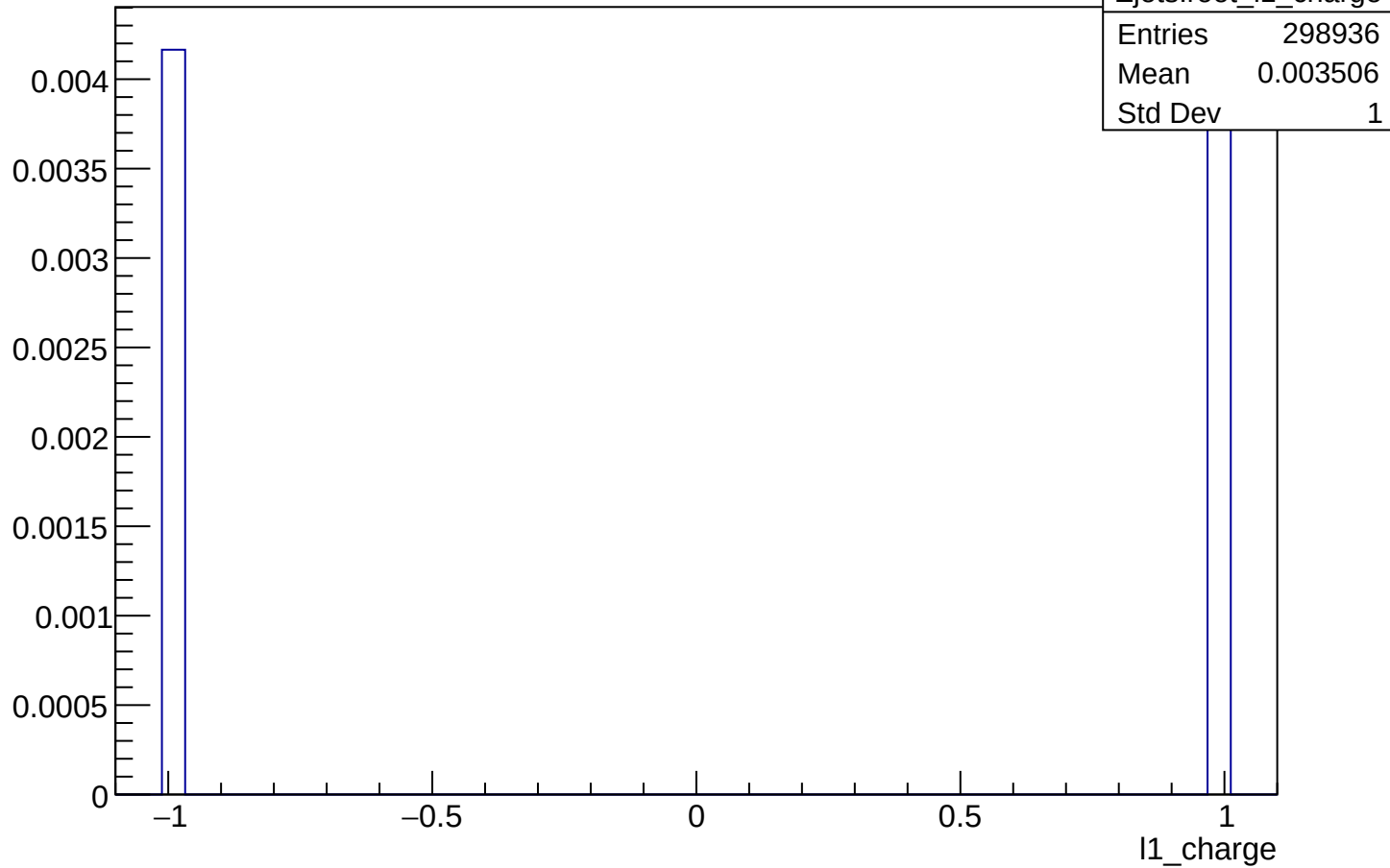
jvt_effSF

Zjets.root_jvt_effSF	
Entries	298936
Mean	0.9984
Std Dev	0.09423

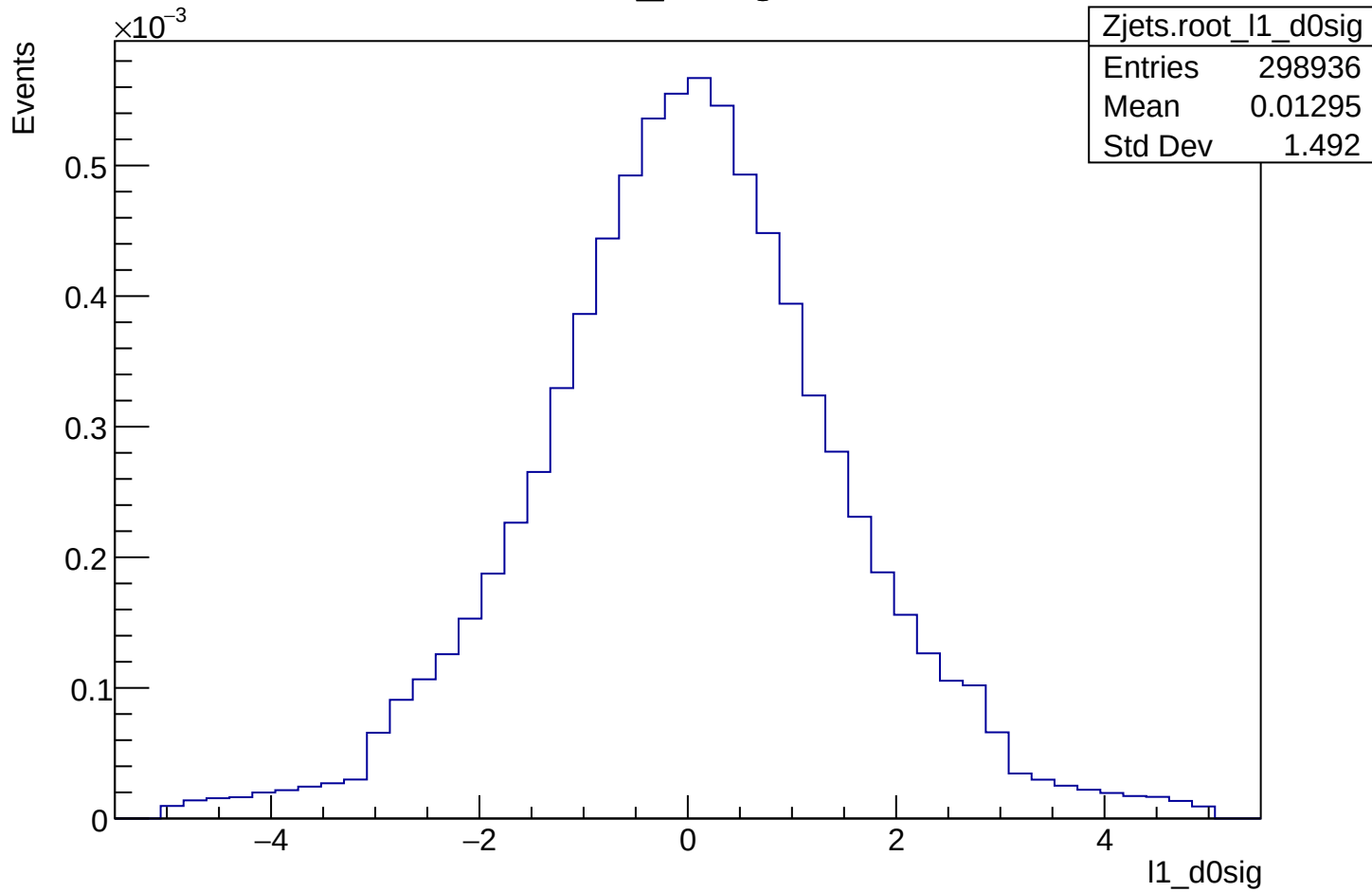


l1_charge

Events

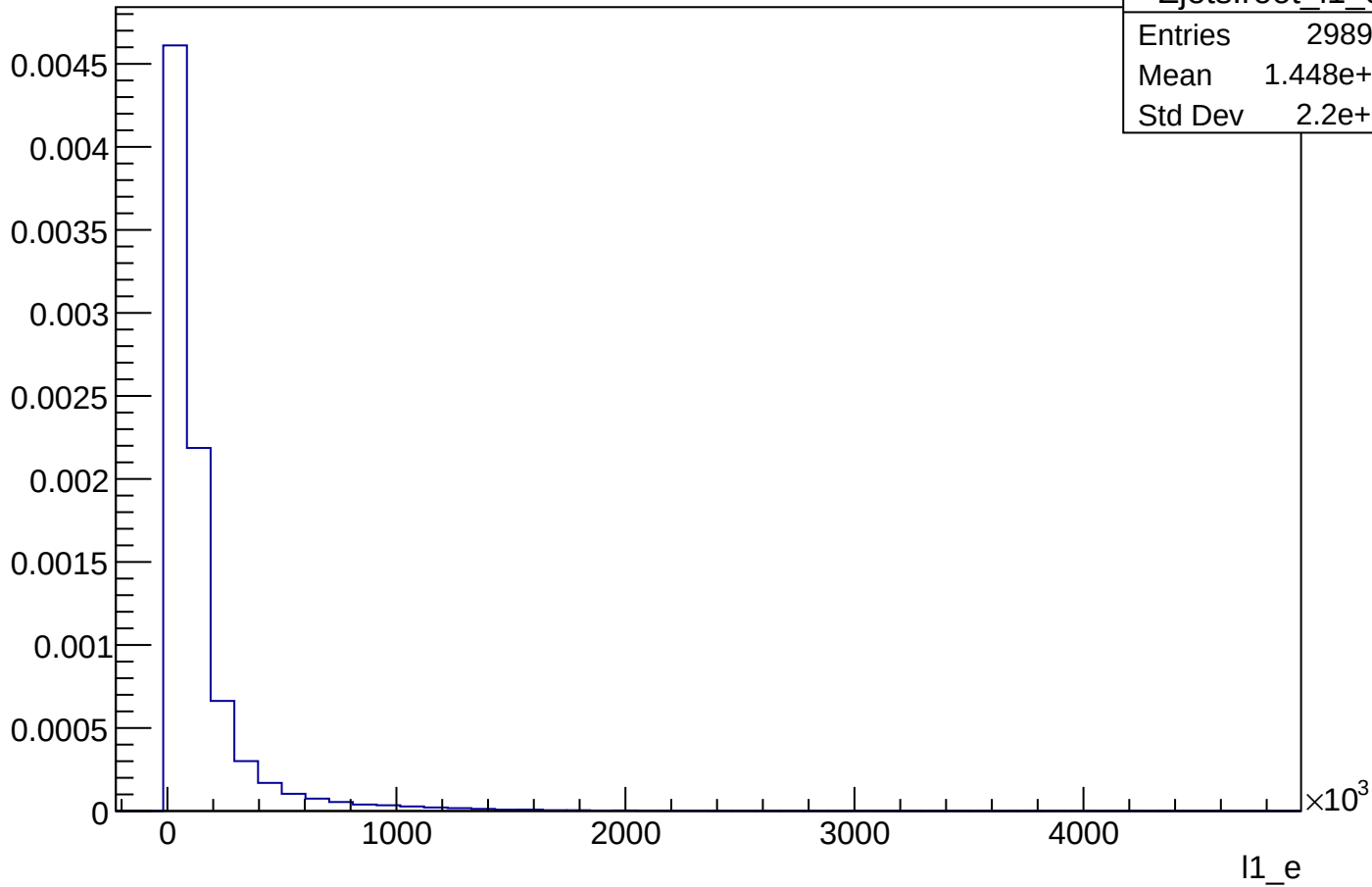


l1_d0sig



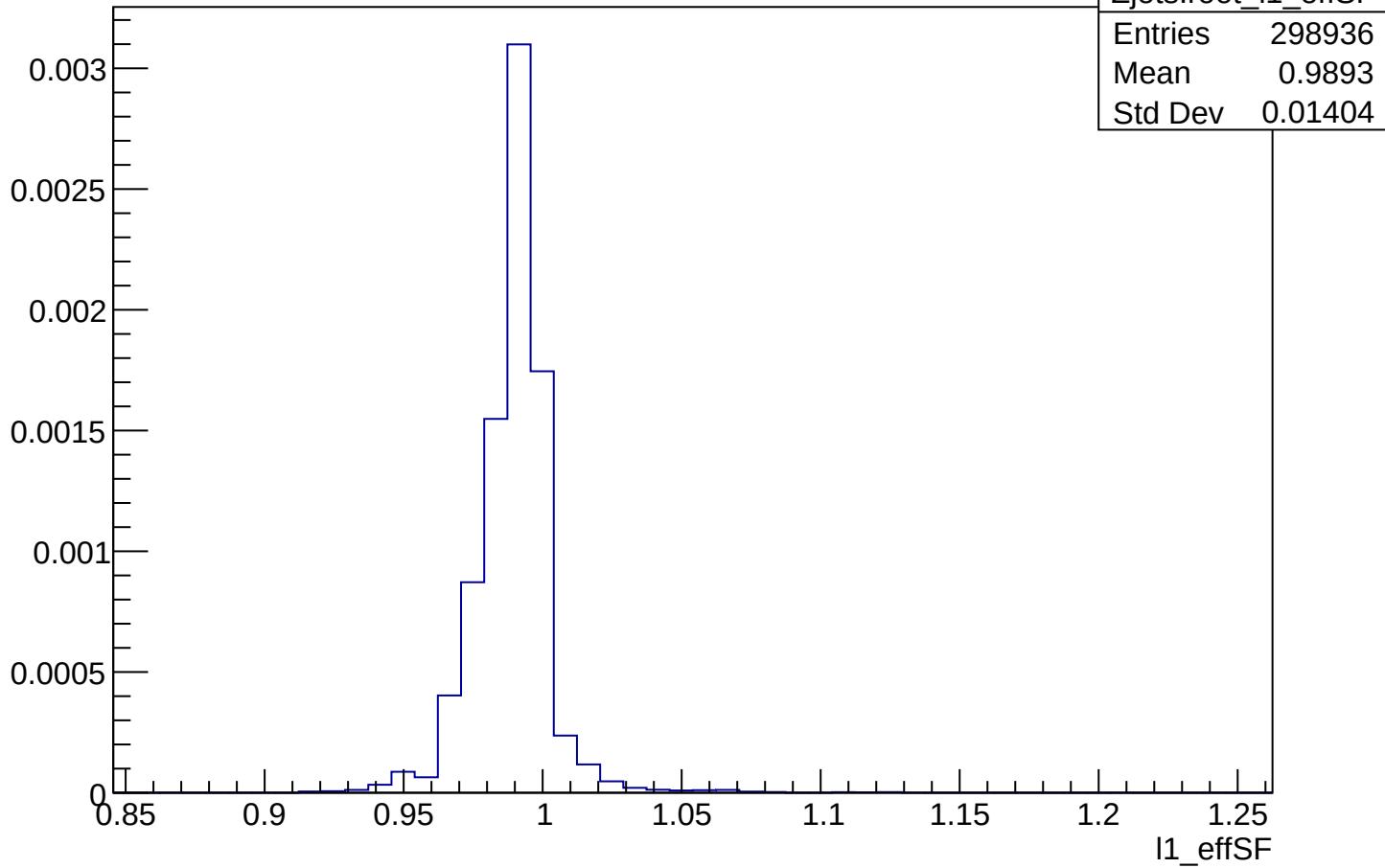
l1_e

Events

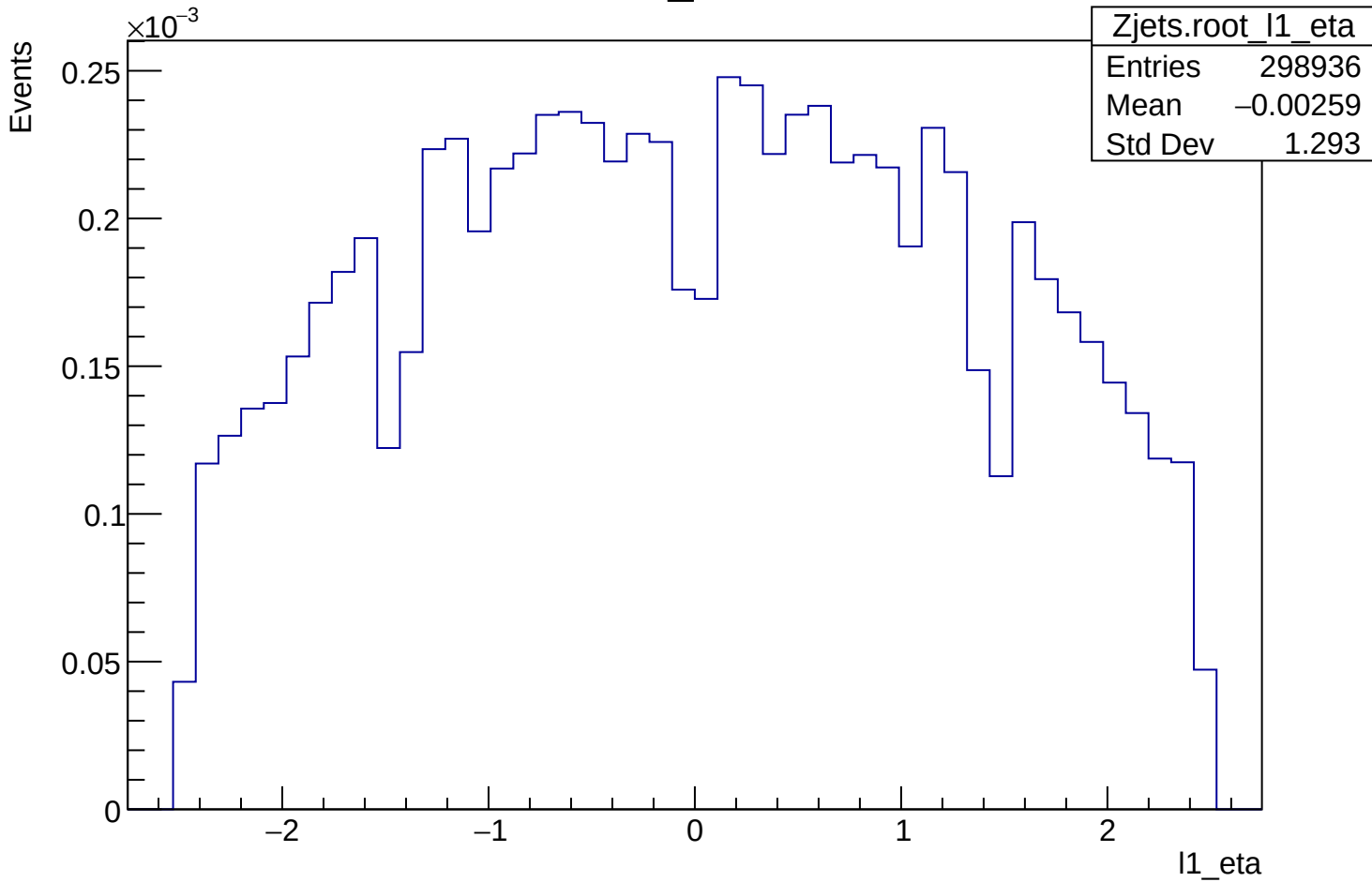


l1_effSF

Events



$l1_eta$



l1_pdg

Events

0.0022
0.002
0.0018
0.0016
0.0014
0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0

-10

-5

0

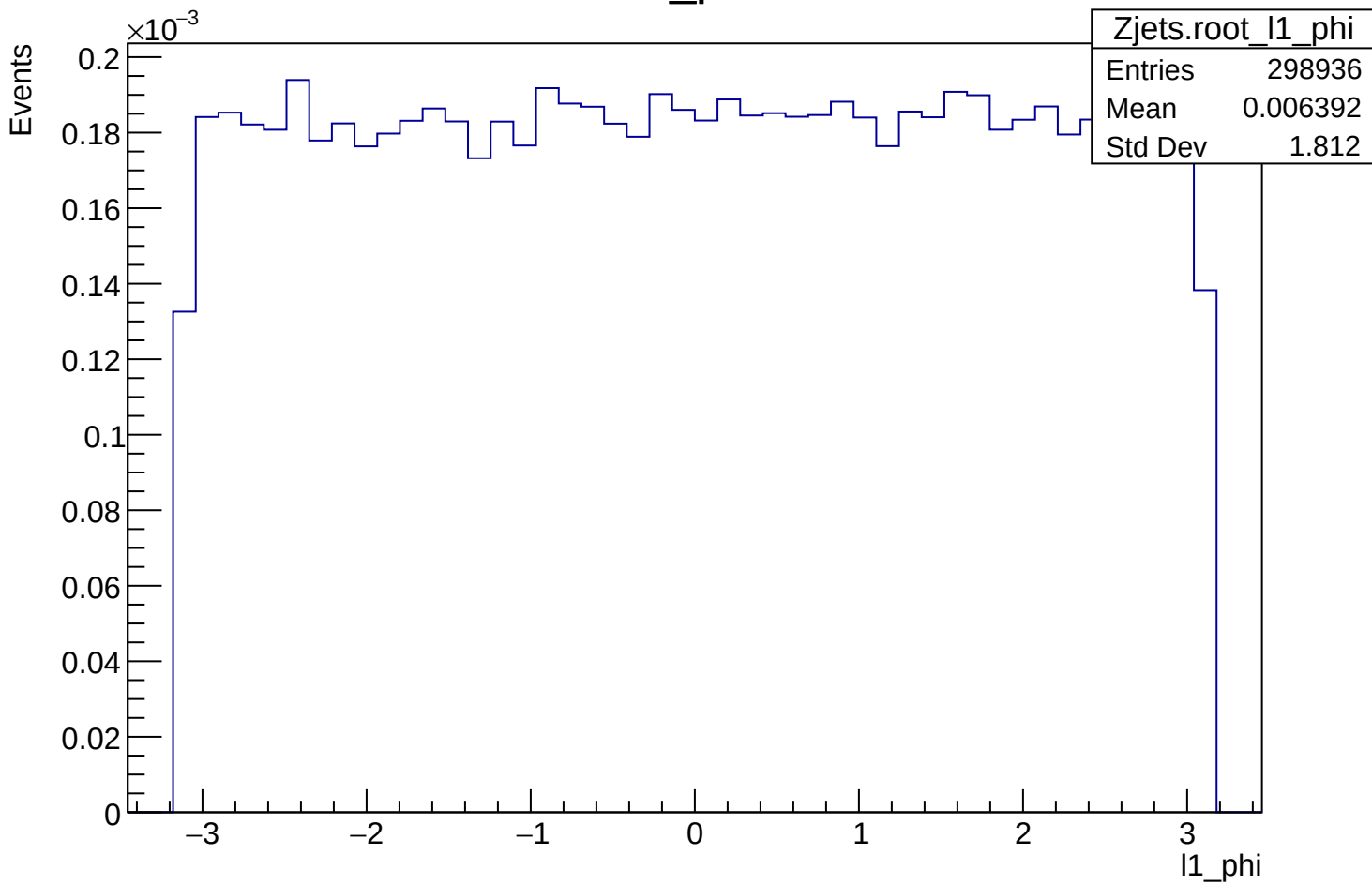
5

10

l1_pdg

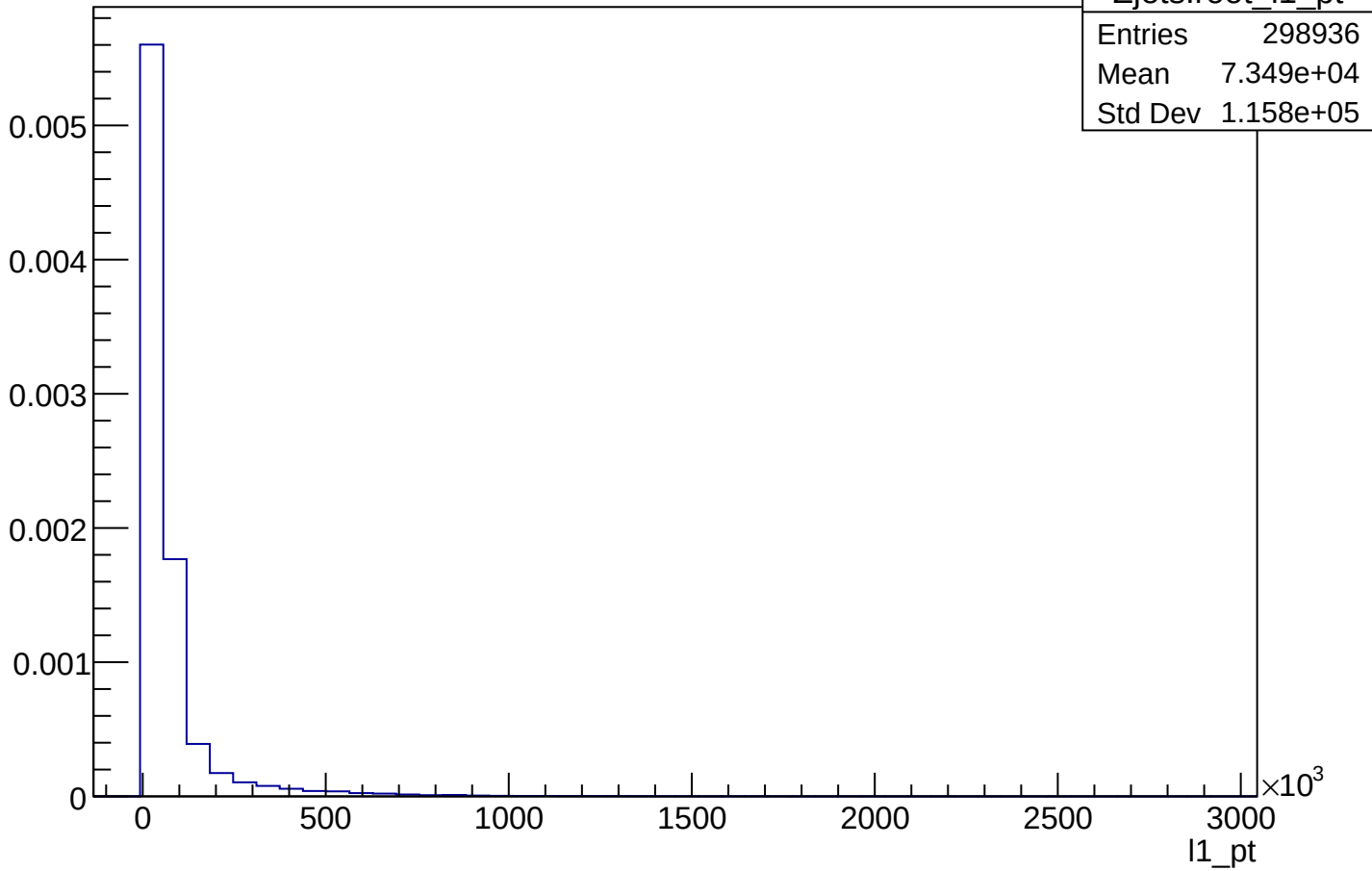
Zjets.root_l1_pdg	
Entries	298936
Mean	-0.03191
Std Dev	12.06

l1_phi



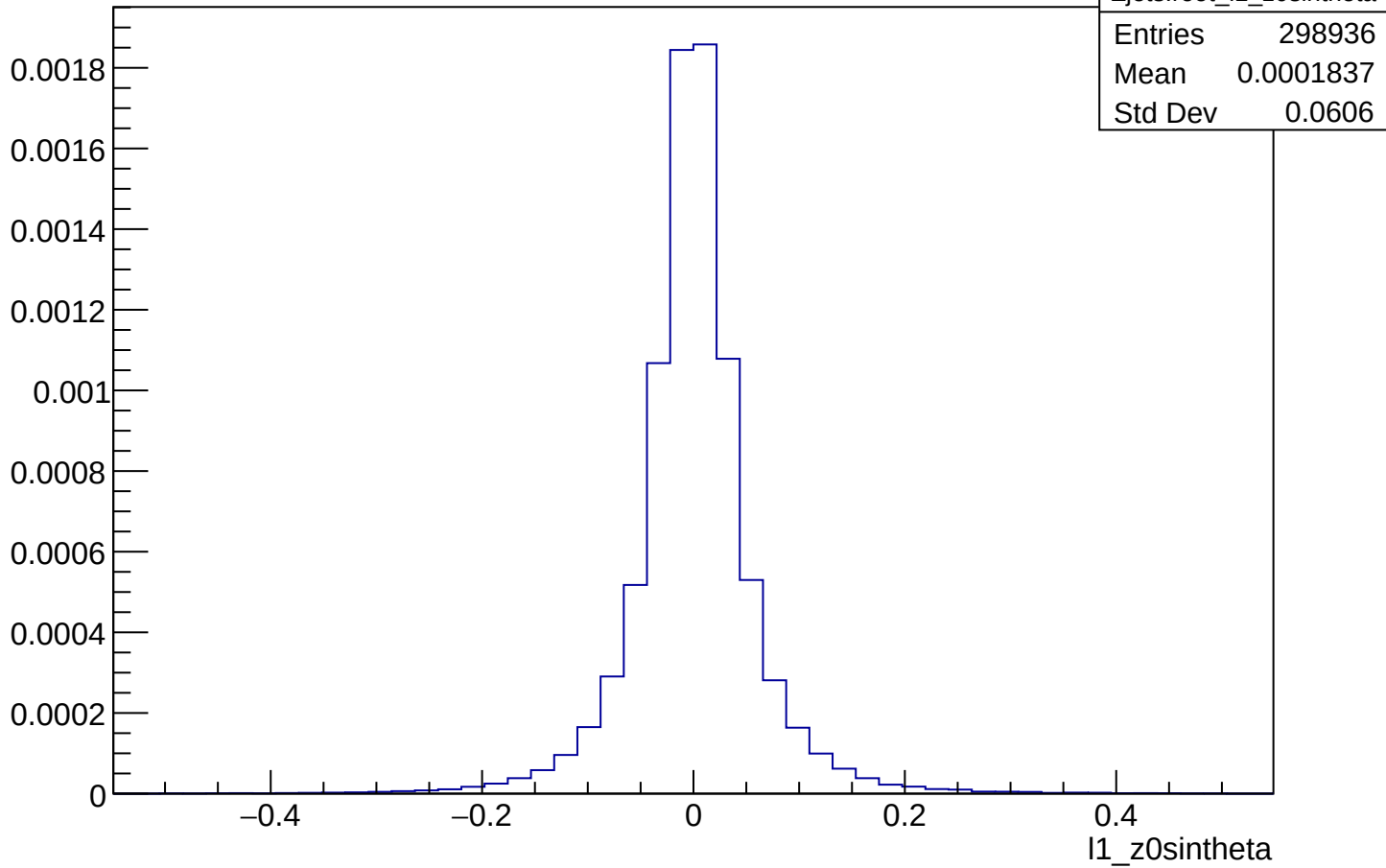
l1_pt

Events



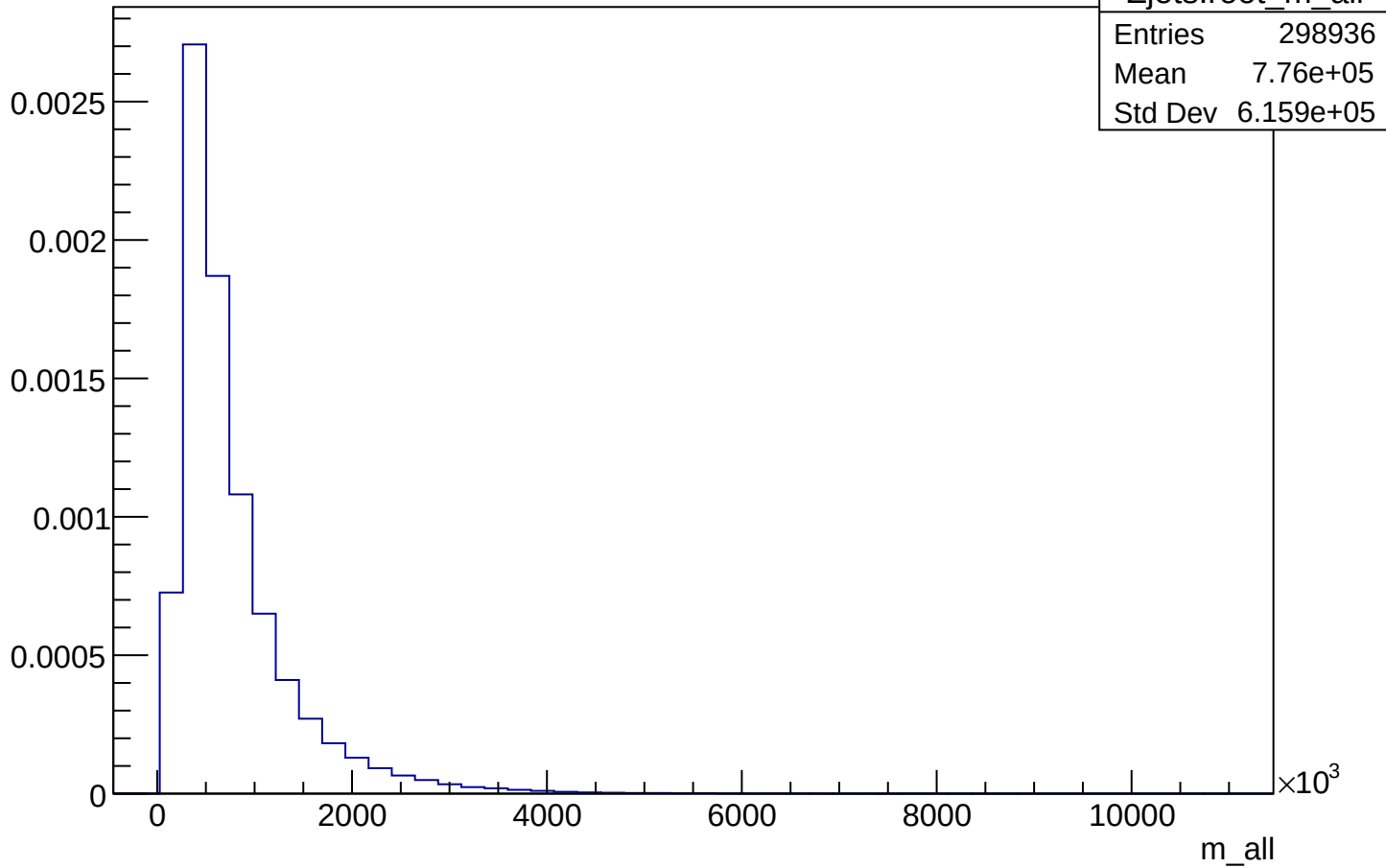
l1_z0sintheta

Events



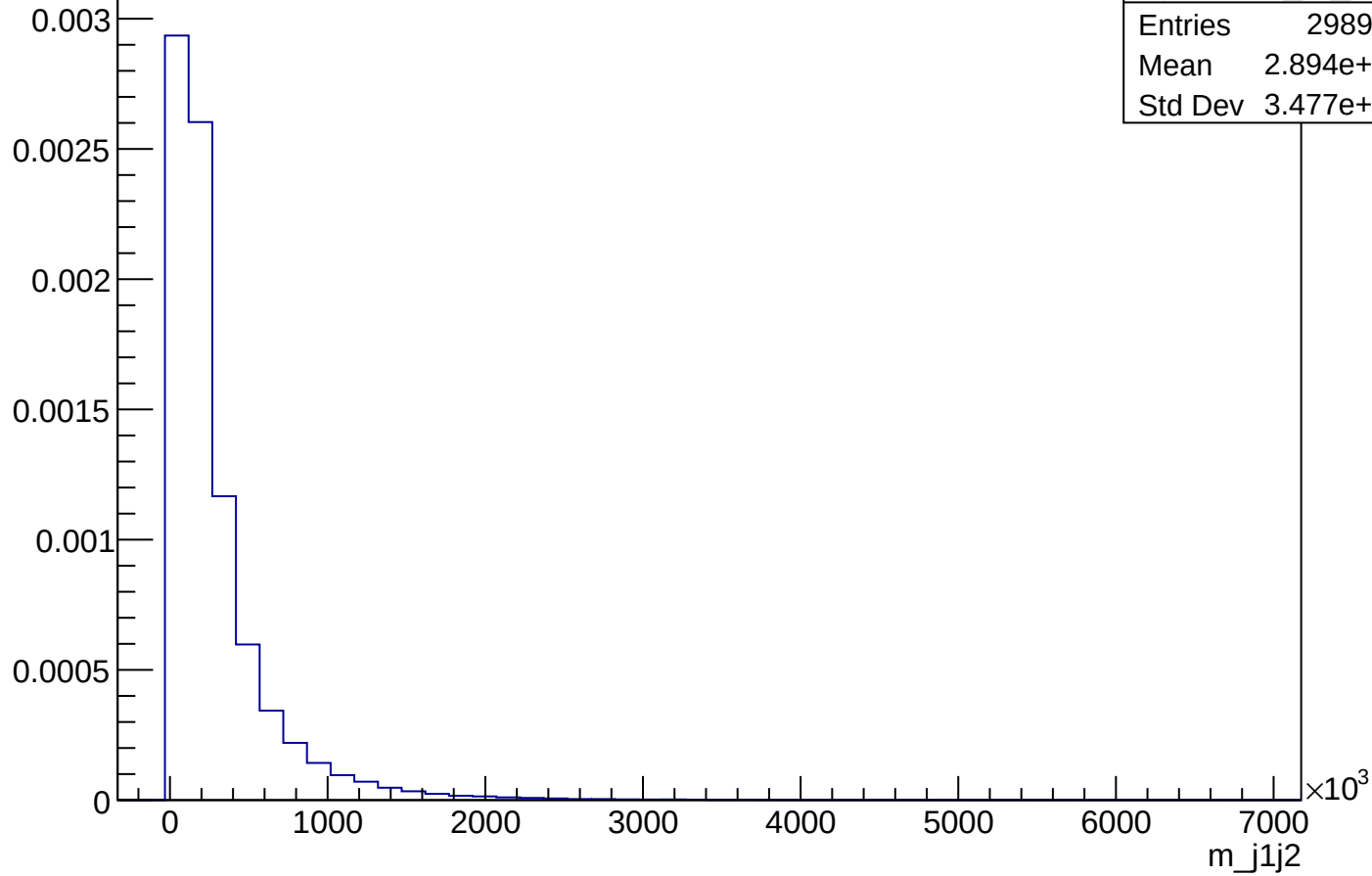
m_all

Events



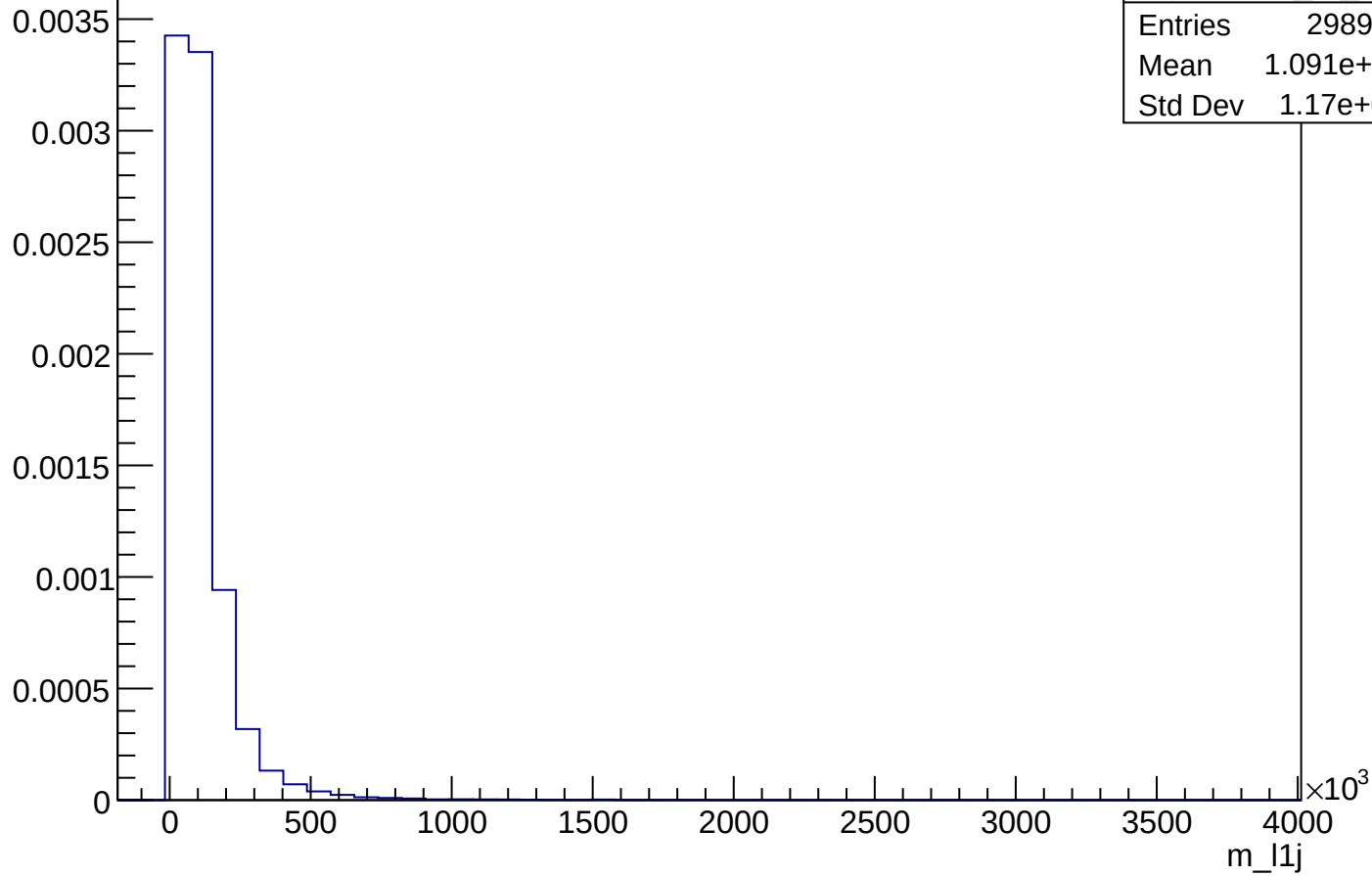
m_j1j2

Events



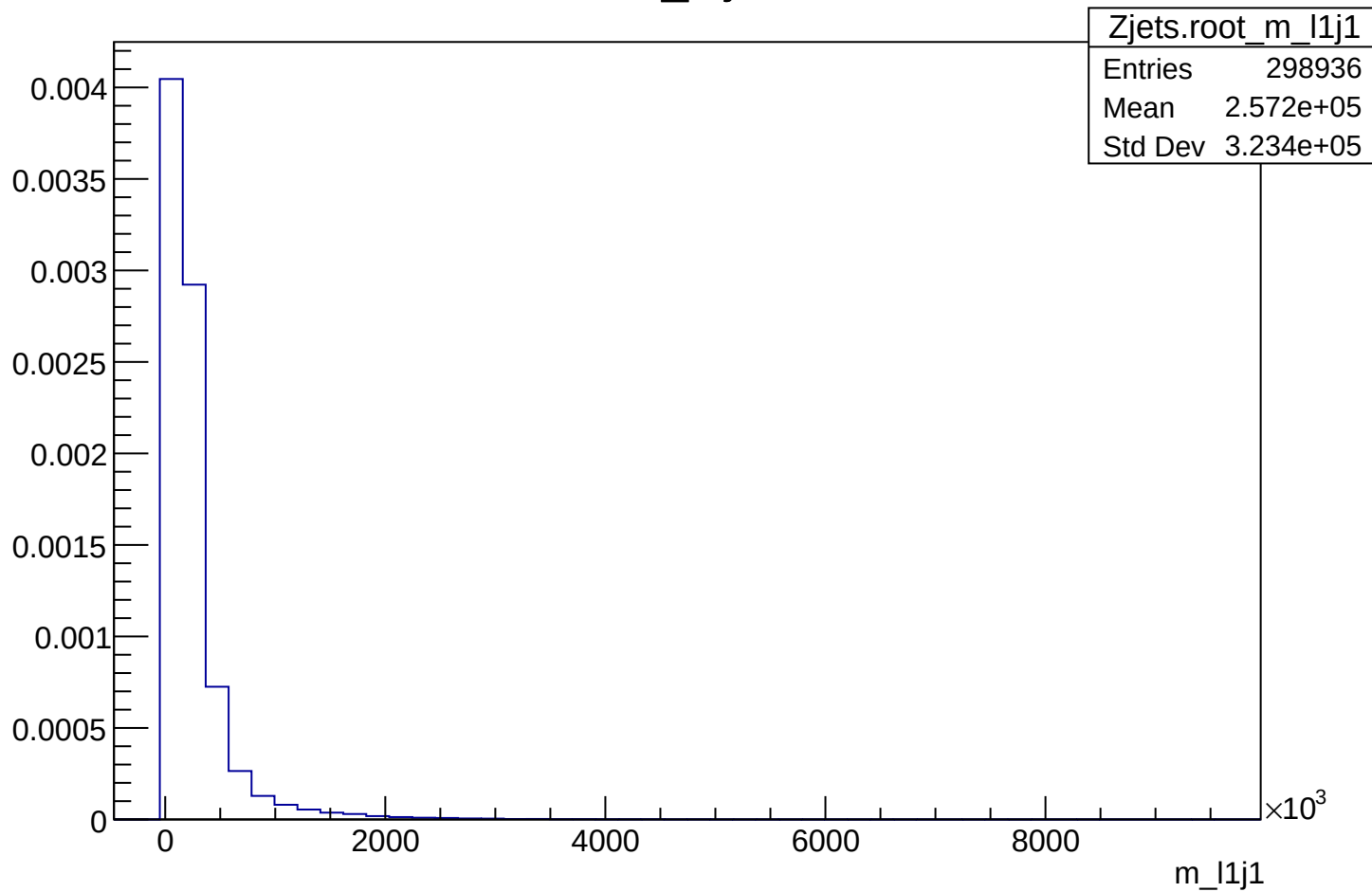
m_l1j

Events



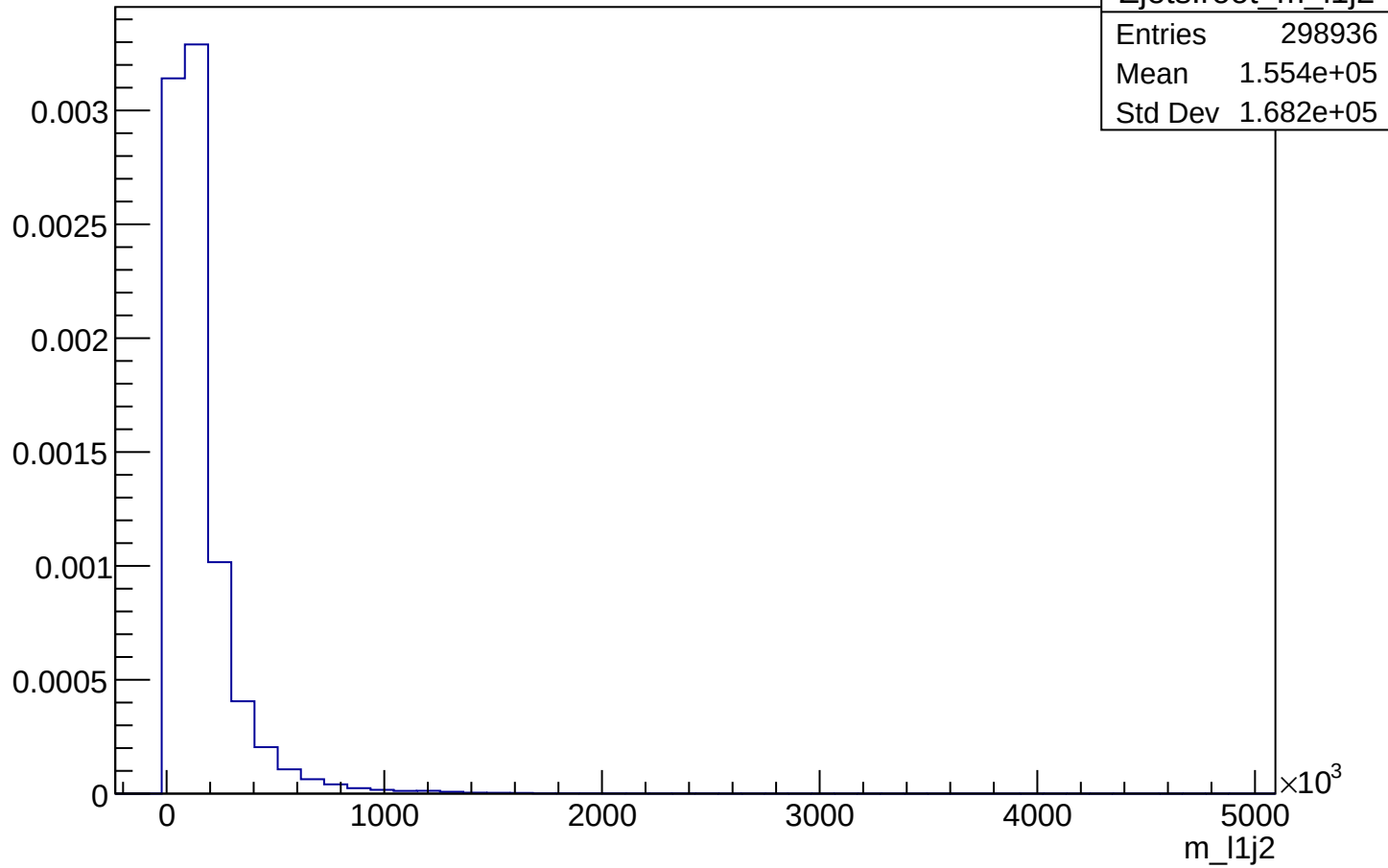
m_l1j1

Events



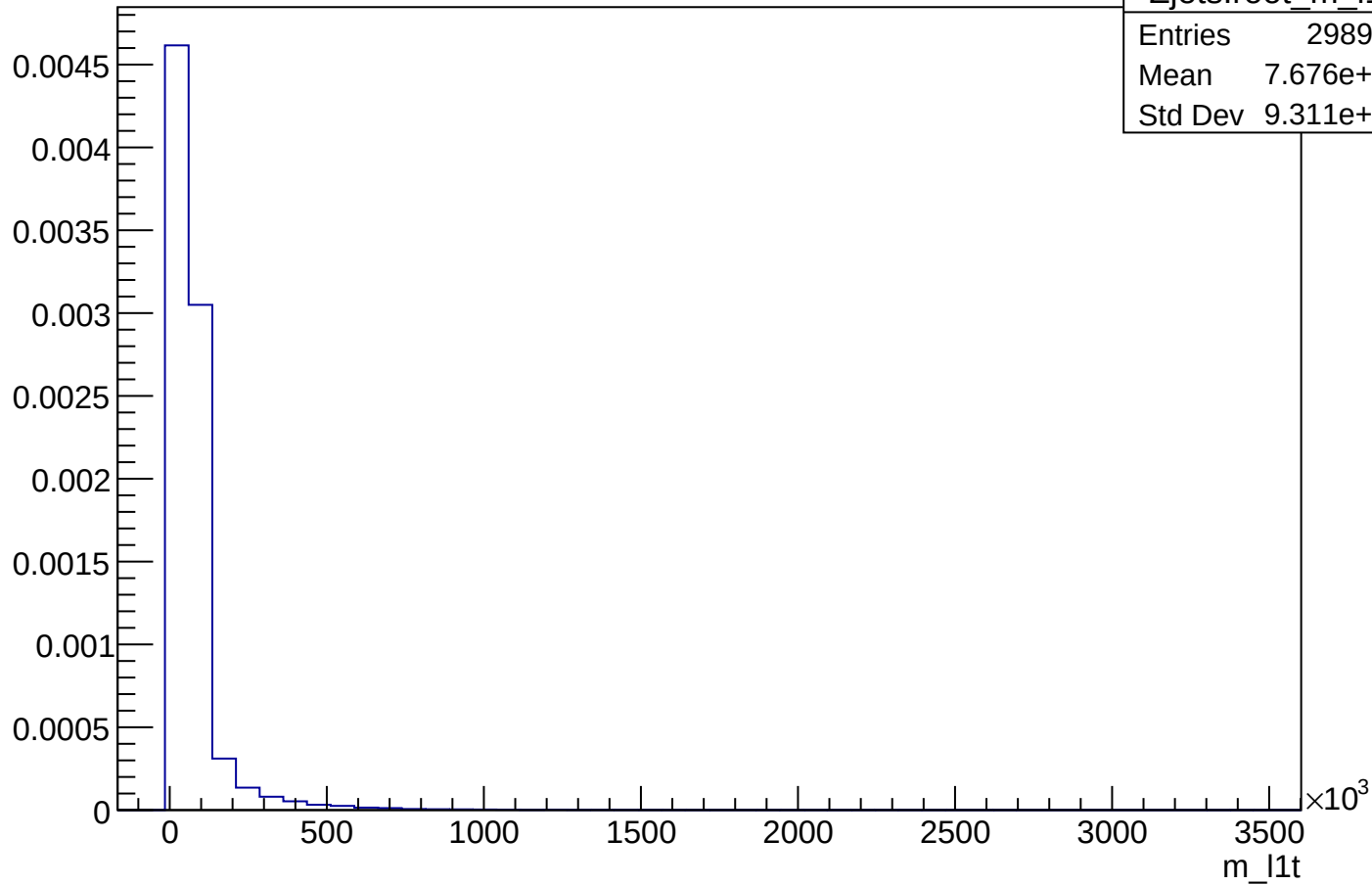
m_l1j2

Events



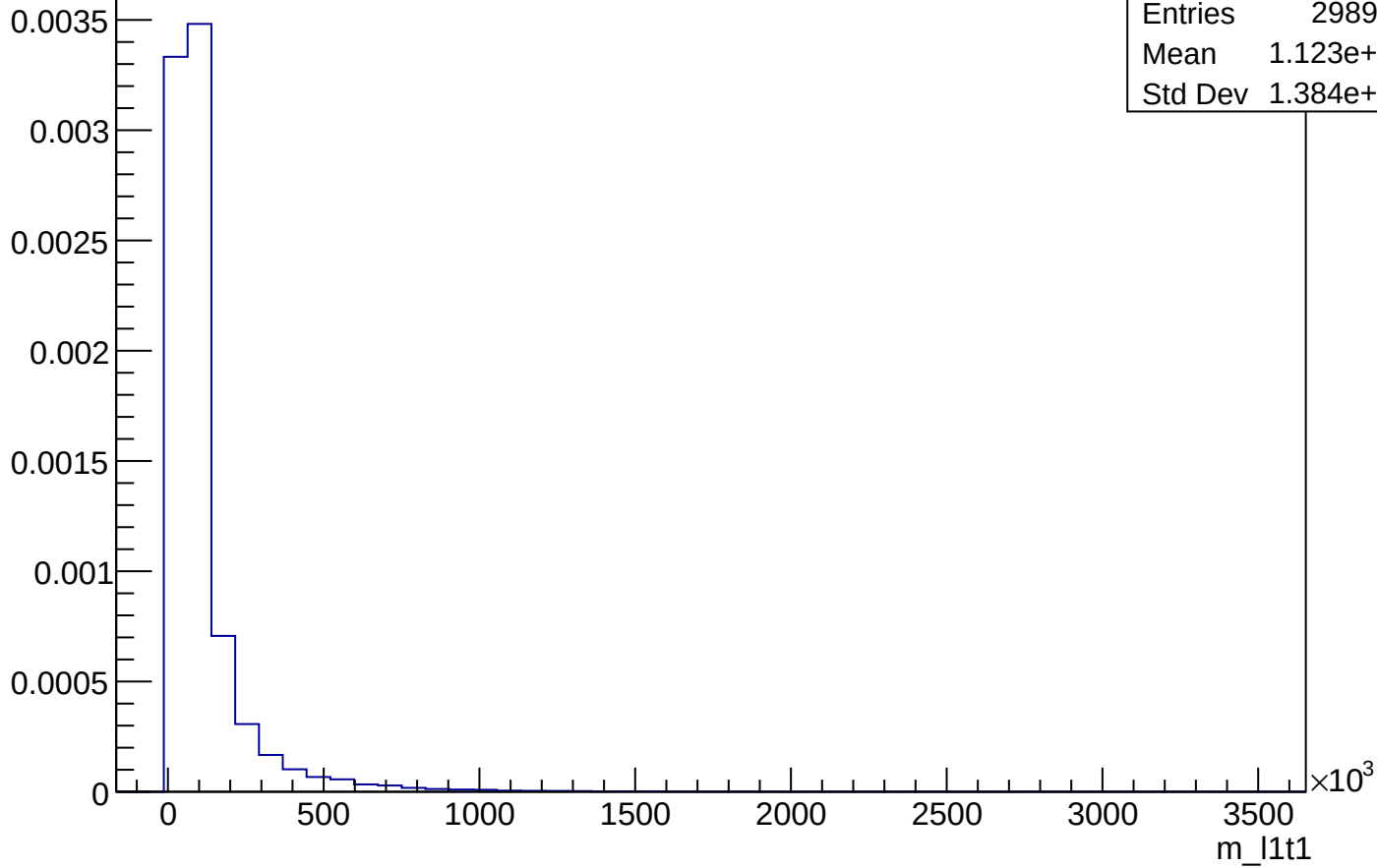
m_l1t

Events



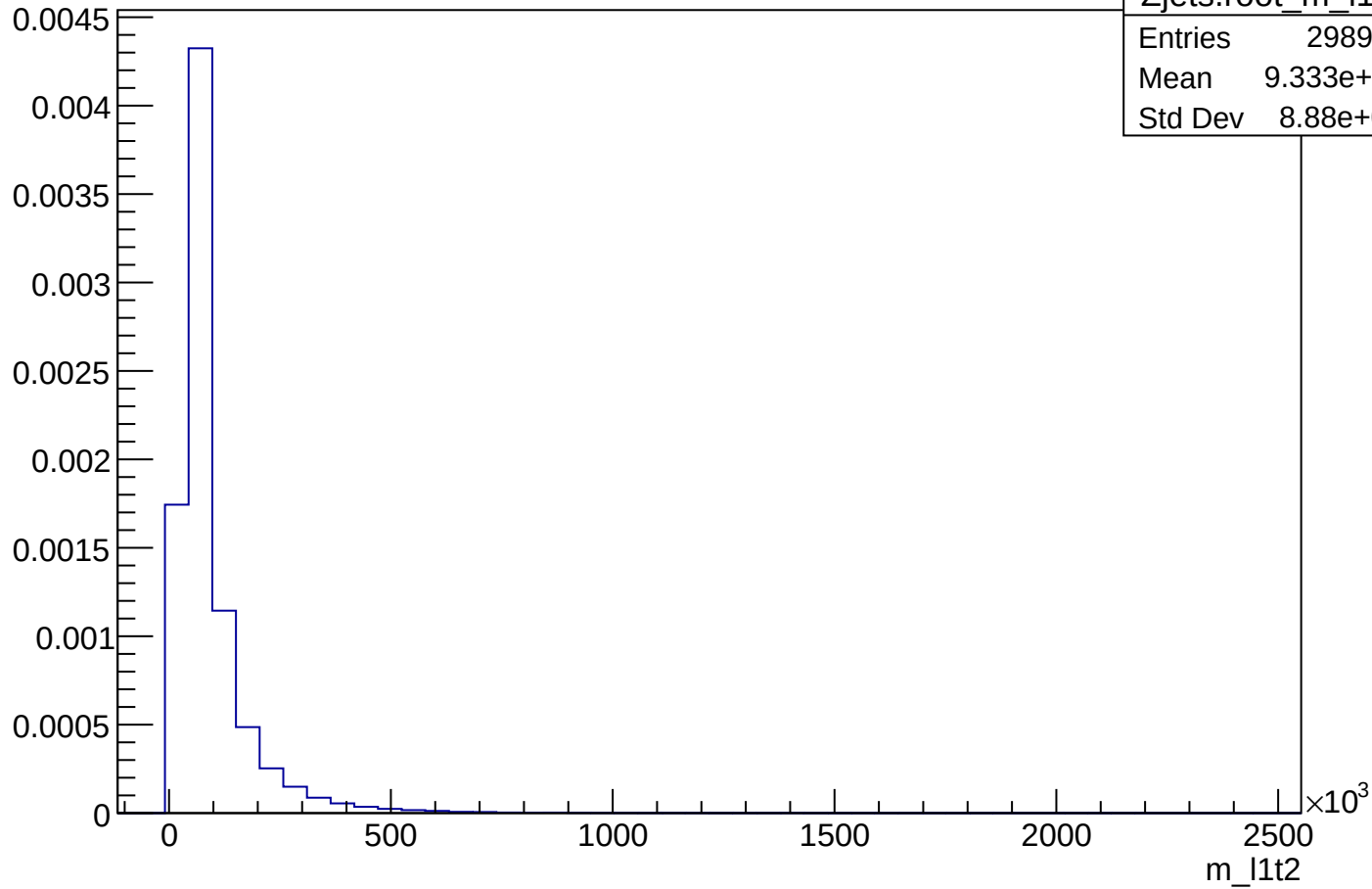
m_l1t1

Events



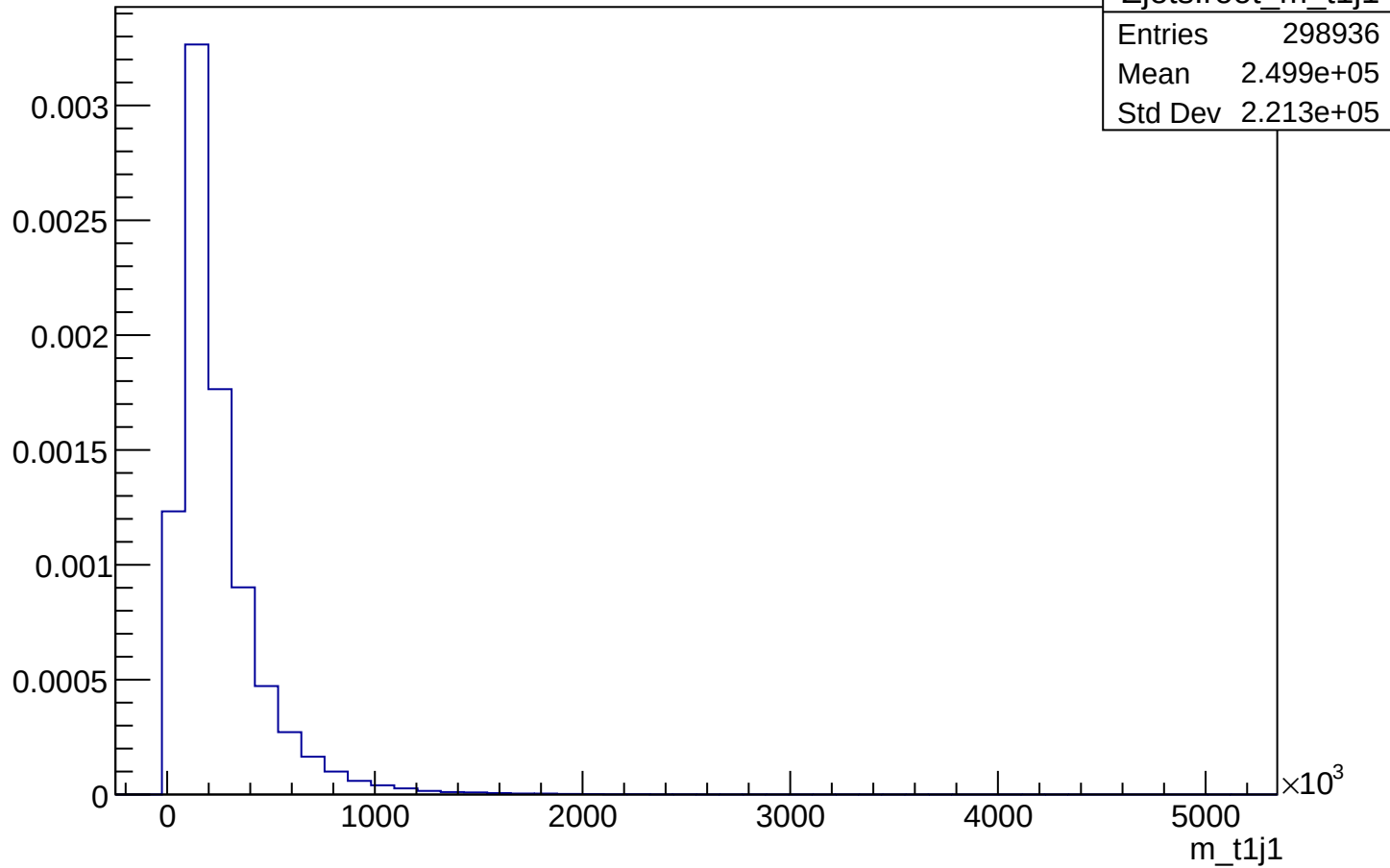
m_l1t2

Events



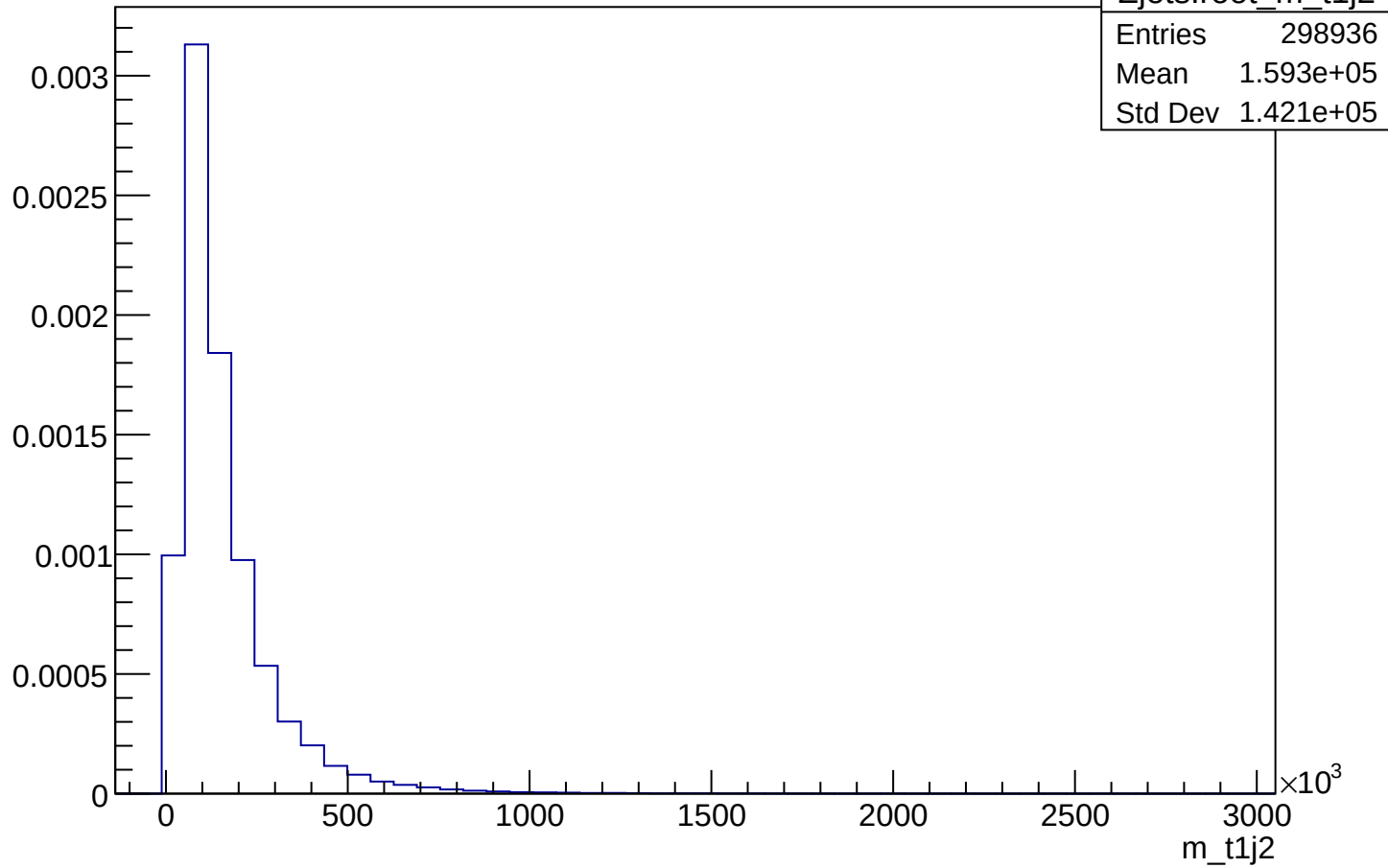
m_t1j1

Events



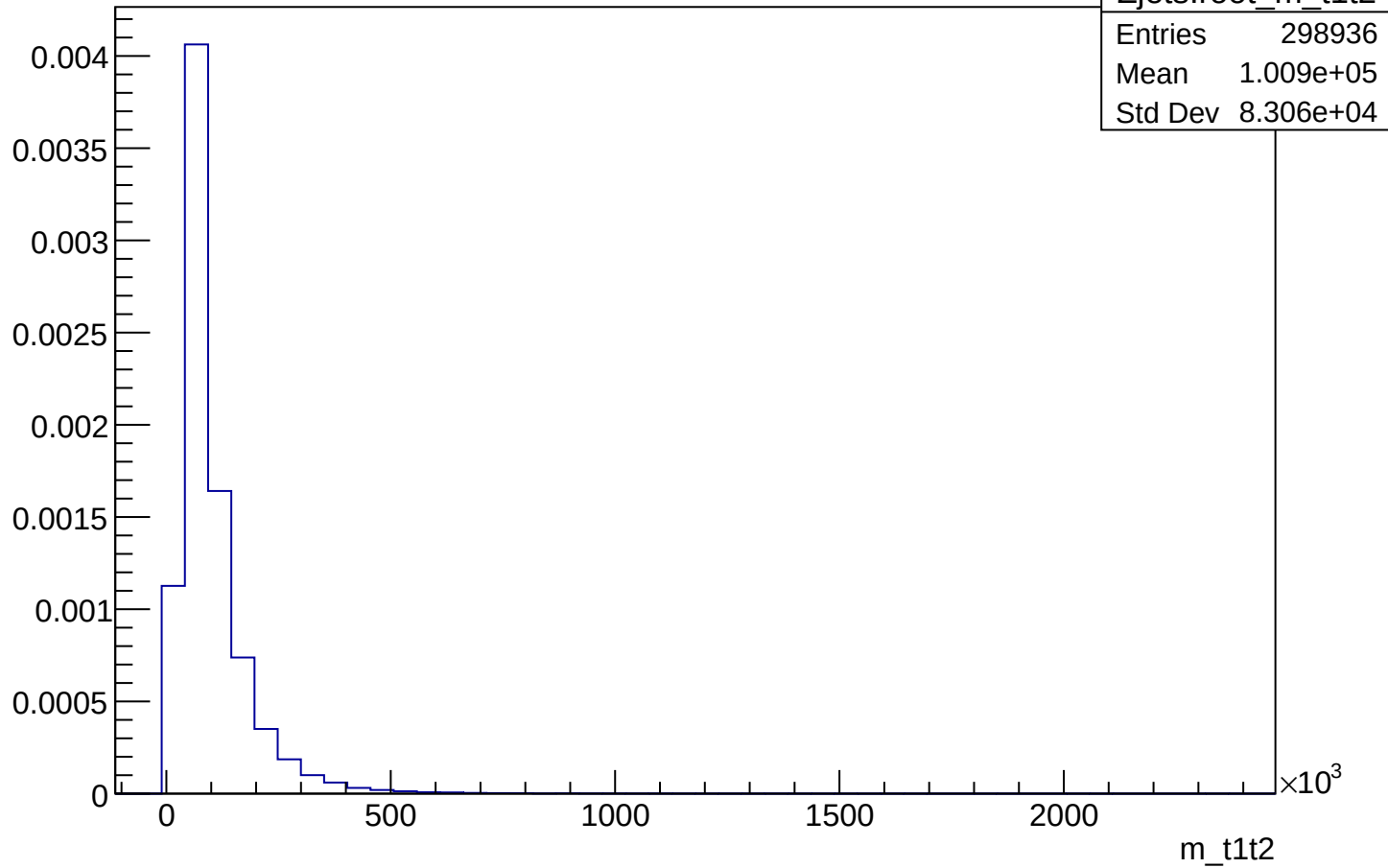
m_t1j2

Events



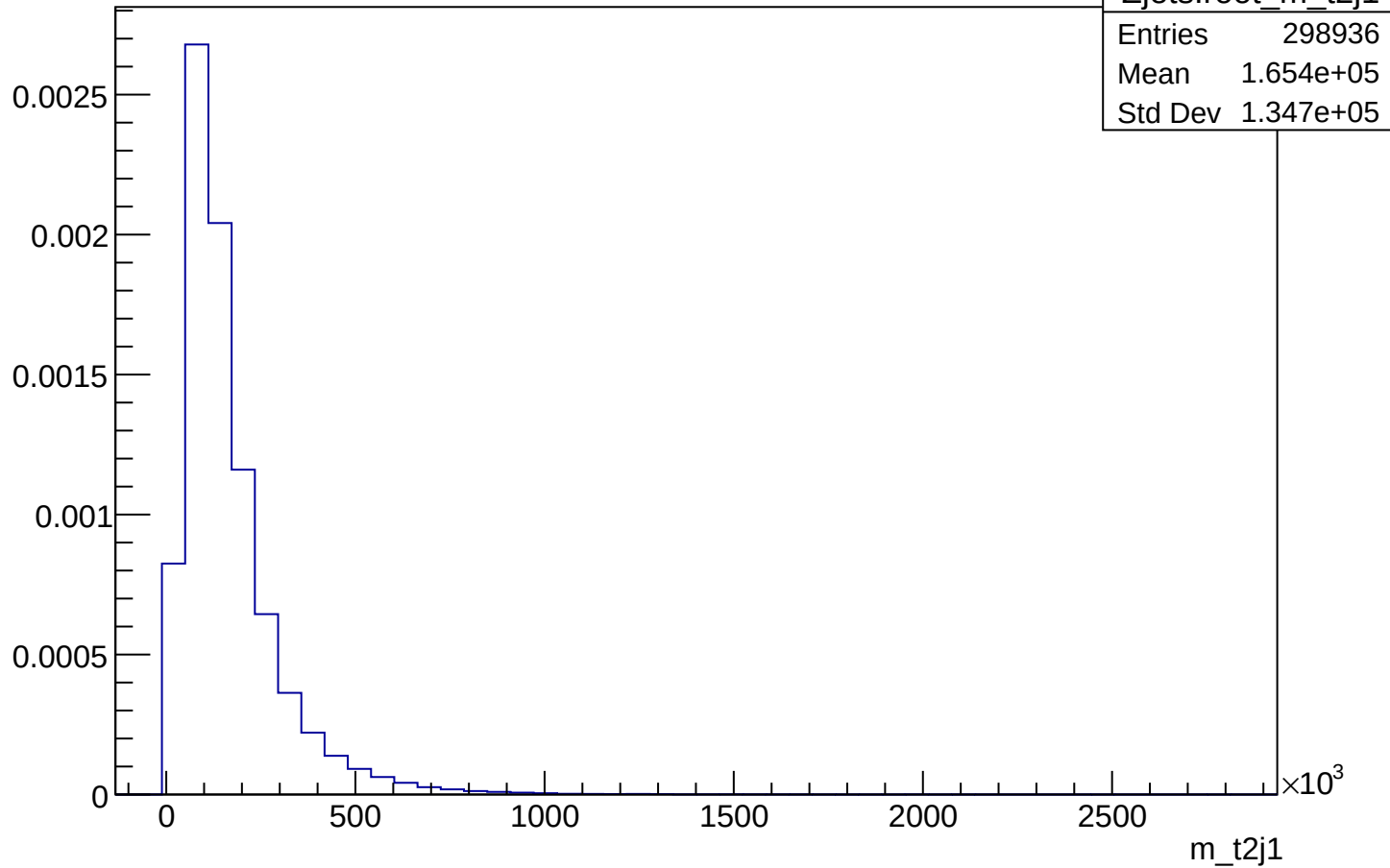
m_t1t2

Events

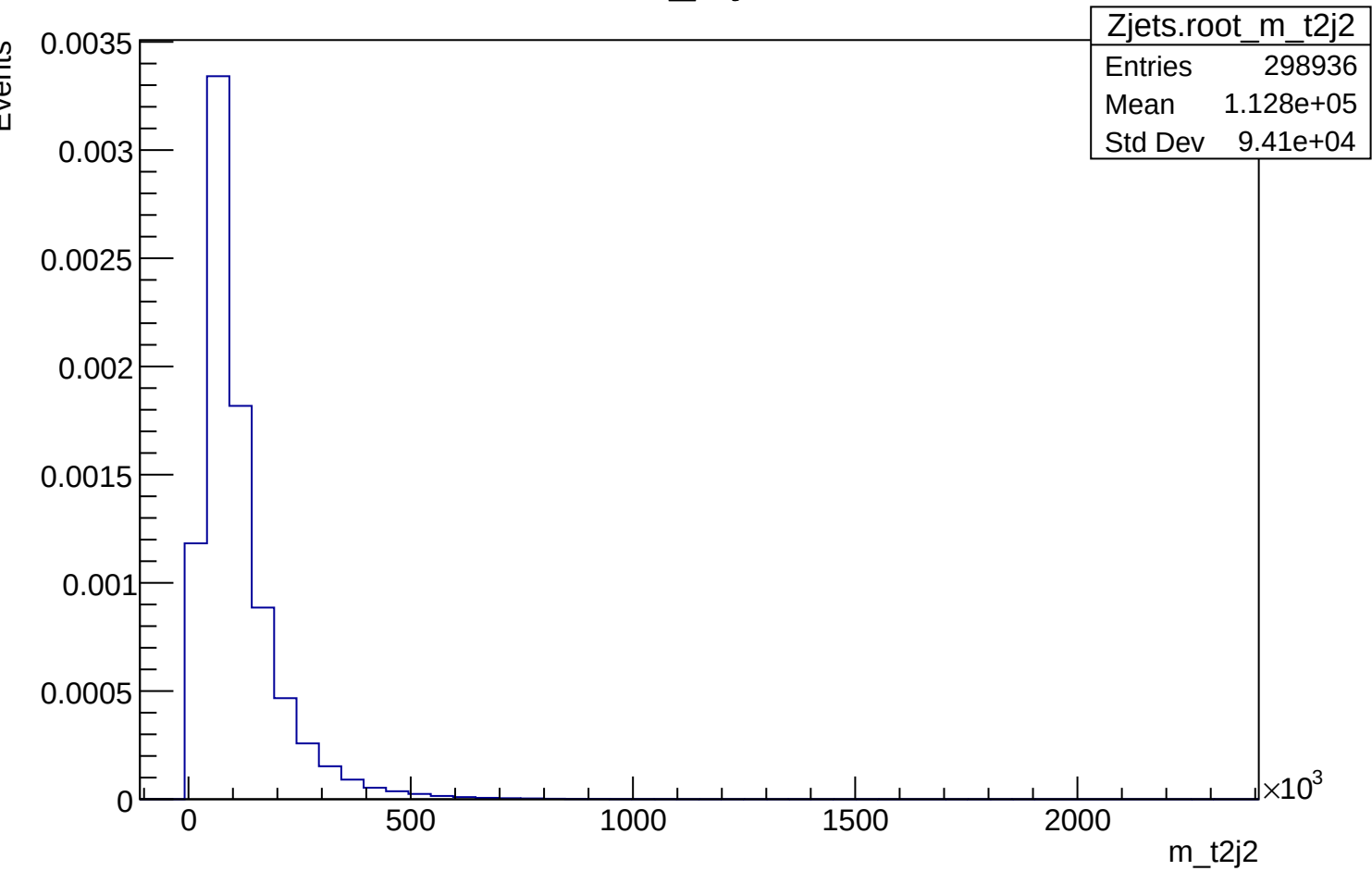


m_t2j1

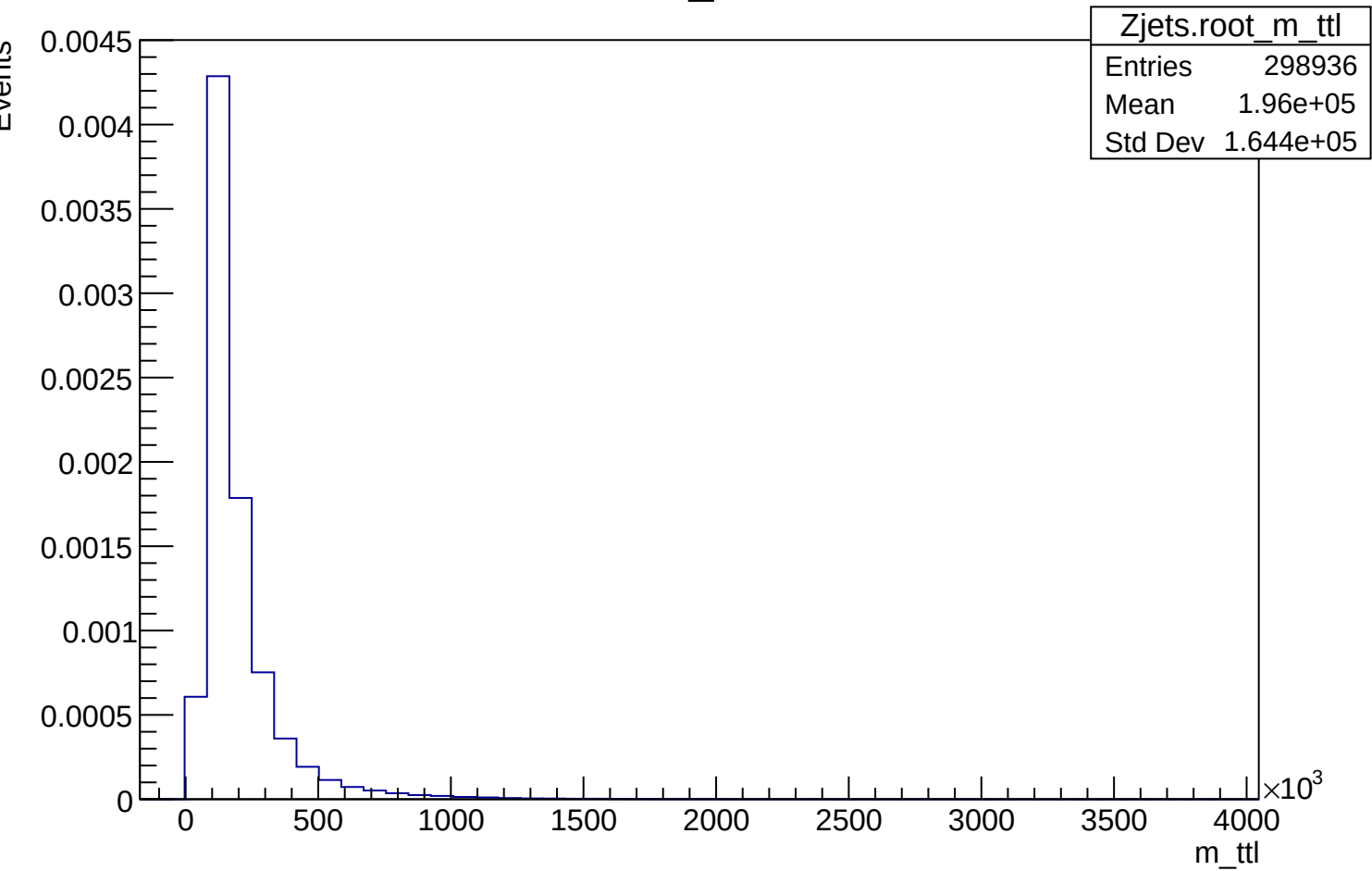
Events



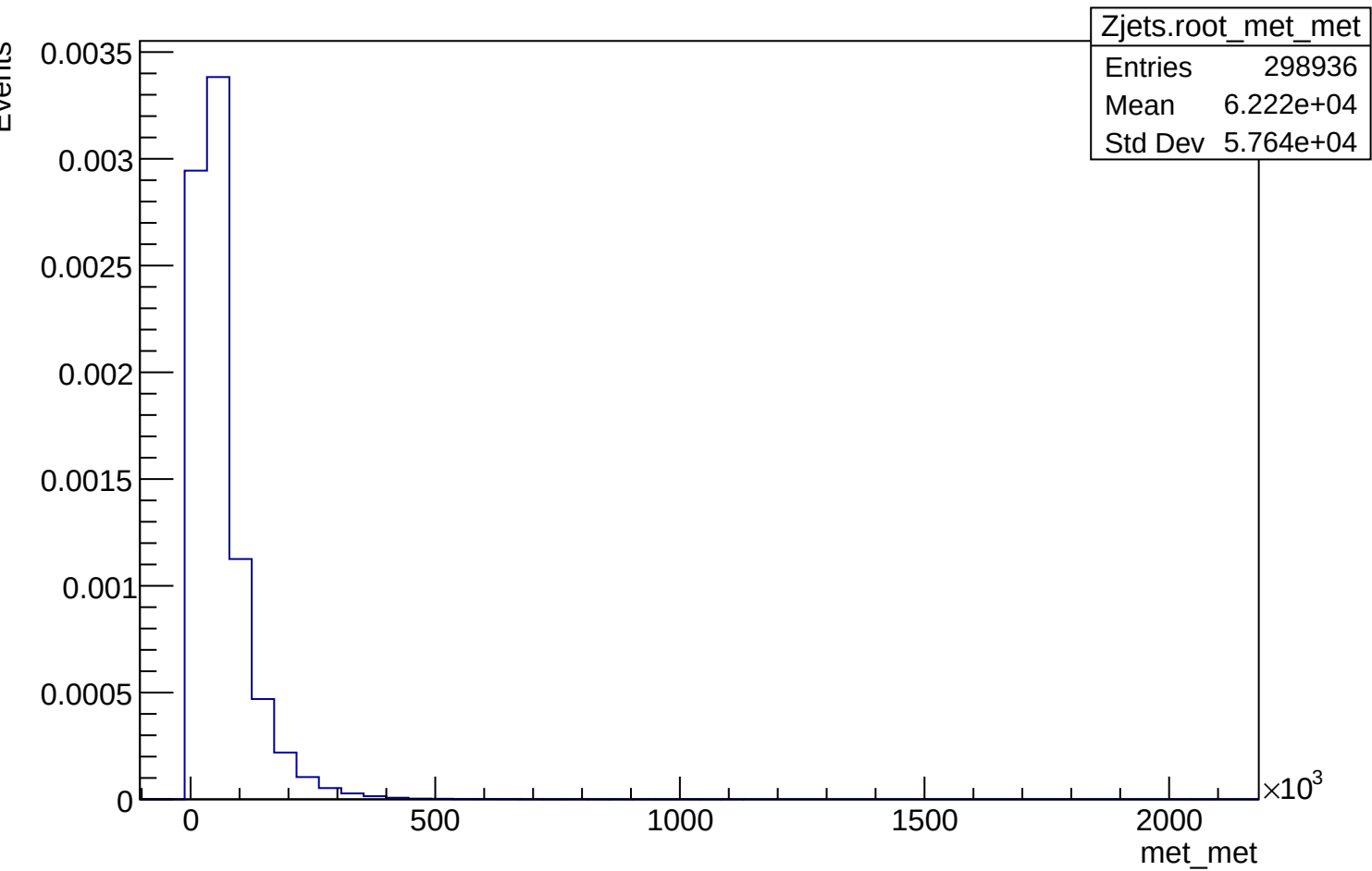
m_t2j2



m_ttl

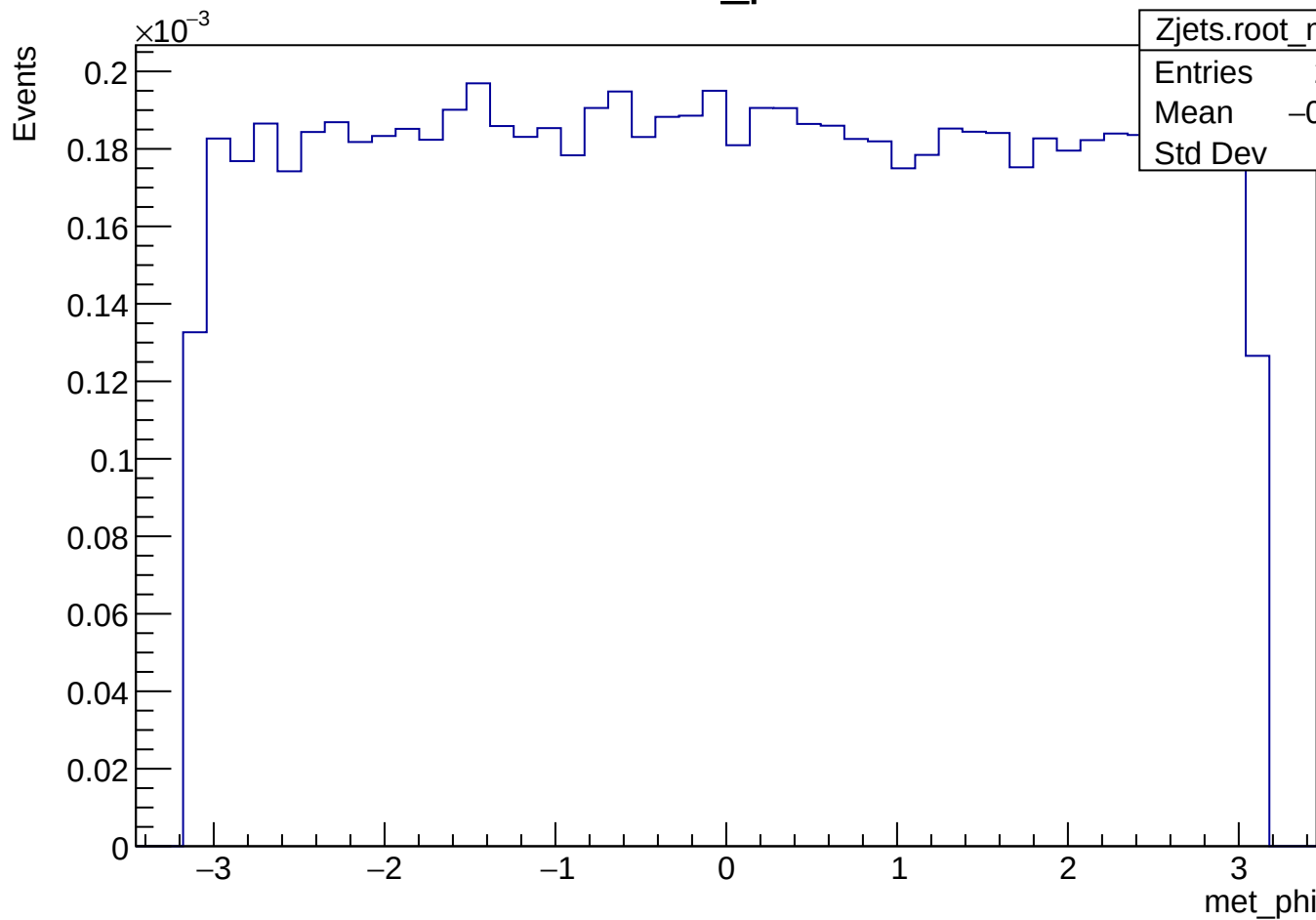


met_met

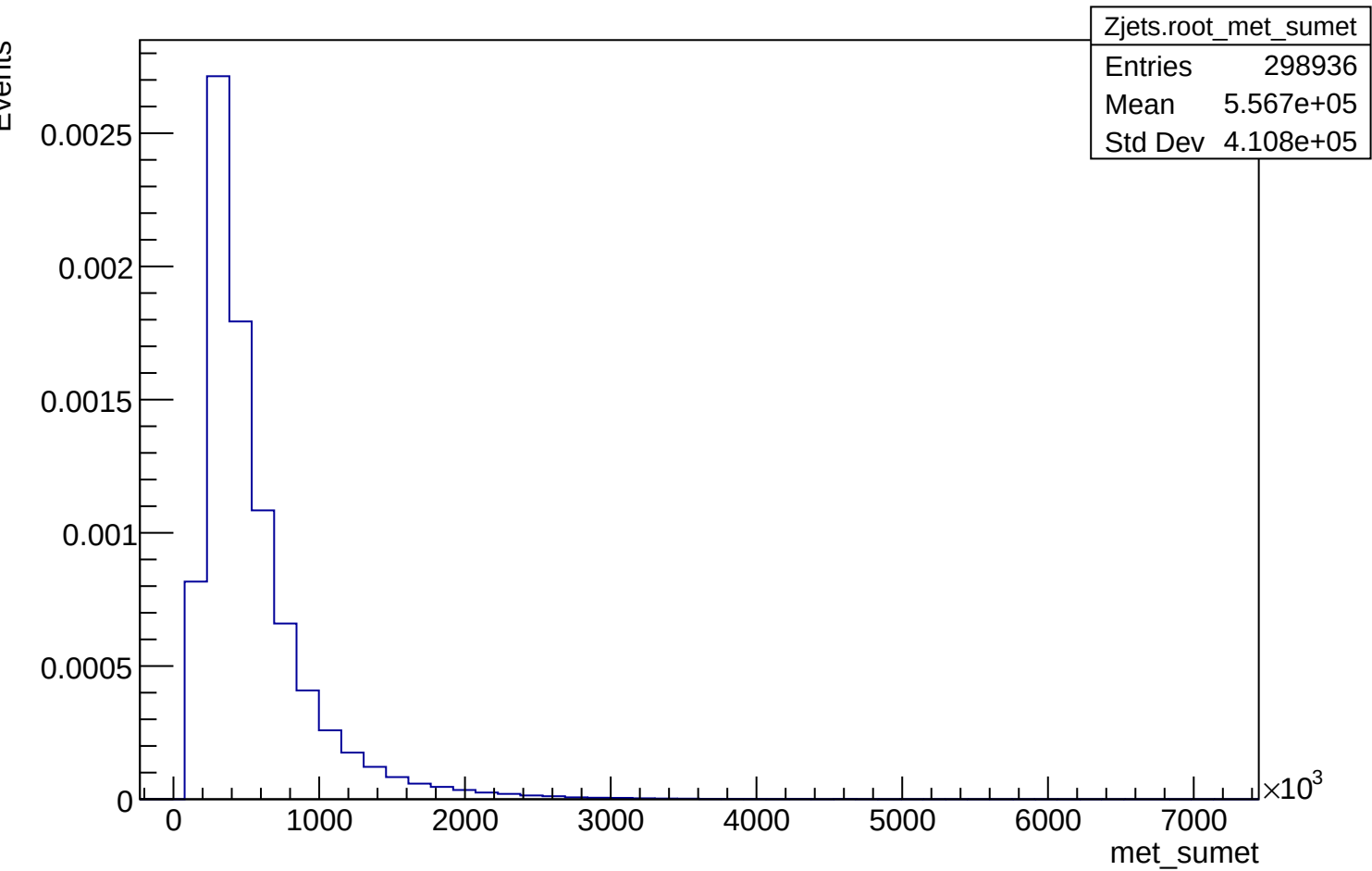


met_phi

Zjets.root_met_phi	
Entries	298936
Mean	-0.01149
Std Dev	1.804

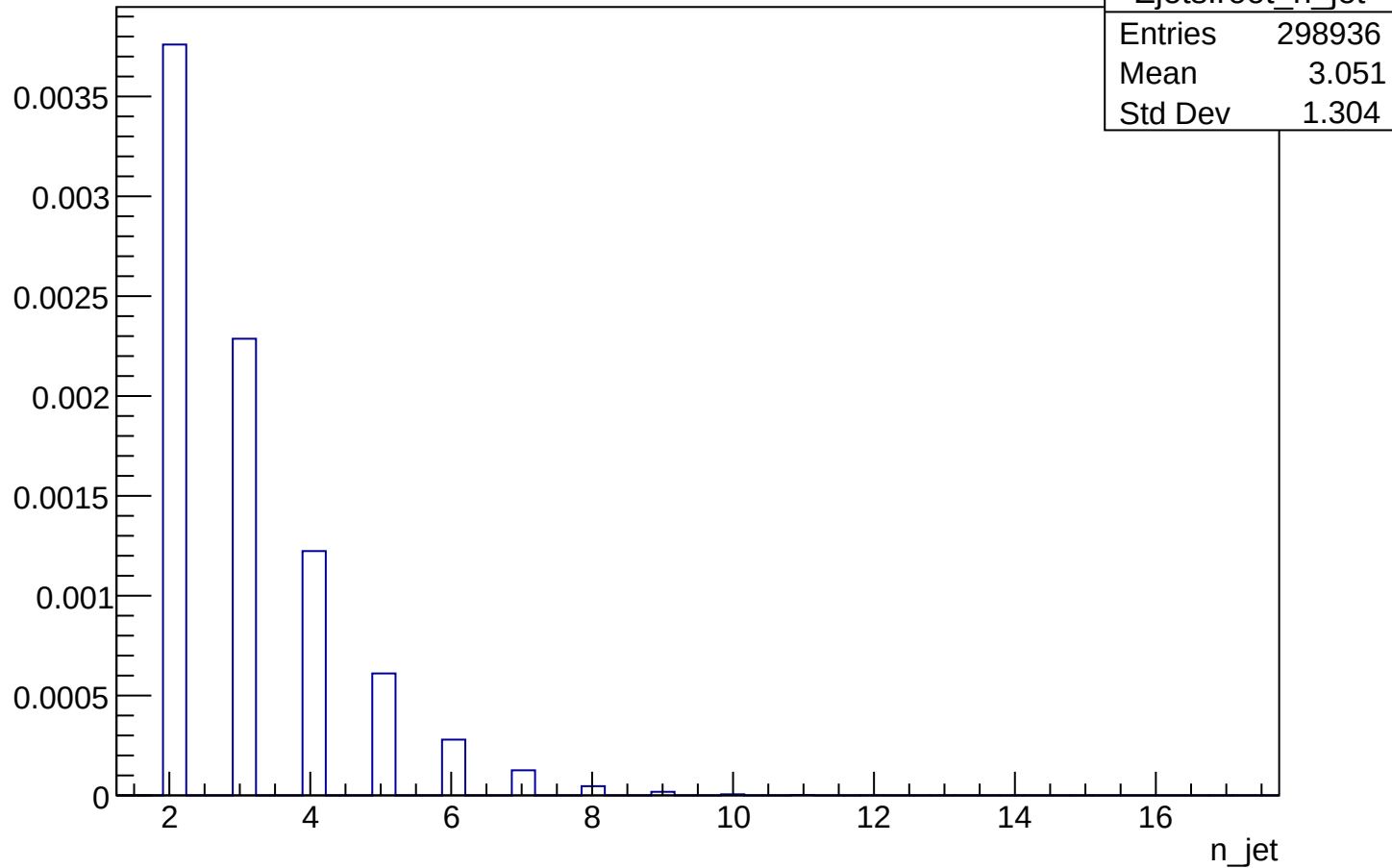


met_sumet



n_jet

Events



pair_isOSleptau

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

0.8

pair_isOSleptau

Zjets.root_pair_isOSleptau	
Entries	298936
Mean	0.7443
Std Dev	0.4363

pair_isOStaus

Events

0.0045
0.004
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0

0

0.2

0.4

0.6

0.8

pair_isOStaus

Zjets.root_pair_isOStaus	
Entries	298936
Mean	0.5491
Std Dev	0.4976

pass_DTT

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

0.8

1

pass_DTT

Zjets.root_pass_DTT	
Entries	298936
Mean	0.092
Std Dev	0.289

pass_LTT

Events

0.005

0.004

0.003

0.002

0.001

0

0

0.2

0.4

0.6

0.8

pass_LTT

Zjets.root_pass_LTT	
Entries	298936
Mean	0.6182
Std Dev	0.4858

pass_SLT

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

0.8

pass_SLT

Zjets.root_pass_SLT	
Entries	298936
Mean	0.7146
Std Dev	0.4516

pass_STT

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

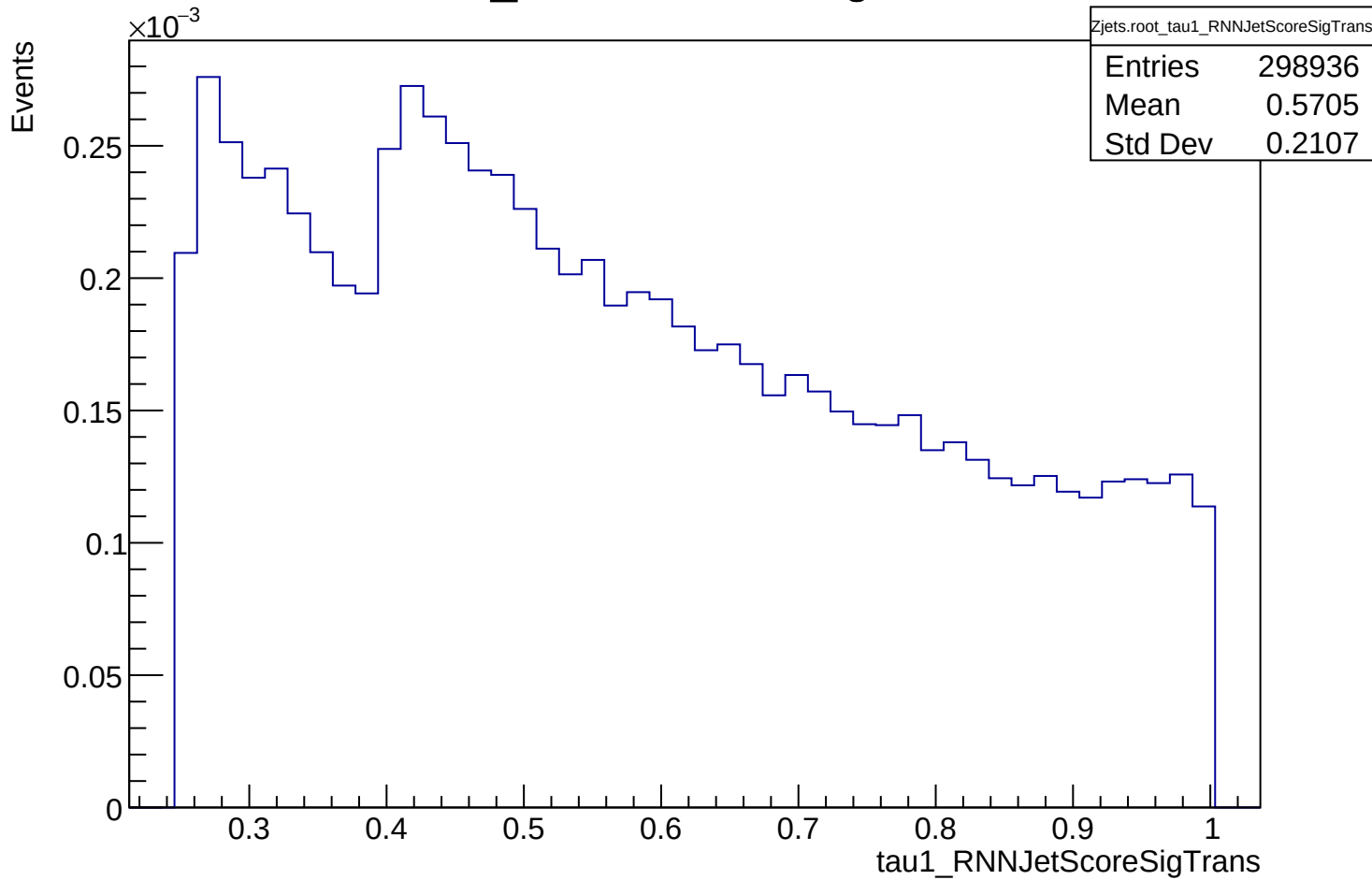
0.8

1

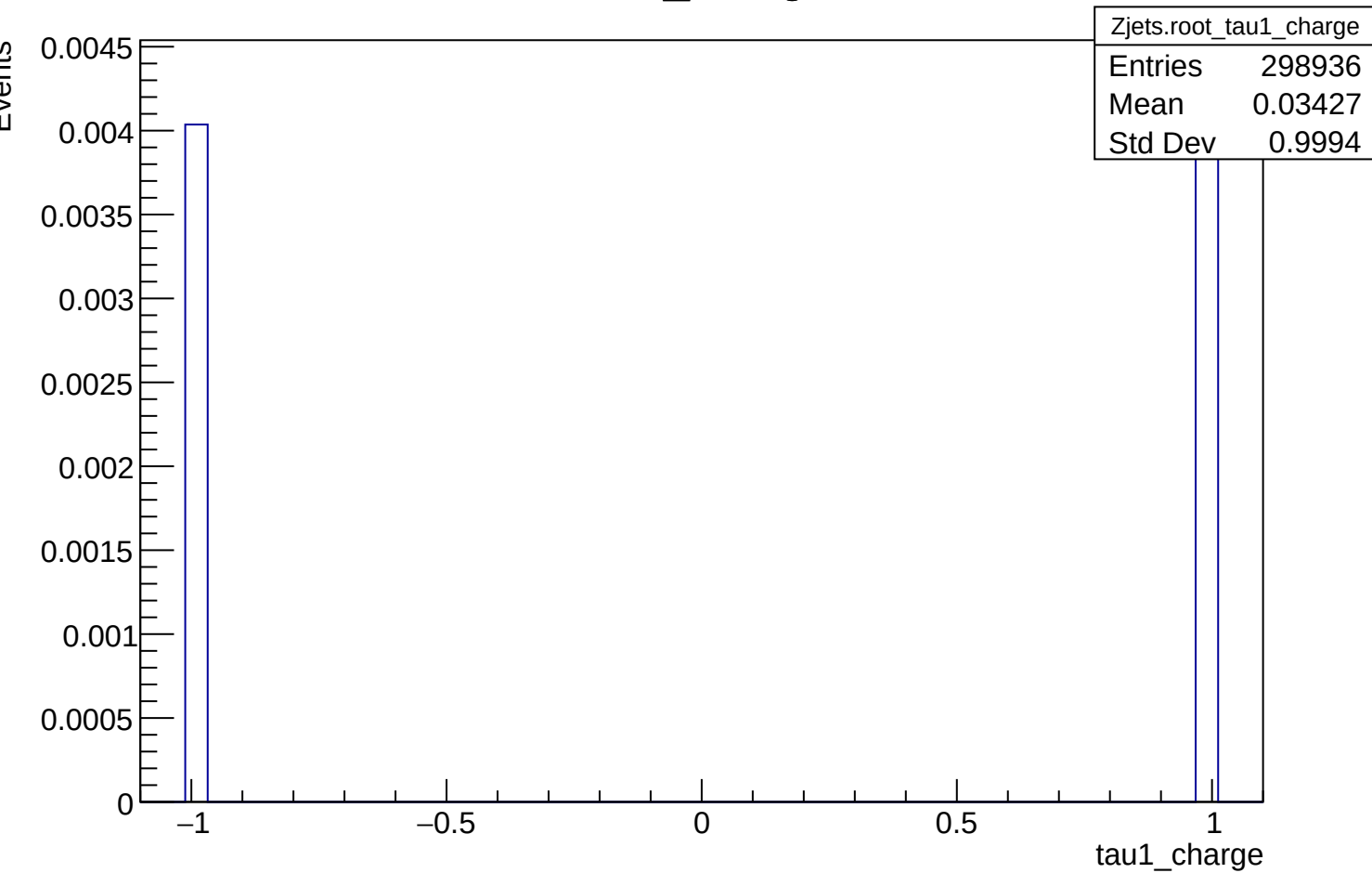
pass_STT

Zjets.root_pass_STT	
Entries	298936
Mean	0.09371
Std Dev	0.2914

tau1_RNNJetScoreSigTrans

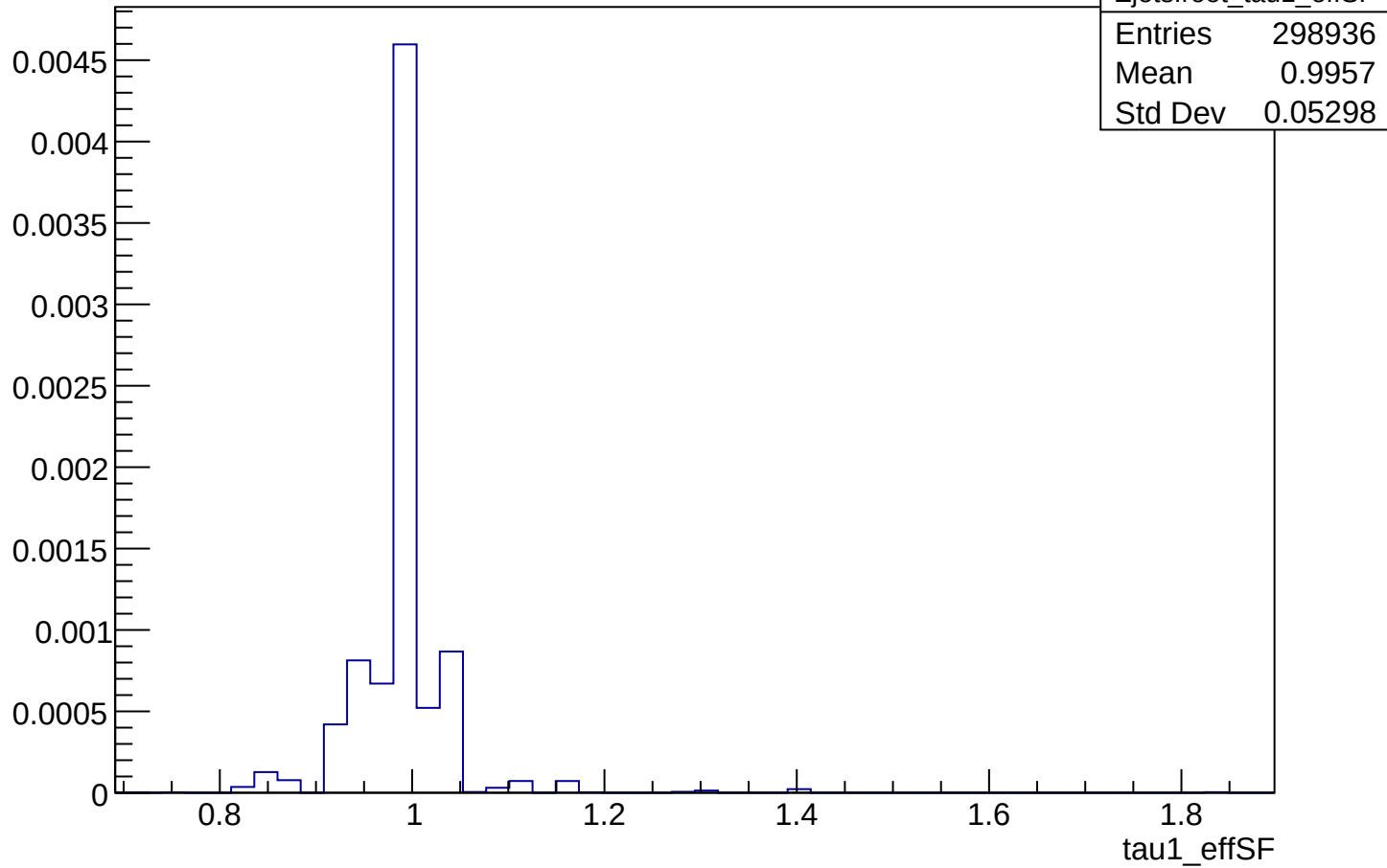


tau1_charge

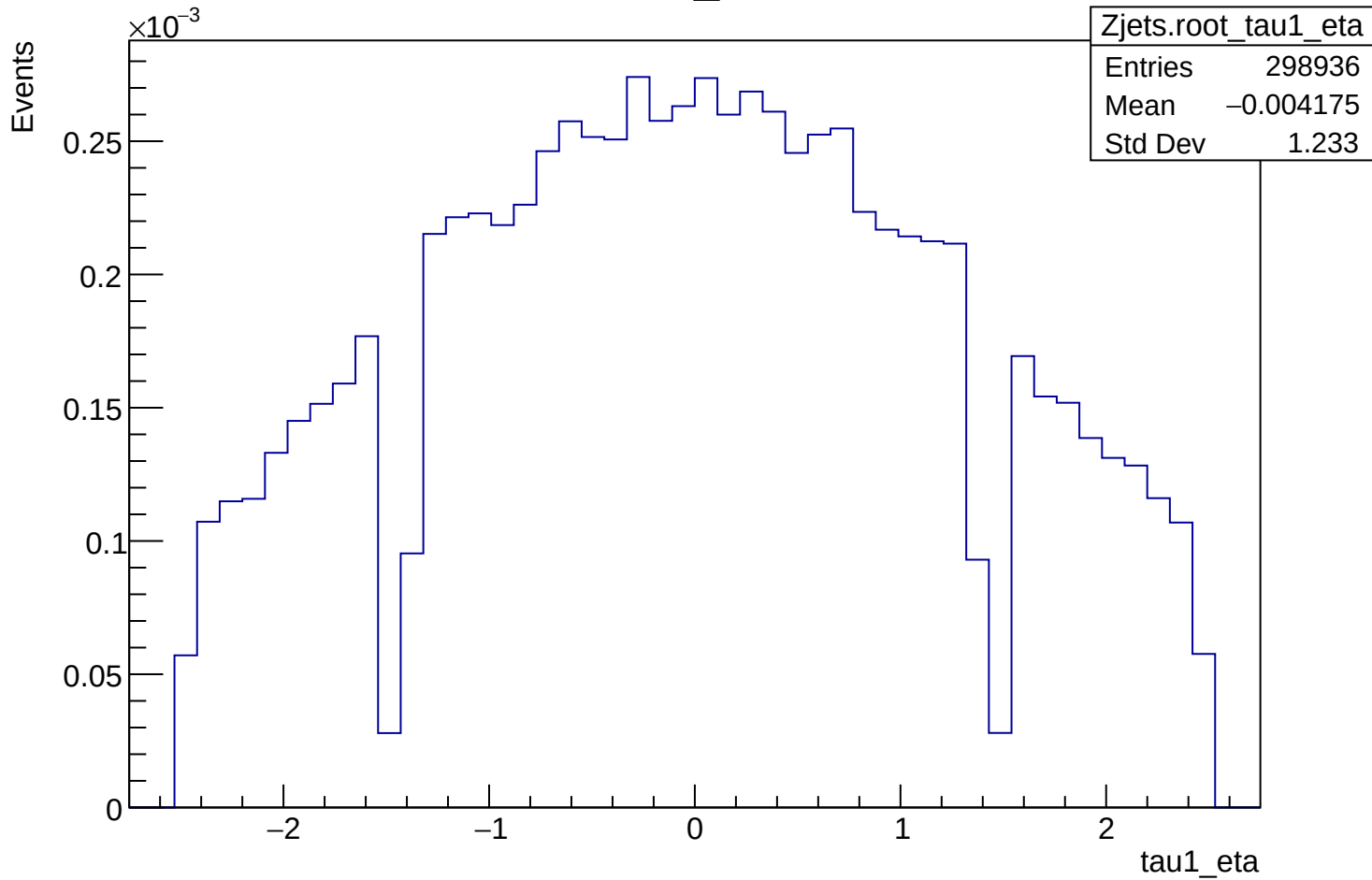


tau1_effSF

Events



tau1_eta



tau1_nprong

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

1

1.5

2

2.5

3

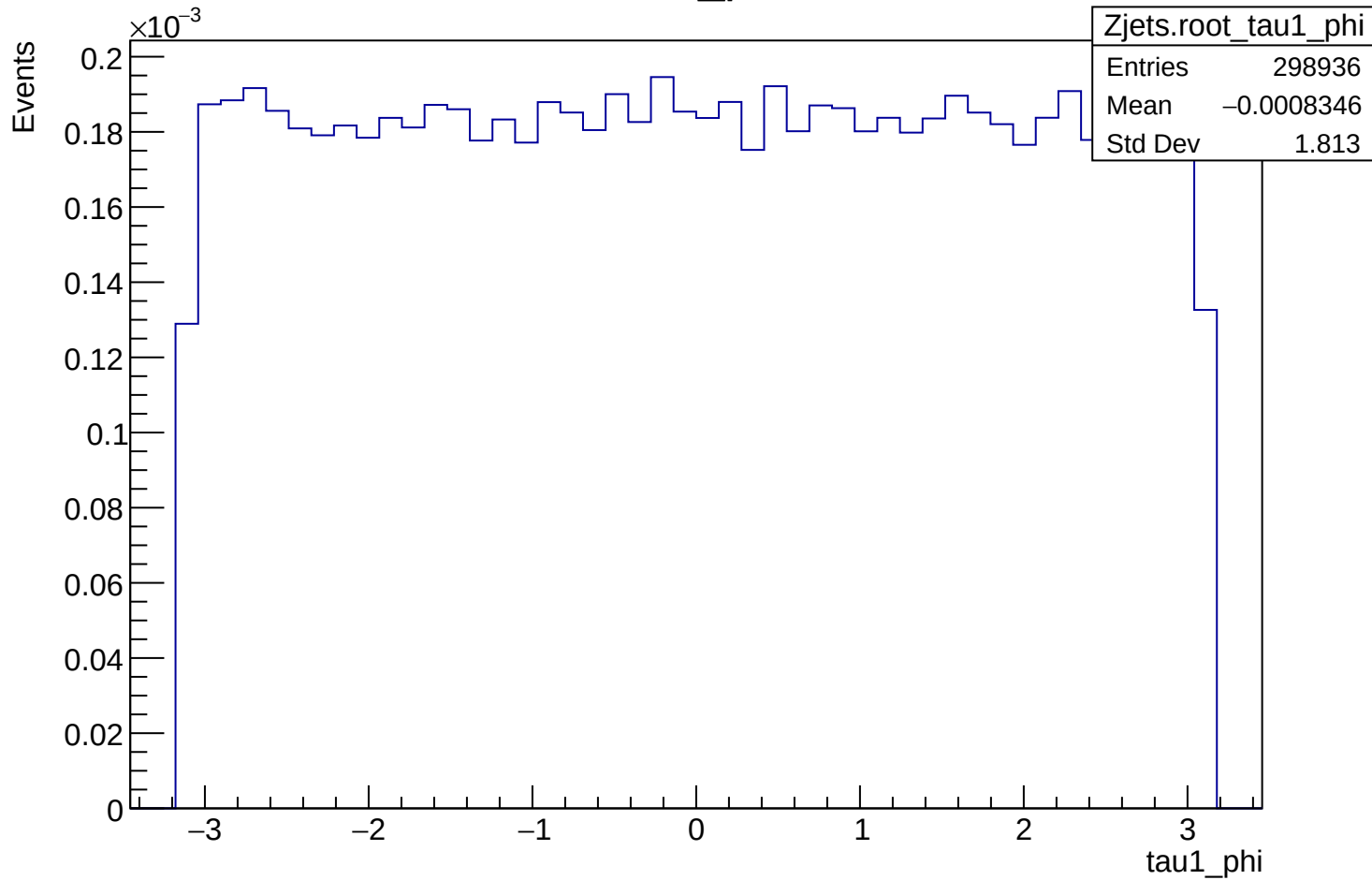
tau1_nprong

Zjets.root_tau1_nprong	
Entries	298936
Mean	1.402
Std Dev	0.8014

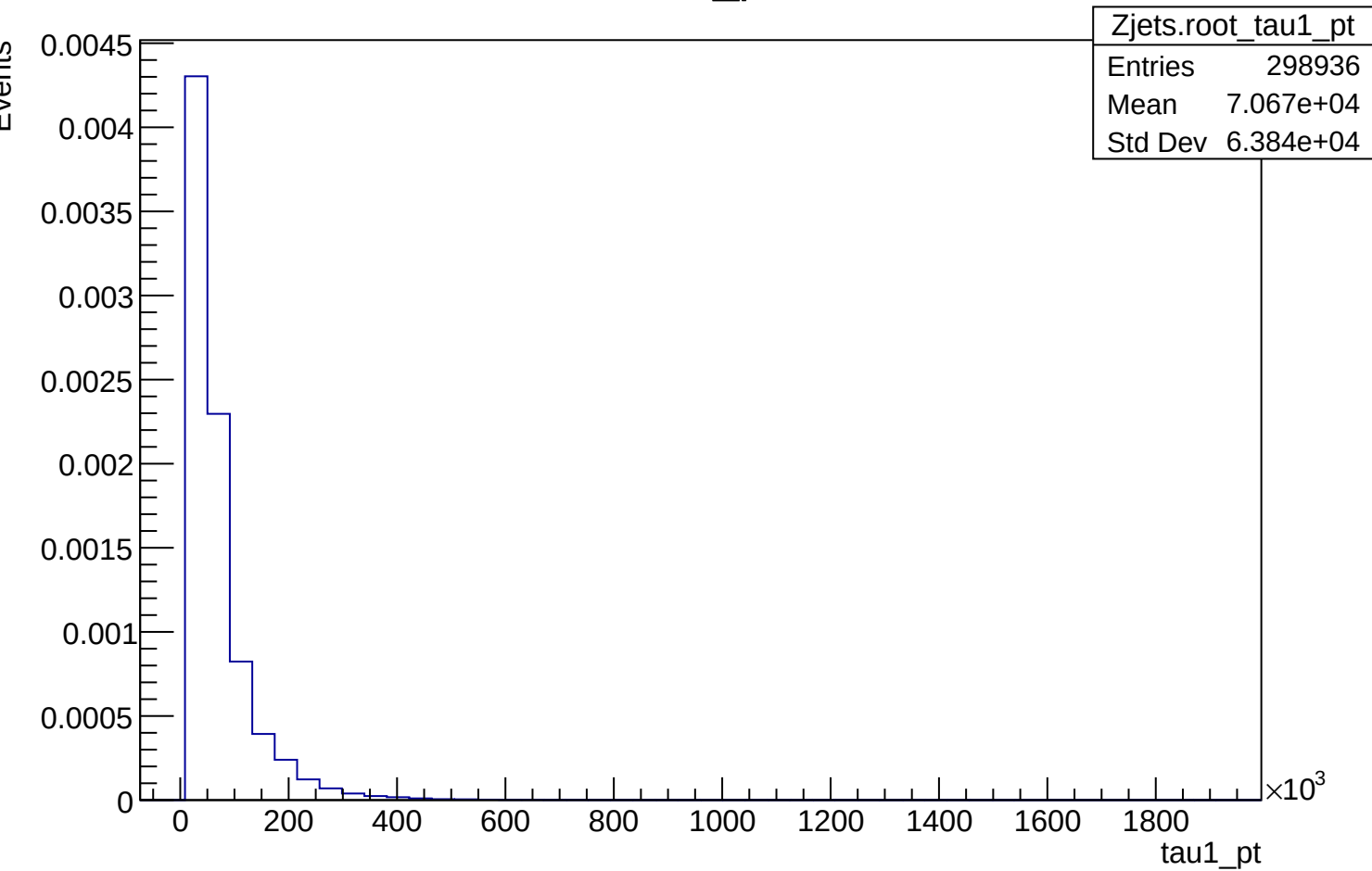
0.0067

0.0017

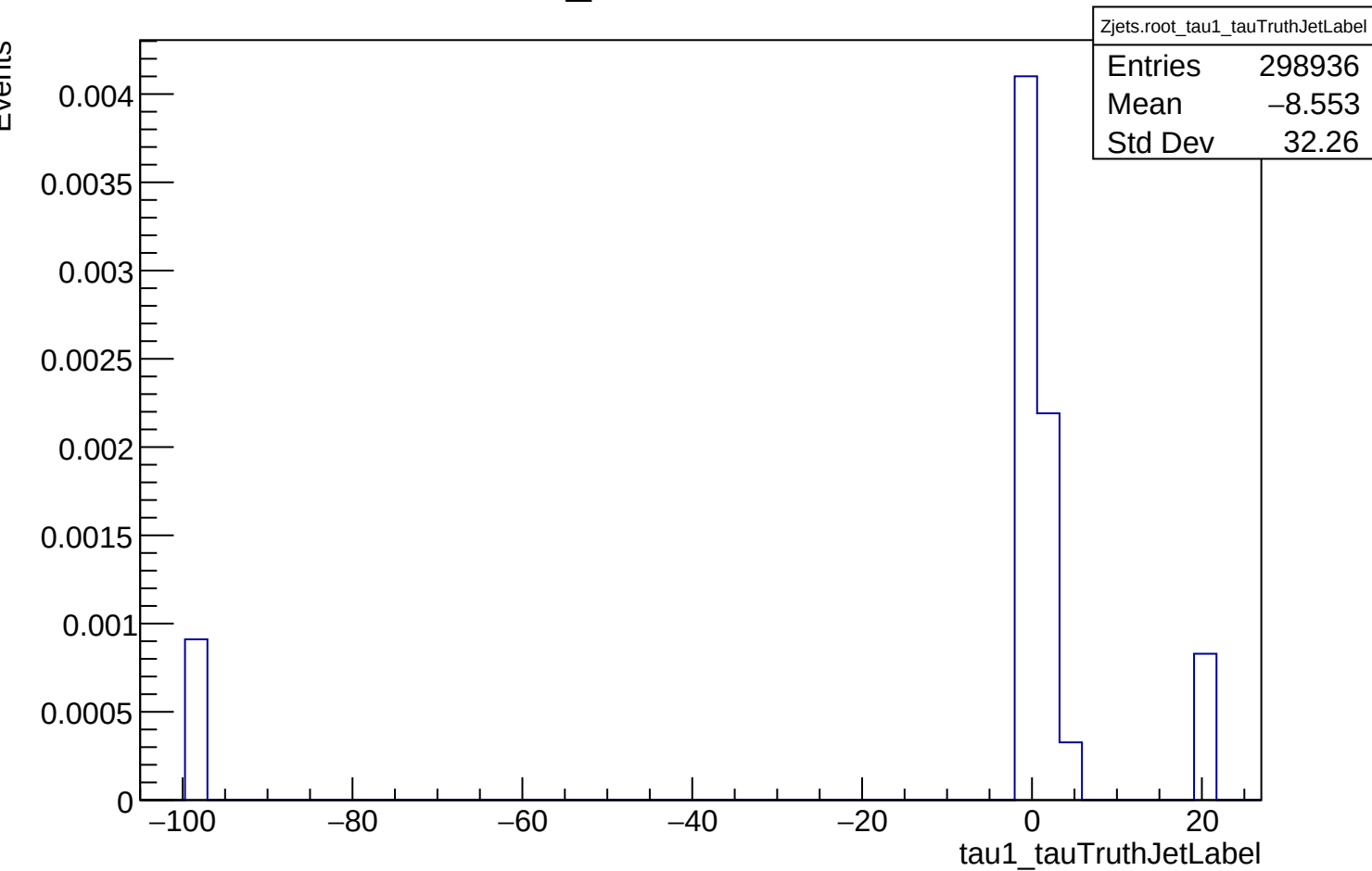
tau1_phi



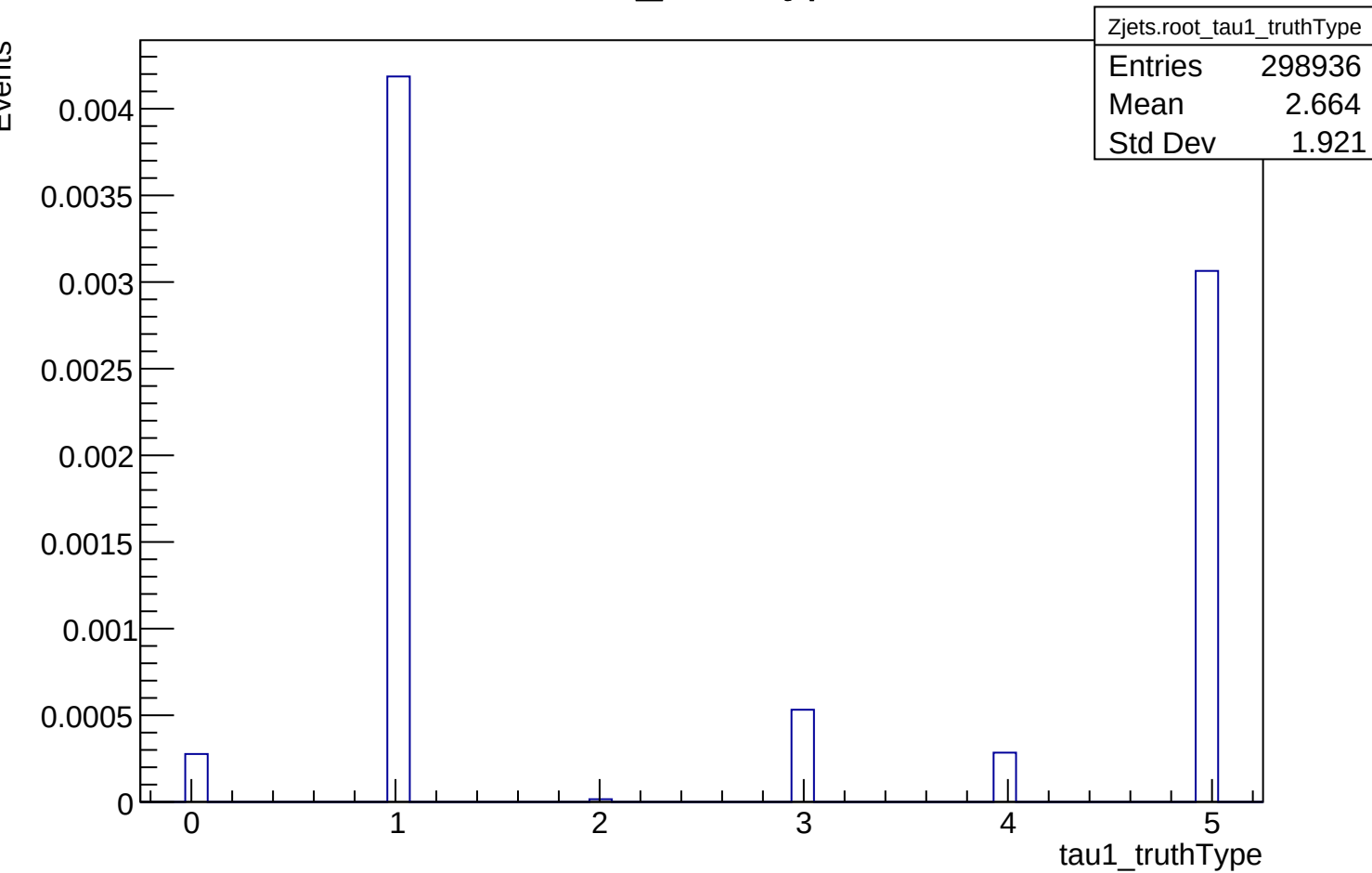
tau1_pt



tau1_tauTruthJetLabel

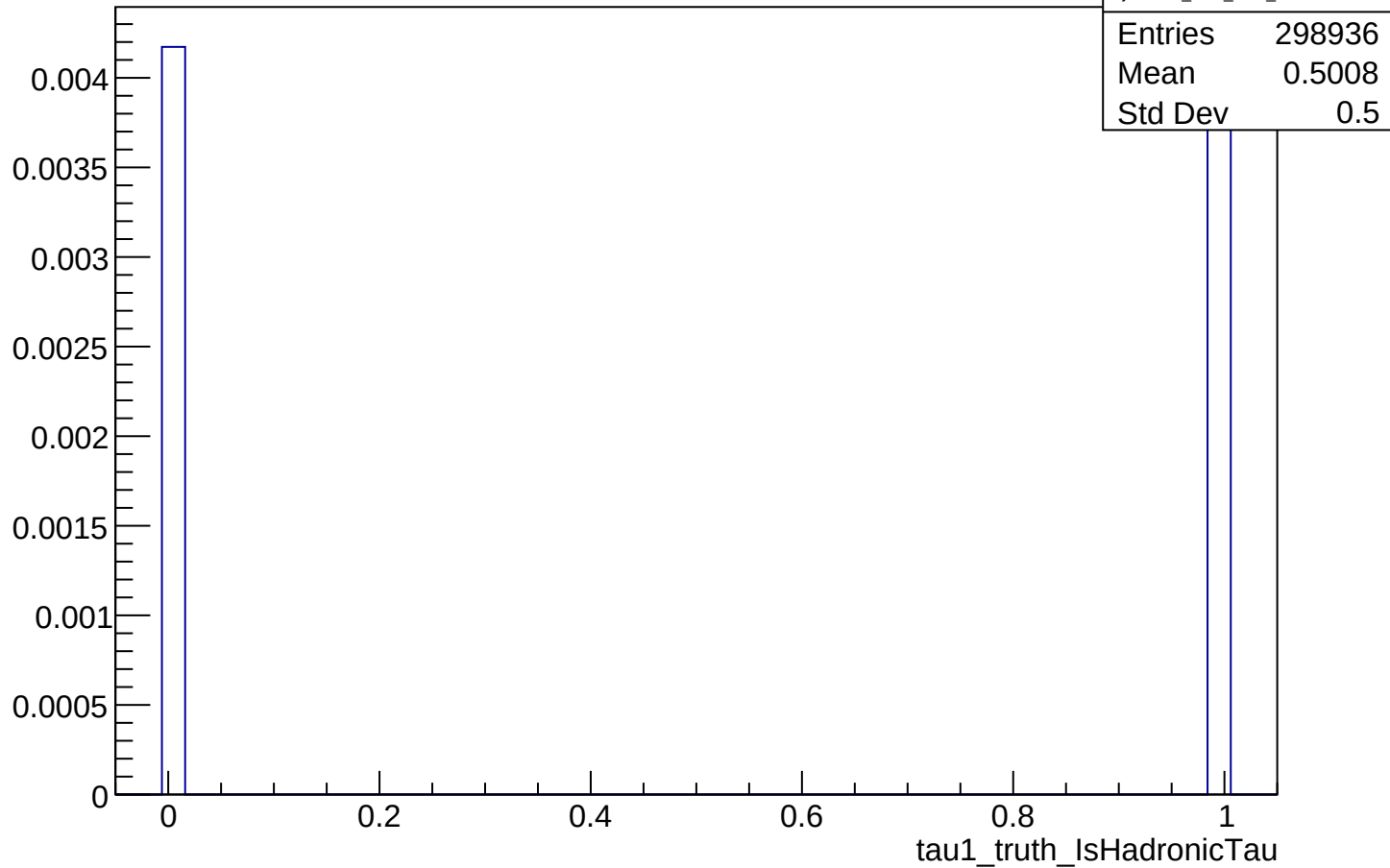


tau1_truthType

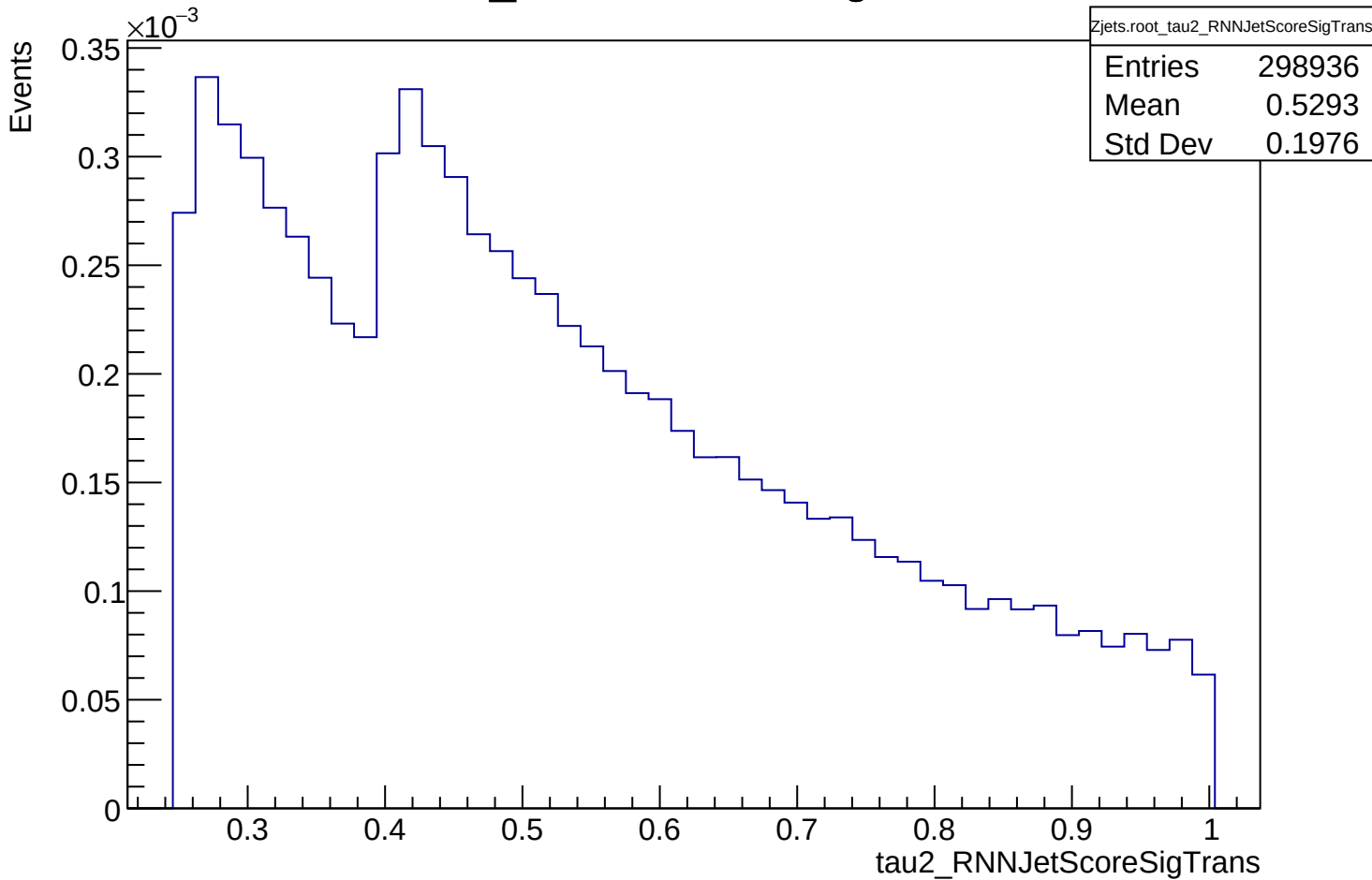


tau1_truth_IsHadronicTau

Events



tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

Events

0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

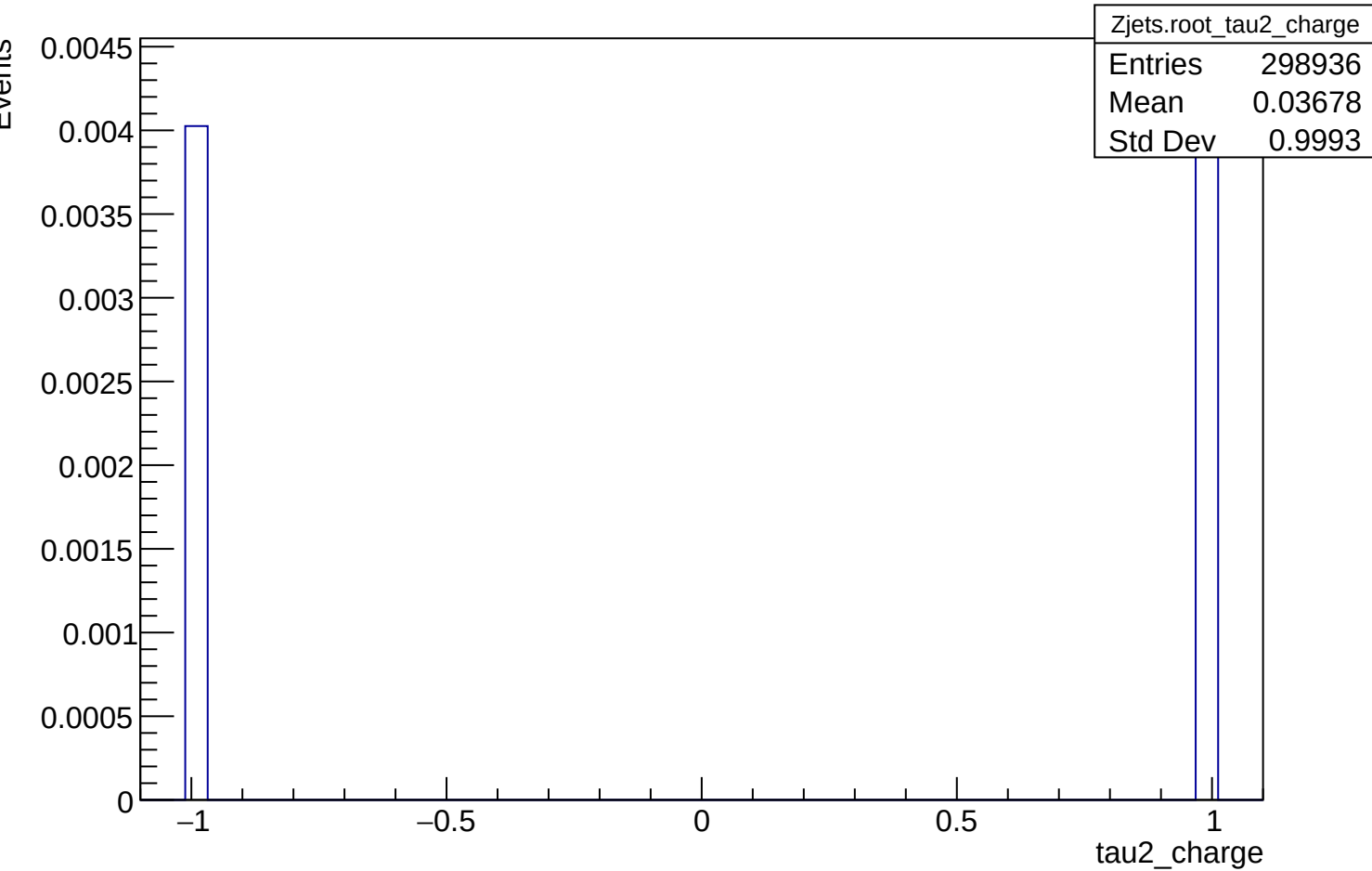
0.8

1

tau2_baseline_RNNTight

Zjets.root_tau2_baseline_RNNTight	
Entries	298936
Mean	0.5944
Std Dev	0.491

tau2_charge



tau2_effSF

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0.8

1.0

1.2

1.4

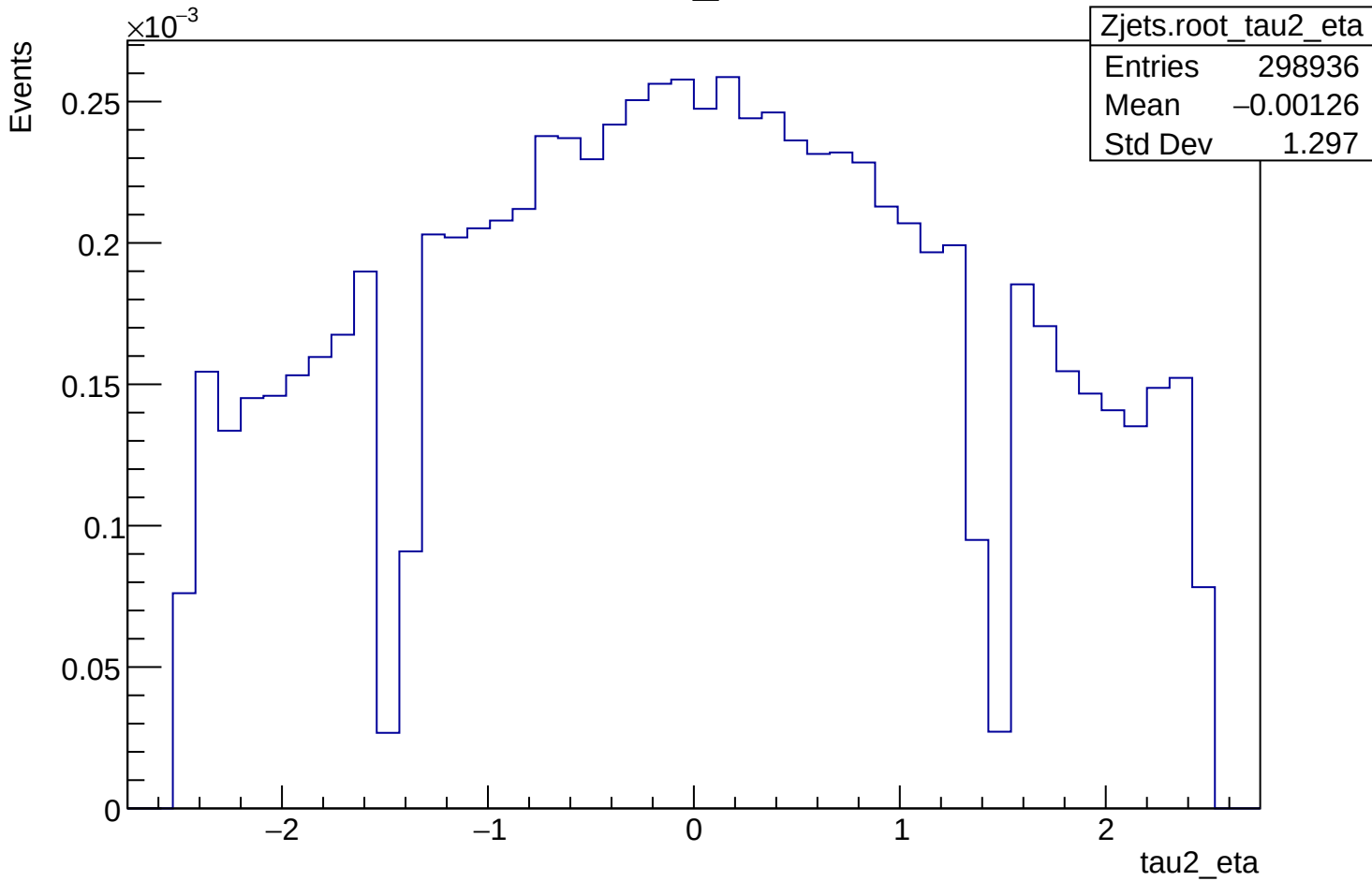
1.6

1.8

tau2_effSF

Zjets.root_tau2_effSF	
Entries	298936
Mean	0.9915
Std Dev	0.05603

tau2_eta



tau2_nprong

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

1

1.5

2

2.5

3

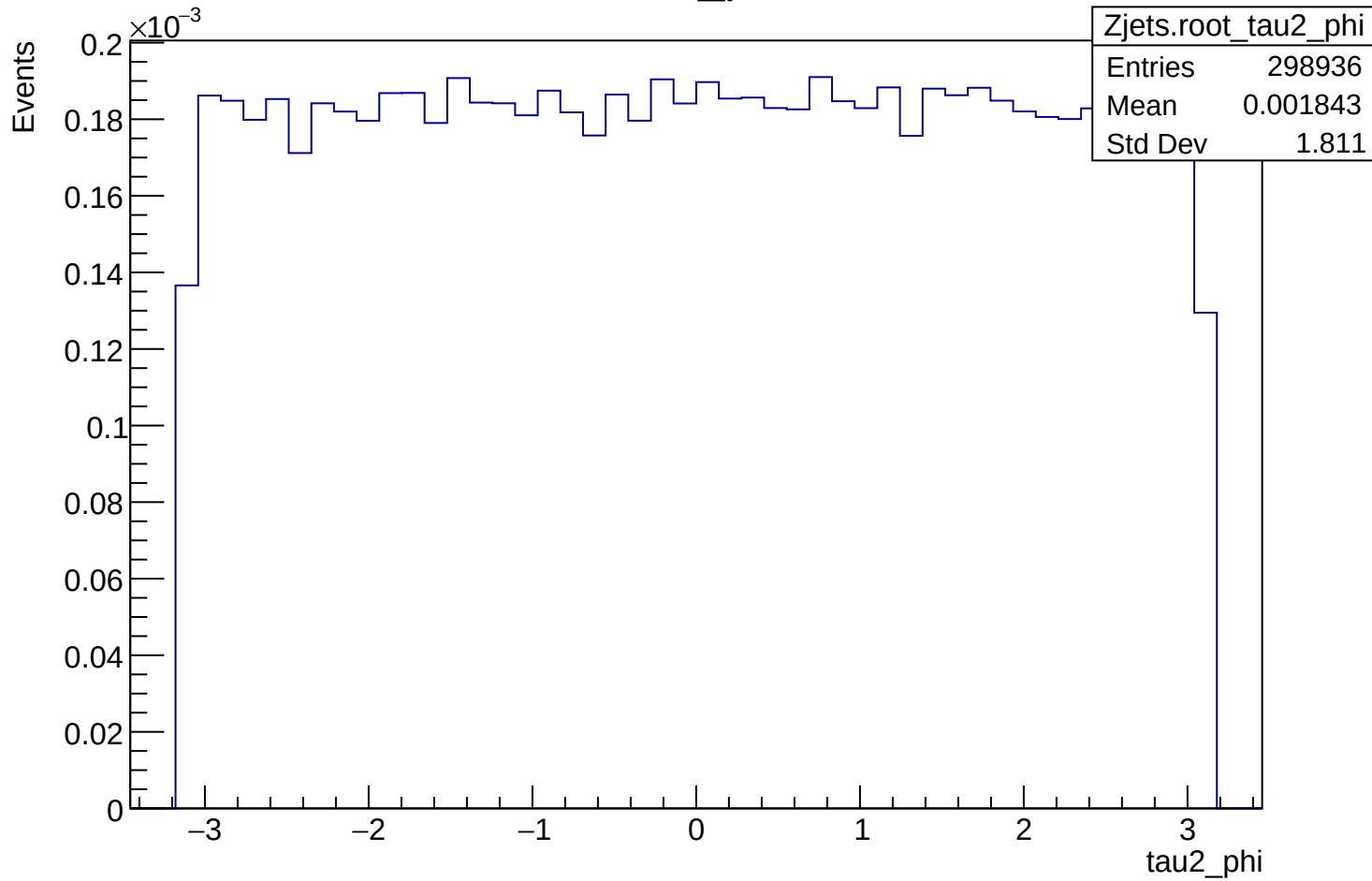
tau2_nprong

Zjets.root_tau2_nprong	
Entries	298936
Mean	1.393
Std Dev	0.7945

1

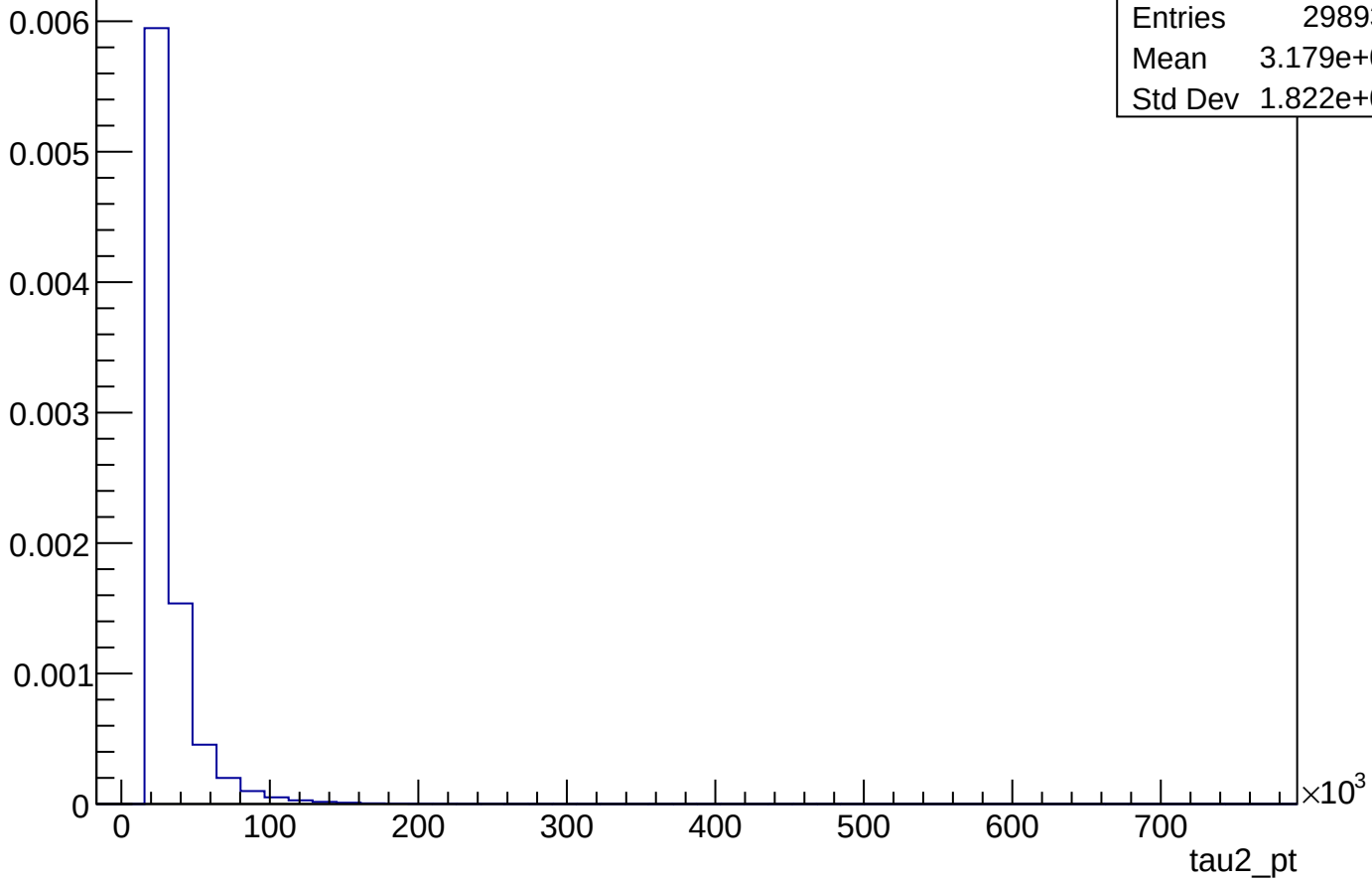
3

tau2_phi

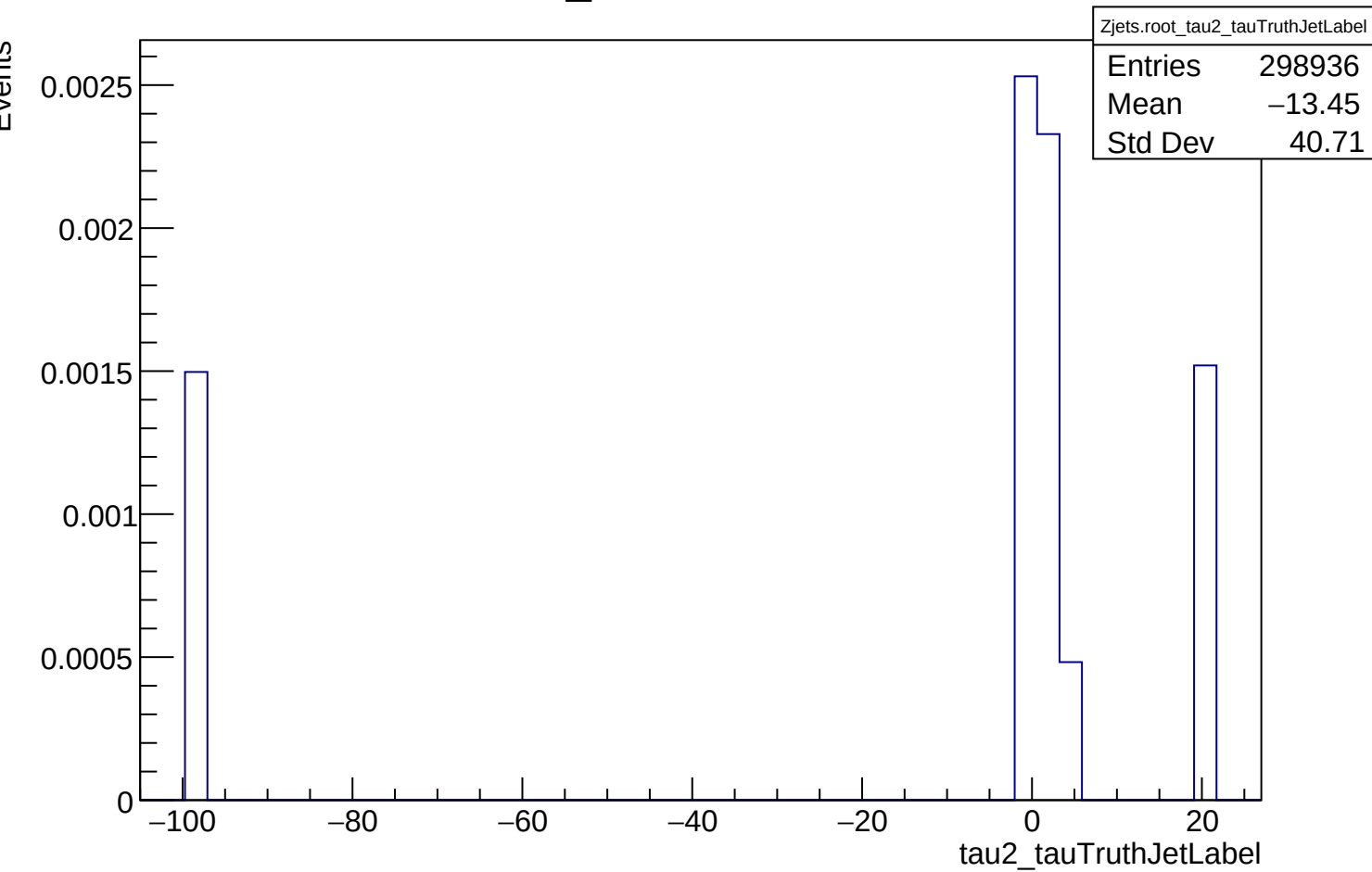


tau2_pt

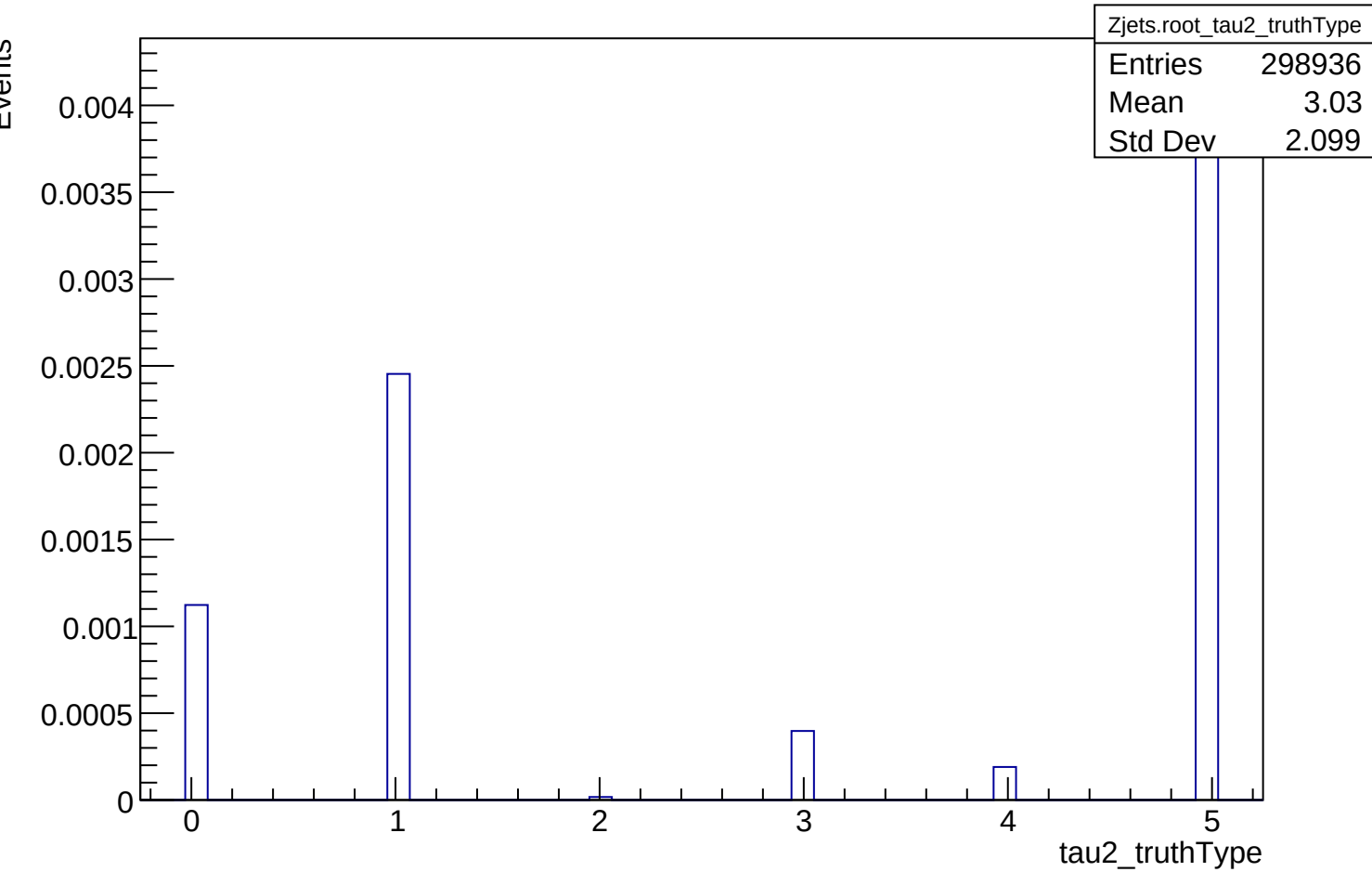
Events



tau2_tauTruthJetLabel



tau2_truthType



tau2_truth_IsHadronicTau

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

0.2

0.4

0.6

0.8

1

tau2_truth_IsHadronicTau

Zjets.root_tau2_truth_IsHadronicTau	
Entries	298936
Mean	0.2935
Std Dev	0.4554

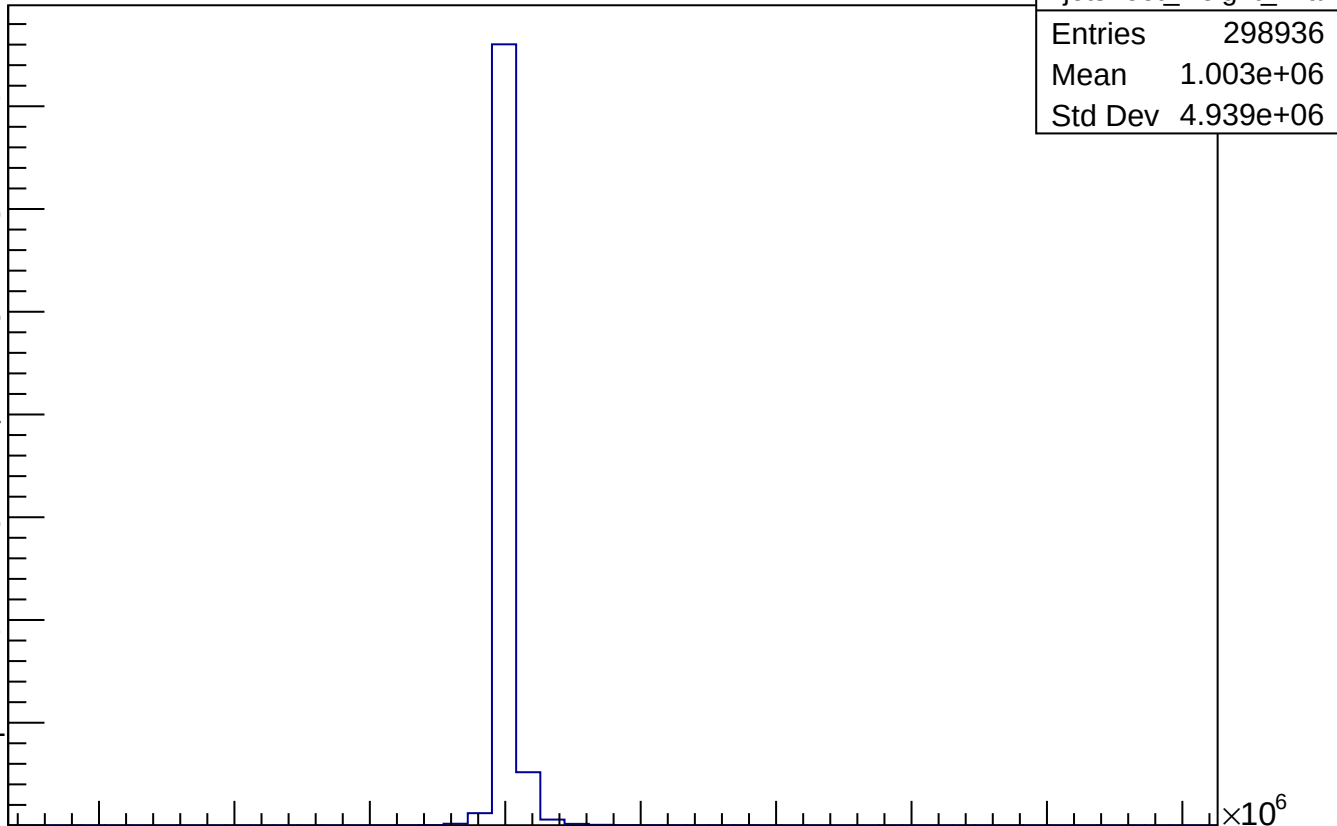
weight_final

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

Zjets.root_weight_final	
Entries	298936
Mean	1.003e+06
Std Dev	4.939e+06

-150 -100 -50 0 50 100 150 200 250 $\times 10^6$
weight_final



weights

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

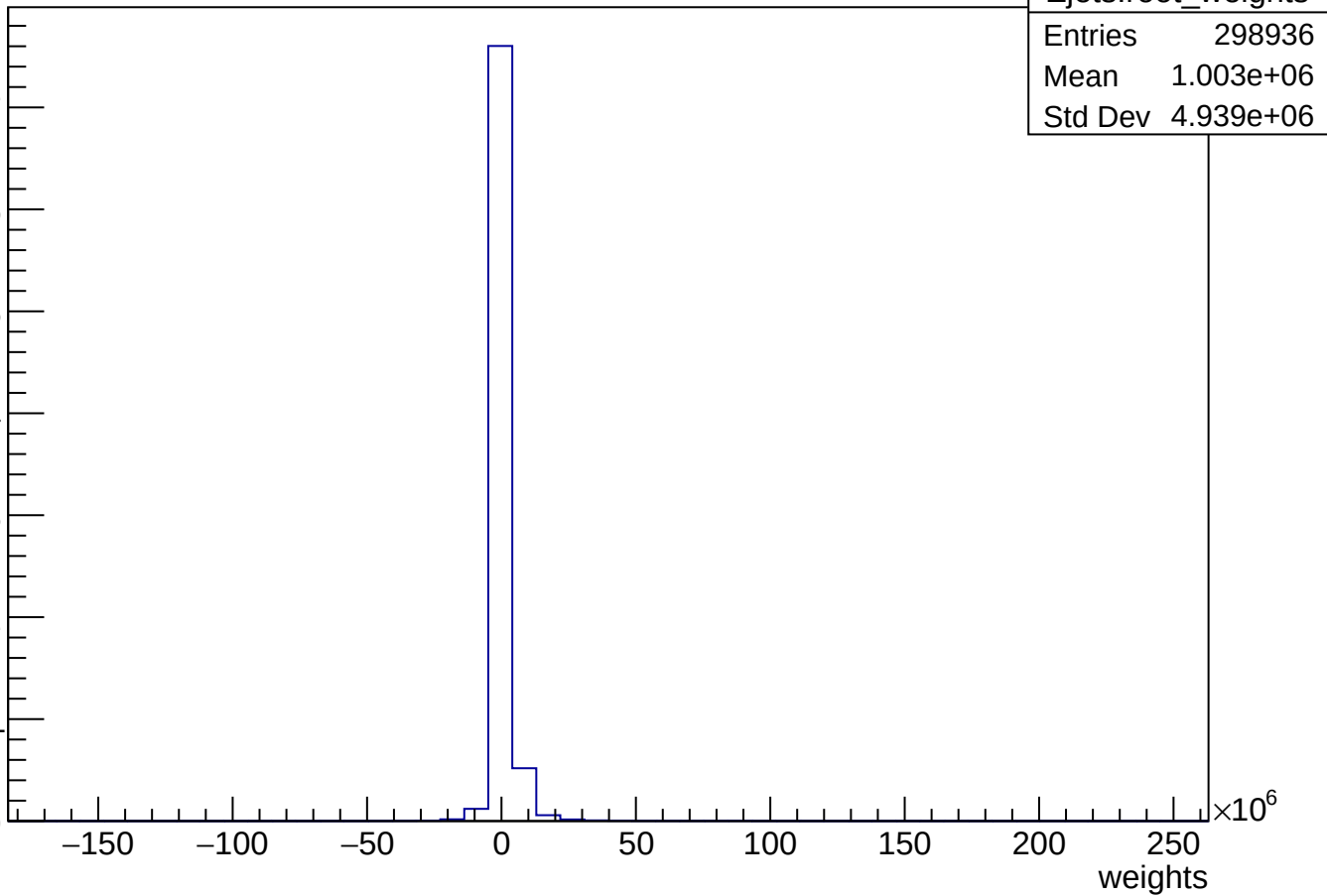
-150 -100 -50 0 50 100 150 200 250 $\times 10^6$
weights

Zjets.root_weights

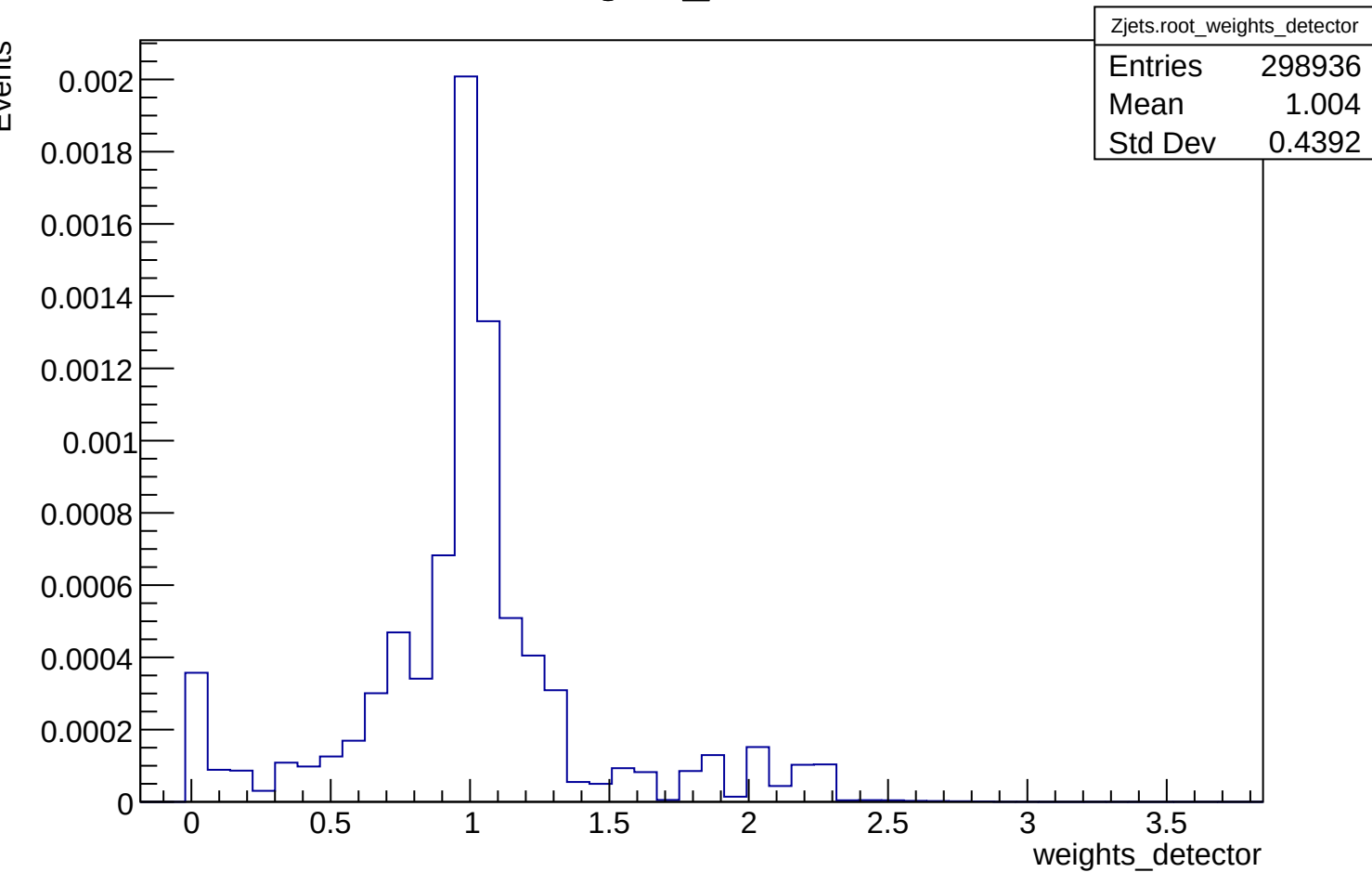
Entries 298936

Mean 1.003e+06

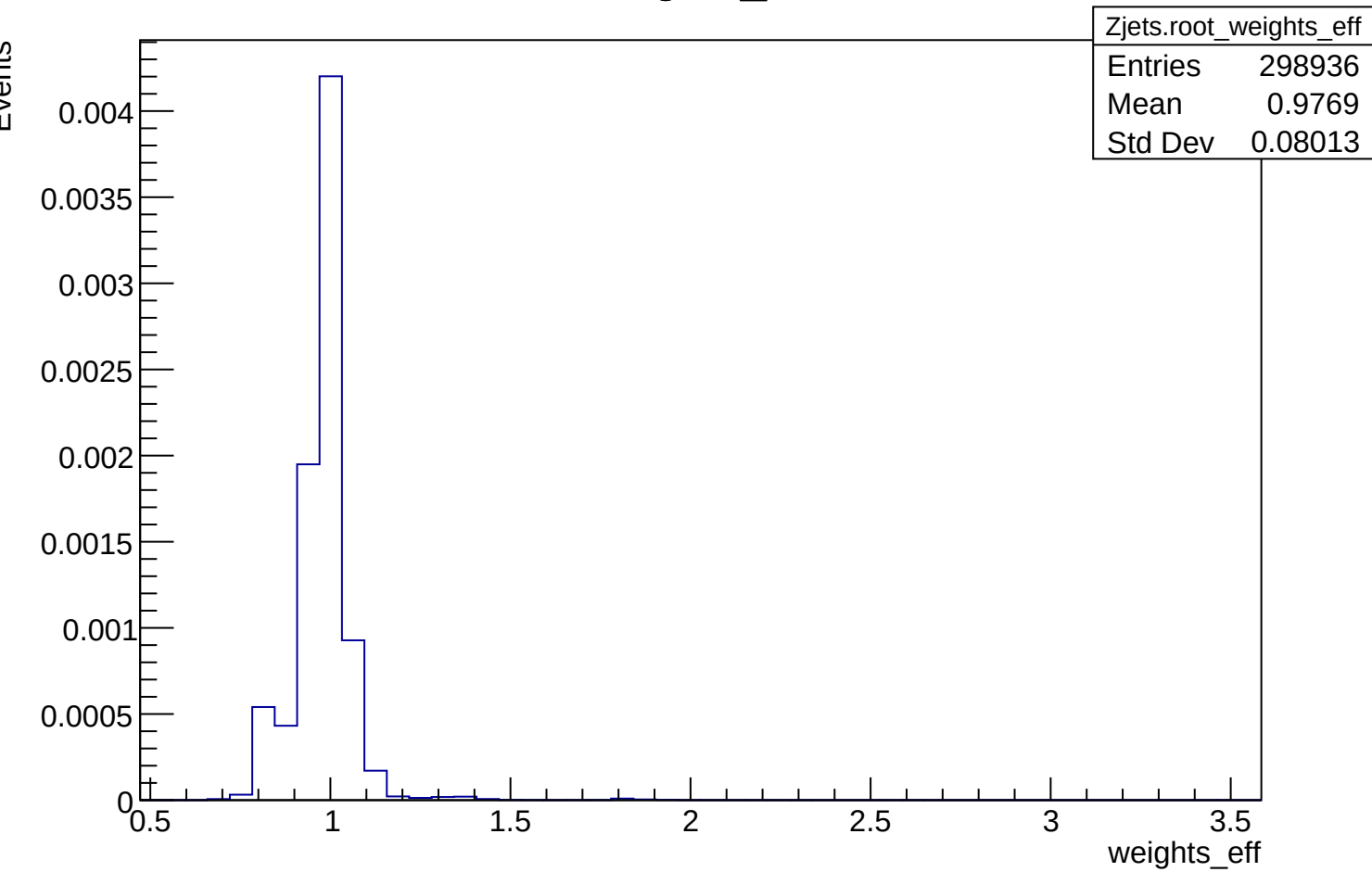
Std Dev 4.939e+06



weights_detector



weights_eff



weights_mc

Events

0.006

0.005

0.004

0.003

0.002

0.001

0

-150

-100

-50

0

50

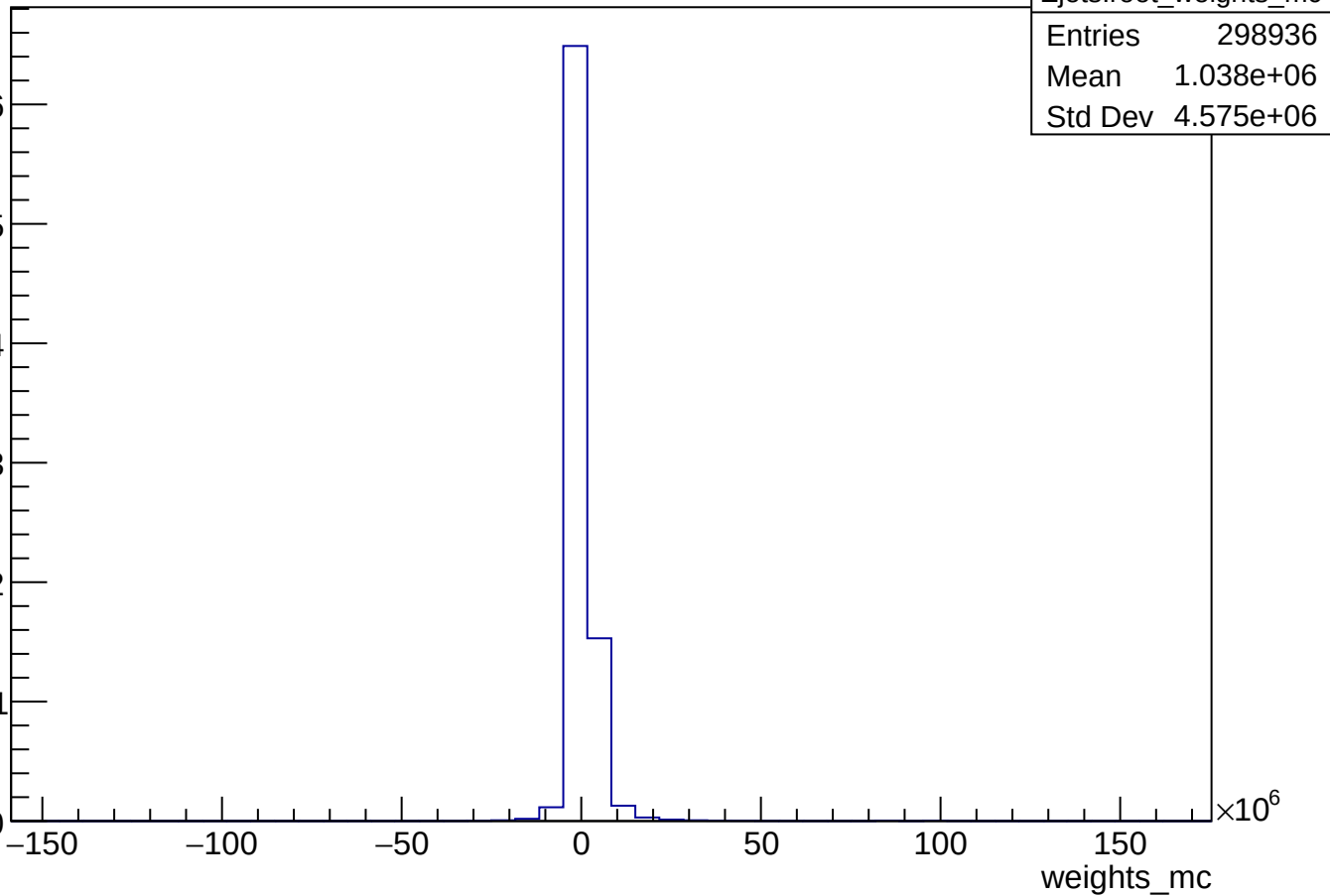
100

150

weights_mc

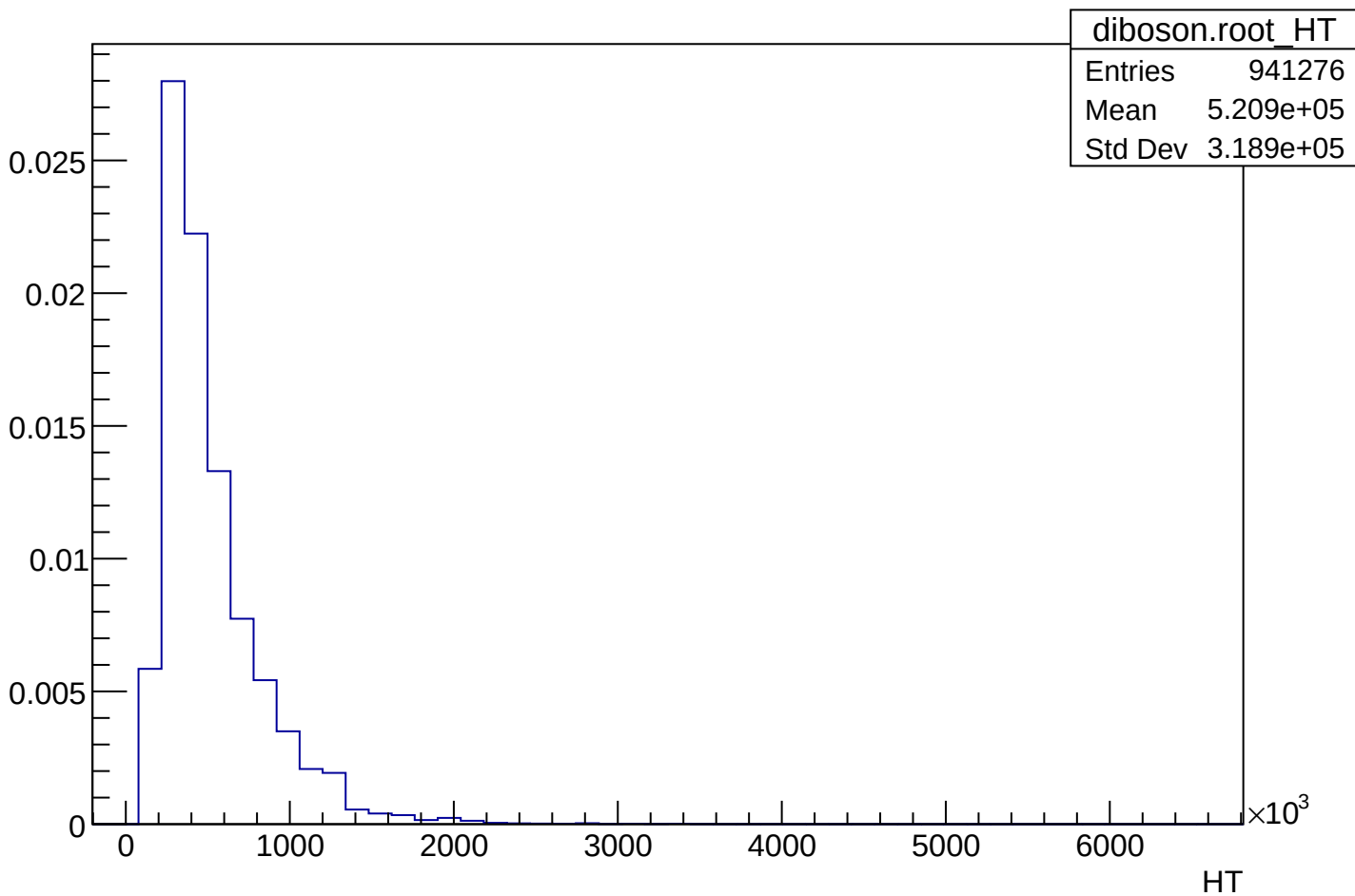
$\times 10^6$

Zjets.root_weights_mc	
Entries	298936
Mean	1.038e+06
Std Dev	4.575e+06



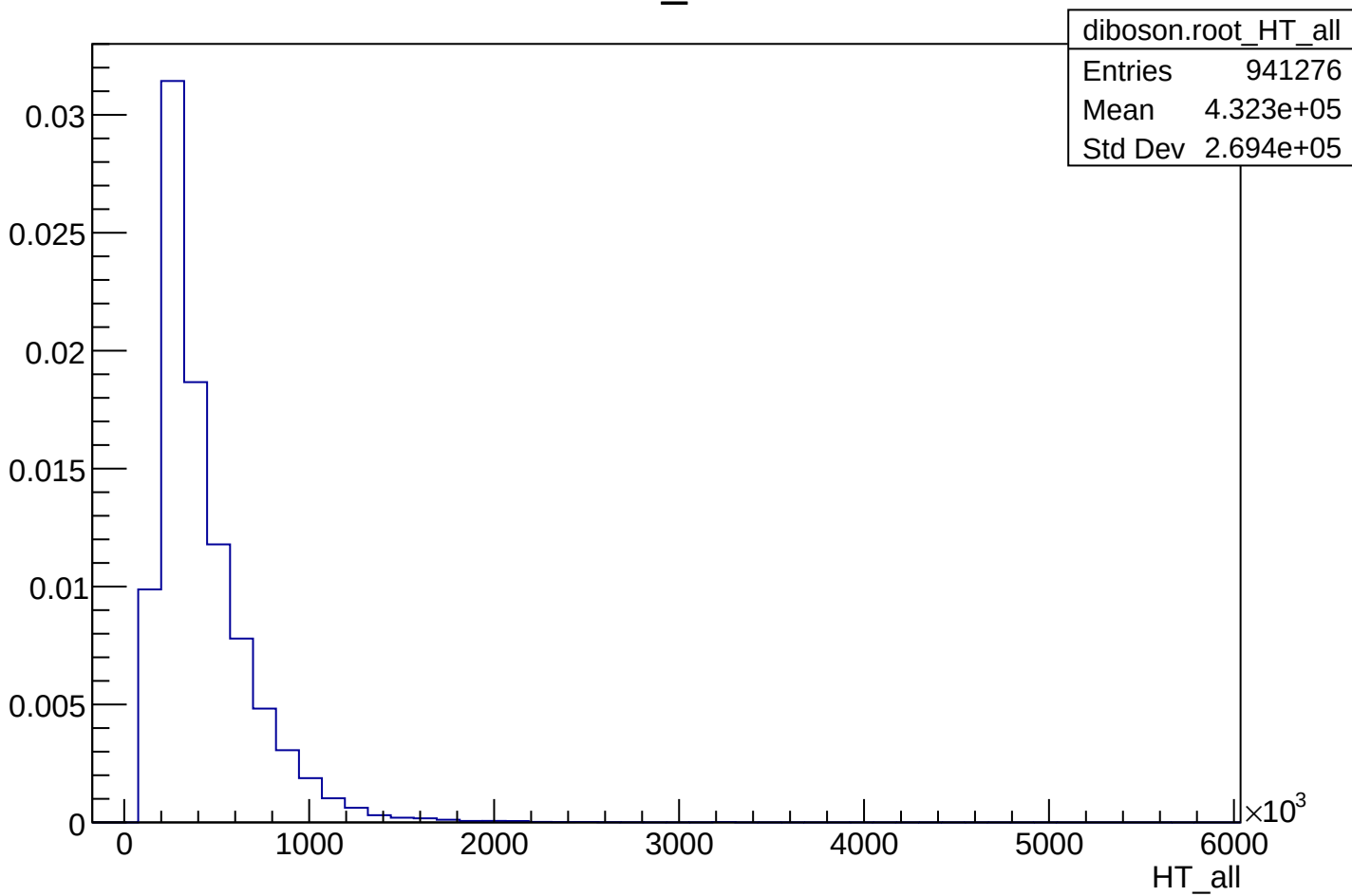
HT

Events



HT_all

Events



HT_jet

Events

0.03

0.025

0.02

0.015

0.01

0.005

0

0

1000

2000

3000

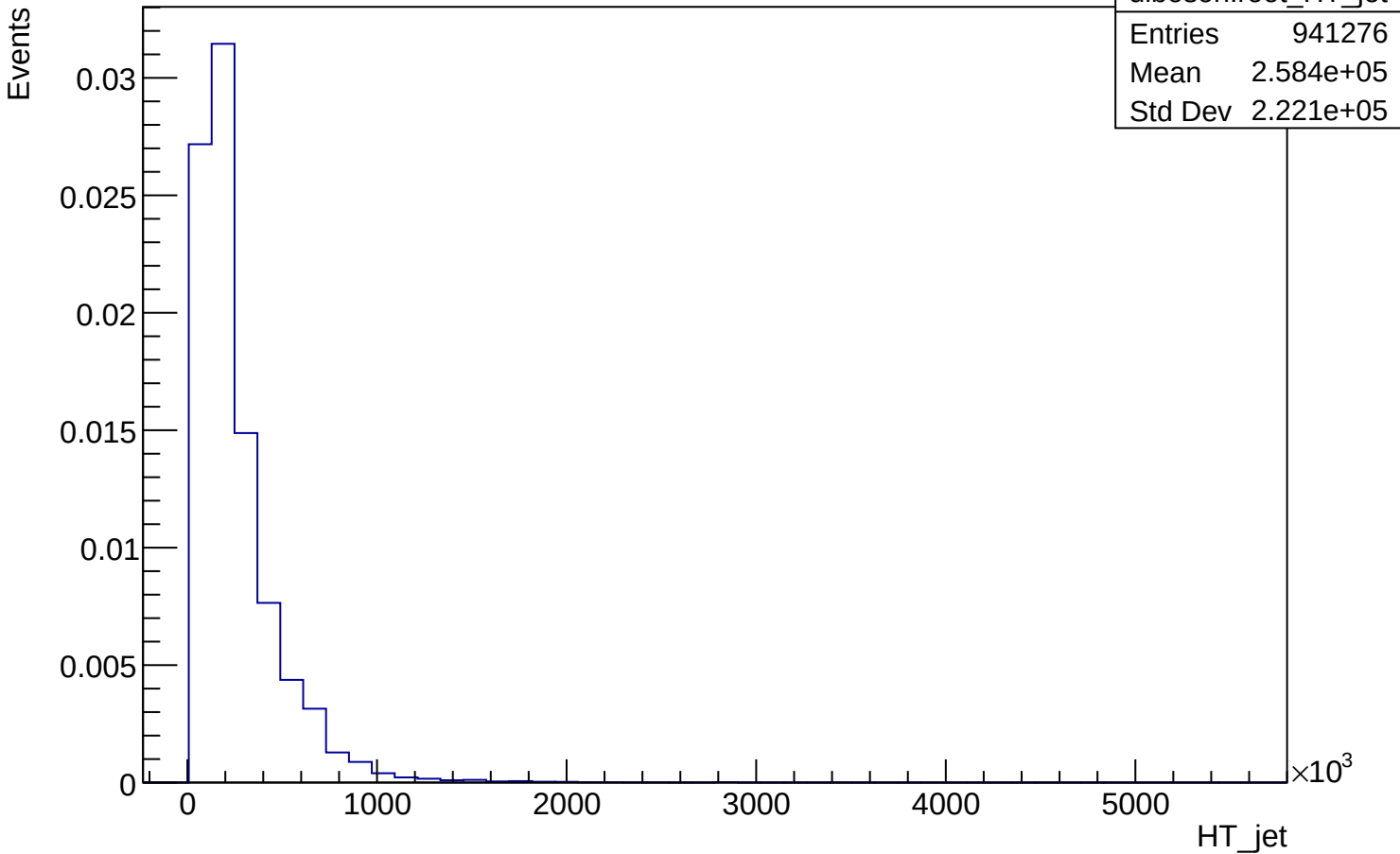
4000

5000

HT_jet

$\times 10^3$

diboson.root_HT_jet	
Entries	941276
Mean	2.584e+05
Std Dev	2.221e+05



HT_lep

Events

0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0

500

1000

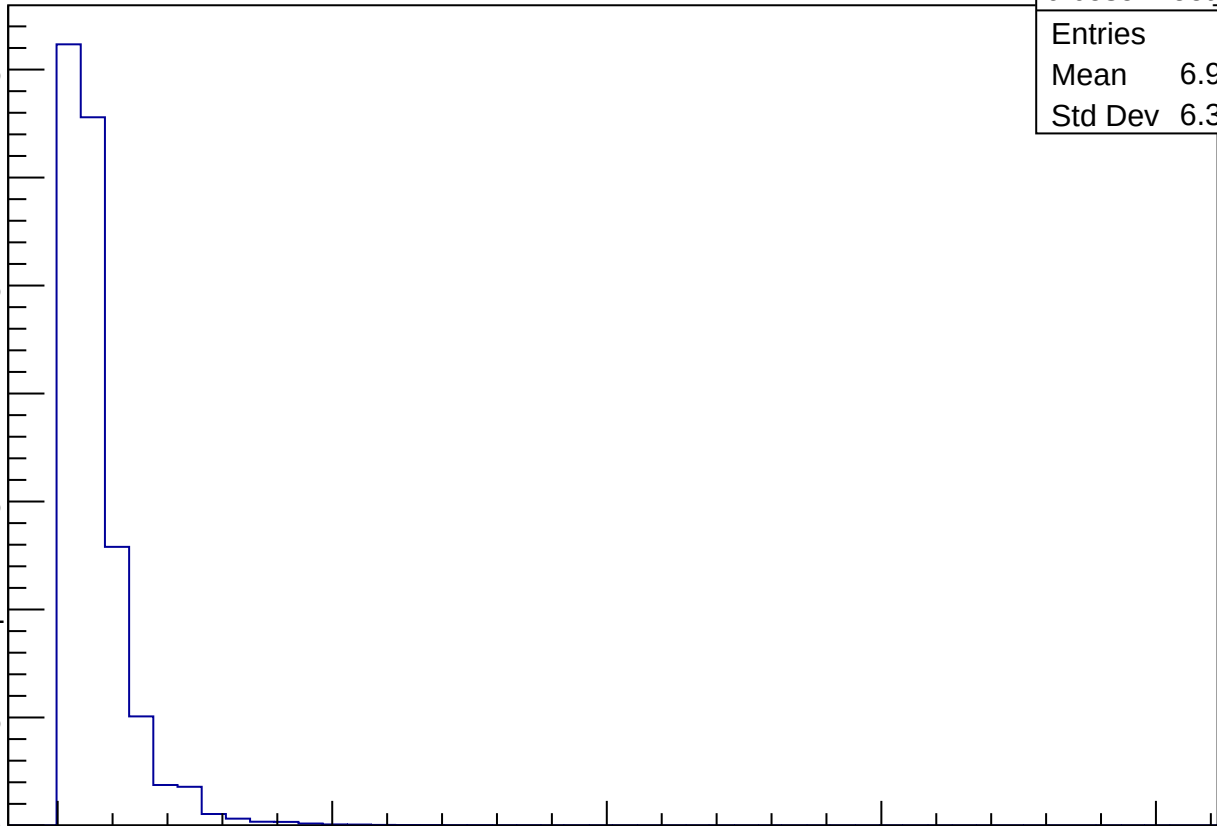
1500

2000

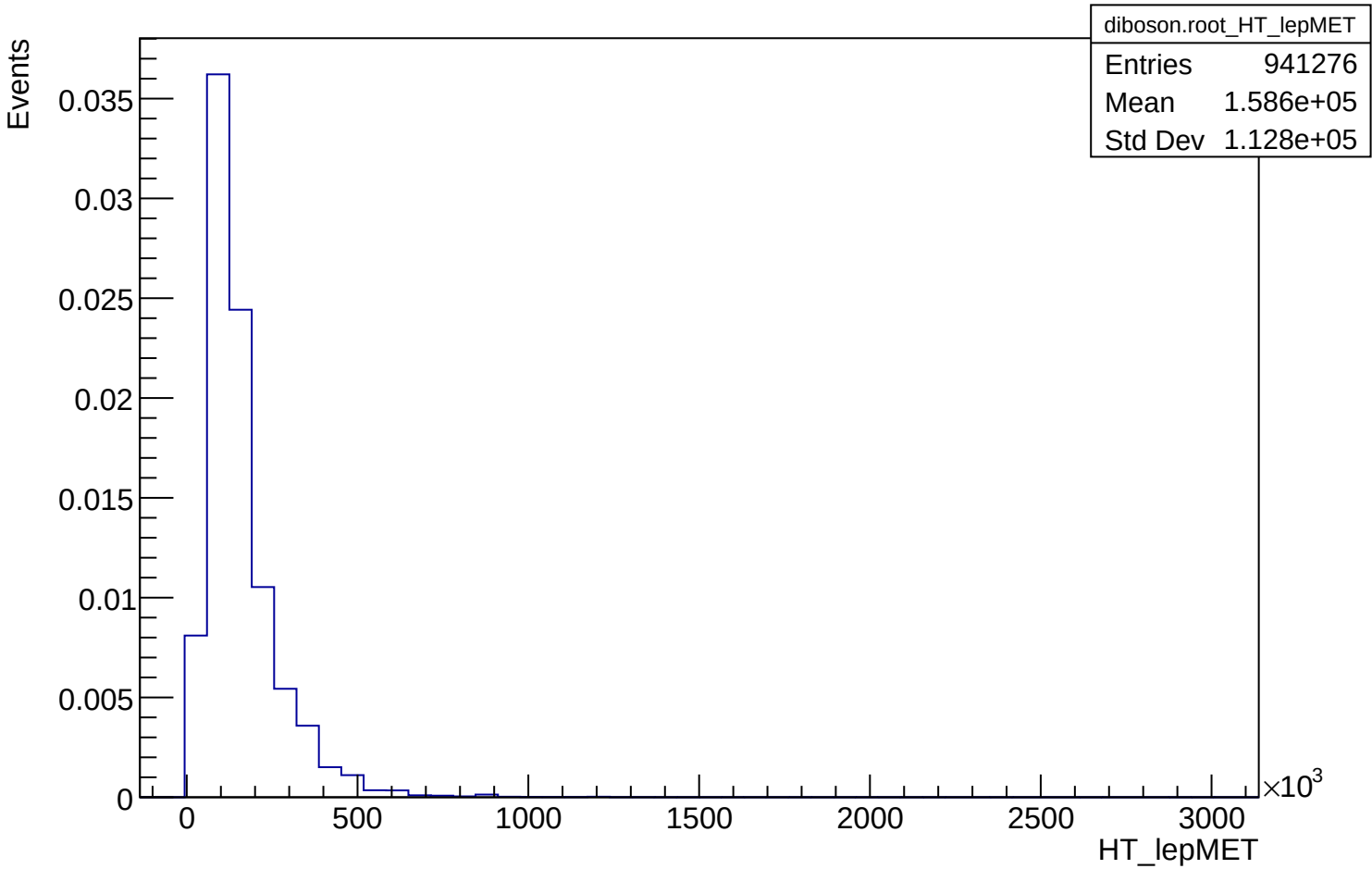
HT_lep

$\times 10^3$

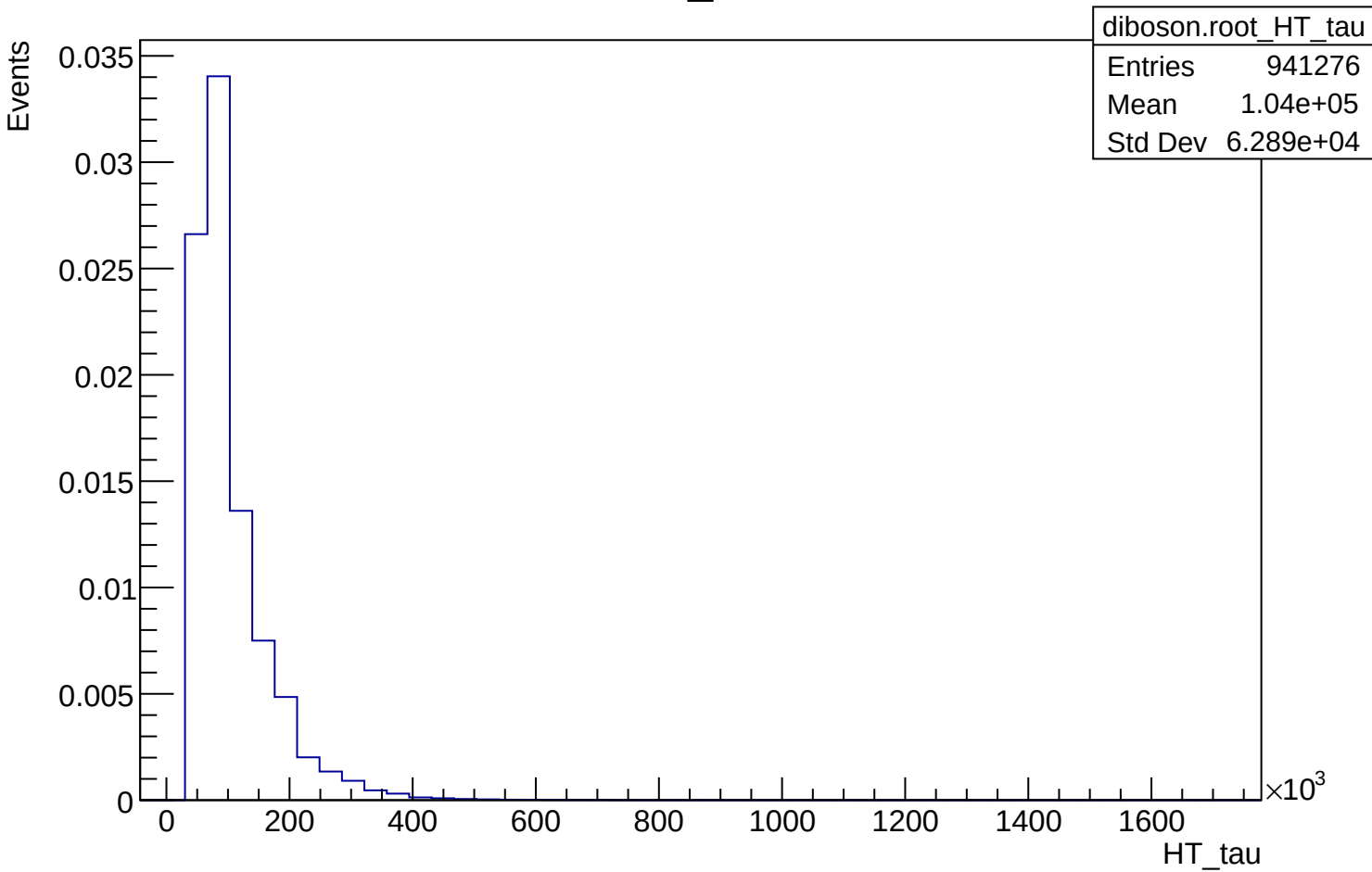
diboson.root_HT_lep	
Entries	941276
Mean	6.992e+04
Std Dev	6.389e+04



HT_lepMET

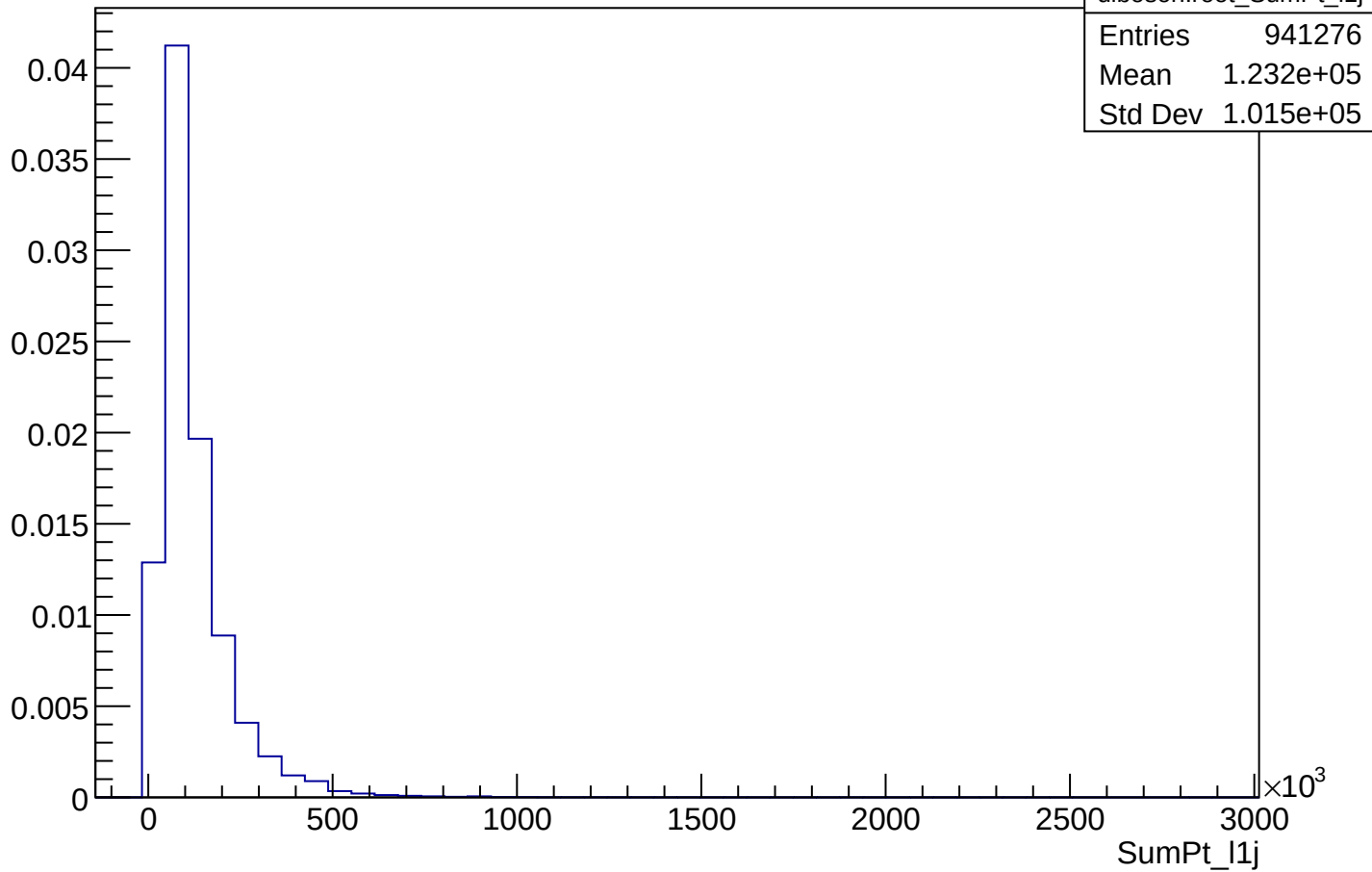


HT_tau

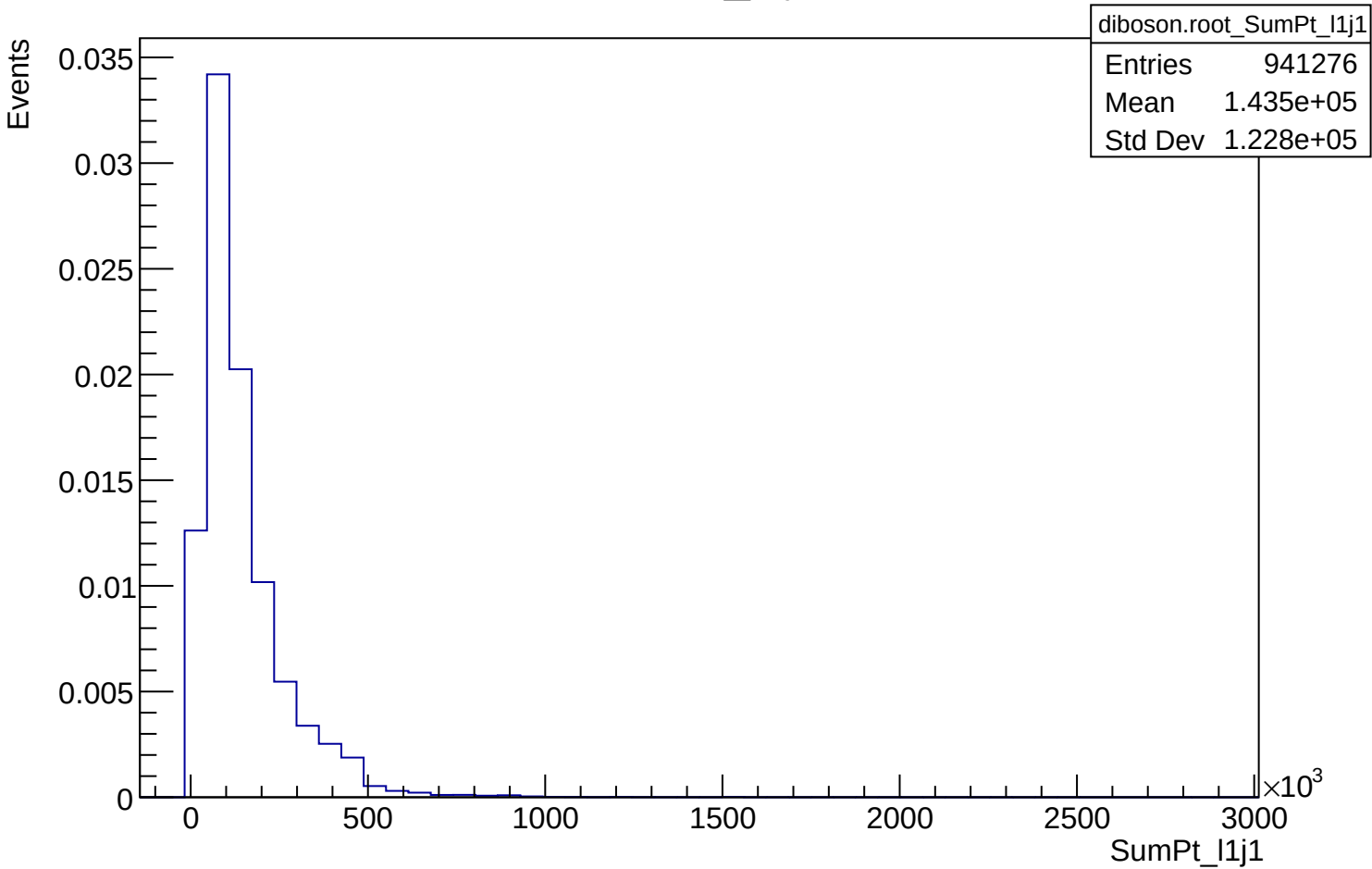


SumPt_l1j

Events

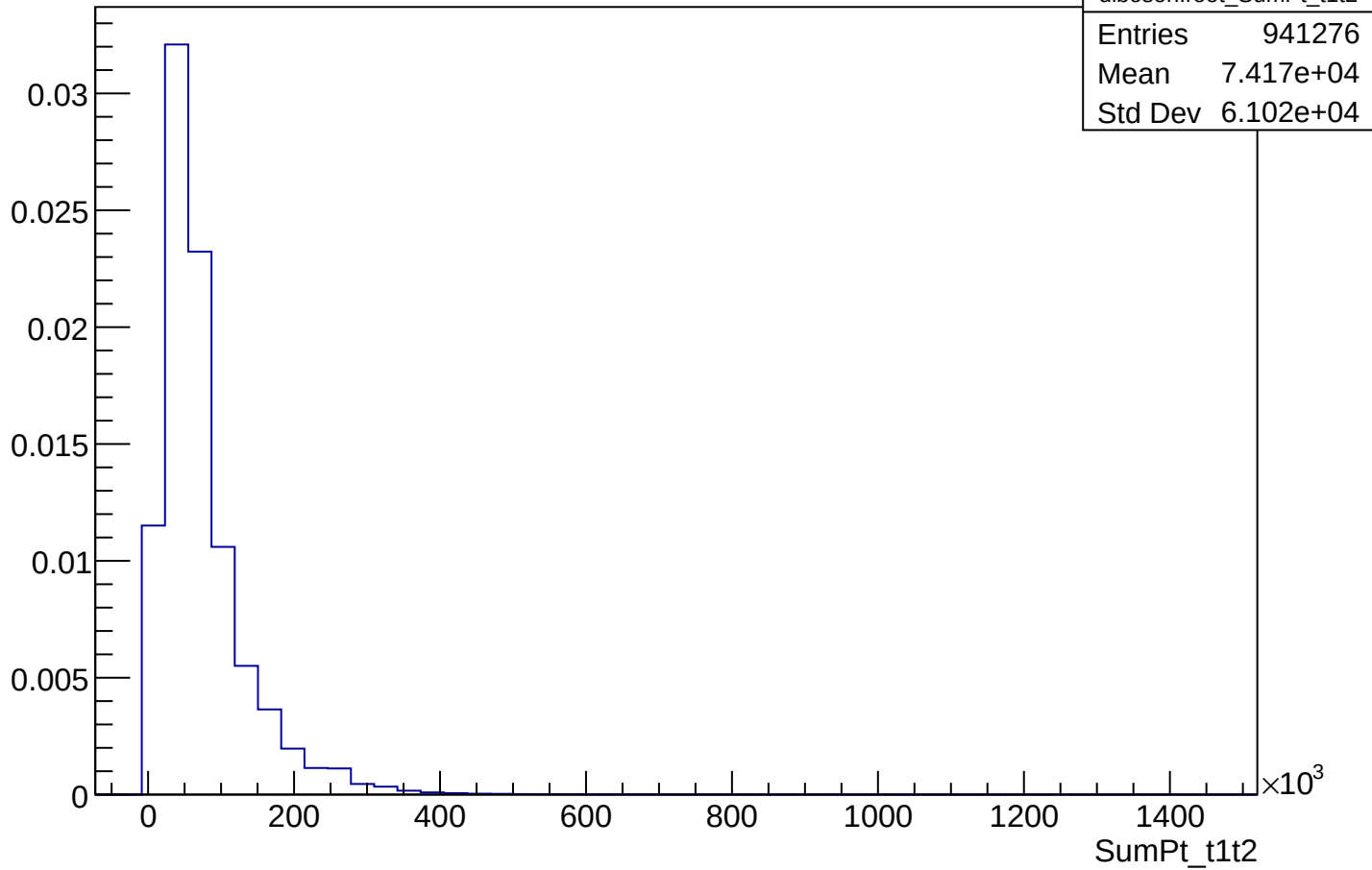


SumPt_l1j1

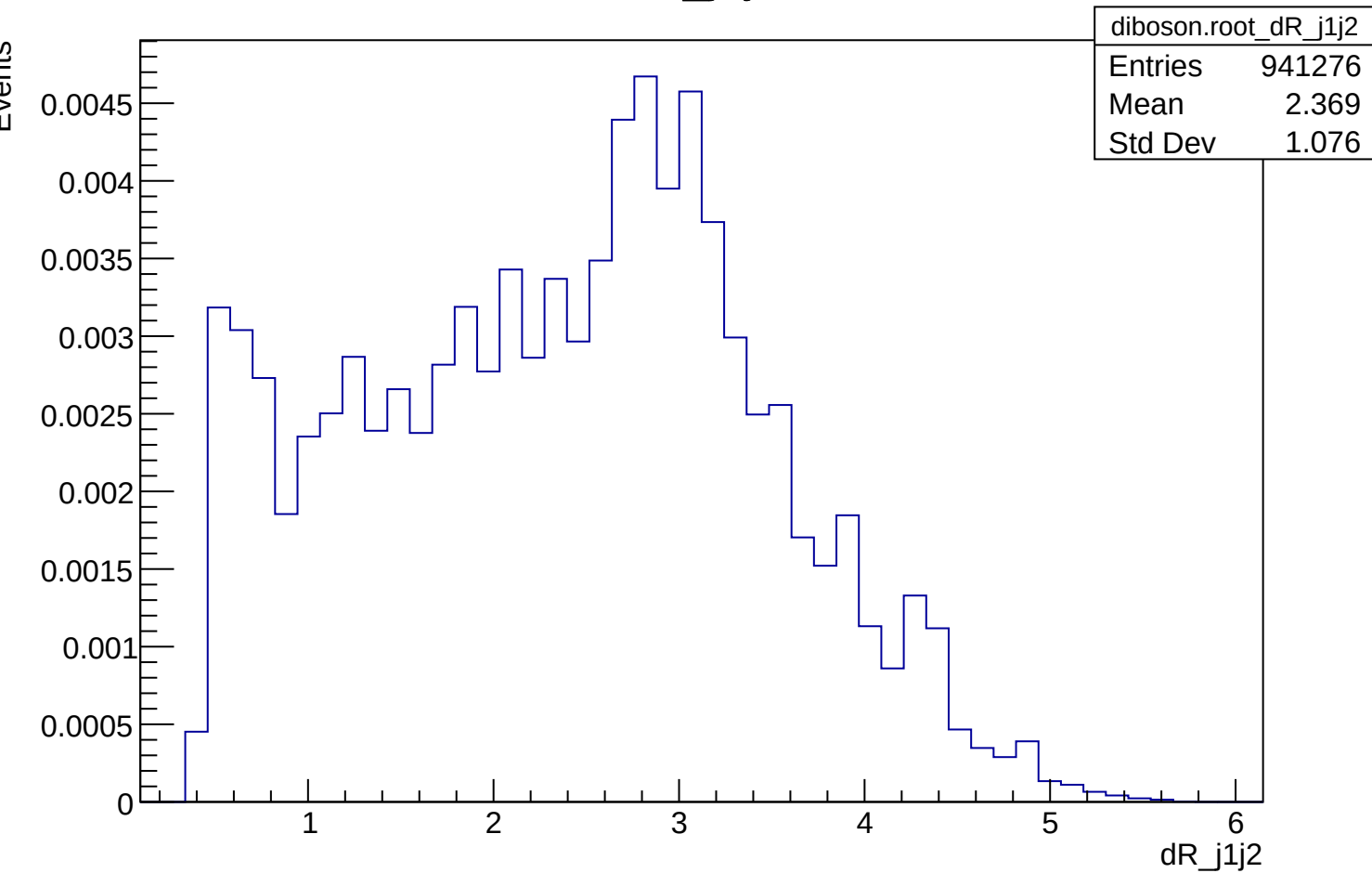


SumPt_t1t2

Events



dR_j1j2



dR_l1j1

Events

0.006

0.005

0.004

0.003

0.002

0.001

0

1

2

3

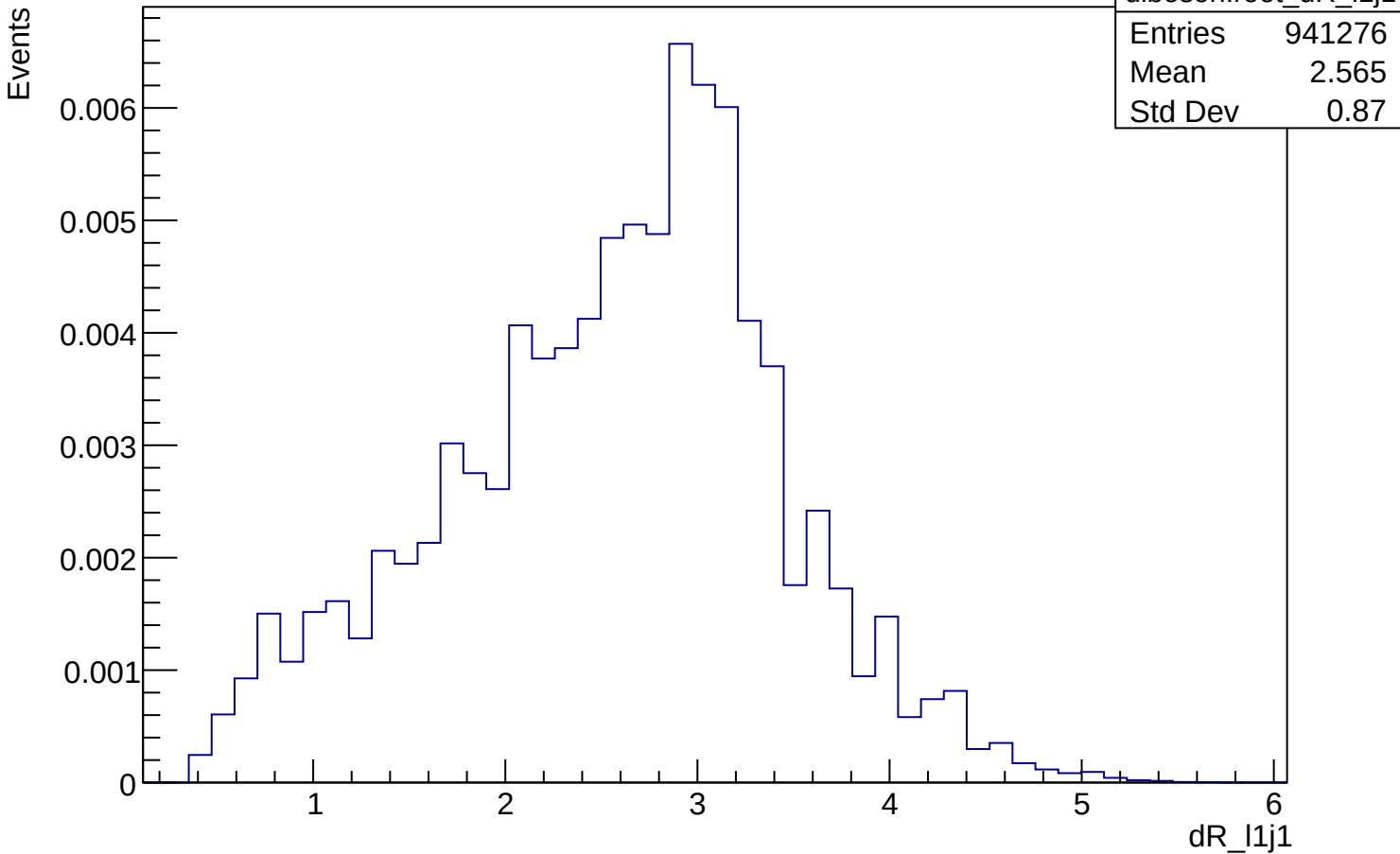
4

5

6

dR_l1j1

diboson.root_dR_l1j1	
Entries	941276
Mean	2.565
Std Dev	0.87



dR_l1j2

Events

0.005
0.004
0.003
0.002
0.001
0

diboson.root_dR_l1j2	
Entries	941276
Mean	2.392
Std Dev	0.9575

dR_l1j2

dR_l1t

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

1

2

3

4

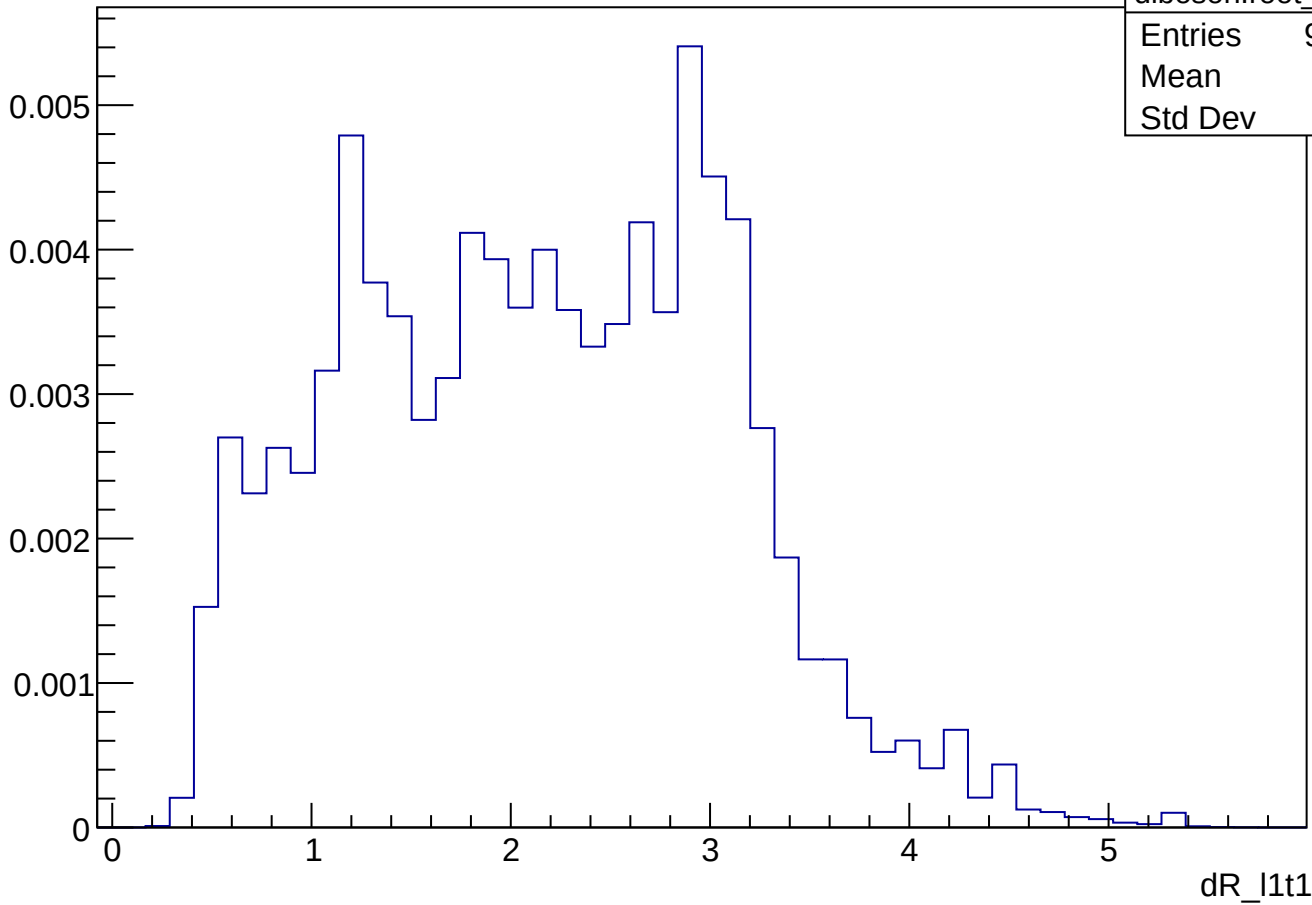
5

dR_l1t

diboson.root_dR_l1t	
Entries	941276
Mean	1.69
Std Dev	0.7799

dR_l1t1

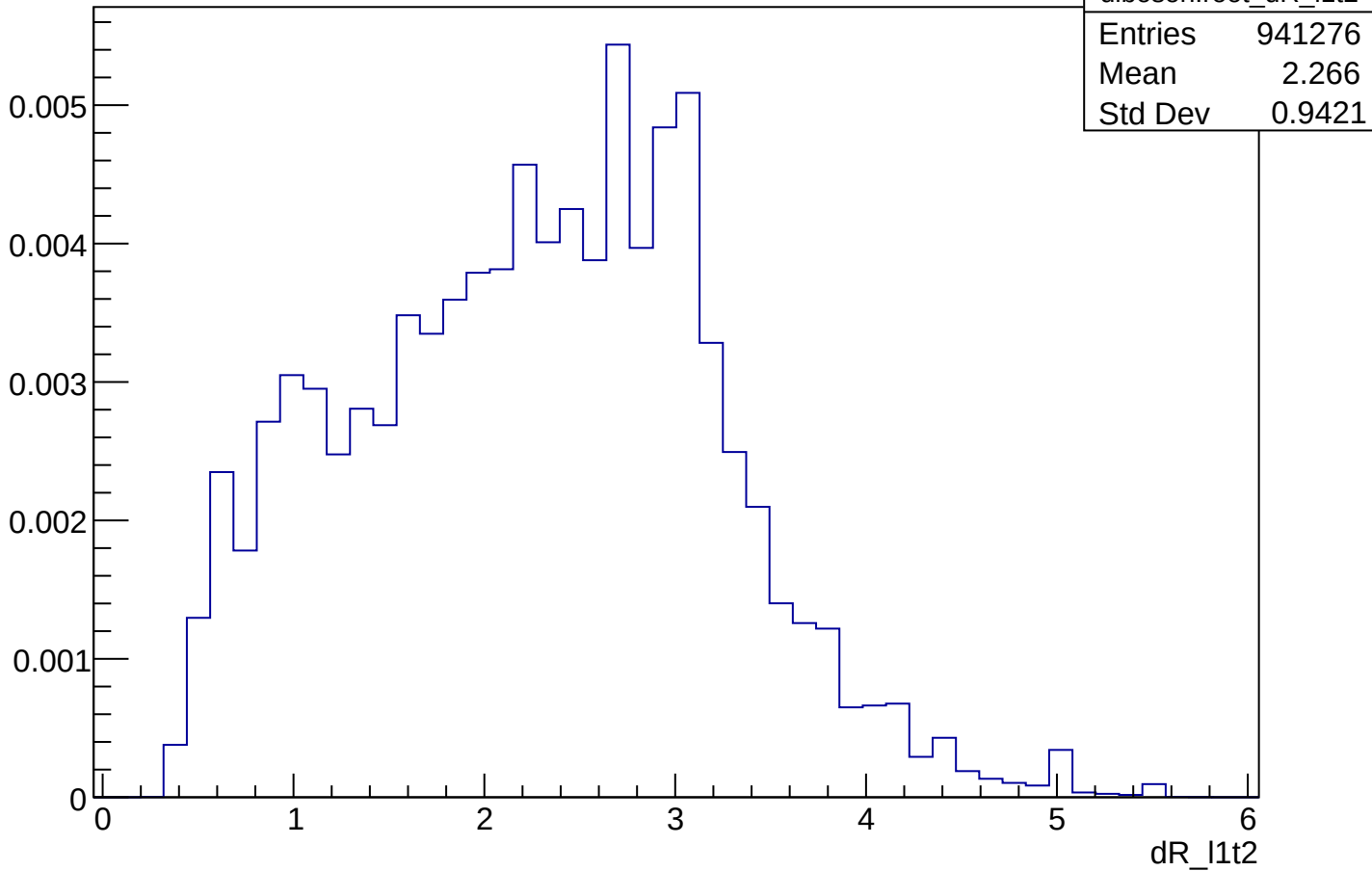
Events



diboson.root_dR_l1t1	
Entries	941276
Mean	2.151
Std Dev	0.9387

dR_l1t2

Events



dR_t1j1

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

0

1

2

3

4

5

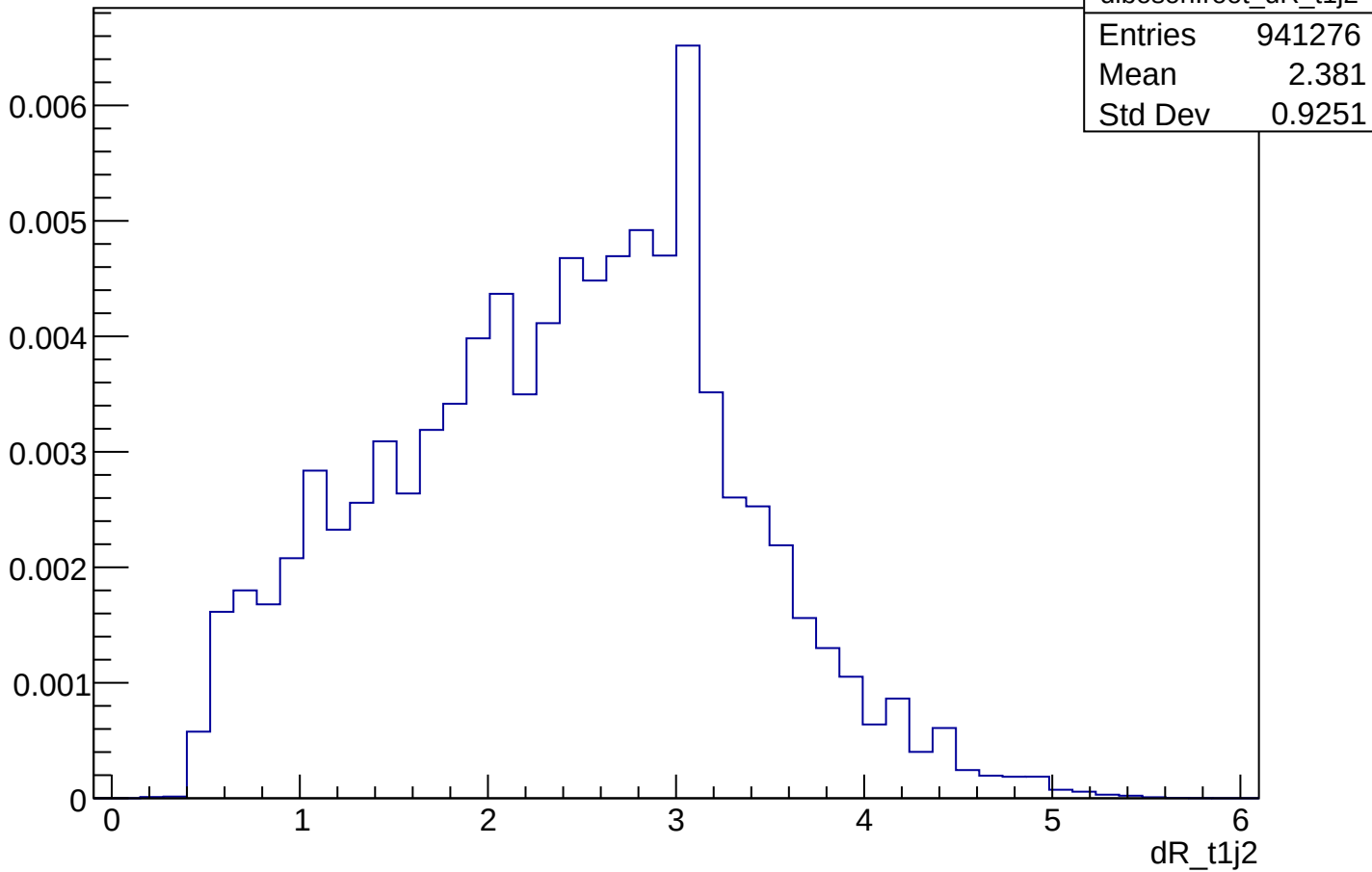
6

dR_t1j1

diboson.root_dR_t1j1	
Entries	941276
Mean	2.58
Std Dev	0.8765

dR_t1j2

Events



dR_t1t2

Events

0.005
0.004
0.003
0.002
0.001
0

0

1

2

3

4

5

6

dR_t1t2

diboson.root_dR_t1t2	
Entries	941276
Mean	2.189
Std Dev	0.9424

dR_t1t2_l1

Events

0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

0

2

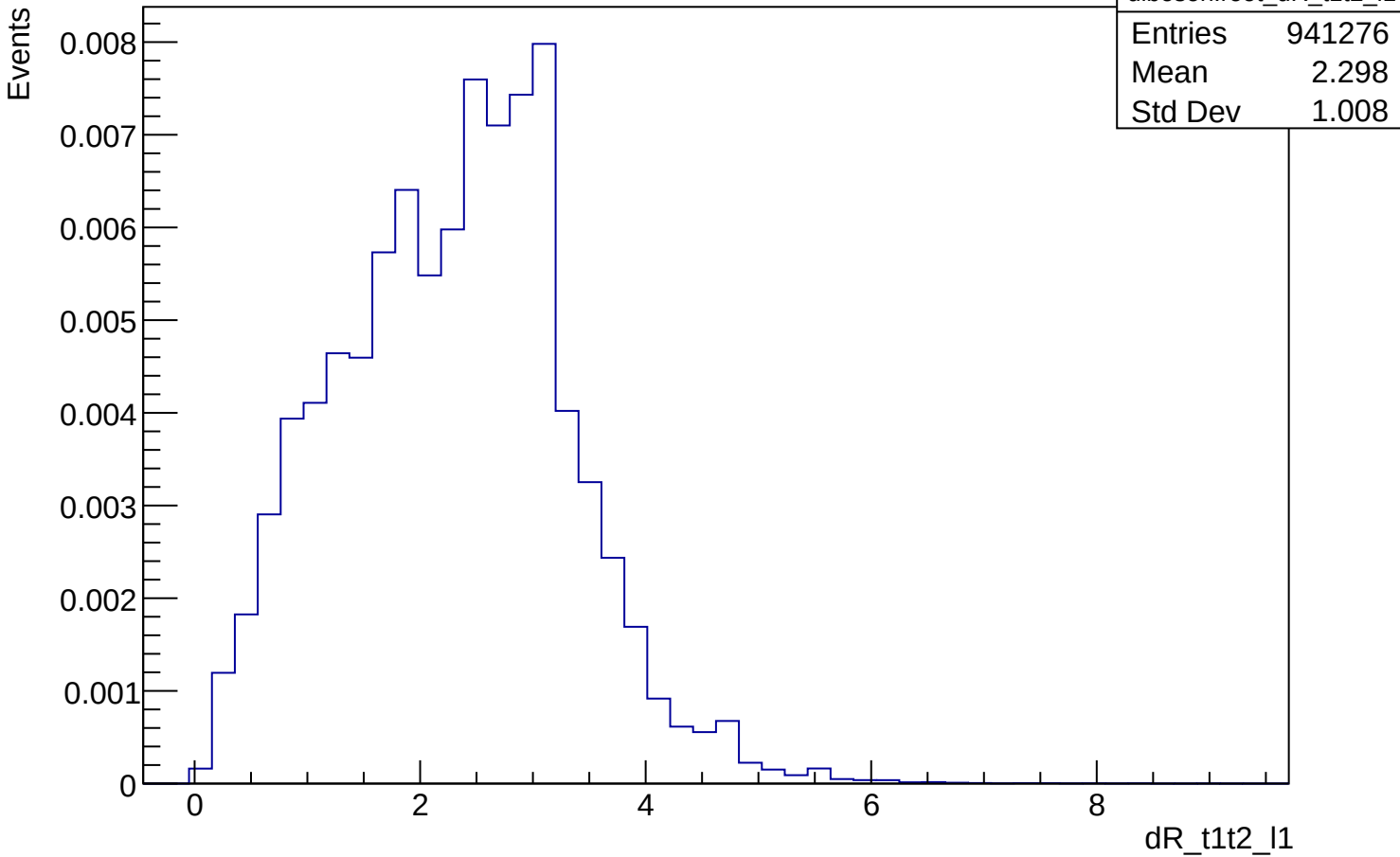
4

6

8

diboson.root_dR_t1t2_l1	
Entries	941276
Mean	2.298
Std Dev	1.008

dR_t1t2_l1



dR_t2j1

Events

0.006
0.005
0.004
0.003
0.002
0.001
0

0

1

2

3

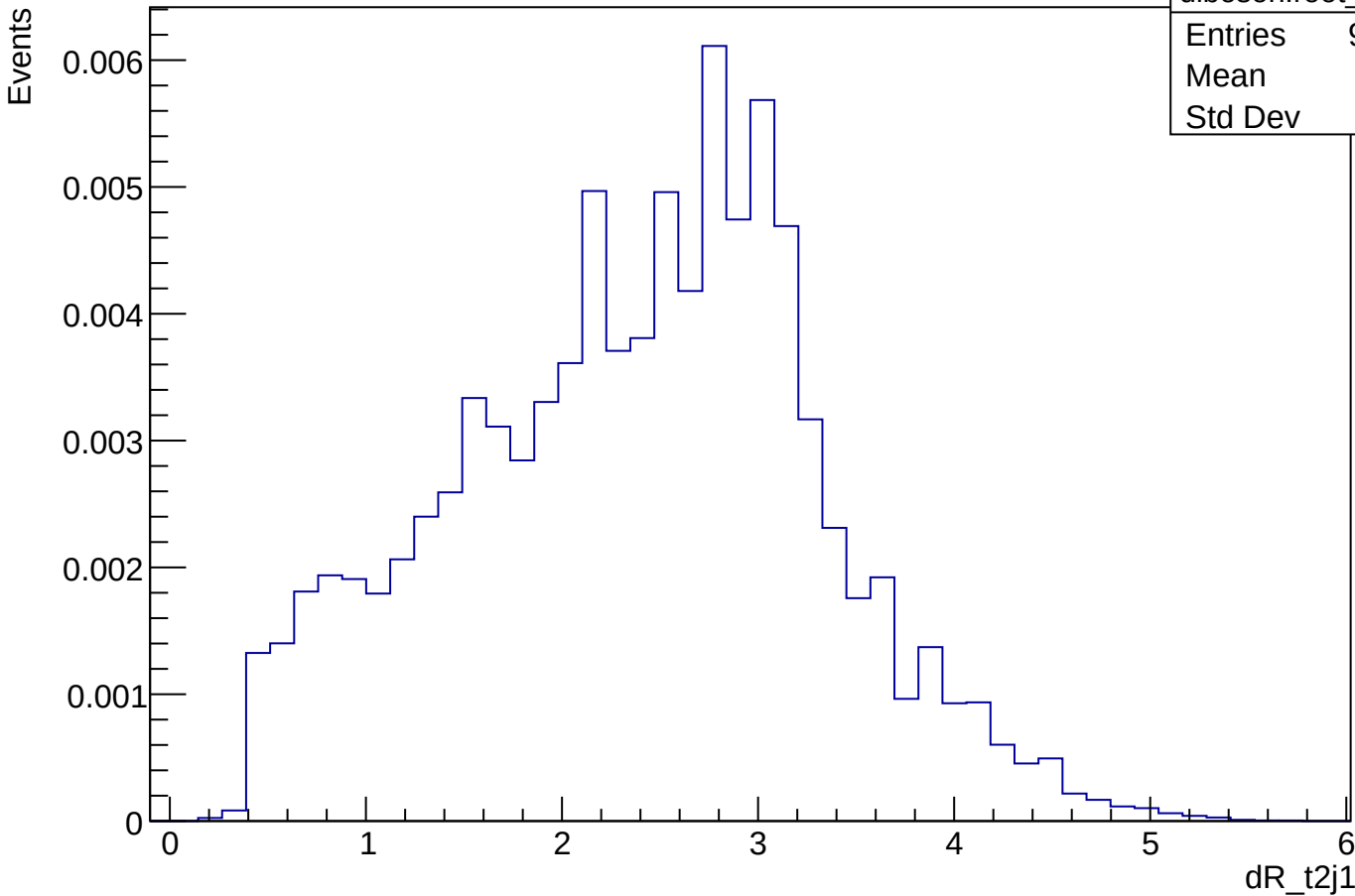
4

5

6

dR_t2j1

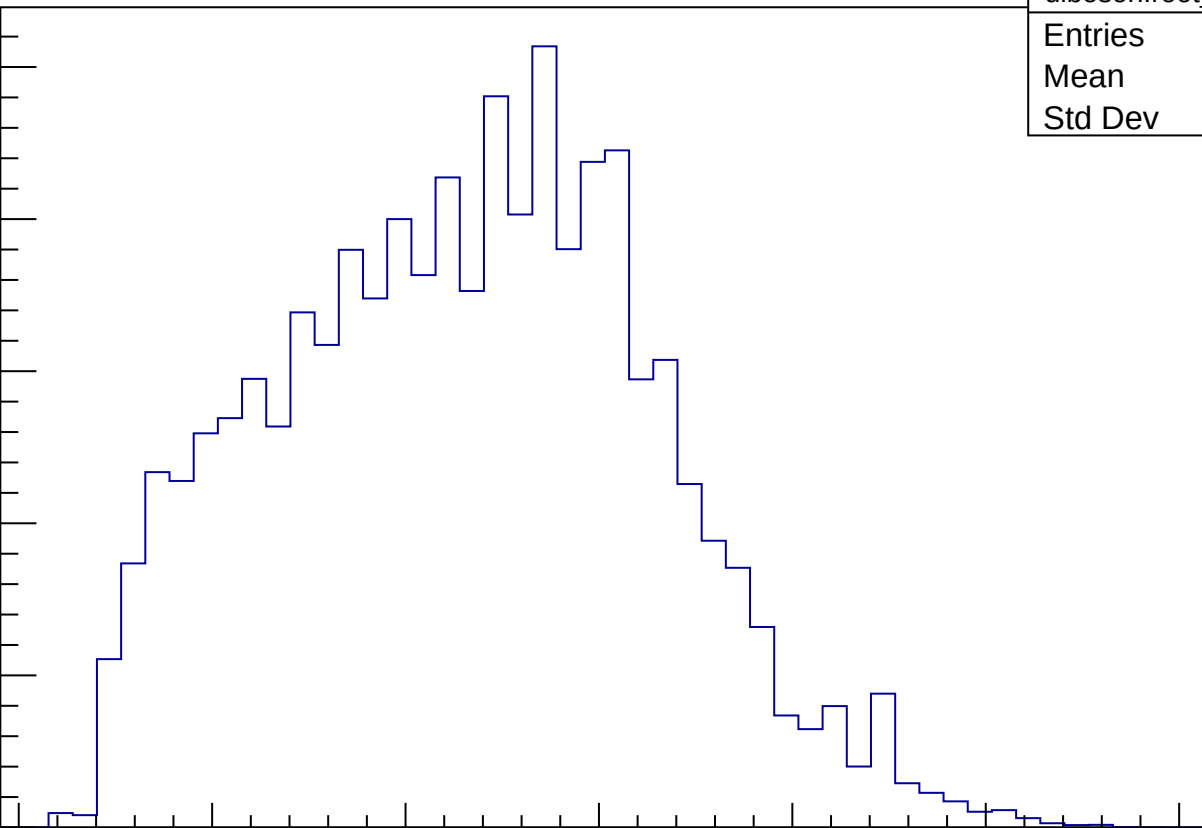
diboson.root_dR_t2j1	
Entries	941276
Mean	2.384
Std Dev	0.9319



dR_t2j2

Events

0.005
0.004
0.003
0.002
0.001
0



diboson.root_dR_t2j2	
Entries	941276
Mean	2.316
Std Dev	0.9701

dR_t2j2

fjvt_effSF

diboson.root_fjvt_effSF	
Entries	941276
Mean	0.9976
Std Dev	0.03184

Events

0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0.4

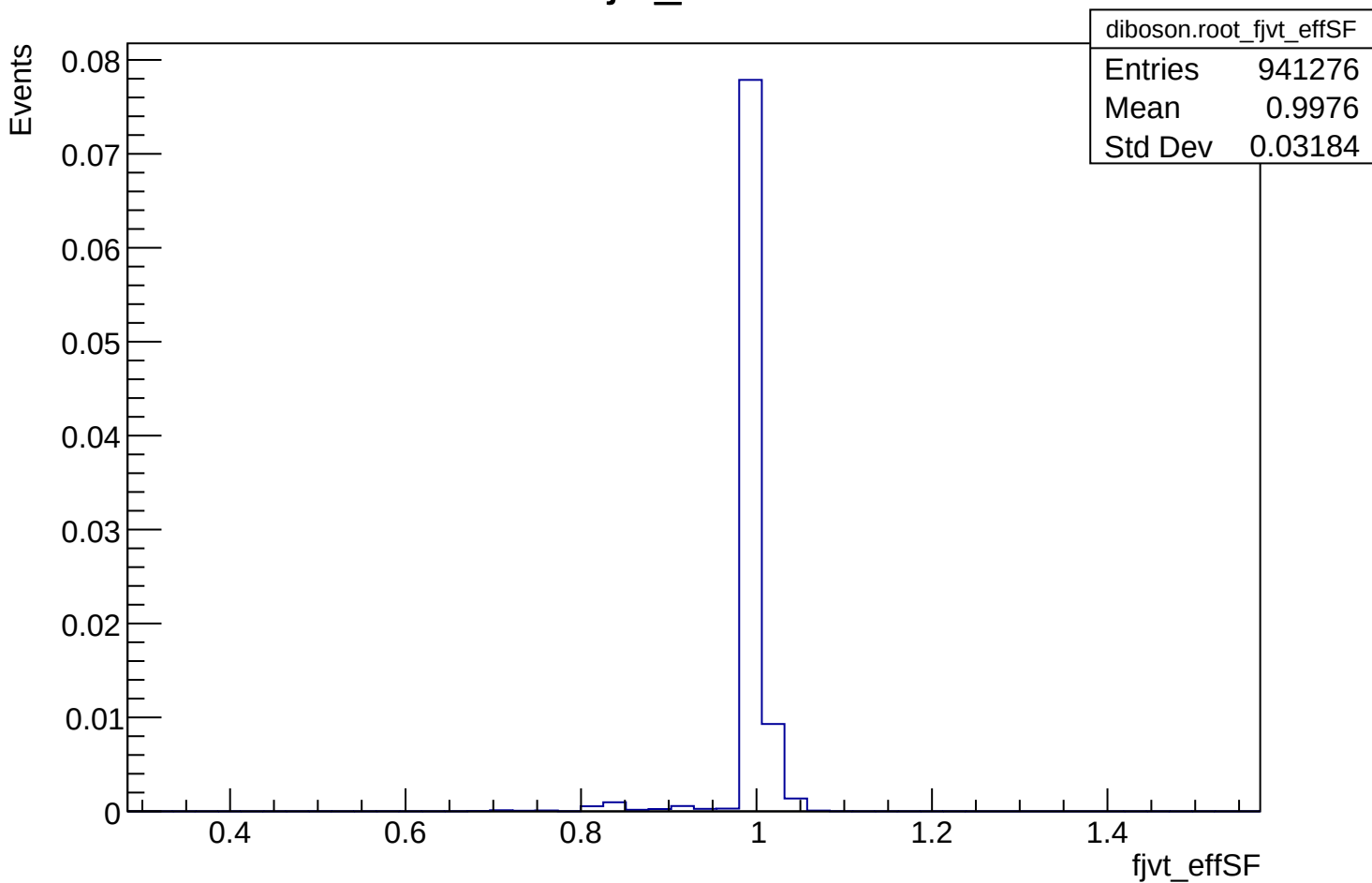
0.6

0.8

1.2

1.4

fjvt_effSF



ftag_effSF_GN2v01_Continuous

diboson.root_ftag_effSF_GN2v01_Continuous	
Entries	941276
Mean	0.9996
Std Dev	0.09133

Events

0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0.5

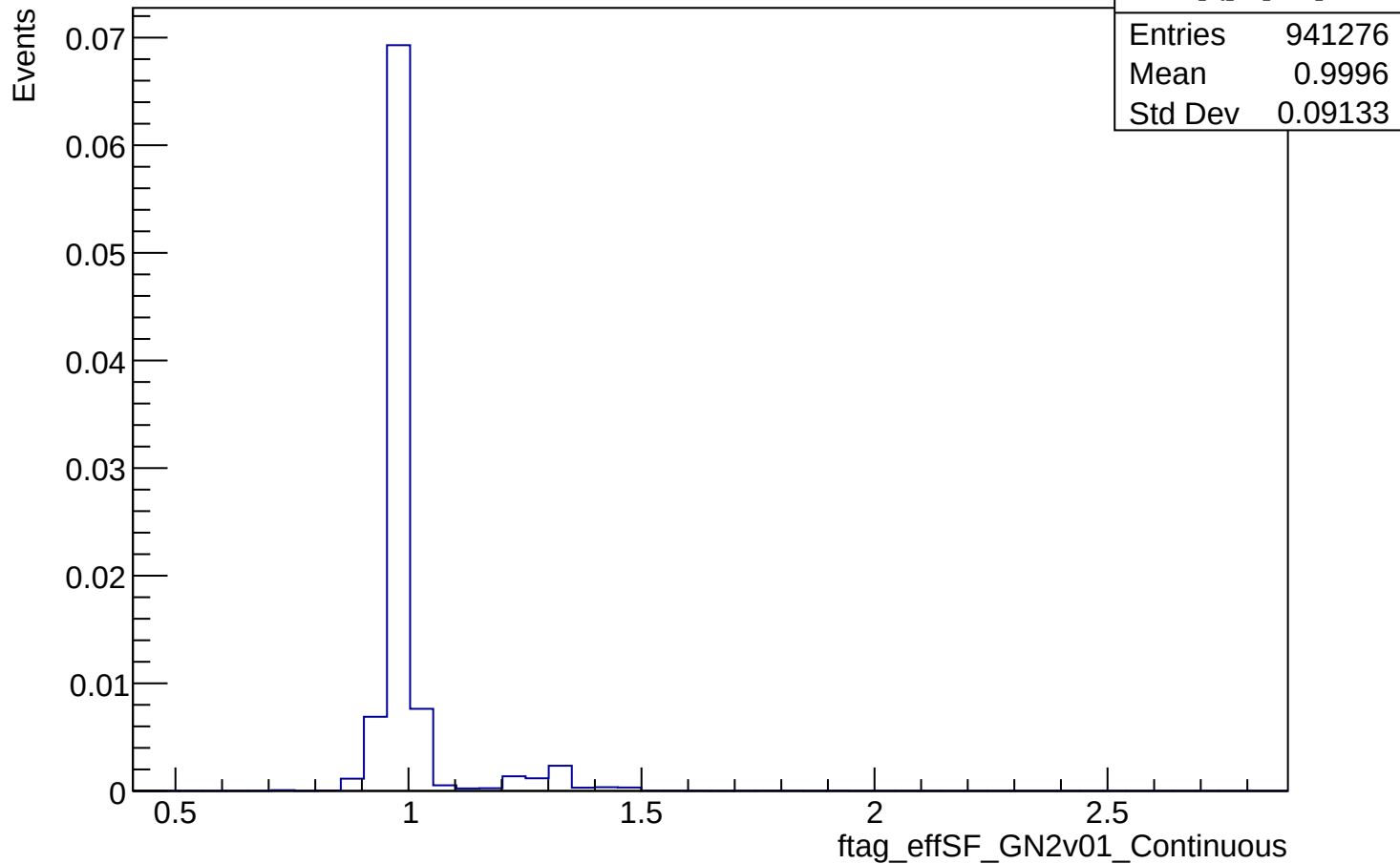
1

1.5

2

2.5

ftag_effSF_GN2v01_Continuous



globalTriggerEffSF

Events

0.05

0.04

0.03

0.02

0.01

0

0

0.2

0.4

0.6

0.8

globalTriggerEffSF

diboson.root_globalTriggerEffSF	
Entries	941276
Mean	0.942
Std Dev	0.1043

hhml_subChannelFlavor

Events

0.05
0.04
0.03
0.02
0.01
0

1

1.2

1.4

1.6

1.8

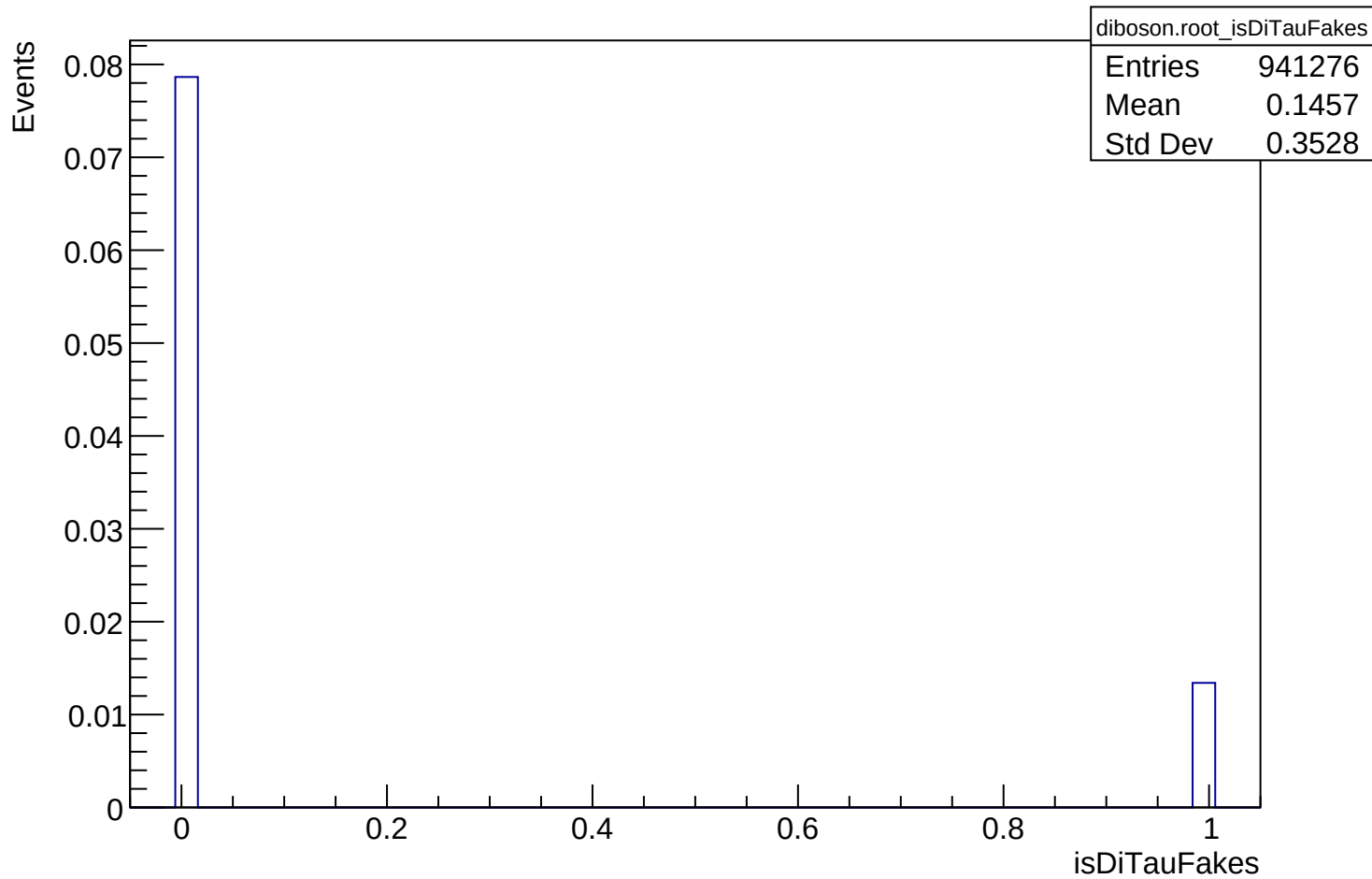
2

hhml_subChannelFlavor

diboson.root_hhml_subChannelFlavor

Entries	941276
Mean	1.521
Std Dev	0.4996

isDiTauFakes



isSingleFakeTau

diboson.root_isSingleFakeTau	
Entries	941276
Mean	0.5821
Std Dev	0.4932

Events

0.05

0.04

0.03

0.02

0.01

0

0

0.2

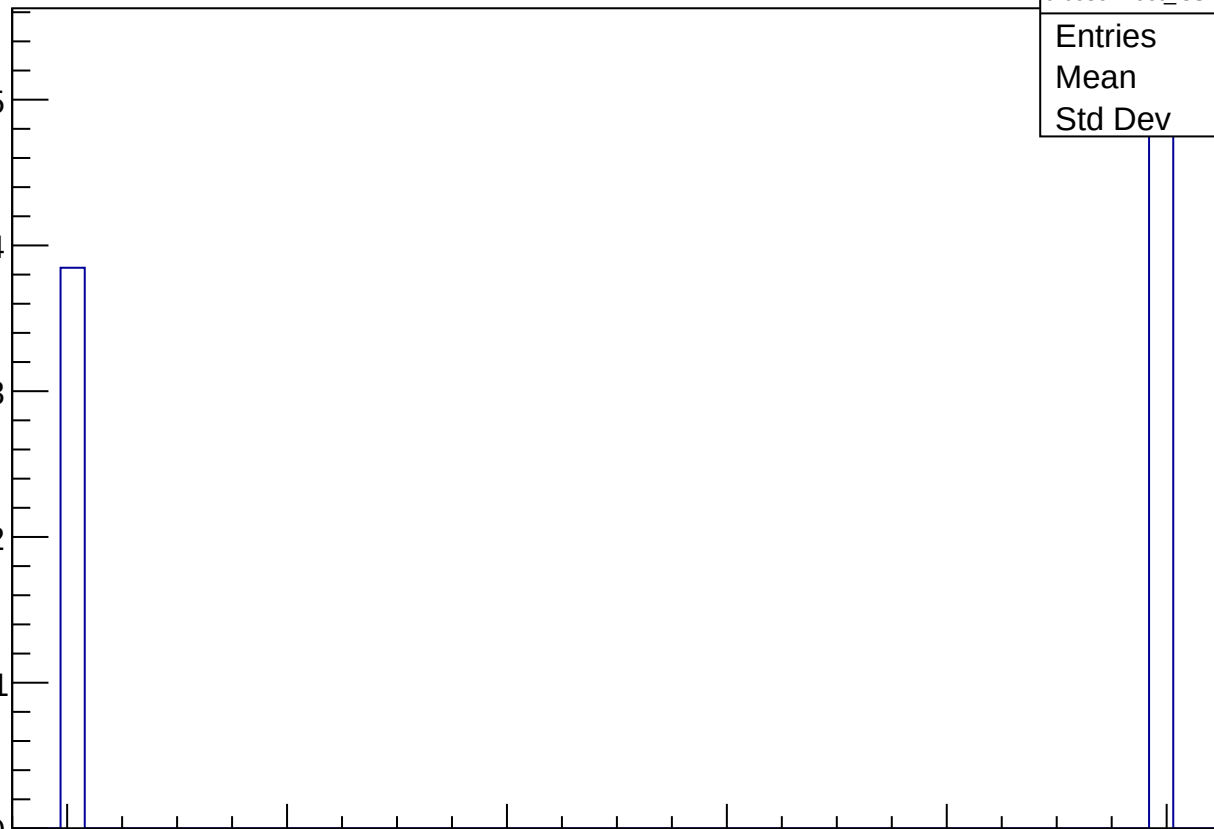
0.4

0.6

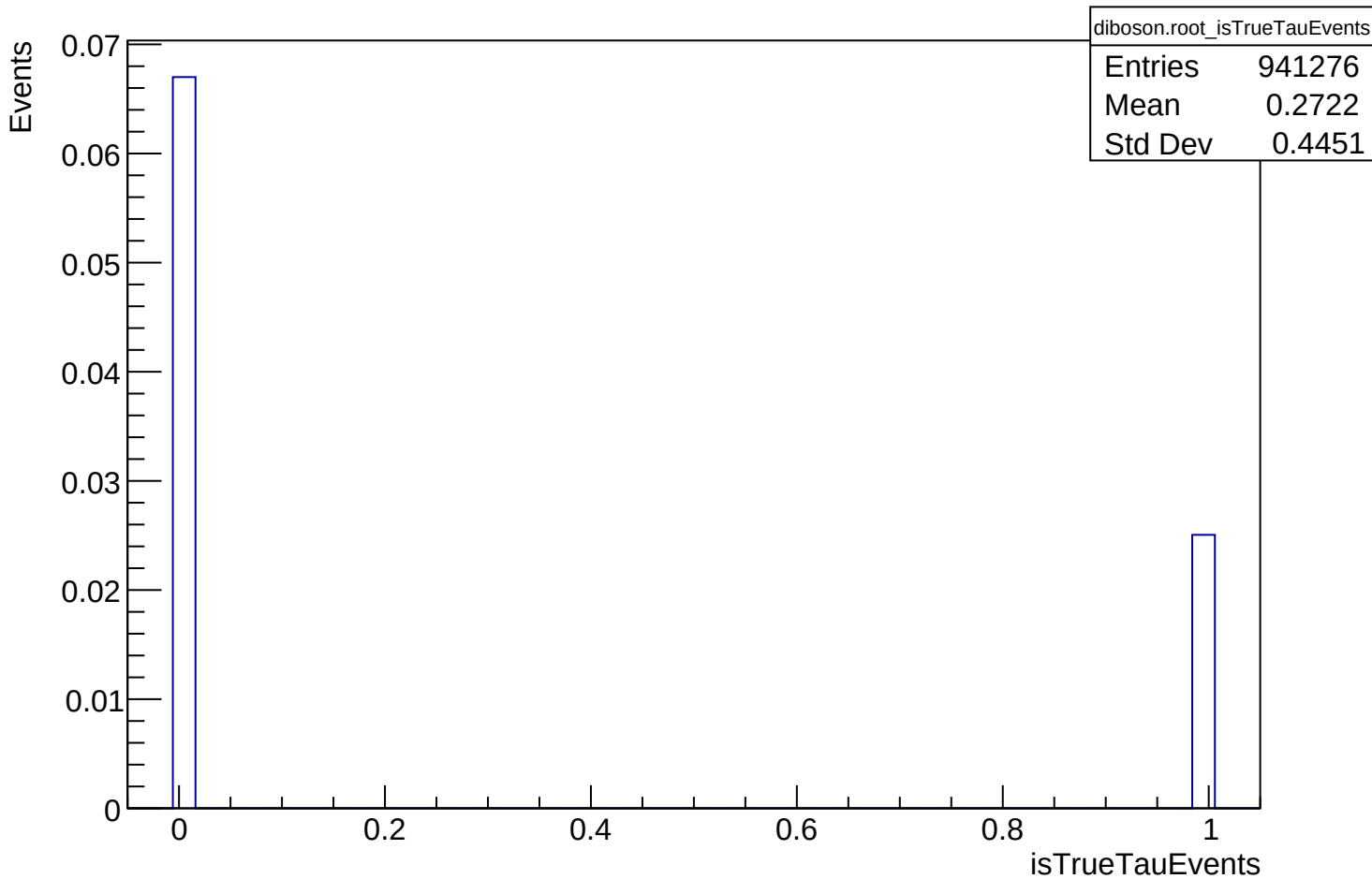
0.8

isSingleFakeTau

1

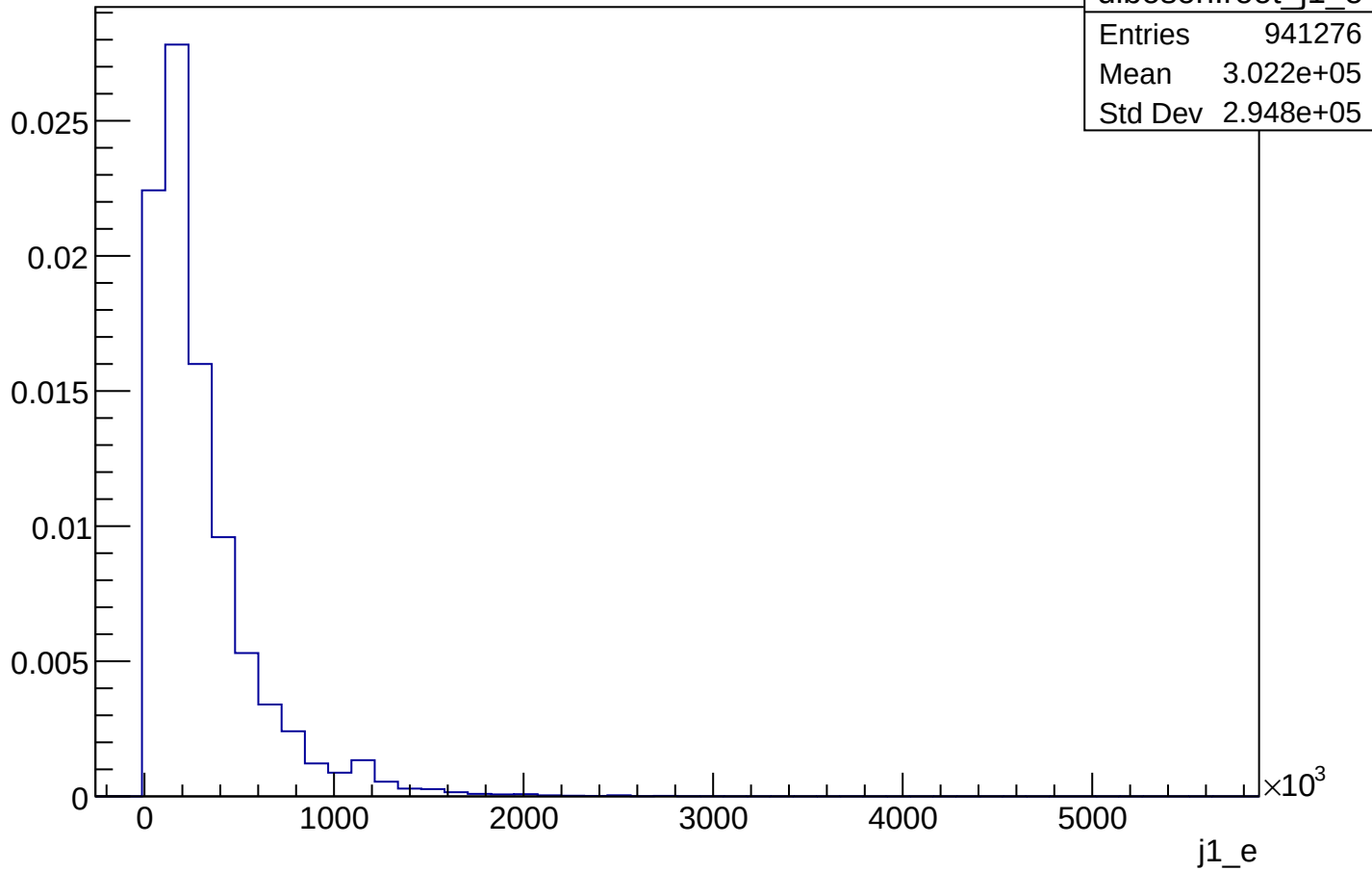


isTrueTauEvents



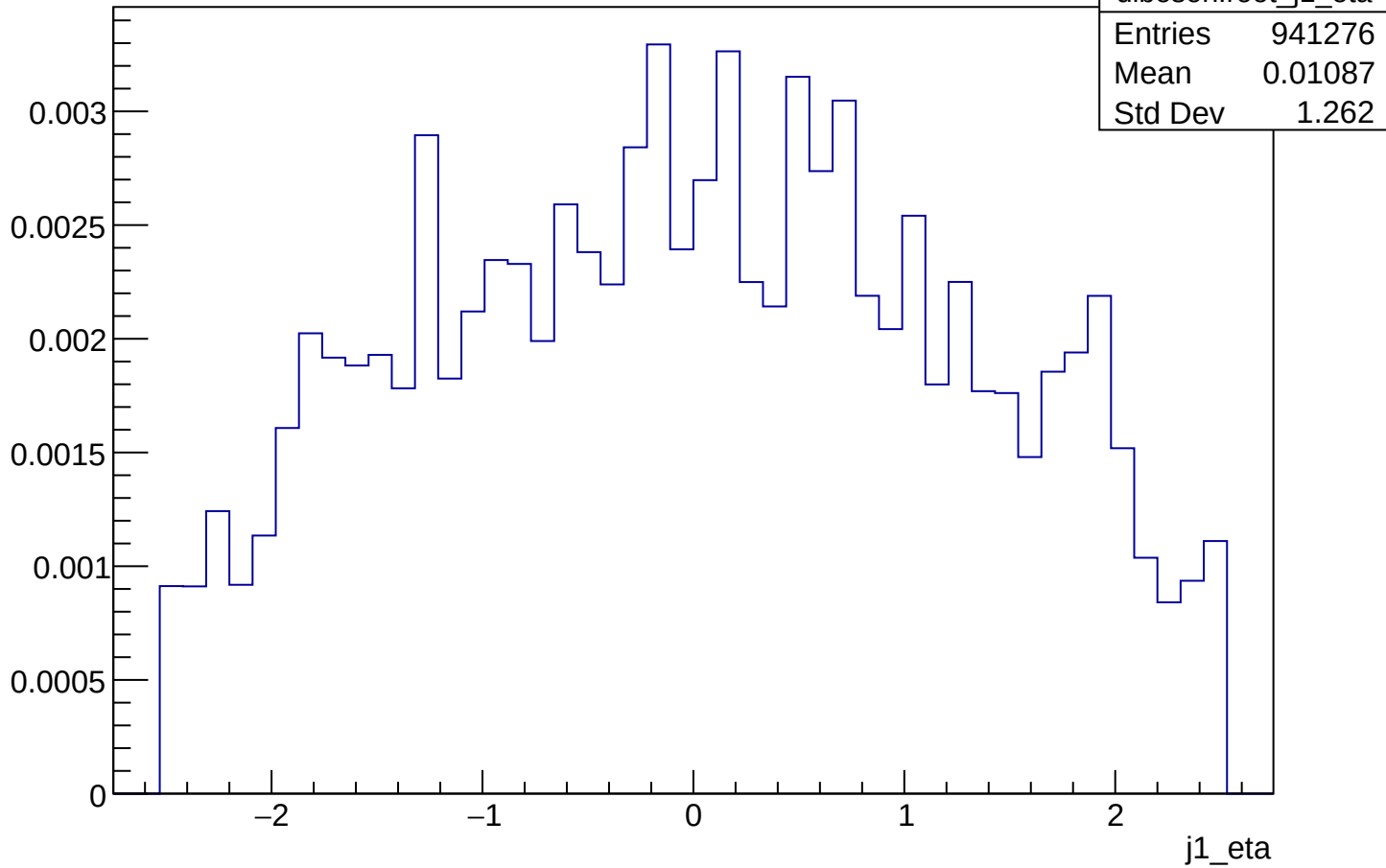
j1_e

Events



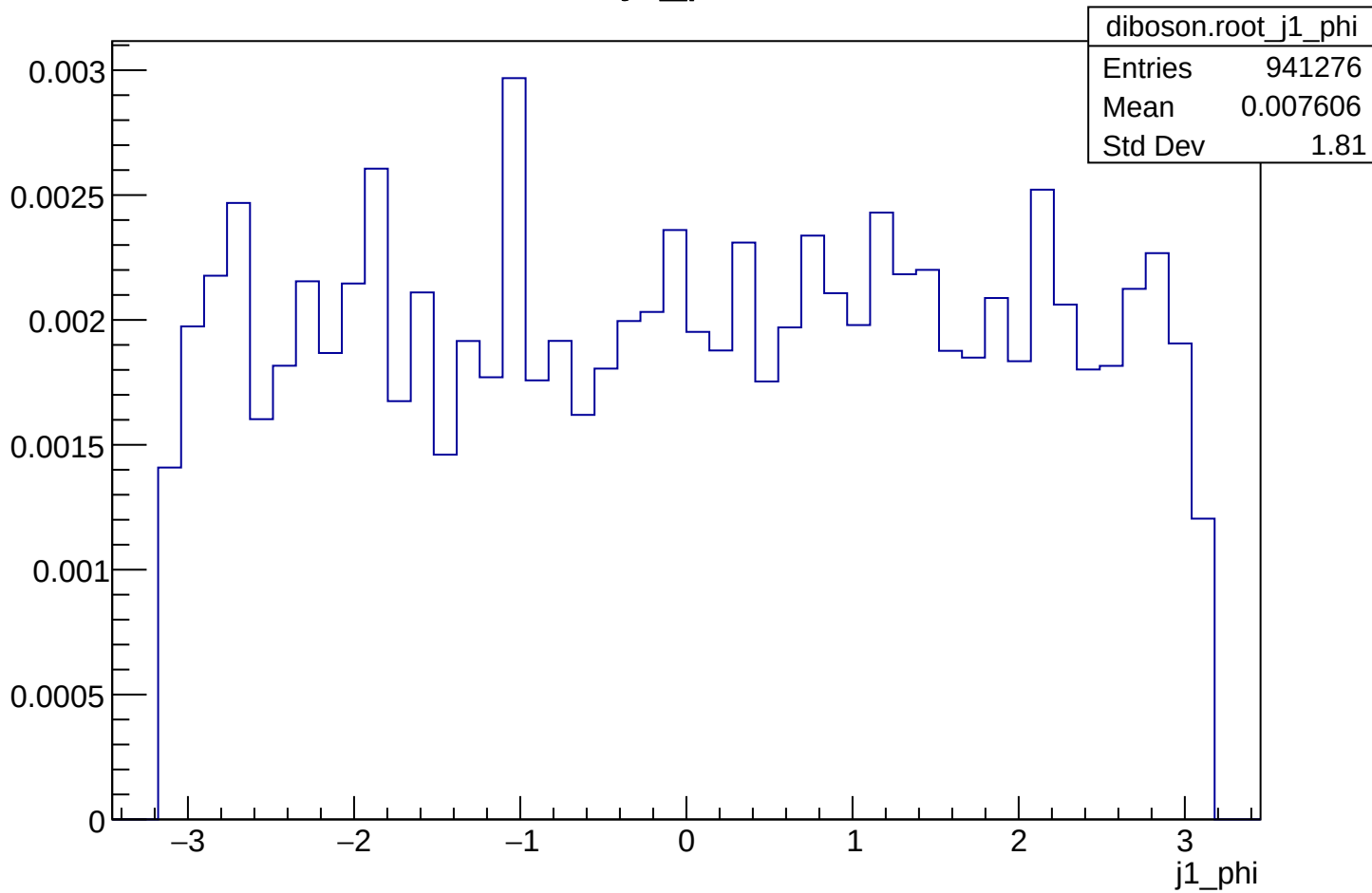
j1_eta

Events



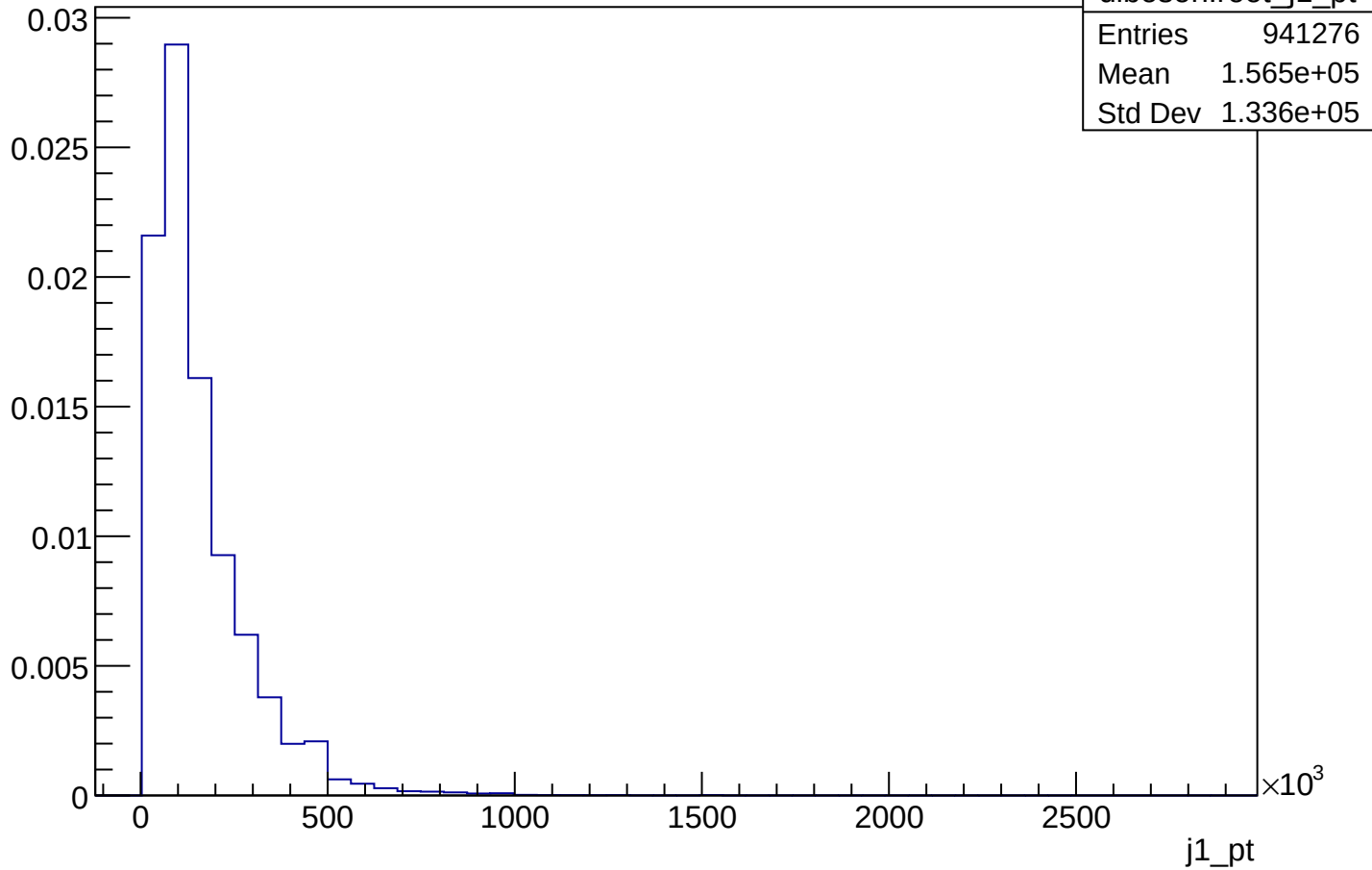
j1_phi

Events



j1_pt

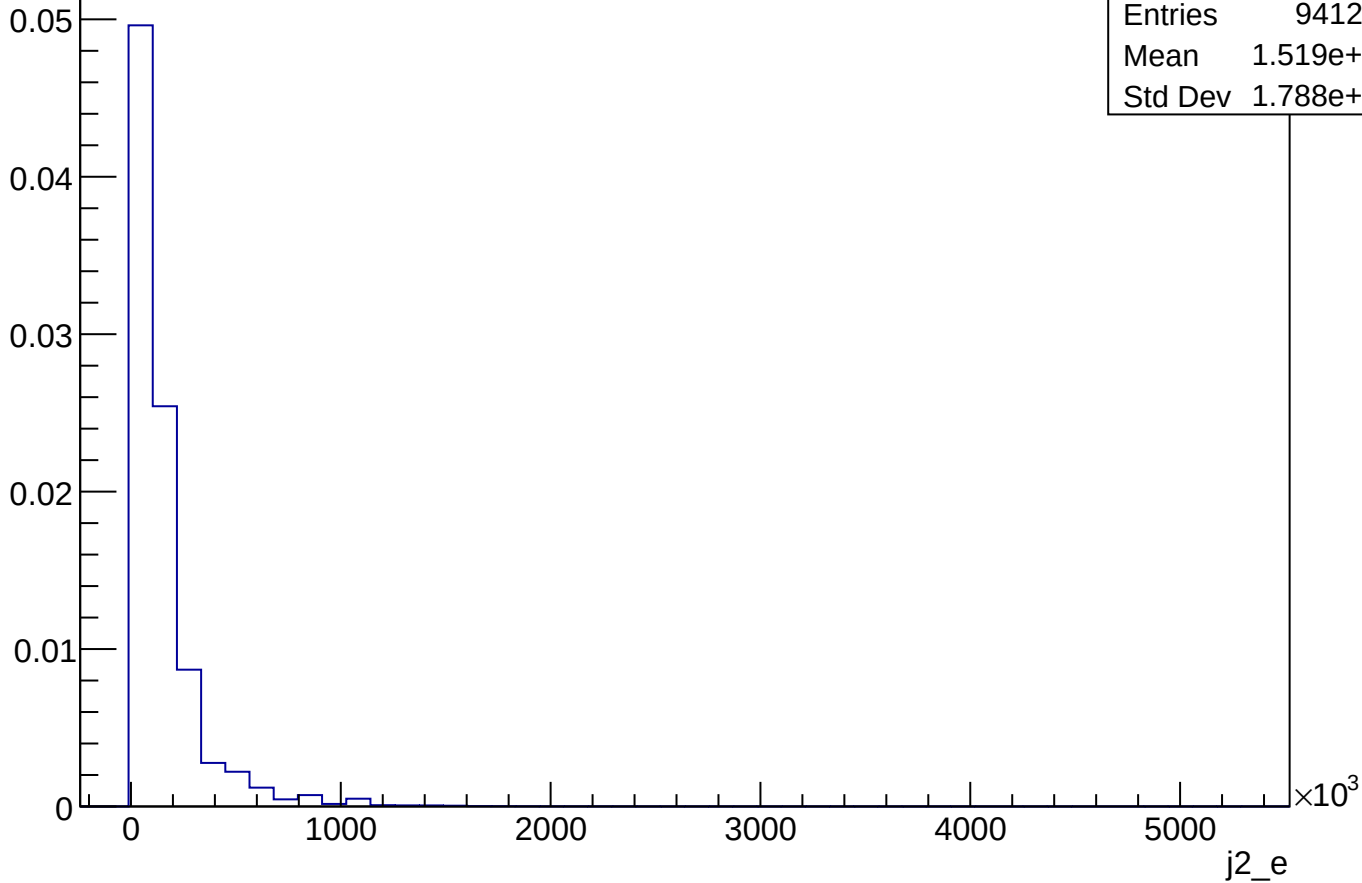
Events



j2_e

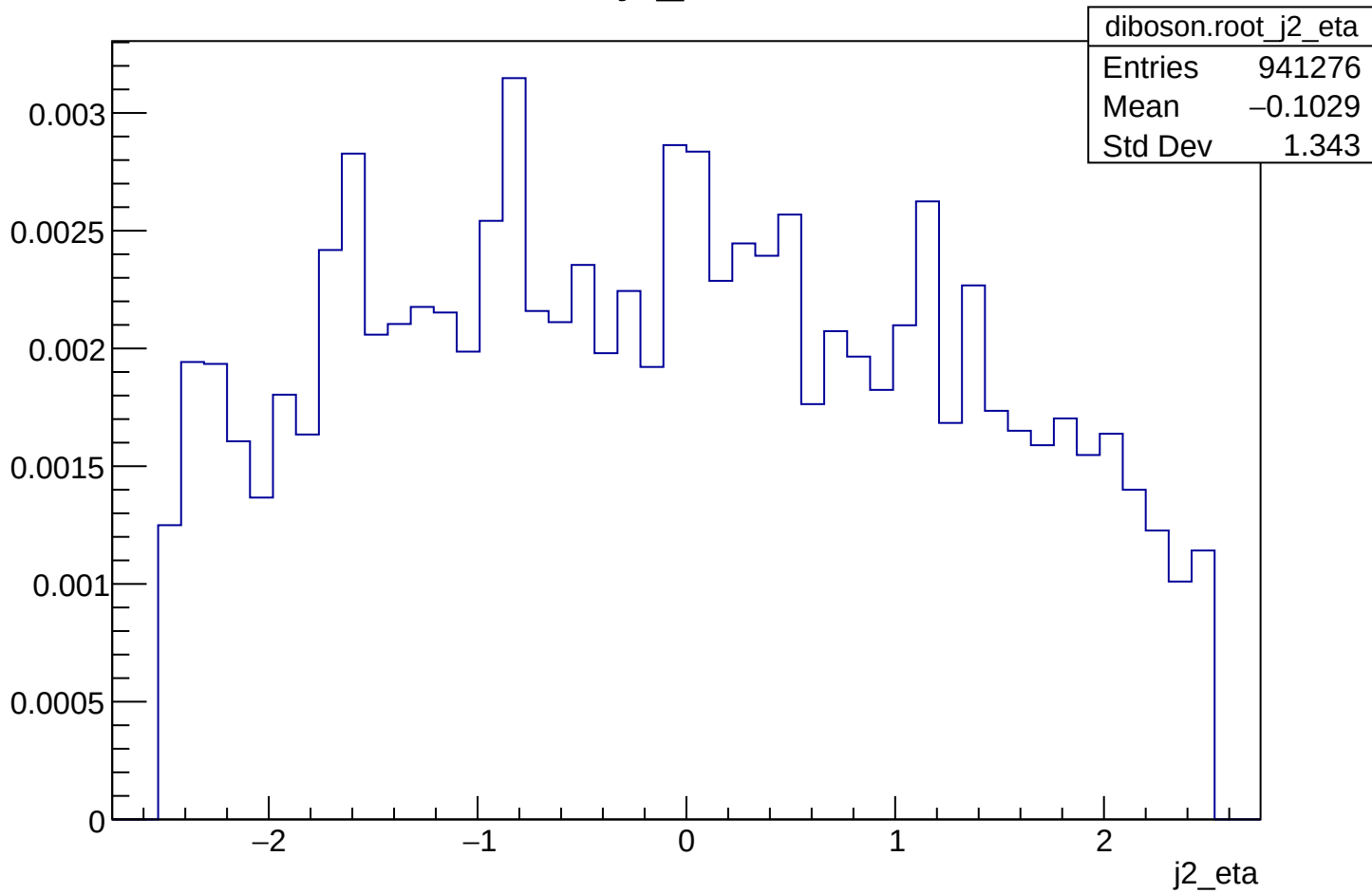
diboson.root_j2_e	
Entries	941276
Mean	1.519e+05
Std Dev	1.788e+05

Events



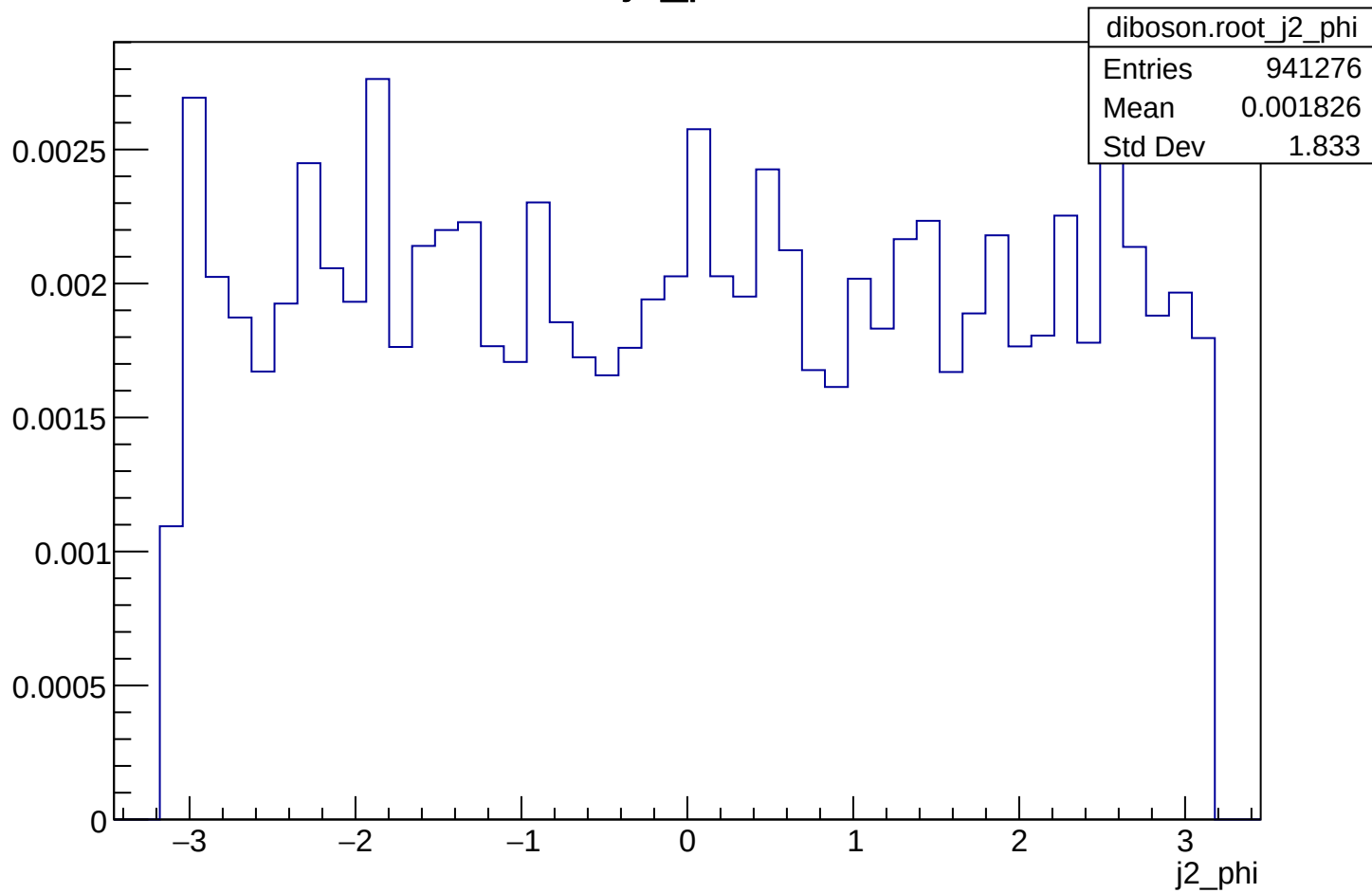
j2_eta

Events



j2_phi

Events



j2_pt

diboson.root_j2_pt	
Entries	941276
Mean	7.16e+04
Std Dev	7.422e+04

Events

0.05

0.04

0.03

0.02

0.01

0

0

500

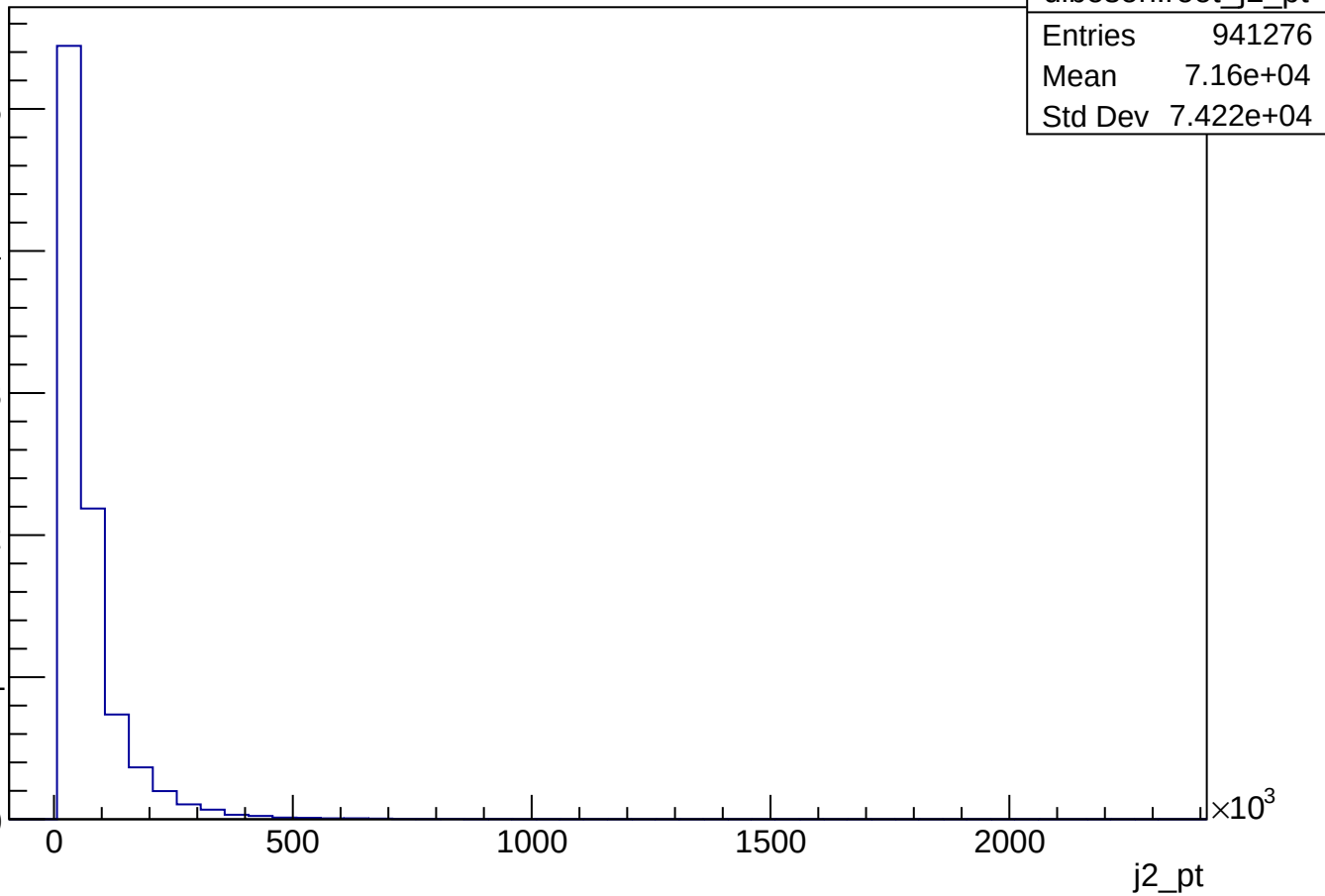
1000

1500

2000

j2_pt

$\times 10^3$



jvt_effSF

diboson.root_jvt_effSF	
Entries	941276
Mean	0.9996
Std Dev	0.09133

Events

0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0.5

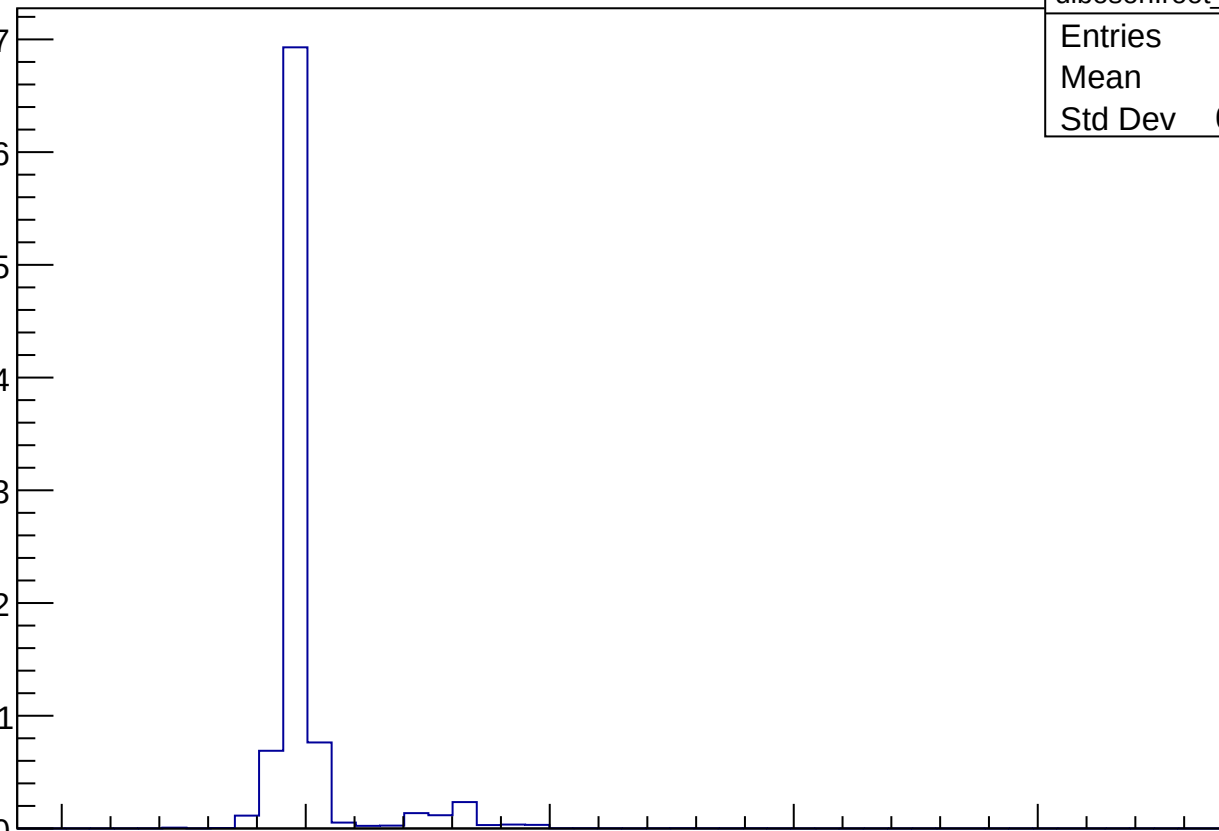
1

1.5

2

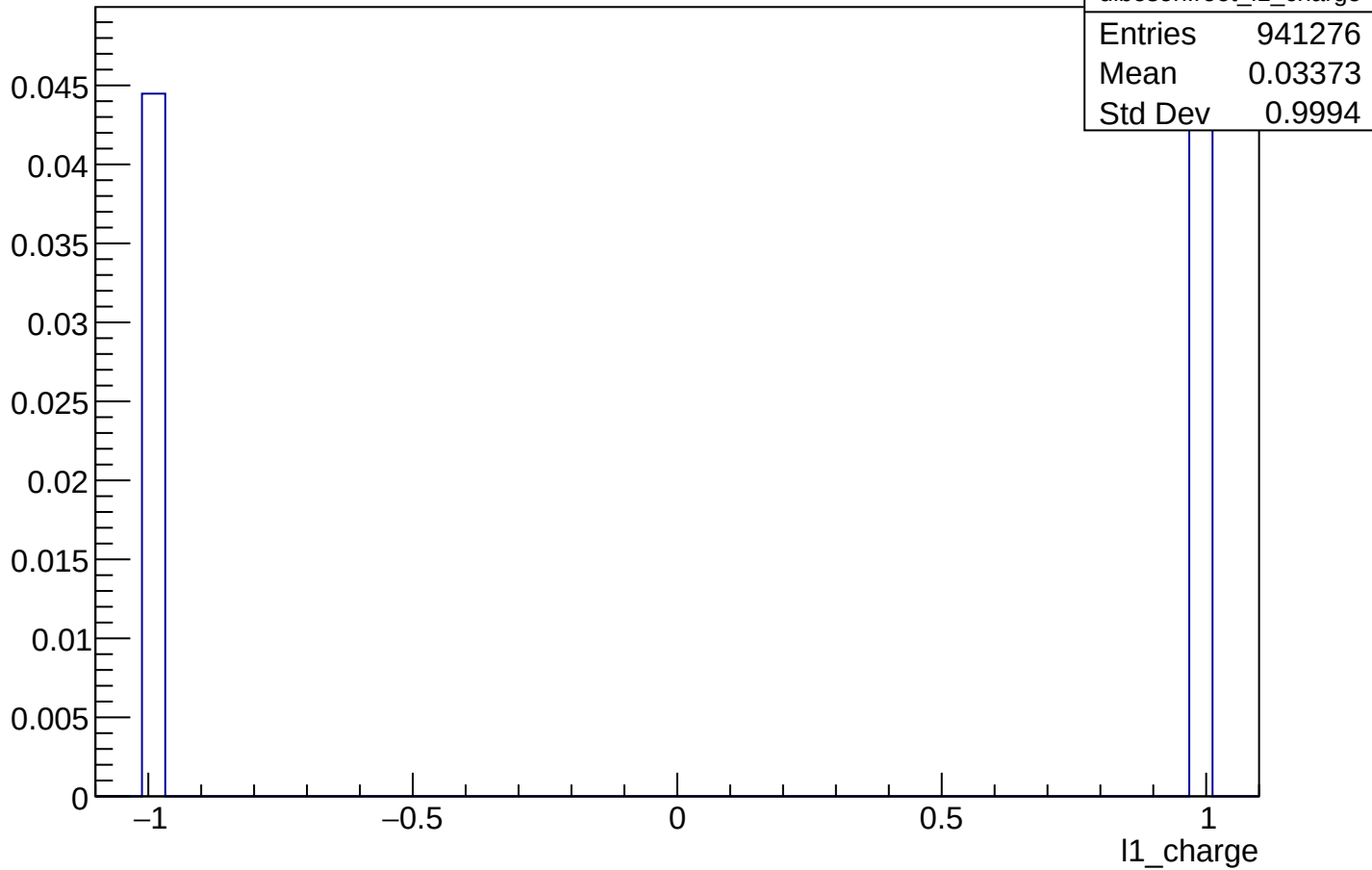
2.5

jvt_effSF



l1_charge

Events



l1_d0sig

Events

0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

-4

-2

0

2

4

l1_d0sig

diboson.root_l1_d0sig	
Entries	941276
Mean	-0.02835
Std Dev	1.252

l1_e

Events

0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0 500 1000 1500 2000 2500 3000 3500 4000 $\times 10^3$
l1_e

diboson.root_l1_e	
Entries	941276
Mean	1.337e+05
Std Dev	1.424e+05

l1_effSF

Events

0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0.85

0.9

0.95

0.9881

1.0

1.05

1.1

1.15

1.2

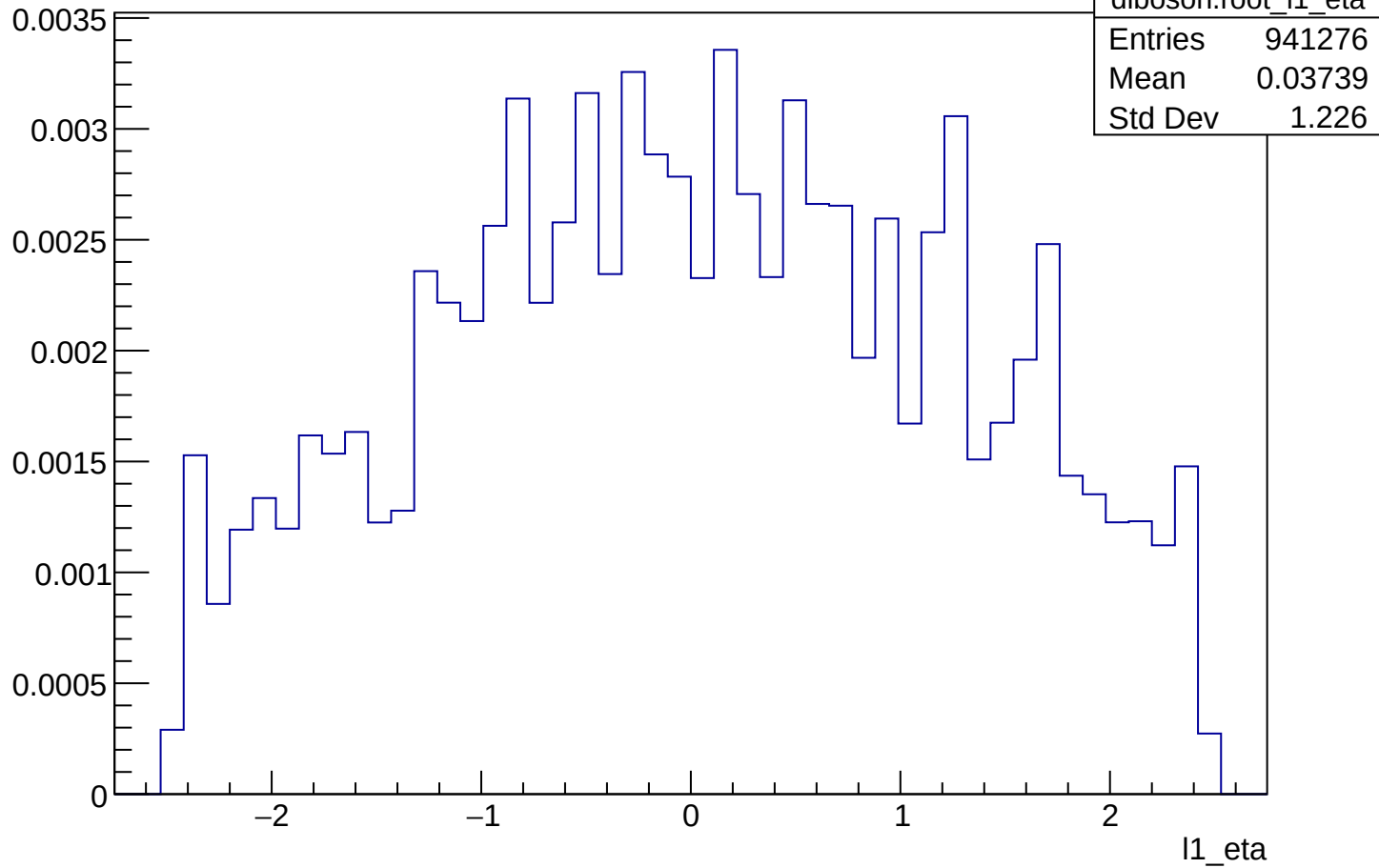
1.25

l1_effSF

diboson.root_l1_effSF	
Entries	941276
Mean	0.9881
Std Dev	0.01298

$l1_eta$

Events



l1_pdg

Events

0.025
0.02
0.015
0.01
0.005
0

-10

-5

0

5

10

l1_pdg

diboson.root_l1_pdg	
Entries	941276
Mean	-0.3971
Std Dev	12.08

0.021

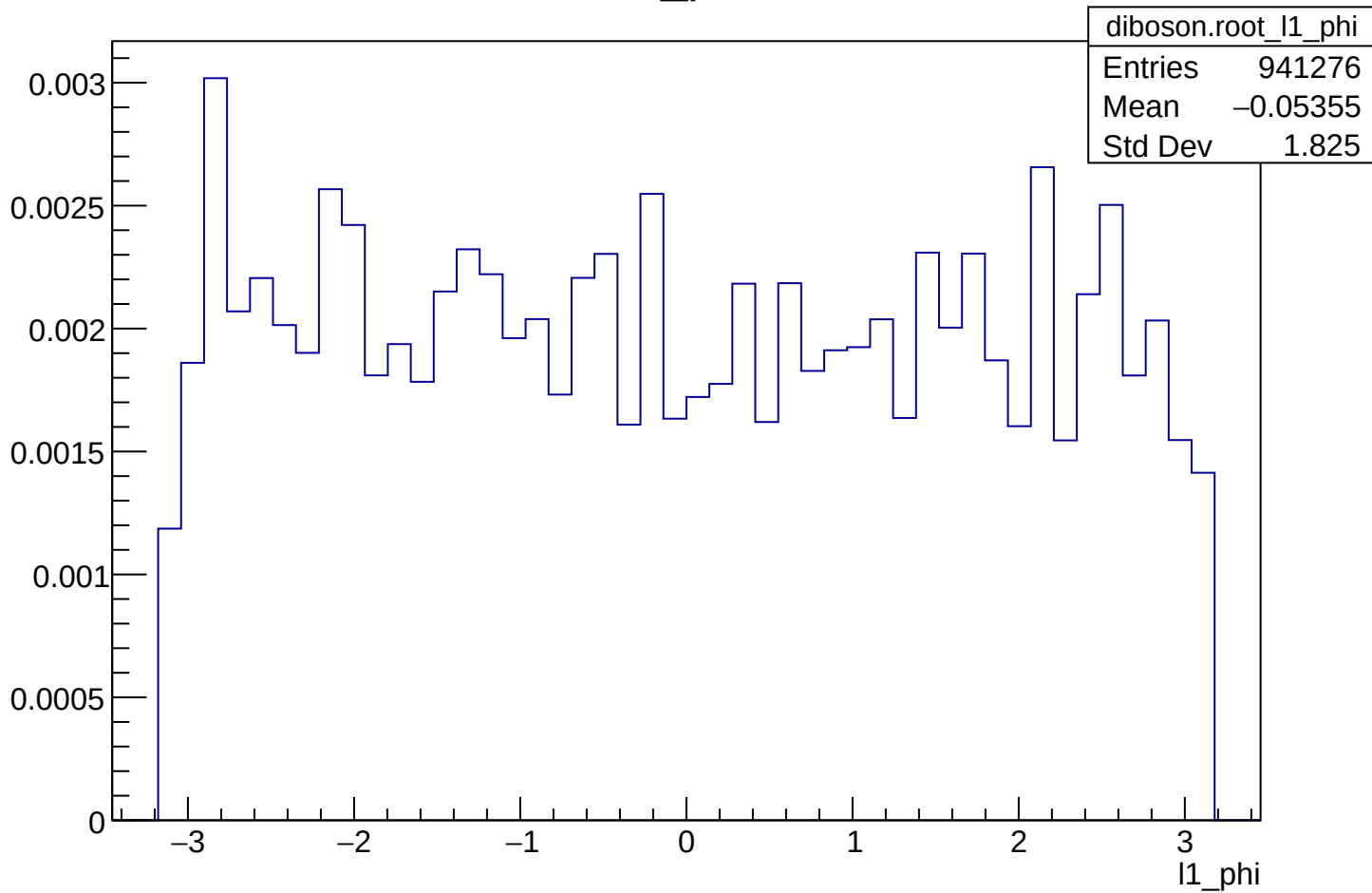
0.022

0.023

0.024

l1_phi

Events



l1_pt

Events

0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0

500

1000

1500

2000

$\times 10^3$
l1_pt

diboson.root_l1_pt	
Entries	941276
Mean	6.992e+04
Std Dev	6.389e+04

l1_z0sintheta

Events

0.024
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0

-0.4

-0.2

0

0.2

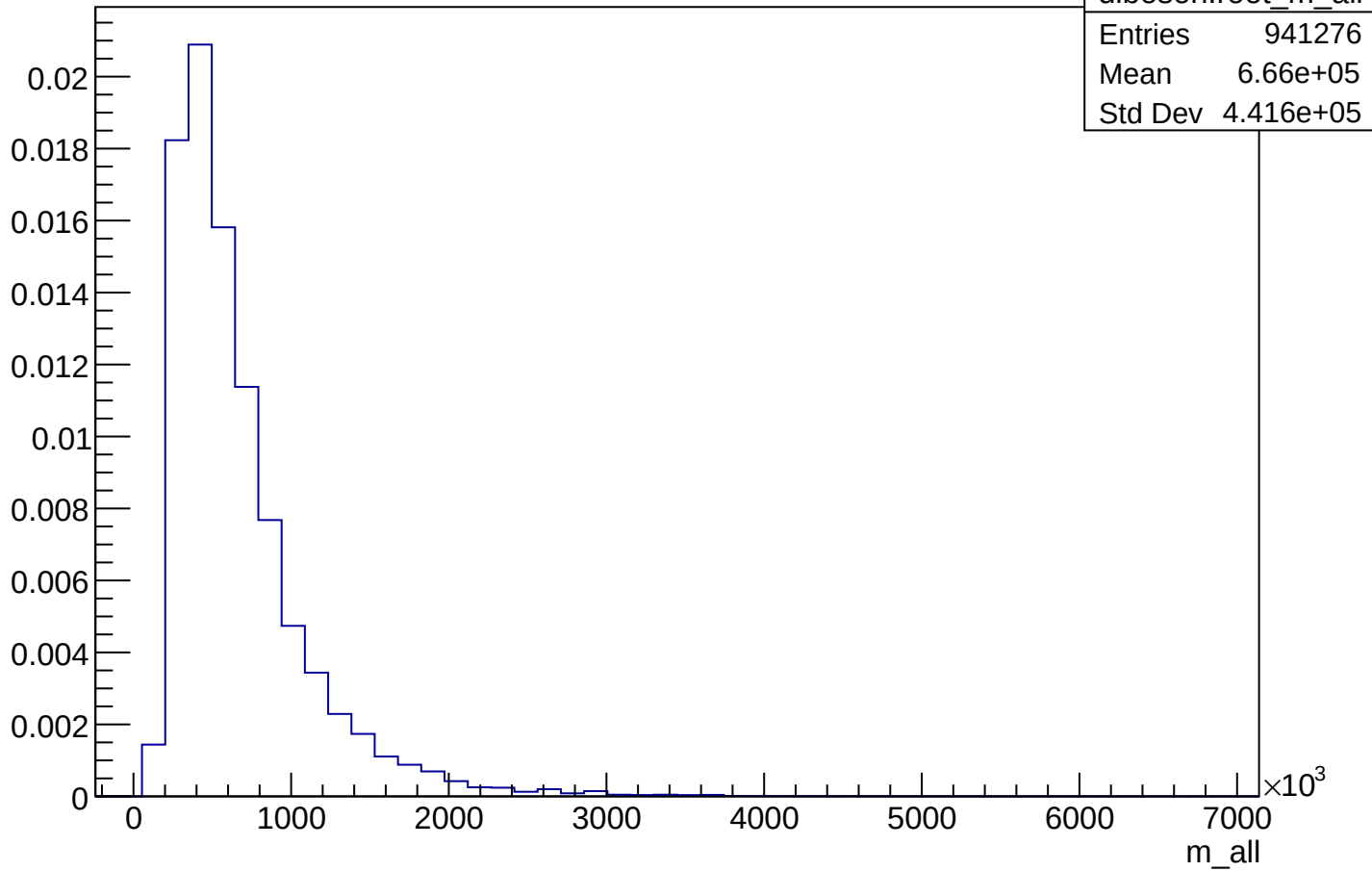
0.4

l1_z0sintheta

diboson.root_l1_z0sintheta	
Entries	941276
Mean	-0.0007184
Std Dev	0.0486

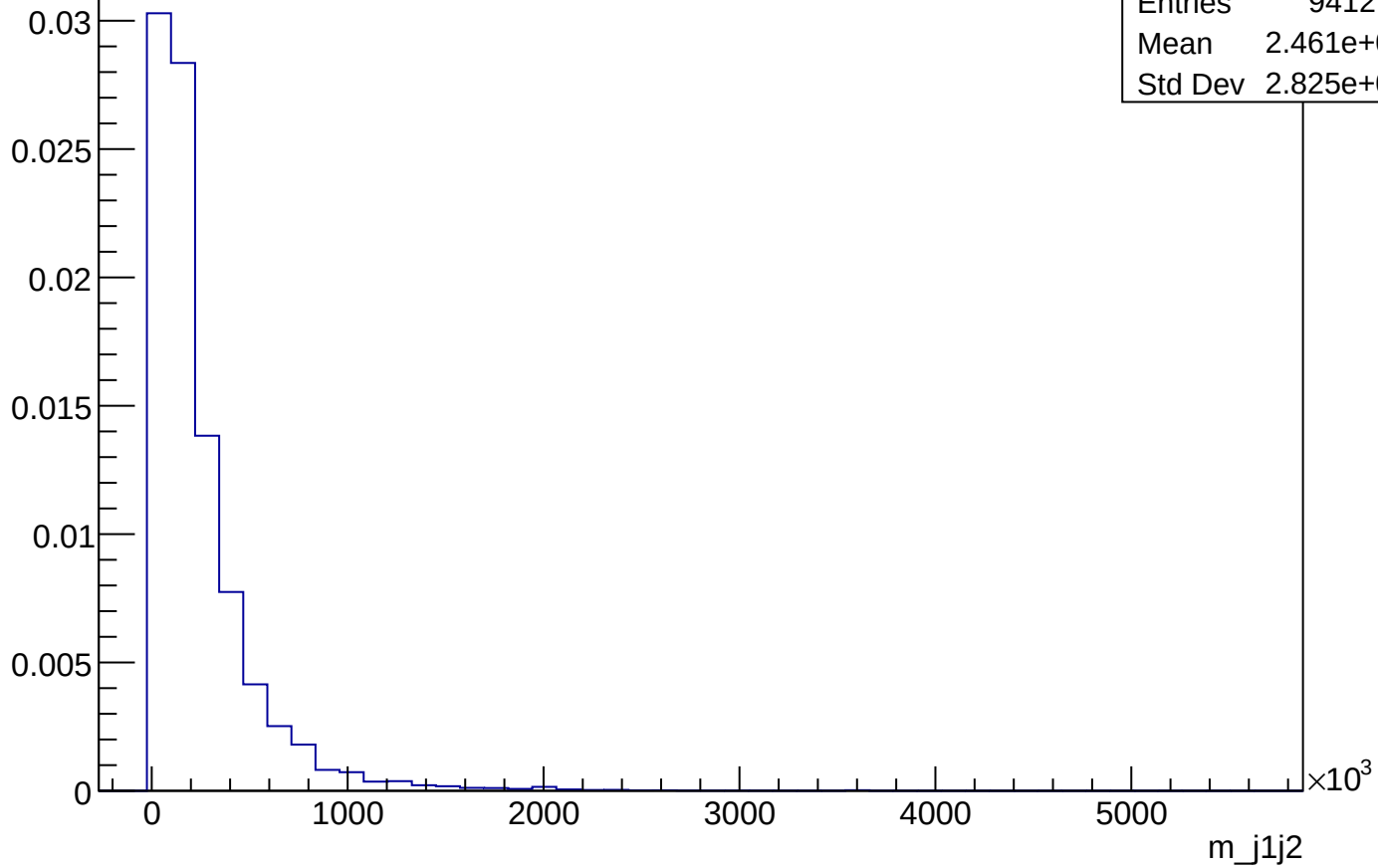
m_all

Events



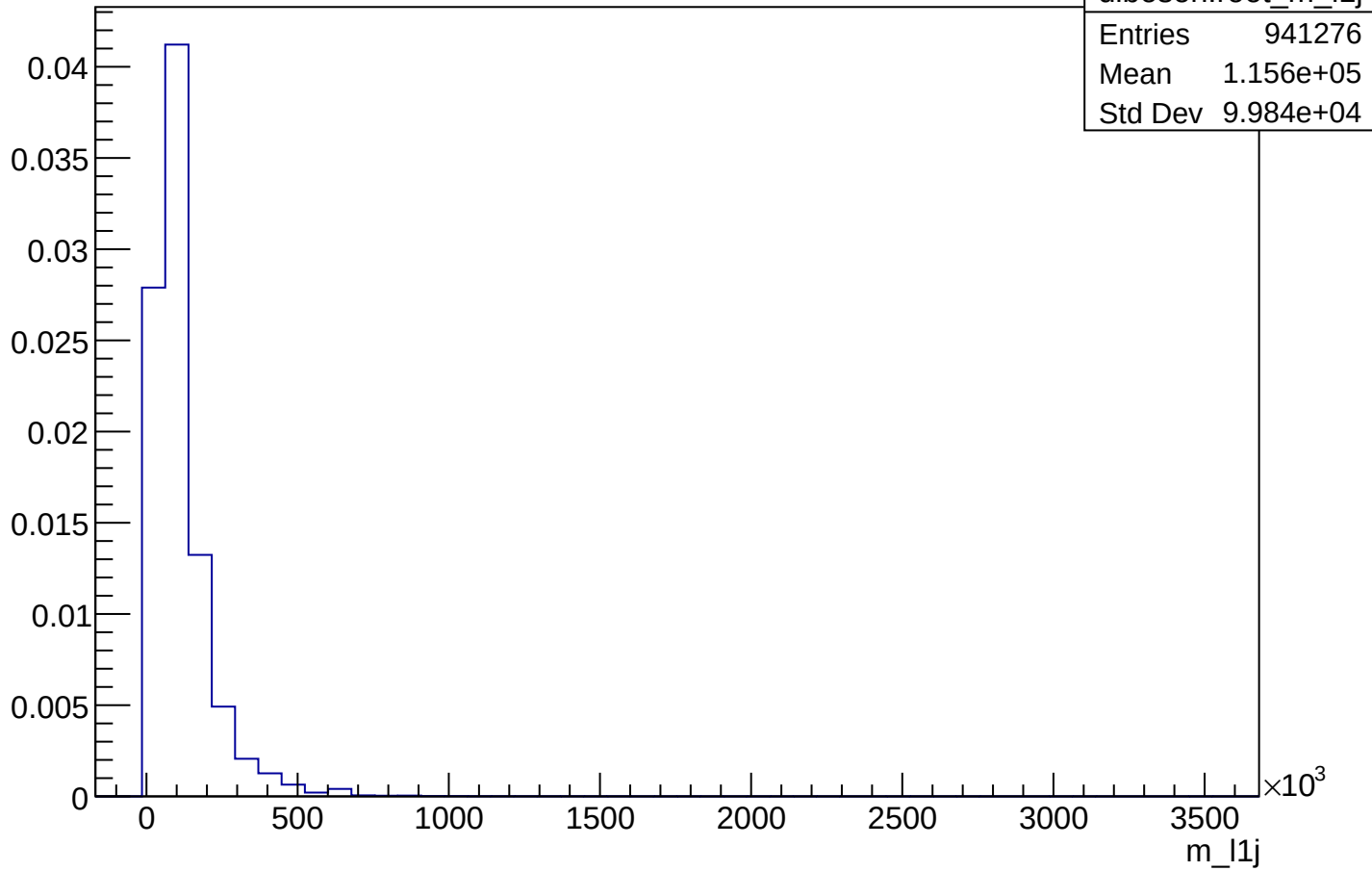
m_j1j2

Events



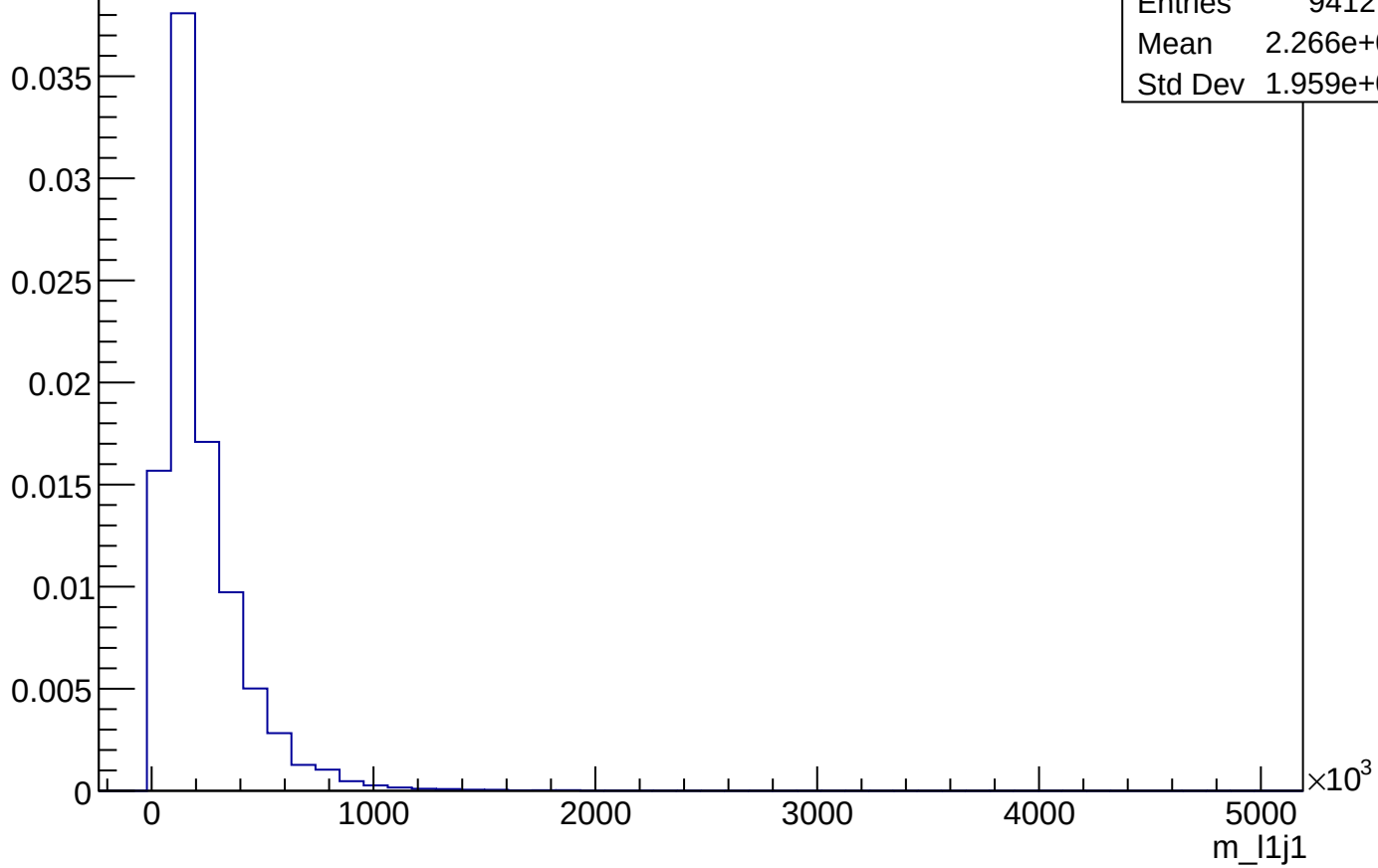
m_l1j

Events



m_l1j1

Events



m_l1j2

Events

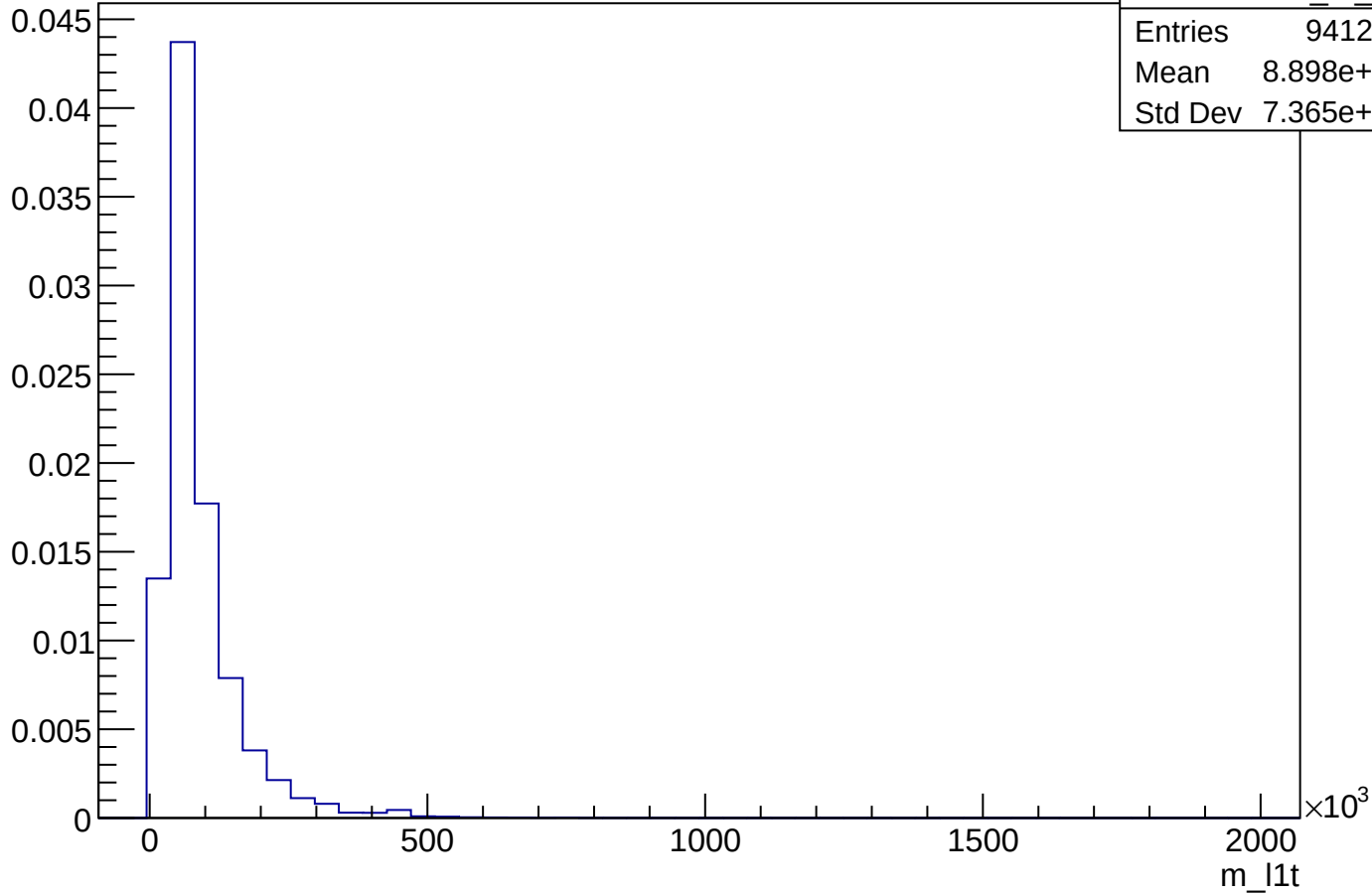
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0 500 1000 1500 2000 2500 3000 3500 4000 $\times 10^3$
m_l1j2

diboson.root_m_l1j2	
Entries	941276
Mean	1.441e+05
Std Dev	1.24e+05

m_l1t

Events



m_l1t1

Events

0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0 500 1000 1500 2000 2500 3000 3500 $\times 10^3$
m_l1t1

diboson.root_m_l1t1	
Entries	941276
Mean	1.271e+05
Std Dev	1.039e+05

m_l1t2

Events

0.025

0.02

0.015

0.01

0.005

0

0

200

400

600

800

1000

1200

1400

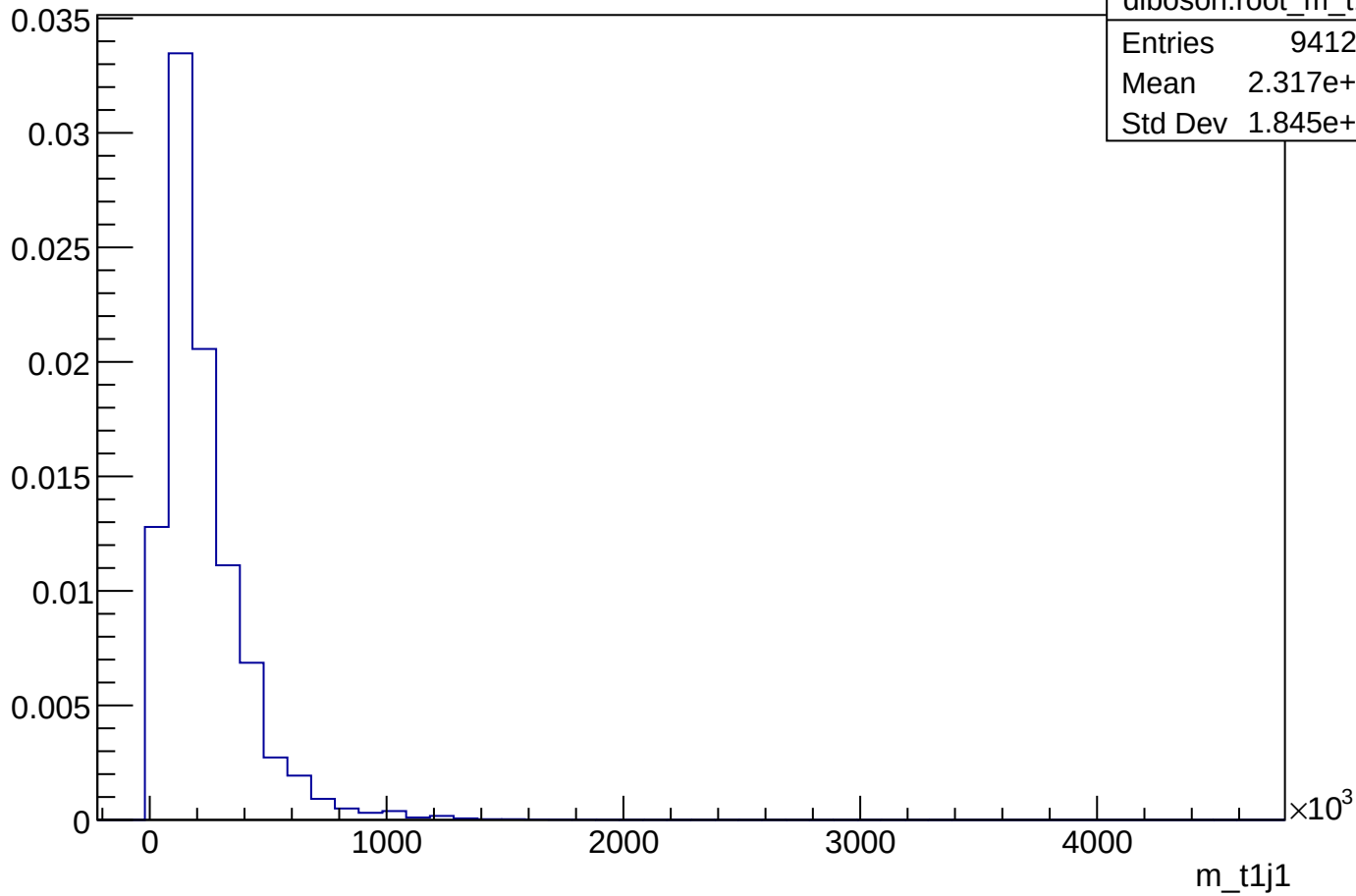
m_l1t2

$\times 10^3$

diboson.root_m_l1t2	
Entries	941276
Mean	9.671e+04
Std Dev	7.225e+04

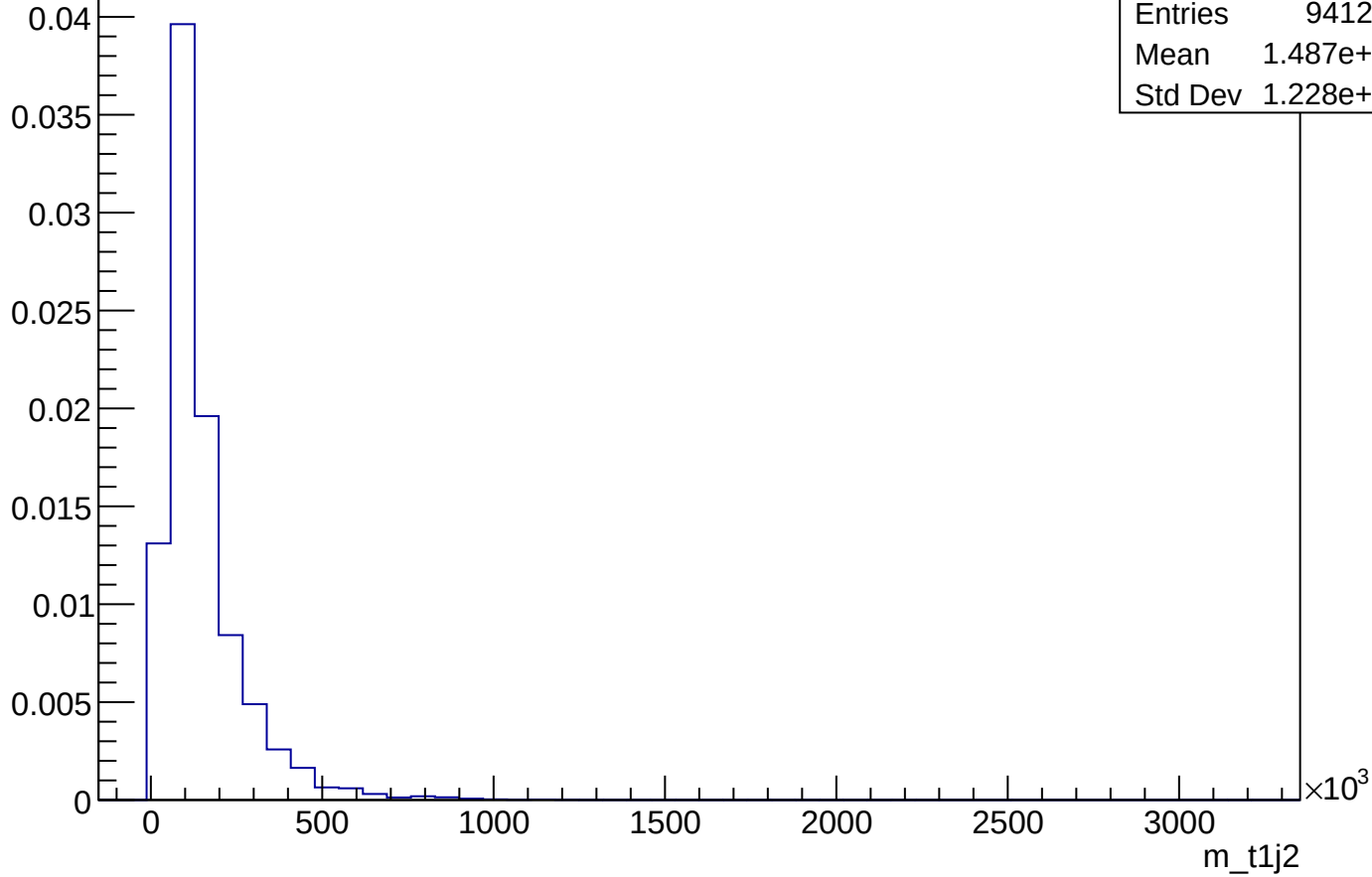
m_t1j1

Events



m_t1j2

Events



m_t1t2

Events

0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

0 200 400 600 800 1000 1200 1400 1600 $\times 10^3$
m_t1t2

diboson.root_m_t1t2	
Entries	941276
Mean	9.364e+04
Std Dev	6.888e+04

m_t2j1

Events

0.025

0.02

0.015

0.01

0.005

0

0

500

1000

1500

2000

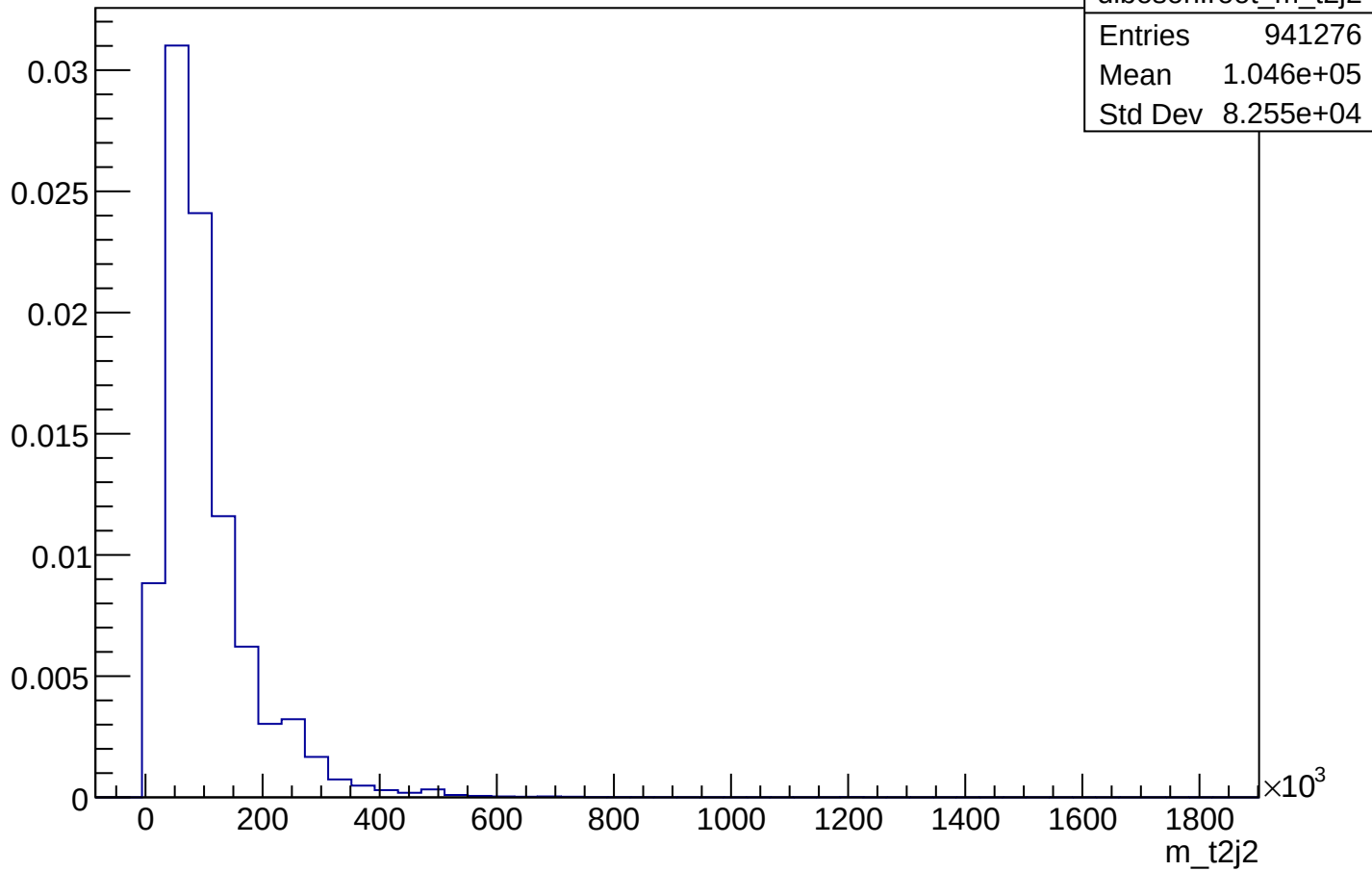
m_t2j1

$\times 10^3$

diboson.root_m_t2j1	
Entries	941276
Mean	1.557e+05
Std Dev	1.194e+05

m_t2j2

Events



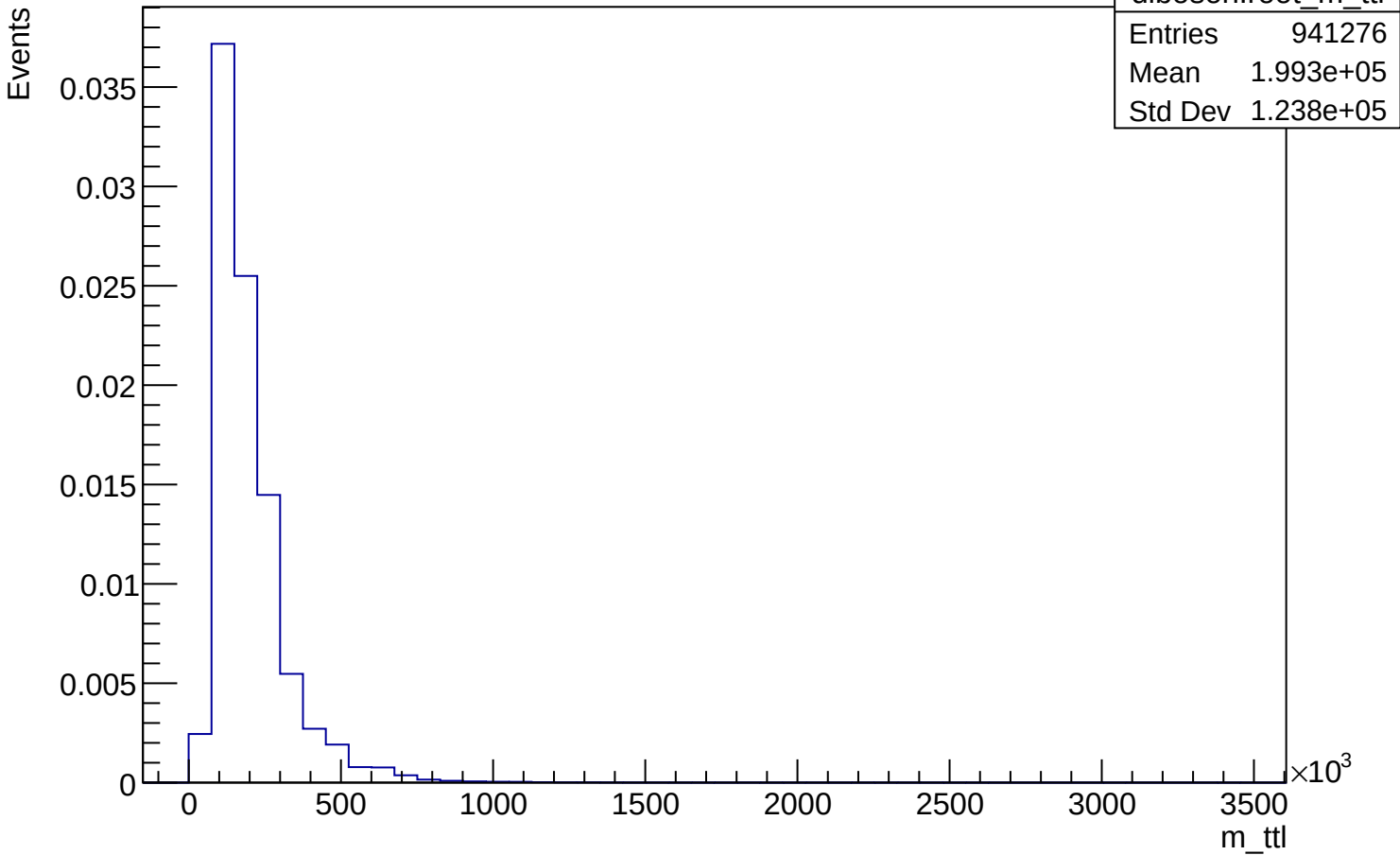
m_ttl

diboson.root_m_ttl

Entries	941276
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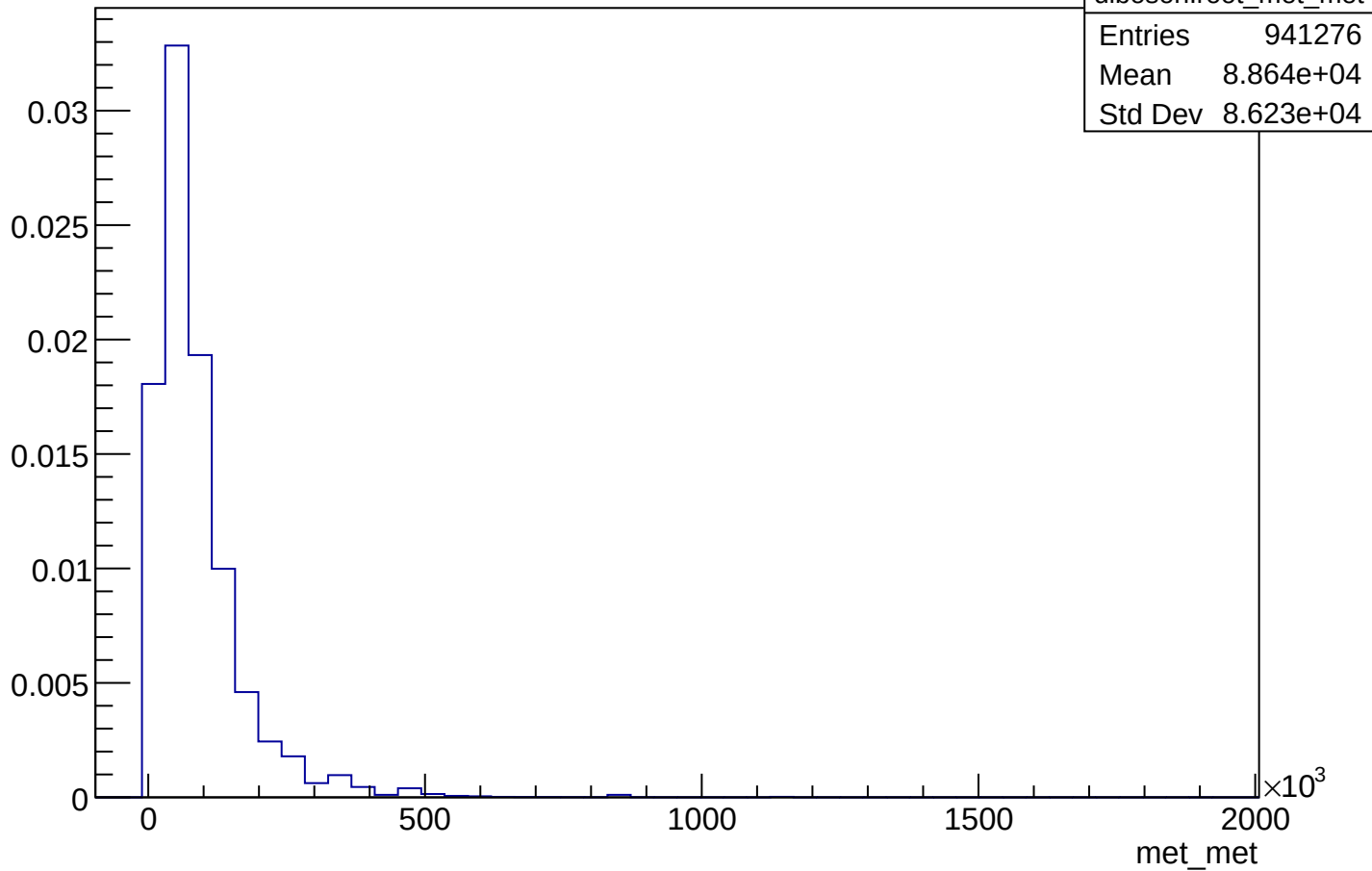
Mean	1.993e+05
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Std Dev	1.238e+05
---------	-----------

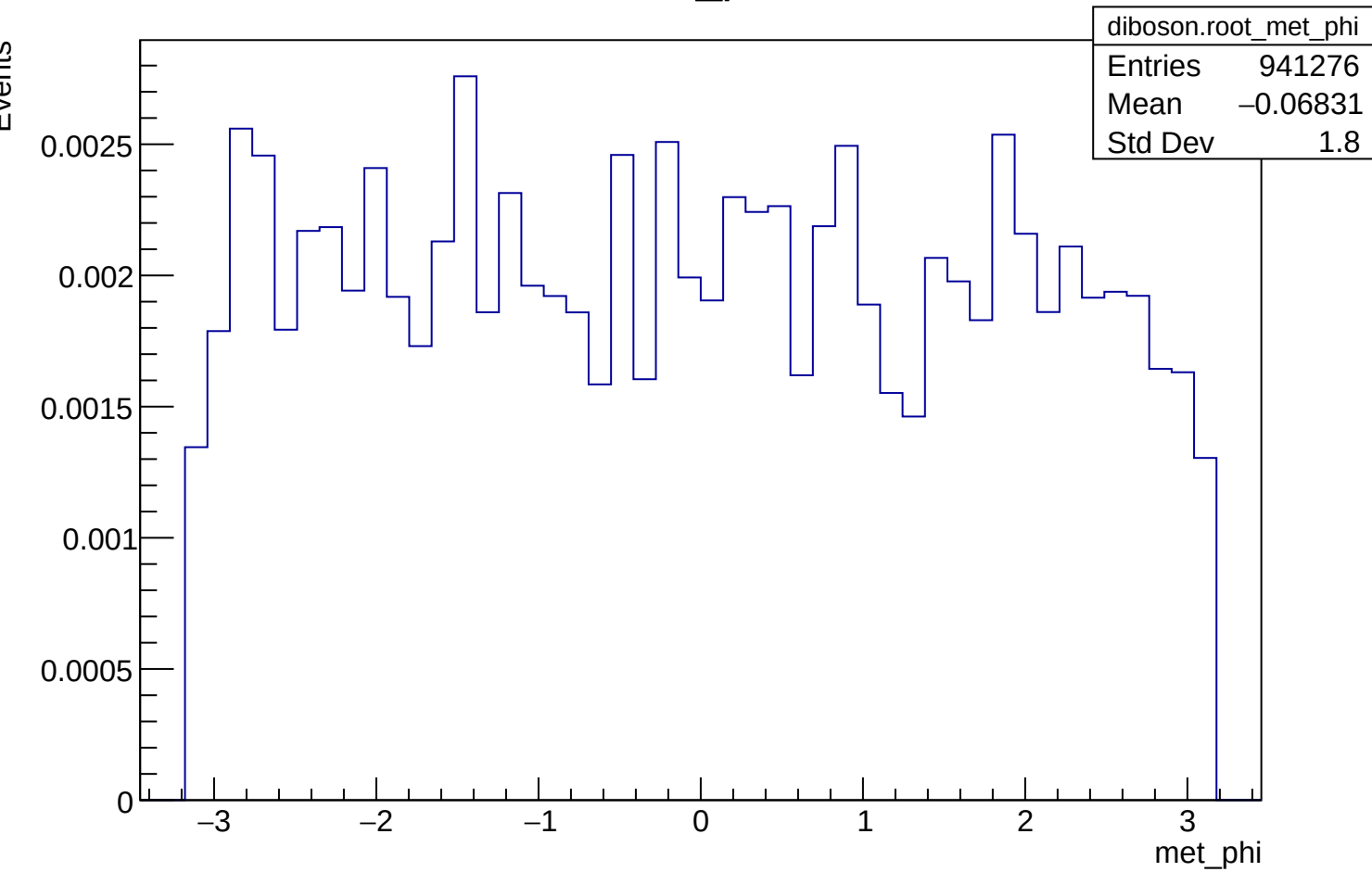


met_met

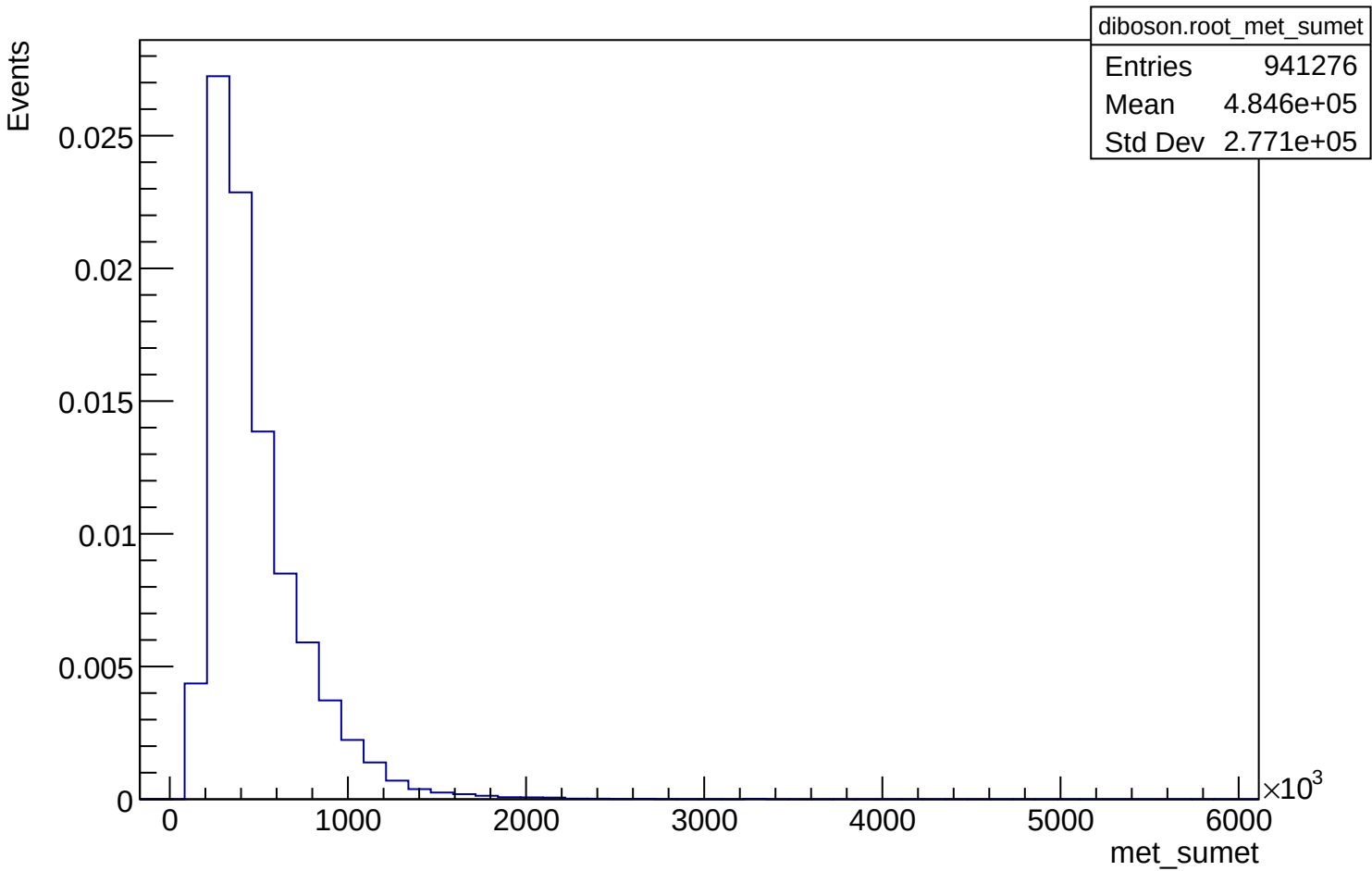
Events



met_phi



met_sumet



n_jet

diboson.root_n_jet	
Entries	941276
Mean	2.735
Std Dev	1.03

Events

0.05
0.04
0.03
0.02
0.01
0

2

4

6

8

10

12

14

n_jet

2

4

6

8

10

12

14

pair_isOSleptau

Events

0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

pair_isOSleptau

diboson.root_pair_isOSleptau	
Entries	941276
Mean	0.6174
Std Dev	0.486

pair_isOStaus

Events

0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

pair_isOStaus

diboson.root_pair_isOStaus	
Entries	941276
Mean	0.6841
Std Dev	0.4649

pass_DTT

Events

0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

1

pass_DTT

diboson.root_pass_DTT	
Entries	941276
Mean	0.09623
Std Dev	0.2949

pass_LTT

diboson.root_pass_LTT	
Entries	941276
Mean	0.6627
Std Dev	0.4728

Events

0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

1

pass_LTT

pass_SLT

diboson.root_pass_SLT	
Entries	941276
Mean	0.8085
Std Dev	0.3935

Events

0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

pass_SLT

1

pass_STT

diboson.root_pass_STT	
Entries	941276
Mean	0.06913
Std Dev	0.2537

Events

0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

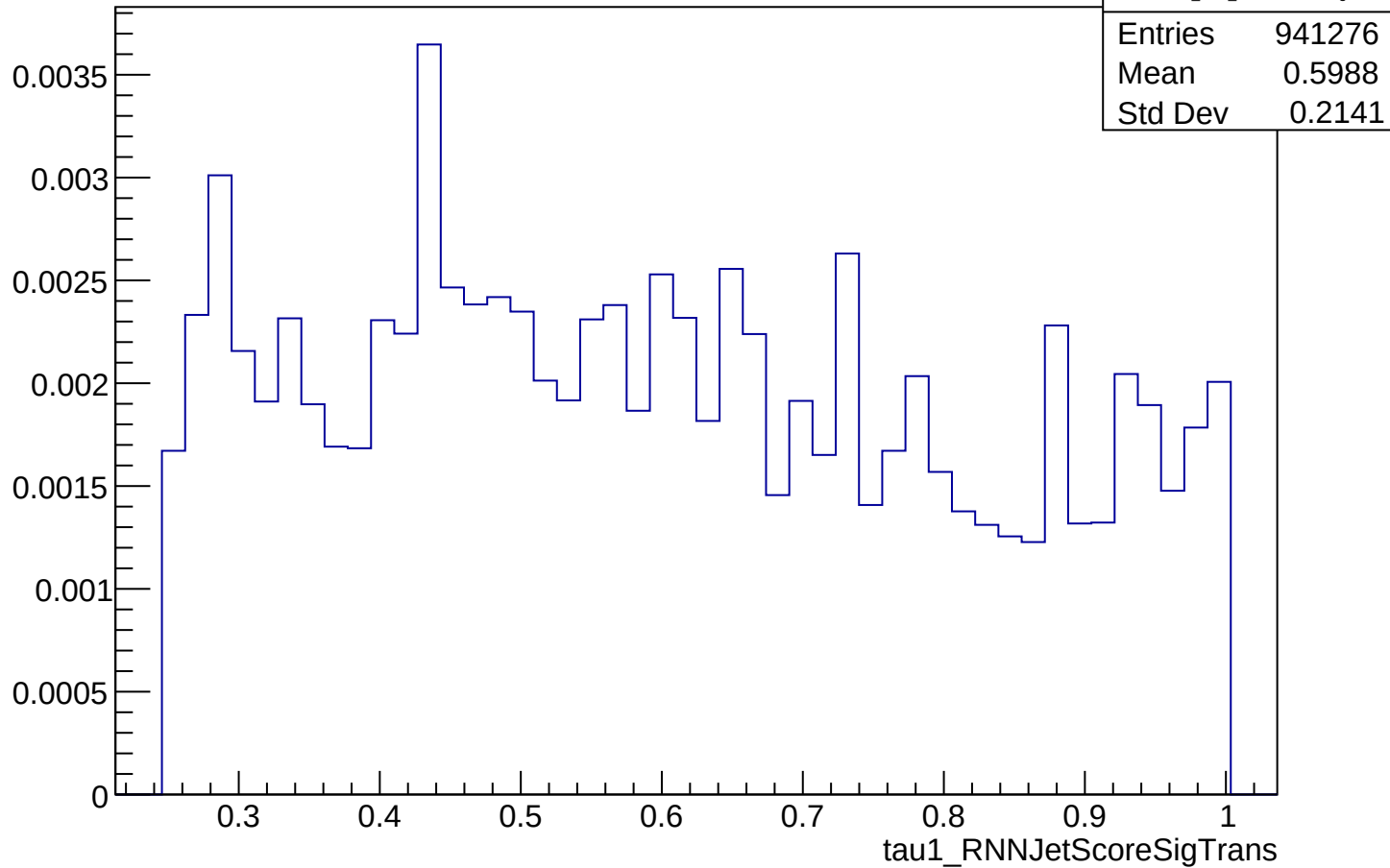
0.8

1

pass_STT

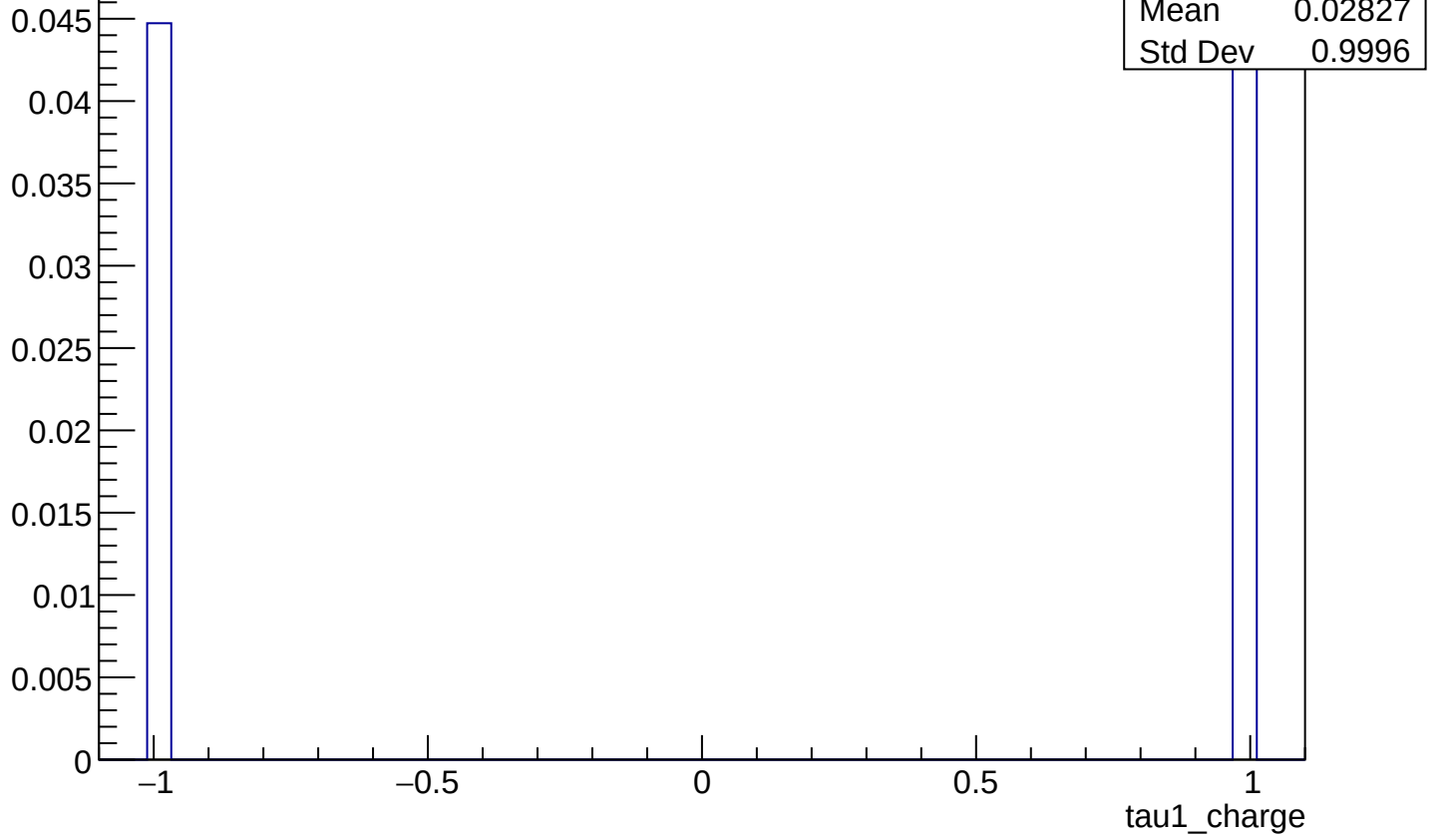
tau1_RNNJetScoreSigTrans

Events



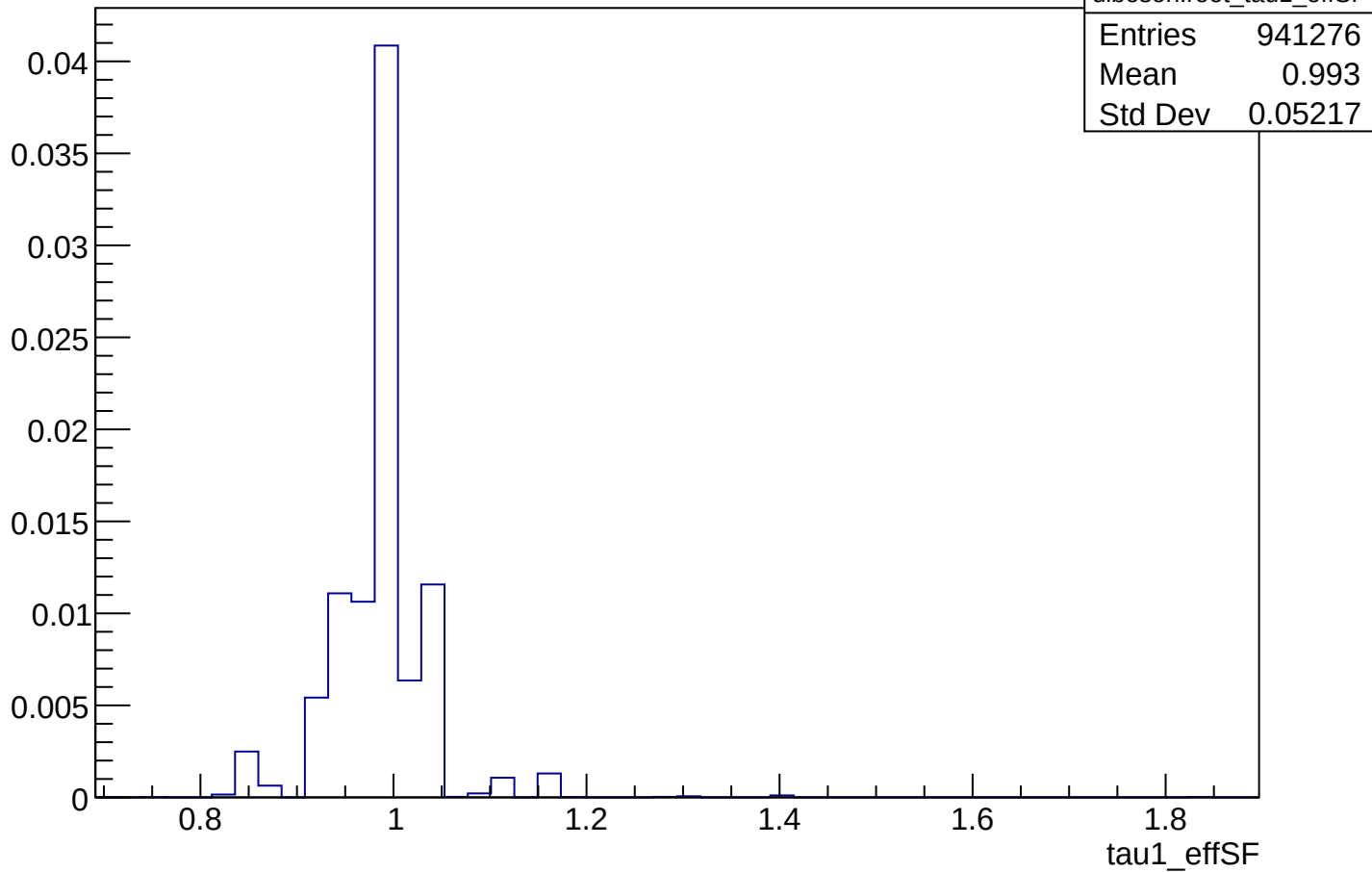
tau1_charge

Events

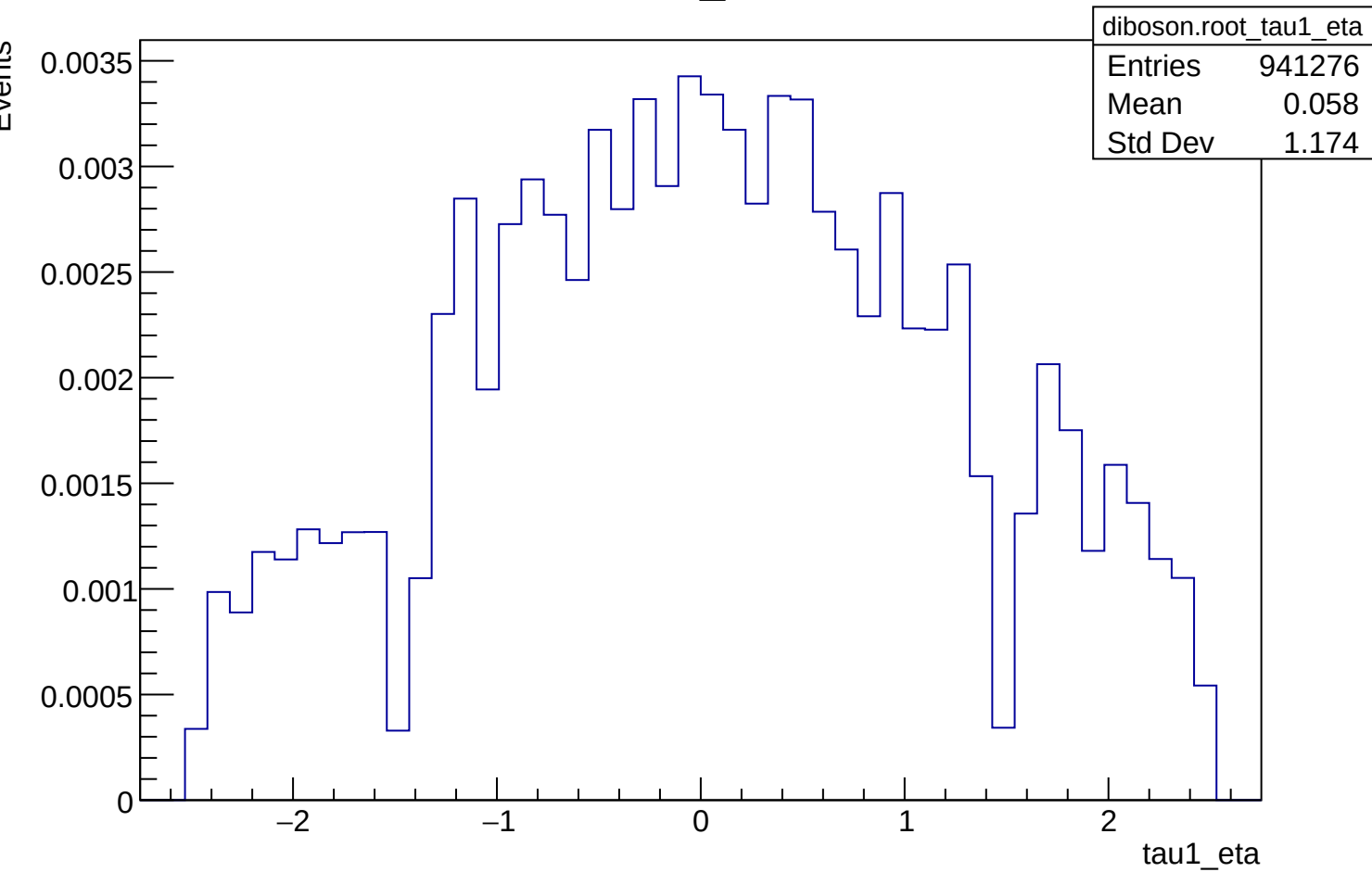


tau1_effSF

Events



tau1_eta



tau1_nprong

Events

0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

1

1.5

2

2.5

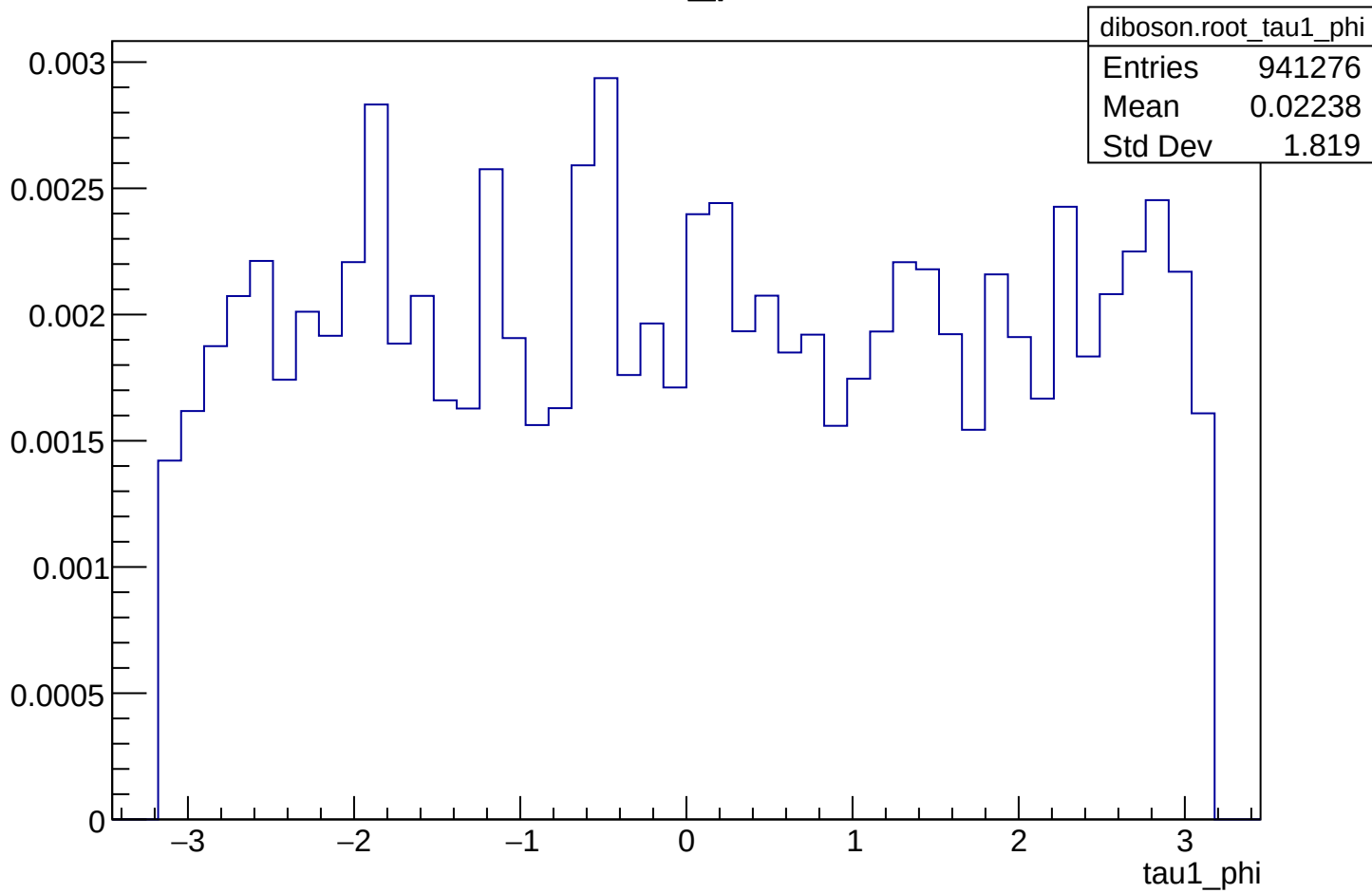
3

tau1_nprong

diboson.root_tau1_nprong	
Entries	941276
Mean	1.452
Std Dev	0.8362

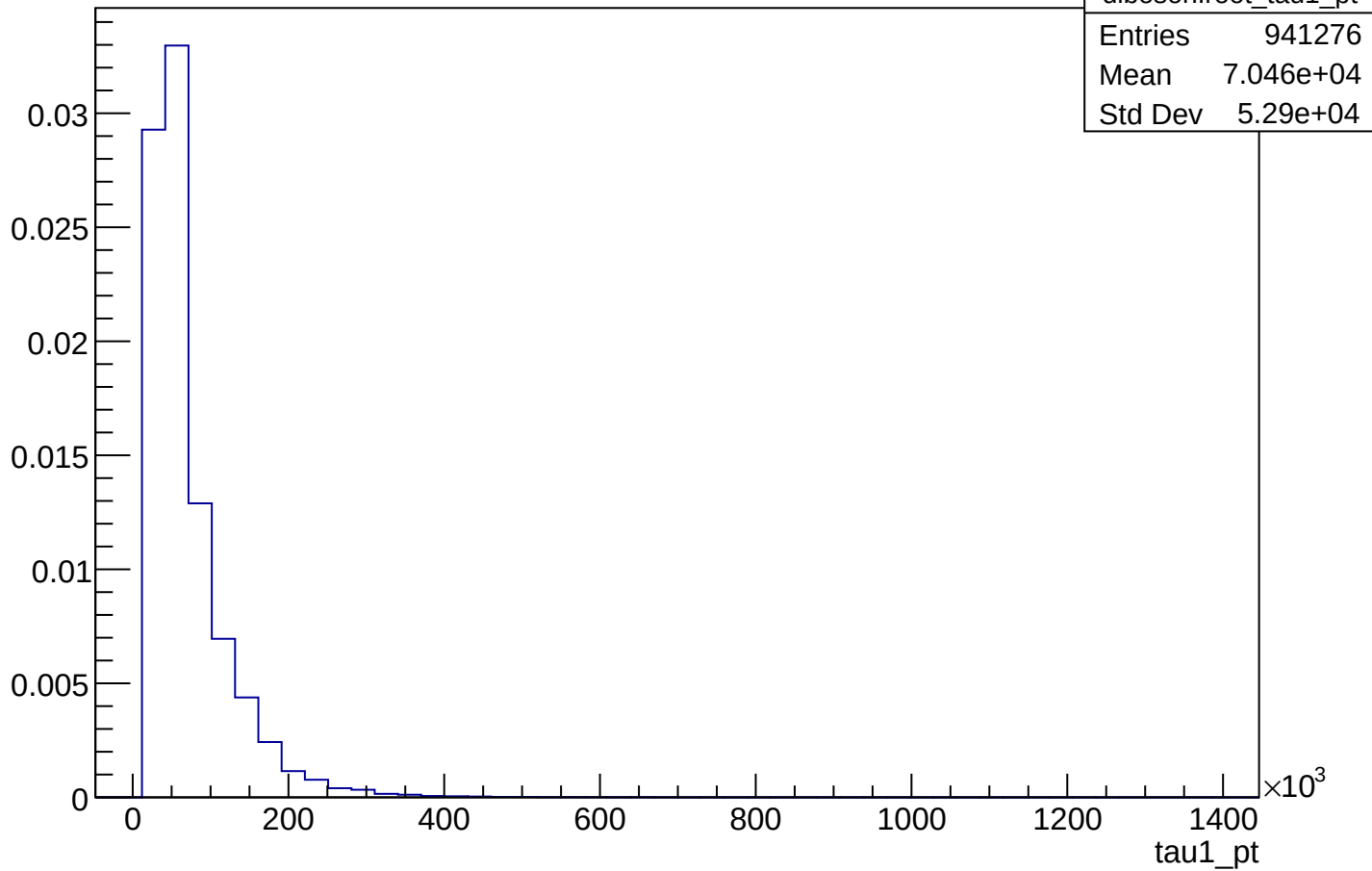
tau1_phi

Events



tau1_pt

Events



tau1_tauTruthJetLabel

Events

0.05
0.04
0.03
0.02
0.01
0

diboson.root_tau1_tauTruthJetLabel	
Entries	941276
Mean	-3.468
Std Dev	21.66

-100

-80

-60

-40

-20

tau1_tauTruthJetLabel

0

20

tau1_truthType

diboson.root_tau1_truthType	
Entries	941276
Mean	2.207
Std Dev	1.814

Events

0.06
0.05
0.04
0.03
0.02
0.01
0

0

1

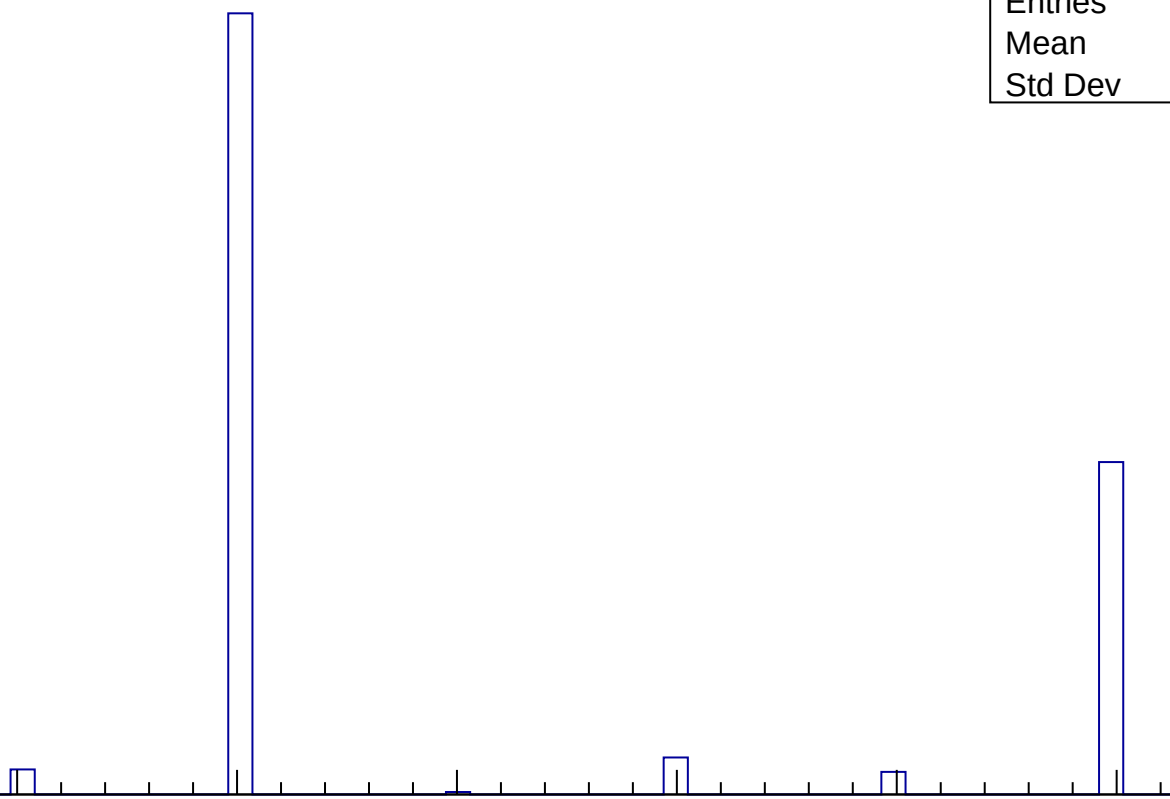
2

3

4

5

tau1_truthType



tau1_truth_IsHadronicTau

Events

0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

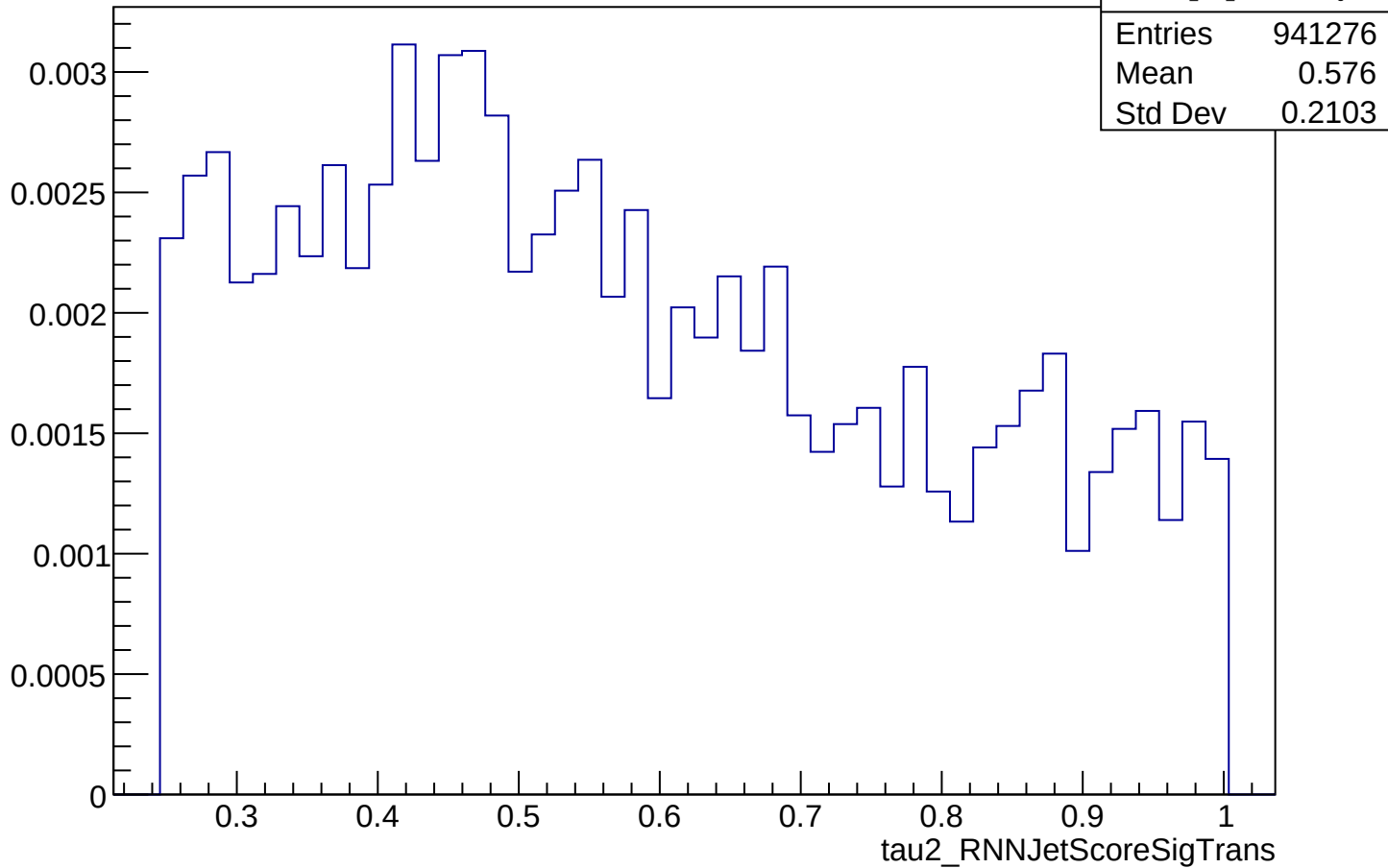
1

tau1_truth_IsHadronicTau

diboson.root_tau1_truth_IsHadronicTau	
Entries	941276
Mean	0.6506
Std Dev	0.4768

tau2_RNNJetScoreSigTrans

Events



tau2_baseline_RNNTight

diboson.root_tau2_baseline_RNNTight	
Entries	941276
Mean	0.664
Std Dev	0.4723

Events

0.06
0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

0.8

1

tau2_baseline_RNNTight

tau2_charge

diboson.root_tau2_charge	
Entries	941276
Mean	0.03773
Std Dev	0.9993

Events

0.05
0.04
0.03
0.02
0.01
0

-1

-0.5

0

0.5

1

tau2_charge

tau2_effSF

Events

0.05
0.04
0.03
0.02
0.01
0

0.8

1.0

1.2

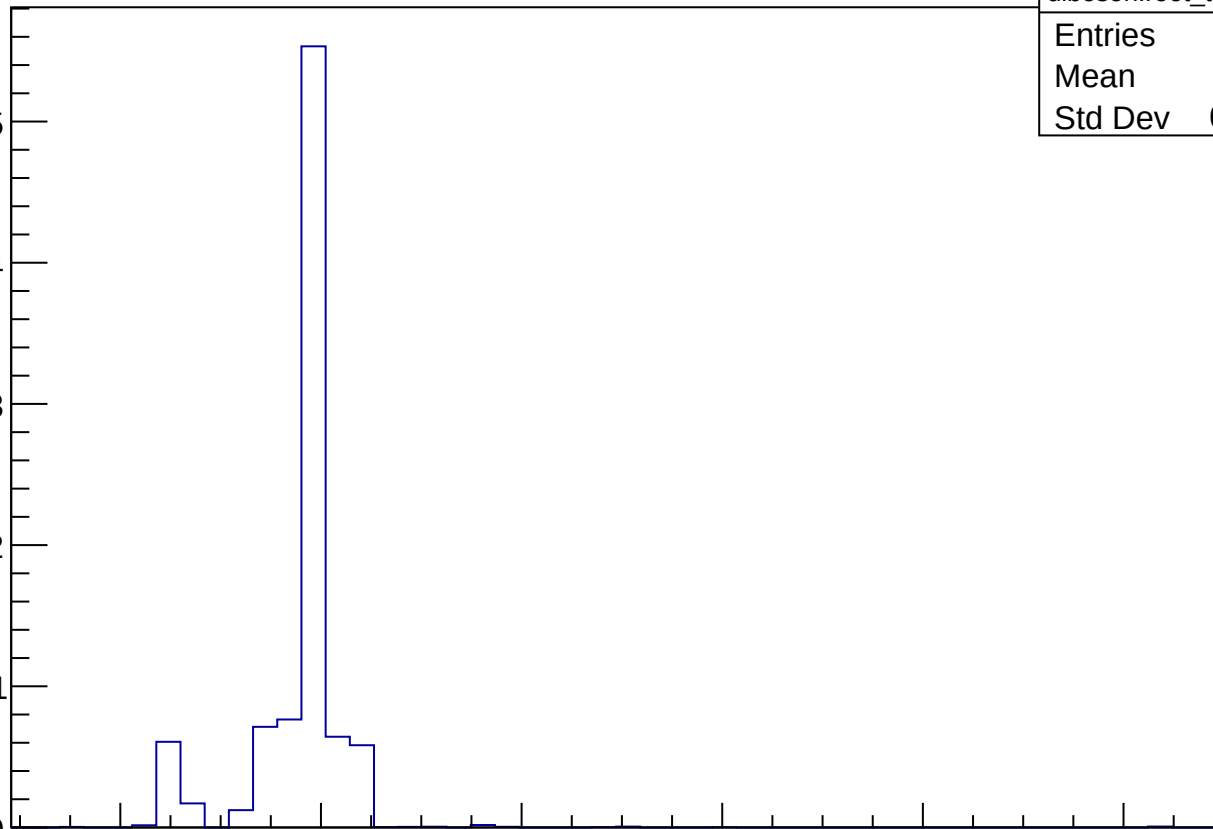
1.4

1.6

1.8

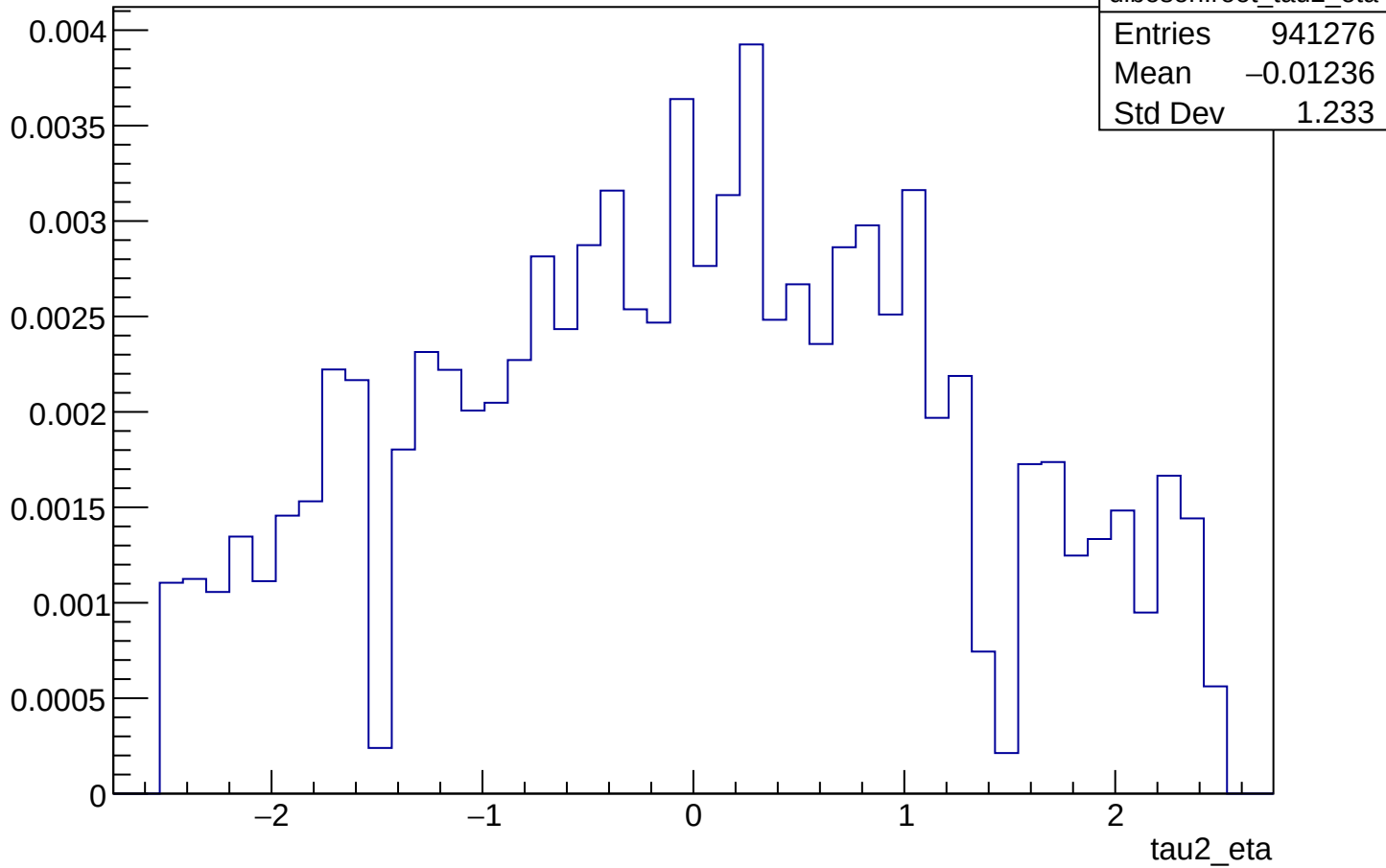
tau2_effSF

diboson.root_tau2_effSF	
Entries	941276
Mean	0.9856
Std Dev	0.05407



tau2_eta

Events



tau2_nprong

Events

0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

1

1.5

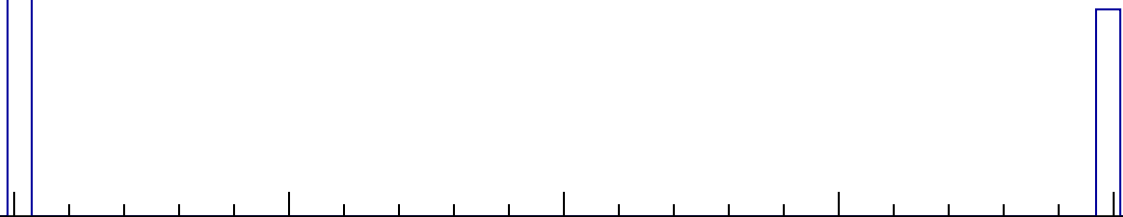
2

2.5

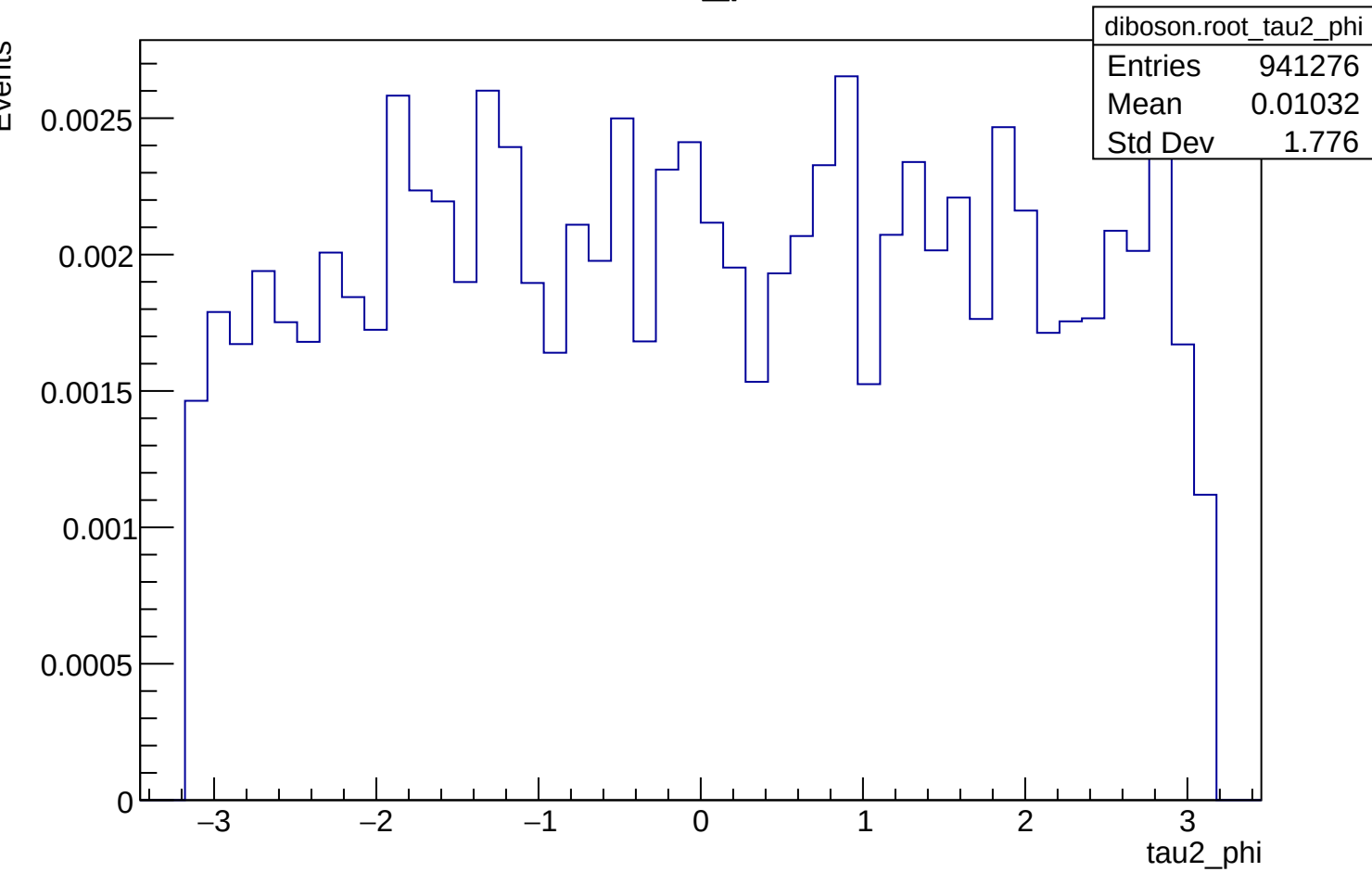
3

tau2_nprong

diboson.root_tau2_nprong	
Entries	941276
Mean	1.419
Std Dev	0.814



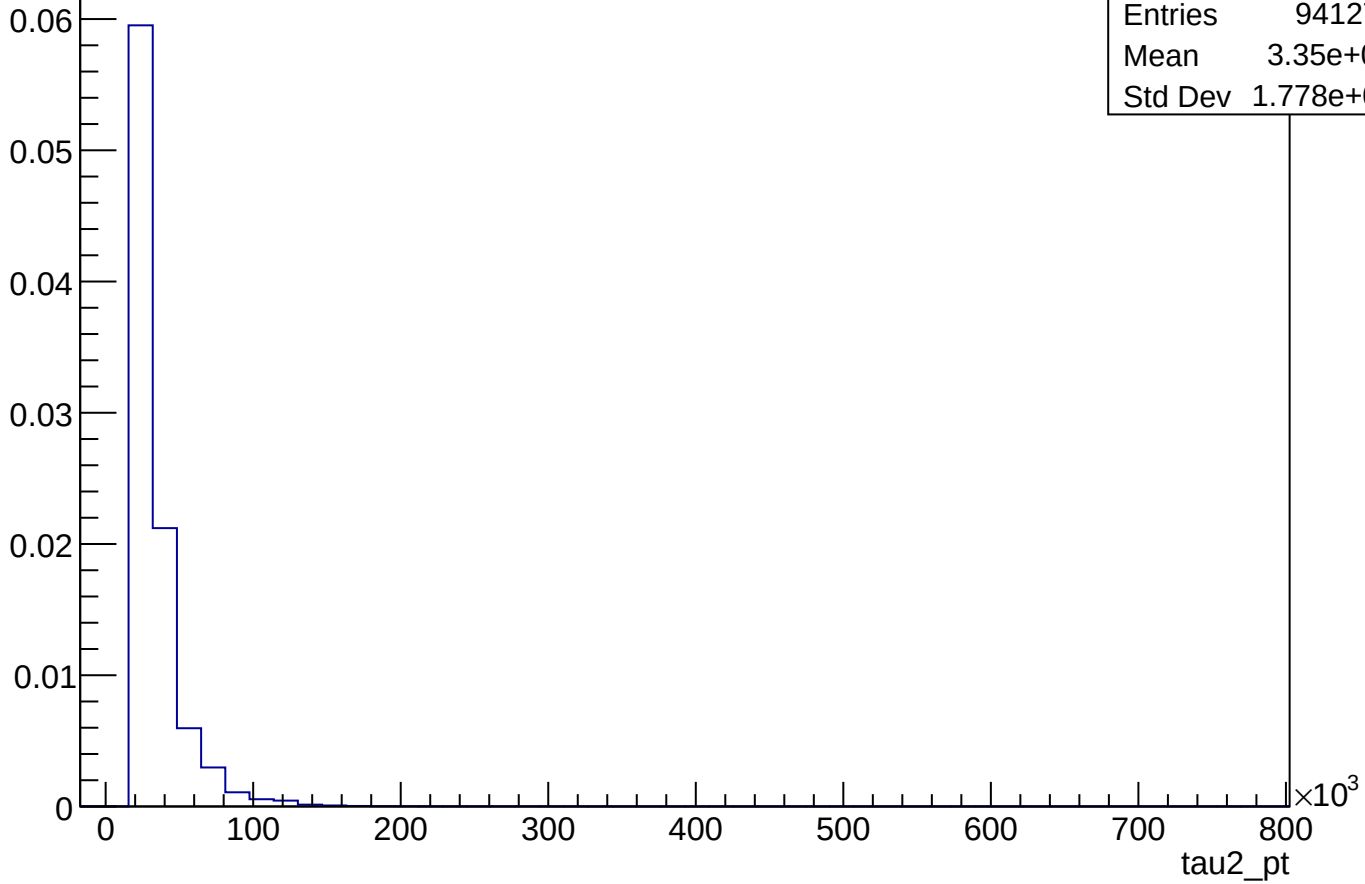
tau2_phi



tau2_pt

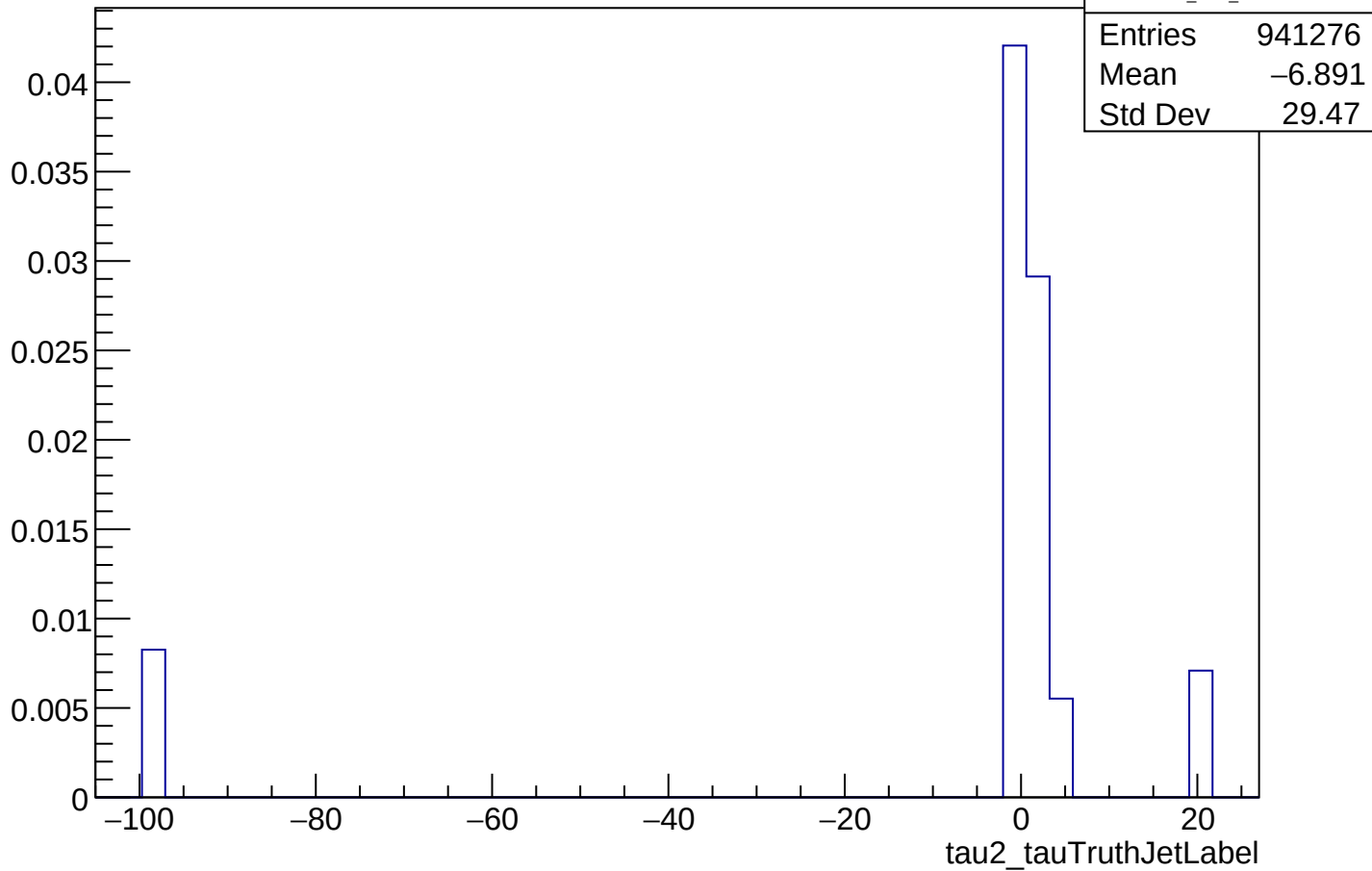
diboson.root_tau2_pt	
Entries	941276
Mean	3.35e+04
Std Dev	1.778e+04

Events

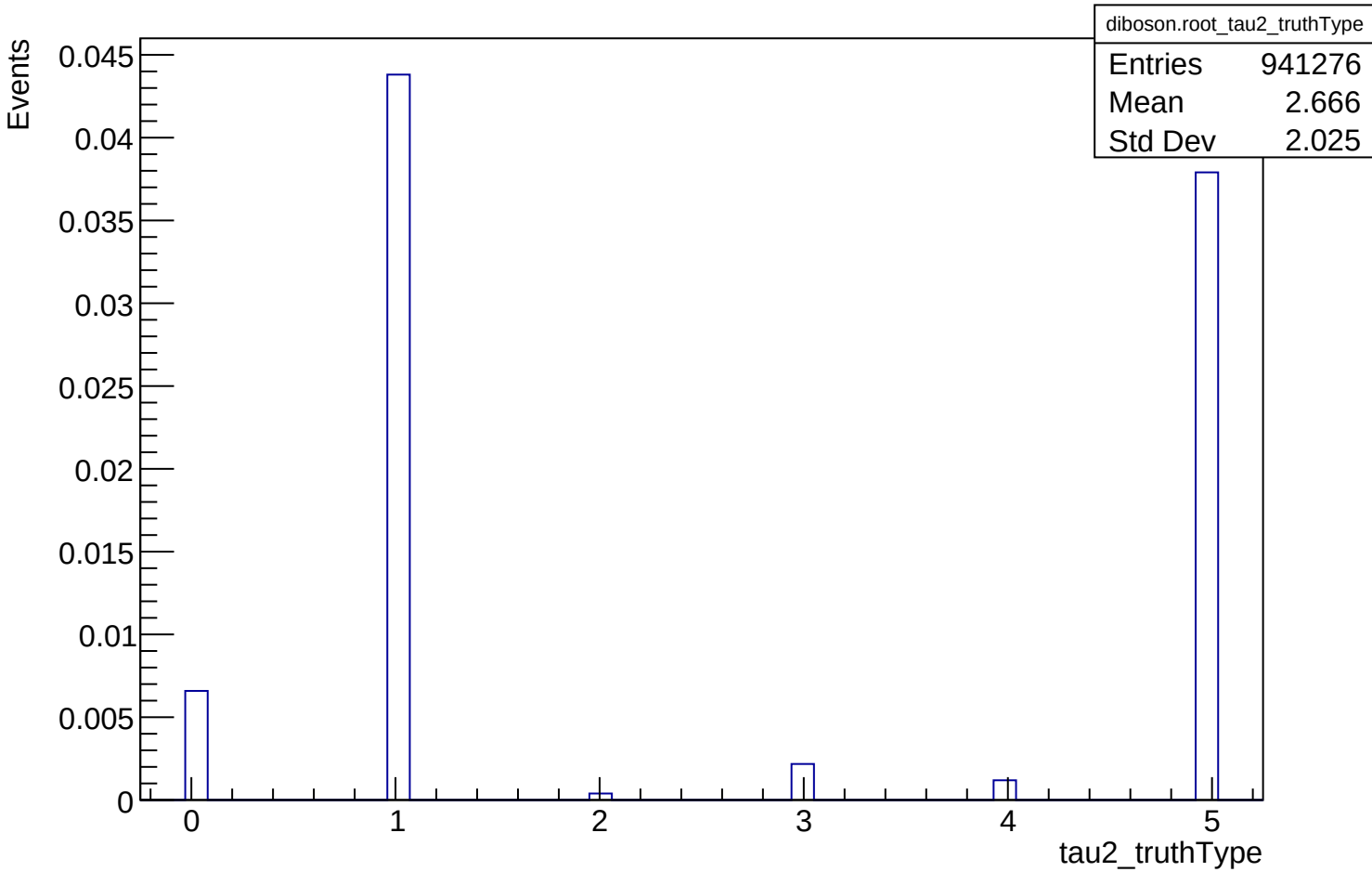


tau2_tauTruthJetLabel

Events



tau2_truthType



tau2_truth_IsHadronicTau

Events

0.05
0.04
0.03
0.02
0.01
0

0

0.2

0.4

0.6

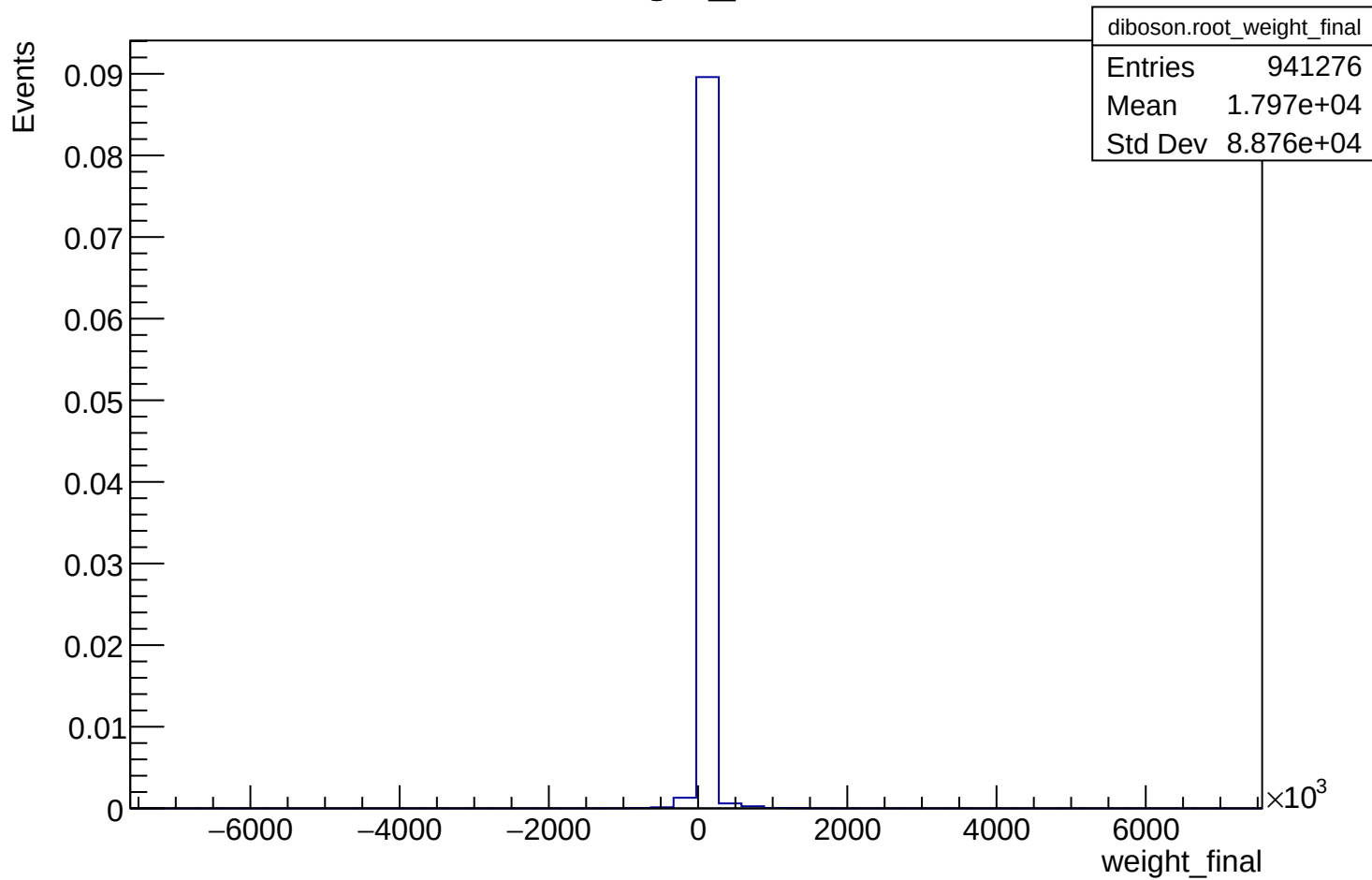
0.8

1

tau2_truth_IsHadronicTau

diboson.root_tau2_truth_IsHadronicTau	
Entries	941276
Mean	0.4759
Std Dev	0.4994

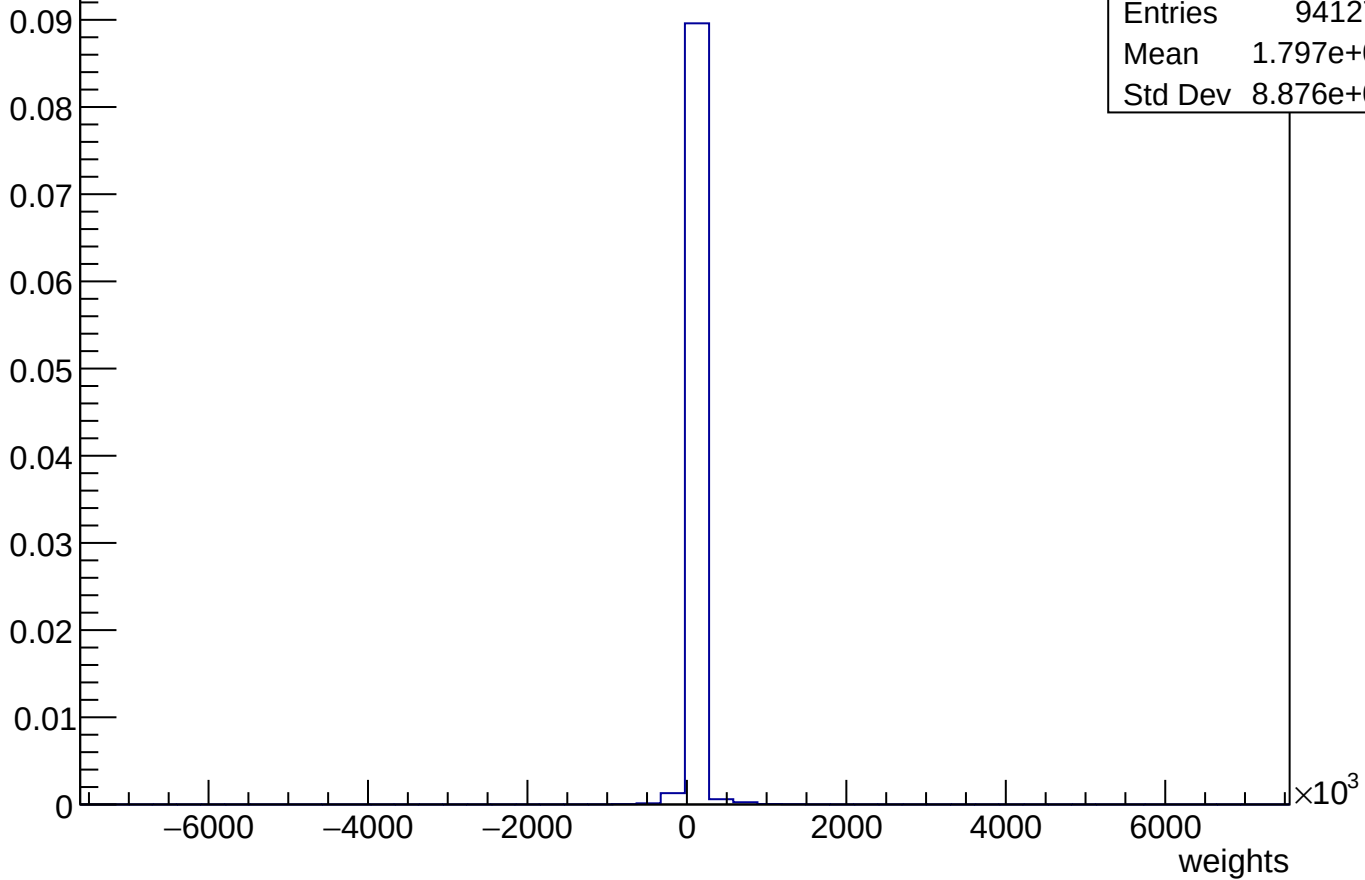
weight_final



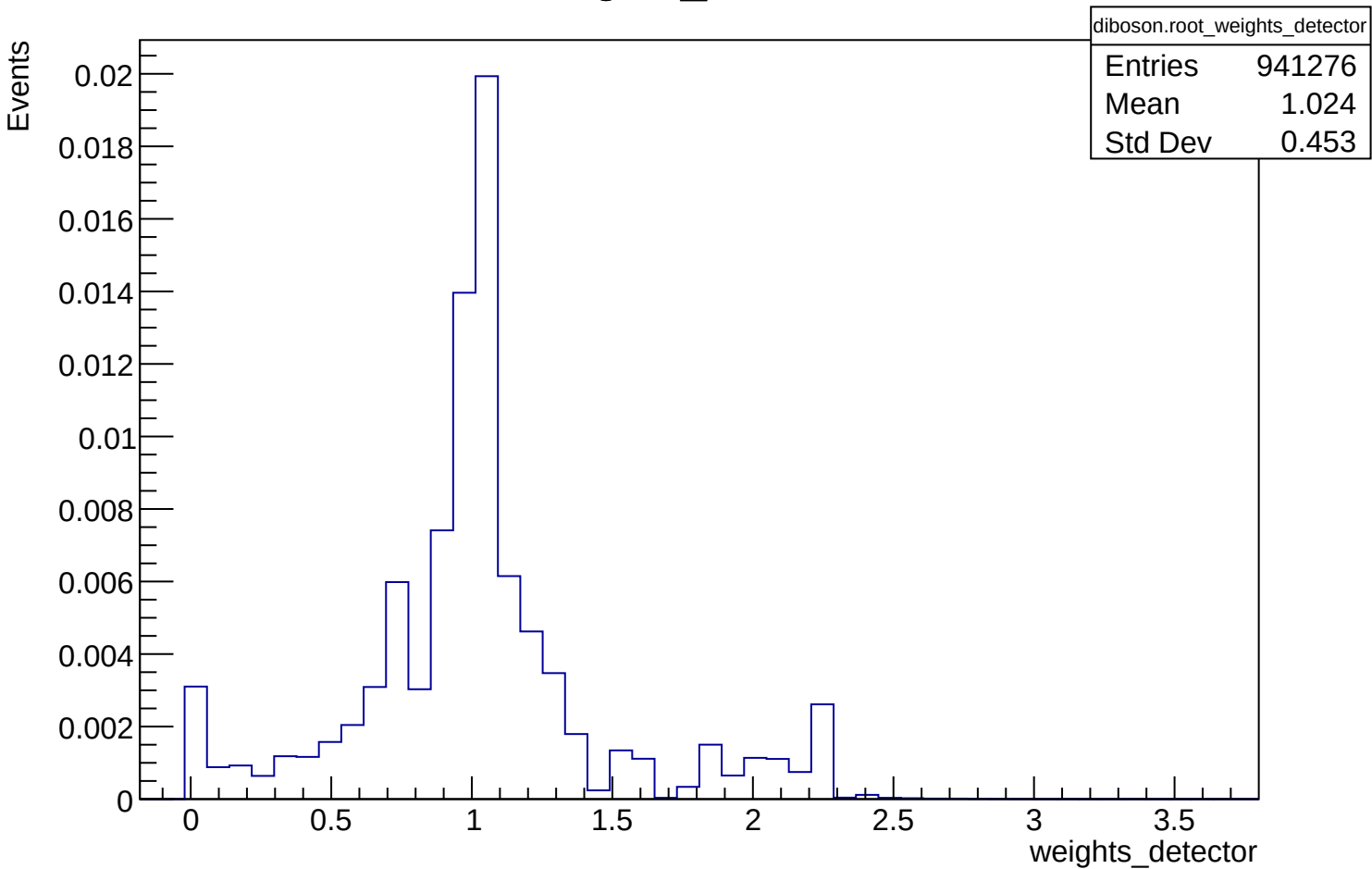
weights

diboson.root_weights	
Entries	941276
Mean	1.797e+04
Std Dev	8.876e+04

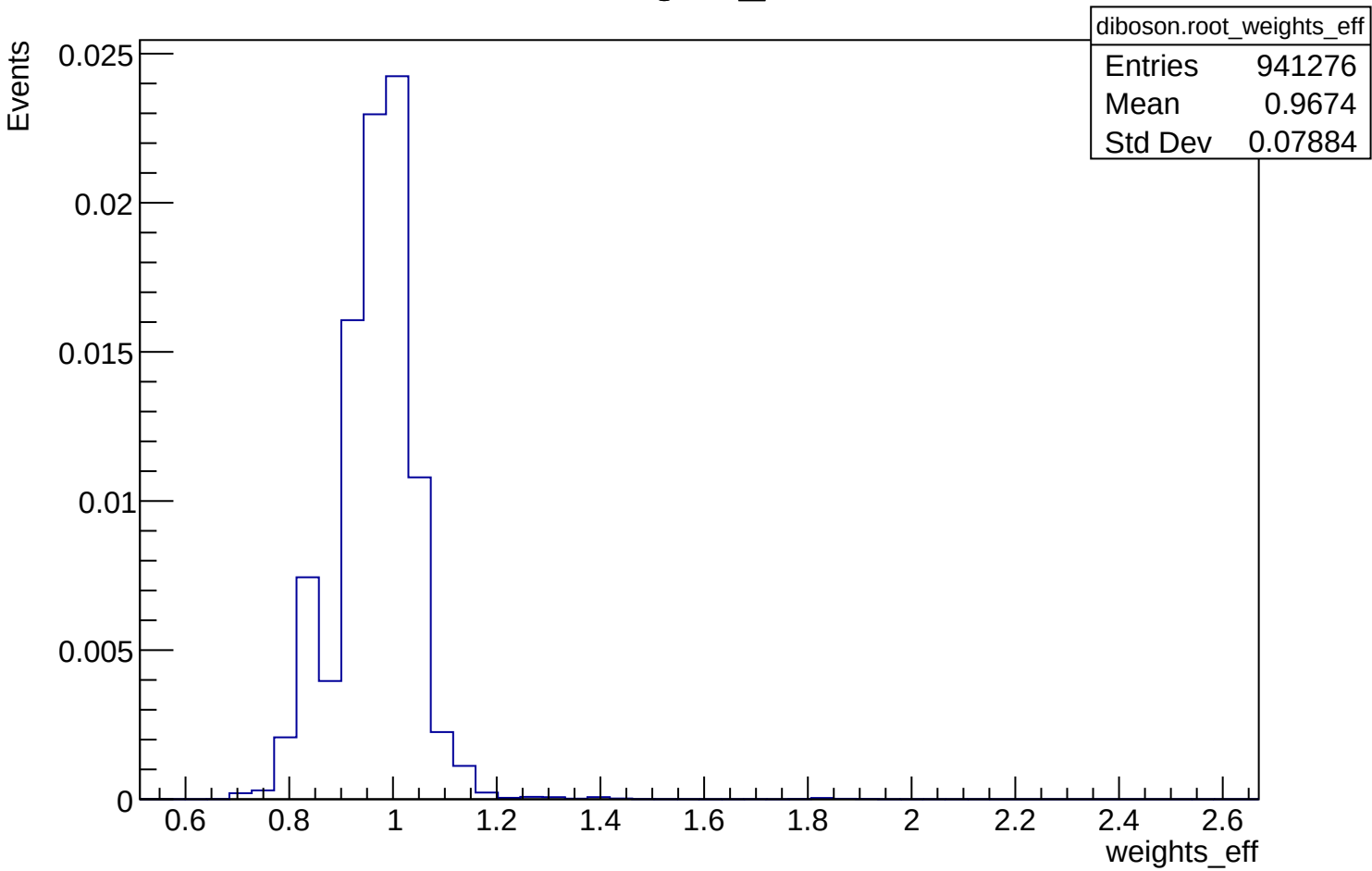
Events



weights_detector



weights_eff



weights_mc

Events

0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0

-4000

-2000

0

2000

4000

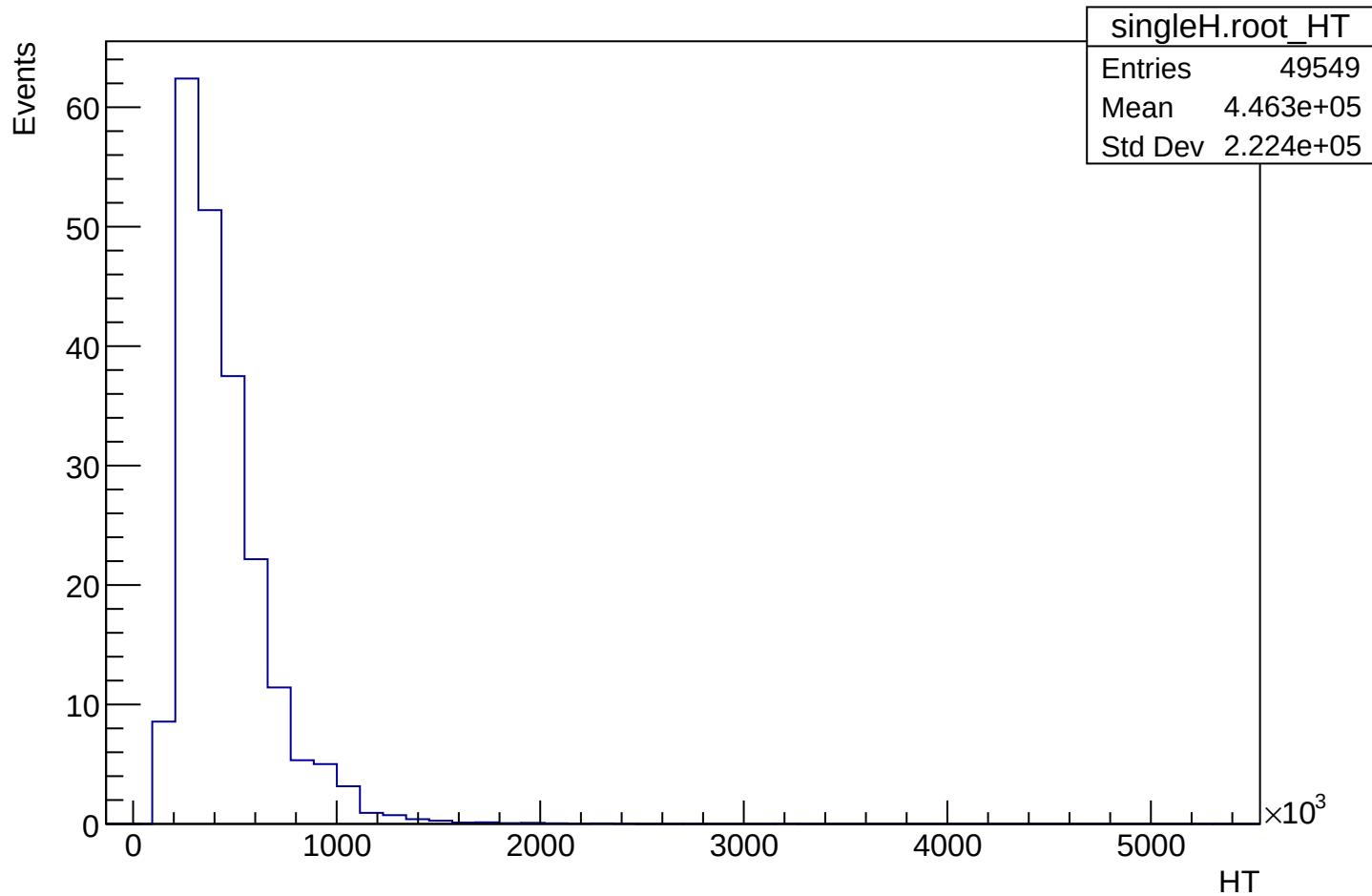
6000

$\times 10^3$

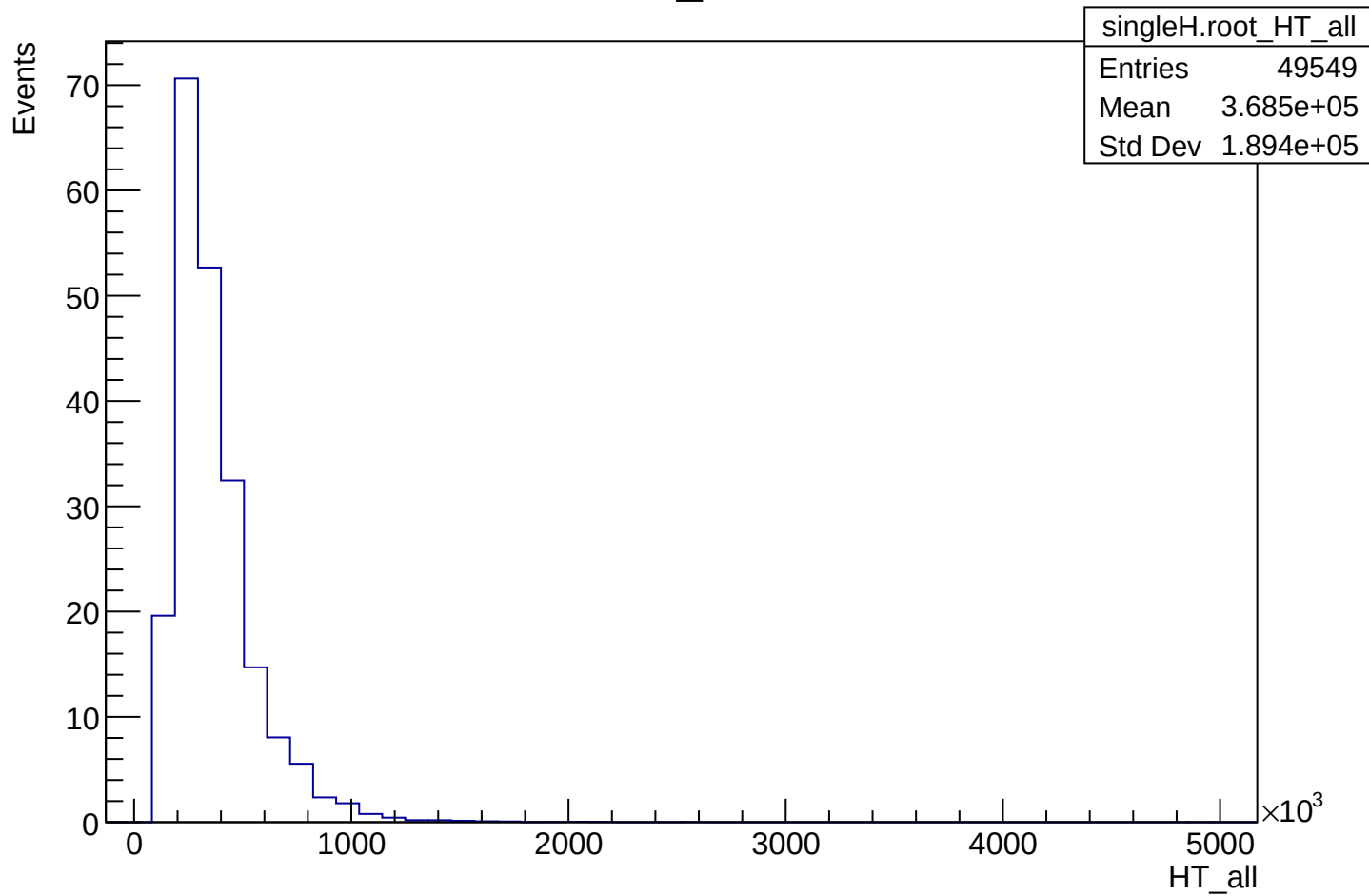
diboson.root_weights_mc	
Entries	941276
Mean	1.865e+04
Std Dev	8.267e+04

weights_mc

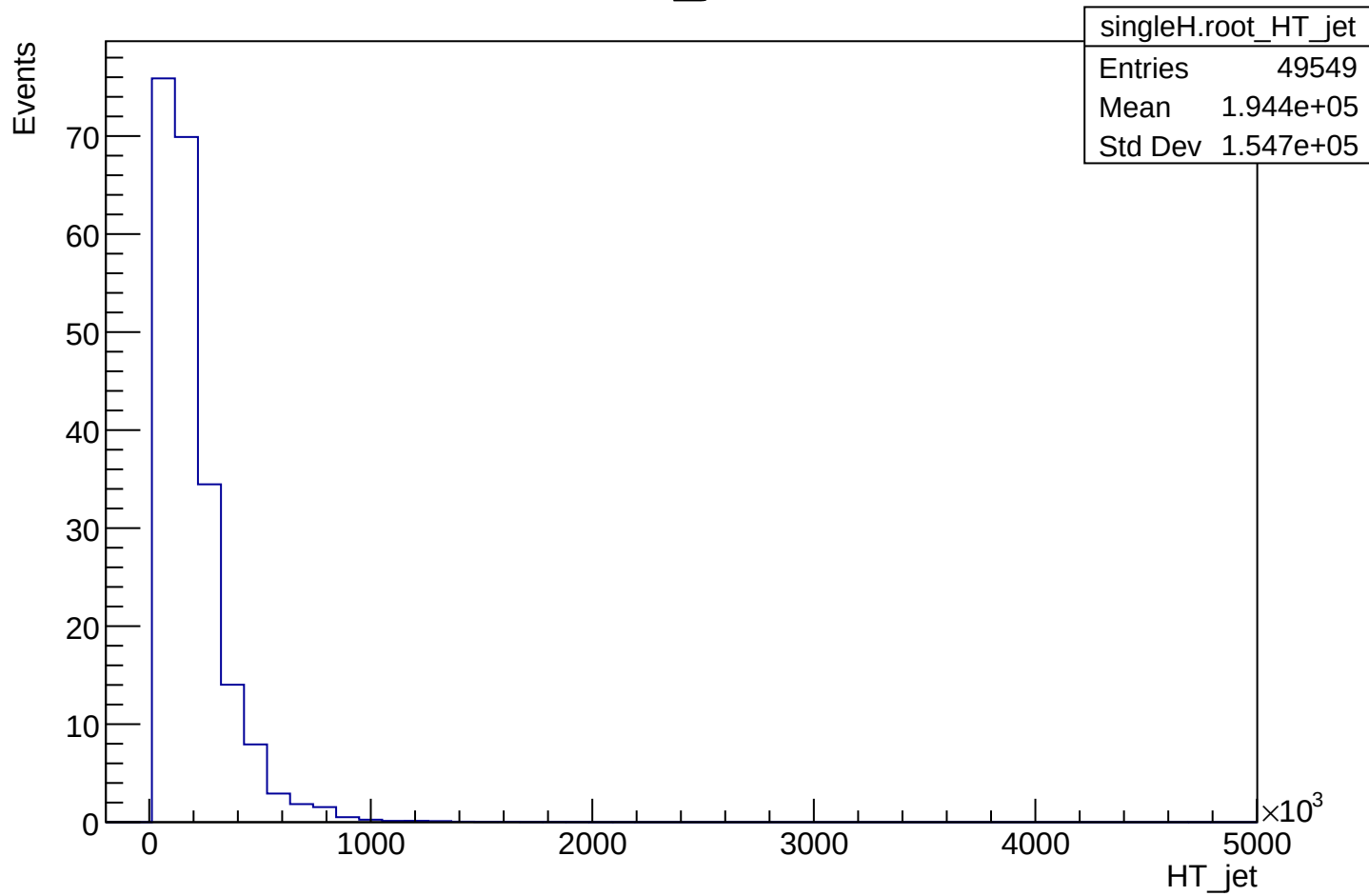
HT



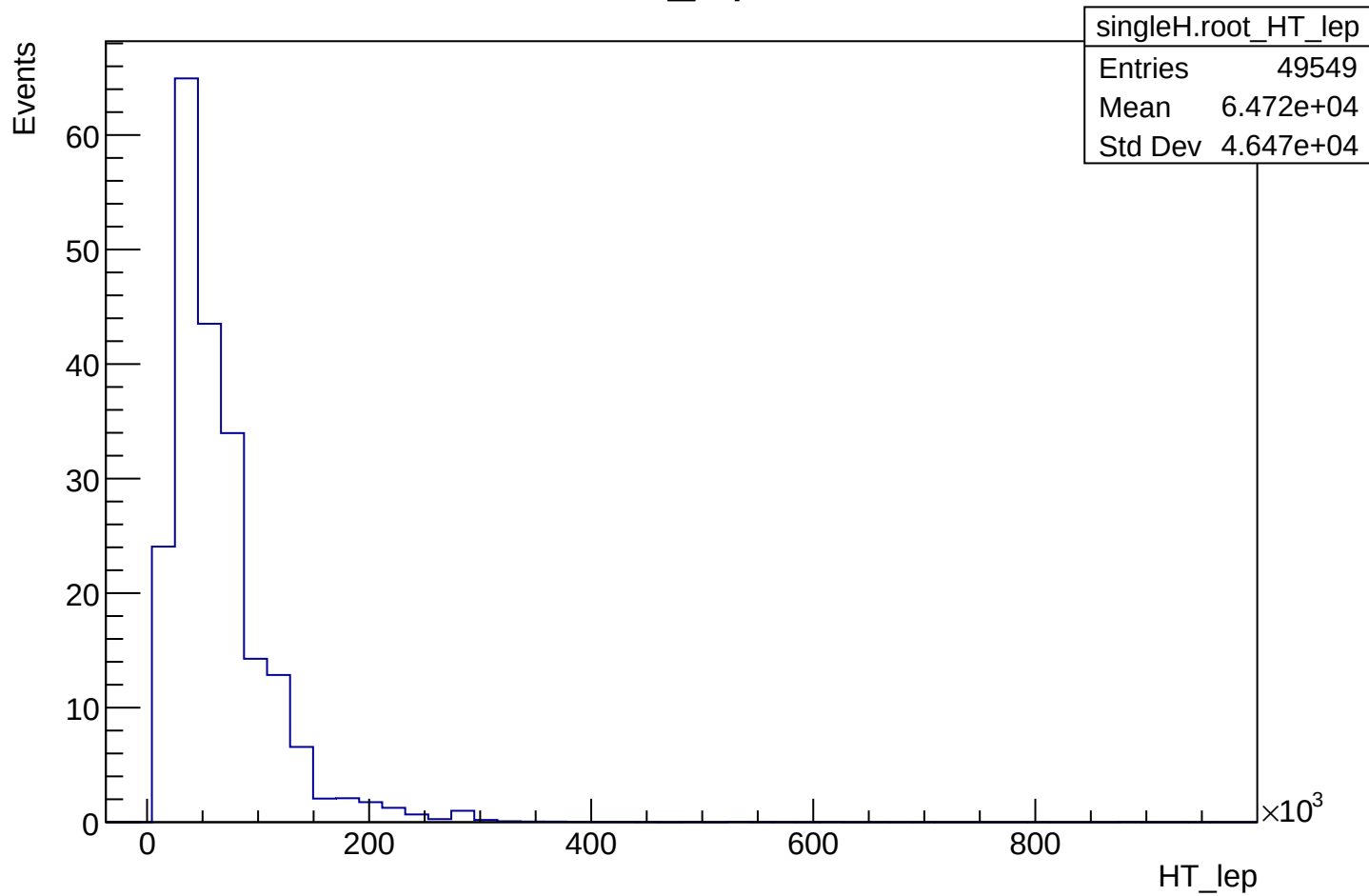
HT_all



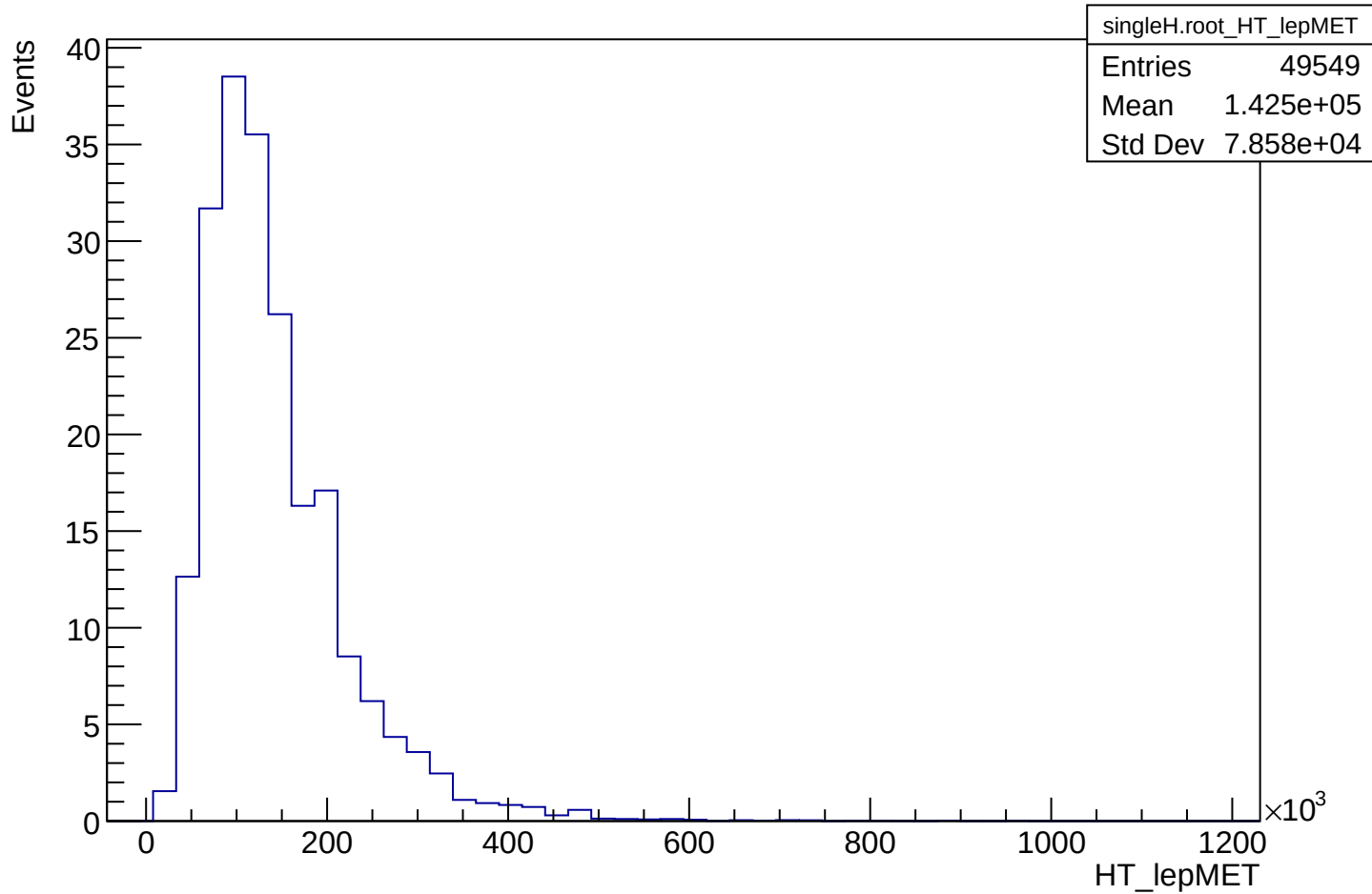
HT_jet



HT_lep



HT_lepMET



HT_tau

Events

35
30
25
20
15
10
5
0

100

200

300

400

500

600

700

HT_tau

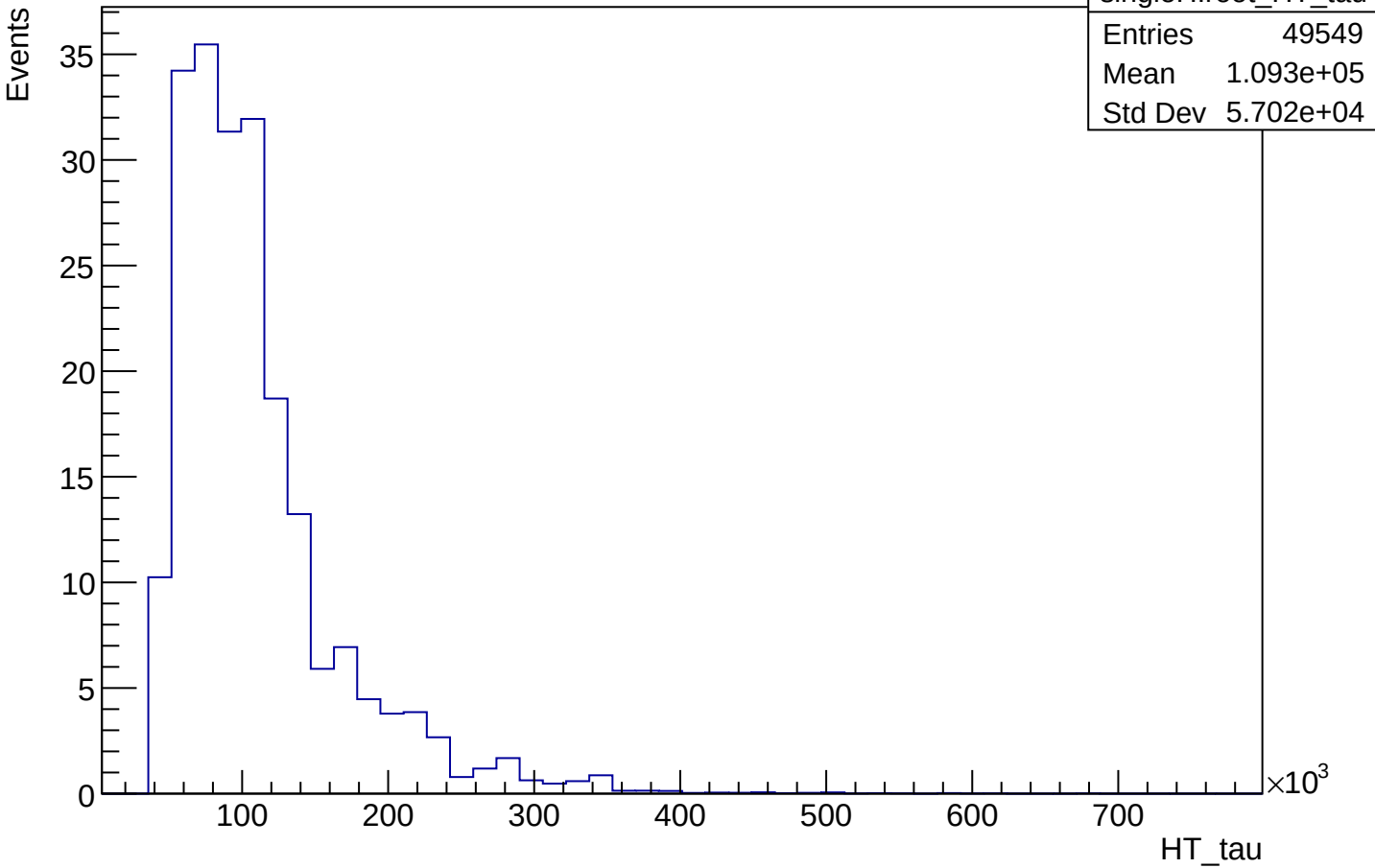
$\times 10^3$

singleH.root_HT_tau

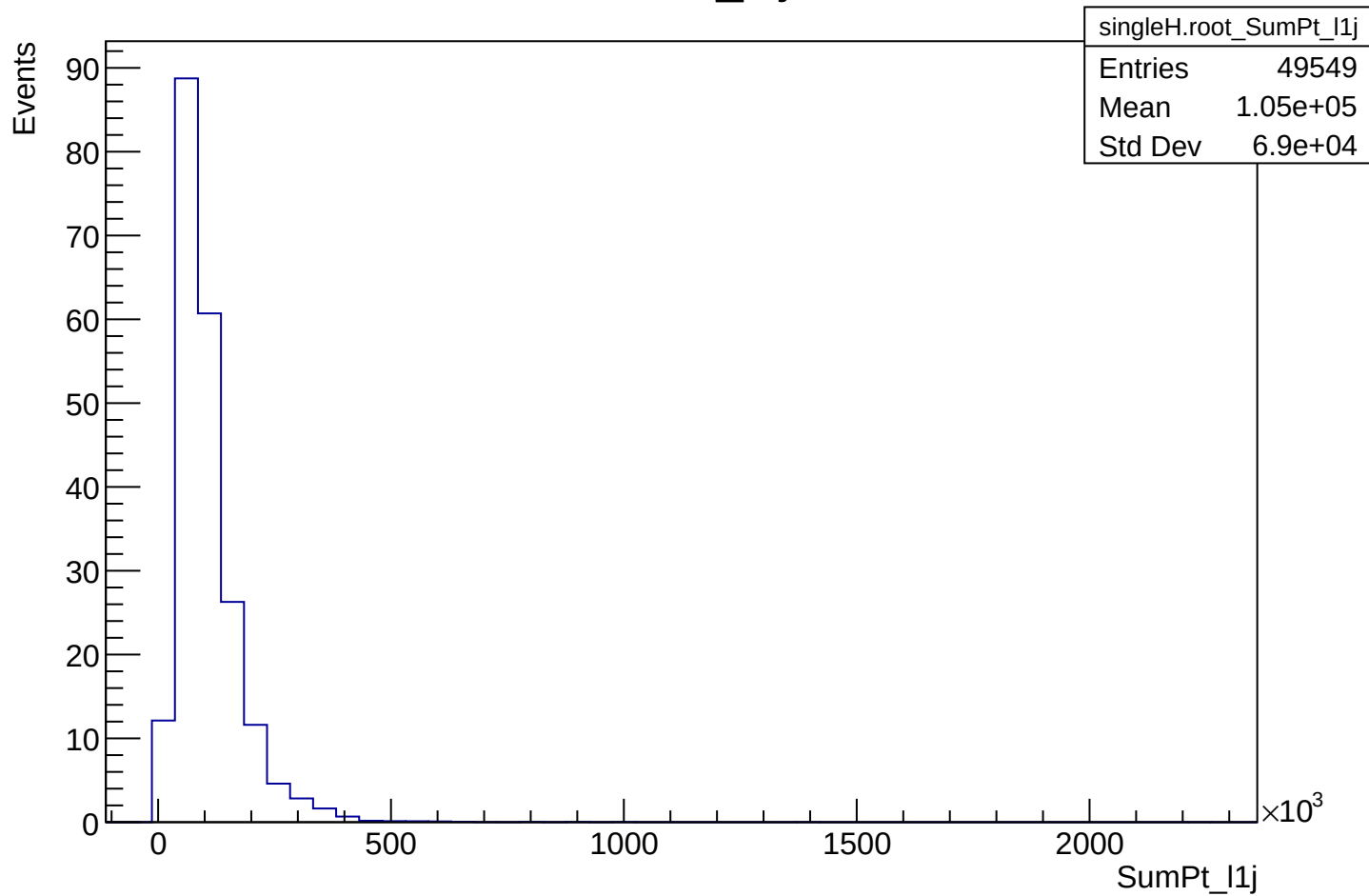
Entries 49549

Mean 1.093e+05

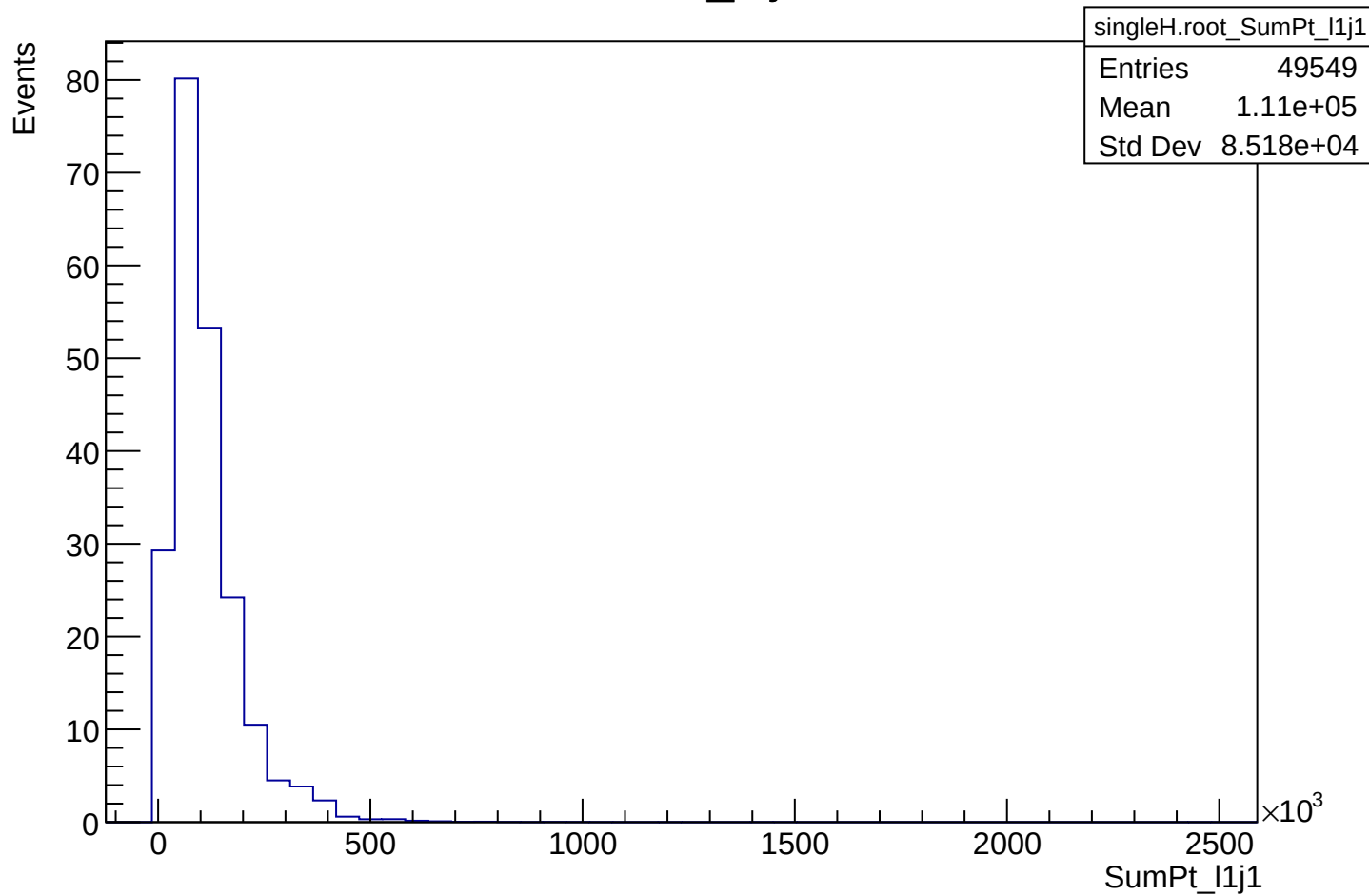
Std Dev 5.702e+04



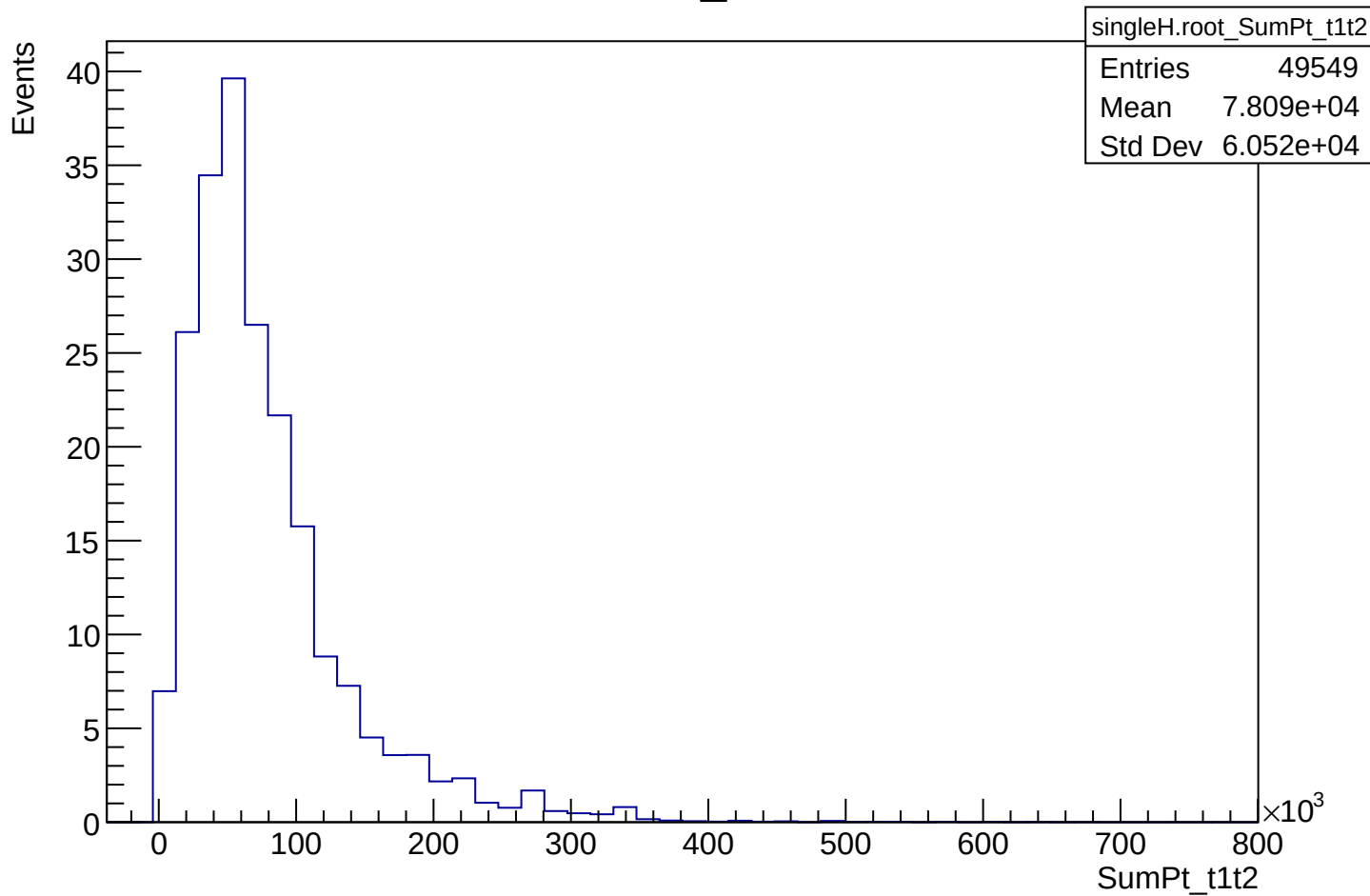
SumPt_l1j



SumPt_l1j1

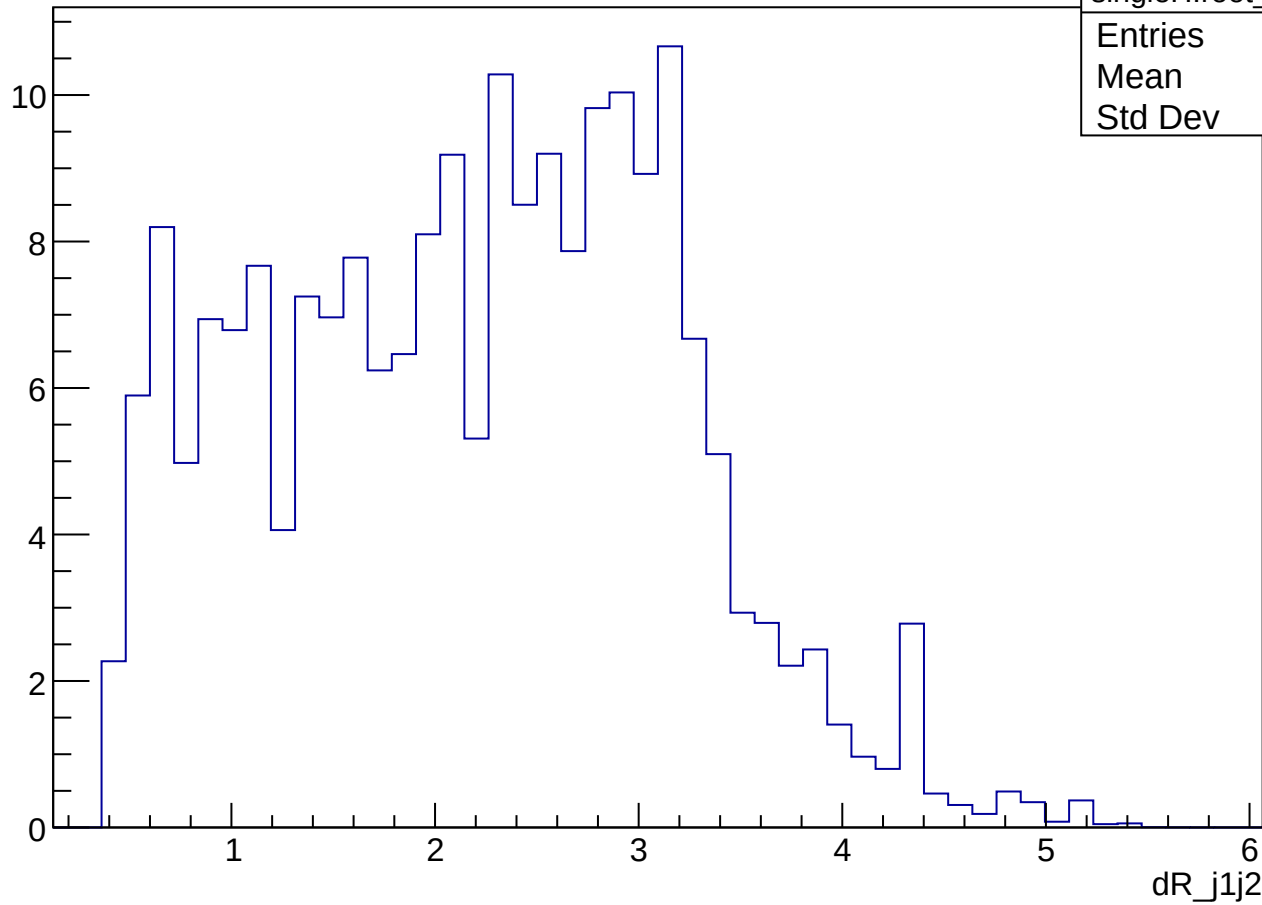


SumPt_t1t2



dR_j1j2

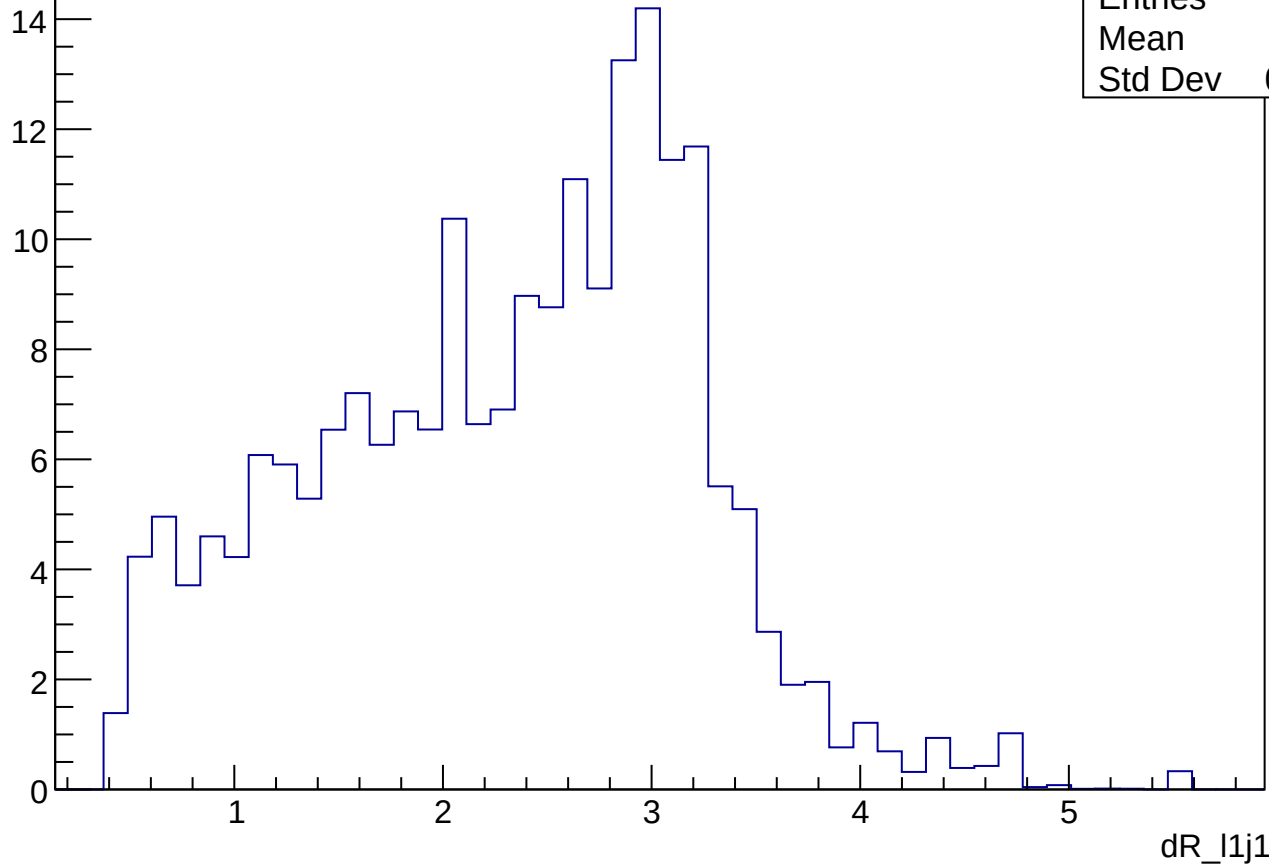
Events



singleH.root_dR_j1j2	
Entries	49549
Mean	2.206
Std Dev	0.9929

dR_l1j1

Events



singleH.root_dR_l1j1

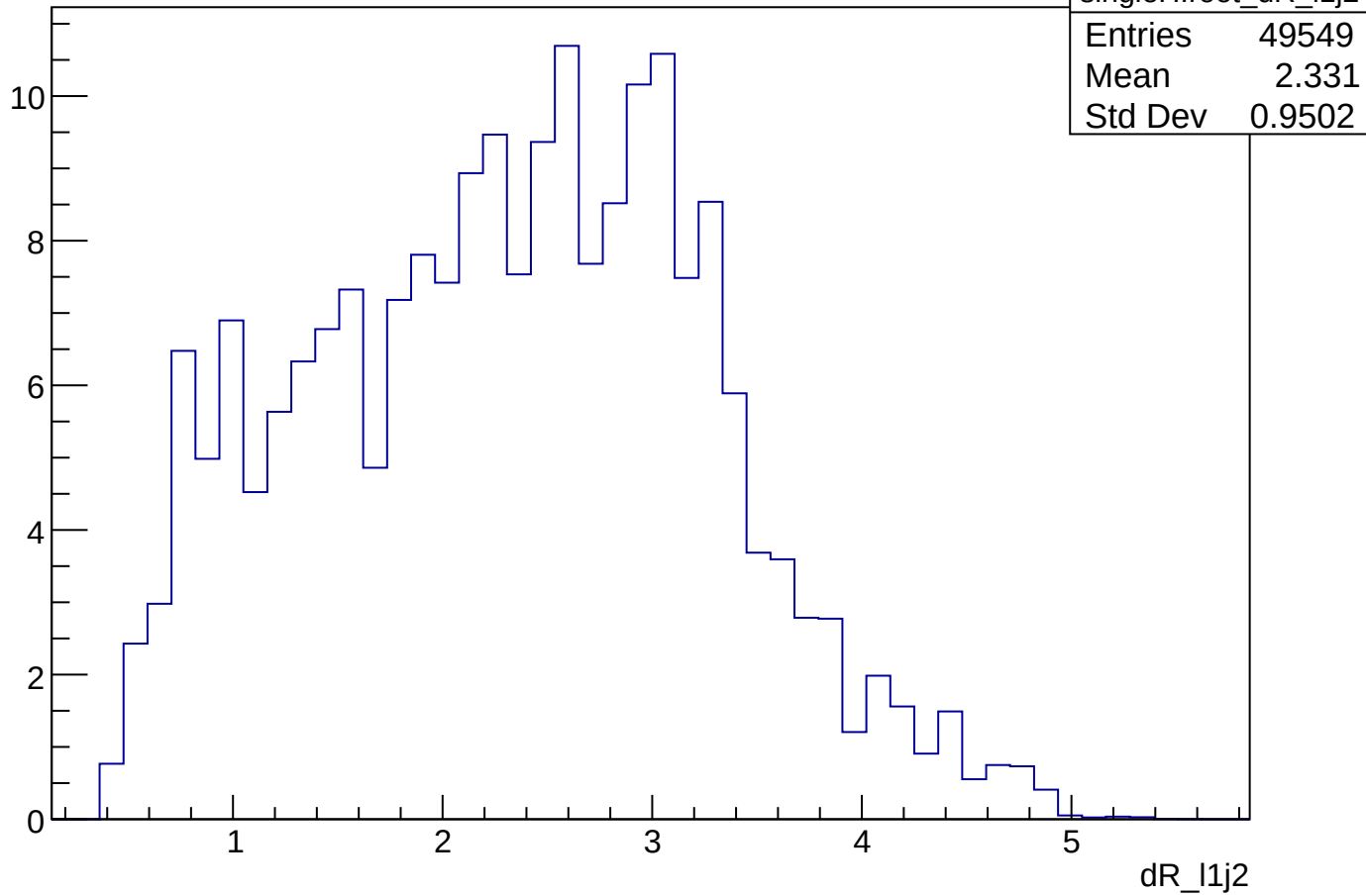
Entries 49549

Mean 2.318

Std Dev 0.9076

dR_l1j2

Events



dR_l1t

Events

12
10
8
6
4
2
0

1

2

3

4

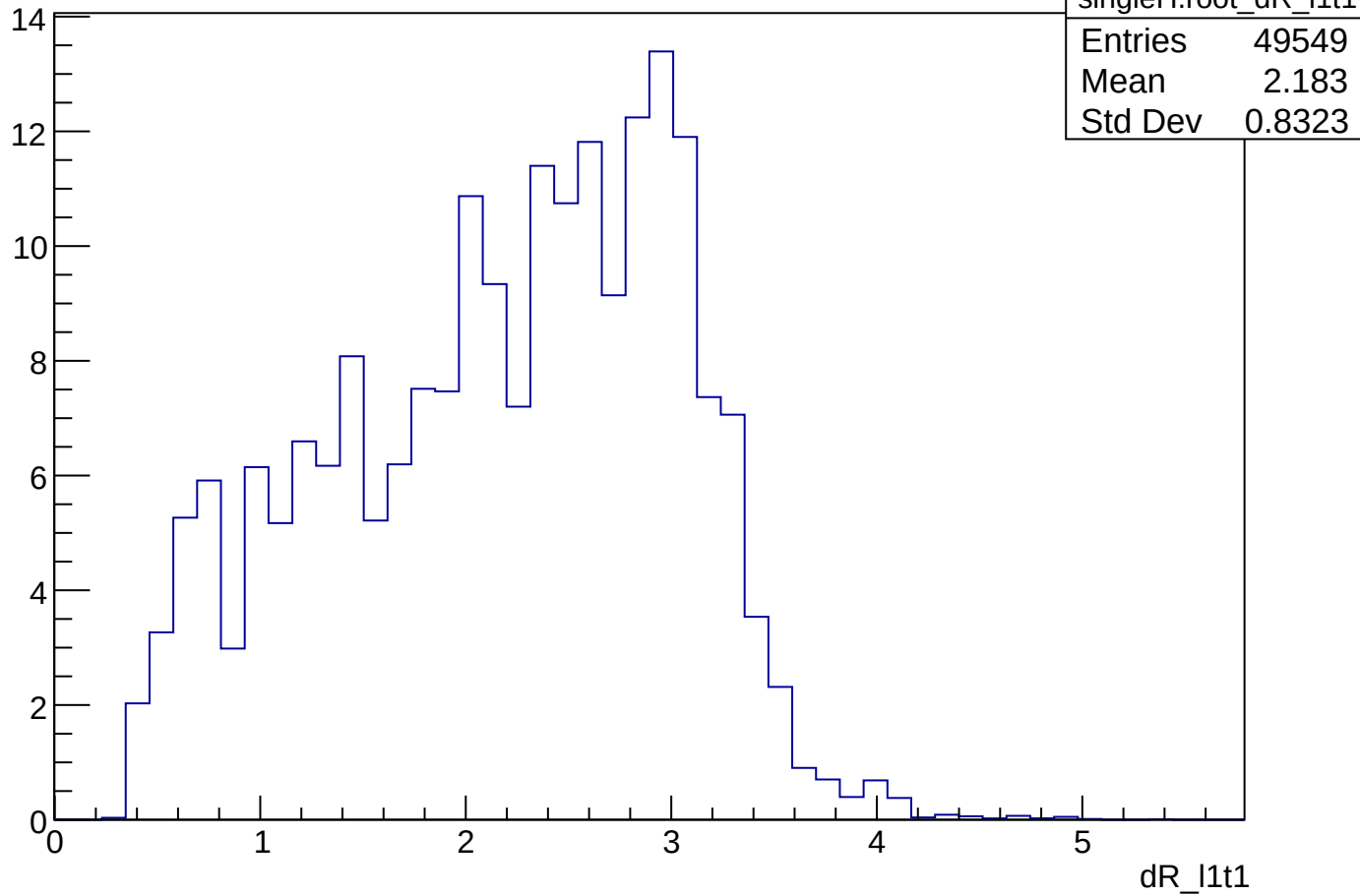
5

dR_l1t

singleH.root_dR_l1t	
Entries	49549
Mean	1.673
Std Dev	0.7092

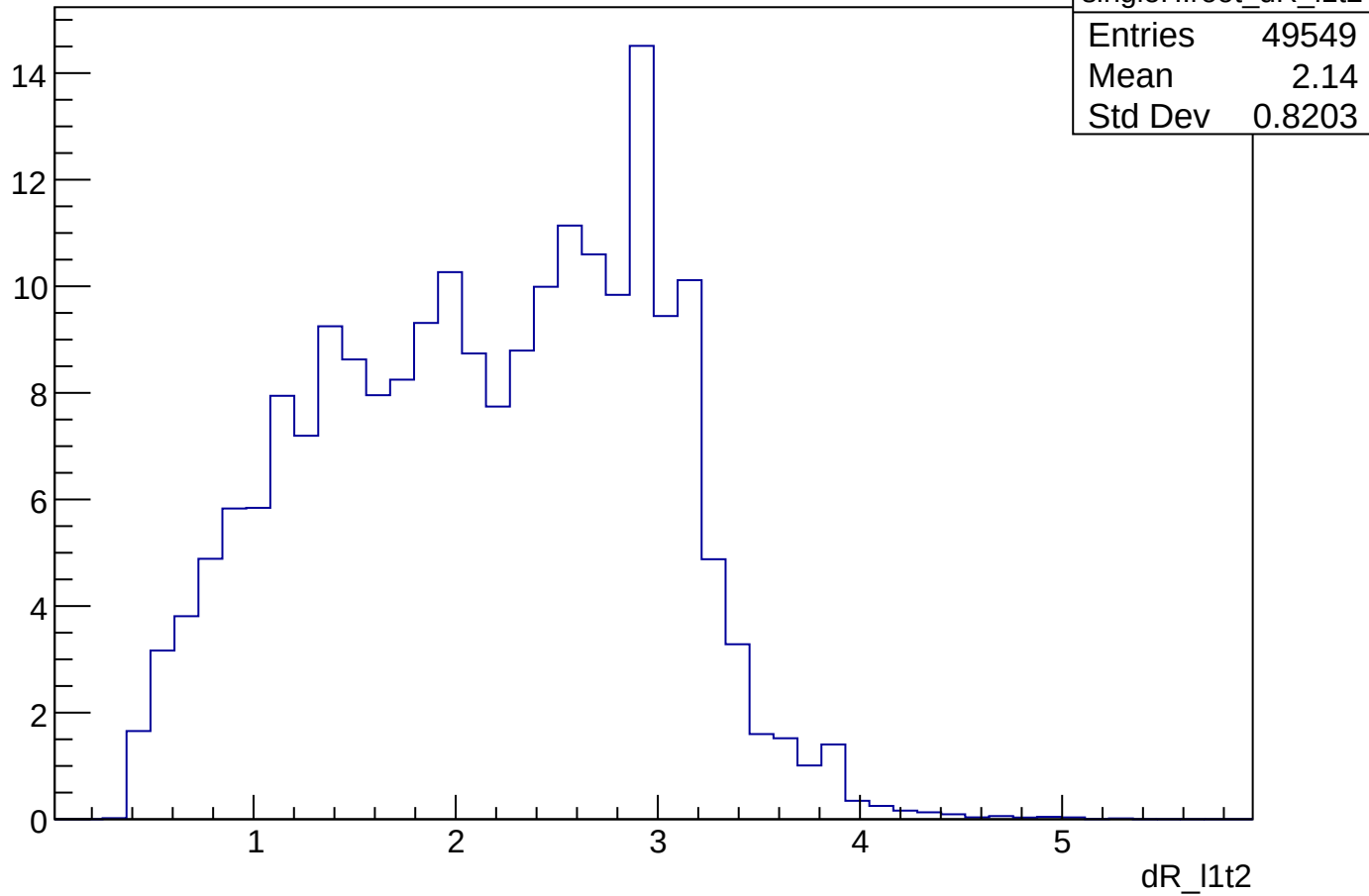
dR_l1t1

Events

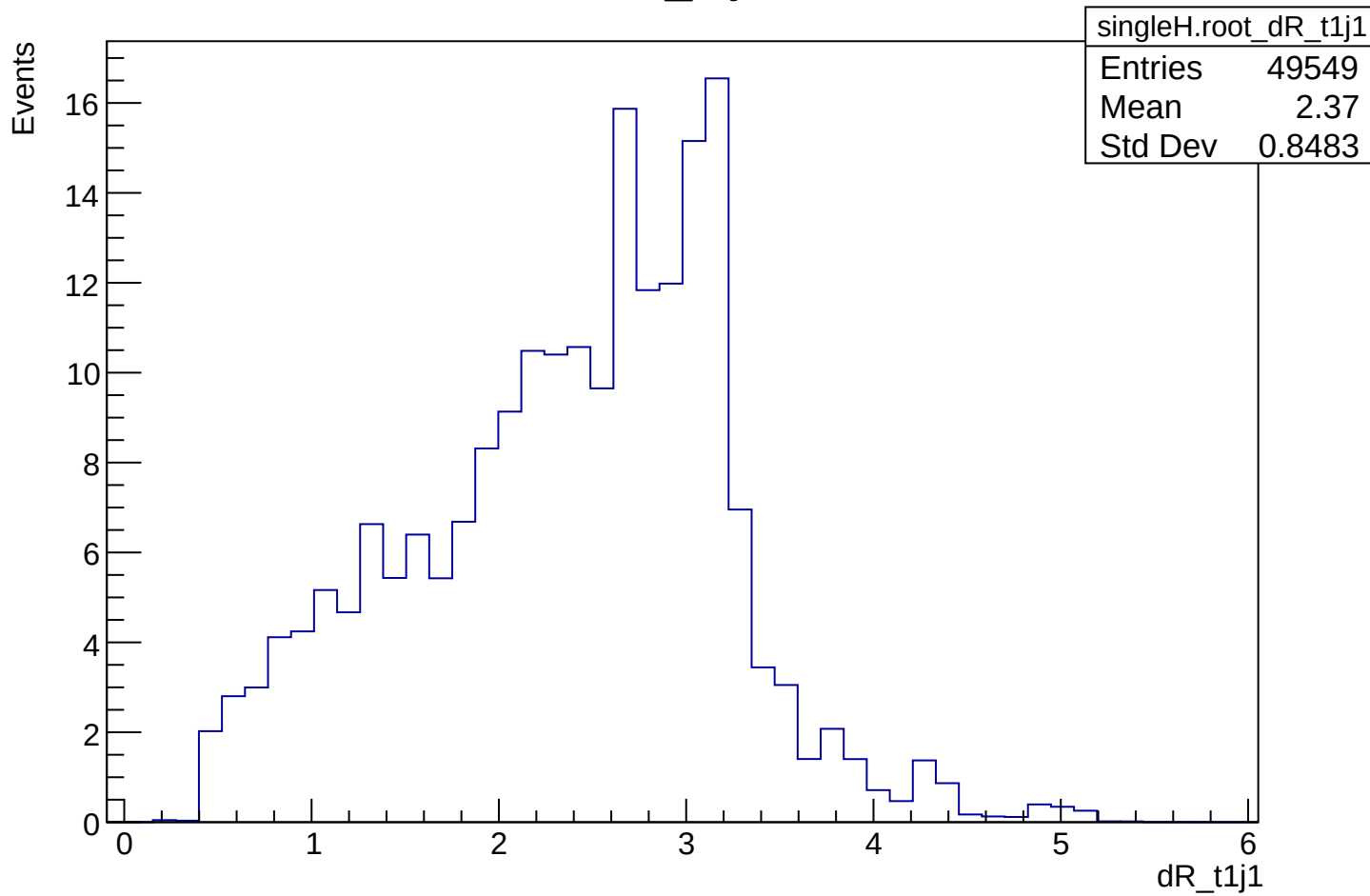


dR_l1t2

Events



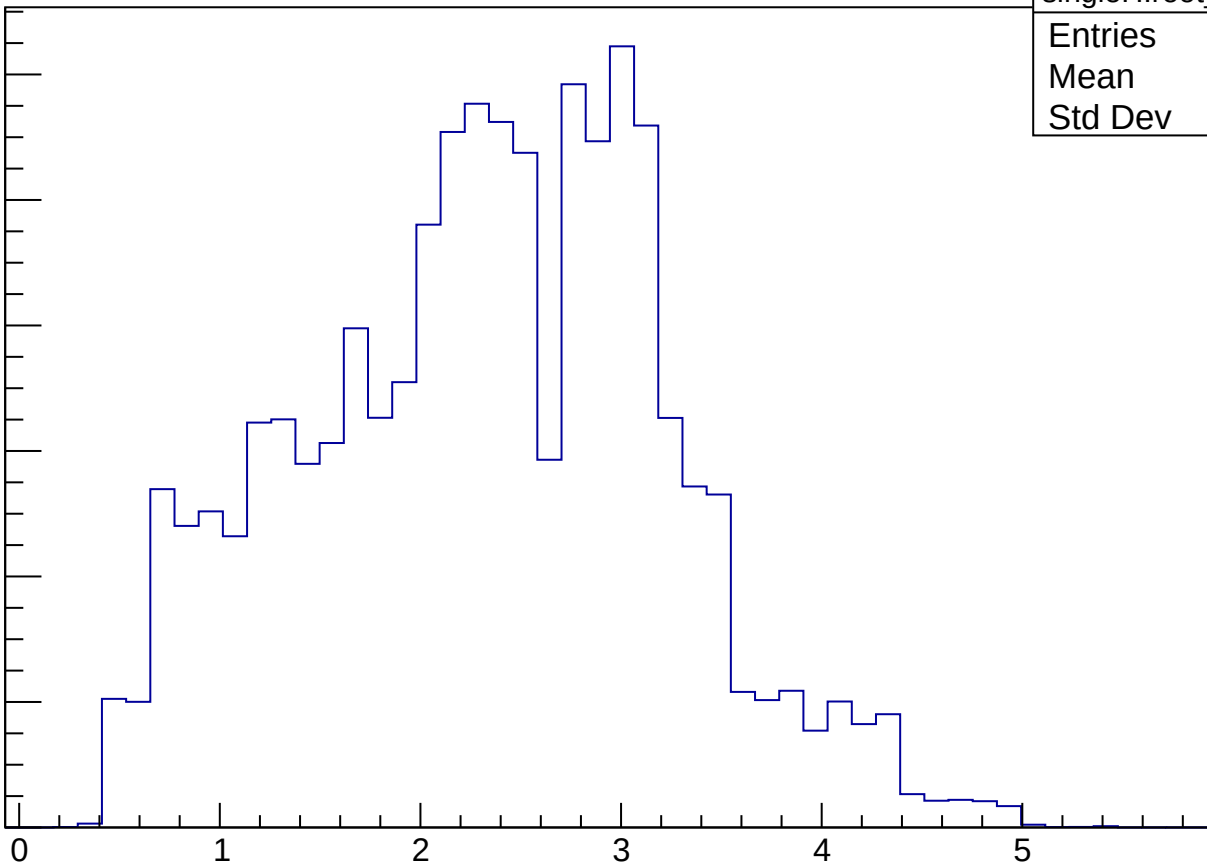
dR_t1j1



dR_t1j2

Events

12
10
8
6
4
2
0

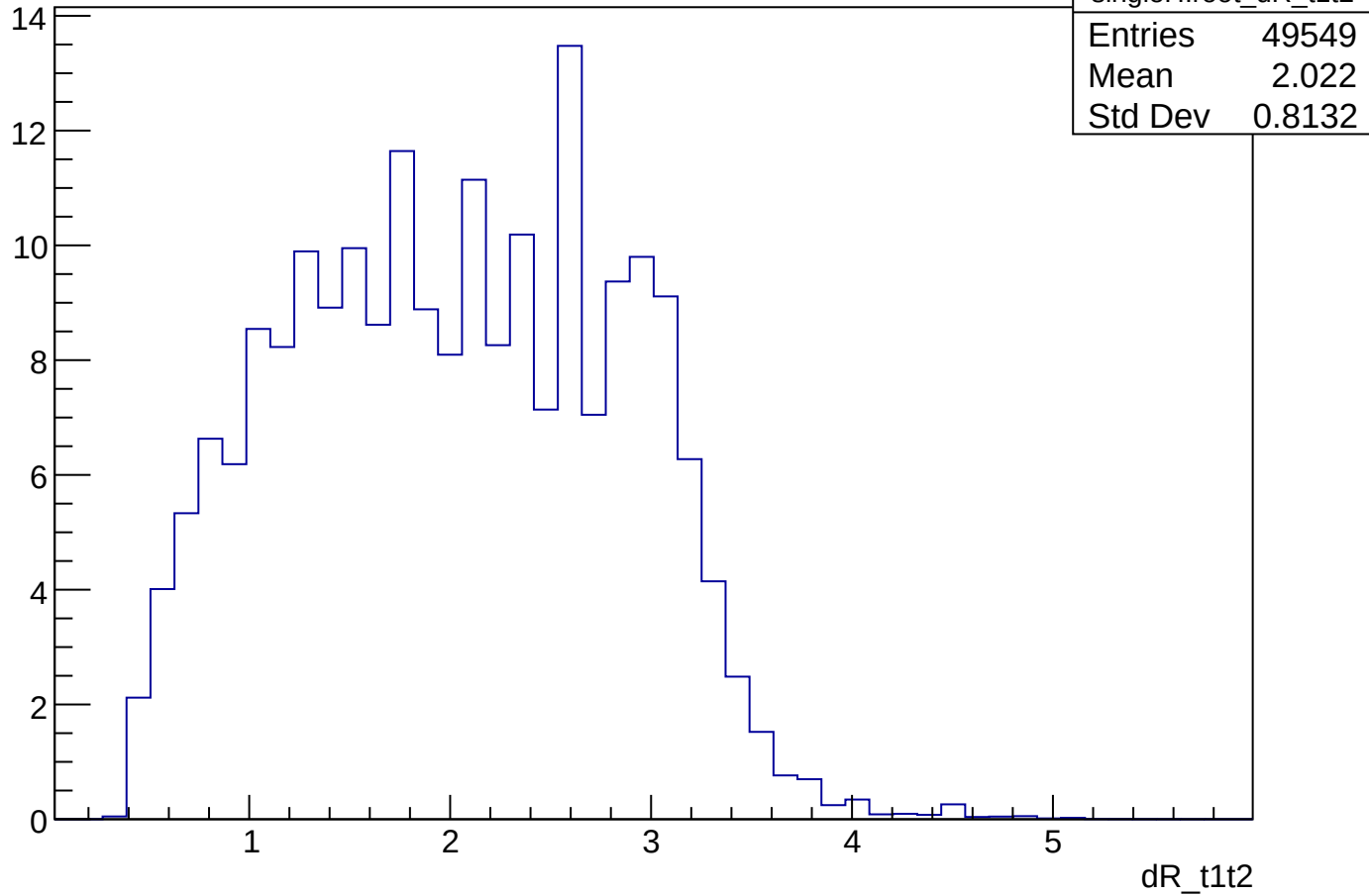


singleH.root_dR_t1j2	
Entries	49549
Mean	2.335
Std Dev	0.9053

dR_t1j2

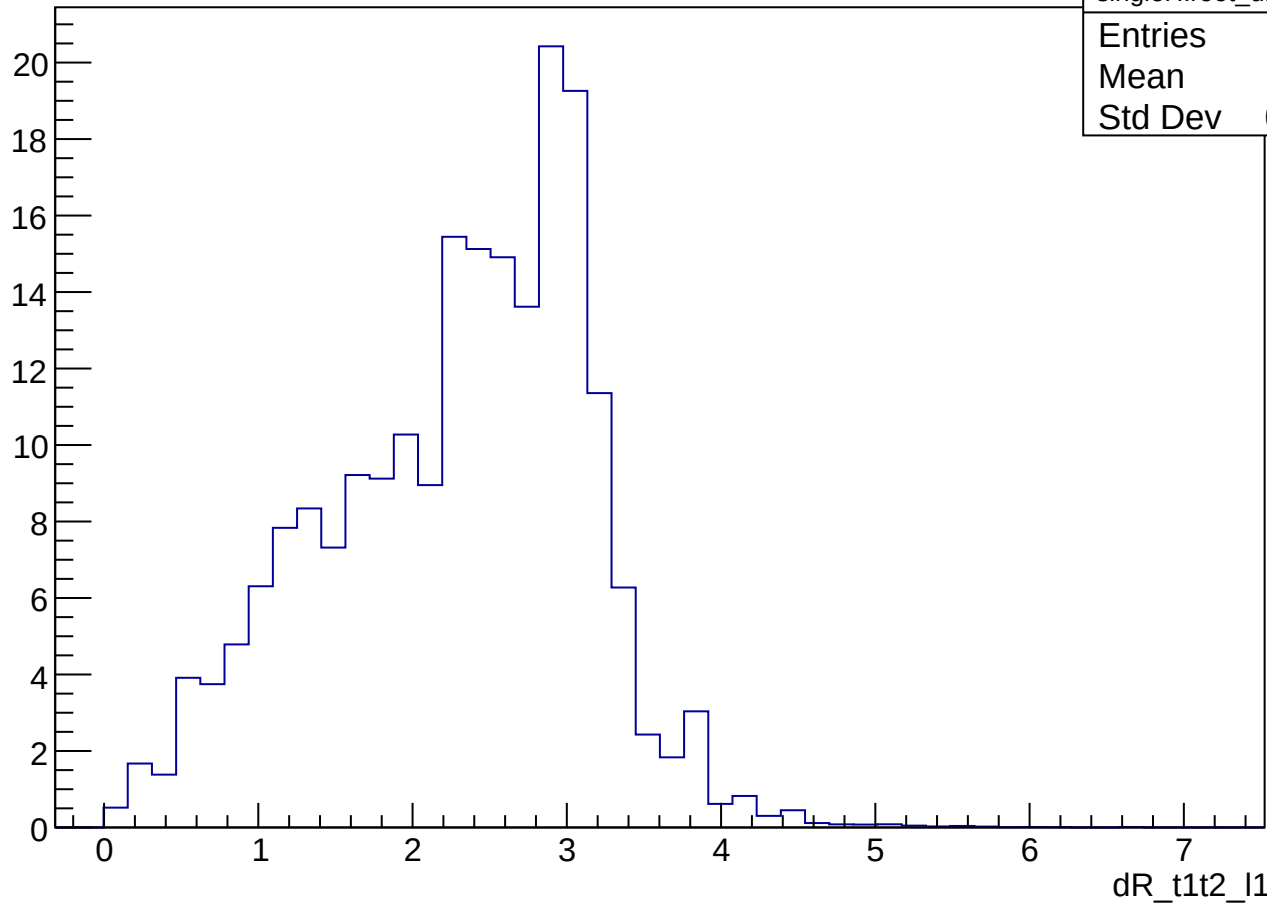
dR_t1t2

Events



dR_t1t2_l1

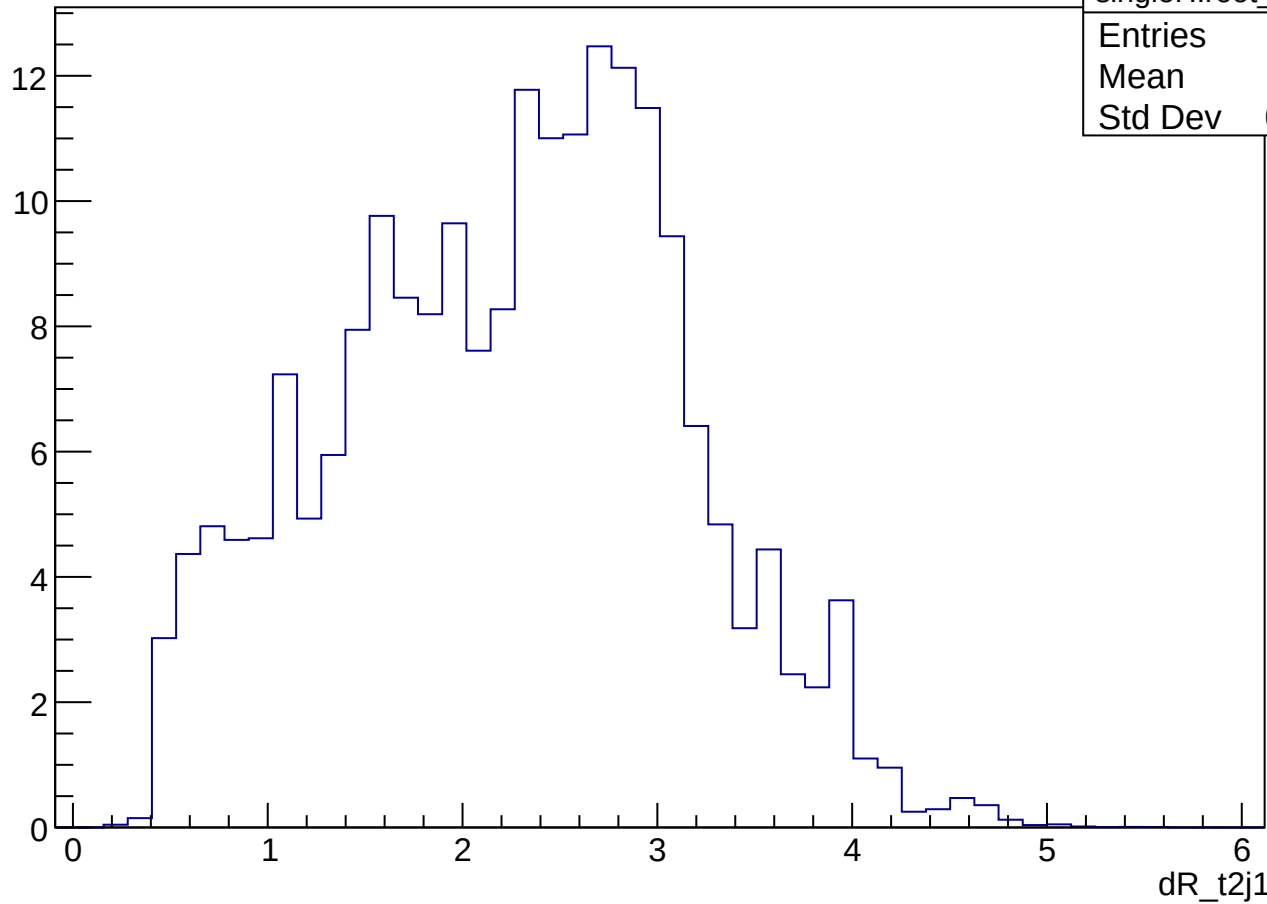
Events



singleH.root_dR_t1t2_l1	
Entries	49549
Mean	2.289
Std Dev	0.8535

dR_t2j1

Events



singleH.root_dR_t2j1

Entries 49549

Mean 2.251

Std Dev 0.8893

dR_t2j2

Events

12
10
8
6
4
2
0

0

1

2

3

4

5

6

dR_t2j2

singleH.root_dR_t2j2	
Entries	49549
Mean	2.307
Std Dev	0.8904

fjvt_effSF

Events

160
140
120
100
80
60
40
20
0

0.4

0.6

0.8

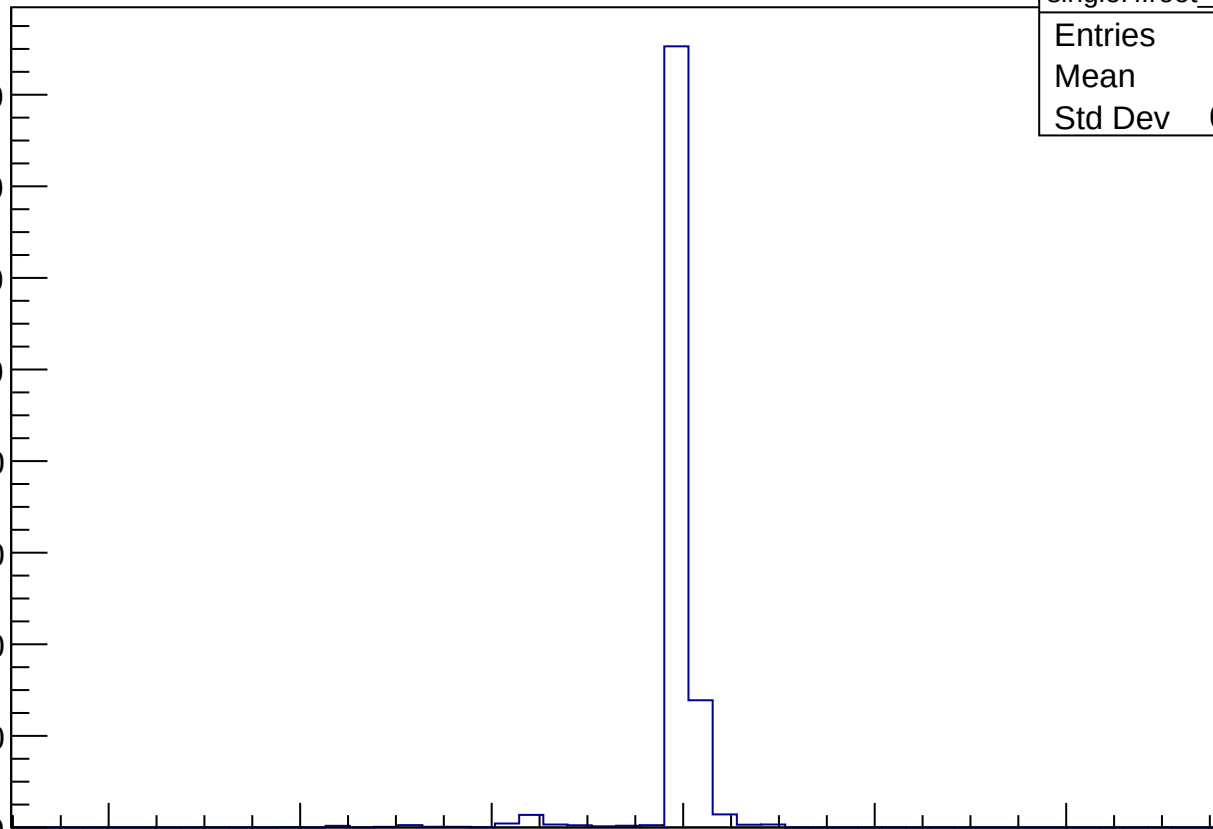
1

1.2

1.4

fjvt_effSF

singleH.root_fjvt_effSF	
Entries	49549
Mean	0.9977
Std Dev	0.03698



ftag_effSF_GN2v01_Continuous

Events

100
80
60
40
20
0

0.5

1

1.5

2

2.5

ftag_effSF_GN2v01_Continuous

singleH.root_ftag_effSF_GN2v01_Continuous	
Entries	49549
Mean	1.011
Std Dev	0.1098

globalTriggerEffSF

Events

100

80

60

40

20

0

0.2

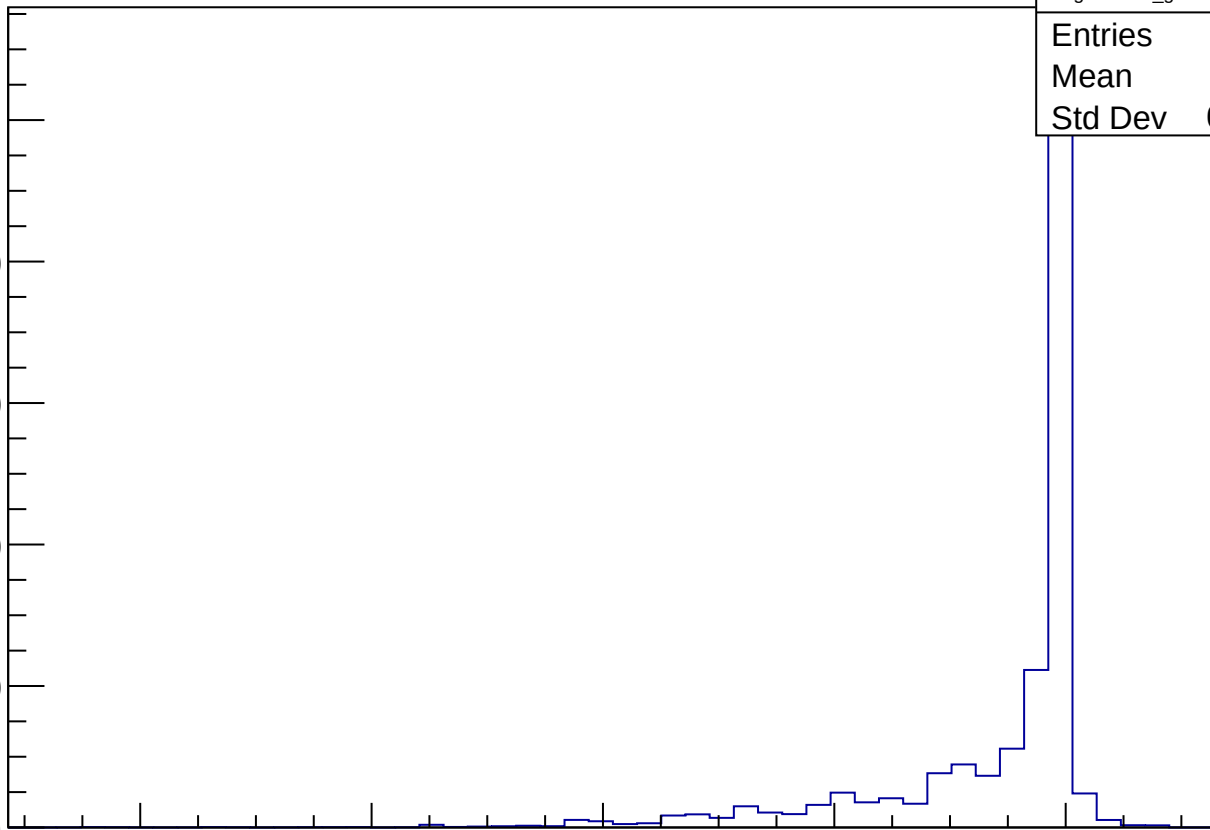
0.4

0.6

0.8

globalTriggerEffSF

singleH.root_globalTriggerEffSF	
Entries	49549
Mean	0.943
Std Dev	0.09894



hhml_subChannelFlavor

Events

100

80

60

40

20

0

1

1.2

1.4

1.6

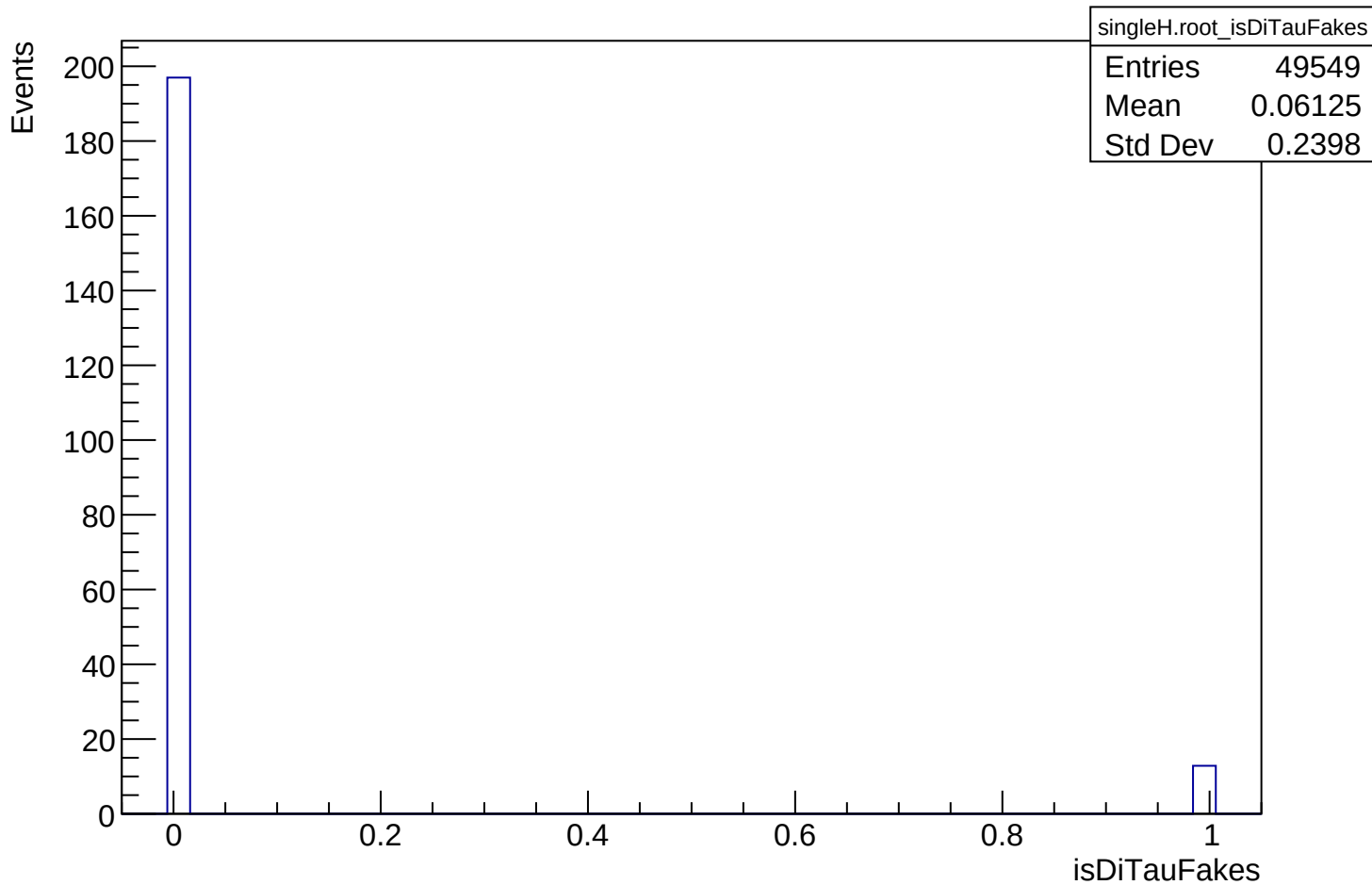
1.8

2

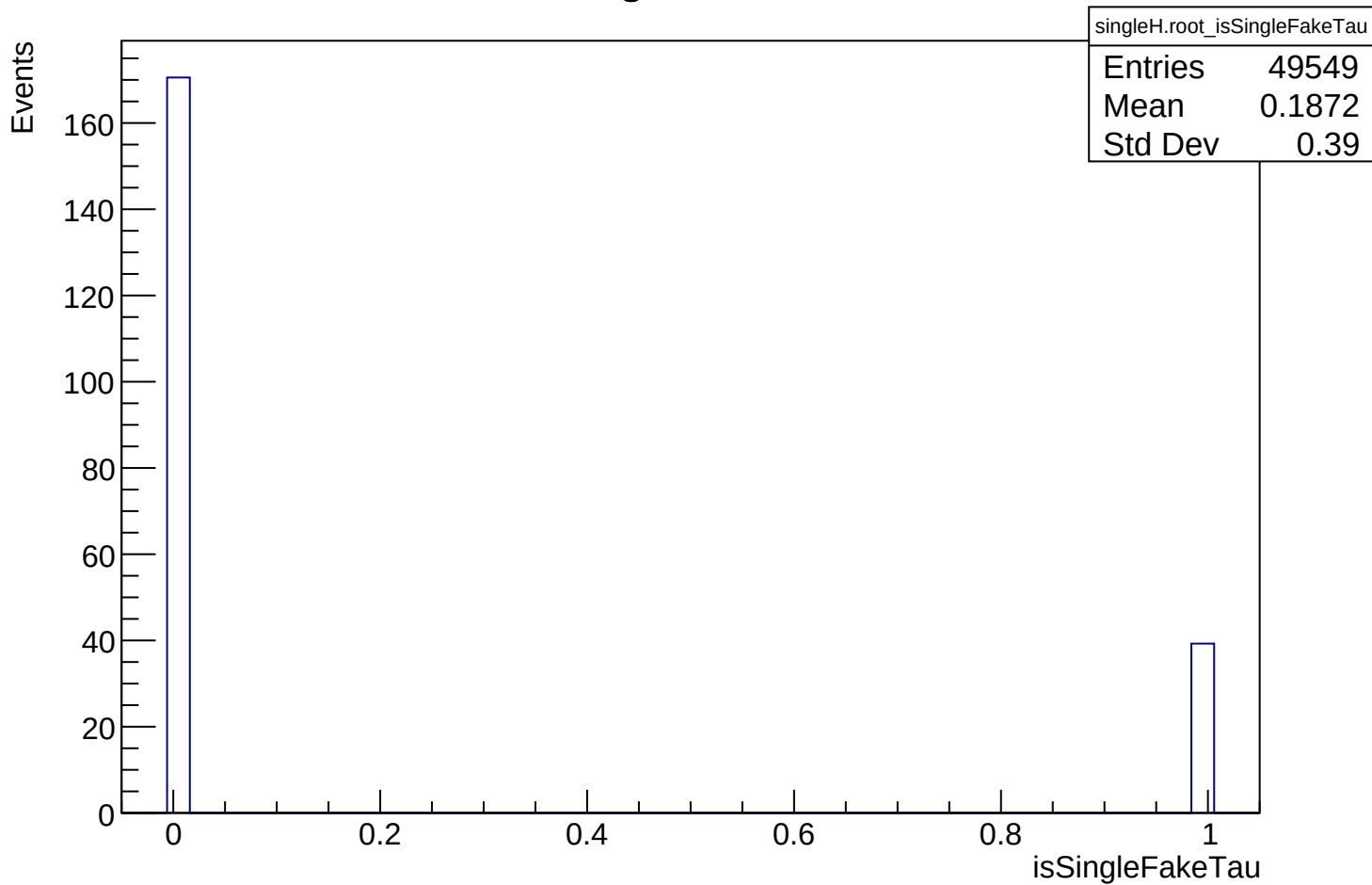
hhml_subChannelFlavor

singleH.root_hhml_subChannelFlavor	
Entries	49549
Mean	1.498
Std Dev	0.5

isDiTauFakes



isSingleFakeTau



isTrueTauEvents

Events

160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

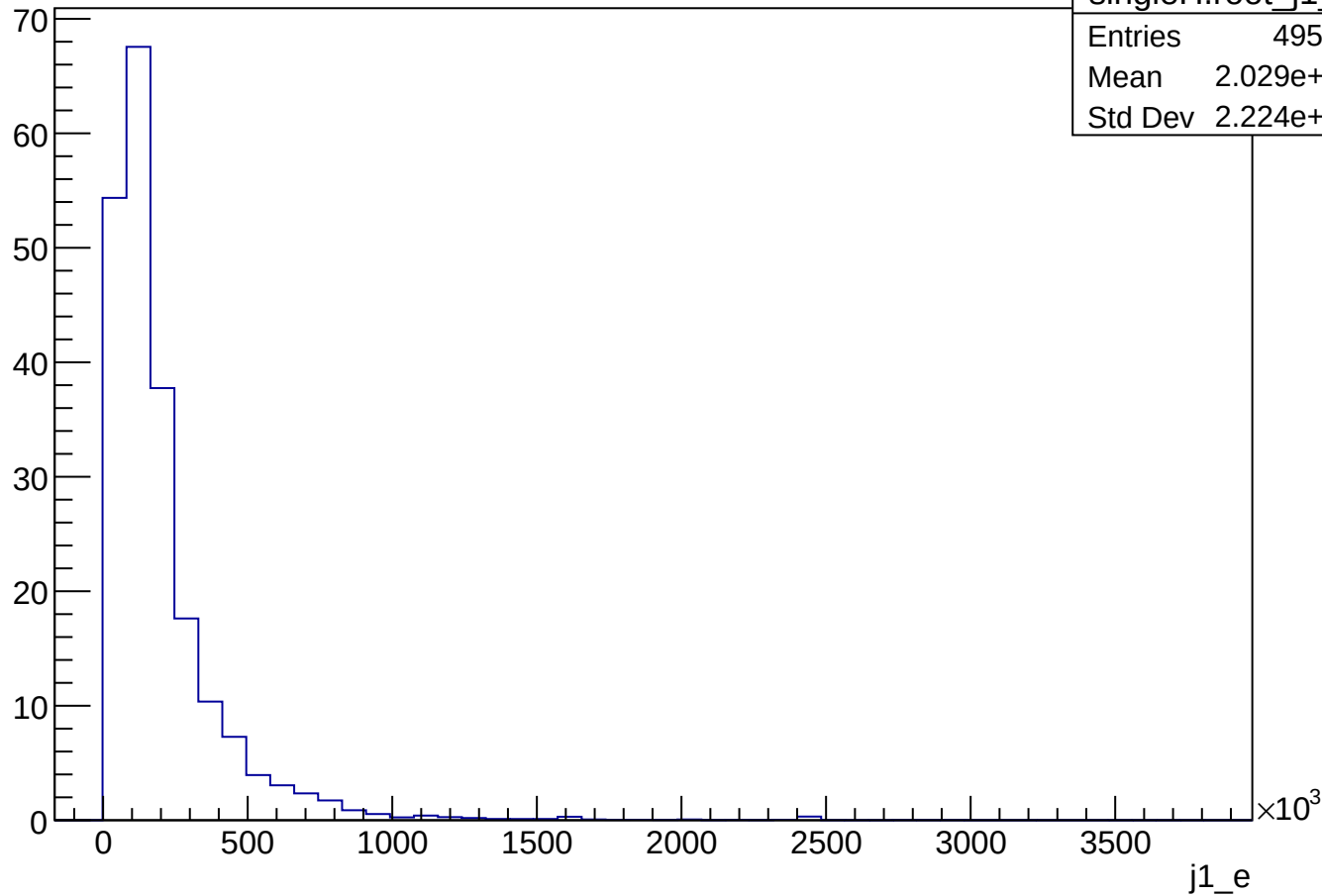
1

isTrueTauEvents

singleH.root_isTrueTauEvents	
Entries	49549
Mean	0.7516
Std Dev	0.4321

j1_e

Events



j1_eta

Events

10
8
6
4
2
0

-2

-1

0

1

2

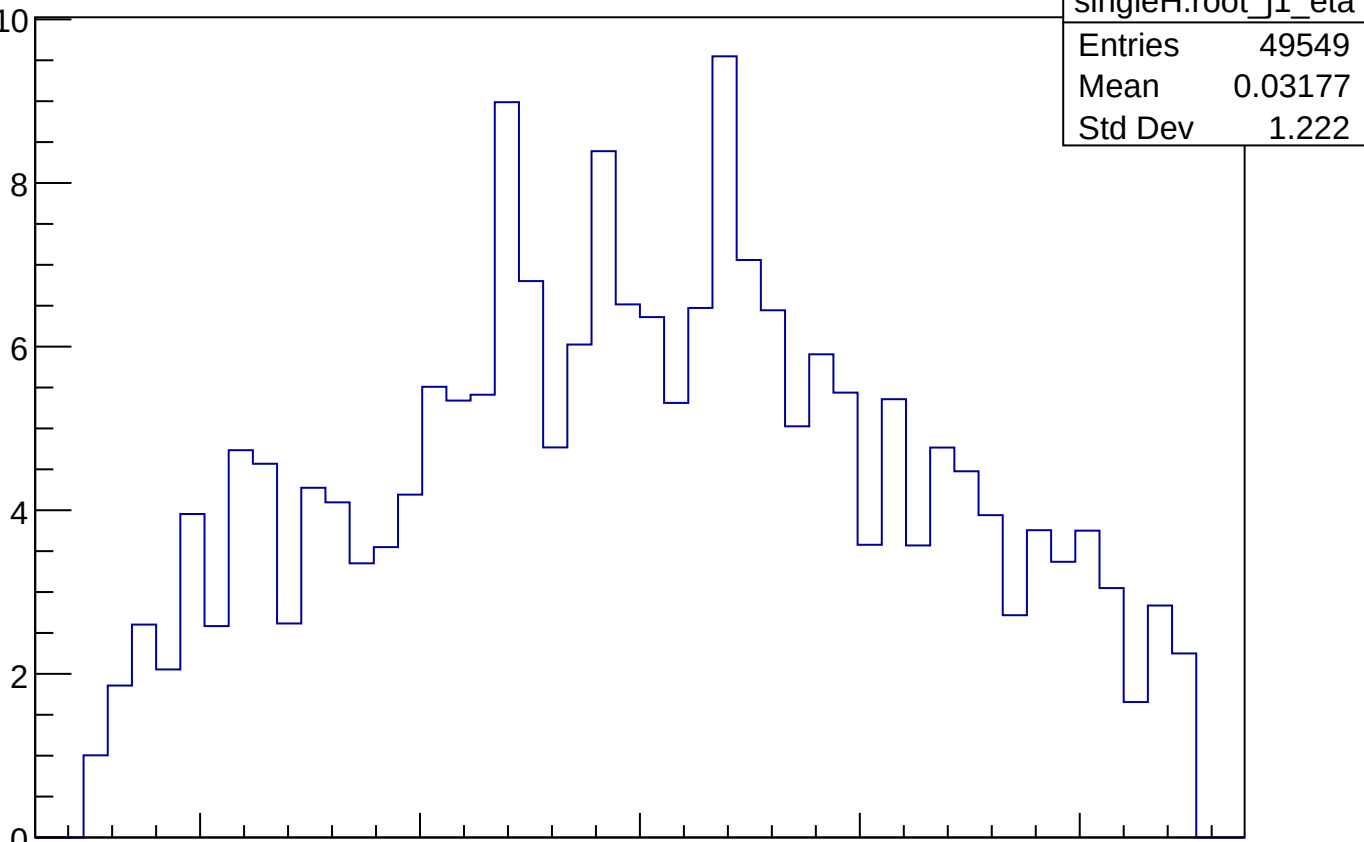
j1_eta

singleH.root_j1_eta

Entries 49549

Mean 0.03177

Std Dev 1.222



j1_phi

Events

singleH.root_j1_phi

Entries 49549

Mean 0.06162

Std Dev 1.808

7
6
5
4
3
2
1
0

-3

-2

-1

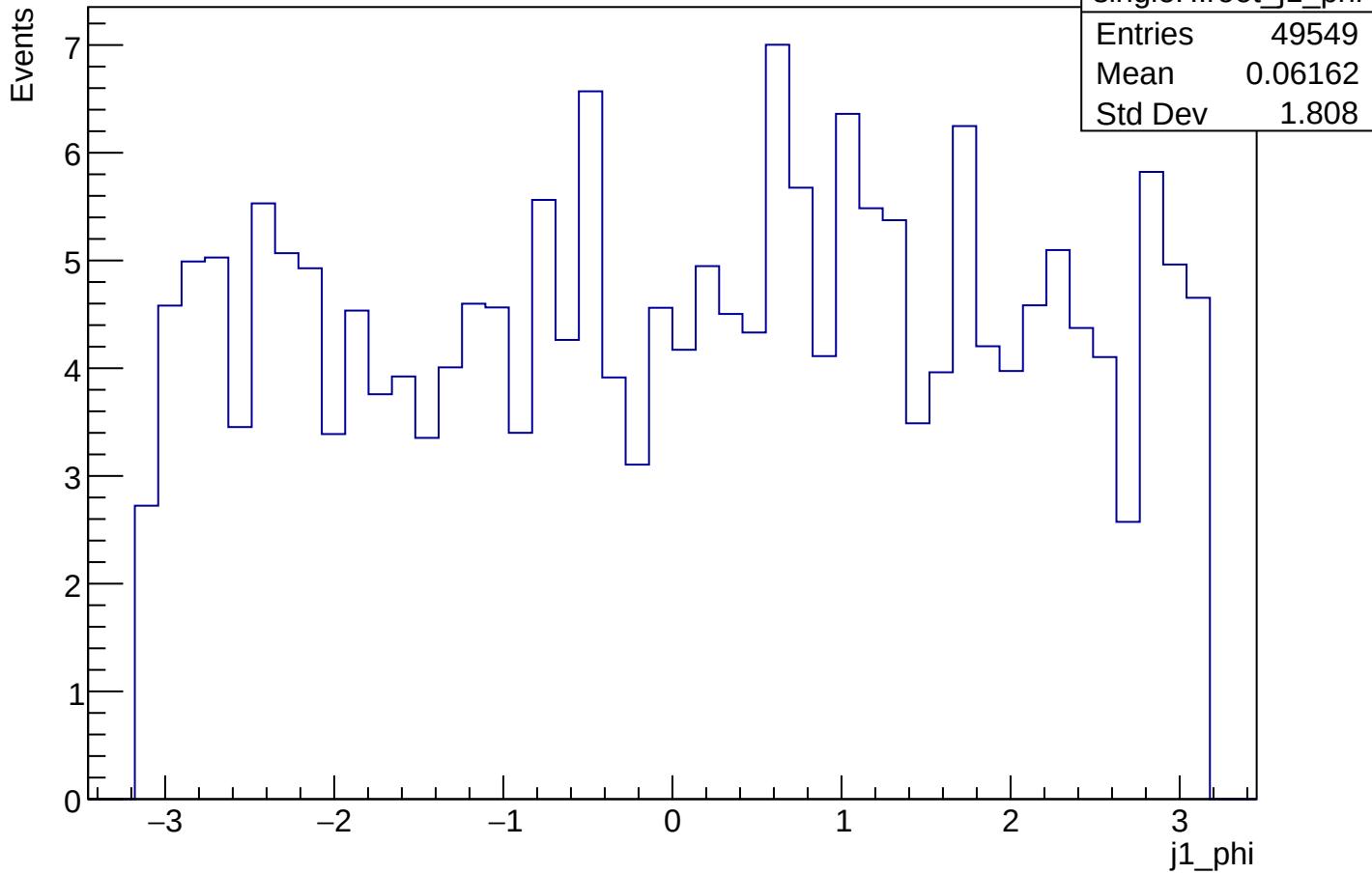
0

1

2

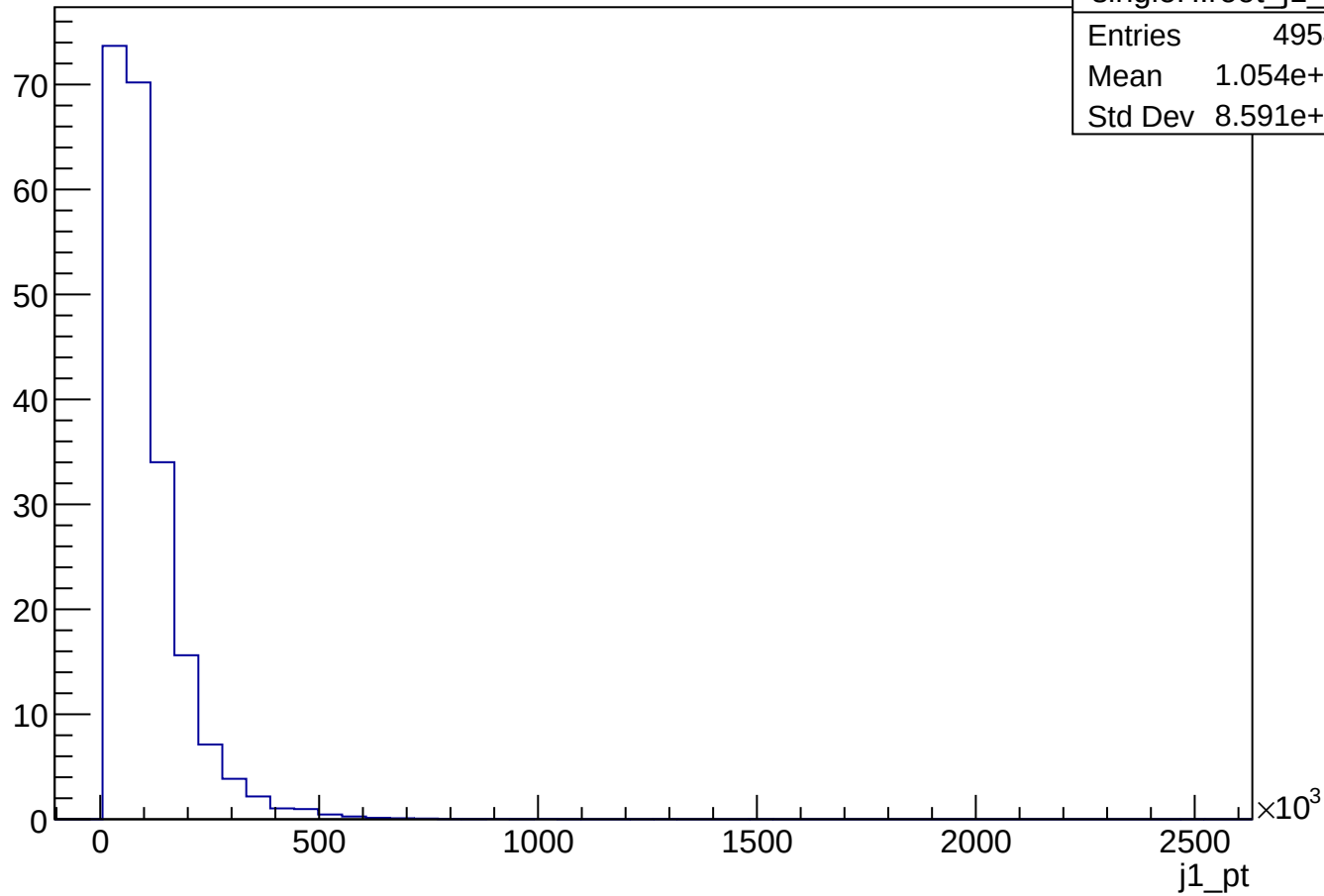
3

j1_phi



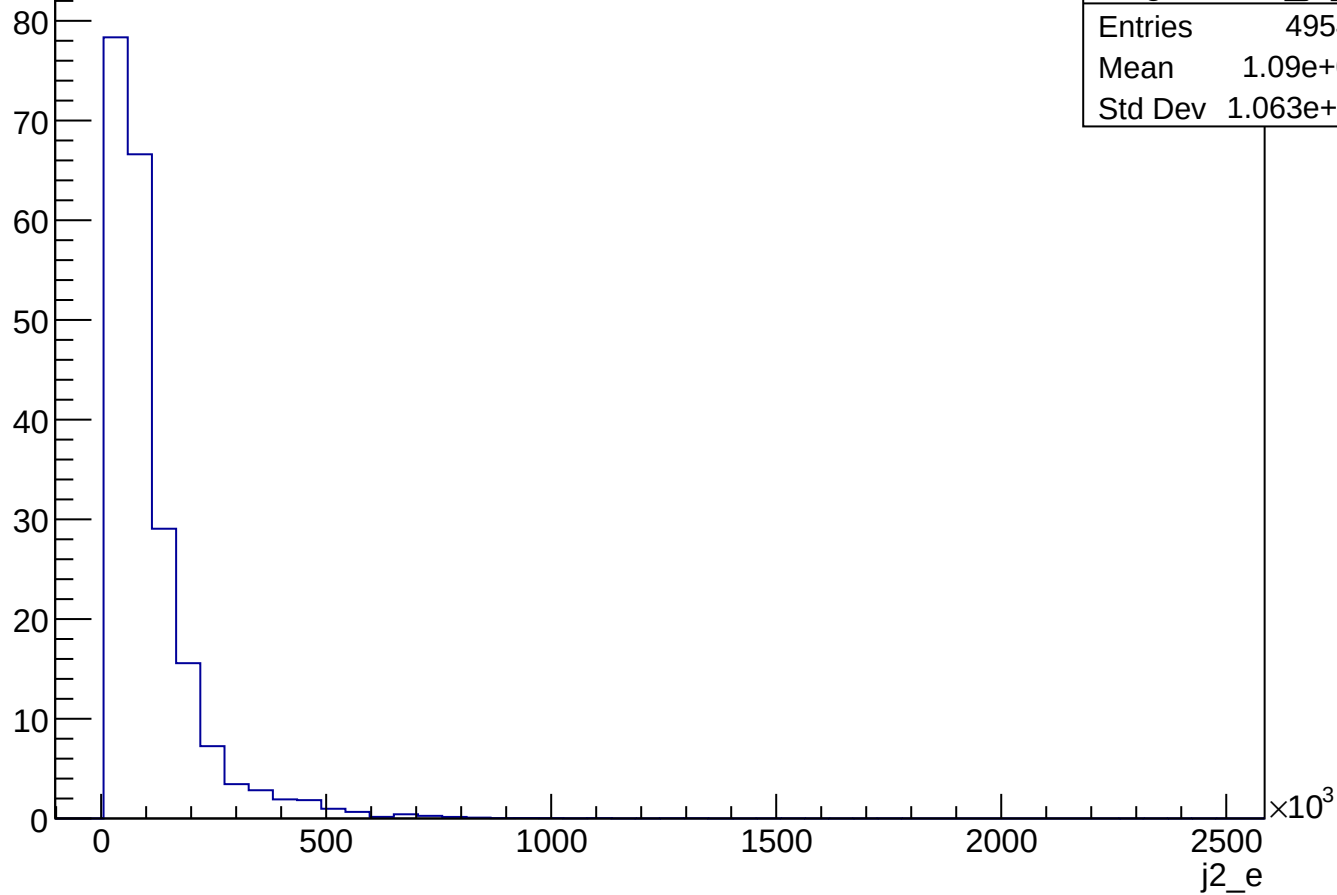
j1_pt

Events



j2_e

Events



j2_eta

Events

singleH.root_j2_eta

Entries 49549

Mean 0.006143

Std Dev 1.298

7
6
5
4
3
2
1
0

-2

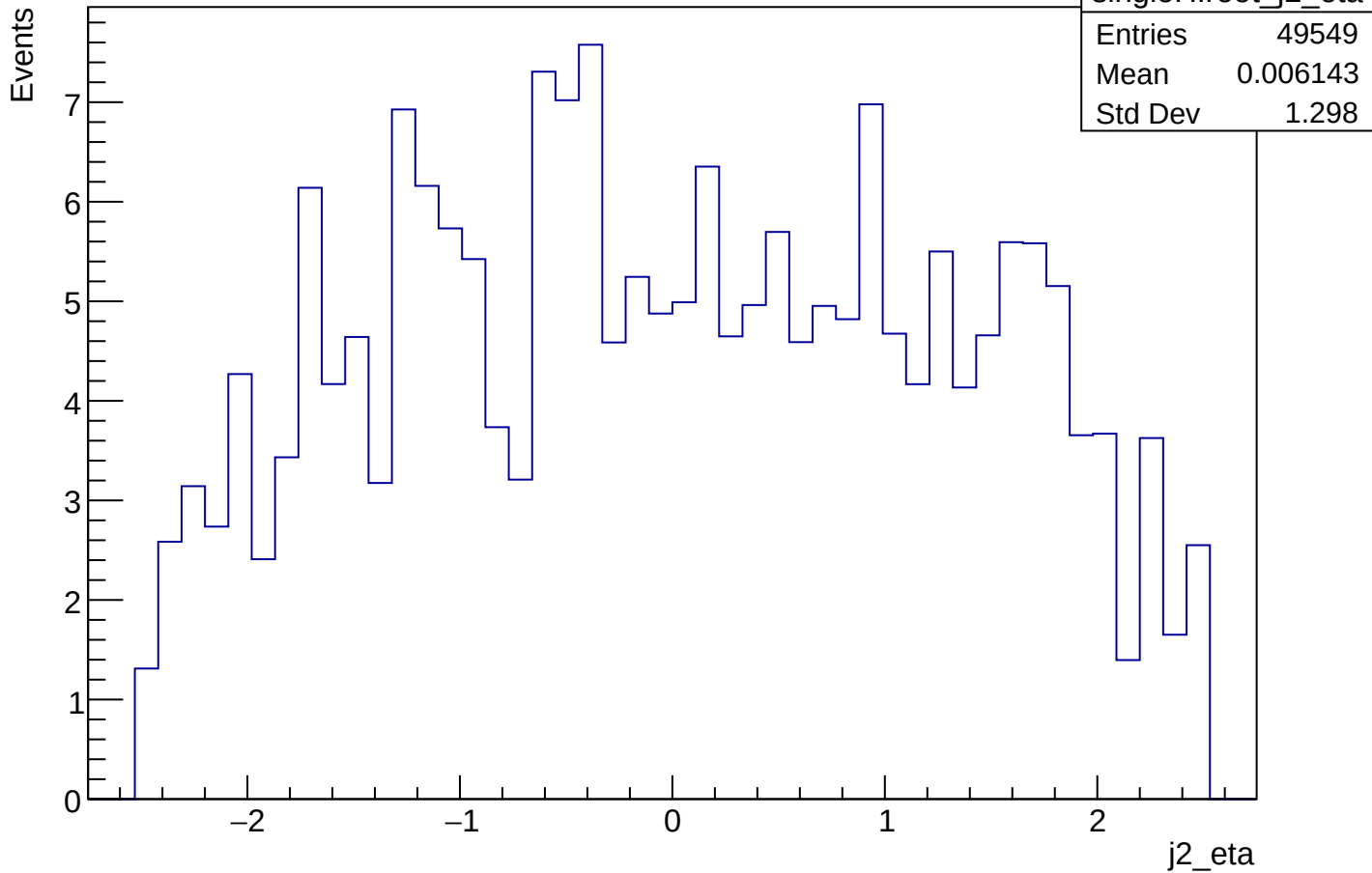
-1

0

1

2

j2_eta



j2_phi

Events

singleH.root_j2_phi

Entries 49549

Mean 0.002456

Std Dev 1.828

7
6
5
4
3
2
1
0

-3

-2

-1

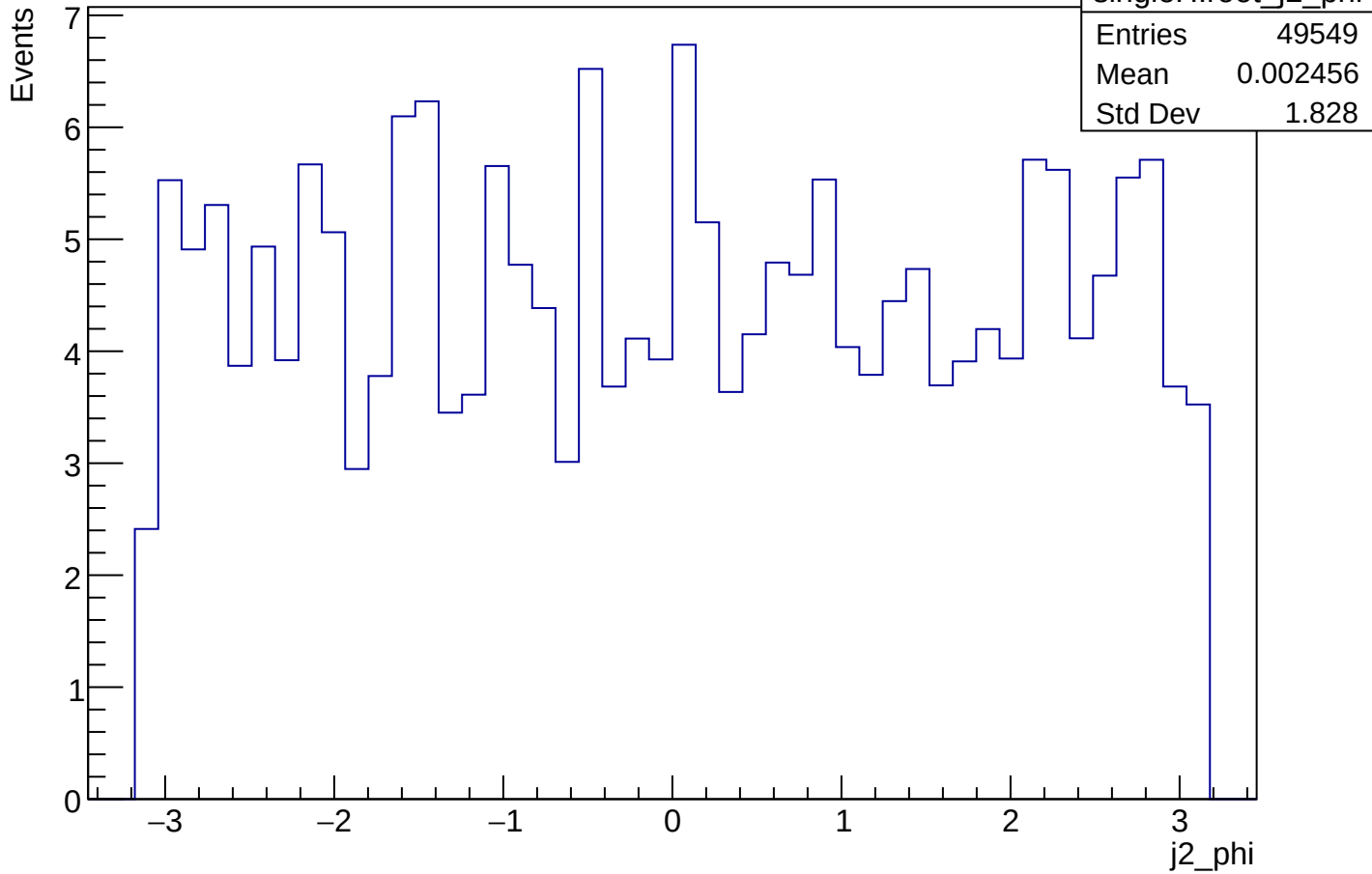
0

1

2

3

j2_phi



j2_pt

Events

140
120
100
80
60
40
20
0

0

500

1000

1500

2000

j2_pt

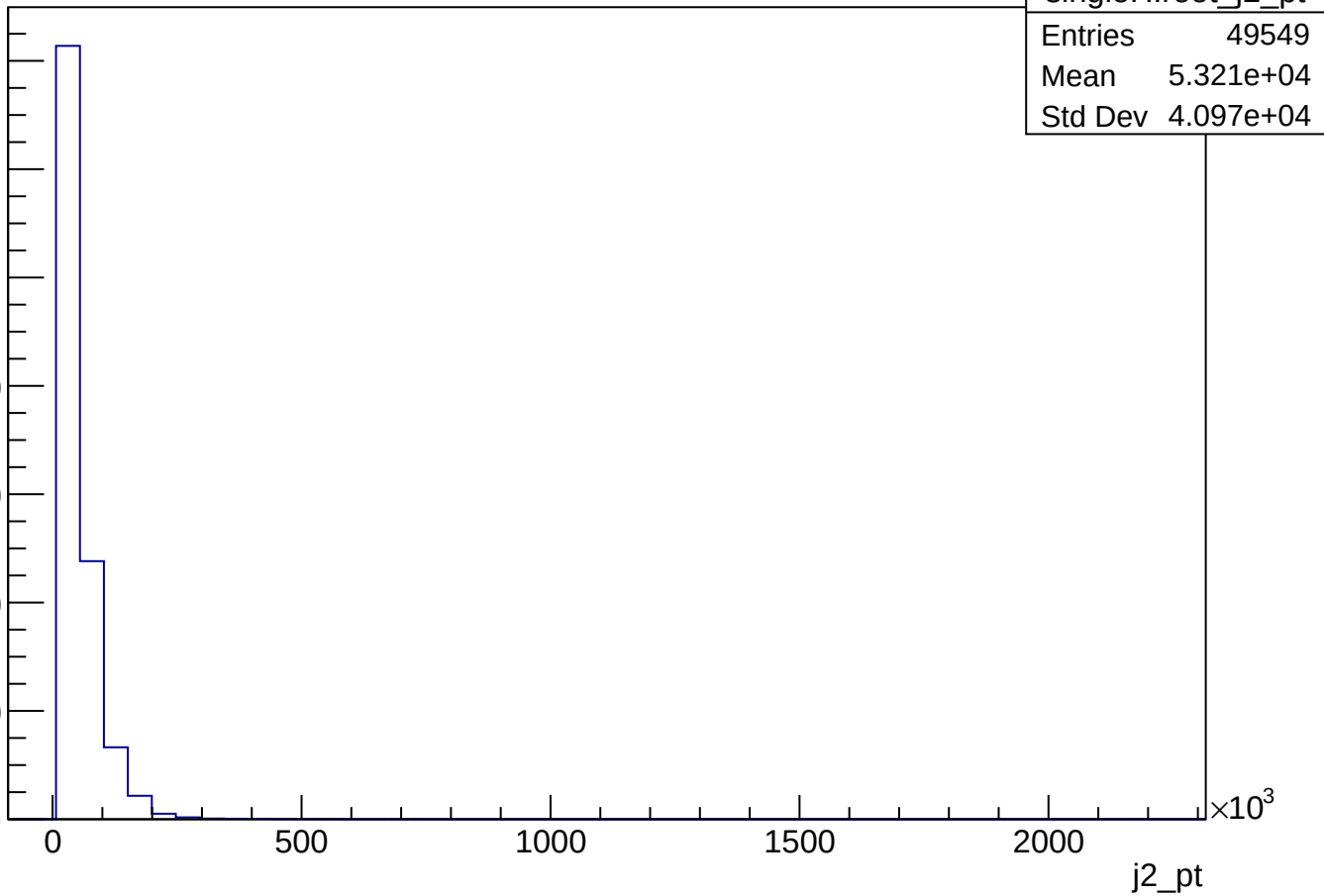
$\times 10^3$

singleH.root_j2_pt

Entries 49549

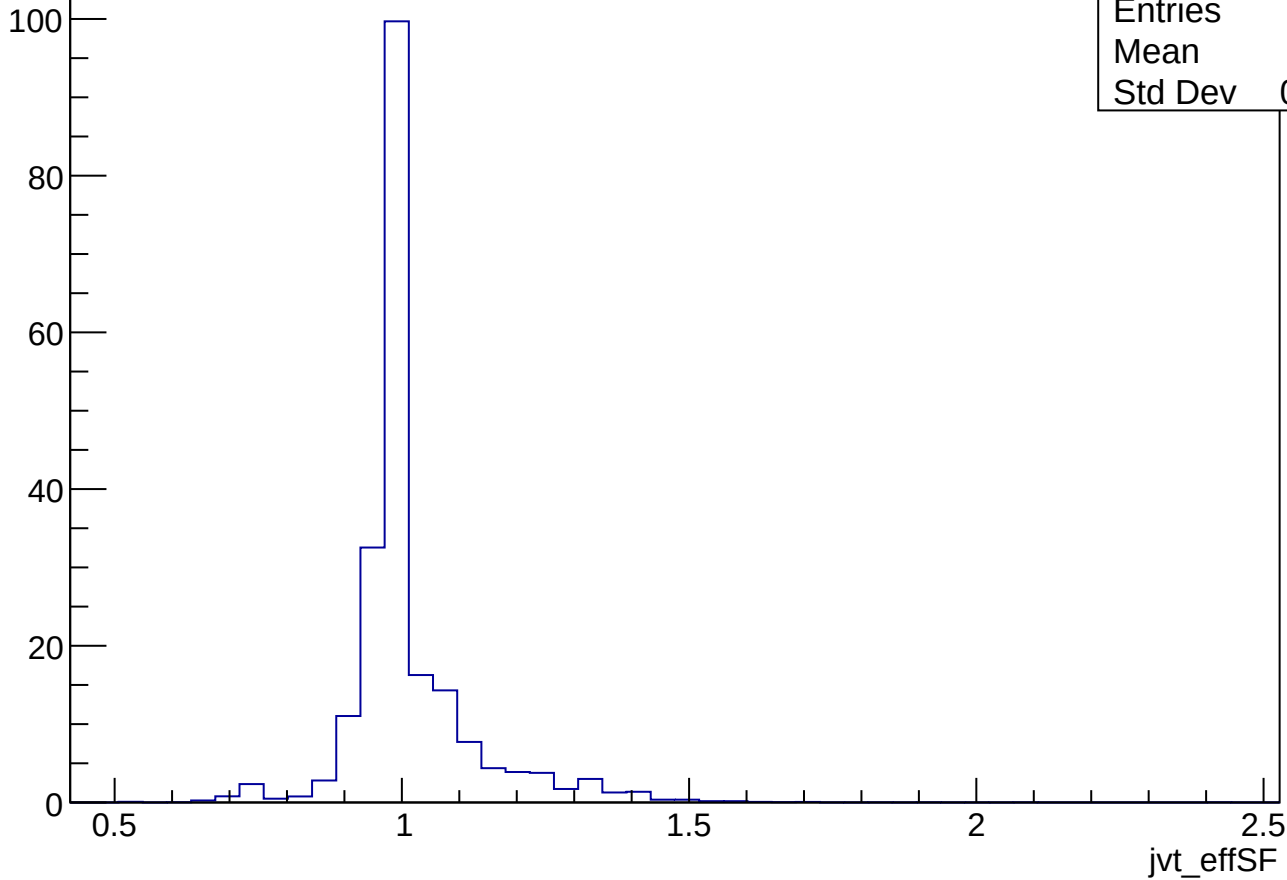
Mean 5.321e+04

Std Dev 4.097e+04



jvt_effSF

Events



l1_charge

Events

100

80

60

40

20

0

-1

-0.5

0

0.5

1

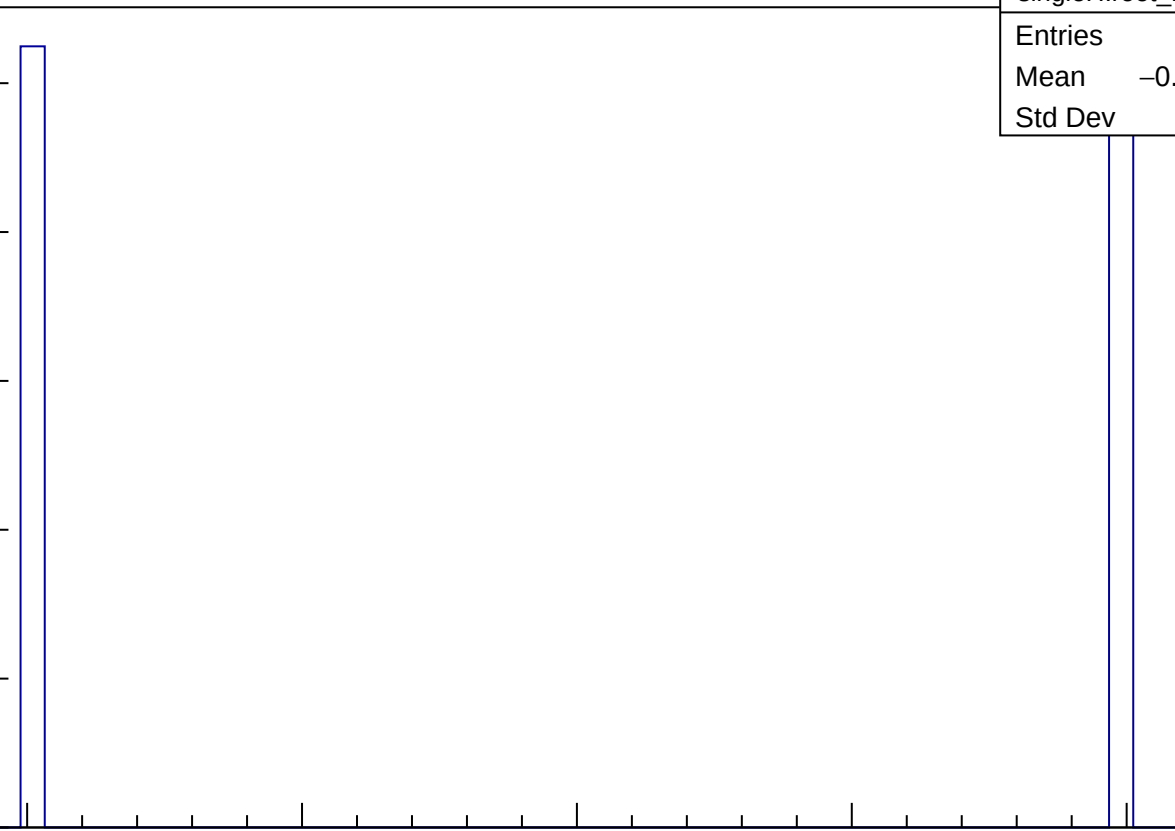
singleH.root_l1_charge

Entries 49549

Mean -0.0003858

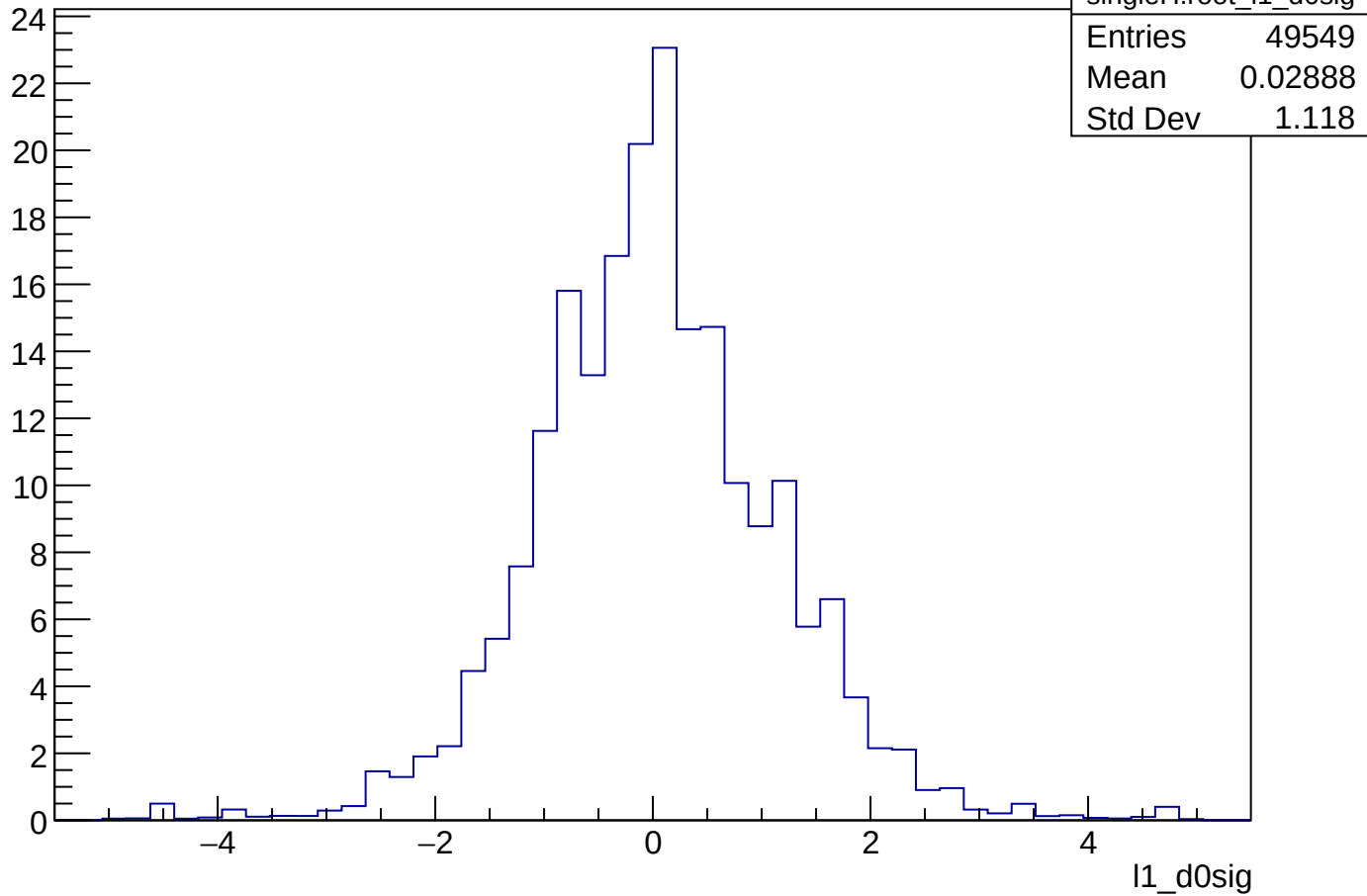
Std Dev 1

l1_charge



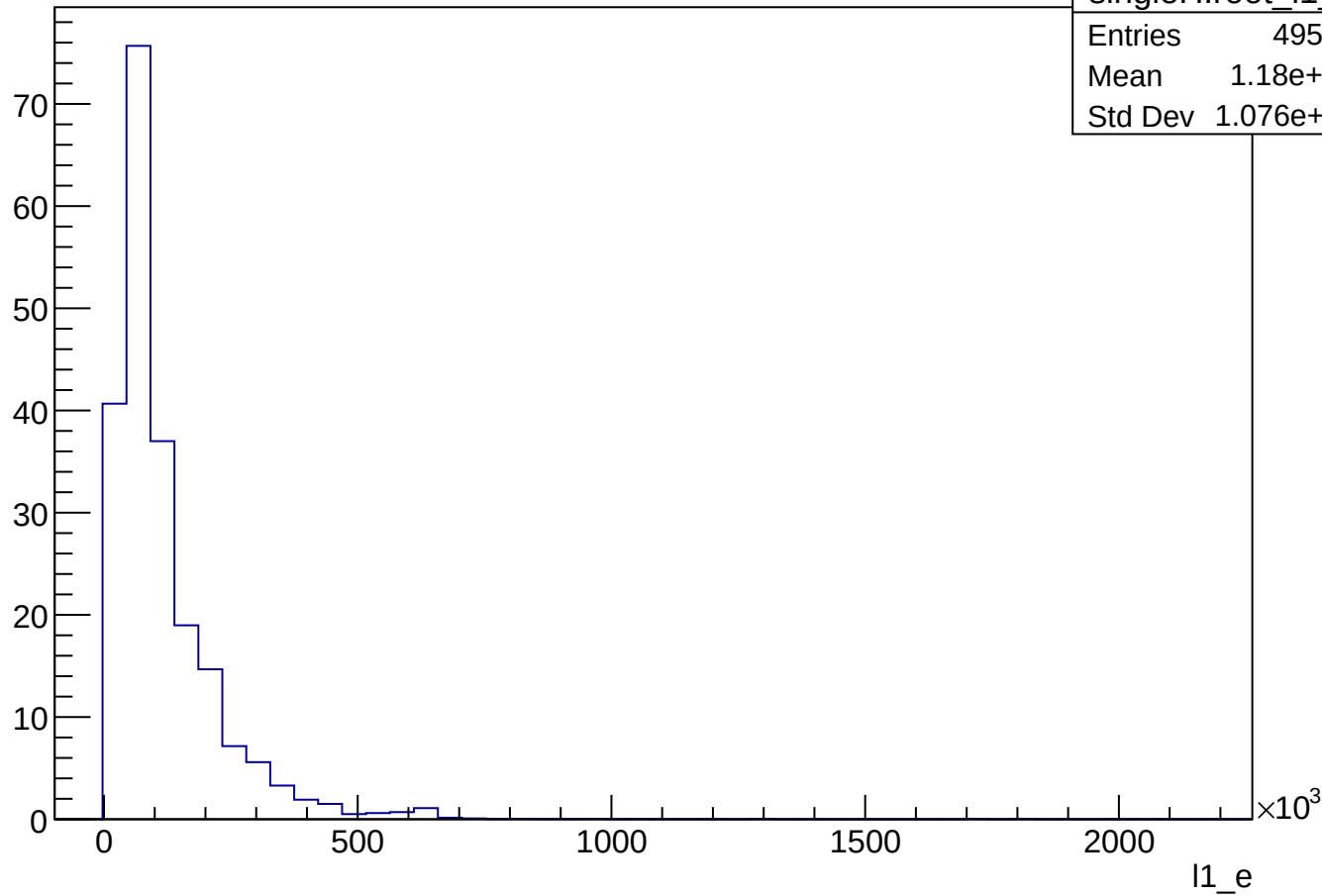
l1_d0sig

Events

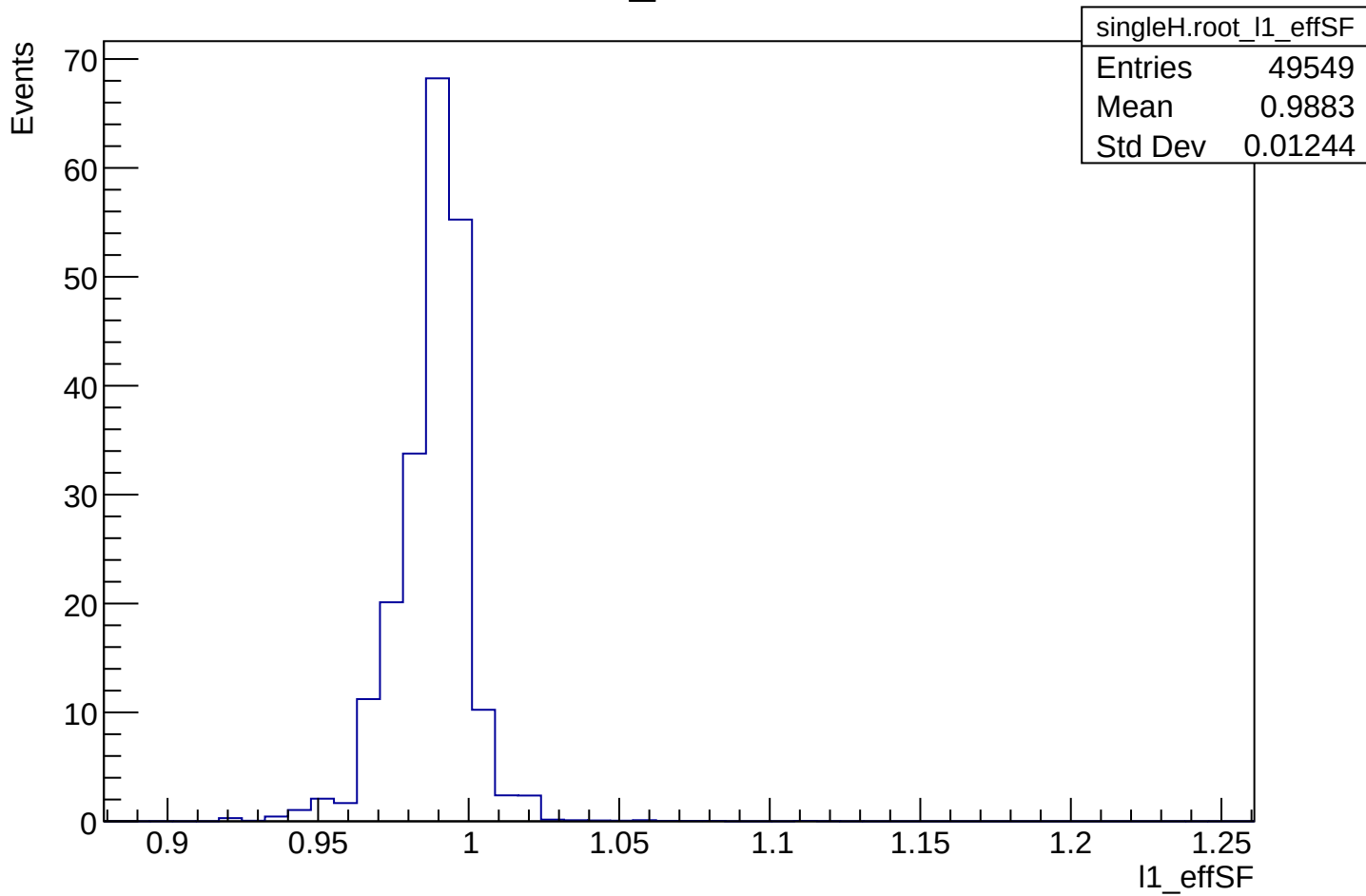


l1_e

Events



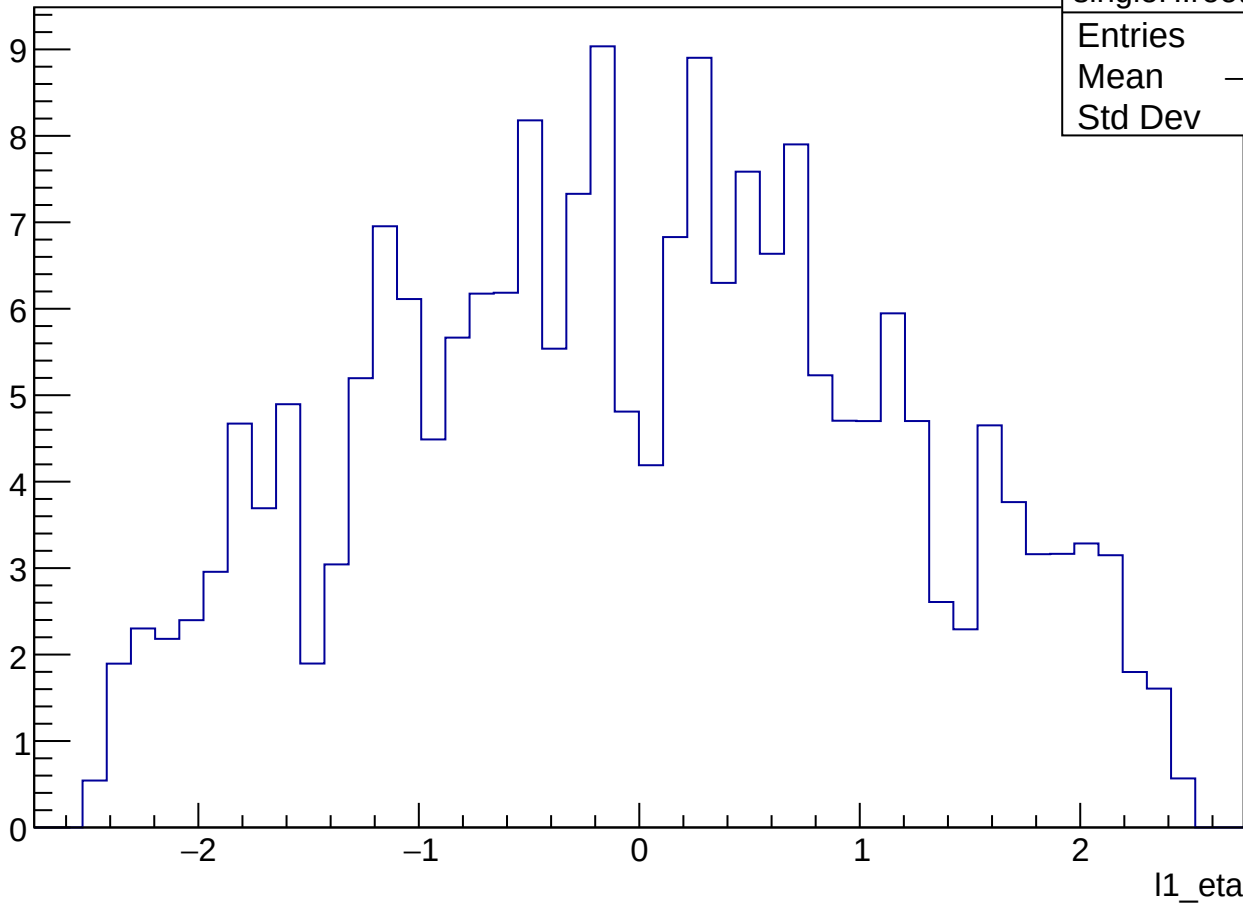
l1_effSF



$l1_eta$

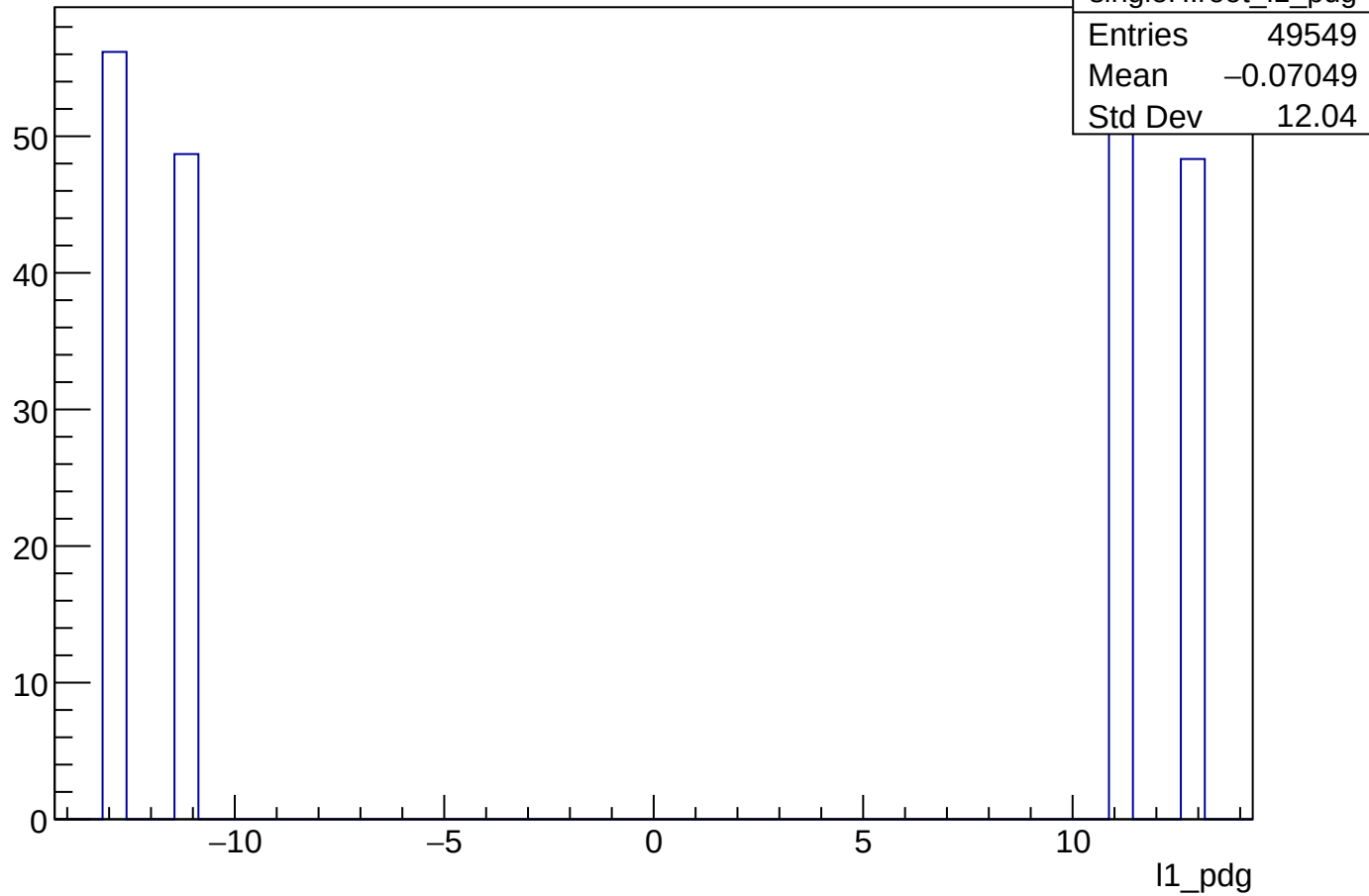
Events

singleH.root_l1_eta	
Entries	49549
Mean	-0.0135
Std Dev	1.17

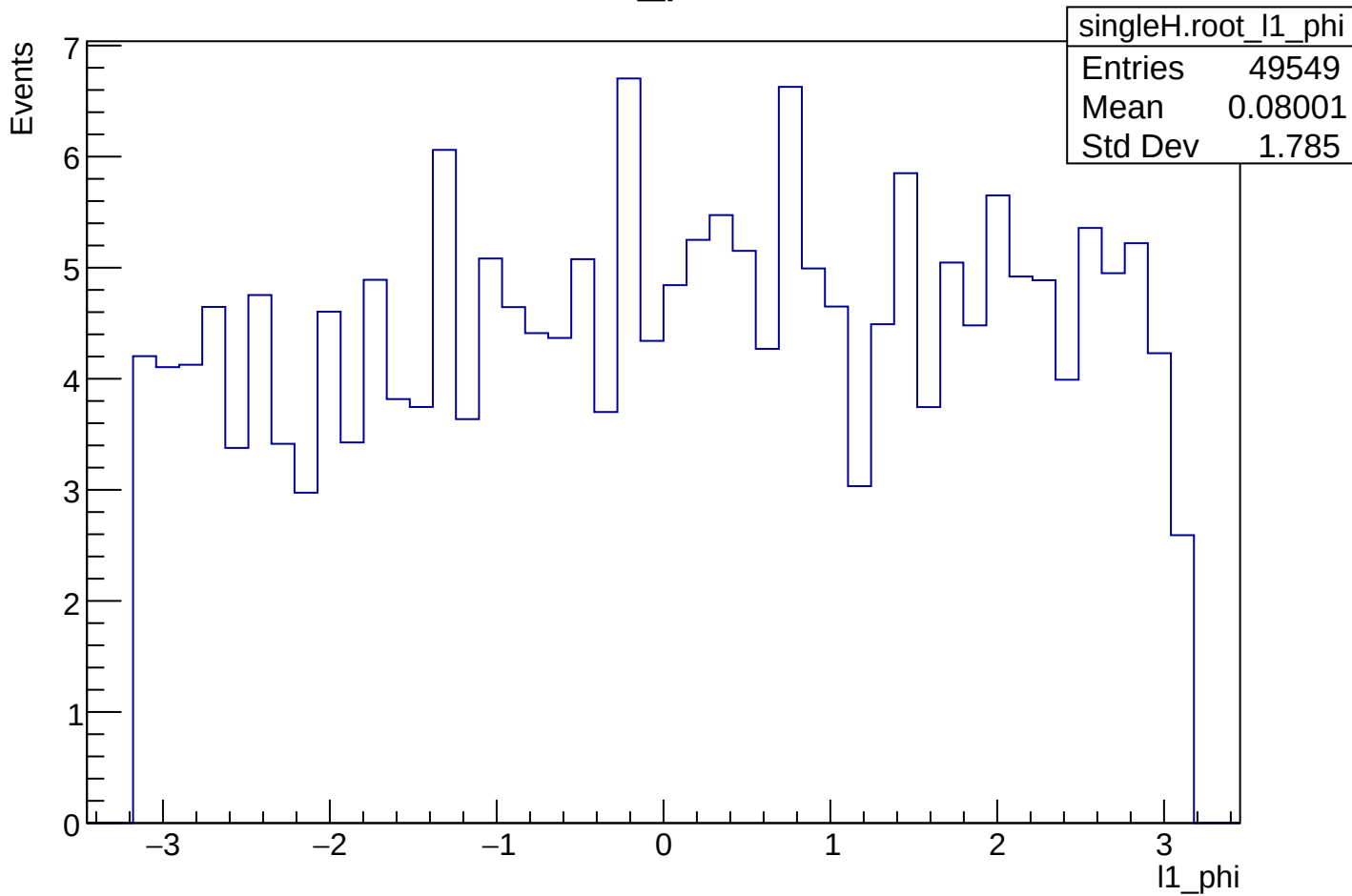


l1_pdg

Events

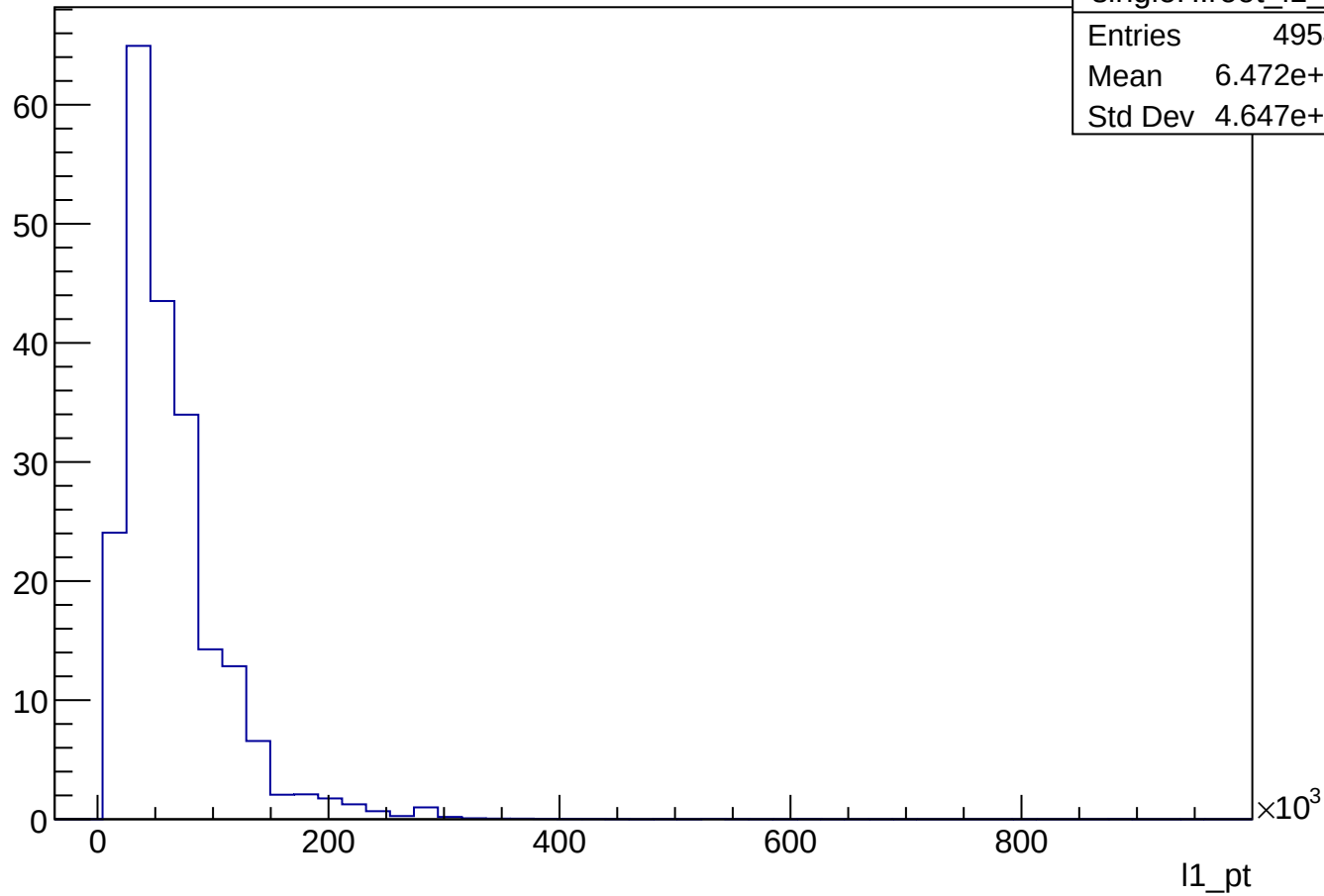


l1_phi



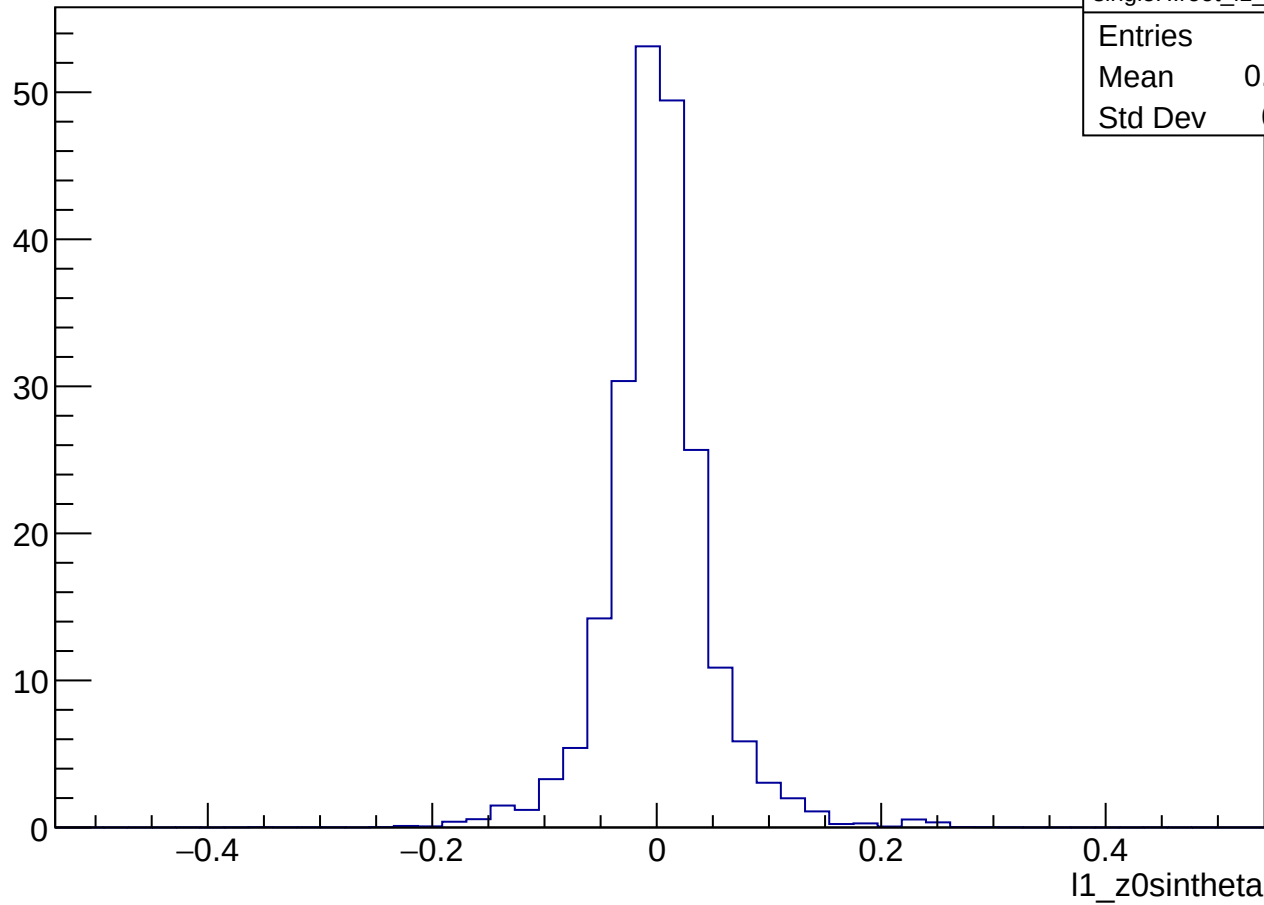
l1_pt

Events



l1_z0sintheta

Events



singleH.root_l1_z0sintheta	
Entries	49549
Mean	0.001705
Std Dev	0.04792

m_all

Events

50

40

30

20

10

0

0

1000

2000

3000

4000

5000

6000

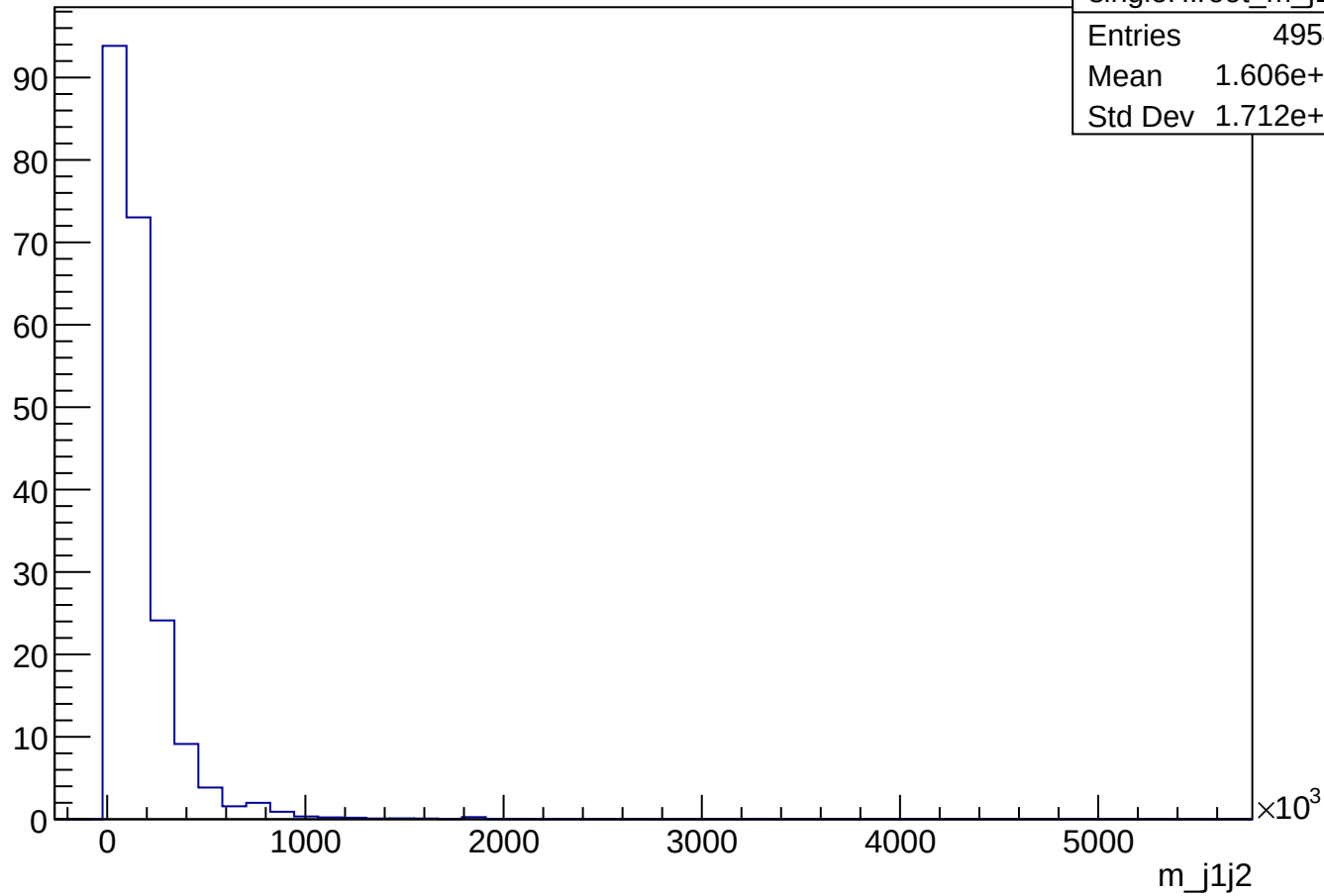
m_all

$\times 10^3$

singleH.root_m_all	
Entries	49549
Mean	5.314e+05
Std Dev	3.31e+05

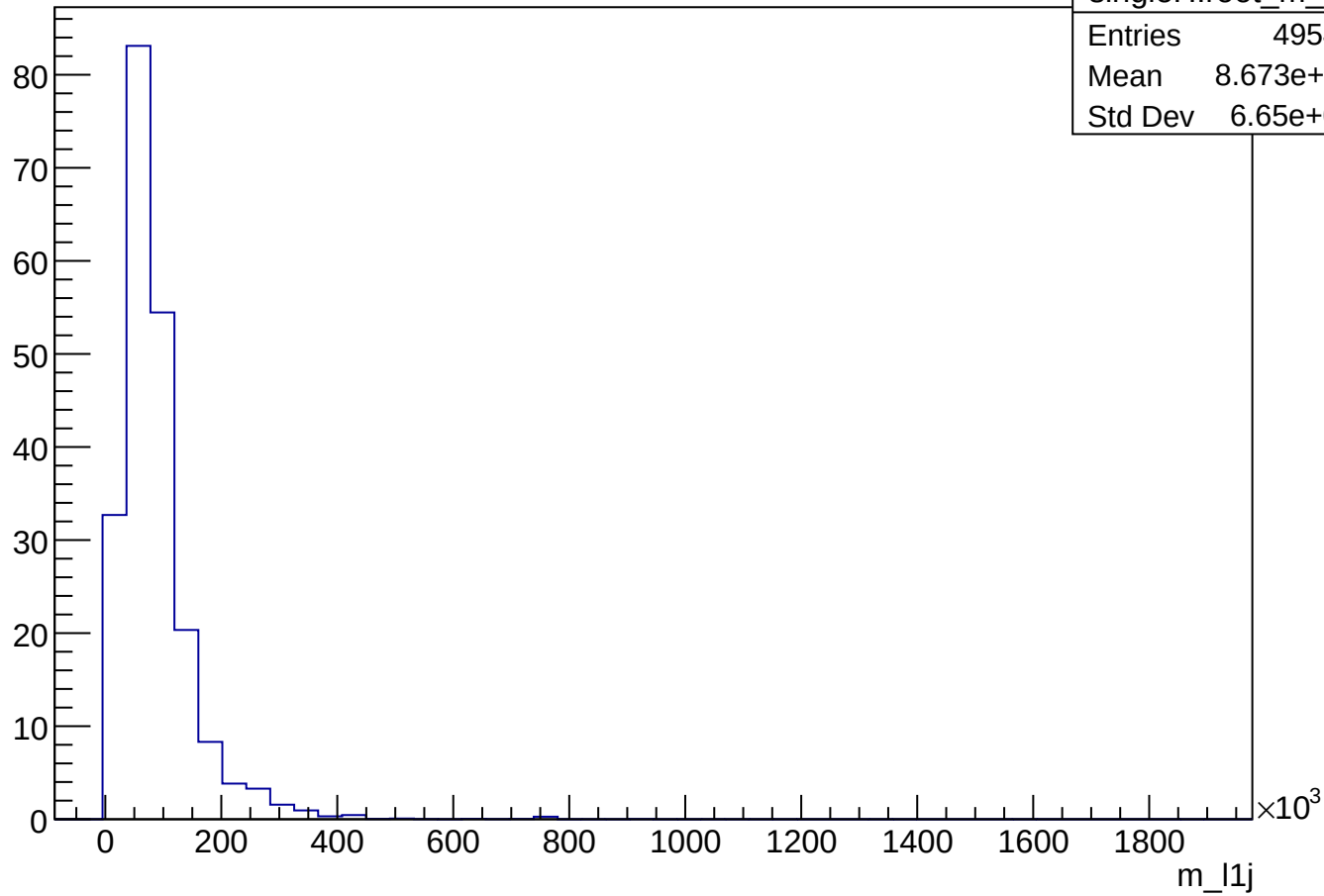
m_j1j2

Events



m_l1j

Events



m_l1j1

Events

50

40

30

20

10

0

0

500

1000

1500

2000

m_l1j1

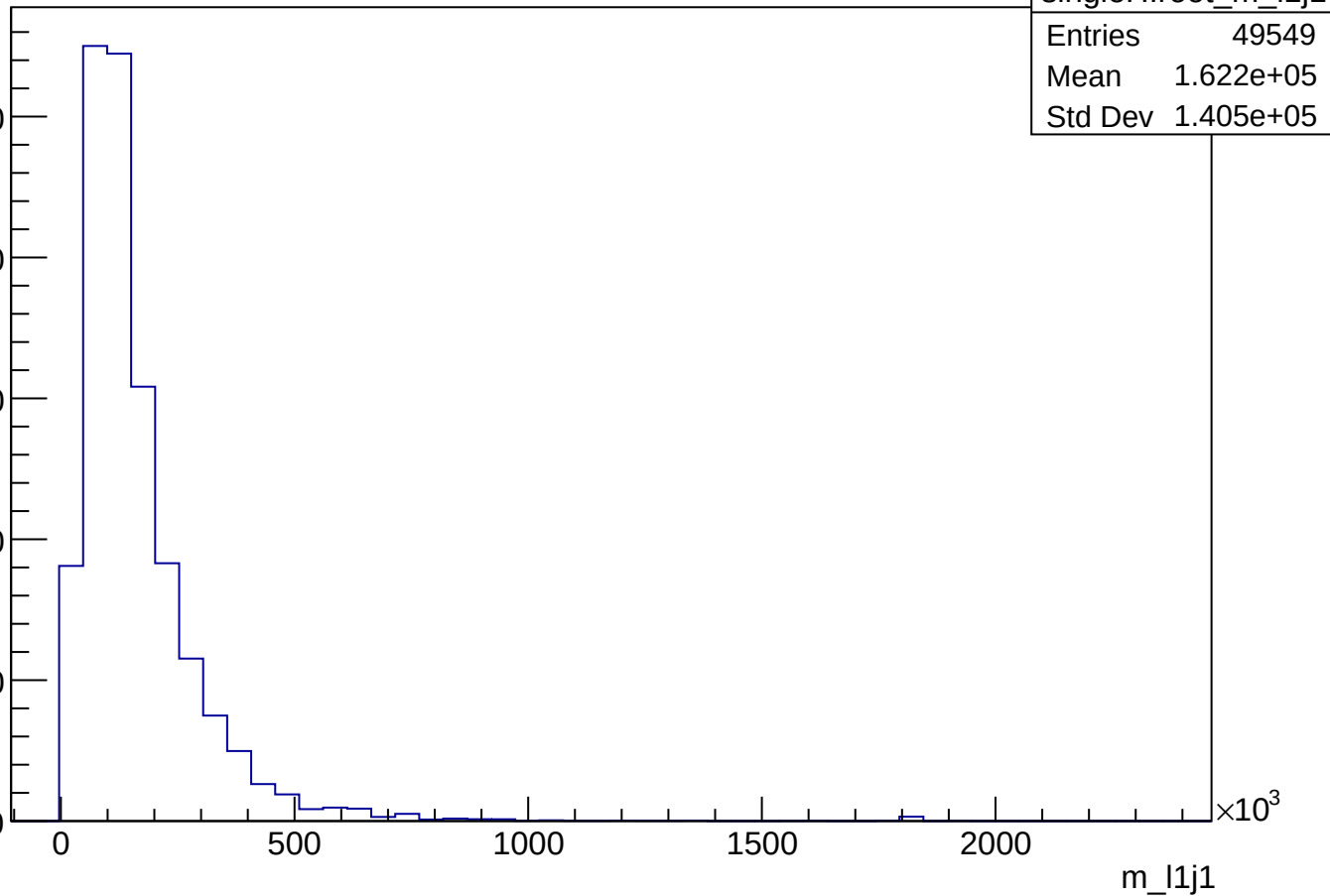
$\times 10^3$

singleH.root_m_l1j1

Entries 49549

Mean 1.622e+05

Std Dev 1.405e+05



m_l1j2

Events

80
70
60
50
40
30
20
10
0

0

500

1000

1500

2000

m_l1j2

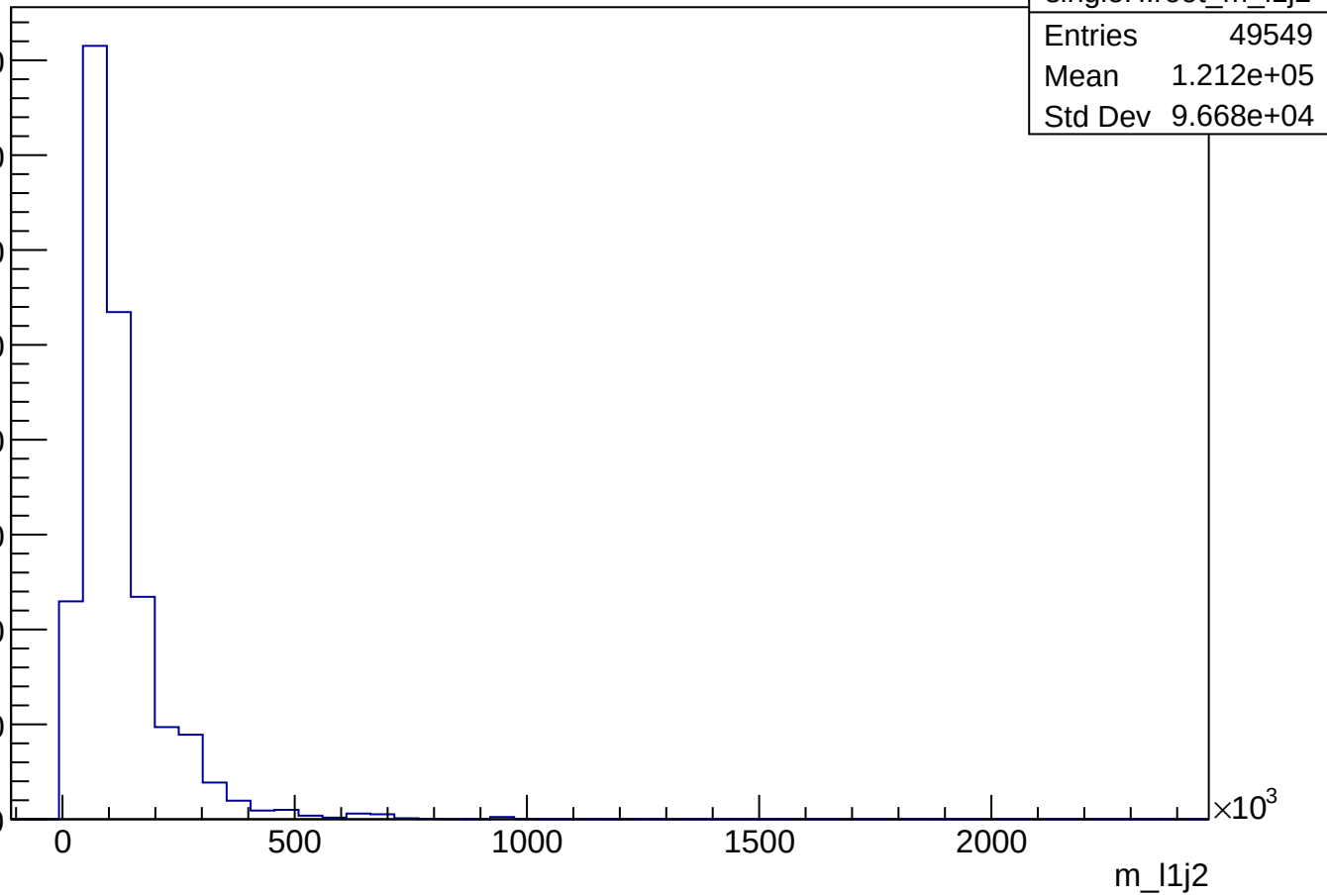
$\times 10^3$

singleH.root_m_l1j2

Entries 49549

Mean 1.212e+05

Std Dev 9.668e+04



m_l1t

Events

70
60
50
40
30
20
10
0

0

0

200

400

600

800

1000

1200

1400

1600

1800

2000

2200

2400

2600

2800

3000

3200

3400

3600

3800

4000

4200

4400

$\times 10^3$

m_l1t

singleH.root_m_l1t

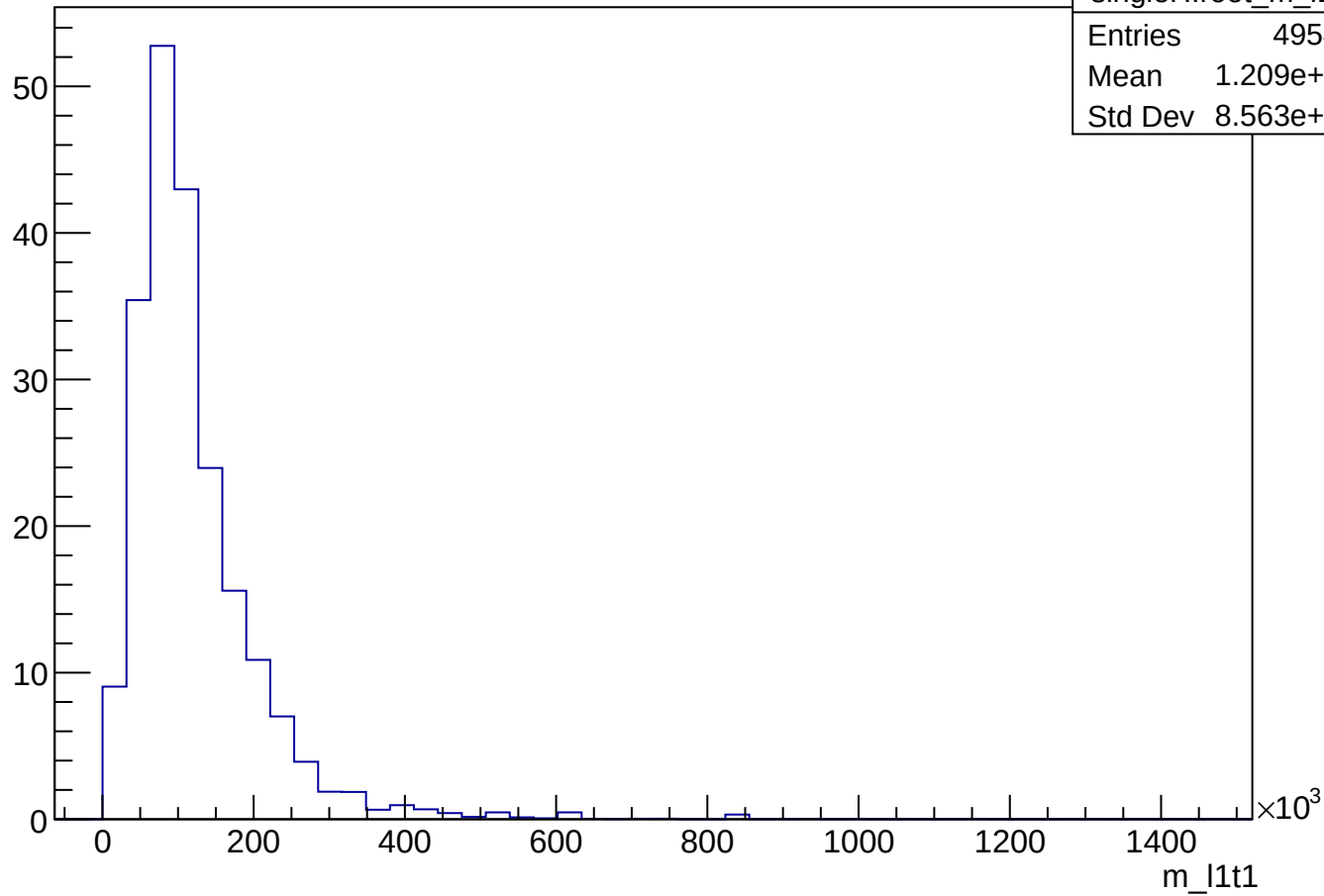
Entries 49549

Mean 8.43e+04

Std Dev 5.946e+04

m_l1t1

Events



m_l1t2

Events

50
40
30
20
10
0

0

200

400

600

800

1000

m_l1t2

$\times 10^3$

singleH.root_m_l1t2	
Entries	49549
Mean	8.681e+04
Std Dev	5.281e+04

m_t1j1

Events

60
50
40
30
20
10
0

0

500

1000

1500

2000

2500

m_t1j1

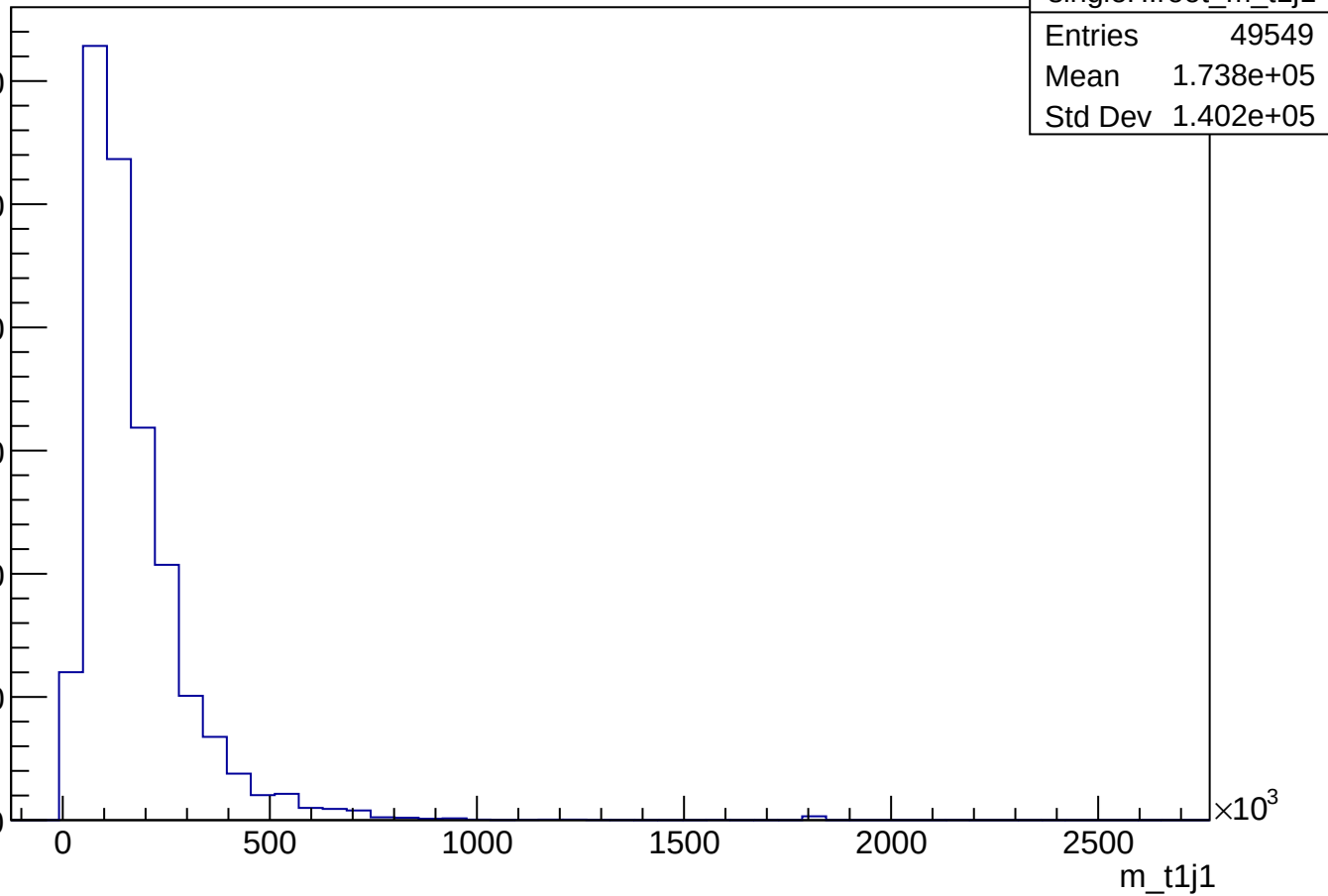
$\times 10^3$

singleH.root_m_t1j1

Entries 49549

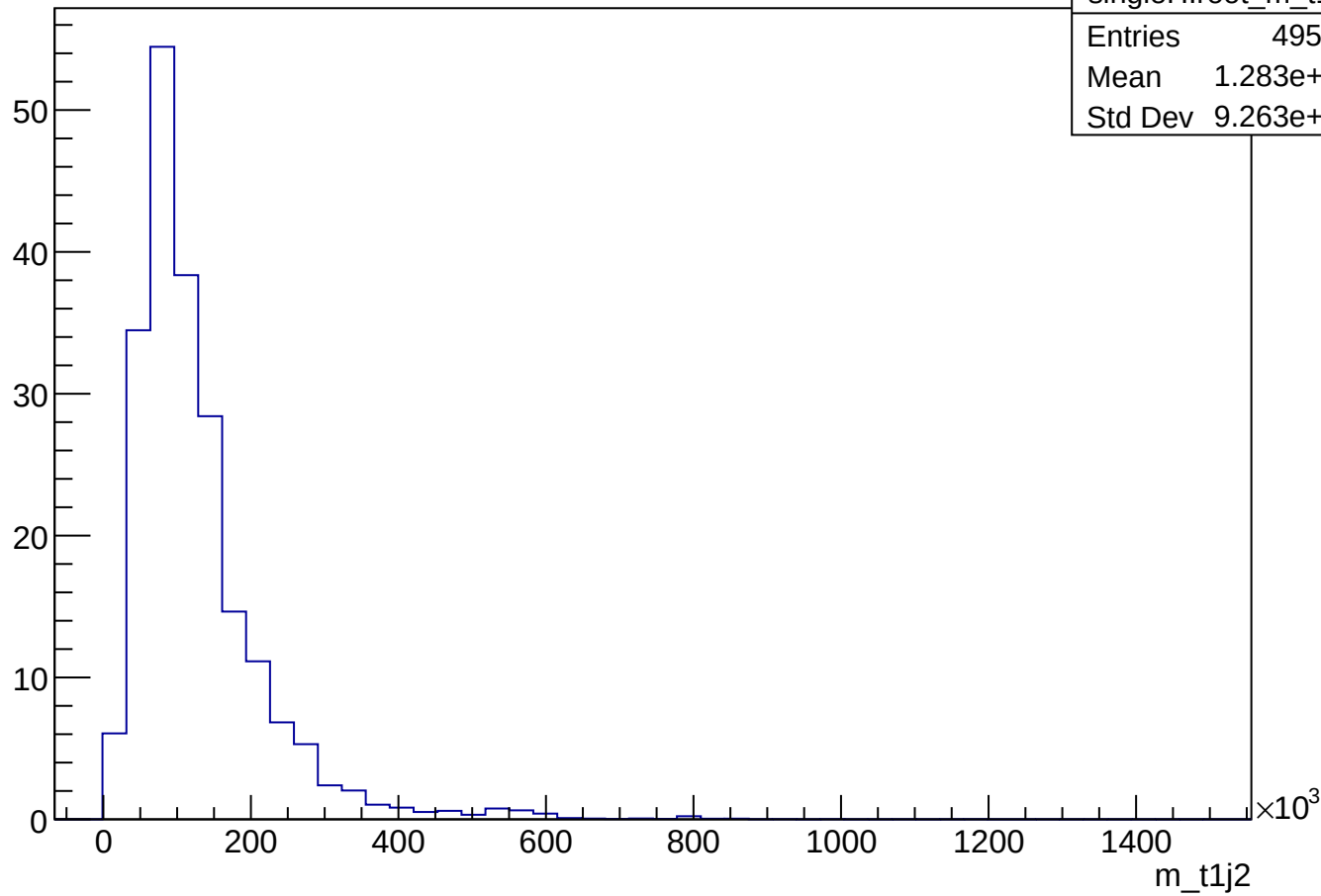
Mean 1.738e+05

Std Dev 1.402e+05



m_t1j2

Events



m_t1t2

Events

70
60
50
40
30
20
10
0

0

200

400

600

800

1000

1200

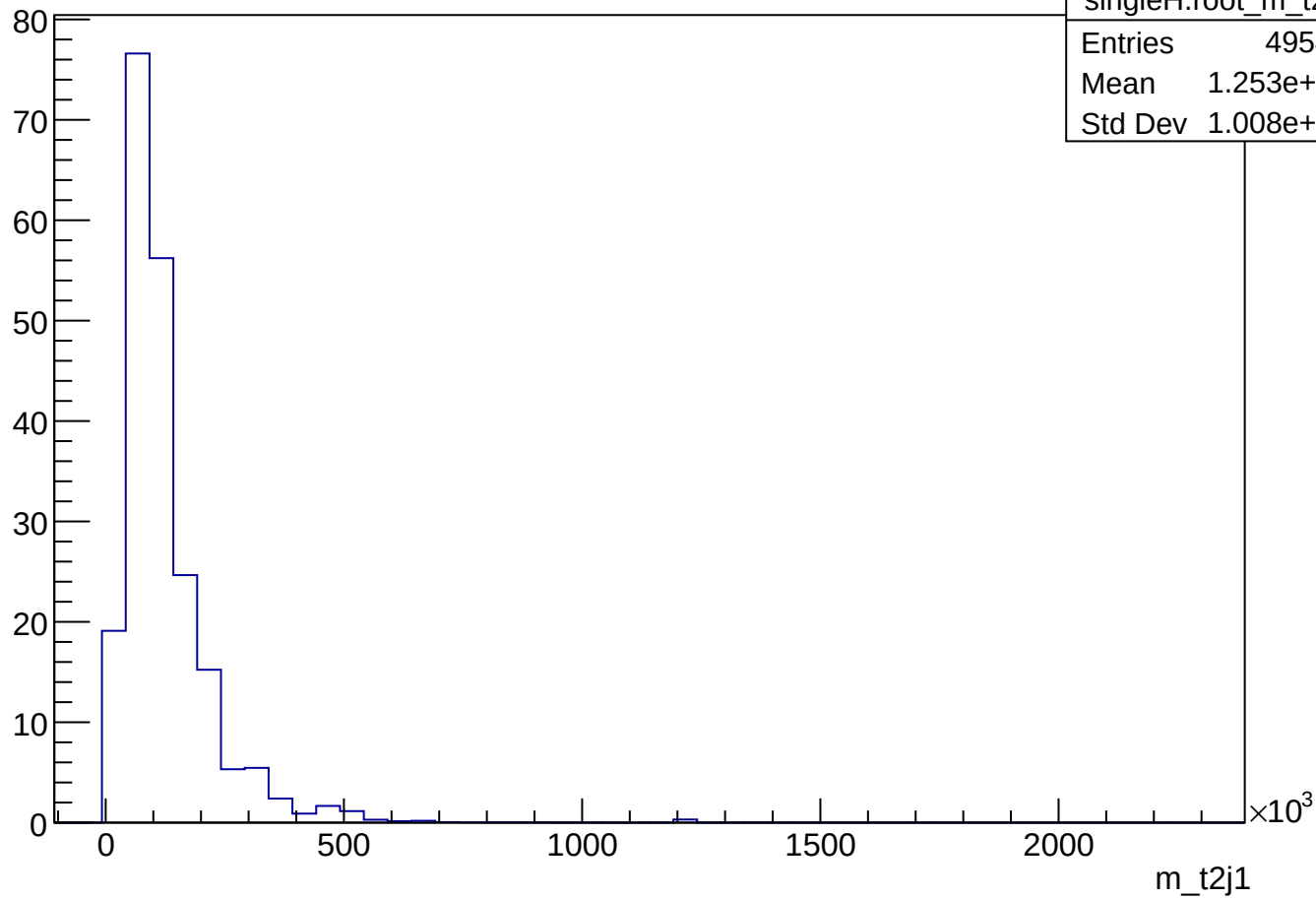
m_t1t2

$\times 10^3$

singleH.root_m_t1t2	
Entries	49549
Mean	8.453e+04
Std Dev	4.153e+04

m_t2j1

Events



m_t2j2

Events

60
50
40
30
20
10
0

0

0

200

400

600

800

1000

1200

1400

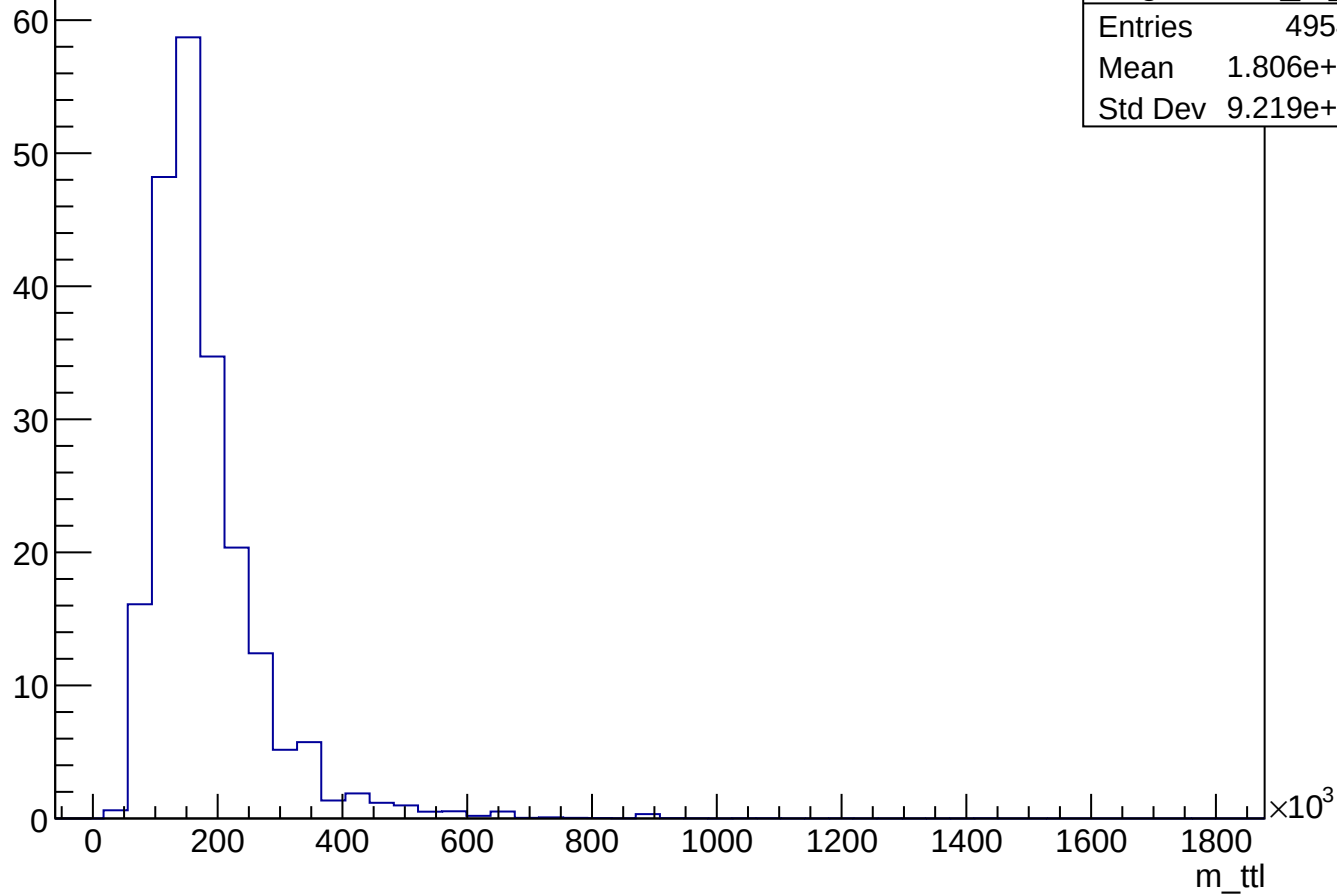
m_t2j2

$\times 10^3$

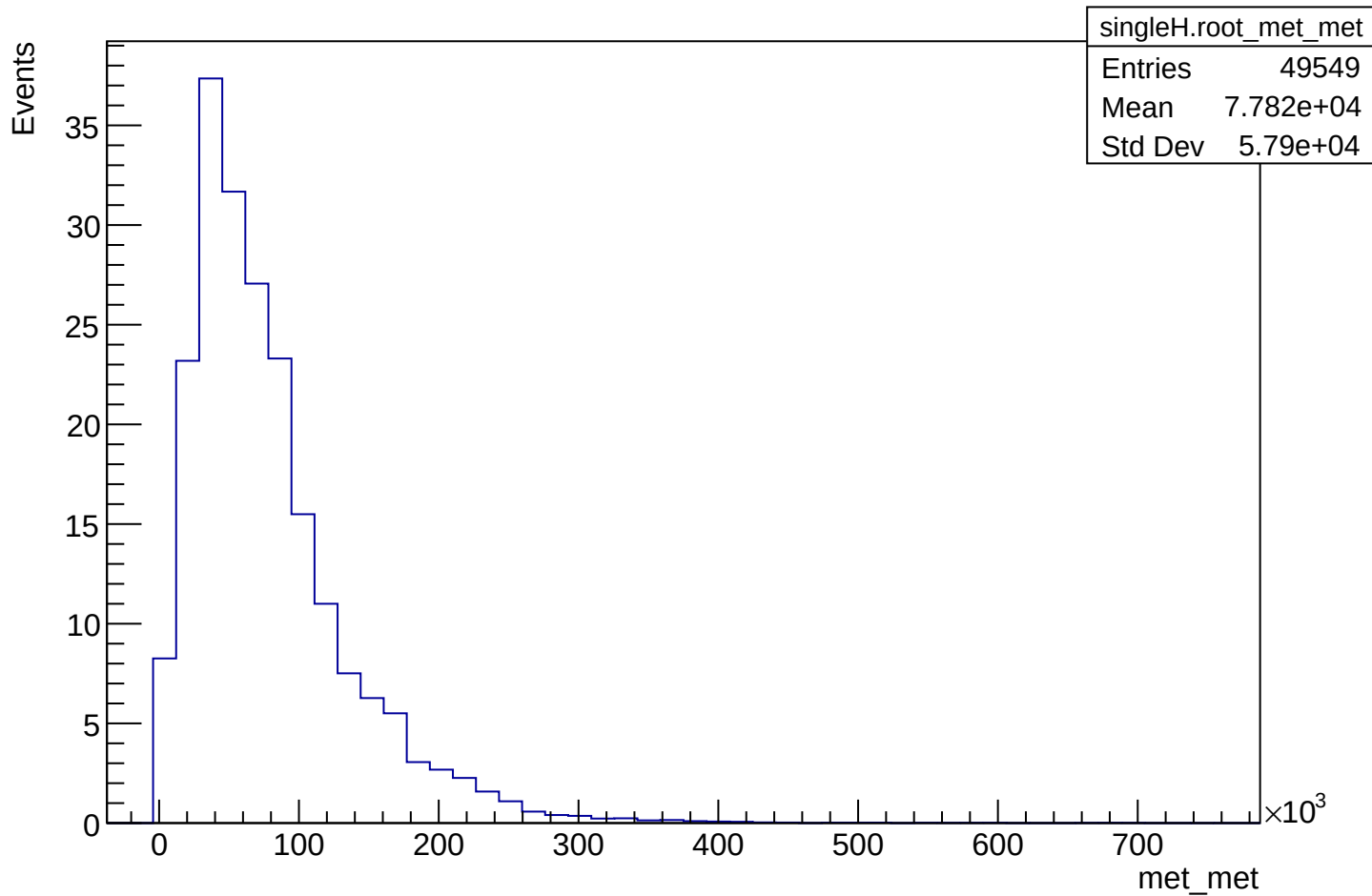
singleH.root_m_t2j2	
Entries	49549
Mean	9.391e+04
Std Dev	6.501e+04

m_ttl

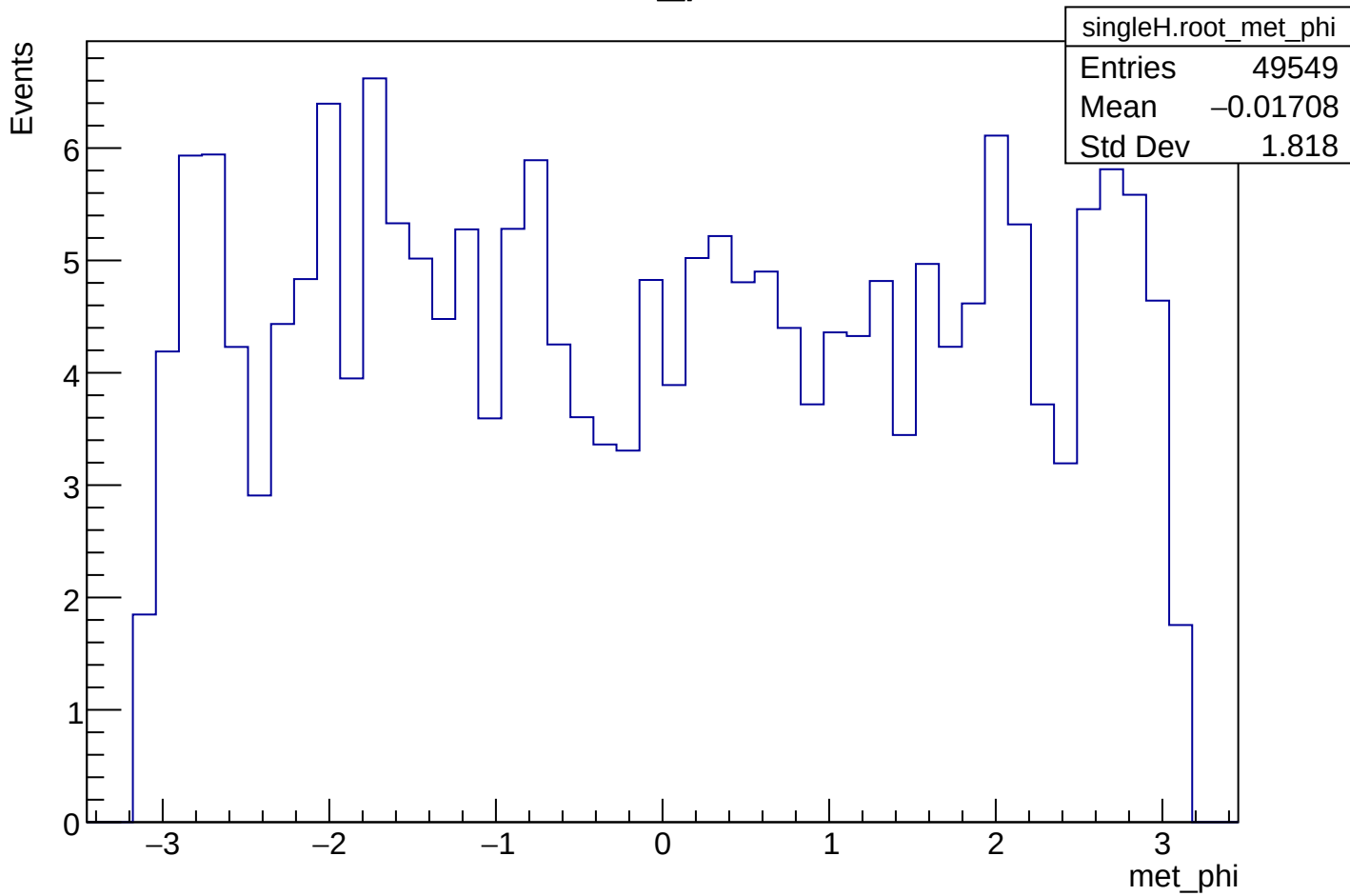
Events



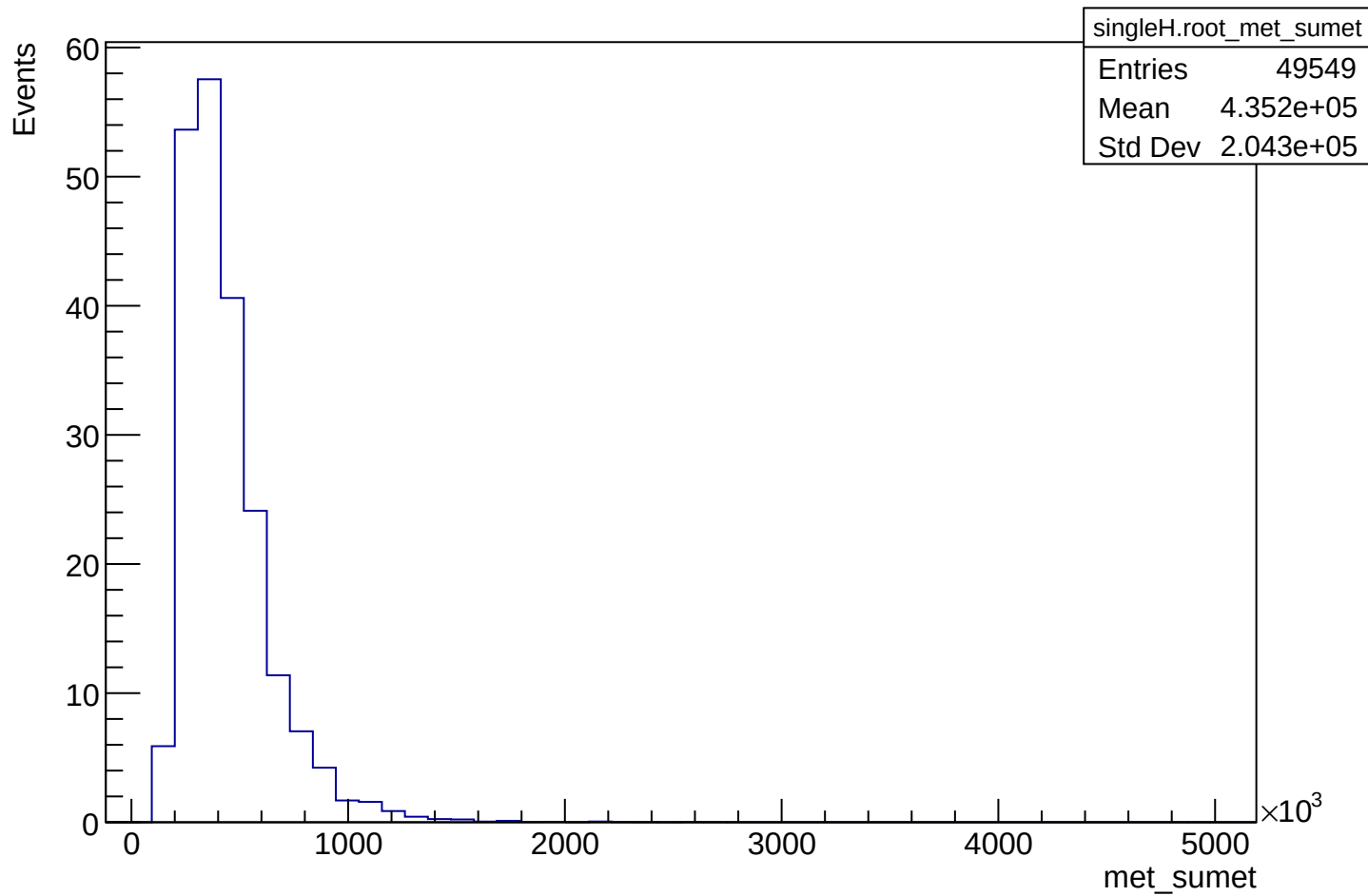
met_met



met_phi



met_sumet



n_jet

Events

100

80

60

40

20

0

2

4

6

8

10

12

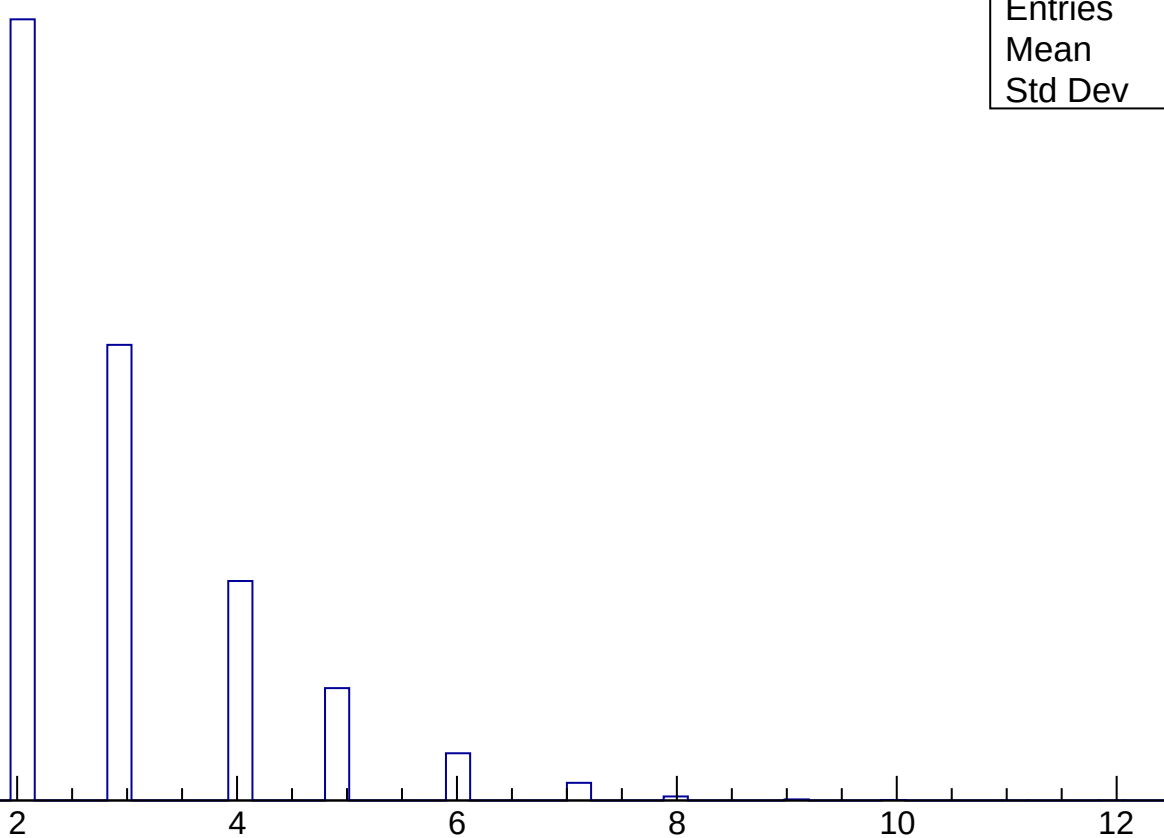
n_jet

singleH.root_n_jet

Entries 49549

Mean 2.943

Std Dev 1.19



pair_isOSleptau

Events

120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

pair_isOSleptau

singleH.root_pair_isOSleptau	
Entries	49549
Mean	0.6119
Std Dev	0.4873

pair_isOStaus

Events

160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

pair_isOStaus

singleH.root_pair_isOStaus	
Entries	49549
Mean	0.759
Std Dev	0.4277

pass_DTT

Events

180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

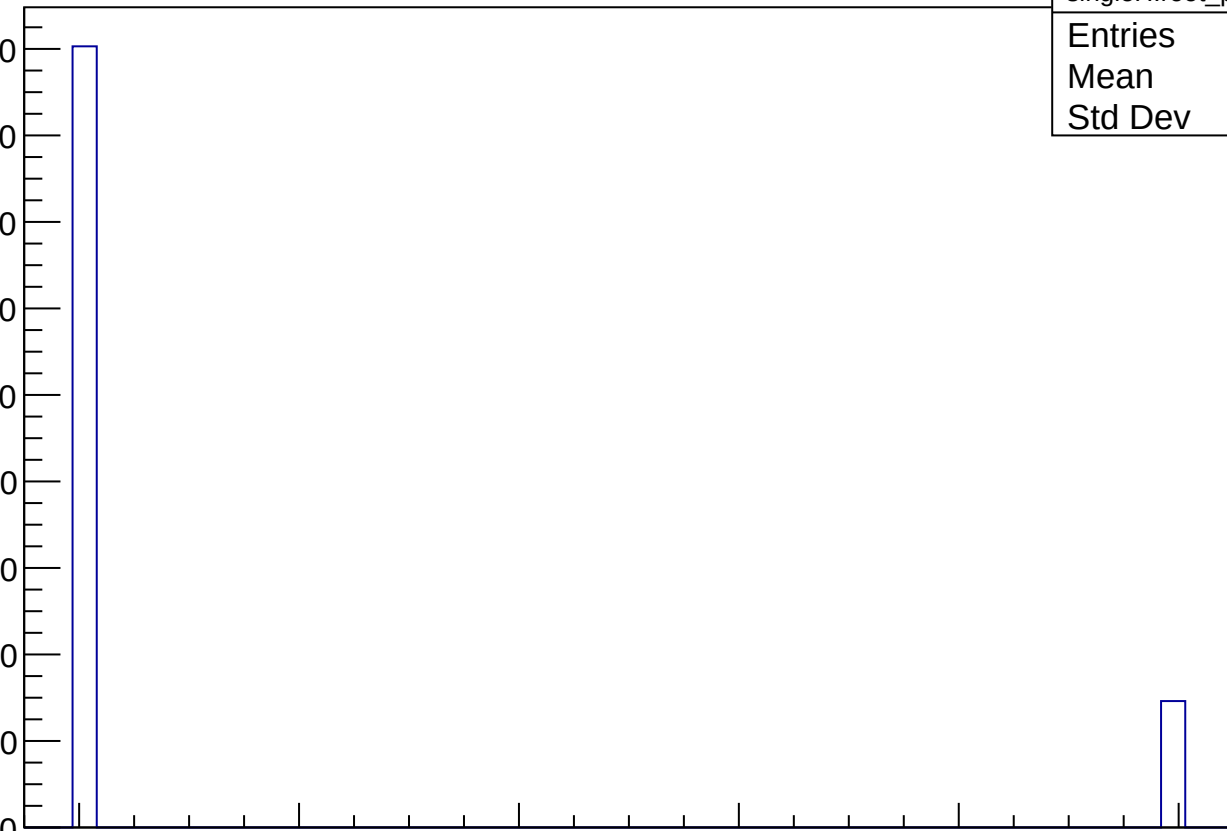
0.6

0.8

1

pass_DTT

singleH.root_pass_DTT	
Entries	49549
Mean	0.1393
Std Dev	0.3462



pass_LTT

Events

160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

pass_LTT

singleH.root_pass_LTT	
Entries	49549
Mean	0.7278
Std Dev	0.4451

pass_SLT

Events

180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

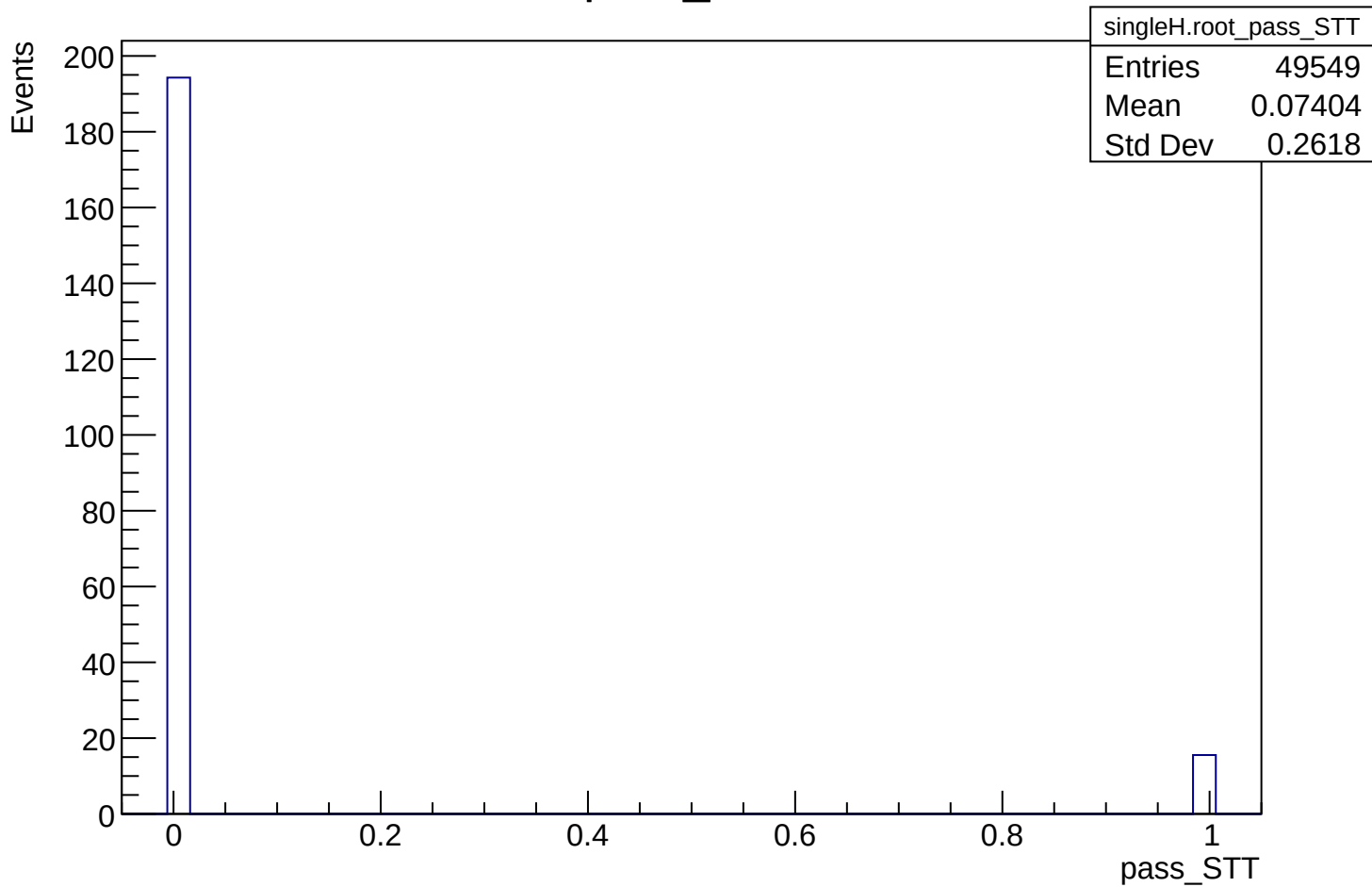
0.8

1

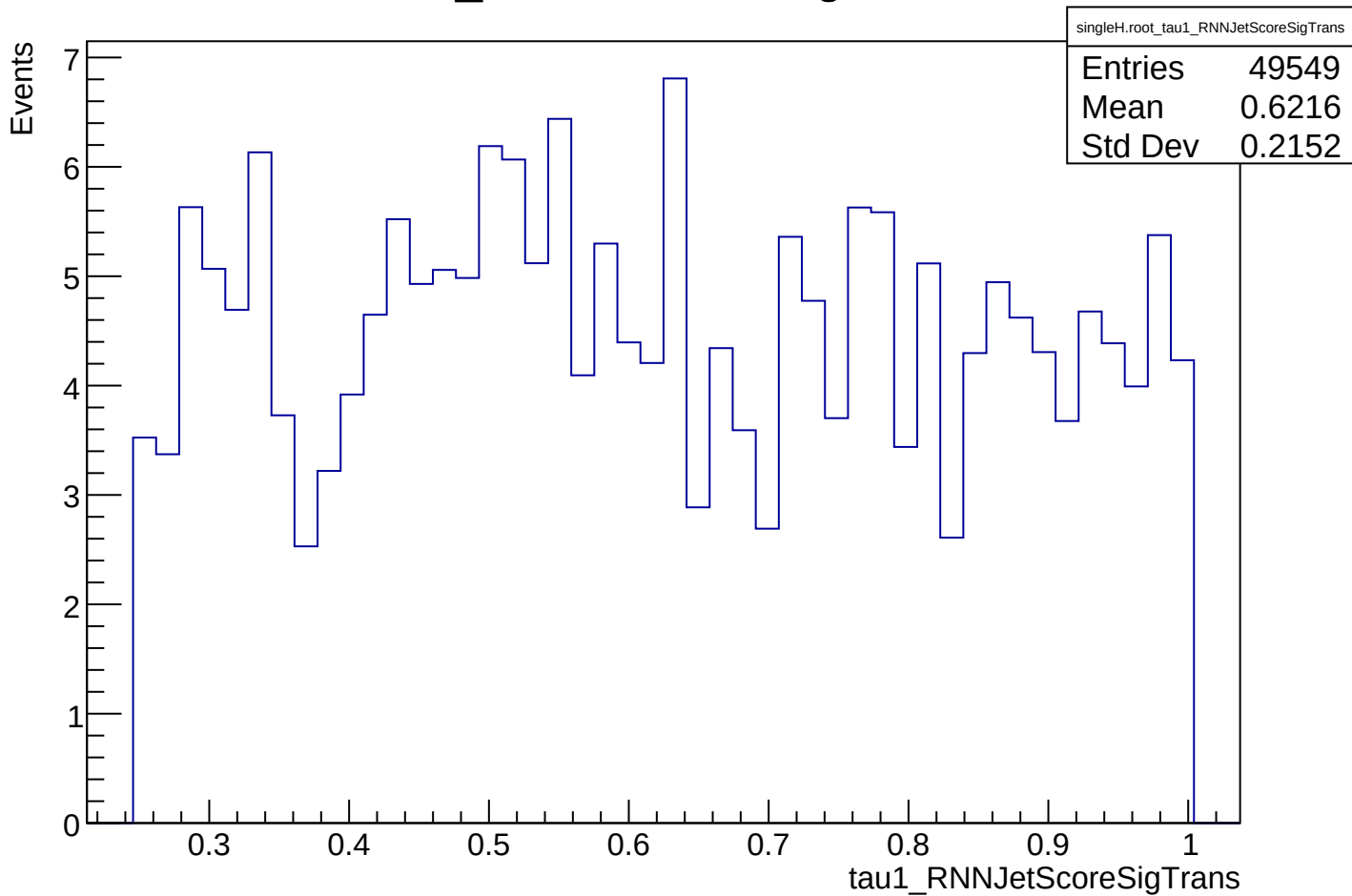
pass_SLT

singleH.root_pass_SLT	
Entries	49549
Mean	0.8323
Std Dev	0.3736

pass_STT

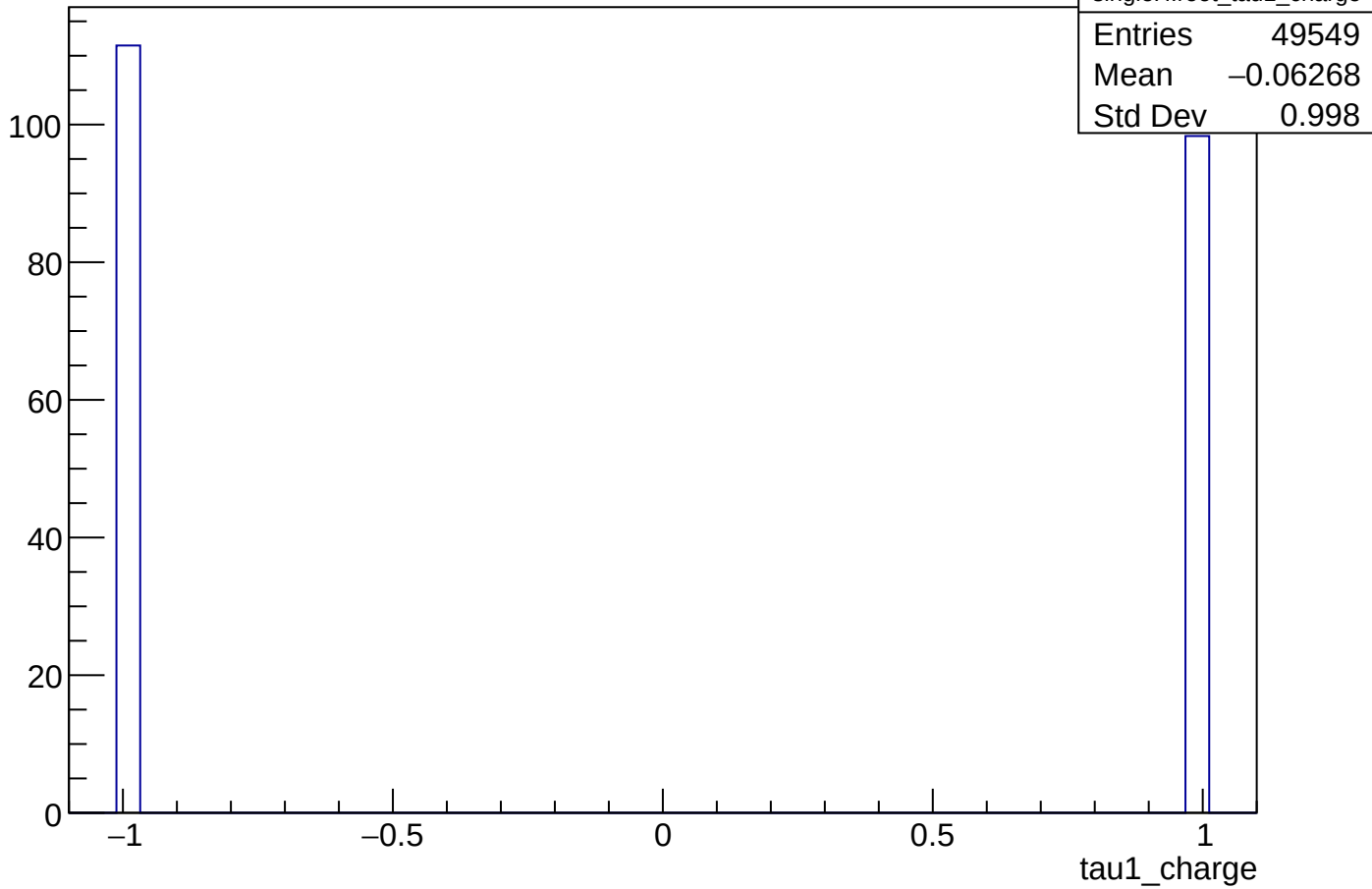


tau1_RNNJetScoreSigTrans

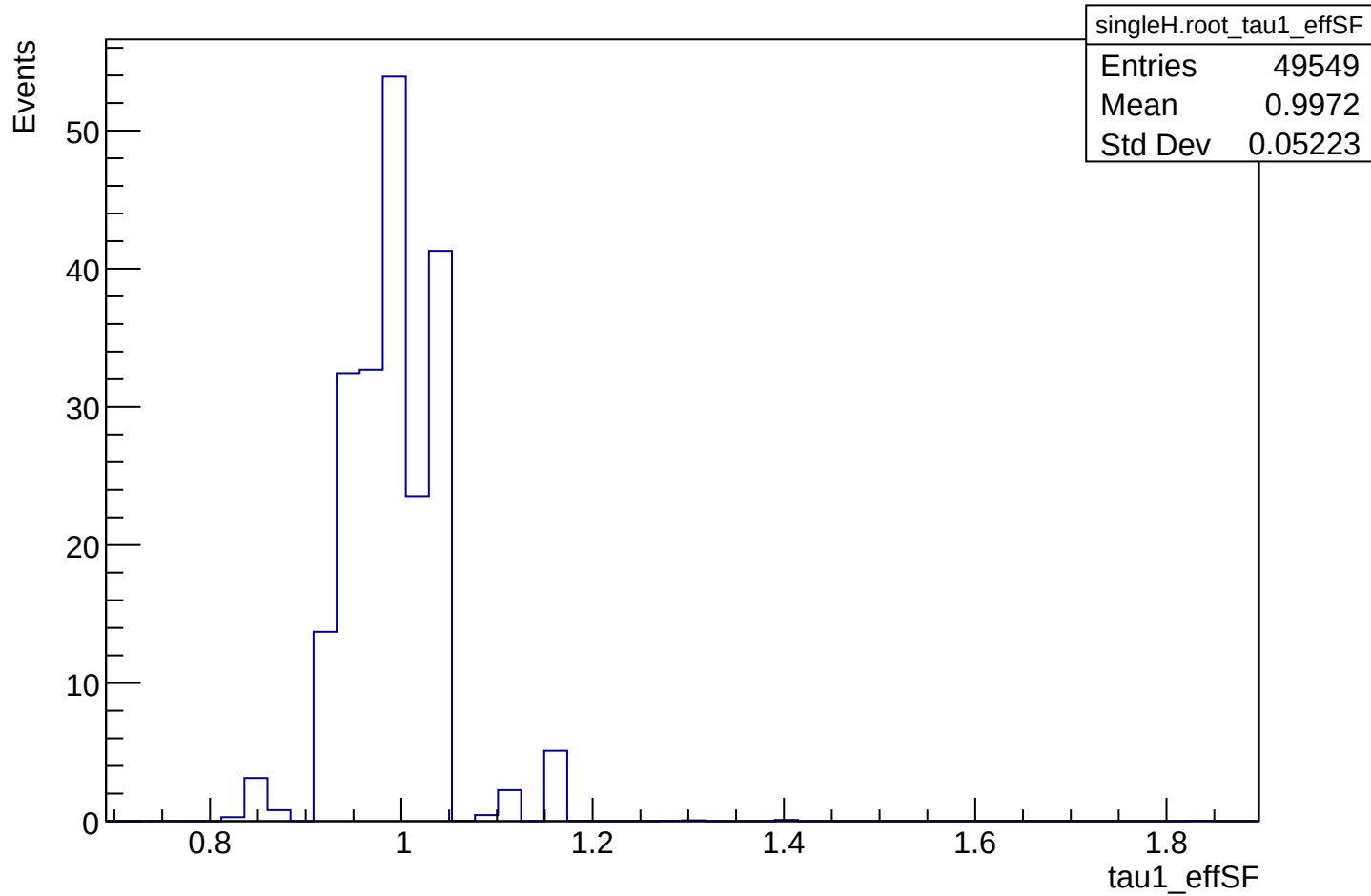


tau1_charge

Events

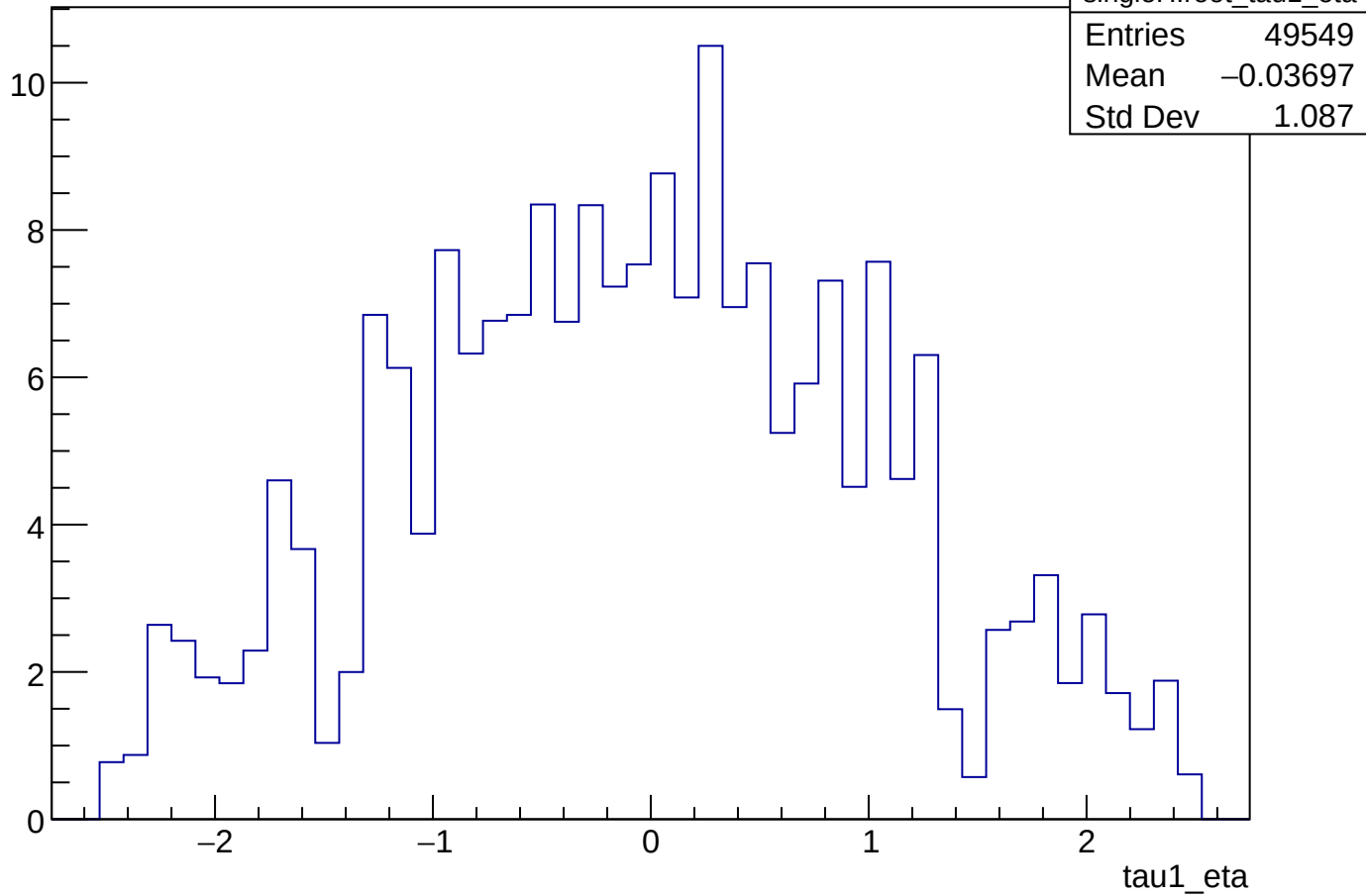


tau1_effSF



tau1_eta

Events



tau1_nprong

Events

160
140
120
100
80
60
40
20
0

1

1.5

2

2.5

3

tau1_nprong

singleH.root_tau1_nprong

Entries 49549

Mean 1.433

Std Dev 0.8237

1

1.5

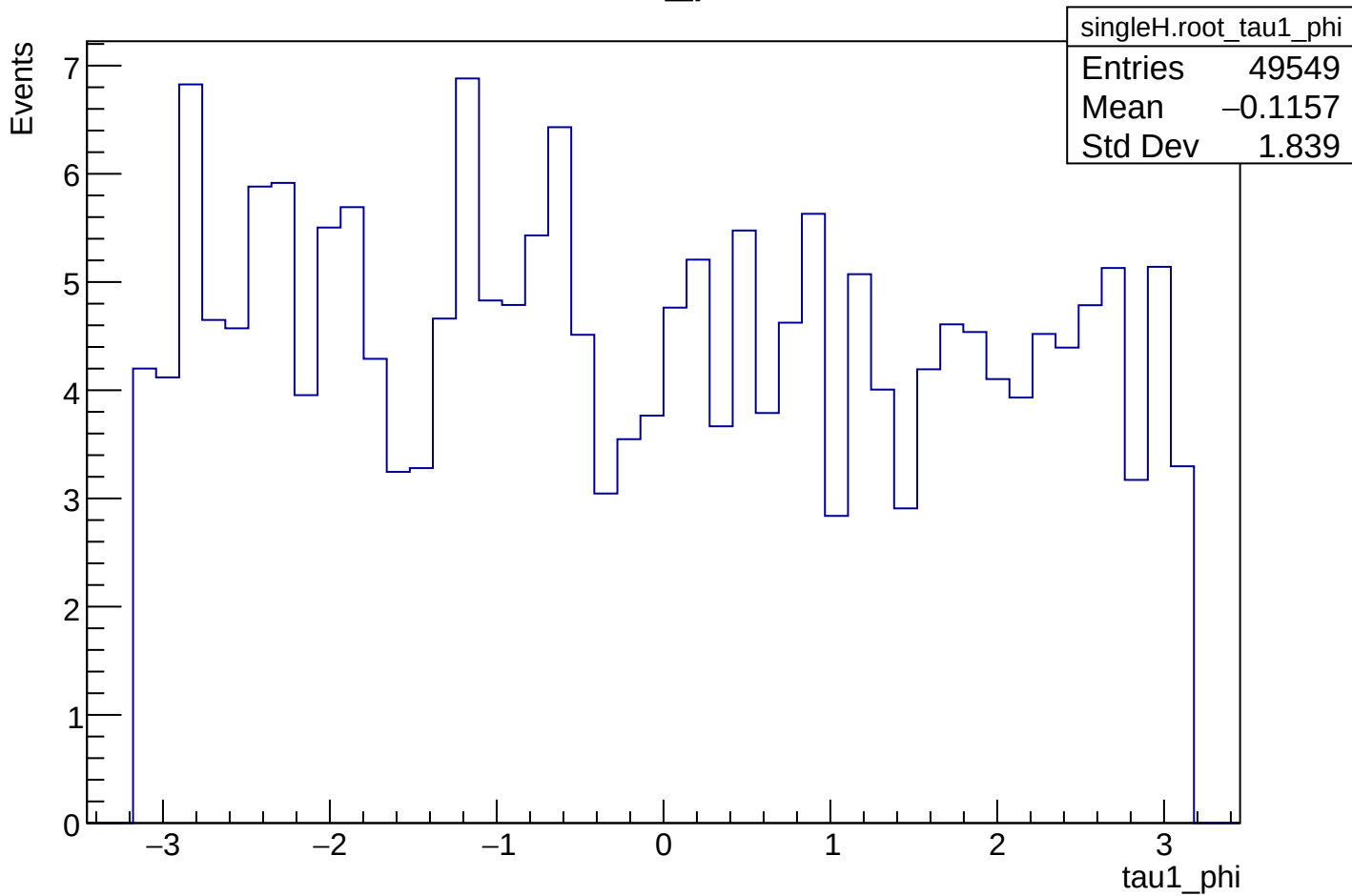
2

2.5

3

tau1_nprong

tau1_phi



tau1_pt

Events

50
40
30
20
10
0

0

100

200

300

400

500

600

700

tau1_pt

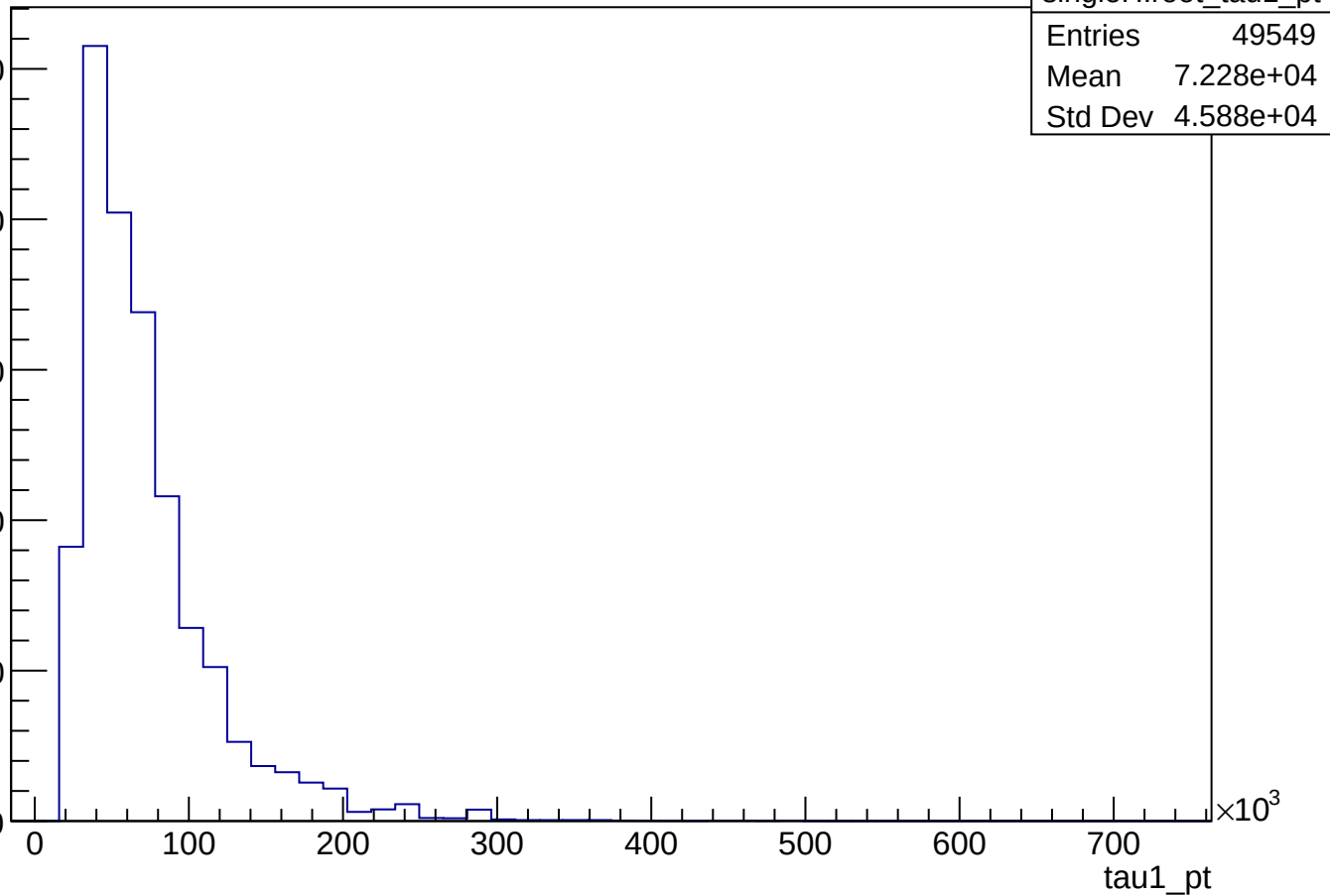
$\times 10^3$

singleH.root_tau1_pt

Entries 49549

Mean 7.228e+04

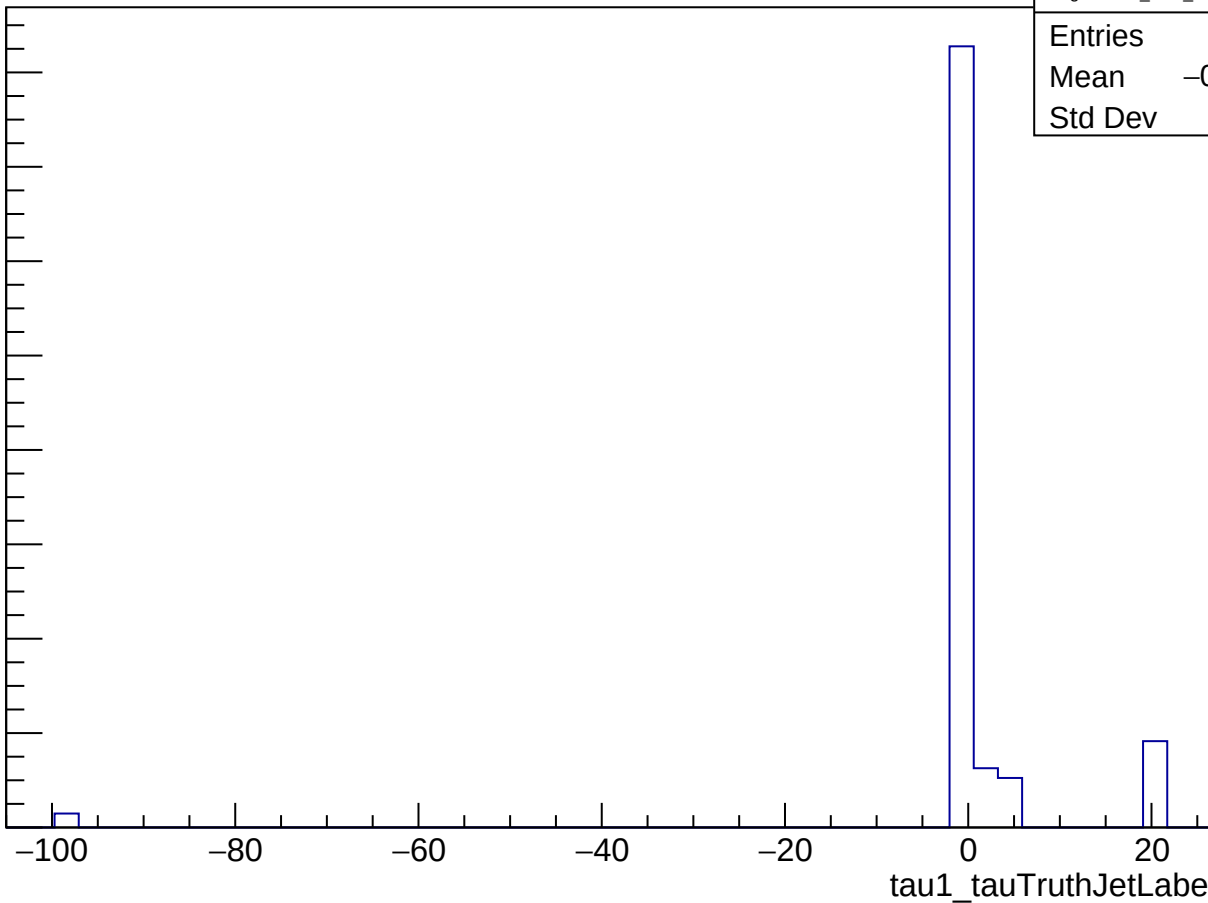
Std Dev 4.588e+04



tau1_tauTruthJetLabel

Events

160
140
120
100
80
60
40
20
0



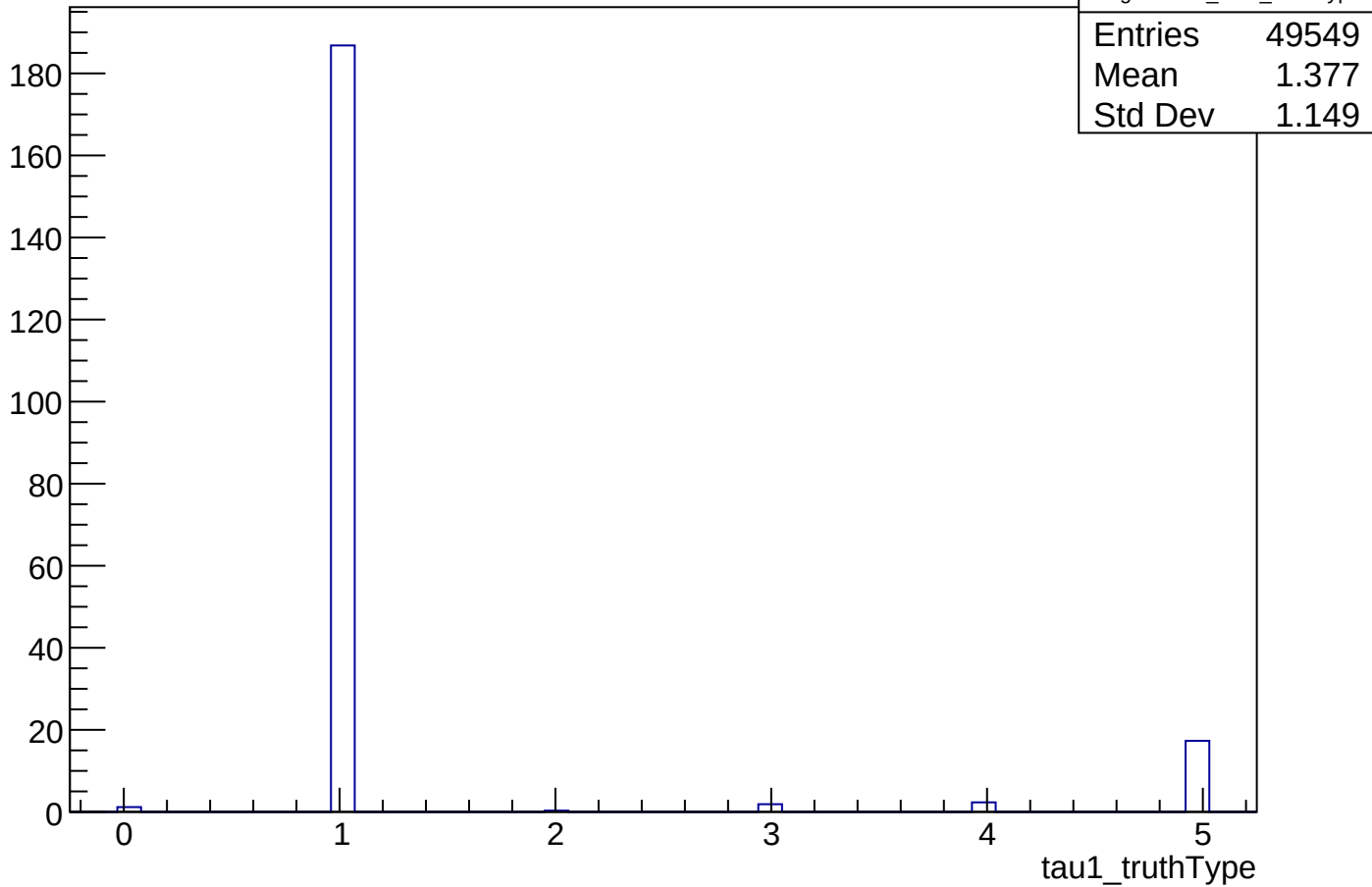
singleH.root_tau1_tauTruthJetLabel

Entries	49549
Mean	-0.009881
Std Dev	13.36

tau1_tauTruthJetLabel

tau1_truthType

Events



tau1_truth_IsHadronicTau

Events

180
160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

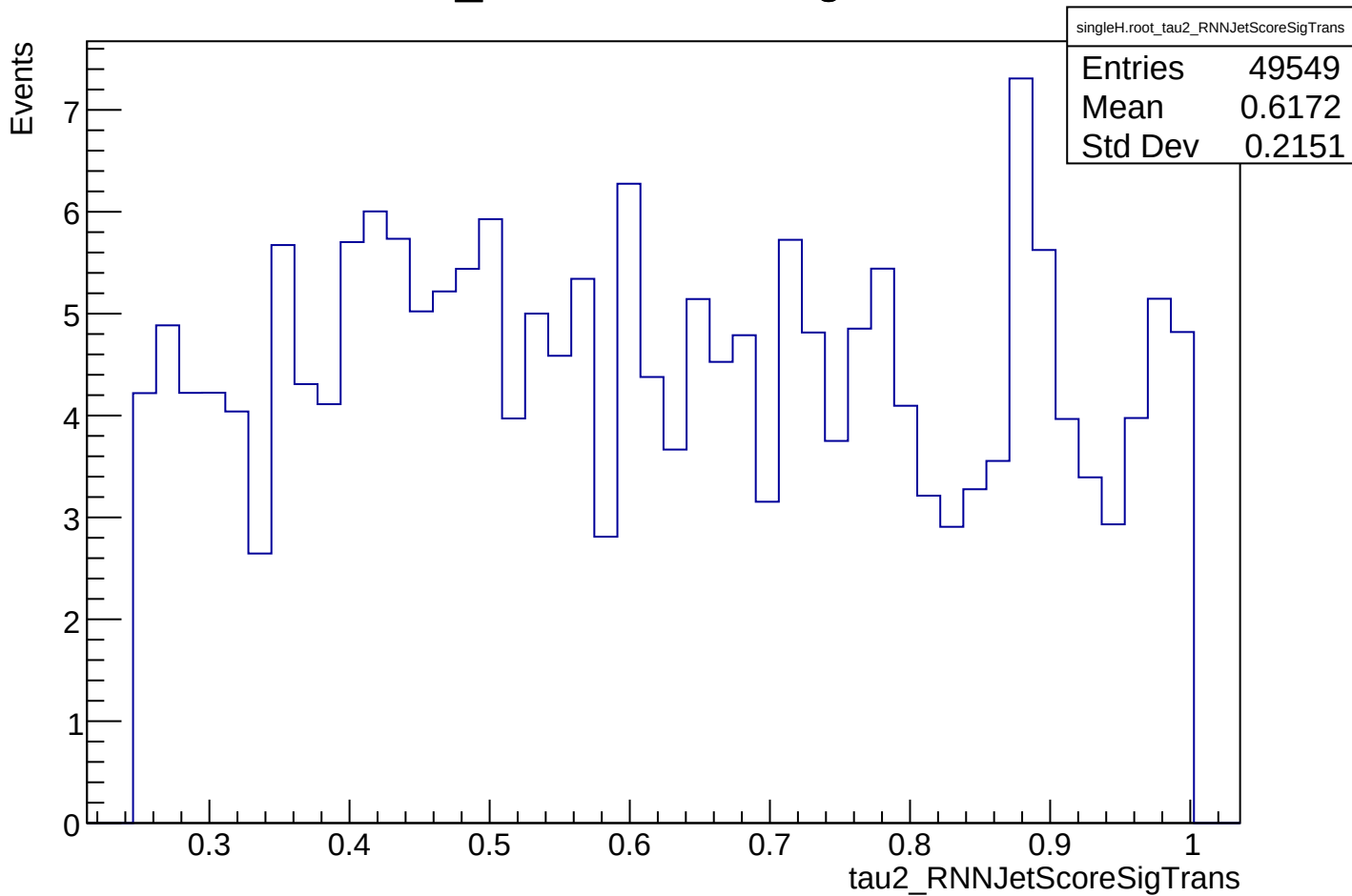
0.8

1

tau1_truth_IsHadronicTau

singleH.root_tau1_truth_IsHadronicTau	
Entries	49549
Mean	0.8903
Std Dev	0.3125

tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

Events

160
140
120
100
80
60
40
20
0

0

0.2

0.4

0.6

0.8

1

tau2_baseline_RNNTight

singleH.root_tau2_baseline_RNNTight	
Entries	49549
Mean	0.7437
Std Dev	0.4366

tau2_charge

Events

100

80

60

40

20

0

-1

-0.5

0

0.5

1

tau2_charge

singleH.root_tau2_charge	
Entries	49549
Mean	0.03752
Std Dev	0.9993

tau2_effSF

Events

70
60
50
40
30
20
10
0

0.8

1.2

1.4

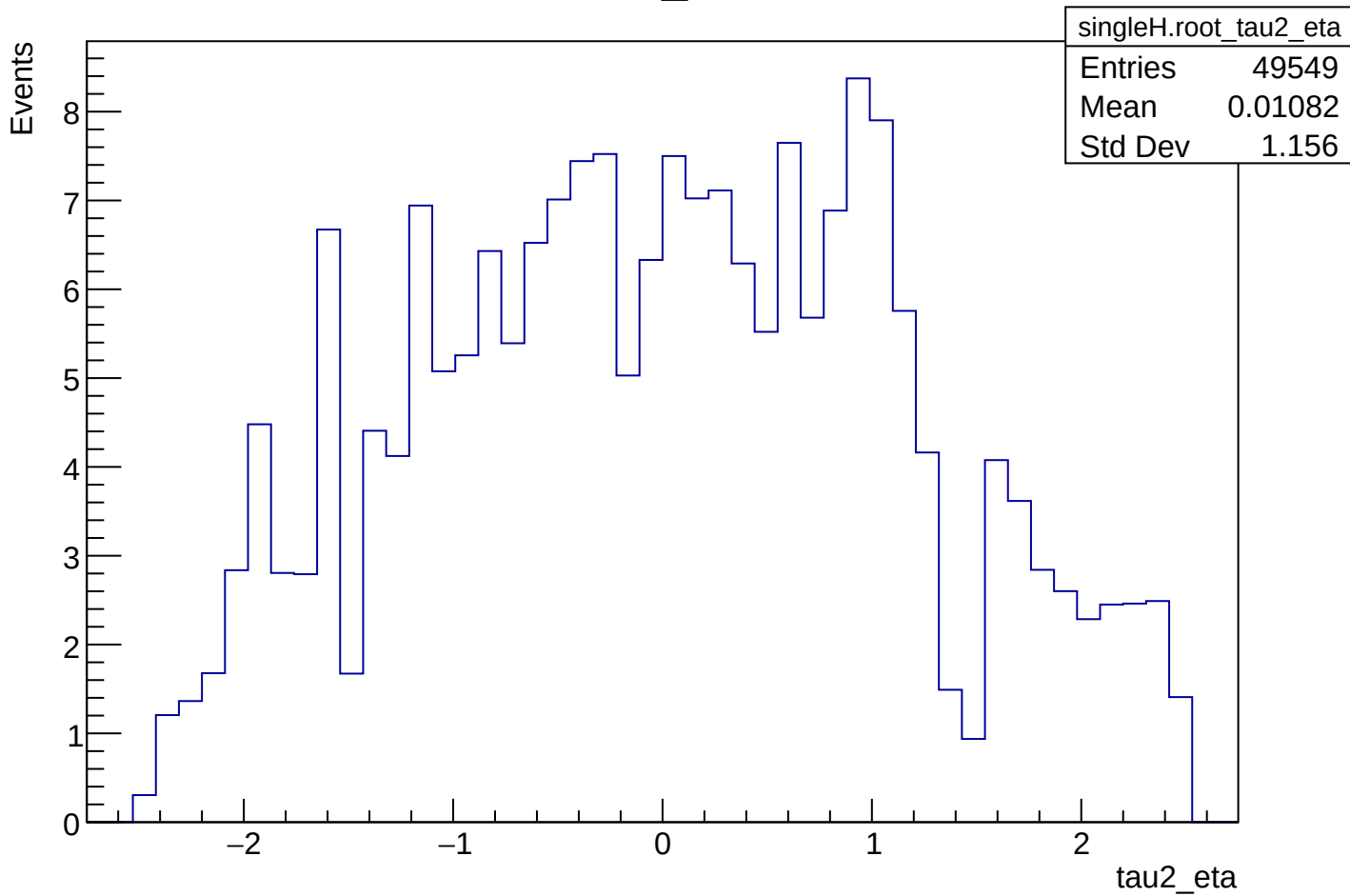
1.6

1.8

tau2_effSF

singleH.root_tau2_effSF	
Entries	49549
Mean	0.9774
Std Dev	0.05936

tau2_eta



tau2_nprong

Events

160
140
120
100
80
60
40
20
0

1

1.5

2

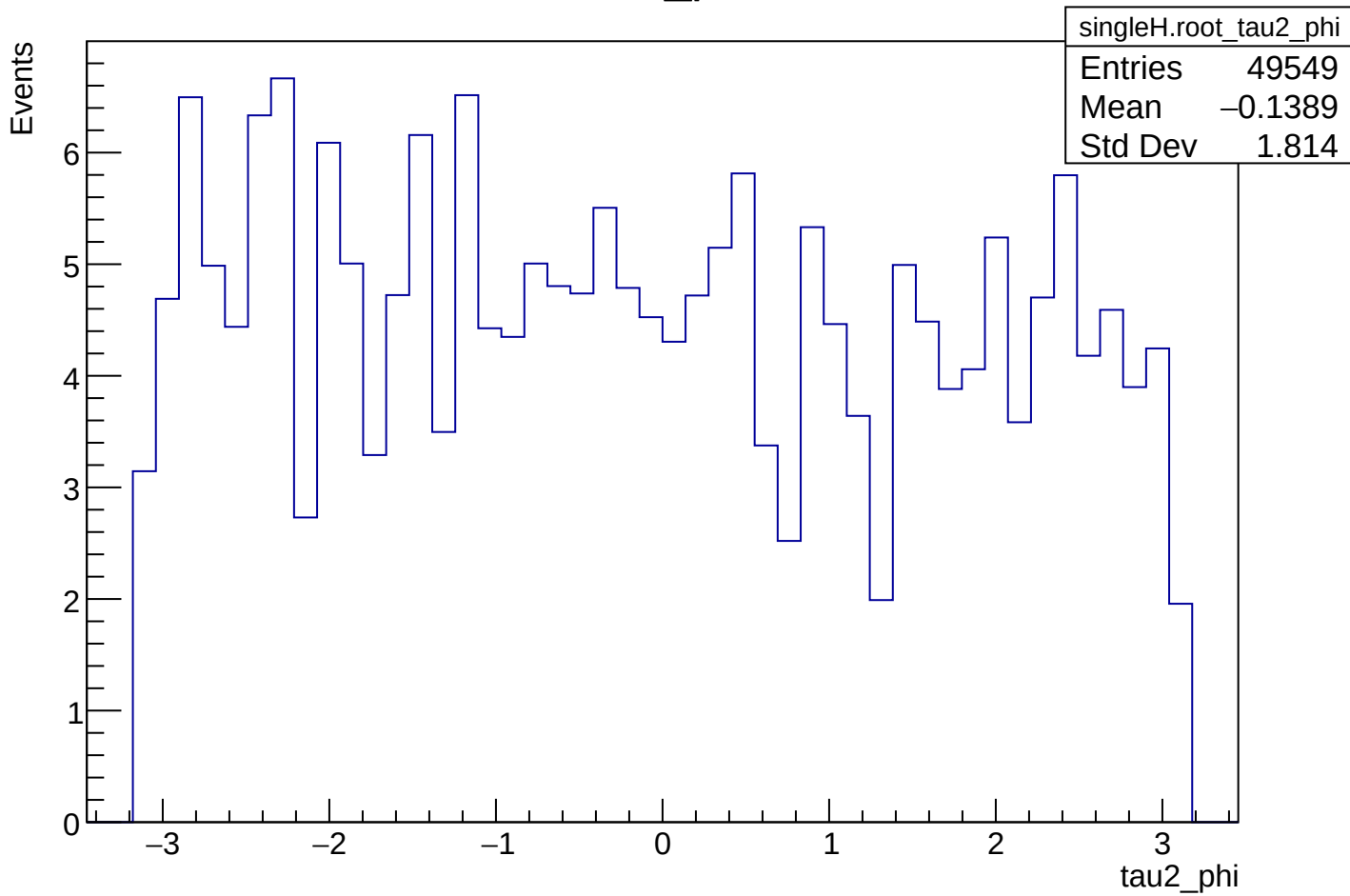
2.5

3

tau2_nprong

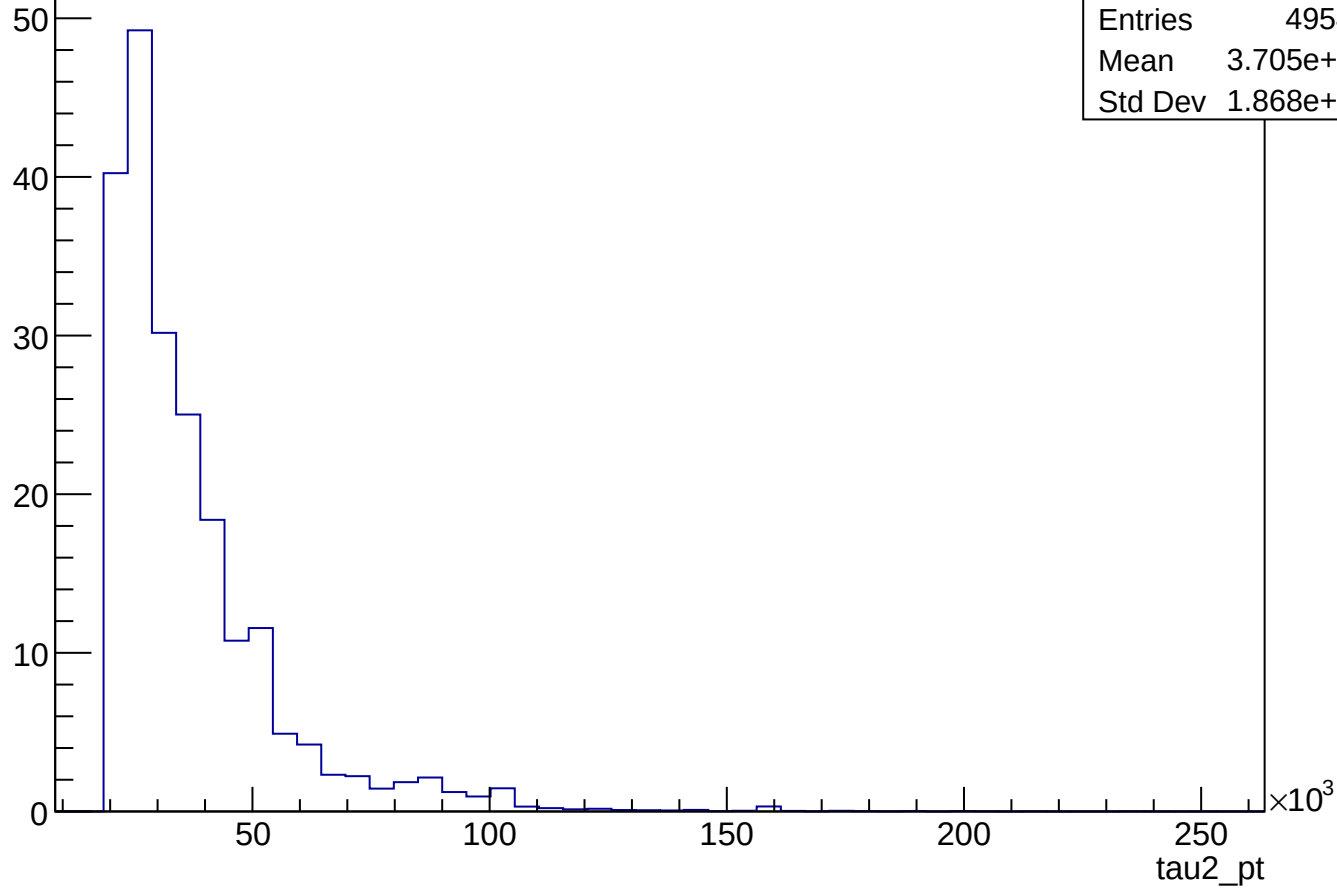
singleH.root_tau2_nprong	
Entries	49549
Mean	1.419
Std Dev	0.8142

tau2_phi



tau2_pt

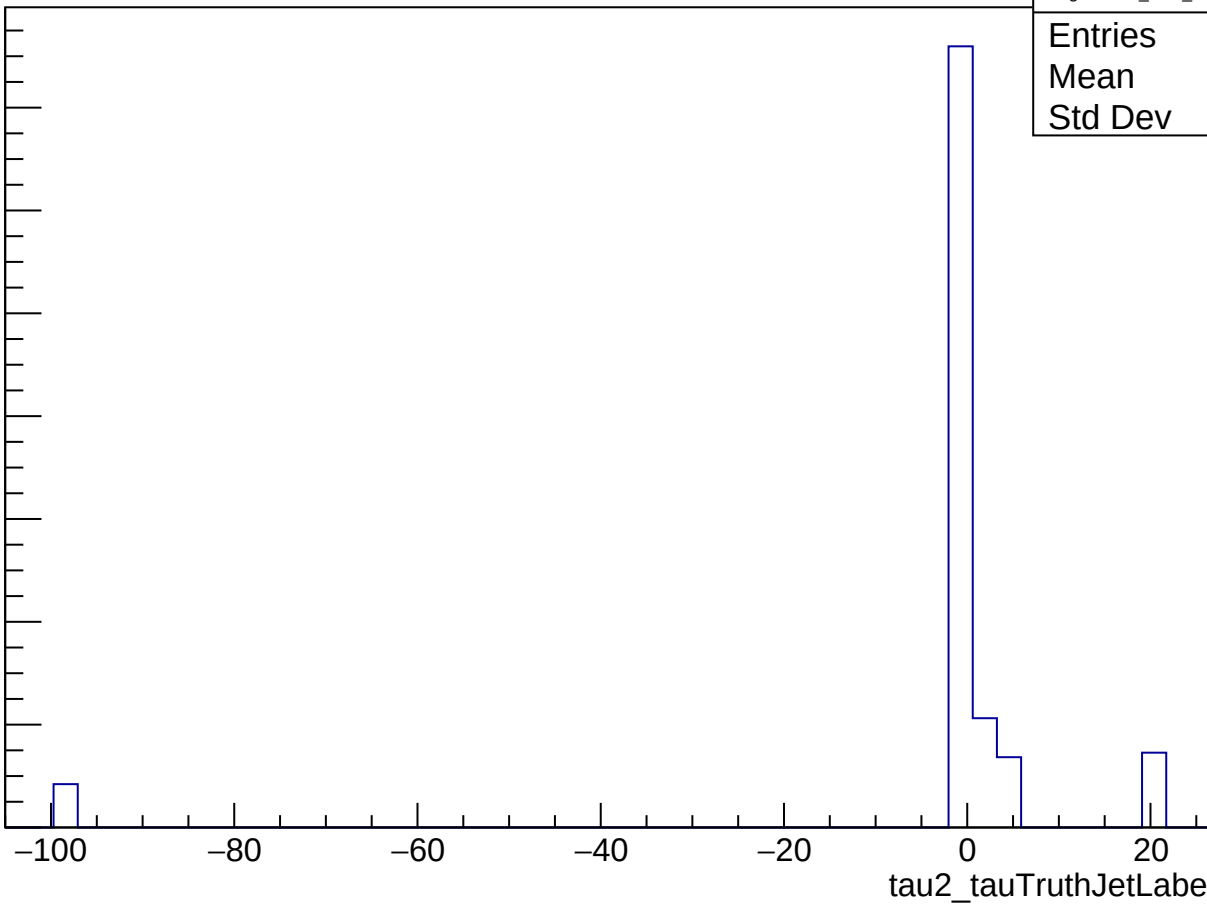
Events



tau2_tauTruthJetLabel

Events

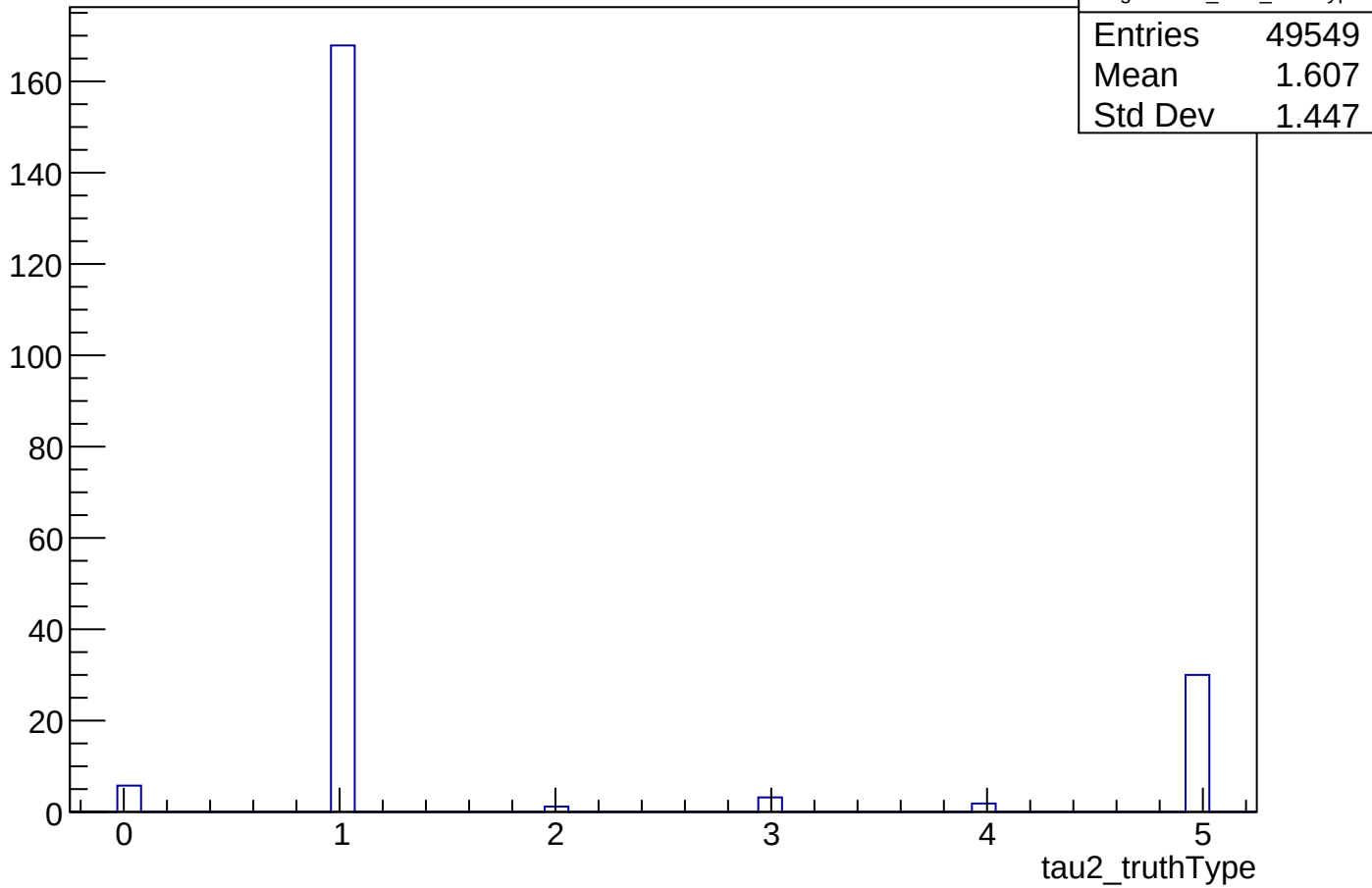
140
120
100
80
60
40
20
0



singleH.root_tau2_tauTruthJetLabel	
Entries	49549
Mean	-2.737
Std Dev	20.48

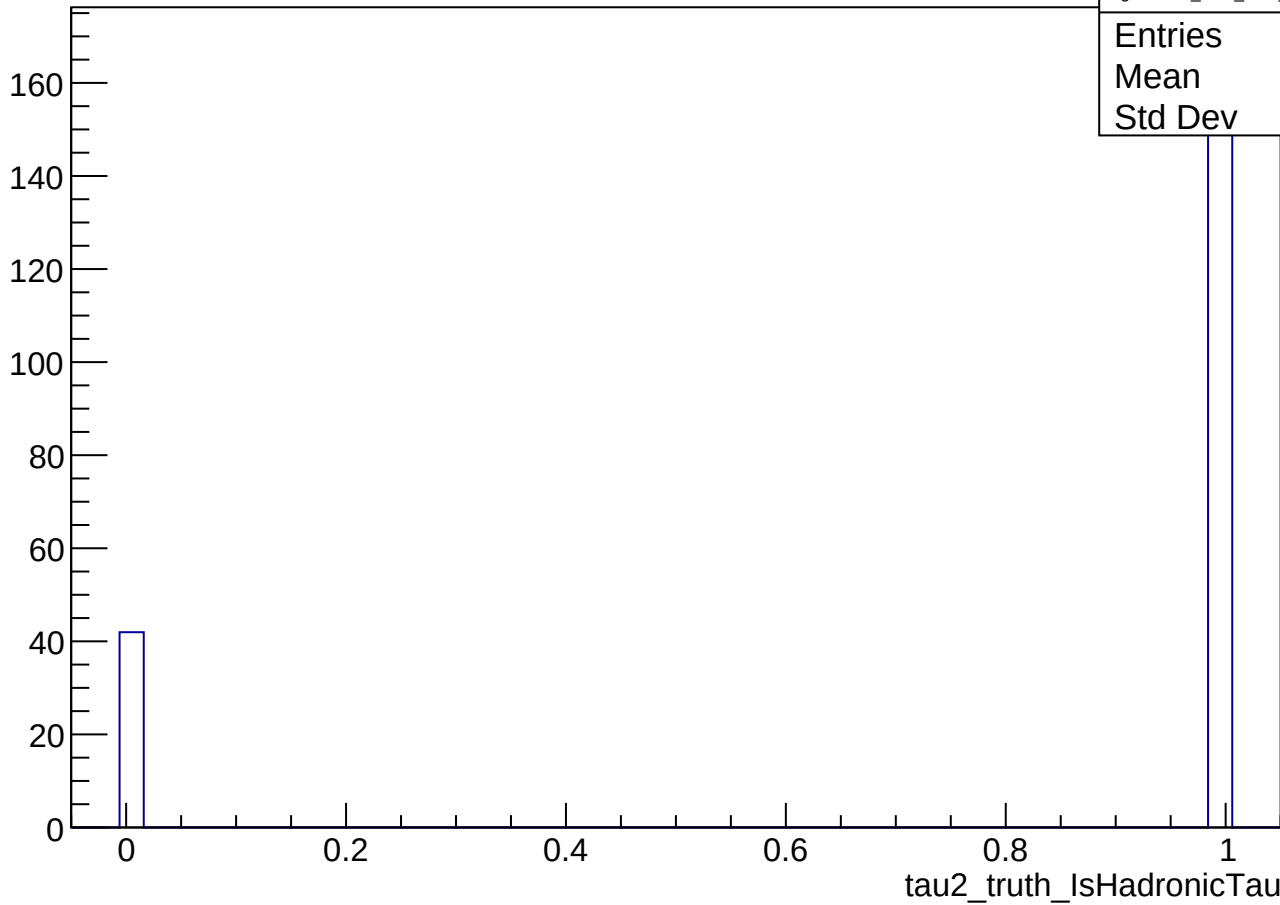
tau2_truthType

Events



tau2_truth_IsHadronicTau

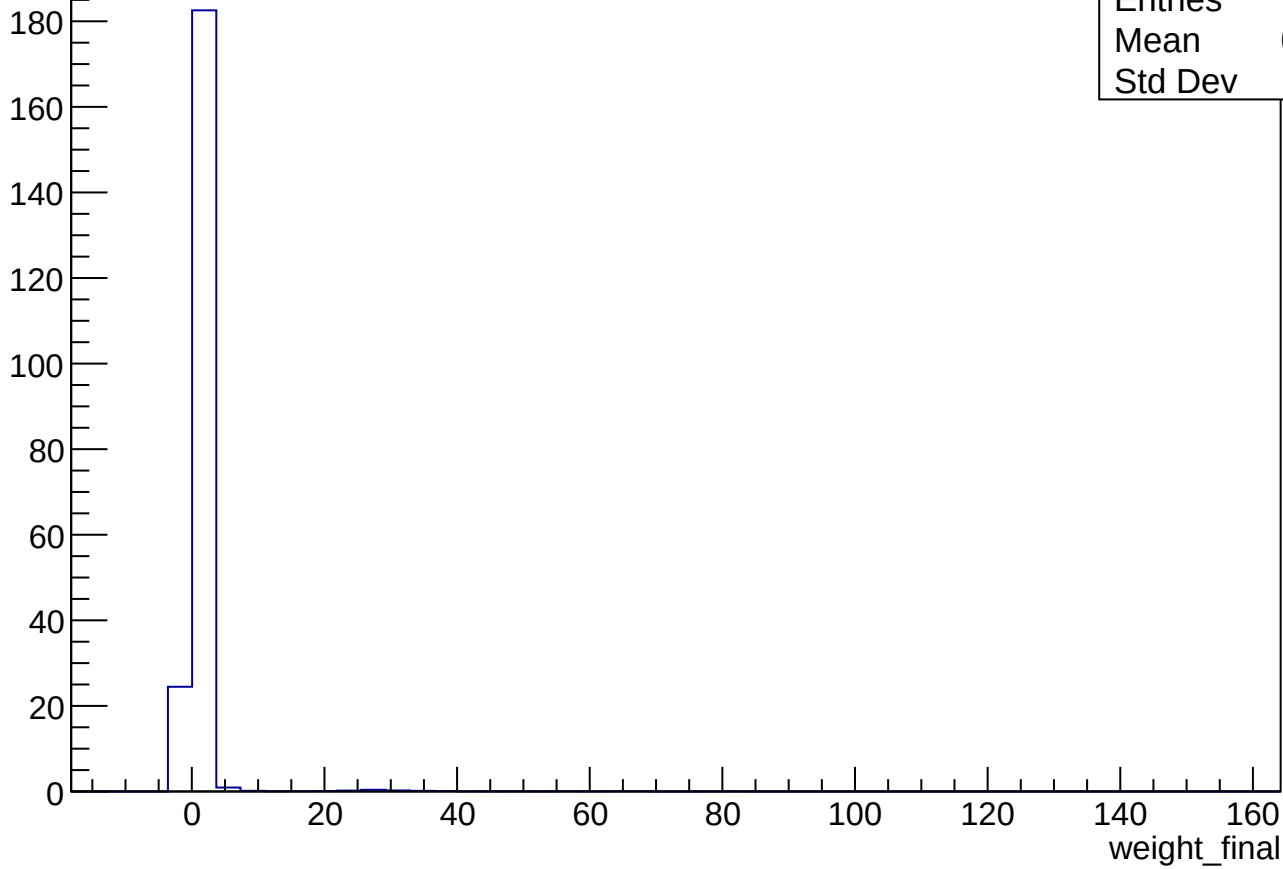
Events



singleH.root_tau2_truth_IsHadronicTau	
Entries	49549
Mean	0.8
Std Dev	0.4

weight_final

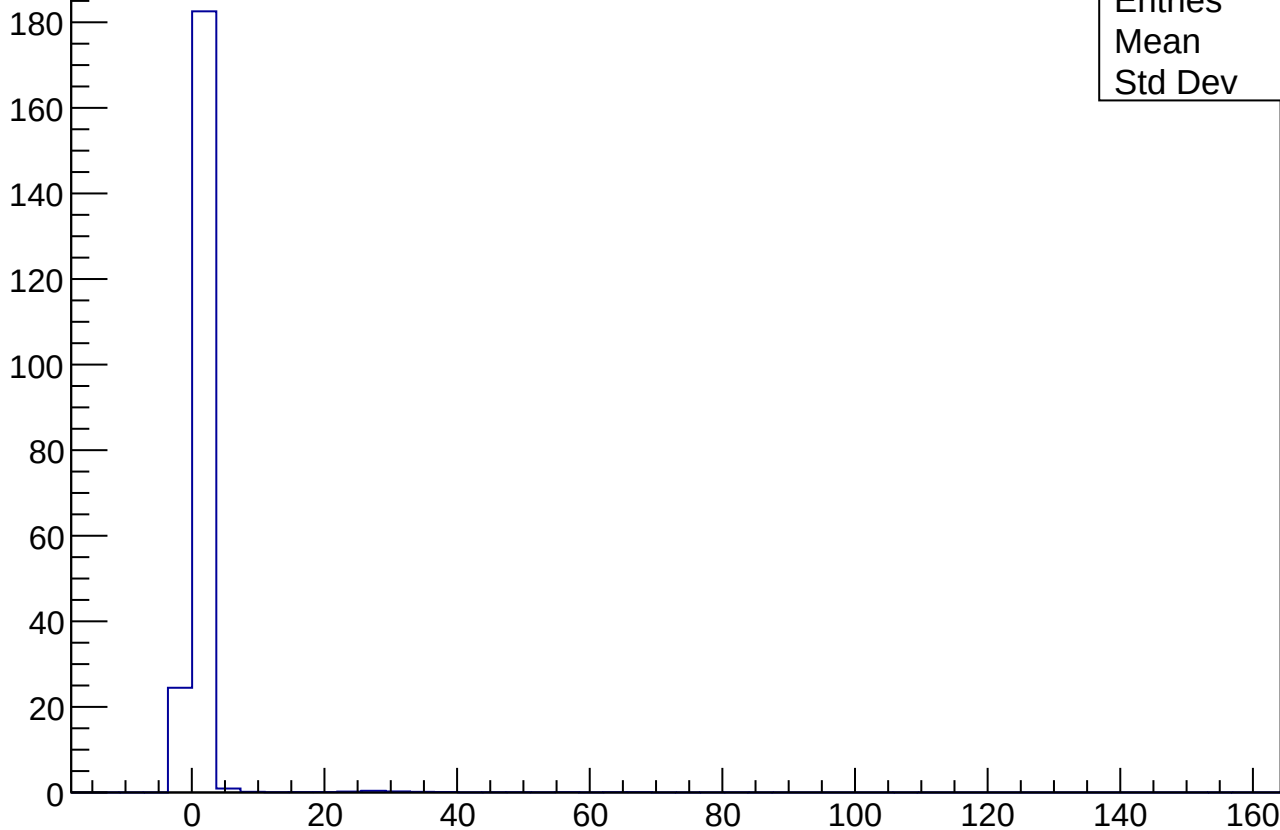
Events



singleH.root_weight_final	
Entries	49549
Mean	0.7677
Std Dev	3.406

weights

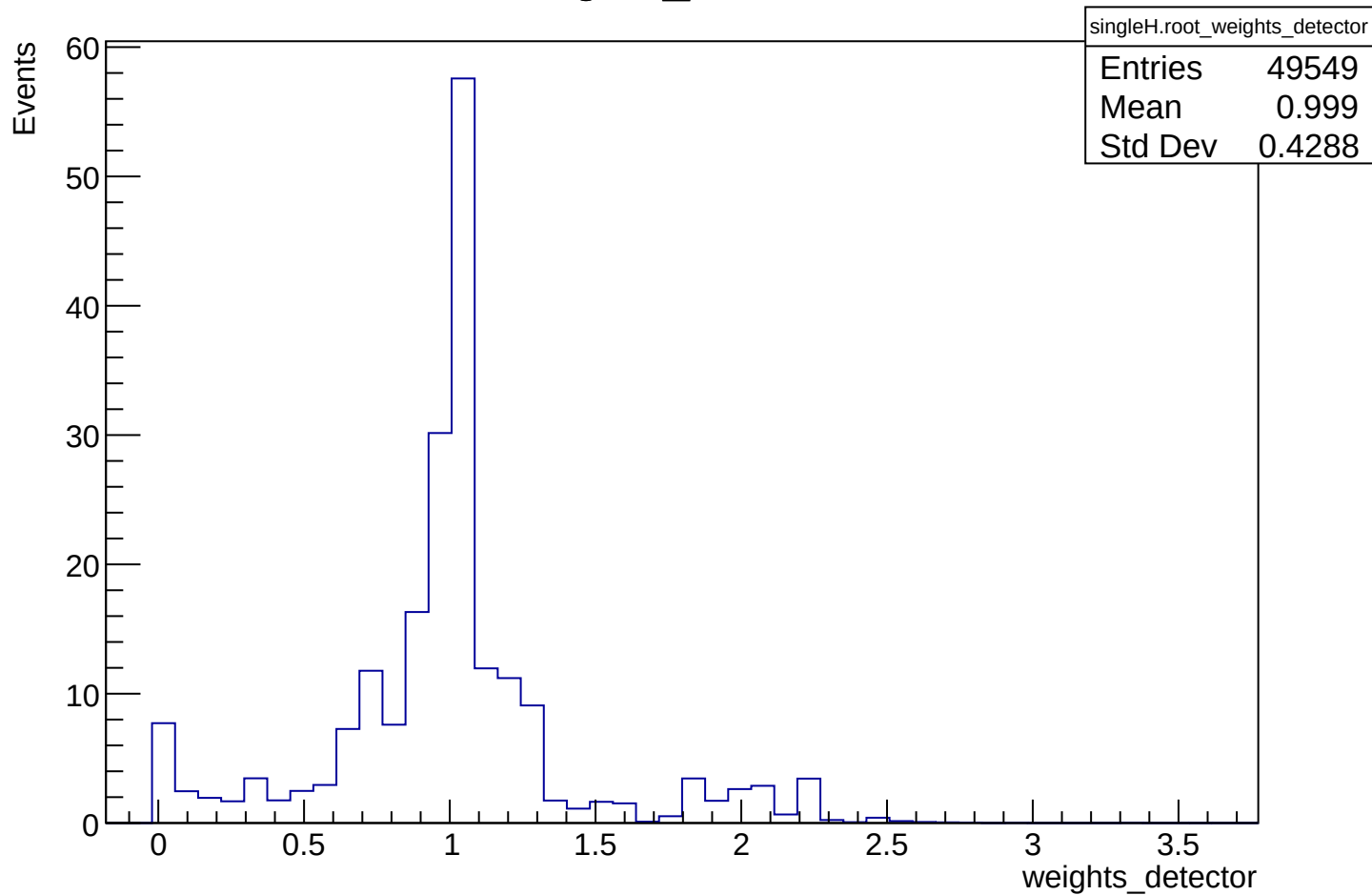
Events



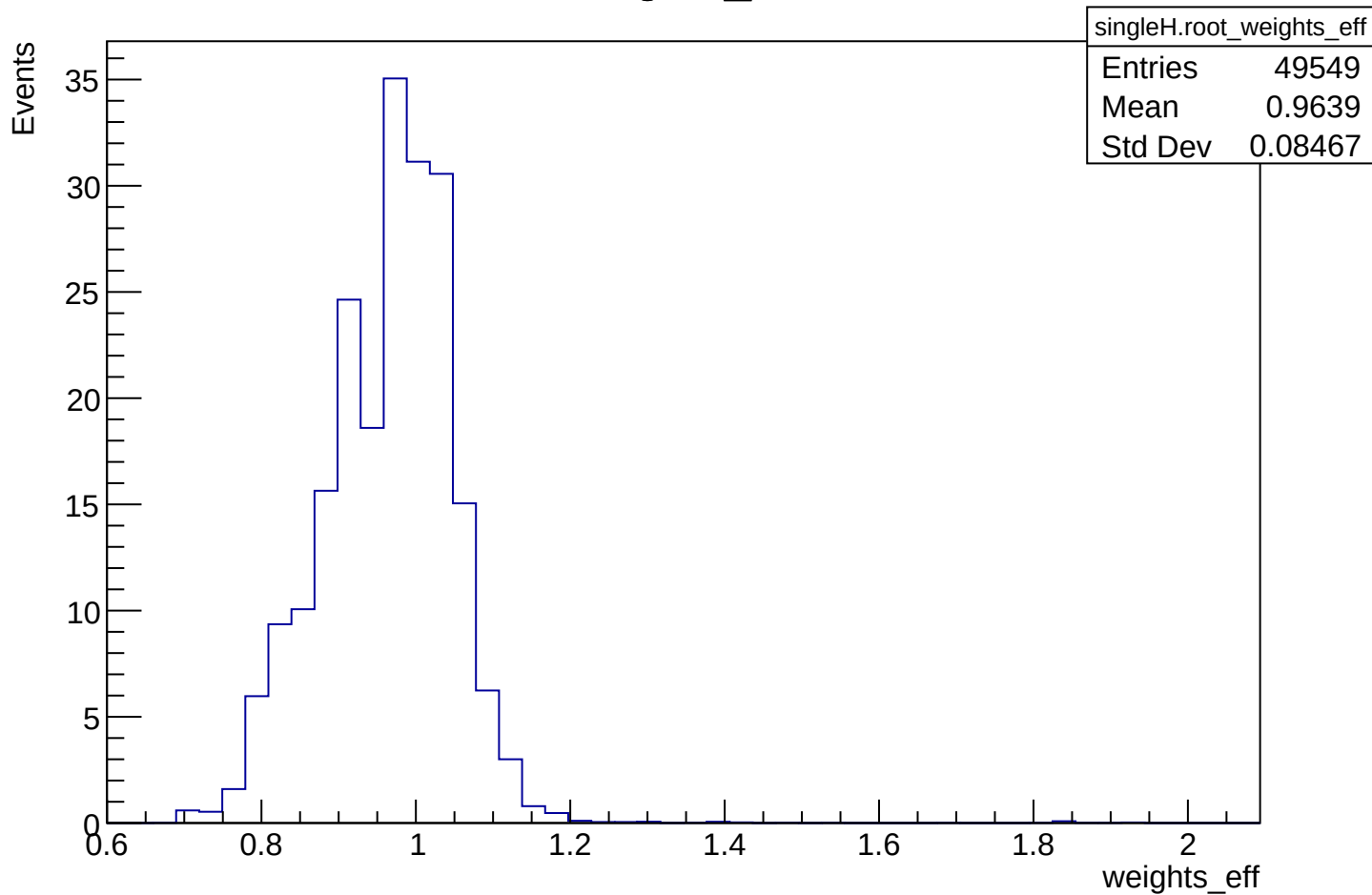
singleH.root_weights	
Entries	49549
Mean	0.7677
Std Dev	3.406

weights

weights_detector



weights_eff



weights_mc

Events

200
180
160
140
120
100
80
60
40
20
0

0

10

20

30

40

50

60

70

80

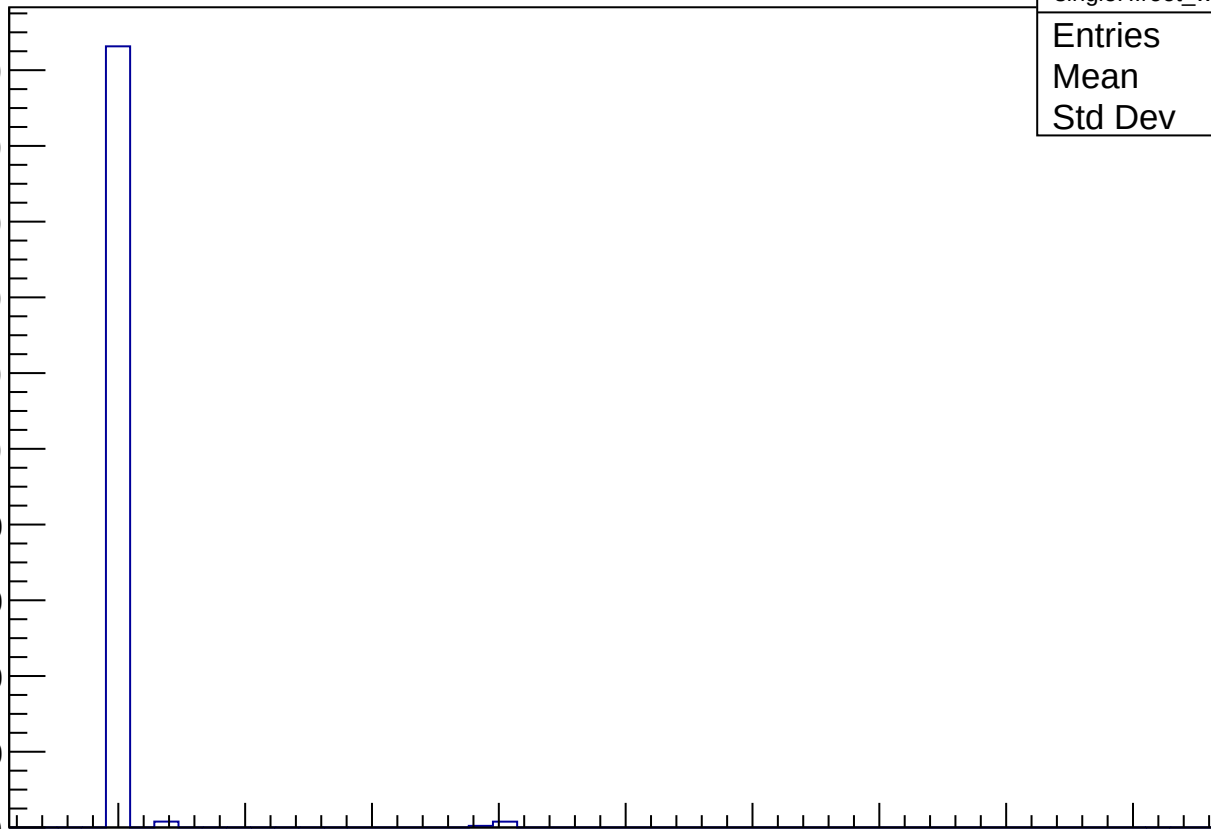
weights_mc

singleH.root_weights_mc

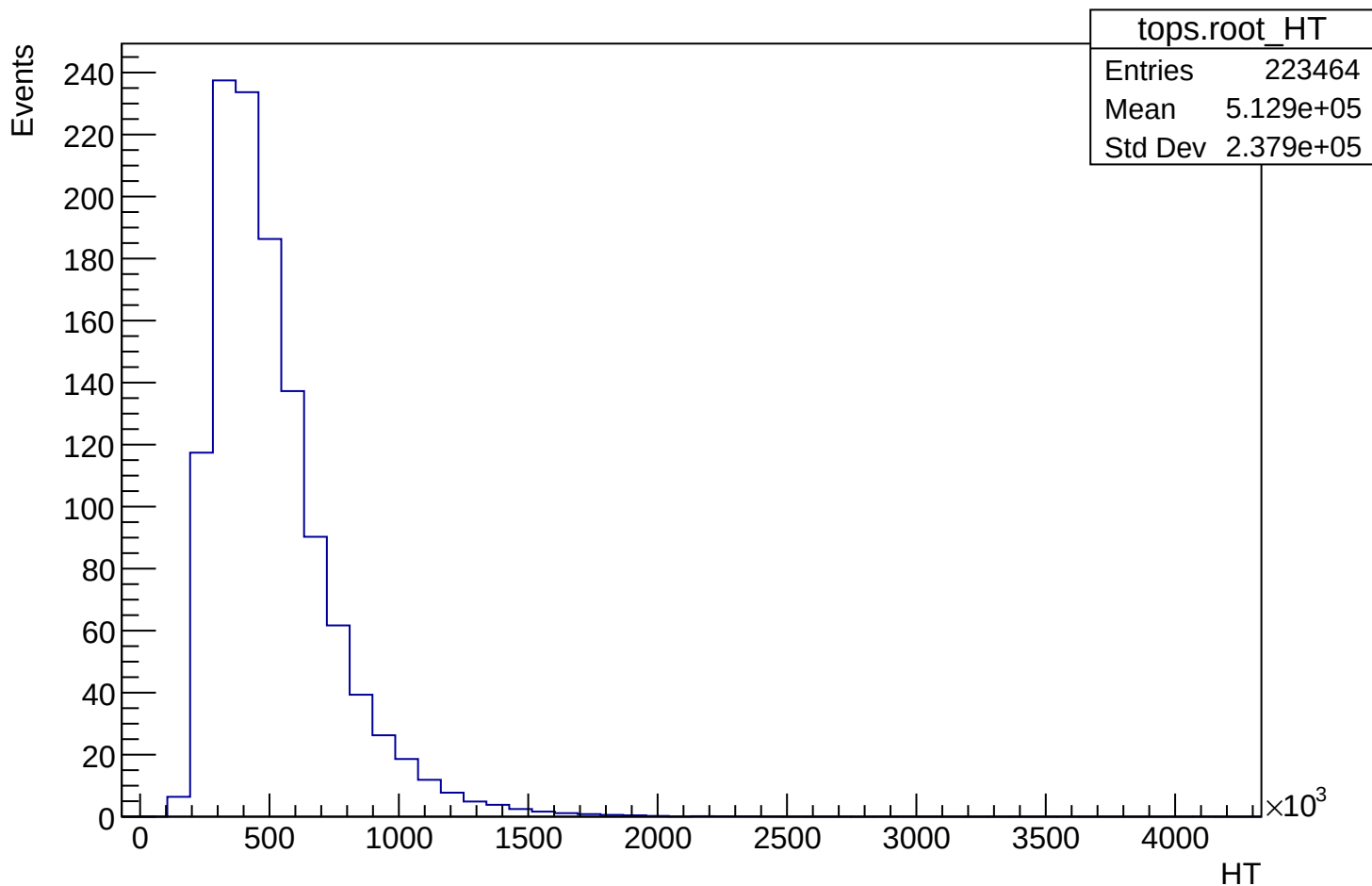
Entries 49549

Mean 0.7856

Std Dev 2.908



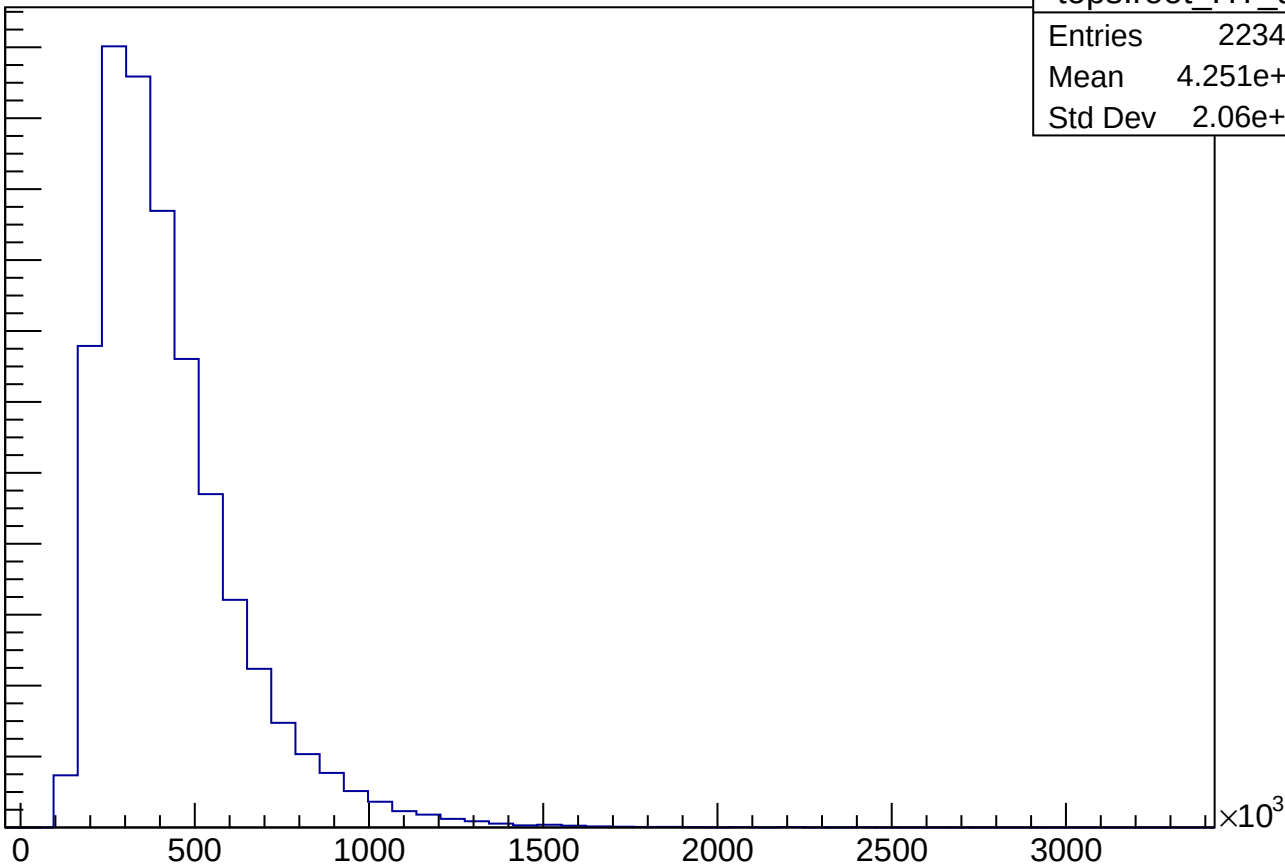
HT



HT_all

Events

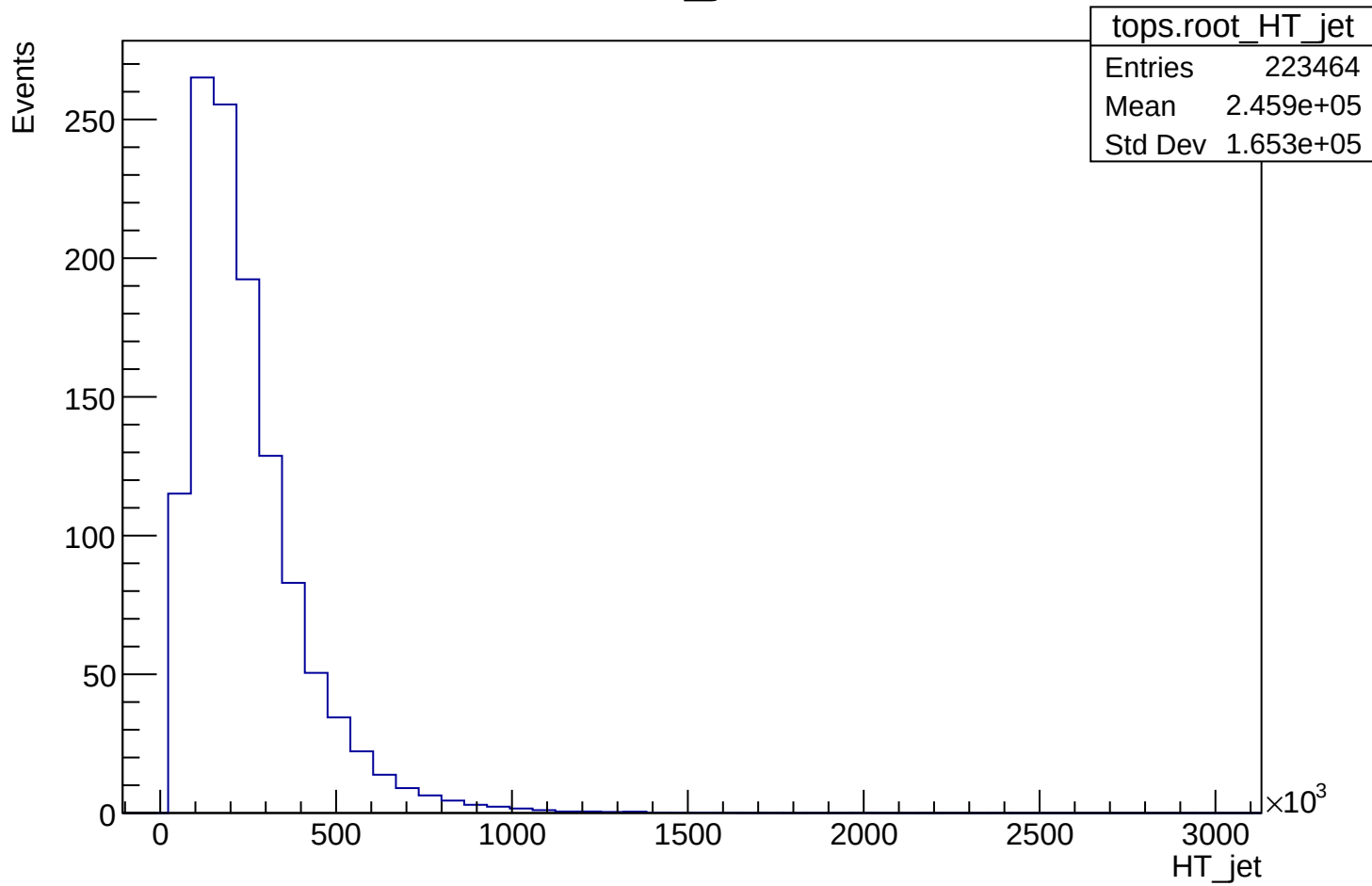
220
200
180
160
140
120
100
80
60
40
20
0



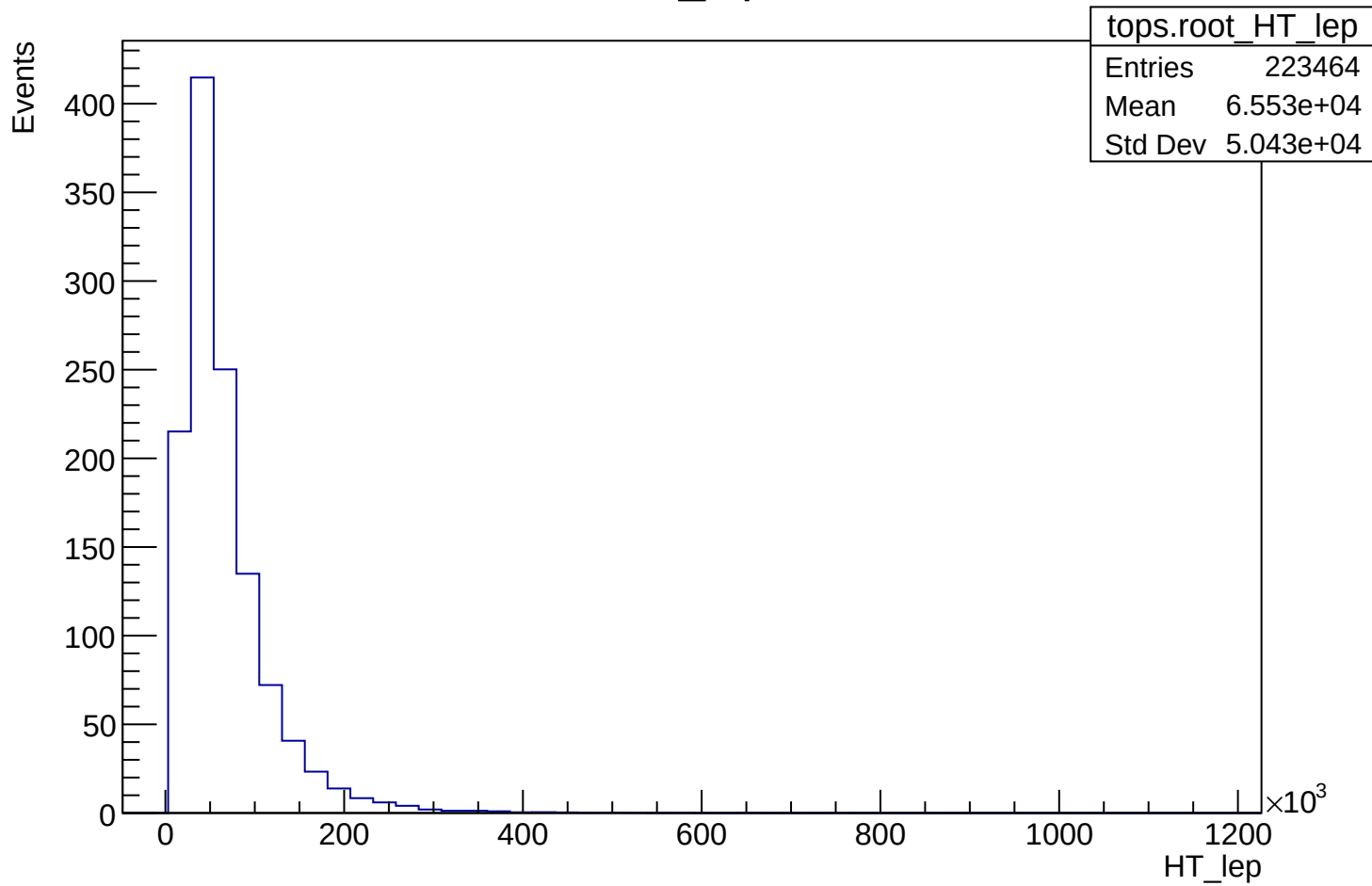
HT_all

$\times 10^3$

HT_jet

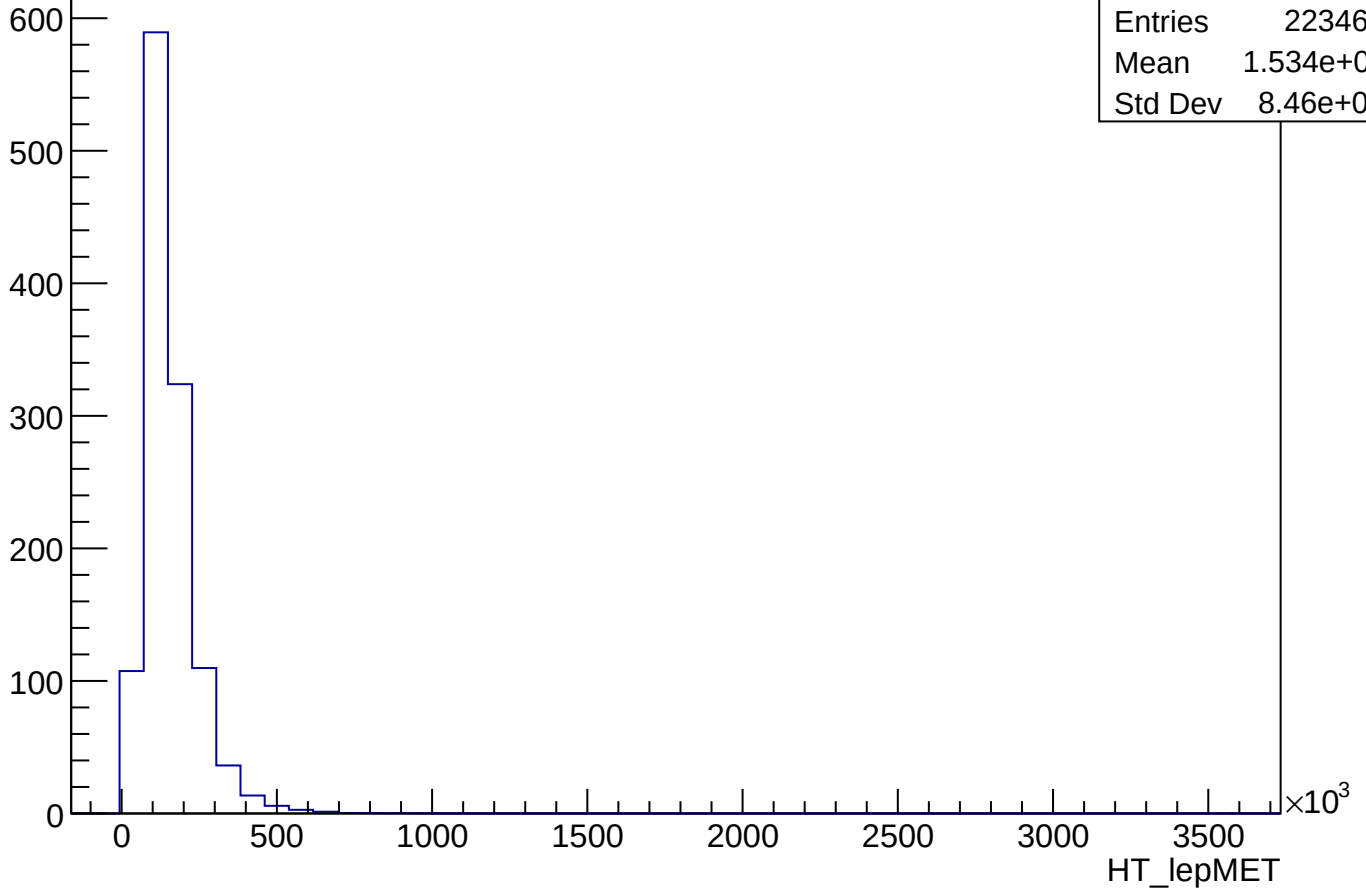


HT_lep



HT_lepMET

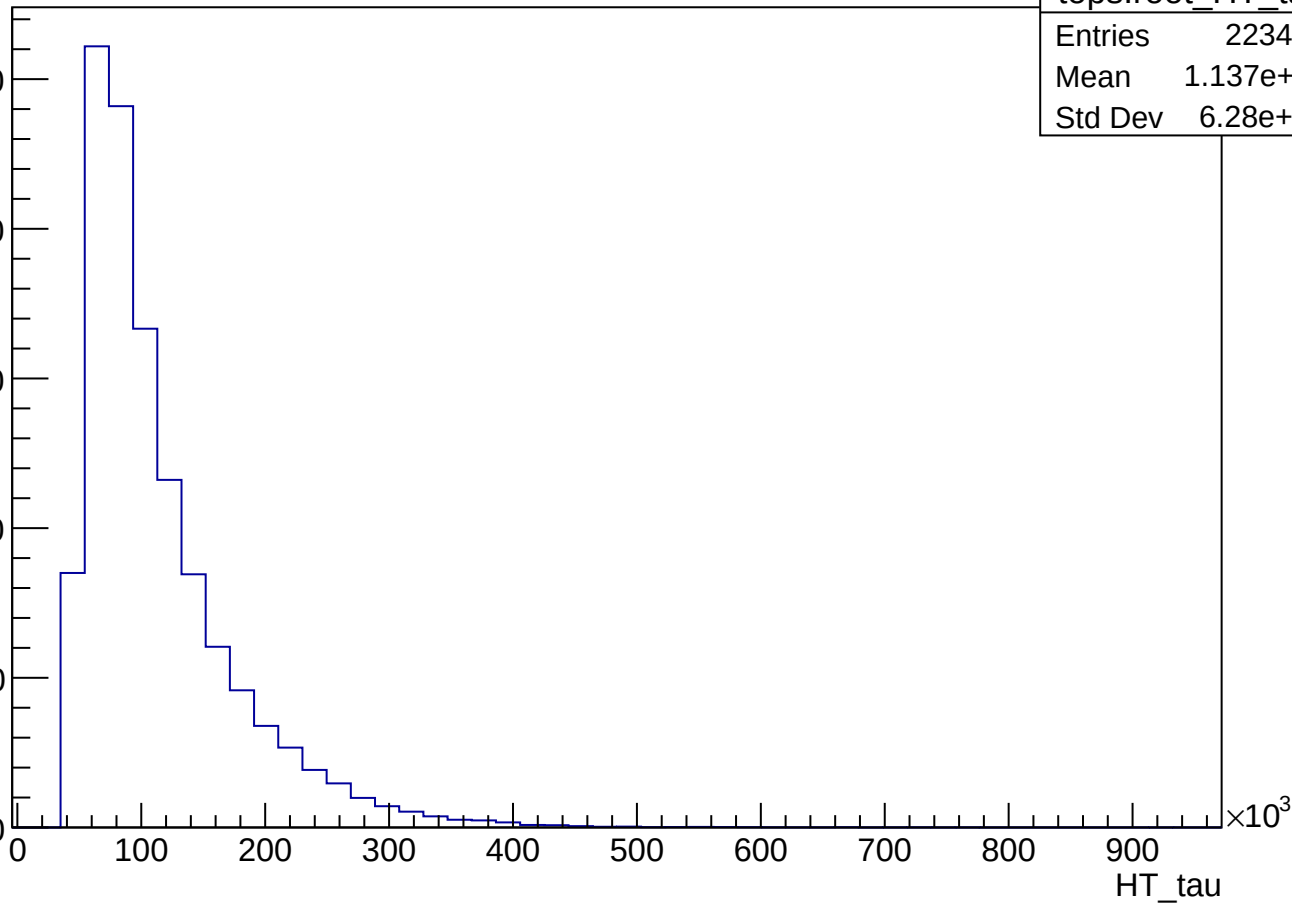
Events



HT_tau

Events

250
200
150
100
50
0



HT_tau

$\times 10^3$

SumPt_l1j

Events

300
250
200
150
100
50
0

0

0

200

400

600

800

1000

1200

1400

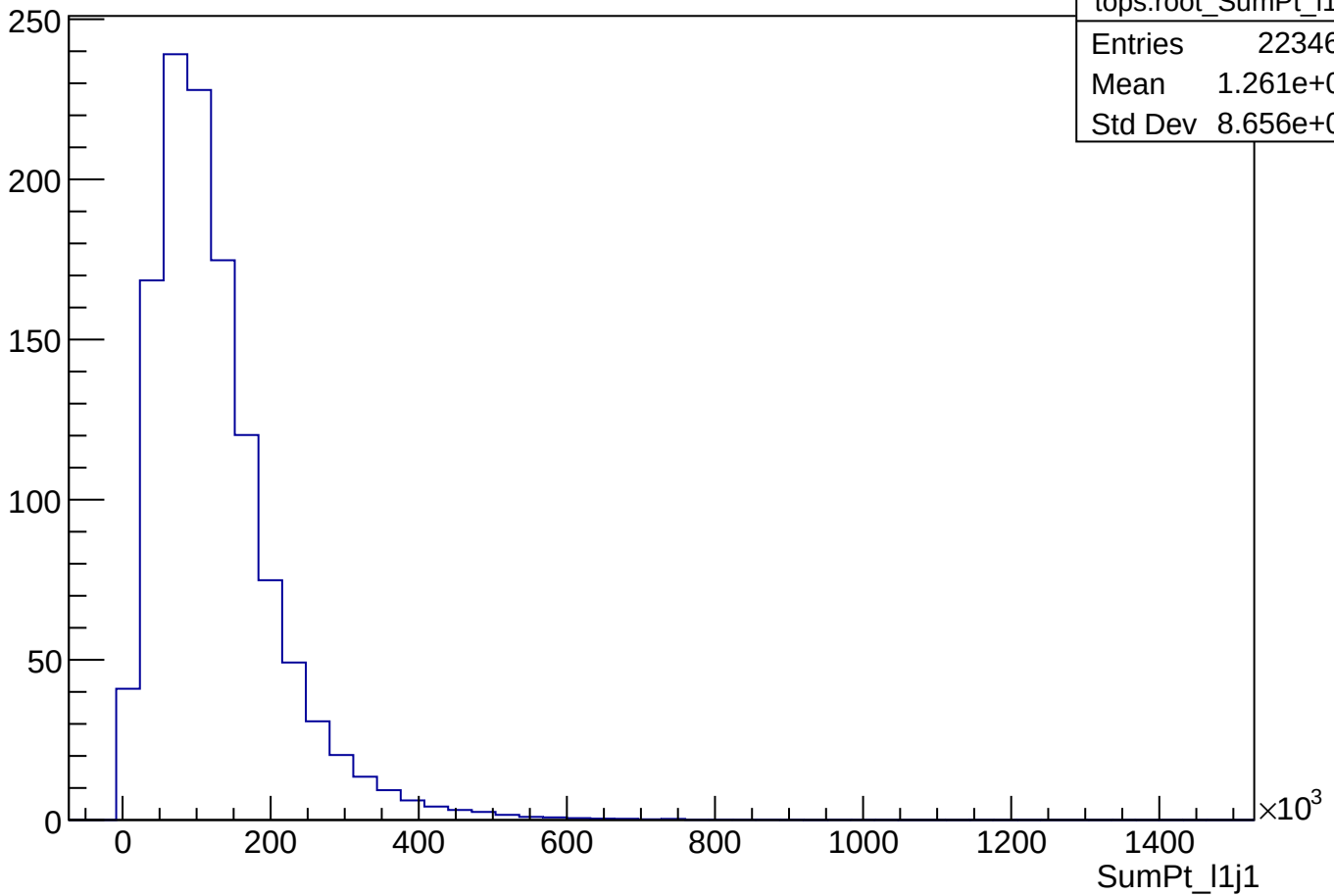
tops.root_SumPt_l1j	
Entries	223464
Mean	1.128e+05
Std Dev	7.155e+04

SumPt_l1j

$\times 10^3$

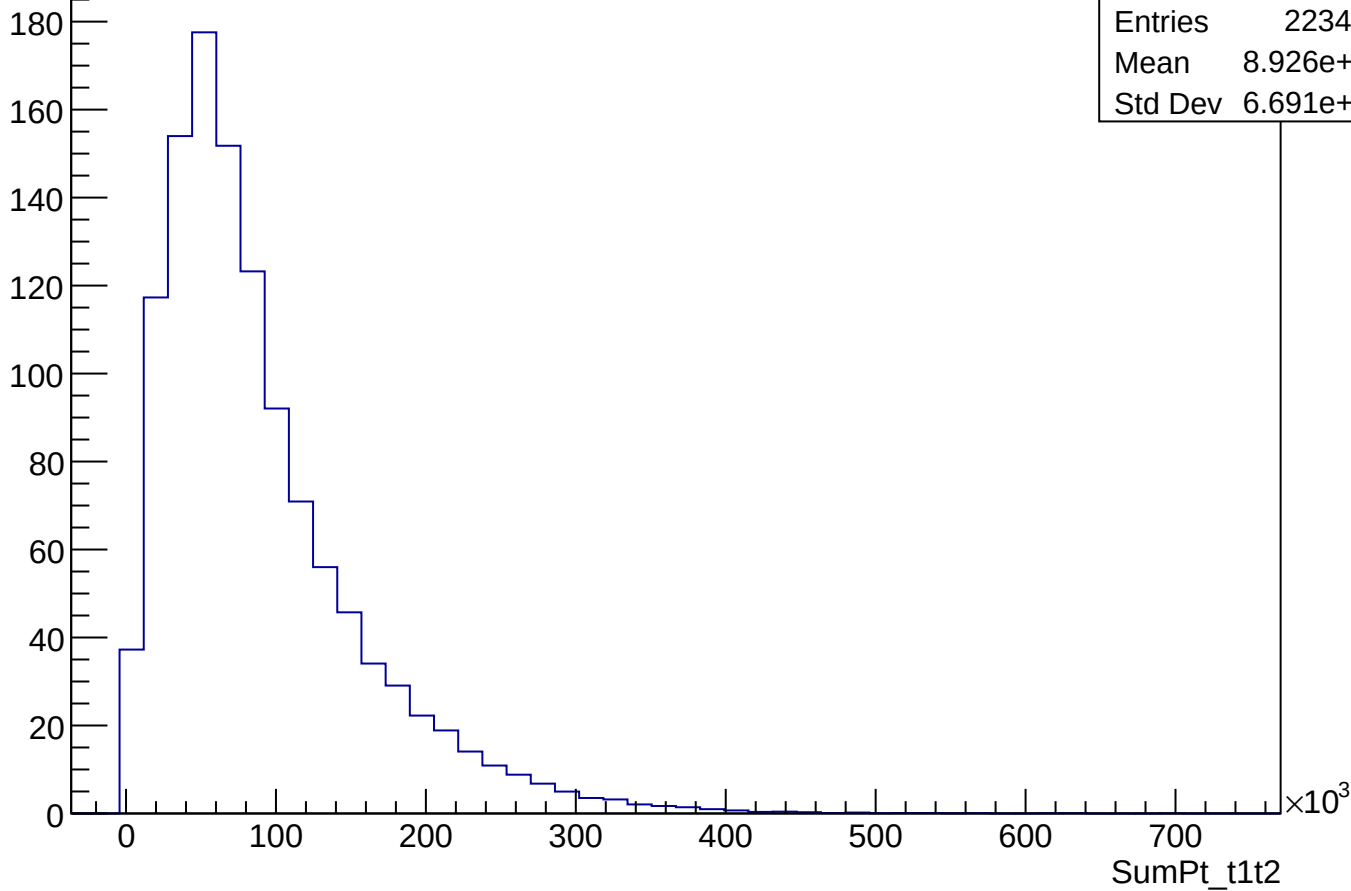
SumPt_l1j1

Events

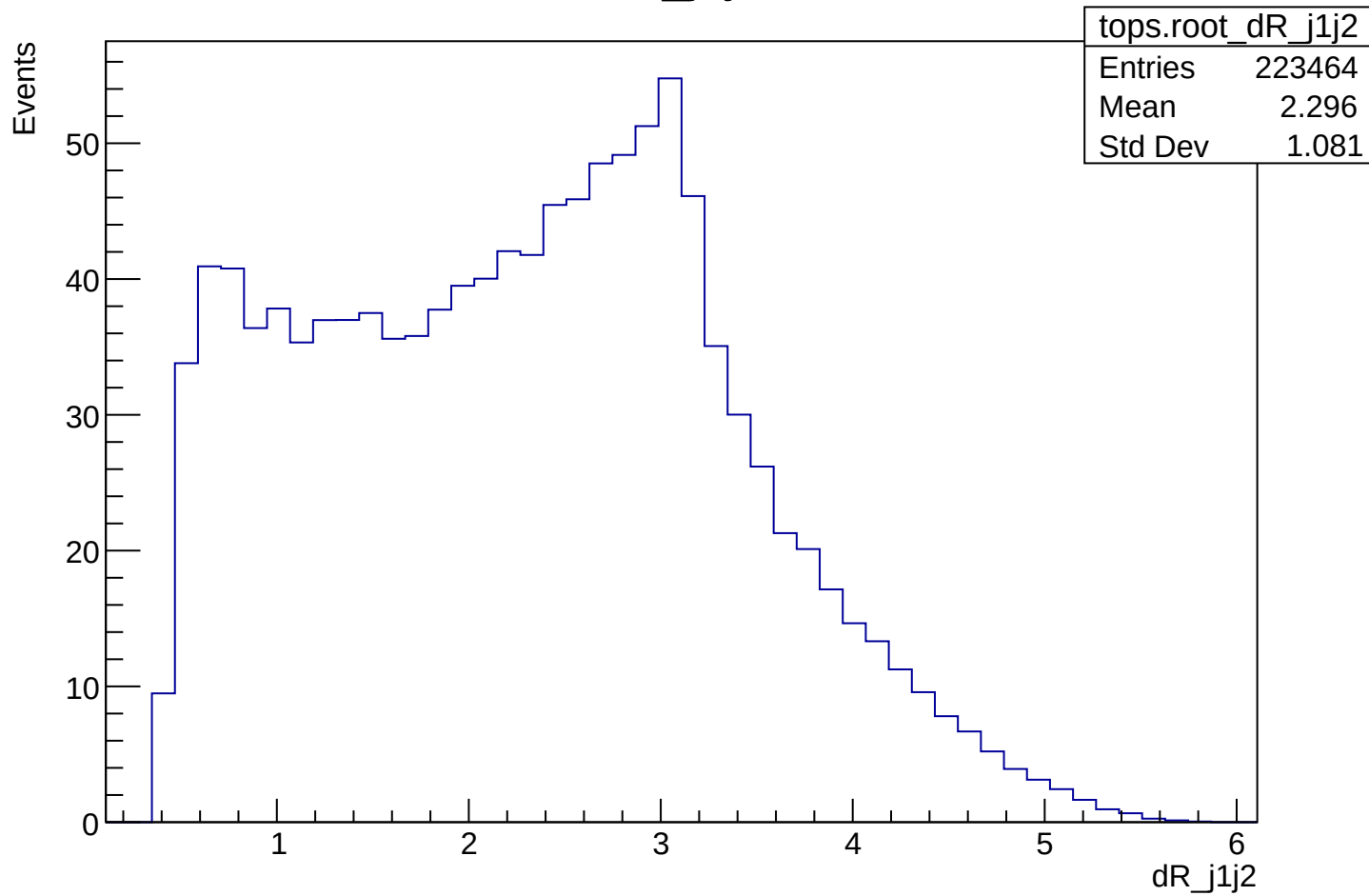


SumPt_t1t2

Events

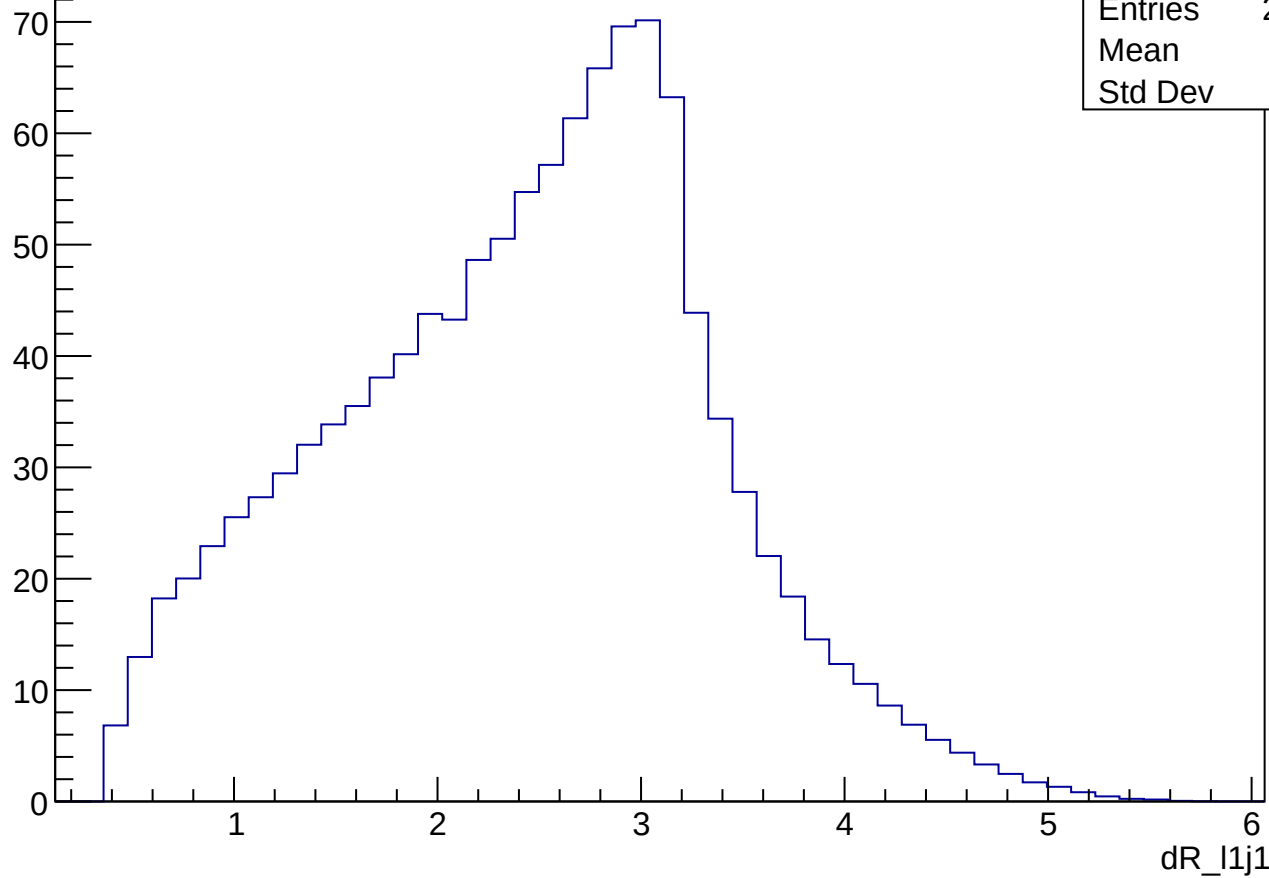


dR_j1j2



dR_l1j1

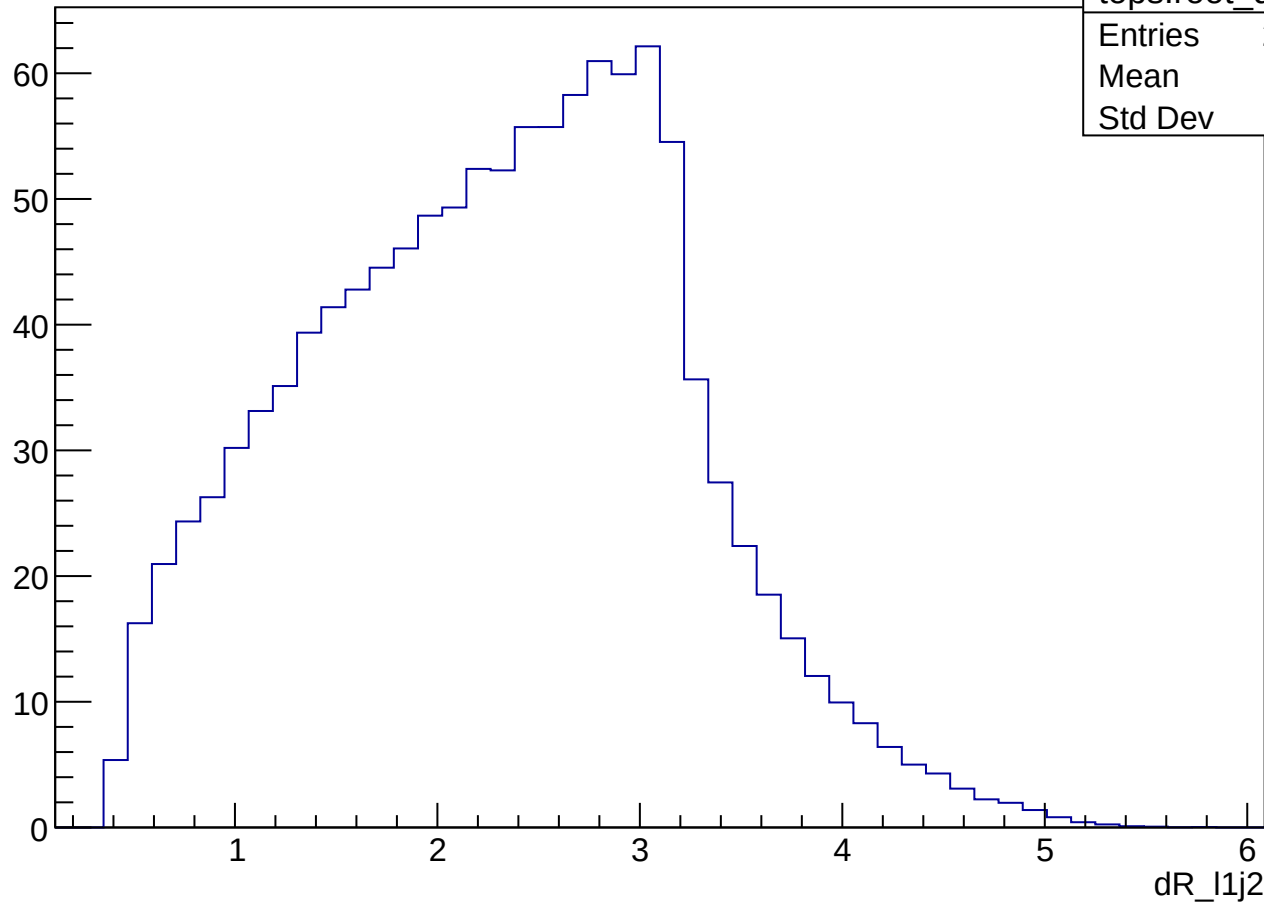
Events



tops.root_dR_l1j1	
Entries	223464
Mean	2.438
Std Dev	0.9233

dR_l1j2

Events



tops.root_dR_l1j2	
Entries	223464
Mean	2.311
Std Dev	0.9128

dR_l1t

Events

60
50
40
30
20
10
0

0

1

2

3

4

5

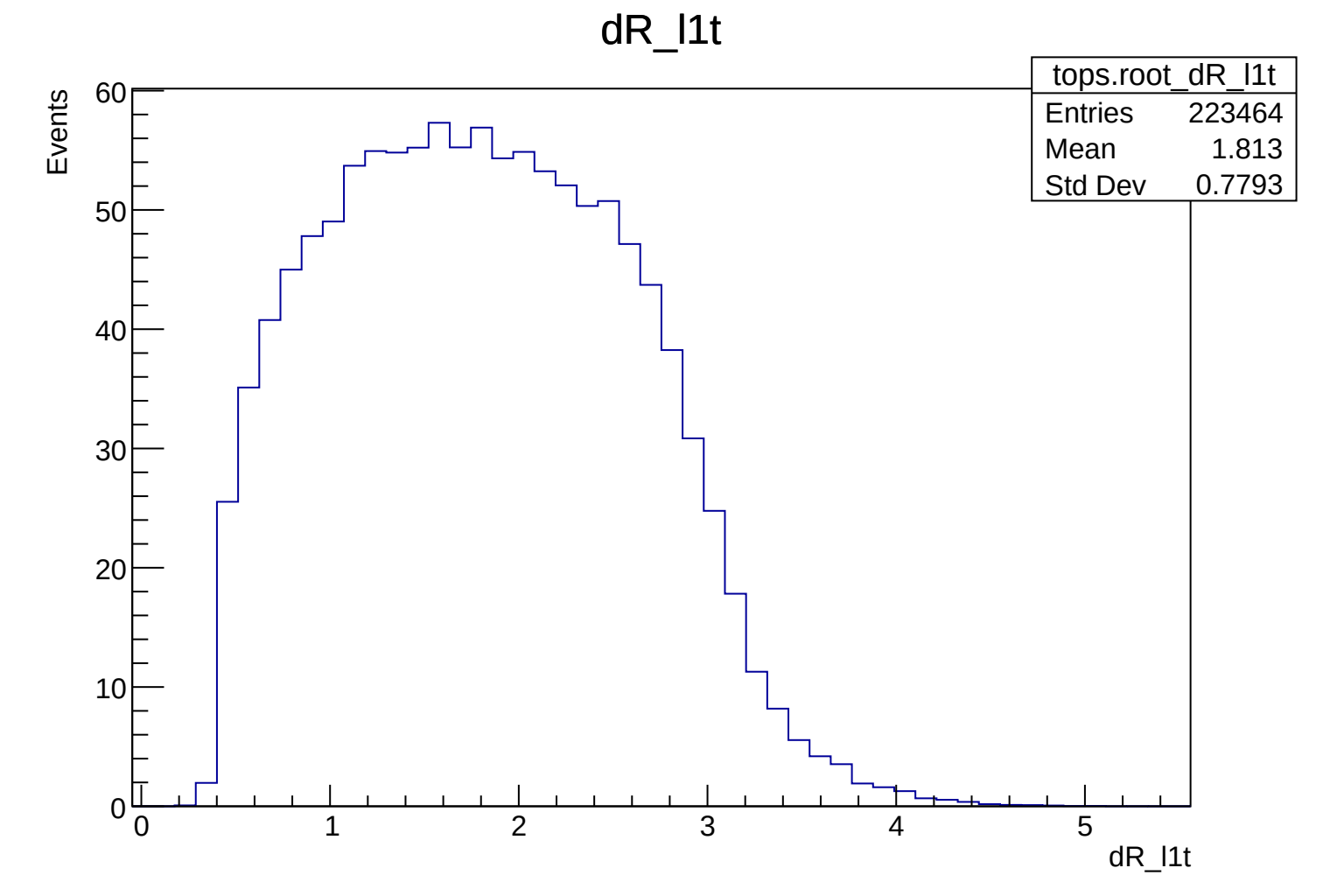
dR_l1t

tops.root_dR_l1t

Entries 223464

Mean 1.813

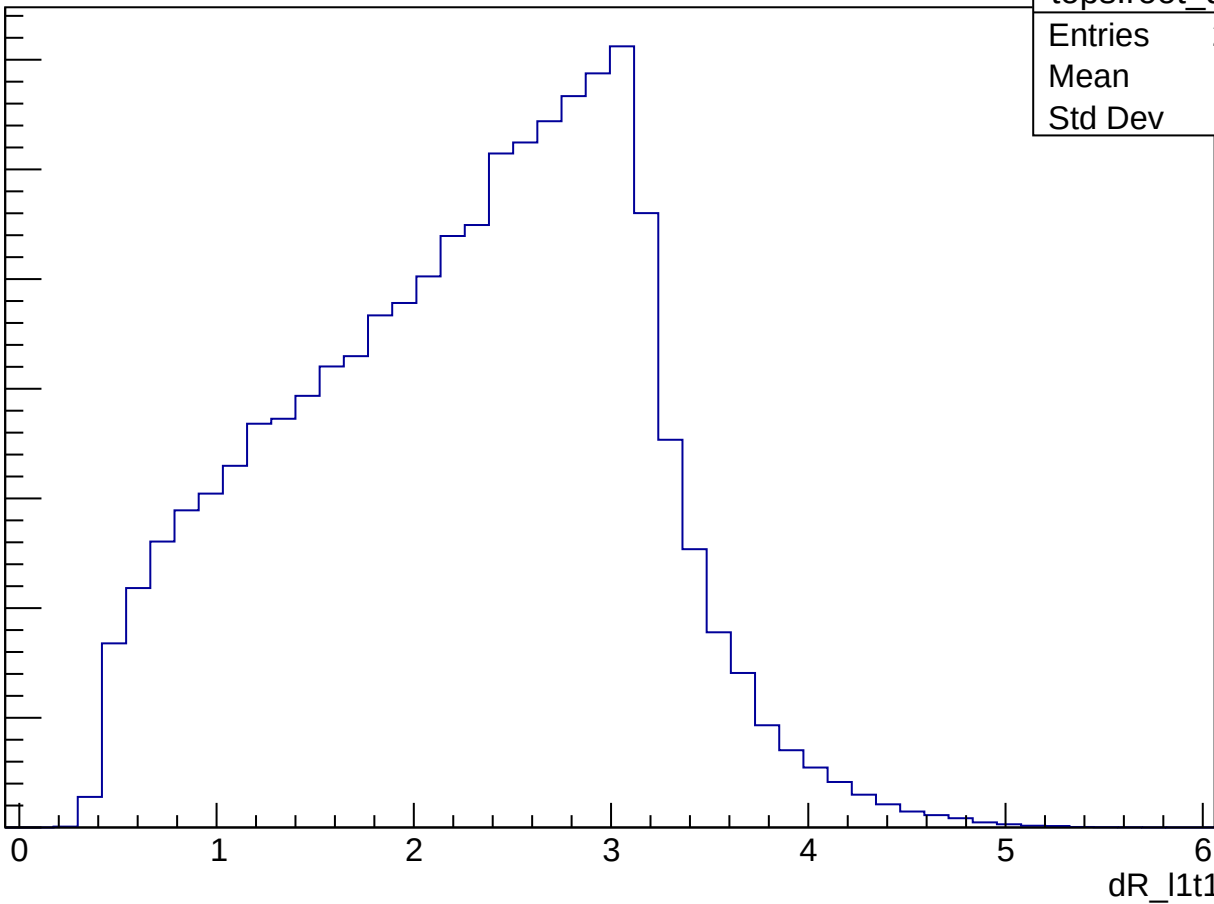
Std Dev 0.7793



dR_l1t1

Events

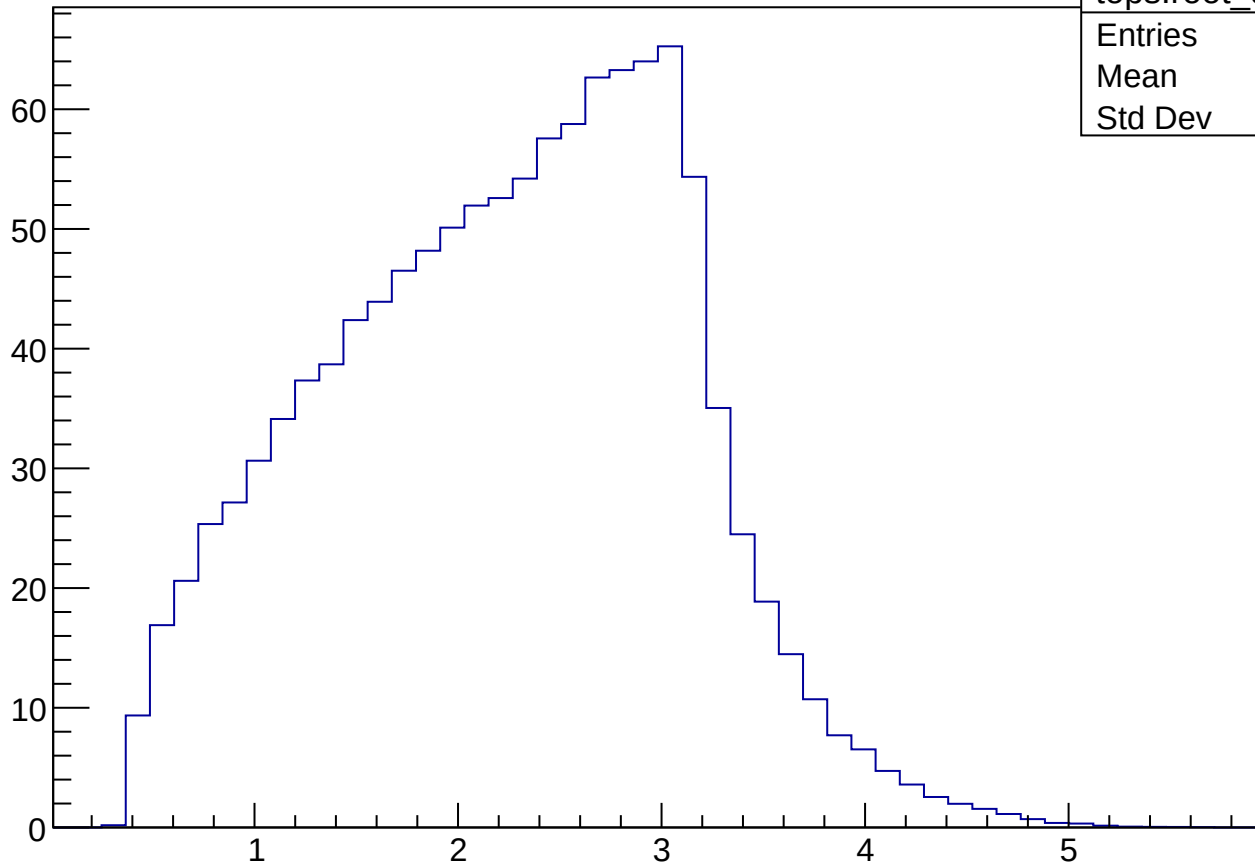
70
60
50
40
30
20
10
0



tops.root_dR_l1t1	
Entries	223464
Mean	2.254
Std Dev	0.8681

dR_l1t2

Events



tops.root_dR_l1t2

Entries 223464

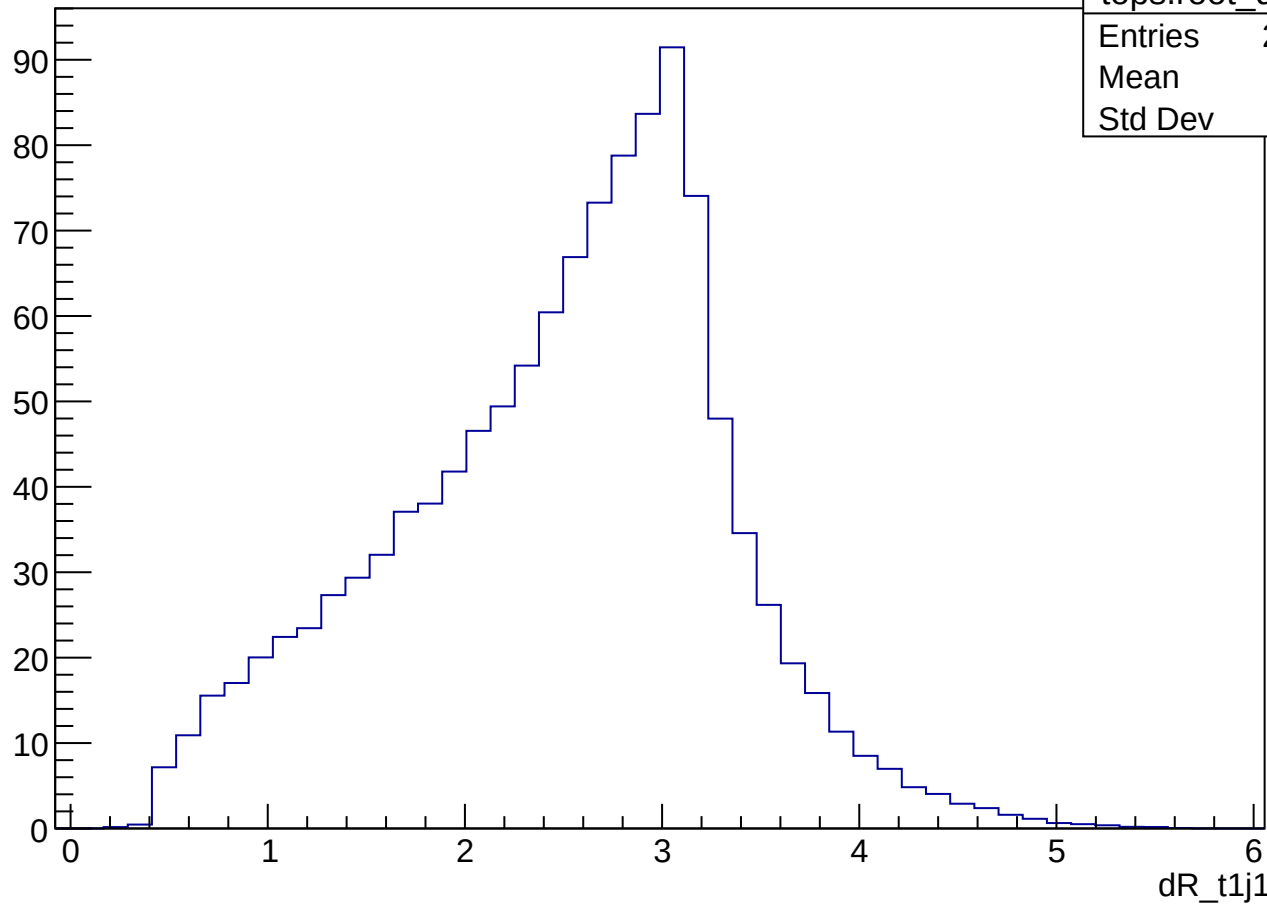
Mean 2.249

Std Dev 0.8608

dR_l1t2

dR_t1j1

Events



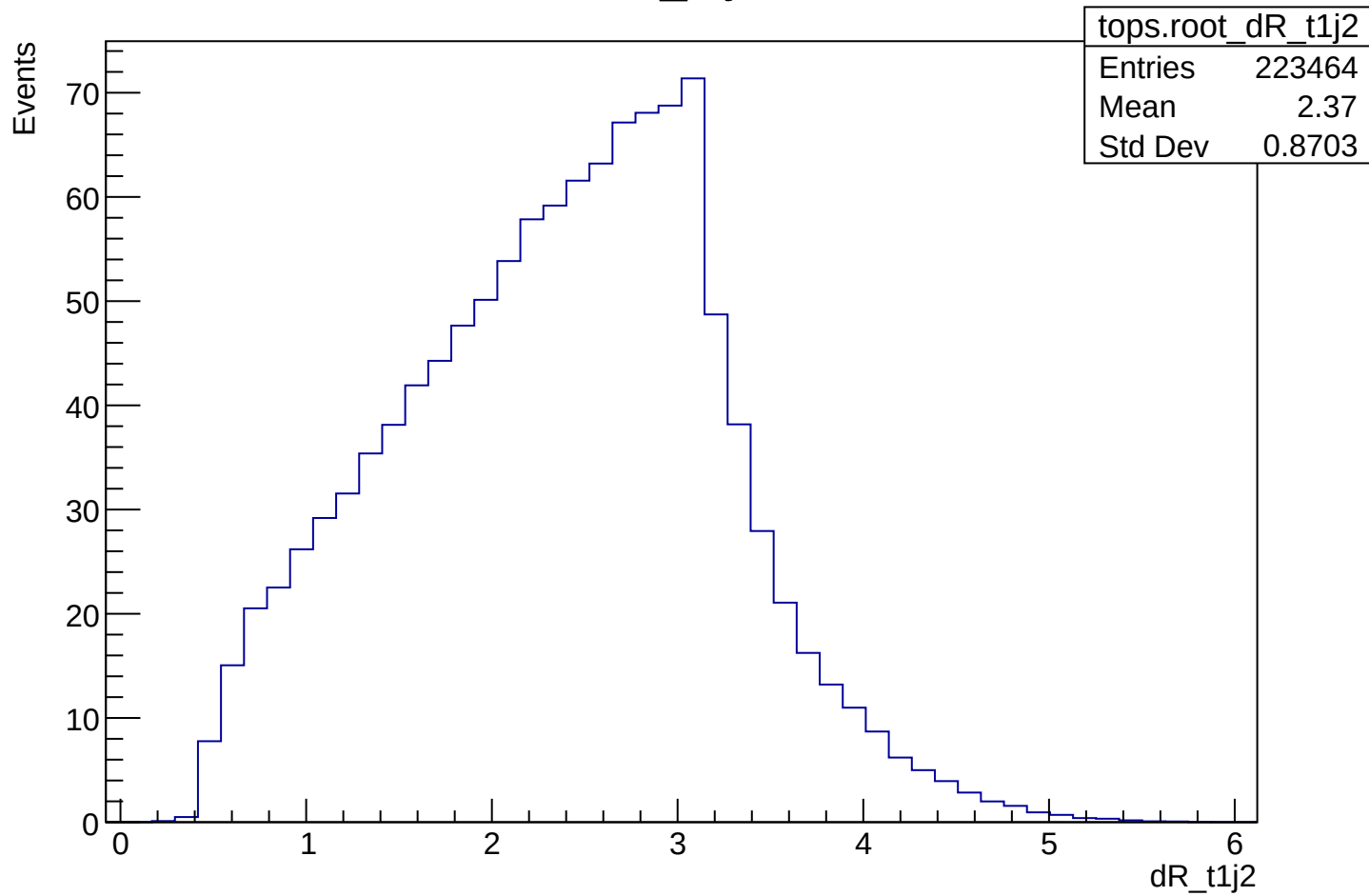
tops.root_dR_t1j1

Entries 223464

Mean 2.493

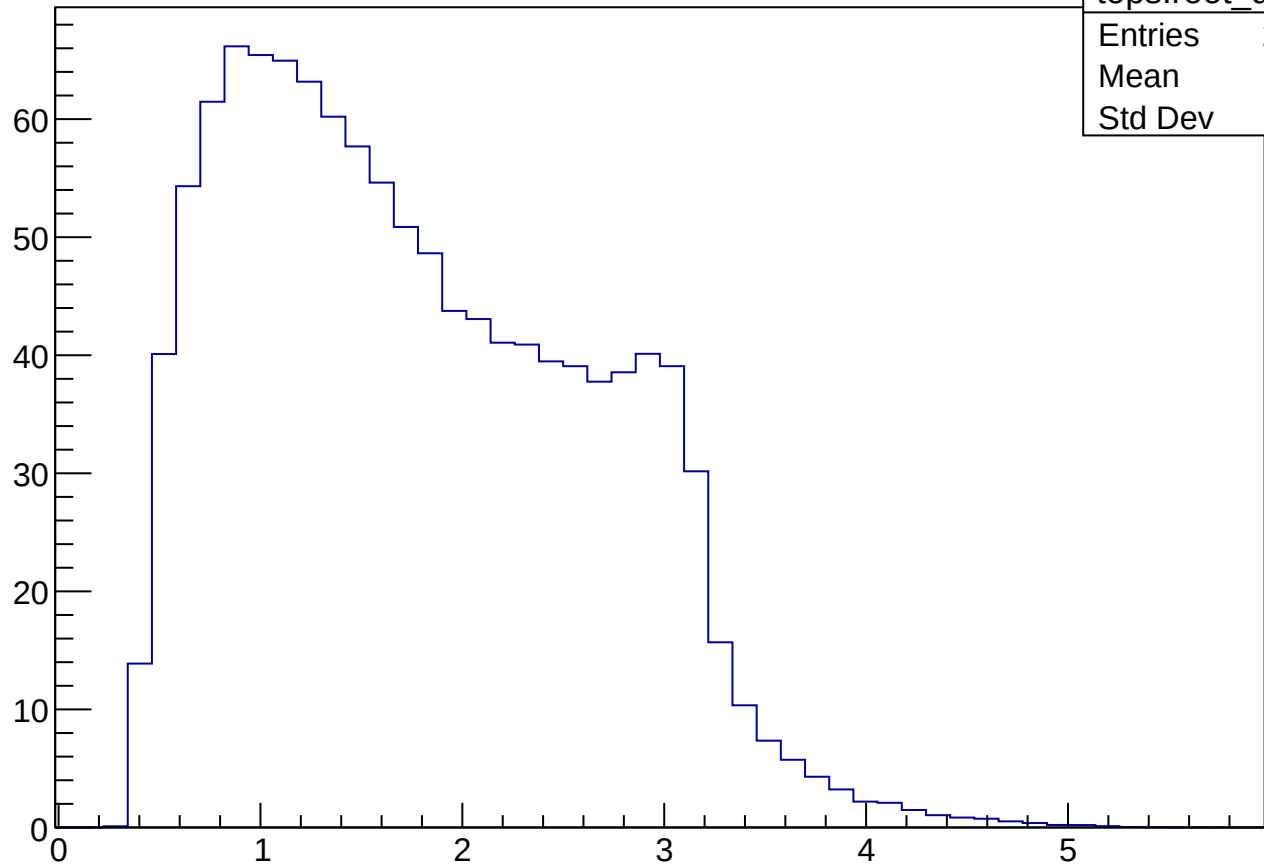
Std Dev 0.8364

dR_t1j2



dR_t1t2

Events



tops.root_dR_t1t2

Entries 223464

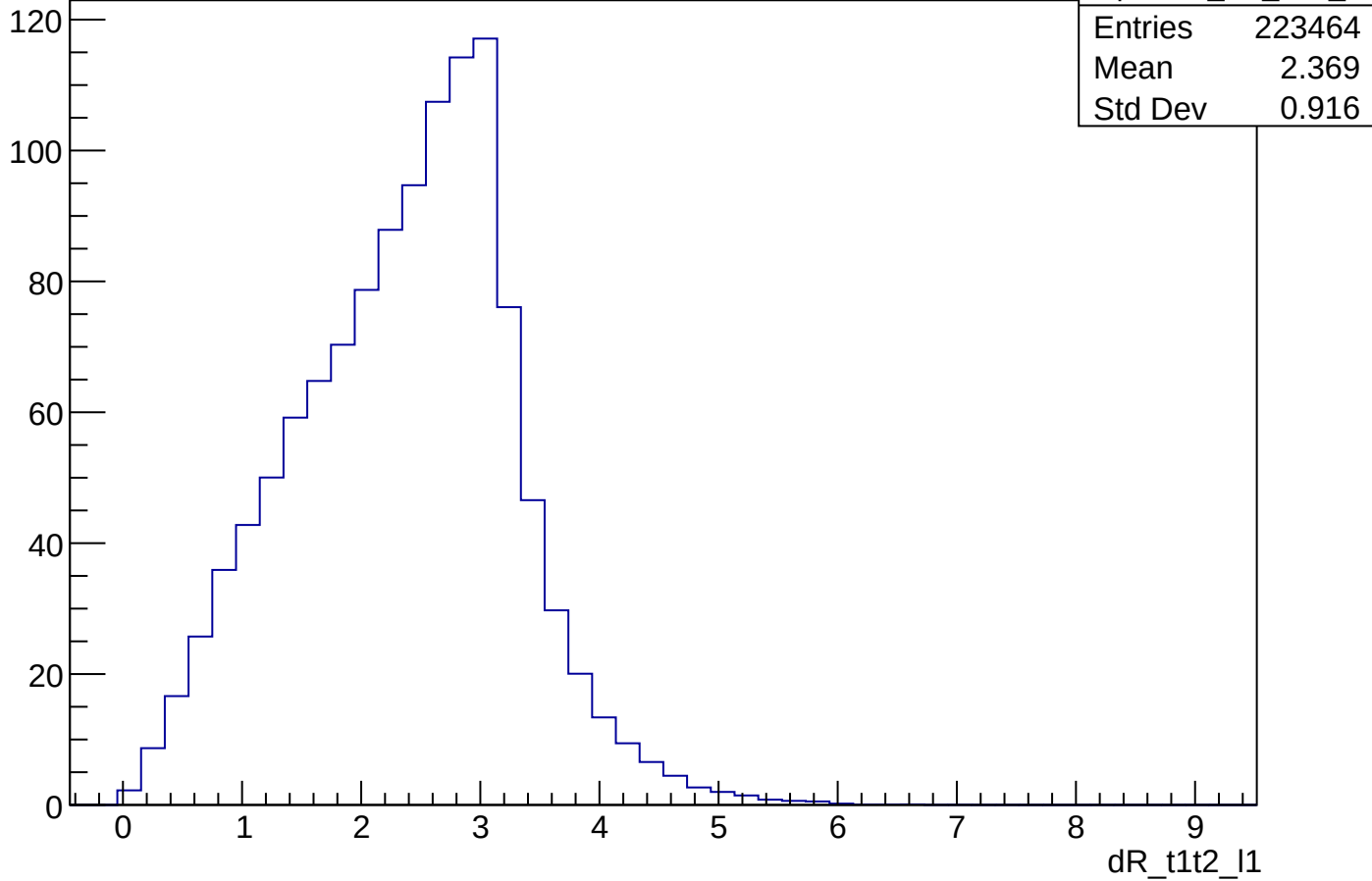
Mean 1.777

Std Dev 0.8629

dR_t1t2

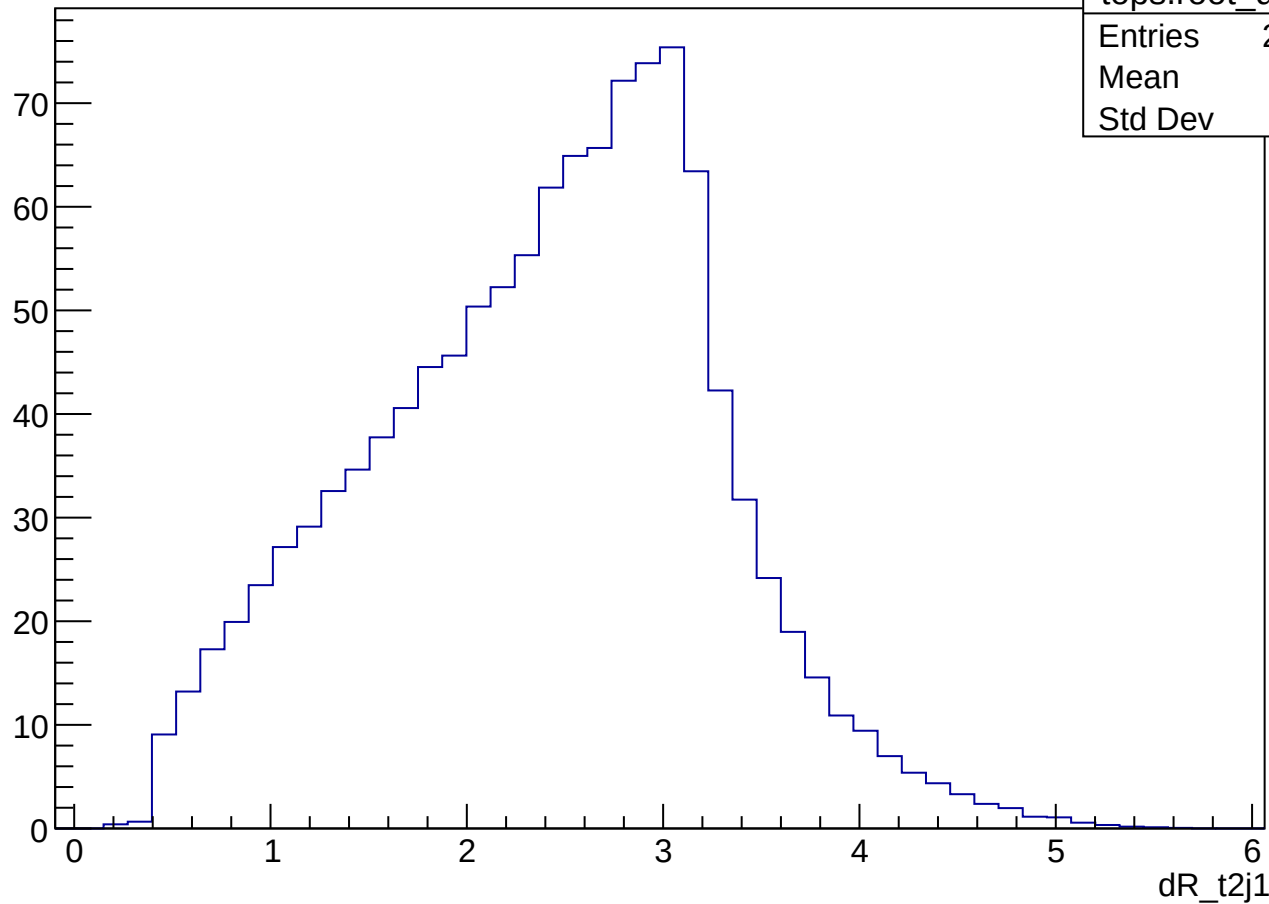
dR_t1t2_l1

Events



dR_t2j1

Events



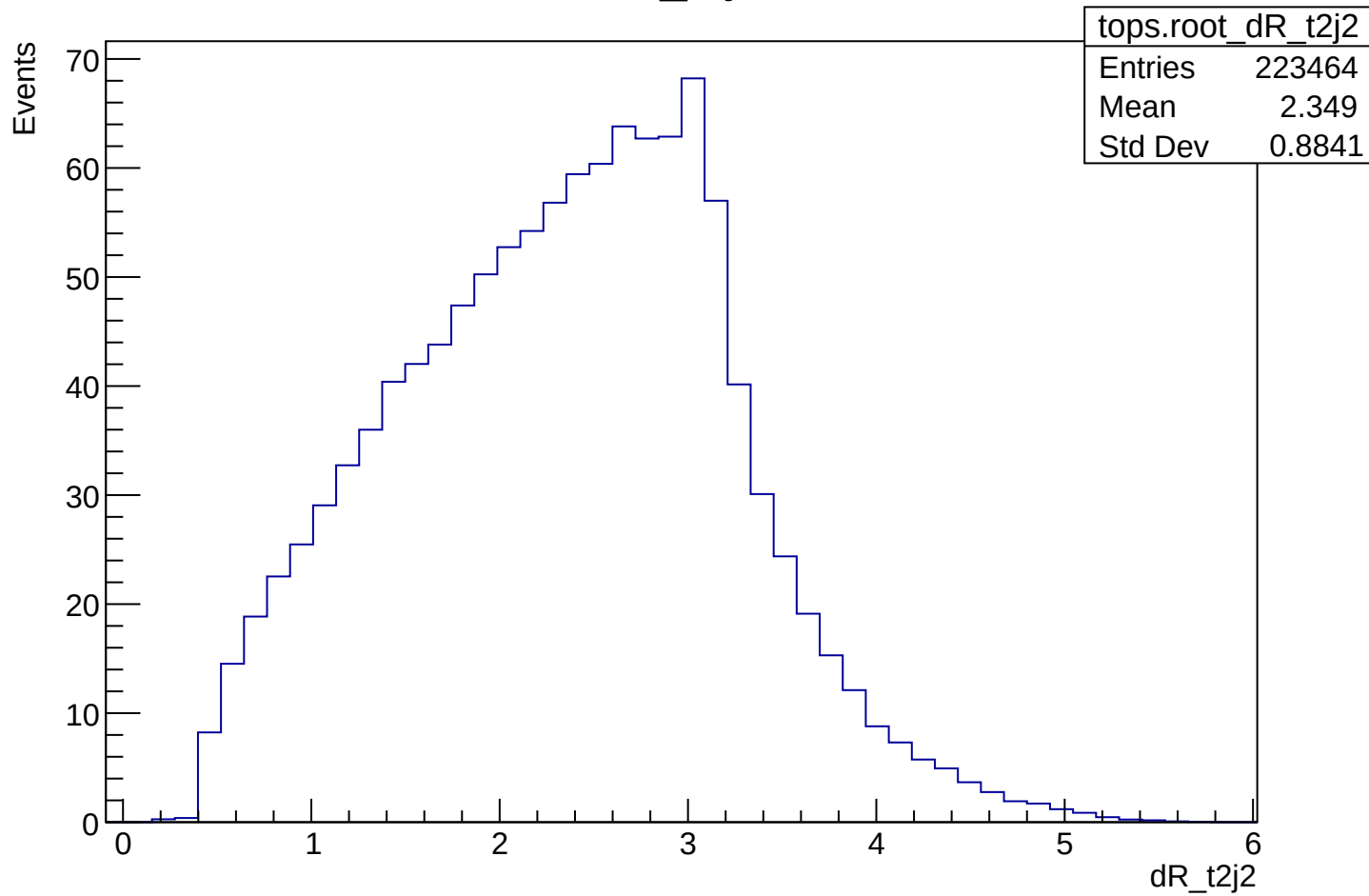
tops.root_dR_t2j1

Entries 223464

Mean 2.405

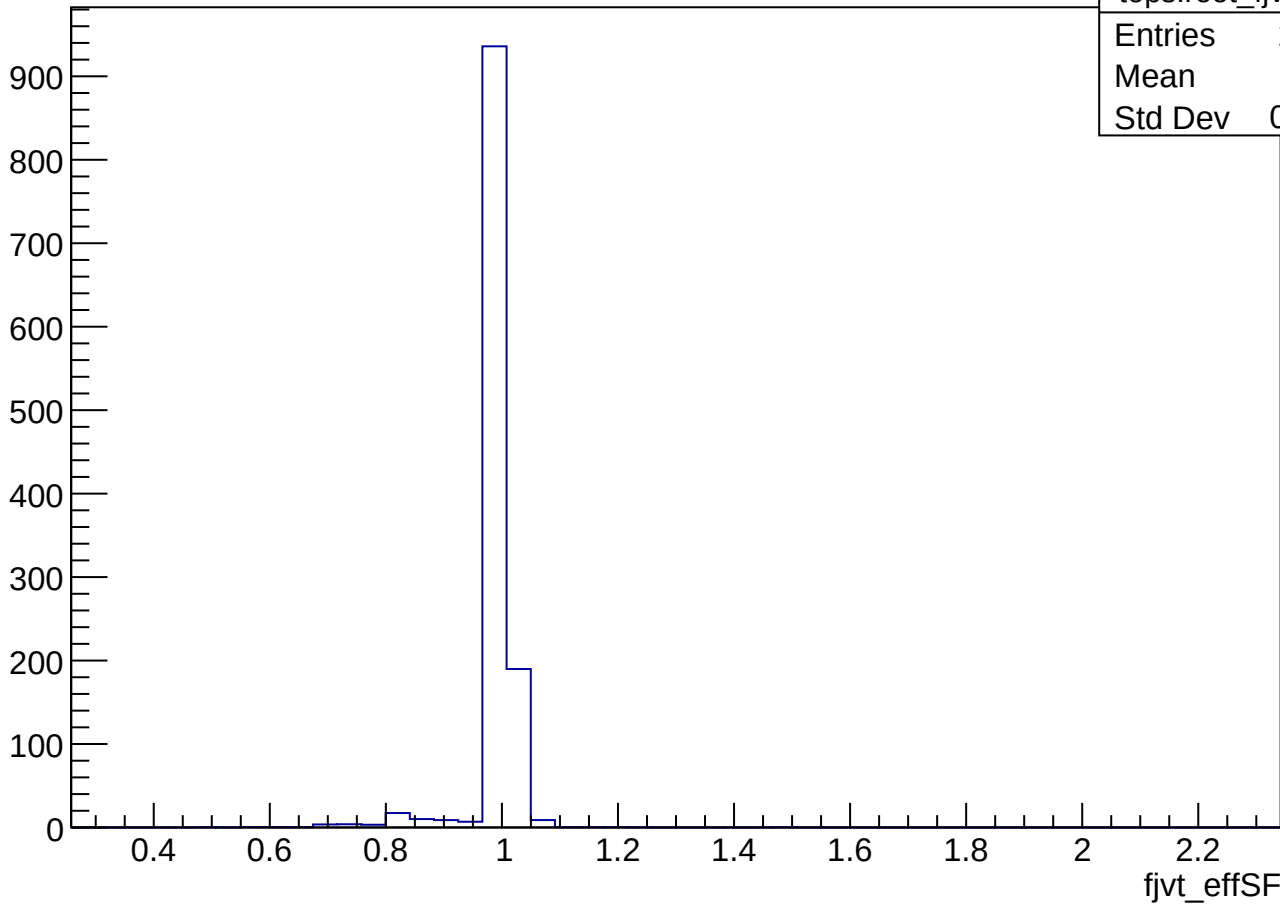
Std Dev 0.8706

dR_t2j2



fjvt_effSF

Events



tops.root_fjvt_effSF	
Entries	223464
Mean	0.9971
Std Dev	0.04185

ftag_effSF_GN2v01_Continuous

Events

400
350
300
250
200
150
100
50
0

0.6

0.8

1.0

1.2

1.4

1.6

1.8

2.0

2.2

2.4

ftag_effSF_GN2v01_Continuous

tops.root_ftag_effSF_GN2v01_Continuous

Entries	223464
Mean	1.024
Std Dev	0.1233

globalTriggerEffSF

Events

600
500
400
300
200
100
0

0

0.2

0.4

0.6

0.8

1

globalTriggerEffSF

tops.root_globalTriggerEffSF	
Entries	223464
Mean	0.9402
Std Dev	0.1069

hhml_subChannelFlavor

Events

600
500
400
300
200
100
0

1

1.2

1.4

1.6

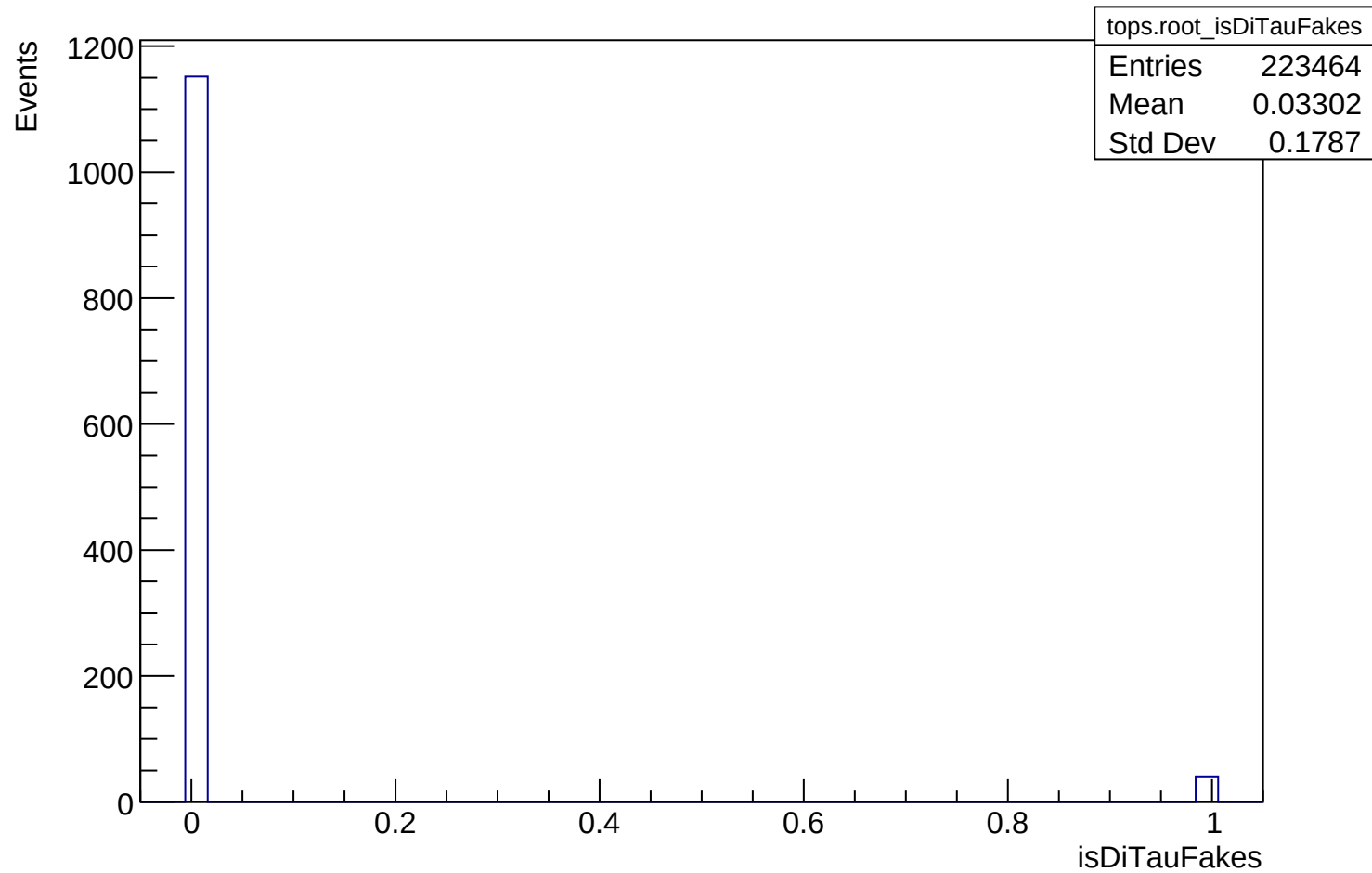
1.8

2

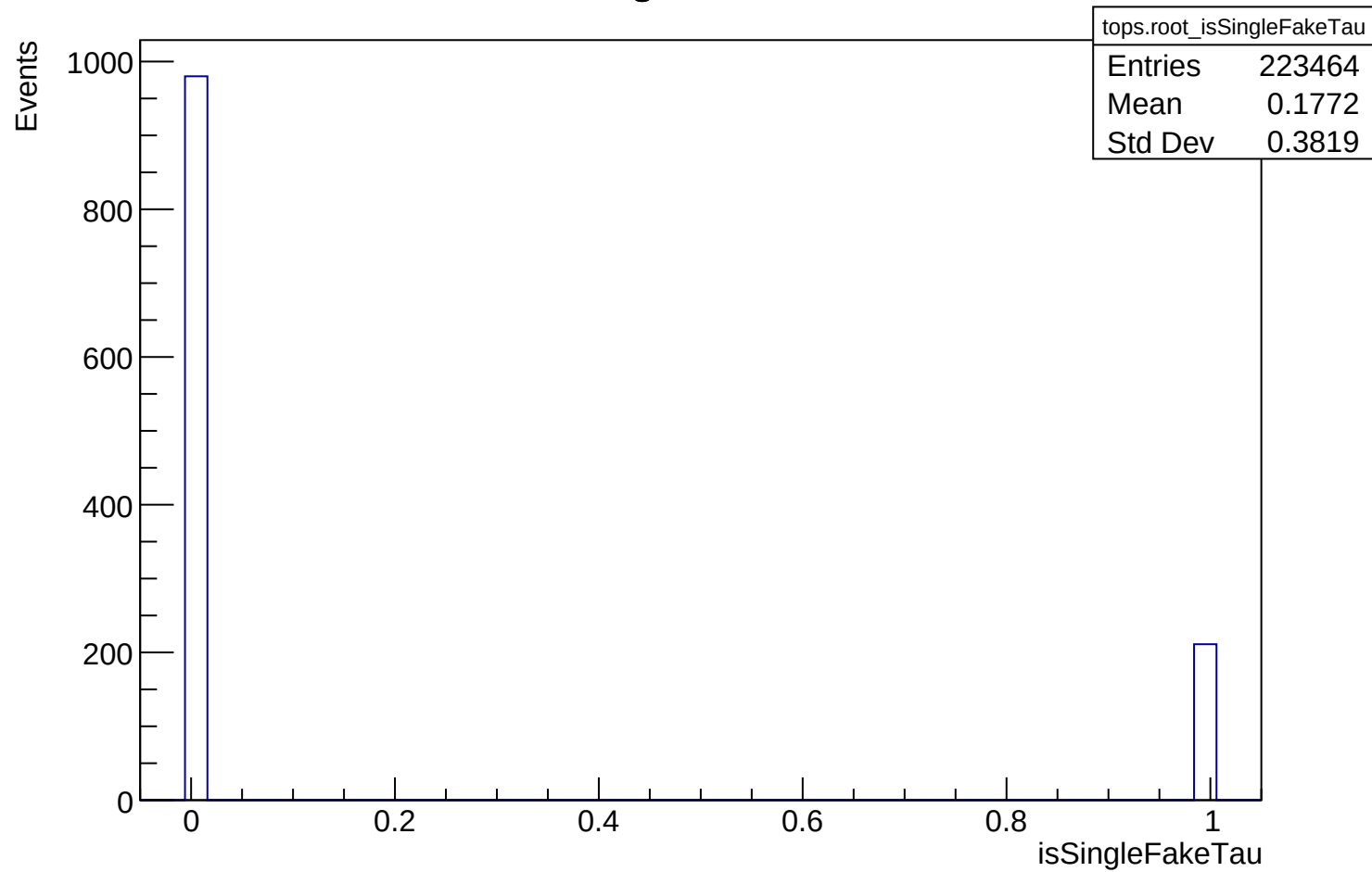
hhml_subChannelFlavor

tops.root_hhml_subChannelFlavor	
Entries	223464
Mean	1.52
Std Dev	0.4996

isDiTauFakes

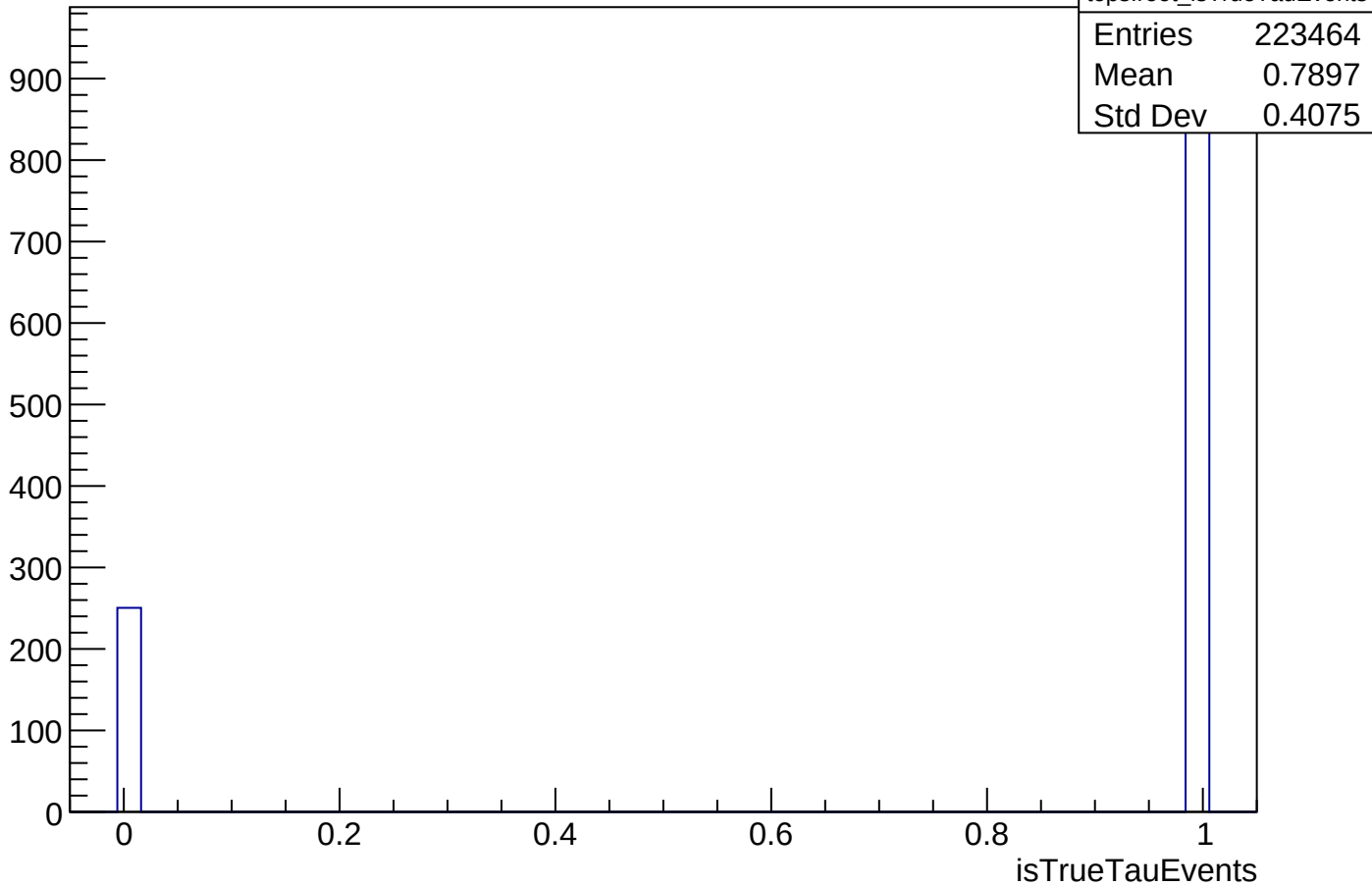


isSingleFakeTau



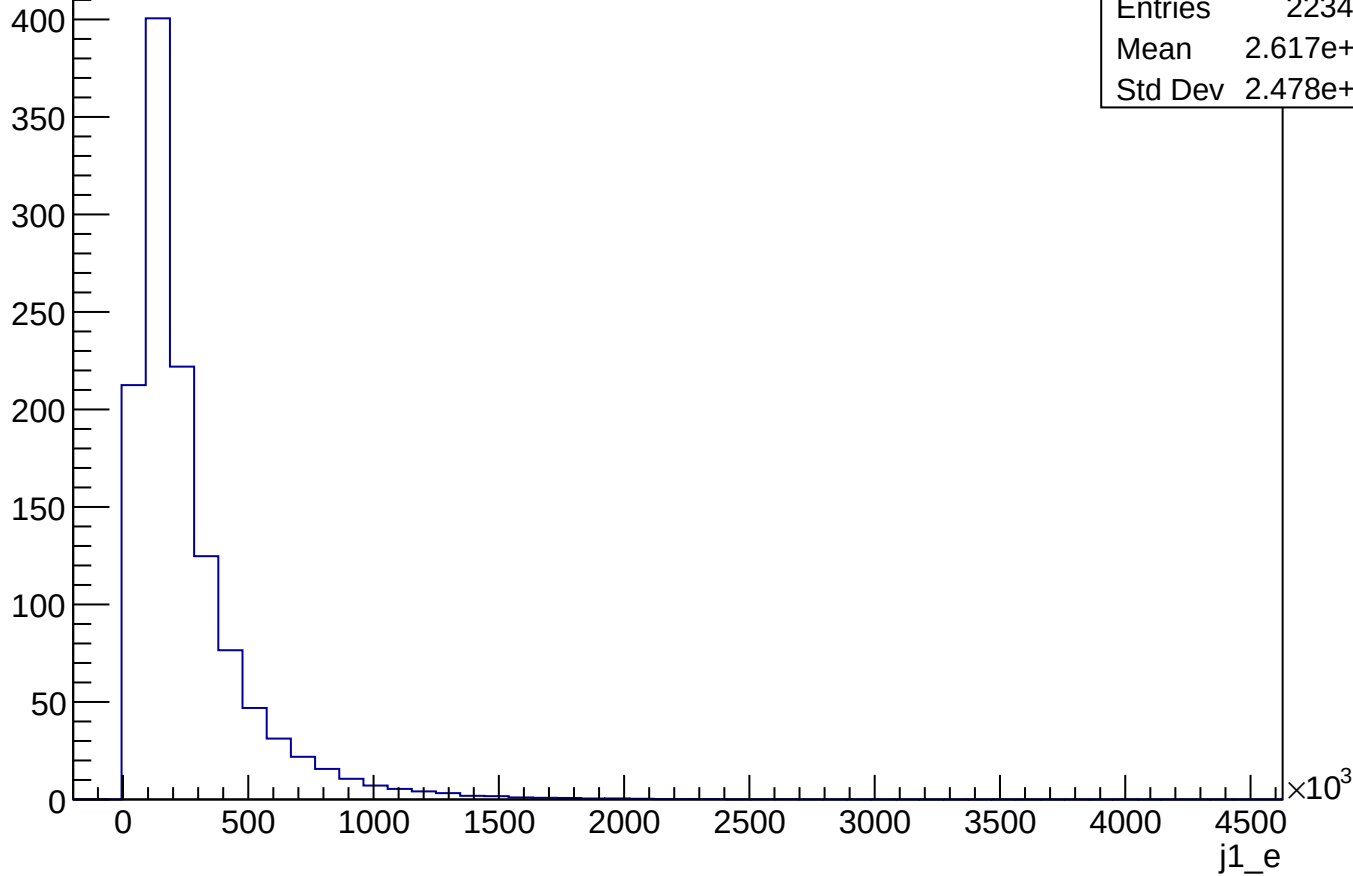
isTrueTauEvents

Events



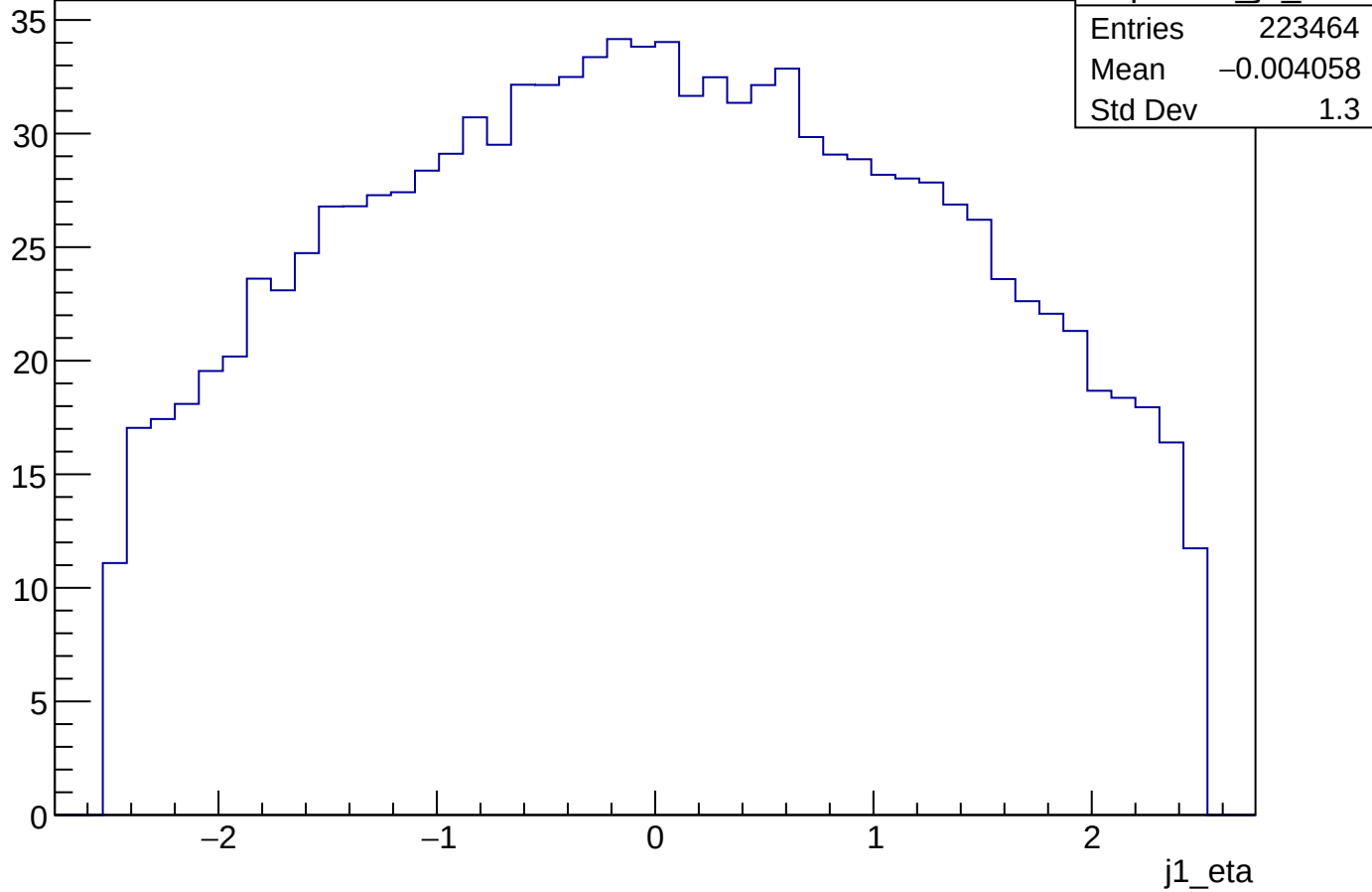
j1_e

Events



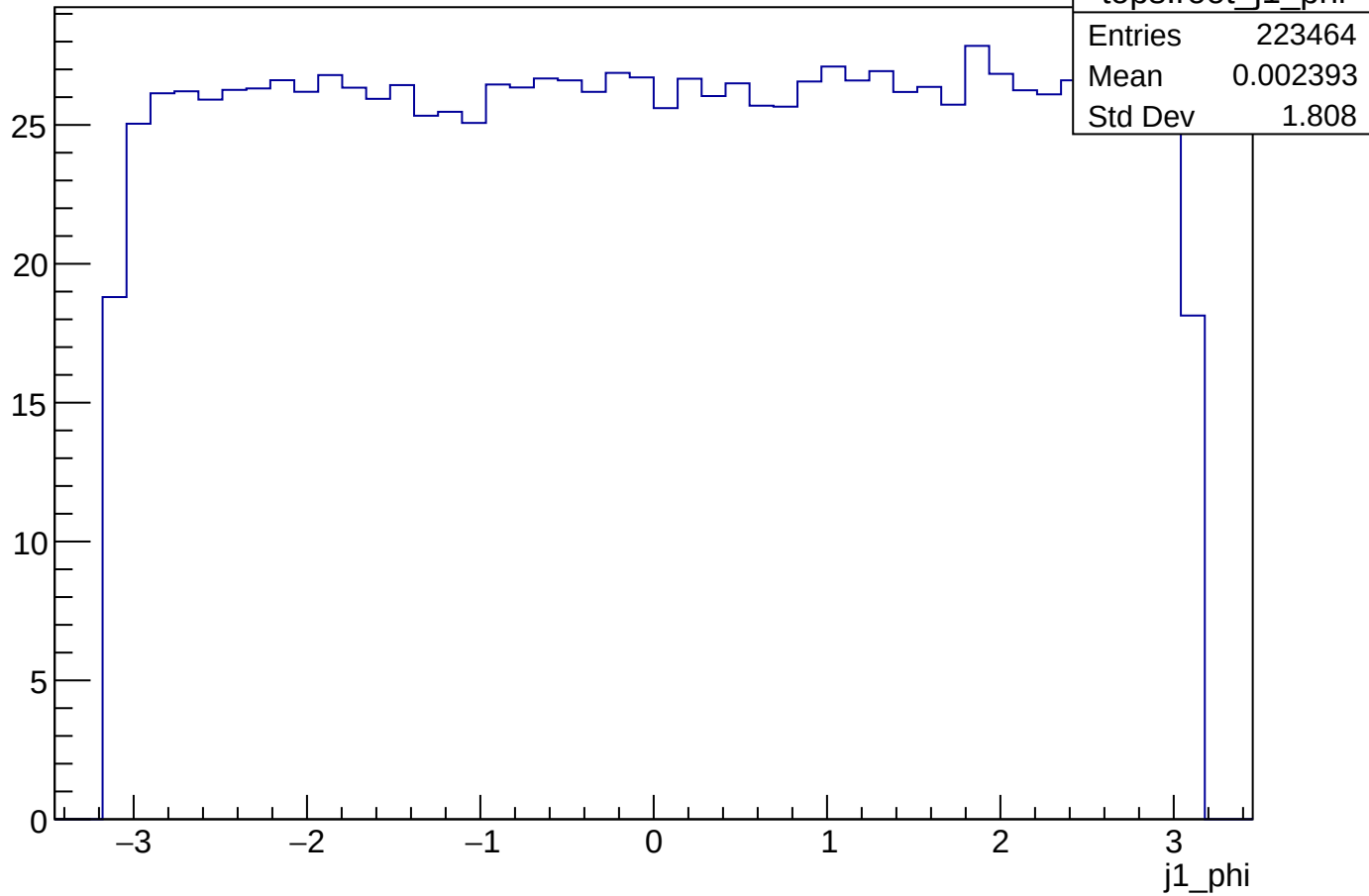
j1_eta

Events



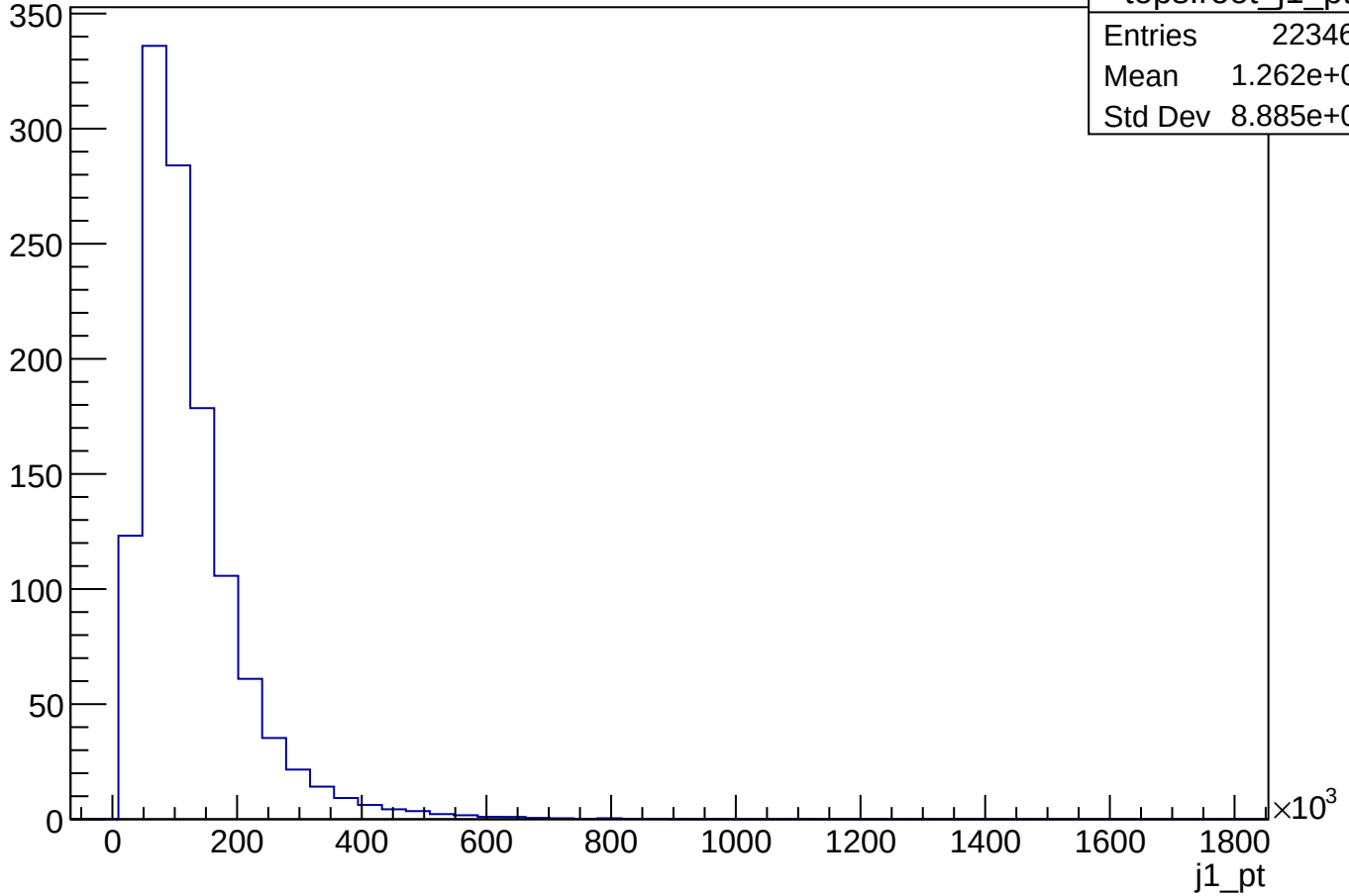
j1_phi

Events



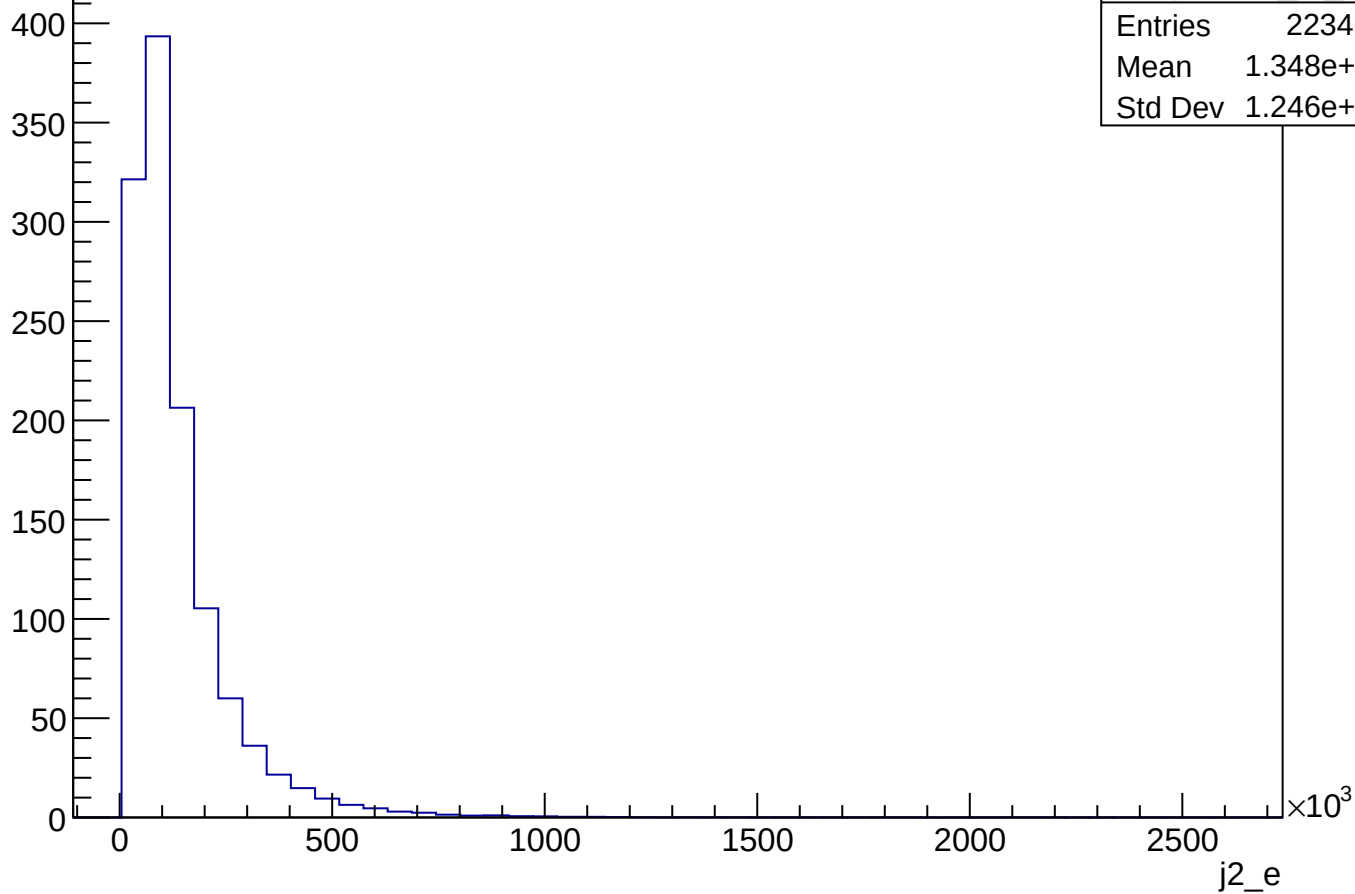
j1_pt

Events



j2_e

Events



j2_eta

Events

35
30
25
20
15
10
5
0

-2

-1

0

1

2

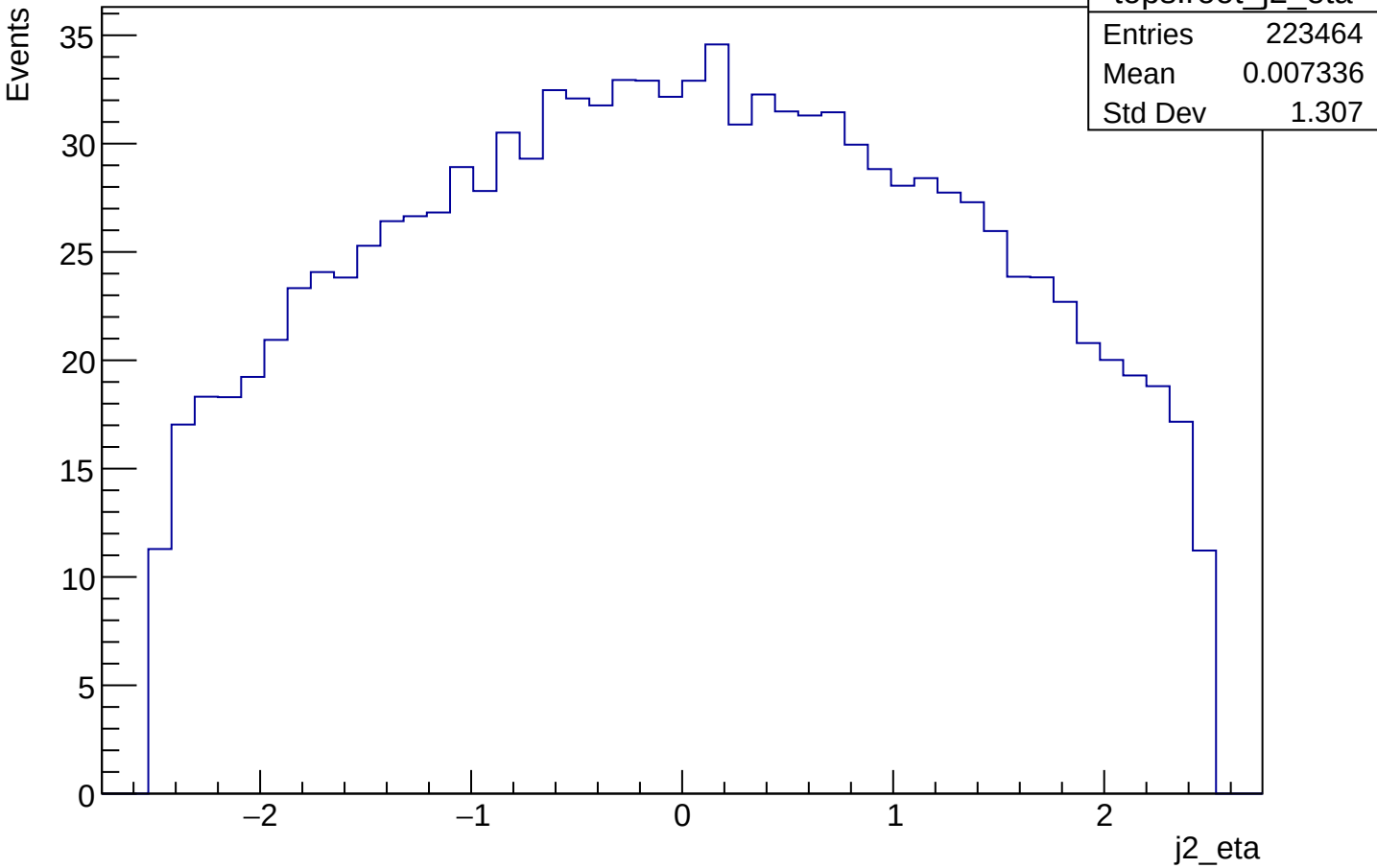
j2_eta

tops.root_j2_eta

Entries 223464

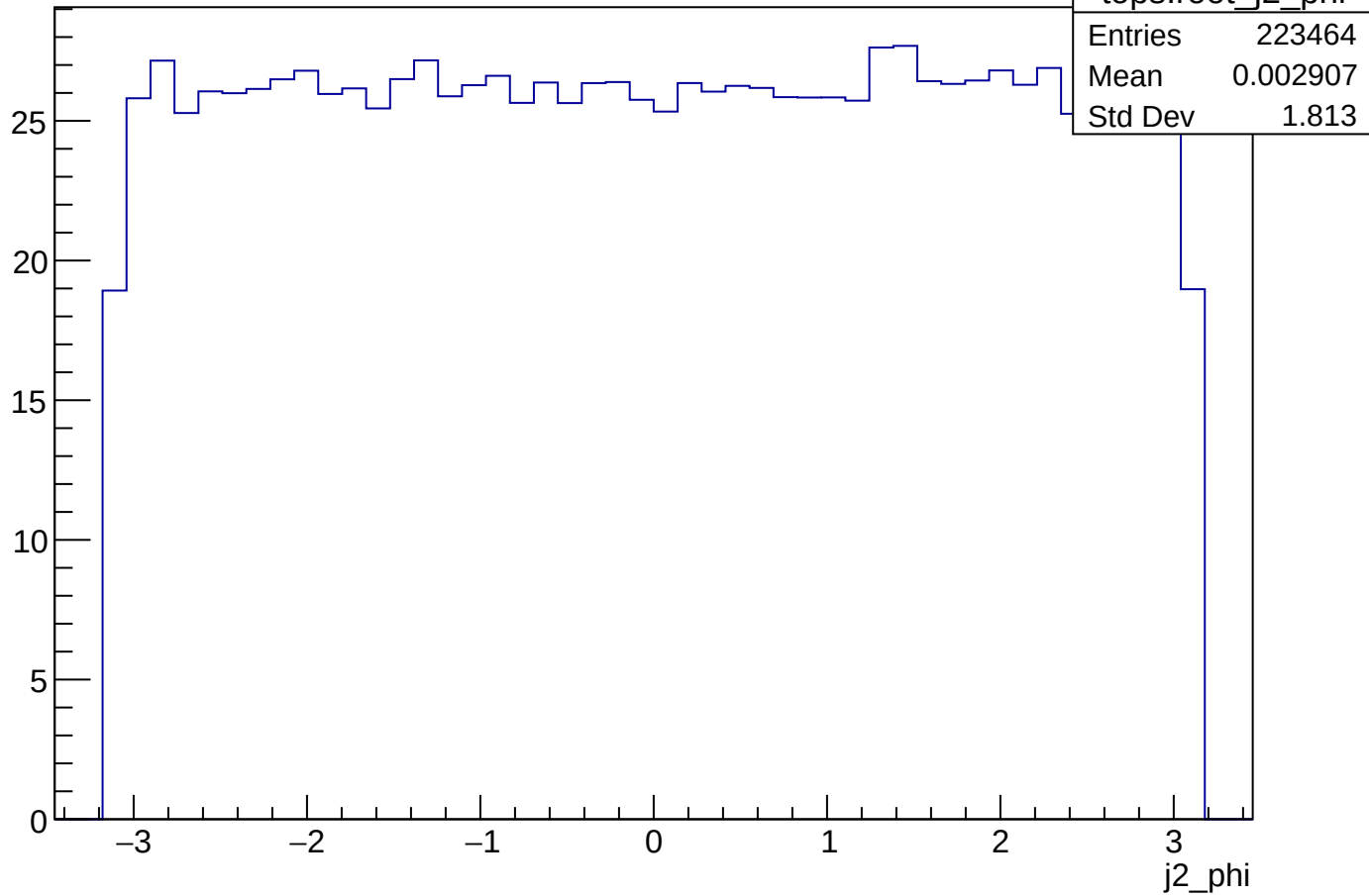
Mean 0.007336

Std Dev 1.307



j2_phi

Events



j2_pt

Events

400
350
300
250
200
150
100
50
0

0

200

400

600

800

1000

1200

j2_pt

$\times 10^3$

tops.root_j2_pt

Entries 223464

Mean 6.547e+04

Std Dev 4.495e+04



jvt_effSF

Events

400
350
300
250
200
150
100
50
0

0.6

0.8

1.0

1.2

1.4

1.6

1.8

2.0

2.2

2.4

jvt_effSF

tops.root_jvt_effSF	
Entries	223464
Mean	1.024
Std Dev	0.1233

l1_charge

Events

600
500
400
300
200
100
0

-1

-0.5

0

0.5

1

l1_charge

tops.root_l1_charge	
Entries	223464
Mean	-0.009823
Std Dev	1

l1_d0sig

Events

100
80
60
40
20
0

-4

-2

0

2

4

l1_d0sig

tops.root_l1_d0sig	
Entries	223464
Mean	0.01261
Std Dev	1.146

l1_e

Events

400
350
300
250
200
150
100
50
0

0

500

1000

1500

2000

2500

$\times 10^3$

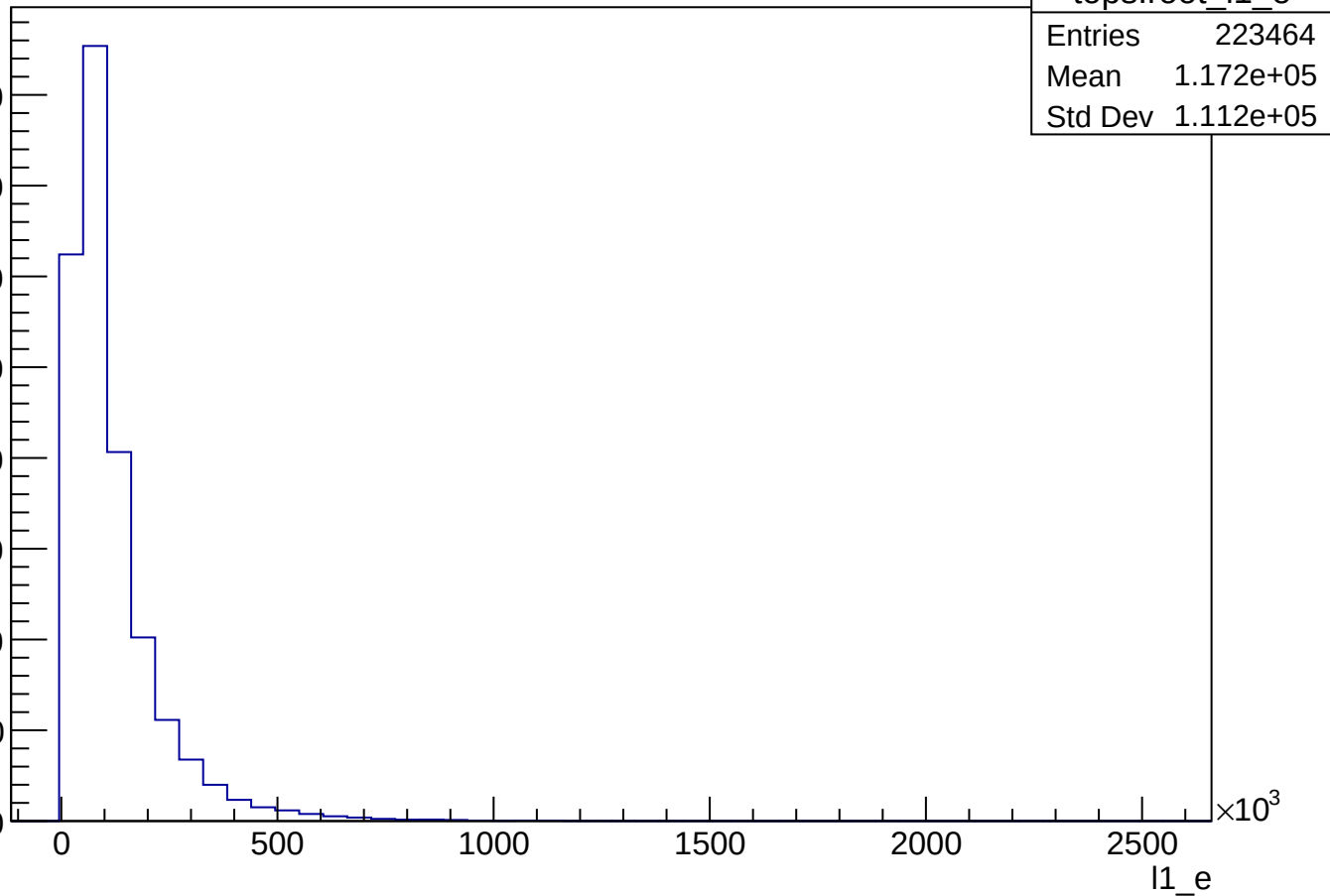
tops.root_l1_e

Entries 223464

Mean 1.172e+05

Std Dev 1.112e+05

l1_e



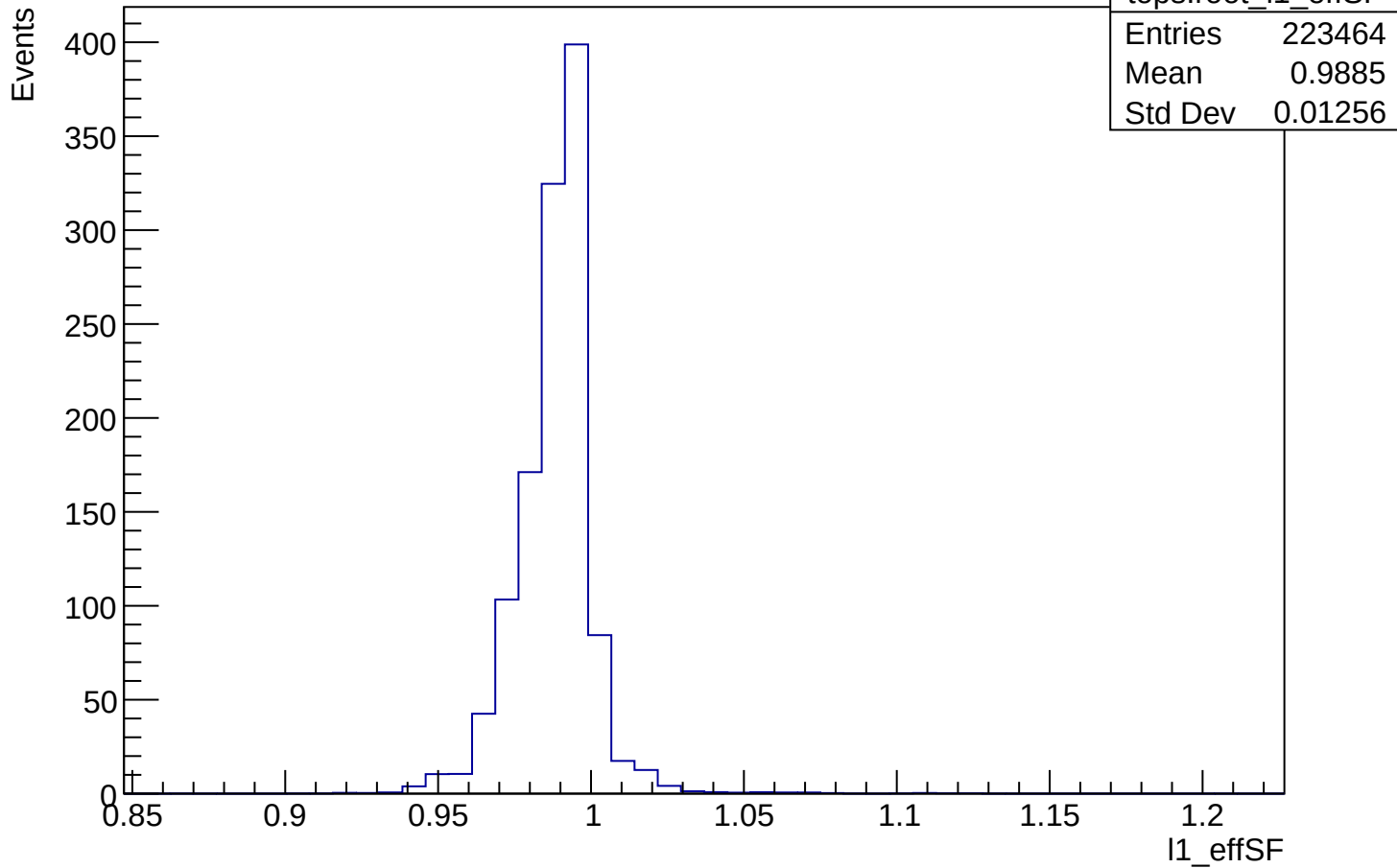
l1_effSF

Events

400
350
300
250
200
150
100
50
0

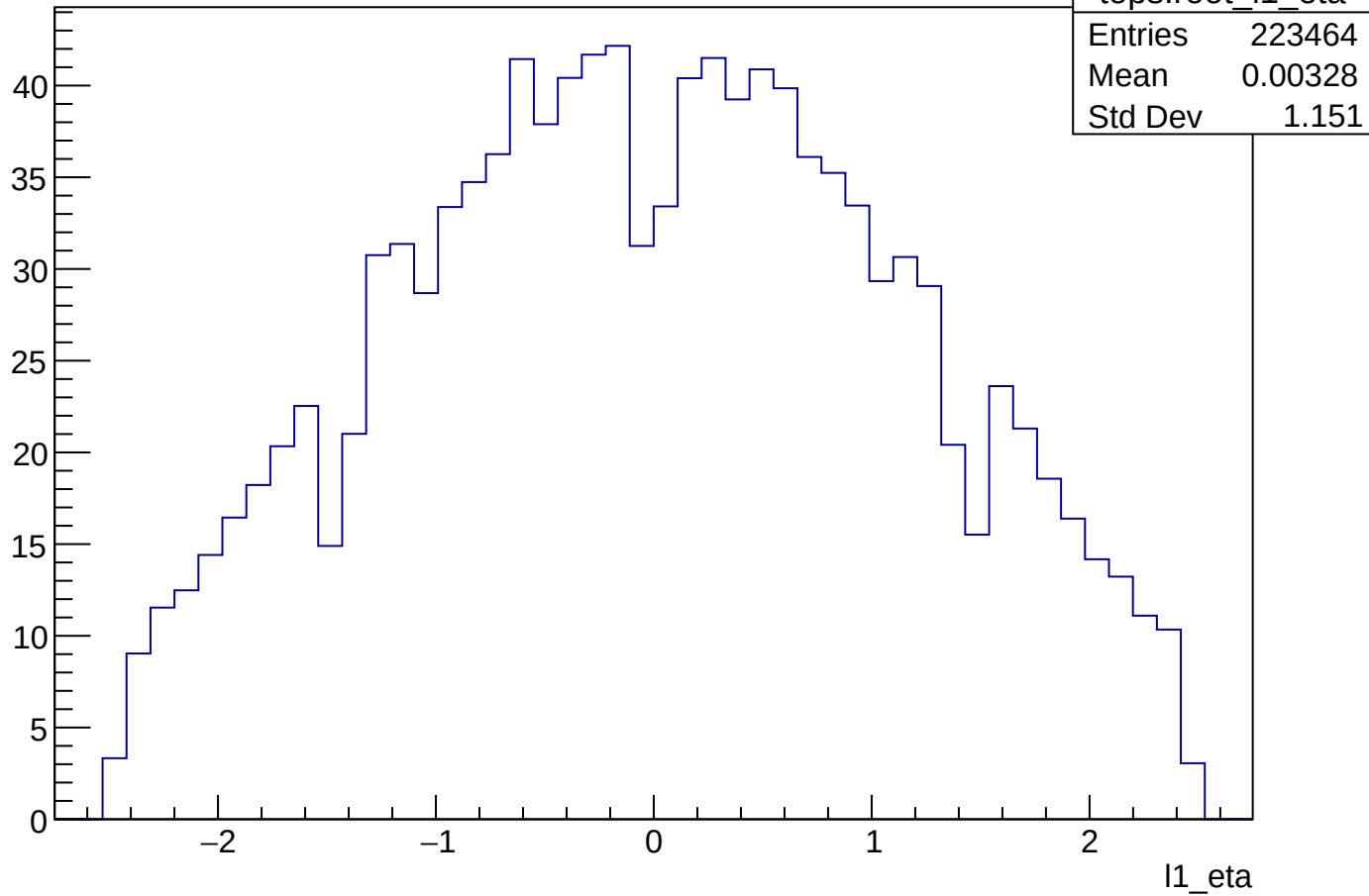
tops.root_l1_effSF	
Entries	223464
Mean	0.9885
Std Dev	0.01256

0.85 0.9 0.95 1 1.05 1.1 1.15 1.2
l1_effSF



$l1_eta$

Events



l1_pdg

Events

300
250
200
150
100
50
0

-10

-5

0

5

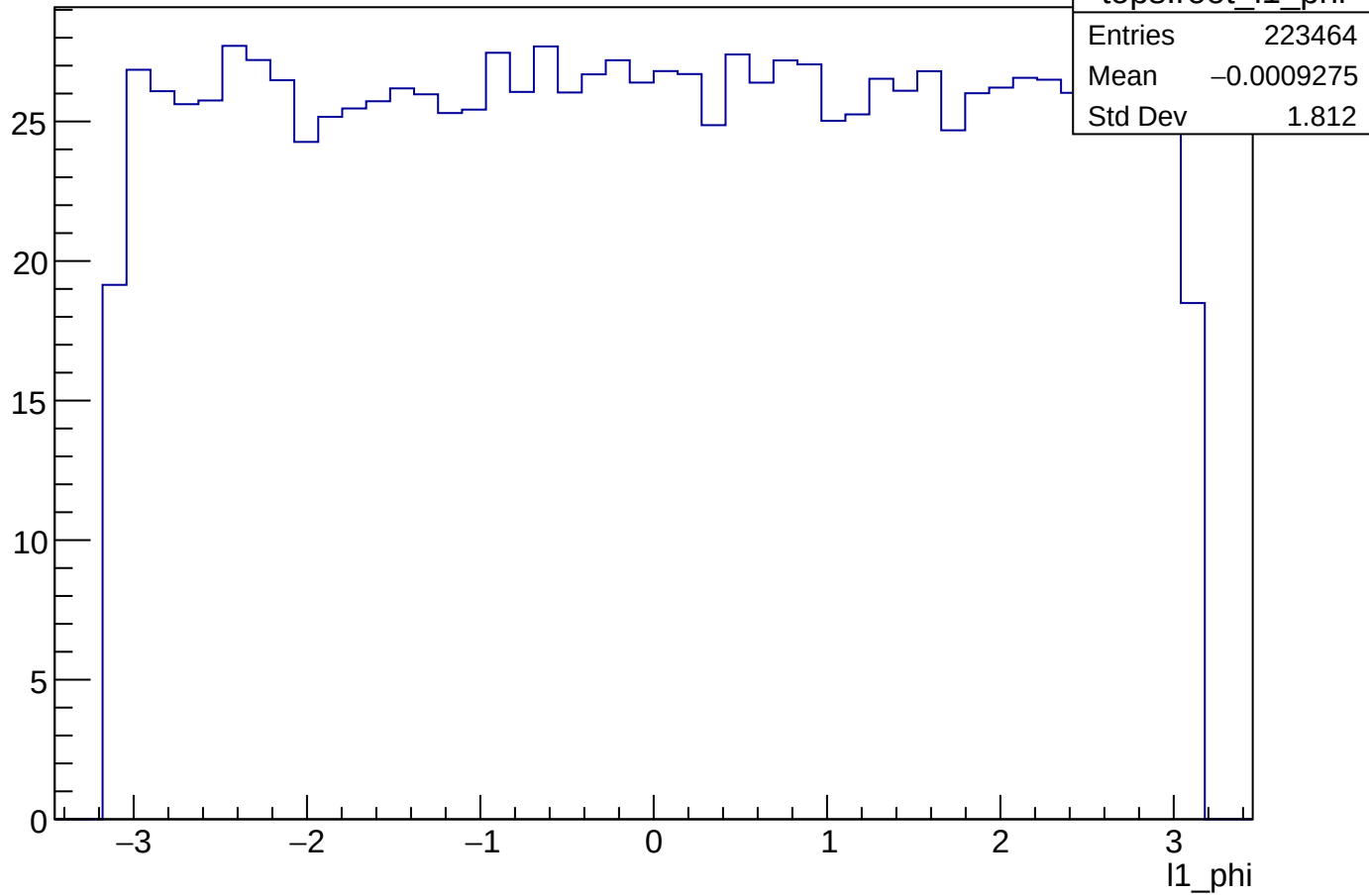
10

l1_pdg

tops.root_l1_pdg	
Entries	223464
Mean	0.1231
Std Dev	12.08

l1_phi

Events



l1_pt

Events

400
350
300
250
200
150
100
50
0

0

200

400

600

800

1000

1200

l1_pt

$\times 10^3$

tops.root_l1_pt

Entries 223464

Mean 6.553e+04

Std Dev 5.043e+04



l1_z0sintheta

Events

300
250
200
150
100
50
0

-0.4

-0.2

0

0.2

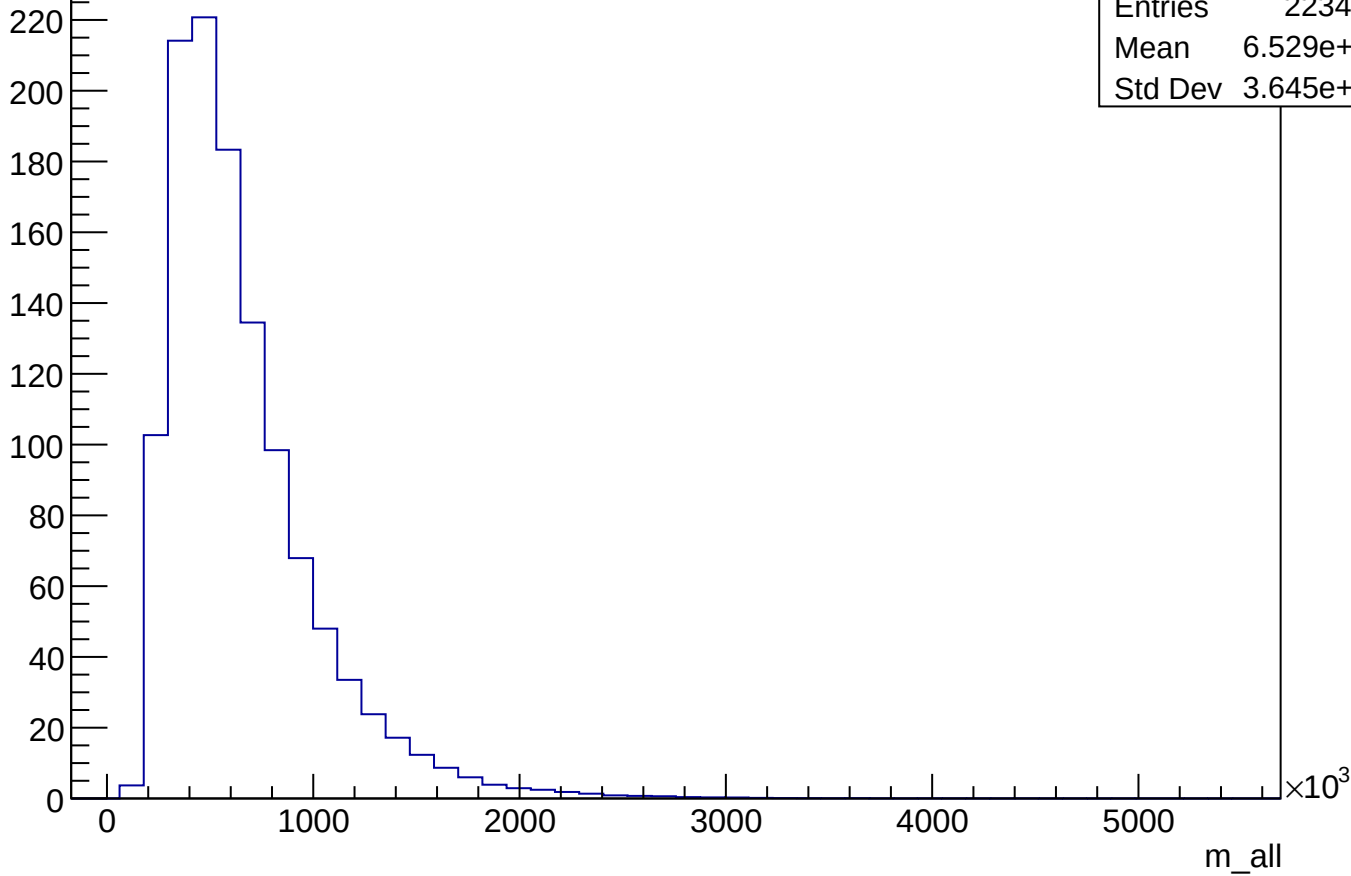
0.4

l1_z0sintheta

tops.root_l1_z0sintheta	
Entries	223464
Mean	2.801e-05
Std Dev	0.04808

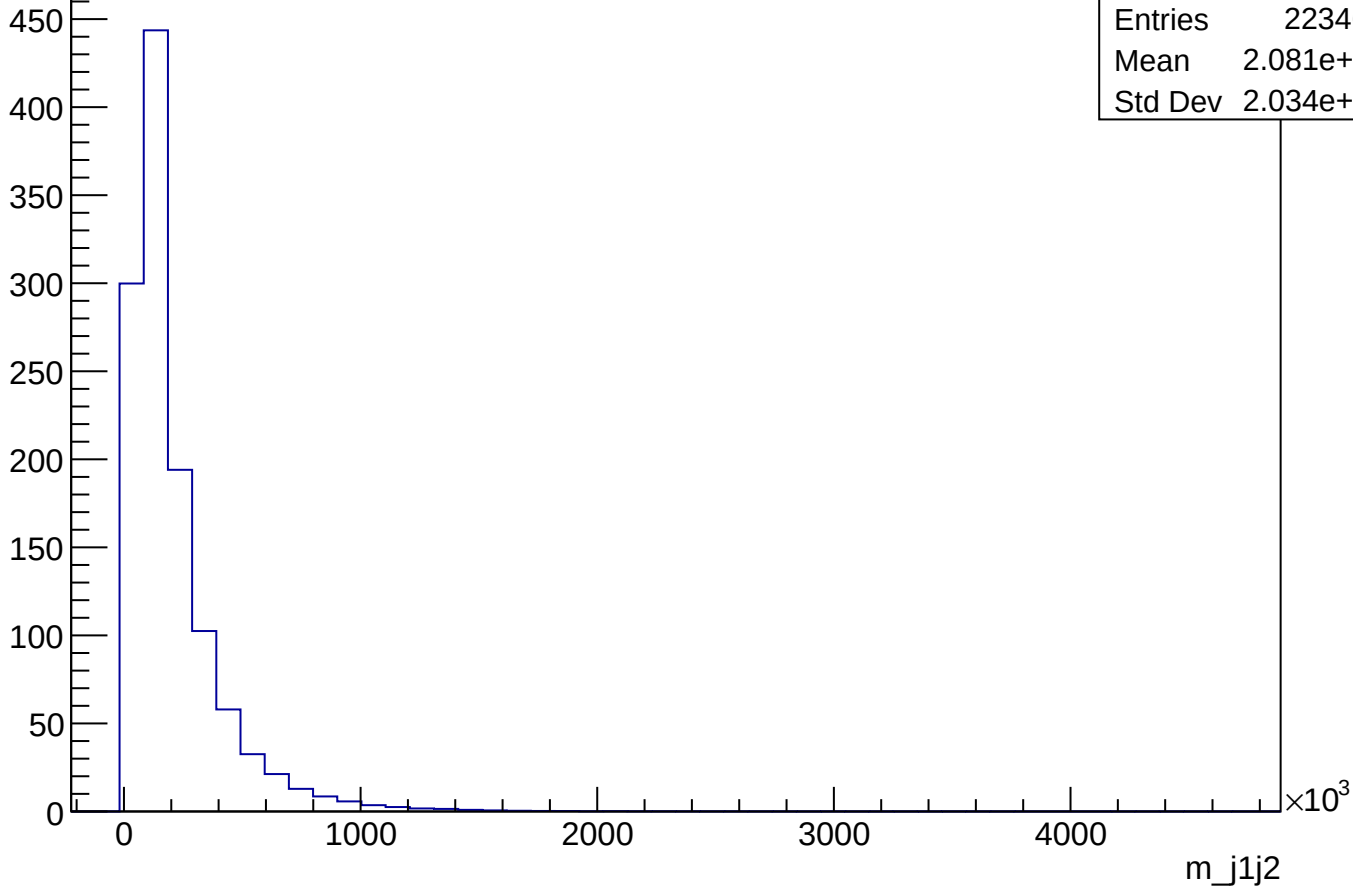
m_all

Events



m_j1j2

Events



m_l1j

Events

500
400
300
200
100
0

0

500

1000

1500

2000

m_l1j

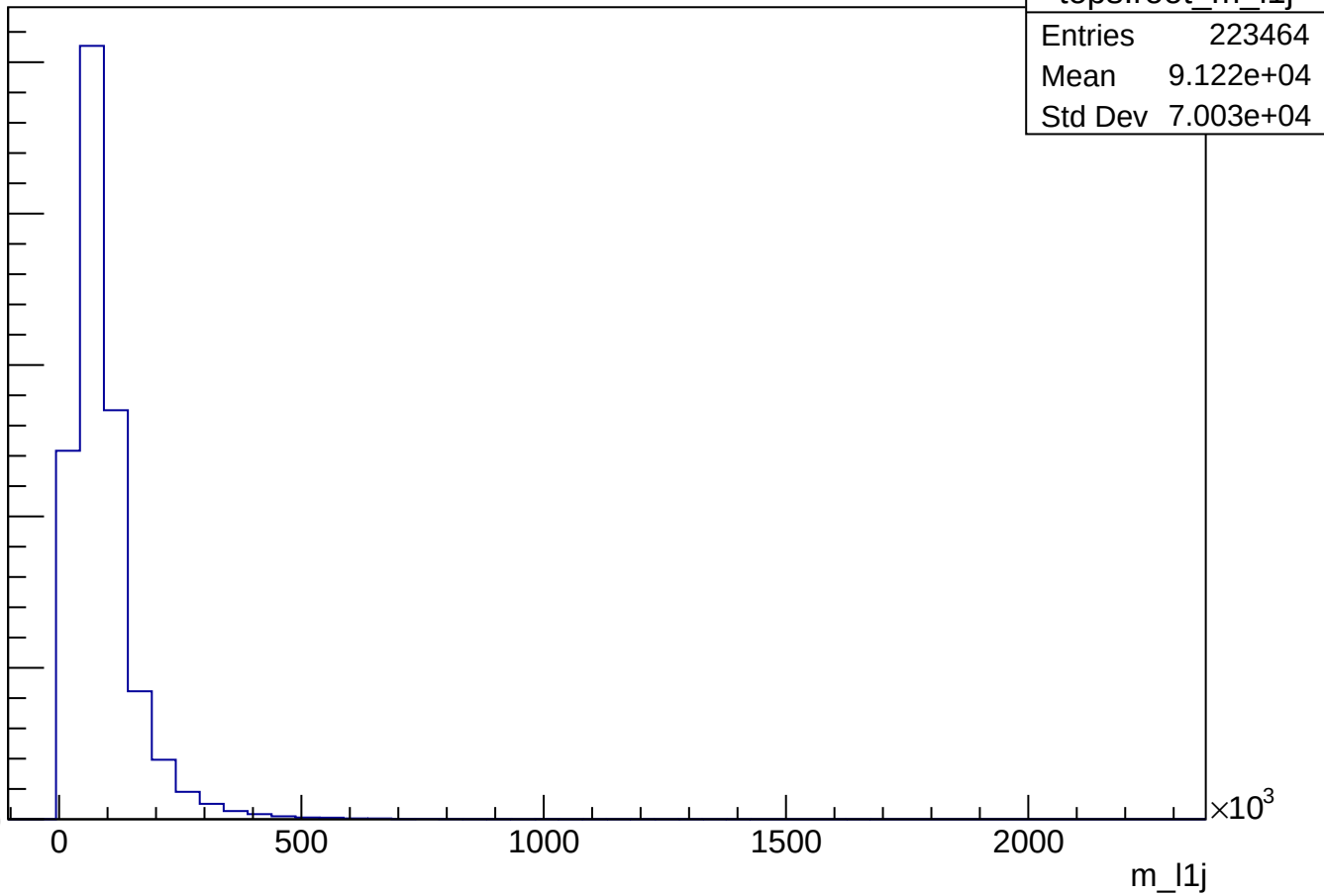
$\times 10^3$

tops.root_m_l1j

Entries 223464

Mean 9.122e+04

Std Dev 7.003e+04



m_l1j1

Events

350
300
250
200
150
100
50
0

0

500

1000

1500

2000

2500

3000

$\times 10^3$

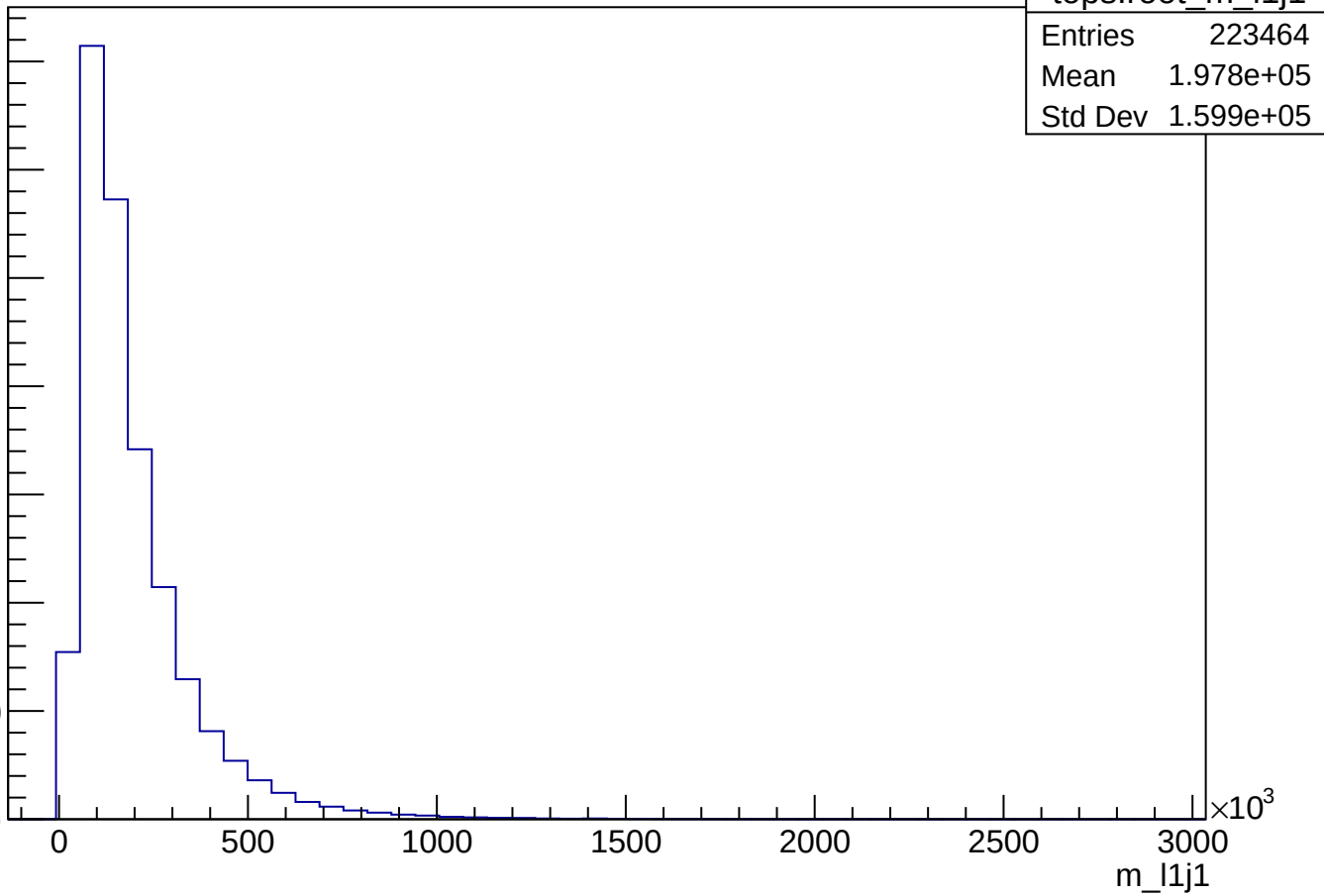
m_l1j1

tops.root_m_l1j1

Entries 223464

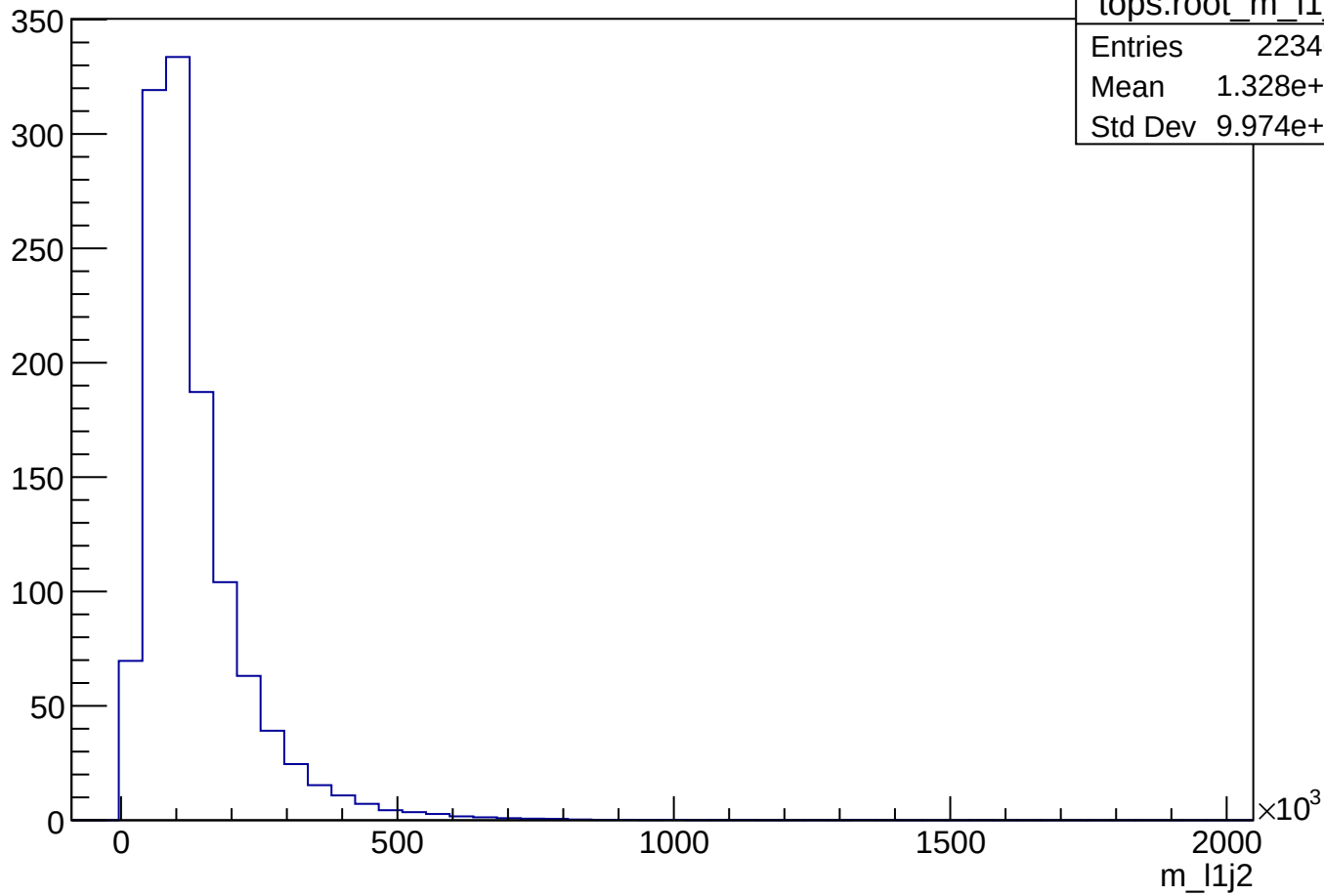
Mean 1.978e+05

Std Dev 1.599e+05



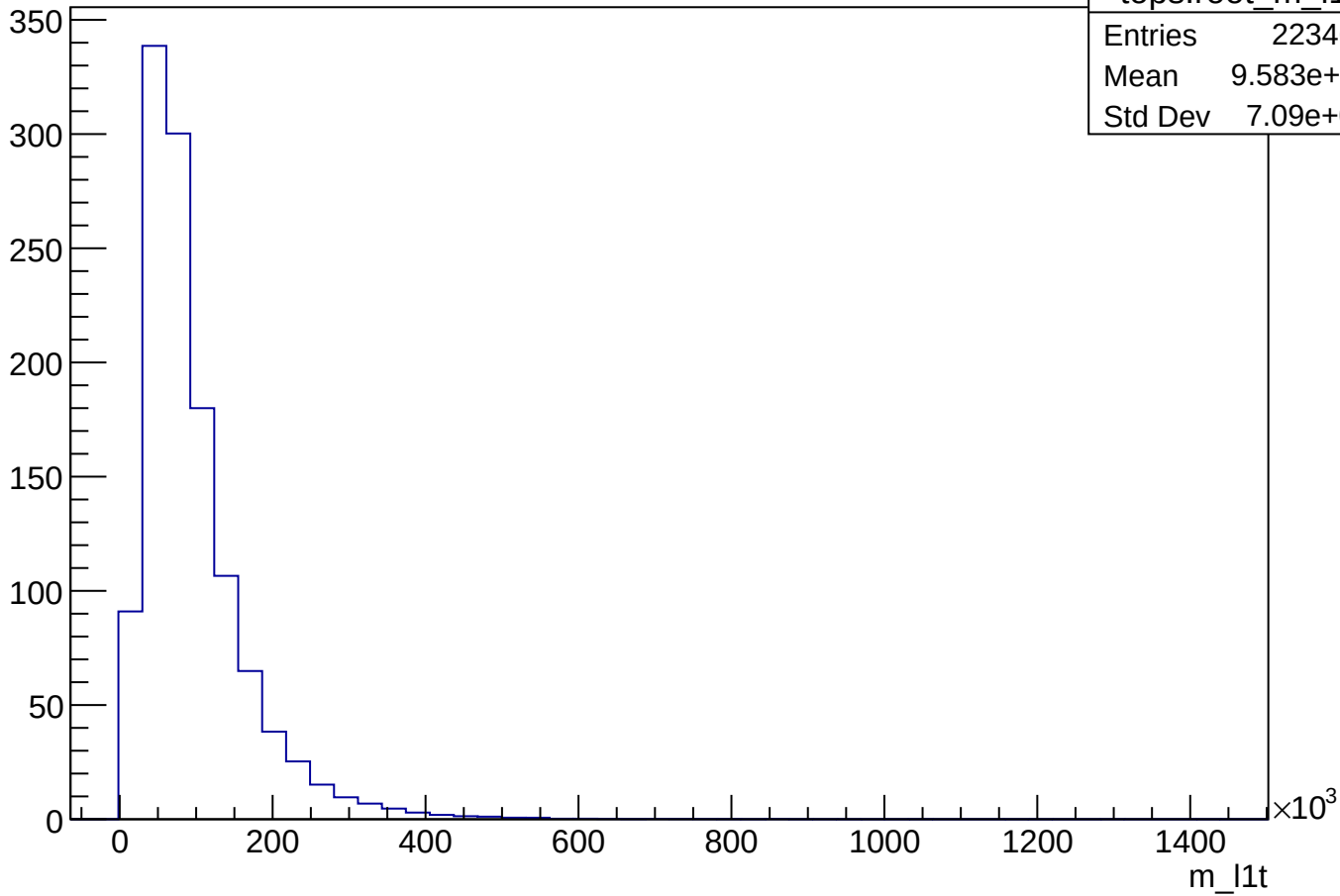
m_l1j2

Events



m_l1t

Events



m_l1t1

Events

250
200
150
100
50
0

0

0

200

400

600

800

1000

1200

1400

m_l1t1

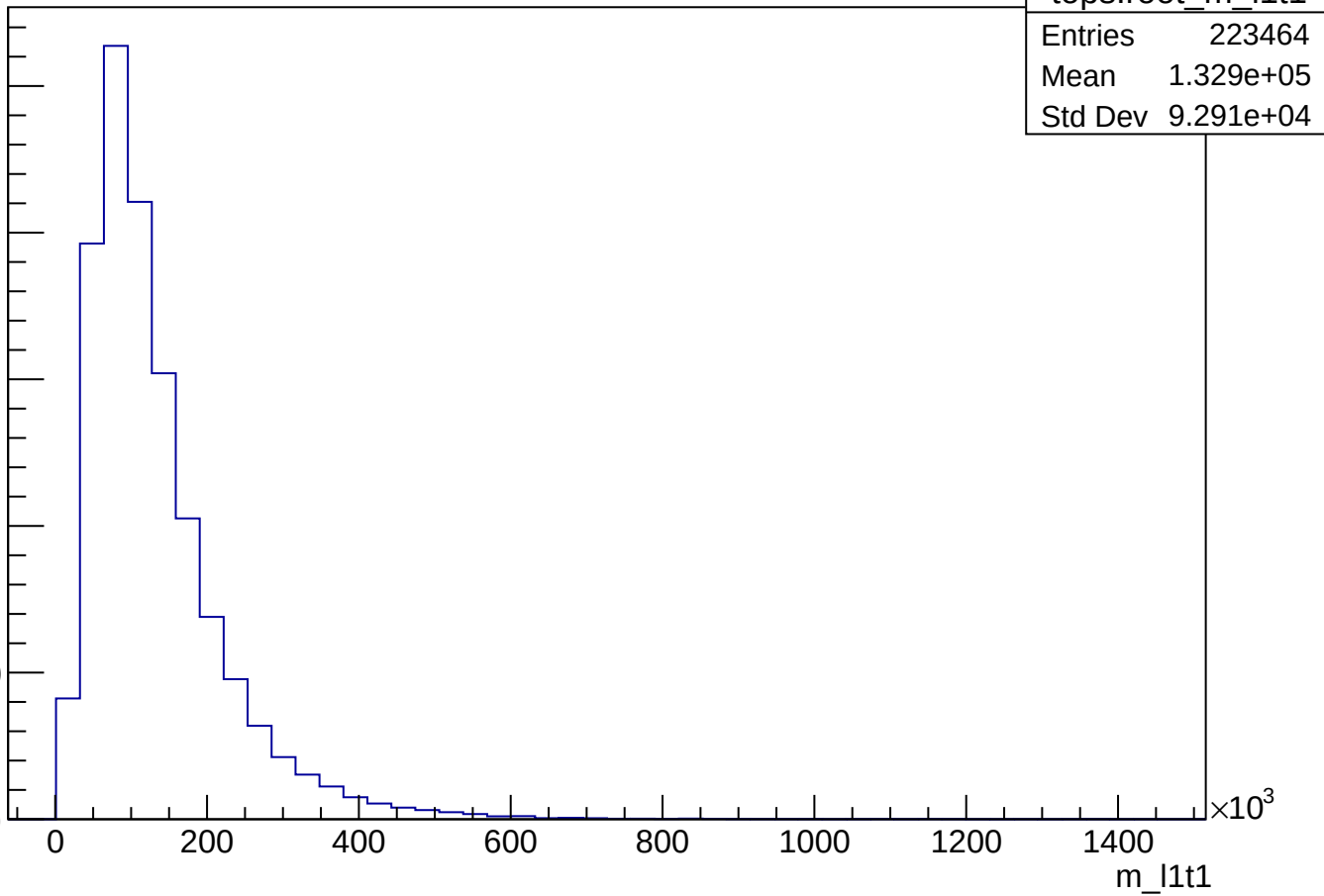
$\times 10^3$

tops.root_m_l1t1

Entries 223464

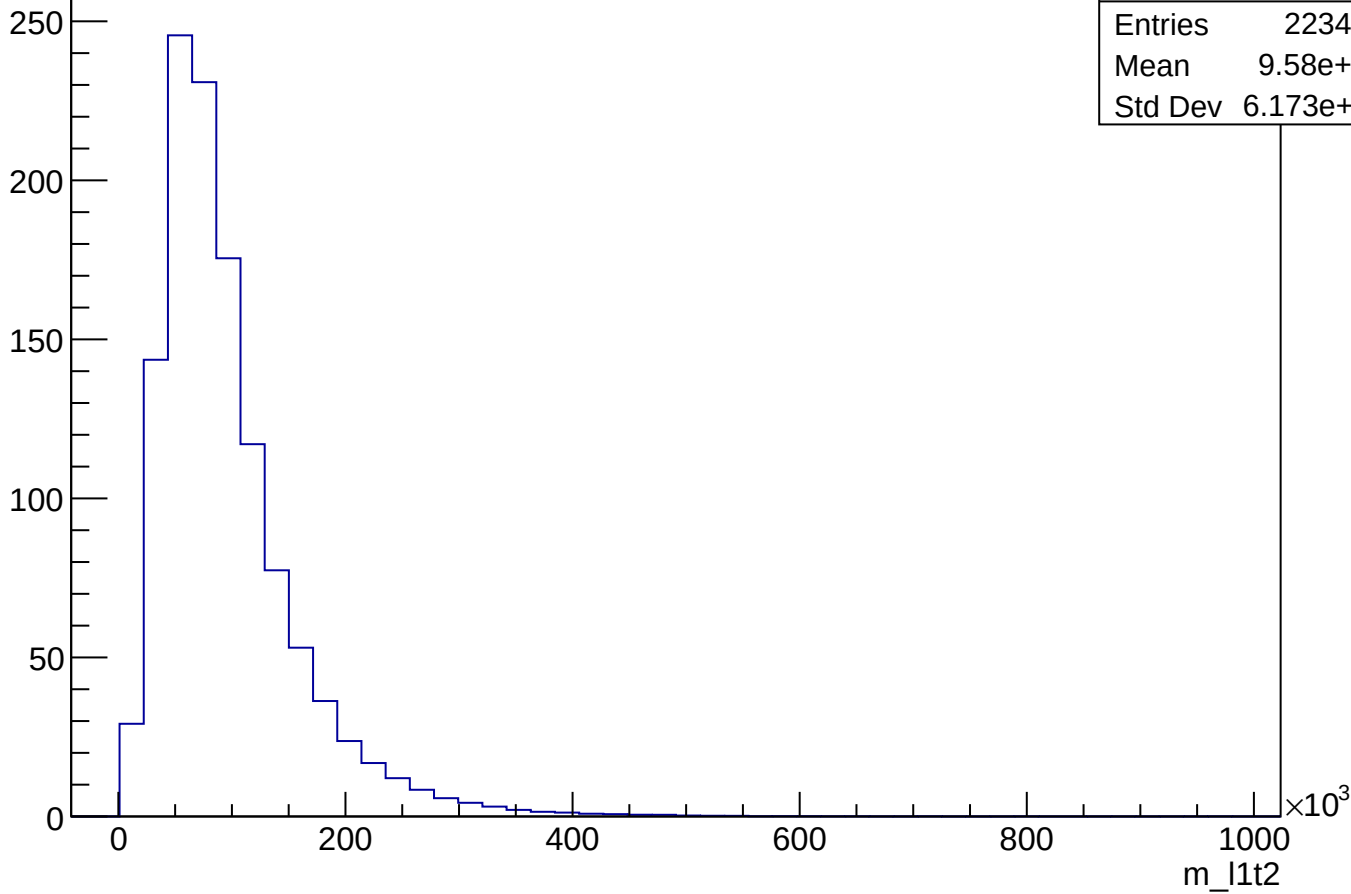
Mean 1.329e+05

Std Dev 9.291e+04



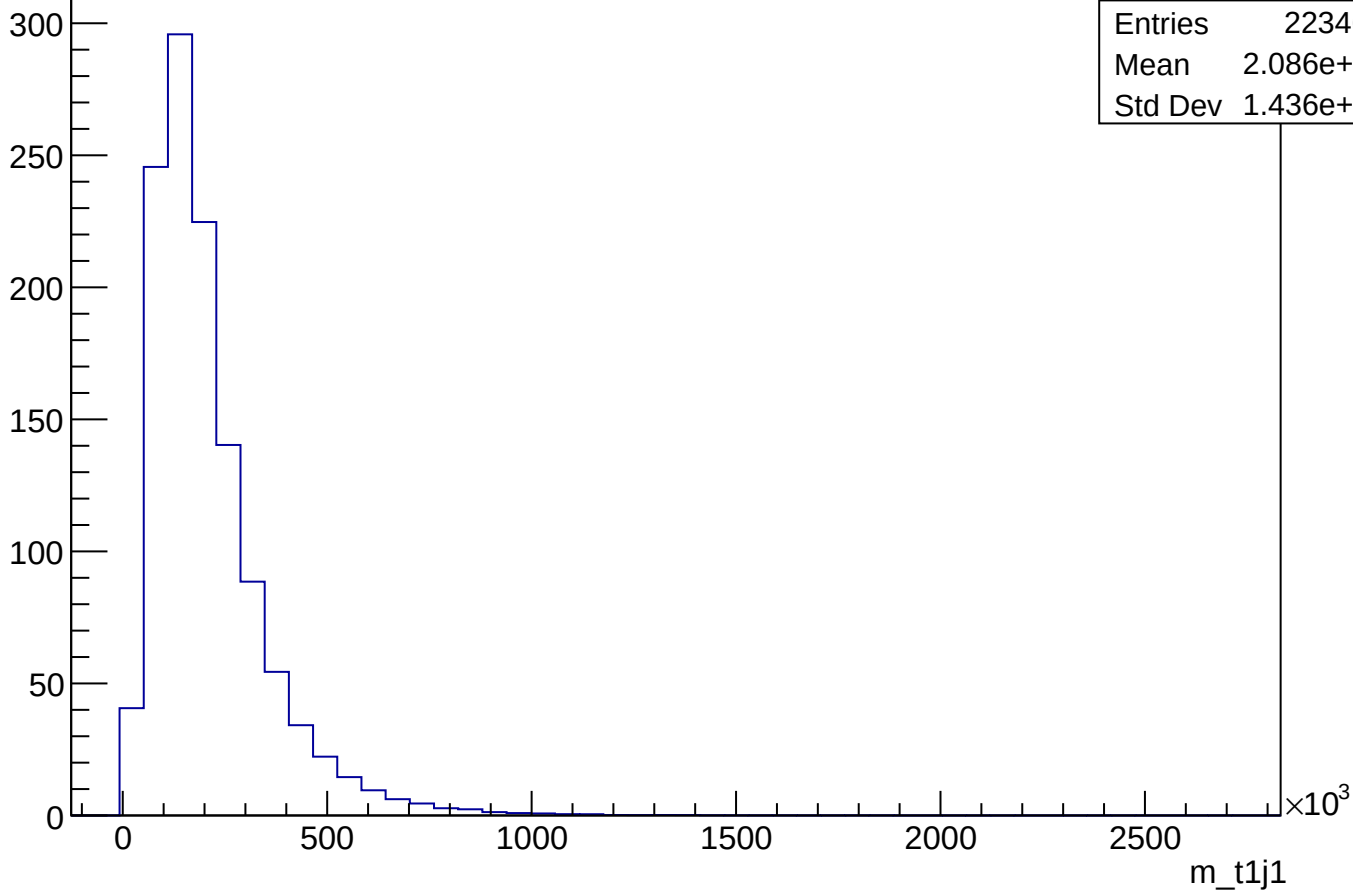
m_l1t2

Events



m_t1j1

Events



m_t1j2

Events

300

250

200

150

100

50

0

0

500

1000

1500

2000

m_t1j2

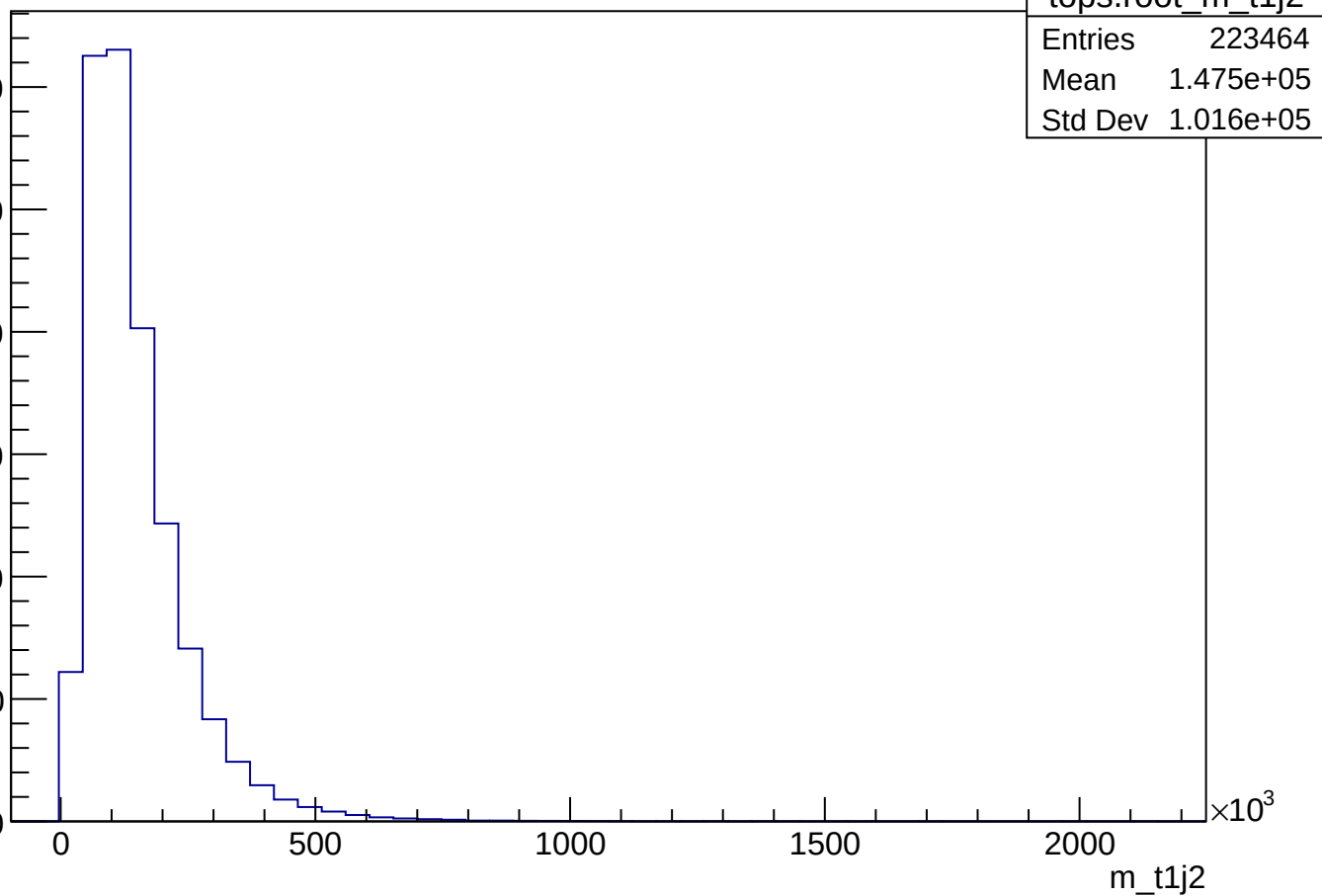
$\times 10^3$

tops.root_m_t1j2

Entries 223464

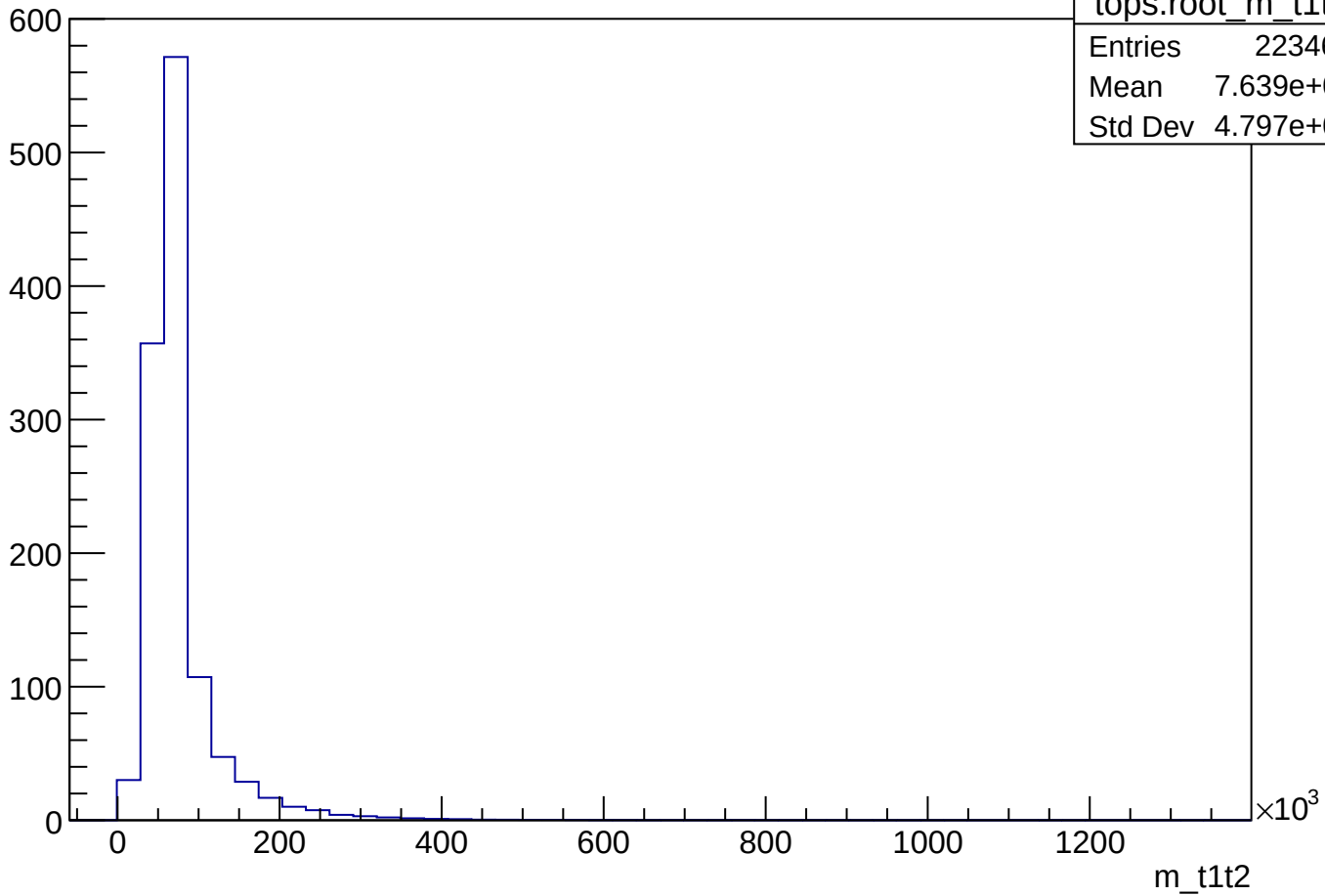
Mean 1.475e+05

Std Dev 1.016e+05



m_t1t2

Events



m_t2j1

Events

300

250

200

150

100

50

0

0

500

1000

1500

2000

m_t2j1

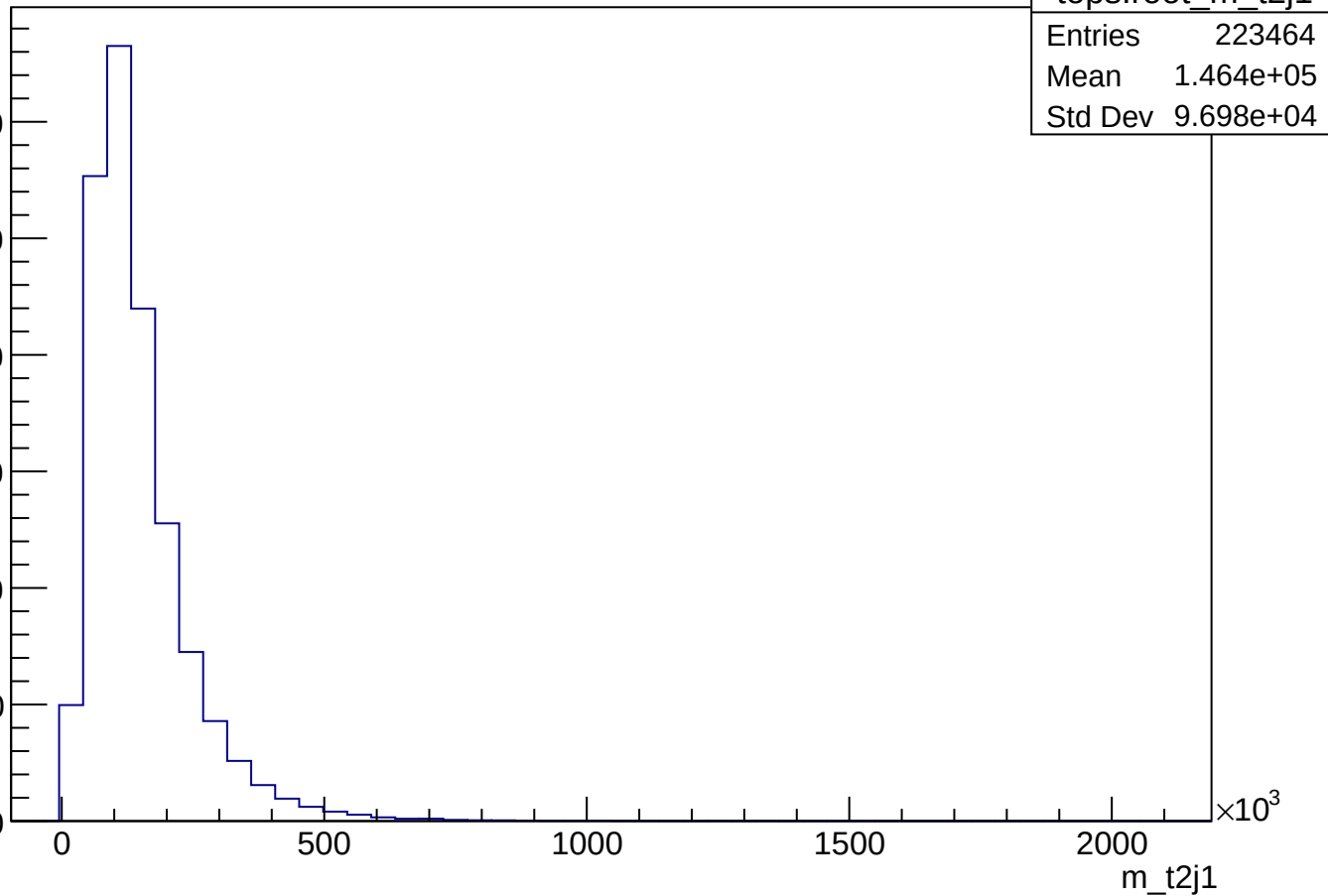
$\times 10^3$

tops.root_m_t2j1

Entries 223464

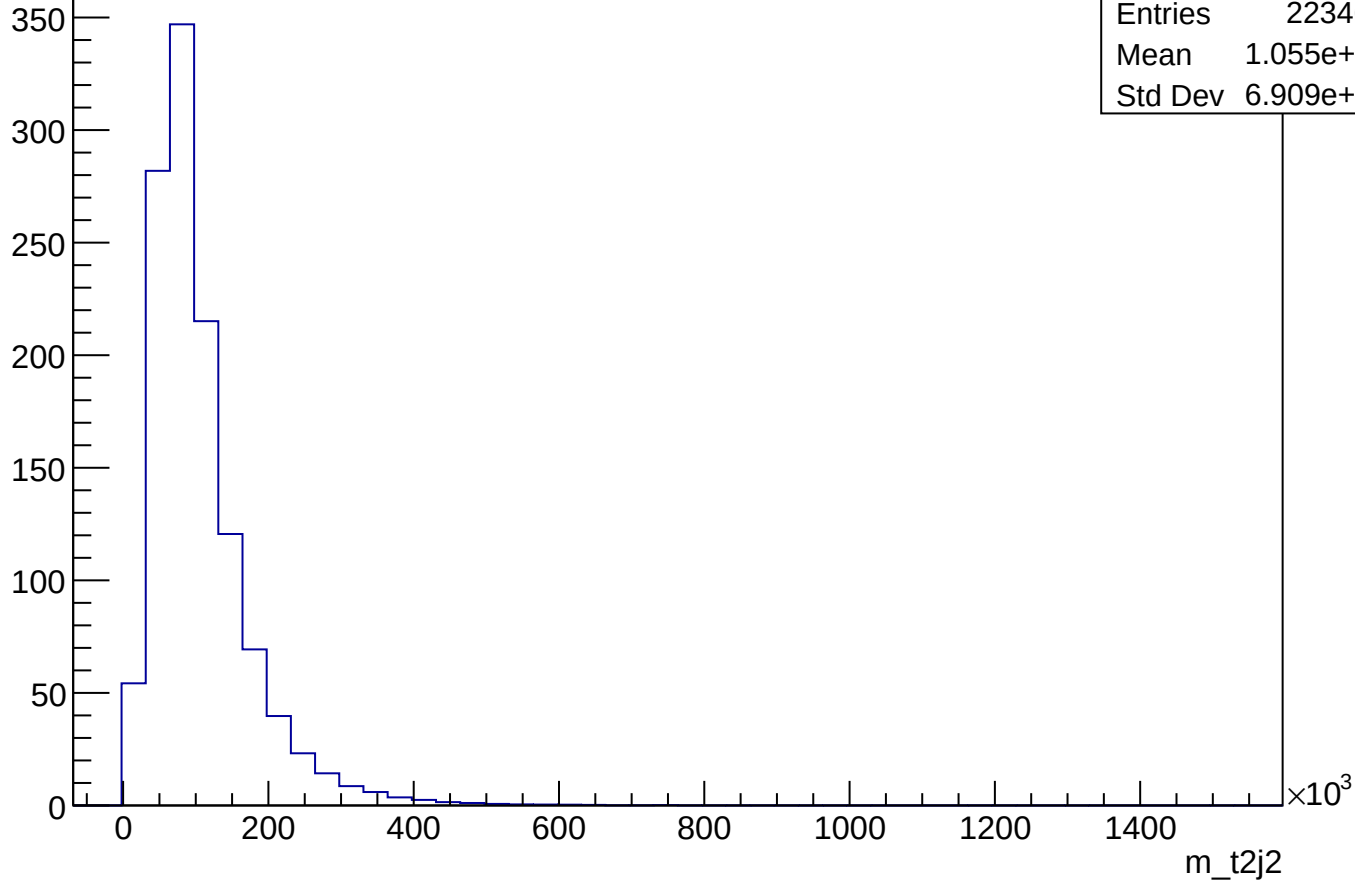
Mean 1.464e+05

Std Dev 9.698e+04



m_t2j2

Events



m_ttl

Events

250
200
150
100
50
0

0

0

200

400

600

800

1000

1200

1400

1600

1800

2000

$\times 10^3$

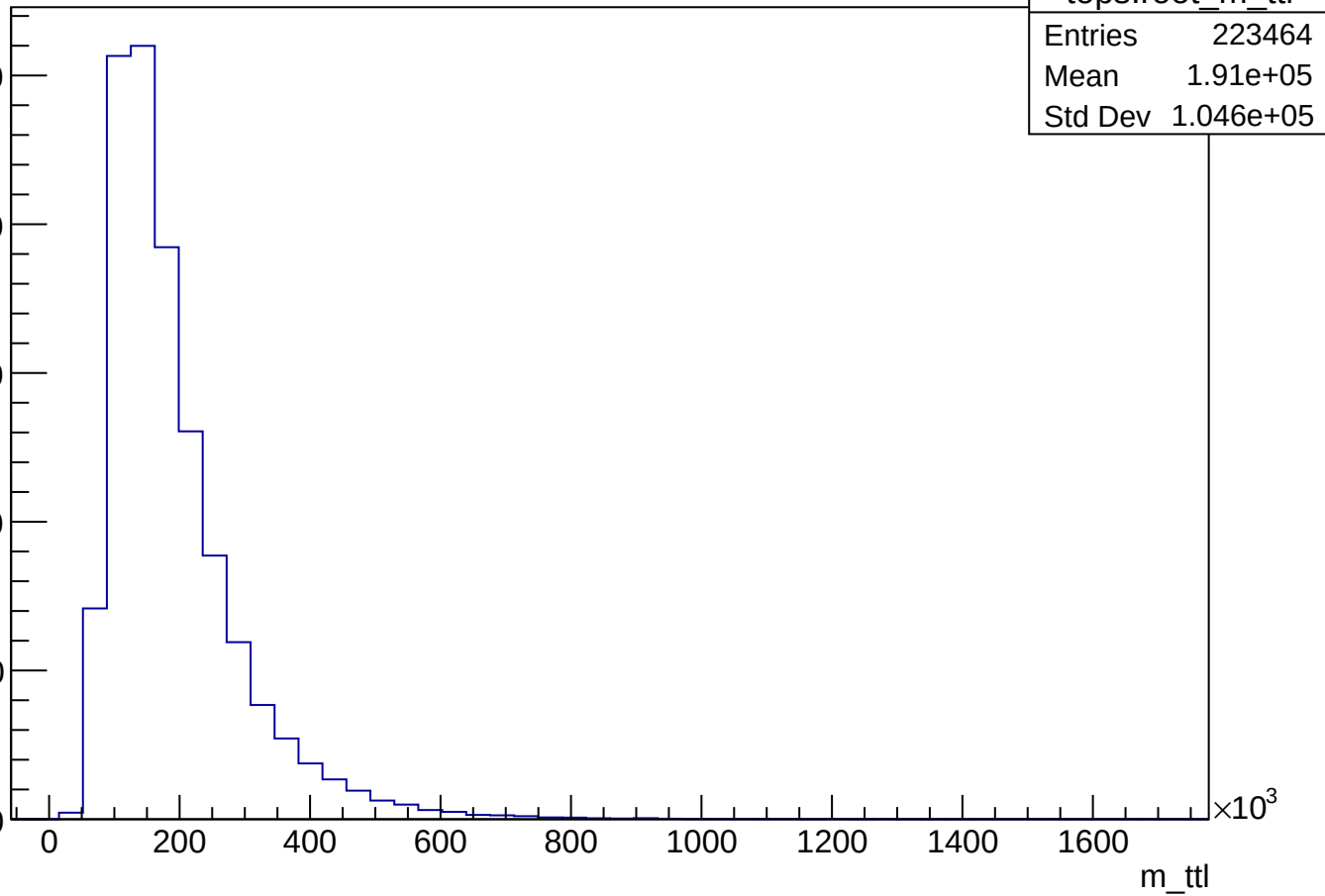
m_ttl

tops.root_m_ttl

Entries 223464

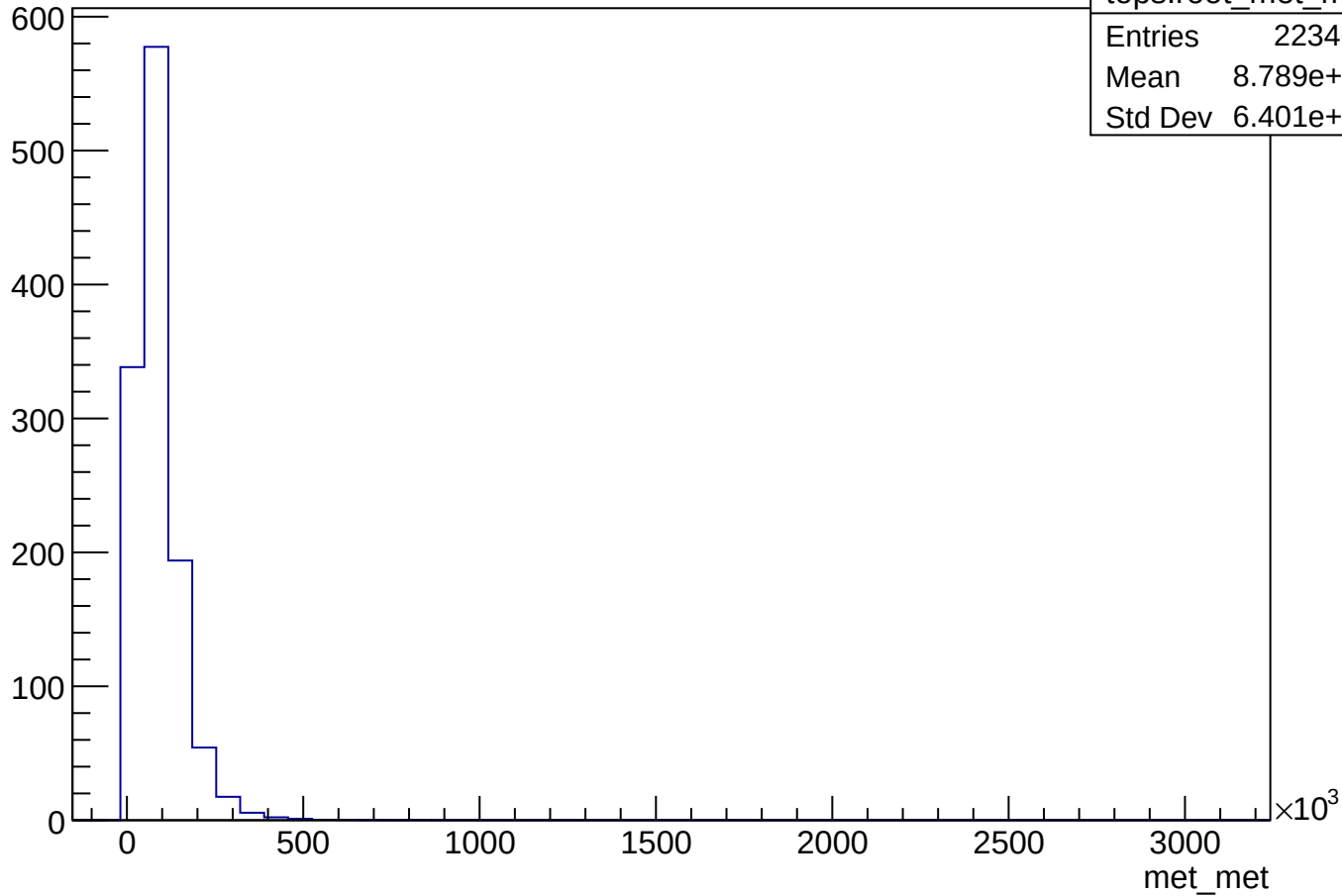
Mean 1.91e+05

Std Dev 1.046e+05



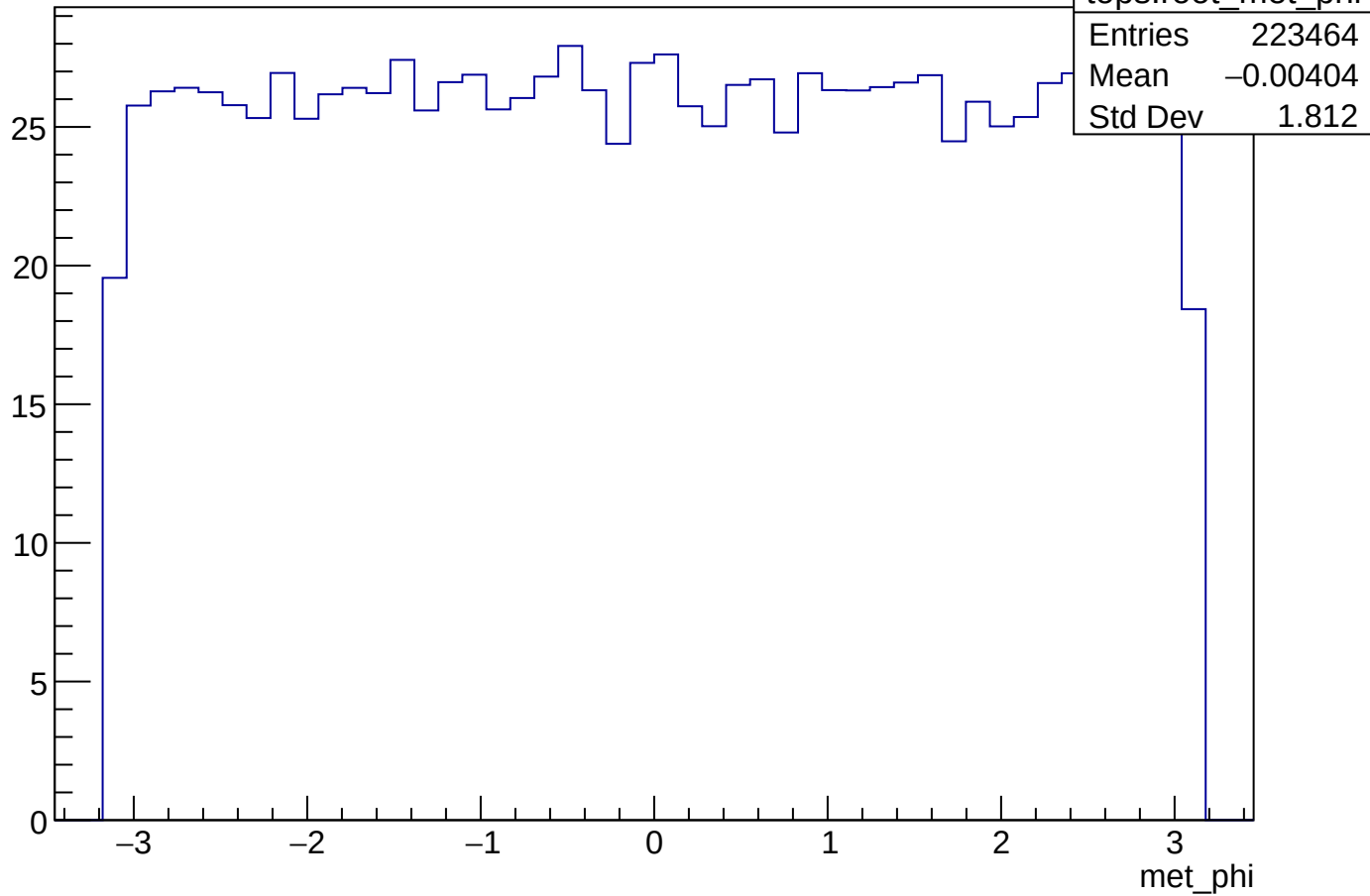
met_met

Events



met_phi

Events



met_sumet

Events

250
200
150
100
50
0

0

500

1000

1500

2000

2500

3000

3500

4000

4500

5000

5500

6000

6500

7000

7500

8000

8500

9000

9500

10000

10500

11000

11500

12000

12500

13000

13500

14000

14500

15000

15500

16000

16500

17000

17500

18000

18500

19000

19500

20000

20500

21000

21500

22000

22500

23000

23500

24000

24500

25000

25500

26000

26500

27000

27500

28000

28500

29000

29500

30000

30500

31000

31500

32000

32500

33000

33500

34000

34500

35000

35500

36000

36500

37000

37500

38000

38500

39000

39500

40000

40500

41000

41500

42000

42500

43000

43500

44000

44500

45000

45500

46000

46500

47000

47500

48000

48500

49000

49500

50000

50500

51000

51500

52000

52500

53000

53500

54000

54500

55000

55500

56000

56500

57000

57500

58000

58500

59000

59500

60000

60500

61000

61500

62000

62500

63000

63500

64000

64500

65000

65500

66000

66500

67000

67500

68000

68500

69000

69500

70000

70500

71000

71500

72000

72500

73000

73500

74000

74500

75000

75500

76000

76500

77000

77500

78000

78500

79000

79500

80000

80500

81000

81500

82000

82500

83000

83500

84000

84500

85000

85500

86000

86500

87000

87500

88000

88500

89000

89500

90000

90500

91000

91500

92000

92500

93000

93500

94000

94500

95000

95500

96000

96500

97000

97500

98000

98500

99000

99500

100000

100500

101000

101500

102000

102500

103000

103500

104000

104500

105000

105500

106000

106500

107000

107500

108000

108500

109000

109500

110000

110500

111000

111500

112000

112500

113000

113500

114000

114500

115000

115500

116000

116500

117000

117500

118000

118500

119000

119500

120000

120500

121000

121500

122000

122500

123000

123500

124000

124500

125000

125500

126000

126500

127000

127500

128000

128500

129000

129500

130000

130500

131000

131500

132000

132500

133000

133500

134000

134500

135000

135500

136000

136500

137000

137500

138000

138500

139000

139500

140000

140500

141000

141500

142000

142500

143000

143500

144000

144500

145000

145500

146000

146500

147000

147500

148000

148500

149000

149500

150000

150500

151000

151500

152000

152500

153000

153500

154000

154500

155000

155500

156000

156500

157000

157500

158000

158500

159000

159500

160000

160500

161000

161500

162000

162500

163000

163500

n_jet

Events

400
350
300
250
200
150
100
50
0

2

4

6

8

10

12

14

n_jet

tops.root_n_jet

Entries 223464

Mean 3.335

Std Dev 1.347

2

4

6

8

10

12

14

n_jet

pair_isOSleptau

Events

600
500
400
300
200
100
0

0

0.2

0.4

0.6

0.8

1

pair_isOSleptau

tops.root_pair_isOSleptau	
Entries	223464
Mean	0.559
Std Dev	0.4965

pair_isOStaus

Events

1000
800
600
400
200
0

0

0.2

0.4

0.6

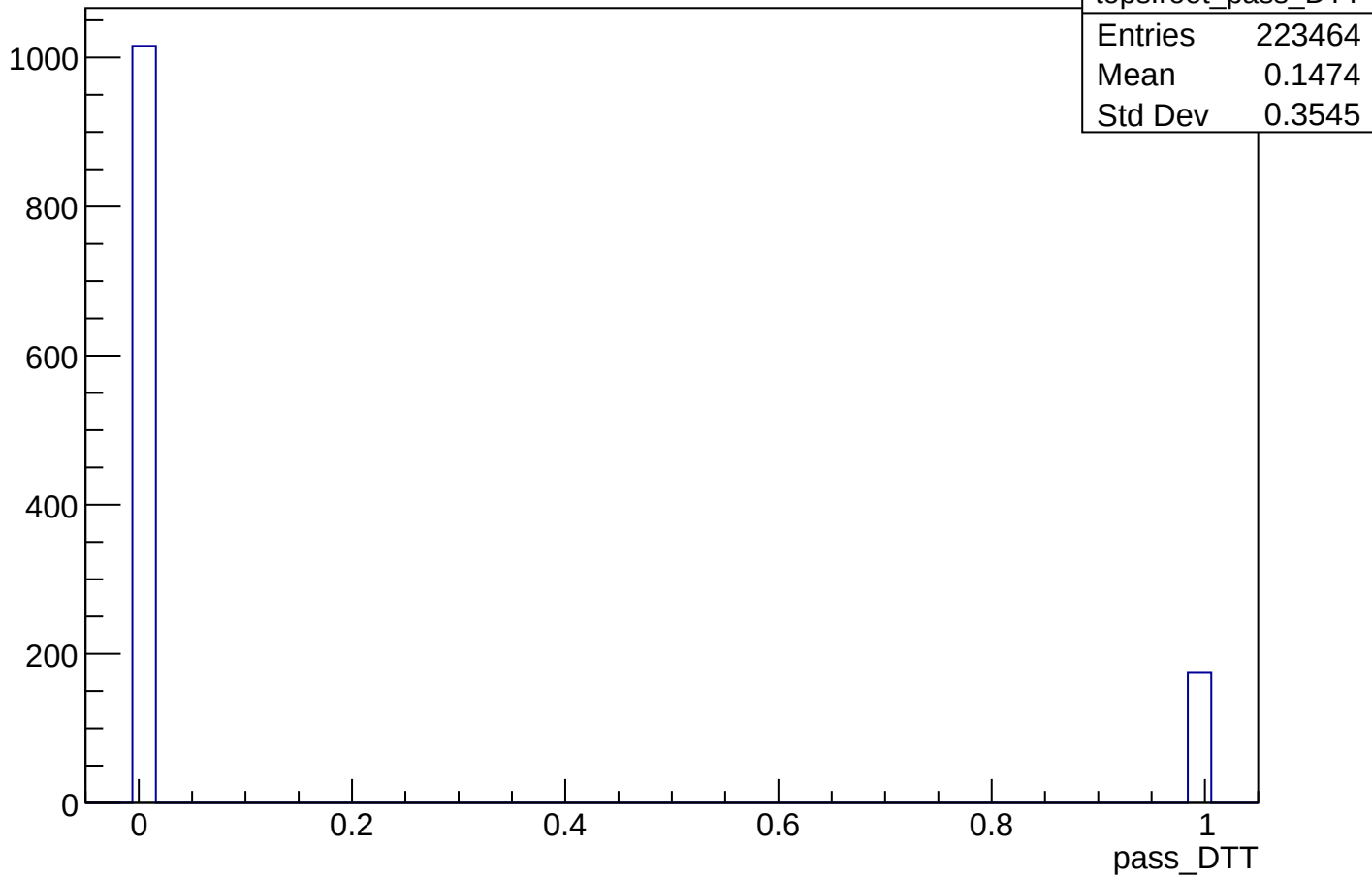
0.8

pair_isOStaus

tops.root_pair_isOStaus	
Entries	223464
Mean	0.8341
Std Dev	0.372

pass_DTT

Events



pass_LTT

Events

900
800
700
600
500
400
300
200
100
0

0

0.2

0.4

0.6

0.8

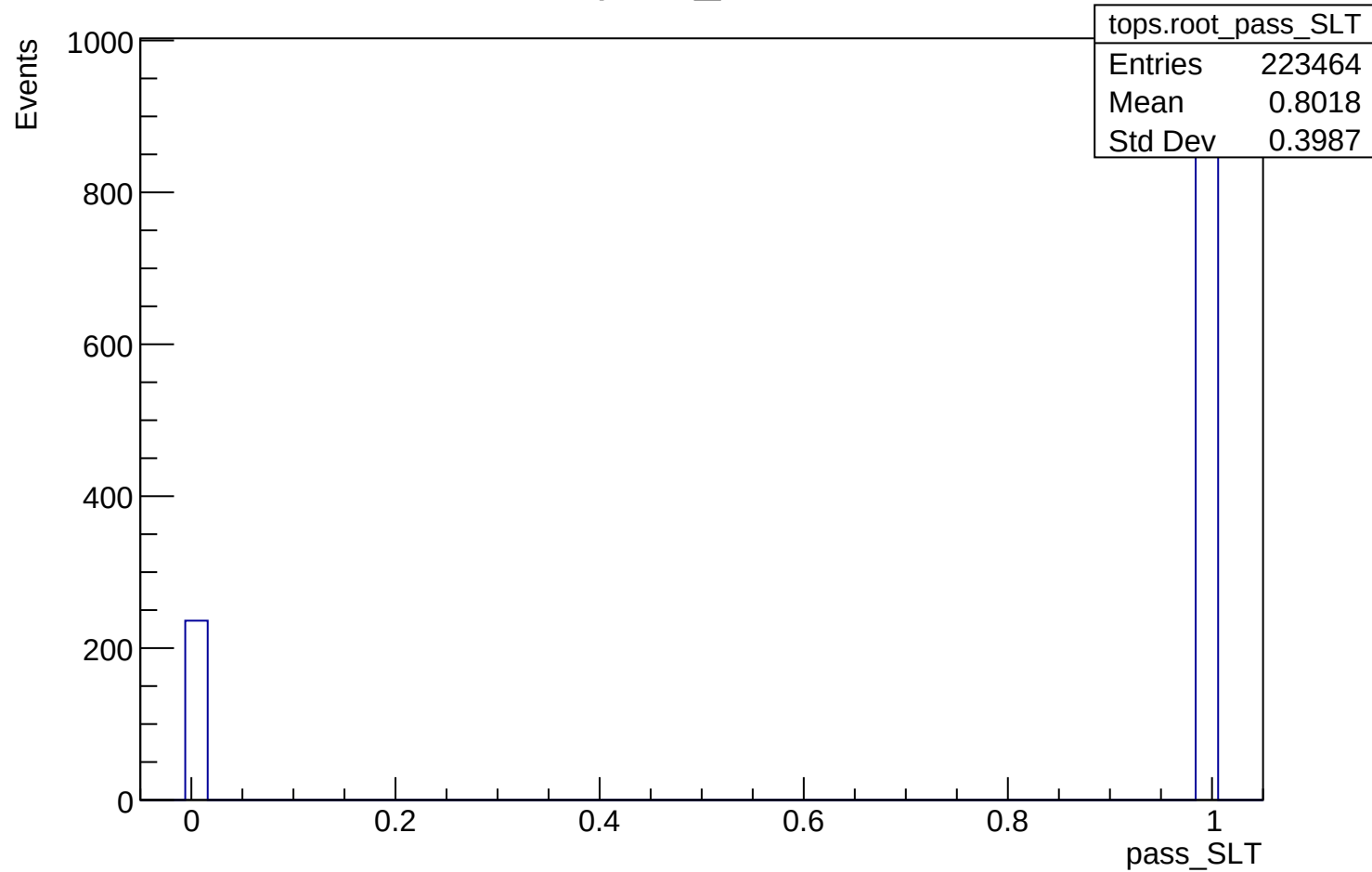
1

pass_LTT

tops.root_pass_LTT	
Entries	223464
Mean	0.7199
Std Dev	0.449

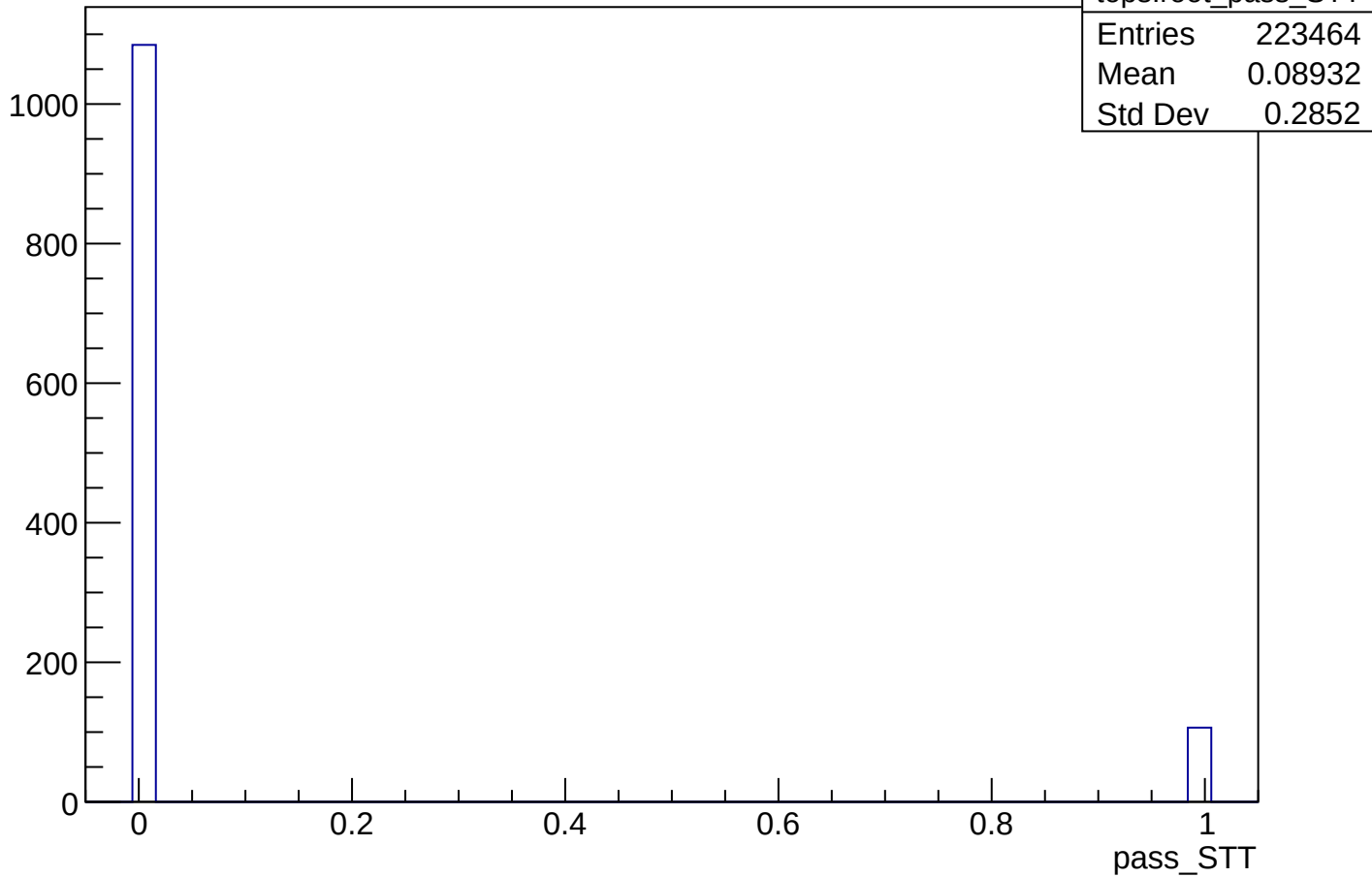


pass_SLT

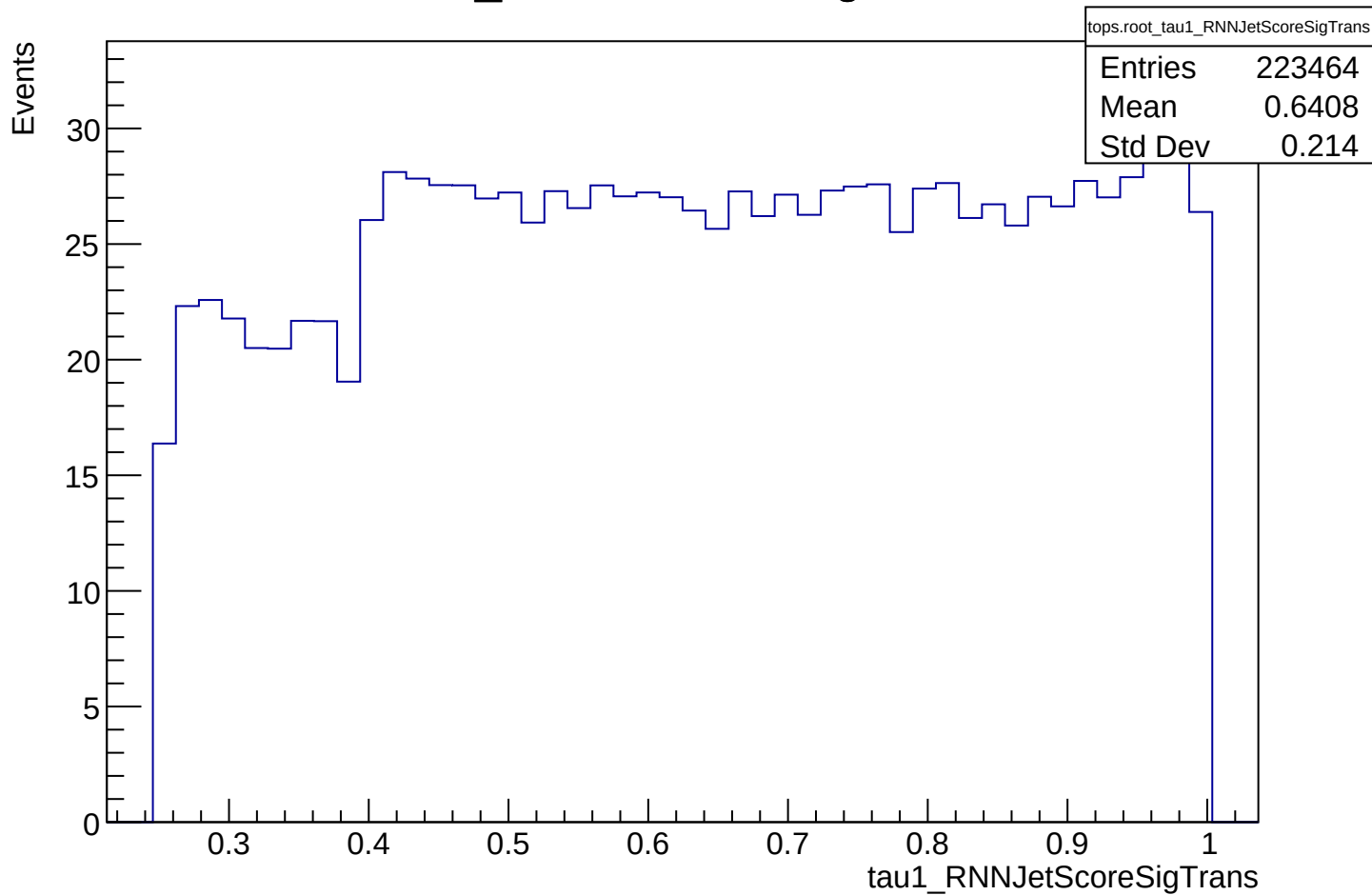


pass_STT

Events



tau1_RNNJetScoreSigTrans



tau1_charge

Events

600
500
400
300
200
100
0

-1

-0.5

0

0.5

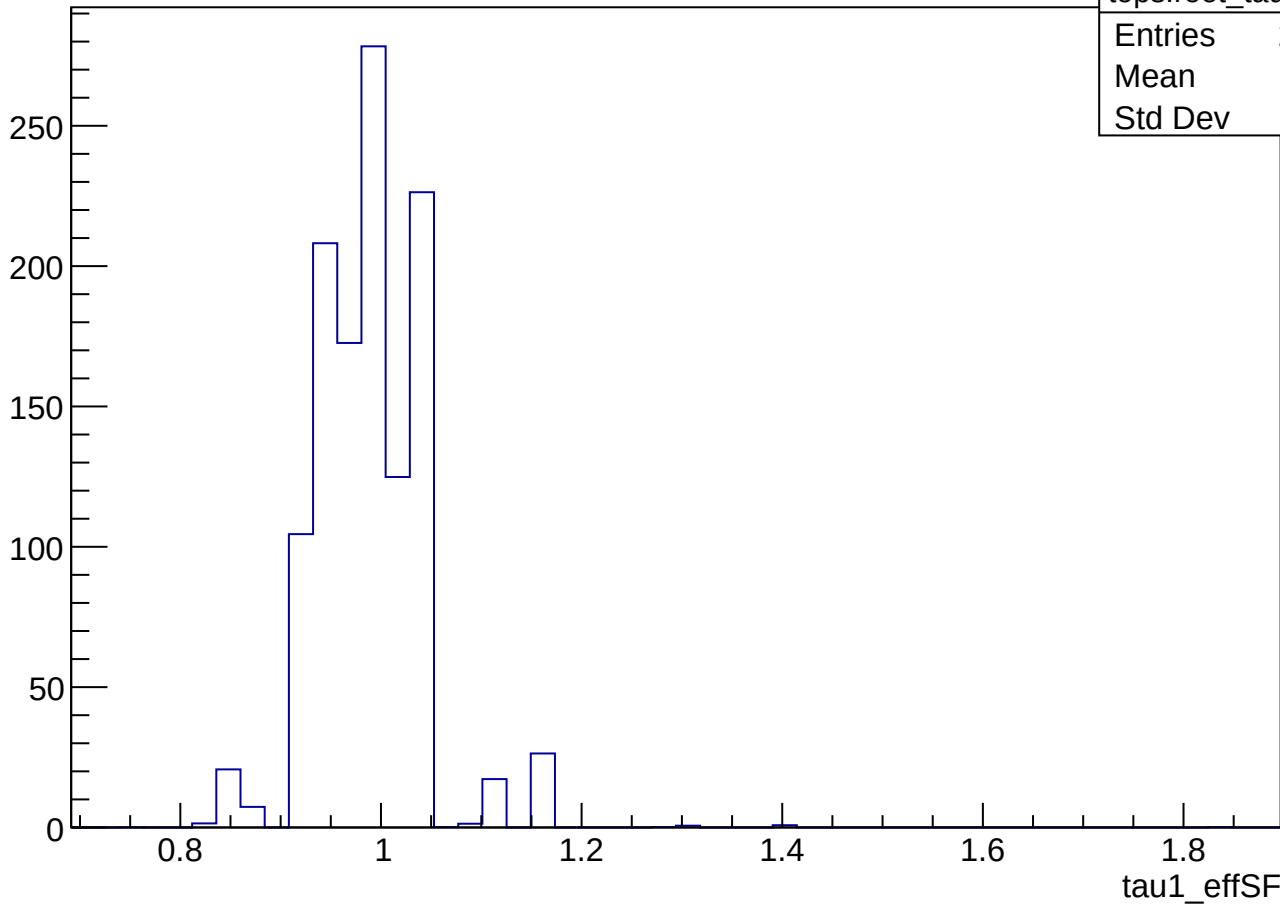
1

tau1_charge

tops.root_tau1_charge	
Entries	223464
Mean	-0.004223
Std Dev	1

tau1_effSF

Events



tops.root_tau1_effSF	
Entries	223464
Mean	0.9938
Std Dev	0.0548

tau1_eta

Events

50
40
30
20
10
0

-2

-1

0

1

2

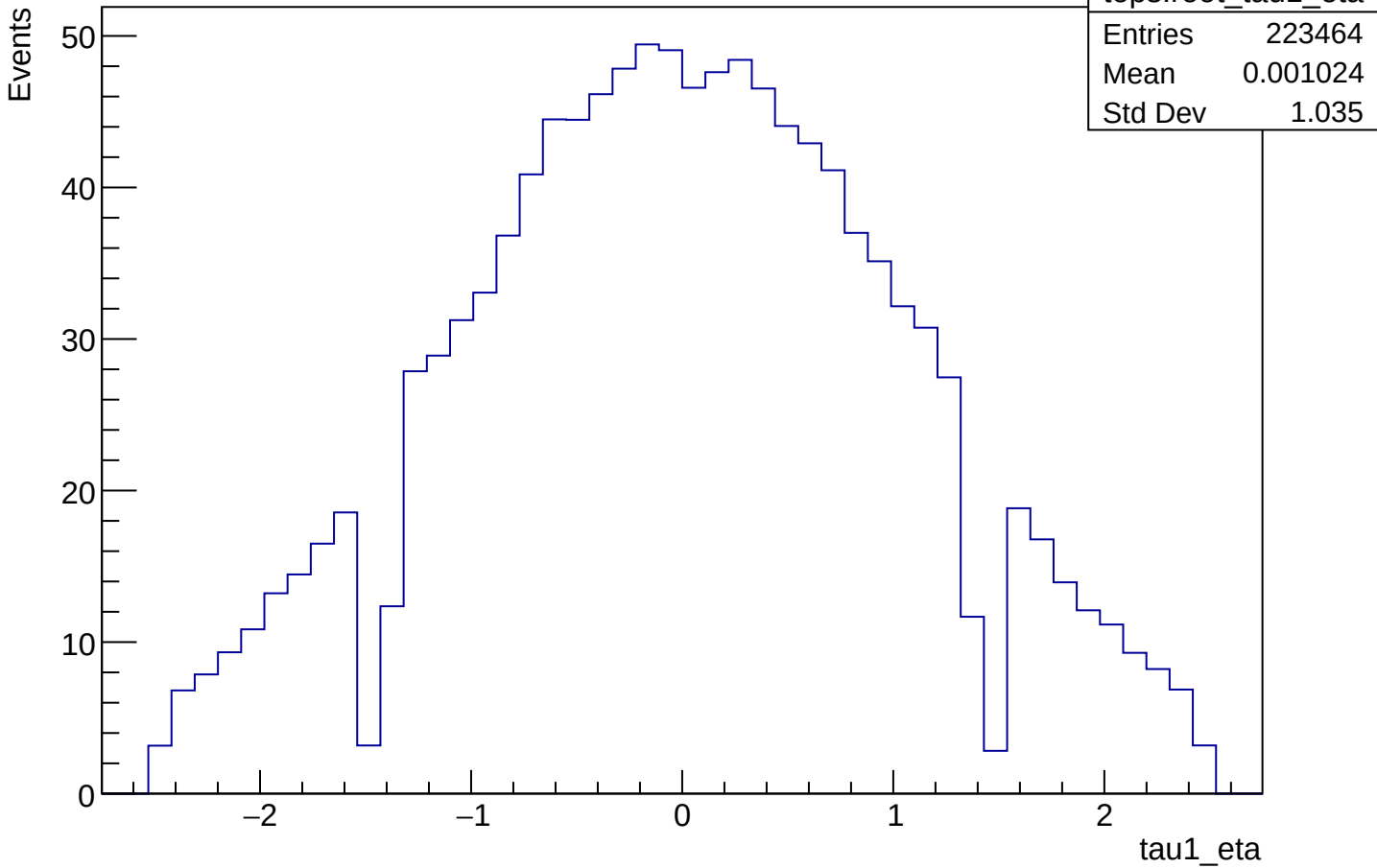
tau1_eta

tops.root_tau1_eta

Entries 223464

Mean 0.001024

Std Dev 1.035



tau1_nprong

Events

900
800
700
600
500
400
300
200
100
0

1

1.5

2

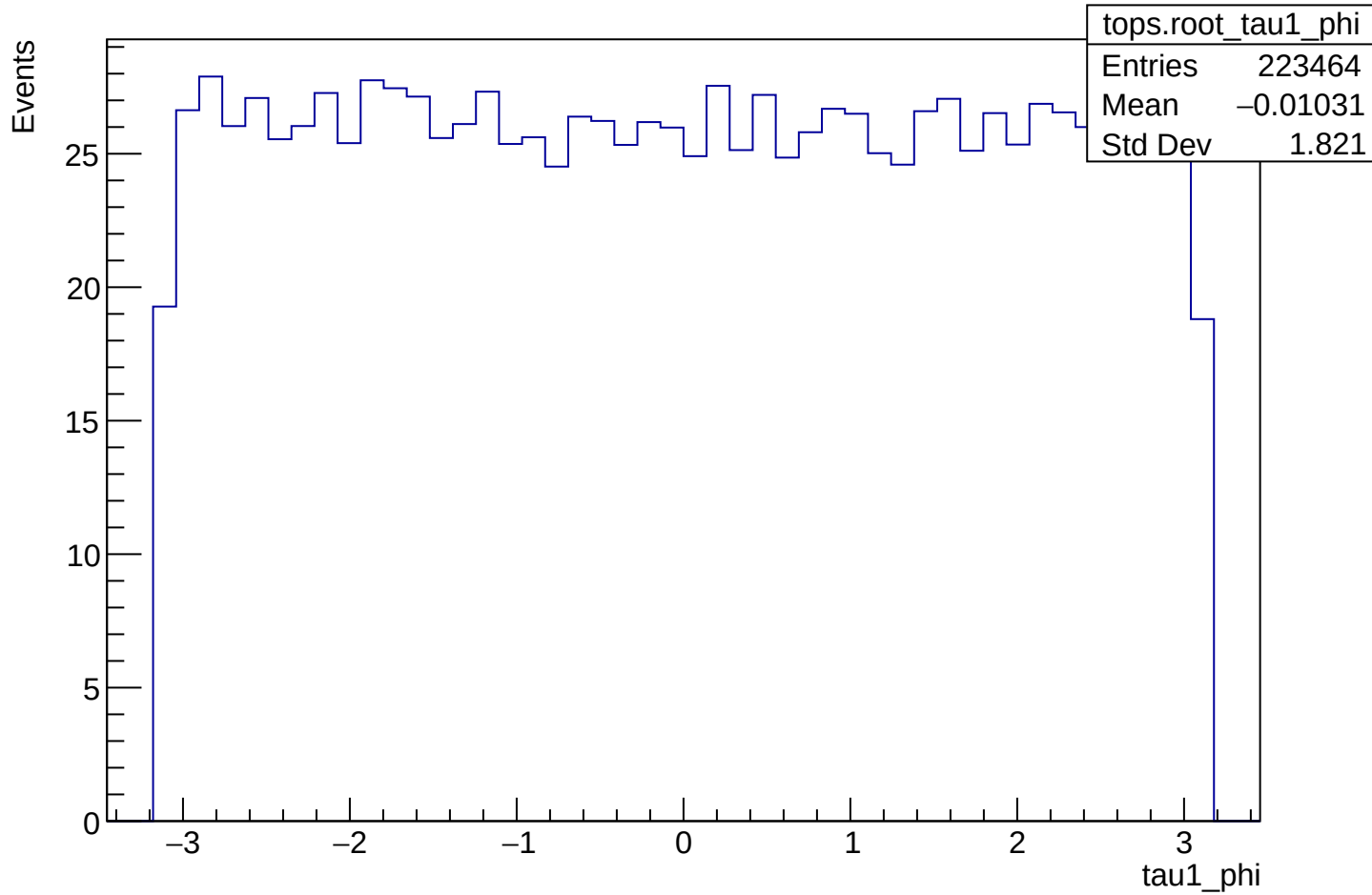
2.5

3

tau1_nprong

tops.root_tau1_nprong	
Entries	223464
Mean	1.439
Std Dev	0.8279

tau1_phi



tau1_pt

Events

250
200
150
100
50
0

0

100

200

300

400

500

600

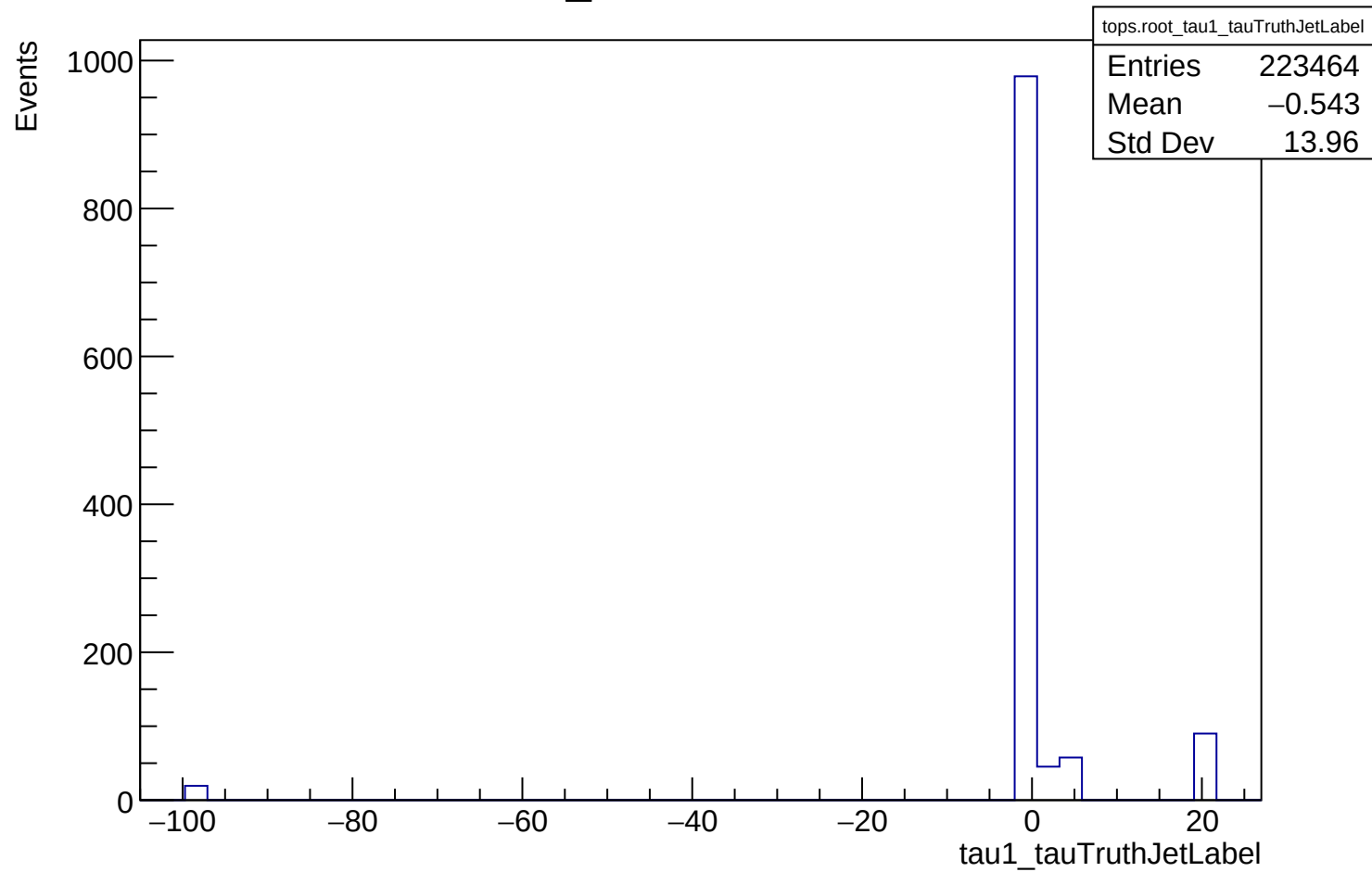
700

tau1_pt

$\times 10^3$

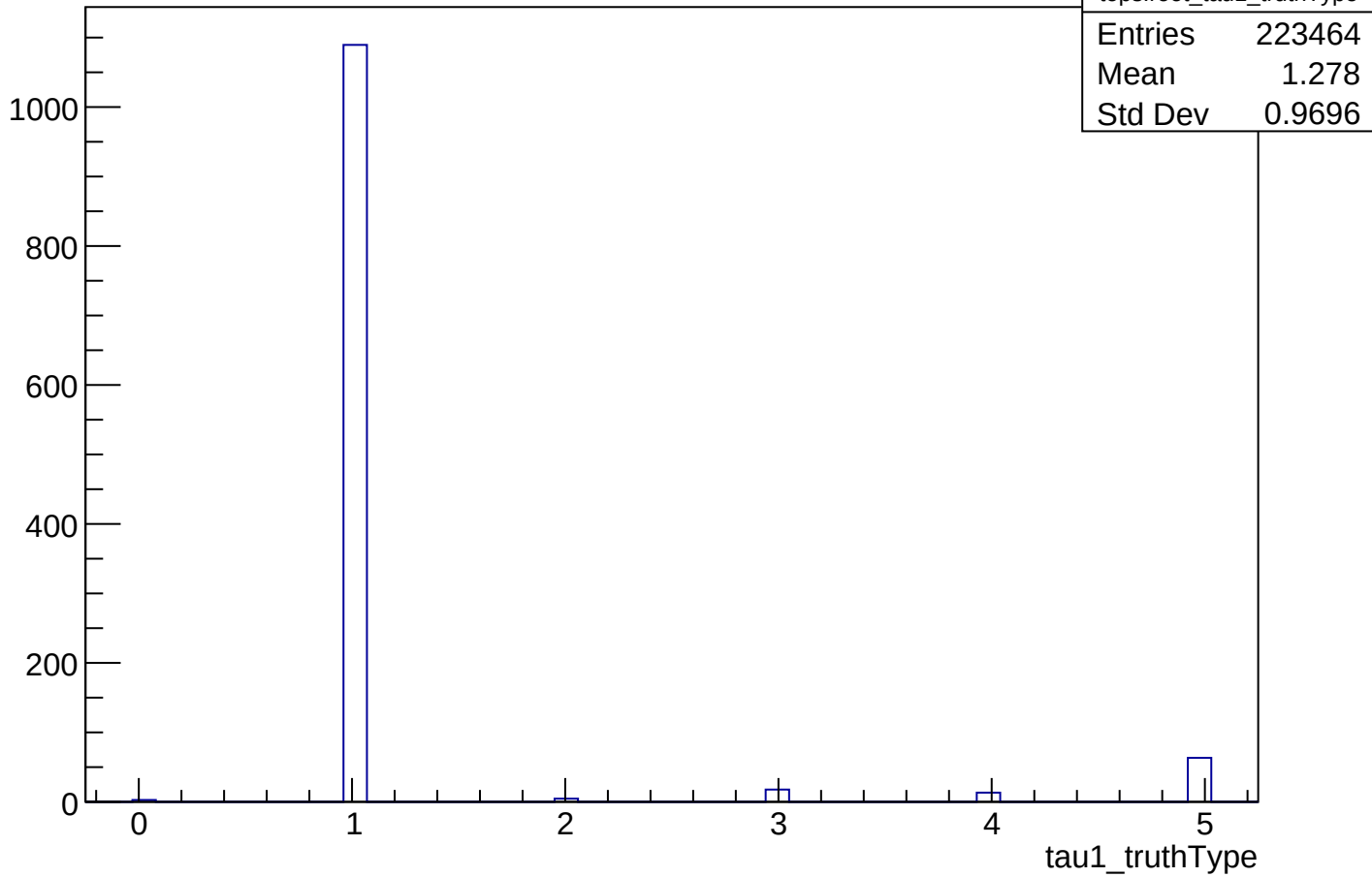
tops.root_tau1_pt	
Entries	223464
Mean	7.605e+04
Std Dev	5.093e+04

tau1_tauTruthJetLabel

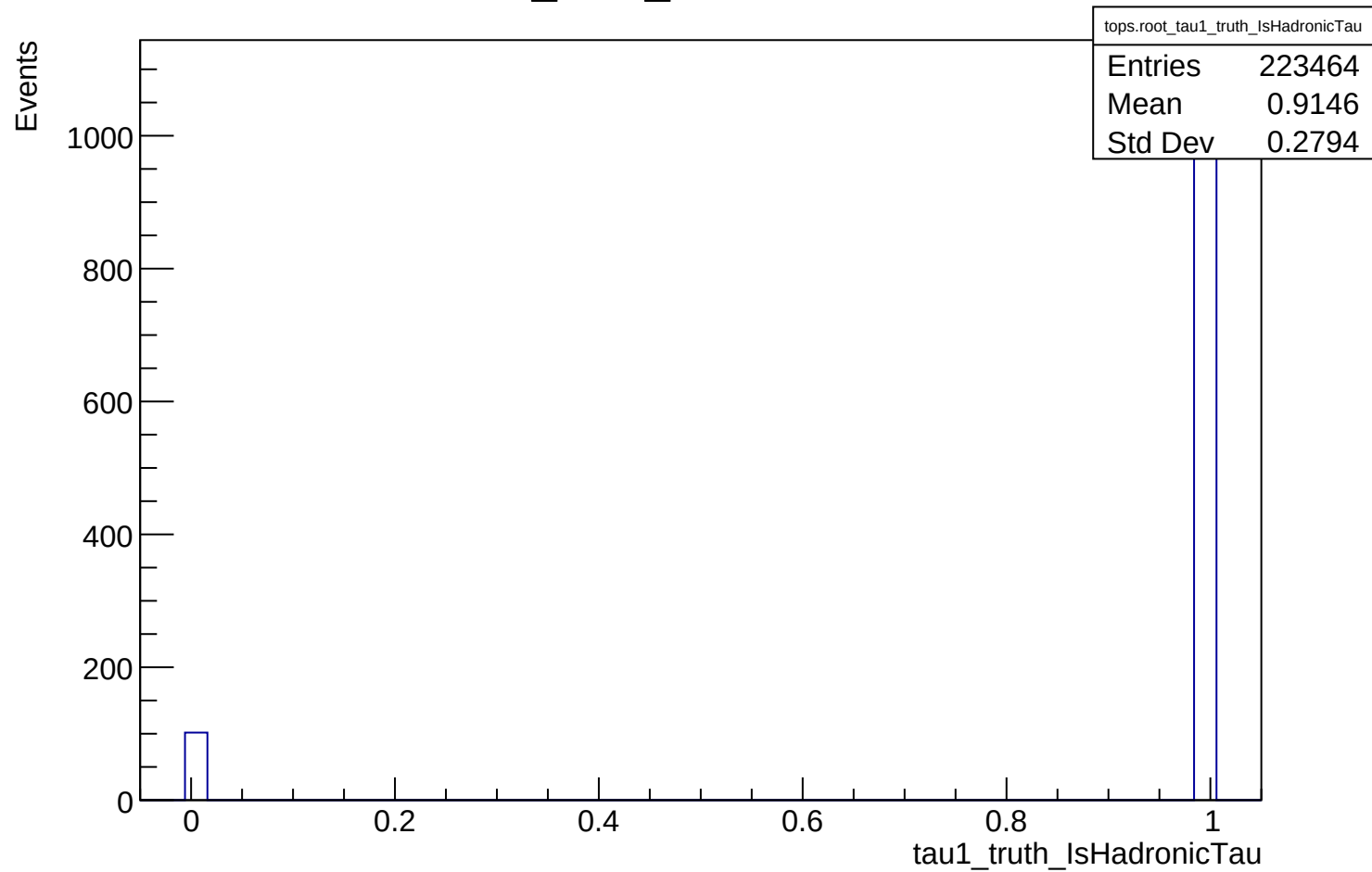


tau1_truthType

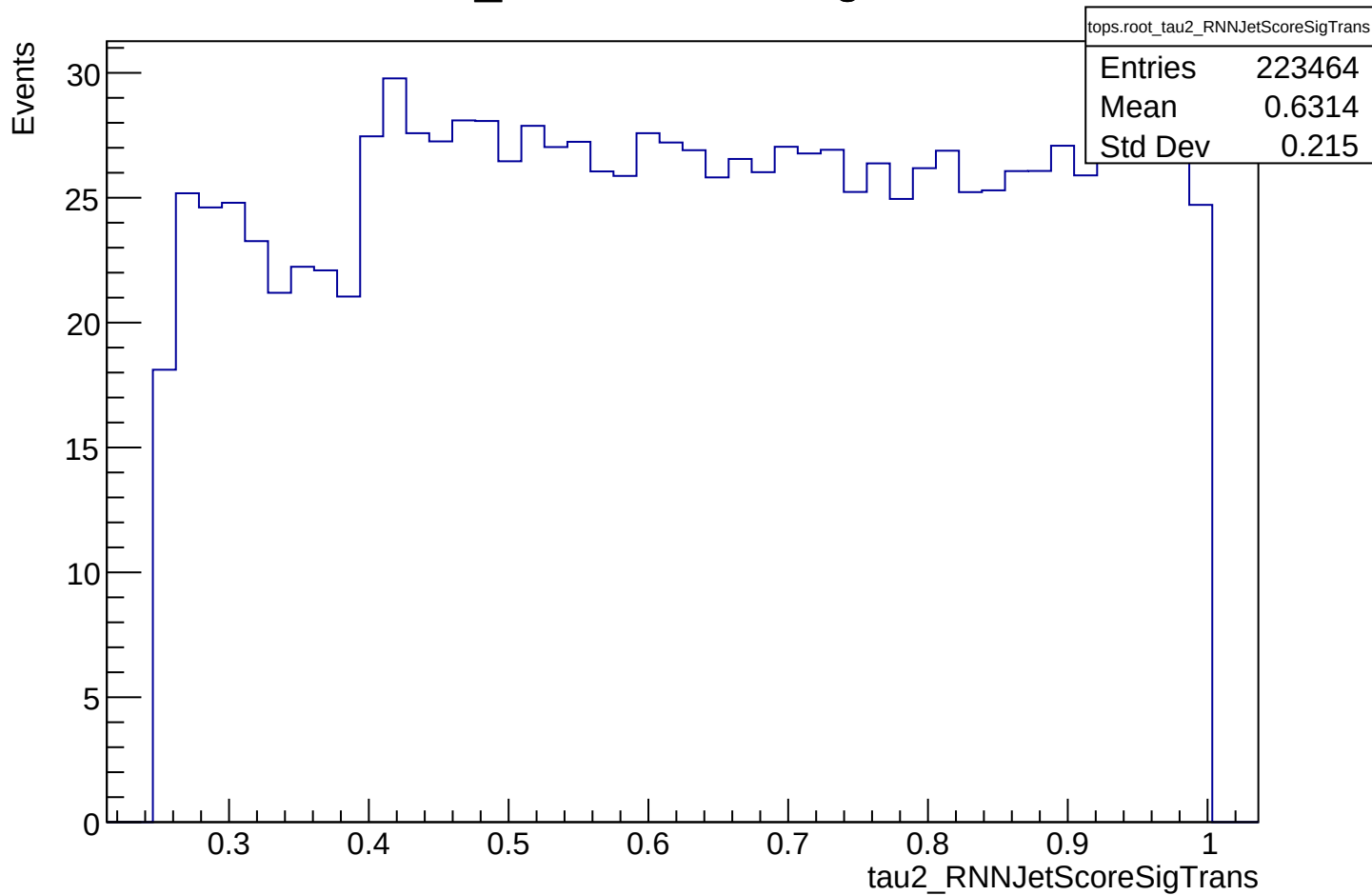
Events



tau1_truth_IsHadronicTau



tau2_RNNJetScoreSigTrans



tau2_baseline_RNNTight

Events

900
800
700
600
500
400
300
200
100
0

0

0.2

0.4

0.6

0.8

1

tau2_baseline_RNNTight

tops.root_tau2_baseline_RNNTight	
Entries	223464
Mean	0.76
Std Dev	0.4271

tau2_charge

Events

600
500
400
300
200
100
0

-1

-0.5

0

0.5

1

tau2_charge

tops.root_tau2_charge	
Entries	223464
Mean	0.007786
Std Dev	1

tau2_effSF

Events

350
300
250
200
150
100
50
0

0.8

1.0

1.2

1.4

1.6

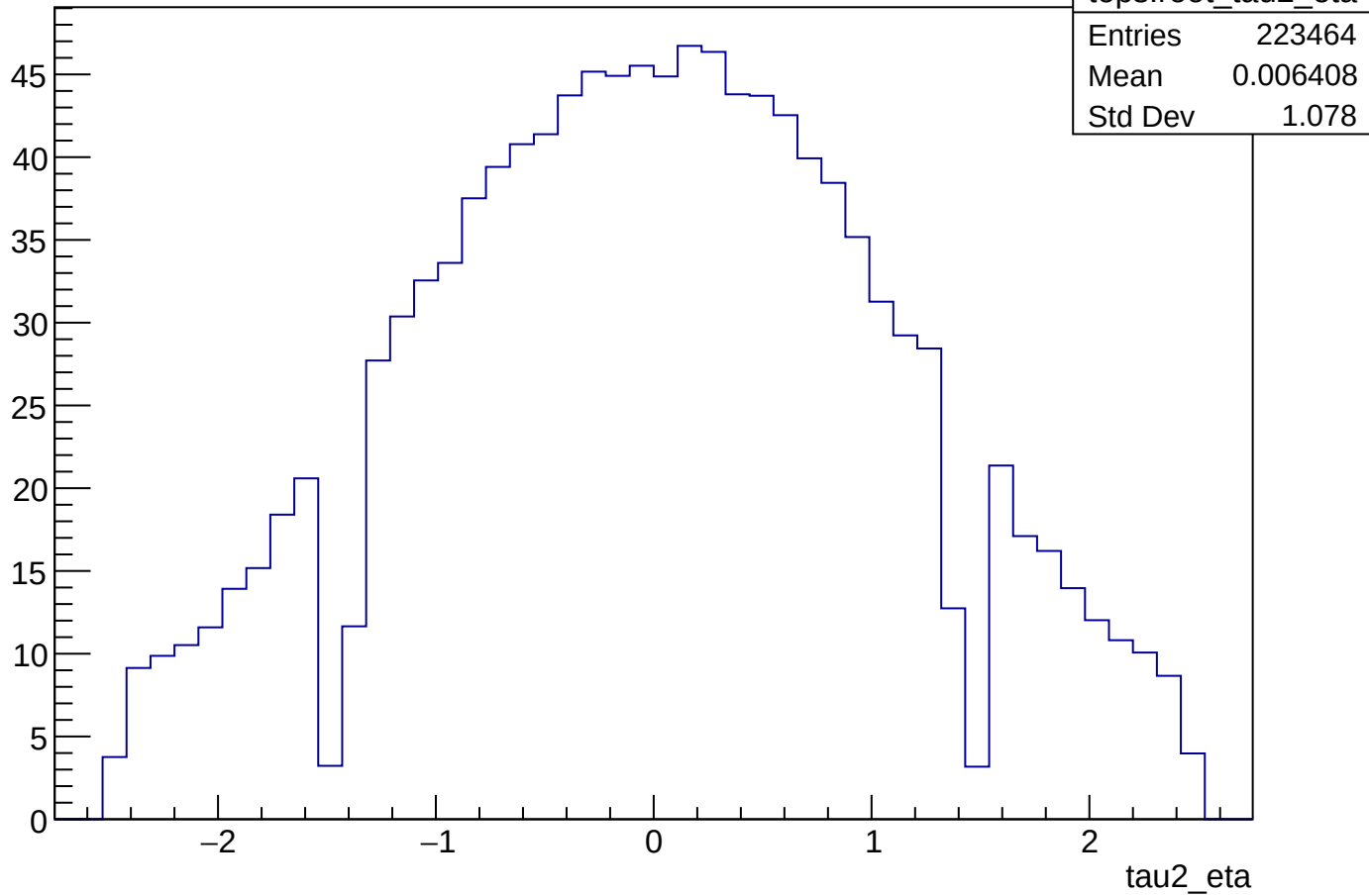
1.8

tau2_effSF

tops.root_tau2_effSF	
Entries	223464
Mean	0.9749
Std Dev	0.06499

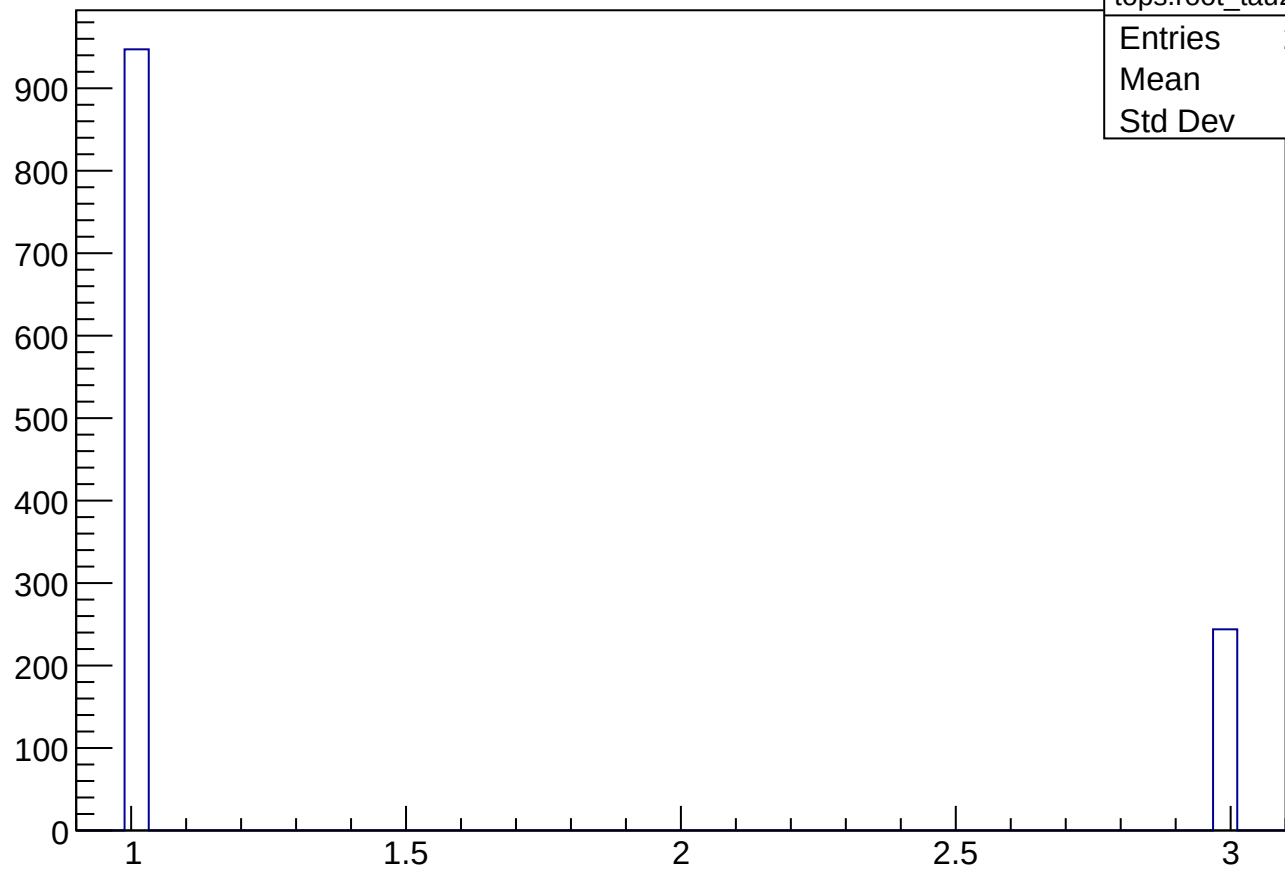
tau2_eta

Events



tau2_nprong

Events

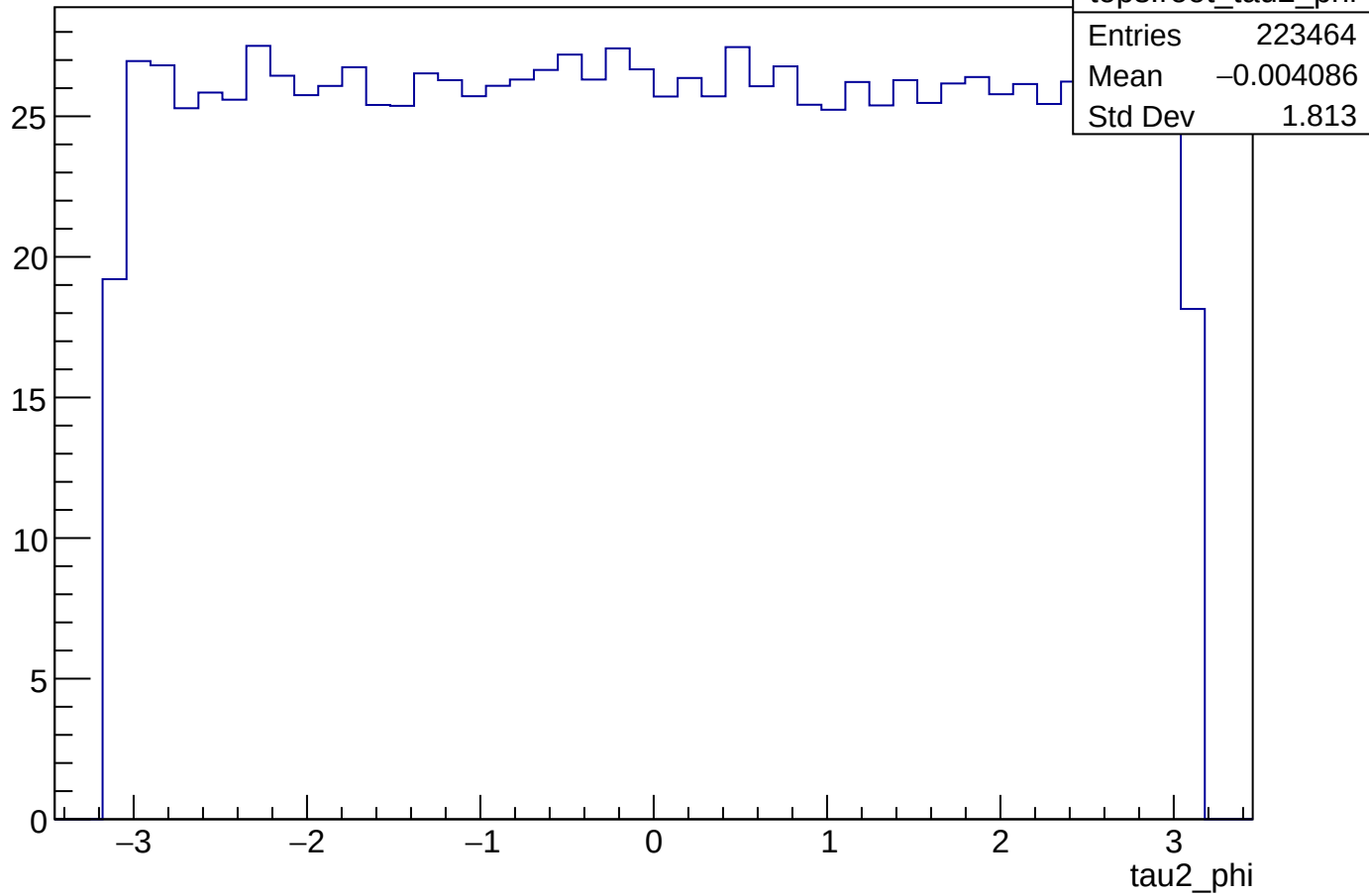


tops.root_tau2_nprong	
Entries	223464
Mean	1.41
Std Dev	0.8071

tau2_nprong

tau2_phi

Events



tau2_pt

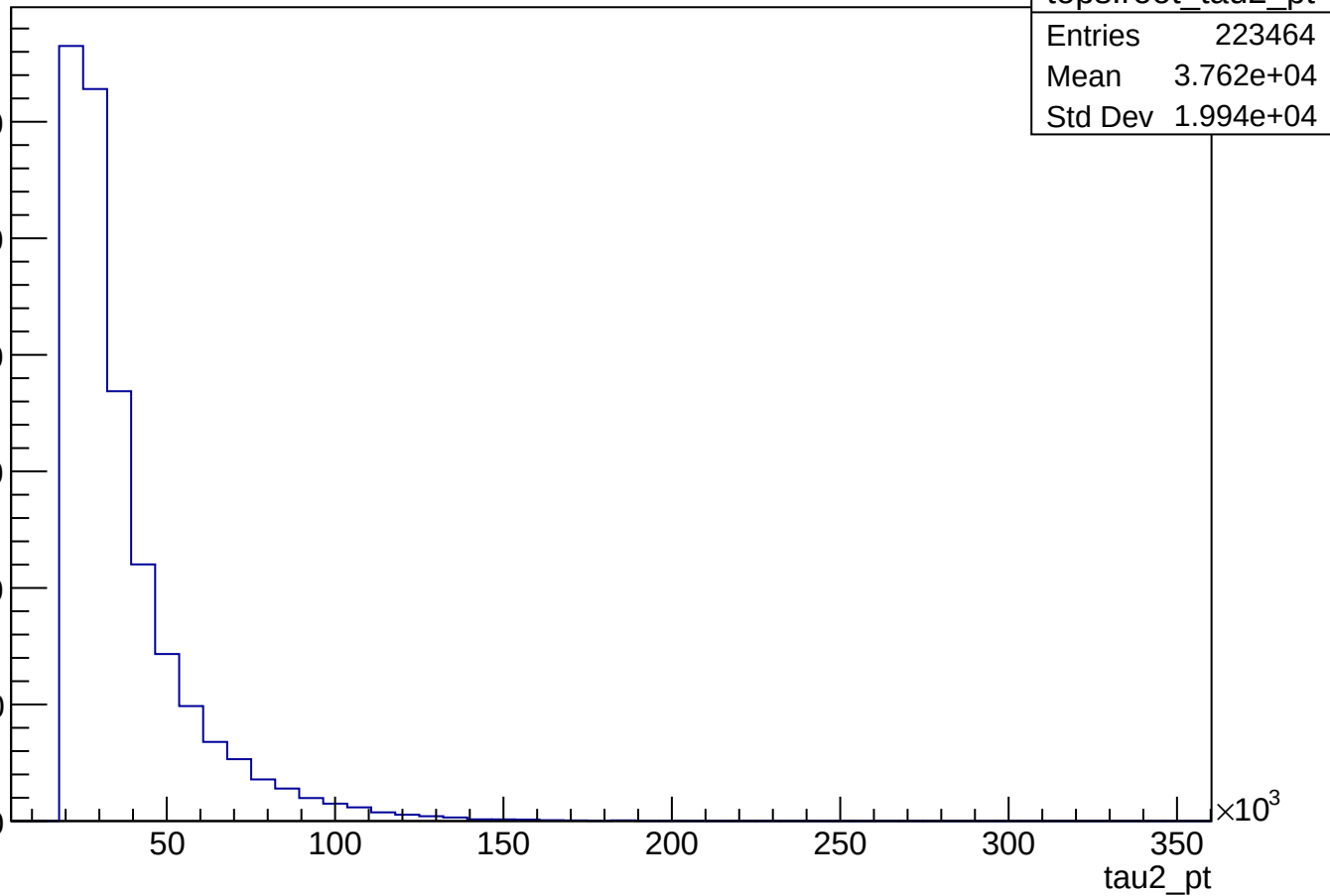
Events

300
250
200
150
100
50
0

tops.root_tau2_pt	
Entries	223464
Mean	3.762e+04
Std Dev	1.994e+04

$\times 10^3$

tau2_pt



tau2_tauTruthJetLabel

Events

900
800
700
600
500
400
300
200
100
0

-100

-80

-60

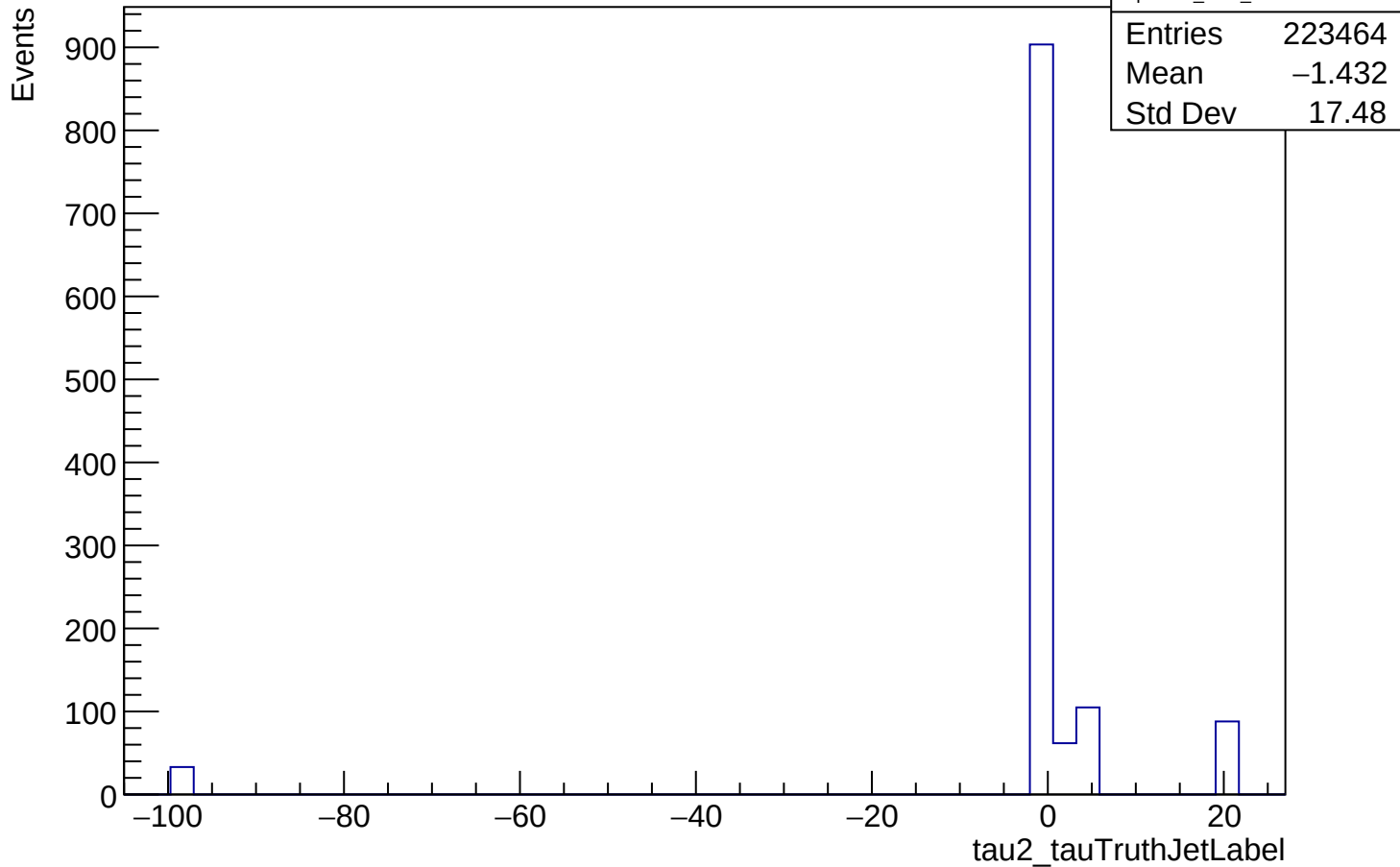
-40

-20

tau2_tauTruthJetLabel

tops.root_tau2_tauTruthJetLabel	
Entries	223464
Mean	-1.432
Std Dev	17.48

20



tau2_truthType

Events

1000
800
600
400
200
0

0

1

2

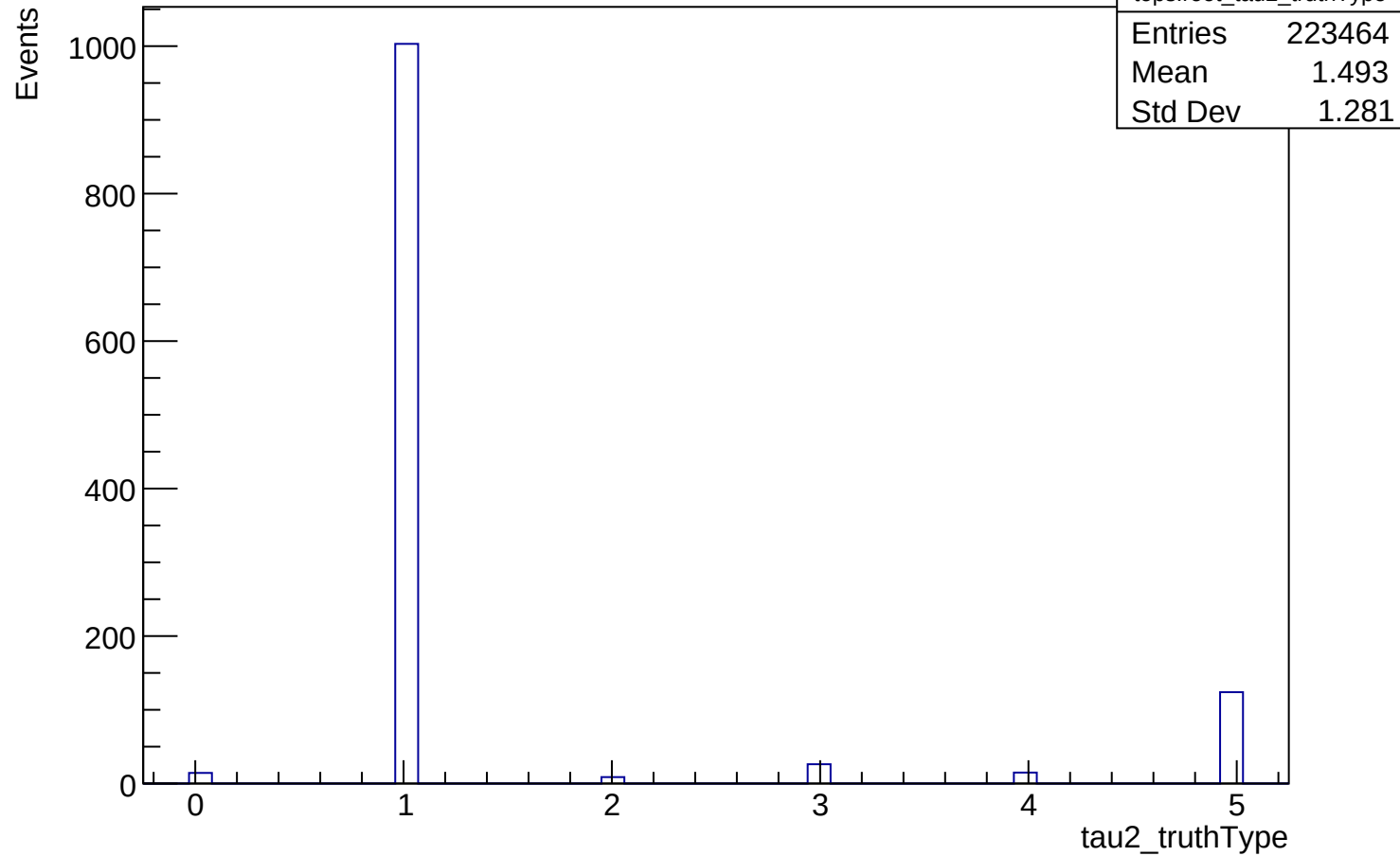
3

4

5

tau2_truthType

tops.root_tau2_truthType	
Entries	223464
Mean	1.493
Std Dev	1.281



tau2_truth_IsHadronicTau

Events

1000
800
600
400
200
0

0

0.2

0.4

0.6

0.8

1

tau2_truth_IsHadronicTau

tops.root_tau2_truth_IsHadronicTau	
Entries	223464
Mean	0.8421
Std Dev	0.3646

weight_final

Events

1200
1000
800
600
400
200
0

-20000

-10000

0

10000

20000

30000

40000

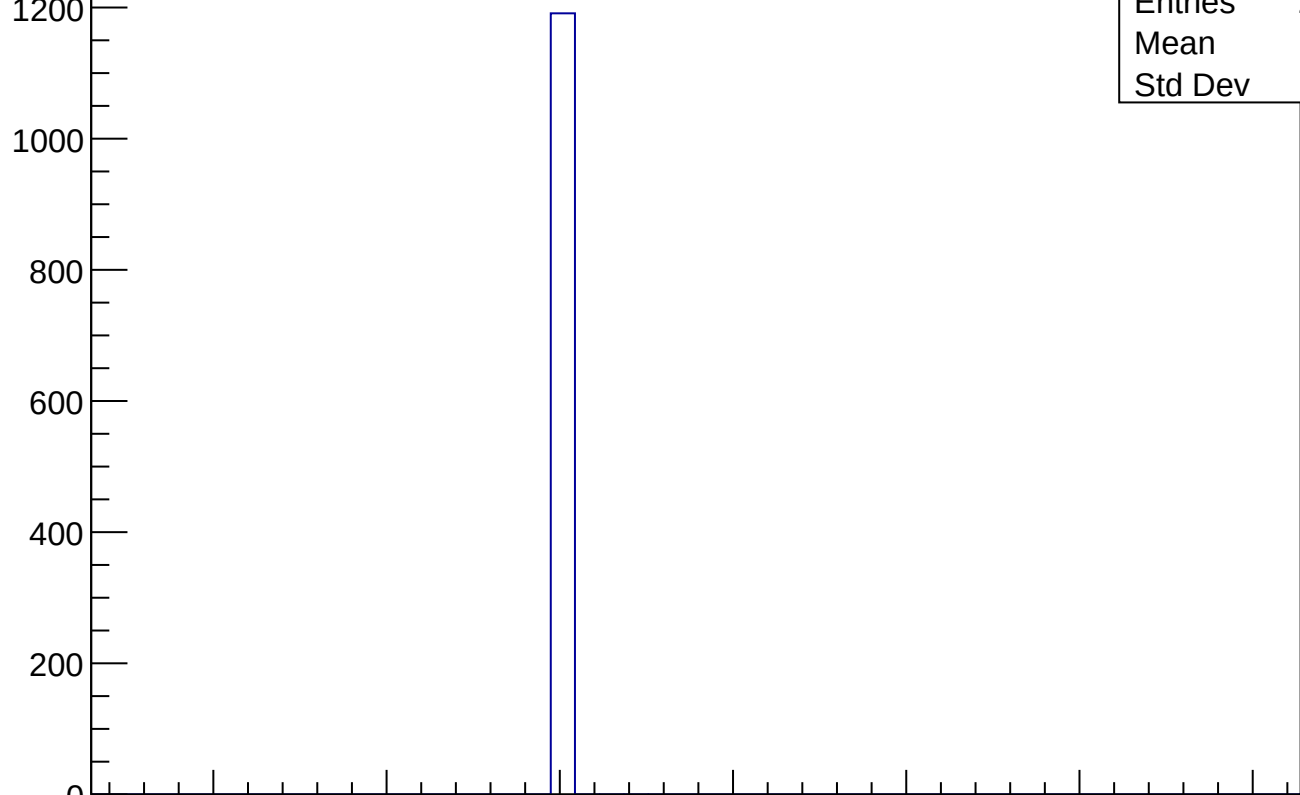
weight_final

tops.root_weight_final

Entries 223464

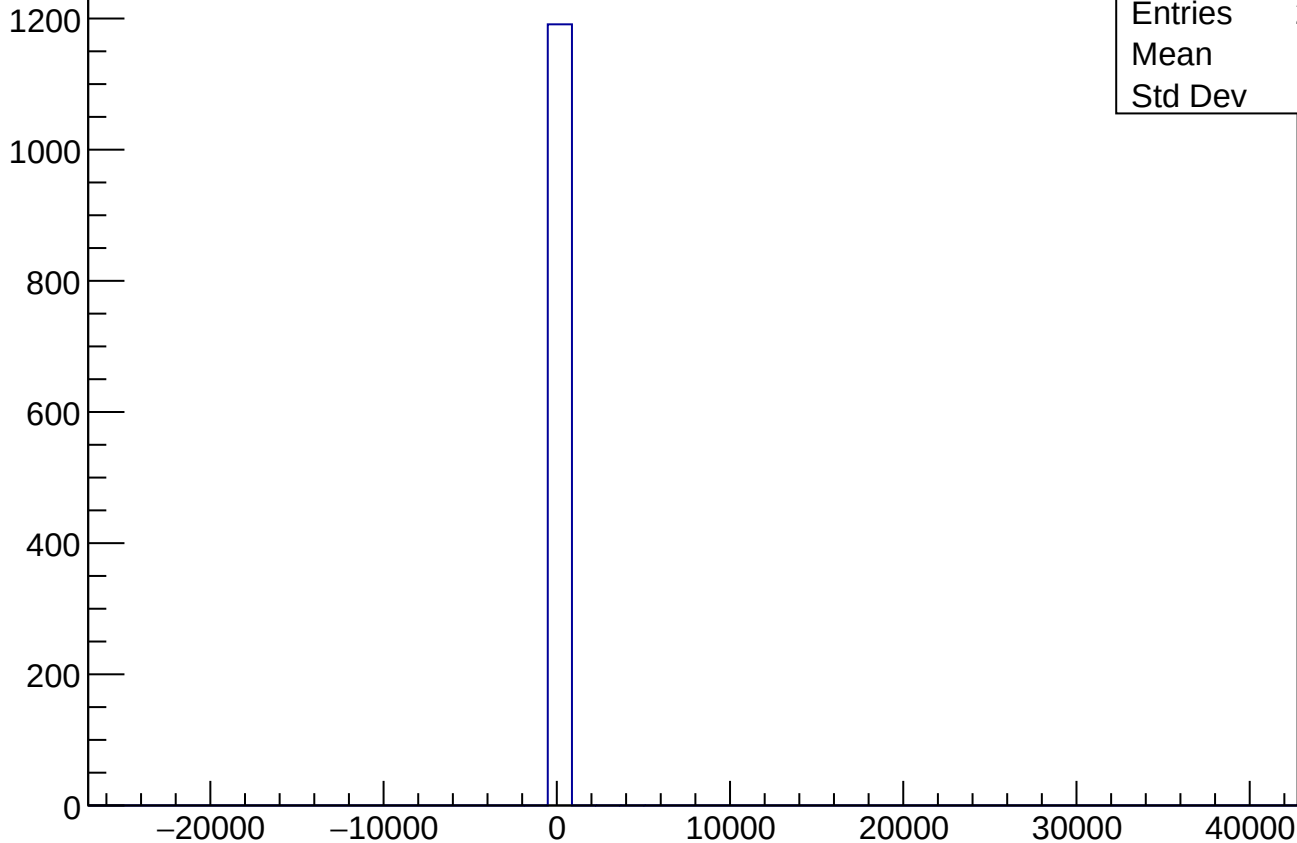
Mean 0.3023

Std Dev 6.849



weights

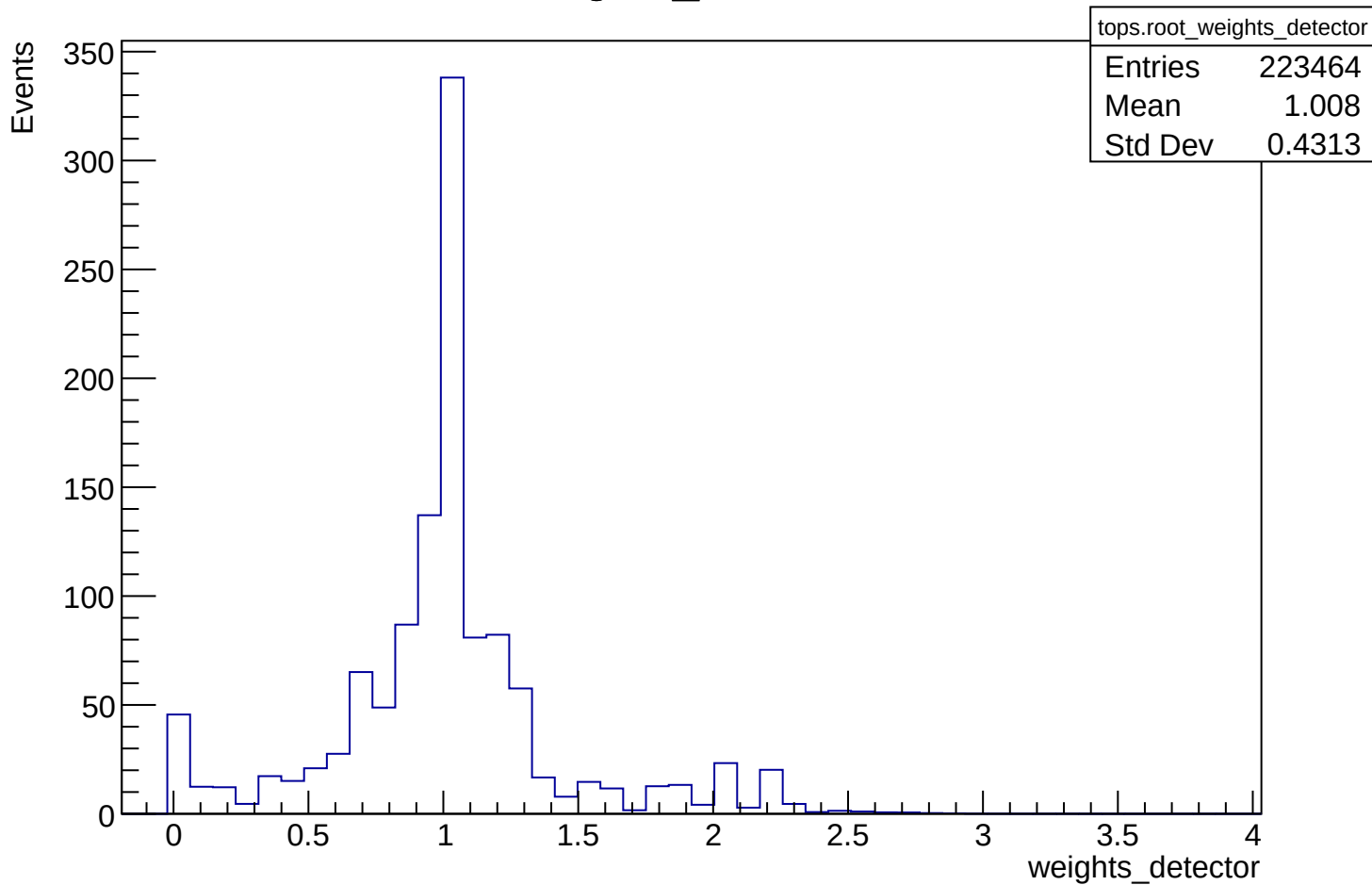
Events



tops.root_weights	
Entries	223464
Mean	0.3023
Std Dev	6.849

weights

weights_detector



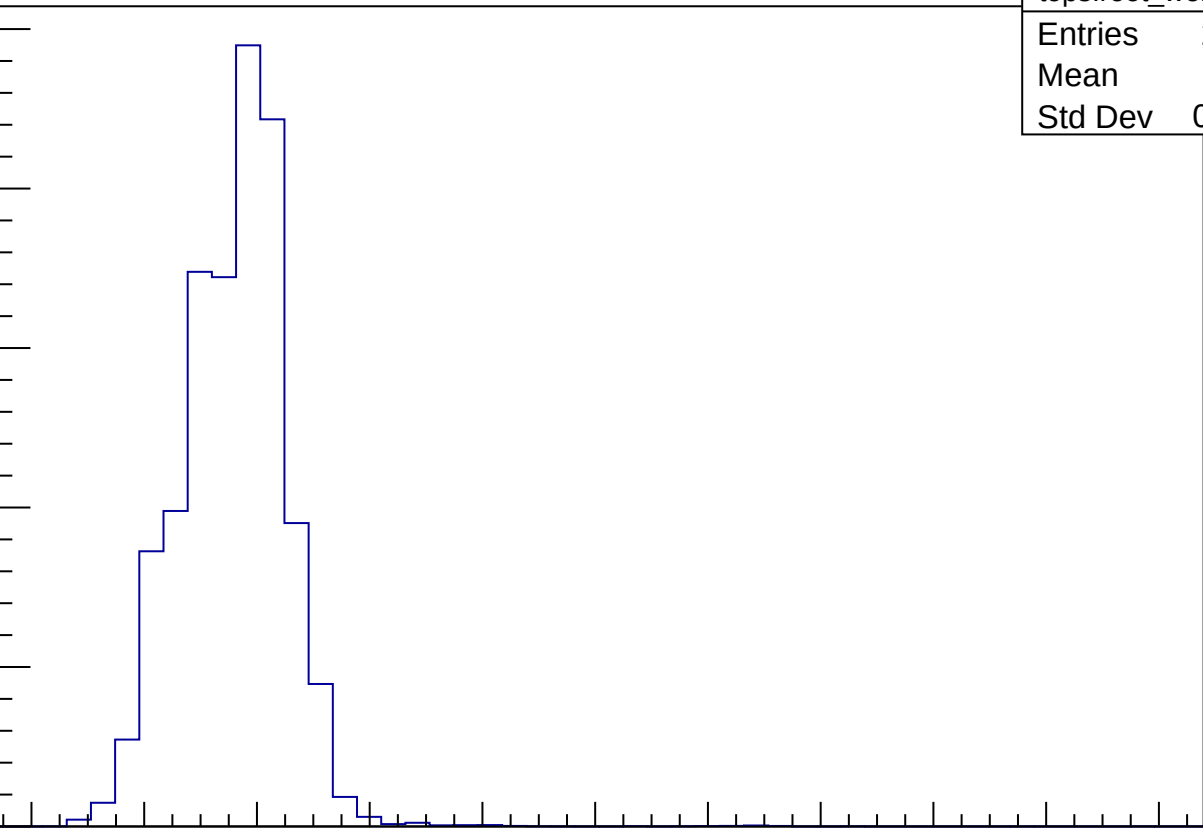
weights_eff

Events

250
200
150
100
50
0

0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6
weights_eff

tops.root_weights_eff	
Entries	223464
Mean	0.9585
Std Dev	0.09232



weights_mc

Events

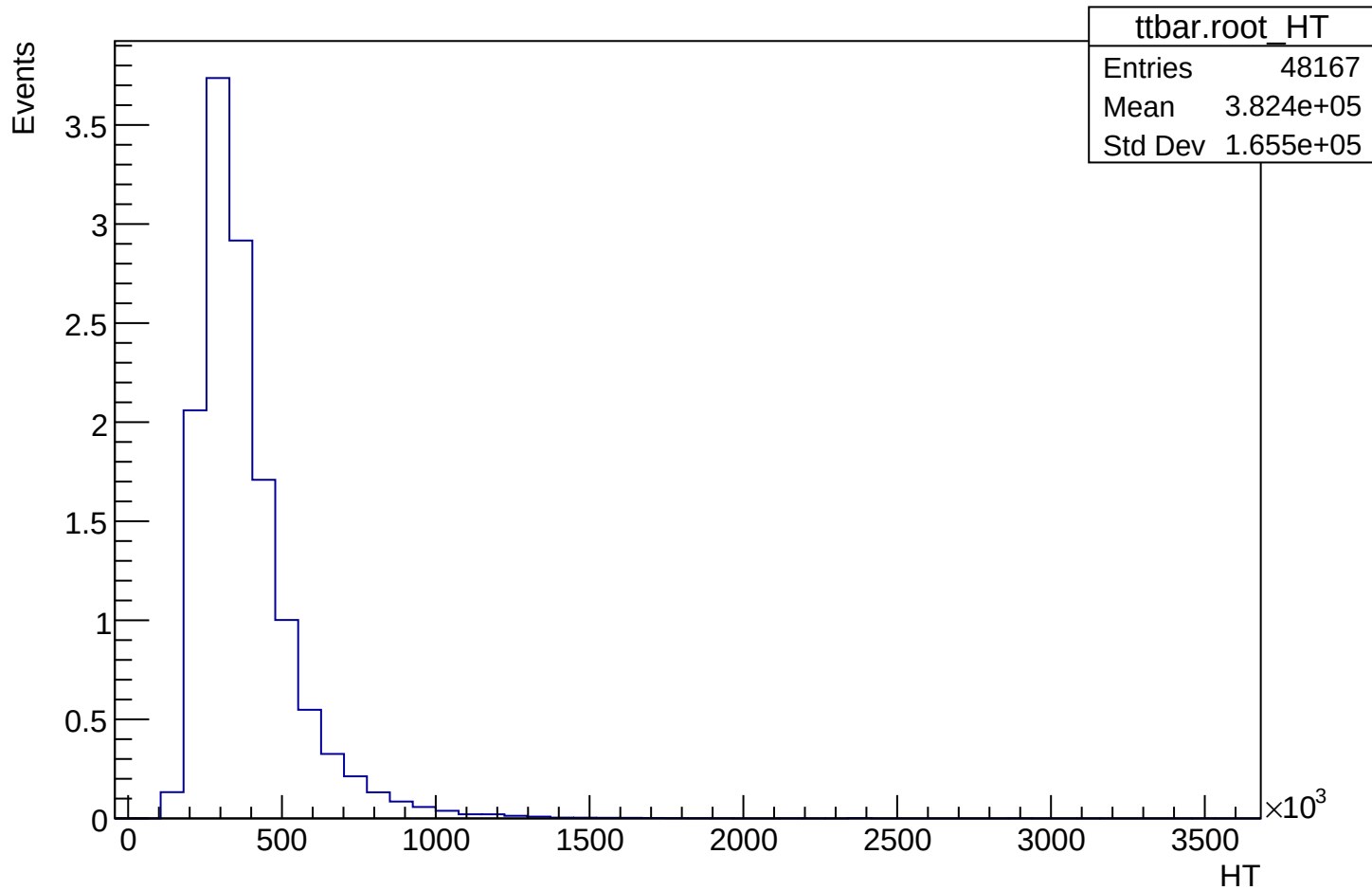
1200
1000
800
600
400
200
0

-15000 -10000 -5000 0 5000 10000 15000 20000

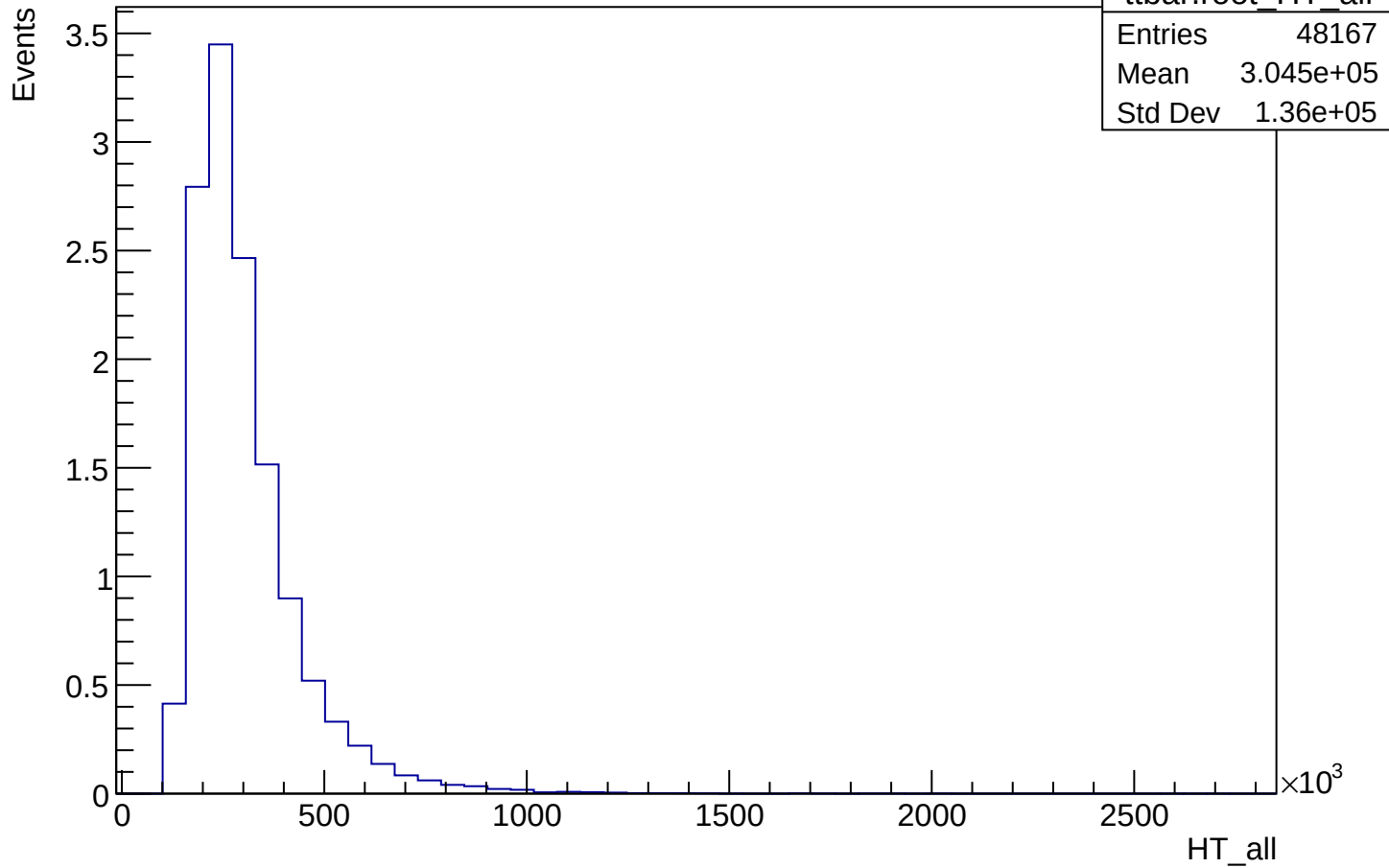
tops.root_weights_mc	
Entries	223464
Mean	0.3116
Std Dev	6.217

weights_mc

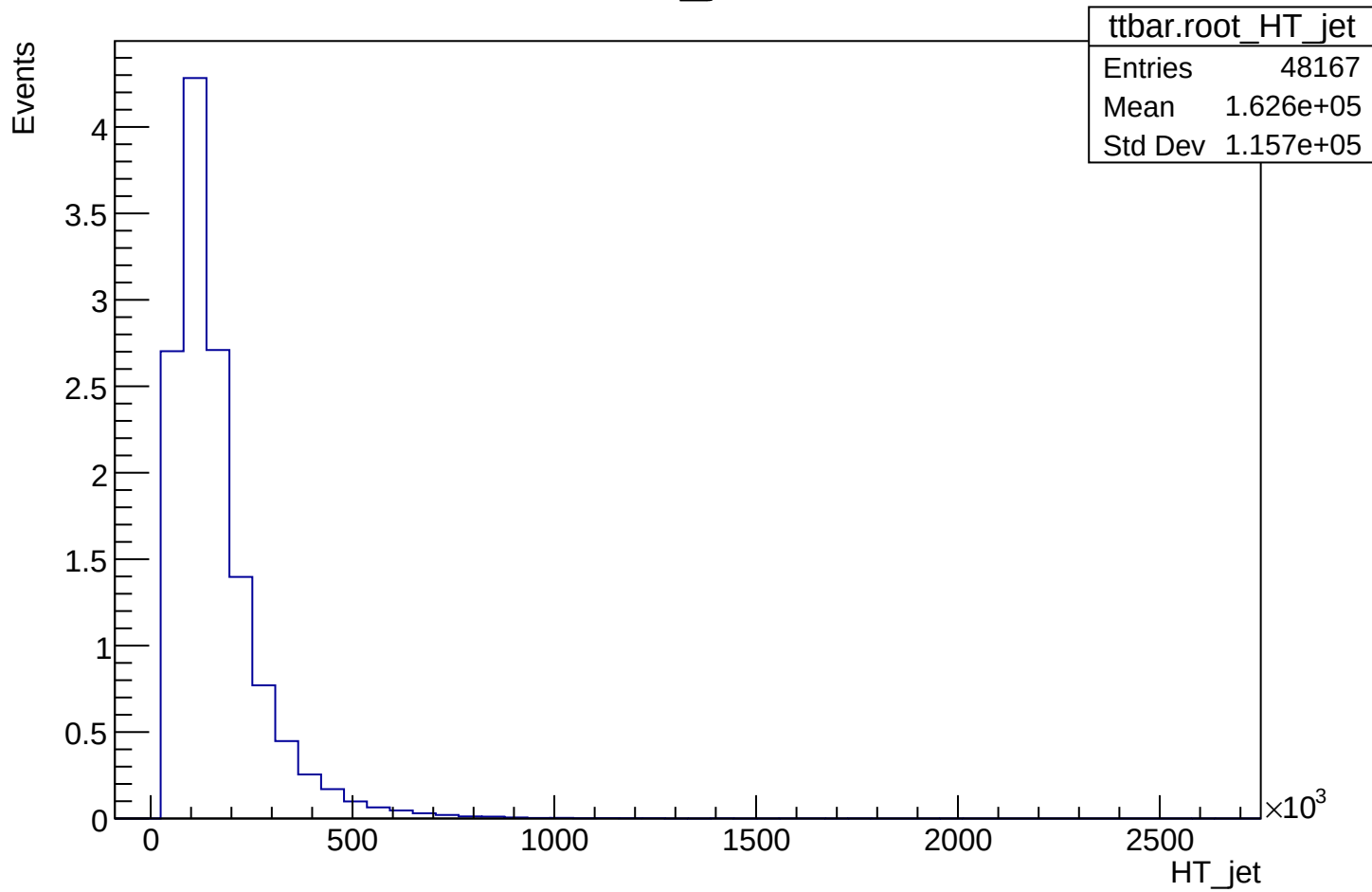
HT



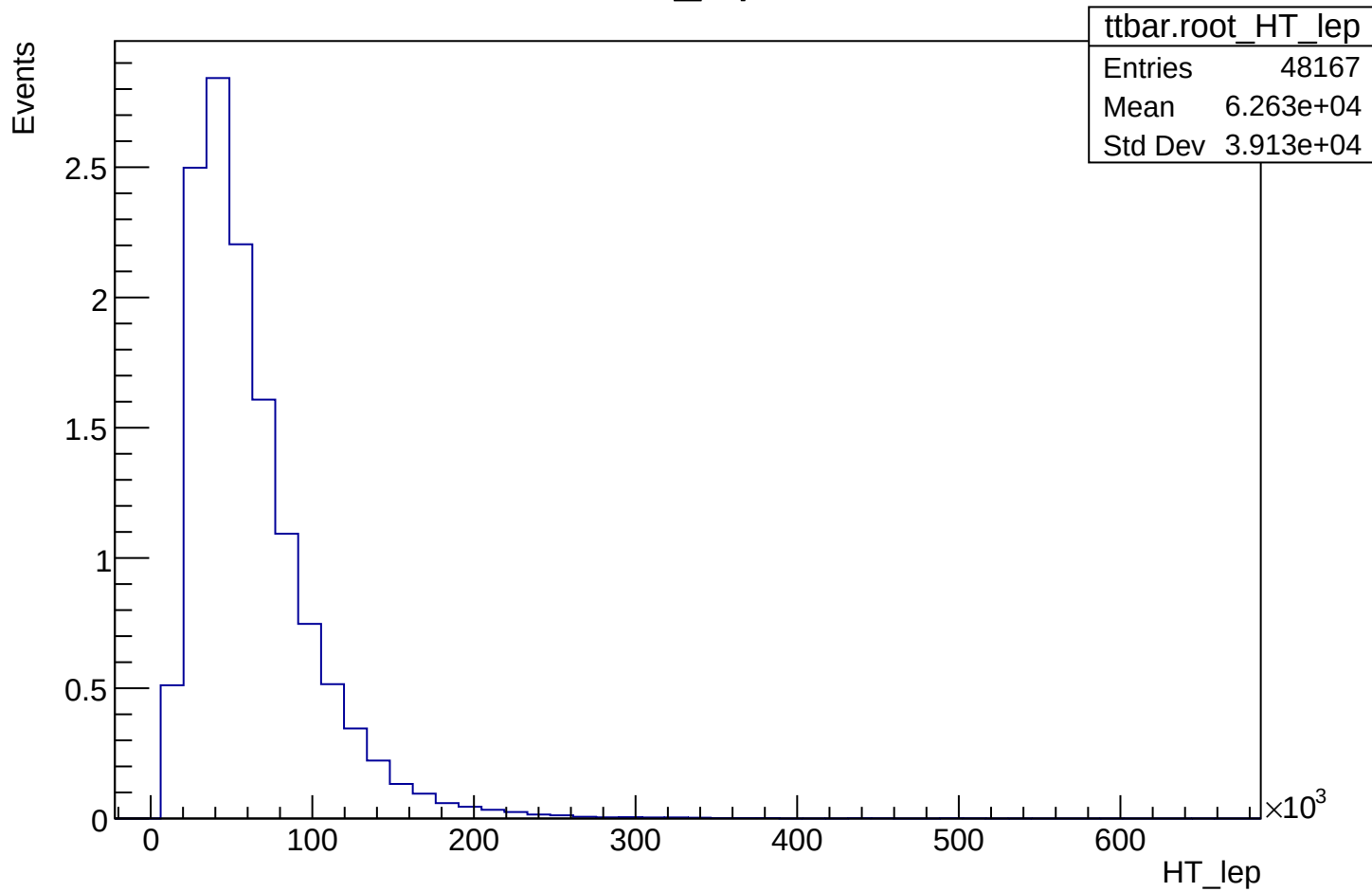
HT_all



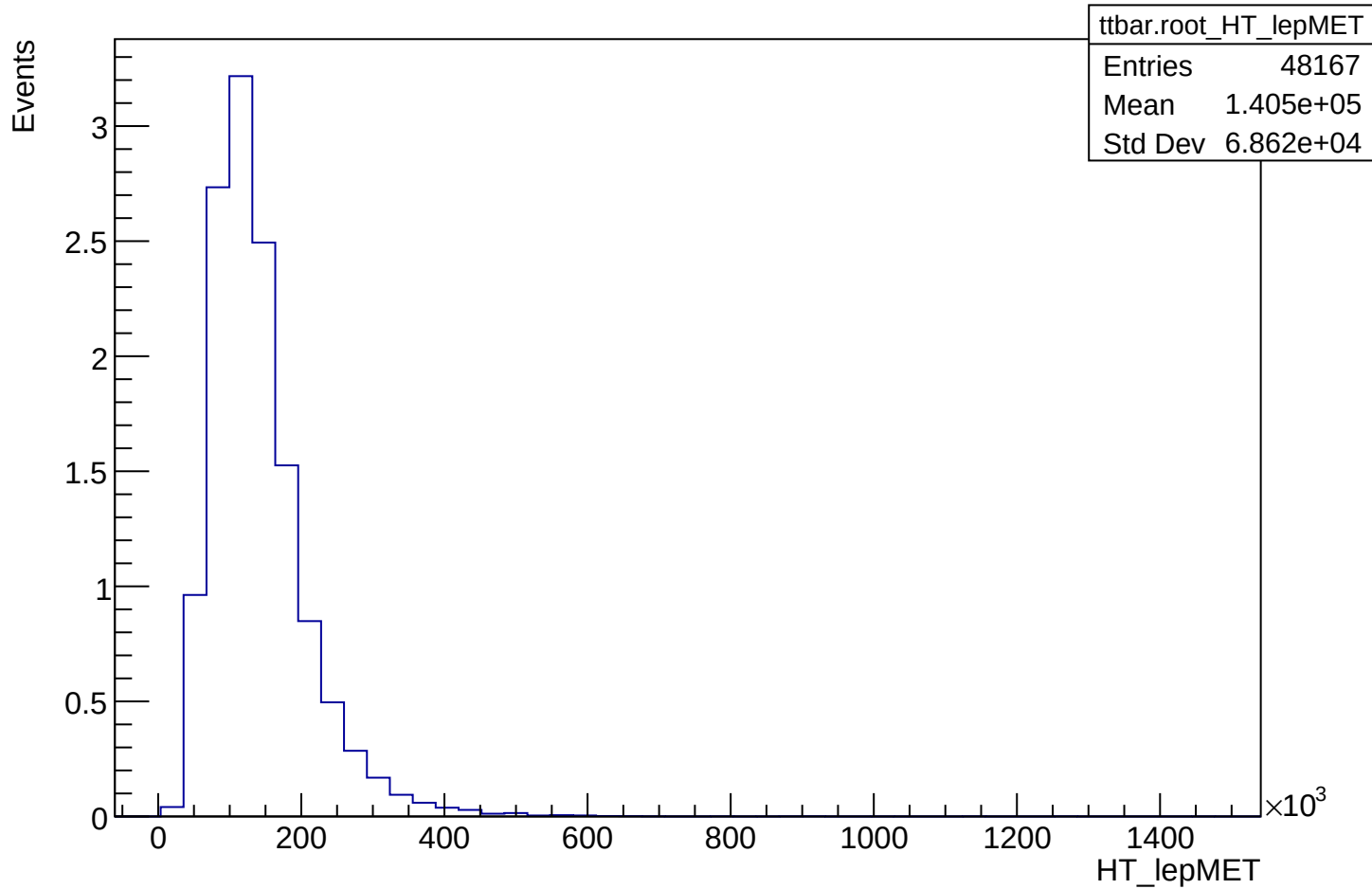
HT_jet



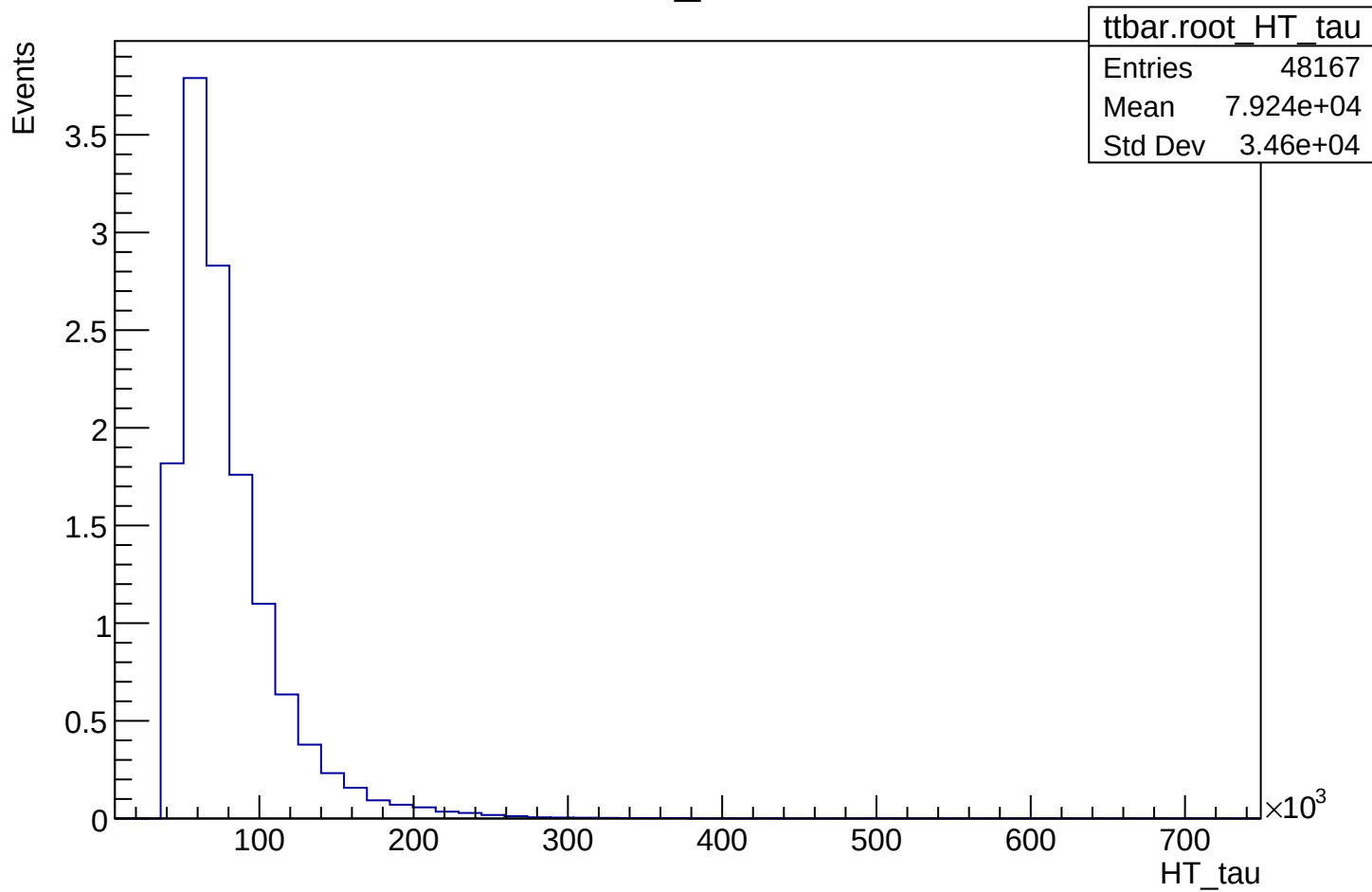
HT_lep



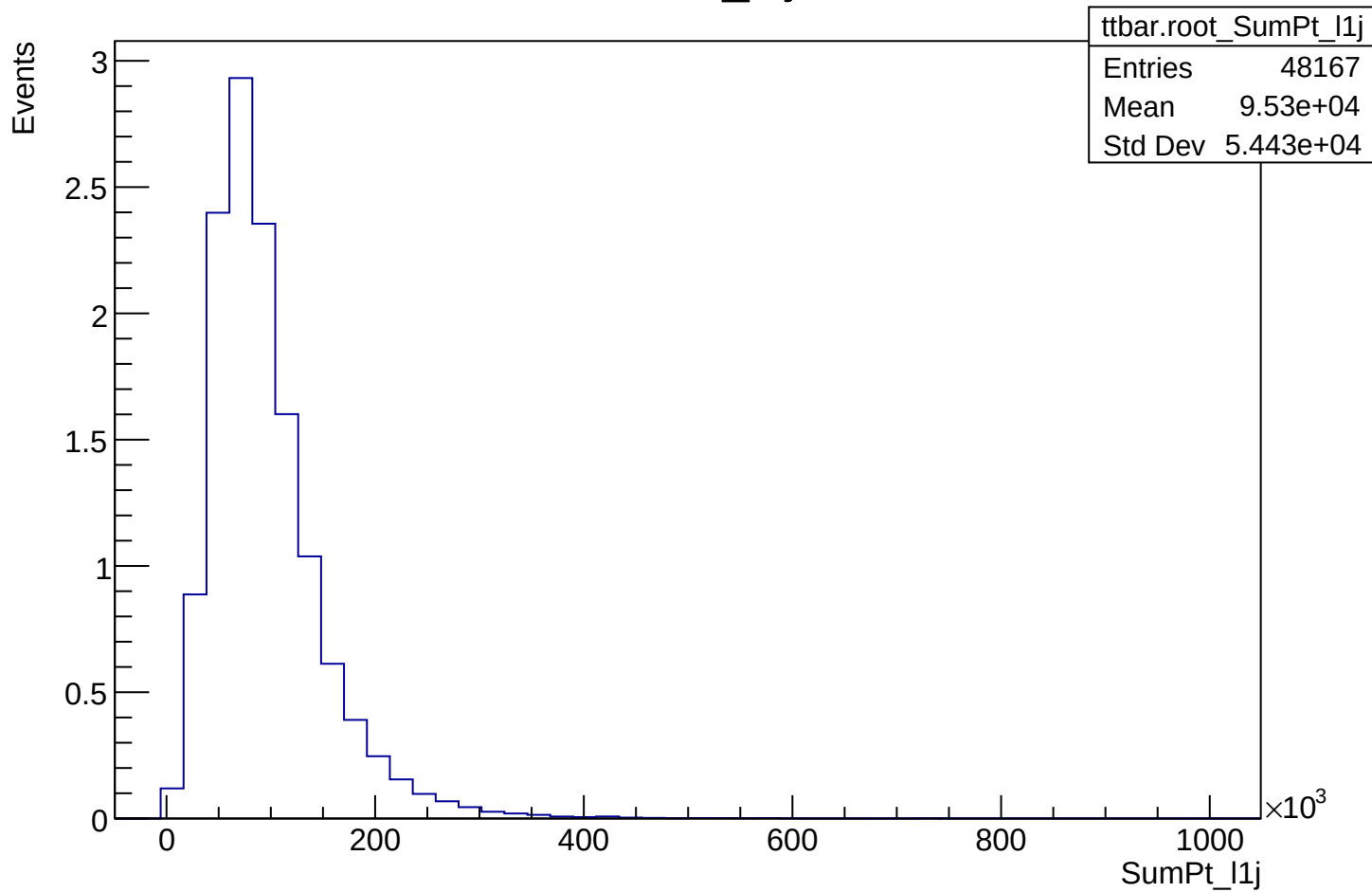
HT_lepMET



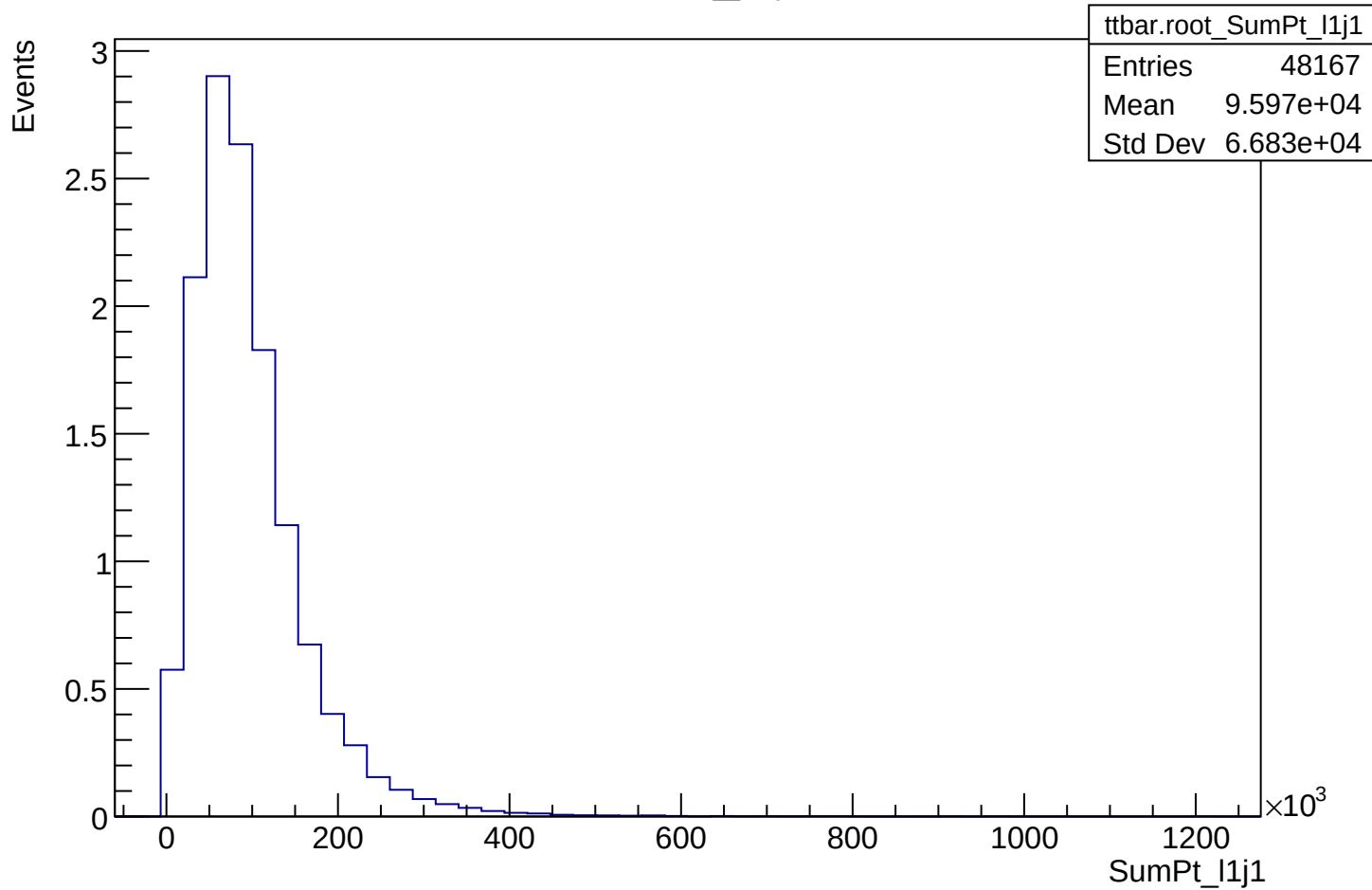
HT_tau



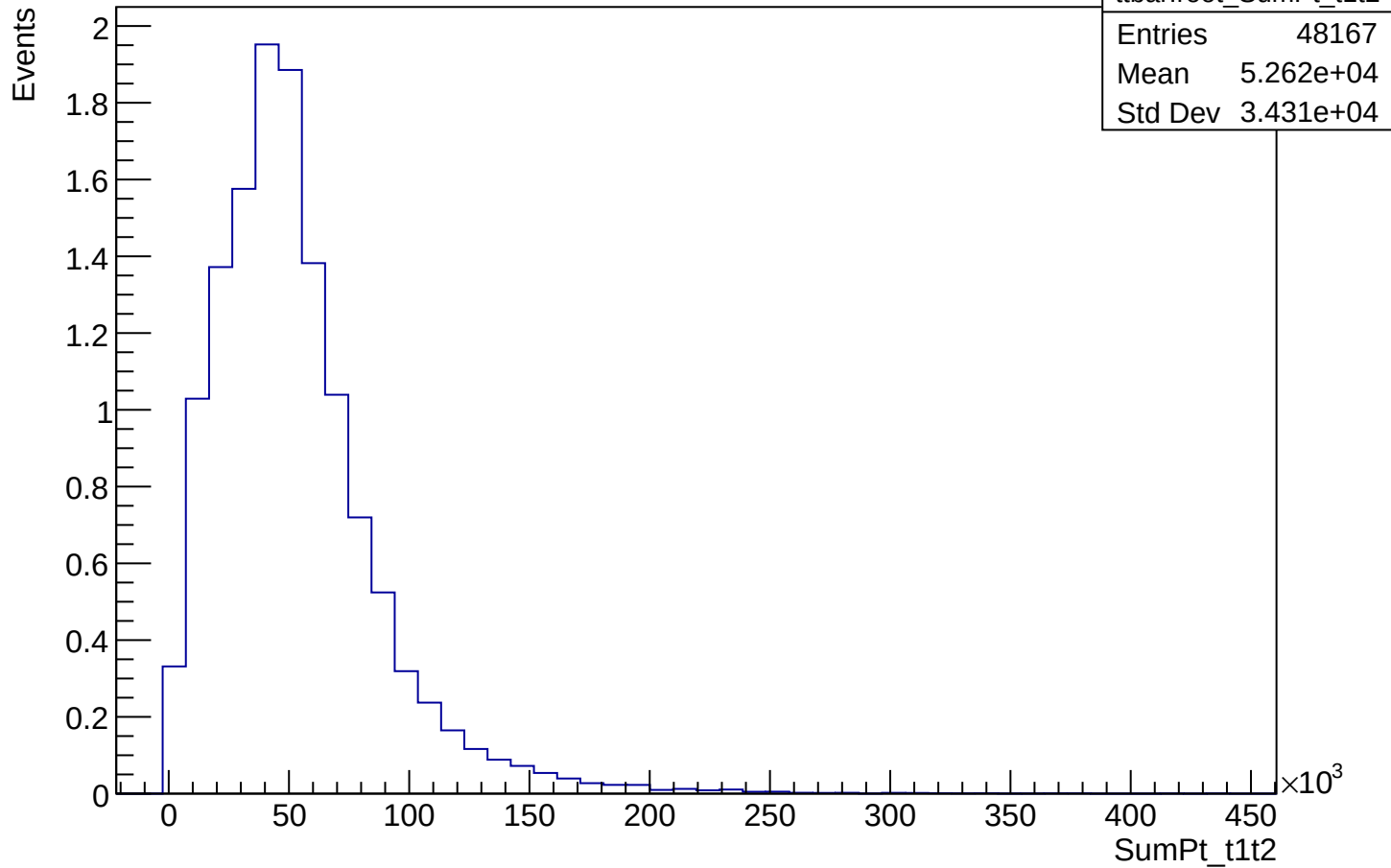
SumPt_l1j



SumPt_l1j1

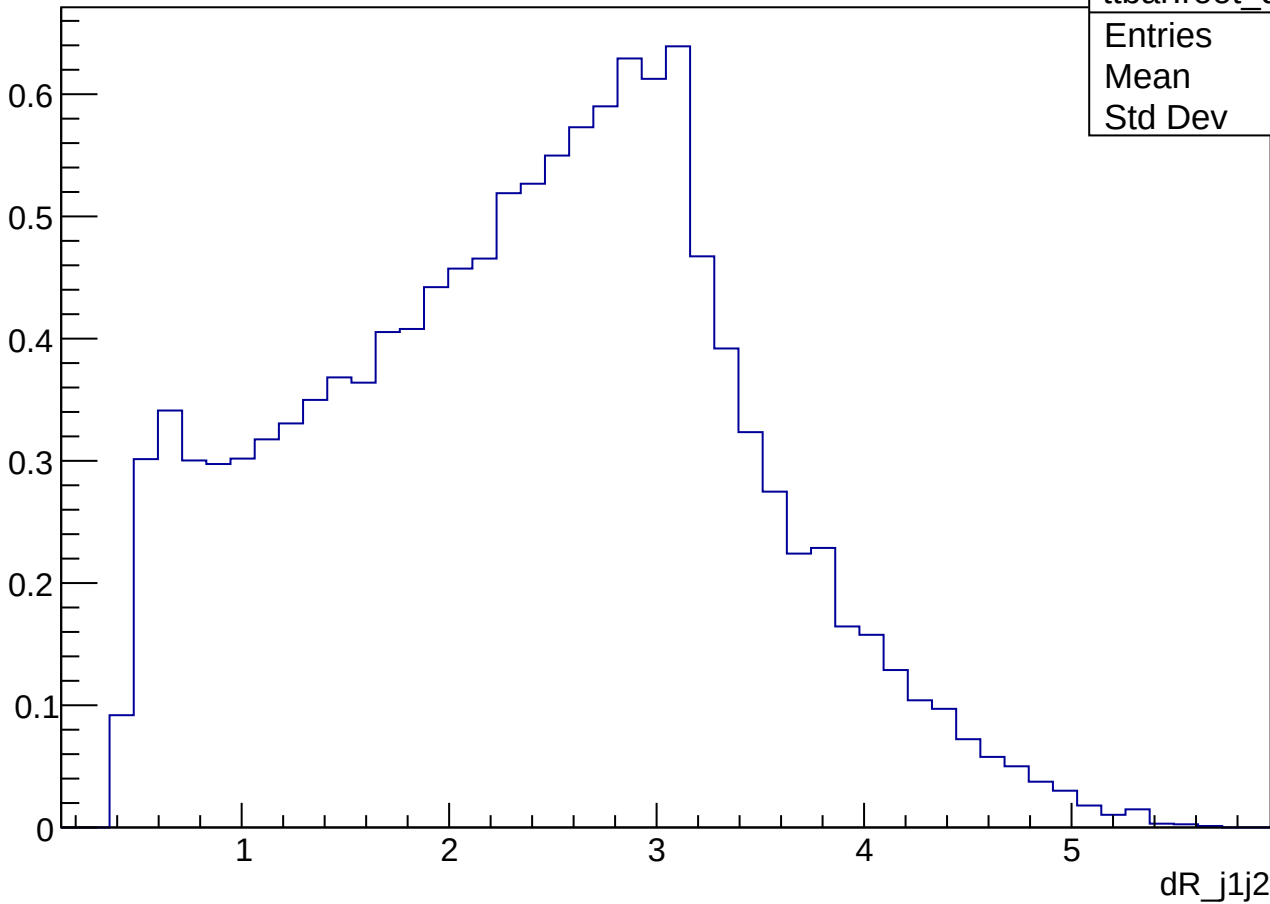


SumPt_t1t2



dR_j1j2

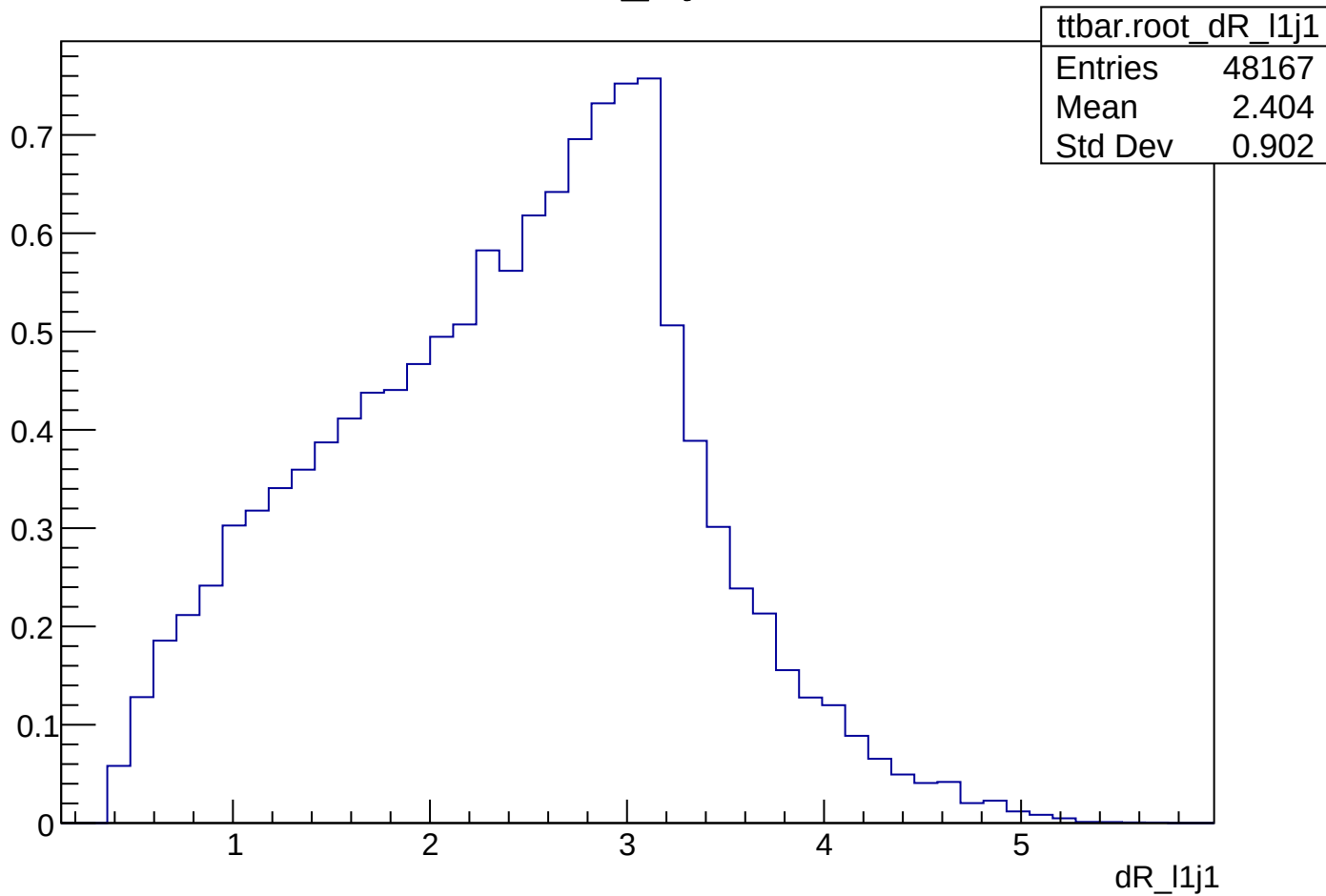
Events



ttbar.root_dR_j1j2	
Entries	48167
Mean	2.387
Std Dev	1.02

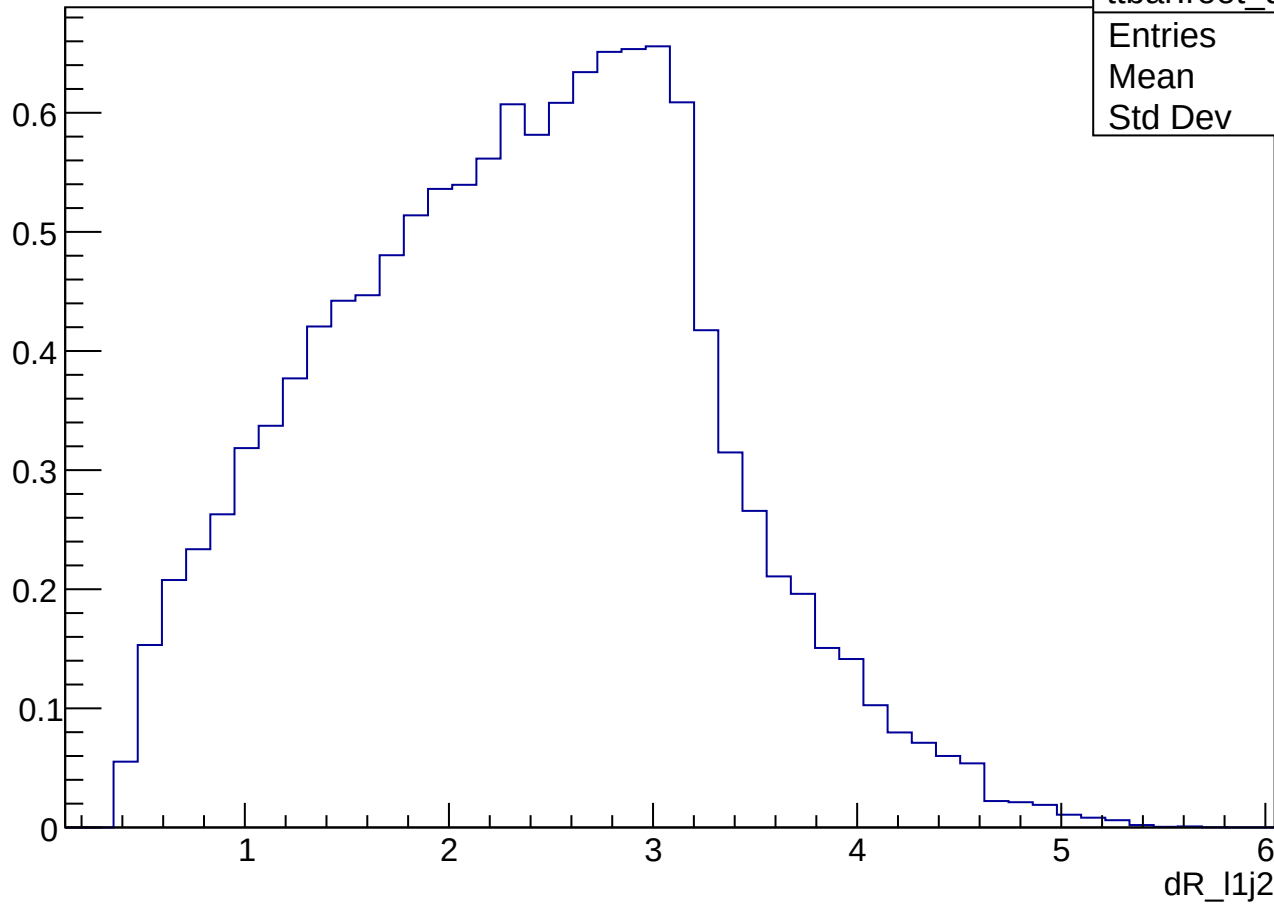
dR_l1j1

Events



dR_l1j2

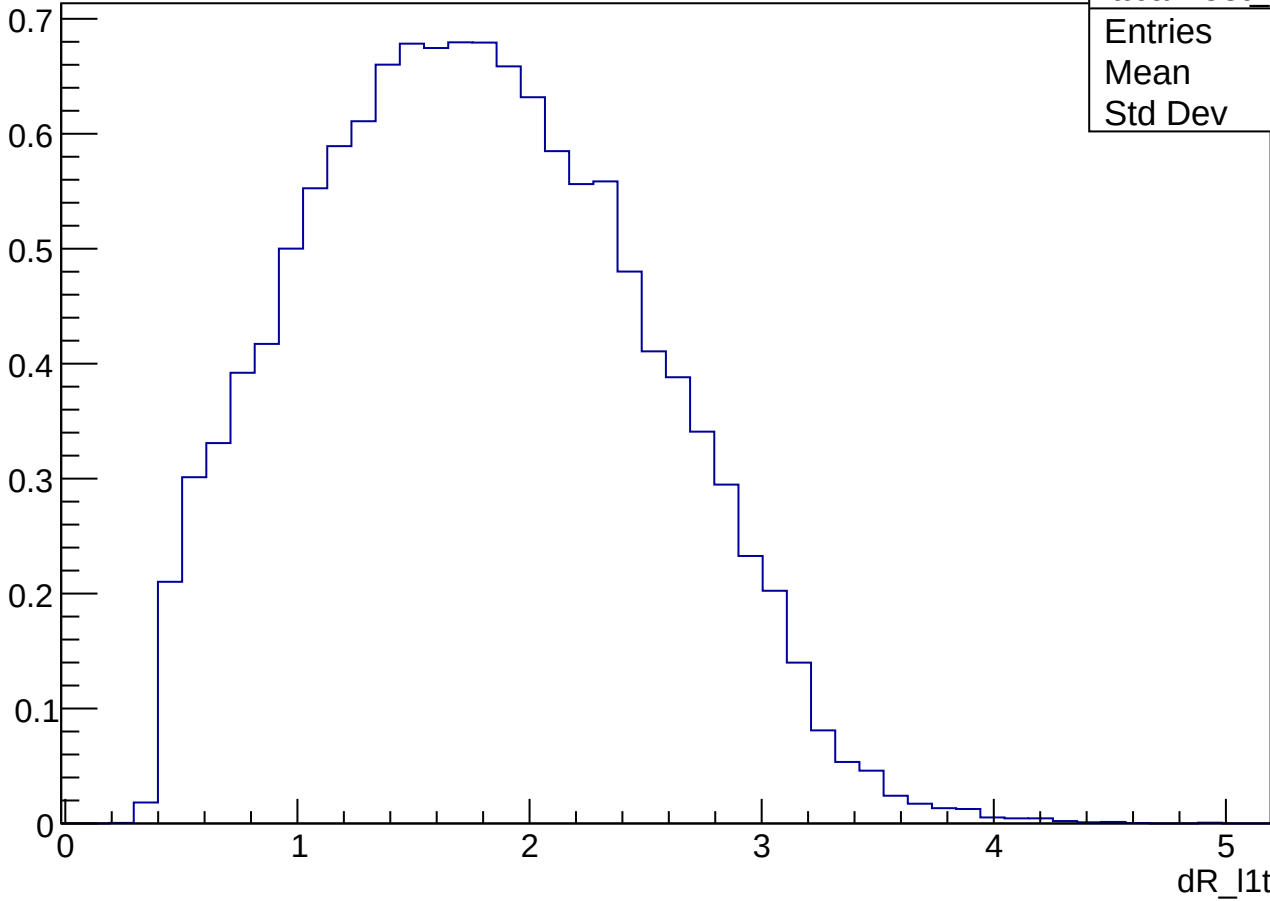
Events



tbar.root_dR_l1j2	
Entries	48167
Mean	2.348
Std Dev	0.9173

dR_l1t

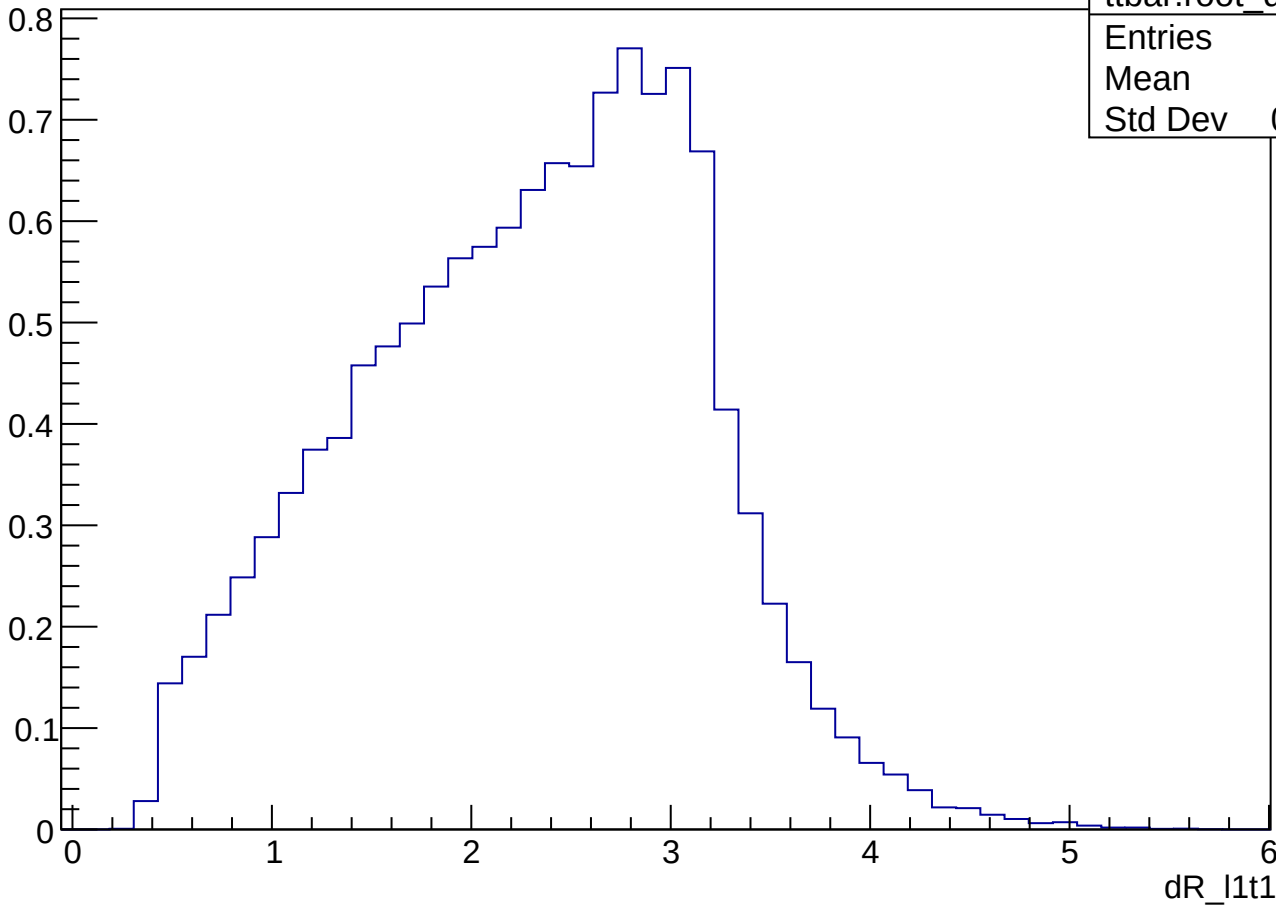
Events



ttbar.root	dR_l1t
Entries	48167
Mean	1.771
Std Dev	0.7145

dR_l1t1

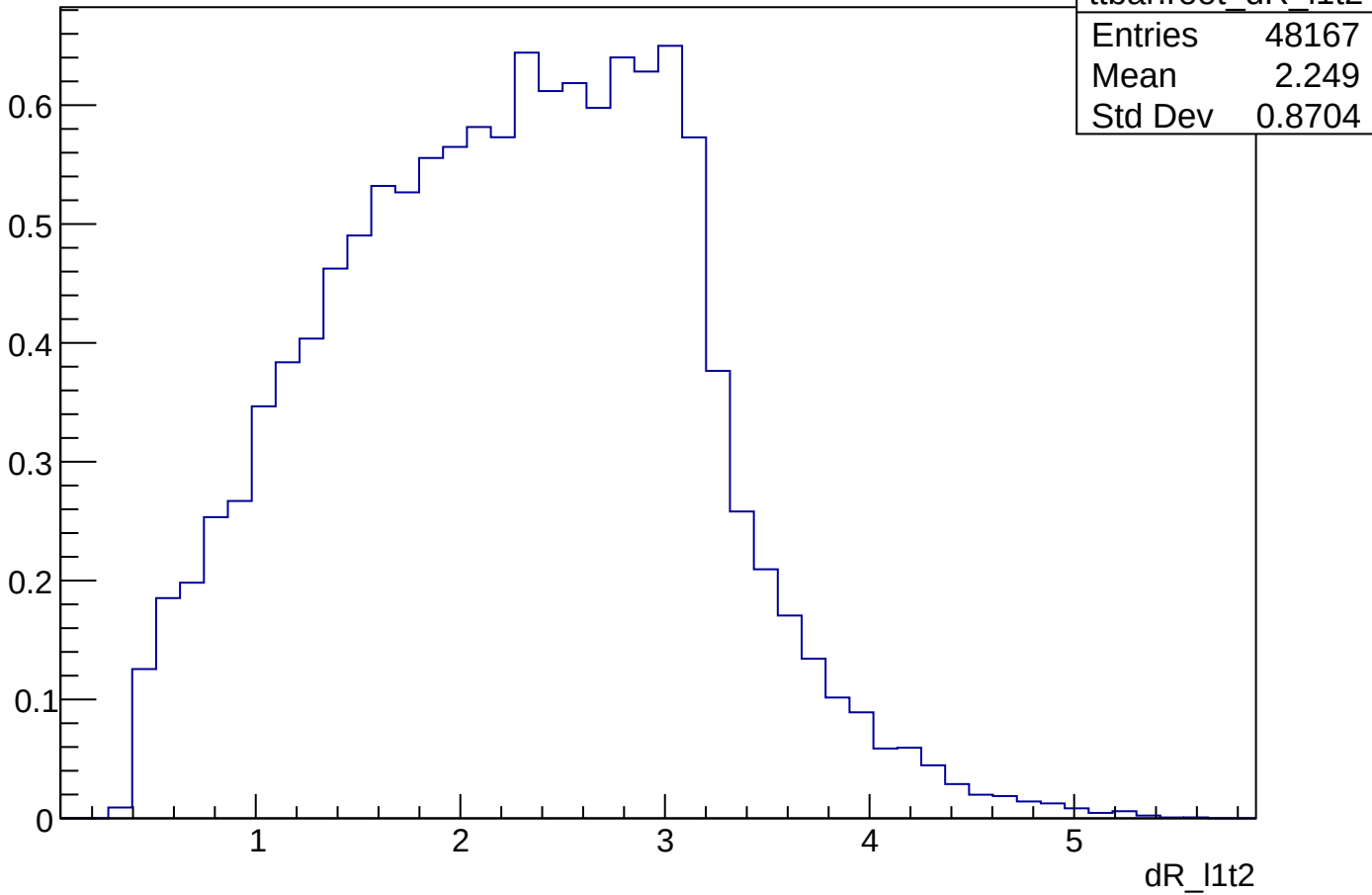
Events



ttbar.root	dR_l1t1
Entries	48167
Mean	2.3
Std Dev	0.8435

dR_l1t2

Events



dR_t1j1

Events

0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0

0

1

2

3

4

5

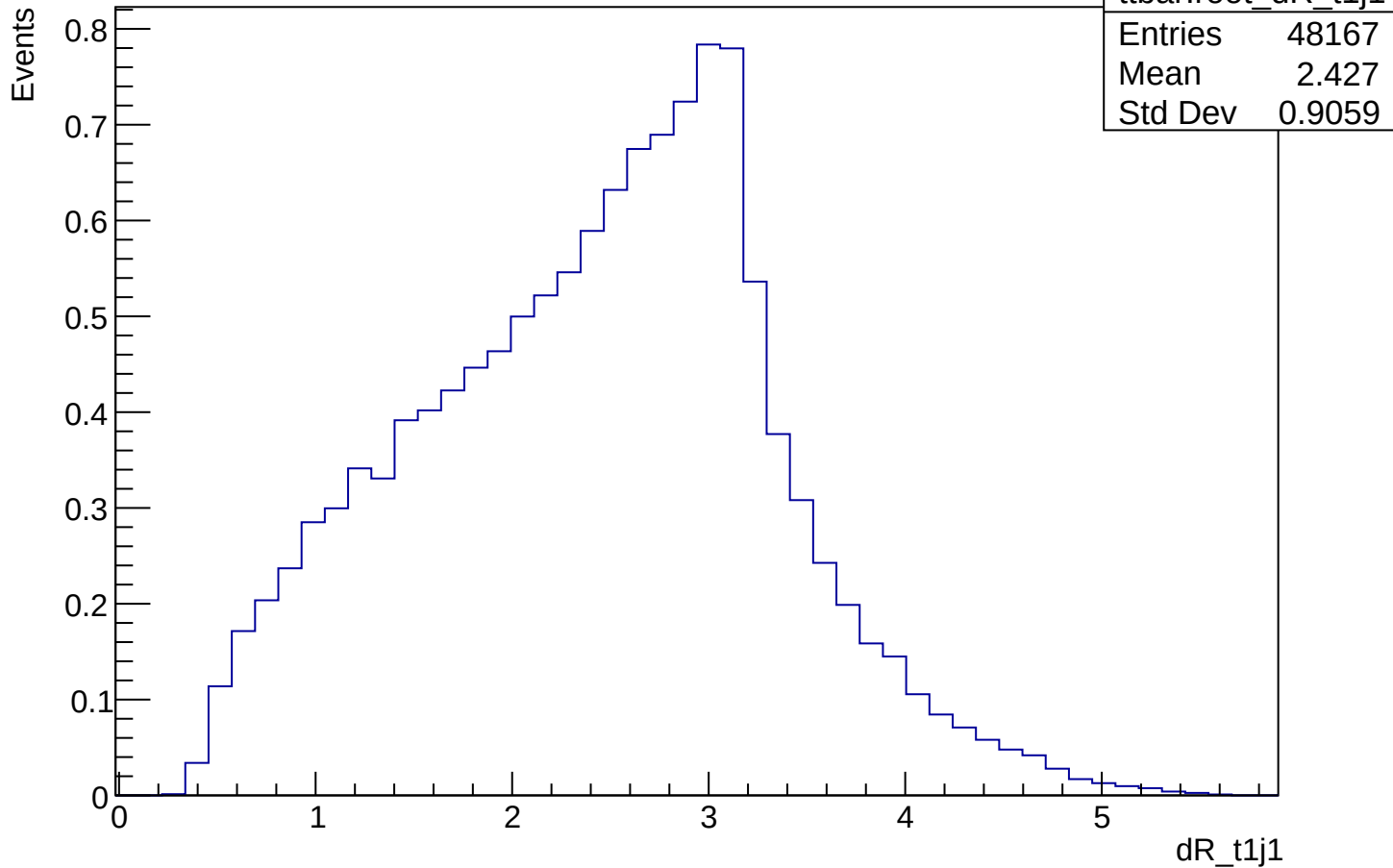
dR_t1j1

ttbar.root_dR_t1j1

Entries 48167

Mean 2.427

Std Dev 0.9059



dR_t1j2

Events

0.7
0.6
0.5
0.4
0.3
0.2
0.1
0

0

1

2

3

4

5

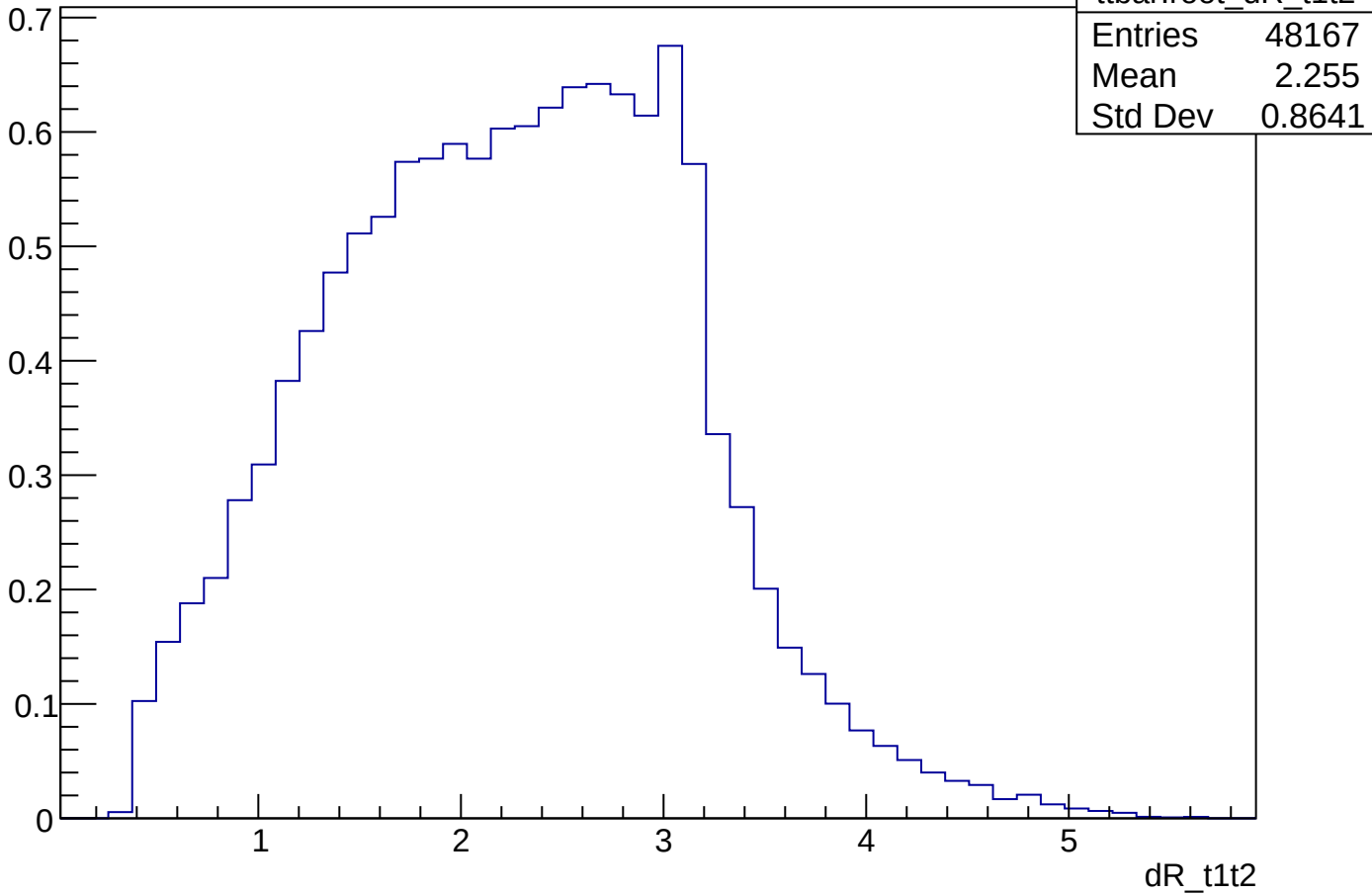
6

dR_t1j2

ttbar.root_dR_t1j2	
Entries	48167
Mean	2.363
Std Dev	0.9258

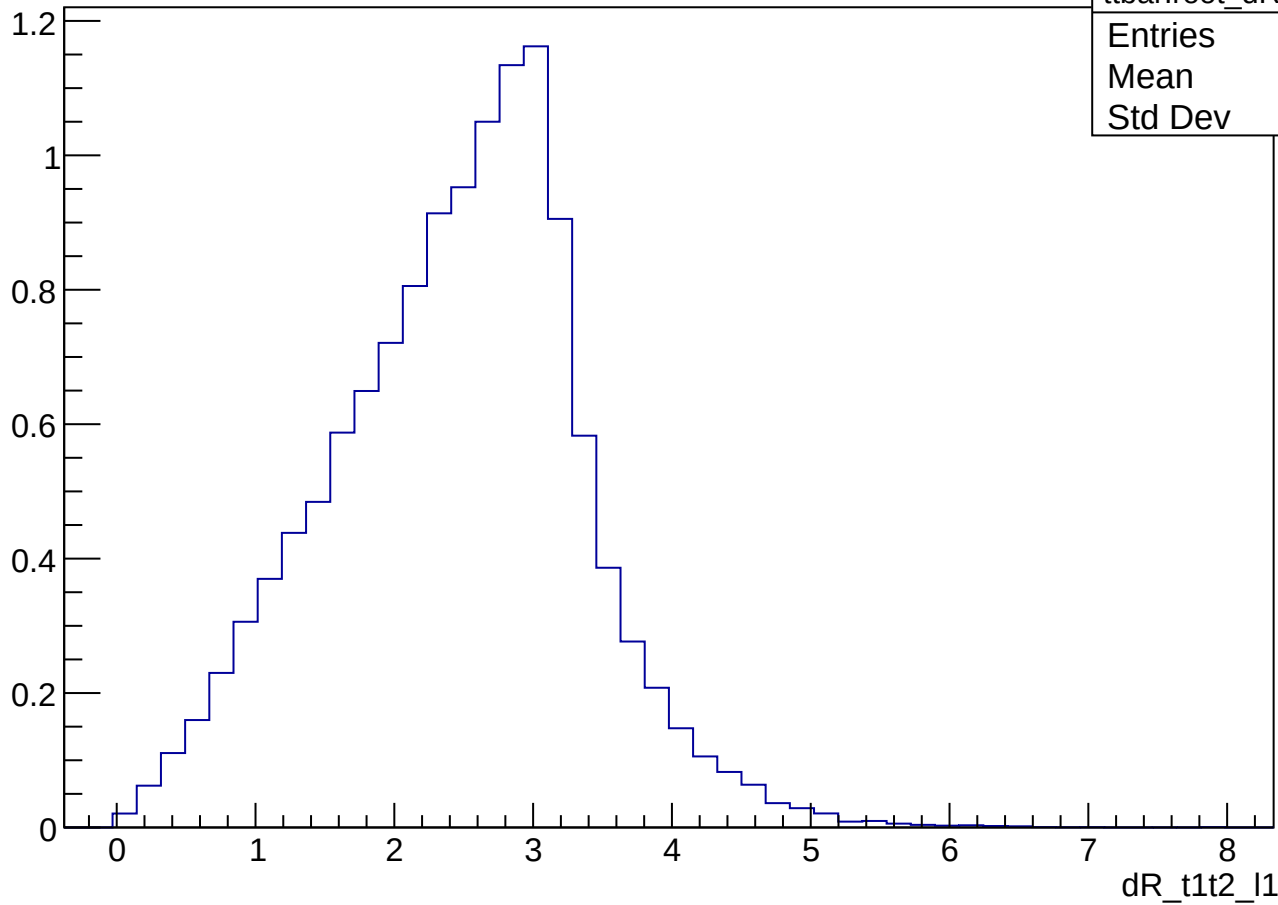
dR_t1t2

Events



dR_t1t2_l1

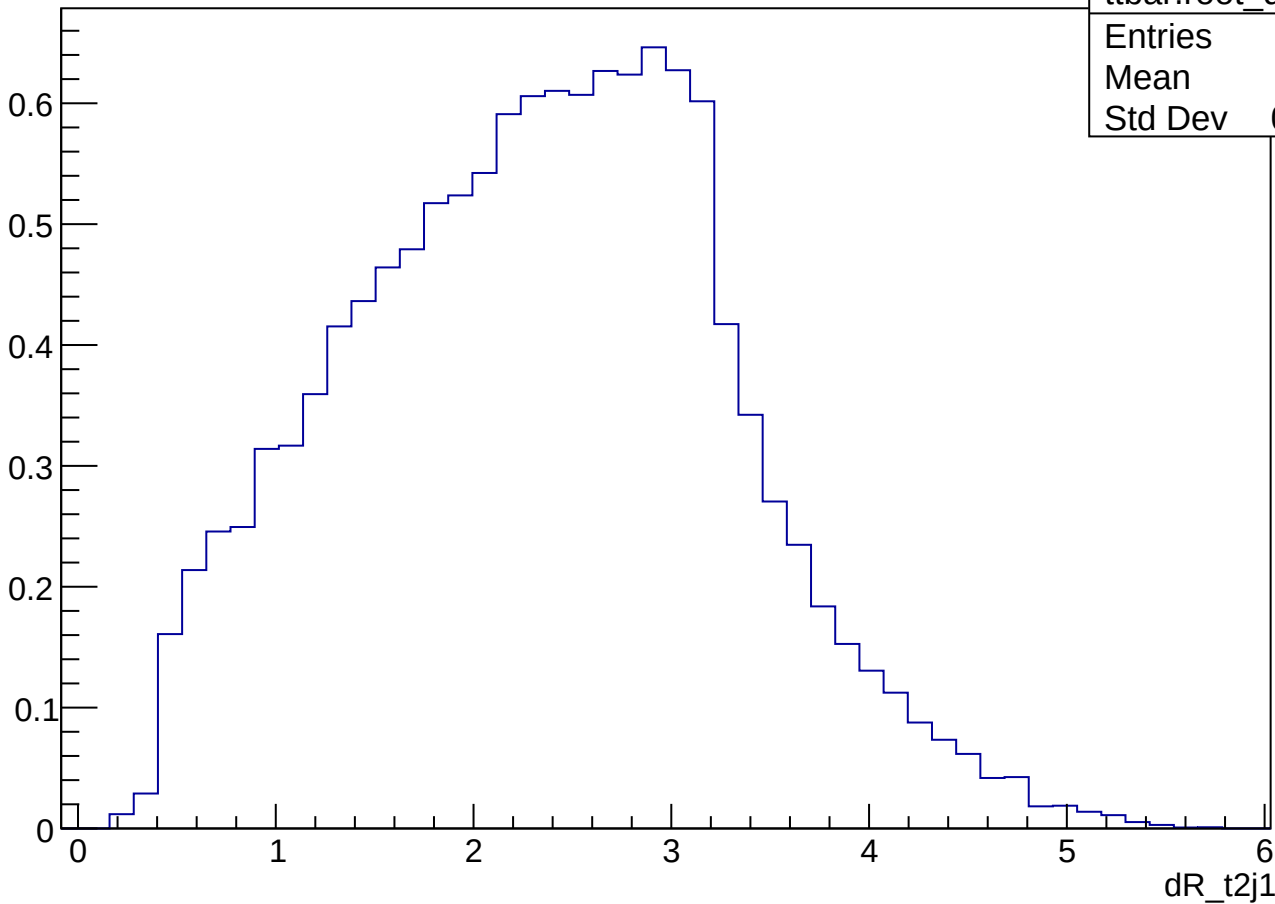
Events



tbar.root_dR_t1t2_l1	
Entries	48167
Mean	2.455
Std Dev	0.905

dR_t2j1

Events



ttbar.root_dR_t2j1

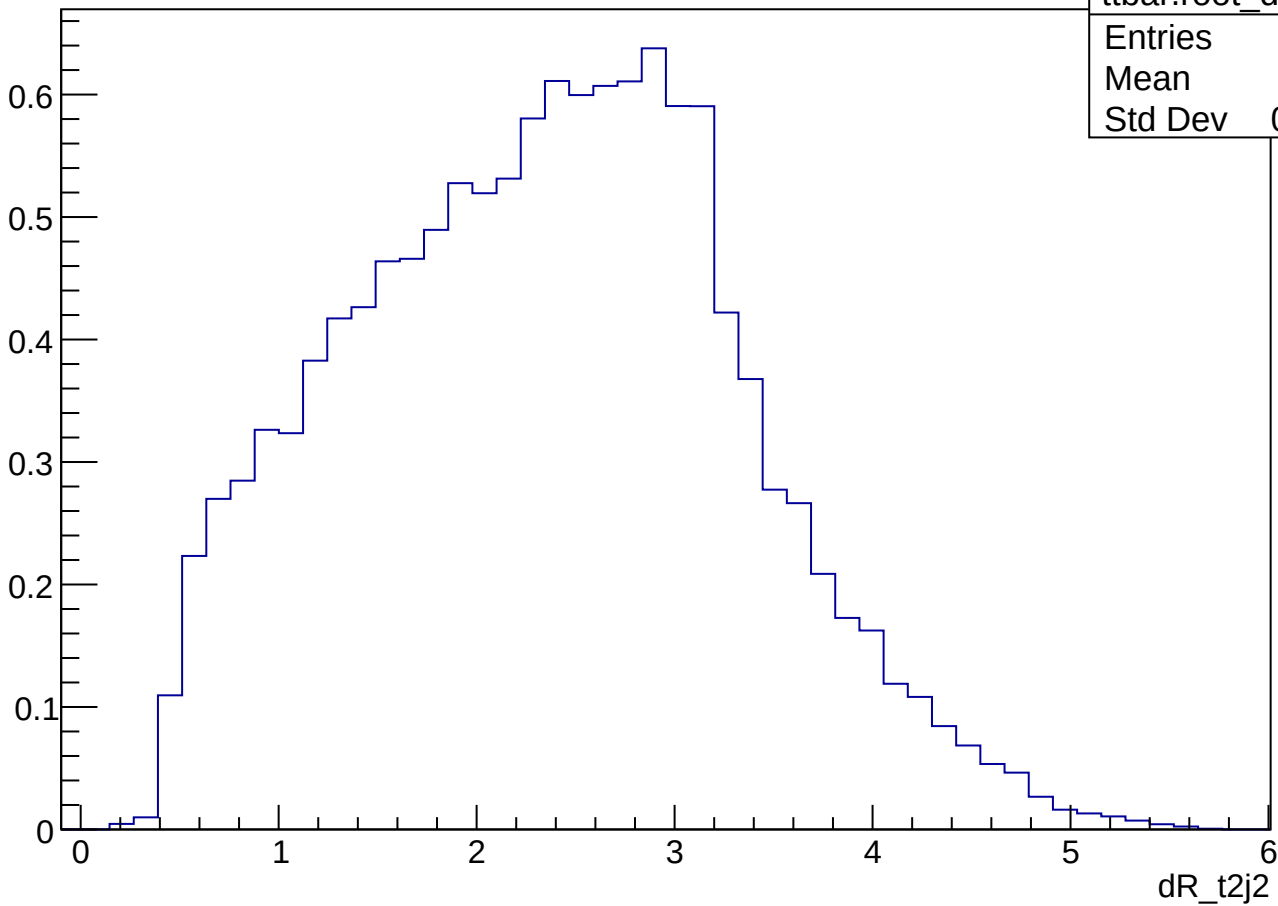
Entries 48167

Mean 2.346

Std Dev 0.9508

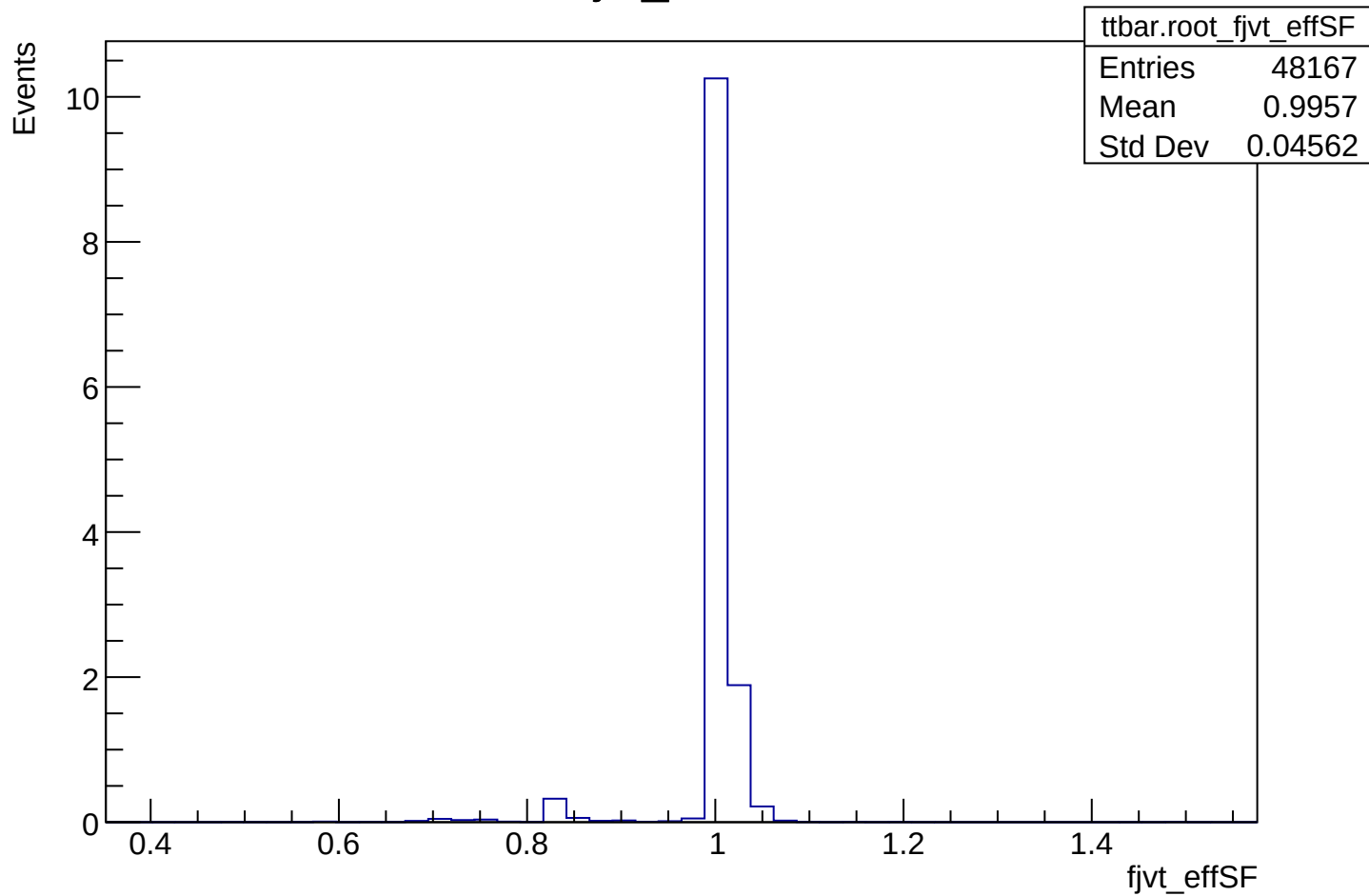
dR_t2j2

Events

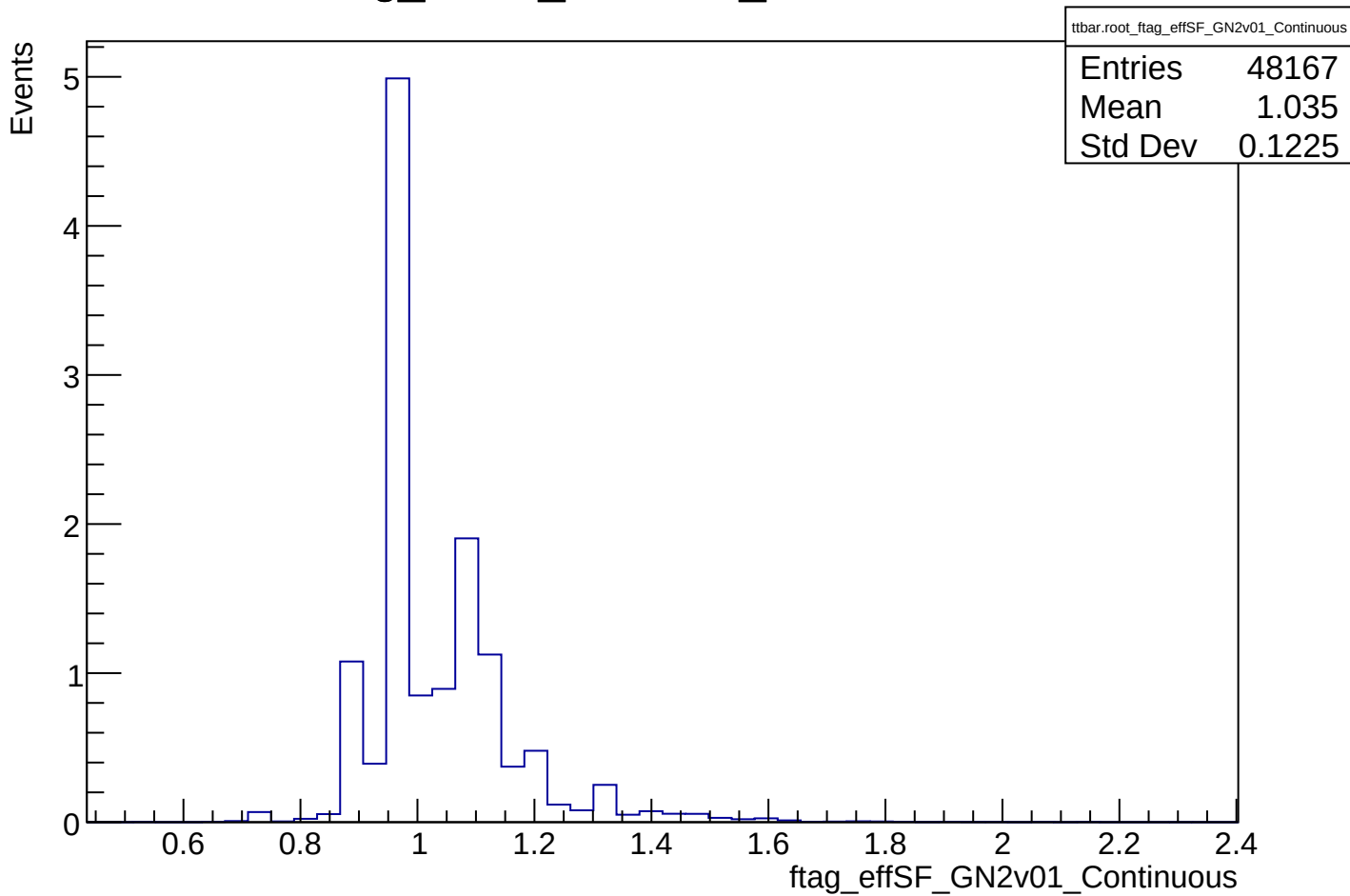


ttbar.root_dR_t2j2	
Entries	48167
Mean	2.355
Std Dev	0.9694

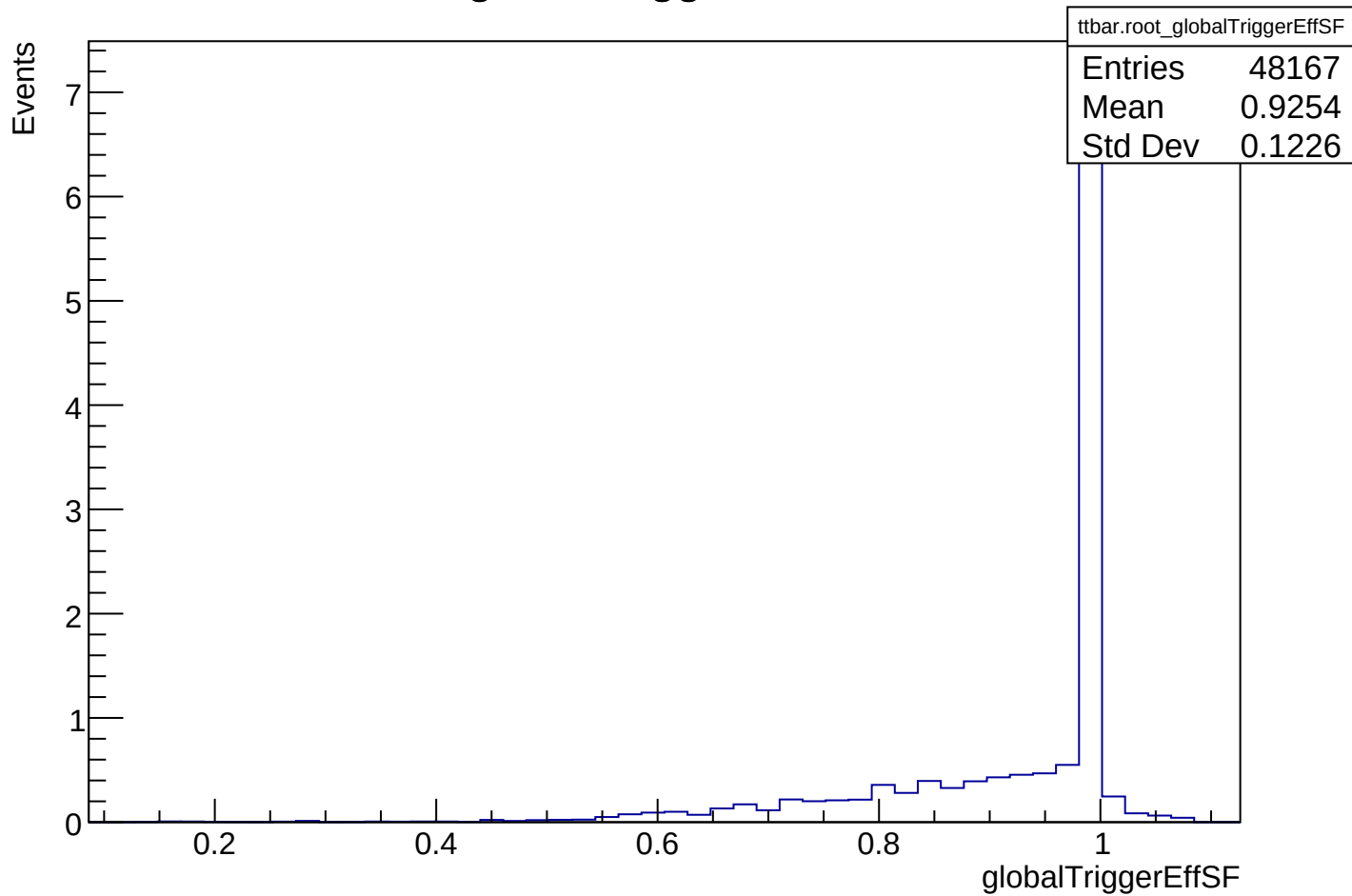
fjvt_effSF



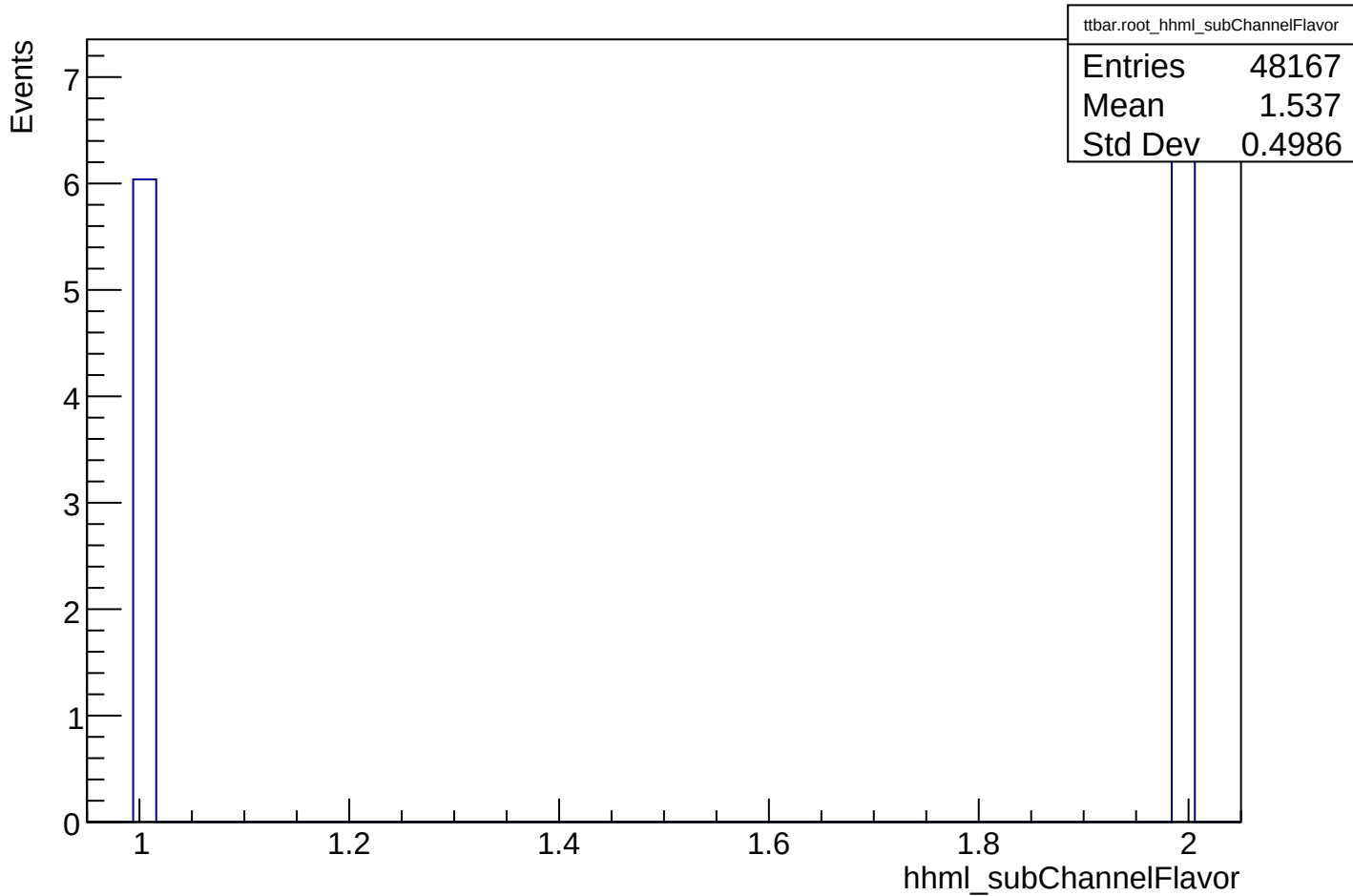
ftag_effSF_GN2v01_Continuous



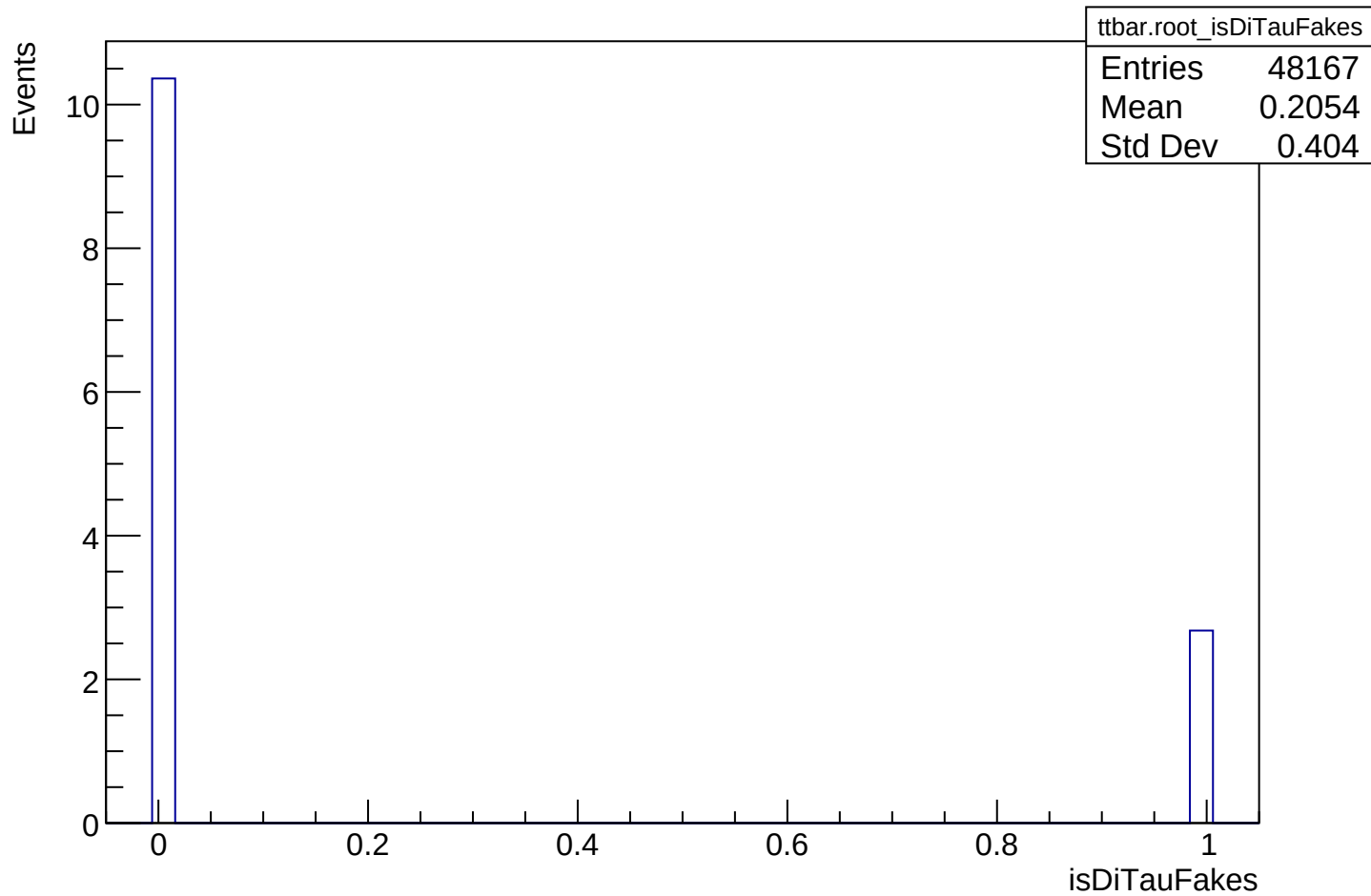
globalTriggerEffSF



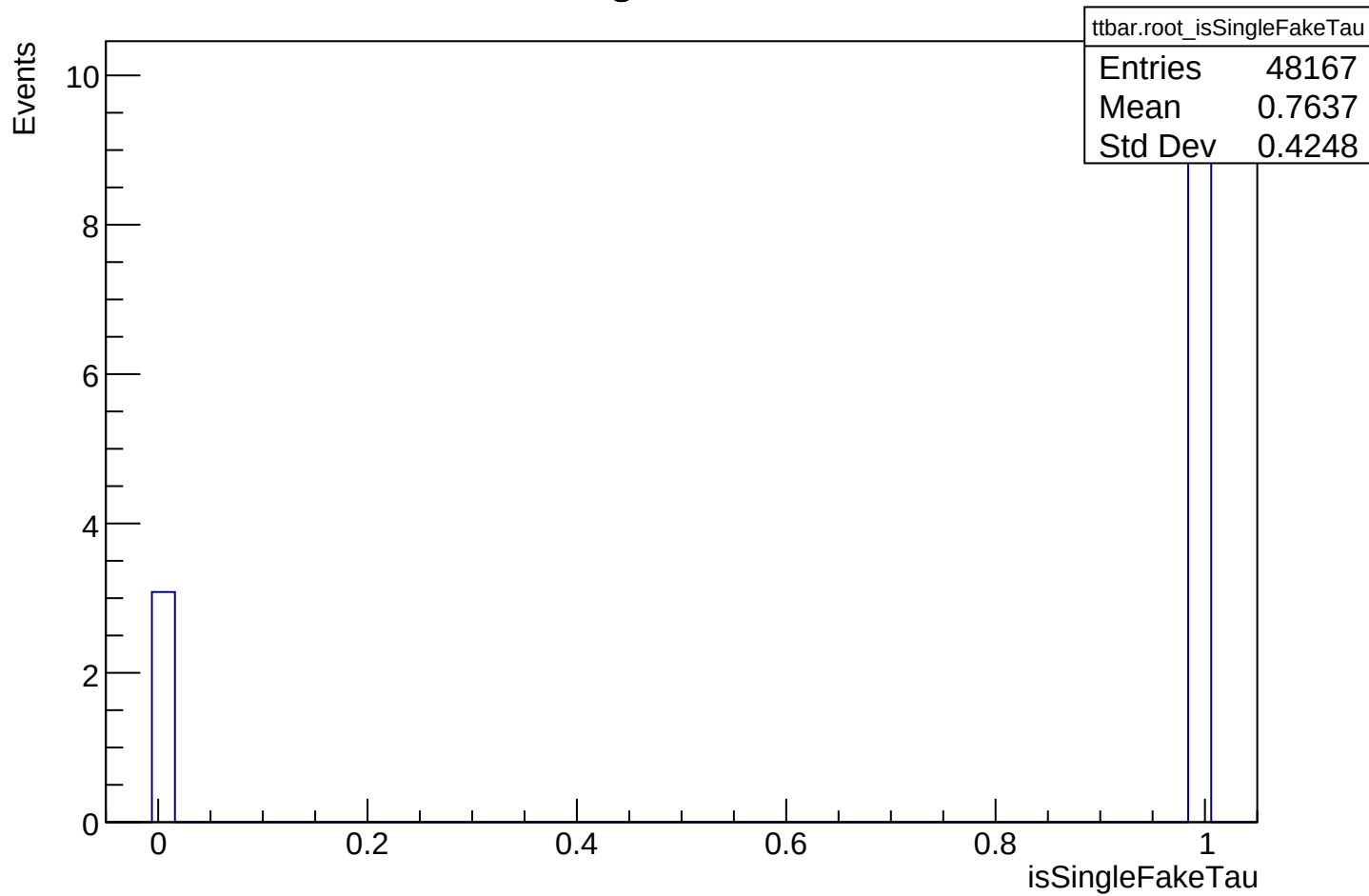
hhml_subChannelFlavor



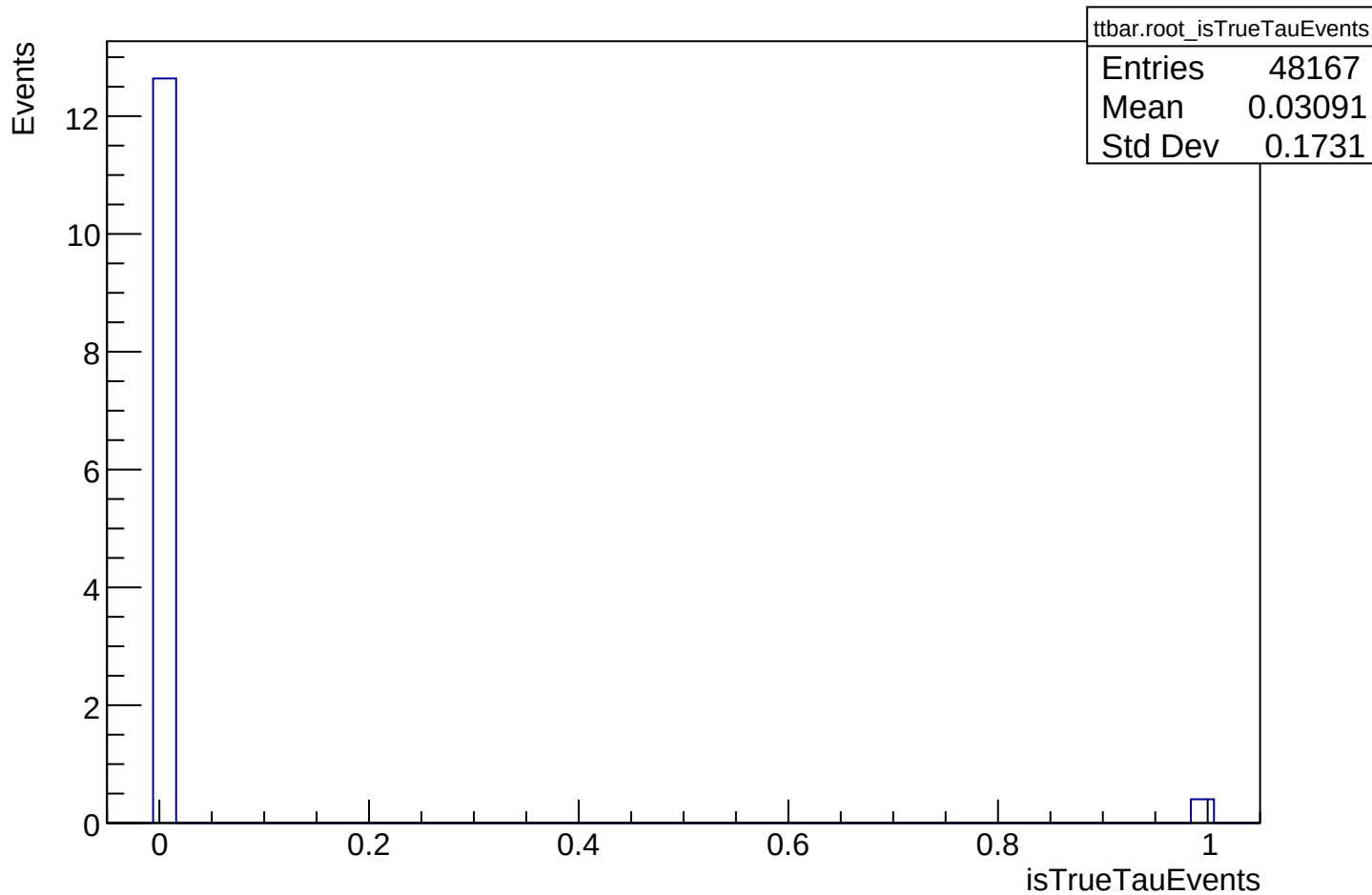
isDiTauFakes



isSingleFakeTau

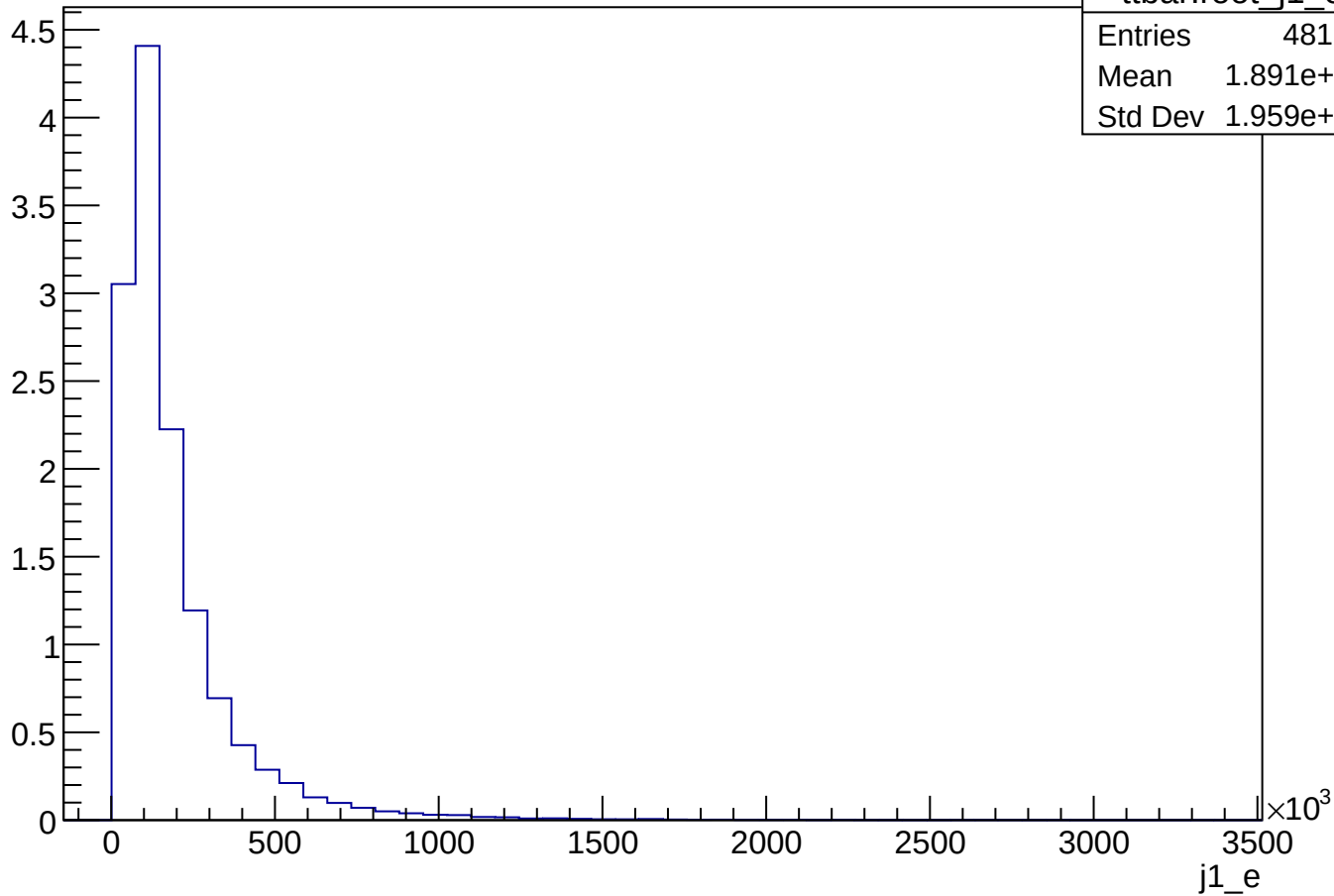


isTrueTauEvents



j1_e

Events



j1_eta

ttbar.root_j1_eta

Entries 48167

Mean -0.002722

Std Dev 1.285

Events

0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0

-2

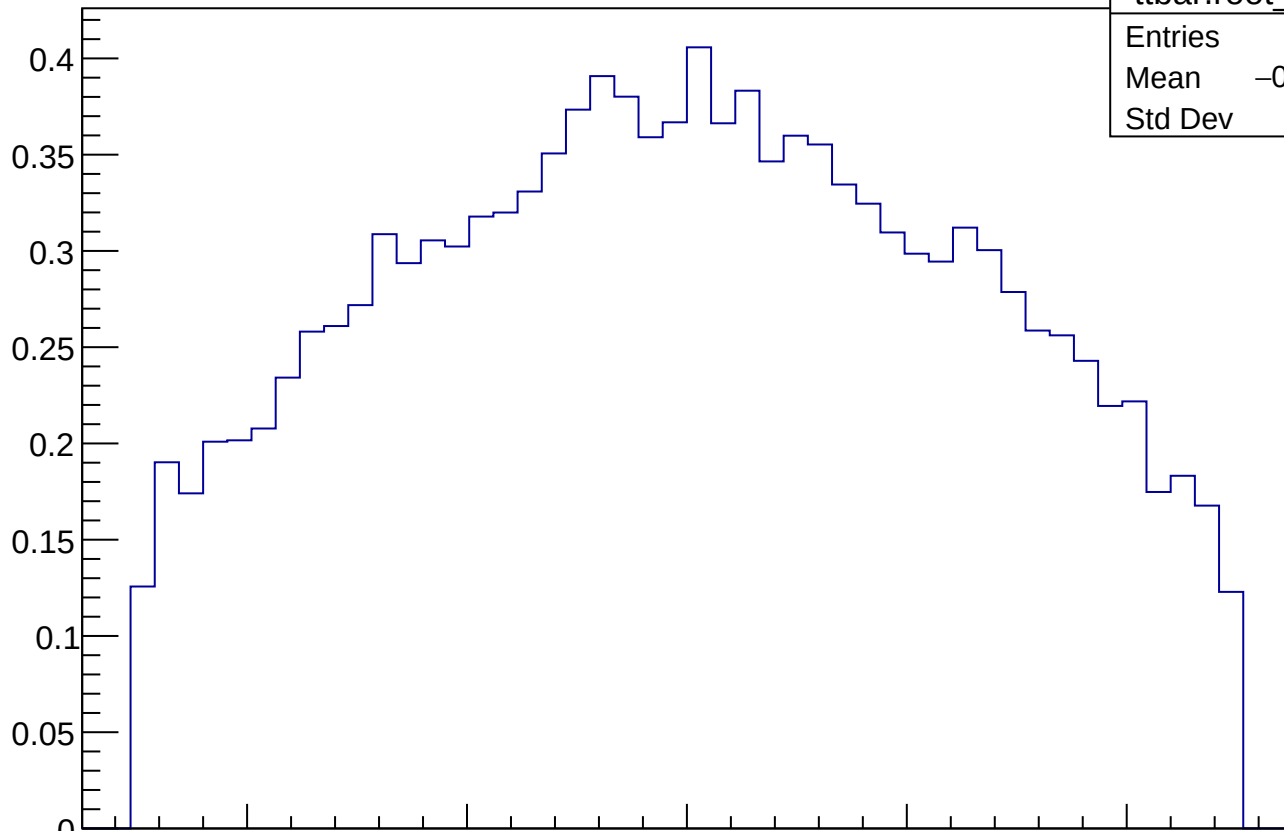
-1

0

1

2

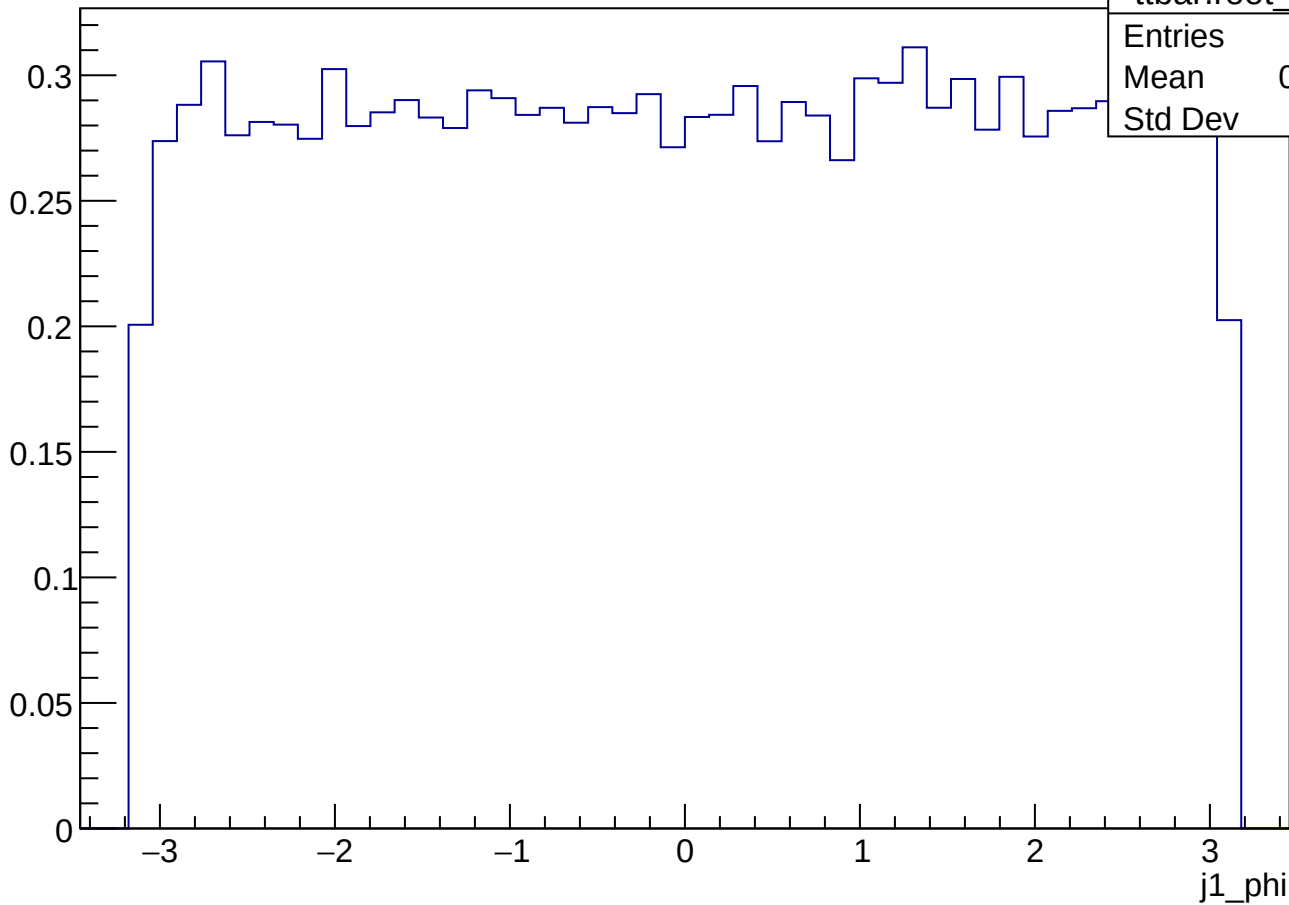
j1_eta



j1_phi

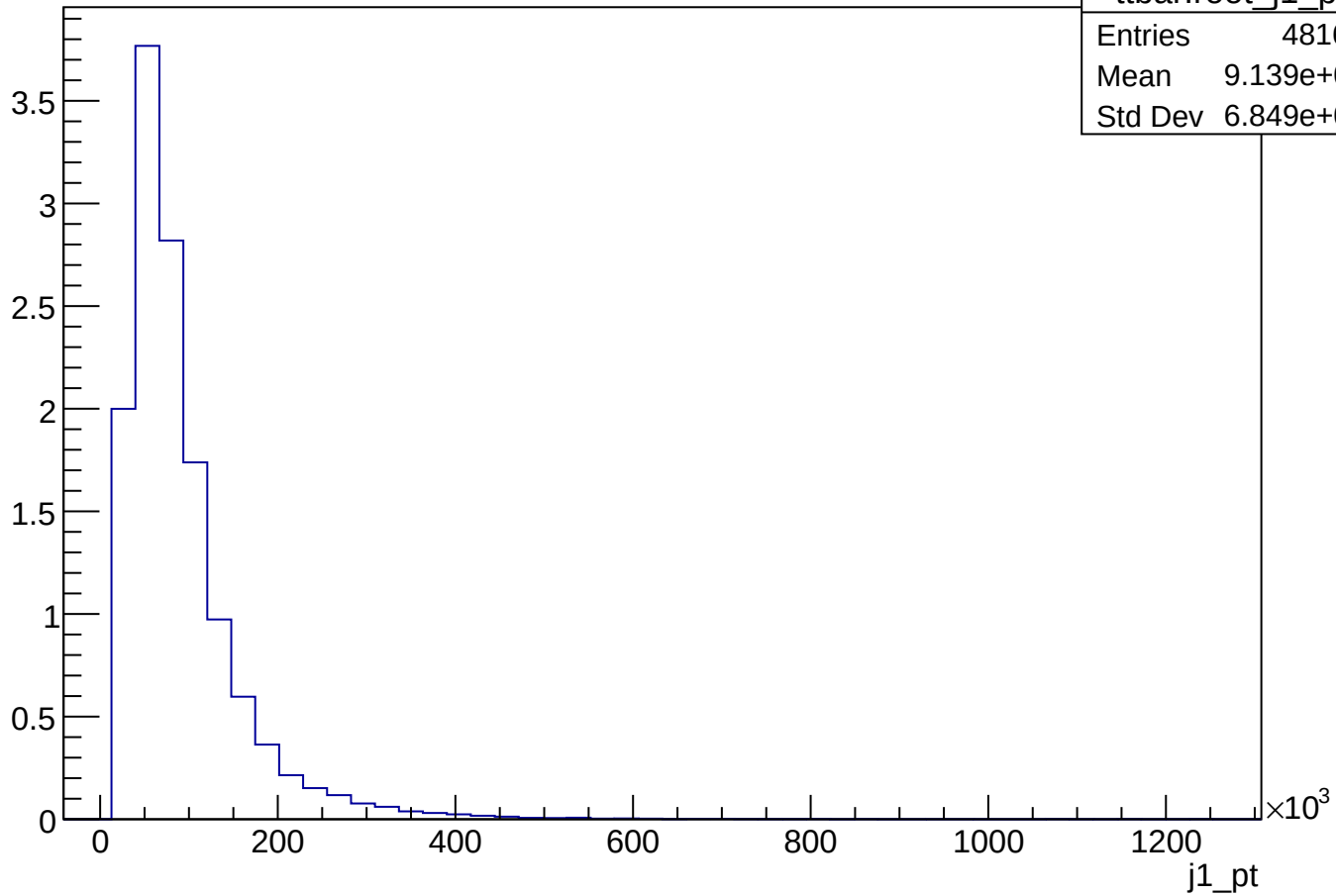
ttbar.root_j1_phi	
Entries	48167
Mean	0.01523
Std Dev	1.815

Events



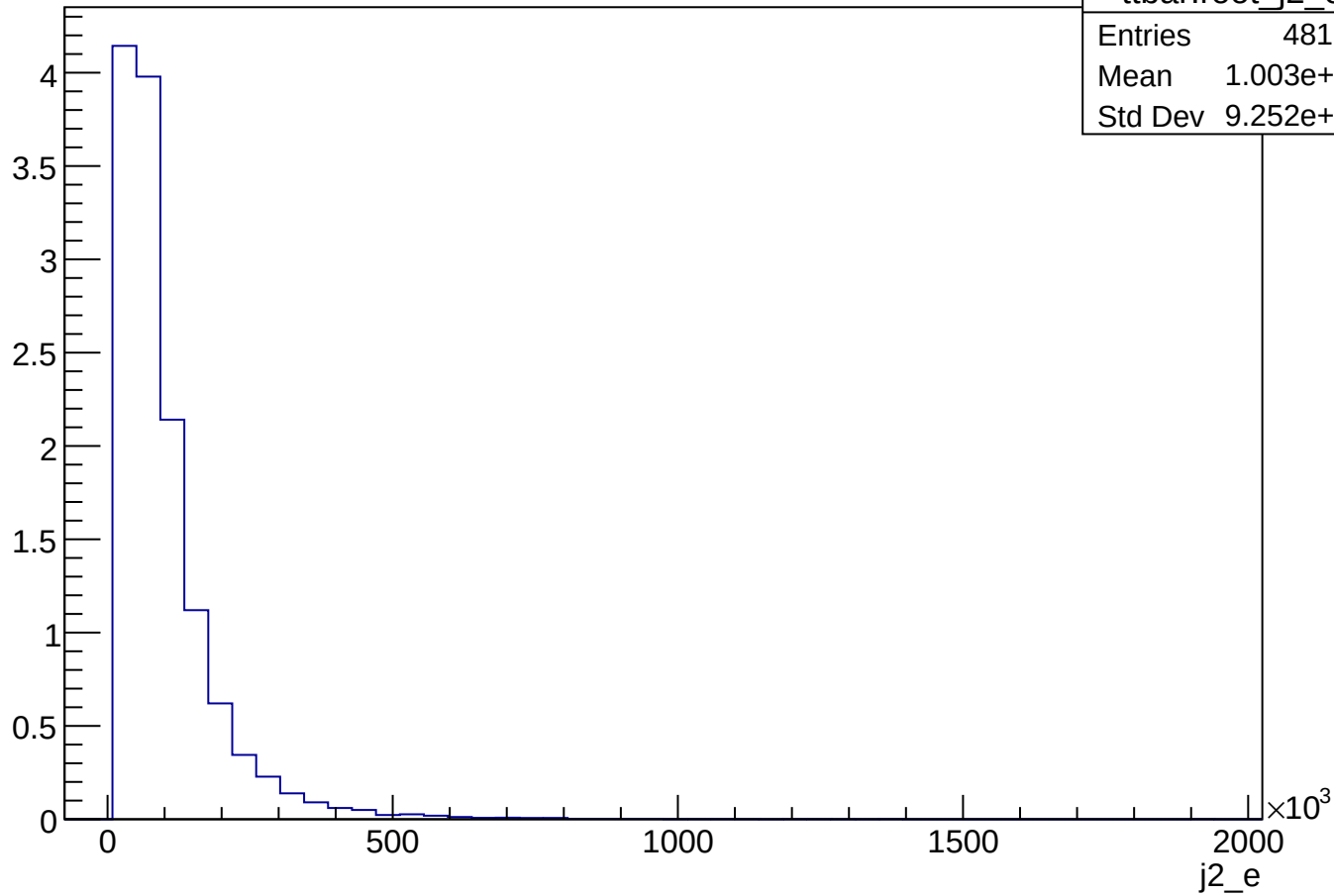
j1_pt

Events



j2_e

Events



j2_eta

ttbar.root_j2_eta	
Entries	48167
Mean	-0.005411
Std Dev	1.347

Events

0.35
0.3
0.25
0.2
0.15
0.1
0.05
0

-2

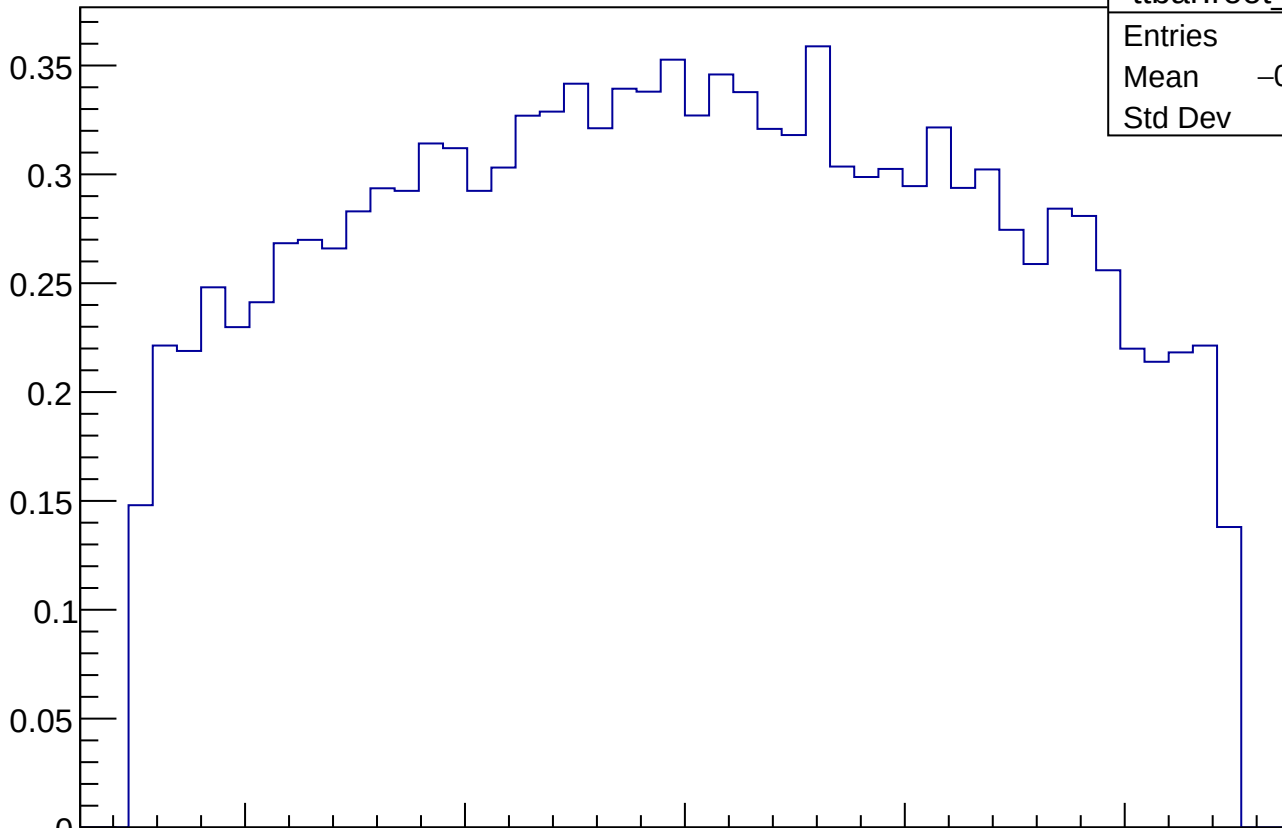
-1

0

1

2

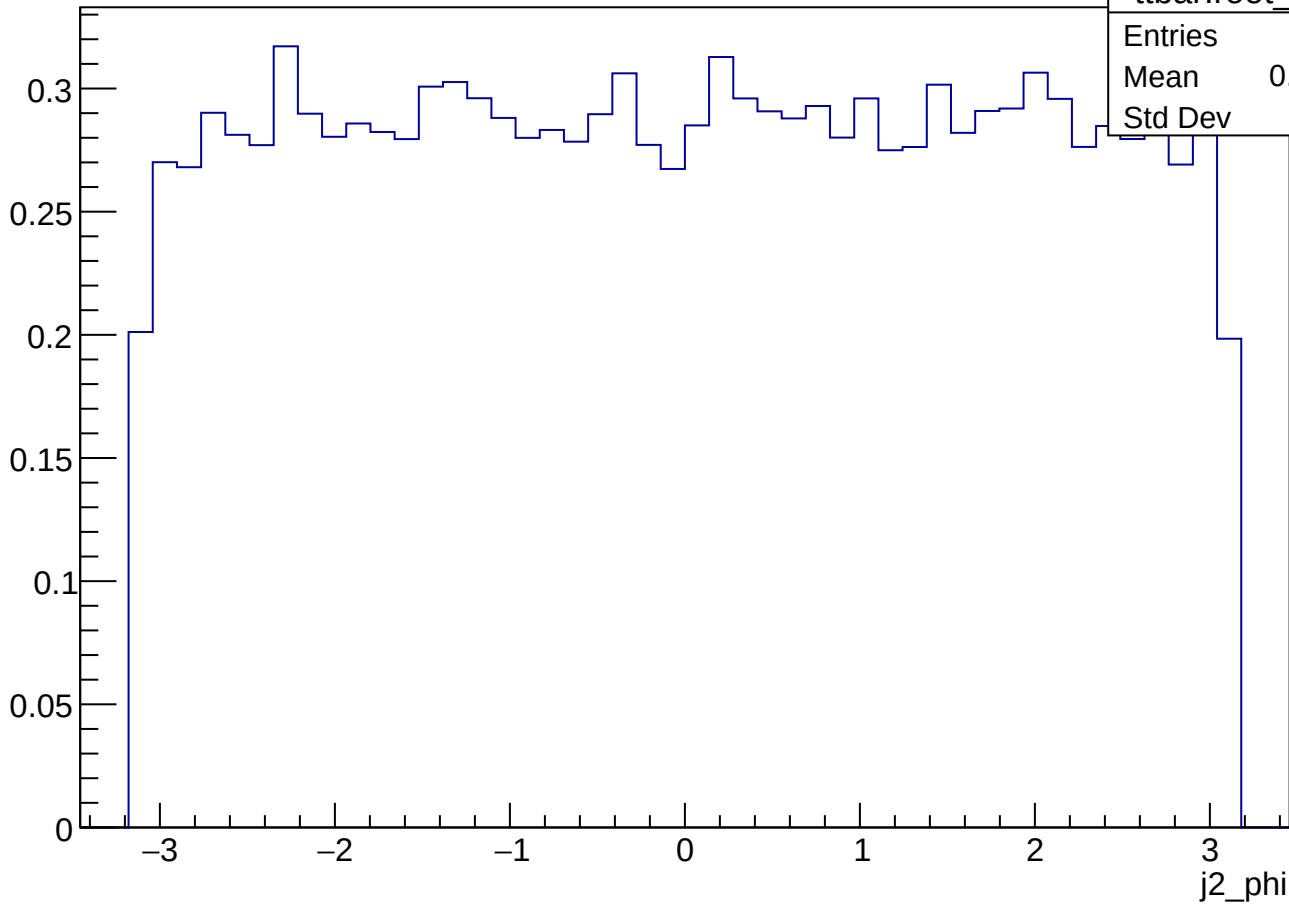
j2_eta



j2_phi

ttbar.root_j2_phi	
Entries	48167
Mean	0.002847
Std Dev	1.804

Events



j2_pt

Events

ttbar.root_j2_pt

Entries 48167

Mean 4.652e+04

Std Dev 3.083e+04

5

4

3

2

1

0

0

200

300

400

500

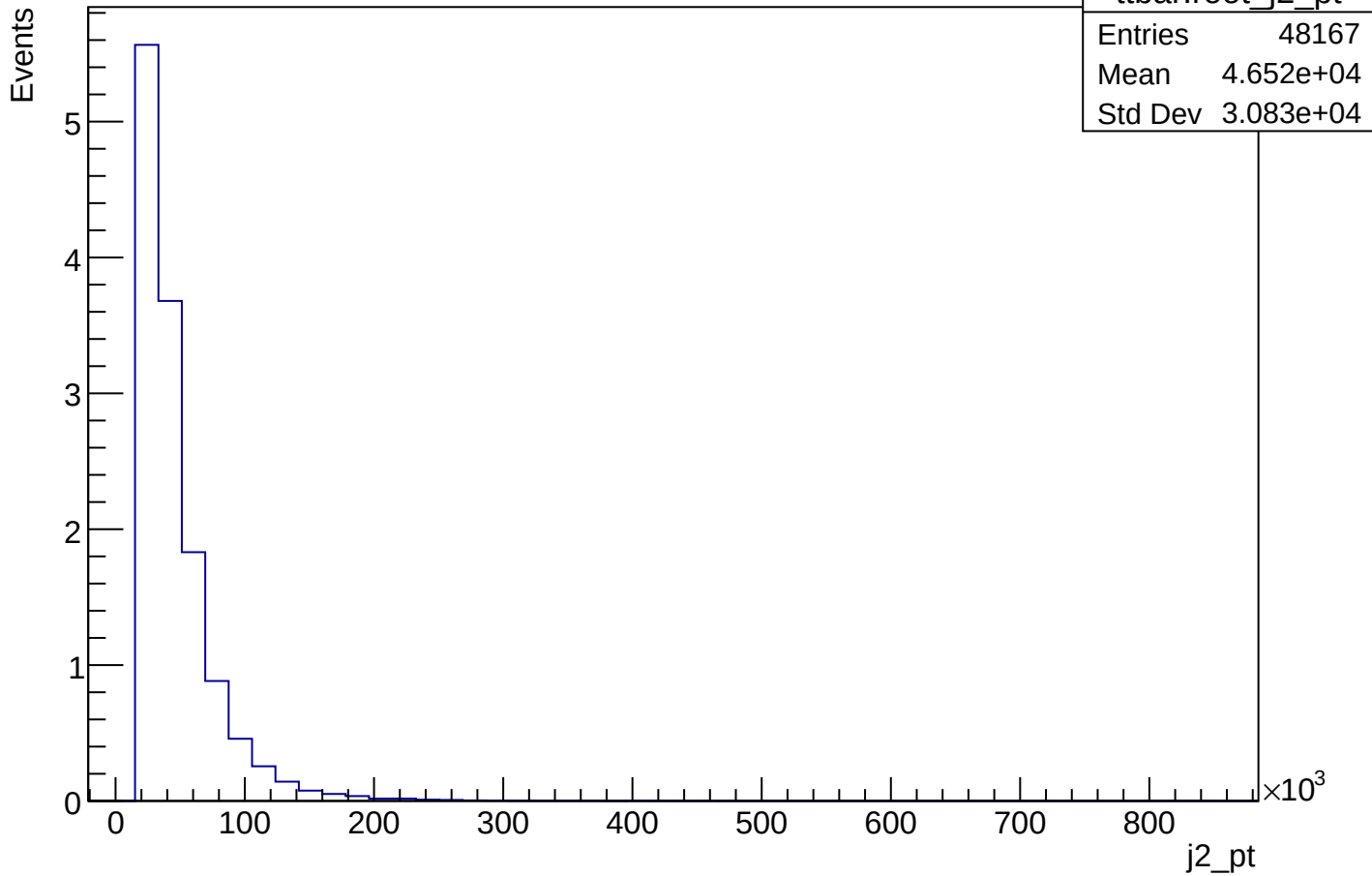
600

700

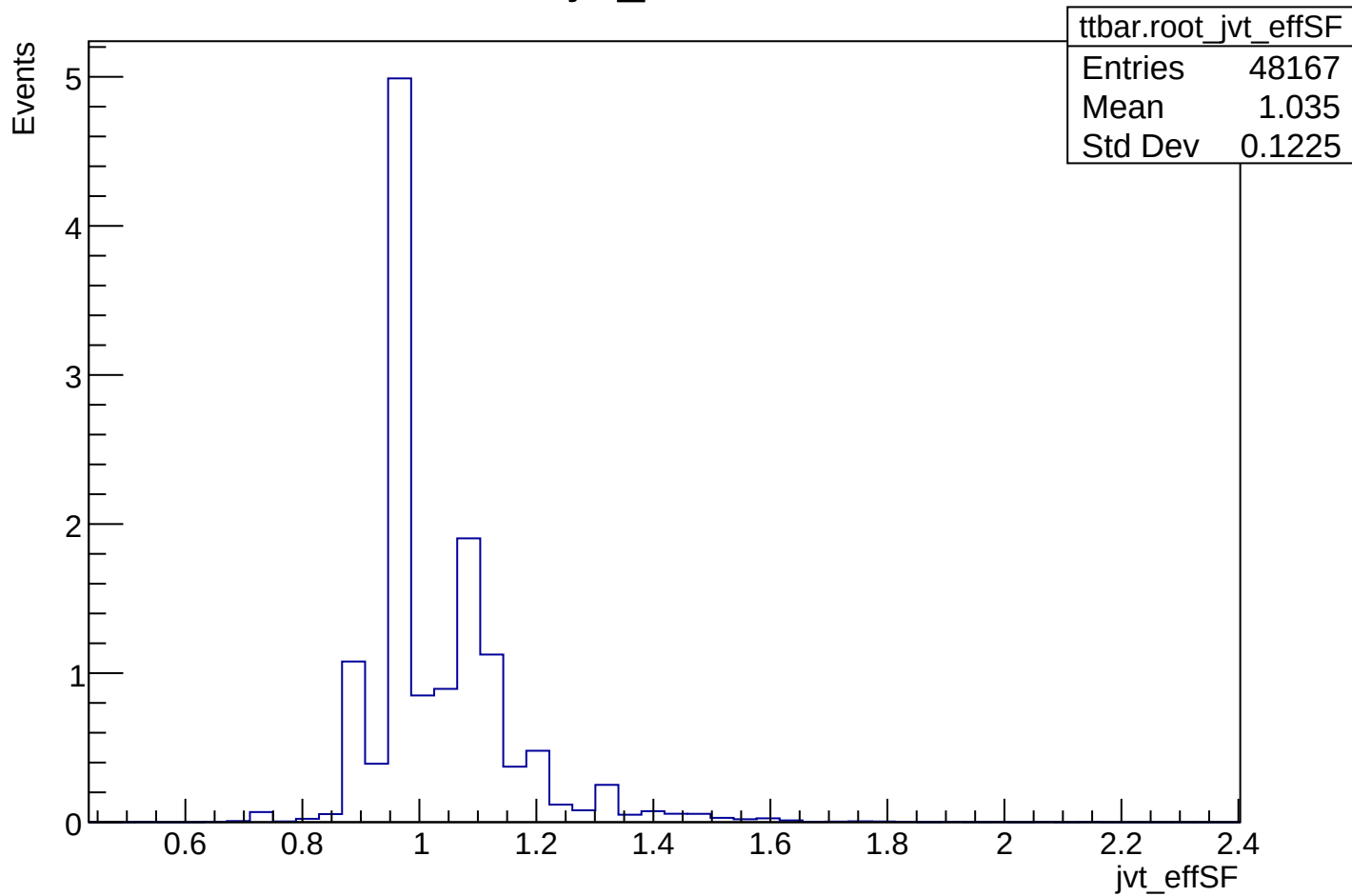
800

j2_pt

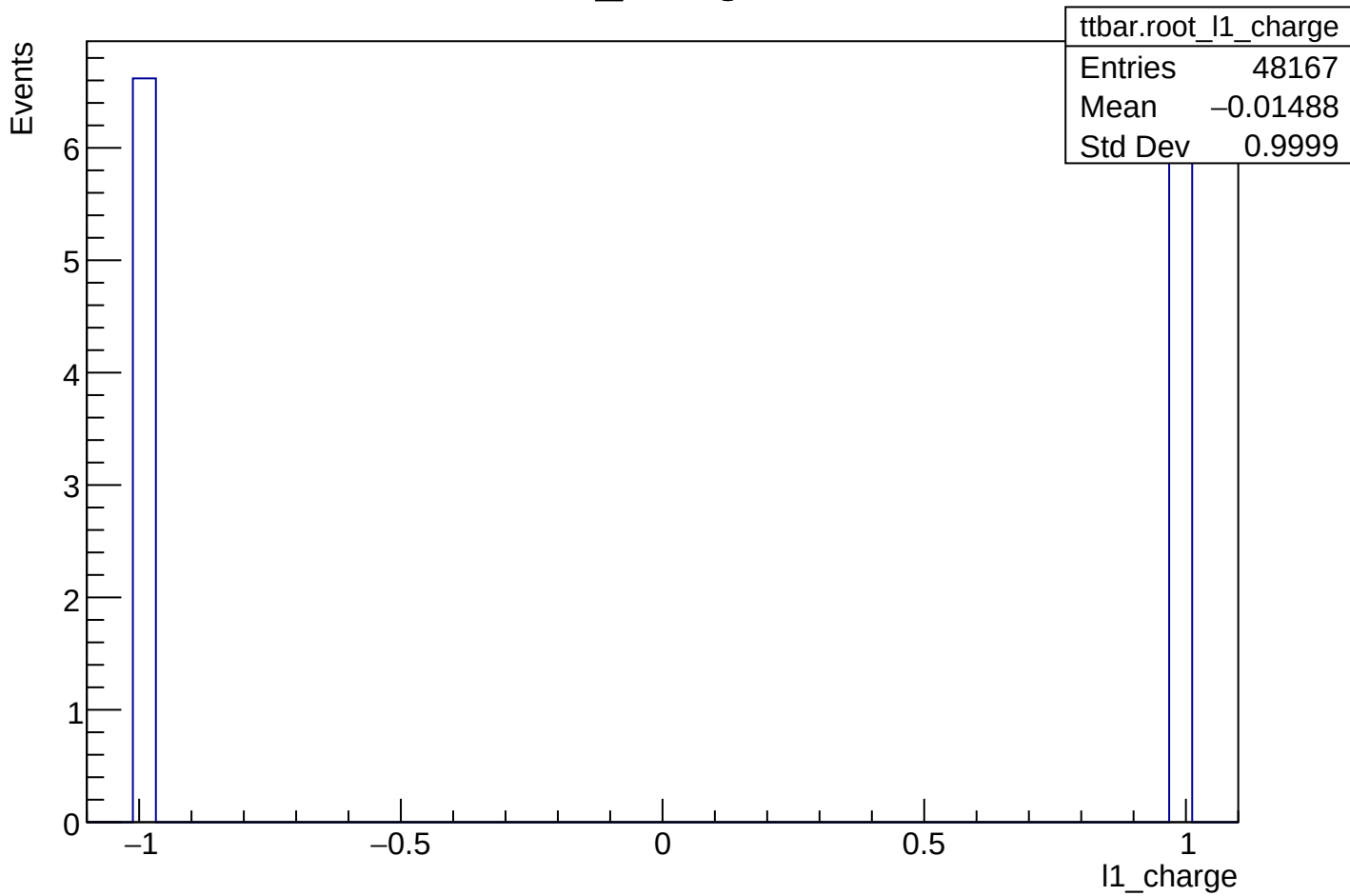
$\times 10^3$



jvt_effSF

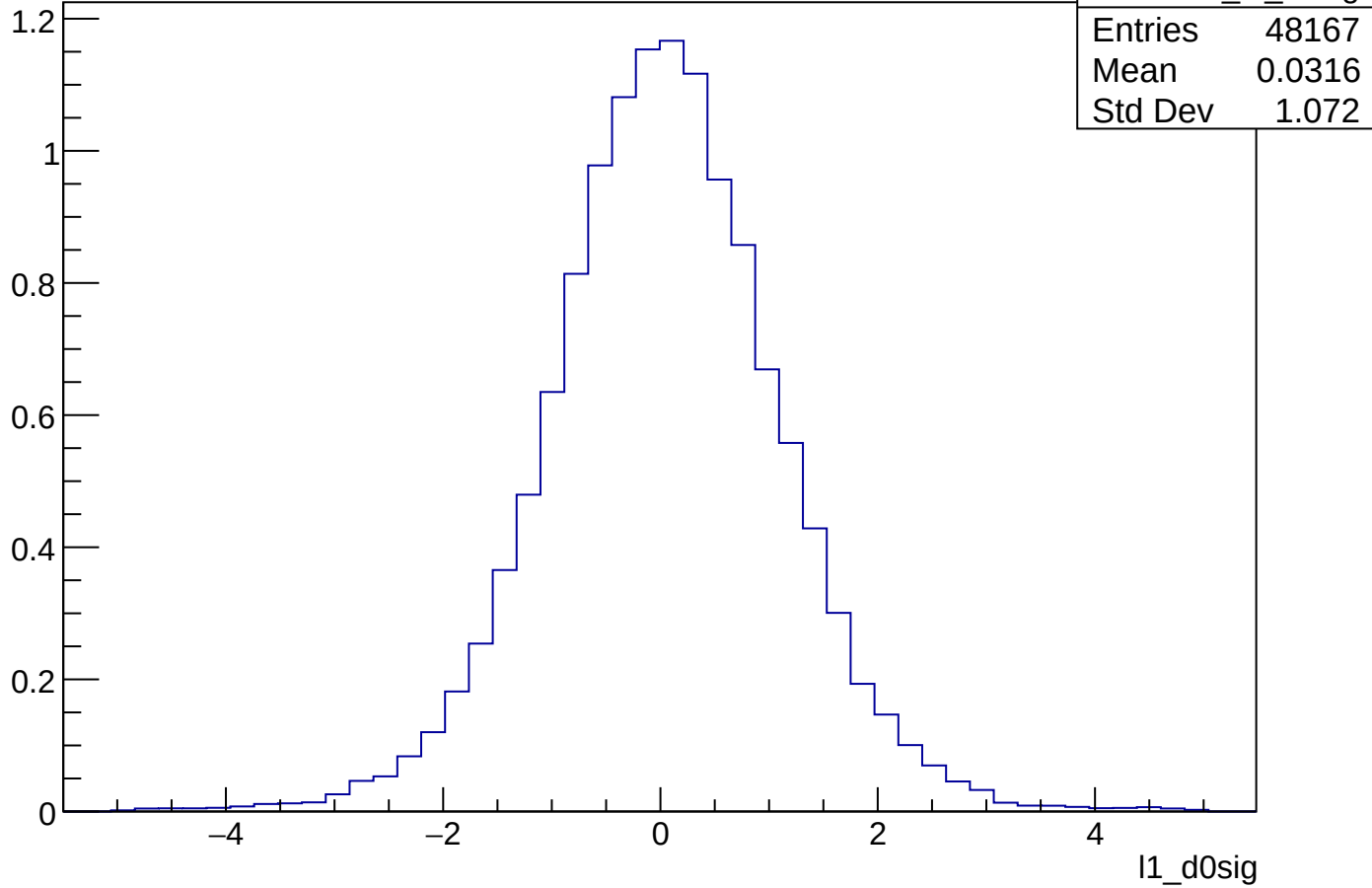


l1_charge



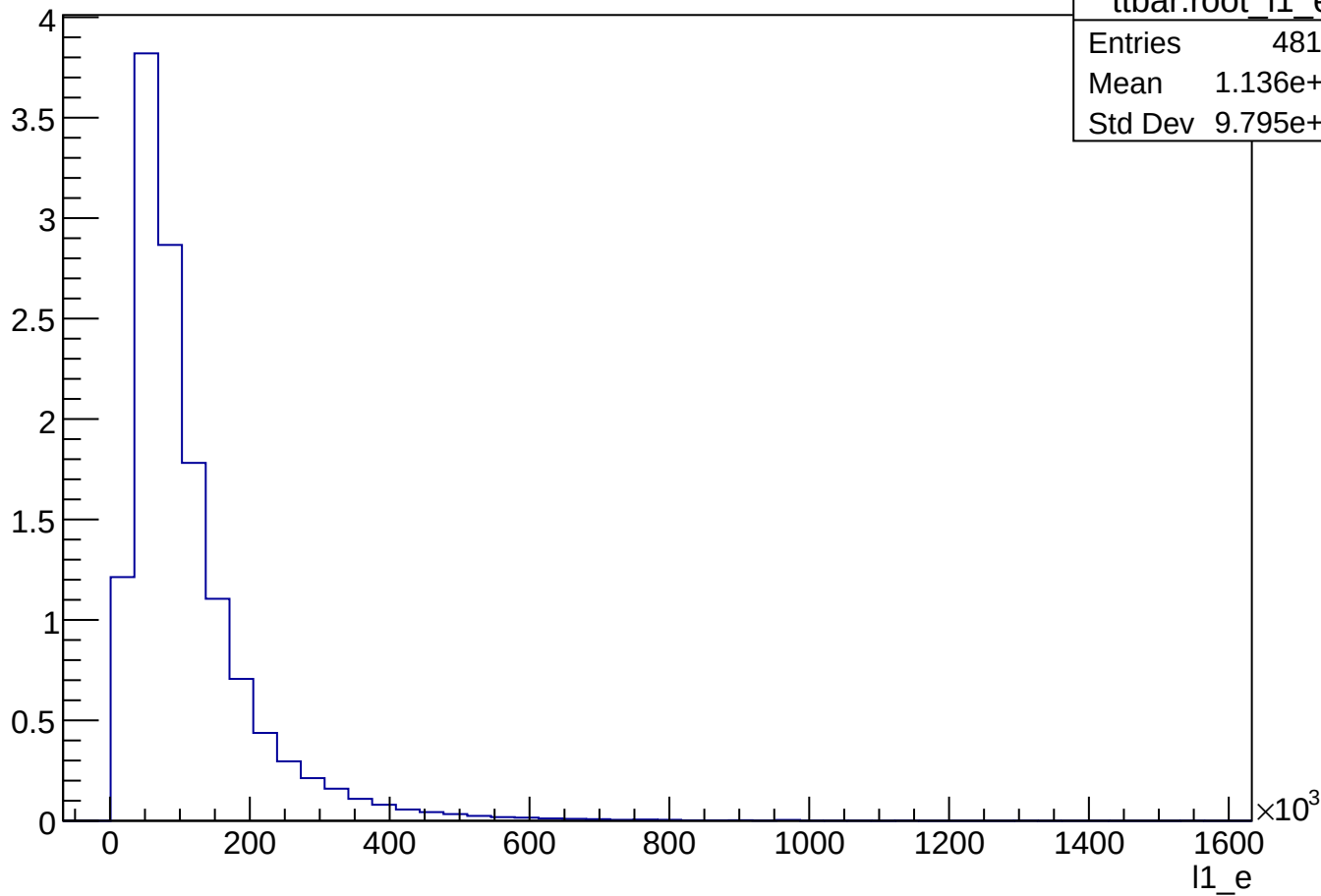
l1_d0sig

Events



l1_e

Events



l1_effSF

Events

3.5
3
2.5
2
1.5
1
0.5
0

0.92 0.94 0.96 0.98 1 1.02 1.04 1.06 1.08 1.1 1.12

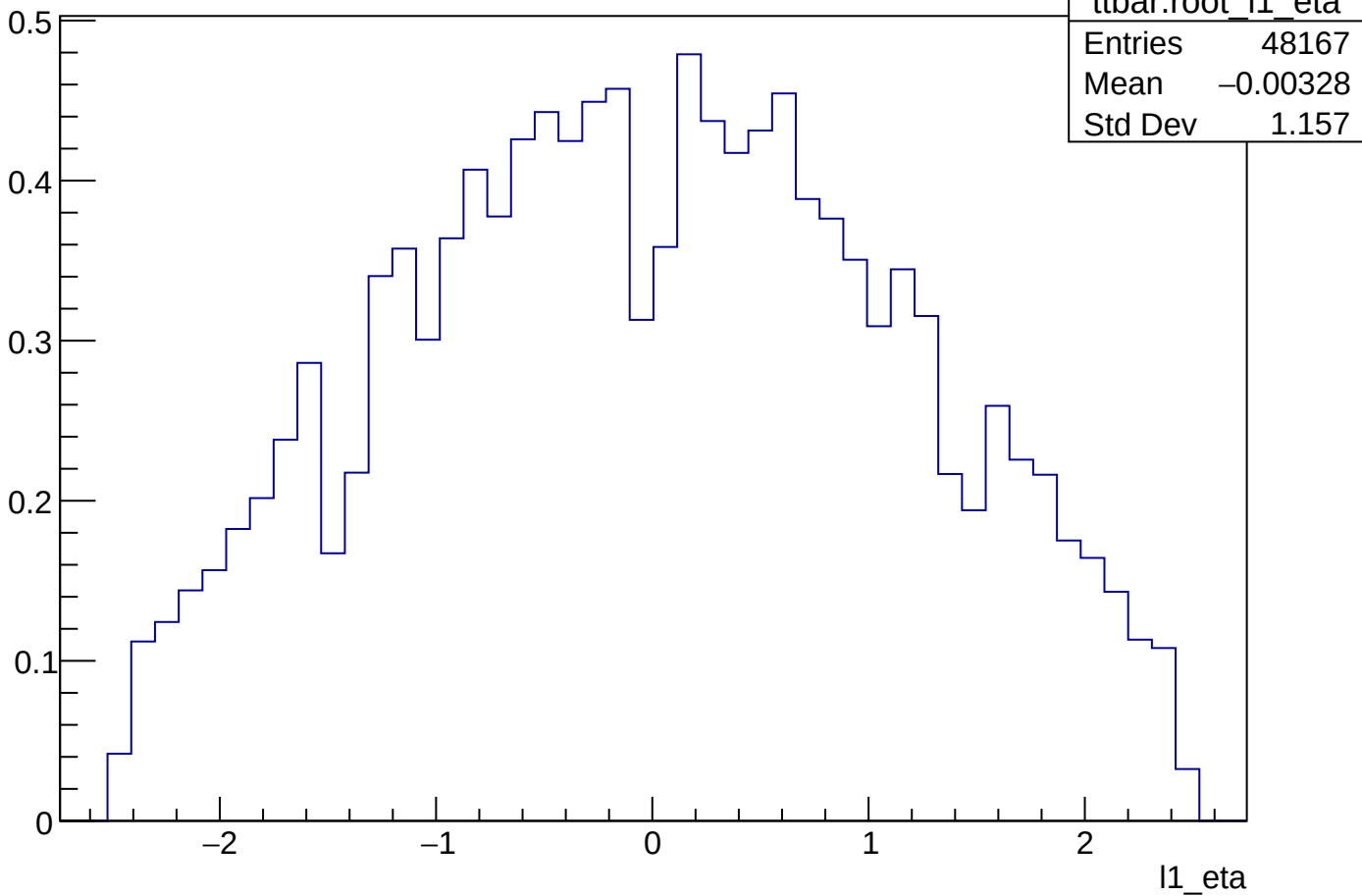
l1_effSF

ttbar.root	l1_effSF
Entries	48167
Mean	0.985
Std Dev	0.01119



$l1_eta$

Events



l1_pdg

Events

3.5
3
2.5
2
1.5
1
0.5
0

-10

-5

0

5

10

l1_pdg

ttbar.root_l1_pdg	
Entries	48167
Mean	0.1917
Std Dev	12.11

3

3

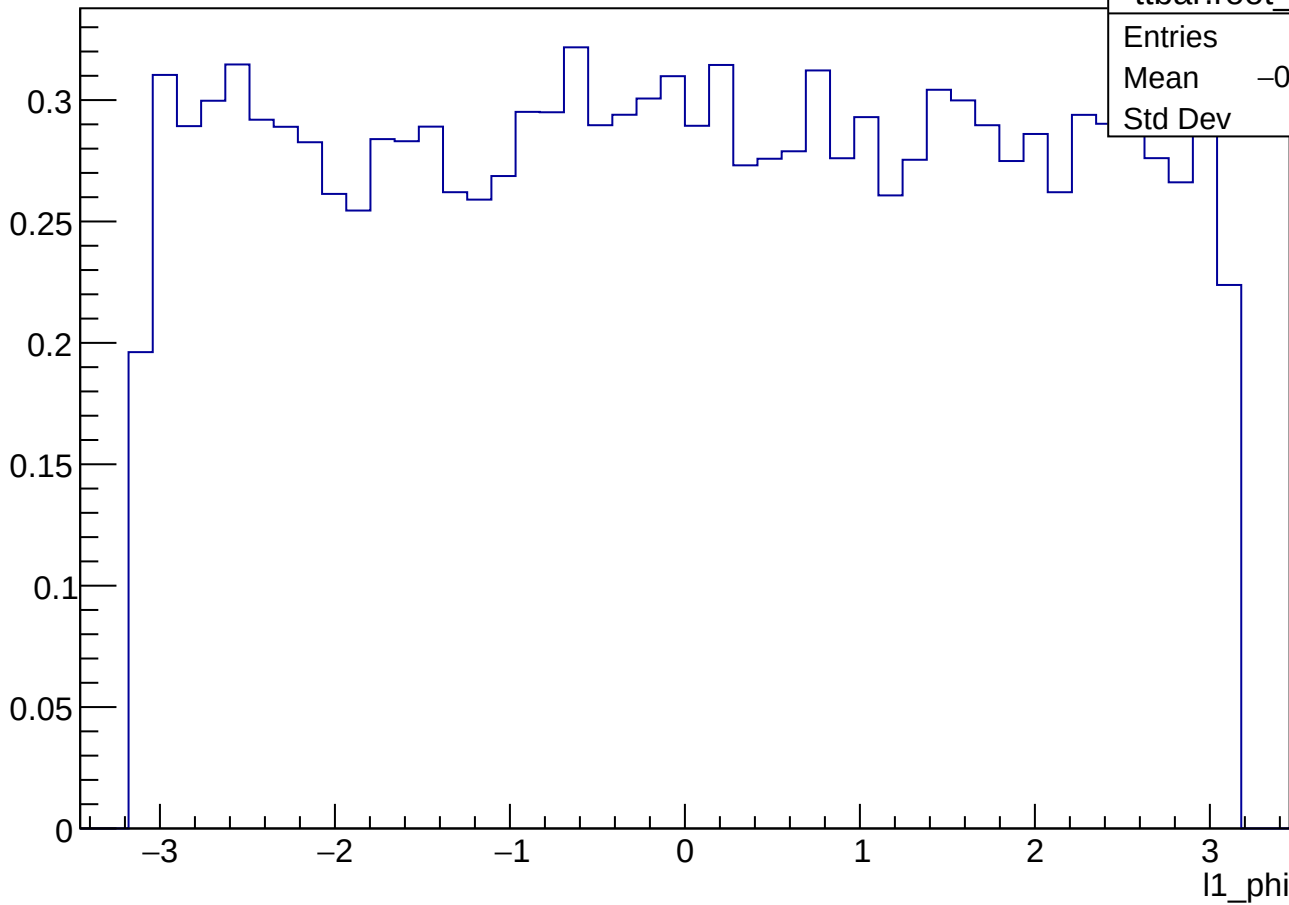
3

3

$l1_phi$

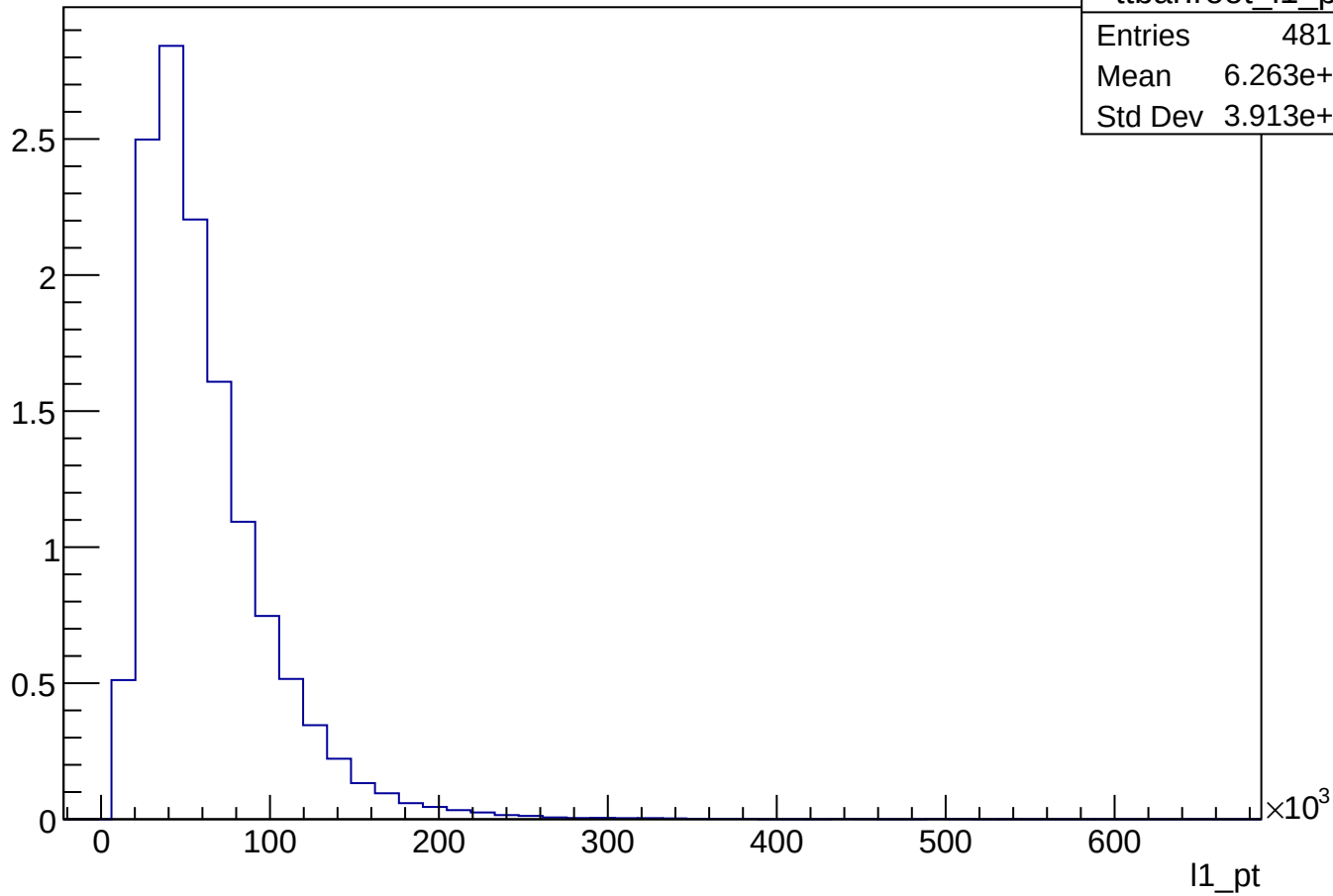
ttbar.root_l1_phi	
Entries	48167
Mean	-0.003331
Std Dev	1.814

Events



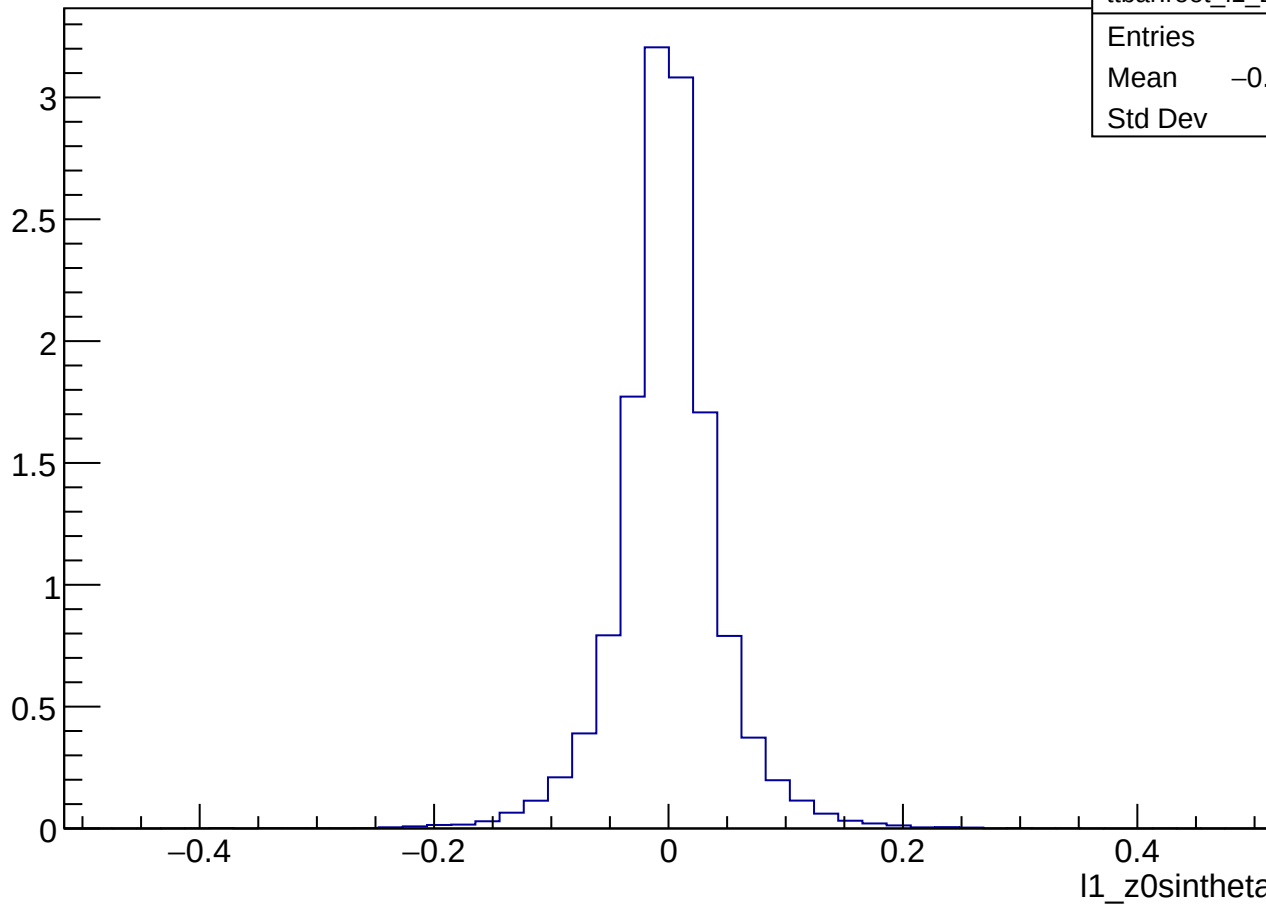
l1_pt

Events



l1_z0sintheta

Events



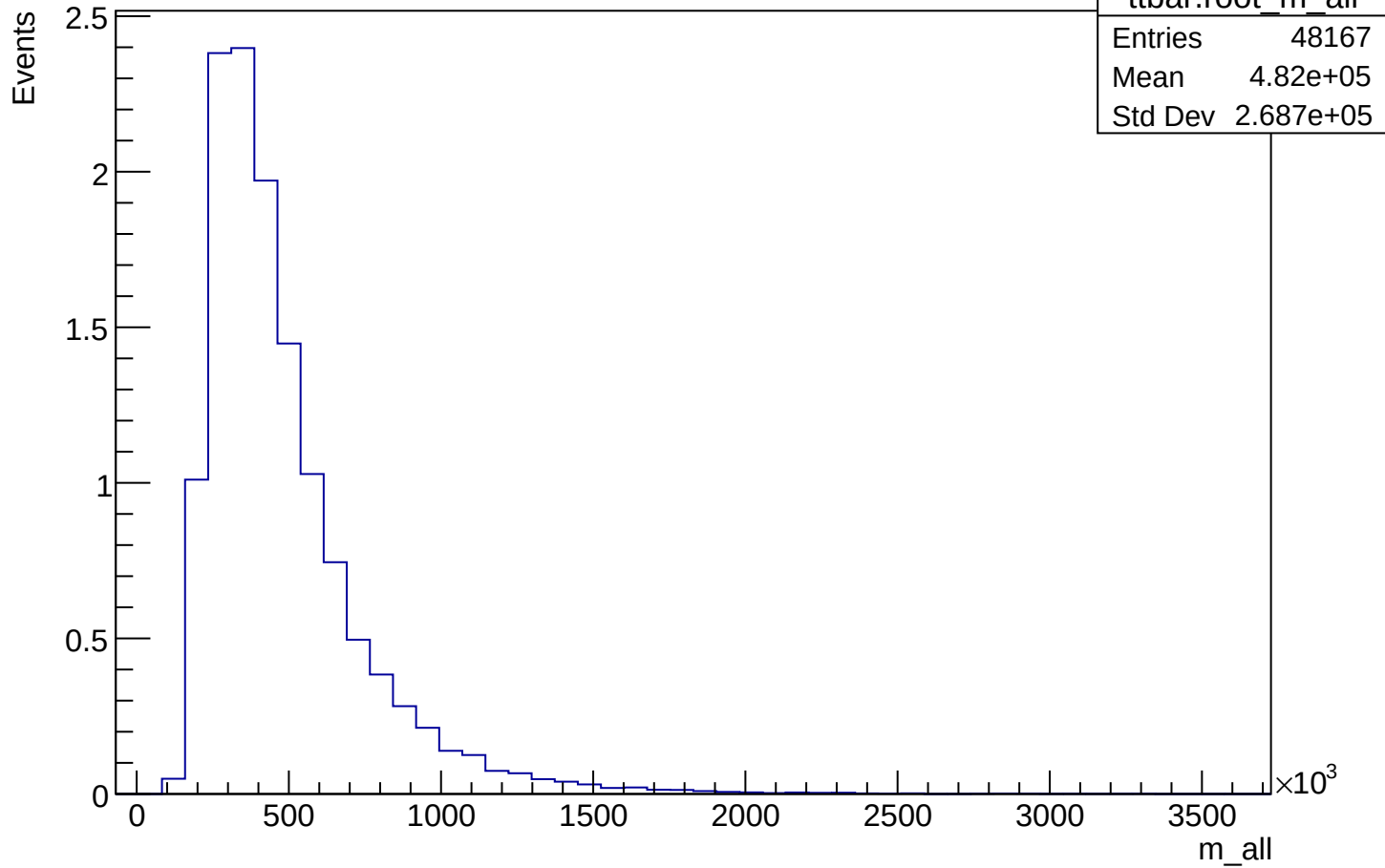
ttbar.root_l1_z0sintheta

Entries 48167

Mean -0.0002033

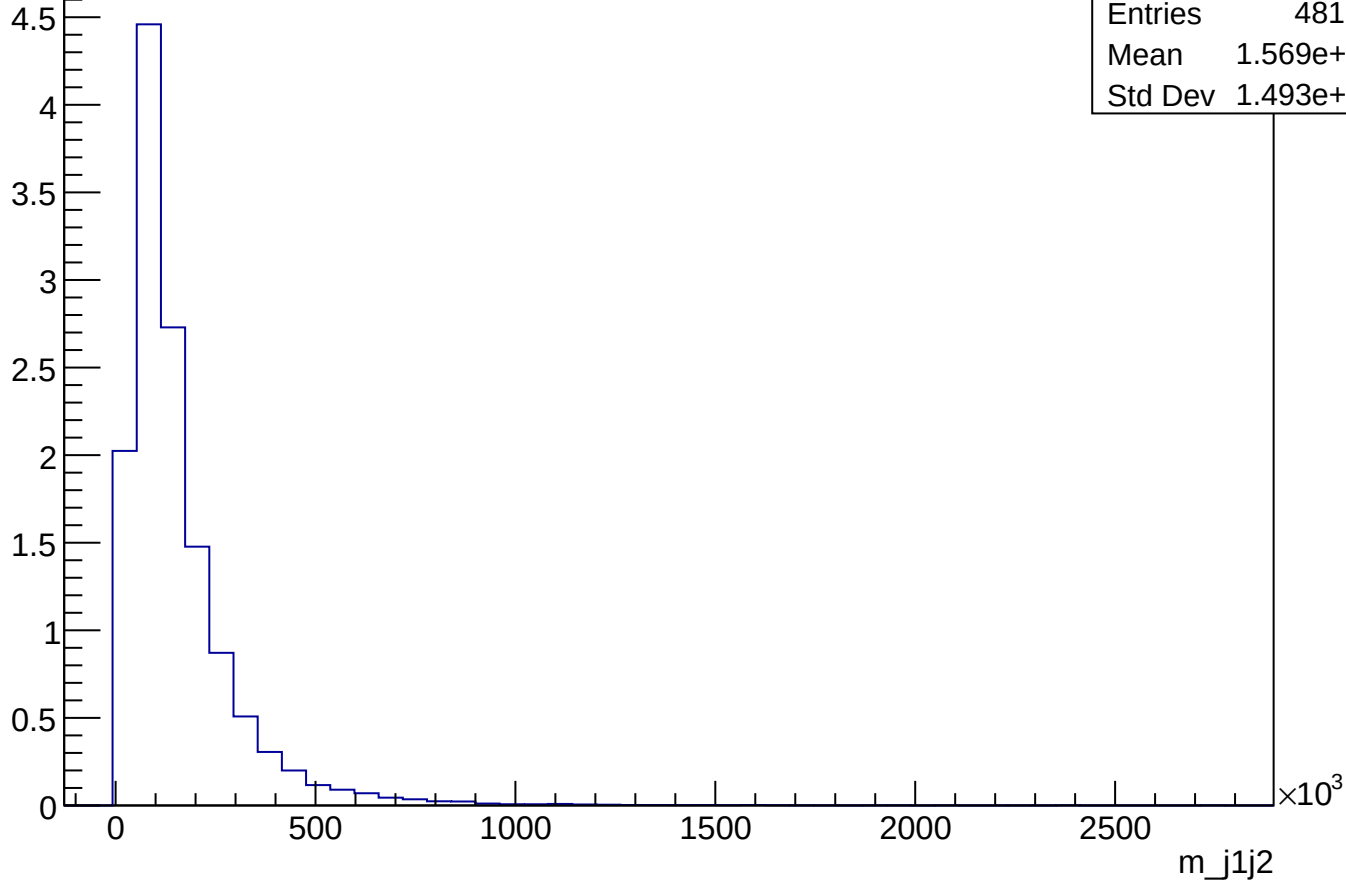
Std Dev 0.04576

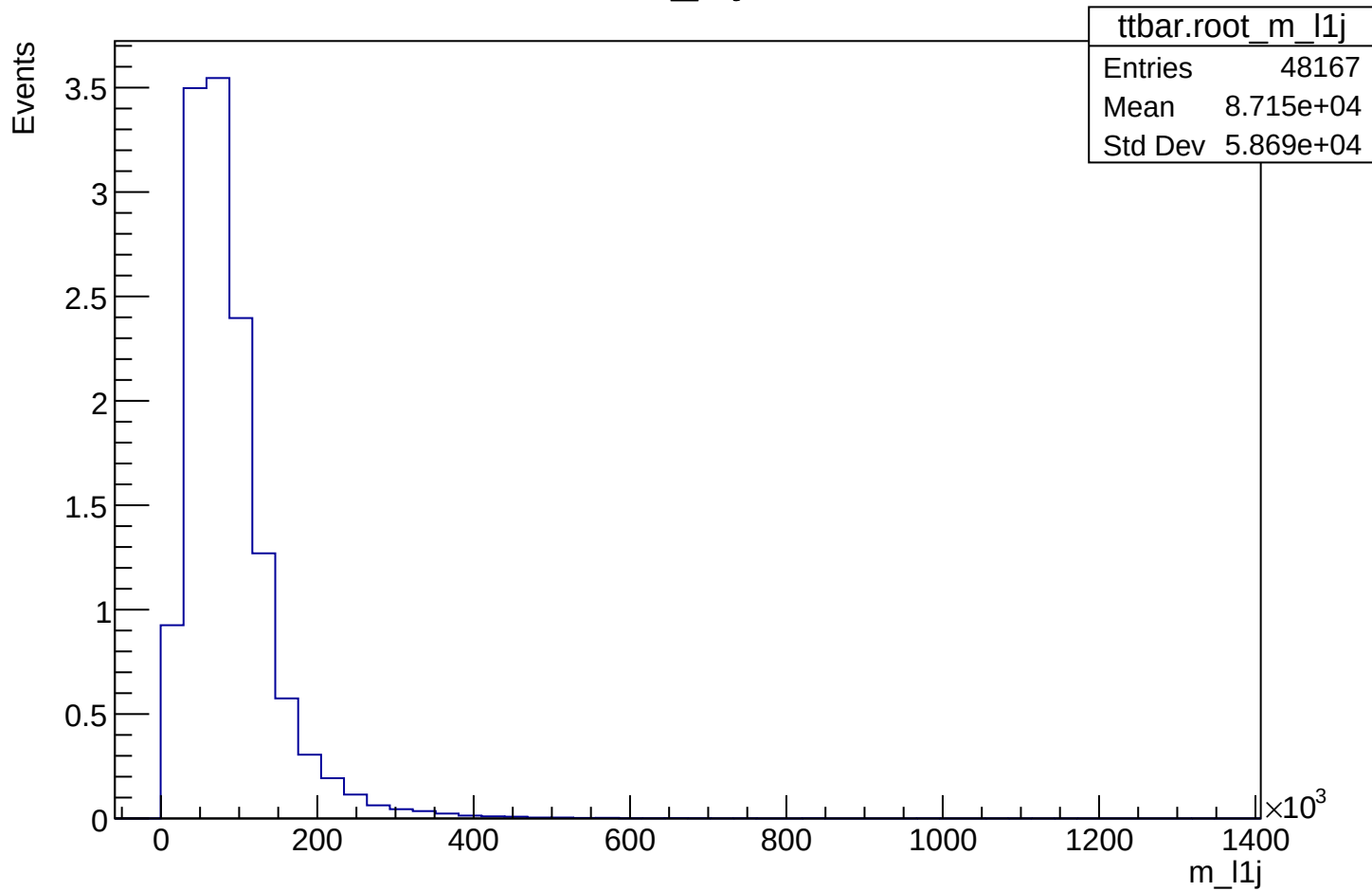
m_all



m_j1j2

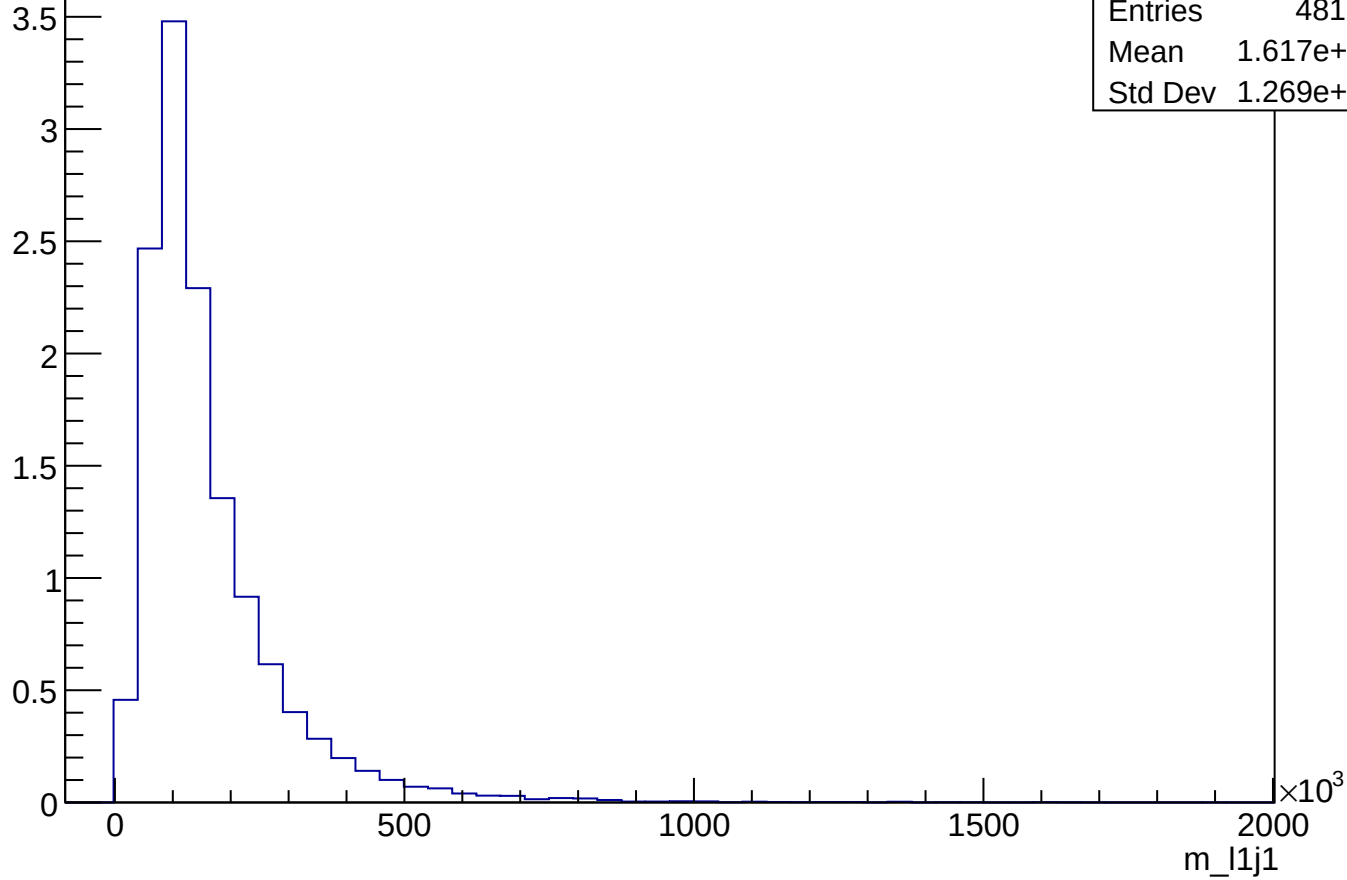
Events



$$m_{l1j}$$


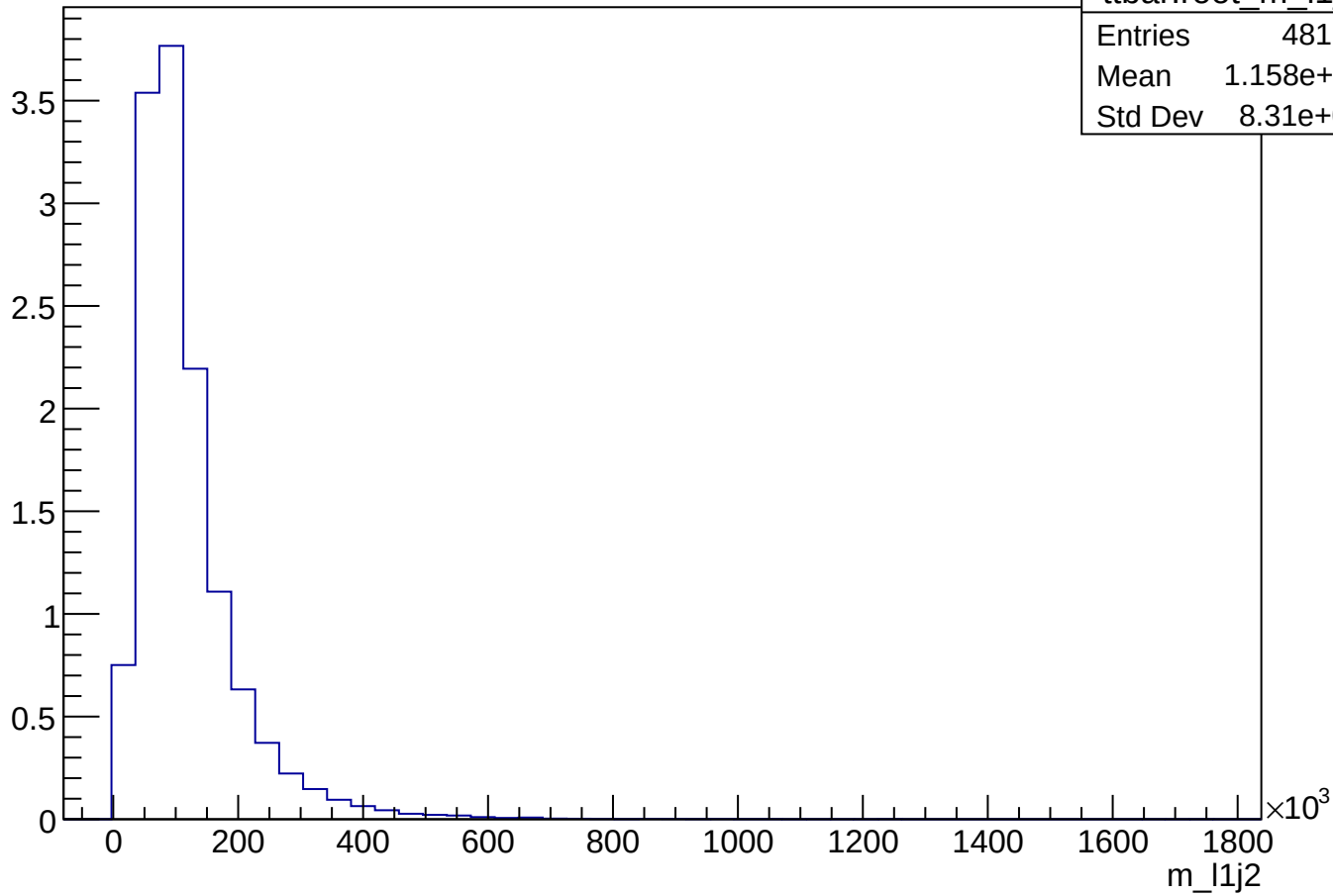
m_l1j1

Events



m_l1j2

Events



m_l1t

Events

2.5

2

1.5

1

0.5

0

0

100

200

300

400

500

600

700

800

m_l1t

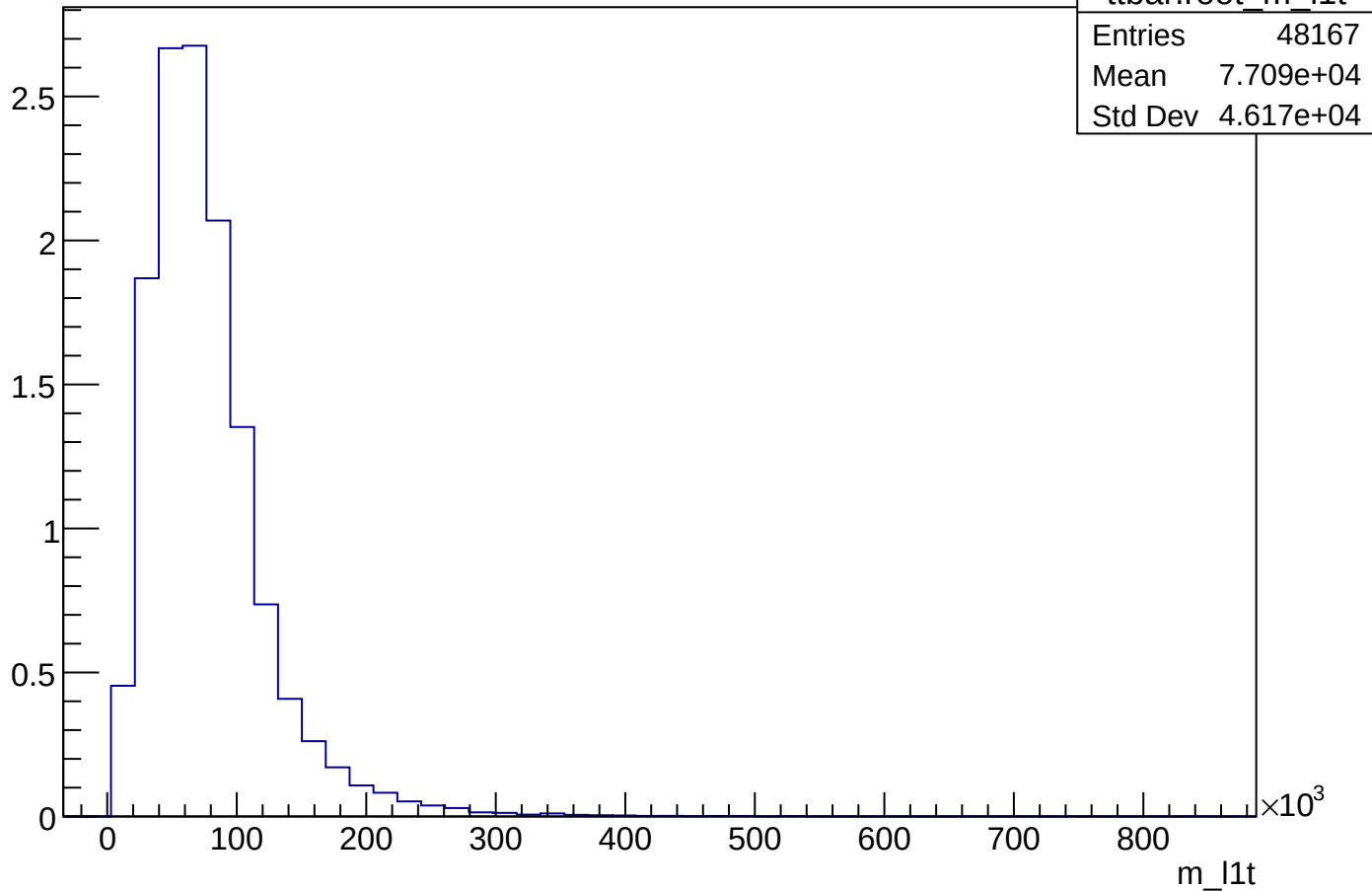
$\times 10^3$

ttbar.root_m_l1t

Entries 48167

Mean 7.709e+04

Std Dev 4.617e+04



m_l1t1

Events

2.5

2

1.5

1

0.5

0

0

200

400

600

800

1000

m_l1t1

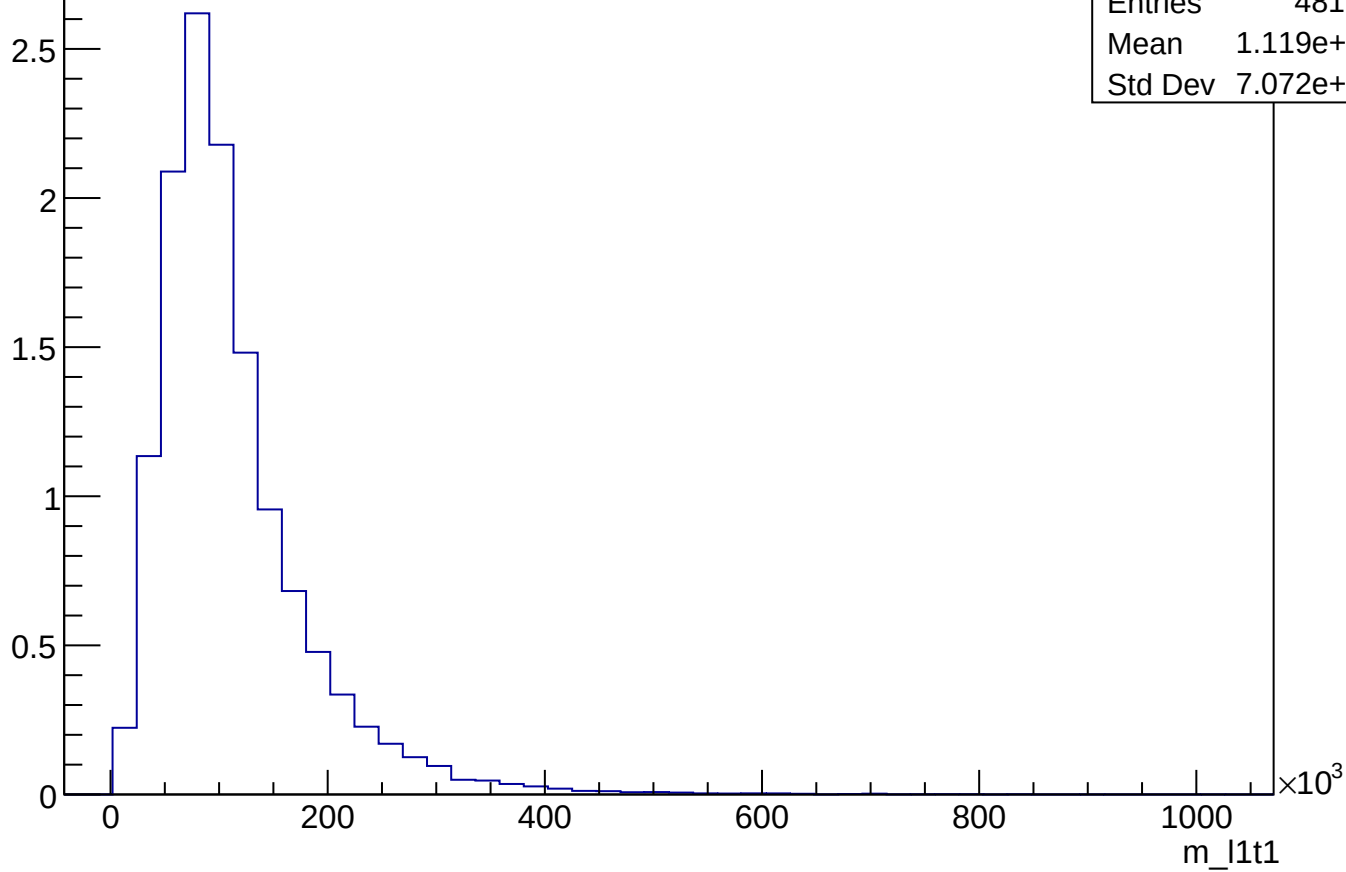
$\times 10^3$

ttbar.root m_l1t1

Entries 48167

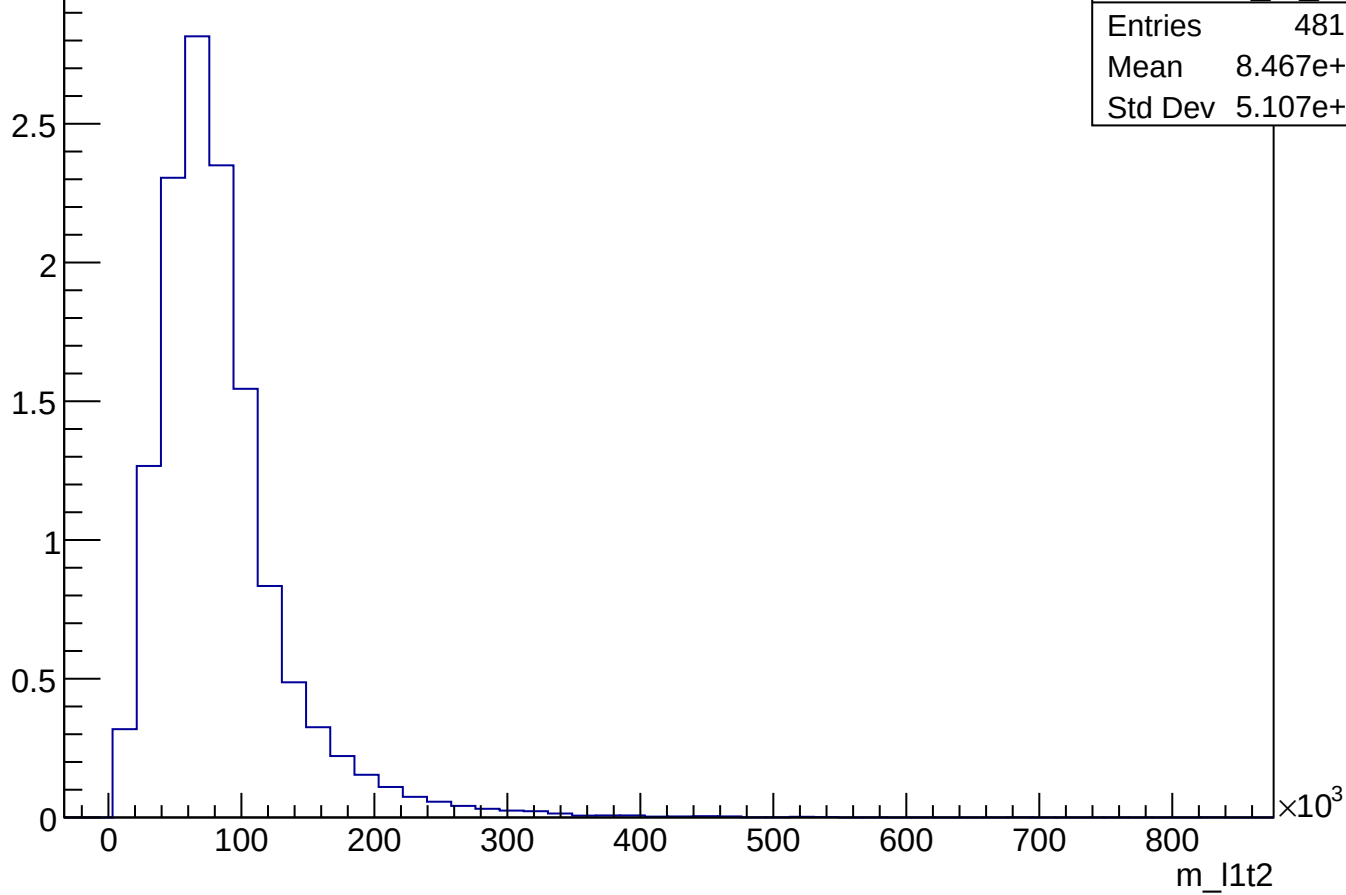
Mean 1.119e+05

Std Dev 7.072e+04



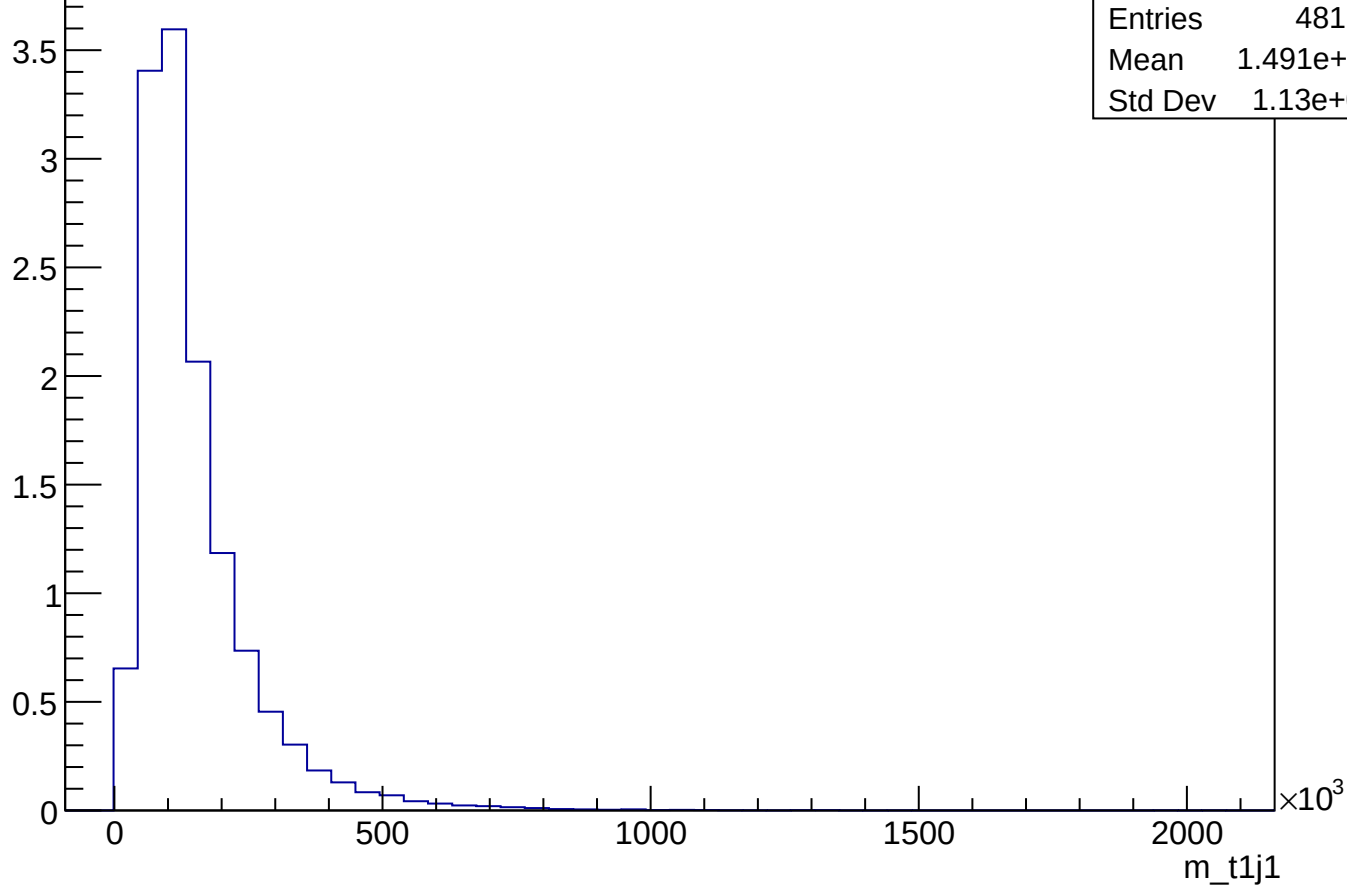
m_l1t2

Events



m_t1j1

Events



m_t1j2

Events

ttbar.root_m_t1j2

Entries 48167

Mean 1.069e+05

Std Dev 7.621e+04

6

5

4

3

2

1

0

0

500

1000

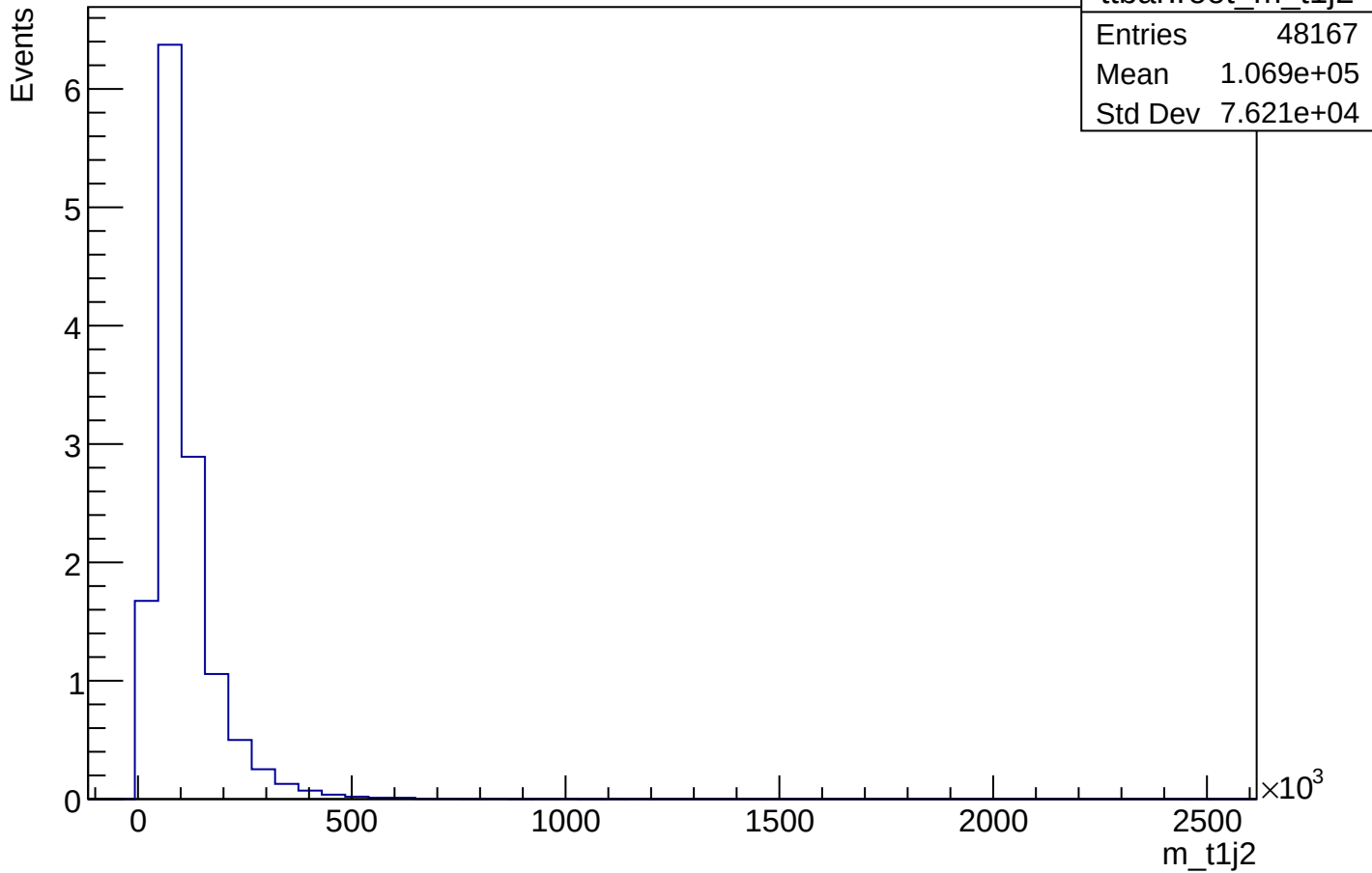
1500

2000

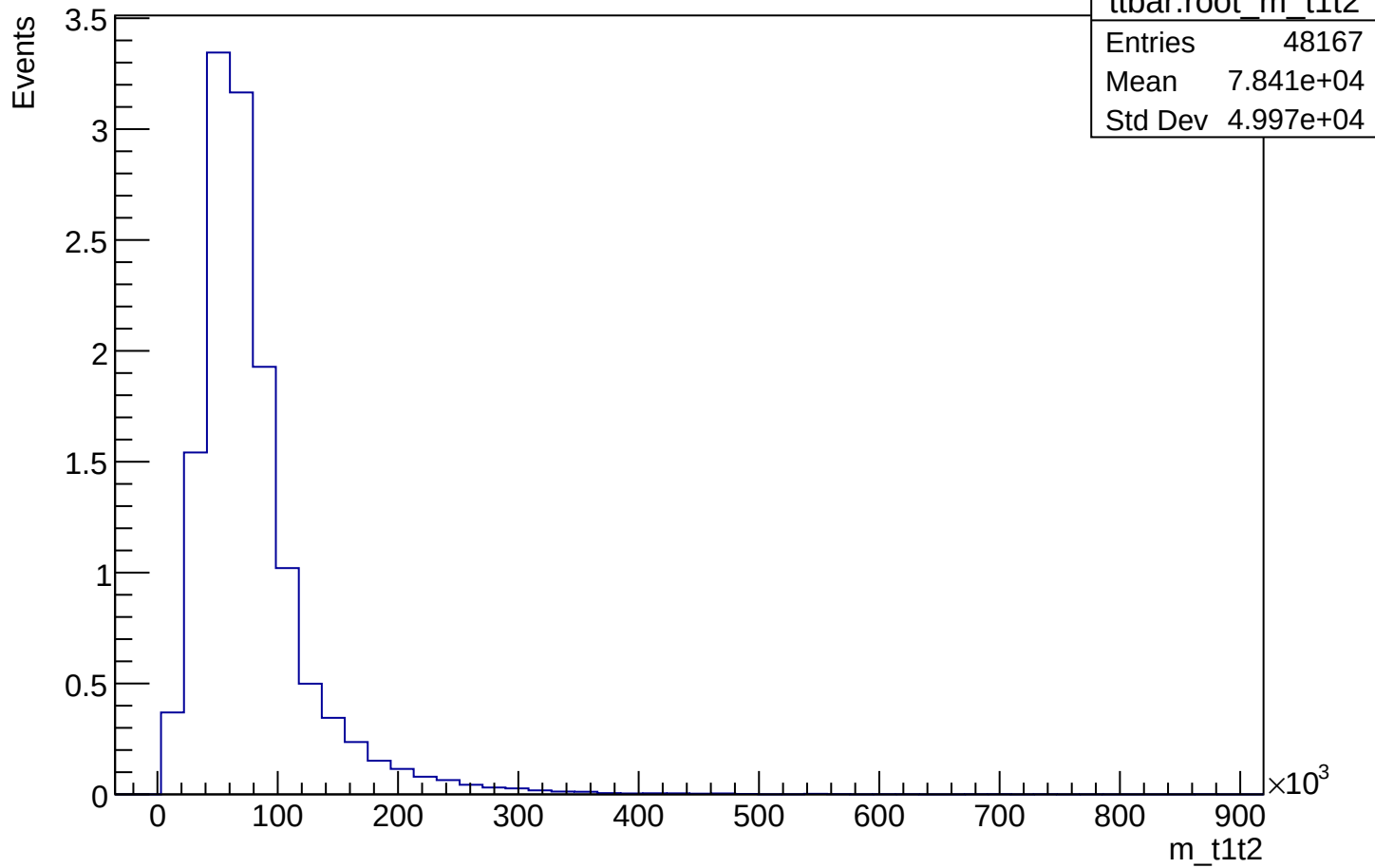
2500

m_t1j2

$\times 10^3$



m_t1t2



m_t2j1

Events

2.5

2

1.5

1

0.5

0

0

200

400

600

800

1000

m_t2j1

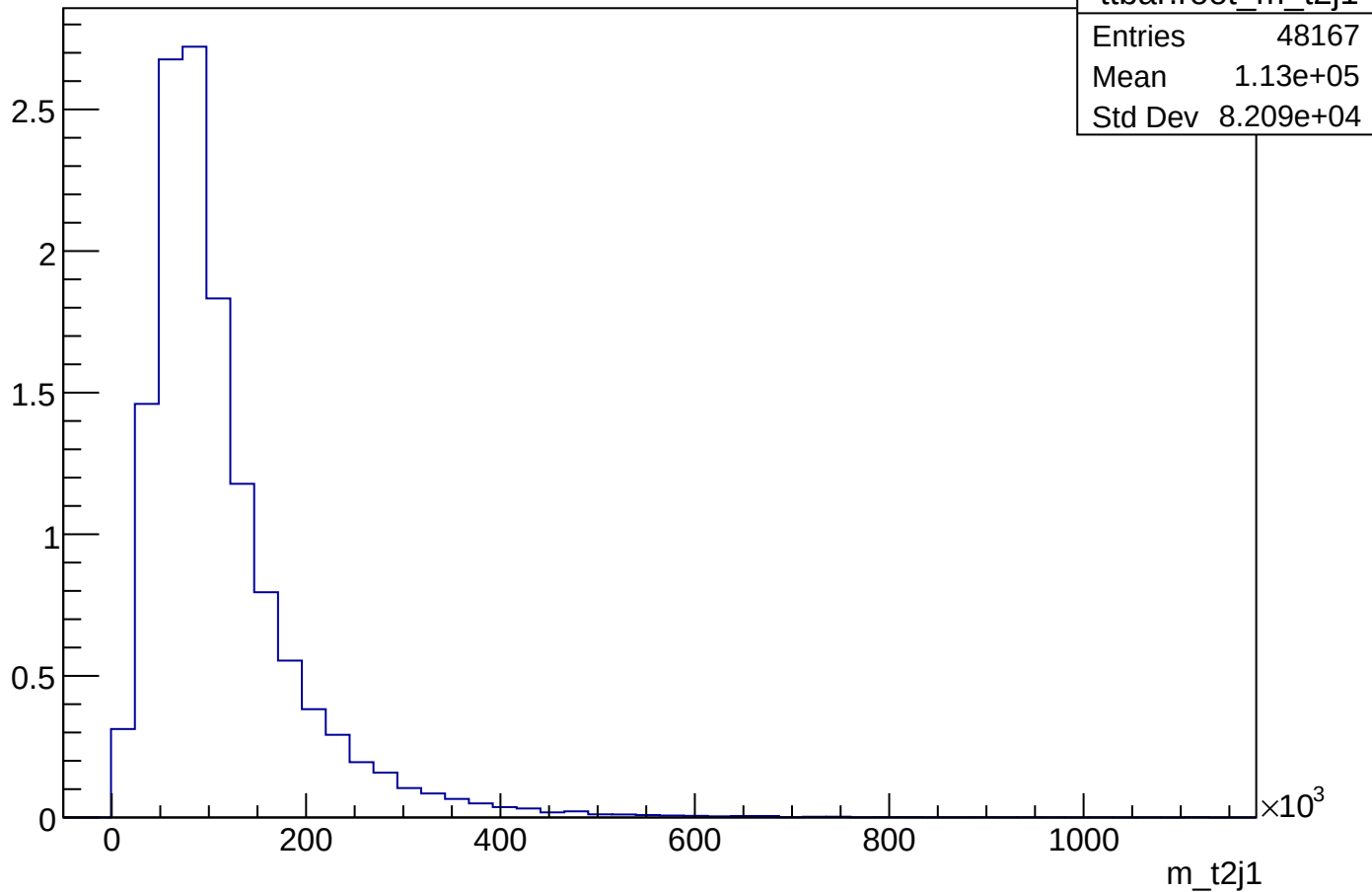
$\times 10^3$

ttbar.root_m_t2j1

Entries 48167

Mean $1.13\text{e}+05$

Std Dev $8.209\text{e}+04$



m_t2j2

Events

2.5
2
1.5
1
0.5
0

0

100

200

300

400

500

600

700

m_t2j2

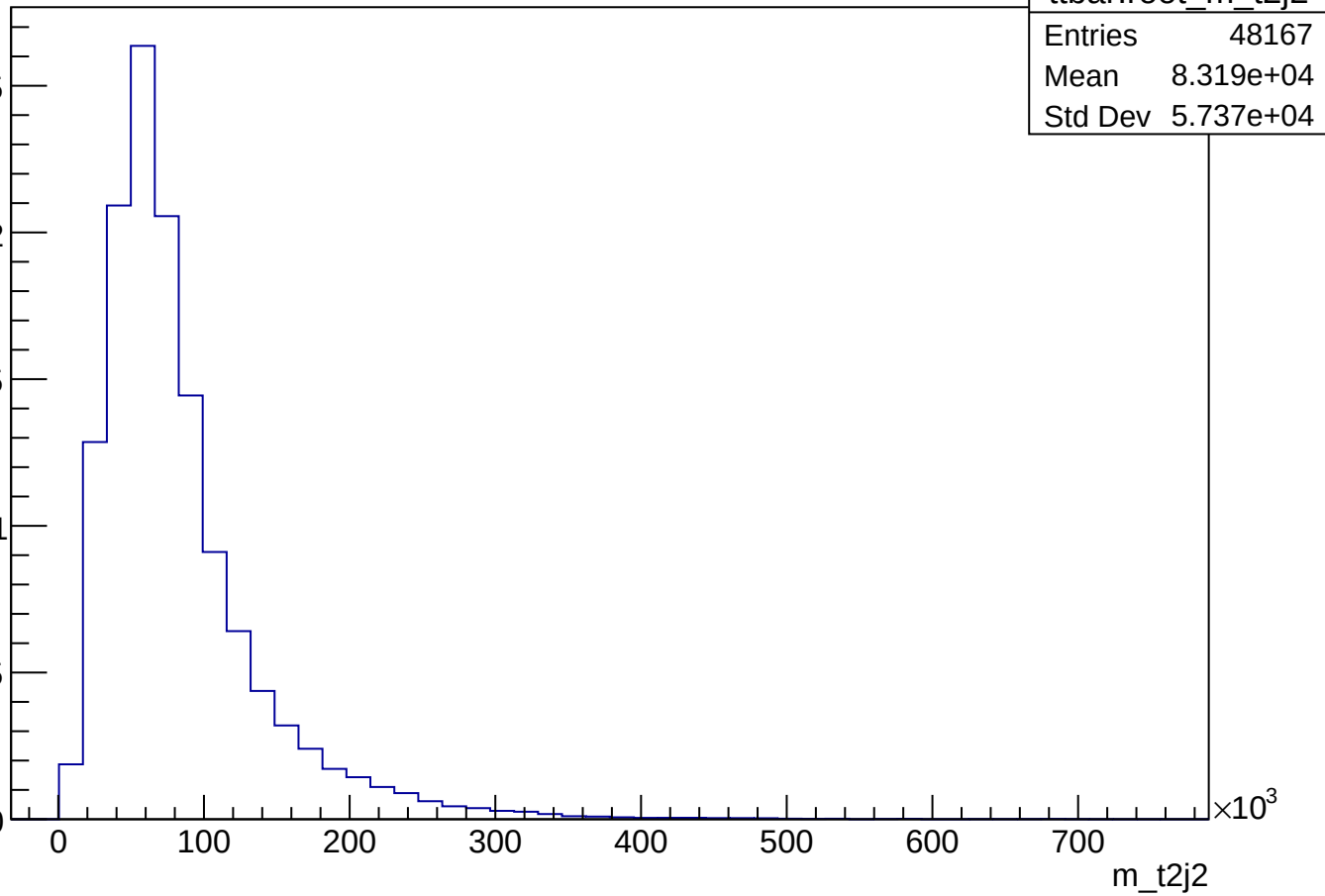
$\times 10^3$

ttbar.root_m_t2j2

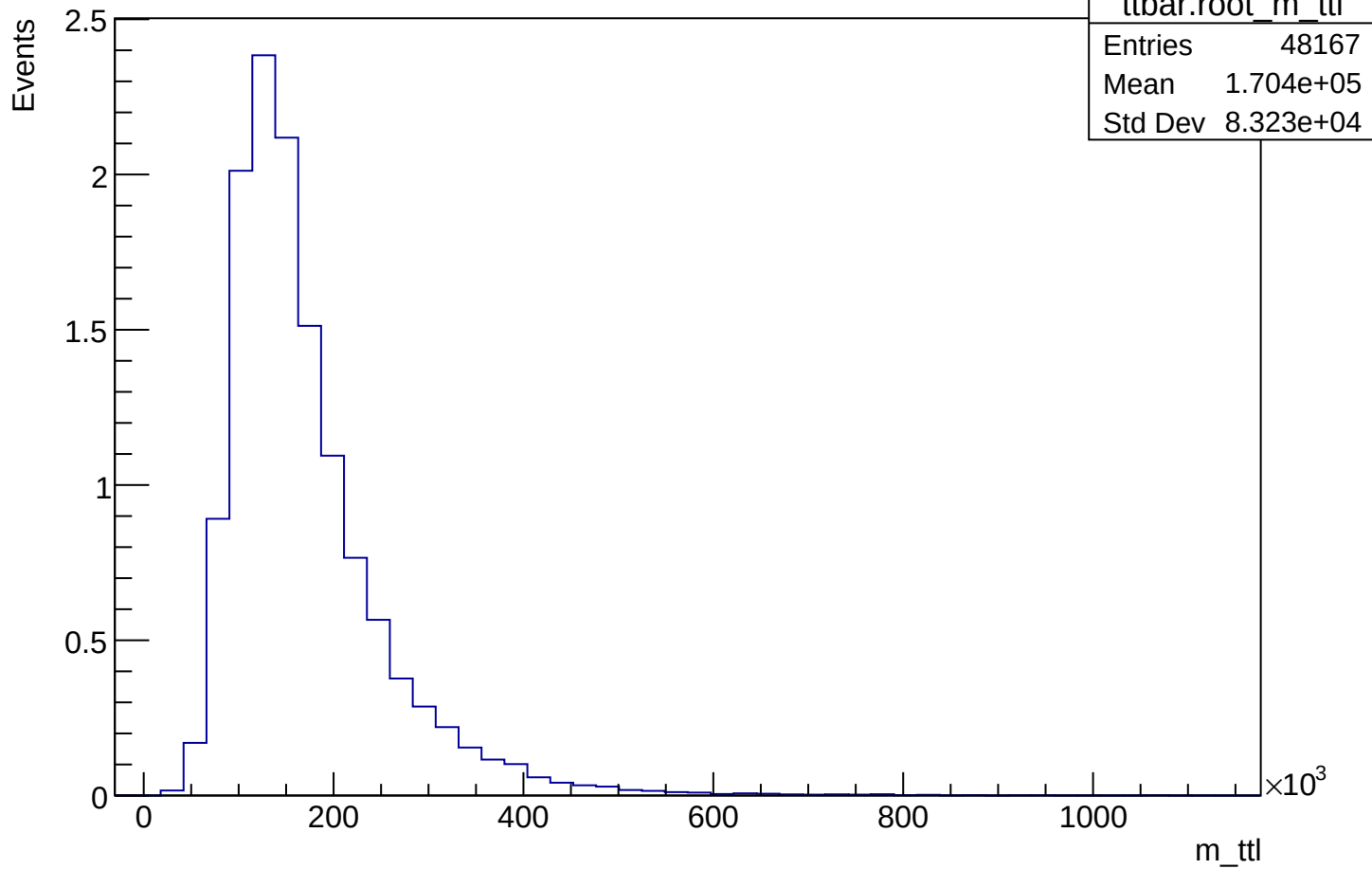
Entries 48167

Mean 8.319e+04

Std Dev 5.737e+04



m_ttl



met_met

Events

3
2.5
2
1.5
1
0.5
0

0

200

400

600

800

1000

$\times 10^3$

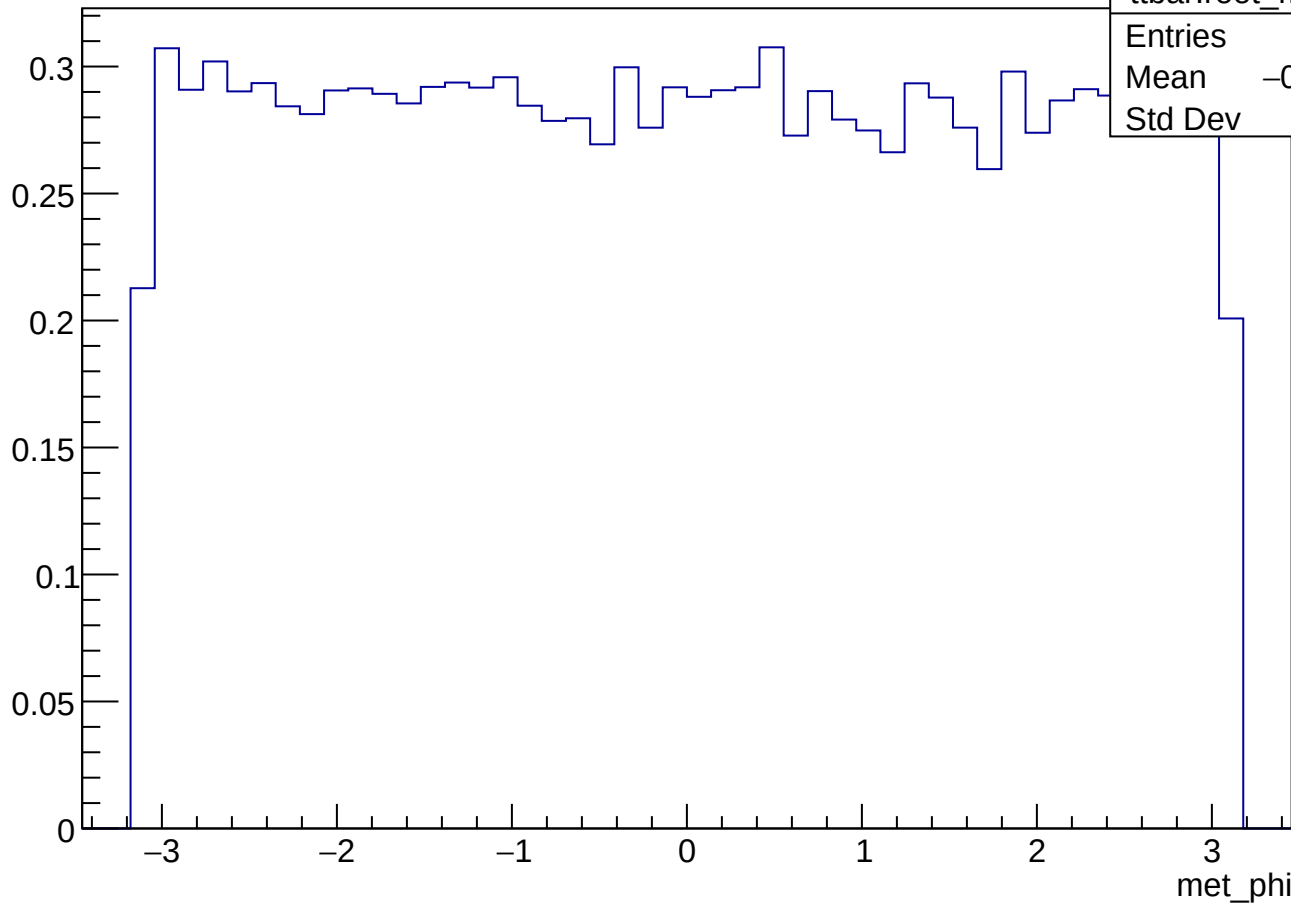
met_met

ttbar.root_met_met	
Entries	48167
Mean	7.786e+04
Std Dev	5.195e+04

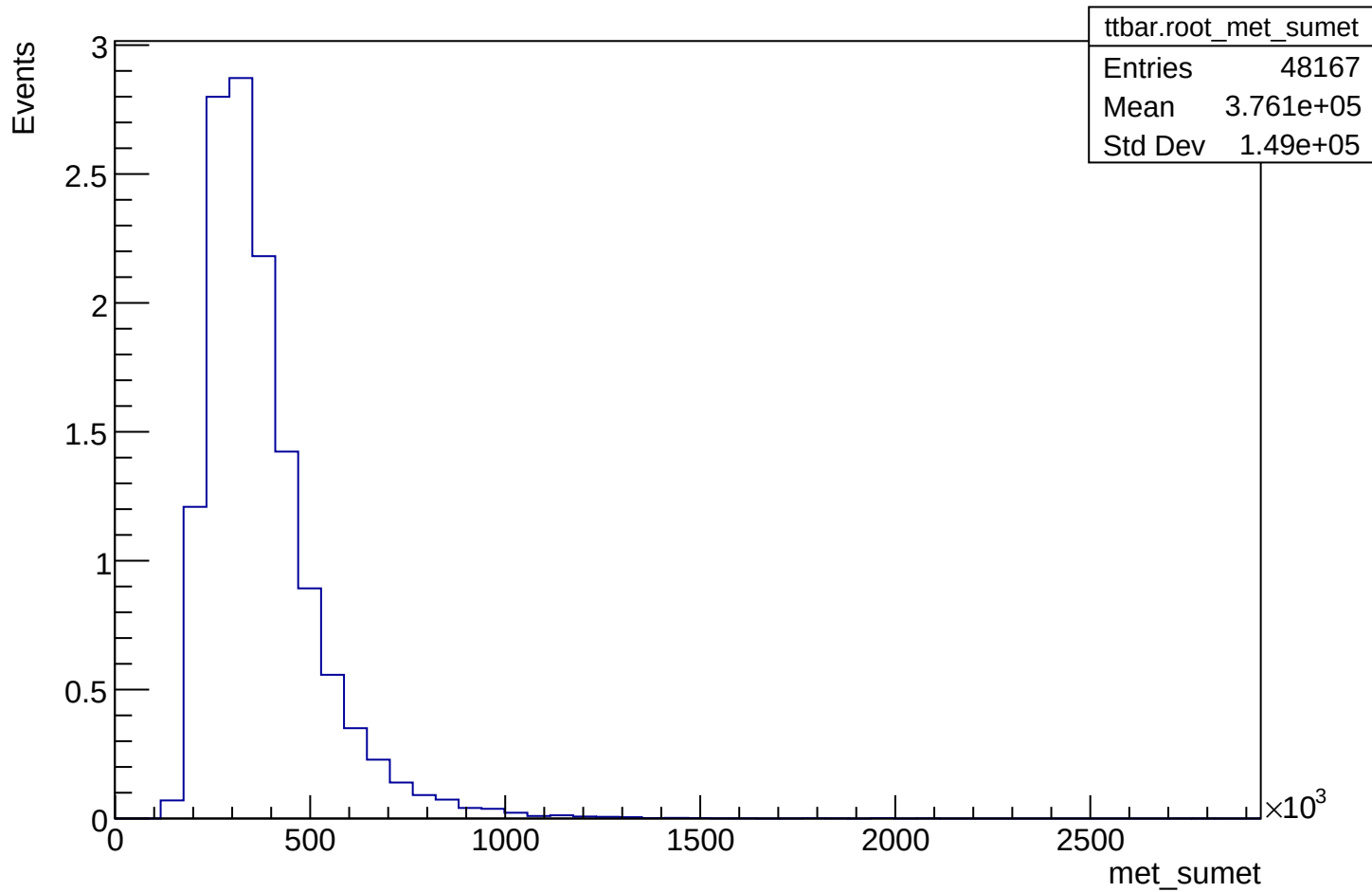
met_phi

ttbar.root_met_phi	
Entries	48167
Mean	-0.01924
Std Dev	1.82

Events



met_sumet



n_jet

Events

ttbar.root_n_jet	
Entries	48167
Mean	2.73
Std Dev	0.9937

7
6
5
4
3
2
1
0

2

3

4

5

6

7

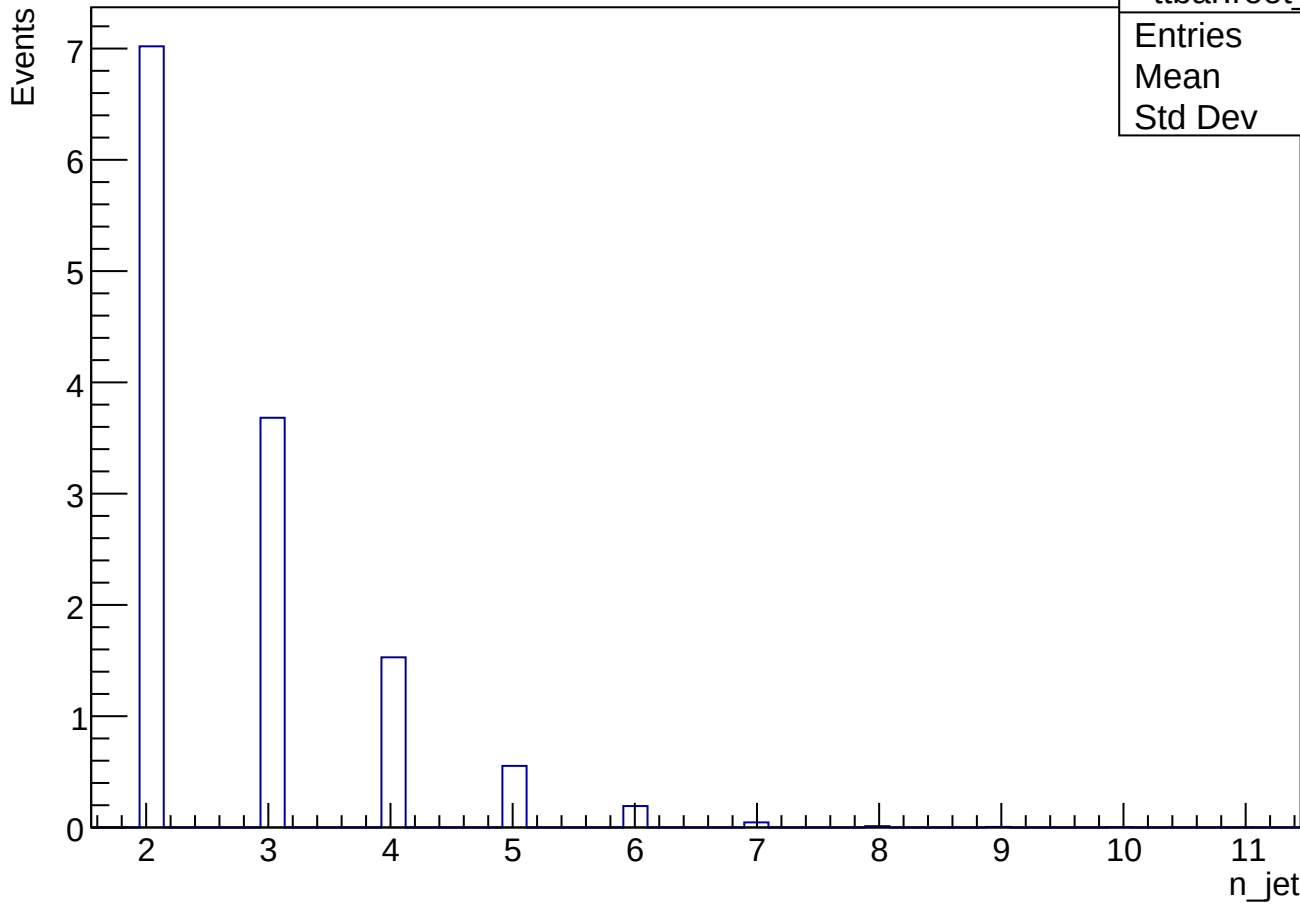
8

9

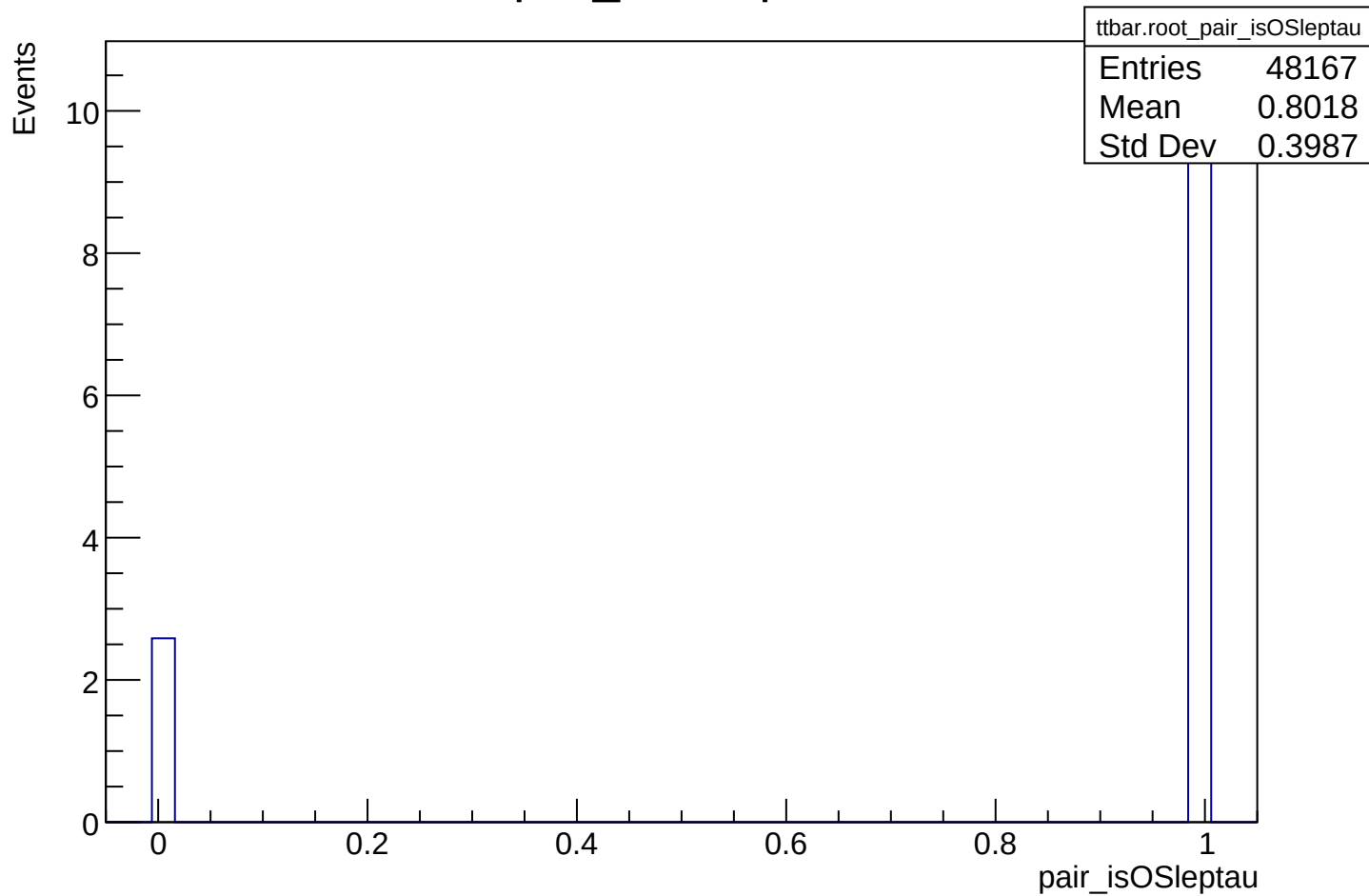
10

11

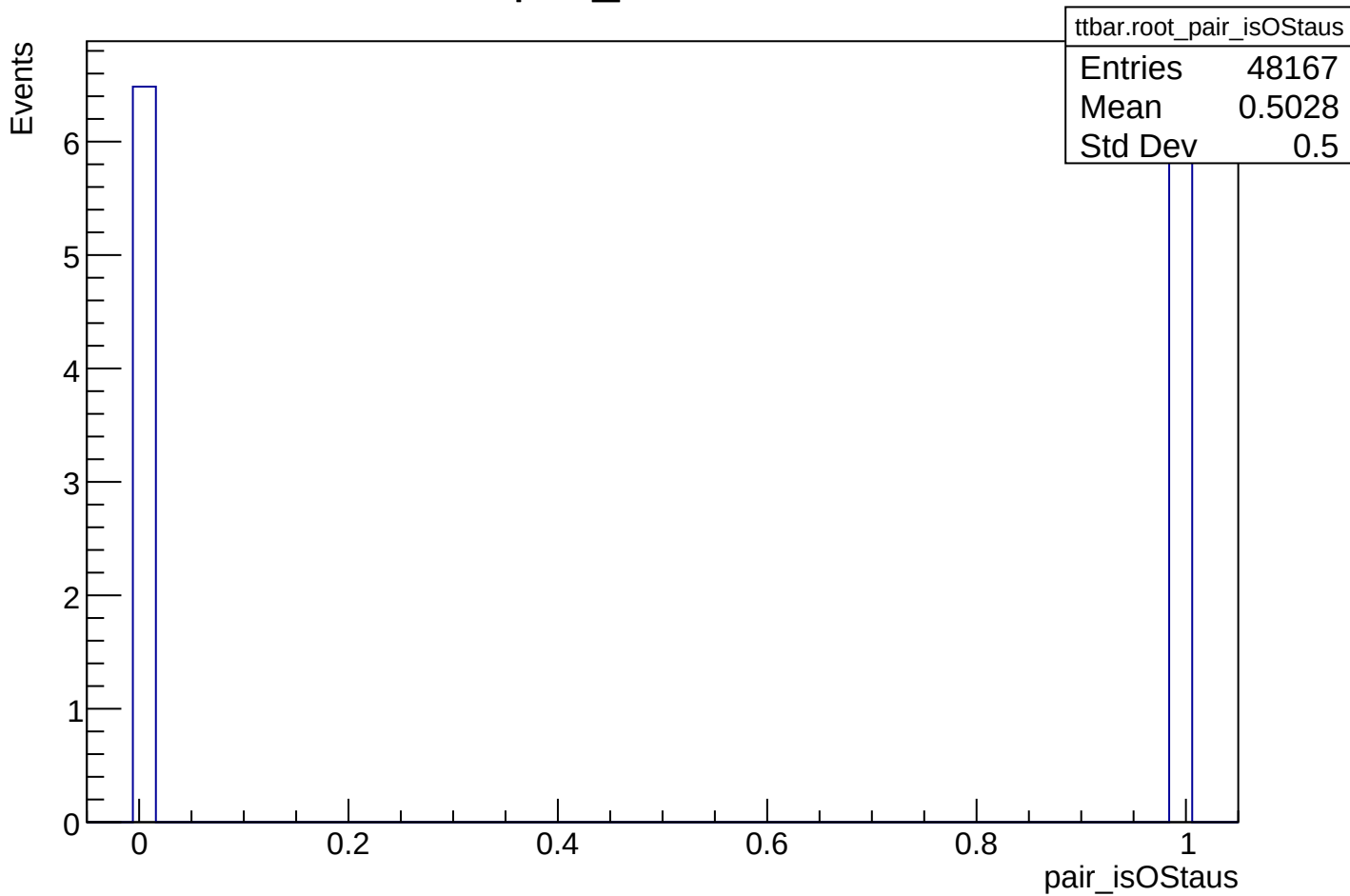
n_jet



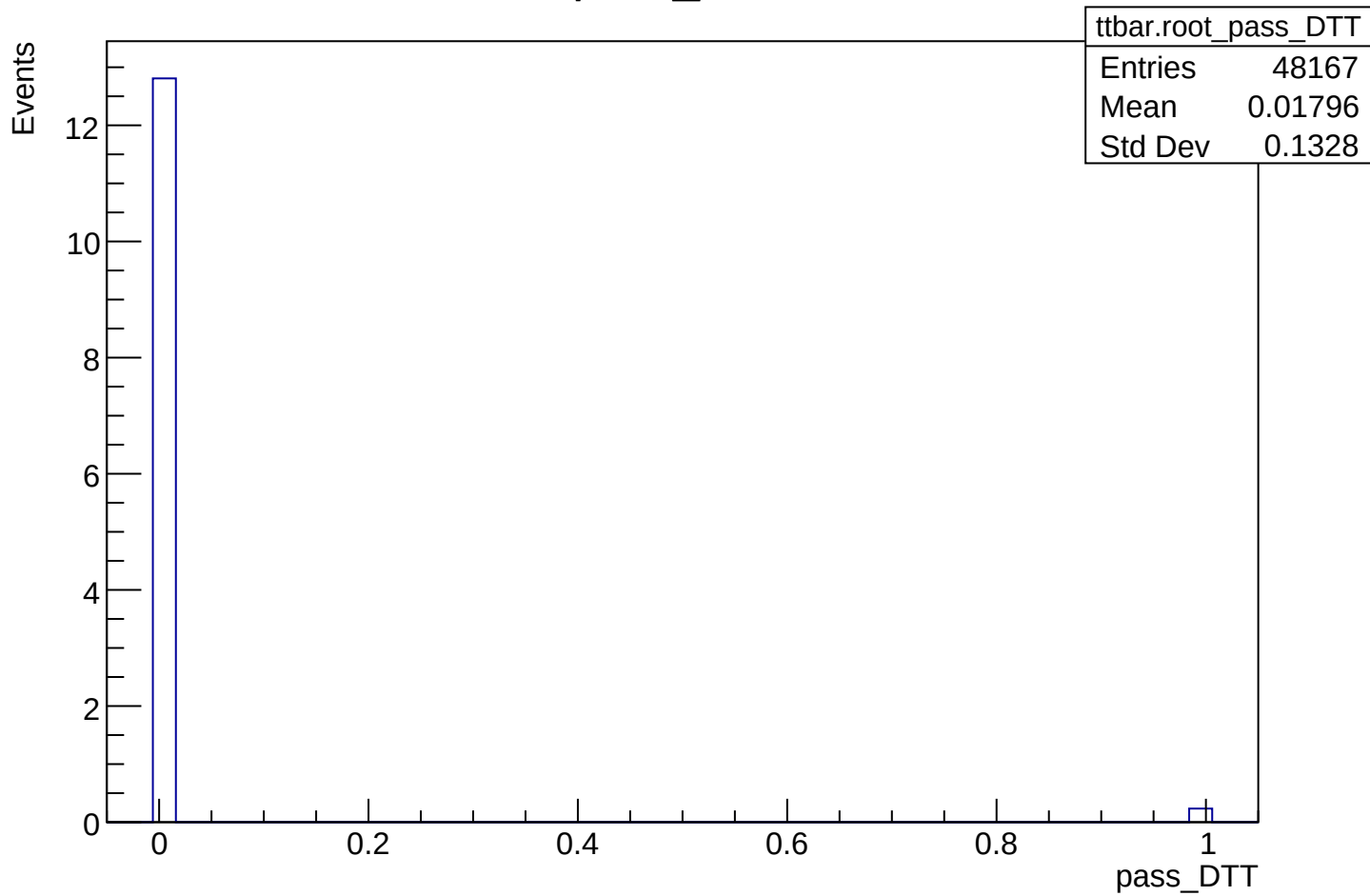
pair_isOSleptau



pair_isOStaus



pass_DTT



pass_LTT

Events

7
6
5
4
3
2
1
0

0

0.2

0.4

0.6

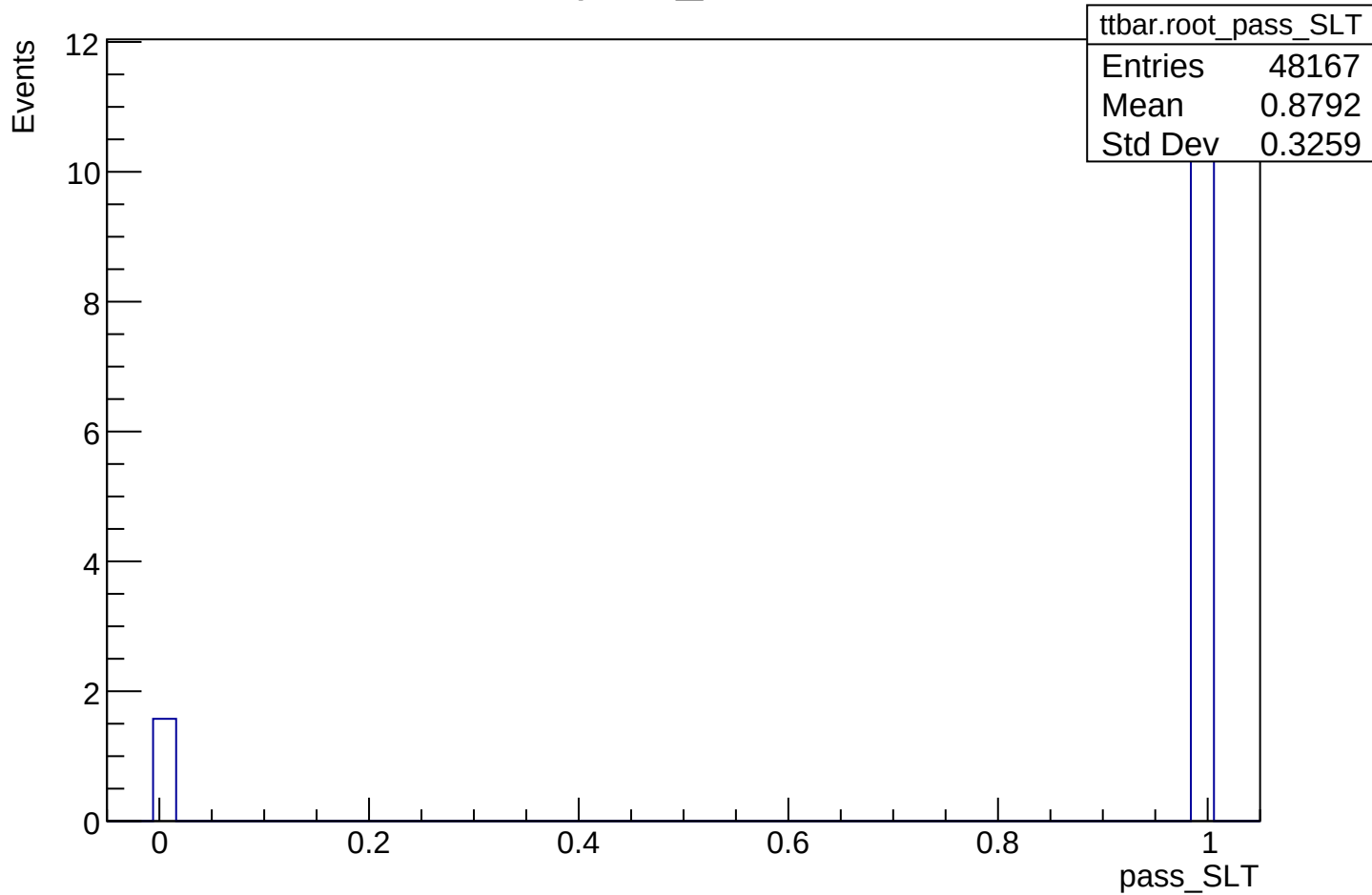
0.8

1

pass_LTT

tbar.root_pass_LTT	
Entries	48167
Mean	0.5664
Std Dev	0.4956

pass_SLT



pass_STT

Events

12

10

8

6

4

2

0

0

0.2

0.4

0.6

0.8

1

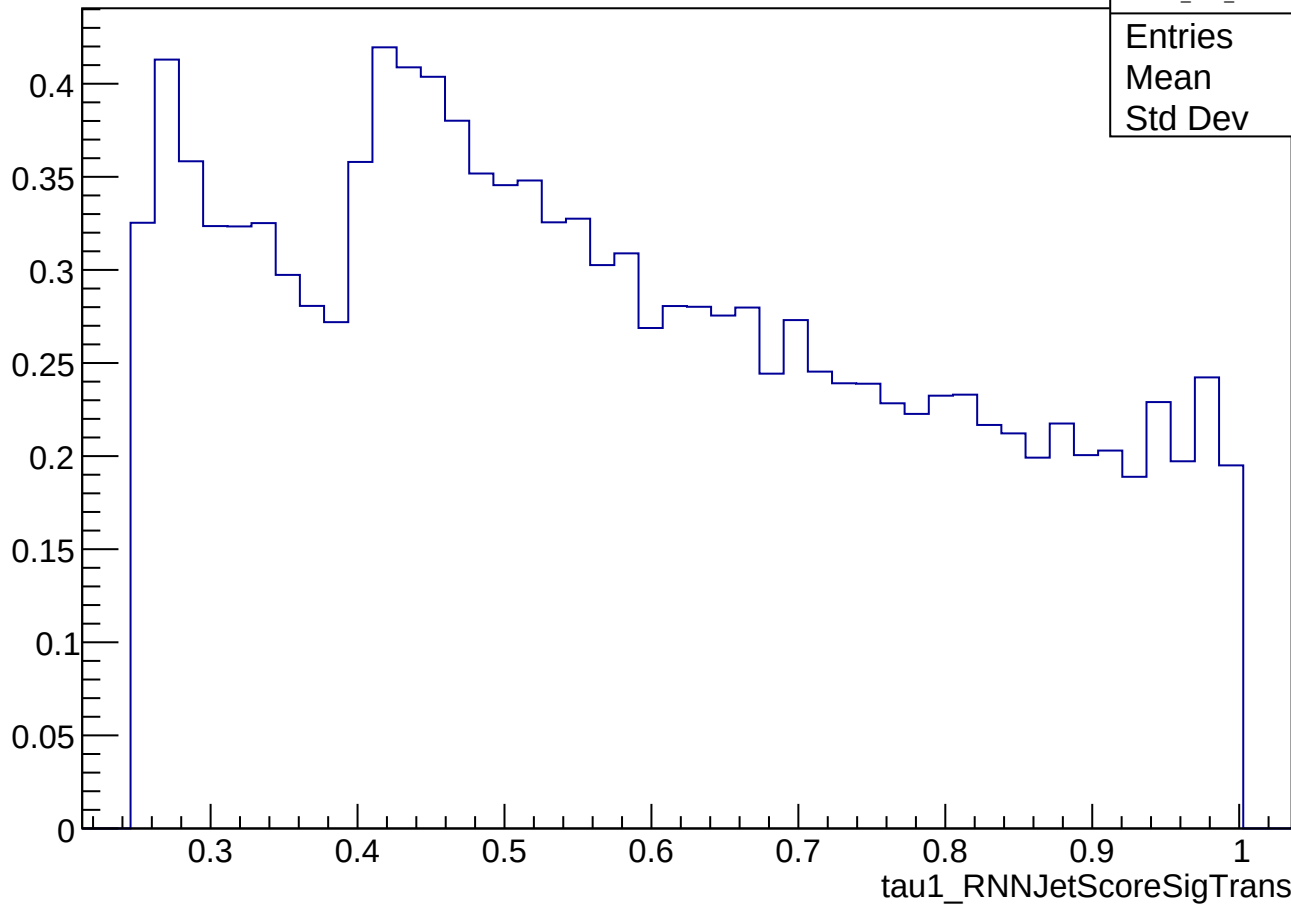
pass_STT

tbar.root_pass_STT	
Entries	48167
Mean	0.01235
Std Dev	0.1104

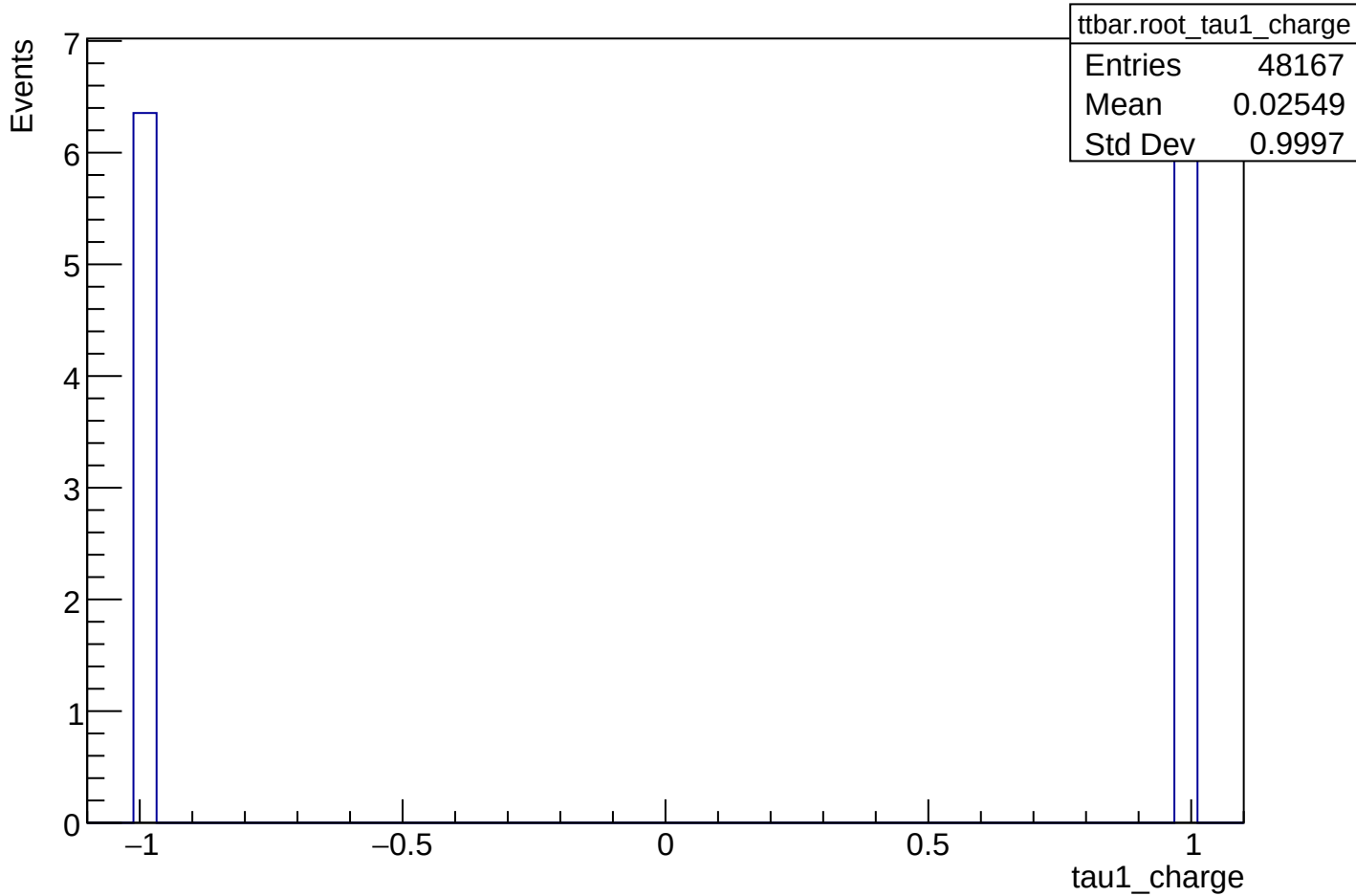
tau1_RNNJetScoreSigTrans

ttbar.root_tau1_RNNJetScoreSigTrans	
Entries	48167
Mean	0.5821
Std Dev	0.2122

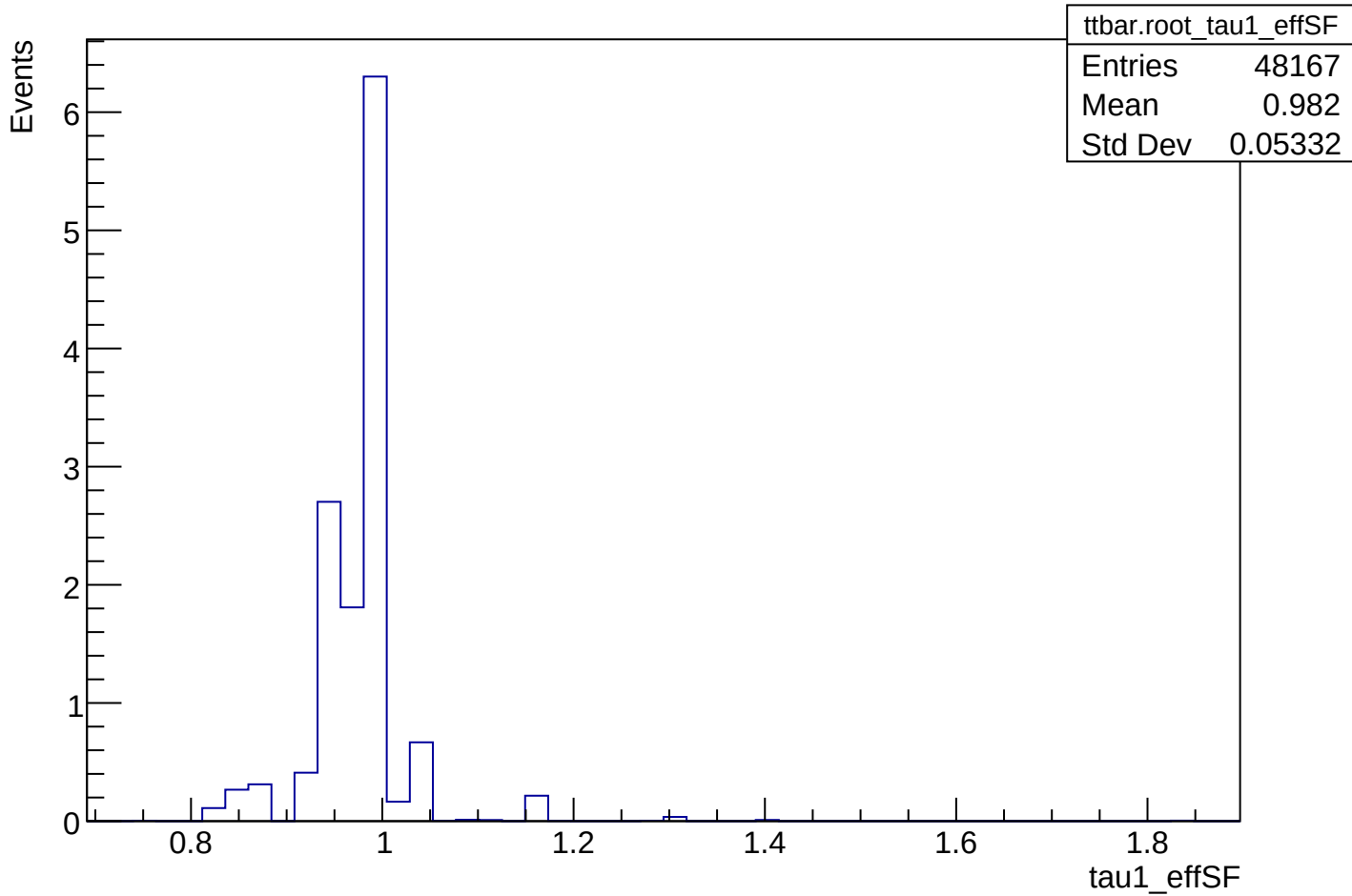
Events



tau1_charge



tau1_effSF



tau1_eta

Events

0.5
0.4
0.3
0.2
0.1
0

-2

-1

0

1

2

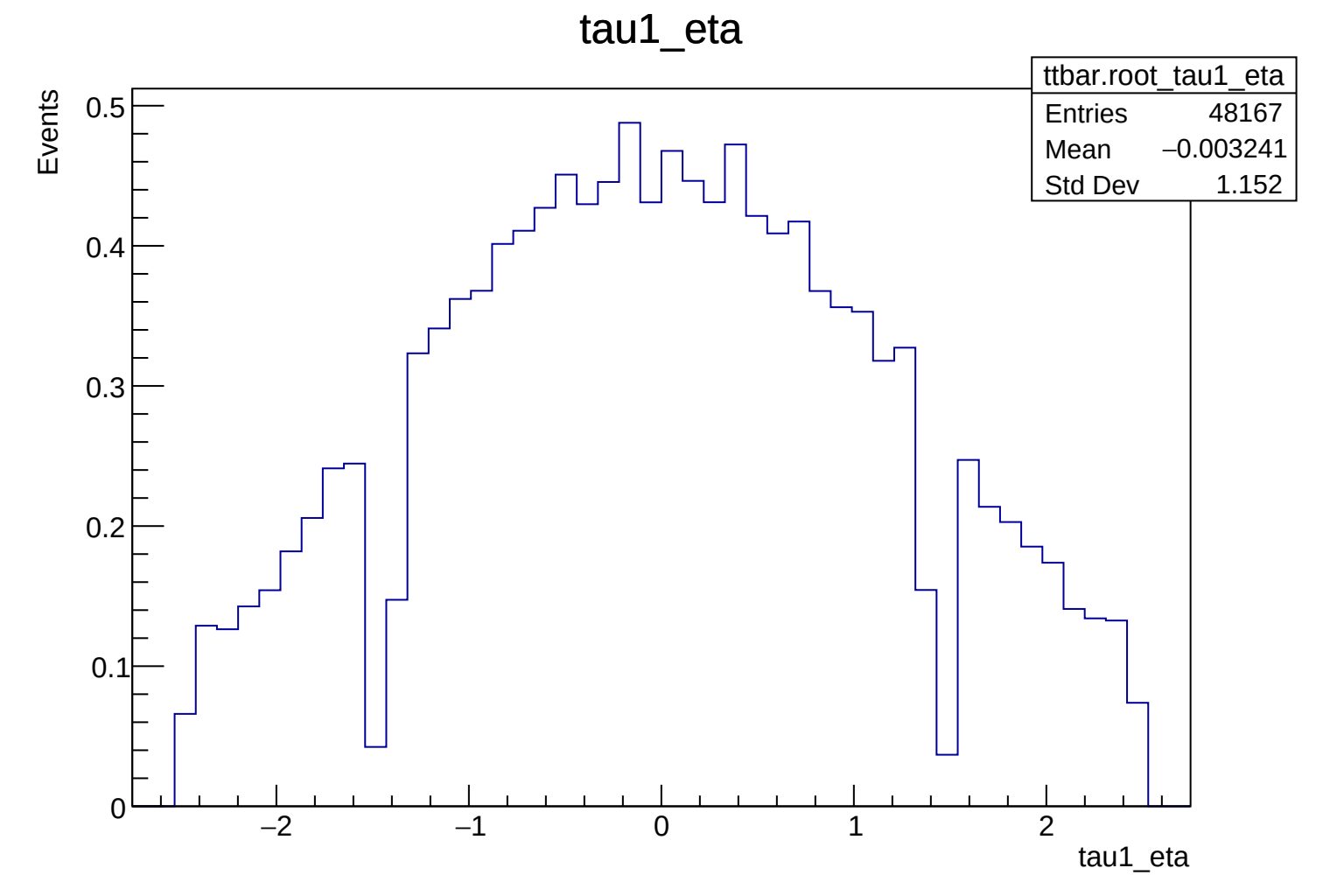
tau1_eta

tbar.root_tau1_eta

Entries 48167

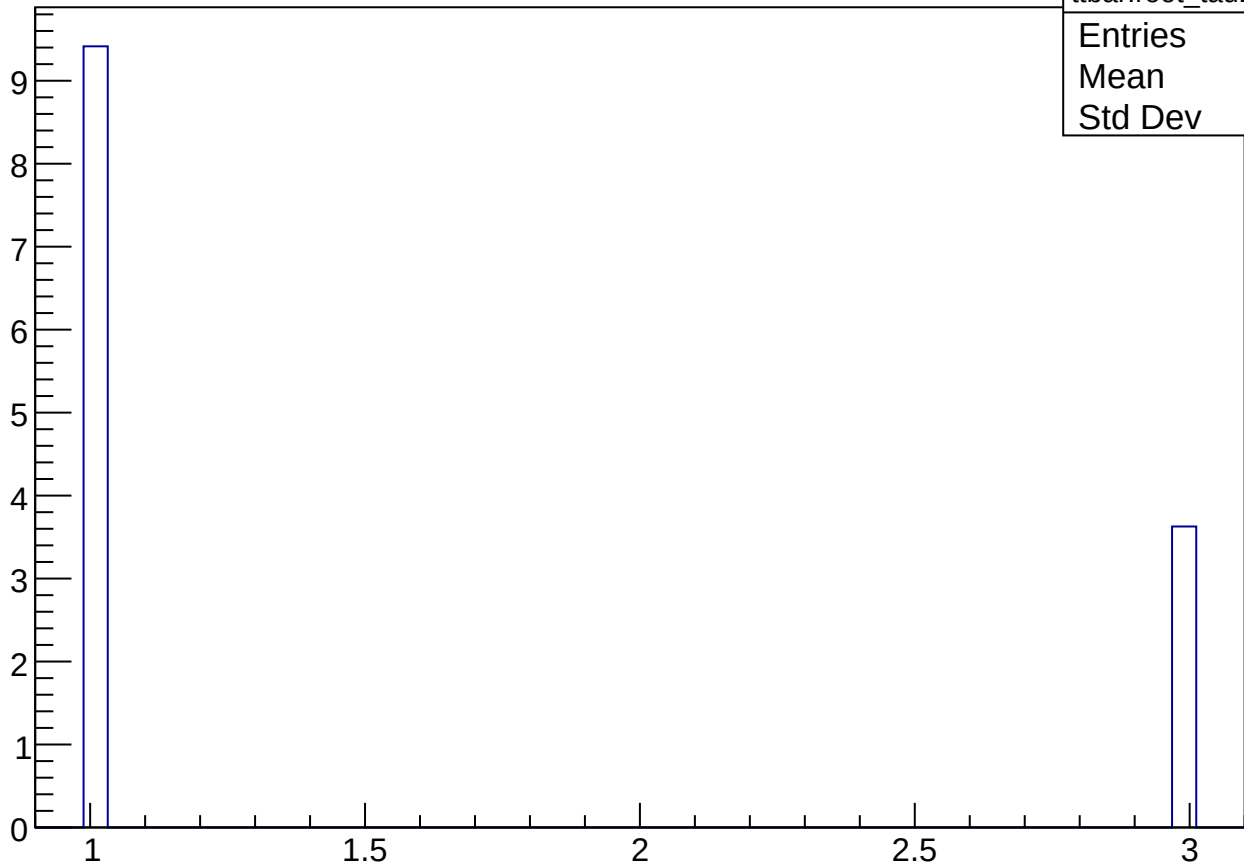
Mean -0.003241

Std Dev 1.152



tau1_nprong

Events



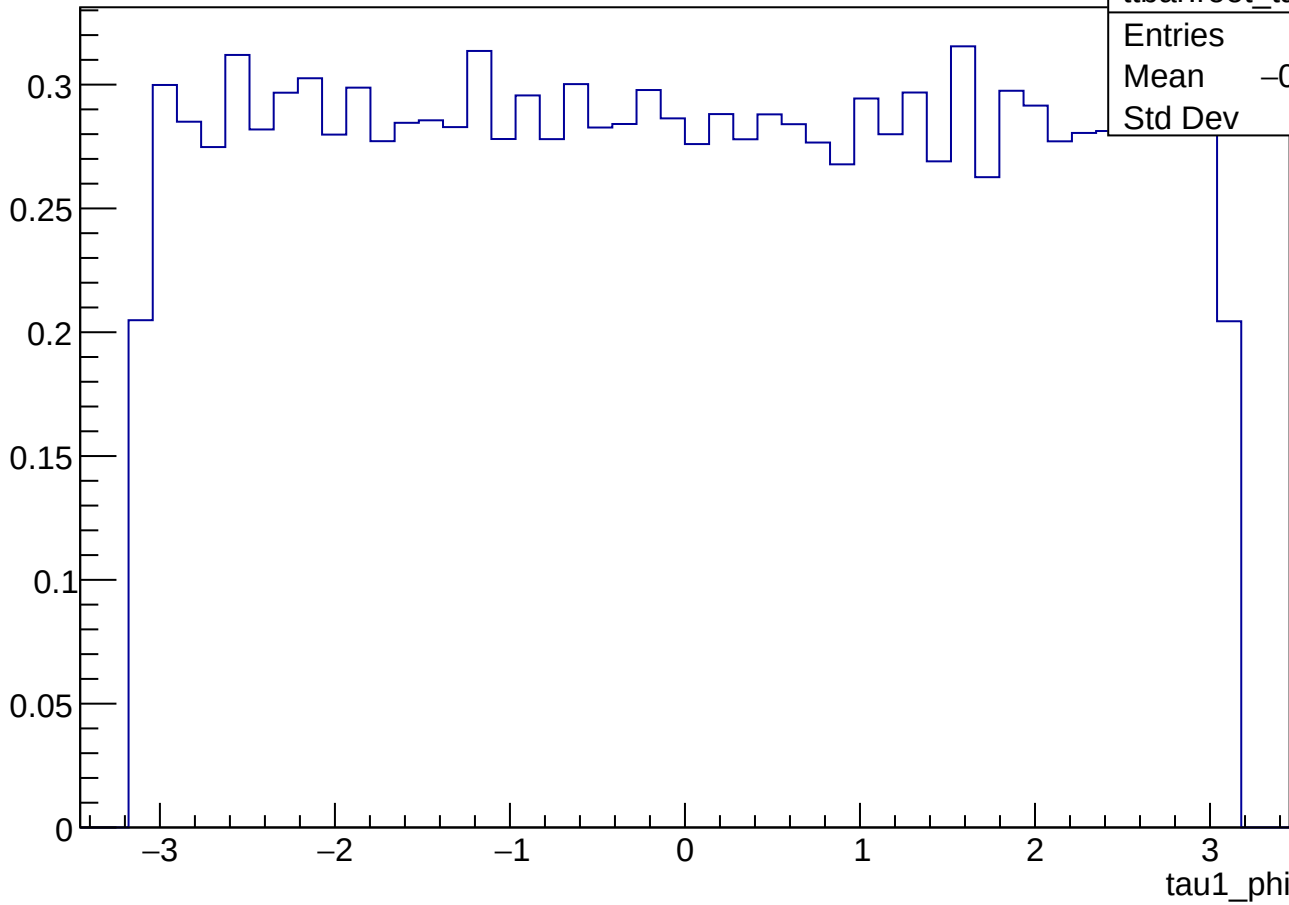
ttbar.root_tau1_nprong	
Entries	48167
Mean	1.556
Std Dev	0.8962

tau1_nprong

tau1_phi

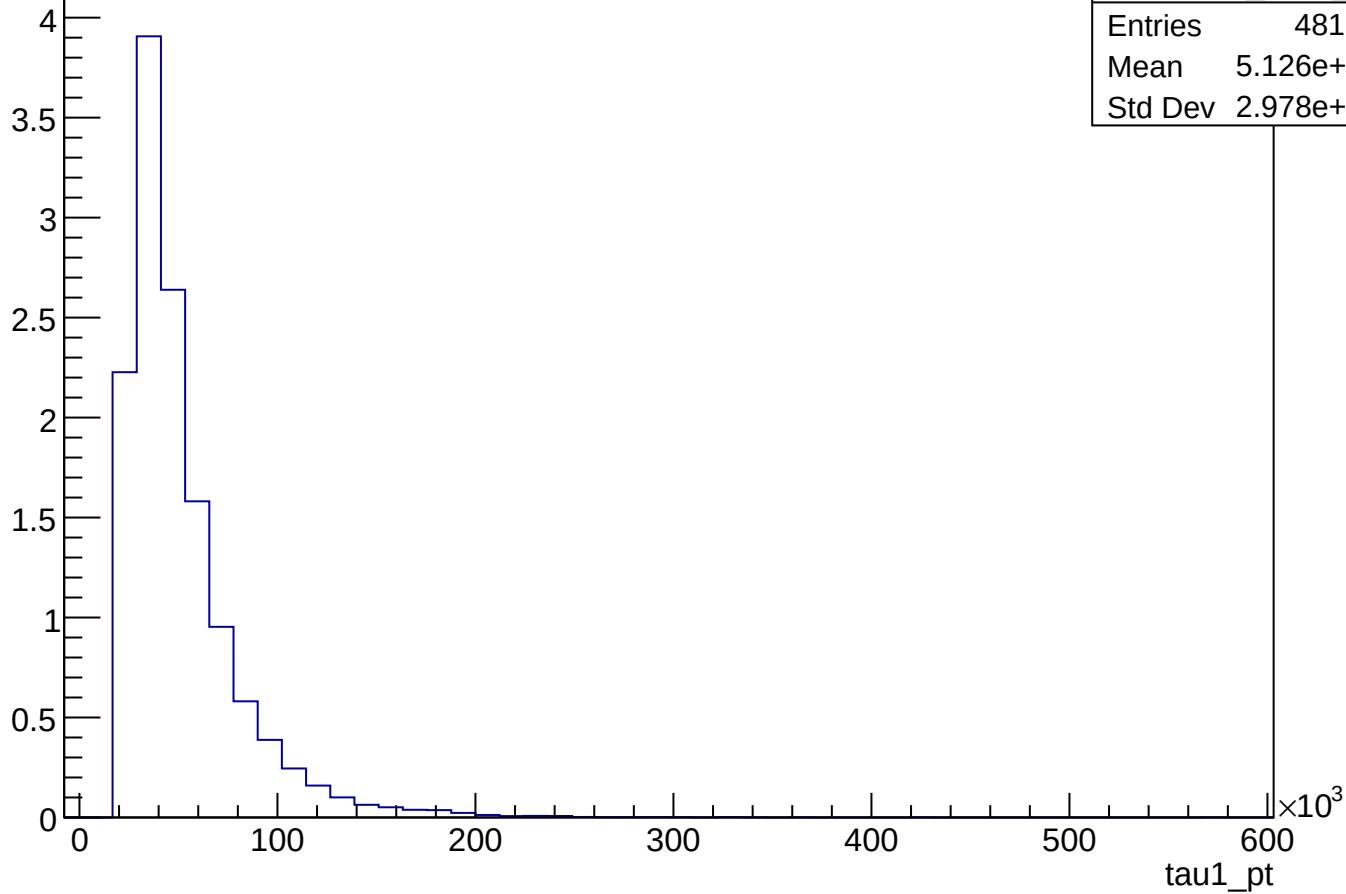
ttbar.root_tau1_phi	
Entries	48167
Mean	-0.01336
Std Dev	1.816

Events

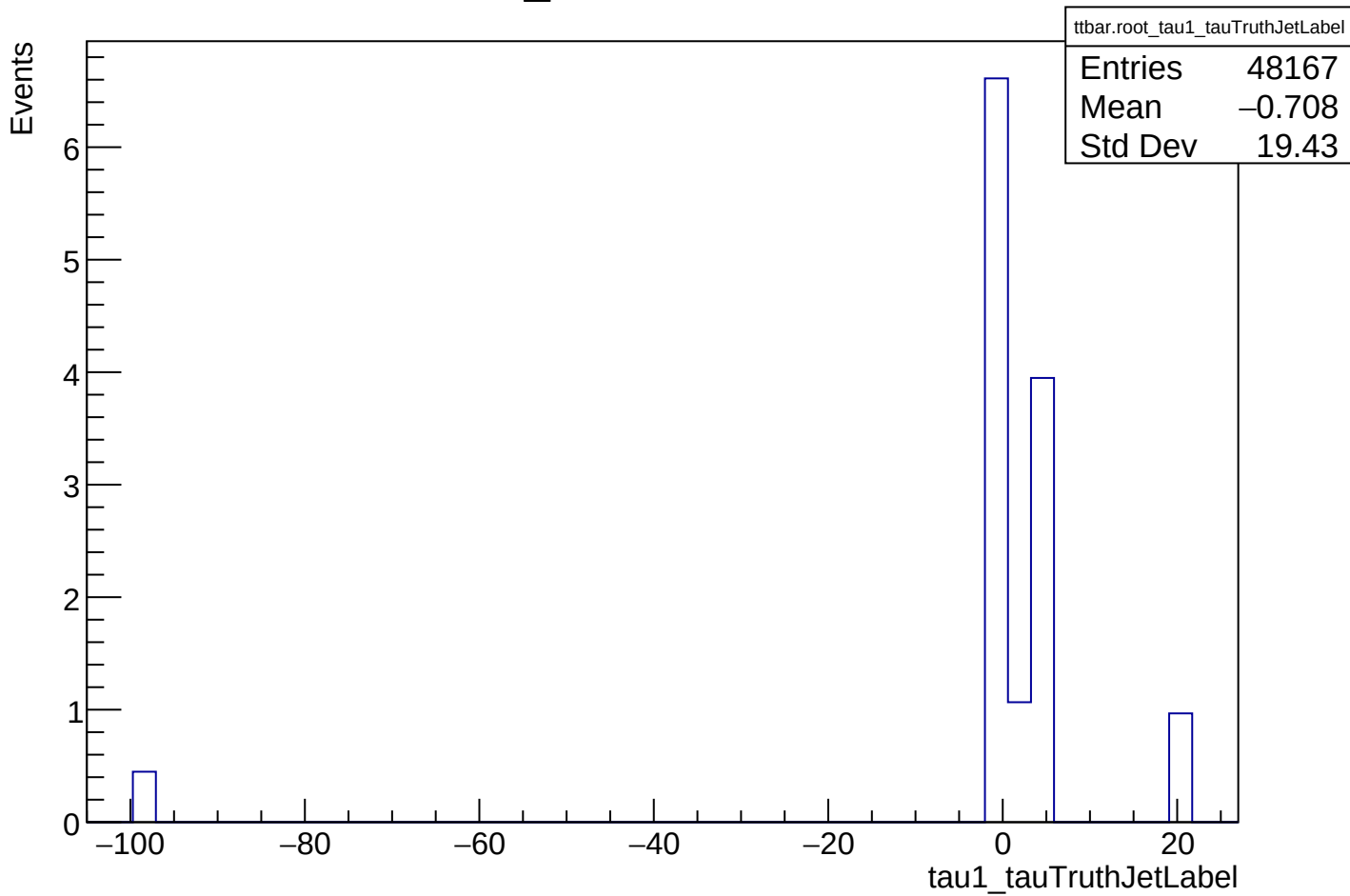


tau1_pt

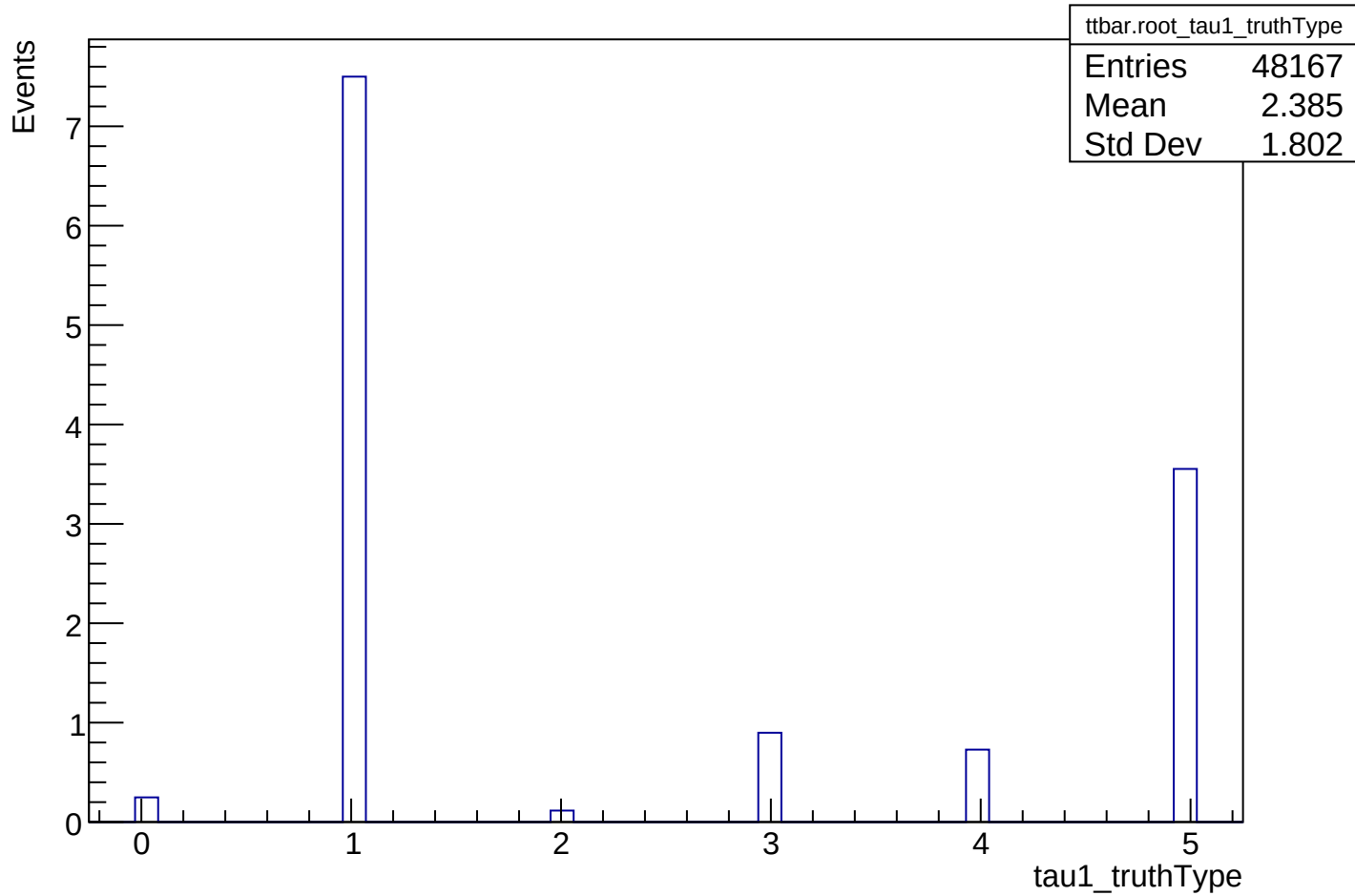
Events



tau1_tauTruthJetLabel



tau1_truthType



tau1_truth_IsHadronicTau

Events

7
6
5
4
3
2
1
0

0

0.2

0.4

0.6

0.8

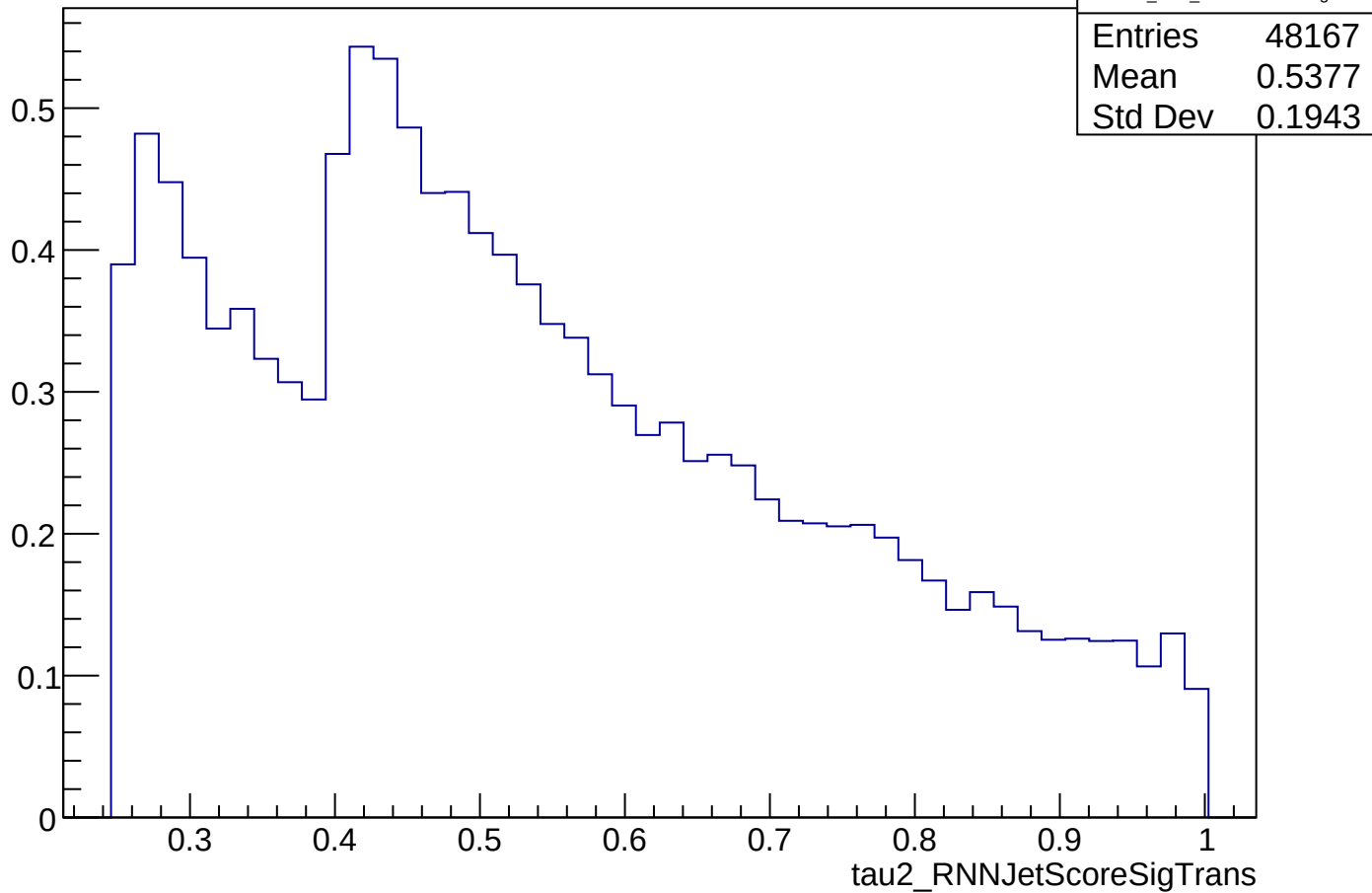
1

tau1_truth_IsHadronicTau

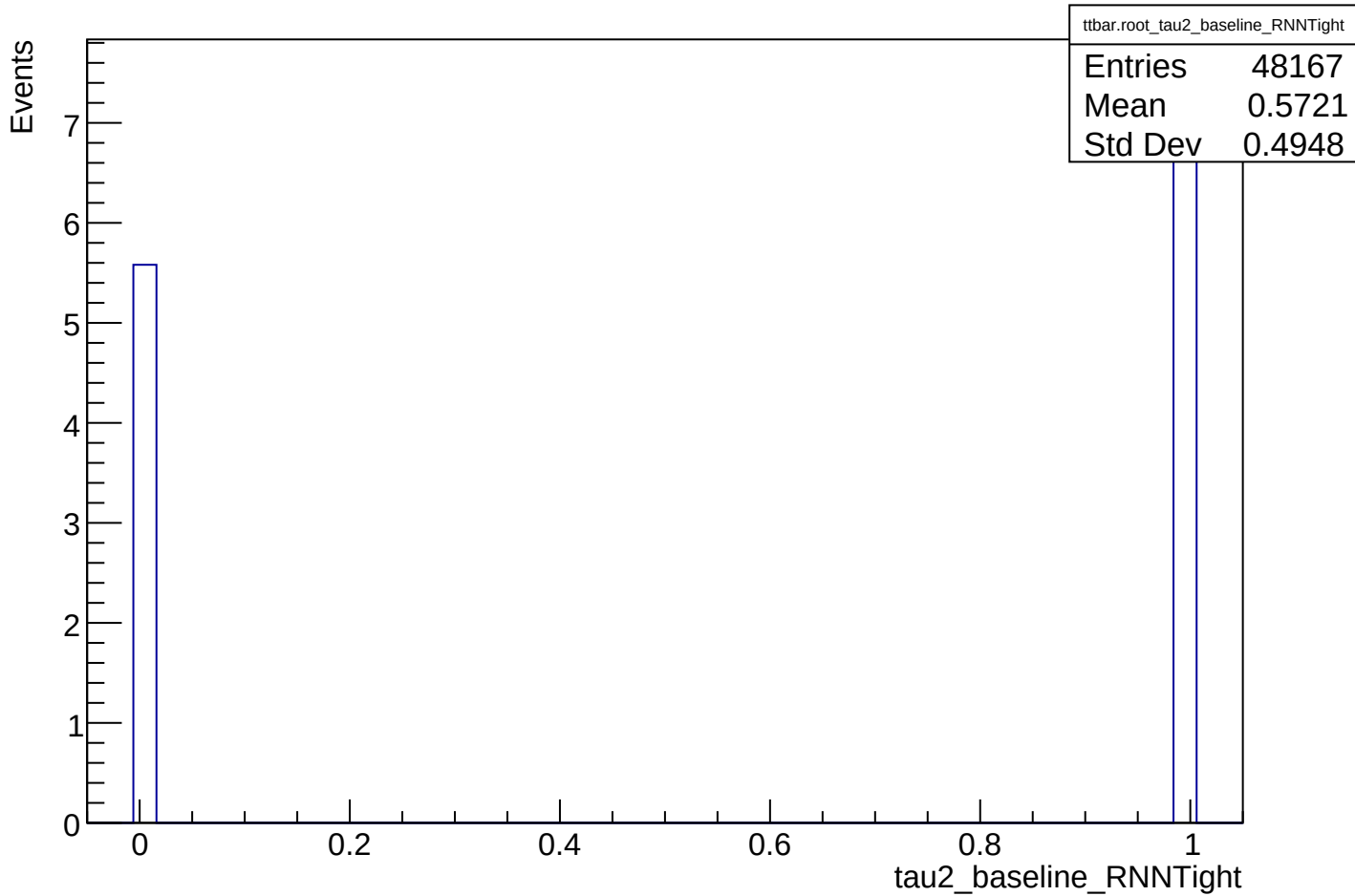
ttbar.root_tau1_truth_IsHadronicTau	
Entries	48167
Mean	0.575
Std Dev	0.4943

tau2_RNNJetScoreSigTrans

Events

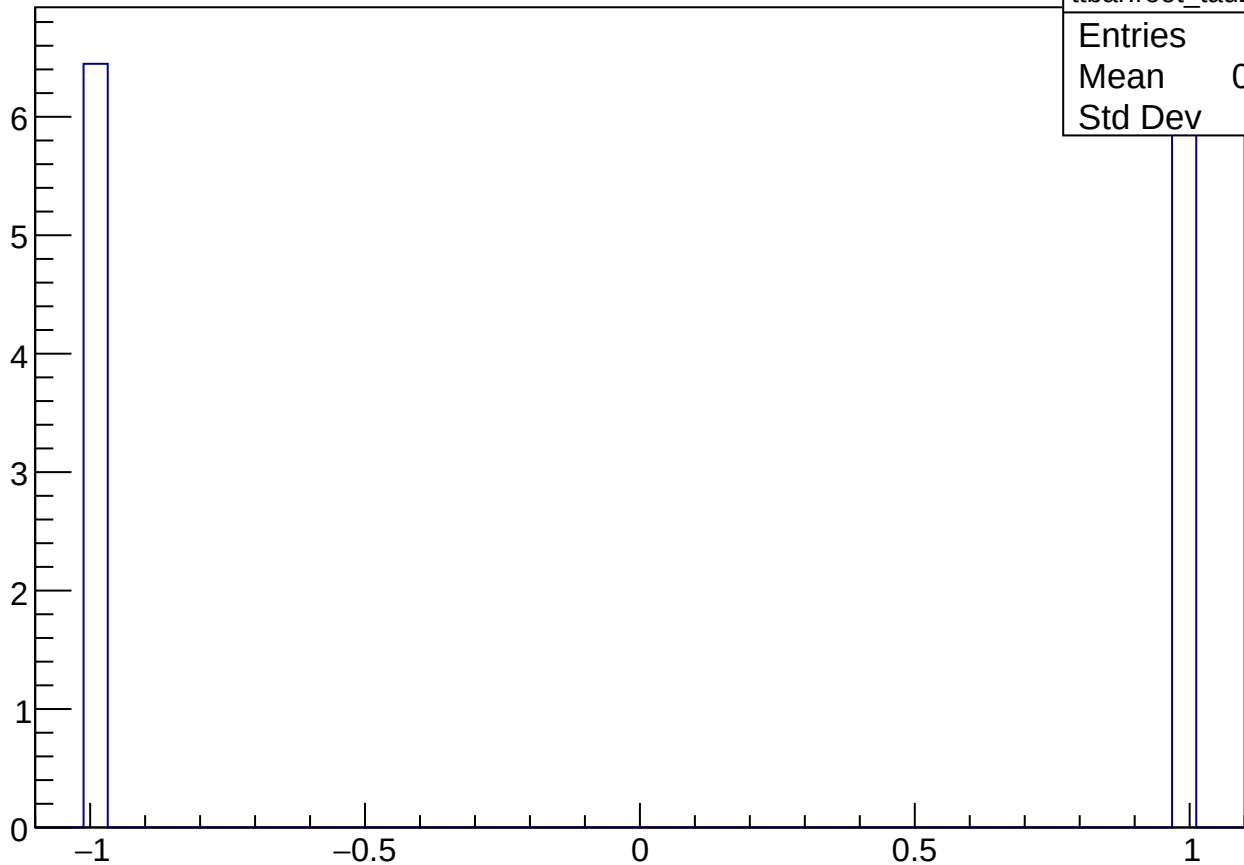


tau2_baseline_RNNTight



tau2_charge

Events

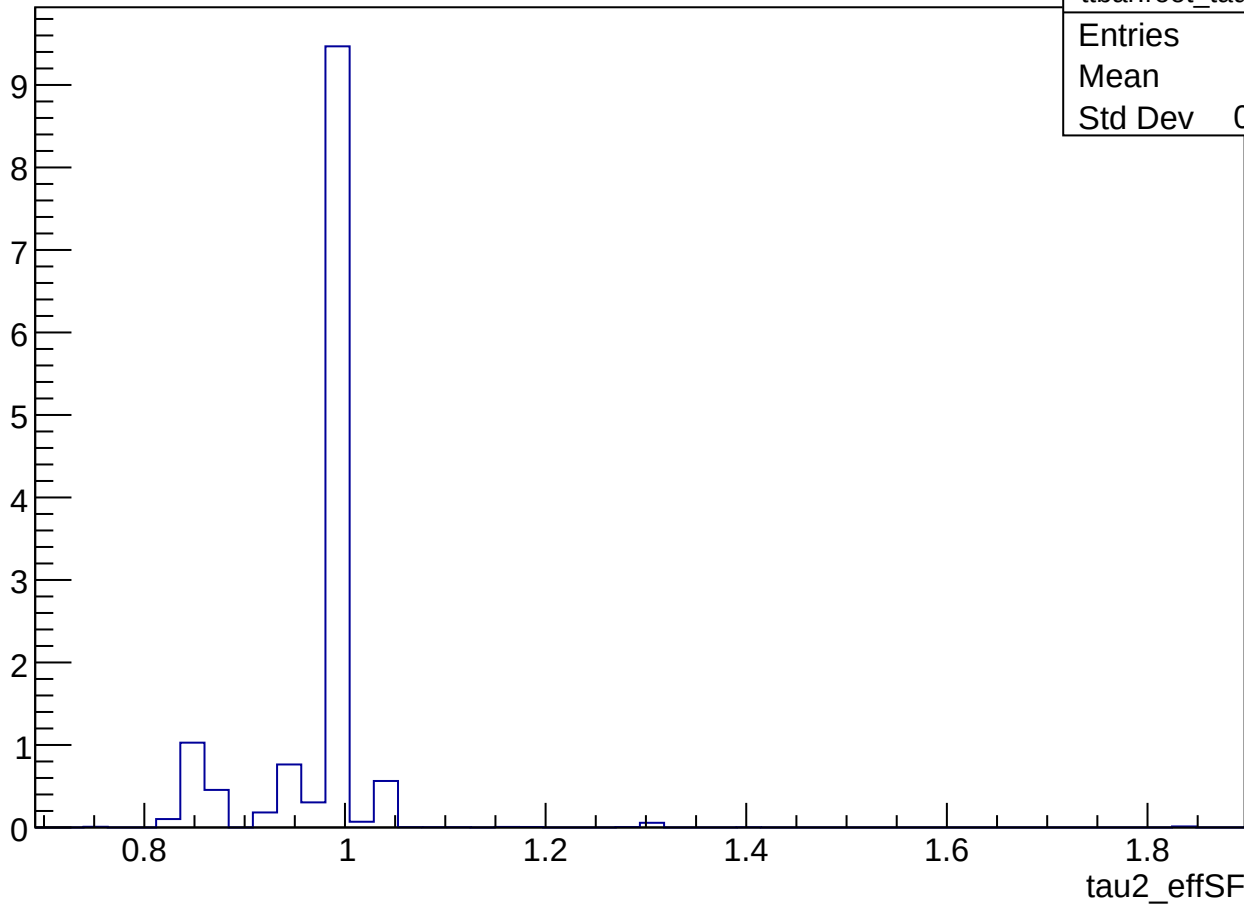


ttbar.root_tau2_charge	
Entries	48167
Mean	0.01131
Std Dev	0.9999

τ_{2_charge}

tau2_effSF

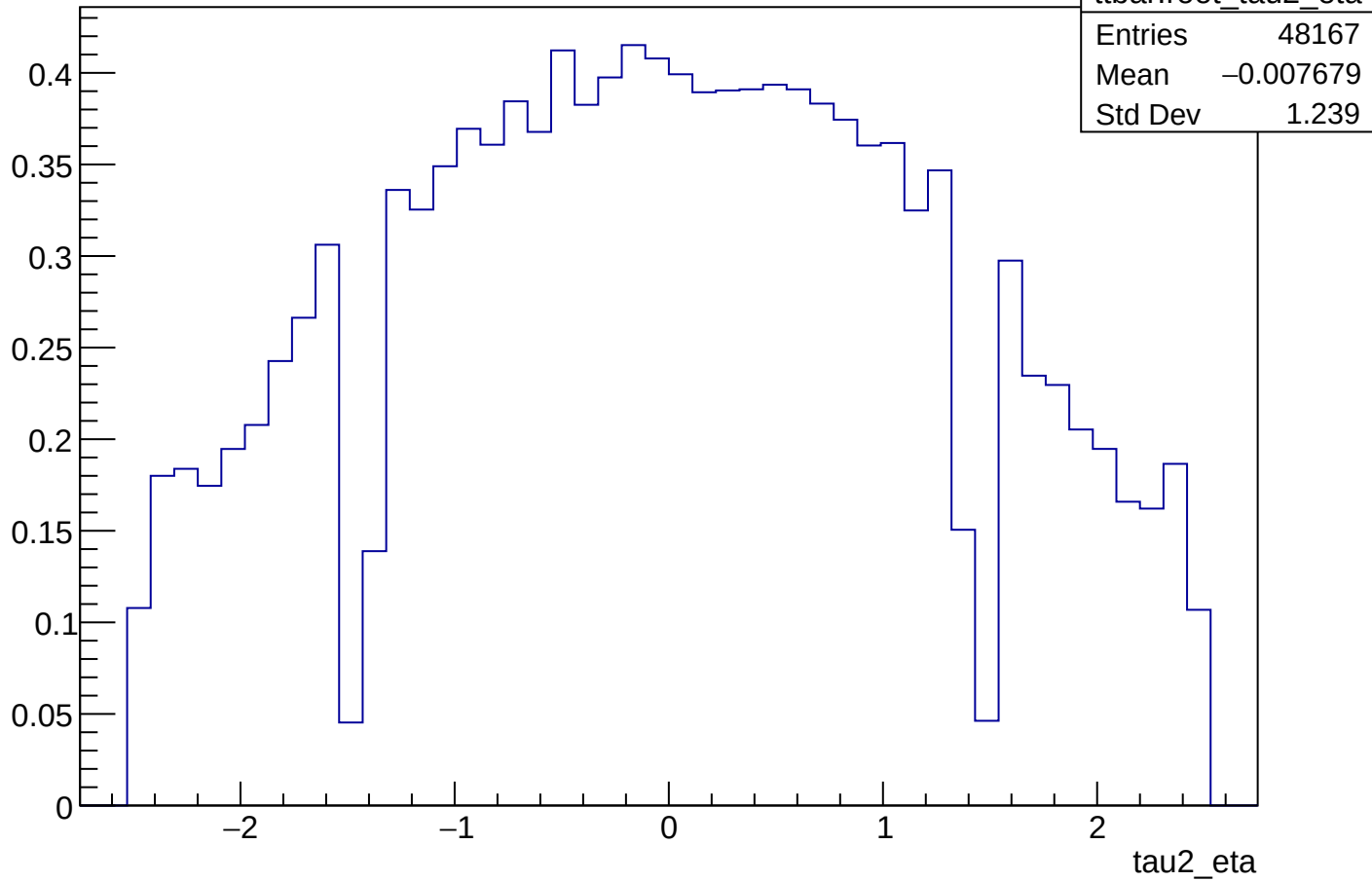
Events



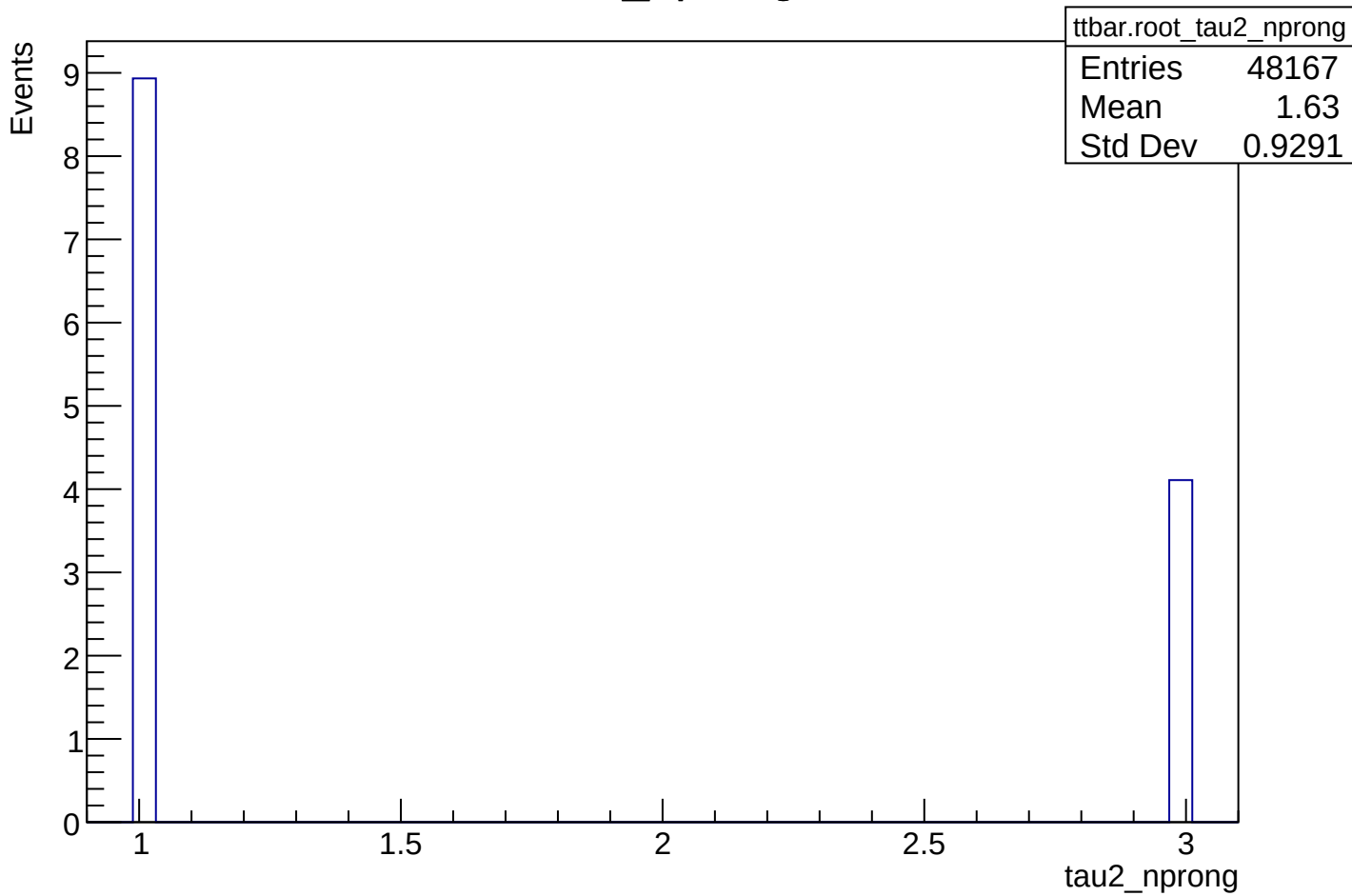
ttbar.root_tau2_effSF	
Entries	48167
Mean	0.9819
Std Dev	0.06239

tau2_eta

Events



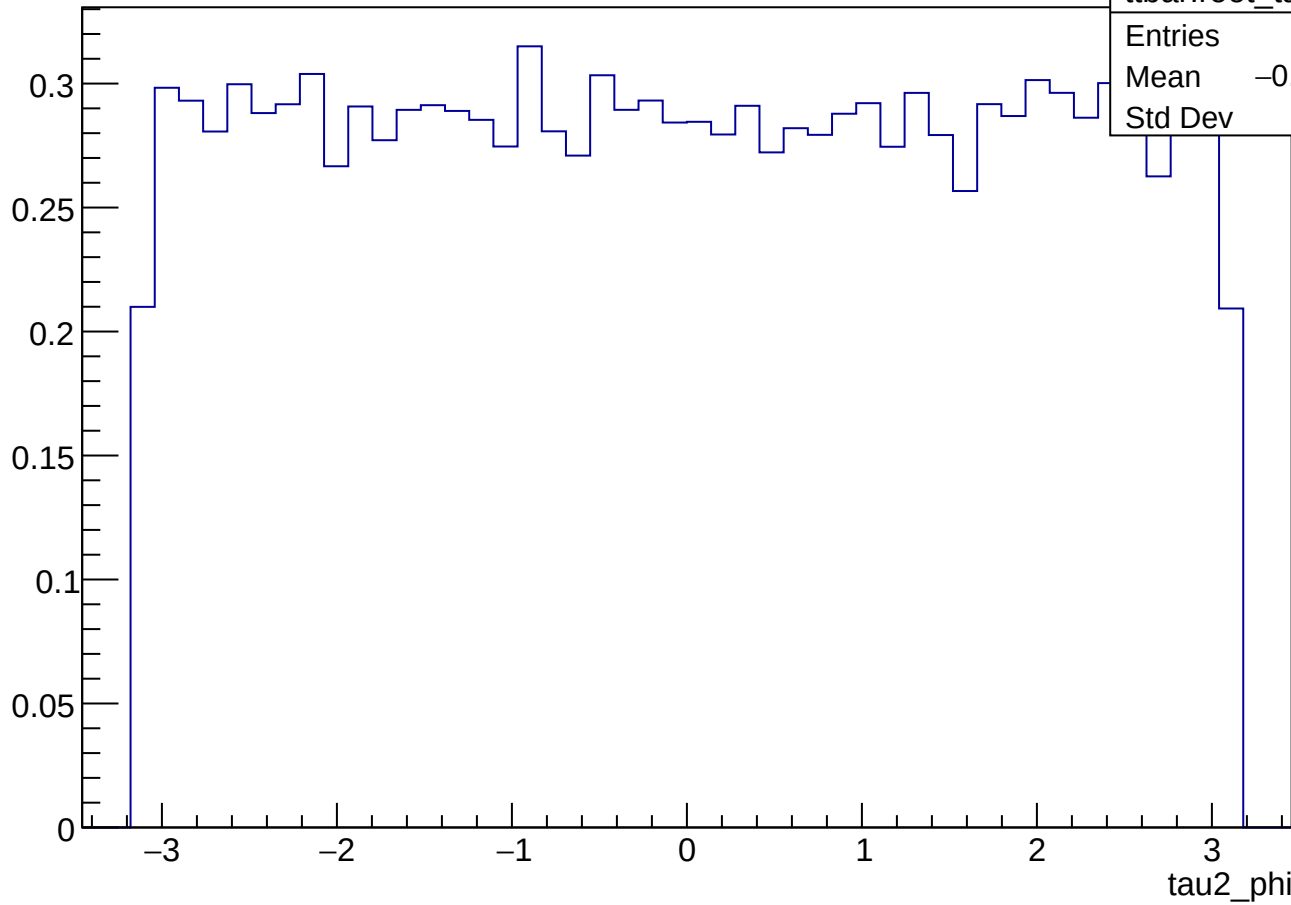
tau2_nprong



tau2_phi

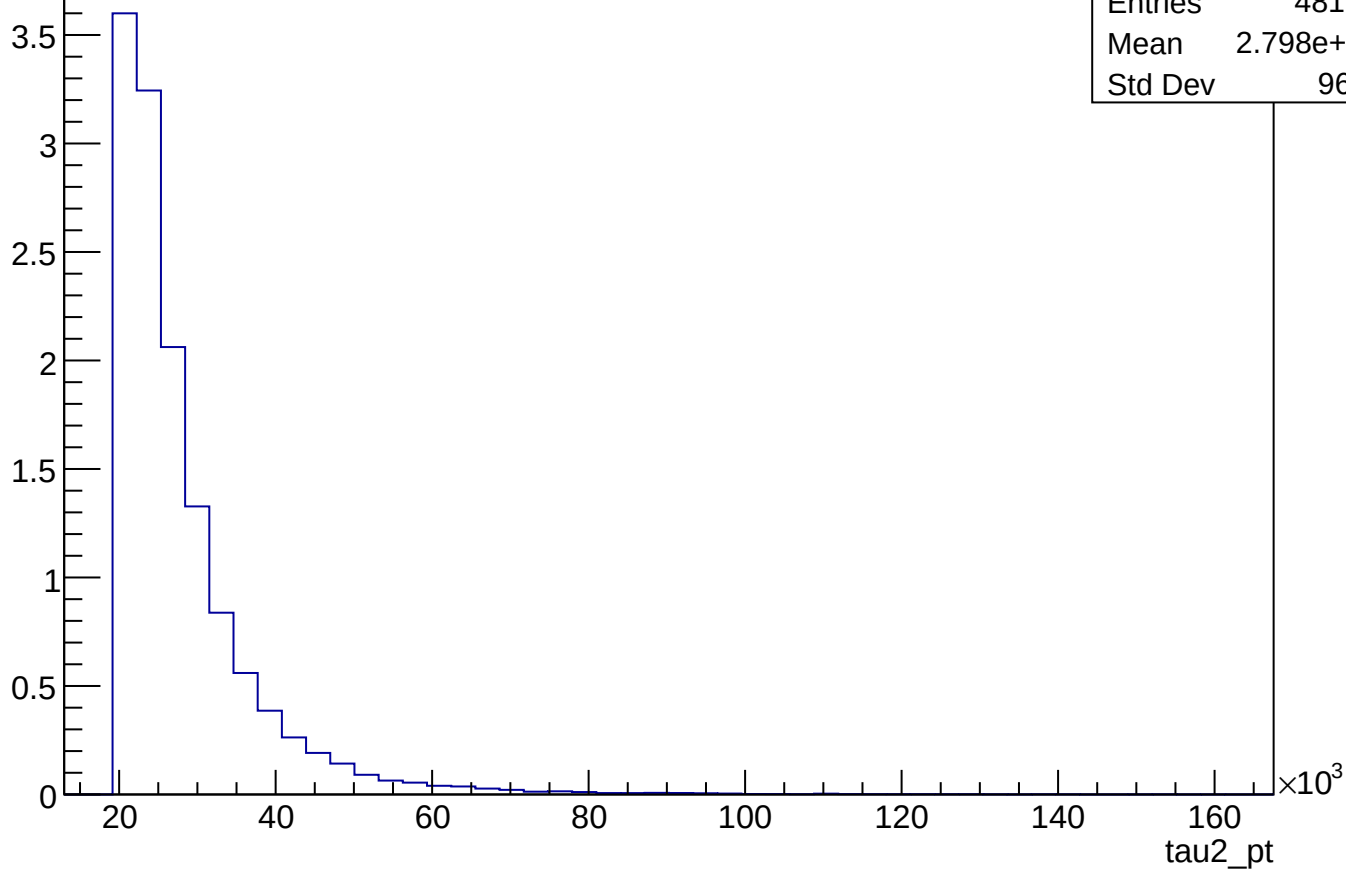
ttbar.root_tau2_phi	
Entries	48167
Mean	-0.009404
Std Dev	1.818

Events

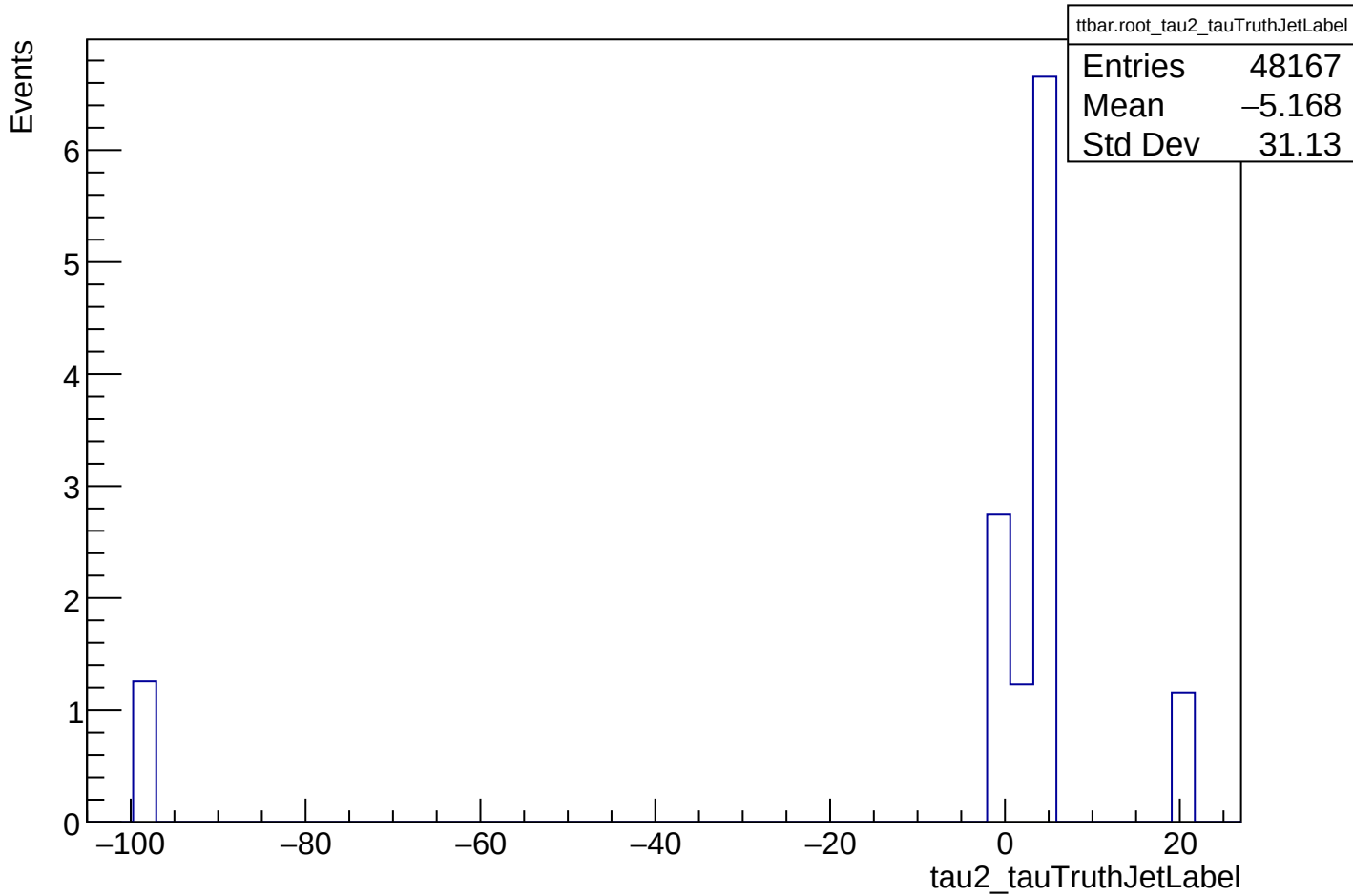


tau2_pt

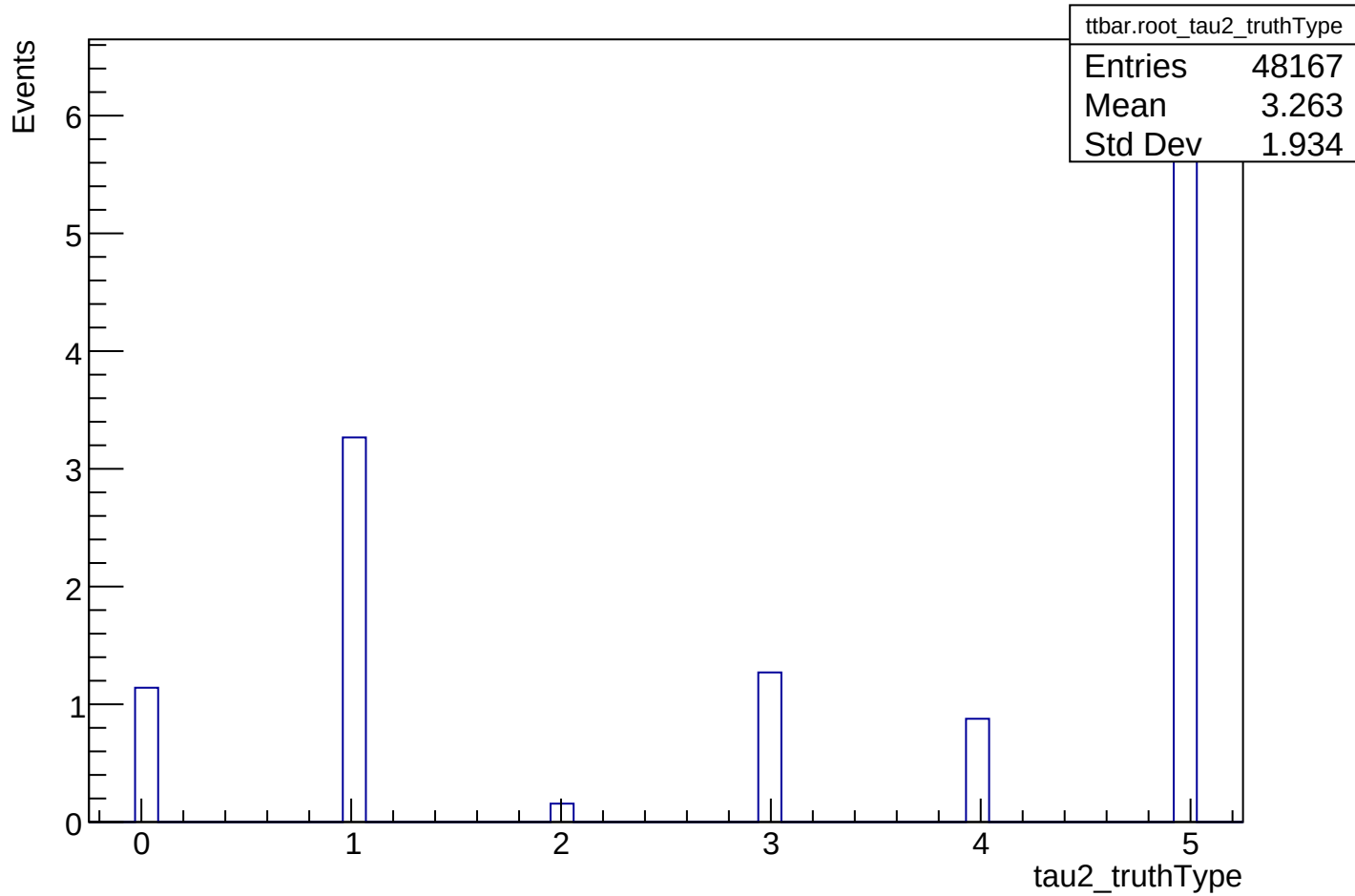
Events



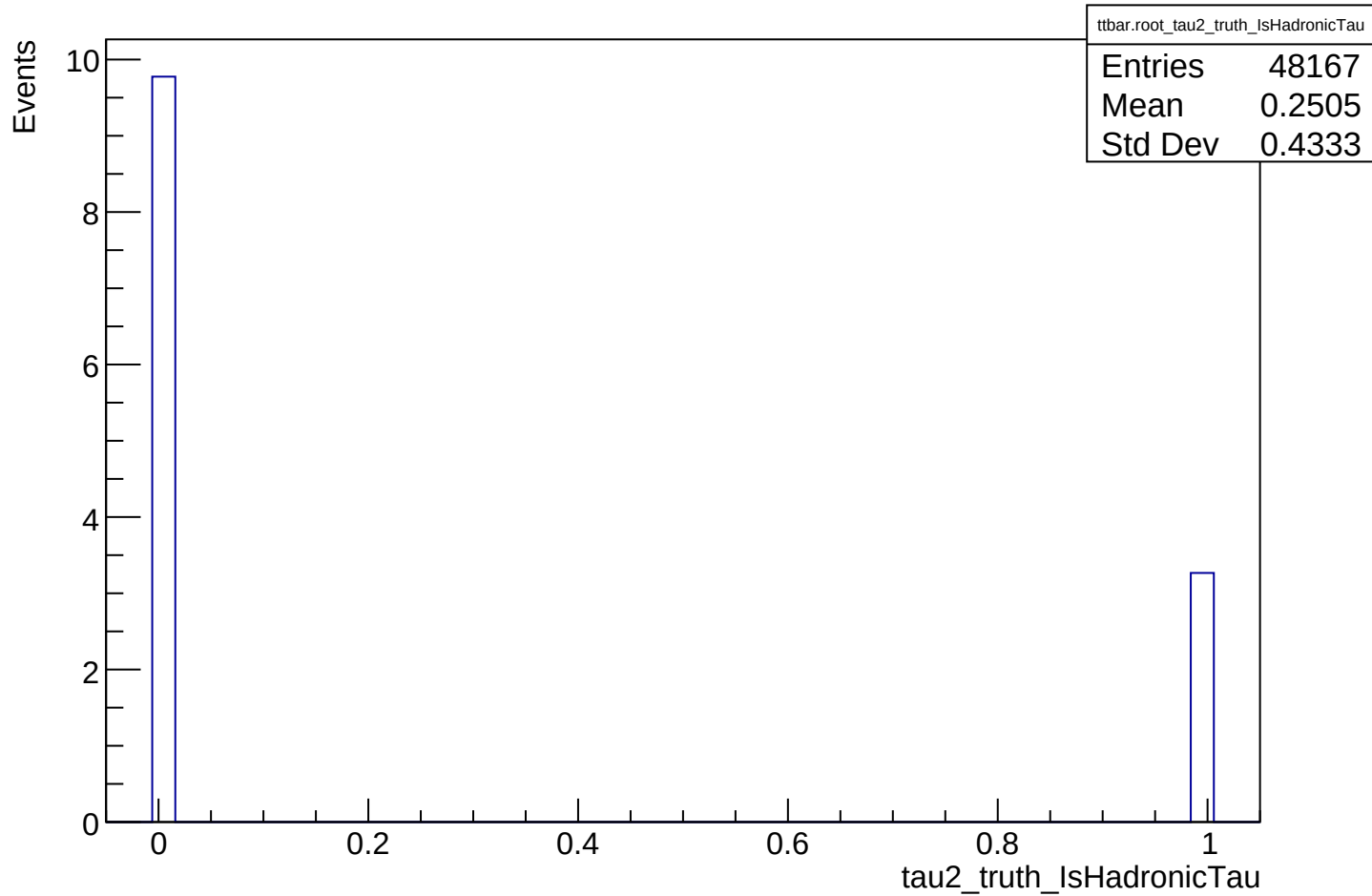
tau2_tauTruthJetLabel



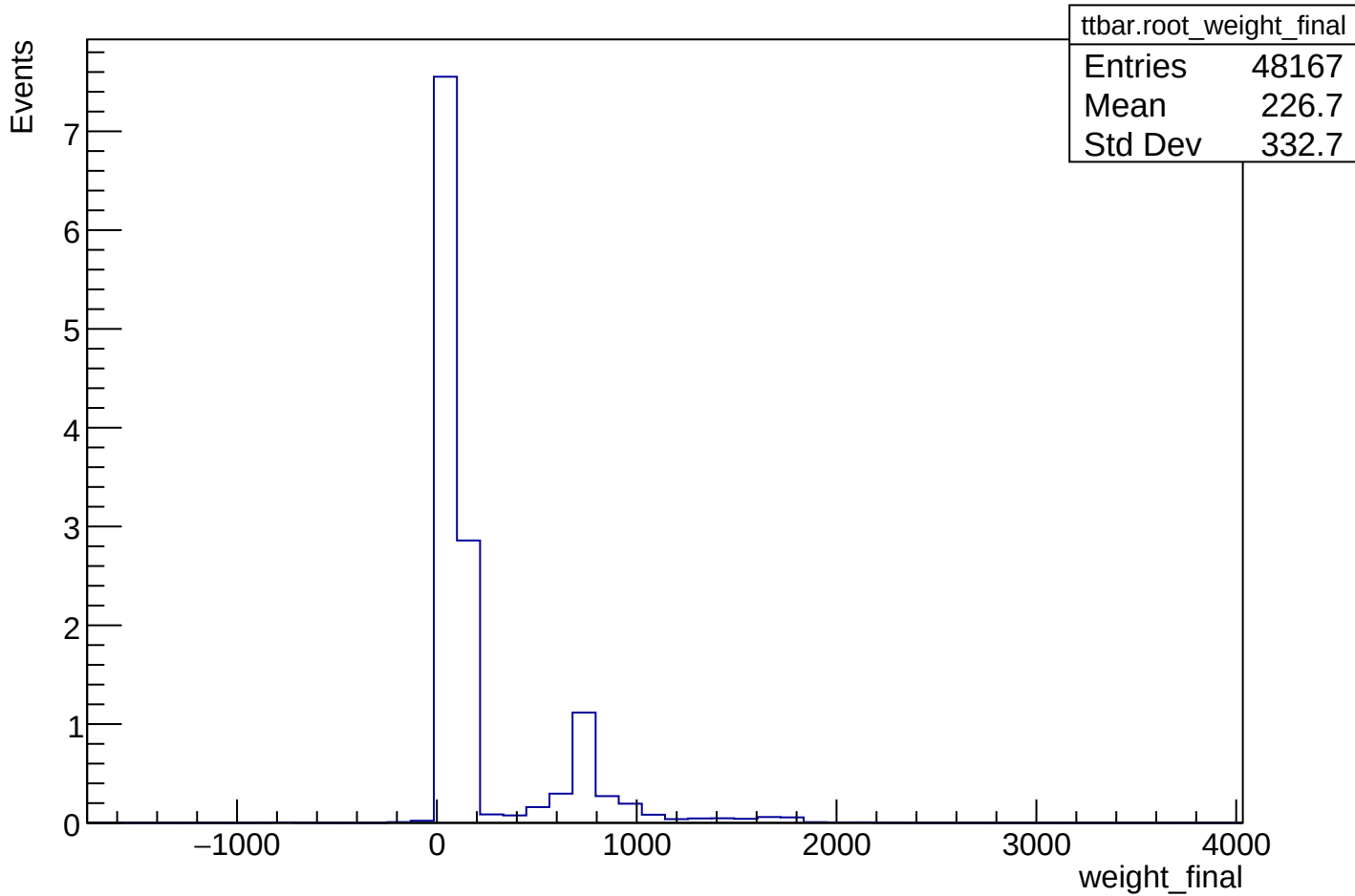
tau2_truthType



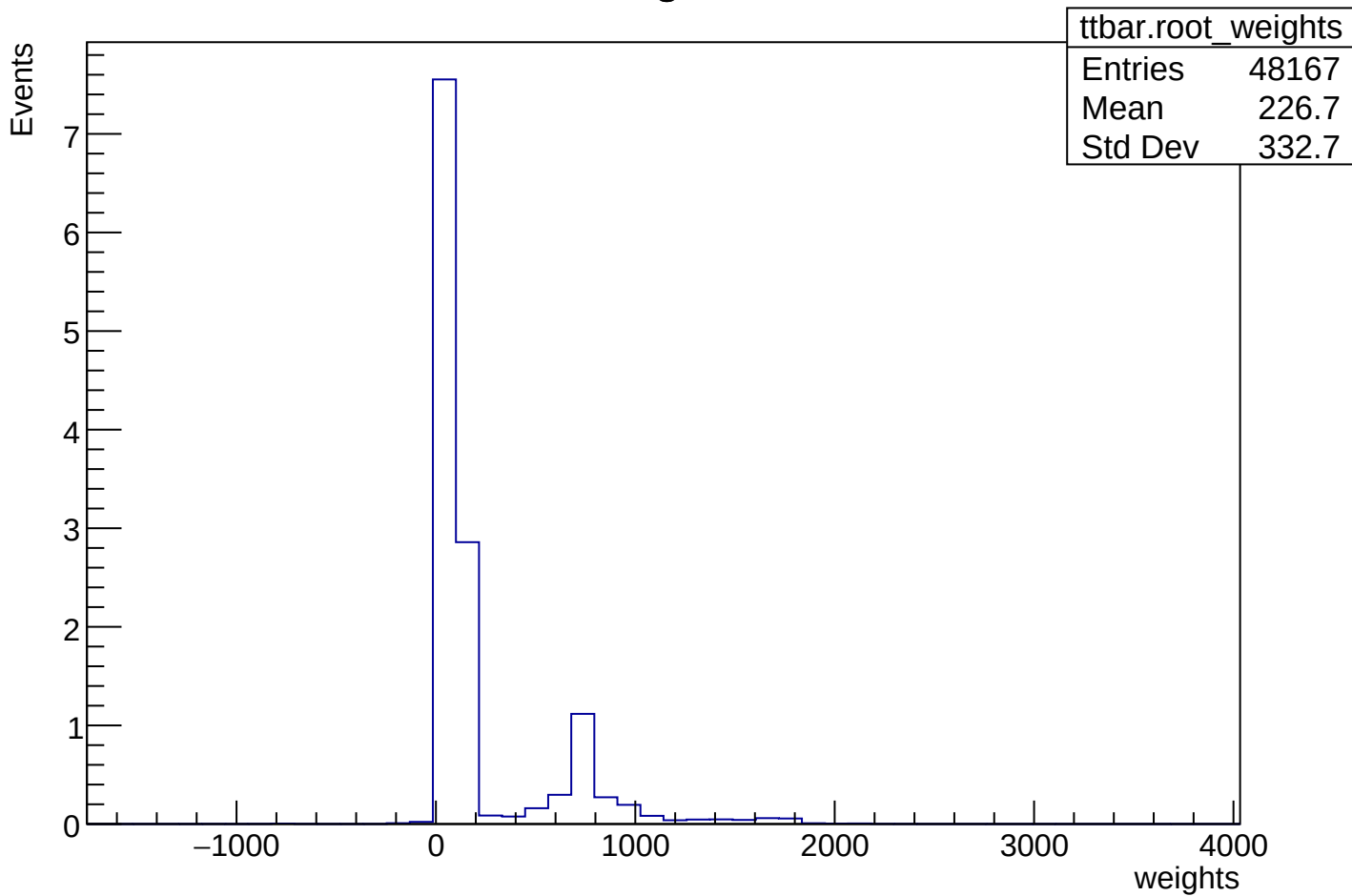
tau2_truth_IsHadronicTau



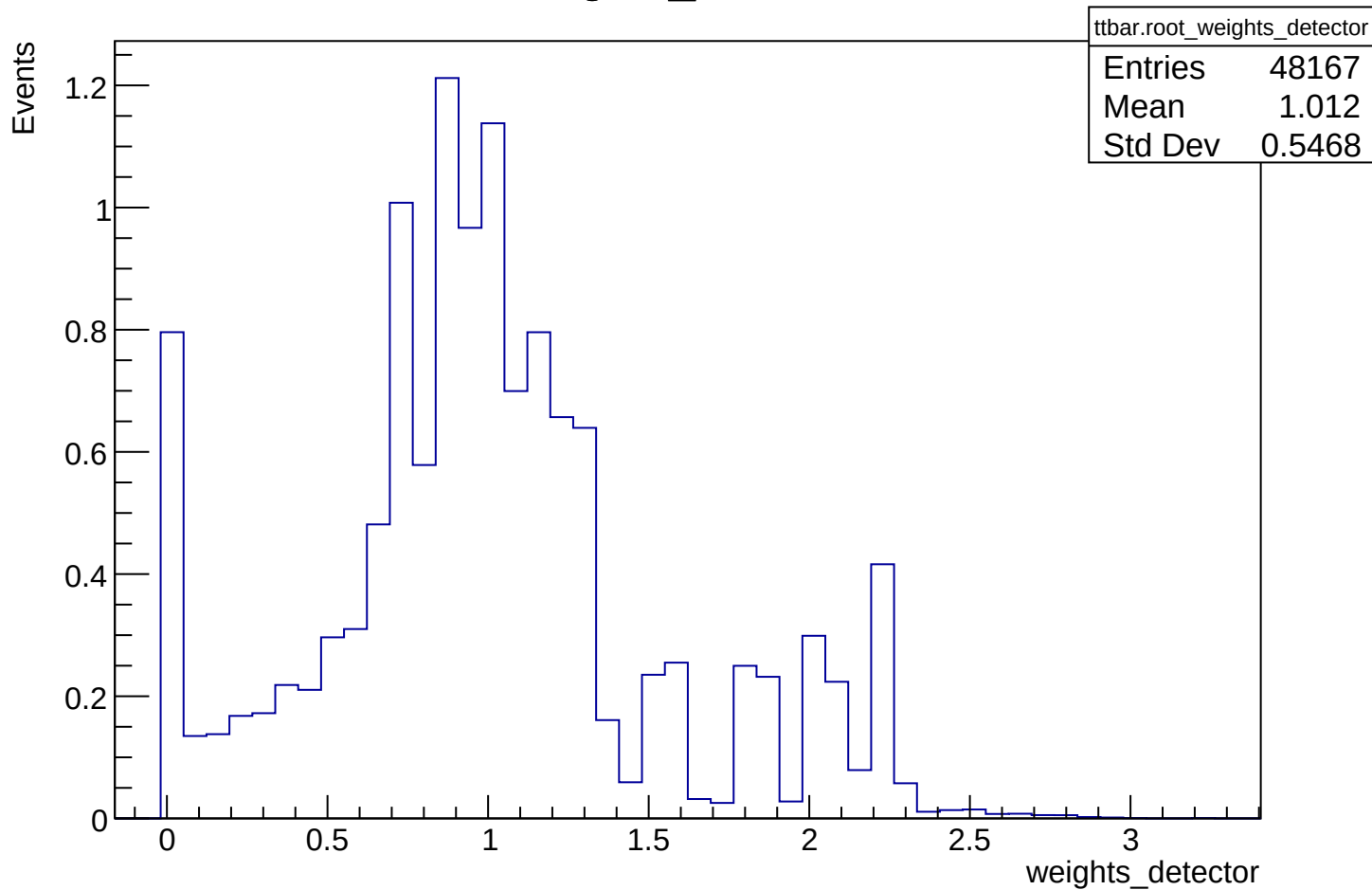
weight_final



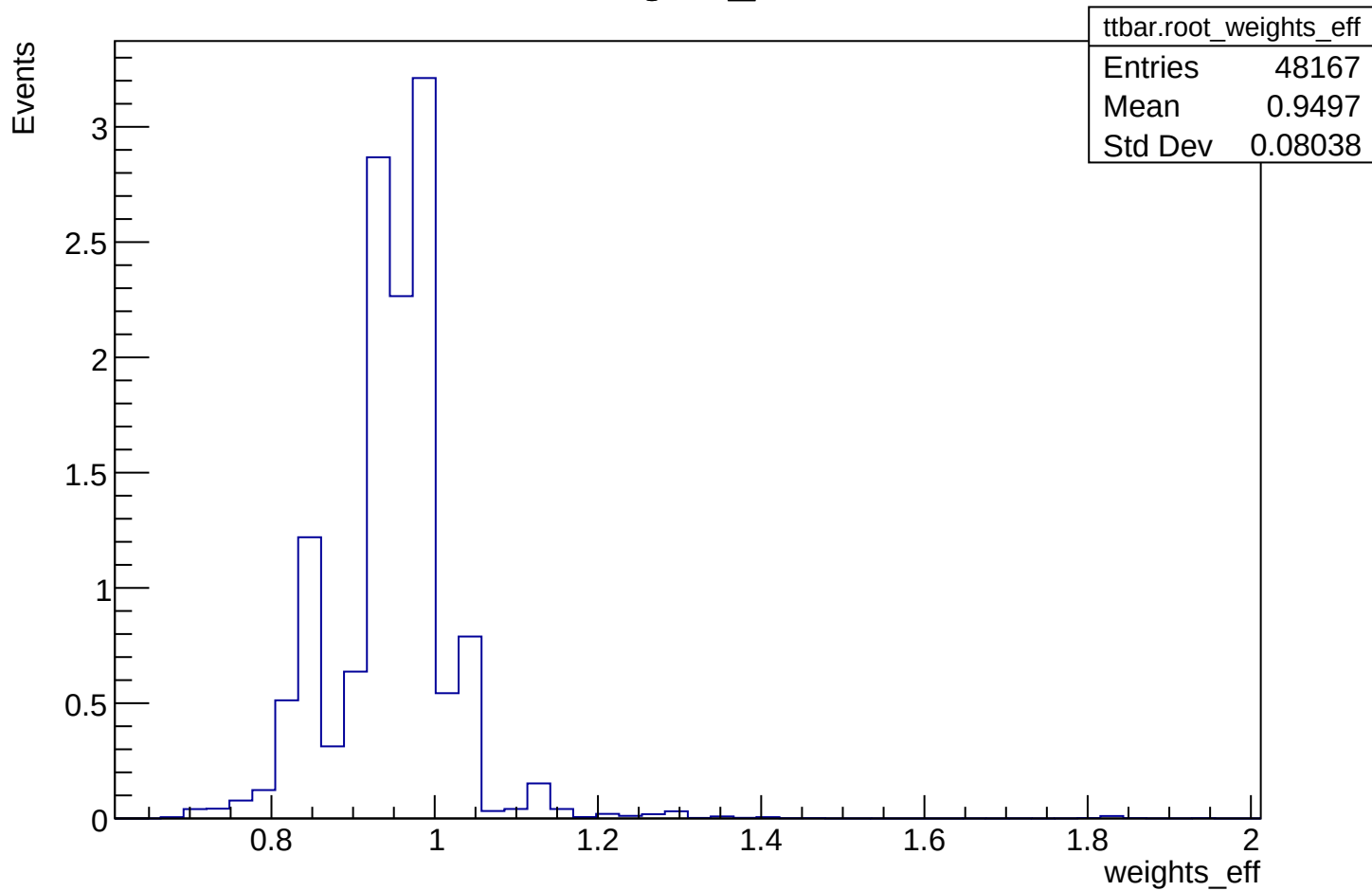
weights



weights_detector



weights_eff



weights_mc

Events

10

8

6

4

2

0

-1000

0

1000

2000

3000

weights_mc

tbar.root_weights_mc	
Entries	48167
Mean	229.5
Std Dev	283.8

